

# Nuclear Structure and Decay Data for A=35 Isobars\*

LIJIE SUN AND JUN CHEN

*Facility for Rare Isotope Beams, Michigan State University, East Lansing, MI 48824, USA*

**Abstract:** This work presents a comprehensive and critical evaluation of experimental nuclear spectroscopic data from reactions and decays for all 11 known nuclides with mass number 35 (Ne, Na, Mg, Al, Si, P, S, Cl, Ar, K, Ca). Recommended values are produced for level energies, spins and parities, half-lives, and radiation properties including energies, branching ratios, and multipolarities of  $\gamma$  rays, as well as characteristics of  $\beta$  radiation decays, based on a rigorous assessment of all available experimental data. Discrepancies among existing results are carefully addressed. This work supersedes earlier full evaluations of A=35 published by [2011Ch48](#), [1990En08](#) (also [1998En04](#) update) and [1978En02](#).

$^{35}\text{Cl}$  remains the most extensively studied nuclide, investigated through 28 different reactions and decays, followed by  $^{35}\text{S}$ ,  $^{35}\text{P}$ , and  $^{35}\text{Ar}$ . For  $^{35}\text{Na}$ , only two excited states have been observed, and its  $\beta^-$ -decay scheme has not yet been established. For  $^{35}\text{Mg}$ , no  $\gamma$  rays have been placed with certainty, and no decay dataset exists on  $^{35}\text{Mg} \rightarrow ^{35}\text{Al}$ .  $^{35}\text{Ar}$  resonances of astrophysical interest have been studied via transfer reactions and  $^{35}\text{K}$   $\varepsilon + \beta^+$  decays, but their properties are yet to be determined. For  $^{35}\text{K}$ , many excited states discovered via  $^{35}\text{Ca}$   $\varepsilon + \beta^+$  decay are unresolved. For  $^{35}\text{Ca}$ , no  $\gamma$  rays have been observed and only one excited state has been identified. Notably,  $^{35}\text{Ca}$  is the only nuclide in this mass chain that decays by  $\varepsilon + \beta^+$ -delayed two protons, although the emission mechanism remains unclear.

**Cutoff Date:** Literature available up to September 30, 2025 has been consulted, with the help of the primary bibliographic source, the NSR database ([2014Pr09](#)) available at Brookhaven National Laboratory web page: [www.nndc.bnl.gov/nsr/](http://www.nndc.bnl.gov/nsr/).

**General Policies and Organization of Material:** See the April 2026 issue of the *Nuclear Data Sheets* or <https://www.nndc.bnl.gov/nds/docs/NDSPolicies.pdf>.

**General Comments:** The statistical analysis of  $\gamma$ -ray data and deduced level schemes is carried out through a suite of current versions of computer codes available at the IAEA-NDS web page of ENSDF codes ([https://www-nds.iaea.org/public/ensdf\\_pgm](https://www-nds.iaea.org/public/ensdf_pgm)) such as GLSC, Java-RULER, BrIccMixing, and ConsistencyCheck. The direct feedings to excited states in  $\beta^-$  and  $\varepsilon$  decays have generally been computed from  $I(\gamma + ce)$  intensity balances at each level; the associated  $\log ft$  values are calculated using the BetaShape code. Theoretical total conversion coefficients are calculated using version v2.3e of BrIcc code ([2008Ki07](#)) for frozen-orbit option with an implicit uncertainty of 1.4% when not stated. Q values and particle separation energies have been adopted from [2021Wa16](#) (AME2020). Static magnetic dipole moments are taken from [2019StZV](#) (long-lived states) and [2020StZV](#) (short-lived states) evaluations, and static electric quadrupole moments from [2021StZZ](#) evaluation, when available in these publications. Nuclear rms charge radii have been adopted from [2013An02](#) evaluation. In cases where weighted averaging procedures have been used, the assigned uncertainty in the result is generally not lower than the lowest uncertainty in the data points used in the procedure.

**Acknowledgements:** Earlier evaluations ([2011Ch48](#), [1998En04](#) update, [1990En08](#), [1978En02](#), [1973EnVA](#), [1967En05](#), [1962En05](#), [1957En01](#), [1954En19](#)), and compilations available in the XUNDL database provided valuable resources for the present work. The evaluators are grateful to the reviewers at different stages of this work, and to the personnel at the National Nuclear Data Center (NNDC) at Brookhaven National Laboratory for facilitating this work.

Contact: [sunli@frib.msu.edu](mailto:sunli@frib.msu.edu), [chenj@frib.msu.edu](mailto:chenj@frib.msu.edu)

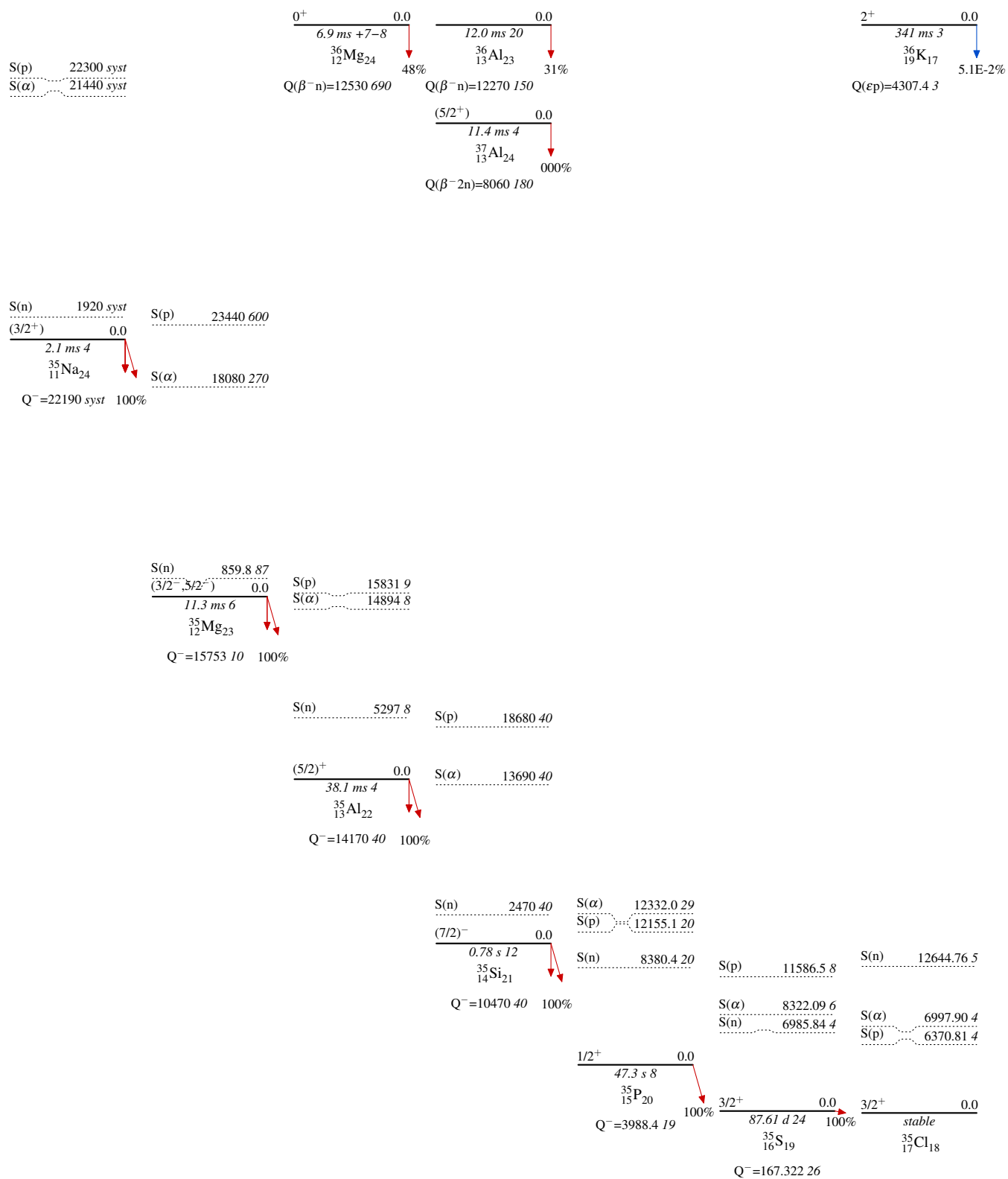
\* Work supported by the Office of Nuclear Physics, Office of Science, U.S. Department of Energy, through DE-SC0016948 grant at Michigan State University

## Index for A=35

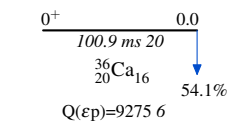
| Nuclide                    | Data Type                                                                                | Page | Nuclide                                                                             | Data Type                                             | Page |
|----------------------------|------------------------------------------------------------------------------------------|------|-------------------------------------------------------------------------------------|-------------------------------------------------------|------|
|                            | Skeleton Scheme for A=35                                                                 | 4    |                                                                                     | $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{S}\gamma)$ | 140  |
| $^{35}_{10}\text{Ne}_{25}$ | Adopted Levels: not observed                                                             | 6    | $^{35}_{17}\text{Cl}_{18}$                                                          | Adopted Levels, Gammas                                | 143  |
| $^{35}_{11}\text{Na}_{24}$ | Adopted Levels, Gammas                                                                   | 7    | $^{35}\text{S}$                                                                     | $\beta^-$ decay (87.61 d)                             | 230  |
|                            | $^9\text{Be}(^{48}\text{Ca}, ^{35}\text{Na})$                                            | 10   | $^{35}\text{Ar}$                                                                    | $\varepsilon$ decay (1.7755 s)                        | 231  |
|                            | $\text{C}(^{36}\text{Mg}, ^{35}\text{Na}\gamma)$                                         | 11   | $^{36}\text{K}$                                                                     | $\varepsilon\text{p}$ decay (341 ms)                  | 236  |
| $^{35}_{12}\text{Mg}_{23}$ | Adopted Levels, Gammas                                                                   | 13   | $^1\text{H}(^{34}\text{S}, ^{35}\text{Cl}\gamma)$                                   |                                                       | 239  |
|                            | $^{35}\text{Na}$ $\beta^-$ decay (2.1 ms)                                                | 16   | $^2\text{H}(^{34}\text{S}, \text{n}\gamma)$                                         |                                                       | 240  |
|                            | $^9\text{Be}(^{38}\text{Si}, ^{35}\text{Mg}\gamma)$                                      | 17   | $^{12}\text{C}(^{28}\text{Si}, \alpha\text{p}\gamma)$                               |                                                       | 241  |
|                            | $\text{C}(^{36}\text{Mg}, ^{35}\text{Mg}\gamma), (^{37}\text{Al}, ^{35}\text{Mg}\gamma)$ | 19   | $^{16}\text{O}(^{24}\text{Mg}, \alpha\text{p}\gamma)$                               |                                                       | 247  |
| $^{35}_{13}\text{Al}_{22}$ | Adopted Levels, Gammas                                                                   | 21   | $^{24}\text{Mg}(^{16}\text{O}, \alpha\text{p}\gamma)$                               |                                                       | 249  |
|                            | $^{36}\text{Mg}$ $\beta^-$ n decay (6.9 ms)                                              | 25   | $^{27}\text{Al}(^{14}\text{N}, \alpha\text{p}\text{n}\gamma)$                       |                                                       | 255  |
|                            | $\text{He}(^{35}\text{Al}, ^{35}\text{Al}'\gamma)$                                       | 27   | $^{31}\text{P}(\alpha, \text{p}), (\alpha, \text{n}): \text{resonances}$            |                                                       | 258  |
|                            | $^9\text{Be}(^{36}\text{Si}, ^{35}\text{Al}\gamma)$                                      | 28   | $^{31}\text{P}(^6\text{Li}, \text{d})$                                              |                                                       | 262  |
|                            | $\text{Pb}(^{35}\text{Al}, ^{34}\text{Al}\text{n}\gamma)$                                | 30   | $^{32}\text{S}(\alpha, \text{p})$                                                   |                                                       | 263  |
|                            | Coulomb excitation                                                                       | 31   | $^{32}\text{S}(\alpha, \text{p}\gamma)$                                             |                                                       | 266  |
| $^{35}_{14}\text{Si}_{21}$ | Adopted Levels, Gammas                                                                   | 32   | $^{33}\text{S}(\alpha, \text{d})$                                                   |                                                       | 271  |
|                            | $^{35}\text{Al}$ $\beta^-$ decay (38.1 ms)                                               | 36   | $^{34}\text{S}(\text{p}, \gamma)$                                                   |                                                       | 272  |
|                            | $^{36}\text{Al}$ $\beta^-$ n decay (12.0 ms)                                             | 40   | $^{34}\text{S}(\text{p}, \text{p}), (\text{p}, \text{p}'\gamma): \text{resonances}$ |                                                       | 339  |
|                            | $^{37}\text{Al}$ $\beta^-$ 2n decay (11.4 ms)                                            | 42   | $^{34}\text{S}(\text{d}, \text{n})$                                                 |                                                       | 342  |
|                            | $^2\text{H}(^{34}\text{Si}, ^{35}\text{Si}\gamma)$                                       | 44   | $^{34}\text{S}(^3\text{He}, \text{d})$                                              |                                                       | 343  |
|                            | $^9\text{Be}(^{36}\text{Si}, ^{35}\text{Si}\gamma)$                                      | 46   | $^{35}\text{Cl}(\gamma, \gamma')$                                                   |                                                       | 346  |
|                            | $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{Si}\gamma)$                                   | 49   | $^{35}\text{Cl}(\text{n}, \text{n}'\gamma)$                                         |                                                       | 348  |
| $^{35}_{15}\text{P}_{20}$  | Adopted Levels, Gammas                                                                   | 51   | $^{35}\text{Cl}(\text{p}, \text{p}'\gamma)$                                         |                                                       | 350  |
|                            | $^{35}\text{Si}$ $\beta^-$ decay (0.78 s)                                                | 57   | $^{37}\text{Cl}(\text{p}, \text{t})$                                                |                                                       | 355  |
|                            | $^1\text{H}(^{34}\text{Si}, \text{p}): \text{resonances}$                                | 61   | $^{36}\text{Ar}(\text{n}, \text{d})$                                                |                                                       | 356  |
|                            | $^2\text{H}(^{34}\text{Si}, ^{35}\text{P}\gamma)$                                        | 62   | $^{36}\text{Ar}(\text{pol d}, ^3\text{He})$                                         |                                                       | 357  |
|                            | $^2\text{H}(^{36}\text{S}, ^3\text{He})$                                                 | 64   | $^{36}\text{Ar}(\text{d}, ^3\text{He})$                                             |                                                       | 359  |
|                            | $^9\text{Be}(^{36}\text{S}, ^{35}\text{P}\gamma)$                                        | 65   | $^{39}\text{K}(\gamma, \alpha)$                                                     |                                                       | 360  |
|                            | $^{34}\text{S}(^{18}\text{O}, ^{17}\text{F})$                                            | 68   | $^{40}\text{Ca}(\mu^-, \text{v}\alpha\text{n}\gamma)$                               |                                                       | 361  |
|                            | $^{36}\text{S}(\text{d}, ^3\text{He})$                                                   | 69   | Coulomb excitation                                                                  |                                                       | 363  |
|                            | $^{36}\text{S}(\text{pol d}, ^3\text{He})$                                               | 70   | $^{35}_{18}\text{Ar}_{17}$                                                          | Adopted Levels, Gammas                                | 365  |
|                            | $^{37}\text{Cl}(^{11}\text{B}, ^{13}\text{N})$                                           | 71   | $^{35}\text{K}$                                                                     | $\varepsilon + \beta^+$ decay (175 ms)                | 372  |
|                            | $^{160}\text{Gd}(^{37}\text{Cl}, \text{X}\gamma)$                                        | 72   | $^{36}\text{Ca}$                                                                    | $\varepsilon\text{p}$ decay (100.9 ms)                | 376  |
|                            | $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$                                    | 73   | $^1\text{H}(^{36}\text{Ar}, \text{d})$                                              |                                                       | 379  |
| $^{35}_{16}\text{S}_{19}$  | Adopted Levels, Gammas                                                                   | 77   | $^{16}\text{O}(^{24}\text{Mg}, \alpha\text{n}\gamma)$                               |                                                       | 380  |
|                            | $^{35}\text{P}$ $\beta^-$ decay (47.3 s)                                                 | 98   | $^{24}\text{Mg}(^{16}\text{O}, \alpha\text{n}\gamma)$                               |                                                       | 382  |
|                            | $^2\text{H}(^{34}\text{S}, \text{p}\gamma)$                                              | 100  | $^{32}\text{S}(\alpha, \text{n})$                                                   |                                                       | 386  |
|                            | $^9\text{Be}(^{36}\text{S}, ^{35}\text{S}\gamma)$                                        | 101  | $^{33}\text{S}(^3\text{He}, \text{n}\gamma)$                                        |                                                       | 387  |
|                            | $^{24}\text{Mg}(^{14}\text{N}, 3\text{p}\gamma)$                                         | 104  | $^{35}\text{Cl}(^3\text{He}, \text{t})$                                             |                                                       | 389  |
|                            | $^{26}\text{Mg}(^{18}\text{O}, 2\alpha\text{n}\gamma)$                                   | 108  | $^{36}\text{Ar}(\text{p}, \text{d})$                                                |                                                       | 390  |
|                            | $^{34}\text{S}(\text{n}, \gamma)$ E=thermal                                              | 112  | $^{36}\text{Ar}(\text{d}, \text{t})$                                                |                                                       | 392  |
|                            | $^{34}\text{S}(\text{n}, \text{n}), (\text{n}, \gamma): \text{resonances}$               | 118  | $^{36}\text{Ar}(^3\text{He}, \alpha)$                                               |                                                       | 393  |
|                            | $^{34}\text{S}(\text{pol d}, \text{p})$                                                  | 120  | $^{35}_{19}\text{K}_{16}$                                                           | Adopted Levels                                        | 395  |
|                            | $^{34}\text{S}(\text{d}, \text{p})$                                                      | 121  | $^{35}\text{Ca}$                                                                    | $\varepsilon + \beta^+$ decay (25.7 ms)               | 397  |
|                            | $^{34}\text{S}(\text{d}, \text{p}\gamma)$                                                | 125  | $^9\text{Be}(^{36}\text{Ca}, ^{35}\text{K})$                                        |                                                       | 399  |
|                            | $^{36}\text{S}(\text{p}, \text{d})$                                                      | 131  | $^{40}\text{Ca}(^3\text{He}, ^8\text{Li})$                                          |                                                       | 400  |
|                            | $^{37}\text{Cl}(\text{p}, ^3\text{He})$                                                  | 134  | $^{35}_{20}\text{Ca}_{15}$                                                          | Adopted Levels                                        | 401  |
|                            | $^{37}\text{Cl}(\text{d}, \alpha\gamma)$                                                 | 136  | $^1\text{H}(^{37}\text{Ca}, \text{t})$                                              |                                                       | 402  |
|                            | $^{160}\text{Gd}(^{34}\text{S}, \text{X}\gamma), (^{37}\text{Cl}, \text{X}\gamma)$       | 139  | $^9\text{Be}(^{36}\text{Ca}, ^{35}\text{Ca})$                                       |                                                       | 403  |
|                            |                                                                                          |      | $^9\text{Be}(^{37}\text{Ca}, \text{X})$                                             |                                                       | 404  |
|                            |                                                                                          |      | References                                                                          |                                                       | 405  |



Skeleton Scheme for A=35

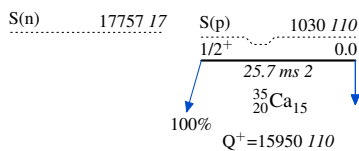


## Skeleton Scheme for A=35 (continued)

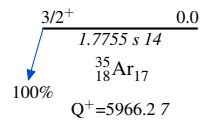
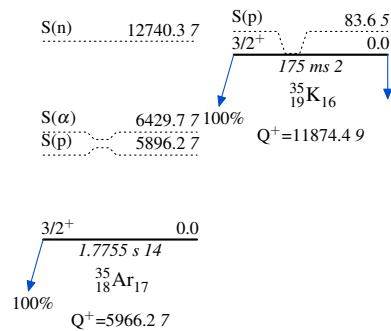


S(n) 17770 syst

S( $\alpha$ ) 8940 110



S( $\alpha$ ) 6563.3



Ground-State and Isomeric-Level Properties

| Nuclide          | Level | $J\pi$           | $T_{1/2}$   | Decay Mode                                                                                          |
|------------------|-------|------------------|-------------|-----------------------------------------------------------------------------------------------------|
| $^{35}\text{Na}$ | 0.0   | $(3/2^+)$        | 2.1 ms 4    | $\% \beta^- = 100$ ; $\% \beta^- n > 0$ ; $\% \beta^- 2n = ?$ ...                                   |
| $^{35}\text{Mg}$ | 0.0   | $(3/2^-, 5/2^-)$ | 11.3 ms 6   | $\% \beta^- = 100$ ; $\% \beta^- n = 52.46$ ; $\% \beta^- 2n = ?$                                   |
| $^{35}\text{Al}$ | 0.0   | $(5/2^+)$        | 38.1 ms 4   | $\% \beta^- = 100$ ; $\% \beta^- n = 37.4$ ; $\% \beta^- 2n = ?$                                    |
| $^{35}\text{Si}$ | 0.0   | $(7/2^-)$        | 0.78 s 12   | $\% \beta^- = 100$ ; $\% \beta^- n < 5$                                                             |
| $^{35}\text{P}$  | 0.0   | $1/2^+$          | 47.3 s 8    | $\% \beta^- = 100$                                                                                  |
| $^{35}\text{S}$  | 0.0   | $3/2^+$          | 87.61 d 24  | $\% \beta^- = 100$                                                                                  |
| $^{35}\text{Cl}$ | 0.0   | $3/2^+$          | stable      |                                                                                                     |
| $^{35}\text{Ar}$ | 0.0   | $3/2^+$          | 1.7755 s 14 | $\% \epsilon + \% \beta^+ = 100$                                                                    |
| $^{35}\text{K}$  | 0.0   | $3/2^+$          | 175 ms 2    | $\% \epsilon + \% \beta^+ = 100$ ; $\% \epsilon\text{p} = 0.37.15$                                  |
| $^{35}\text{Ca}$ | 0.0   | $1/2^+$          | 25.7 ms 2   | $\% \epsilon + \% \beta^+ = 100$ ; $\% \epsilon\text{p} = 95.8.3$ ; $\% \epsilon 2\text{p} = 4.2.3$ |
| $^{36}\text{Mg}$ | 0.0   | $0^+$            | 6.9 ms +7-8 | $\% \beta^- n = 48.12$                                                                              |
| $^{36}\text{Al}$ | 0.0   |                  | 12.0 ms 20  | $\% \beta^- n < 31$                                                                                 |
| $^{37}\text{Al}$ | 0.0   | $(5/2^+)$        | 11.4 ms 4   | $\% \beta^- 2n \geq 000$                                                                            |
| $^{36}\text{K}$  | 0.0   | $2^+$            | 341 ms 3    | $\% \epsilon\text{p} = 5.1 \times 10^{-2}$                                                          |
| $^{36}\text{Ca}$ | 0.0   | $0^+$            | 100.9 ms 20 | $\% \epsilon\text{p} = 54.1.12$                                                                     |

Adopted Levels: not observed

$Q(\beta^-)=24430$  *calc*;  $S(n)=-2370$  *calc*;  $S(p)=29480$  *calc*    [2019Mo01](#)

$S(2n)=-470$  ([2019Mo01](#), FRDM).

[2019Ah07](#):  $^9\text{Be}(^{48}\text{Ca}, X)$  using a 345-MeV/nucleon, 450-pnA  $^{48}\text{Ca}$  beam provided by the cascade operation of the RIBF accelerator complex at RIKEN and impinging on a 20-mm-thick beryllium target. Projectile fragments were separated and identified using  $\Delta E$ -tof- $B\rho$  by the large-acceptance two-stage separator BigRIPS. Tof was measured using two thin plastic scintillators placed at the intermediate and final foci of the second stage of BigRIPS.  $B\rho$  was measured from position measurement at the intermediate focus using the plastic scintillator.  $\Delta E$  was measured using a stack of six silicon detectors installed at the final focus. Optimum settings of  $B\rho$  were tuned to transmit  $^{33}\text{F}$  for 14 hours and  $^{36}\text{Ne}+^{39}\text{Na}$  for 7.8 hours. The Be target was irradiated with  $1.4\times 10^{17}$  and  $7.8\times 10^{16}$   $^{48}\text{Ca}$  ions, respectively. Measured Z vs A/Z particle-identification plot. No  $^{35}\text{Ne}$  events were observed in either setting. Under  $^{33}\text{F}$  setting, the expected  $^{35}\text{Ne}$  yields obtained from LISE++ are 177 53 using the production  $\sigma=37.8$  fb from EPAX 2.15 systematics and 69 17 using the production  $\sigma=14.8$  fb 36 from  $Q_g$  systematics.

[2022Ah02](#): Same experimental setup as [2019Ah07](#) with 540-pnA  $^{48}\text{Ca}$  beam. Optimum settings of  $B\rho$  were tuned to transmit  $^{39}\text{Na}$  for 46.1 hours and  $^{36}\text{Ne}$  for 25.3 hours. Measured Z vs A/Z particle-identification plot. No  $^{35}\text{Ne}$  events were observed in either setting. Also see [2025Ku15](#).

[2023Ra22](#): Relativistic Hartree-Bogoliubov model with nuclear energy density functional (EDF) calculations of binding energies.

[2020Mi15](#): VS-IMSRG ab initio calculations of ground-state energies and  $S(2n)$ .

 $^{35}\text{Ne}$  Levels

| E(level) | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0?       | <p><math>\%0n=?</math>; <math>\%2n=?</math></p> <p>Evaluators estimate the probability of not observing <math>^{35}\text{Ne}</math> events by chance is <math>2.6\times 10^{-23}</math> using the lowest expected yield of 52 events (<a href="#">2019Ah07</a>) and Poisson probability distributions. <math>^{35}\text{Ne}</math> is determined to be unbound at a confidence level of <math>1-2.6\times 10^{-23}</math>. The heaviest bound neon isotope is <math>^{34}\text{Ne}</math>.</p> <p><math>J^\pi</math>: <math>5/2^-</math> calculated projection of the odd-neutron angular momentum along the symmetry axis and parity of the wave function (<a href="#">2019Mo01</a>).</p> <p><math>T_{1/2}</math>: 2.7 ms calculated with respect to Gamow-Teller QRPA transitions and phenomenological first-forbidden contributions (<a href="#">2019Mo01</a>).</p> |

Adopted Levels, Gammas

$Q(\beta^-)=22190$  syst;  $S(n)=1920$  syst;  $S(p)=22300$  syst;  $Q(\alpha)=-21440$  syst [2021Wa16](#)

$\Delta Q(\beta^-)=720$ ,  $\Delta S(n)=300$ ,  $\Delta S(p)=840$ ,  $\Delta Q(\alpha)=860$  (syst,[2021Wa16](#)).

$S(2n)=2090$  810,  $Q(\beta^-n)=21440$  670 (syst,[2021Wa16](#)).

Isotope discovery ([2012Th10](#)): Ir(p,X) $^{35}\text{Na}$  at CERN ([1983La12](#)).

$^{35}\text{Na}$  production:

[2025Ku15](#),[2022Ah02](#),[2019Ah07](#):  $^9\text{Be}(^{48}\text{Ca},X)$  at RIKEN.

[2002LuZT](#): Ta( $^{48}\text{Ca},X$ ) at GANIL.

$^{35}\text{Na}$  decay measurements:

[2022Cr03](#):  $^9\text{Be}(^{48}\text{Ca},X)$  at FRIB. Measured  $T_{1/2}$ .

[2013StZY](#):  $^9\text{Be}(^{48}\text{Ca},X)$  at RIKEN. Measured  $T_{1/2}$  and  $\beta^-$ -delayed  $\gamma$  rays.

[1983La12](#),[1984La03](#): Ir(p,X) at CERN. Measured  $T_{1/2}$  and  $\beta^-$ -delayed neutrons.

$^{35}\text{Na}$  mass measurements: None.

Theoretical calculations (binding energies, deformation, quadrupole moments, radii, levels, J,  $\pi$ , mass,  $T_{1/2}$ , etc.): [2024Lu06](#),  
[2023Zh15](#), [2023Ba27](#), [2022Ot01](#), [2020Ch21](#), [2020Ts03](#), [2013Li39](#), [2013Sh05](#), [2009Ly01](#), [2004Ge02](#), [2004Lu10](#), [2002Sa08](#),  
[1997Mo25](#), [1991Pa19](#), [1991Pa21](#), [1989Ly01](#), [1987SaZQ](#), [1985Ly02](#), [1975Ca27](#).

 $^{35}\text{Na}$  LevelsCross Reference (XREF) Flags

**A**  $^9\text{Be}(^{48}\text{Ca},^{35}\text{Na})$   
**B**  $\text{C}(^{36}\text{Mg},^{35}\text{Na}\gamma)$

| E(level)            | $J^\pi^\dagger$ | $T_{1/2}$ | XREF      | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------|-----------------|-----------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0.0 $^\ddagger$     | (3/2 $^+$ )     | 2.1 ms 4  | <b>AB</b> | $\% \beta^- = 100$ ; $\% \beta^- n > 0$ ; $\% \beta^- 2n = ?$ ; $\% \beta^- 3n = ?$ ; $\% \beta^- 4n = ?$<br>$^{35}\text{Na}$ $\beta^-$ -delayed neutrons have been observed by <a href="#">1983La12</a> . Experimental $\% \beta^- n$ values are unknown.<br>Theoretical $\% \beta^- 0n = 1.4$ , $\% \beta^- 1n = 73.5$ , $\% \beta^- 2n = 20.1$ , $\% \beta^- 3n = 4.8$ ( <a href="#">2021Mi17</a> ).<br>Theoretical $\% \beta^- 0n = 14.0$ , $\% \beta^- 1n = 73.0$ , $\% \beta^- 2n = 10.0$ , $\% \beta^- 3n = 3.0$ ( <a href="#">2019Mo01</a> ).<br>$T_{1/2}$ : weighted average of 2.4 ms 3 (stat) 2 (syst) ( <a href="#">2022Cr03</a> , implant- $\beta$ correlation), 2.4 ms 3 (stat) 6 (syst) ( <a href="#">2013StZY</a> , implant- $\beta$ correlation), and 1.5 ms 5 ( <a href="#">1983La12</a> , <a href="#">1984La03</a> , decay curve of $\beta n$ -coin). |
| 373 $^\ddagger$ 5   | (5/2 $^+$ )     |           | <b>B</b>  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 1014 $^\ddagger$ 17 | (7/2 $^+$ )     |           | <b>B</b>  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

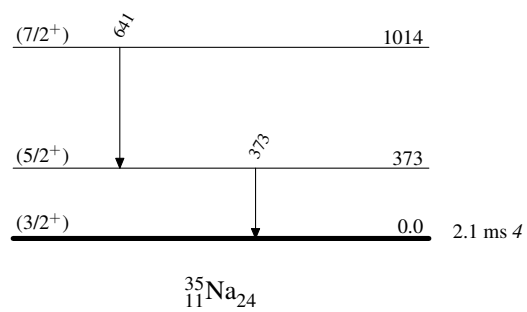
$^\dagger$  From Monte-Carlo shell-model calculations using the SPDF-M effective interaction ([2014Do05](#)).

$^\ddagger$  Band(A):  $K^\pi=(3/2^+)$  rotational band predicted by the shell model ([2014Do05](#)).

 $\gamma(^{35}\text{Na})$ 

| $E_i(\text{level})$ | $J_i^\pi$   | $E_\gamma^\dagger$ | $E_f$ | $J_f^\pi$   |
|---------------------|-------------|--------------------|-------|-------------|
| 373                 | (5/2 $^+$ ) | 373 5              | 0.0   | (3/2 $^+$ ) |
| 1014                | (7/2 $^+$ ) | 641 16             | 373   | (5/2 $^+$ ) |

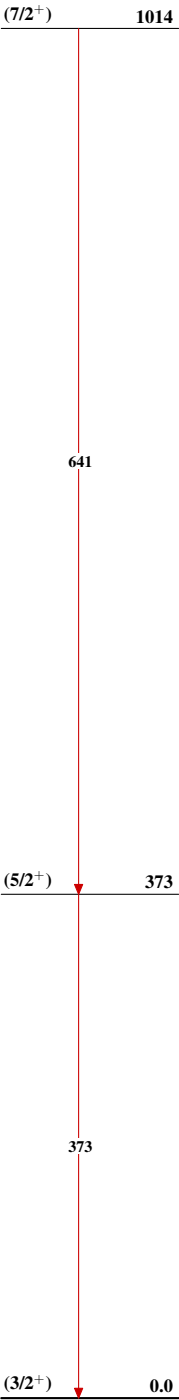
$^\dagger$  From  $\text{C}(^{36}\text{Mg},^{35}\text{Na}\gamma)$ .

**Adopted Levels, Gammas**Level Scheme



Adopted Levels, Gammas

Band(A):  $K^\pi=(3/2^+)$   
rotational band  
predicted by the shell  
model (2014Do05)



<sup>35</sup>Na<sub>24</sub>

$^9\text{Be}(^{48}\text{Ca}, ^{35}\text{Na})$  2013StZY,2022Cr03

**2013StZY:**  $^{35}\text{Na}$  was produced via the projectile fragmentation of a 345-MeV/nucleon, 70-pnA  $^{48}\text{Ca}^{20+}$  primary beam from the linear accelerator RILAC and the four cyclotrons RRC, fRC, IRC, and SRC at RIKEN impinging on a 15-mm-thick  $^9\text{Be}$  target. The secondary cocktail beam was selected by the BigRIPS separator and the zero-degree spectrometer (ZDS) using the  $\text{B}\rho\text{-}\Delta\text{E-ToF}$  method, and implanted into the Cylindrical Active Implantation Target for Exotic Nuclei (CAITEN) consisting of a segmented movable hollow-cylindrical-shaped plastic scintillator and a stationary ring of 24 position-sensitive photomultiplier tubes (PSPMTs) arranged on a ring inside the scintillator at the height of the beam line. To reduce background buildup, the scintillator barrel was rotated quickly and axially-moved slowly in vertical direction, resulting in a helix-shaped motion.  $\beta$  particles were detected by the CAITEN and  $\gamma$  rays were detected using three HPGe clover detectors. Measured  $\text{E}\gamma$ ,  $\beta\gamma$ -coin, and implant- $\beta$  correlation, and deduced the  $T_{1/2}$  of  $^{35}\text{Na}$ . Comparisons with QRPA and shell-model calculations.

**2022Cr03:**  $^{35}\text{Na}$  was produced via the projectile fragmentation of a 172.3-MeV/nucleon, 120-pnA  $^{48}\text{Ca}^{20+}$  primary beam from the FRIB linac impinging on an 8.89-mm-thick  $^9\text{Be}$  target. The secondary cocktail beam centered around  $^{42}\text{Si}$  was selected by the ARIS separator and implanted into a 5-mm-thick YSO segmented scintillator sandwiched between two plastic scintillator veto detectors. Surrounding the implantation array were 11 HPGe clover detectors and 15 fast-timing  $\text{LaBr}_3$  detectors, and the VANDLE array of 88 neutron detectors. Two Si PIN detectors and a plastic scintillator were placed 1.5 m upstream in the last diagnostic detector box of the beamline. Ions were identified event-by-event by energy loss in the upstream PIN detector (Z determination) and by time-of-flight between a plastic timing scintillator at the start of stage 3 of ARIS and the timing scintillator in the diagnostic detector box (A/Q determination) over a flight path of 33.5 m. Measured implant- $\beta$  correlation and deduced  $T_{1/2}$  of  $^{35}\text{Na}$ . Comparisons with QRPA and shell-model calculations.

 $^{35}\text{Na}$  Levels

| <u>E(level)</u> | <u><math>T_{1/2}</math></u> | <u>Comments</u>                                                                                                                                                                                                                             |
|-----------------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0.0             | 2.1 ms 4                    | $T_{1/2}$ : From the Adopted Levels of $^{35}\text{Na}$ . Values from this dataset: 2.4 ms 3 (stat) 2 (syst) ( <b>2022Cr03</b> , implant- $\beta$ correlation); 2.4 ms 3 (stat) 6 (syst) ( <b>2013StZY</b> , implant- $\beta$ correlation). |

$\text{C}(^{36}\text{Mg}, ^{35}\text{Na}\gamma)$  2014Do05

**2014Do05:** A 345-MeV/nucleon  $^{48}\text{Ca}$  primary beam was accelerated by the superconducting ring cyclotron (SRC) at RIKEN. A 236-MeV/nucleon  $^{36}\text{Mg}$  secondary beam was produced by the fragmentation of  $^{48}\text{Ca}$  on a Be target and separated by the BigRIPS separator using the  $\text{B}\rho$ - $\Delta\text{E}$ - $\text{B}\rho$  method. The secondary reaction targets were 2.54 g/cm<sup>2</sup> thick carbon and 2.13 g/cm<sup>2</sup> thick CH<sub>2</sub> polyethylene. The secondary reaction products were identified by the magnetic spectrometer ZeroDegree using the  $\text{B}\rho$ - $\Delta\text{E}$ - $\text{B}\rho$  method.  $\gamma$  rays in coincidence with  $^{35}\text{Na}$  were detected using the DALI2 array of 186 large NaI(Tl) detectors. Measured  $E_\gamma$  with Doppler correction,  $I_\gamma$ , and  $\gamma\gamma$ -coin. Deduced levels,  $J$ ,  $\pi$ , and bands. Compared with shell-model calculations using the SPDF-M effective interaction.

 $^{35}\text{Na}$  Levels

| <u><math>E(\text{level})^\dagger</math></u> | <u><math>J^\pi^\ddagger</math></u> |
|---------------------------------------------|------------------------------------|
| 0 <sup>#</sup>                              | (3/2 <sup>+</sup> )                |
| 373 <sup>#</sup> 5                          | (5/2 <sup>+</sup> )                |
| 1014 <sup>#</sup> 17                        | (7/2 <sup>+</sup> )                |

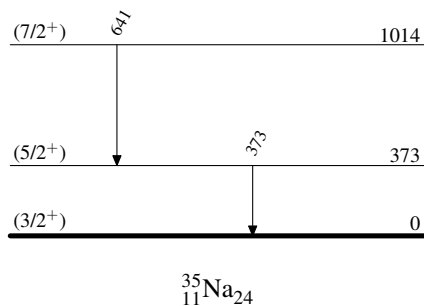
<sup>†</sup> From a least-squares fit to  $\gamma$ -ray energies.

<sup>‡</sup> From shell-model calculations.

<sup>#</sup> Band(A):  $K^\pi=(3/2^+)$  rotational band predicted by the shell model.

 $\gamma(^{35}\text{Na})$ 

| <u><math>E_\gamma</math></u> | <u><math>E_i(\text{level})</math></u> | <u><math>J_i^\pi</math></u> | <u><math>E_f</math></u> | <u><math>J_f^\pi</math></u> |
|------------------------------|---------------------------------------|-----------------------------|-------------------------|-----------------------------|
| 373 5                        | 373                                   | (5/2 <sup>+</sup> )         | 0                       | (3/2 <sup>+</sup> )         |
| 641 16                       | 1014                                  | (7/2 <sup>+</sup> )         | 373                     | (5/2 <sup>+</sup> )         |

 $\text{C}(^{36}\text{Mg}, ^{35}\text{Na}\gamma)$  2014Do05Level Scheme

C( $^{36}\text{Mg}, ^{35}\text{Na}\gamma$ ) 2014Do05

**Band(A):  $K^\pi=(3/2^+)$   
rotational band  
predicted by the shell  
model**

( $7/2^+$ )      **1014**

641

( $5/2^+$ )      **373**

373

( $3/2^+$ )      **0**

$^{35}_{11}\text{Na}_{24}$

Adopted Levels, Gammas

$Q(\beta^-)=15753$  10;  $S(n)=859.8$  87;  $S(p)=2.344\times 10^4$  60;  $Q(\alpha)=-1.808\times 10^4$  27

$Q(\beta^-), S(n), S(p), Q(\alpha)$ : Deduced by the evaluators using mass excesses of 15529.5 71 for  $^{35}\text{Mg}$  measured by 2025Ly01, and 8318 5 for  $^{34}\text{Mg}$ ; a weighted average of 8323 7 (2019As04) and 8315 5 (2025Ly01);  $-224$  7 for  $^{35}\text{Al}$ , 31680 600 for  $^{34}\text{Na}$ , and 31180 270 for  $^{31}\text{Ne}$  from 2021Wa16. Values from 2021Wa16:  $Q(\beta^-)=15860$  270,  $S(n)=750$  270,  $S(p)=23330$  660,  $Q(\alpha)=-17970$  380.

$S(2n)=5576.0$  76,  $Q(\beta^-n)=10455.8$  74, from mass excesses of 15529.5 71 for  $^{35}\text{Mg}$  measured by 2025Ly01; 4962.9 27 for  $^{33}\text{Mg}$  and  $-2997.6$  21 for  $^{34}\text{Al}$  from 2021Wa16. Values from 2021Wa16:  $S(2n)=5470$  270,  $Q(\beta^-n)=10570$  270.

$S(2p)=45070$  660 (syst) (2021Wa16).

Isotope discovery (2012Th10):  $\text{Ta}(^{48}\text{Ca}, X)$  projectile fragmentation at GANIL (1989Gu03, 1991Or01).

$^{35}\text{Mg}$  production:

2012Kw02:  $^{35}\text{Mg}$  produced by  $^{\text{nat}}\text{Ni}(^{40}\text{Ar}, X)$  at  $E(^{40}\text{Ar})=140$  MeV/nucleon at NSCL. Measured fragmentation cross sections, parallel momentum transfers, and widths. Compared with empirical formula EPAX, and predictions from internuclear cascade and deep inelastic models using Monte Carlo ISABEL-GEMINI and DIT-GEMINI codes.

2011FuZZ:  $^{35}\text{Mg}$  produced by  $^9\text{Be}(^{48}\text{Ca}, X)$  fragmentation at  $E(^{48}\text{Ca})=345$  MeV/nucleon at RIKEN. Measured thick target fragmentation, deduced production cross sections, and compared with EPAX-2.15 systematics.

2007Ts09: Analyzed fragmentation  $\sigma$  of n-rich Na, Mg isotopes (including  $^{35}\text{Mg}$ ) from  $^9\text{Be}, ^{181}\text{Ta}(^{48}\text{Ca}, X)$ . Used systematics based on average binding energy to extrapolate towards drip line, predicting  $\sigma$  for  $^{40}\text{Mg}$  and discussing  $^{39}\text{Na}$ . Compared different extrapolation models.

$^{35}\text{Mg}$  decay measurements:

2013StZY:  $^9\text{Be}(^{48}\text{Ca}, X)$  at RIKEN. Measured  $T_{1/2}$ .

1999YoZW:  $^9\text{Be}(^{48}\text{Ca}, X)$  and  $^{181}\text{Ta}(^{48}\text{Ca}, X)$  at RIKEN. Measured  $T_{1/2}$  and  $\% \beta^- n$ .

$^{35}\text{Mg}$  radius measurements:

2011Ka01:  $^{35}\text{Mg}$  produced by  $^9\text{Be}(^{48}\text{Ca}, X)$  fragmentation at GSI. Measured interaction cross sections with C and  $\text{CH}_2$  targets at 900 MeV/nucleon. Deduced rms matter radii.

2006Kh08:  $^{35}\text{Mg}$  produced by  $^{181}\text{Ta}(^{48}\text{Ca}, X)$  fragmentation at  $E(^{48}\text{Ca})=60.3$  MeV/nucleon at GANIL. Measured energy-integrated reaction cross sections at 30-65 MeV/nucleon using a silicon telescope as both active target and detector. Deduced reduced strong absorption radii, isospin dependence, and possible halo structure or large deformation.

$^{35}\text{Mg}$  mass measurements: 2025Ly01, 2007Ju03, 2001Sa72, 2000Sa21, 1991Or01.

Theoretical calculations (binding energies, deformation, quadrupole moments, radii, levels, J,  $\pi$ , mass,  $T_{1/2}$ , etc): 2023Ra22,

2021Ka07, 2020Mi15, 2016Ba59, 2016Sa46, 2016Sh05, 2015Sh21, 2014Ga13, 2014Wa14, 2013Ch31, 2013Li39, 2013Sh05, 2012Fo27, 2012Ho19, 2007Ha53, 2006Zh19, 2005Ch71, 2004Kh16, 1996Re10, 1991Pa19, 1991Pa21.

The level scheme is based on that of 2011Ga15 and 2017Mo26.

 $^{35}\text{Mg}$  LevelsCross Reference (XREF) Flags

- A  $^{35}\text{Na} \beta^-$  decay (2.1 ms)  
 B  $^9\text{Be}(^{38}\text{Si}, ^{35}\text{Mg}\gamma)$   
 C  $\text{C}(^{36}\text{Mg}, ^{35}\text{Mg}\gamma), (^{37}\text{Al}, ^{35}\text{Mg}\gamma)$

| $E(\text{level})^\dagger$ | $J^\pi$          | $T_{1/2}$ | XREF | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------|------------------|-----------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0                         | $(3/2^-, 5/2^-)$ | 11.3 ms 6 | BC   | $\% \beta^- = 100$ ; $\% \beta^- n = 52$ 46; $\% \beta^- 2n = ?$<br>$\% \beta^- n$ : from 1995ReZZ, 2008ReZZ. Other: 52 11 (1999YoZW, preliminary).<br>Theoretical $\% \beta^- 0n = 29$ , $\% \beta^- 1n = 66$ , $\% \beta^- 2n = 5$ (2021Mi17).<br>Theoretical $\% \beta^- 0n = 65$ , $\% \beta^- 1n = 32$ , $\% \beta^- 2n = 3$ (2019Mo01).<br>$J^\pi$ : $3/2^-$ from shell-model calculations with the SDPF-M and SDPF-M+2p $_{1/2}$ interactions (2017Mo26). Nearly degenerate 30-keV $3/2^-$ and $5/2^-$ g.s. from Monte Carlo shell-model calculations with the SDPF-M interaction (2011Ga15), and $3/2^-$ g.s. from shell-model calculations with the SDPF-U interaction (2011Ga15). $3/2^-$ from projection of the odd-neutron angular momentum along the symmetry axis and parity of the wave function (2019Mo01). Others: $3/2^+$ from antisymmetrized molecular dynamics (AMD) calculations with the Gogny DIS force (2017Mo26). |

Continued on next page (footnotes at end of table)

**Adopted Levels, Gammas (continued)** $^{35}\text{Mg}$  Levels (continued)

| <u>E(level)<sup>†</sup></u> | <u>J<sup>π</sup></u>                   | <u>XREF</u> | <u>Comments</u>                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------|----------------------------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             |                                        |             | T <sub>1/2</sub> : 11.3 ms 5 (stat) 4 (syst) (2013StZY, implant-β correlation). Other: 72 ms 43 (2008ReZZ, 1995ReZZ) and ≈9 ms (1999YoZW, implant-β correlation, preliminary).<br>Reduced strong absorption radius r <sub>0</sub> <sup>2</sup> =1.64 fm <sup>2</sup> 15 from the energy-integrated σ of Si( <sup>35</sup> Mg,X) (2006Kh08). The rms matter radius R <sub>rms</sub> <sup>m</sup> =3.40 fm 24 from the interaction σ of C,CH <sub>2</sub> ( <sup>35</sup> Mg,X) (2011Ka01). |
| 0+x                         |                                        | BC          | E(level): x≤80 keV (2011Ga15 detection threshold); x≤200 keV (2017Mo26 detection threshold).<br>J <sup>π</sup> : Shell-model calculations with the SDPF-M+2p <sub>1/2</sub> interaction predict a 1/2 <sup>-</sup> level at 141 keV (2017Mo26). Shell-model calculations with the SDPF-M interaction predict a 5/2 <sup>-</sup> level at 84 keV (2017Mo26). Monte Carlo shell-model calculations with the SDPF-M interaction predict a 3/2 <sup>-</sup> level at 30 keV (2011Ga15).       |
| 0+y?                        |                                        | C           | E(level): y≤200 keV (2017Mo26 detection threshold). 2017Mo26 suggested a low-lying level from the observed 42(1)% neutron knockout from the l=3 orbit in <sup>36</sup> Mg in the inclusive parallel momentum distribution.                                                                                                                                                                                                                                                                |
| 206+x 8                     |                                        | C           | J <sup>π</sup> : 2017Mo26 states that based on the observed weak γ-ray intensity, this level is not the 1/2 <sup>-</sup> level at 141 keV predicted by shell-model calculations with the SDPF-M+2p <sub>1/2</sub> interaction.                                                                                                                                                                                                                                                            |
| 445+x 5                     | (3/2 <sup>+</sup> , 5/2 <sup>+</sup> ) | BC          | J <sup>π</sup> : L( <sup>36</sup> Mg, <sup>35</sup> Mg)=(2) from 0 <sup>+</sup> . 3/2 <sup>+</sup> from shell-model calculations with the SDPF-M+2p <sub>1/2</sub> interaction (2017Mo26).                                                                                                                                                                                                                                                                                                |
| 619+x 7                     | (1/2 <sup>-</sup> , 3/2 <sup>-</sup> ) | BC          | J <sup>π</sup> : L( <sup>36</sup> Mg, <sup>35</sup> Mg)=(1) from 0 <sup>+</sup> . 3/2 <sup>-</sup> from shell-model calculations with the SDPF-M+2p <sub>1/2</sub> interaction (2017Mo26).                                                                                                                                                                                                                                                                                                |
| 670+x 8                     |                                        | BC          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

<sup>†</sup> From E<sub>γ</sub> data in (<sup>38</sup>Si, <sup>35</sup>Mgγ) and (<sup>36</sup>Mg, <sup>35</sup>Mgγ), (<sup>37</sup>Al, <sup>35</sup>Mgγ).

γ(<sup>35</sup>Mg)

| <u>E<sub>i</sub>(level)</u> | <u>J<sub>i</sub><sup>π</sup></u>       | <u>E<sub>γ</sub></u> | <u>I<sub>γ</sub></u> | <u>E<sub>f</sub></u> | <u>Comments</u>                                                                                                                                                                                           |
|-----------------------------|----------------------------------------|----------------------|----------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 206+x                       |                                        | 206 8                | 100                  | 0+x                  | E <sub>γ</sub> : From 2017Mo26.                                                                                                                                                                           |
| 445+x                       | (3/2 <sup>+</sup> , 5/2 <sup>+</sup> ) | 445 5                | 100                  | 0+x                  | E <sub>γ</sub> : weighted average of 443 7 (2017Mo26) and 446 5 (2011Ga15).                                                                                                                               |
| 619+x                       | (1/2 <sup>-</sup> , 3/2 <sup>-</sup> ) | 619 7                | 100                  | 0+x                  | E <sub>γ</sub> : weighted average of 616 8 (2017Mo26) and 621 7 (2011Ga15).                                                                                                                               |
| 670+x                       |                                        | 670 <sup>†</sup> 8   | 100                  | 0+x                  | E <sub>γ</sub> : From 2011Ga15, as this γ is not resolved from the 616γ in 2017Mo26, but its presence is indicated in the fit of the spectrum. 2017Mo26 states that the origin of the 670γ remains vague. |

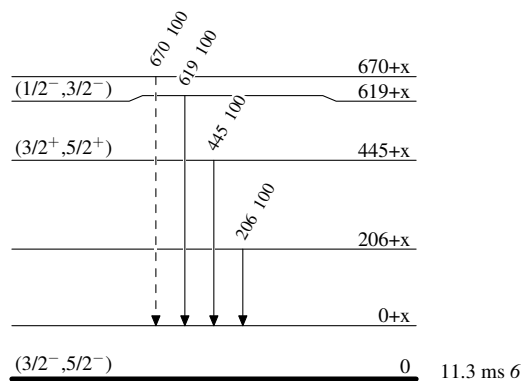
<sup>†</sup> Placement of transition in the level scheme is uncertain.

**Adopted Levels, Gammas**

Legend

Level Scheme

Intensities: Relative photon branching from each level

-----►  $\gamma$  Decay (Uncertain) $^{35}_{12}\text{Mg}_{23}$

**$^{35}\text{Na}$   $\beta^-$  decay (2.1 ms)    2013StZY,2022Cr03,1983La12**

Parent:  $^{35}\text{Na}$ :  $E=0$ ;  $J^\pi=(3/2^+)$ ;  $T_{1/2}=2.1$  ms 4;  $Q(\beta^-)=22190$  syst;  $\% \beta^-$  decay=100

$^{35}\text{Na}$ - $J^\pi$ ,  $T_{1/2}$ : From the Adopted Levels of  $^{35}\text{Na}$ .

$^{35}\text{Na}$ - $Q(\beta^-)$ : 22190 720 (syst, 2021Wa16).

$^{35}\text{Na}$ - $\% \beta^-$  decay:  $\% \beta^- = 100$ ,  $\% \beta^- n > 0$ .

2013StZY:  $^{35}\text{Na}$  was produced via the projectile fragmentation of a 345-MeV/nucleon, 70-pnA  $^{48}\text{Ca}^{20+}$  primary beam from the linear accelerator RILAC and the four cyclotrons RRC, fRC, IRC, and SRC at RIKEN impinging on a 15-mm-thick  $^9\text{Be}$  target. The secondary cocktail beam was selected by the BigRIPS separator and the zero-degree spectrometer (ZDS) using the  $B\rho$ - $\Delta E$ -ToF method, and implanted into the Cylindrical Active Implantation Target for Exotic Nuclei (CAITEN) consisting of a segmented movable hollow-cylindrical-shaped plastic scintillator and a stationary ring of 24 position-sensitive photomultiplier tubes (PSPMTs) arranged on a ring inside the scintillator at the height of the beam line. To reduce background buildup, the scintillator barrel was rotated quickly and axially-moved slowly in vertical direction, resulting in a helix-shaped motion.  $\beta$  particles were detected by the CAITEN and  $\gamma$  rays were detected using three HPGe clover detectors. Measured  $E_\gamma$ ,  $\beta\gamma$ -coin, and implant- $\beta$  correlation, and deduced the  $T_{1/2}$  of  $^{35}\text{Na}$ . Comparisons with QRPA and shell-model calculations.

2022Cr03:  $^{35}\text{Na}$  was produced via the projectile fragmentation of a 172.3-MeV/nucleon, 120-pnA  $^{48}\text{Ca}^{20+}$  primary beam from the FRIB linac impinging on an 8.89-mm-thick  $^9\text{Be}$  target. The secondary cocktail beam centered around  $^{42}\text{Si}$  was selected by the ARIS separator and implanted into a 5-mm-thick YSO segmented scintillator sandwiched between two plastic scintillator veto detectors. Surrounding the implantation array were 11 HPGe clover detectors and 15 fast-timing  $\text{LaBr}_3$  detectors, and the VANDLE array of 88 neutron detectors. Two Si PIN detectors and a plastic scintillator were placed 1.5 m upstream in the last diagnostic detector box of the beamline. Ions were identified event by event by energy loss in the upstream PIN detector (Z determination) and time of flight between a plastic timing scintillator at the start of stage 3 of ARIS and the timing scintillator in the diagnostic detector box (A/Q determination) over a flight path of 33.5 m. Measured implant- $\beta$  correlation and deduced  $T_{1/2}$  of  $^{35}\text{Na}$ .

Comparisons with QRPA and shell-model calculations.

1983La12,1984La03:  $^{35}\text{Na}$  was produced via the fragmentation of a 30 g/cm<sup>2</sup> iridium target by 10-GeV protons from the CERN synchrotron. The recoiling nuclear fragments are thermalized in heated graphite. The alkali elements selectively diffuse out of the target, are surface ionized, and analyzed by the online mass spectrometer, then focused at the bottom of a well-shaped NE213 liquid scintillator neutron detector.  $\beta$  particles were detected using a well-shaped thin plastic scintillator. Measured  $\beta$  and neutron counts. Deduced the  $T_{1/2}$  of  $^{35}\text{Na}$ .  $P_n$  values were deduced for other isotopes but not for  $^{35}\text{Na}$ .

$^{35}\text{Na}$   $\beta^-$ -delayed neutrons have been observed by 1983La12. Experimental  $\% \beta^- n$  values are unknown.

 $\gamma(^{35}\text{Mg})$ 

| $E_\gamma$       | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <sup>x</sup> 661 | 2013StZY observed this $\gamma$ ray in the background-subtracted $\gamma$ -ray spectrum in a time window of 0-10 ms after the implantation of $^{35}\text{Na}$ but did not observe it in later time windows. 2013StZY proposed that it could either be the deexcitation of the first $2^+$ to $0^+$ g.s. in $^{34}\text{Mg}$ or the deexcitation of an excited state in $^{35}\text{Mg}$ . Based on theoretical calculations (2019Mo01,2021Mi17), the $\beta^- 1n$ should be the dominant branch, implying that this $\gamma$ ray is more likely from $^{34}\text{Mg}$ . |

<sup>x</sup>  $\gamma$  ray not placed in level scheme.



$^9\text{Be}(^{38}\text{Si}, ^{35}\text{Mg}\gamma)$  2011Ga15 $^{38}\text{Si} \rightarrow 2\text{p} + 1\text{n} + ^{35}\text{Mg}$ .

**2011Ga15:** A  $^{38}\text{Si}$  secondary beam was produced via the projectile fragmentation of a 140-MeV/nucleon  $^{48}\text{Ca}$  primary beam impinging on a  $^9\text{Be}$  target and selected by the A1900 separator at NSCL, MSU.  $E(^{38}\text{Si})=83$  MeV/nucleon in the middle of the 376(4)-mg/cm $^2$   $^9\text{Be}$  secondary target. The reaction leading to  $^{35}\text{Mg}$  from  $^{38}\text{Si}$  is likely dominated by the two-proton knockout into the continuum of  $^{36}\text{Mg}$  and subsequent neutron emission. The reaction residues were selected and identified by the S800 spectrograph using the  $\Delta E$ -ToF method. The  $\gamma$  rays in coincidence with reaction residues were detected using a 32-fold segmented high-purity germanium detector array (SeGA) at 90° and 37°. Measured Doppler-corrected  $E_\gamma(>80$  keV) and deduced levels. Results were compared with Monte Carlo shell-model calculations using the SDPF-M effective interaction that allows for unrestricted mixing of neutron particle-hole configurations across the  $N = 20$  gap, and with conventional shell-model calculations with the SDPF-U effective interaction that does not include neutron intruder configurations in the model space.

 $^{35}\text{Mg}$  Levels

| $E(\text{level})^\dagger$ | $J^\pi$     | Comments                                                             |
|---------------------------|-------------|----------------------------------------------------------------------|
| 0                         | (5/2 $^-$ ) | $J^\pi$ : From shell-model calculations with the SDPF-M interaction. |
| 0+x                       |             |                                                                      |
| 0+y                       |             |                                                                      |
| 446+x 5                   |             |                                                                      |
| 621+y 7                   |             |                                                                      |
| 670+y 8                   |             |                                                                      |

$^\dagger$  From measured  $E_\gamma$ .  $x \leq 80$  keV and  $y \leq 80$  keV (detection threshold). Shell-model calculations with the SDPF-M interaction predicts a 3/2 $^-$  level at 30 keV.

 $\gamma(^{35}\text{Mg})$ 

**2011Ga15** states that the  $\gamma$ -ray transitions they observed were not in coincidence and depopulate excited states at  $\approx 500$  and  $\approx 700$  keV to the ground state or the alleged near-degenerate excited state below 100-keV excitation energy. **2017Mo26** states that the difference of the  $\gamma$ -ray emission between 1n-knockout and 1p1n-removal reaction channels suggests that the 621 and 670-keV  $\gamma$  rays were emitted from two different excited states. Another possible placement of the 621 and 670-keV  $\gamma$  rays: **2011Ga15** states that it might be possible that the two transitions originate from the same level and that their energy difference of 49(11) keV corresponds to the energy spacing of the alleged (5/2 $^-$ , 3/2 $^-$ ) doublet near the ground state. In Adopted Gammas, all  $\gamma$  rays in this dataset are placed to feed the same final levels based on suggestions by **2017Mo26**.

| $E_\gamma$       | $E_i(\text{level})$ | $E_f$ | Comments                                                                                                                                                                                                                                                                                                                                       |
|------------------|---------------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 446 5            | 446+x               | 0+x   | $E_\gamma$ : strong $\gamma$ transition observed in <b>2011Ga15</b> ; could correspond to a transition to the g.s. or the shell-model predicted low-lying excited state from any of the shell-model predicted levels at 350 keV with $J^\pi=7/2^-$ ; 360 keV with $J^\pi=3/2^+$ , 400 keV with $J^\pi=1/2^-$ , or 560 keV with $J^\pi=7/2^-$ . |
| 621 $^\dagger$ 7 | 621+y               | 0+y   |                                                                                                                                                                                                                                                                                                                                                |
| 670 $^\dagger$ 8 | 670+y               | 0+y   |                                                                                                                                                                                                                                                                                                                                                |

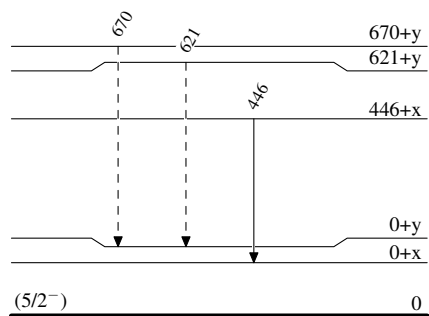
$^\dagger$  Placement of transition in the level scheme is uncertain.

$^9\text{Be}(^{38}\text{Si}, ^{35}\text{Mg}\gamma)$  2011Ga15

Legend

Level Scheme

-----►  $\gamma$  Decay (Uncertain)



$^{35}_{12}\text{Mg}_{23}$

$\text{C}(^{36}\text{Mg}, ^{35}\text{Mg}\gamma), (^{37}\text{Al}, ^{35}\text{Mg}\gamma)$  **2017Mo26**

$^{36}\text{Mg} \rightarrow 1\text{n} + ^{35}\text{Mg}$  from  $J^\pi=0^+$   $^{36}\text{Mg}$  ground state.

$^{37}\text{Al} \rightarrow 1\text{p}1\text{n} + ^{35}\text{Mg}$  from  $J^\pi=(5/2^+)$   $^{37}\text{Al}$  ground state.

**2017Mo26:** A secondary beam composed of  $^{36}\text{Mg}$  and  $^{37}\text{Al}$  was produced via the projectile fragmentation of a 345-MeV/nucleon  $^{48}\text{Ca}$  primary beam impinging on a  $^9\text{Be}$  target and selected by the BigRIPS separator at RIKEN.  $E(^{36}\text{Mg})=235$  MeV/nucleon and  $E(^{37}\text{Al})=246$  MeV/nucleon in front of the 2.54 g/cm<sup>2</sup> carbon secondary target. The reactions leading to  $^{35}\text{Mg}$  from  $^{36}\text{Mg}$  and  $^{37}\text{Al}$  are likely 1n-knockout and 1p1n-removal reactions, respectively. The reaction residues were selected and identified by the Zero Degree spectrometer using the B $\rho$ - $\Delta E$ -ToF method. The  $\gamma$  rays in coincidence with  $^{35}\text{Mg}$  residues were detected using the DALI2 array of 186 NaI(Tl) crystals at 20°–150°. Measured  $E_\gamma(>200\text{ keV})$ ,  $I_\gamma$ ,  $(^{35}\text{Mg})\gamma$ -coin, the inclusive one-neutron knockout cross section and exclusive  $\gamma$ -ray emission cross sections, and parallel momentum distributions of  $^{35}\text{Mg}$  in coincidence with  $\gamma$  rays. Deduced levels, L-transfers, J, and  $\pi$ . Compared with shell-model calculations using the SDPF-M interaction in the sd shell with  $\nu 1f_{7/2}$  and  $\nu 2p_{3/2}$  orbits, and the SDPF-M interaction in a model space up to  $\nu 2p_{1/2}$ , and antisymmetrized molecular dynamics (AMD) model calculations using the Gogny D1S force.

 $^{35}\text{Mg}$  Levels

| $E(\text{level})^\dagger$ | $J^\pi^\ddagger$      | $L^\ddagger$ | Comments                                                                                                                                                                                                           |
|---------------------------|-----------------------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0                         |                       |              | $J^\pi$ : $3/2^-$ from shell-model calculations with the SDPF-M and SDPF-M+ $2p_{1/2}$ interactions; $3/2^+$ from AMD with the Gogny D1S force ( <b>2017Mo26</b> ).                                                |
| 0+x                       |                       |              | $E(\text{level})$ : <200 keV; the detection threshold in <b>2017Mo26</b> . The original placement in <b>2017Mo26</b> of all $\gamma$ rays is to the ground state, which cannot be confirmed with existing data.    |
| 0+y?                      | ( $5/2^-$ , $7/2^-$ ) |              | $E(\text{level})$ : <200 keV; the detection threshold. <b>2017Mo26</b> suggested a low-lying level from the observed 42(1)% neutron knockout from the $l=3$ orbit in the inclusive parallel momentum distribution. |
| 206+x 8                   |                       |              | $J^\pi$ : $\gamma$ -ray intensity is too low to be assigned to the $1/2^-$ level at 141 keV from shell-model calculations with the SDPF-M+ $2p_{1/2}$ interaction ( <b>2017Mo26</b> ).                             |
| 443+x 7                   | ( $3/2^+$ , $5/2^+$ ) | (2)          |                                                                                                                                                                                                                    |
| 616+x 8                   | ( $1/2^-$ , $3/2^-$ ) | (1)          |                                                                                                                                                                                                                    |
| 670+x 8                   |                       |              |                                                                                                                                                                                                                    |

<sup>†</sup> From  $E_\gamma$  data. **2017Mo26** suggested that all the four observed  $\gamma$  rays were emitted independently and fed either the ground state or a low-lying excited state below 200 keV. But the suggested experimental level scheme in FIG.4 of **2017Mo26** (with no  $\gamma$  rays shown) indicates all four  $\gamma$  rays feeding the ground state, which is not adopted by the evaluators since it cannot be confirmed with existing data.

<sup>‡</sup> **2017Mo26** deduced L by comparing the measured and eikonal-calculated parallel momentum distributions of  $^{35}\text{Mg}$  residues.  $J^\pi$  options are deduced accordingly.

 $\gamma(^{35}\text{Mg})$ 

**2017Mo26** states that no clear  $\gamma\gamma$  coincidence was observed.

It is possible that the final levels of 206 $\gamma$ , 443 $\gamma$ , 616 $\gamma$ , and 670 $\gamma$  are not the same level from authors' comparisons with shell-model calculations with different interactions.

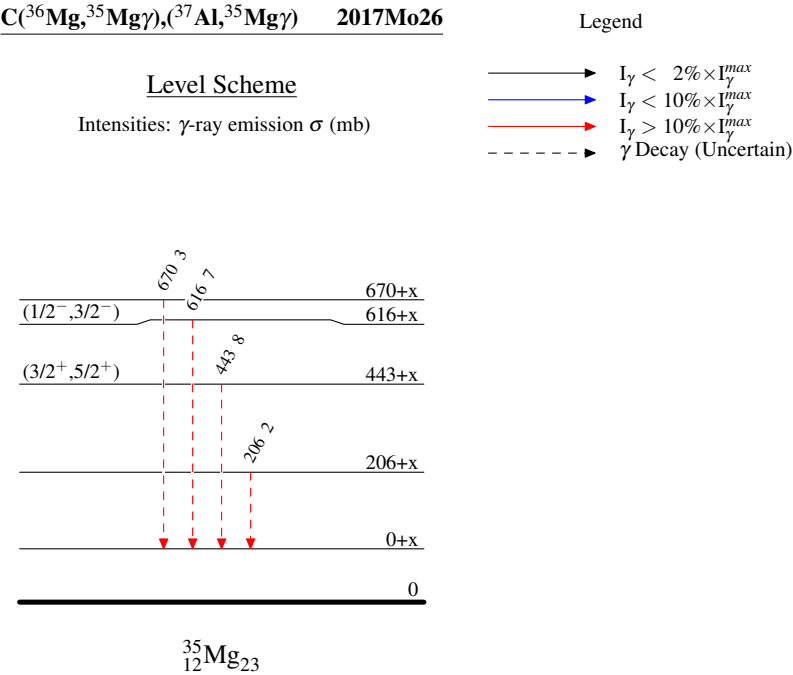
| $E_\gamma$        | $I_\gamma^\dagger$ | $E_i(\text{level})$ | $J^\pi_i$             | $E_f$ | Comments                                                                                                                                                                                                                                            |
|-------------------|--------------------|---------------------|-----------------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 206 $^\ddagger$ 8 | 2 1                | 206+x               |                       | 0+x   |                                                                                                                                                                                                                                                     |
| 443 $^\ddagger$ 7 | 8 1                | 443+x               | ( $3/2^+$ , $5/2^+$ ) | 0+x   | <b>2017Mo26</b> assigned this $\gamma$ as the $3/2^+$ at 788 keV to $3/2^-$ g.s. transition in shell-model calculations with the SDPF-M+ $2p_{1/2}$ interaction ( <b>2017Mo26</b> ).                                                                |
| 616 $^\ddagger$ 8 | 7 1                | 616+x               | ( $1/2^-$ , $3/2^-$ ) | 0+x   | <b>2017Mo26</b> assigned this $\gamma$ as the $3/2^-$ at 664 keV to $1/2^-$ at 141 keV transition in shell-model calculations with the SDPF-M+ $2p_{1/2}$ interaction ( <b>2017Mo26</b> ).                                                          |
| 670 $^\ddagger$ 8 | 3 1                | 670+x               |                       | 0+x   | $E_\gamma$ : From <b>2011Ga15</b> , as this $\gamma$ is not resolved from the 616 $\gamma$ in <b>2017Mo26</b> , but its presence is indicated in the fit of the spectrum. <b>2017Mo26</b> states that the origin of the 670 $\gamma$ remains vague. |

Continued on next page (footnotes at end of table)

$\text{C}(^{36}\text{Mg}, ^{35}\text{Mg}\gamma), (^{37}\text{Al}, ^{35}\text{Mg}\gamma)$  2017Mo26 (continued)

$\gamma(^{35}\text{Mg})$  (continued)

<sup>†</sup>  $\gamma$ -ray emission  $\sigma$  (mb).  
<sup>‡</sup> Placement of transition in the level scheme is uncertain.



Adopted Levels, Gammas

$Q(\beta^-)=14170$  40;  $S(n)=5297$  8;  $S(p)=15831$  9;  $Q(\alpha)=-14894$  8 [2021Wa16](#)

$S(p), Q(\alpha)$ : Deduced by evaluators using mass excesses of 8318 5 for  $^{34}\text{Mg}$ : a weighted average of 8323 7 ([2019As04](#)) and 8315 5 ([2025Ly01](#)), and 12245.3 26 for  $^{31}\text{Na}$  measured by [2025Ly01](#);  $-224$  7 for  $^{35}\text{Al}$  from [2021Wa16](#). Values from [2021Wa16](#):  $S(p)=15836$  10,  $Q(\alpha)=-14895$  16.

$S(2p)=38400.5$  85,  $Q(\epsilon)=-15753$  10, from mass excesses of 23598.8 43 for  $^{33}\text{Na}$  and 15529.5 71 for  $^{35}\text{Mg}$  measured by [2025Ly01](#);  $-224$  7 for  $^{35}\text{Al}$  from [2021Wa16](#). Values from [2021Wa16](#):  $S(2p)=38580$  450,  $Q(\epsilon)=-15860$  270.

$S(2n)=7869$  10,  $Q(\beta^-n)=11697$  7 ([2021Wa16](#)).

Isotope discovery ([2012Th10](#)):  $\text{C}(^{40}\text{Ar}, \text{X})$  projectile fragmentation at Berkeley ([1979Sy01](#)).

$^{35}\text{Al}$  production:

[2015Mo17](#):  $^9\text{Be}(^{40}\text{Ar}, \text{X})$  at  $E(^{40}\text{Ar})=95$  MeV/nucleon at RIKEN. Measured angular distributions and transverse momentum distributions of fragments. Deduced formulation for the width of transverse momentum distribution as a function of fragment velocity.

[2012Kw02](#):  $E(^{40}\text{Ar})=140$  MeV/nucleon impinged on  $^9\text{Be}$ , natural nickel, and  $^{181}\text{Ta}$  targets at NSCL. Measured fragmentation cross sections, parallel momentum transfers, and widths. Compared with empirical formula EPAX, and predictions from internuclear cascade and deep inelastic models using Monte Carlo ISABEL-GEMINI and DIT-GEMINI codes.

[2012Zh06](#):  $^9\text{Be}(^{40}\text{Ar}, \text{X})$  at  $E(^{40}\text{Ar})=57$  MeV/nucleon at HIRFL. Measured momentum distributions and production cross sections of fragments. Observed competition between projectile fragmentation and other mechanisms. Compared with EPAX, abrasion-ablation, and HIPSE models. Studied target dependence of fragment cross sections.

[2007No13](#):  $^9\text{Be}(^{40}\text{Ar}, \text{X})$  at  $E(^{40}\text{Ar})=100$  MeV/nucleon at RIKEN. Measured fragment momentum distributions and production cross sections.

$^{35}\text{Al}$  decay measurements:

[2017Ha23](#):  $^9\text{Be}(^{40}\text{Ar}, \text{X})$  at HIRFL. Measured  $T_{1/2}$ .

[2005Ti11](#), [2006AnZW](#):  $(^{36}\text{S}, \text{X})$  at GANIL. Measured  $T_{1/2}$ ,  $\beta^-$ -delayed  $\gamma$  and neutron spectroscopy.

[2001Nu01](#), [2002Nu02](#):  $\text{U}(p, \text{X})$  at CERN. Measured  $T_{1/2}$ ,  $\beta^-$ -delayed  $\gamma$  and neutron spectroscopy.

[1999YoZW](#):  $^9\text{Be}(^{48}\text{Ca}, \text{X})$  and  $^{181}\text{Ta}(^{48}\text{Ca}, \text{X})$  at RIKEN. Measured  $T_{1/2}$  and  $\% \beta^-n$ .

[1995ReZZ](#), [2008ReZZ](#):  $^{232}\text{Th}(p, \text{X})$  at LAMPF. Measured  $T_{1/2}$  and  $\% \beta^-n$ , and average  $E_n$ .

[1989Le16](#), [1989MuZU](#):  $^{181}\text{Ta}(^{48}\text{Ca}, \text{X})$  at GANIL. Measured  $T_{1/2}$  and  $\% \beta^-n$ .

[1988Mu08](#), [1988MuZY](#), [1988BaYZ](#):  $^{181}\text{Ta}(^{86}\text{Kr}, \text{X})$  at GANIL.  $T_{1/2}$  and  $\% \beta^-n$ .

$^{35}\text{Al}$  radius measurement:

[2006Kh08](#):  $^{35}\text{Al}$  produced by  $^{181}\text{Ta}(^{48}\text{Ca}, \text{X})$  fragmentation at  $E(^{48}\text{Ca})=60.3$  MeV/nucleon at GANIL. Measured energy-integrated reaction cross sections at 30-65 MeV/nucleon using a silicon telescope as both active target and detector. Deduced reduced strong absorption radii, isospin dependence, and possible halo structure or large deformation.

$^{35}\text{Al}$  knockout-reaction measurements:

[2012No05](#):  $^{33,34,35,36}\text{Al}$  produced by  $\text{Be}(^{48}\text{Ca}, \text{X})$  fragmentation at  $E(^{48}\text{Ca})=900$  MeV/nucleon at GSI. Measured  $1n$  removal cross sections and longitudinal momentum distributions of the residues. Deduced single-particle occupancies in the ground states of  $^{33,34,35}\text{Al}$ .  $\sigma(^{35}\text{Al} \rightarrow ^{34}\text{Al})=75$  mb 4 and  $\sigma(^{36}\text{Al} \rightarrow ^{35}\text{Al})=95$  mb 5.

[2010Ro23](#):  $^{35}\text{Al}$  produced by  $^9\text{Be}(^{40}\text{Ar}, \text{X})$  fragmentation at  $E(^{40}\text{Ar})=700$  MeV/nucleon at GSI. Measured  $1n$  knockout cross sections and longitudinal momentum distributions of the residues.  $\sigma(^{35}\text{Al} \rightarrow ^{34}\text{Al})=65$  mb 18.

$^{35}\text{Al}$  in-beam  $\gamma$  spectroscopy:

[2006FuZX](#):  $^{35}\text{Al}$  produced by  $\text{Be}, \text{C}(^{40}\text{Ar}, \text{X})$  and observed one 760.1-keV 21  $\gamma$  ray from  $\text{He}(^{35}\text{Al}, ^{35}\text{Al}\gamma)$  without placing it in the level scheme.

$^{35}\text{Al}$  mass measurements: [2017Ga20](#), [2007Ju03](#), [1991Or01](#), [1991Zh24](#), [1987Gi05](#).

Theoretical calculations (binding energies, deformation, quadrupole moments, radii, levels,  $J$ ,  $\pi$ , mass,  $T_{1/2}$ , etc): [2016Sa46](#), [2014Ca21](#), [2013Li39](#), [2013Sh05](#), [2011Ki12](#), [2009Yo05](#), [2004Kh16](#), [1994Po05](#).

 $^{35}\text{Al}$  LevelsCross Reference (XREF) Flags

|          |                                                     |          |                                                    |
|----------|-----------------------------------------------------|----------|----------------------------------------------------|
| <b>A</b> | $^{36}\text{Mg} \beta^-n$ decay (6.9 ms)            | <b>D</b> | $\text{Pb}(^{35}\text{Al}, ^{34}\text{Al}n\gamma)$ |
| <b>B</b> | $^9\text{Be}(^{36}\text{Si}, ^{35}\text{Al}\gamma)$ | <b>E</b> | Coulomb excitation                                 |
| <b>C</b> | $\text{He}(^{35}\text{Al}, ^{35}\text{Al}'\gamma)$  |          |                                                    |

Continued on next page (footnotes at end of table)

**Adopted Levels, Gammas (continued)** $^{35}\text{Al}$  Levels (continued)

| $E(\text{level})^\dagger$ | $J^\pi$                 | $T_{1/2}$ | XREF  | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------------|-------------------------|-----------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0                         | $(5/2)^+$               | 38.1 ms 4 | AB DE | $\% \beta^- = 100$ ; $\% \beta^- n = 37$ 4; $\% \beta^- 2n = ?$<br>$J^\pi$ : From shell-model calculations and $L(^{36}\text{Si}, ^{35}\text{Al}) = 2$ from $0^+$ .<br>$T_{1/2}$ : weighted average of 38.4 ms 3 (2017Ha23, implant- $\beta$ correlation), 36.8 ms 5 (2005Ti11, 2006AnZW, implant- $\beta$ correlation), 38.6 ms 4 (2001Nu01, 2002Nu02, three $\beta$ -counting rates and one $\gamma$ -counting rate), 30 ms 4 (1995ReZZ, 2008ReZZ, implant- $\beta/n$ correlation), 170 ms +90-50 (1989Le16, 1989MuZU, $\beta$ -decay curve), 30 ms 10 (1988DuZT, beam pulsations), and 130 ms +100-50 (1988Mu08, 1988BaYZ, $\beta$ counting rate).<br>$\% \beta^- n$ : Unweighted average of 41 2 (2005Ti11, 2006AnZW), 41 13 (2001Nu01, 2002Nu02), 26 4 (1995ReZZ, 2008ReZZ), and 40 10 (1989Le16, 1989MuZU). Others: 43 9 (1999YoZW, preliminary), 87 +37-25 (1988Mu08, 1988BaYZ) with a non-physical upper bound. Note that the original value reported in 2005Ti11 is 38 2 (2006AnZW report 38 3) and the evaluators have added intensities of $\gamma$ rays deexciting the 4379 (0.9% 2) and 4255 (2.0% 8) levels in $^{34}\text{Si}$ not included in their reported $\% \beta^- n$ value, as the authors state that these levels are fed by either neutrons with energy below their detection threshold or by weak transitions that cannot be identified in the neutron spectrum.<br>$\% \beta^- 2n$ : No evidence was found for any $A=33$ activity and $2n$ emission was thus found negligible (2001Nu01).<br>Reduced strong absorption radius $r_0^2 = 1.188 \text{ fm}^2$ 14 from the energy-integrated $\sigma$ of $\text{Si}(^{35}\text{Al}, X)$ (2006Kh08).<br>Major configurations for $J^\pi = 5/2^+$ of $^{35}\text{Al}$ g.s. from $\text{Pb}(^{35}\text{Al}, ^{34}\text{Aln}\gamma)$ : (g.s., $4^-$ in $^{34}\text{Al}$ ) $\otimes$ $\nu p_{3/2}$ , $S=0.36$ 9 (2017Ch36); (46 keV, $1^+$ in $^{34}\text{Al}$ ) $\otimes$ $\nu d_{3/2}$ , $S=1.47$ 22 (2017Ch36); (1.4 MeV, $2^+$ in $^{34}\text{Al}$ ) $\otimes$ $\nu s_{1/2}$ , $S=0.16$ 1 (2021Bh12); (2.5 MeV, $3^-$ in $^{34}\text{Al}$ ) $\otimes$ $\nu p_{3/2}$ , $S=1.48$ 18 (2021Bh12). |
| 802 3                     |                         |           | AB    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 1007 4                    |                         |           | B E   | XREF: $E(1020)$<br>$B(E2)^\ddagger = 0.0142$ 52 from Coulomb excitation assuming E2 excitation mode (1999Ib01). Others: $B(E1)^\ddagger \leq 0.00020$ 9, $B(E2)^\ddagger \leq 0.0125$ 56, $B(M1)^\ddagger \leq 0.0024$ 11.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 1866 4                    |                         |           | B     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 1975 4                    | $3/2^+, 5/2^+^\ddagger$ |           | B     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 2734 7                    |                         |           | B     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 3245 5                    | $3/2^+, 5/2^+^\ddagger$ |           | B     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 4275? 9                   | $3/2^+, 5/2^+^\ddagger$ |           | B     | XREF: B(?)<br>Level exists assuming that the 4275 $\gamma$ feeds the ground state of $^{35}\text{Al}$ (2014St18).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

 $^\dagger$  From a least-squares fit to  $\gamma$ -ray energies. $^\ddagger$  From  $L(^{36}\text{Si}, ^{35}\text{Al}) = 2$  of parallel momentum distributions of  $^{35}\text{Al}$  residues (2014St18). $\gamma(^{35}\text{Al})$ 

| $E_i(\text{level})$ | $J_i^\pi$      | $E_\gamma^\dagger$ | $I_\gamma^\dagger$ | $E_f$ | $J_f^\pi$ | Comments                                                                                                                  |
|---------------------|----------------|--------------------|--------------------|-------|-----------|---------------------------------------------------------------------------------------------------------------------------|
| 802                 |                | 802 4              | 100                | 0     | $(5/2)^+$ |                                                                                                                           |
| 1007                |                | 1006 6             | 100                | 0     | $(5/2)^+$ | $E_\gamma$ : weighted average of 1003 4 from $(^{36}\text{Si}, ^{35}\text{Al}\gamma)$ and 1020 9 from Coulomb excitation. |
| 1866                |                | 859 4              | 100 8              | 1007  |           |                                                                                                                           |
|                     |                | 1064 4             | 22 6               | 802   |           |                                                                                                                           |
| 1975                | $3/2^+, 5/2^+$ | 968 4              | 59 4               | 1007  |           |                                                                                                                           |
|                     |                | 1174 5             | 37 4               | 802   |           |                                                                                                                           |
|                     |                | 1972 6             | 100 7              | 0     | $(5/2)^+$ |                                                                                                                           |

Continued on next page (footnotes at end of table)

**Adopted Levels, Gammas (continued)** $\gamma(^{35}\text{Al})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$      | $E_\gamma^\dagger$ | $I_\gamma^\dagger$ | $E_f$ | $J_f^\pi$ | Comments                                                                                                                                                                                                                                                                                |
|---------------------|----------------|--------------------|--------------------|-------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2734                |                | 1932 6             | 100                | 802   |           |                                                                                                                                                                                                                                                                                         |
| 3245                | $3/2^+, 5/2^+$ | 2237 6             | 100 8              | 1007  |           |                                                                                                                                                                                                                                                                                         |
|                     |                | 2440 7             | 18.0 26            | 802   |           |                                                                                                                                                                                                                                                                                         |
|                     |                | 3250 8             | 42 5               | 0     | $(5/2)^+$ |                                                                                                                                                                                                                                                                                         |
| 4275?               | $3/2^+, 5/2^+$ | 4275 $^\ddagger$ 9 | 100                | 0     | $(5/2)^+$ | <a href="#">2014St18</a> states that the $4275\gamma$ transition could populate the ground state or one of the lowest two excited states. In the absence of any observed $\gamma\gamma$ coincidences, they tentatively place the 4275-keV $\gamma$ transition feeding the ground state. |

$^\dagger$  From  $^9\text{Be}(^{36}\text{Si}, ^{35}\text{Al}\gamma)$ , unless otherwise noted.

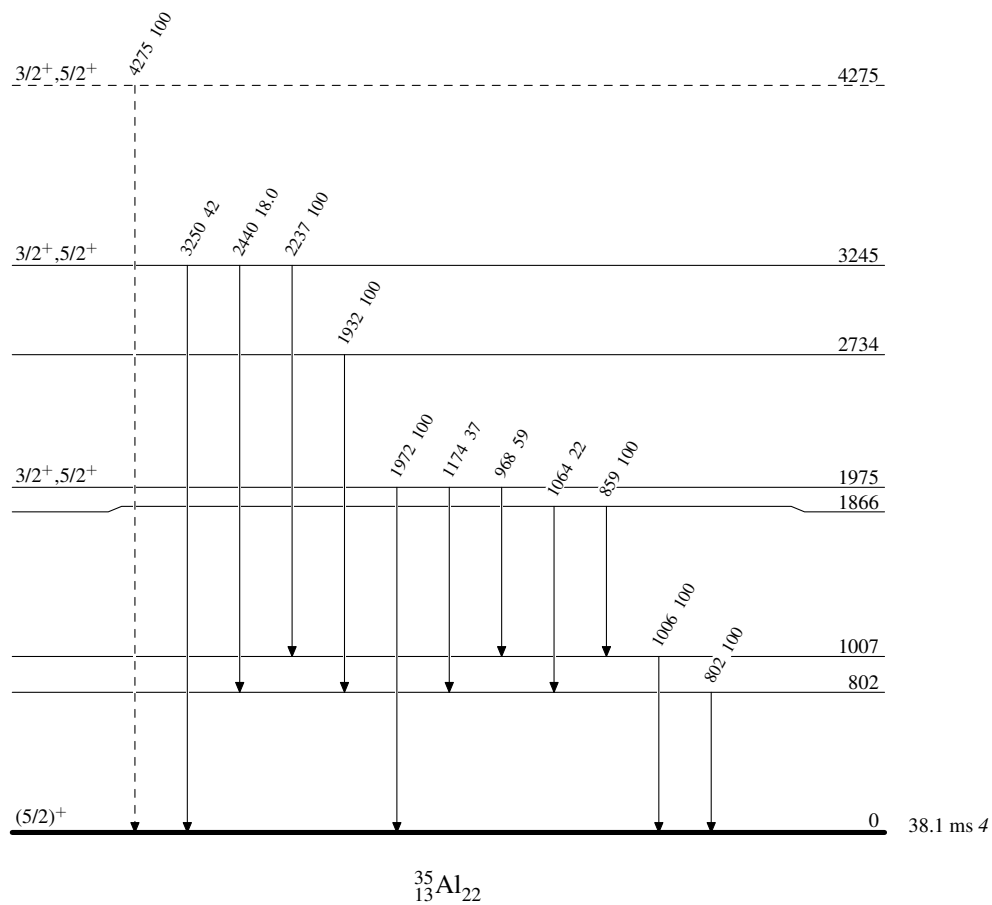
$^\ddagger$  Placement of transition in the level scheme is uncertain.

Adopted Levels, Gammas

Legend

Level Scheme

Intensities: Relative photon branching from each level

-----►  $\gamma$  Decay (Uncertain)



$^{36}\text{Mg}$   $\beta^-$ -n decay (6.9 ms) 2023Lu07

Parent:  $^{36}\text{Mg}$ :  $E=0$ ;  $J^\pi=0^+$ ;  $T_{1/2}=6.9$  ms  $+7-8$ ;  $Q(\beta^-n)=1.253\times 10^4$  69;  $\%\beta^-n$  decay=48 12

$^{36}\text{Mg}$ - $T_{1/2}$ : Weighted average of 3.9 ms 13 (2004Gr20,2003Gr22, implant- $\beta$  correlation), 7.6 ms  $+5-8$  (2013StZY, implant- $\beta$  correlation, original  $T_{1/2}=7.6$  ms 1 (stat)  $+5-8$  (syst)), 7.2 ms 12 (2022Cr03, implant- $\beta$  correlation, original  $T_{1/2}=7.2$  ms 1 (stat) 12 (syst)), and 6.8 ms 10 (2023Lu07, implant- $\beta\gamma$  correlation). Other:  $\approx 5$  ms (1999YoZW, implant- $\beta$  correlation, preliminary).

$^{36}\text{Mg}$ - $Q(\beta^-n)$ : From 2021Wa16.

$^{36}\text{Mg}$ - $\%\beta^-n$  decay: From 1999YoZW (preliminary; not adopted in Adopted Levels of  $^{36}\text{Mg}$ ).

2023Lu07: Exp 1:  $^{36}\text{Mg}$  and  $^{36}\text{Al}$  were produced via the projectile fragmentation of a 140-MeV/nucleon, 80-pnA  $^{48}\text{Ca}$  primary beam from the NSCL cyclotrons impinging on a 642-mg/cm<sup>2</sup>-thick  $^9\text{Be}$  target. The secondary cocktail beam centered around  $^{33}\text{Na}$  was selected by the A1900 separator and implanted into a CeBr<sub>3</sub> scintillator sandwiched between two plastic scintillator veto detectors. Surrounding the implantation array were the SeGA array of 16 segmented Ge detectors and 15 LaBr<sub>3</sub> detectors. Exp 2:  $^{36}\text{Mg}$  and  $^{36}\text{Al}$  were produced via the projectile fragmentation of a 172.3-MeV/nucleon, 120-pnA  $^{48}\text{Ca}$  primary beam from the FRIB linac impinging on an 8.89-mm-thick  $^9\text{Be}$  target. The secondary cocktail beam centered around  $^{42}\text{Si}$  was selected by the ARIS separator and implanted into a 5-mm-thick YSO segmented scintillator sandwiched between two plastic scintillator veto detectors. Surrounding the implantation array were 11 HPGe clover detectors and 15 fast-timing LaBr<sub>3</sub> detectors, and the VANDLE array of 88 neutron detectors. Measured  $E_\gamma$ ,  $I_\gamma$ ,  $\beta\gamma$ -coin,  $\gamma\gamma$ -coin, implant- $\beta\gamma$  correlation. Deduced  $T_{1/2}$  of  $^{36}\text{Mg}$  g.s.,  $^{36}\text{Al}$  g.s. and a  $^{36}\text{Al}$  isomer. Comparisons with FSU shell-model calculations.

2013StZY:  $^{36}\text{Mg}$  was produced via the projectile fragmentation of a 345-MeV/nucleon, 70-pnA  $^{48}\text{Ca}^{20+}$  primary beam from the linear accelerator RILAC and the four cyclotrons RRC, fRC, IRC, and SRC at RIKEN impinging on an 15-mm-thick  $^9\text{Be}$  target. The secondary cocktail beam was selected by the BigRIPS separator and the zero-degree spectrometer (ZDS) using the  $B\rho$ - $\Delta E$ -ToF method, and implanted into the Cylindrical Active Implantation Target for Exotic Nuclei (CAITEN) consisting of a segmented movable hollow-cylindrical-shaped plastic scintillator and a stationary ring of 24 position-sensitive photomultiplier tubes (PSPMTs) arranged on a ring inside the scintillator at the height of the beam line. To reduce background buildup, the scintillator barrel was rotated quickly and axially-moved slowly in vertical direction, resulting in a helix-shaped motion.  $\beta$  particles were detected by the CAITEN and  $\gamma$  rays were detected using three HPGe clover detectors. Measured  $E_\gamma$ ,  $\beta\gamma$ -coin, and implant- $\beta$  correlation. Deduced parent  $T_{1/2}$ . Comparisons with QRPA and shell-model calculations.

 $^{35}\text{Al}$  Levels

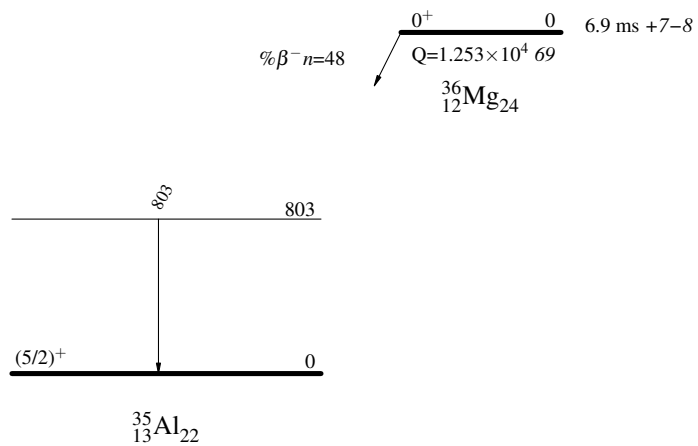
| <u>E(level)</u> | <u><math>J^\pi</math></u> | <u>Comments</u>                    |
|-----------------|---------------------------|------------------------------------|
| 0<br>803        | (5/2) <sup>+</sup>        | $J^\pi$ : From the Adopted Levels. |

 $\gamma(^{35}\text{Al})$ 

| <u><math>E_\gamma</math></u> | <u><math>E_i(\text{level})</math></u> | <u><math>E_f</math></u> | <u><math>J_f^\pi</math></u> | <u>Comments</u>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------|---------------------------------------|-------------------------|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 803                          | 803                                   | 0                       | (5/2) <sup>+</sup>          | <p><math>E_\gamma</math>: From 2023Lu07.</p> <p>2023Lu07 observed the 803<math>\gamma</math> within the 100-ms window following the arrival of a <math>^{36}\text{Mg}</math> ion in both NSCL and FRIB experiments.</p> <p>2013StZY observed a 804-keV <math>\gamma</math> ray that was not in coincidence with other <math>\gamma</math> rays. 2013StZY states that the 804<math>\gamma</math> is a prompt line that predominantly appears in the time window 0-10 ms after an implantation of <math>^{36}\text{Mg}</math>, so it must originate from a deexcitation in one of the daughter nuclei <math>^{34,35,36}\text{Al}</math>.</p> |

$^{36}\text{Mg} \beta^- n$  decay (6.9 ms) 2023Lu07

Decay Scheme



$\text{He}(^{35}\text{Al}, ^{35}\text{Al}'\gamma)$     **2006FuZX**

**2006FuZX:**  $^{35}\text{Al}$  was produced via the projectile fragmentation of a 63-MeV/nucleon  $^{40}\text{Ar}$  primary beam impinging on 1-mm-thick C and Be targets and selected by the RIKEN Projectile-Fragment Separator (RIPS). The secondary cocktail beam was directed onto a 150-mg/cm<sup>2</sup> liquid He target and the outgoing particles were identified by energy loss, kinetic energy, and time-of-flight measured using a Si detector, a NaI calorimeter, and a couple of plastic scintillators. The positions and directions of incident and outgoing particles were measured using PPACs. Only outgoing particles were identified so that the excited states could be populated by any allowed reactions. Deexcited  $\gamma$  rays emitted from outgoing particle moving at  $v/c \approx 0.3$  were detected by the Gamma-Ray detector Array with Position and Energy sensitivity (GRAPE) consisting of segmented Ge detectors surrounding the liquid He target.

 $\gamma(^{35}\text{Al})$ 

$E_{\gamma}^{\dagger}$   
<sup>x</sup>760.1 21

<sup>†</sup> From **2006FuZX**.

<sup>x</sup>  $\gamma$  ray not placed in level scheme.

$^9\text{Be}(^{36}\text{Si}, ^{35}\text{Al})$  **2014St18**

$^{36}\text{Si} \rightarrow \text{p} + ^{35}\text{Al}$  from  $J^\pi = 0^+$   $^{36}\text{Si}$  ground state.

**2014St18:** A  $^{36}\text{Si}$  secondary beam was produced via the projectile fragmentation of a 140-MeV/nucleon  $^{48}\text{Ca}$  primary beam impinging on a  $^9\text{Be}$  target at NSCL, MSU and was selected by the A1900 separator. The states of  $^{35}\text{Al}$  and  $^{35}\text{Si}$  were populated by the one-proton/neutron knockout reactions, respectively, from the  $^{36}\text{Si}$  beam at a midtarget energy of 97.7 MeV/nucleon 5 on a 287-mg/cm<sup>2</sup>  $^9\text{Be}$  secondary target. Knockout residues were identified from their energy loss measured by an ionization chamber at the focal plane of the S800 spectrometer and from their ToF measured between two scintillators at the object position and at the focal plane of the S800 spectrometer. The position and angle of the residues were measured using two cathode-readout drift chambers. Prompt  $\gamma$  rays from the deexcitation of the residues were detected by the GRETINA Ge array. Measured Doppler-corrected  $E_\gamma$ ,  $I_\gamma$ , ( $^{35}\text{Al}$ ) $\gamma$ -coin,  $\gamma\gamma$ -coin, the parallel momentum distributions of populated states in  $^{35}\text{Al}$  residues. Deduced levels, J,  $\pi$ , L-transfers, inclusive and exclusive knockout cross section for producing  $^{35}\text{Al}$  from  $^{36}\text{Si}$ . Calculations using eikonal model and shell model calculations with SDPF-U and SDPF-MU interactions.

 $^{35}\text{Al}$  Levels

Total knockout  $\sigma = 22$  mb / for producing  $^{35}\text{Al}$  from  $^{36}\text{Si}$ .

| E(level) <sup>†</sup> | $J^\pi$ <sup>‡</sup>                | L <sup>‡</sup> | Comments                                                                                                                                         |
|-----------------------|-------------------------------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| 0                     | (5/2) <sup>+</sup>                  | 2              | $J^\pi$ : interpreted as the 1d <sub>5/2</sub> proton removal from $^{36}\text{Si}$ ( <b>2014St18</b> ).<br>Partial knockout $\sigma = 13$ mb 2. |
| 801 3                 |                                     |                | Partial knockout $\sigma = 1.0$ mb 7.                                                                                                            |
| 1005 3                |                                     |                | Partial knockout $\sigma = 0.8$ mb 9.                                                                                                            |
| 1865 4                |                                     |                | Partial knockout $\sigma = 1.0$ mb 2.                                                                                                            |
| 1973 4                | 3/2 <sup>+</sup> , 5/2 <sup>+</sup> | 2              | Partial knockout $\sigma = 3.2$ mb 5.                                                                                                            |
| 2733 7                |                                     |                | Partial knockout $\sigma = 0.5$ mb 1.                                                                                                            |
| 3244 5                | 3/2 <sup>+</sup> , 5/2 <sup>+</sup> | 2              | Partial knockout $\sigma = 2.6$ mb 3.                                                                                                            |
| 4275? 9               | 3/2 <sup>+</sup> , 5/2 <sup>+</sup> | 2              | Partial knockout $\sigma = 0.5$ mb 1.                                                                                                            |

<sup>†</sup> From a least-squares fit to  $\gamma$ -ray energies.

<sup>‡</sup> **2014St18** deduced L by comparing the measured and eikonal-calculated parallel momentum distributions of residuals.  $J^\pi$  options are deduced accordingly.

 $\gamma(^{35}\text{Al})$ 

| $E_\gamma$          | $I_\gamma$ | $E_i(\text{level})$ | $J^\pi_i$                           | $E_f$ | $J^\pi_f$          | Comments                                                                                                                                                                                                                                                                        |
|---------------------|------------|---------------------|-------------------------------------|-------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 802 4               | 10 1       | 801                 |                                     | 0     | (5/2) <sup>+</sup> |                                                                                                                                                                                                                                                                                 |
| 859 4               | 3.6 3      | 1865                |                                     | 1005  |                    |                                                                                                                                                                                                                                                                                 |
| 968 4               | 4.4 3      | 1973                | 3/2 <sup>+</sup> , 5/2 <sup>+</sup> | 1005  |                    |                                                                                                                                                                                                                                                                                 |
| 1003 4              | 19 1       | 1005                |                                     | 0     | (5/2) <sup>+</sup> |                                                                                                                                                                                                                                                                                 |
| 1064 4              | 0.8 2      | 1865                |                                     | 801   |                    |                                                                                                                                                                                                                                                                                 |
| 1174 5              | 2.8 3      | 1973                | 3/2 <sup>+</sup> , 5/2 <sup>+</sup> | 801   |                    |                                                                                                                                                                                                                                                                                 |
| <sup>x</sup> 1473 5 | 1.1 2      |                     |                                     |       |                    |                                                                                                                                                                                                                                                                                 |
| 1932 6              | 2.5 3      | 2733                |                                     | 801   |                    |                                                                                                                                                                                                                                                                                 |
| 1972 6              | 7.5 5      | 1973                | 3/2 <sup>+</sup> , 5/2 <sup>+</sup> | 0     | (5/2) <sup>+</sup> |                                                                                                                                                                                                                                                                                 |
| 2237 6              | 7.8 6      | 3244                | 3/2 <sup>+</sup> , 5/2 <sup>+</sup> | 1005  |                    |                                                                                                                                                                                                                                                                                 |
| 2440 7              | 1.4 2      | 3244                | 3/2 <sup>+</sup> , 5/2 <sup>+</sup> | 801   |                    |                                                                                                                                                                                                                                                                                 |
| <sup>x</sup> 3060 8 | 1.6 4      |                     |                                     |       |                    |                                                                                                                                                                                                                                                                                 |
| 3250 8              | 3.3 4      | 3244                | 3/2 <sup>+</sup> , 5/2 <sup>+</sup> | 0     | (5/2) <sup>+</sup> |                                                                                                                                                                                                                                                                                 |
| 4275 <sup>†</sup> 9 | 3 1        | 4275?               | 3/2 <sup>+</sup> , 5/2 <sup>+</sup> | 0     | (5/2) <sup>+</sup> | <b>2014St18</b> states that the 4275 $\gamma$ transition could populate the ground state or one of the lowest two excited states. In the absence of any observed $\gamma\gamma$ coincidences, they tentatively place the 4275-keV $\gamma$ transition feeding the ground state. |

Continued on next page (footnotes at end of table)

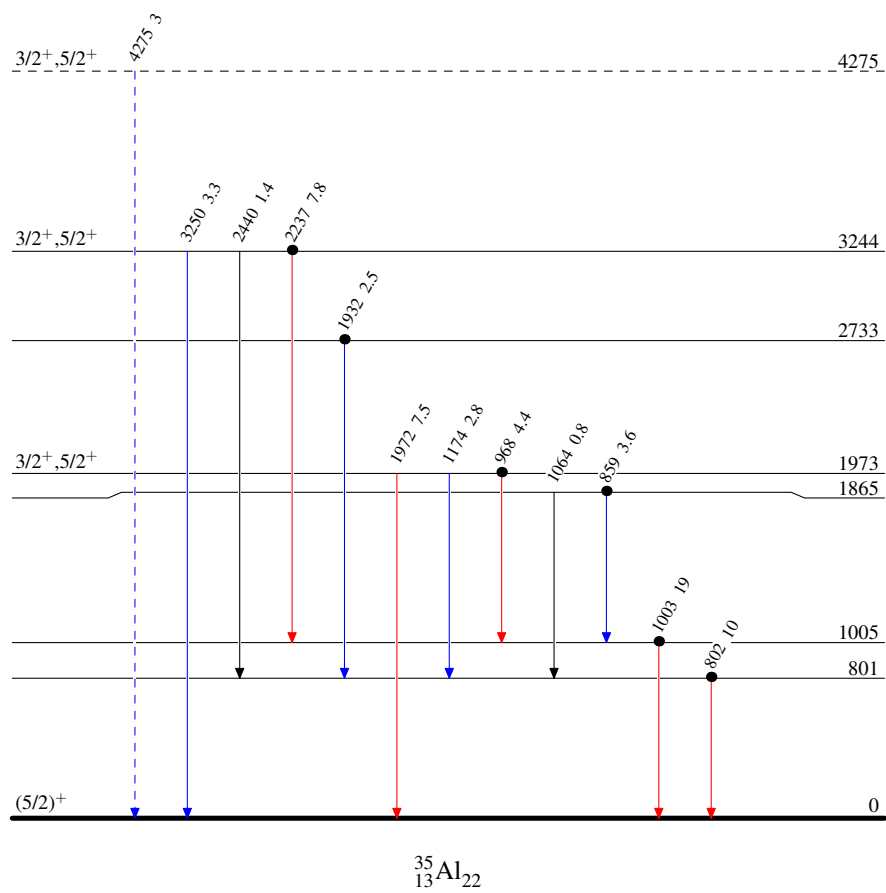
$^9\text{Be}(^{36}\text{Si}, ^{35}\text{Al}\gamma)$  2014St18 (continued) $\gamma(^{35}\text{Al})$  (continued)<sup>†</sup> Placement of transition in the level scheme is uncertain.<sup>x</sup>  $\gamma$  ray not placed in level scheme. $^9\text{Be}(^{36}\text{Si}, ^{35}\text{Al}\gamma)$  2014St18

## Level Scheme

Intensities: Yield/100 ions

## Legend

- $I_\gamma < 2\% \times I_\gamma^{\max}$
- $I_\gamma < 10\% \times I_\gamma^{\max}$
- $I_\gamma > 10\% \times I_\gamma^{\max}$
- - - - -  $\gamma$  Decay (Uncertain)
- Coincidence

 $^{35}_{13}\text{Al}_{22}$

**Pb( $^{35}\text{Al}, ^{34}\text{Al}n\gamma$ )    2017Ch36,2021Bh12**

Coulomb dissociation of  $^{35}\text{Al}$  on Pb target.

**2017Ch36,2014ChZZ:**  $^{35}\text{Al}$  was produced via the projectile fragmentation of a 531-MeV/nucleon  $^{40}\text{Ar}$  primary beam from the Heavy Ion Synchrotron (SIS18) at GSI. The secondary cocktail beam was separated by the FRS separator and impinged on a 2 g/cm<sup>2</sup> Pb target and a 0.93 g/cm<sup>2</sup> C target. Projectiles and reaction fragments were detected using 8 DSSDs, separated by a large-area dipole magnet (ALADIN) and tracked using two large scintillator fiber detectors (GFIs). Neutrons from the excited projectiles were detected using the high-efficiency Large Area Neutron Detector (LAND).  $\gamma$  rays from the deexcited projectile and projectile-like fragments were detected using a spherical  $4\pi$  Crystal Ball detector array of 162 NaI(Tl) crystals. Measured  $E(\text{fragment})$ ,  $E_n$ ,  $E_\gamma$ , Coulomb dissociation cross sections. Deduced relative populations of  $^{34}\text{Al}$ ,  $^{35}\text{Al}$  g.s. configuration. Comparisons with shell-model calculations with the SDPF-M interaction. The measured inclusive differential CD cross section (integrated up to 5.0 MeV relative energy) for  $^{35}\text{Al} \rightarrow ^{34}\text{Al}+n$  using a Pb target is 78 mb 13.

**2021Bh12:** A further analysis of the data from **2017Ch36**. The  $^{35}\text{Al}(\gamma,n)^{34}\text{Al}$  photoabsorption cross section was obtained from fitting the direct breakup model to the measured differential Coulomb dissociation cross section of  $^{35}\text{Al}$  breaking up into  $^{34}\text{Al}$  core excited states. The  $^{34}\text{Al}(n,\gamma)^{35}\text{Al}$  neutron capture cross sections were obtained from the photoabsorption cross sections using the detailed balance theorem.

 **$^{35}\text{Al}$  Levels**

| <u>E(level)</u> | <u><math>J^\pi</math></u>                                 | <u>Comments</u>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0               | (5/2 <sup>+</sup> , 3/2 <sup>+</sup> , 1/2 <sup>+</sup> ) | <p><math>J^\pi</math>: From comparisons of measured differential Coulomb dissociation cross section of <math>^{35}\text{Al}</math> breaking up into <math>^{34}\text{Al}</math> in its g.s. and/or 46-keV isomer with theoretical calculations from the direct breakup model using the plane-wave approximation assuming the valence neutron at different orbitals. <b>2017Ch36</b> states that the differential CD cross section of <math>^{35}\text{Al} \rightarrow ^{34}\text{Al}+n</math> has been interpreted in the light of a direct breakup model, and it suggests that the possible ground-state spin and parity of <math>^{35}\text{Al}</math> could be, tentatively, 1/2<sup>+</sup> or 3/2<sup>+</sup> or 5/2<sup>+</sup>.</p> <p>Major configurations and spectroscopic factor for <math>J^\pi=5/2^+</math> of <math>^{35}\text{Al}</math> g.s.: (g.s., 4<sup>-</sup> in <math>^{34}\text{Al}</math>)<math>\otimes</math>vp<sub>3/2</sub>, S=0.36 9 (<b>2017Ch36</b>); (46 keV, 1<sup>+</sup> in <math>^{34}\text{Al}</math>)<math>\otimes</math>vd<sub>3/2</sub>, S=1.47 22 (<b>2017Ch36</b>); (1.4 MeV, 2<sup>+</sup> in <math>^{34}\text{Al}</math>)<math>\otimes</math>vs<sub>1/2</sub>, S=0.16 1 (<b>2021Bh12</b>); (2.5 MeV, 3<sup>-</sup> in <math>^{34}\text{Al}</math>)<math>\otimes</math>vp<sub>3/2</sub>, S=1.48 18 (<b>2021Bh12</b>).</p> <p>Other configurations for <math>J^\pi=1/2^+, 3/2^+</math> of <math>^{35}\text{Al}</math> g.s.: (g.s., 4<sup>-</sup> in <math>^{34}\text{Al}</math>)<math>\otimes</math>vf<sub>7/2</sub>, S=1.03 43 and (46 keV, 1<sup>+</sup> in <math>^{34}\text{Al}</math>)<math>\otimes</math>vs<sub>1/2</sub>, S=0.62 7; (46 keV, 1<sup>+</sup> in <math>^{34}\text{Al}</math>)<math>\otimes</math>vs<sub>1/2</sub>, S=0.72 8; (46 keV, 1<sup>+</sup> in <math>^{34}\text{Al}</math>)<math>\otimes</math>vs<sub>1/2</sub>, S=0.45 7 and (46 keV, 1<sup>+</sup> in <math>^{34}\text{Al}</math>)<math>\otimes</math>vd<sub>5/2</sub>, S=0.94 22.</p> |

**Coulomb excitation** [1999Ib01,2000PrZX](#)

[1999Ib01](#):  $^{197}\text{Au}(^{35}\text{Al}, ^{35}\text{Al}')$  Nuclei of interest were produced via the projectile fragmentation of a 70-MeV/nucleon,  $^{48}\text{Ca}$  primary beam from the NSCL K1200 cyclotron impinging on a 285-mg/cm<sup>2</sup>-thick  $^9\text{Be}$  target. The secondary cocktail beam was selected by the A1200 separator and impinged on a 532 mg/cm<sup>2</sup>  $^{197}\text{Au}$  target. The position and direction of each incident beam particle were measured using two upstream parallel-plate avalanche counters (PPAC). Scattered beam particles  $\theta_{\text{lab}} < 3.8^\circ$  were detected using a downstream position-sensitive plastic phoswich detector in coincidence with  $\gamma$  rays detected using an array of 38 cylindrical NaI(Tl) detectors centered around the  $^{197}\text{Au}$  target. Measured Doppler-corrected  $E_\gamma$ ,  $I_\gamma$ , and excitation cross sections. Deduced levels and E2 transition probabilities for  $^{35}\text{Al}$ ,  $^{37}\text{Si}$ ,  $^{39}\text{P}$ ,  $^{41}\text{S}$ ,  $^{43}\text{S}$ , and  $^{45}\text{Cl}$ .

[2000PrZX](#):  $^{197}\text{Au}(^{35}\text{Al}, ^{35}\text{Al}')$  The same experimental setup as [1999Ib01](#) with an 80-MeV/nucleon  $^{48}\text{Ca}$  primary beam. Scattered beam particles  $\theta_{\text{c.m.}} < 3.3^\circ$  were detected.

 $^{35}\text{Al}$  Levels

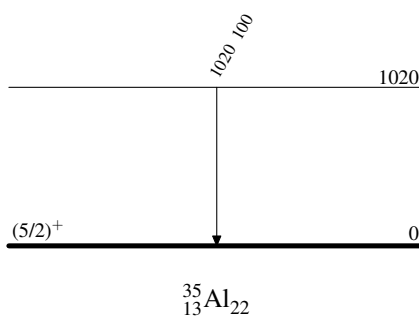
| <u>E(level)</u> | <u><math>J^\pi</math></u> | <u>Comments</u>                                                                                                                                                                                                                                                                                                                                |
|-----------------|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0               | $(5/2)^+$                 | $J^\pi$ : From the Adopted Levels.                                                                                                                                                                                                                                                                                                             |
| 1020 9          |                           | E(level): From measured $E_\gamma$ .<br>B(E2) $\uparrow$ =0.0142 52 from <a href="#">1999Ib01</a> , assuming E2 excitation mode. Others: B(E1) $\uparrow$ $\leq$ 0.00020 9,<br>B(E2) $\uparrow$ $\leq$ 0.0125 56, B(M1) $\uparrow$ $\leq$ 0.0024 11 ( <a href="#">2000PrZX</a> ).<br>Measured $\sigma$ =30 mb 14 ( <a href="#">2000PrZX</a> ). |

 $\gamma(^{35}\text{Al})$ 

| <u><math>E_i(\text{level})</math></u> | <u><math>E_\gamma</math></u> | <u><math>I_\gamma</math></u> | <u><math>E_f</math></u> | <u><math>J_f^\pi</math></u> | <u>Comments</u>                                                                                                                                                                                                       |
|---------------------------------------|------------------------------|------------------------------|-------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1020                                  | 1020 9                       | 100                          | 0                       | $(5/2)^+$                   | $E_\gamma$ : Weighted average of 1006 19 ( <a href="#">1999Ib01</a> ) and 1023 9 ( <a href="#">2000PrZX</a> ). <a href="#">2000PrZX</a> reports $E_\gamma$ =1023 8 in Table 4.9 and $E_\gamma$ =1023 9 in Table 4.10. |

**Coulomb excitation** [1999Ib01,2000PrZX](#)Level Scheme

Intensities: % photon branching from each level



Adopted Levels, Gammas

$Q(\beta^-)=10470$  40;  $S(n)=2470$  40;  $S(p)=18680$  40;  $Q(\alpha)=-13690$  40 2021Wa16

$S(2n)=10020$  40,  $S(2p)=33930$  40,  $Q(\beta^-n)=2090$  40 (2021Wa16).

Isotope discovery (2012Th10):  $^{232}\text{Th}(^{40}\text{Ar},X)$  at Dubna (1971Ar32).

$^{35}\text{Si}$  production:

2015Mo17:  $^9\text{Be}(^{40}\text{Ar},X)$  at  $E(^{40}\text{Ar})=95$  MeV/nucleon at RIKEN. Measured angular distributions and transverse momentum distributions of fragments. Deduced formulation for the width of transverse momentum distribution as a function of fragment velocity.

2012Kw02:  $E(^{40}\text{Ar})=140$  MeV/nucleon impinged on  $^9\text{Be}$ , natural nickel, and  $^{181}\text{Ta}$  targets at NSCL. Measured fragmentation cross sections, parallel momentum transfers, and widths. Compared with empirical formula EPAX, and predictions from internuclear cascade and deep inelastic models using Monte Carlo ISABEL-GEMINI and DIT-GEMINI codes.

2012Zh06:  $^9\text{Be}, ^{181}\text{Ta}(^{40}\text{Ar},X)$  at  $E(^{40}\text{Ar})=57$  MeV/nucleon at HIRFL. Measured momentum distributions and production cross sections of fragments. Observed competition between projectile fragmentation and other mechanisms. Compared with EPAX, abrasion-ablation, and HIPSE models. Studied target dependence of fragment cross sections.

2007No13:  $^9\text{Be}(^{40}\text{Ar},X)$  at  $E(^{40}\text{Ar})=100$  MeV/nucleon at RIKEN. Measured fragment momentum distributions and production cross sections.

2006Ro34:  $^2\text{H}(^{42}\text{S},X)$  at  $E(^{42}\text{S})=99.8$  MeV/nucleon at NSCL. Measured production cross sections.

1997Fo01:  $^{208}\text{Pb}(^{37}\text{Cl},X)$  at  $E(^{37}\text{Cl})=230$  MeV at Legnaro. Measured yields.

$^{35}\text{Si}$  decay measurements:

1986Du07, 1986HuZW, 1987DuZU, 1988DuZS, 1988DuZT:  $^9\text{Be}(^{40}\text{Ar},X)$  at GANIL. Measured  $T_{1/2}$  and  $\beta^-$ -delayed  $\gamma$  rays.

2007Ne14: Polarized  $^{35}\text{Si}$  from  $^9\text{Be}(^{36}\text{S},X)$  1n pickup at GANIL.  $^{35}\text{Si}$  g.s. magnetic moment and  $g$ -factor using  $\beta$ -NMR.

$^{35}\text{Si}$  radius measurements:

2006Kh08:  $^{35}\text{Si}$  produced by  $^{181}\text{Ta}(^{48}\text{Ca},X)$  fragmentation at  $E(^{48}\text{Ca})=60.3$  MeV/nucleon at GANIL. Measured energy-integrated reaction cross sections at 30-65 MeV/nucleon using a silicon telescope as both active target and detector. Deduced reduced strong absorption radii, isospin dependence, and possible halo structure or large deformation.

1999Ai02:  $\text{Si}(^{35}\text{Si},X)$  at NSCL. Measured energy-integrated reaction cross sections at  $E=38$ -80 MeV/ nucleon. Deduced strong absorption radii.

$^{35}\text{Si}$  mass measurements: 1986Fi06, 1986Sm05, 1984Ma49.

Theoretical calculations (binding energies, deformation, quadrupole moments, radii, levels,  $J^\pi$ , etc.): 2011Ka03, 2009No01, 2008Wi11, 2007Ch82, 2004Kh16, 1999Du05, 1994Mo37, 1994Po05, 1987Wa10, 1986Wo02.

 $^{35}\text{Si}$  LevelsCross Reference (XREF) Flags

|   |                                              |   |                                                        |
|---|----------------------------------------------|---|--------------------------------------------------------|
| A | $^{35}\text{Al}$ $\beta^-$ decay (38.1 ms)   | D | $^2\text{H}(^{34}\text{Si}, ^{35}\text{Si}\gamma)$     |
| B | $^{36}\text{Al}$ $\beta^-n$ decay (12.0 ms)  | E | $^9\text{Be}(^{36}\text{Si}, ^{35}\text{Si}\gamma)$    |
| C | $^{37}\text{Al}$ $\beta^-2n$ decay (11.4 ms) | F | $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{Si}\gamma)$ |

| $E(\text{level})^\dagger$ | $J^\pi$   | $T_{1/2}$ | XREF   | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------|-----------|-----------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0                         | $(7/2)^-$ | 0.78 s 12 | ABCDEF | $\% \beta^- = 100$ ; $\% \beta^- n < 5$ (1995ReZZ, 2008ReZZ)<br>$\mu = (-)1.639$ 4 (2007Ne14, 2019StZV)<br>$\mu$ : From original $(-)1.638$ 4 using $\beta$ -NMR (2007Ne14) with corrections by 2019StZV.<br>$J^\pi$ : $L=3$ from $0^+$ in $^2\text{H}(^{34}\text{Si}, ^{35}\text{Si})$ and $^9\text{Be}(^{36}\text{Si}, ^{35}\text{Si})$ and $\nu f_{7/2}$ configuration from shell model.<br>$T_{1/2}$ : From $\beta\gamma(t)$ (1988DuZS, 1988DuZT). Other: 0.87 s 17 by the same group (1986Du07).<br>Reduced strong absorption radius $r_0^2 = 1.261$ fm <sup>2</sup> 35 from the energy-integrated $\sigma$ of $\text{Si}(^{35}\text{Si}, X)$ (2006Kh08) and $r_0^2 = 1.258$ fm <sup>2</sup> 92 from the energy-integrated $\sigma$ of $\text{Si}(^{35}\text{Si}, X)$ (1999Ai02). |
| 909.95 23                 | $(3/2)^-$ | 55 ps 14  | ABCDEF | $J^\pi$ : $L=1$ from $0^+$ in $^2\text{H}(^{34}\text{Si}, ^{35}\text{Si})$ and $^9\text{Be}(^{36}\text{Si}, ^{35}\text{Si})$ and $\nu p_{3/2}$ configuration from shell model.<br>$T_{1/2}$ : From analysis of broadened line shapes in $^9\text{Be}(^{36}\text{Si}, ^{35}\text{Si}\gamma)$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

Continued on next page (footnotes at end of table)



**Adopted Levels, Gammas (continued)** $^{35}\text{Si}$  Levels (continued)

| E(level) <sup>†</sup> | J <sup>π</sup>                                                         | T <sub>1/2</sub> | XREF | Comments                                                                                                                                                                                                                                                                        |
|-----------------------|------------------------------------------------------------------------|------------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 973.88 18             | (3/2 <sup>+</sup> )                                                    | 5.9 ns 6         | A EF | J <sup>π</sup> : $\nu d_{3/2}$ configuration from shell model and 715 $\gamma$ from 1689, 1/2 <sup>+</sup> .<br>T <sub>1/2</sub> : From $\beta\gamma(t)$ in $^{35}\text{Al}$ $\beta^-$ decay.                                                                                   |
| 1689.4 28             | 1/2 <sup>+</sup>                                                       |                  | E    | J <sup>π</sup> : L=0 from 0 <sup>+</sup> in $^9\text{Be}(^{36}\text{Si}, ^{35}\text{Si})$ .                                                                                                                                                                                     |
| 1970 6                |                                                                        |                  | E    |                                                                                                                                                                                                                                                                                 |
| 2044 5                | (1/2) <sup>-</sup>                                                     |                  | DE   | J <sup>π</sup> : L=1 from 0 <sup>+</sup> in $^2\text{H}(^{34}\text{Si}, ^{35}\text{Si})$ and $\nu p_{1/2}$ configuration from shell model.                                                                                                                                      |
| 2168.2 4              | (5/2 <sup>+</sup> )                                                    |                  | A E  | J <sup>π</sup> : From shell model and its possible isobaric analog state in $^{35}\text{P}$ with L=2 from 0 <sup>+</sup> in $^1\text{H}(^{34}\text{Si}, p)$ from R-matrix analysis in 2012Im01.<br>L=2,3 from 0 <sup>+</sup> in $^9\text{Be}(^{36}\text{Si}, ^{35}\text{Si})$ . |
| 2275 6                |                                                                        |                  | E    |                                                                                                                                                                                                                                                                                 |
| 2377 7                |                                                                        |                  | E    |                                                                                                                                                                                                                                                                                 |
| 3140                  | (3/2 <sup>+</sup> , 5/2 <sup>+</sup> , 7/2 <sup>+</sup> ) <sup>‡</sup> |                  | A    |                                                                                                                                                                                                                                                                                 |
| 3450                  | (3/2 <sup>+</sup> , 5/2 <sup>+</sup> , 7/2 <sup>+</sup> ) <sup>‡</sup> |                  | A    |                                                                                                                                                                                                                                                                                 |
| 3611? 8               |                                                                        |                  | E    | XREF: E(?)<br>Level exists assuming that the 3611 $\gamma$ feeds the ground state of $^{35}\text{Si}$ (2014St18).                                                                                                                                                               |
| 3770                  | (3/2 <sup>+</sup> , 5/2 <sup>+</sup> , 7/2 <sup>+</sup> ) <sup>‡</sup> |                  | A    |                                                                                                                                                                                                                                                                                 |
| 5190                  | (3/2 <sup>+</sup> , 5/2 <sup>+</sup> , 7/2 <sup>+</sup> ) <sup>‡</sup> |                  | A    |                                                                                                                                                                                                                                                                                 |
| ≈5500                 | (5/2) <sup>-</sup>                                                     |                  | D    | E(level): A broad level at ≈5500 deduced from E <sub>p</sub> in $^2\text{H}(^{34}\text{Si}, ^{35}\text{Si})$ .<br>J <sup>π</sup> : L=3 from 0 <sup>+</sup> in $^2\text{H}(^{34}\text{Si}, ^{35}\text{Si})$ and $\nu f_{5/2}$ configuration from shell model.                    |
| 5760                  | (3/2 <sup>+</sup> , 5/2 <sup>+</sup> , 7/2 <sup>+</sup> ) <sup>‡</sup> |                  | A    |                                                                                                                                                                                                                                                                                 |
| 6330                  | (3/2 <sup>+</sup> , 5/2 <sup>+</sup> , 7/2 <sup>+</sup> ) <sup>‡</sup> |                  | A    |                                                                                                                                                                                                                                                                                 |
| 7360                  | (3/2 <sup>+</sup> , 5/2 <sup>+</sup> , 7/2 <sup>+</sup> ) <sup>‡</sup> |                  | A    |                                                                                                                                                                                                                                                                                 |
| 7690                  | (3/2 <sup>+</sup> , 5/2 <sup>+</sup> , 7/2 <sup>+</sup> ) <sup>‡</sup> |                  | A    |                                                                                                                                                                                                                                                                                 |

<sup>†</sup> From a least-squares fit to  $\gamma$ -ray energies for levels connected with  $\gamma$  transitions; from  $^{35}\text{Al}$   $\beta^-$ -delayed neutron decays for other levels, unless otherwise noted.

<sup>‡</sup> Possible allowed  $\beta^-$  feeding from (5/2)<sup>+</sup> parent in  $^{35}\text{Al}$ .

 $\gamma(^{35}\text{Si})$ 

| E <sub>i</sub> (level) | J <sup>π</sup> <sub>i</sub> | E <sub><math>\gamma</math></sub> <sup>†</sup> | I <sub><math>\gamma</math></sub> <sup>†</sup> | E <sub>f</sub>   | J <sup>π</sup> <sub>f</sub>               | Mult.        | $\alpha^{\#}$                       | Comments                                                                                                                                                                                                                                                                  |
|------------------------|-----------------------------|-----------------------------------------------|-----------------------------------------------|------------------|-------------------------------------------|--------------|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 909.95                 | (3/2) <sup>-</sup>          | 910.11 30                                     | 100                                           | 0                | (7/2) <sup>-</sup>                        | [E2]         | 4.13×10 <sup>-5</sup> 6             | B(E2)(W.u.)=2.4 +8-5<br>E <sub><math>\gamma</math></sub> : others: 910 3 from $^2\text{H}(^{34}\text{Si}, ^{35}\text{Si}\gamma)$ , 908 4 from $^9\text{Be}(^{36}\text{Si}, ^{35}\text{Si}\gamma)$ , and 910 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{Si}\gamma)$ . |
| 973.88                 | (3/2 <sup>+</sup> )         | 64.1 3<br>973.78 20                           | 100<br>11.8 24                                | 909.95<br>0      | (3/2) <sup>-</sup><br>(7/2) <sup>-</sup>  | [E1]<br>[M2] | 0.0368 8<br>5.05×10 <sup>-5</sup> 7 | B(E1)(W.u.)=3.52×10 <sup>-4</sup> +41-34<br>B(M2)(W.u.)=0.057 +13-12<br>E <sub><math>\gamma</math></sub> : other: 974 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{Si}\gamma)$ .                                                                                       |
| 1689.4                 | 1/2 <sup>+</sup>            | 715 <sup>‡</sup> 4<br>780 <sup>‡</sup> 4      | 14.6 <sup>‡</sup> 16<br>100 <sup>‡</sup> 8    | 973.88<br>909.95 | (3/2 <sup>+</sup> )<br>(3/2) <sup>-</sup> |              |                                     |                                                                                                                                                                                                                                                                           |
| 1970                   |                             | 1970 <sup>‡</sup> 6                           | 100                                           | 0                | (7/2) <sup>-</sup>                        |              |                                     |                                                                                                                                                                                                                                                                           |
| 2044                   | (1/2) <sup>-</sup>          | 1134 <sup>‡</sup> 5                           | 100                                           | 909.95           | (3/2) <sup>-</sup>                        |              |                                     | E <sub><math>\gamma</math></sub> : Other: 1134 6 from $^2\text{H}(^{34}\text{Si}, ^{35}\text{Si}\gamma)$ .                                                                                                                                                                |
| 2168.2                 | (5/2 <sup>+</sup> )         | 1194.2 4                                      | 35 8                                          | 973.88           | (3/2 <sup>+</sup> )                       |              |                                     |                                                                                                                                                                                                                                                                           |

Continued on next page (footnotes at end of table)

**Adopted Levels, Gammas (continued)** $\gamma(^{35}\text{Si})$  (continued)

| $E_i(\text{level})$ | $E_\gamma^\dagger$ | $I_\gamma^\dagger$ | $E_f$ | $J_f^\pi$ | Comments                                                                                                                                                                                                                                                                                                                                                 |
|---------------------|--------------------|--------------------|-------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2168.2              | 2168.2 $6^\dagger$ | 100 20             | 0     | $(7/2)^-$ | $E_\gamma$ : Other: 2164 6 from $^9\text{Be}(^{36}\text{Si}, ^{35}\text{Si}\gamma)$ .                                                                                                                                                                                                                                                                    |
| 2275                | 2275 $6^\ddagger$  | 100                | 0     | $(7/2)^-$ |                                                                                                                                                                                                                                                                                                                                                          |
| 2377                | 2377 $7^\ddagger$  | 100                | 0     | $(7/2)^-$ |                                                                                                                                                                                                                                                                                                                                                          |
| 3611?               | 3611 $8^\ddagger@$ | 100                | 0     | $(7/2)^-$ | <a href="#">2014St18</a> observed this surprising $\gamma$ transition at 3611 keV, which must depopulate a state that is unbound to neutron emission by at least 1.1 MeV, considering $S(n)=2470$ 40 ( <a href="#">2021Wa16</a> ). <a href="#">2014St18</a> proposes two possible explanations involving hindered neutron emission or multistep process. |

$^\dagger$  From  $^{35}\text{Al}$   $\beta^-$  decay, unless otherwise noted.

$^\ddagger$  From  $^9\text{Be}(^{36}\text{Si}, ^{35}\text{Si}\gamma)$ .

# Total theoretical internal conversion coefficients, calculated using the BrIcc code ([2008Ki07](#)) with “Frozen Orbitals” approximation based on  $\gamma$ -ray energies, assigned multipolarities, and mixing ratios, unless otherwise specified.

@ Placement of transition in the level scheme is uncertain.

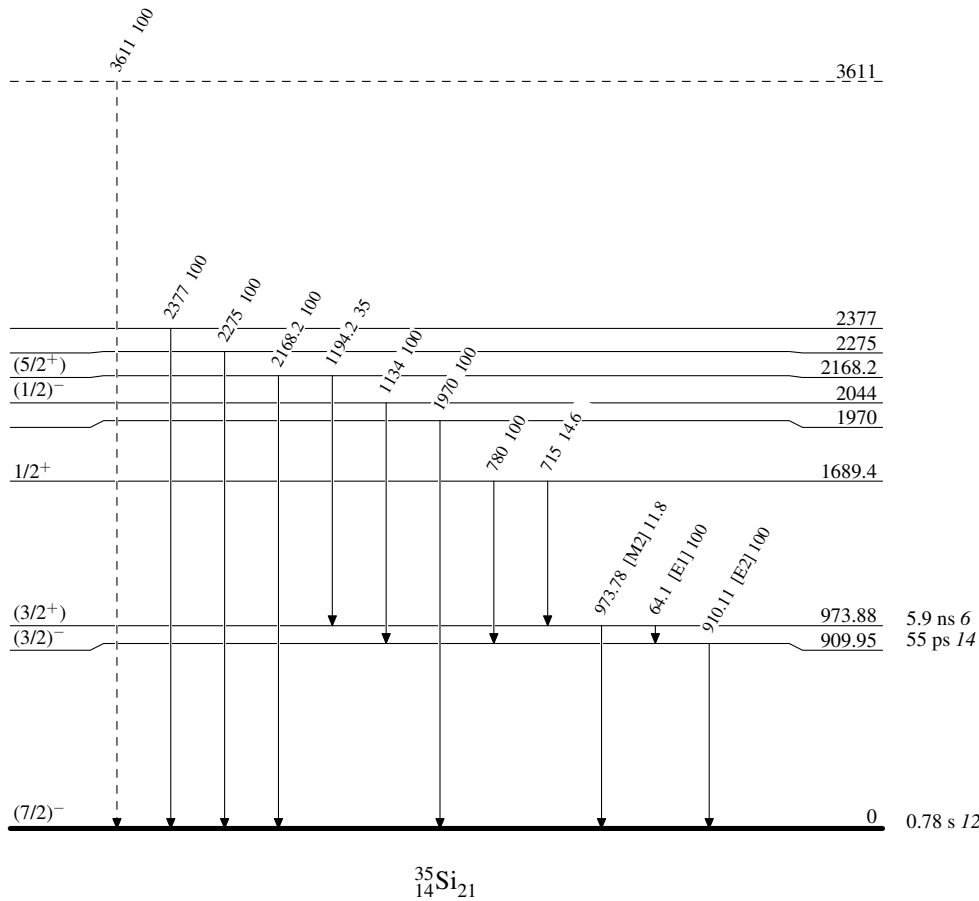
Adopted Levels, Gammas

Legend

Level Scheme

Intensities: Relative photon branching from each level

-----► γ Decay (Uncertain)



**$^{35}\text{Al}$   $\beta^-$  decay (38.1 ms)    2005Ti11,2001Nu01**

Parent:  $^{35}\text{Al}$ :  $E=0$ ;  $J^\pi=(5/2)^+$ ;  $T_{1/2}=38.1$  ms 4;  $Q(\beta^-)=14170$  40;  $\% \beta^-$  decay=100

$^{35}\text{Al}$ - $J^\pi$ ,  $T_{1/2}$ : From the Adopted Levels of  $^{35}\text{Al}$ .

$^{35}\text{Al}$ - $Q(\beta^-)$ : From 2021Wa16.

2005Ti11,2006AnZW: A  $^{35}\text{Al}$  secondary beam at  $\approx 2$  pps was produced via the fragmentation of a 78-MeV/nucleon  $^{36}\text{S}$  primary beam and selected by the LISE3 spectrometer at GANIL. A total of  $3.46 \times 10^5$   $^{35}\text{Al}$  ions were continuously implanted into an NE102A plastic scintillator also for detecting  $\beta$ . The implantation detector was sandwiched between two silicon detectors for monitoring beam and for veto, respectively. Neutrons were detected using the TONNERRE array consisting of 19 plastic scintillator modules.  $\gamma$  rays were detected using two EXOGAM clover modules and a LEPS detector. Measured  $E_\gamma$ ,  $I_\gamma$ ,  $E_n$ ,  $I_n$ ,  $\beta\gamma$ -coin,  $\beta n$ -coin, and  $\beta n\gamma$ -coin. Deduced the decay scheme consisting of  $^{35}\text{Si}$  and  $^{34}\text{Si}$  levels,  $^{35}\text{Al}$   $T_{1/2}$ , decay branching ratios,  $\log ft$ ,  $B(\text{GT})$ , and  $\beta$ -delayed neutron emission probability. Comparisons with shell-model calculations.

2001Nu01,2002Nu02: Exp 1: A  $^{35}\text{Al}$  secondary beam at 8 pps was produced via the fragmentation of a UC target with 1.4 GeV protons at ISOLDE, CERN with subsequent surface-ionization and mass separation.  $^{35}\text{Al}$  ions were collected onto a moving tape.  $\beta$  particles were detected using a thin cylindrical plastic scintillator,  $\gamma$  rays were detected using two Ge detectors, and neutrons were detected using eight low-threshold plastic scintillators. Measured  $E_\gamma$ ,  $I_\gamma$ ,  $E_n$ ,  $I_n$ ,  $\beta\gamma$ -coin,  $\gamma\gamma$ -coin,  $\beta n$ -coin. Deduced the decay scheme consisting of  $^{35}\text{Si}$  and  $^{34}\text{Si}$  levels,  $^{35}\text{Al}$   $T_{1/2}$ , decay branching ratios,  $\log ft$ , and  $\beta$ -delayed neutron emission probability. Comparisons with shell-model calculations. Exp 2: A lifetime measurement for the 974-keV level in  $^{35}\text{Si}$  used a thin plastic scintillator for detecting  $\beta$  and a  $\text{BaF}_2$  detector for detecting  $\gamma$ .

Other experimental studies on  $^{35}\text{Al}$   $T_{1/2}$  and  $\beta$ -delayed neutron emission probability: 2017Ha23, 1999YoZW,

1995ReZZ/2008ReZZ, 1989MuZU, 1989Le16, 1988MuZY, 1988Mu08, 1988DuZT, 1988BaYZ, 1987DuZU, 1987BaZI.

Theoretical studies involving  $^{35}\text{Al}$  decay: 2018Yo06, 2013Li39.

 **$^{35}\text{Si}$  Levels**

$E(n)$  under comments are deduced from  $E(n)_{\text{c.m.}} = E(\text{level})(^{35}\text{Si}) - S(n)(^{35}\text{Si}) - E(\text{level})(^{34}\text{Si})$ , with  $E(\text{level})(^{35}\text{Si})$  reported in 2005Ti11 based on the neutron time-of-flight spectrum. 2005Ti11 used  $E(\text{level})(^{34}\text{Si})=3326$  for  $E(n1)$ . 2005Ti11 used  $S(n)(^{35}\text{Si})=2474$  43, consistent with  $S(n)(^{35}\text{Si})=2470$  40 from 2021Wa16.

| $E(\text{level})^\dagger$ | $J^\pi^\ddagger$        | $T_{1/2}^\ddagger$ | Comments                                                                                                                                                                                                                          |
|---------------------------|-------------------------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0                         | $(7/2)^-$               | 0.78 s 12          |                                                                                                                                                                                                                                   |
| 909.95 23                 | $(3/2)^-$               | 55 ps 14           | 2001Nu01 reports a $\%I\beta^- < 0.9$ from 2001Nu01 without considering the conversion electron of the transition from 974 level to 910 level. The net feeding is deduced to be $-1.9$ 9 from the $\gamma$ +ce intensity balance. |
| 973.88 18                 | $(3/2^+)$               | 5.9 ns 6           | $T_{1/2}$ : lifetime=8.5 ns 9 from $\beta\gamma(t)$ in 2001Nu01, also adopted in the Adopted Levels.                                                                                                                              |
| 2168.16 36                | $(5/2^+)$               |                    |                                                                                                                                                                                                                                   |
| 3140                      | $(3/2^+, 5/2^+, 7/2^+)$ |                    | Deduced $E(n0)_{\text{c.m.}}=666$ .                                                                                                                                                                                               |
| 3450                      | $(3/2^+, 5/2^+, 7/2^+)$ |                    | Deduced $E(n0)_{\text{c.m.}}=976$ .                                                                                                                                                                                               |
| 3770                      | $(3/2^+, 5/2^+, 7/2^+)$ |                    | Deduced $E(n0)_{\text{c.m.}}=1296$ .                                                                                                                                                                                              |
| 5190                      | $(3/2^+, 5/2^+, 7/2^+)$ |                    | Deduced $E(n0)_{\text{c.m.}}=2716$ .                                                                                                                                                                                              |
| 5760                      | $(3/2^+, 5/2^+, 7/2^+)$ |                    | Deduced $E(n0)_{\text{c.m.}}=3286$ .                                                                                                                                                                                              |
| 6330                      | $(3/2^+, 5/2^+, 7/2^+)$ |                    | Deduced $E(n0)_{\text{c.m.}}=3856$ ; $E(n1)_{\text{c.m.}}=530$ .                                                                                                                                                                  |
| 7360                      | $(3/2^+, 5/2^+, 7/2^+)$ |                    | Deduced $E(n1)_{\text{c.m.}}=1560$ .                                                                                                                                                                                              |
| 7690                      | $(3/2^+, 5/2^+, 7/2^+)$ |                    | Deduced $E(n1)_{\text{c.m.}}=1890$ .                                                                                                                                                                                              |

$^\dagger$  From a least-squares fit to  $\gamma$ -ray energies for levels connected with  $\gamma$  transitions. For unbound levels, from 2005Ti11 based on the neutron time-of-flight spectrum.

$^\ddagger$  From the Adopted Levels of  $^{35}\text{Si}$ .

$^{35}\text{Al}$   $\beta^-$  decay (38.1 ms) [2005Ti11](#), [2001Nu01](#) (continued) $\beta^-$  radiations

The decay scheme is incomplete due to possible unobserved levels in a large gap of about 6.5 MeV between Q-value=14.17 MeV and the highest populated level at E=7.69 MeV.

| E(decay)                  | E(level) | $I\beta^-$ <sup>†‡</sup> | Log $ft$ <sup>‡</sup> | Comments                                                                                                                                                                                                                                                                                                                                     |
|---------------------------|----------|--------------------------|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (6.48×10 <sup>3</sup> 4)  | 7690     | 2.7 2                    | 4.5                   |                                                                                                                                                                                                                                                                                                                                              |
| (6.81×10 <sup>3</sup> 4)  | 7360     | 2.6 2                    | 4.6                   |                                                                                                                                                                                                                                                                                                                                              |
| (7.84×10 <sup>3</sup> 4)  | 6330     | 6.8 3                    | 4.5                   |                                                                                                                                                                                                                                                                                                                                              |
| (8.41×10 <sup>3</sup> 4)  | 5760     | 4.5 2                    | 4.8                   |                                                                                                                                                                                                                                                                                                                                              |
| (8.98×10 <sup>3</sup> 4)  | 5190     | 8.9 3                    | 4.6                   |                                                                                                                                                                                                                                                                                                                                              |
| (1.040×10 <sup>4</sup> 4) | 3770     | 3.2 2                    | 5.4                   |                                                                                                                                                                                                                                                                                                                                              |
| (1.072×10 <sup>4</sup> 4) | 3450     | 6.0 3                    | 5.2                   |                                                                                                                                                                                                                                                                                                                                              |
| (1.103×10 <sup>4</sup> 4) | 3140     | 3.3 2                    | 5.5                   |                                                                                                                                                                                                                                                                                                                                              |
| (1.200×10 <sup>4</sup> 4) | 2168.16  | 9.5                      | 5.2                   | $I\beta^-$ : 9.2 19 from <a href="#">2001Nu01</a> ; 6.7 9 from <a href="#">2005Ti11</a> with only one branch 2168 $\gamma$ observed.                                                                                                                                                                                                         |
| (1.320×10 <sup>4</sup> 4) | 973.88   | 52                       | 4.7                   | $I\beta^-$ : 48 9 from <a href="#">2001Nu01</a> ; 50 3 from <a href="#">2005Ti11</a> .                                                                                                                                                                                                                                                       |
| (1.417×10 <sup>4</sup> 4) | 0        | 3.0 10                   | 6.1 2                 | $I\beta^-$ : from <a href="#">2001Nu01</a> , which states that the $\beta$ branch to the $^{35}\text{Si}$ ground state was evaluated by comparing the total $\gamma$ intensity due to the deexcitation of excited states of $^{35}\text{Si}$ with the decay of $^{35}\text{Si}$ activity and assuming no direct $^{35}\text{Si}$ production. |

<sup>†</sup> For the two bound excited states,  $I\beta^-$  is from the  $\gamma$ +ce intensity balance at each state. Quoted  $I\beta^-$  values without uncertainties are considered upper limits due to the incomplete decay scheme, and the associated log  $ft$  values are considered lower limits. For unbound excited states (>2470),  $I\beta^-$  is from the absolute  $I\beta^-$  in [2005Ti11](#) based on neutron intensities.

<sup>‡</sup> For unbound levels, [2005Ti11](#) did not report uncertainties for neutron energies or level energies. The evaluators estimate the uncertainties of log  $ft$  are <0.1, assuming a 200-keV uncertainty in level energies.

<sup>#</sup> Absolute intensity per 100 decays.

 $\gamma(^{35}\text{Si})$ 

$I\gamma$  normalization: 0.47 4 from  $\Sigma I(\gamma\text{+ce to g.s.})=60$  4, deduced from  $100-\% \beta^- \text{ n}-\% I\beta^- (\text{g.s.})$ , where  $\% \beta^- \text{ n}=37$  4 from the Adopted Levels of  $^{35}\text{Al}$  and  $\% I\beta^- (\text{g.s.})=3.0$  10 from [2001Nu01](#). The deduced normalization factor should be considered an upper limit due to potential missing  $\gamma$  transitions from unobserved levels in the gap to the ground state. Other: 0.45 from [2001Nu01](#), based on  $\% \beta^- \text{ n}=41$  13.

| $E_\gamma$ <sup>†</sup> | $I_\gamma$ <sup>†‡</sup> | $E_i(\text{level})$ | $J_i^\pi$           | $E_f$  | $J_f^\pi$           | Mult. | $\alpha$ <sup>#</sup>   | Comments                                                                                                                                                                                                              |
|-------------------------|--------------------------|---------------------|---------------------|--------|---------------------|-------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 64.1 3                  | 100                      | 973.88              | (3/2 <sup>+</sup> ) | 909.95 | (3/2) <sup>-</sup>  | [E1]  | 0.0368 8                | $\% I\gamma=47$                                                                                                                                                                                                       |
| 910.11 30               | 99.7 19                  | 909.95              | (3/2) <sup>-</sup>  | 0      | (7/2) <sup>-</sup>  | [E2]  | 4.13×10 <sup>-5</sup> 6 | $\% I\gamma=47$                                                                                                                                                                                                       |
| 973.78 20               | 11.8 24                  | 973.88              | (3/2 <sup>+</sup> ) | 0      | (7/2) <sup>-</sup>  | [M2]  | 5.05×10 <sup>-5</sup> 7 | $\% I\gamma=5.6$                                                                                                                                                                                                      |
| <sup>x</sup> 1130.4 4   | 3.2 9                    |                     |                     |        |                     |       |                         | $\% I\gamma=1.5$<br><a href="#">2014St18</a> suggests this $\gamma$ is likely the 1134 $\gamma$ from 2042 to 908 transition observed in the $^{36}\text{Si} \rightarrow \text{n} + ^{35}\text{Si}$ knockout reaction. |
| 1194.2 4                | 5.3 12                   | 2168.16             | (5/2 <sup>+</sup> ) | 973.88 | (3/2 <sup>+</sup> ) |       |                         | $\% I\gamma=2.5$                                                                                                                                                                                                      |
| 2168.2 6                | 15 3                     | 2168.16             | (5/2 <sup>+</sup> ) | 0      | (7/2) <sup>-</sup>  |       |                         | $\% I\gamma=7.1$                                                                                                                                                                                                      |
| <sup>x</sup> 5629 3     | 2.4 12                   |                     |                     |        |                     |       |                         | $\% I\gamma=1.1$                                                                                                                                                                                                      |

<sup>†</sup> From [2001Nu01](#).

<sup>‡</sup> For absolute intensity per 100 decays, multiply by 0.47.

Continued on next page (footnotes at end of table)

---

$^{35}\text{Al} \beta^-$  decay (38.1 ms)    [2005Ti11](#), [2001Nu01](#) (continued)

---

$\gamma(^{35}\text{Si})$  (continued)

# Total theoretical internal conversion coefficients, calculated using the BrIcc code ([2008Ki07](#)) with “Frozen Orbitals” approximation based on  $\gamma$ -ray energies, assigned multipolarities, and mixing ratios, unless otherwise specified.

<sup>x</sup>  $\gamma$  ray not placed in level scheme.

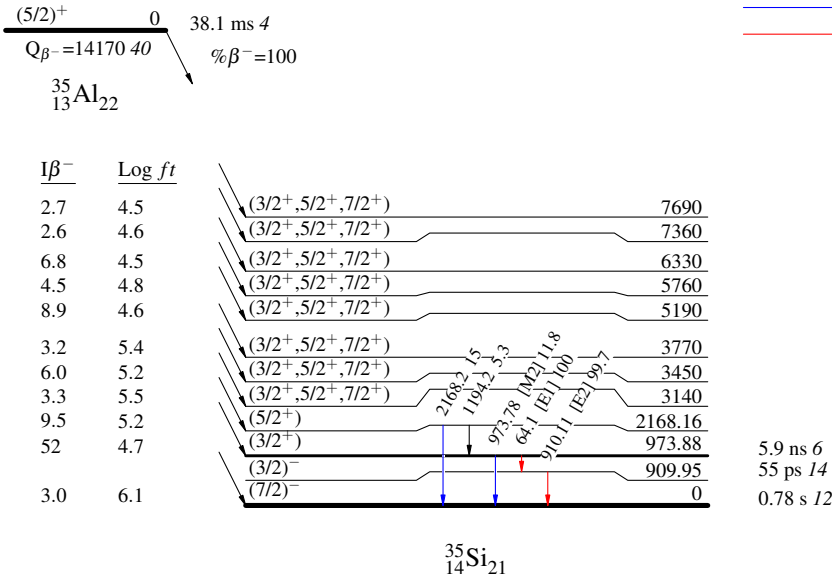
<sup>35</sup>Al β<sup>-</sup> decay (38.1 ms) 2005Ti11,2001Nu01

Decay Scheme

Intensities: Relative I<sub>γ</sub>

Legend

- I<sub>γ</sub> < 2% × I<sub>γ</sub><sup>max</sup>
- I<sub>γ</sub> < 10% × I<sub>γ</sub><sup>max</sup>
- I<sub>γ</sub> > 10% × I<sub>γ</sub><sup>max</sup>



$^{36}\text{Al} \beta^- \text{n} \text{ decay (12.0 ms) } 2023\text{Lu07}$ 

Parent:  $^{36}\text{Al}$ :  $E=0$ ;  $T_{1/2}=12.0 \text{ ms } 20$ ;  $Q(\beta^- \text{n})=1.227 \times 10^4 \text{ 15}$ ;  $\% \beta^- \text{n} \text{ decay} < 31$

$^{36}\text{Al}$ - $T_{1/2}$ : From implant- $\beta\gamma$  correlation ([2023Lu07](#)). Other: 14.7 ms *10* ([2023Lu07](#), implant- $\beta$  correlation), 94 ms *37* ([1995ReZZ/2008ReZZ](#), implant- $\beta$  correlation),  $\approx 15 \text{ ms}$  ([1999YoZW](#), implant- $\beta$  correlation, preliminary). [2023Lu07](#) adopted 12.0 ms *20* instead of 14.7 ms *10* based on the cleanest determination with the  $\gamma$ -ray gating.

$^{36}\text{Al}$ - $Q(\beta^- \text{n})$ : From [2021Wa16](#).

$^{36}\text{Al}$ - $\% \beta^- \text{n} \text{ decay}$ : From [1995ReZZ/2008ReZZ](#).

[2023Lu07](#): Exp 1:  $^{36}\text{Mg}$  and  $^{36}\text{Al}$  were produced via the projectile fragmentation of a 140-MeV/nucleon, 80-pnA  $^{48}\text{Ca}$  primary beam from the NSCL cyclotrons impinging on a 642-mg/cm<sup>2</sup>-thick  $^9\text{Be}$  target. The secondary cocktail beam centered around  $^{33}\text{Na}$  was selected by the A1900 separator and implanted into a CeBr<sub>3</sub> scintillator sandwiched between two plastic scintillator veto detectors. Surrounding the implantation array were the SeGA array of 16 segmented Ge detectors and 15 LaBr<sub>3</sub> detectors. Exp 2:  $^{36}\text{Mg}$  and  $^{36}\text{Al}$  were produced via the projectile fragmentation of a 172.3-MeV/nucleon, 120-pnA  $^{48}\text{Ca}$  primary beam from the FRIB linac impinging on an 8.89-mm-thick  $^9\text{Be}$  target. The secondary cocktail beam centered around  $^{42}\text{Si}$  was selected by the ARIS separator and implanted into a 5-mm-thick YSO segmented scintillator sandwiched between two plastic scintillator veto detectors. Surrounding the implantation array were 11 HPGe clover detectors and 15 fast-timing LaBr<sub>3</sub> detectors, and the VANDLE array of 88 neutron detectors. Both the NSCL and FRIB experiments in [2023Lu07](#) measured  $E_\gamma$ ,  $I_\gamma$ ,  $\beta\gamma$ -coin,  $\gamma\gamma$ -coin, implant- $\beta\gamma$  correlation and deduced  $T_{1/2}$  of  $^{36}\text{Mg}$  g.s.,  $^{36}\text{Al}$  g.s. and a  $^{36}\text{Al}$  isomer. Comparisons with FSU shell-model calculations.

 $^{35}\text{Si}$  Levels

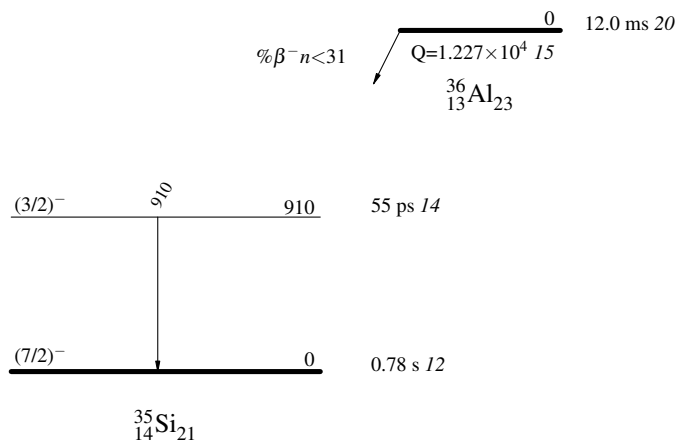
| <u><math>E(\text{level})</math></u> | <u><math>J^\pi^\dagger</math></u> | <u><math>T_{1/2}^\dagger</math></u> |
|-------------------------------------|-----------------------------------|-------------------------------------|
| 0                                   | $(7/2)^-$                         | 0.78 s <i>12</i>                    |
| 910                                 | $(3/2)^-$                         | 55 ps <i>14</i>                     |

<sup>†</sup> From the Adopted Levels of  $^{35}\text{Si}$ .

 $\gamma(^{35}\text{Si})$ 

| <u><math>E_\gamma</math></u> | <u><math>E_i(\text{level})</math></u> | <u><math>J_i^\pi</math></u> | <u><math>E_f</math></u> | <u><math>J_f^\pi</math></u> | <u>Comments</u>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------|---------------------------------------|-----------------------------|-------------------------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 910                          | 910                                   | $(3/2)^-$                   | 0                       | $(7/2)^-$                   | <p><math>E_\gamma</math>: From <a href="#">2023Lu07</a>.</p> <p><a href="#">2023Lu07</a> observed the 910<math>\gamma</math> within the 100-ms window following the arrival of a <math>^{36}\text{Al}</math> ion in both NSCL and FRIB experiments. <a href="#">2023Lu07</a> also observed the 910<math>\gamma</math> within the 100-ms window following the arrival of a <math>^{36}\text{Mg}</math> ion in both NSCL and FRIB experiments.</p> <p><a href="#">2013StZY</a> observed the 910<math>\gamma</math> in <math>^{36}\text{Mg}</math> decay with its maximum intensity in the 20-30 ms time window after the implantation of <math>^{36}\text{Mg}</math>, indicating its origin from the granddaughter generation.</p> |



$^{36}\text{Al}$   $\beta^-n$  decay (12.0 ms)    2023Lu07Decay Scheme

$^{37}\text{Al}$   $\beta^-2n$  decay (11.4 ms)    [2013StZY](#)

Parent:  $^{37}\text{Al}$ :  $E=0$ ;  $J^\pi=(5/2^+)$ ;  $T_{1/2}=11.4$  ms *4*;  $Q(\beta^-2n)=8.06\times 10^3$  *18*;  $\% \beta^-2n$  decay  $\geq 000$

$^{37}\text{Al}$ - $J^\pi$ : From shell-model predictions ([2013StZY](#)).

$^{37}\text{Al}$ - $T_{1/2}$ : Weighted average of 10.7 ms *13* ([2004Gr20,2003Gr22](#), implant- $\beta$  correlation), 11.5 ms *4* ([2013StZY](#), implant- $\beta\gamma$  correlation). Other: 11.8 ms *1* (stat) +22–34 (syst) ([2013StZY](#), implant- $\beta$  correlation).

$^{37}\text{Al}$ - $Q(\beta^-2n)$ : Deduced from mass excesses in [2021Wa16](#).

$^{37}\text{Al}$ - $\% \beta^-2n$  decay:  $\% \beta^-2n \geq 1$  *1* ([2013StZY](#)).

[2013StZY](#):  $^{37}\text{Al}$  was produced via the projectile fragmentation of a 345-MeV/nucleon, 70-pnA  $^{48}\text{Ca}^{20+}$  primary beam from the linear accelerator RILAC and the four cyclotrons RRC, fRC, IRC, and SRC at RIKEN impinging on an 15-mm-thick  $^9\text{Be}$  target. The secondary cocktail beam was selected by the BigRIPS separator and the zero-degree spectrometer (ZDS) using the  $B\rho$ - $\Delta E$ -ToF method, and implanted into the Cylindrical Active Implantation Target for Exotic Nuclei (CAITEN) consisting of a segmented movable hollow-cylindrical-shaped plastic scintillator and a stationary ring of 24 position-sensitive photomultiplier tubes (PSPMTs) arranged on a ring inside the scintillator at the height of the beam line. To reduce background buildup, the scintillator barrel was rotated quickly and axially-moved slowly in vertical direction, resulting in a helix-shaped motion.  $\beta$  particles were detected by the CAITEN and  $\gamma$  rays were detected using three HPGe clover detectors. Measured  $E_\gamma$ ,  $\beta\gamma$ -coin, and implant- $\beta$  correlation, and deduced  $T_{1/2}$ . Comparisons with QRPA and shell-model calculations.

 $^{35}\text{Si}$  Levels

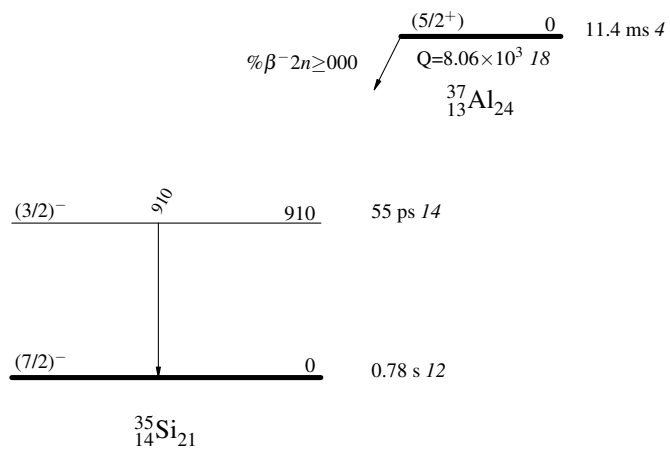
| <u><math>E(\text{level})</math></u> | <u><math>J^\pi^\dagger</math></u> | <u><math>T_{1/2}^\dagger</math></u> |
|-------------------------------------|-----------------------------------|-------------------------------------|
| 0                                   | $(7/2)^-$                         | 0.78 s <i>12</i>                    |
| 910                                 | $(3/2)^-$                         | 55 ps <i>14</i>                     |

$^\dagger$  From the Adopted Levels of  $^{35}\text{Si}$ .

 $\gamma(^{35}\text{Si})$ 

| <u><math>E_\gamma</math></u> | <u><math>E_i(\text{level})</math></u> | <u><math>J_i^\pi</math></u> | <u><math>E_f</math></u> | <u><math>J_f^\pi</math></u> | <u>Comments</u>                                                                                                                                           |
|------------------------------|---------------------------------------|-----------------------------|-------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 910                          | 910                                   | $(3/2)^-$                   | 0                       | $(7/2)^-$                   | $I_\gamma$ : <a href="#">2013StZY</a> reports 2 <i>3</i> relative to $I_\gamma=100$ for the 156 $\gamma$ in $^{37}\text{Si}$ from $^{37}\text{Al}$ decay. |

---

 $^{37}\text{Al}$   $\beta^- 2n$  decay (11.4 ms) 2013StZYDecay Scheme

$^2\text{H}(^{34}\text{Si}, ^{35}\text{Si}\gamma)$  **2014Bu01**

$^{34}\text{Si}(\text{d},\text{p})^{35}\text{Si}$  from  $J^\pi=0^+$   $^{34}\text{Si}$  g.s. in inverse kinematics.

**2014Bu01**: A 20.5-MeV/nucleon,  $1.1 \times 10^5$  pps, and 95% pure  $^{34}\text{Si}$  secondary beam was produced via the fragmentation of a 55-MeV/nucleon  $^{36}\text{S}^{16+}$  primary beam impinging on a 1075- $\mu\text{m}$ -thick Be target, separated by the LISE3 spectrometer at GANIL, and incident on a 2.6- mg/cm<sup>2</sup>  $^1\text{CD}_2$  secondary target. Incoming beam ions were tracked using two position-sensitive multiwire proportional chambers upstream of the target. Outgoing ions were identified using an ionization chamber (IC) downstream of target for energy loss and a plastic scintillator behind the IC for TOF measurements. Protons from the (d,p) reaction were detected using four modules of the MUST2 array placed 10 cm from the target covering polar angles ranging from 105° to 150° with respect to the beam direction and a 16-strip annular Si detector at a distance of 11.3 cm covering polar angles from 156° to 168°.  $\gamma$  rays were detected using four segmented Ge detectors from the EXOGAM array perpendicular to the beam axis at a mean distance of 5 cm, and 9 cm downstream from the target with efficiency  $\varepsilon=3.8\%$  at 1 MeV. Measured  $\sigma(E_p, \theta)$ , Doppler-corrected  $E_\gamma$ ,  $I_\gamma$ , ( $^{34}\text{Si}$ )p-coin. Deduced levels,  $J$ ,  $\pi$ , L-transfer and spectroscopic factors. Comparisons with shell-model calculations.

**2007GeZX**: 30-AMeV  $^{34}\text{Si}$  beam on 30-mg/cm<sup>2</sup>  $^1\text{CD}_2$  secondary target at GANIL. Heavy ions produced in reactions were identified by the VAMOS spectrometer.  $\gamma$  rays were detected using the EXOGAM germanium clover array. Measured Doppler-corrected  $E_\gamma$ ,  $I_\gamma$ ,  $\gamma\gamma$ -coin, and ( $^{35}\text{Si}$ ) $\gamma$ -coin. Deduced levels,  $J$ ,  $\pi$ . Compared with shell-model calculations.

 $^{35}\text{Si}$  Levels

| $E(\text{level})^\dagger$ | $J^\pi^\ddagger$ | $L^\#$ | $S^\#$  | Comments                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------|------------------|--------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0                         | $7/2^-$          | 3      | 0.56 6  | Interpreted as the $1f_{7/2}$ neutron on top of the $^{34}\text{Si}$ core.                                                                                                                                                                                                                                                                                   |
| 910 3                     | $3/2^-$          | 1      | 0.69 10 | Interpreted as the $2p_{3/2}$ neutron on top of the $^{34}\text{Si}$ core.<br>E(level): Other: 906 32 from measured $E_p$ . <b>2014Bu01</b> deduced that a contamination of the proton spectrum at $E(\text{level})=906$ 32 due to transfer to the $3/2^+$ level at 970 keV is less than 30% of the $3/2^-$ component with a confidence limit of $3\sigma$ . |
| 2044 7                    | $1/2^-$          | 1      | 0.73 10 | Interpreted as the $2p_{1/2}$ neutron on top of the $^{34}\text{Si}$ core.<br>E(level): Other: 2060 50 from measured $E_p$ .<br>$J^\pi$ : <b>2014Bu01</b> states that $J^\pi$ is likely to be $1/2^-$ as its large spectroscopic factor value excludes another large $L=1$ , $3/2^-$ component.                                                              |
| $\approx 5500$            | $5/2^-$          | 3      | 0.32 3  | Interpreted as the $1f_{5/2}$ neutron on top of the $^{34}\text{Si}$ core.<br>S: <b>2014Bu01</b> reports $S=0.32$ 4 in Fig. 2, $S=0.32$ 2 in text on page 3, $S=0.32$ 3 in text on page 4, and a full error bar $\approx 0.05$ in Fig. 3.                                                                                                                    |

<sup>†</sup> From a least-squares fit to  $\gamma$ -ray energies, except for a broad level observed at  $\approx 5500$  keV from  $E_p$ . Another broad structure is observed at 3330 keV 120 from  $E_p$  and likely corresponds to the elastic deuteron break-up process, the cross section of which was estimated to be 0.1 mb/MeV (**2014Bu01**).

<sup>‡</sup> As given in **2014Bu01** based on L-transfers and shell-model predictions.

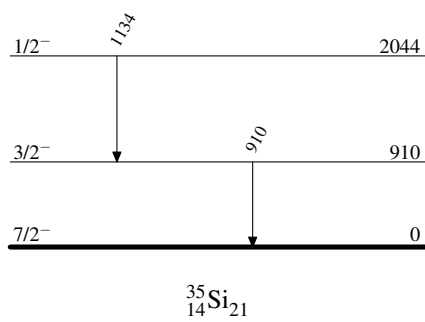
<sup>#</sup> From TWOFN-ADWA analysis of measured proton angular distributions (**2014Bu01**). Additional uncertainties of  $\approx 15\%$  in spectroscopic factors due to global potentials in the ADWA calculation are not included.

 $\gamma(^{35}\text{Si})$ 

| $E_\gamma$ | $E_i(\text{level})$ | $J_i^\pi$ | $E_f$ | $J_f^\pi$ | Comments                                                                                                                                                                                       |
|------------|---------------------|-----------|-------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 910 3      | 910                 | $3/2^-$   | 0     | $7/2^-$   | $E_\gamma$ : Other: 905 1 in <b>2007GeZX</b> , with uncertainty likely statistical only considering it is from a $\gamma$ -ray spectrum with a bin size of 4 keV.                              |
| 1134 6     | 2044                | $1/2^-$   | 910   | $3/2^-$   | $E_\gamma$ : Other: 1133 3 in <b>2007GeZX</b> , with uncertainty likely statistical only considering it is a very weak $\gamma$ ray seen in a $\gamma$ -ray spectrum with a bin size of 4 keV. |

$^2\text{H}(^{34}\text{Si}, ^{35}\text{Si})\gamma$  2014Bu01

Level Scheme



$^9\text{Be}(^{36}\text{Si}, ^{35}\text{Si}\gamma)$  2014St18, 2015St06

$^{36}\text{Si} \rightarrow n + ^{35}\text{Si}$  from  $J^\pi=0^+$   $^{36}\text{Si}$  ground state.

**2014St18:** A  $^{36}\text{Si}$  secondary beam was produced via the projectile fragmentation of a 140-MeV/nucleon  $^{48}\text{Ca}$  primary beam impinging on a  $^9\text{Be}$  target at NSCL, MSU and was selected by the A1900 separator. The states of  $^{35}\text{Al}$  and  $^{35}\text{Si}$  were populated by the one-proton/neutron knockout reactions, respectively, from the  $^{36}\text{Si}$  beam at a midtarget energy of 97.7 MeV/nucleon 5 on a 287-mg/cm<sup>2</sup>  $^9\text{Be}$  secondary target. Knockout residues were identified from their energy loss measured by an ionization chamber at the focal plane of the S800 spectrometer and from their ToF measured between two scintillators at the object position and at the focal plane of the S800 spectrometer. The position and angle of the residues were measured using two cathode-readout drift chambers. Prompt  $\gamma$  rays from the deexcitation of the residues were detected by the GRETINA Ge array. Measured Doppler-corrected  $E_\gamma$ ,  $I_\gamma$ ,  $(^{35}\text{Si})\gamma$ -coin,  $\gamma\gamma$ -coin, the parallel momentum distributions of populated states in  $^{35}\text{Si}$  residues. Deduced levels,  $J$ ,  $\pi$ ,  $L$ -transfers, inclusive and exclusive knockout cross section for producing  $^{35}\text{Si}$  from  $^{36}\text{Si}$ . Calculations using eikonal reaction model and shell model spectroscopic strengths with SDPF-U and SDPF-MU interactions.

**2015St06:** Further theoretical discussions on the data from 2014St18.

 $^{35}\text{Si}$  Levels

Total knockout  $\sigma=81$  mb 2 for producing  $^{35}\text{Si}$  from  $^{36}\text{Si}$ .

| <u>E(level)<sup>†</sup></u> | <u><math>J^\pi</math></u> | <u><math>T_{1/2}</math></u> | <u><math>L^{\#}</math></u> | <u>Comments</u>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------|---------------------------|-----------------------------|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0                           | $(7/2^-)$                 |                             | 3                          | Partial knockout $\sigma=52$ mb 4, including an unresolved population of the $3/2^+$ isomer at 973 keV (2014St18).<br>Partial knockout $\sigma=23$ mb 6, resolved by the fitting parallel momentum distribution subtracting all events that decay by prompt $\gamma$ emission with a linear combination of the theoretical distributions for neutron removal from the $f_{7/2}$ and $d_{3/2}$ orbits together via the $\chi^2$ minimization method (2015St06).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 908 4                       | $(3/2^-)$                 | 55 ps 14                    | 1                          | $T_{1/2}$ : lifetime=80 ps 20 extracted from fitting Geant4 simulation to broadened line shapes via the maximum likelihood method (2014St18).<br>Partial knockout $\sigma=8$ mb 3.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 973 7                       | $(3/2^+)$                 |                             |                            | Partial knockout $\sigma=29$ mb 6, resolved by 2015St06.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 1688 6                      | $1/2^+$                   |                             | 0                          | Partial knockout $\sigma=13$ mb 1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 1970 6                      |                           |                             |                            | Partial knockout $\sigma=1.1$ mb 2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 2042 7                      | $(1/2^-)$                 |                             | 0,1                        | Partial knockout $\sigma=1.3$ mb 2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 2164 6                      | $(5/2^+)$                 |                             | 2,3                        | Partial knockout $\sigma=1.1$ mb 2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 2275 6                      |                           |                             |                            | Partial knockout $\sigma=1.6$ mb 2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 2377 7                      |                           |                             |                            | Partial knockout $\sigma=2.1$ mb 2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                             |                           |                             |                            | L: 2014St18 showed $L=0,1,2,3$ fits to the measured parallel momentum distribution but did not specify $L$ . $L=0$ seems unlikely.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 3611? 8                     |                           |                             |                            | 2014St18 observed a surprising $\gamma$ transition at 3611 keV, which must depopulate a state that is unbound to neutron emission by at least 1.1 MeV, considering $S(n)=2470$ 40 (2021Wa16).<br>One possible explanation is this $^{35}\text{Si}$ level is populated by $1n$ knockout from the deeply-bound $1d_{5/2}$ orbit in $^{36}\text{Si}$ and followed by the emission of the other $1d_{5/2}$ neutron, resulting in a $2p-2h$ configuration dominated by two holes in the deeply bound neutron $1d_{5/2}$ orbit. Such a configuration should have a small overlap with the ground state in $^{34}\text{Si}$ that has a large contribution of $0p-0h$ configurations, leading to a significantly hindered neutron decay. The $\gamma$ decay, on the other hand, would be a fast $5/2^+ \rightarrow 7/2^-$ g.s. $E1$ decay and could potentially compete with the neutron emission.<br>A second possible explanation could be that this state is weakly populated in a more complicated multistep reaction pathway. The 3611-keV $\gamma$ results predominantly from events with residue momenta in the tail of the momentum distribution. The low-momentum tail is associated with more inelastic reactions, which would be consistent with a multistep process.<br>Partial knockout $\sigma=0.8$ mb 2. |

<sup>†</sup> From a least-squares fit to  $\gamma$ -ray energies.

$^9\text{Be}(^{36}\text{Si}, ^{35}\text{Si}\gamma)$  2014St18, 2015St06 (continued) $^{35}\text{Si}$  Levels (continued)

$^{\ddagger}$  As given in 2014St18 based on L-transfers and shell-model predictions.

$^{\#}$  2014St18 deduced L by comparing the measured and eikonal-calculated parallel momentum distributions of residuals.

 $\gamma(^{35}\text{Si})$ 

| $E_{\gamma}^{\dagger}$ | $I_{\gamma}^{\dagger}$ | $E_i(\text{level})$ | $J_i^{\pi}$ | $E_f$ | $J_f^{\pi}$ | Comments                                               |
|------------------------|------------------------|---------------------|-------------|-------|-------------|--------------------------------------------------------|
| 715 4                  | 1.9 2                  | 1688                | $1/2^{+}$   | 973   | $(3/2^{+})$ |                                                        |
| 780 4                  | 13 1                   | 1688                | $1/2^{+}$   | 908   | $(3/2^{-})$ |                                                        |
| 908 4                  | 25 2                   | 908                 | $(3/2^{-})$ | 0     | $(7/2^{-})$ | $B(E2)_{\downarrow}=0.0017 \text{ }^{+4-5}$ (2014St18) |
| 1134 5                 | 1.5 2                  | 2042                | $(1/2^{-})$ | 908   | $(3/2^{-})$ |                                                        |
| 1970 6                 | 1.4 2                  | 1970                |             | 0     | $(7/2^{-})$ |                                                        |
| 2164 6                 | 1.4 2                  | 2164                | $(5/2^{+})$ | 0     | $(7/2^{-})$ |                                                        |
| 2275 6                 | 2.0 3                  | 2275                |             | 0     | $(7/2^{-})$ |                                                        |
| 2377 7                 | 2.5 3                  | 2377                |             | 0     | $(7/2^{-})$ |                                                        |
| 3611 $^{\ddagger}$ 8   | 1.0 2                  | 3611?               |             | 0     | $(7/2^{-})$ |                                                        |

$^{\dagger}$  From 2014St18, unless otherwise noted.

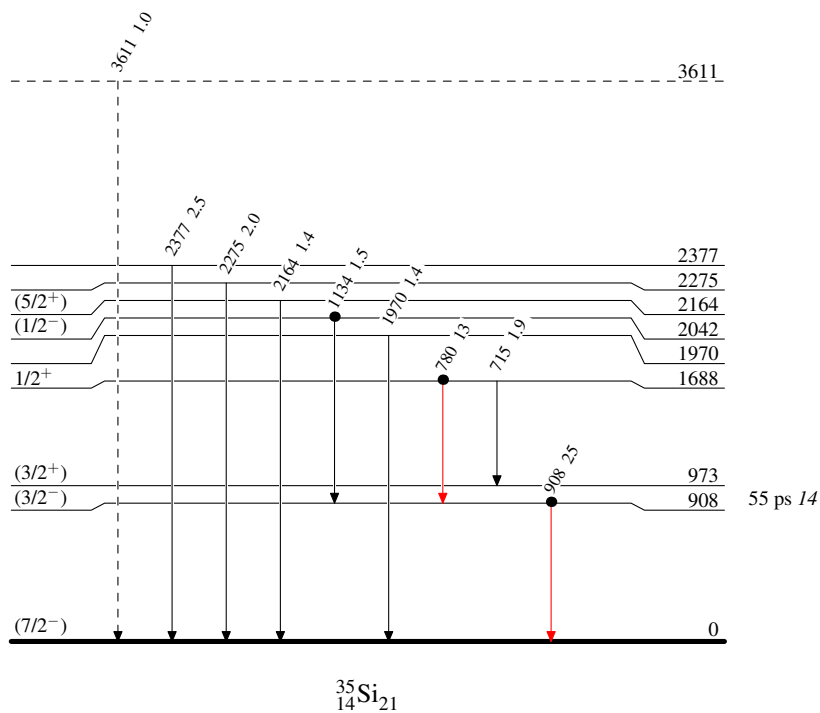
$^{\ddagger}$  Placement of transition in the level scheme is uncertain.

$^9\text{Be}(^{36}\text{Si}, ^{35}\text{Si}\gamma)$  2014St18,2015St06Level Scheme

Intensities: Yield/100 ions

## Legend

- $I_\gamma < 2\% \times I_\gamma^{\max}$
- $I_\gamma < 10\% \times I_\gamma^{\max}$
- $I_\gamma > 10\% \times I_\gamma^{\max}$
- - - - -→  $\gamma$  Decay (Uncertain)
- Coincidence





$^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{Si}\gamma)$  2010WaZT

2010WaZT,2010Wa20: A 215-MeV  $^{36}\text{S}^{9+}$  beam was produced from the XTU-Tandem ALPI-superconducting linear accelerator complex at the INFN Legnaro National Laboratory, Italy. The target was  $300\text{ }\mu\text{g}/\text{cm}^2$  99.7% enriched  $^{208}\text{Pb}$  with  $20\text{ }\mu\text{g}/\text{cm}^2$  carbon backing. Projectile-like fragments produced in binary grazing reactions were separated and identified by the PRISMA spectrometer.  $\gamma$  rays were detected using the CLARA array of 22 escape-suppressed Ge clover detectors covering the azimuthal angles from  $98^\circ$  to  $180^\circ$ . Measured  $E_\gamma$  with Doppler corrections,  $I_\gamma$ , and  $(^{35}\text{Si})\gamma$ -coin. Deduced levels. Compared with shell-model calculations.

 $^{35}\text{Si}$  Levels

| <u><math>E(\text{level})^\dagger</math></u> | <u><math>J^\pi^\ddagger</math></u> |
|---------------------------------------------|------------------------------------|
| 0                                           | $(7/2^-)$                          |
| 910                                         | $(3/2^-)$                          |
| 974                                         | $(3/2^+)$                          |

$^\dagger$  From  $E_\gamma$  data.

$^\ddagger$  As given in 2010WaZT, based on shell-model calculations.

 $\gamma(^{35}\text{Si})$ 

| <u><math>E_\gamma^\dagger</math></u> | <u><math>I_\gamma^\dagger</math></u> | <u><math>E_i(\text{level})</math></u> | <u><math>J_i^\pi</math></u> | <u><math>E_f</math></u> | <u><math>J_f^\pi</math></u> |
|--------------------------------------|--------------------------------------|---------------------------------------|-----------------------------|-------------------------|-----------------------------|
| 910                                  | 100 25                               | 910                                   | $(3/2^-)$                   | 0                       | $(7/2^-)$                   |
| 974 $^\ddagger$                      | 21 10                                | 974                                   | $(3/2^+)$                   | 0                       | $(7/2^-)$                   |

$^\dagger$  From 2010WaZT. The author report  $\Delta E_\gamma=1\text{ keV}$ , but it is very likely statistical only since  $910\gamma$  and  $974\gamma$  are weak with poor statistics in the  $\gamma$ -ray spectrum, and thus is not adopted by the evaluators.

$^\ddagger$  Placement of transition in the level scheme is uncertain.

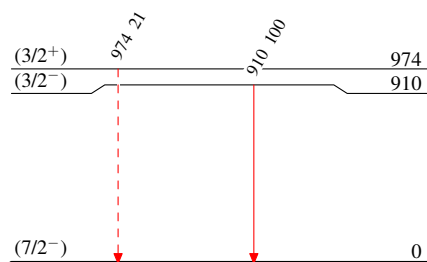
$^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{Si}\gamma)$  2010WaZT

Legend

Level Scheme

 Intensities: Relative  $I_\gamma$ 

- ▶  $I_\gamma < 2\% \times I_\gamma^{\max}$
- ▶  $I_\gamma < 10\% \times I_\gamma^{\max}$
- ▶  $I_\gamma > 10\% \times I_\gamma^{\max}$
- - - - -▶  $\gamma$  Decay (Uncertain)


 $^{35}_{14}\text{Si}_{21}$

Adopted Levels, Gammas

$Q(\beta^-)=3988.4$  19;  $S(n)=8380.4$  20;  $S(p)=12155.1$  20;  $Q(\alpha)=-12332.0$  29 2021Wa16

$S(2n)=14663.1$  22,  $S(2p)=30938$  7 (2021Wa16).

Isotope discovery (2012Th10):  $^{232}\text{Th}(^{40}\text{Ar},X)$  at Dubna (1971Ar32) and  $^{37}\text{Cl}(\gamma,2p)^{35}\text{P}$  at Mainz (1971Gr53).

$^{35}\text{P}$  production:

2012Kw02:  $E(^{40}\text{Ar})=140$  MeV/nucleon impinged on  $^9\text{Be}$  and natural nickel targets at NSCL. Measured fragmentation cross sections, parallel momentum transfers, and widths. Compared with empirical formula EPAX, and predictions from internuclear cascade and deep inelastic models using Monte Carlo ISABEL-GEMINI and DIT-GEMINI codes.

2012Zh06:  $^9\text{Be}, ^{181}\text{Ta}(^{40}\text{Ar},X)$  at  $E(^{40}\text{Ar})=57$  MeV/nucleon at HIRFL. Measured momentum distributions and production cross sections of fragments. Observed competition between projectile fragmentation and other mechanisms. Compared with EPAX, abrasion-ablation, and HIPSE models. Studied target dependence of fragment cross sections.

2007No13:  $^9\text{Be}, ^{181}\text{Ta}(^{40}\text{Ar},X)$  and  $(^{40}\text{Ar},X)$  at  $E(^{40}\text{Ar})=100$  MeV/nucleon at RIKEN. Measured fragment momentum distributions and production cross sections.

1997Vo03:  $^{56}\text{Fe}(p,X)$  at  $E_p=800$  MeV at LANL. Measured  $\gamma$  radiation. Deduced production cross sections.

$^{35}\text{P}$  decay measurements:

1986Wa22:  $^{35}\text{P}$  produced via  $^{36}\text{S}(t,\alpha)^{35}\text{P}$  at BNL. Measured  $\beta$ -delayed  $\gamma$  rays.

1972Ap01:  $^{35}\text{P}$  produced via  $^{37}\text{Cl}(t,\alpha p)^{35}\text{P}$  at LANL. Measured  $T_{1/2}$  and  $\beta$ -delayed  $\gamma$  rays.

1972Go31:  $^{35}\text{P}$  produced via  $^{18}\text{O}(^{19}\text{F},2p)^{35}\text{P}$  and  $^{36}\text{S}(t,\alpha)^{35}\text{P}$  at BNL. Measured  $T_{1/2}$  and  $\beta$ -delayed  $\gamma$  rays.

1971Gr53:  $^{35}\text{P}$  produced via  $^{37}\text{Cl}(\gamma,2p)^{35}\text{P}$  at Mainz. Measured  $T_{1/2}$  and  $\gamma$  rays.

$^{35}\text{P}$  radius measurement:

1999Ai02:  $\text{Si}(^{35}\text{P},X)$  at NSCL. Measured energy-integrated reaction cross sections at  $E=38$ -80 MeV/ nucleon. Deduced strong absorption radii.

$^{35}\text{P}$  mass measurements:

$^{34}\text{S}(^{18}\text{O},^{17}\text{F})$  and  $^{37}\text{Cl}(^{11}\text{B},^{13}\text{N})$  (1988Or01),  $^{36}\text{S}(^6\text{Li},^7\text{Be})$  (1985Dr06),  $^{36}\text{S}(d,^3\text{He})$  (1985Kh04),  $^{36}\text{S}(^{14}\text{C},^{15}\text{N})$  (1984Ma49).

Theoretical calculations (binding energies, dipole moments, quadrupole moments, radii, levels,  $J$ ,  $\pi$ , etc.): 2012BoZT, 2009No01, 2004Kh16, 2003Sm02, 1999Du05, 1988Wa04, 1987Wa10, 1986Wo02, 1983Wi08, 1975JeZX.

 $^{35}\text{P}$  LevelsCross Reference (XREF) Flags

|   |                                                  |   |                                                  |   |                                                      |
|---|--------------------------------------------------|---|--------------------------------------------------|---|------------------------------------------------------|
| A | $^{35}\text{Si} \beta^-$ decay (0.78 s)          | E | $^9\text{Be}(^{36}\text{S},^{35}\text{P}\gamma)$ | I | $^{37}\text{Cl}(^{11}\text{B},^{13}\text{N})$        |
| B | $^1\text{H}(^{34}\text{Si},p):\text{resonances}$ | F | $^{34}\text{S}(^{18}\text{O},^{17}\text{F})$     | J | $^{160}\text{Gd}(^{37}\text{Cl},X\gamma)$            |
| C | $^2\text{H}(^{34}\text{Si},^{35}\text{P}\gamma)$ | G | $^{36}\text{S}(\text{pol } d,^3\text{He})$       | K | $^{208}\text{Pb}(^{36}\text{S},^{35}\text{P}\gamma)$ |
| D | $^2\text{H}(^{36}\text{S},^3\text{He})$          | H | $^{36}\text{S}(d,^3\text{He})$                   |   |                                                      |

| $E(\text{level})^\dagger$ | $J^\pi$            | $T_{1/2}$ or $\Gamma^\#$ | XREF        | Comments                                                                                                                                                                                                                                                                      |
|---------------------------|--------------------|--------------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0                         | $1/2^+$            | 47.3 s 8                 | A CDEFGHIJK | $\% \beta^- = 100$<br>$J^\pi$ : L(pol d, $^3\text{He}$ )=0 from $0^+$ and analyzing power.<br>$T_{1/2}$ : weighted average of 45 s 2 (1971Gr53, $\gamma(t)$ ), 47.7 s 29 (1972Go31, $\gamma(t)$ ), 48.2 s 17 (1972Go31, $\gamma(t)$ ), and 47.4 s 8 (1972Ap01, $\gamma(t)$ ). |
| 2386.9 11                 | $3/2^+$            | <0.69 ps                 | A CDEFG I K | XREF: F(2420)<br>$J^\pi$ : L(pol d, $^3\text{He}$ )=2 from $0^+$ and L-1/2 transfer from analyzing power.                                                                                                                                                                     |
| 3860.4 11                 | $5/2^+$            | <0.69 ps                 | A CDE GHIJK | $J^\pi$ : L(pol d, $^3\text{He}$ )=2 from $0^+$ and L+1/2 transfer from analyzing power.                                                                                                                                                                                      |
| 4101.7 11                 | $(7/2^-)^\ddagger$ | >69 ps                   | A C E JK    |                                                                                                                                                                                                                                                                               |
| 4250 20                   |                    |                          | I           |                                                                                                                                                                                                                                                                               |
| 4382.0 12                 | $(5/2^-)$          |                          | A E K       | XREF: A(?)<br>$J^\pi$ : possibly allowed $\beta^-$ feeding from $7/2^-$ parent; 1994.9 $\gamma$ to $3/2^+$ .                                                                                                                                                                  |
| 4494.1 12                 | $(7/2^-)^\ddagger$ | 2.29 ps 49               | A C E G JK  | XREF: G(4474)<br>$J^\pi$ : L=(3) in $^9\text{Be}(^{36}\text{S},^{35}\text{P})$ from $0^+$ .                                                                                                                                                                                   |
| 4666.2 16                 | $5/2^+$            |                          | DE GHI      | XREF: I(4640)<br>$J^\pi$ : L(pol d, $^3\text{He}$ )=2 from $0^+$ and L+1/2 transfer from analyzing power.                                                                                                                                                                     |
| 4767.0 13                 | $(9/2^-)^\ddagger$ |                          | E K         |                                                                                                                                                                                                                                                                               |

Continued on next page (footnotes at end of table)

**Adopted Levels, Gammas (continued)**

| $^{35}\text{P}$ Levels (continued) |                                              |                                    |        |                                                                                                                                                                                                                                                                                |
|------------------------------------|----------------------------------------------|------------------------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E(level) <sup>†</sup>              | J <sup>π</sup>                               | T <sub>1/2</sub> or Γ <sup>#</sup> | XREF   | Comments                                                                                                                                                                                                                                                                       |
| 4869.6 12                          | (5/2 <sup>-</sup> , 7/2 <sup>-</sup> )       |                                    | A C K  | J <sup>π</sup> : possibly allowed β <sup>-</sup> feeding from 7/2 <sup>-</sup> parent; 1009.7γ to 5/2 <sup>+</sup> .                                                                                                                                                           |
| 4962.8 12                          | (9/2 <sup>-</sup> ) <sup>‡</sup>             |                                    | A E K  | XREF: A(?)                                                                                                                                                                                                                                                                     |
| 5010 20                            |                                              |                                    | I      | 1988Or01 observed a 5010 20 level in ( $^{11}\text{B}, ^{13}\text{N}$ ) and a 5070 40 level in ( $^{18}\text{O}, ^{17}\text{F}$ ) but did not assign them as the same level.                                                                                                   |
| 5090.2 13                          | (11/2 <sup>-</sup> ) <sup>‡</sup>            |                                    | EF K   | XREF: F(5070)                                                                                                                                                                                                                                                                  |
| 5199.3 16                          | 5/2 <sup>+</sup>                             |                                    | DE GHI | XREF: I(5220)<br>J <sup>π</sup> : L(pol d, $^3\text{He}$ )=2 from 0 <sup>+</sup> and L+1/2 transfer from analyzing power.                                                                                                                                                      |
| 5487.9 13                          |                                              |                                    | K      |                                                                                                                                                                                                                                                                                |
| 5561.0 13                          | (5/2 <sup>-</sup> )                          |                                    | A K    | J <sup>π</sup> : possibly allowed β <sup>-</sup> feeding from 7/2 <sup>-</sup> parent; 3173.5γ to 3/2 <sup>+</sup> .                                                                                                                                                           |
| 5709.5 23                          | (1/2 <sup>-</sup> )                          |                                    | DE     | J <sup>π</sup> : L=(1) in $^9\text{Be}(^{36}\text{S}, ^{35}\text{P})$ from 0 <sup>+</sup> ; interpreted as the deeply bound 1p <sub>1/2</sub> proton removal from 0 <sup>+</sup> in $^9\text{Be}(^{36}\text{S}, ^{35}\text{P})$ ; 5709γ to 1/2 <sup>+</sup> .                  |
| 5.86×10 <sup>3</sup> 5             |                                              |                                    | F I    | XREF: F(5890)I(5840)<br>E(level): weighted average of 5890 70 from ( $^{18}\text{O}, ^{17}\text{F}$ ) and 5840 50 from ( $^{11}\text{B}, ^{13}\text{N}$ ).                                                                                                                     |
| 6222.7 13                          | (7/2 <sup>-</sup> , 9/2, 11/2 <sup>-</sup> ) |                                    | K      | J <sup>π</sup> : 1132γ to (11/2 <sup>-</sup> ) and 1729γ to (7/2 <sup>-</sup> ).                                                                                                                                                                                               |
| 6440 60                            |                                              |                                    | F      |                                                                                                                                                                                                                                                                                |
| 7050 60                            |                                              |                                    | F      |                                                                                                                                                                                                                                                                                |
| 7440 60                            |                                              |                                    | Fg     | XREF: g(7520)                                                                                                                                                                                                                                                                  |
| 7526.9 23                          | (1/2 <sup>-</sup> )                          |                                    | E g    | XREF: g(7520)<br>J <sup>π</sup> : L=(1) in $^9\text{Be}(^{36}\text{S}, ^{35}\text{P})$ from 0 <sup>+</sup> ; interpreted as the deeply bound 1p <sub>1/2</sub> proton removal from 0 <sup>+</sup> in $^9\text{Be}(^{36}\text{S}, ^{35}\text{P})$ ; 7526γ to 1/2 <sup>+</sup> . |
| 7590 20                            |                                              |                                    | I      |                                                                                                                                                                                                                                                                                |
| 7920 60                            |                                              |                                    | F      |                                                                                                                                                                                                                                                                                |
| 8390 40                            |                                              |                                    | I      |                                                                                                                                                                                                                                                                                |
| 8.60×10 <sup>3</sup> 10            |                                              |                                    | F      |                                                                                                                                                                                                                                                                                |
| 9290 50                            |                                              |                                    | F      |                                                                                                                                                                                                                                                                                |
| 14938 24                           | 1/2 <sup>+</sup>                             | <12.7 keV                          | B      | J <sup>π</sup> : L=0 in $^1\text{H}(^{34}\text{Si}, \text{p})$ .                                                                                                                                                                                                               |
| 15161 3                            | 5/2 <sup>-</sup> , 7/2 <sup>-</sup>          | <4.4 keV                           | B      | J <sup>π</sup> : L=3 in $^1\text{H}(^{34}\text{Si}, \text{p})$ .                                                                                                                                                                                                               |
| 15306 24                           | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>          | <30.4 keV                          | B      | J <sup>π</sup> : L=2 in $^1\text{H}(^{34}\text{Si}, \text{p})$ .                                                                                                                                                                                                               |
| 15964 18                           | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>          | 84 keV 25                          | B      | J <sup>π</sup> : L=2 in $^1\text{H}(^{34}\text{Si}, \text{p})$ .                                                                                                                                                                                                               |
| 16145 36                           | 1/2 <sup>-</sup> , 3/2 <sup>-</sup>          | 0.35 MeV 9                         | B      | J <sup>π</sup> : L=1 in $^1\text{H}(^{34}\text{Si}, \text{p})$ .                                                                                                                                                                                                               |
| 16605 44                           | 1/2 <sup>+</sup>                             | 0.22 MeV 15                        | B      | J <sup>π</sup> : L=0 in $^1\text{H}(^{34}\text{Si}, \text{p})$ .                                                                                                                                                                                                               |
| 17254 12                           | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>          | <11.6 keV                          | B      | J <sup>π</sup> : L=2 in $^1\text{H}(^{34}\text{Si}, \text{p})$ .                                                                                                                                                                                                               |
| 17355 15                           | 1/2 <sup>-</sup> , 3/2 <sup>-</sup>          | 32 keV 22                          | B      | J <sup>π</sup> : L=1 in $^1\text{H}(^{34}\text{Si}, \text{p})$ .                                                                                                                                                                                                               |

<sup>†</sup> From a least-squares fit to γ-ray energies for levels connected with γ transitions, from particle-transfer reactions for other levels, or from proton elastic scattering for resonances.

<sup>‡</sup> Comparisons with shell-model calculations (2019Gr08).

<sup>#</sup> T<sub>1/2</sub> from the differential recoil-distance method (2019Gr08) in  $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$  and widths from the R-matrix analysis of ( $^{34}\text{Si}, \text{p}$ ) for resonances, unless otherwise noted.

Adopted Levels, Gammas (continued)

| $\gamma(^{35}\text{P})$ |                     |            |                     |        |                     |         |          |                          |                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------|---------------------|------------|---------------------|--------|---------------------|---------|----------|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $E_i(\text{level})$     | $J_i^\pi$           | $E_\gamma$ | $I_\gamma$          | $E_f$  | $J_f^\pi$           | Mult.   | $\delta$ | $\alpha^\#$              | Comments                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 2386.9                  | 3/2 <sup>+</sup>    | 2386.3 6   | 100                 | 0      | 1/2 <sup>+</sup>    | [M1,E2] |          | 0.00046 5                | $E_\gamma$ : weighted average of 2386.4 6 from $^{35}\text{Si } \beta^-$ decay, 2386 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{P}\gamma)$ , and 2386 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$ .<br>B(M1)(W.u.)>0.0023 if M1, B(E2)(W.u.)>1.6 if E2.                                                                                                                                                           |
| 3860.4                  | 5/2 <sup>+</sup>    | 1473.5 5   | 15.6 14             | 2386.9 | 3/2 <sup>+</sup>    | [M1,E2] |          | 8.3×10 <sup>-5</sup> 13  | $E_\gamma$ : weighted average of 1473.4 5 from $^{35}\text{Si } \beta^-$ decay, 1473 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{P}\gamma)$ , and 1474 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$ .<br>$I_\gamma$ : weighted average of 14.1 33 from $^{35}\text{Si } \beta^-$ decay and 15.9 14 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$ .<br>B(M1)(W.u.)>0.0012 if M1, B(E2)(W.u.)>2.1 if E2. |
|                         |                     | 3860.2 10  | 100.0 32            | 0      | 1/2 <sup>+</sup>    | [E2]    |          | 1.12×10 <sup>-3</sup> 2  | B(E2)(W.u.)>0.12<br>$E_\gamma$ : weighted average of 3859.5 10 from $^{35}\text{Si } \beta^-$ decay, 3860 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{P}\gamma)$ , and 3861 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$ .<br>$I_\gamma$ : From $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$ . Other: 100 7 from $^{35}\text{Si } \beta^-$ decay.                                                          |
| 4101.7                  | (7/2 <sup>-</sup> ) | 241.3 5    | 100 <sup>†</sup> 7  | 3860.4 | 5/2 <sup>+</sup>    | [E1]    |          | 0.000665 10              | B(E1)(W.u.)<4.4×10 <sup>-4</sup><br>$E_\gamma$ : weighted average of 241.4 3 from $^{35}\text{Si } \beta^-$ decay, 237 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{P}\gamma)$ , and 241 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$ .<br>$I_\gamma$ : other: 100 4 from $^{35}\text{Si } \beta^-$ decay.                                                                                                           |
|                         |                     | 1714.8 6   | 6.6 <sup>†</sup> 17 | 2386.9 | 3/2 <sup>+</sup>    | [M2]    |          | 7.93×10 <sup>-5</sup> 11 | B(M2)(W.u.)<0.16<br>$E_\gamma$ : weighted average of 1714.7 6 from $^{35}\text{Si } \beta^-$ decay and 1715 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$ .<br>$I_\gamma$ : other: 22 5 from $^{35}\text{Si } \beta^-$ decay.                                                                                                                                                                                          |
|                         |                     | 4101.4 10  | 54 <sup>†</sup> 8   | 0      | 1/2 <sup>+</sup>    | [E3]    |          | 0.000924 13              | B(E3)(W.u.)<4.8<br>$E_\gamma$ : weighted average of 4100.8 10 from $^{35}\text{Si } \beta^-$ decay and 4102 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$ .<br>$I_\gamma$ : other: 135 8 from $^{35}\text{Si } \beta^-$ decay.                                                                                                                                                                                         |
| 4382.0                  | (5/2 <sup>-</sup> ) | 1994.9 6   | 100                 | 2386.9 | 3/2 <sup>+</sup>    |         |          |                          | $E_\gamma$ : weighted average of 1994.8 6 from $^{35}\text{Si } \beta^-$ decay, 1995 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{P}\gamma)$ , and 1995 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$ .<br>Placement by 1988DuZS, 2008Wi09, and 2016Mu03.<br>1988DuZT and 1987Wa10 placed this $\gamma$ as the 6096 to 4101 transition. 1988Or01 placed this $\gamma$ as the 6488 to 4493 transition.                 |
| 4494.1                  | (7/2 <sup>-</sup> ) | 392.3 3    | 100 5               | 4101.7 | (7/2 <sup>-</sup> ) | [M1+E2] | <0.22    | 0.000199 12              | B(M1)(W.u.)=0.117 +42-29<br>$E_\gamma$ : weighted average of 392.3 3 from $^{35}\text{Si } \beta^-$ decay, 391                                                                                                                                                                                                                                                                                                                        |

Adopted Levels, Gammas (continued)

| $\gamma(^{35}\text{P})$ (continued) |                                        |                                                          |                                                      |        |                                                                                 |       |                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------|----------------------------------------|----------------------------------------------------------|------------------------------------------------------|--------|---------------------------------------------------------------------------------|-------|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $E_i(\text{level})$                 | $J_i^\pi$                              | $E_\gamma$                                               | $I_\gamma$                                           | $E_f$  | $J_f^\pi$                                                                       | Mult. | $\alpha^\#$             | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 4494.1                              | (7/2 <sup>-</sup> )                    | 633.6 5                                                  | 34 5                                                 | 3860.4 | 5/2 <sup>+</sup>                                                                | [E1]  | 4.64×10 <sup>-5</sup> 7 | 2 from $^9\text{Be}(^{36}\text{S},^{35}\text{P}\gamma)$ , and 392 1 from $^{208}\text{Pb}(^{36}\text{S},^{35}\text{P}\gamma)$ .<br>$I_\gamma$ : From $^{35}\text{Si}$ $\beta^-$ decay. Other: 100 17 from $^{208}\text{Pb}(^{36}\text{S},^{35}\text{P}\gamma)$ .<br>$\delta$ : deduced by evaluators from RUL=100 for B(E2)(W.u.).<br>B(E1)(W.u.)=2.8×10 <sup>-4</sup> +8-6<br>$E_\gamma$ : weighted average of 633.7 5 from $^{35}\text{Si}$ $\beta^-$ decay, 634 2 from $^9\text{Be}(^{36}\text{S},^{35}\text{P}\gamma)$ , and 633 1 from $^{208}\text{Pb}(^{36}\text{S},^{35}\text{P}\gamma)$ .<br>$I_\gamma$ : weighted average of 38 5 from $^{35}\text{Si}$ $\beta^-$ decay and 27 7 from $^{208}\text{Pb}(^{36}\text{S},^{35}\text{P}\gamma)$ . |
| 4666.2                              | 5/2 <sup>+</sup>                       | 804 $\dagger$ 2<br>2279 $\dagger$ 2<br>4668 $\dagger$ 2  |                                                      | 3860.4 | 5/2 <sup>+</sup><br>2386.9 3/2 <sup>+</sup><br>0 1/2 <sup>+</sup>               |       |                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 4767.0                              | (9/2 <sup>-</sup> )                    | 273 1                                                    | 40.0 $\dagger$ 25                                    | 4494.1 | (7/2 <sup>-</sup> )                                                             |       |                         | $E_\gamma$ : weighted average of 274 2 from $^9\text{Be}(^{36}\text{S},^{35}\text{P}\gamma)$ and 273 1 from $^{208}\text{Pb}(^{36}\text{S},^{35}\text{P}\gamma)$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                     |                                        | 664 1                                                    | 100 $\dagger$ 47                                     | 4101.7 | (7/2 <sup>-</sup> )                                                             |       |                         | $E_\gamma$ : weighted average of 666 2 from $^9\text{Be}(^{36}\text{S},^{35}\text{P}\gamma)$ and 664 1 from $^{208}\text{Pb}(^{36}\text{S},^{35}\text{P}\gamma)$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 4869.6                              | (5/2 <sup>-</sup> , 7/2 <sup>-</sup> ) | 374 $\dagger$ 1<br>487 $\dagger$ 1<br>767.9 4            | 60 $\dagger$ 20<br><40 $\dagger$<br>100 $\dagger$ 20 | 4494.1 | (7/2 <sup>-</sup> )<br>4382.0 (5/2 <sup>-</sup> )<br>4101.7 (7/2 <sup>-</sup> ) |       |                         | $E_\gamma$ : weighted average of 768.0 4 from $^{35}\text{Si}$ $\beta^-$ decay and 767 1 from $^{208}\text{Pb}(^{36}\text{S},^{35}\text{P}\gamma)$ .<br>$I_\gamma$ : other: 100 18 from $^{35}\text{Si}$ $\beta^-$ decay.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                     |                                        | 1009.7 5                                                 | <20 $\dagger$                                        | 3860.4 | 5/2 <sup>+</sup>                                                                |       |                         | $E_\gamma$ : weighted average of 1009.9 5 from $^{35}\text{Si}$ $\beta^-$ decay and 1009 1 from $^{208}\text{Pb}(^{36}\text{S},^{35}\text{P}\gamma)$ .<br>$I_\gamma$ : other: 152 32 from $^{35}\text{Si}$ $\beta^-$ decay.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 4962.8                              | (9/2 <sup>-</sup> )                    | 468.9 4                                                  | 100 $\dagger$ 8                                      | 4494.1 | (7/2 <sup>-</sup> )                                                             |       |                         | $E_\gamma$ : weighted average of 468.9 4 from $^{35}\text{Si}$ $\beta^-$ decay, 469 2 from $^9\text{Be}(^{36}\text{S},^{35}\text{P}\gamma)$ , and 468 2 from $^{208}\text{Pb}(^{36}\text{S},^{35}\text{P}\gamma)$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                     |                                        | 859 $\dagger$ 3                                          | 66 $\dagger$ 9                                       | 4101.7 | (7/2 <sup>-</sup> )                                                             |       |                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 5090.2                              | (11/2 <sup>-</sup> )                   | 128 1                                                    | 50 $\dagger$ 25                                      | 4962.8 | (9/2 <sup>-</sup> )                                                             |       |                         | $E_\gamma$ : weighted average of 127 2 from $^9\text{Be}(^{36}\text{S},^{35}\text{P}\gamma)$ and 128 1 from $^{208}\text{Pb}(^{36}\text{S},^{35}\text{P}\gamma)$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                     |                                        | 322 1                                                    | 100 $\dagger$ 35                                     | 4767.0 | (9/2 <sup>-</sup> )                                                             |       |                         | $E_\gamma$ : weighted average of 321 2 from $^9\text{Be}(^{36}\text{S},^{35}\text{P}\gamma)$ and 322 1 from $^{208}\text{Pb}(^{36}\text{S},^{35}\text{P}\gamma)$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 5199.3                              | 5/2 <sup>+</sup>                       | 1337 $\dagger$ 2<br>2811 $\dagger$ 2<br>5202 $\dagger$ 2 |                                                      | 3860.4 | 5/2 <sup>+</sup><br>2386.9 3/2 <sup>+</sup><br>0 1/2 <sup>+</sup>               |       |                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 5487.9                              |                                        | 993 $\dagger$ 1<br>1387 $\dagger$ 1                      | 100 $\dagger$ 20<br>60 $\dagger$ 20                  | 4494.1 | (7/2 <sup>-</sup> )<br>4101.7 (7/2 <sup>-</sup> )                               |       |                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{P})$  (continued)

| <u>E<sub>i</sub>(level)</u> | <u>J<sub>i</sub><sup><math>\pi</math></sup></u> | <u>E<sub><math>\gamma</math></sub></u> | <u>I<sub><math>\gamma</math></sub></u> | <u>E<sub>f</sub></u> | <u>J<sub>f</sub><sup><math>\pi</math></sup></u> | <u>Comments</u>                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------|-------------------------------------------------|----------------------------------------|----------------------------------------|----------------------|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5561.0                      | (5/2 <sup>-</sup> )                             | 1459.4 7                               | 34 12                                  | 4101.7               | (7/2 <sup>-</sup> )                             | E <sub><math>\gamma</math></sub> : weighted average of 1459.7 5 from <sup>35</sup> Si $\beta^-$ decay and 1458 1 from <sup>208</sup> Pb( <sup>36</sup> S, <sup>35</sup> P $\gamma$ ).<br>I <sub><math>\gamma</math></sub> : From <sup>35</sup> Si $\beta^-$ decay.                                                                                                                                                 |
|                             |                                                 | 3173.5 10                              | 100 17                                 | 2386.9               | 3/2 <sup>+</sup>                                | 3173.5 $\gamma$ is not observed in <sup>208</sup> Pb( <sup>36</sup> S, <sup>35</sup> P $\gamma$ ), but the weaker 1459 $\gamma$ deexciting the same level is observed in <sup>208</sup> Pb( <sup>36</sup> S, <sup>35</sup> P $\gamma$ ). Further experiments are needed to resolve the discrepancy.<br>E <sub><math>\gamma</math></sub> ,I <sub><math>\gamma</math></sub> : From <sup>35</sup> Si $\beta^-$ decay. |
| 5709.5                      | (1/2 <sup>-</sup> )                             | 5709 <sup>‡</sup> 2                    |                                        | 0                    | 1/2 <sup>+</sup>                                |                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 6222.7                      | (7/2 <sup>-</sup> ,9/2,11/2 <sup>-</sup> )      | 1132 <sup>†</sup> 1                    | <25 <sup>†</sup>                       | 5090.2               | (11/2 <sup>-</sup> )                            |                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                             |                                                 | 1260 <sup>†</sup> 1                    | 100 <sup>†</sup> 25                    | 4962.8               | (9/2 <sup>-</sup> )                             |                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                             |                                                 | 1729 <sup>†</sup> 1                    | 100 <sup>†</sup> 25                    | 4494.1               | (7/2 <sup>-</sup> )                             |                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 7526.9                      | (1/2 <sup>-</sup> )                             | 7526 2                                 |                                        | 0                    | 1/2 <sup>+</sup>                                |                                                                                                                                                                                                                                                                                                                                                                                                                    |

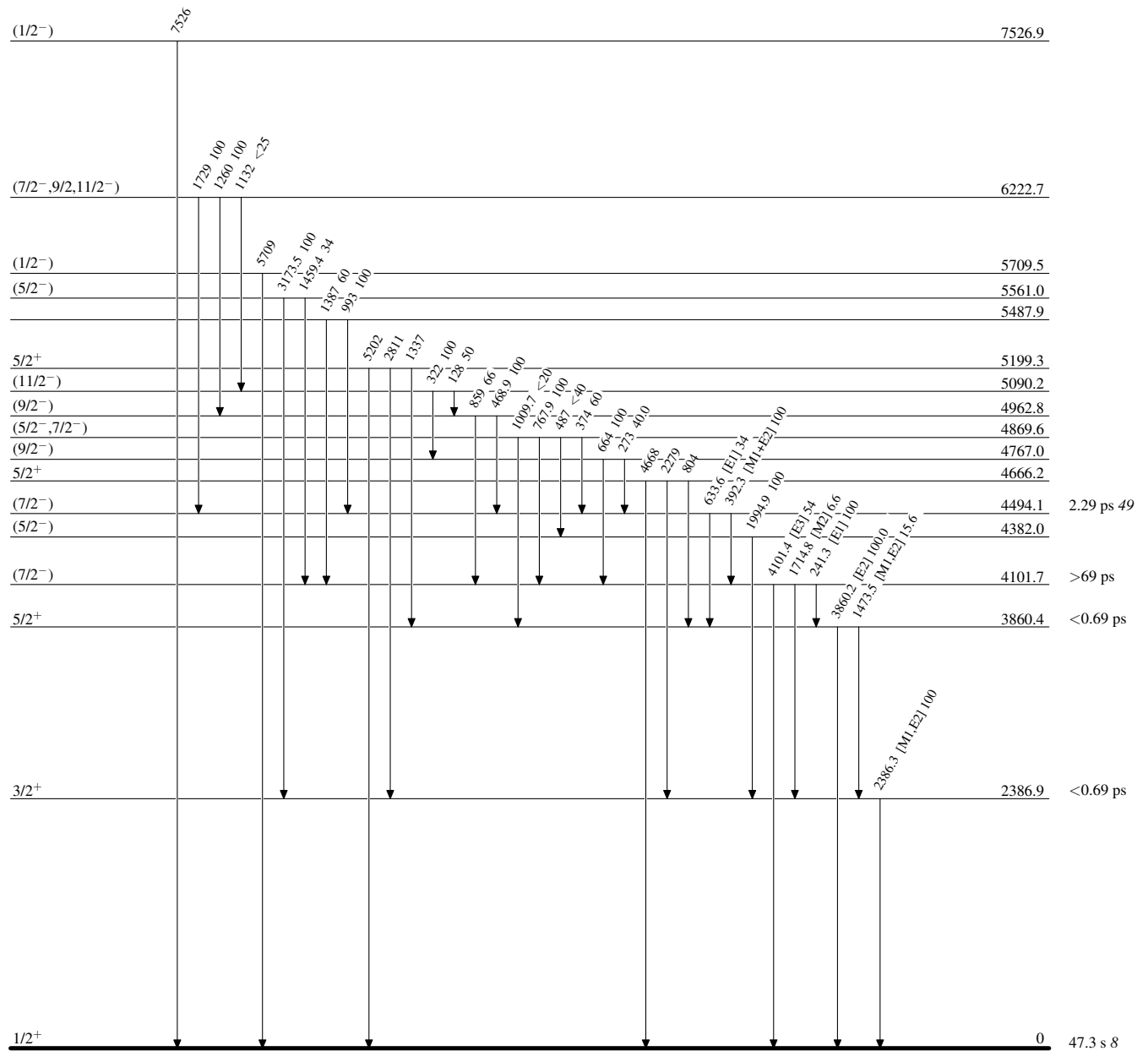
<sup>†</sup> From <sup>208</sup>Pb(<sup>36</sup>S,<sup>35</sup>P $\gamma$ ).

<sup>‡</sup> From <sup>9</sup>Be(<sup>36</sup>S,<sup>35</sup>P $\gamma$ ).

# Total theoretical internal conversion coefficients, calculated using the BrIcc code (2008Ki07) with "Frozen Orbitals" approximation based on  $\gamma$ -ray energies, assigned multipolarities, and mixing ratios, unless otherwise specified.

Adopted Levels, GammasLevel Scheme

Intensities: Relative photon branching from each level





**$^{35}\text{Si}$   $\beta^-$  decay (0.78 s) 1988DuZS,1986Du07,1988DuZT**

Parent:  $^{35}\text{Si}$ :  $E=0$ ;  $J^\pi=(7/2)^-$ ;  $T_{1/2}=0.78$  s 12;  $Q(\beta^-)=10470$  40;  $\% \beta^-$  decay=100

$^{35}\text{Si}$ - $J^\pi$ ,  $T_{1/2}$ : From the Adopted Levels of  $^{35}\text{Si}$ .

$^{35}\text{Si}$ - $Q(\beta^-)$ : From 2021Wa16.

1988DuZS,1986Du07,1988DuZT,1987DuZU:  $^{35}\text{Si}$  was produced by the fragmentation of an  $^{40}\text{Ar}$  beam of  $2 \times 10^{11}$  particles/s at 60 MeV/nucleon on a 190 mg/cm<sup>2</sup> Be target at GANIL. Decay observed with a 1 mm thick plastic scintillator and a 174 cm<sup>3</sup> intrinsic Ge detector with 1.2% absolute efficiency at 1.33 MeV. Measured  $\beta\gamma(t)$ ,  $E\gamma$ ,  $I\gamma$ . Deduced levels,  $J$ ,  $\pi$ , parent  $T_{1/2}$ .

2007Ne14: measured  $^{35}\text{Si}$  ground state magnetic moment and  $g$ -factor using the  $\beta$ -NMR method.

1987Wa10: shell-model calculations for  $^{35}\text{Si}$   $\beta^-$  decay scheme,  $^{35}\text{P}$  levels, decay branching ratios,  $\log ft$ , and Gamow-Teller transition strengths.

The decay scheme is considered incomplete due to a large gap of about 4.9 MeV between the highest observed level at  $E=5561$  and  $Q(\beta^-)$  value=10470 40 (2021Wa16). There may be missing transitions from unobserved levels in the gap.

 $^{35}\text{P}$  Levels

| $E(\text{level})^\dagger$ | $J^\pi^\ddagger$ | $T_{1/2}^\ddagger$ |
|---------------------------|------------------|--------------------|
| 0                         | $1/2^+$          | 47.3 s 8           |
| 2386.5 5                  | $3/2^+$          | <0.69 ps           |
| 3859.7 5                  | $5/2^+$          | <0.69 ps           |
| 4101.2 5                  | $(7/2^-)$        | >69 ps             |
| 4381.3? 8                 | $(5/2^-)$        |                    |
| 4493.5 6                  | $(7/2^-)$        | 2.29 ps 49         |
| 4869.4 6                  | $(5/2^-, 7/2^-)$ |                    |
| 4962.4? 7                 | $(9/2^-)$        |                    |
| 5560.7 7                  | $(5/2^-)$        |                    |

$^\dagger$  From a least-squares fit to  $\gamma$ -ray energies.

$^\ddagger$  From the Adopted Levels.

 $\beta^-$  radiations

| $E(\text{decay})$       | $E(\text{level})$ | $I\beta^-^{\ddagger\ddagger}$ | $\log ft^\dagger$ | Comments                                                                                                                                                                        |
|-------------------------|-------------------|-------------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $(4.91 \times 10^3)$ 4) | 5560.7            | 12.7                          | 4.6               | $I\beta=12.4$ 18 and $\log ft=4.6$ from 1988DuZS.                                                                                                                               |
| $(5.51 \times 10^3)$ 4) | 4962.4?           | 5.1                           | 5.2               | Placed based on the 4959 to 4493 transition observed in $^{208}\text{Pb}(^{36}\text{S}, X\gamma)$ (2008Wi09) and adopted 468.9I $\gamma$ from 1988DuZS to deduce its $I\beta$ . |
| $(5.60 \times 10^3)$ 4) | 4869.4            | 10.8                          | 4.9               | $I\beta=10.8$ 16 and $\log ft=4.9$ from 1988DuZS.                                                                                                                               |
| $(5.98 \times 10^3)$ 4) | 4493.5            | 16.6                          | 4.9               | $I\beta=21.4$ 11 and $\log ft=4.8$ from 1988DuZS. 468.9I $\gamma$ feeding this level is deducted from its $I\beta$ .                                                            |
| $(6.09 \times 10^3)$ 4) | 4381.3?           | 9.7                           | 5.1               | $I\beta=9.4$ 13 and $\log ft=5.1$ from 1988DuZS.                                                                                                                                |
| $(6.37 \times 10^3)$ 4) | 4101.2            | 46.1                          | 4.6               | $I\beta=45.9$ 31 and $\log ft=4.5$ from 1988DuZS.                                                                                                                               |
| $(8.08 \times 10^3)$ 4) | 2386.5            | 1.9                           | 8.6 <sup>1u</sup> |                                                                                                                                                                                 |

$^\dagger$   $\beta$ -feeding from  $\gamma$ -ray intensity balance at each level. Quoted  $I\beta^-$  values are considered upper limits due to the incomplete decay scheme, and the associated  $\log ft$  values are considered lower limits.

$^\ddagger\ddagger$  Absolute intensity per 100 decays.

$\gamma(^{35}\text{P})$ 

I $\gamma$  normalization: From  $\Sigma\%I(\gamma \text{ to g.s.})=100$ . The deduced normalization factor of 0.27 should be considered an upper limit due to potential missing  $\gamma$  transitions from unobserved levels in the gap to the ground state.

| $E_\gamma$ <sup>‡</sup>             | $I_\gamma$ <sup>‡@</sup> | $E_i(\text{level})$ | $J_i^\pi$                              | $E_f$  | $J_f^\pi$           | Mult. <sup>†</sup> | $\alpha$ <sup>&amp;</sup> | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------|--------------------------|---------------------|----------------------------------------|--------|---------------------|--------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 241.4 3                             | 100 4                    | 4101.2              | (7/2 <sup>-</sup> )                    | 3859.7 | 5/2 <sup>+</sup>    | [E1]               | 0.000664 10               | %I $\gamma$ =27                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 392.3 3                             | 58.2 28                  | 4493.5              | (7/2 <sup>-</sup> )                    | 4101.2 | (7/2 <sup>-</sup> ) | [M1+E2]            | 4.4×10 <sup>-4</sup> 25   | %I $\gamma$ =16                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 468.9 <sup>a</sup> 4                | 18.7 25                  | 4962.4?             | (9/2 <sup>-</sup> )                    | 4493.5 | (7/2 <sup>-</sup> ) |                    |                           | %I $\gamma$ =5.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                     |                          |                     |                                        |        |                     |                    |                           | Unplaced $\gamma$ ray in <a href="#">1988DuZS</a> . The placement is based on a 466 $\gamma$ observed in $^{208}\text{Pb}(^{36}\text{S},\text{X}\gamma)$ ( <a href="#">2008Wi09</a> ).                                                                                                                                                                                                                                                                                                                                         |
| 633.7 5                             | 21.9 28                  | 4493.5              | (7/2 <sup>-</sup> )                    | 3859.7 | 5/2 <sup>+</sup>    | [E1]               | 4.64×10 <sup>-5</sup> 7   | %I $\gamma$ =5.9                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 768.0 4                             | 15.9 29                  | 4869.4              | (5/2 <sup>-</sup> , 7/2 <sup>-</sup> ) | 4101.2 | (7/2 <sup>-</sup> ) |                    |                           | %I $\gamma$ =4.3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 1009.9 5                            | 24 5                     | 4869.4              | (5/2 <sup>-</sup> , 7/2 <sup>-</sup> ) | 3859.7 | 5/2 <sup>+</sup>    |                    |                           | %I $\gamma$ =6.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 1459.7 5                            | 12 4                     | 5560.7              | (5/2 <sup>-</sup> )                    | 4101.2 | (7/2 <sup>-</sup> ) |                    |                           | %I $\gamma$ =3.2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 1473.4 5                            | 17 4                     | 3859.7              | 5/2 <sup>+</sup>                       | 2386.5 | 3/2 <sup>+</sup>    | [M1,E2]            | 8.3×10 <sup>-5</sup> 13   | %I $\gamma$ =4.6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 1714.7 6                            | 22 5                     | 4101.2              | (7/2 <sup>-</sup> )                    | 2386.5 | 3/2 <sup>+</sup>    | [M2]               | 7.93×10 <sup>-5</sup> 11  | %I $\gamma$ =5.9                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 1994.8 <sup>a</sup> 6               | 36 6                     | 4381.3?             | (5/2 <sup>-</sup> )                    | 2386.5 | 3/2 <sup>+</sup>    |                    |                           | %I $\gamma$ =9.7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                     |                          |                     |                                        |        |                     |                    |                           | Placement from <a href="#">1988DuZS</a> , consistent with the 1995 $\gamma$ 4381 to 2386 transition observed in $^{208}\text{Pb}(^{36}\text{S},\text{X}\gamma)$ ( <a href="#">2008Wi09</a> ) and the 1995 $\gamma$ 4382 to 2386 transition observed in $^9\text{Be}(^{36}\text{S},^{35}\text{P}\gamma)$ ( <a href="#">2016Mu03</a> ). <a href="#">1988DuZT</a> and <a href="#">1987Wa10</a> placed this $\gamma$ as the 6096 to 4101 transition. <a href="#">1988Or01</a> placed this $\gamma$ as the 6488 to 4493 transition. |
| 2386.4 6                            | 117 7                    | 2386.5              | 3/2 <sup>+</sup>                       | 0      | 1/2 <sup>+</sup>    | [M1,E2]            | 0.00046 5                 | %I $\gamma$ =32                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 3173.5 10                           | 35 6                     | 5560.7              | (5/2 <sup>-</sup> )                    | 2386.5 | 3/2 <sup>+</sup>    |                    |                           | %I $\gamma$ =9.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                     |                          |                     |                                        |        |                     |                    |                           | <a href="#">1988Or01</a> suggested that this $\gamma$ ray cannot be placed into the decay scheme of $^{35}\text{Si}$ , and it could be a transition to the 4101 or 4493 levels based on intensity balances.                                                                                                                                                                                                                                                                                                                    |
| <sup>x</sup> 3349.1 <sup>#</sup> 10 | 46 <sup>#</sup> 6        |                     |                                        |        |                     |                    |                           | %I $\gamma$ =12                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <sup>x</sup> 3590.0 <sup>#</sup> 11 | 60 <sup>#</sup> 7        |                     |                                        |        |                     |                    |                           | %I $\gamma$ =16                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 3859.5 10                           | 121 8                    | 3859.7              | 5/2 <sup>+</sup>                       | 0      | 1/2 <sup>+</sup>    | [E2]               | 1.12×10 <sup>-3</sup> 2   | %I $\gamma$ =33                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 4100.8 10                           | 135 8                    | 4101.2              | (7/2 <sup>-</sup> )                    | 0      | 1/2 <sup>+</sup>    | [E3]               | 0.000924 13               | %I $\gamma$ =36                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

<sup>†</sup> From the Adopted Levels.

<sup>‡</sup> From [1988DuZS](#), unless otherwise noted.

<sup>#</sup> From [1986Du07](#). [1988Or01](#) tentatively suggested that these  $\gamma$  rays deexcite a level at 7450, but this suggestion has not been experimentally confirmed.

<sup>@</sup> For absolute intensity per 100 decays, multiply by 0.27.

<sup>&</sup> Total theoretical internal conversion coefficients, calculated using the BrIcc code ([2008Ki07](#)) with “Frozen Orbitals” approximation based on  $\gamma$ -ray energies, assigned multipolarities, and mixing ratios, unless otherwise specified.

$^{35}\text{Si}$   $\beta^-$  decay (0.78 s)    [1988DuZS](#),[1986Du07](#),[1988DuZT](#) (continued)

$\gamma(^{35}\text{P})$  (continued)

<sup>a</sup> Placement of transition in the level scheme is uncertain.

<sup>x</sup>  $\gamma$  ray not placed in level scheme.

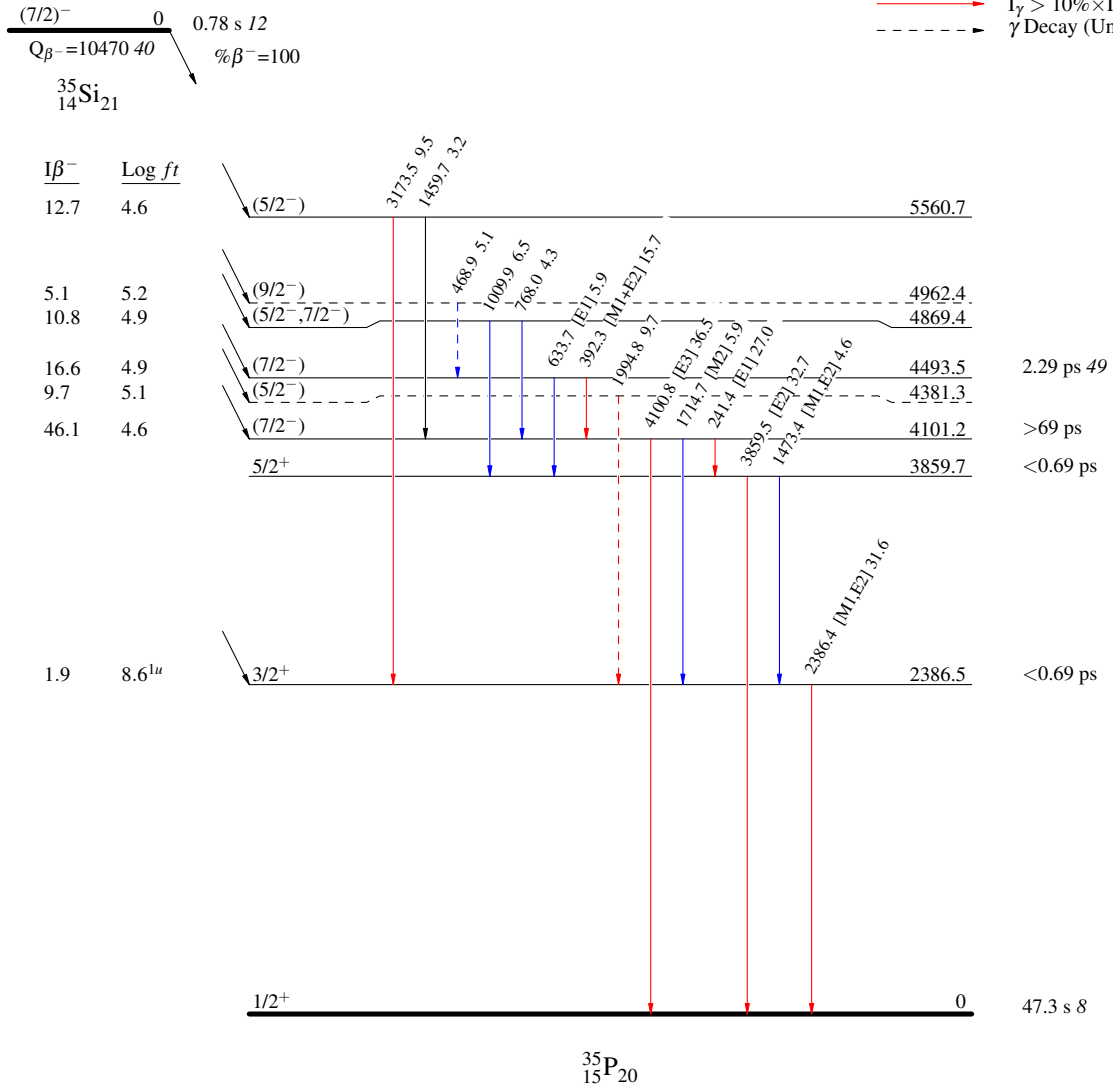
$^{35}\text{Si}$   $\beta^-$  decay (0.78 s) 1988DuZS,1986Du07,1988DuZT

## Decay Scheme

Intensities:  $I_\gamma$  per 100 parent decays

## Legend

- $\longrightarrow$   $I_\gamma < 2\% \times I_\gamma^{\max}$   
 $\longrightarrow$   $I_\gamma < 10\% \times I_\gamma^{\max}$   
 $\longrightarrow$   $I_\gamma > 10\% \times I_\gamma^{\max}$   
 $\cdots$   $\gamma$  Decay (Uncertain)



$^1\text{H}(^{34}\text{Si},\text{p})$ :resonances **2012Im01**

$J^\pi=0^+$  for  $^{34}\text{Si}$  ground state.

**2012Im01**: A  $^{34}\text{Si}$  beam at  $7 \times 10^4$  pps and a purity of 97% was produced by the projectile fragmentation of a 63-MeV/nucleon  $^{40}\text{Ar}$  primary beam and separated by the RIPS separator at RIKEN. The secondary target was a 10.9-mg/cm<sup>2</sup> 5 polyethylene film. An incident energy of 4.4 MeV/nucleon 12 for the  $^{34}\text{Si}$  beam was determined by the timing difference between a plastic scintillator and two PPACs placed upstream of the target. The PPACs also record the positions and angles of the projectiles incident upon the target. Outgoing particles were detected and identified by a three-layer  $\Delta\text{E-E}$  telescope consisting of 0.5-mm DSSD, 1.5-mm silicon, and 1.5-mm silicon detectors mounted at  $0^\circ$  with an  $E_{\text{lab}}$  resolution  $\sigma=130$  keV. Measured excitation functions of proton elastic scattering on  $^{34}\text{Si}$  for  $\theta_{\text{lab}} < 10^\circ$  using thick target inverse kinematics. Deduced  $E_{\text{R}}$ , L-transfer,  $\Gamma_{\text{p}}$ , and  $\Gamma$  from R-matrix analysis for 8 resonances in the highly excited states in  $^{35}\text{P}$ , which are isobaric analog states of  $^{35}\text{Si}$  states.

 $^{35}\text{P}$  Levels

All data are from **2012Im01**.

| <u>E(level)<sup>†</sup></u> | <u><math>\Gamma</math></u> | <u>L</u> | <u><math>S^\ddagger</math></u> | <u>Comments</u>                                                                                                                                                             |
|-----------------------------|----------------------------|----------|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 14938 24                    | <12.7 keV                  | 0        |                                | $\Gamma$ : from original value of 4.6 keV 81 in <b>2012Im01</b> .<br>$E_{\text{R}}=2783$ 24, $\Gamma_{\text{p}}=4.6$ keV 28.                                                |
| 15161 3                     | <4.4 keV                   | 3        | 0.63 16                        | $\Gamma$ : from original value of 1.6 keV 28 in <b>2012Im01</b> .<br>$E_{\text{R}}=3006$ 2, $\Gamma_{\text{p}}=1.6$ keV 4.<br>IAR of the $7/2^-$ g.s. of $^{35}\text{Si}$ . |
| 15306 24                    | <30.4 keV                  | 2        | 0.19 15                        | $\Gamma$ : from original value of 10.4 keV 200 in <b>2012Im01</b> .<br>$E_{\text{R}}=3151$ 24, $\Gamma_{\text{p}}=3.3$ keV 27.                                              |
| 15964 18                    | 84 keV 25                  | 2        | 0.79 20                        | $E_{\text{R}}=3809$ 18, $\Gamma_{\text{p}}=26.7$ keV 69 in <b>2012Im01</b> .                                                                                                |
| 16145 36                    | 354 keV 87                 | 1        | 1.37 32                        | $E_{\text{R}}=3990$ 36, $\Gamma_{\text{p}}=185$ keV 43.                                                                                                                     |
| 16605 44                    | 0.22 MeV 15                | 0        | 0.45 28                        | $E_{\text{R}}=4450$ 44, $\Gamma_{\text{p}}=58.4$ keV 370.                                                                                                                   |
| 17254 12                    | <11.6 keV                  | 2        | 0.04 1                         | $\Gamma$ : from original value of 3.8 keV 78 in <b>2012Im01</b> .<br>$E_{\text{R}}=5099$ 12, $\Gamma_{\text{p}}=3.8$ keV 9.                                                 |
| 17355 15                    | 32 keV 22                  | 1        | 0.12 7                         | $E_{\text{R}}=5200$ 15, $\Gamma_{\text{p}}=20.9$ keV 120.                                                                                                                   |

<sup>†</sup> Excitation energies are deduced by evaluators from  $E_{\text{R}}+S_{\text{p}}(^{35}\text{P})$ , where  $S_{\text{p}}(^{35}\text{P})=12155.1$  20 (**2021Wa16**).  $E_{\text{R}}$  given in **2012Im01** are in the center-of-mass system.

<sup>‡</sup> Spectroscopic factors are derived from  $\Gamma_{\text{p}}$  using the formula from **1968Th07** as described in **2012Im01**.

$^2\text{H}(^{34}\text{Si}, ^{35}\text{P}\gamma)$  2007GeZX

$^{34}\text{Si}(\text{d}, \text{n})^{35}\text{P}$  from  $J^\pi=0^+$   $^{34}\text{Si}$  g.s. in inverse kinematics.

2007GeZX: 30-AMeV  $^{34}\text{Si}$  beam on 30-mg/cm<sup>2</sup> CD<sub>2</sub> secondary target at GANIL. Heavy ions produced in reactions were identified by the VAMOS spectrometer.  $\gamma$  rays were detected using the EXOGAM germanium clover array. Measured Doppler-corrected  $E_\gamma$ ,  $I_\gamma$ ,  $\gamma\gamma$ -coin, and  $(^{35}\text{P})\gamma$ -coin. Deduced levels,  $J$ ,  $\pi$ . Compared with shell-model calculations.

 $^{35}\text{P}$  Levels

$E(\text{level})^\dagger$

0  
2386.5 8  
3859.9 8  
4100.9 13  
4492.9 16  
4868.9 13

<sup>†</sup> From a least-squares fit to  $\gamma$ -ray energies, assuming  $\Delta E_\gamma=1$  keV since uncertainties were not given in 2007GeZX.

 $\gamma(^{35}\text{P})$ 

| <u><math>E_\gamma^\dagger</math></u> | <u><math>E_i(\text{level})</math></u> | <u><math>E_f</math></u> |
|--------------------------------------|---------------------------------------|-------------------------|
| 241                                  | 4100.9                                | 3859.9                  |
| 392                                  | 4492.9                                | 4100.9                  |
| 1009                                 | 4868.9                                | 3859.9                  |
| 1473 <sup>‡</sup>                    | 3859.9                                | 2386.5                  |
| 2386                                 | 2386.5                                | 0                       |
| 3860                                 | 3859.9                                | 0                       |

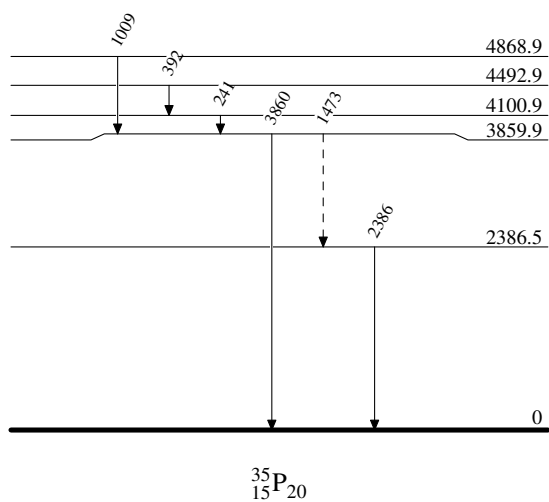
<sup>†</sup> From 2007GeZX.

<sup>‡</sup> Placement of transition in the level scheme is uncertain.

$^2\text{H}(^{34}\text{Si}, ^{35}\text{P}\gamma)$  2007GeZX

Legend

Level Scheme

 - - - - -  $\rightarrow$   $\gamma$  Decay (Uncertain)


$^2\text{H}(^{36}\text{S}, ^3\text{He})$  2020Sa44

**2020Sa44:**  $E(^{36}\text{S})=15.3$  MeV/nucleon was provided by the Argonne Tandem Linear Accelerator System. Targets were 81, 127, 529- $\mu\text{g}/\text{cm}^2$  deuterated plastic. The reaction was studied in inverse kinematics with the HELical Orbit Spectrometer (HELIOS). The  $^3\text{He}$  ions spiralled in a 2.85-T magnetic field and were detected by a position-sensitive silicon array. The heavy-ion products were measured with a 65- $\mu\text{m}$  thick silicon detector installed between the target and the silicon array. Measured  $\sigma(E(^3\text{He}), \theta)$ . Deduced relative spectroscopic factors for 6 levels from the PTOLEMY-DWBA analysis of the angular distributions. A potential  $1/2^+$  neutron 2p2h excitation bandhead in 2.5-3.6 MeV was investigated, and an upper limit for the transfer reaction cross section to populate such an intruder configuration was deduced.

 $^{35}\text{P}$  Levels

**2020Sa44** states that the measurement of the beam current was not made with sufficient accuracy to calculate absolute spectroscopic factors. The  $\text{C}^2\text{S}$  values are relative spectroscopic factors normalized such that the ground-state value is 2.

| <u><math>E(\text{level})^\dagger</math></u> | <u><math>J^\pi^\ddagger</math></u> | <u><math>\text{C}^2\text{S}^\#</math></u> |
|---------------------------------------------|------------------------------------|-------------------------------------------|
| 0 1                                         | $1/2^+$                            | 2.0                                       |
| 2388 13                                     | $3/2^+$                            | 0.33 9                                    |
| 3860 2                                      | $5/2^+$                            | 2.9 10                                    |
| 4666 9                                      | $5/2^+$                            | 0.71 34                                   |
| 5202 8                                      | $5/2^+$                            | 1.1 6                                     |
| 5706 38                                     | $(1/2^-)$                          | 0.23 5                                    |

<sup>†</sup> From 2020Sa44.

<sup>‡</sup> Assumed by 2020Sa44 for deducing  $\text{C}^2\text{S}$ , consistent with the Adopted Levels.

<sup>#</sup> Uncertainties are dominated by systematic components.



$^9\text{Be}(^{36}\text{S}, ^{35}\text{P}\gamma)$  2016Mu03

$J^\pi=0^+$  for  $^{36}\text{S}$  ground state.

**2016Mu03:** A secondary beam of  $^{36}\text{S}$  was produced via the projectile fragmentation of a 140-MeV/nucleon  $^{48}\text{Ca}$  primary beam impinging on an 846-mg/cm<sup>2</sup>  $^9\text{Be}$  target at the coupled cyclotron facility at NSCL, MSU. The  $^{36}\text{S}$  nuclei were selected using the A1900 separator with an intensity of  $8.1 \times 10^5$  pps and a purity of 89.7%. The excited states of  $^{35}\text{P}$  were populated by the one-proton knockout reaction from the  $^{36}\text{S}$  beam at 88 MeV/nucleon (midtarget energy) on a 100-mg/cm<sup>2</sup>  $^9\text{Be}$  secondary target located at the reaction target position of the S800 spectrometer. The projectile-like residues were identified from their energy loss measured by an ionization chamber located at the focal plane of the S800 spectrometer and from their ToF measured between two scintillators situated at the object position and at the focal plane of the S800 spectrometer. Prompt  $\gamma$  rays from the deexcitation of the  $^{35}\text{P}$  residues were detected by seven modules of the GRETINA Ge array. Measured  $E_\gamma$ ,  $I_\gamma$ ,  $(^{35}\text{P})\gamma$ -coin,  $\gamma\gamma$ -coin, the inclusive knockout cross section for producing  $^{35}\text{P}$  from  $^{36}\text{S}$ , the fractional populations and parallel momentum distributions of 8 populated states in  $^{35}\text{P}$  residues. Deduced levels,  $J$ ,  $\pi$ , L-transfers, partial knockout cross sections and spectroscopic factors. Calculated single-particle cross sections ( $\sigma_{\text{sp}}$ ) for proton removal and parallel momentum distributions using the eikonal model. This work is also part of the thesis [2015MuZY](#).

 $^{35}\text{P}$  Levels

$b_f$  values under comments are population fraction in the knock-out reaction,  $\sigma_f^{\text{sp}}$  values are for calculated single-particle cross sections, and  $R_s$  values are quenching factors from a systematics study in [2014To14](#). The partial cross sections to each level can be deduced from  $b_f \times \sigma_{\text{inc}}^{\text{exp}}$ .

The inclusive cross section  $\sigma_{\text{inc}}^{\text{exp}}$  for producing  $^{35}\text{P}$  from  $^{36}\text{S}$  is measured to be 51 mb. 95.3% is attributed to a direct knockout process from the p-sd shell, i.e.,  $\sigma_{\text{inc,KO}}^{\text{exp}} = 0.953 \times \sigma_{\text{inc}}^{\text{exp}}$ . The remaining 4.7% is possibly attributed to a charge-exchange or a two-step process:  $^{36}\text{S}$  core excitation by inelastic scattering, followed by a proton removal. This is proposed based on their shifted parallel momentum distributions, their decay modes to negative parity states, their high excitation energy (around 4.7 MeV), and the fact that they were not observed in the  $^{36}\text{S}(d, ^3\text{He})$  reaction study ([1985Kh04](#)).

| E(level) <sup>†</sup> | $J^\pi$ @                         | L <sup>#</sup> | C <sup>2</sup> S& | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------|-----------------------------------|----------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0.0                   | 1/2 <sup>+</sup>                  | 0              | 2.2 7             | $b_f=31.4\%$ 72, $\sigma_f^{\text{sp}}=13.5$ mb, $R_s=0.55$ 11.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 2388 1                | 3/2 <sup>+</sup>                  | 2              | 0.7 3             | $b_f=7.8\%$ 36, $\sigma_f^{\text{sp}}=10.2$ mb, $R_s=0.52$ 10.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 3862 1                | 5/2 <sup>+</sup>                  | 2              | 2.7 8             | $b_f=27.3\%$ 58, $\sigma_f^{\text{sp}}=10.6$ mb, $R_s=0.49$ 10.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 4101 2                | (7/2 <sup>-</sup> ) <sup>‡</sup>  |                |                   | $b_f=-0.7\%$ 7.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 4383 3                | (5/2 <sup>-</sup> ) <sup>‡</sup>  |                |                   | $b_f=2.0\%$ 4.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 4494 2                | (7/2 <sup>-</sup> )               | (3)            | 0.30 7            | L: 3 reported in <a href="#">2016Mu03</a> . The parentheses are added by evaluators based on the statement in <a href="#">2016Mu03</a> that the parallel momentum distribution has limited statistics and seems more consistent with L=3.<br>$b_f=2.9\%$ 5, $\sigma_f^{\text{sp}}=10.7$ mb, $R_s=0.48$ 10.<br>$b_f=9.5\%$ 10, $\sigma_f^{\text{sp}}=10.3$ mb, $R_s=0.47$ 9.                                                                                                                                                                                      |
| 4667 2                | 5/2 <sup>+</sup>                  | 2              | 1.0 2             | $b_f=0.9\%$ 3.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 4768 2                | (9/2 <sup>-</sup> ) <sup>‡</sup>  |                |                   | $b_f=0.3\%$ 2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 4962 3                | (9/2 <sup>-</sup> ) <sup>‡</sup>  |                |                   | $b_f=1.5\%$ 2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5089 3                | (11/2 <sup>-</sup> ) <sup>‡</sup> |                |                   | $b_f=14.5\%$ 14, $\sigma_f^{\text{sp}}=10.2$ mb, $R_s=0.47$ 9.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5200 2                | 5/2 <sup>+</sup>                  | 2              | 1.5 3             | $b_f=1.4\%$ 10, $\sigma_f^{\text{sp}}=10.8$ mb, $R_s=0.47$ 9.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 5710 2                | (1/2 <sup>-</sup> )               | (1)            | 0.21 16           | $J^\pi$ : Interpreted as the deeply bound $1p_{1/2}$ proton removal.<br>L: 1 reported in <a href="#">2016Mu03</a> . The parentheses are added by evaluators based on the parallel momentum distribution (Fig. 3) in <a href="#">2016Mu03</a> .<br>$b_f=1.2\%$ 3, $\sigma_f^{\text{sp}}=10.2$ mb, $R_s=0.44$ 9.<br>$J^\pi$ : Interpreted as the deeply bound $1p_{1/2}$ proton removal.<br>L: 1 reported in <a href="#">2016Mu03</a> . The parentheses are added by evaluators based on the parallel momentum distribution (Fig. 3) in <a href="#">2016Mu03</a> . |
| 7527 2                | (1/2 <sup>-</sup> )               | (1)            | 0.20 6            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

<sup>†</sup> From a least-squares fit to  $\gamma$ -ray energies.

<sup>‡</sup> Populated by a two-step reaction mechanism. [2016Mu03](#) states that they are likely to have  $J \geq 5/2$  and negative parities.

$^9\text{Be}(^{36}\text{S}, ^{35}\text{P}\gamma)$  2016Mu03 (continued) $^{35}\text{P}$  Levels (continued)

# 2016Mu03 deduced L by comparing the measured and eikonal-calculated parallel momentum distributions of residuals.

@ From the Adopted Levels, also assumed by 2016Mu03 for deducing  $\text{C}^2\text{S}$ .

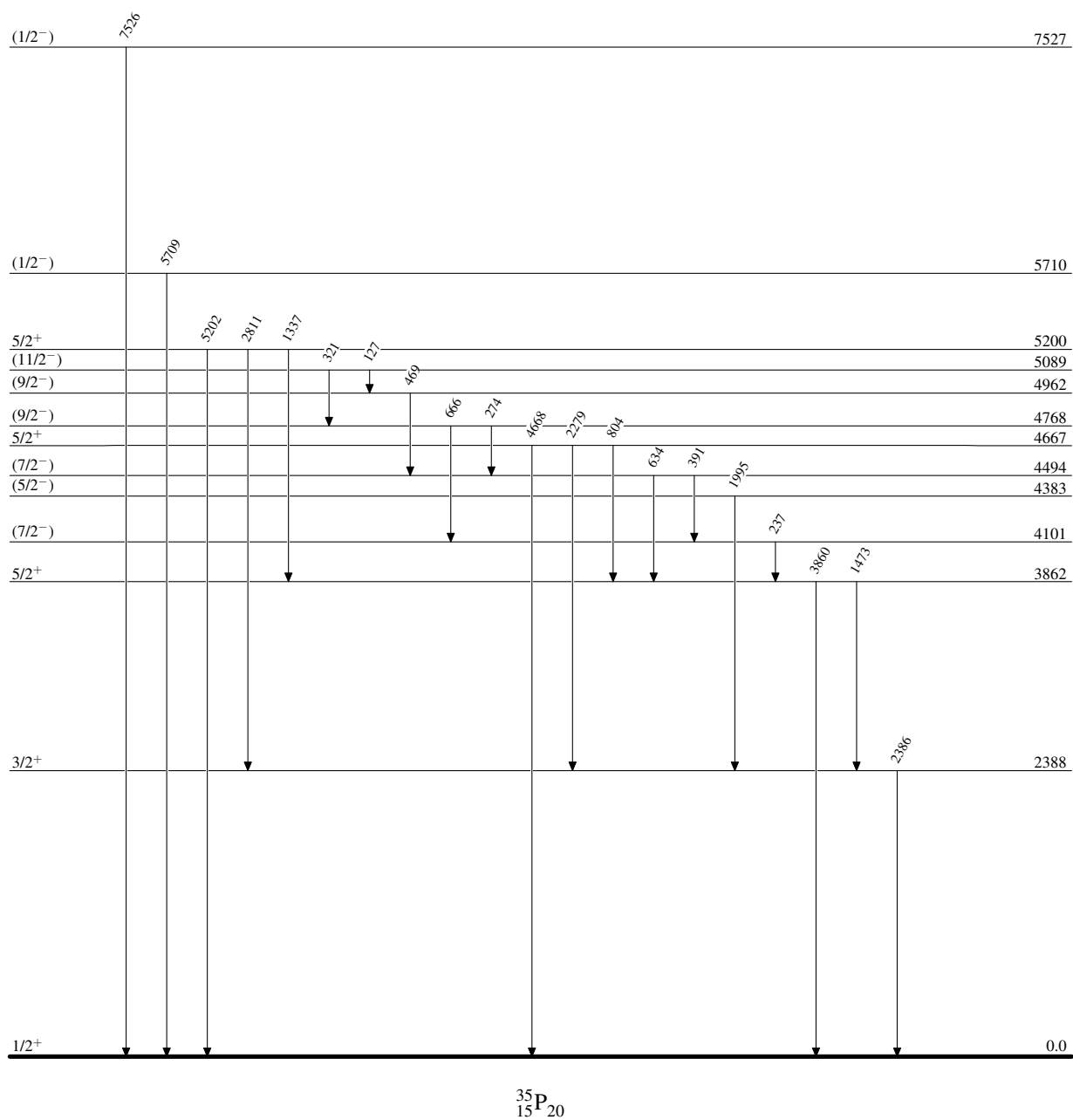
& The spectroscopic factor  $\text{C}^2\text{S} = b_f \sigma_{\text{inc}}^{\text{exp}} / R_s \sigma_f^{\text{sp}}$ , where  $b_f$  is the population fraction,  $\sigma_{\text{inc}}^{\text{exp}}$  is the measured total inclusive cross section,  $R_s$  is the quenching factor from a systematics study 2014To14, and  $\sigma_f^{\text{sp}}$  is the eikonal model calculated single-particle cross section.

 $\gamma(^{35}\text{P})$ 

| $E_\gamma$ | $E_i(\text{level})$ | $J_i^\pi$            | $E_f$ | $J_f^\pi$           | $E_\gamma$ | $E_i(\text{level})$ | $J_i^\pi$           | $E_f$ | $J_f^\pi$        |
|------------|---------------------|----------------------|-------|---------------------|------------|---------------------|---------------------|-------|------------------|
| 127 2      | 5089                | (11/2 <sup>-</sup> ) | 4962  | (9/2 <sup>-</sup> ) | 1473 2     | 3862                | 5/2 <sup>+</sup>    | 2388  | 3/2 <sup>+</sup> |
| 237 2      | 4101                | (7/2 <sup>-</sup> )  | 3862  | 5/2 <sup>+</sup>    | 1995 2     | 4383                | (5/2 <sup>-</sup> ) | 2388  | 3/2 <sup>+</sup> |
| 274 2      | 4768                | (9/2 <sup>-</sup> )  | 4494  | (7/2 <sup>-</sup> ) | 2279 2     | 4667                | 5/2 <sup>+</sup>    | 2388  | 3/2 <sup>+</sup> |
| 321 2      | 5089                | (11/2 <sup>-</sup> ) | 4768  | (9/2 <sup>-</sup> ) | 2386 2     | 2388                | 3/2 <sup>+</sup>    | 0.0   | 1/2 <sup>+</sup> |
| 391 2      | 4494                | (7/2 <sup>-</sup> )  | 4101  | (7/2 <sup>-</sup> ) | 2811 2     | 5200                | 5/2 <sup>+</sup>    | 2388  | 3/2 <sup>+</sup> |
| 469 2      | 4962                | (9/2 <sup>-</sup> )  | 4494  | (7/2 <sup>-</sup> ) | 3860 2     | 3862                | 5/2 <sup>+</sup>    | 0.0   | 1/2 <sup>+</sup> |
| 634 2      | 4494                | (7/2 <sup>-</sup> )  | 3862  | 5/2 <sup>+</sup>    | 4668 2     | 4667                | 5/2 <sup>+</sup>    | 0.0   | 1/2 <sup>+</sup> |
| 666 2      | 4768                | (9/2 <sup>-</sup> )  | 4101  | (7/2 <sup>-</sup> ) | 5202 2     | 5200                | 5/2 <sup>+</sup>    | 0.0   | 1/2 <sup>+</sup> |
| 804 2      | 4667                | 5/2 <sup>+</sup>     | 3862  | 5/2 <sup>+</sup>    | 5709 2     | 5710                | (1/2 <sup>-</sup> ) | 0.0   | 1/2 <sup>+</sup> |
| 1337 2     | 5200                | 5/2 <sup>+</sup>     | 3862  | 5/2 <sup>+</sup>    | 7526 2     | 7527                | (1/2 <sup>-</sup> ) | 0.0   | 1/2 <sup>+</sup> |

$^9\text{Be}(^{36}\text{S}, ^{35}\text{P}\gamma)$  2016Mu03

## Level Scheme



---

 $^{34}\text{S}(^{18}\text{O}, ^{17}\text{F})$  **1988Or01**

---

**1988Or01:** A 108-MeV  $^{18}\text{O}^{7+}$  beam was produced from the ANU 14UD Pelletron accelerator. Targets were enriched  $\text{Ag}_2\text{S}$ .

Reaction products were momentum-analyzed with an Enge split-pole spectrometer (FWHM=250 keV at 5.75-10.25°) and detected with a multi-element gas-filled detector at the focal plane. Measured  $\sigma(E(^{17}\text{F}))$ . Deduced levels and ground state mass excess (−24.87 MeV *4*). Comparisons with shell-model calculations. Proposed a decay scheme of  $^{35}\text{Si}$  for the  $\gamma$ -ray transitions observed but not placed in a decay scheme by **1986Du07**.

 $^{35}\text{P}$  Levels

| <u>E(level)<sup>†</sup></u>  |
|------------------------------|
| 0                            |
| 2420 <i>40</i>               |
| 5070 <i>40</i>               |
| 5890 <i>70</i>               |
| 6440 <i>60</i>               |
| 7050 <i>60</i>               |
| 7440 <i>60</i>               |
| 7920 <i>60</i>               |
| $8.60 \times 10^3$ <i>10</i> |
| 9290 <i>50</i>               |

<sup>†</sup> From **1988Or01**.

$^{36}\text{S}(\text{d}, ^3\text{He})$  1984Th08

$J^\pi=0^+$  for  $^{36}\text{S}$  ground state.

1984Th08: A 30-MeV deuteron beam was produced from the BNL Double MP tandem facility. Targets were 20.1 and 23.0  $\mu\text{g}/\text{cm}^2$  sulphur with  $^{36}\text{S}$  enriched to 81.1%. Reaction products were momentum-analyzed with the BNL Q3D magnetic spectrometer (FWHM $\approx$ 32 keV) and detected with a multi-wire proportional counter backed by a topping plastic scintillator. Measured  $\sigma(\text{E}(^3\text{He}), \theta)$ . Deduced levels, J,  $\pi$ , L-transfers, spectroscopic factors from the finite-range DWUCK4-DWBA analysis of the angular distributions.

 $^{35}\text{P}$  Levels

Spectroscopic factor  $\text{C}^2\text{S}=(2j+1)\times\sigma(\theta)_{\text{exp}}/\sigma(\theta)_{\text{DWBA}}/\text{N}$ , where j denotes the total angular momentum of the transferred nucleon.

N=29.5 is a normalization factor adopted by 1984Th08. The discrepancy from the original N=2.95 in 1966Ba54 likely arises from different units of cross sections.

| <u>E(level)<sup>†</sup></u> | <u>L<sup>‡</sup></u> | <u>C<sup>2</sup>S<sup>‡</sup></u> |
|-----------------------------|----------------------|-----------------------------------|
| 0                           | 0                    | 2.3 12                            |
| 3864 10                     | 2                    | 1.45, 1.10 <sup>#</sup>           |
| 4664 10                     | 2                    | 0.53, 0.41 <sup>#</sup>           |
| 5202 10                     | 2                    | 0.40, 0.30 <sup>#</sup>           |

<sup>†</sup> From 1984Th08.

<sup>‡</sup> From DWBA analysis of measured  $\sigma(\theta)$ . The uncertainty of C<sup>2</sup>S is estimated to be 50% by 1984Th08.

<sup>#</sup> Quoted values are for j=L-1/2 (1d<sub>3/2</sub>) and L+1/2 (1d<sub>5/2</sub>), respectively.

$^{36}\text{S}(\text{pol d}, ^3\text{He})$  1985Kh04

$J^\pi=0^+$  for  $^{36}\text{S}$  ground state.

1985Kh04: 52-MeV unpolarized and vector-polarized deuteron beams of 100 nA were produced from the Karlsruhe cyclotron. The target was a 1 mg/cm<sup>2</sup>  $^{208}\text{Pb}$  sulfide with 81.1% enriched  $^{36}\text{S}$  on  $^{12}\text{C}$  backing. Reaction products were detected using two 300- $\mu\text{m}$  and 1500- $\mu\text{m}$  thick  $\Delta\text{E-E}$  surface-barrier detector telescopes (FWHM $\approx$ 90 keV). Measured angular distributions of differential cross sections  $\sigma(\text{E}(^3\text{He}), \theta)$  and angular distributions of analyzing powers  $i\text{T}_{11}(\text{E}(^3\text{He}), \theta)$ . Deduced mass excess, levels, J,  $\pi$ , L-transfers, and spectroscopic factors from standard local, zero-range JULIE-DWBA analysis of  $\sigma(\text{E}(^3\text{He}), \theta)$  and  $i\text{T}_{11}(\text{E}(^3\text{He}), \theta)$ .

 $^{35}\text{P}$  Levels

Spectroscopic factor  $\text{C}^2\text{S}=\sigma(\theta)_{\text{exp}}/\sigma(\theta)_{\text{DWBA}}/\text{N}$ , where  $\text{N}=2.95$  is a normalization factor adopted by 1985Kh04 from 1974Ma34, originally from 1966Ba54.

| $\text{E}(\text{level})^\dagger$ | $\text{J}^\pi$ | $\text{L}^\ddagger$ | $\text{C}^2\text{S}^\ddagger$ | Comments                                                                                                                                                                       |
|----------------------------------|----------------|---------------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0                                | $1/2^+$        | 0                   | 1.63                          |                                                                                                                                                                                |
| 2386 6                           | $3/2^+^\#$     | 2                   | $0.31^\#$                     |                                                                                                                                                                                |
| 3857 2                           | $5/2^+@$       | 2                   | $2.91@$                       |                                                                                                                                                                                |
| 4474 21                          |                |                     | <0.2                          | $1\text{d}_{3/2}$ proton transfer assumed in DWBA calculations, but 2016Mu03 proposed $1\text{f}_{7/2}$ ( $\text{L}=3$ ) based on the measured parallel momentum distribution. |
| 4665 3                           | $5/2^+@$       | 2                   | $1.06@$                       |                                                                                                                                                                                |
| 5189 13                          | $5/2^+@$       | 2                   | $1.38@$                       |                                                                                                                                                                                |
| 7520 30                          |                |                     | <0.4                          |                                                                                                                                                                                |

$^\dagger$  From 1985Kh04.

$^\ddagger$  From DWBA analysis of measured  $\sigma(\theta)$ .

$^\#$  L-1/2 transfer from analyzing power measurements;  $1\text{d}_{3/2}$  proton transfer assumed in DWBA calculations.

$@$  L+1/2 transfer from analyzing power measurements;  $1\text{d}_{5/2}$  proton transfer assumed in DWBA calculations.

---

 $^{37}\text{Cl}(^{11}\text{B},^{13}\text{N})$  **1988Or01**

---

**1988Or01:** An 81-MeV  $^{11}\text{B}^{5+}$  beam was produced from the ANU 14UD Pelletron accelerator. Targets were enriched  $\text{BaCl}_2$ .

Reaction products were momentum-analyzed with an Enge split-pole spectrometer (FWHM=200 keV at 5.75-10.25°) and detected with a multi-element gas-filled detector at the focal plane. Measured  $\sigma(E(^{13}\text{N}))$ . Deduced levels. Comparisons with shell-model and DWBA calculations. Proposed a decay scheme of  $^{35}\text{Si}$  based on the  $\gamma$  transitions observed by **1986Du07**.

---

 $^{35}\text{P}$  Levels

---

| E(level) <sup>†</sup> | Comments                                      |
|-----------------------|-----------------------------------------------|
| 0                     | $d\sigma/d\Omega=120 \mu\text{b/sr}$ 30.      |
| 2389 4                | $d\sigma/d\Omega=200 \mu\text{b/sr}$ 50.      |
| 3860 10               | $d\sigma/d\Omega=35 \mu\text{b/sr}$ 9.        |
| 4250 20               |                                               |
| 4640 20               | $d\sigma/d\Omega=30 \mu\text{b/sr}$ 8.        |
| 5010 20               |                                               |
| 5220 40               | $d\sigma/d\Omega\approx 10 \mu\text{b/sr}$ 3. |
| 5840 50               |                                               |
| 7590 20               | $d\sigma/d\Omega=45 \mu\text{b/sr}$ 11.       |
| 8390 40               |                                               |

<sup>†</sup> From **1988Or01**.

$^{160}\text{Gd}(^{37}\text{Cl},\text{X}\gamma)$  1994Fo04

Possibly  $^{160}\text{Gd} + ^{37}\text{Cl} \rightarrow ^{160}\text{Dy} + ^{35}\text{P} + 2\text{n}$ .

1994Fo04: A 167-MeV  $^{37}\text{Cl}$  beam was produced from the Argonne Tandem Linac Accelerator System (ATLAS). Targets were 1 mg/cm<sup>2</sup> 98.1% enriched  $^{160}\text{Gd}$  backed by 15 mg/cm<sup>2</sup> gold.  $\gamma$  rays were detected using the Argonne-Notre Dame BGO  $\gamma$ -ray facility consisting of 12 Compton-suppressed Ge detectors and a 50-element bismuth germanate (BGO) array. Measured  $E_\gamma$ ,  $I_\gamma$ ,  $\gamma\gamma$ -coin. Deduced levels.

 $^{35}\text{P}$  Levels

$E(\text{level})^\dagger$

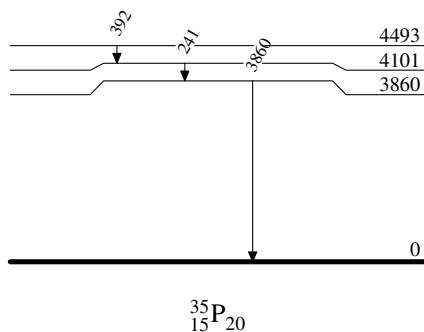
0  
3860  
4101  
4493

<sup>†</sup> From  $E_\gamma$  data.

 $\gamma(^{35}\text{P})$ 

| <u><math>E_\gamma^\dagger</math></u> | <u><math>E_i(\text{level})</math></u> | <u><math>E_f</math></u> |
|--------------------------------------|---------------------------------------|-------------------------|
| 241                                  | 4101                                  | 3860                    |
| 392                                  | 4493                                  | 4101                    |
| 3860                                 | 3860                                  | 0                       |

<sup>†</sup> From 1994Fo04.

 $^{160}\text{Gd}(^{37}\text{Cl},\text{X}\gamma)$  1994Fo04Level Scheme



**$^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$  2008Wi09,2015Ch56,2019Gr08**

**2008Wi09,2010WiZZ:** A 230-MeV  $^{36}\text{S}$  beam was produced by the Argonne Tandem Linac Accelerator System (ATLAS) with an intensity of 1.5 pA on a 0.5 mg/cm<sup>2</sup>  $^{208}\text{Pb}$  target and an intensity of 0.3 pA on a 44 mg/cm<sup>2</sup>  $^{208}\text{Pb}$  target. In the thin-target run, binary transfer products were detected using a heavy-ion parallel-plate avalanche counter (PPAC) array (CHICO) (Time resolution  $\approx 0.7$  ns). The polar angle covered was 12°–85° with respect to the beam.  $\gamma$  rays were detected by Gammasphere consisting of 101 HPGe detectors with FWHM=2–10 keV at  $E_\gamma=1$  MeV. Event-by-event Doppler shift correction was applied. In the thick-target run, binary transfer products were stopped in the target.  $\gamma$  rays were detected by Gammasphere consisting of 95 HPGe detectors with FWHM=2–3 keV at  $E_\gamma=1$  MeV. Measured  $E_\gamma$ ,  $I_\gamma$ ,  $\gamma\gamma$ -coin. Deduced levels. Comparisons with shell-model calculations. Branching-ratio limits were reported for predicted transitions to the  $2\hbar\omega$  bandheads in  $^{35}\text{P}$  and  $^{34}\text{Si}$ . An e-mail reply from Mathis Wiedeking in April, 2010 (**2010WiZZ**) provides relative  $\gamma$ -ray intensities, supplementing **2008Wi09**.

**2015Ch56:** A 215-MeV  $^{36}\text{S}$  beam was produced using the combination of XTU tandem Van de Graaff accelerator and ALPI superconducting linear accelerator at the INFN Legnaro National Laboratory. The target was 300- $\mu\text{g}/\text{cm}^2$  99.7% enriched  $^{208}\text{Pb}$  on a 20  $\mu\text{g}/\text{cm}^2$  carbon backing. Projectile-like fragments produced in multinucleon binary grazing reactions were separated and identified by the PRISMA spectrometer.  $\gamma$  rays were detected using the CLARA array of 22 EUROBALL escape-suppressed HPGe clover detectors. Doppler corrections of  $\gamma$ -ray energies were performed event by event. Measured  $E_\gamma$ ,  $I_\gamma$ ,  $(^{35}\text{P})\gamma$ -coin, and  $\gamma\gamma$ -coin. **2015Ch56** also revisited the  $\gamma\gamma\gamma$ -coin of  $^{36}\text{S}+^{176}\text{Yb}$  deep-inelastic data by J. Ollier Ph.D. thesis, University of Paisley (2004) to strengthen the evidence for  $\gamma$ -ray placements (see Ref. [39] in **2015Ch56**). Deduced levels,  $J$ ,  $\pi$ . Comparisons with shell-model calculations.

**2019Gr08:** A 225-MeV  $^{36}\text{S}$  beam was provided by Tandem-ALPI accelerator complex at the INFN Legnaro National Laboratory. The target was 1 mg/cm<sup>2</sup> 99.7% enriched  $^{208}\text{Pb}$  with 1 mg/cm<sup>2</sup> Nb backing and mounted onto the Cologne differential plunger. Projectile-like fragments produced in binary grazing reactions were separated and identified by the PRISMA spectrometer.  $\gamma$  rays were detected using the AGATA demonstrator array of five triple cluster modules of 36-fold segmented Ge crystals covering backward angles from 135° to 175°. Doppler corrections of  $\gamma$ -ray energies were performed event by event. Measured  $E_\gamma$ ,  $(^{35}\text{P})\gamma$ -coin, and level lifetimes using the differential recoil-distance method (DRDM). Comparison with shell-model calculations.

 $^{35}\text{P}$  Levels

| $E(\text{level})^\dagger$ | $J^\pi^\ddagger$         | $T_{1/2}^\#$ | Comments                                                                                                                                                                       |
|---------------------------|--------------------------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0                         | $1/2^+$                  |              |                                                                                                                                                                                |
| 2386.7 7                  | $3/2^+$                  | <0.69 ps     | $T_{1/2}$ : estimated mean lifetime $\tau < 1$ ps ( <b>2019Gr08</b> ).                                                                                                         |
| 3860.8 7                  | $5/2^+$                  | <0.69 ps     | $T_{1/2}$ : estimated mean lifetime $\tau < 1$ ps ( <b>2019Gr08</b> ).                                                                                                         |
| 4102.1 7                  | $(7/2^-)$                | >69 ps       | $J^\pi$ : $7/2^-$ proposed by <b>2019Gr08</b> based on comparisons with shell-model calculations.<br>$T_{1/2}$ : estimated mean lifetime $\tau > 100$ ps ( <b>2019Gr08</b> ).  |
| 4381.9 10                 | $(5/2^-)$                |              |                                                                                                                                                                                |
| 4494.2 8                  | $(7/2^-)$                | 2.29 ps 49   | $J^\pi$ : $7/2^-$ proposed by <b>2019Gr08</b> based on comparisons with shell-model calculations.<br>$T_{1/2}$ : measured mean lifetime $\tau = 3.3$ ps 7 ( <b>2019Gr08</b> ). |
| 4767.1 10                 | $(9/2^-)$                |              | $J^\pi$ : $9/2^-$ proposed by <b>2019Gr08</b> based on comparisons with shell-model calculations.                                                                              |
| 4869.0 8                  | $(5/2^-, 7/2^-)$         |              |                                                                                                                                                                                |
| 4962.1 12                 | $(9/2^-)$                |              | $J^\pi$ : $9/2^-$ proposed by <b>2019Gr08</b> based on comparisons with shell-model calculations.                                                                              |
| 5089.8 11                 | $(11/2^-)$               |              | $J^\pi$ : $11/2^-$ proposed by <b>2019Gr08</b> based on comparisons with shell-model calculations.                                                                             |
| 5488.2 10                 |                          |              |                                                                                                                                                                                |
| 5560.1 12                 | $(5/2^-)$                |              |                                                                                                                                                                                |
| 6222.4 11                 | $(7/2^-, 9/2^-, 11/2^-)$ |              |                                                                                                                                                                                |

<sup>†</sup> From a least-squares fit to  $\gamma$ -ray energies.

<sup>‡</sup> From the Adopted Levels.

<sup>#</sup> From differential recoil-distance method (DRDM) (**2019Gr08**).

$^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$  **2008Wi09,2015Ch56,2019Gr08 (continued)** $\gamma(^{35}\text{P})$ 

| $E_\gamma$                 | $I_\gamma$              | $E_i(\text{level})$ | $J_i^\pi$                                    | $E_f$  | $J_f^\pi$            | Comments                                                                                                                                                                                                           |
|----------------------------|-------------------------|---------------------|----------------------------------------------|--------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 128 <i>I</i>               | 10 5                    | 5089.8              | (11/2 <sup>-</sup> )                         | 4962.1 | (9/2 <sup>-</sup> )  | $E_\gamma$ : weighted average of 128 <i>I</i> (2008Wi09) and 127 <i>I</i> (2015Ch56).<br>$I_\gamma$ : unweighted average of 14 2 (2008Wi09) and 5.2 6 (2015Ch56).                                                  |
| 241 <i>I</i>               | 61 4                    | 4102.1              | (7/2 <sup>-</sup> )                          | 3860.8 | 5/2 <sup>+</sup>     | $E_\gamma$ : From 2008Wi09 and 2015Ch56.<br>$I_\gamma$ : From 2008Wi09. Other: 32.6 9 (2015Ch56).                                                                                                                  |
| 273 <i>I</i>               | 12.8 8                  | 4767.1              | (9/2 <sup>-</sup> )                          | 4494.2 | (7/2 <sup>-</sup> )  | $E_\gamma$ : From 2008Wi09 and 2015Ch56.<br>$I_\gamma$ : weighted average of 12 2 (2008Wi09) and 12.9 8 (2015Ch56).                                                                                                |
| 322 <i>I</i>               | 20 7                    | 5089.8              | (11/2 <sup>-</sup> )                         | 4767.1 | (9/2 <sup>-</sup> )  | $E_\gamma$ : weighted average of 321 <i>I</i> (2008Wi09) and 323 <i>I</i> (2015Ch56).<br>$I_\gamma$ : unweighted average of 27 3 (2008Wi09) and 12.9 8 (2015Ch56).                                                 |
| 374 <sup>†</sup> <i>I</i>  | 3 <sup>†</sup> <i>I</i> | 4869.0              | (5/2 <sup>-</sup> , 7/2 <sup>-</sup> )       | 4494.2 | (7/2 <sup>-</sup> )  | $E_\gamma$ : weighted average of 391 <i>I</i> (2008Wi09) and 392 <i>I</i> (2015Ch56).<br>$I_\gamma$ : unweighted average of 35 3 (2008Wi09) and 24.9 11 (2015Ch56).                                                |
| 392 <i>I</i>               | 30 5                    | 4494.2              | (7/2 <sup>-</sup> )                          | 4102.1 | (7/2 <sup>-</sup> )  |                                                                                                                                                                                                                    |
| 468 2                      | 16.2 12                 | 4962.1              | (9/2 <sup>-</sup> )                          | 4494.2 | (7/2 <sup>-</sup> )  | $E_\gamma$ : unweighted average of 466 <i>I</i> (2008Wi09) and 469 <i>I</i> (2015Ch56).<br>$I_\gamma$ : weighted average of 14 2 (2008Wi09) and 16.8 11 (2015Ch56).                                                |
| 487 <sup>†</sup> <i>I</i>  | <2 <sup>†</sup>         | 4869.0              | (5/2 <sup>-</sup> , 7/2 <sup>-</sup> )       | 4381.9 | (5/2 <sup>-</sup> )  | $I_\gamma$ : 1 <i>I</i> from 2008Wi09.<br>$E_\gamma$ : weighted average of 632 <i>I</i> (2008Wi09) and 633 <i>I</i> (2015Ch56).<br>$I_\gamma$ : unweighted average of 6 <i>I</i> (2008Wi09) and 10.4 9 (2015Ch56). |
| 633 <i>I</i>               | 8.2 22                  | 4494.2              | (7/2 <sup>-</sup> )                          | 3860.8 | 5/2 <sup>+</sup>     |                                                                                                                                                                                                                    |
| 664 <i>I</i>               | 32 15                   | 4767.1              | (9/2 <sup>-</sup> )                          | 4102.1 | (7/2 <sup>-</sup> )  | $E_\gamma$ : weighted average of 663 <i>I</i> (2008Wi09) and 665 <i>I</i> (2015Ch56).<br>$I_\gamma$ : unweighted average of 47 4 (2008Wi09) and 17.8 10 (2015Ch56).                                                |
| 767 <sup>†</sup> <i>I</i>  | 5 <sup>†</sup> <i>I</i> | 4869.0              | (5/2 <sup>-</sup> , 7/2 <sup>-</sup> )       | 4102.1 | (7/2 <sup>-</sup> )  | $E_\gamma$ : unweighted average of 856 <i>I</i> (2008Wi09) and 861 <i>I</i> (2015Ch56).<br>$I_\gamma$ : weighted average of 13 2 (2008Wi09) and 9.8 12 (2015Ch56).                                                 |
| 859 3                      | 10.7 14                 | 4962.1              | (9/2 <sup>-</sup> )                          | 4102.1 | (7/2 <sup>-</sup> )  |                                                                                                                                                                                                                    |
| 993 <sup>†</sup> <i>I</i>  | 5 <sup>†</sup> <i>I</i> | 5488.2              |                                              | 4494.2 | (7/2 <sup>-</sup> )  | $E_\gamma, I_\gamma$ : From 2015Ch56.                                                                                                                                                                              |
| 1009 <sup>†</sup> <i>I</i> | <1 <sup>†</sup>         | 4869.0              | (5/2 <sup>-</sup> , 7/2 <sup>-</sup> )       | 3860.8 | 5/2 <sup>+</sup>     |                                                                                                                                                                                                                    |
| 1132 <sup>†</sup> <i>I</i> | <1 <sup>†</sup>         | 6222.4              | (7/2 <sup>-</sup> , 9/2, 11/2 <sup>-</sup> ) | 5089.8 | (11/2 <sup>-</sup> ) |                                                                                                                                                                                                                    |
| 1260 <sup>†</sup> <i>I</i> | 4 <sup>†</sup> <i>I</i> | 6222.4              | (7/2 <sup>-</sup> , 9/2, 11/2 <sup>-</sup> ) | 4962.1 | (9/2 <sup>-</sup> )  |                                                                                                                                                                                                                    |
| <sup>x</sup> 1353 <i>I</i> | 9.2 11                  |                     |                                              |        |                      |                                                                                                                                                                                                                    |
| 1387 <sup>†</sup> <i>I</i> | 3 <sup>†</sup> <i>I</i> | 5488.2              |                                              | 4102.1 | (7/2 <sup>-</sup> )  | $E_\gamma$ : weighted average of 1473 <i>I</i> (2008Wi09) and 1474 <i>I</i> (2015Ch56).<br>$I_\gamma$ : weighted average of 15 2 (2008Wi09) and 16.4 14 (2015Ch56).<br>$E_\gamma, I_\gamma$ : From 2015Ch56.       |
| 1458 <sup>†</sup> <i>I</i> | 7 <sup>†</sup> 2        | 5560.1              | (5/2 <sup>-</sup> )                          | 4102.1 | (7/2 <sup>-</sup> )  |                                                                                                                                                                                                                    |
| 1474 <i>I</i>              | 15.9 14                 | 3860.8              | 5/2 <sup>+</sup>                             | 2386.7 | 3/2 <sup>+</sup>     |                                                                                                                                                                                                                    |
| <sup>x</sup> 1592 <i>I</i> | 7.7 10                  |                     |                                              |        |                      |                                                                                                                                                                                                                    |
| 1715 <sup>†</sup> <i>I</i> | 4 <sup>†</sup> <i>I</i> | 4102.1              | (7/2 <sup>-</sup> )                          | 2386.7 | 3/2 <sup>+</sup>     | $E_\gamma$ : weighted average of 1995 <i>I</i> (2008Wi09) and 1994 <i>I</i> (2015Ch56, unplaced).                                                                                                                  |
| 1729 <sup>†</sup> <i>I</i> | 4 <sup>†</sup> <i>I</i> | 6222.4              | (7/2 <sup>-</sup> , 9/2, 11/2 <sup>-</sup> ) | 4494.2 | (7/2 <sup>-</sup> )  |                                                                                                                                                                                                                    |
| 1995 <i>I</i>              | 8 6                     | 4381.9              | (5/2 <sup>-</sup> )                          | 2386.7 | 3/2 <sup>+</sup>     |                                                                                                                                                                                                                    |

Continued on next page (footnotes at end of table)

**$^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$  2008Wi09,2015Ch56,2019Gr08 (continued)** $\gamma(^{35}\text{P})$  (continued)

| $E_\gamma$          | $I_\gamma$        | $E_i(\text{level})$ | $J_i^\pi$ | $E_f$ | $J_f^\pi$ | Comments                                                                                                                                                                                                                                                                                                    |
|---------------------|-------------------|---------------------|-----------|-------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2386 1              | 30 4              | 2386.7              | $3/2^+$   | 0     | $1/2^+$   | $I_\gamma$ : unweighted average of 2 1 (2008Wi09) and 14.2 11 (2015Ch56).<br>$E_\gamma$ : From 2008Wi09 and 2015Ch56.<br>$I_\gamma$ : From 2008Wi09. Other: 99.2 28 (2015Ch56). Shell-model calculations indicate a small occupancy of the proton $1d_{3/2}$ orbit in the ground state of $^{36}\text{S}$ . |
| 3861 1              | 100.0 32          | 3860.8              | $5/2^+$   | 0     | $1/2^+$   | $E_\gamma$ : weighted average of 3861 1 (2008Wi09) and 3860 1 (2015Ch56).<br>$I_\gamma$ : From 2015Ch56. Other: 100 (2008Wi09).                                                                                                                                                                             |
| 4102 <sup>†</sup> 1 | 33 <sup>†</sup> 5 | 4102.1              | $(7/2^-)$ | 0     | $1/2^+$   | $E_\gamma$ : other: 4101 (2015Ch56).                                                                                                                                                                                                                                                                        |

<sup>†</sup> From 2008Wi09.<sup>x</sup>  $\gamma$  ray not placed in level scheme.

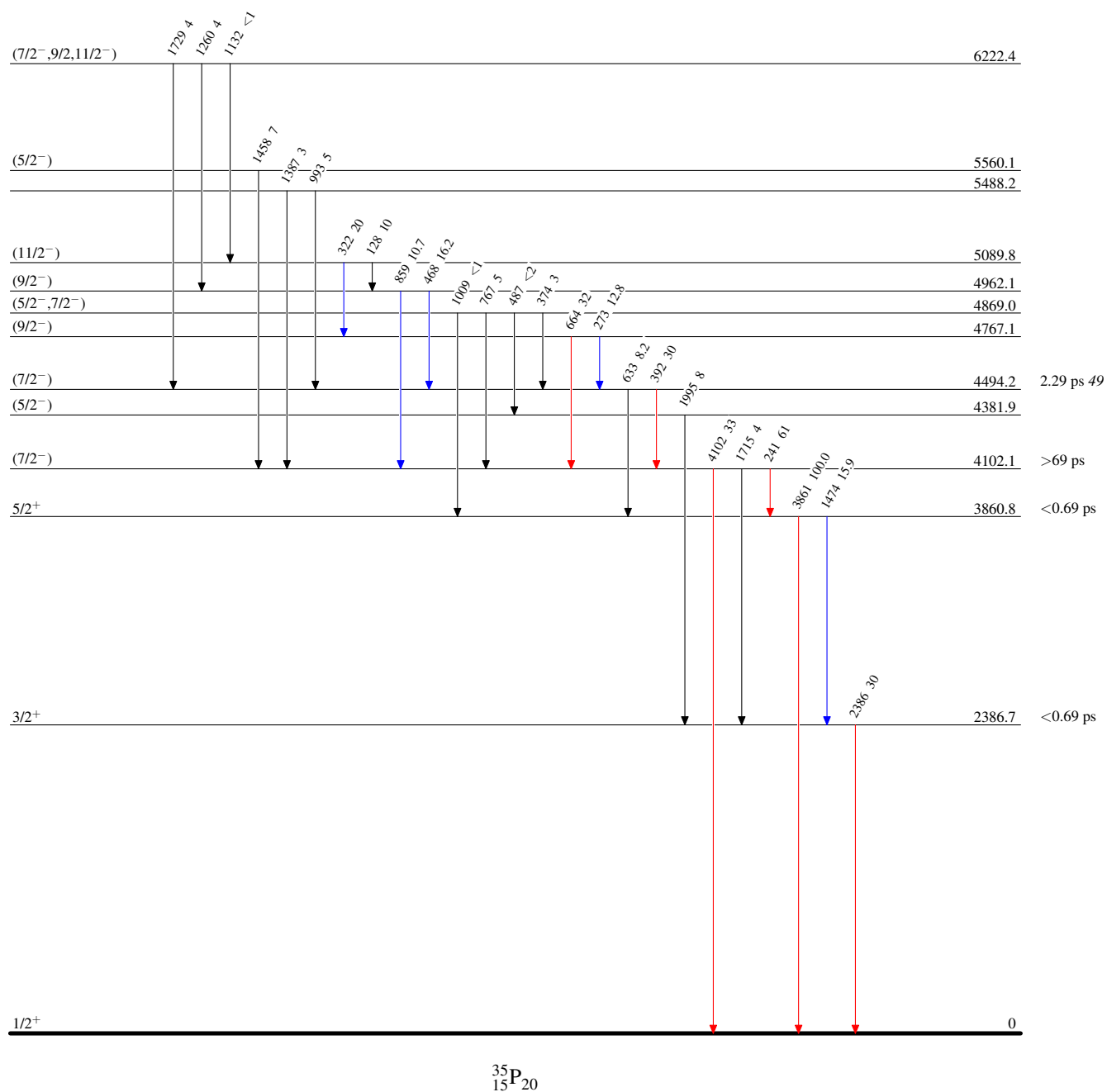
$^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$  2008Wi09,2015Ch56,2019Gr08

## Level Scheme

Intensities: Relative  $I_\gamma$ 

## Legend

- $\longrightarrow$   $I_\gamma < 2\% \times I_\gamma^{\text{max}}$   
 $\longrightarrow$   $I_\gamma < 10\% \times I_\gamma^{\text{max}}$   
 $\longrightarrow$   $I_\gamma > 10\% \times I_\gamma^{\text{max}}$



Adopted Levels, Gammas

$Q(\beta^-)=167.322\ 26$ ;  $S(n)=6985.84\ 4$ ;  $S(p)=11586.5\ 8$ ;  $Q(\alpha)=-8322.09\ 6$  [2021Wa16](#)

$S(2n)=18402.99\ 4$ ,  $S(2p)=22909.8\ 7$  ([2021Wa16](#)).

Isotope discovery ([2012Th10](#)):  $\text{Cl}(n,X)$  reaction at Physical Institute of Aarhus University ([1936An01](#)).

$^{35}\text{S}$  decay measurements:

[2014Si21](#), [2000Ho13](#), [1999Pa18](#), [1999Oh09](#).

$^{35}\text{S}$  cross-section measurements:

$^{35}\text{Cl}(n,p)^{35}\text{S}$ : [2024Ha18](#), [2023De34](#), [2021WaZY](#), [2020Ku17](#), [2019Ba16](#), [2000Ka30](#), [1999GI09](#), [1999GI02](#);

Projectile fragmentation  $^9\text{Be}(^{40}\text{Ar},X)$  and  $^{181}\text{Ta}(^{40}\text{Ar},X)$ : [2012Zh06](#), [2007No13](#);

Spallation  $^{136}\text{Xe}(p,X)$ : [2007Na31](#);

$^{34}\text{S}(n,\gamma)$   $E=\text{thermal}$ : [2023KoZS](#), [2008FiZZ](#).

 $^{35}\text{S}$  LevelsCross Reference (XREF) Flags

|          |                                                  |          |                                                   |          |                                                                   |
|----------|--------------------------------------------------|----------|---------------------------------------------------|----------|-------------------------------------------------------------------|
| <b>A</b> | $^{35}\text{P}\ \beta^-$ decay (47.3 s)          | <b>F</b> | $^{34}\text{S}(n,\gamma)$ $E=\text{thermal}$      | <b>K</b> | $^{36}\text{S}(p,d)$                                              |
| <b>B</b> | $^2\text{H}(^{34}\text{S},p\gamma)$              | <b>G</b> | $^{34}\text{S}(n,n),(n,\gamma):\text{resonances}$ | <b>L</b> | $^{37}\text{Cl}(p,^3\text{He})$                                   |
| <b>C</b> | $^9\text{Be}(^{36}\text{S},^{35}\text{S}\gamma)$ | <b>H</b> | $^{34}\text{S}(\text{pol } d,p)$                  | <b>M</b> | $^{37}\text{Cl}(d,\alpha\gamma)$                                  |
| <b>D</b> | $^{24}\text{Mg}(^{14}\text{N},3p\gamma)$         | <b>I</b> | $^{34}\text{S}(d,p)$                              | <b>N</b> | $^{160}\text{Gd}(^{34}\text{S},X\gamma),(^{37}\text{Cl},X\gamma)$ |
| <b>E</b> | $^{26}\text{Mg}(^{18}\text{O},2\alpha n\gamma)$  | <b>J</b> | $^{34}\text{S}(d,p\gamma)$                        | <b>O</b> | $^{208}\text{Pb}(^{36}\text{S},^{35}\text{S}\gamma)$              |

| <u><math>E(\text{level})^\dagger</math></u> | <u><math>J^\pi^\ddagger</math></u> | <u><math>T_{1/2}</math></u> | <u>XREF</u>                                                                                    | <u>Comments</u>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------------|------------------------------------|-----------------------------|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0.0                                         | $3/2^+$                            | 87.61 d 24                  | <a href="#">ABCDEF</a> <a href="#">HIJKLMNO</a>                                                | $\% \beta^- = 100$<br>$\mu = (+)1.00\ 4$ ( <a href="#">1954Bu05</a> , <a href="#">2019StZV</a> )<br>$Q = 0.0483\ 3$ ( <a href="#">2018Py01</a> , <a href="#">2021StZZ</a> )<br>$J^\pi$ : $L=2$ from $0^+$ in $^{34}\text{S}(\text{pol } d,p)$ , $^{34}\text{S}(d,p)$ , $^{36}\text{S}(p,d)$ , and $^9\text{Be}(^{36}\text{S},^{35}\text{S})$ and $L-1/2$ transfer from analyzing power. $J=3/2$ also from microwave spectroscopy ( <a href="#">1949Co12</a> , <a href="#">1951We11</a> ).<br>$T_{1/2}$ : unweighted average of 88 d 3 ( <a href="#">1941Ka01</a> ), 87.1 d 12 ( <a href="#">1943He01</a> ), 86.35 d 17 ( <a href="#">1959Co56</a> ), 87.16 d 10 ( <a href="#">1958Se49</a> ), 88.8 d 10 ( <a href="#">1959Ca12</a> ), 89.0 d 5 ( <a href="#">1961Wy01</a> ), 87.1 d 9 ( <a href="#">1961Oz01</a> ), 87.9 d 3 ( <a href="#">1965FI02</a> ), 87.39 d 10 ( <a href="#">1968Wo06</a> ), 87.5 d 4 ( <a href="#">1969La34</a> ), and 87.38 d 3 ( <a href="#">1999Pa18</a> ).<br>$\mu$ : +1.00 4 or -1.07 4 from <a href="#">1954Bu05</a> using the microwave absorption method, and the negative value is less likely from systematics of odd-A S isotopes.<br>$Q$ : from <a href="#">2018Py01</a> using the microwave absorption method. -0.483 3 is given in <a href="#">2021StZZ</a> . Other: 0.045 from <a href="#">1954Bi40</a> using the microwave absorption method. |
| 1572.369 9                                  | $1/2^+$                            | 2.29 ps 14                  | <a href="#">ABCD</a> <a href="#">F</a> <a href="#">HIJKLM</a> <a href="#">O</a>                | $J^\pi$ : $L=0$ from $0^+$ in $^{34}\text{S}(\text{pol } d,p)$ , $^{34}\text{S}(d,p)$ , $^{36}\text{S}(p,d)$ , and $^9\text{Be}(^{36}\text{S},^{35}\text{S})$ .<br>$T_{1/2}$ : lifetime $\tau=3.3$ ps 2 in $^{208}\text{Pb}(^{36}\text{S},^{35}\text{S}\gamma)$ from <a href="#">2022Gr07</a> with DRDM. Others: 3.3 ps 5 in $^{34}\text{S},p\gamma$ from <a href="#">1973Wa10</a> with DSAM, >1.8 ps in $^{34}\text{S}(d,p\gamma)$ from <a href="#">1970Bu18</a> with DSAM, and >1.6 ps in $^{34}\text{S}(d,p\gamma)$ from <a href="#">1972Fr11</a> with DSAM.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 1991.27 4                                   | $7/2^-$                            | 1.03 ns 5                   | <a href="#">CDEF</a> <a href="#">HIJKLMNO</a>                                                  | $J^\pi$ : $L=3$ from $0^+$ in $^{36}\text{S}(p,d)$ and $^{34}\text{S}(\text{pol } d,p)$ , and $L+1/2$ transfer from analyzing power.<br>$T_{1/2}$ : lifetime $\tau=1.48$ ns 7: weighted average of 1.7 ns 3 in $^{34}\text{S}(d,p\gamma)$ from <a href="#">2024Co04</a> with $p\gamma$ -delayed coin and 1.47 ns 7 in $^{34}\text{S}(d,p\gamma)$ from <a href="#">1971Pr11</a> with $p\gamma$ -delayed coin. Other: >4.5 ps in $^{34}\text{S}(d,p\gamma)$ from <a href="#">1970Bu18</a> with DSAM.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 2347.779 10                                 | $3/2^-$                            | 0.78 ps 12                  | <a href="#">BCD</a> <a href="#">F</a> <a href="#">HIJK</a> <a href="#">M</a> <a href="#">O</a> | $J^\pi$ : $L=1$ from $0^+$ in $^{34}\text{S}(\text{pol } d,p)$ , $^{34}\text{S}(d,p)$ , $^{36}\text{S}(p,d)$ , and $^9\text{Be}(^{36}\text{S},^{35}\text{S})$ and $L+1/2$ transfer from analyzing power.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

Continued on next page (footnotes at end of table)

**Adopted Levels, Gammas (continued)** $^{35}\text{S}$  Levels (continued)

| E(level) <sup>†</sup>   | J <sup>π</sup> <sup>‡</sup>                               | T <sub>1/2</sub> | XREF          | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------|-----------------------------------------------------------|------------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2717.06 8               | 5/2 <sup>+</sup>                                          | 69 fs 24         | CD F HIJKLM O | T <sub>1/2</sub> : lifetime $\tau=1.12$ ps 18; weighted average of 1.3 ps 2 in ( $^{34}\text{S},\text{p}\gamma$ ) from 1973Wa10 with DSAM, 0.8 ps 2 in ( $^{34}\text{S}(\text{d},\text{p}\gamma)$ ) from 1970Bu18 with DSAM, and 1.22 ps 18 in ( $^{34}\text{S}(\text{d},\text{p}\gamma)$ ) from 1972Fr11 with DSAM.<br>J <sup>π</sup> : L=2 from 0 <sup>+</sup> in $^{34}\text{S}(\text{pol d,p})$ , $^{34}\text{S}(\text{d,p})$ , $^{36}\text{S}(\text{p,d})$ , and $^9\text{Be}(^{36}\text{S},^{35}\text{S})$ ; L+1/2 transfer from J-dependence in $^{36}\text{S}(\text{p,d})$ and L+1/2 transfer from analyzing power.<br>T <sub>1/2</sub> : lifetime $\tau=100$ fs 35 in ( $^{34}\text{S}(\text{d},\text{p}\gamma)$ ) from 1972Fr11 with DSAM. |
| 2938.63 5               | 3/2 <sup>+</sup>                                          |                  | A C F HIJKLM  | J <sup>π</sup> : L=2 from 0 <sup>+</sup> in $^{34}\text{S}(\text{d,p})$ , $^{36}\text{S}(\text{p,d})$ , $^{34}\text{S}(\text{pol d,p})$ , and L=(2) in $^9\text{Be}(^{36}\text{S},^{35}\text{S})$ ; L-1/2 transfer from J-dependence in $^{36}\text{S}(\text{p,d})$ ; possibly allowed $\beta^-$ feeding from 1/2 <sup>+</sup> parent with log $f_t=5.0$ .                                                                                                                                                                                                                                                                                                                                                                                           |
| 3421.2 20               | 5/2 <sup>+</sup>                                          | <69 fs           | C HIJKLM O    | J <sup>π</sup> : L=2 from 0 <sup>+</sup> in $^{34}\text{S}(\text{pol d,p})$ , $^{34}\text{S}(\text{d,p})$ , $^{36}\text{S}(\text{p,d})$ , and $^9\text{Be}(^{36}\text{S},^{35}\text{S})$ ; L+1/2 transfer from J-dependence in $^{36}\text{S}(\text{p,d})$ and L+1/2 transfer from analyzing power.<br>T <sub>1/2</sub> : lifetime $\tau<100$ fs in ( $^{34}\text{S}(\text{d},\text{p}\gamma)$ ) from 1972Fr11 with DSAM.                                                                                                                                                                                                                                                                                                                            |
| 3558.080 28             | (5/2 <sup>+</sup> )                                       |                  | F HIJK M O    | J <sup>π</sup> : L=(2) from 0 <sup>+</sup> in $^{36}\text{S}(\text{p,d})$ ; 1986 $\gamma$ to 1/2 <sup>+</sup> , 1572.370 and 1566.7 $\gamma$ to 7/2 <sup>-</sup> , 1991.27.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 3594.62 24              | (7/2 <sup>+</sup> )                                       |                  | CDE IJKLM O   | XREF: C(3595)I(3596)K(3592)L(3598)<br>J <sup>π</sup> : 3594.5 $\gamma$ Q, $\Delta J=2$ to 3/2 <sup>+</sup> , g.s.; L=2+4 from 3/2 <sup>+</sup> in (p, $^3\text{He}$ ): L=2 suggests (1/2 <sup>+</sup> to 9/2 <sup>+</sup> ) and L=4 suggests (3/2 <sup>+</sup> to 13/2 <sup>+</sup> ); L=(2) from 0 <sup>+</sup> in $^{36}\text{S}(\text{p,d})$ and $^9\text{Be}(^{36}\text{S},^{35}\text{S})$ . 7/2 <sup>+</sup> is suggested by shell model (2014Ay01,2021Go09). If 3594.5 $\gamma$ is consistent with $\Delta J=0$ to 3/2 <sup>+</sup> , the discrepancy between the in-beam $\gamma$ -ray spectroscopy ( $^{14}\text{N},3\text{p}\gamma$ ) and transfer reactions may be resolved.                                                               |
| 3675? 10<br>3801.952 16 | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> )<br>3/2 <sup>-</sup> | 25 fs 18         | I<br>F HIJKLM | J <sup>π</sup> : L=(1) from 0 <sup>+</sup> in $^{34}\text{S}(\text{d,p})$ .<br>XREF: I(3811)<br>J <sup>π</sup> : L=1 from 0 <sup>+</sup> in $^{34}\text{S}(\text{d,p})$ , $^{36}\text{S}(\text{p,d})$ , and $^{34}\text{S}(\text{pol d,p})$ and L+1/2 transfer from analyzing power.<br>T <sub>1/2</sub> : lifetime $\tau=36$ fs 26 in ( $^{34}\text{S}(\text{d},\text{p}\gamma)$ ) from 1972Fr11 with DSAM.                                                                                                                                                                                                                                                                                                                                         |
| 3816.01 21              | (9/2 <sup>-</sup> )                                       | 0.28 ps 3        | CDE H J LM O  | XREF: I(3811)<br>J <sup>π</sup> : 1824.7 $\gamma$ M1+E2 to 7/2 <sup>-</sup> , 1991.27; 2061.6 $\gamma$ D, $\Delta J=1$ from (11/2 <sup>+</sup> ), 5877.70. L=(3) from 0 <sup>+</sup> in $^9\text{Be}(^{36}\text{S},^{35}\text{S}\gamma)$ is inconsistent but the fit on the parallel momentum distribution is poor.                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 3886.28                 | (5/2)                                                     |                  | CD IJK M      | T <sub>1/2</sub> : in ( $^{14}\text{N},3\text{p}\gamma$ ) from 2014Ay01 with DSAM.<br>XREF: C(3890)M(3889)<br>J <sup>π</sup> : 1894.9 $\gamma$ D, $\Delta J=1$ to 7/2 <sup>-</sup> , 1991.27; 3886 $\gamma$ to 3/2 <sup>+</sup> , g.s.; L=3 from 0 <sup>+</sup> in $^{36}\text{S}(\text{p,d})$ . L=(0) from 0 <sup>+</sup> in $^9\text{Be}(^{36}\text{S},^{35}\text{S}\gamma)$ is inconsistent.                                                                                                                                                                                                                                                                                                                                                      |
| 4023.24 21              | (11/2 <sup>-</sup> )                                      | 0.32 ps 3        | DE MNO        | J <sup>π</sup> : 2032.0 $\gamma$ E2, $\Delta J=2$ to 7/2 <sup>-</sup> , 1991.27; 207.0 $\gamma$ to (9/2 <sup>-</sup> ), 3816.01.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 4027.2 14               | (3/2 <sup>+</sup> ,5/2 <sup>+</sup> )                     |                  | C IJKLM       | T <sub>1/2</sub> : in ( $^{14}\text{N},3\text{p}\gamma$ ) from 2014Ay01 with DSAM.<br>XREF: I(4025)K(4023)<br>J <sup>π</sup> : L=(2) from 0 <sup>+</sup> in $^{36}\text{S}(\text{p,d})$ and $^9\text{Be}(^{36}\text{S},^{35}\text{S}\gamma)$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 4105.6 8                | (5/2 <sup>+</sup> )                                       | <55 fs           | F IJKLM       | J <sup>π</sup> : L=(2) from 0 <sup>+</sup> in $^{36}\text{S}(\text{p,d})$ ; 2533.1 $\gamma$ to 1/2 <sup>+</sup> , 1572.370 and 2114.3 $\gamma$ to 7/2 <sup>-</sup> , 1991.27.<br>T <sub>1/2</sub> : lifetime $\tau<80$ fs in ( $^{34}\text{S}(\text{d},\text{p}\gamma)$ ) from 1972Fr11 with DSAM.                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

Continued on next page (footnotes at end of table)

**Adopted Levels, Gammas (continued)** $^{35}\text{S}$  Levels (continued)

| <u>E(level)<sup>†</sup></u> | <u>J<sup>π</sup></u>                  | <u>T<sub>1/2</sub></u> | <u>XREF</u>      | <u>Comments</u>                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------|---------------------------------------|------------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4180 3                      | (1/2,3/2,5/2 <sup>+</sup> )           |                        | <b>LM</b>        | XREF: l(4186)<br>J <sup>π</sup> : 2608γ to 1/2 <sup>+</sup> , 1572.370.                                                                                                                                                                                                                                                                                                                                                                                      |
| 4189.240 33                 | 1/2 <sup>-</sup>                      | <35 fs                 | <b>F HIJKLM</b>  | XREF: K(4182)l(4186)M(4187)<br>J <sup>π</sup> : L=1 from 0 <sup>+</sup> in $^{34}\text{S}(\text{d,p})$ , $^{36}\text{S}(\text{p,d})$ , and $^{34}\text{S}(\text{pol d,p})$ and L-1/2 transfer from analyzing power.<br>T <sub>1/2</sub> : lifetime τ<50 fs in $^{34}\text{S}(\text{d,p}γ)$ from <a href="#">1972Fr11</a> with DSAM.                                                                                                                          |
| 4302.6 7                    | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) |                        | <b>I KLM</b>     | XREF: L(4290)<br>J <sup>π</sup> : L=(1) from 0 <sup>+</sup> in $^{36}\text{S}(\text{p,d})$ . L=(2) from 3/2 <sup>+</sup> in (p, $^3\text{He}$ ) suggests (1/2 <sup>+</sup> to 9/2 <sup>+</sup> ).                                                                                                                                                                                                                                                            |
| 4477.63                     | (5/2 <sup>+</sup> )                   | <62 fs                 | <b>F I J LM</b>  | XREF: i(4481)L(4489)<br>J <sup>π</sup> : 2905.1γ to 1/2 <sup>+</sup> , 1572.370 and 2486.27γ to 7/2 <sup>-</sup> , 1991.27. L=2 from 3/2 <sup>+</sup> in (p, $^3\text{He}$ ) suggests (1/2 <sup>+</sup> to 9/2 <sup>+</sup> ).<br>L=2 component of L=2,3 doublet from 0 <sup>+</sup> in $^{34}\text{S}(\text{d,p})$ .<br>T <sub>1/2</sub> : lifetime τ<90 fs in $^{34}\text{S}(\text{d,p}γ)$ from <a href="#">1972Fr11</a> with DSAM.                        |
| 4486 7                      | 7/2 <sup>-</sup>                      |                        | <b>Hi K</b>      | XREF: H(4482)i(4481)<br>E(level): from $^{36}\text{S}(\text{p,d})$ .<br>J <sup>π</sup> : L=3 from 0 <sup>+</sup> in $^{36}\text{S}(\text{p,d})$ and $^{34}\text{S}(\text{pol d,p})$ , and L+1/2 transfer from analyzing power. L=3 component of L=2,3 doublet from 0 <sup>+</sup> in $^{34}\text{S}(\text{d,p})$ .                                                                                                                                           |
| 4575 5                      | 3/2 <sup>+</sup>                      |                        | <b>HI KL</b>     | E(level): weighted average of 4575 8 from $^{34}\text{S}(\text{d,p})$ , 4577 10 from (p, $^3\text{He}$ ), and 4574 5 from $^{36}\text{S}(\text{p,d})$ .<br>J <sup>π</sup> : L=2 from 0 <sup>+</sup> and L-1/2 transfer from J-dependence in $^{36}\text{S}(\text{p,d})$ . L=0+2 from 3/2 <sup>+</sup> in (p, $^3\text{He}$ ): L=0 suggests (1/2 <sup>+</sup> ,3/2 <sup>+</sup> ,5/2 <sup>+</sup> ) and L=2 suggests (1/2 <sup>+</sup> to 9/2 <sup>+</sup> ). |
| 4615 5                      | 3/2 <sup>+</sup> ,5/2 <sup>+</sup>    |                        | <b>KL</b>        | E(level): weighted average of 4614 5 from $^{36}\text{S}(\text{p,d})$ and 4617 10 from (p, $^3\text{He}$ ).<br>J <sup>π</sup> : L=2 from 0 <sup>+</sup> in $^{36}\text{S}(\text{p,d})$ .                                                                                                                                                                                                                                                                     |
| 4822.79 24                  | (9/2 <sup>+</sup> )                   |                        | <b>DE</b>        | J <sup>π</sup> : 1228.1γ D, ΔJ=1 to (7/2 <sup>+</sup> ), 3594.62; 1055.1γ D, ΔJ=1 from (11/2 <sup>+</sup> ), 5877.70; 9/2 <sup>+</sup> suggested by shell-model calculations ( <a href="#">2021Go09</a> ).                                                                                                                                                                                                                                                   |
| 4839 5                      | (1/2 <sup>+</sup> )                   |                        | <b>HI KL</b>     | E(level): weighted average of 4837 8 from $^{34}\text{S}(\text{d,p})$ , 4838 5 from $^{36}\text{S}(\text{p,d})$ , and 4843 10 from (p, $^3\text{He}$ ). Other: 4837 from (pol d,p).<br>J <sup>π</sup> : L=(0) from 0 <sup>+</sup> in $^{36}\text{S}(\text{p,d})$ ; L=2 from 3/2 <sup>+</sup> in (p, $^3\text{He}$ ) suggests (1/2 <sup>+</sup> to 9/2 <sup>+</sup> ).                                                                                        |
| 4899.82 27                  | (9/2 <sup>+</sup> )                   |                        | <b>DE</b>        | J <sup>π</sup> : 2908.3γ D, ΔJ=1 to 7/2 <sup>-</sup> , 1991.27; 977.9γ D+Q, ΔJ=1 from (11/2 <sup>+</sup> ), 5877.70; 9/2 <sup>+</sup> suggested by shell-model calculations ( <a href="#">2021Go09</a> ).                                                                                                                                                                                                                                                    |
| 4903.371 16                 | 1/2 <sup>+</sup>                      |                        | <b>F HI K</b>    | J <sup>π</sup> : L=0 from 0 <sup>+</sup> in $^{36}\text{S}(\text{p,d})$ . Other: L=1 from 0 <sup>+</sup> in $^{34}\text{S}(\text{d,p})$ and $^{34}\text{S}(\text{pol d,p})$ and L-1/2 transfer from analyzing power. <a href="#">2026Jo01</a> in $^{36}\text{S}(\text{p,d})$ acknowledges this discrepancy and states that the J <sup>π</sup> is confronted with strong evidence.                                                                            |
| 4963.082 25                 | 5/2 <sup>+</sup>                      |                        | <b>C F HI KL</b> | XREF: C(4948)K(4955)<br>J <sup>π</sup> : L=2 from 0 <sup>+</sup> and L+1/2 transfer from J-dependence in $^{36}\text{S}(\text{p,d})$ . Other: L=1 from 0 <sup>+</sup> in $^{34}\text{S}(\text{d,p})$ and $^{34}\text{S}(\text{pol d,p})$ and L+1/2 transfer from analyzing power. <a href="#">2026Jo01</a> in $^{36}\text{S}(\text{p,d})$ acknowledges this discrepancy and states that the J <sup>π</sup> is confronted with strong evidence.               |
| 4990 10                     | ( <sup>+</sup> )                      |                        | <b>L</b>         | J <sup>π</sup> : L=0+2 from 3/2 <sup>+</sup> in (p, $^3\text{He}$ ): L=0 suggests (1/2 <sup>+</sup> ,3/2 <sup>+</sup> ,5/2 <sup>+</sup> ) and L=2 suggests (1/2 <sup>+</sup> to 9/2 <sup>+</sup> ).                                                                                                                                                                                                                                                          |
| 5010.05 29                  | (11/2 <sup>-</sup> )                  | 0.45 ps 8              | <b>DE</b>        | J <sup>π</sup> : 3018.5γ E2, ΔJ=2 to 7/2 <sup>-</sup> , 1991.27; 986.8γ to (11/2 <sup>-</sup> ,                                                                                                                                                                                                                                                                                                                                                              |

Continued on next page (footnotes at end of table)

**Adopted Levels, Gammas (continued)** $^{35}\text{S}$  Levels (continued)

| E(level) <sup>†</sup> | J <sup>π</sup> <sup>‡</sup>            | T <sub>1/2</sub> | XREF  | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------|----------------------------------------|------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       |                                        |                  |       | 4023.24.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 5058 8                | 7/2 <sup>-</sup>                       |                  | HI    | T <sub>1/2</sub> : in ( $^{14}\text{N}, 3\text{p}\gamma$ ) from 2014Ay01 with DSAM.<br>E(level): from $^{34}\text{S}(\text{d}, \text{p})$ .<br>J <sup>π</sup> : L=3 from 0 <sup>+</sup> in $^{34}\text{S}(\text{d}, \text{p})$ and $^{34}\text{S}(\text{pol d}, \text{p})$ and L+1/2 transfer from analyzing power.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 5123 5                | 1/2 <sup>+</sup>                       |                  | HI KL | E(level): weighted average of 5126 11 from $^{34}\text{S}(\text{d}, \text{p})$ , 5121 5 from $^{36}\text{S}(\text{p}, \text{d})$ , and 5127 10 from ( $\text{p}, ^3\text{He}$ ).<br>J <sup>π</sup> : L=0 from 0 <sup>+</sup> in $^{36}\text{S}(\text{p}, \text{d})$ ; L=2 from 3/2 <sup>+</sup> in ( $\text{p}, ^3\text{He}$ ) suggests (1/2 <sup>+</sup> to 9/2 <sup>+</sup> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 5286 5                | (1/2 <sup>-</sup> , 3/2 <sup>-</sup> ) |                  | K     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5343 8                | ( <sup>-</sup> )                       |                  | C I L | E(level): weighted average of 5342 8 from $^{34}\text{S}(\text{d}, \text{p})$ and 5345 10 from ( $\text{p}, ^3\text{He}$ ). Other: 5349 8 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{S}\gamma)$ .<br>J <sup>π</sup> : L=3 from 3/2 <sup>+</sup> in ( $\text{p}, ^3\text{He}$ ) suggests (1/2 <sup>-</sup> to 11/2 <sup>-</sup> ).<br>J <sup>π</sup> : 1389.0γ D, ΔJ=1 to (11/2) <sup>-</sup> , 4023.24; 1525.7γ to (5/2), 3886.28; 9/2 <sup>+</sup> suggested by shell model (2014Ay01, 2021Go09).<br>XREF: I(5475?)                                                                                                                                                                                                                                                                                                                                                                           |
| 5412.28 25            | (9/2 <sup>+</sup> )                    |                  | DE    | J <sup>π</sup> : L=2 from 0 <sup>+</sup> in $^{36}\text{S}(\text{p}, \text{d})$ .<br>XREF: I(5542?)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 5476 5                | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>    |                  | I K   | J <sup>π</sup> : L=3 from 3/2 <sup>+</sup> in ( $\text{p}, ^3\text{He}$ ): L=3 suggests (1/2 <sup>-</sup> to 11/2 <sup>-</sup> ).<br>J <sup>π</sup> : L=2 from 0 <sup>+</sup> in $^{36}\text{S}(\text{p}, \text{d})$ .<br>XREF: I(5740)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5550 10               |                                        |                  | I L   | J <sup>π</sup> : L=3 from 0 <sup>+</sup> in $^{34}\text{S}(\text{d}, \text{p})$ .<br>E(level): weighted average of 5766 5 from $^{36}\text{S}(\text{p}, \text{d})$ and 5771 10 from ( $\text{p}, ^3\text{He}$ ). Other: 5775 4 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{S}\gamma)$ .<br>J <sup>π</sup> : L=2 from 0 <sup>+</sup> and L+1/2 transfer from J-dependence in $^{36}\text{S}(\text{p}, \text{d})$ ; L=2 from 3/2 <sup>+</sup> in ( $\text{p}, ^3\text{He}$ ) suggests (1/2 <sup>+</sup> to 9/2 <sup>+</sup> ).<br>J <sup>π</sup> : L=(2) from 0 <sup>+</sup> in $^{36}\text{S}(\text{p}, \text{d})$ .<br>J <sup>π</sup> : 465.4γ D, ΔJ=1 to (9/2 <sup>+</sup> ), 5412.28; 867.6γ D, ΔJ=0 to (11/2) <sup>-</sup> , 5010.05; strong 2283.0γ to (7/2 <sup>+</sup> ), 3594.62.<br>XREF: L(5915?)<br>J <sup>π</sup> : L=2 from 0 <sup>+</sup> in $^{34}\text{S}(\text{d}, \text{p})$ . |
| 5630 5                | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>    |                  | K     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5752.5 8              | 5/2 <sup>-</sup> , 7/2 <sup>-</sup>    |                  | F I   | J <sup>π</sup> : L=1 from 0 <sup>+</sup> in $^{34}\text{S}(\text{d}, \text{p})$ .<br>E(level): weighted average of 6121 5 from $^{36}\text{S}(\text{p}, \text{d})$ and 6129 10 from ( $\text{p}, ^3\text{He}$ ).<br>J <sup>π</sup> : L=(2) from 0 <sup>+</sup> in $^{36}\text{S}(\text{p}, \text{d})$ ; L=0+2 from 3/2 <sup>+</sup> in ( $\text{p}, ^3\text{He}$ ): L=0 suggests (1/2 <sup>+</sup> , 3/2 <sup>+</sup> , 5/2 <sup>+</sup> ) and L=2 suggests (1/2 <sup>+</sup> to 9/2 <sup>+</sup> ).<br>J <sup>π</sup> : L=(3) from 0 <sup>+</sup> in $^{36}\text{S}(\text{p}, \text{d})$ .<br>XREF: I(6292)                                                                                                                                                                                                                                                                                    |
| 5767 5                | 5/2 <sup>+</sup>                       |                  | C KL  | E(level): other: 6292 8 from $^{34}\text{S}(\text{d}, \text{p})$ .<br>J <sup>π</sup> : 887.0γ, ΔJ=1 to (9/2 <sup>+</sup> ), 5412; 2275.9γ to (11/2) <sup>-</sup> , 4023.24.<br>E(level): weighted average of 6334 8 from $^{34}\text{S}(\text{d}, \text{p})$ and 6338 5 from $^{36}\text{S}(\text{p}, \text{d})$ .<br>J <sup>π</sup> : L=0 from 0 <sup>+</sup> in $^{36}\text{S}(\text{p}, \text{d})$ .<br>E(level): weighted average of 6344 8 from $^{34}\text{S}(\text{d}, \text{p})$ and 6347 10 from ( $\text{p}, ^3\text{He}$ ).<br>J <sup>π</sup> : L=(2) from 3/2 <sup>+</sup> in ( $\text{p}, ^3\text{He}$ ) suggests (1/2 <sup>+</sup> to 9/2 <sup>+</sup> ).<br>J <sup>π</sup> : 2535.8γ E2, ΔJ=2 to (9/2) <sup>-</sup> , 3816.01; 2329.5γ (M1), ΔJ=1 to (11/2) <sup>-</sup> , 4023.24.<br>T <sub>1/2</sub> : in ( $^{14}\text{N}, 3\text{p}\gamma$ ) from 2014Ay01 with DSAM.       |
| 5841.482 31           | (3/2 <sup>+</sup> , 5/2 <sup>+</sup> ) |                  | F K   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5877.70 22            | (11/2 <sup>+</sup> )                   |                  | DE    | J <sup>π</sup> : L=2 from 0 <sup>+</sup> in $^{36}\text{S}(\text{p}, \text{d})$ .<br>XREF: L(5915?)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 5890 20               | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>    |                  | I L   | J <sup>π</sup> : L=2 from 0 <sup>+</sup> in $^{34}\text{S}(\text{d}, \text{p})$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5980 10               |                                        |                  | I     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 6018.8 6              |                                        |                  | F     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 6078.475 33           | 1/2 <sup>-</sup> , 3/2 <sup>-</sup>    |                  | F I   | J <sup>π</sup> : L=1 from 0 <sup>+</sup> in $^{34}\text{S}(\text{d}, \text{p})$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 6123 5                | (3/2 <sup>+</sup> , 5/2 <sup>+</sup> ) |                  | KL    | E(level): weighted average of 6121 5 from $^{36}\text{S}(\text{p}, \text{d})$ and 6129 10 from ( $\text{p}, ^3\text{He}$ ).<br>J <sup>π</sup> : L=(2) from 0 <sup>+</sup> in $^{36}\text{S}(\text{p}, \text{d})$ ; L=0+2 from 3/2 <sup>+</sup> in ( $\text{p}, ^3\text{He}$ ): L=0 suggests (1/2 <sup>+</sup> , 3/2 <sup>+</sup> , 5/2 <sup>+</sup> ) and L=2 suggests (1/2 <sup>+</sup> to 9/2 <sup>+</sup> ).<br>J <sup>π</sup> : L=(3) from 0 <sup>+</sup> in $^{36}\text{S}(\text{p}, \text{d})$ .<br>XREF: I(6292)                                                                                                                                                                                                                                                                                                                                                                         |
| 6218 5                | (5/2 <sup>-</sup> , 7/2 <sup>-</sup> ) |                  | K     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 6293.93 6             |                                        |                  | F I   | E(level): other: 6292 8 from $^{34}\text{S}(\text{d}, \text{p})$ .<br>J <sup>π</sup> : 887.0γ, ΔJ=1 to (9/2 <sup>+</sup> ), 5412; 2275.9γ to (11/2) <sup>-</sup> , 4023.24.<br>E(level): weighted average of 6334 8 from $^{34}\text{S}(\text{d}, \text{p})$ and 6338 5 from $^{36}\text{S}(\text{p}, \text{d})$ .<br>J <sup>π</sup> : L=0 from 0 <sup>+</sup> in $^{36}\text{S}(\text{p}, \text{d})$ .<br>E(level): weighted average of 6344 8 from $^{34}\text{S}(\text{d}, \text{p})$ and 6347 10 from ( $\text{p}, ^3\text{He}$ ).<br>J <sup>π</sup> : L=(2) from 3/2 <sup>+</sup> in ( $\text{p}, ^3\text{He}$ ) suggests (1/2 <sup>+</sup> to 9/2 <sup>+</sup> ).<br>J <sup>π</sup> : 2535.8γ E2, ΔJ=2 to (9/2) <sup>-</sup> , 3816.01; 2329.5γ (M1), ΔJ=1 to (11/2) <sup>-</sup> , 4023.24.<br>T <sub>1/2</sub> : in ( $^{14}\text{N}, 3\text{p}\gamma$ ) from 2014Ay01 with DSAM.       |
| 6299.4 4              | (11/2)                                 |                  | D     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 6337 5                | 1/2 <sup>+</sup>                       |                  | I K   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 6345 8                | ( <sup>+</sup> )                       |                  | I L   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 6352.2 5              | (13/2) <sup>-</sup>                    | 50 fs 10         | D     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 6354.89 23            |                                        |                  | F     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

Continued on next page (footnotes at end of table)



**Adopted Levels, Gammas (continued)** $^{35}\text{S}$  Levels (continued)

| E(level) <sup>†</sup> | J <sup>π</sup> <sup>‡</sup>              | T <sub>1/2</sub> | XREF          | Comments                                                                                                                                                                                                                                                                                                                                |
|-----------------------|------------------------------------------|------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6419.9 11             |                                          |                  | <b>F</b>      |                                                                                                                                                                                                                                                                                                                                         |
| 6441 5                | (1/2 <sup>+</sup> )                      |                  | <b>I K</b>    | E(level): weighted average of 6446 8 from $^{34}\text{S}(\text{d,p})$ and 6439 5 from $^{36}\text{S}(\text{p,d})$ .<br>J <sup>π</sup> : L=(0) from 0 <sup>+</sup> in $^{36}\text{S}(\text{p,d})$ .                                                                                                                                      |
| 6496 8                |                                          |                  | <b>I</b>      |                                                                                                                                                                                                                                                                                                                                         |
| 6537.7 14             |                                          |                  | <b>I</b>      |                                                                                                                                                                                                                                                                                                                                         |
| 6545.1 13             | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>      |                  | <b>I K</b>    | XREF: K(6550)<br>J <sup>π</sup> : L=2 from 0 <sup>+</sup> in $^{36}\text{S}(\text{p,d})$ .                                                                                                                                                                                                                                              |
| 6584 10               |                                          |                  | <b>I</b>      |                                                                                                                                                                                                                                                                                                                                         |
| 6629.43 9             |                                          |                  | <b>F</b>      |                                                                                                                                                                                                                                                                                                                                         |
| 6635.2 13             | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>      |                  | <b>I K</b>    | XREF: K(6639)<br>J <sup>π</sup> : L=2 from 0 <sup>+</sup> in $^{36}\text{S}(\text{p,d})$ .                                                                                                                                                                                                                                              |
| 6654 10               | ( <sup>-</sup> )                         |                  | <b>L</b>      | J <sup>π</sup> : L=(3) from 3/2 <sup>+</sup> in (p, $^3\text{He}$ ) suggests (1/2 <sup>-</sup> to 11/2 <sup>-</sup> ).                                                                                                                                                                                                                  |
| 6683 5                | 1/2 <sup>+</sup>                         |                  | <b>I KL</b>   | E(level): weighted average of 6677 8 from $^{34}\text{S}(\text{d,p})$ , 6682 5 from $^{36}\text{S}(\text{p,d})$ , and 6696 10 from (p, $^3\text{He}$ ).<br>J <sup>π</sup> : L=0 from 0 <sup>+</sup> in $^{36}\text{S}(\text{p,d})$ ; L=2 from 3/2 <sup>+</sup> in (p, $^3\text{He}$ ) suggests (1/2 <sup>+</sup> to 9/2 <sup>+</sup> ). |
| 6761.0 12             |                                          |                  | <b>F</b>      |                                                                                                                                                                                                                                                                                                                                         |
| 6790 5                | (3/2 <sup>+</sup> , 5/2 <sup>+</sup> )   |                  | <b>K</b>      | J <sup>π</sup> : L=(2) from 0 <sup>+</sup> in $^{36}\text{S}(\text{p,d})$ .                                                                                                                                                                                                                                                             |
| 6891.3 14             | ( <sup>-</sup> )                         |                  | <b>I</b>      | J <sup>π</sup> : L=(1,3) from 0 <sup>+</sup> in $^{34}\text{S}(\text{d,p})$ .                                                                                                                                                                                                                                                           |
| 6962 5                | (1/2 <sup>-</sup> , 3/2 <sup>-</sup> )   |                  | <b>K</b>      | J <sup>π</sup> : L=(1) from 0 <sup>+</sup> in $^{36}\text{S}(\text{p,d})$ .                                                                                                                                                                                                                                                             |
| (6986.097 25)         | 1/2 <sup>+</sup>                         |                  | <b>F</b>      | J <sup>π</sup> : s-wave resonance capture state.                                                                                                                                                                                                                                                                                        |
| 7018.89 4             | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>      |                  | <b>G I K</b>  | XREF: I(7021)K(7019)<br>E(level): from (n,n),(n,γ):resonances.<br>J <sup>π</sup> : L=2 from 0 <sup>+</sup> in (n,n),(n,γ):resonances. Others: L=(2,3) from 0 <sup>+</sup> in $^{34}\text{S}(\text{d,p})$ for 7021 10; L=(1) from 0 <sup>+</sup> in $^{36}\text{S}(\text{p,d})$ for 7019 5.                                              |
| 7072.66 5             |                                          | 0.14 eV 4        | <b>G</b>      |                                                                                                                                                                                                                                                                                                                                         |
| 7074.33 6             |                                          | 0.084 eV 3       | <b>G</b>      |                                                                                                                                                                                                                                                                                                                                         |
| 7097.72 30            | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>      | 0.009 eV 3       | <b>G</b>      |                                                                                                                                                                                                                                                                                                                                         |
| 7099.85 20            | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>      | 0.033 eV 5       | <b>G</b>      |                                                                                                                                                                                                                                                                                                                                         |
| 7100.73 4             | 3/2 <sup>-</sup>                         |                  | <b>G K</b>    | J <sup>π</sup> : other: L=(1) from 0 <sup>+</sup> in $^{36}\text{S}(\text{p,d})$ .                                                                                                                                                                                                                                                      |
| 7143.27 11            | (5/2 <sup>-</sup> , 7/2 <sup>-</sup> )   | 0.42 eV 1        | <b>G I KL</b> | XREF: I(7150)K(7151)L(7151?)<br>J <sup>π</sup> : L=(3) from 0 <sup>+</sup> in $^{36}\text{S}(\text{p,d})$ . Others: L=(0,3) from 0 <sup>+</sup> in $^{34}\text{S}(\text{d,p})$ ; L=(4) from 3/2 <sup>+</sup> in (p, $^3\text{He}$ ) suggests (3/2 <sup>+</sup> to 13/2 <sup>+</sup> ).                                                  |
| 7179.74 32            | (15/2 <sup>+</sup> )                     | 3.1 ps 12        | <b>DE</b>     | J <sup>π</sup> : 1302.0γ E2, ΔJ=2 to (11/2 <sup>+</sup> ), 5877.70; 827.8γ to (13/2 <sup>-</sup> ), 6352.2.<br>T <sub>1/2</sub> : in ( $^{14}\text{N}, 3\text{p}\gamma$ ) from 2014Ay01 with DSAM.                                                                                                                                      |
| 7210.42 5             | 1/2 <sup>-</sup> , 3/2, 5/2 <sup>+</sup> |                  | <b>G i k</b>  | XREF: i(7205)k(7215)<br>J <sup>π</sup> : others: L=(1,2) from 0 <sup>+</sup> in $^{34}\text{S}(\text{d,p})$ for 7205 35; L=2 from 0 <sup>+</sup> in $^{36}\text{S}(\text{p,d})$ for 7215 5.                                                                                                                                             |
| 7218.0 4              |                                          | 0.33 eV 2        | <b>G i k</b>  | XREF: i(7205)k(7215)                                                                                                                                                                                                                                                                                                                    |
| 7234.0 5              |                                          | 0.19 eV 1        | <b>G i</b>    | XREF: i(7205)                                                                                                                                                                                                                                                                                                                           |
| 7239.8 6              |                                          | 0.06 eV 2        | <b>G i</b>    | XREF: i(7205)                                                                                                                                                                                                                                                                                                                           |
| 7253.2 6              | (3/2 <sup>+</sup> , 5/2 <sup>+</sup> )   | 0.35 eV 2        | <b>G K</b>    | XREF: K(7249)<br>J <sup>π</sup> : L=(2) from 0 <sup>+</sup> in $^{36}\text{S}(\text{p,d})$ .                                                                                                                                                                                                                                            |
| 7275.93 5             | 1/2 <sup>+</sup>                         |                  | <b>G k</b>    | XREF: k(7279)                                                                                                                                                                                                                                                                                                                           |
| 7279.13?              |                                          | 0.10 eV 3        | <b>G k</b>    | J <sup>π</sup> : other: L=(0) from 0 <sup>+</sup> in $^{36}\text{S}(\text{p,d})$ for 7279 5.                                                                                                                                                                                                                                            |
| 7289.8 8              |                                          | 0.31 eV 3        | <b>G</b>      | XREF: G(?)k(7279)                                                                                                                                                                                                                                                                                                                       |
| 7294.19 6             | 3/2 <sup>-</sup>                         |                  | <b>G</b>      |                                                                                                                                                                                                                                                                                                                                         |
| 7331.00 5             | 1/2 <sup>+</sup>                         |                  | <b>G K</b>    | XREF: K(7326)                                                                                                                                                                                                                                                                                                                           |

Continued on next page (footnotes at end of table)

**Adopted Levels, Gammas (continued)** $^{35}\text{S}$  Levels (continued)

| E(level) <sup>†</sup> | J <sup>π</sup> <sup>‡</sup>              | T <sub>1/2</sub> | XREF | Comments                                                                                                                                 |
|-----------------------|------------------------------------------|------------------|------|------------------------------------------------------------------------------------------------------------------------------------------|
|                       |                                          |                  |      | J <sup>π</sup> : other: L=0 from 0 <sup>+</sup> in $^{36}\text{S}(\text{p},\text{d})$ .                                                  |
| 7338.18 20            | 1/2 <sup>-</sup> , 3/2, 5/2 <sup>+</sup> |                  | G    |                                                                                                                                          |
| 7343.91 20            |                                          | 0.72 eV 3        | G k  | XREF: k(7349)                                                                                                                            |
| 7347.2 5              |                                          | 0.44 eV 6        | G k  | XREF: k(7349)                                                                                                                            |
| 7347.9 5              |                                          | 0.30 eV 5        | G k  | XREF: k(7349)                                                                                                                            |
| 7353.9 5              |                                          | 0.44 eV 4        | G k  | XREF: k(7349)                                                                                                                            |
| 7357.6 5              |                                          | 0.16 eV 4        | G    |                                                                                                                                          |
| 7370.56 6             | 1/2 <sup>-</sup>                         |                  | G 1  | XREF: l(7375)                                                                                                                            |
| 7371.97?              |                                          | 0.12 eV          | G 1  | XREF: G(?)l(7375)                                                                                                                        |
| 7396.0 6              | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>      | 0.44 eV 33       | G    |                                                                                                                                          |
| 7404.6 6              |                                          | 0.21 eV 5        | G    |                                                                                                                                          |
| 7408.68 11            | 3/2 <sup>-</sup>                         |                  | G    |                                                                                                                                          |
| 7411.55 6             |                                          |                  | G    |                                                                                                                                          |
| 7416.50 4             |                                          |                  | G    |                                                                                                                                          |
| 7429.1 6              |                                          | 0.12 eV 3        | G    |                                                                                                                                          |
| 7433.56 5             |                                          |                  | G    |                                                                                                                                          |
| 7436.4 4              | 1/2 <sup>-</sup> , 3/2, 5/2 <sup>+</sup> |                  | G    |                                                                                                                                          |
| 7442.14 5             | 1/2 <sup>+</sup>                         |                  | G K  | XREF: K(7441)                                                                                                                            |
|                       |                                          |                  |      | J <sup>π</sup> : other: L=0 from 0 <sup>+</sup> in $^{36}\text{S}(\text{p},\text{d})$ .                                                  |
| 7462.2 6              |                                          | 0.33 eV 4        | G    |                                                                                                                                          |
| 7481.15 7             | 1/2 <sup>-</sup>                         |                  | G I  | E(level): weighted average of 7481.15 6 from (n,n),(n,γ):resonances and 7482.7 13 from $^{34}\text{S}(\text{d},\text{p})$ .              |
|                       |                                          |                  |      | J <sup>π</sup> : other: L=(1,2) from 0 <sup>+</sup> in $^{34}\text{S}(\text{d},\text{p})$ .                                              |
| 7494.5 6              | 1/2 <sup>-</sup> , 3/2 <sup>-</sup>      | 0.17 eV 4        | G K  | XREF: K(7490)                                                                                                                            |
|                       |                                          |                  |      | J <sup>π</sup> : L=1 from 0 <sup>+</sup> in $^{36}\text{S}(\text{p},\text{d})$ .                                                         |
| 7542.3 6              |                                          | 0.21 eV 3        | G i  | XREF: i(7570)                                                                                                                            |
| 7604.0 6              |                                          | 0.90 eV 8        | G i  | XREF: i(7570)                                                                                                                            |
| 7609.26 5             | 3/2 <sup>-</sup>                         |                  | G    |                                                                                                                                          |
| 7648.9 6              |                                          | 0.21 eV 3        | G i  | XREF: i(7640)                                                                                                                            |
| 7655.7 6              |                                          | 0.80 eV 5        | G i  | XREF: i(7640)                                                                                                                            |
| 7663.91 11            | 1/2 <sup>-</sup> , 3/2, 5/2 <sup>+</sup> |                  | G i  | XREF: i(7640)                                                                                                                            |
| 7678.3 7              |                                          | 0.45 eV 7        | G i  | XREF: i(7640)                                                                                                                            |
| 7731.0 7              |                                          | 0.31 eV 13       | G L  | XREF: L(7712?)                                                                                                                           |
| 7749.7 4              | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>      |                  | G K  | XREF: K(7753)                                                                                                                            |
|                       |                                          |                  |      | J <sup>π</sup> : L=2 from 0 <sup>+</sup> in $^{36}\text{S}(\text{p},\text{d})$ .                                                         |
| 7761.46 7             | 3/2 <sup>-</sup>                         |                  | G 1  | XREF: l(7770)                                                                                                                            |
| 7776.17 11            | 1/2 <sup>-</sup>                         |                  | G 1  | XREF: l(7770)                                                                                                                            |
| 7797.99 9             | 1/2 <sup>+</sup>                         |                  | G    |                                                                                                                                          |
| 7812.24 7             | 3/2 <sup>+</sup>                         |                  | G    |                                                                                                                                          |
| 7853.25 7             | 3/2 <sup>-</sup>                         |                  | G    |                                                                                                                                          |
| 7861.8 12             |                                          | 1.13 eV 19       | G    |                                                                                                                                          |
| 7880.8 12             |                                          | 0.67 eV 17       | G k  | XREF: k(7885)                                                                                                                            |
|                       |                                          |                  |      | J <sup>π</sup> : L=(2) from 0 <sup>+</sup> in $^{36}\text{S}(\text{p},\text{d})$ for 7885 5.                                             |
| 7889.0 12             |                                          | 0.57 eV 25       | G k  | XREF: k(7885)                                                                                                                            |
|                       |                                          |                  |      | J <sup>π</sup> : L=(2) from 0 <sup>+</sup> in $^{36}\text{S}(\text{p},\text{d})$ for 7885 5.                                             |
| 7894.63 7             | 3/2 <sup>-</sup>                         |                  | G    |                                                                                                                                          |
| 7899.70 30            |                                          |                  | G    |                                                                                                                                          |
| 7934.1 15             |                                          | 0.5 eV 3         | G    |                                                                                                                                          |
| 7939.5 15             |                                          | 0.89 eV 4        | G    |                                                                                                                                          |
| 7954.96 11            | 3/2 <sup>-</sup>                         |                  | G    |                                                                                                                                          |
| 7974.19 11            | 3/2 <sup>-</sup>                         |                  | G K  | XREF: K(7980)                                                                                                                            |
|                       |                                          |                  |      | J <sup>π</sup> : other: L=1 from 0 <sup>+</sup> in $^{36}\text{S}(\text{p},\text{d})$ .                                                  |
| 8019.5 2              | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>      |                  | G    |                                                                                                                                          |
| 8024.0 7              | (17/2 <sup>+</sup> )                     | 0.15 ps 4        | DE   | J <sup>π</sup> : 844.2γ (M1), ΔJ=1 to (15/2 <sup>+</sup> ), 7179.74. 17/2 <sup>+</sup> suggested by shell-model calculations (2021Go09). |

Continued on next page (footnotes at end of table)

**Adopted Levels, Gammas (continued)**

| $^{35}\text{S}$ Levels (continued) |                                        |                  |      |                                                                                                                                                                                                                                 |
|------------------------------------|----------------------------------------|------------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E(level) <sup>†</sup>              | J <sup>π</sup> <sup>‡</sup>            | T <sub>1/2</sub> | XREF | Comments                                                                                                                                                                                                                        |
| 8041.20 20                         | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>    |                  | G    | T <sub>1/2</sub> : in ( <sup>14</sup> N, 3pγ) from <a href="#">2014Ay01</a> with DSAM.                                                                                                                                          |
| 8076.46 20                         | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>    |                  | G k  | XREF: k(8073)                                                                                                                                                                                                                   |
| 8077.91 30                         | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>    |                  | G k  | J <sup>π</sup> : other: L=(2) from 0 <sup>+</sup> in <sup>36</sup> S(p,d) for 8073 5.<br>XREF: k(8073)                                                                                                                          |
| 8093.0 5                           | 1/2 <sup>-</sup>                       |                  | G KL | J <sup>π</sup> : other: L=(2) from 0 <sup>+</sup> in <sup>36</sup> S(p,d) for 8073 5.<br>XREF: K(8103)L(8103)                                                                                                                   |
| 8143.66 20                         | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>    |                  | G    | E(level): from (n,n),(n,γ):resonances. Others: 8103 5 from <sup>36</sup> S(p,d) and<br>8103 10 from (p, <sup>3</sup> He).                                                                                                       |
| 8160 10                            |                                        |                  |      | J <sup>π</sup> : other: L=1 from 0 <sup>+</sup> in <sup>36</sup> S(p,d); L=(1+3) from 3/2 <sup>+</sup> in (p, <sup>3</sup> He).                                                                                                 |
| 8187.17 30                         | 1/2 <sup>-</sup>                       |                  | G    | J <sup>π</sup> : L=(1) from 3/2 <sup>+</sup> in (p, <sup>3</sup> He) suggests (1/2 <sup>-</sup> to 7/2 <sup>-</sup> ).                                                                                                          |
| 8211.25 30                         |                                        |                  | G    |                                                                                                                                                                                                                                 |
| 8224.2 4                           | 1/2 <sup>-</sup>                       |                  | G K  | XREF: K(8221)                                                                                                                                                                                                                   |
| 8228.3 4                           | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>    |                  | G    | J <sup>π</sup> : other: L=1 from 0 <sup>+</sup> in <sup>36</sup> S(p,d).                                                                                                                                                        |
| 8243.98 30                         | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>    |                  | G    |                                                                                                                                                                                                                                 |
| 8256.31 20                         | 3/2 <sup>-</sup>                       |                  | G    |                                                                                                                                                                                                                                 |
| 8262.4 4                           | 5/2 <sup>+</sup>                       |                  | G    |                                                                                                                                                                                                                                 |
| 8273.3 4                           | (1/2 <sup>-</sup> )                    |                  | G K  | XREF: K(8270)                                                                                                                                                                                                                   |
| 8298.27 30                         | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>    |                  | G    | J <sup>π</sup> : other: L=1 from 0 <sup>+</sup> in <sup>36</sup> S(p,d).                                                                                                                                                        |
| 8333.8 4                           | 3/2 <sup>-</sup>                       |                  | G    |                                                                                                                                                                                                                                 |
| 8335.9 4                           | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>    |                  | G    |                                                                                                                                                                                                                                 |
| 8391.3 4                           | 3/2 <sup>-</sup>                       |                  | G    |                                                                                                                                                                                                                                 |
| 8393.64 30                         | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>    |                  | G    |                                                                                                                                                                                                                                 |
| 8406.55 30                         | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>    |                  | G k  | XREF: k(8410)                                                                                                                                                                                                                   |
| 8418.8 4                           | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>    |                  | G kL | XREF: k(8410)L(8430)                                                                                                                                                                                                            |
| 8465 7                             | (3/2 <sup>+</sup> , 5/2 <sup>+</sup> ) |                  | K    | J <sup>π</sup> : others: L=(2) from 0 <sup>+</sup> in <sup>36</sup> S(p,d); L=2 from 3/2 <sup>+</sup> in (p, <sup>3</sup> He)<br>suggests (1/2 <sup>+</sup> to 9/2 <sup>+</sup> ).                                              |
| 8509 5                             | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>    |                  | K    |                                                                                                                                                                                                                                 |
| 8557 7                             | (5/2 <sup>-</sup> , 7/2 <sup>-</sup> ) |                  | K    |                                                                                                                                                                                                                                 |
| 8602 5                             | (5/2 <sup>-</sup> , 7/2 <sup>-</sup> ) |                  | K    |                                                                                                                                                                                                                                 |
| 8649 7                             | (5/2 <sup>-</sup> , 7/2 <sup>-</sup> ) |                  | K    |                                                                                                                                                                                                                                 |
| 8707 5                             | (3/2 <sup>+</sup> , 5/2 <sup>+</sup> ) |                  | K    |                                                                                                                                                                                                                                 |
| 8755.9 11                          | (17/2)                                 | <1 ps            | E    | J <sup>π</sup> : 1576.3γ D, ΔJ=1 to (15/2 <sup>+</sup> ), 7179.74.<br>T <sub>1/2</sub> : in ( <sup>18</sup> O, 2αγ) from <a href="#">2021Go09</a> with residual Doppler energy<br>shifts observed for the 732γ and 1576γ peaks. |
| 8761 7                             | (3/2 <sup>+</sup> , 5/2 <sup>+</sup> ) |                  | K    |                                                                                                                                                                                                                                 |
| 8822 5                             | (1/2 <sup>-</sup> , 3/2 <sup>-</sup> ) |                  | K    |                                                                                                                                                                                                                                 |
| 8876 5                             | (1/2 <sup>-</sup> , 3/2 <sup>-</sup> ) |                  | K    |                                                                                                                                                                                                                                 |
| 8947 7                             | (1/2 <sup>-</sup> , 3/2 <sup>-</sup> ) |                  | K    |                                                                                                                                                                                                                                 |
| 9032 15                            | (1/2 <sup>-</sup> , 3/2 <sup>-</sup> ) |                  | K    |                                                                                                                                                                                                                                 |
| 9065 15                            | (1/2 <sup>-</sup> , 3/2 <sup>-</sup> ) |                  | K    |                                                                                                                                                                                                                                 |
| 9126 7                             | 1/2 <sup>+</sup>                       |                  | K1   | T=5/2<br>XREF: l(9155)                                                                                                                                                                                                          |
| 9180 5                             |                                        |                  | K1   | J <sup>π</sup> , T: isobaric analog state of 1/2 <sup>+</sup> , T=5/2, g.s. of <sup>35</sup> P ( <a href="#">2026Jo01</a> ).<br>XREF: K(9180)l(9155)                                                                            |
| 9219 15                            | (1/2 <sup>-</sup> , 3/2 <sup>-</sup> ) |                  | K    | J <sup>π</sup> : L=2 from 3/2 <sup>+</sup> in (p, <sup>3</sup> He) suggests (1/2 <sup>+</sup> to 9/2 <sup>+</sup> ).                                                                                                            |
| 9294 8                             | (1/2 <sup>-</sup> , 3/2 <sup>-</sup> ) |                  | K    |                                                                                                                                                                                                                                 |
| 9339 8                             | (1/2 <sup>-</sup> , 3/2 <sup>-</sup> ) |                  | K    |                                                                                                                                                                                                                                 |

Continued on next page (footnotes at end of table)

**Adopted Levels, Gammas (continued)** $^{35}\text{S}$  Levels (continued)

| E(level) <sup>†</sup>   | $J^\pi$ <sup>‡</sup>                  | XREF | Comments                                                                               |
|-------------------------|---------------------------------------|------|----------------------------------------------------------------------------------------|
| 9395 5                  | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | K    |                                                                                        |
| 9446 7                  | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | K    |                                                                                        |
| 9510 10                 | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | K    |                                                                                        |
| 9565 20                 | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | K    |                                                                                        |
| 9615 10                 | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | K    |                                                                                        |
| 9666 10                 | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | K    |                                                                                        |
| 9716 10                 | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | K    |                                                                                        |
| 9774 10                 | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | K    |                                                                                        |
| 9849 10                 | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | K    |                                                                                        |
| 9911 10                 | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | K    |                                                                                        |
| 9989 10                 | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | K    |                                                                                        |
| 10048.6 9               | (19/2 <sup>+</sup> )                  | E    | $J^\pi$ : 2868.7 $\gamma$ Q, $\Delta J=2$ to (15/2 <sup>+</sup> ), 7179.74.            |
| 1.0049 $\times 10^4$ 10 | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | K    |                                                                                        |
| 1.0166 $\times 10^4$ 10 | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | K    |                                                                                        |
| 1.0225 $\times 10^4$ 10 | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | K    |                                                                                        |
| 1.0277 $\times 10^4$ 10 | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | K    |                                                                                        |
| 1.0350 $\times 10^4$ 10 | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | K    |                                                                                        |
| 1.0421 $\times 10^4$ 10 | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | K    |                                                                                        |
| 1.0493 $\times 10^4$ 10 | (5/2 <sup>-</sup> ,7/2 <sup>-</sup> ) | K    |                                                                                        |
| 1.0563 $\times 10^4$ 10 | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | K    |                                                                                        |
| 1.0663 $\times 10^4$ 10 | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | K    |                                                                                        |
| 1.0746 $\times 10^4$ 10 | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | K    |                                                                                        |
| 1.0810 $\times 10^4$ 10 | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | K    |                                                                                        |
| 1.147 $\times 10^4$ 2   | (3/2 <sup>+</sup> )                   | K    | T=5/2                                                                                  |
| 12469.3 14              | (21/2)                                | E    | $J^\pi$ ,T: IAS of 3/2 <sup>+</sup> , T=5/2, 2387 level in $^{35}\text{P}$ (2026Jo01). |
| 1.288 $\times 10^4$ 10  | (5/2 <sup>+</sup> )                   | K    | $J^\pi$ : 2420.6 $\gamma$ (D), $\Delta J=(1)$ to (19/2 <sup>+</sup> ), 10048.6.        |
|                         |                                       |      | T=5/2                                                                                  |
| 1.368 $\times 10^4$ 10  | (5/2 <sup>+</sup> )                   | K    | $J^\pi$ ,T: IAS of 5/2 <sup>+</sup> , T=5/2, 3860 level in $^{35}\text{P}$ (2026Jo01). |
|                         |                                       |      | T=5/2                                                                                  |
| 1.422 $\times 10^4$ 10  | (5/2 <sup>+</sup> )                   | K    | $J^\pi$ ,T: IAS of 5/2 <sup>+</sup> , T=5/2, 4666 level in $^{35}\text{P}$ (2026Jo01). |
|                         |                                       |      | T=5/2                                                                                  |
| 1.470 $\times 10^4$ 10  |                                       | K    | $J^\pi$ ,T: IAS of 5/2 <sup>+</sup> , T=5/2, 5199 level in $^{35}\text{P}$ (2026Jo01). |

<sup>†</sup> From a least-squares fit to  $\gamma$ -ray energies with uncertainties for levels connected with  $\gamma$  transitions; from transfer reactions or  $^{34}\text{S}(n,n),(n,\gamma)$ :resonances for other levels, unless otherwise noted.

<sup>‡</sup> For high-spin states, it is assumed that spin increases as the excitation energy increases. For resonances (7018-8418 keV),  $J^\pi$  is from  $^{34}\text{S}(n,n),(n,\gamma)$ :resonances, unless otherwise noted; for resonances above 8418 keV,  $J^\pi$  is from  $^{36}\text{S}(p,d)$ , unless otherwise noted.

Adopted Levels, Gammas (continued)

| $\gamma(^{35}\text{S})$ |                  |                         |                    |                     |                                      |                    |                   |                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------|------------------|-------------------------|--------------------|---------------------|--------------------------------------|--------------------|-------------------|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $E_i(\text{level})$     | $J_i^\pi$        | $E_\gamma^\dagger$      | $I_\gamma^\dagger$ | $E_f$               | $J_f^\pi$                            | Mult. <sup>‡</sup> | $\delta^\ddagger$ | $\alpha^\#$                                        | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 1572.369                | 1/2 <sup>+</sup> | 1572.327 11             | 100                | 0.0                 | 3/2 <sup>+</sup>                     | (M1)               |                   | 0.0001007 14                                       | B(M1)(W.u.)=0.00248 16<br>E <sub>γ</sub> : weighted average of 1572.252 29 from <sup>35</sup> P β <sup>-</sup> decay, 1572.4 3 from ( <sup>14</sup> N,3pγ), 1572.333 8 from (n,γ) E=thermal, 1572.24 17 from <sup>34</sup> S(d,pγ), and 1572 1 from <sup>208</sup> Pb( <sup>36</sup> S, <sup>35</sup> Sγ).                                                                                                                                                                                                                                                  |
| 1991.27                 | 7/2 <sup>-</sup> | 1991.26 5               | 100                | 0.0                 | 3/2 <sup>+</sup>                     | M2+E3              | +0.17 6           | 0.0001476 22                                       | Mult.: D from ( <sup>14</sup> N,3pγ); Δπ=no from level scheme.<br>B(M2)(W.u.)=0.0874 +46-48; B(E3)(W.u.)=3.7 +29-21<br>E <sub>γ</sub> : weighted average of 1991.3 2 from ( <sup>14</sup> N,3pγ), 1990.5 4 from ( <sup>18</sup> O,2αnγ), 1991.27 5 from (n,γ) E=thermal, and 1991 1 from <sup>208</sup> Pb( <sup>36</sup> S, <sup>35</sup> Sγ).                                                                                                                                                                                                             |
| 2347.779                | 3/2 <sup>-</sup> | 356.66 @ 9<br>775.398 6 | <0.084<br>35.3 27  | 1991.27<br>1572.369 | 7/2 <sup>-</sup><br>1/2 <sup>+</sup> | [E2]<br>(E1)       |                   | 1.15×10 <sup>-3</sup> 2<br>3.54×10 <sup>-5</sup> 5 | Mult.,δ: from <sup>34</sup> S(d,pγ). D+Q or Q+O from γ(θ) in <sup>34</sup> S(d,pγ); E2+M3 is ruled out by RUL; D+Q is inconsistent with level scheme from 7/2 <sup>-</sup> to 3/2 <sup>+</sup> .<br>E <sub>γ</sub> ,I <sub>γ</sub> : from (n,γ) E=thermal.<br>B(E1)(W.u.)=4.5×10 <sup>-4</sup> +9-7<br>E <sub>γ</sub> : from (n,γ) E=thermal. Other: 775.4 4 from ( <sup>14</sup> N,3pγ).<br>I <sub>γ</sub> : weighted average of 37 5 from ( <sup>14</sup> N,3pγ), 35 4 from (n,γ) E=thermal, 37.0 28 from <sup>34</sup> S(d,pγ), and 33.3 27 from (d,αγ). |
|                         |                  | 2347.700 19             | 100.0 27           | 0.0                 | 3/2 <sup>+</sup>                     | (E1)               |                   | 0.000880 12                                        | Mult.: D(+Q) from γ(θ) in <sup>34</sup> S(d,pγ); Δπ=yes from level scheme; M2 ruled out by RUL.<br>B(E1)(W.u.)=4.6×10 <sup>-5</sup> +8-6<br>E <sub>γ</sub> : weighted average of 2347.8 3 from ( <sup>14</sup> N,3pγ), 2347.700 15 from (n,γ) E=thermal, and 2347 2 from <sup>208</sup> Pb( <sup>36</sup> S, <sup>35</sup> Sγ).                                                                                                                                                                                                                             |
| 2717.06                 | 5/2 <sup>+</sup> | 368.7 5                 | 14 8               | 2347.779            | 3/2 <sup>-</sup>                     | [E1]               |                   | 0.0002294 33                                       | I <sub>γ</sub> : from (d,αγ). Others: 100 8 from ( <sup>14</sup> N,3pγ), 100 8 from (n,γ) E=thermal, and 100.0 28 from <sup>34</sup> S(d,pγ).<br>Mult.: D(+Q) from γ(θ) in <sup>34</sup> S(d,pγ); Δπ=yes from level scheme; M2 component is less likely based on RUL.<br>B(E1)(W.u.)=0.022 +20-10<br>E <sub>γ</sub> : weighted average of 368.5 4 from (n,γ) E=thermal and 370 1 from <sup>208</sup> Pb( <sup>36</sup> S, <sup>35</sup> Sγ).                                                                                                                |
|                         |                  | 725.78<br>1144.67       | <5<br><5           | 1991.27<br>1572.369 | 7/2 <sup>-</sup><br>1/2 <sup>+</sup> | [E1]<br>[E2]       |                   | 4.10×10 <sup>-5</sup> 6<br>3.79×10 <sup>-5</sup> 5 | I <sub>γ</sub> : unweighted average of 6.7 23 from (n,γ) E=thermal and 22 4 from <sup>208</sup> Pb( <sup>36</sup> S, <sup>35</sup> Sγ).                                                                                                                                                                                                                                                                                                                                                                                                                     |

Adopted Levels, Gammas (continued)

| $\gamma(^{35}\text{S})$ (continued) |                     |                                                        |                                    |                                                              |                                                                                                                      |                                                       |                   |                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------|---------------------|--------------------------------------------------------|------------------------------------|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $E_i(\text{level})$                 | $J_i^\pi$           | $E_\gamma^\dagger$                                     | $I_\gamma^\dagger$                 | $E_f$                                                        | $J_f^\pi$                                                                                                            | Mult. <sup>‡</sup>                                    | $\delta^\ddagger$ | $\alpha^\#$                                                                                                                | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 2717.06                             | 5/2 <sup>+</sup>    | 2716.99 15                                             | 100 10                             | 0.0                                                          | 3/2 <sup>+</sup>                                                                                                     | M1+E2                                                 | -0.39 5           | 0.000568 9                                                                                                                 | B(M1)(W.u.)=0.012 +7-3; B(E2)(W.u.)=0.9 +6-3<br>E <sub>γ</sub> : weighted average of 2717.0 4 from ( <sup>14</sup> N,3pγ),<br>2716.99 16 from (n,γ) E=thermal, and 2717 2<br>from <sup>208</sup> Pb( <sup>36</sup> S, <sup>35</sup> Sγ).<br>I <sub>γ</sub> : from <sup>208</sup> Pb( <sup>36</sup> S, <sup>35</sup> Sγ). Other: 100 17 from<br>(n,γ) E=thermal.<br>Mult.,δ: D+Q from γ(θ) in <sup>34</sup> S(d,pγ) with δ=-0.39<br>5 or -15.0 34 for J <sub>i</sub> =5/2 (1975VaYG); -15.0 34 is<br>unlikely based on RUL; E1+M2 is ruled out by<br>RUL.                                                                                                                                                                                                                                                                                                            |
| 2938.63                             | 3/2 <sup>+</sup>    | 590.85<br>947.35<br>1366.23<br>2938.56 11              | <10<br><20<br><15<br>100           | 2347.779<br>1991.27<br>1572.369<br>0.0                       | 3/2 <sup>-</sup><br>7/2 <sup>-</sup><br>1/2 <sup>+</sup><br>3/2 <sup>+</sup>                                         |                                                       |                   |                                                                                                                            | E <sub>γ</sub> : weighted average of 2938.29 40 from <sup>35</sup> P β <sup>-</sup><br>decay and 2938.58 11 from (n,γ) E=thermal.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 3421.2                              | 5/2 <sup>+</sup>    | 482.6<br>704.1<br>1073.4<br>1429.9<br>1848.8<br>3421 2 | <4<br><6<br><7<br><30<br><7<br>100 | 2938.63<br>2717.06<br>2347.779<br>1991.27<br>1572.369<br>0.0 | 3/2 <sup>+</sup><br>5/2 <sup>+</sup><br>3/2 <sup>-</sup><br>7/2 <sup>-</sup><br>1/2 <sup>+</sup><br>3/2 <sup>+</sup> | [M1,E2]<br>[M1,E2]<br>[E1]<br>[E1]<br>[E2]<br>[M1,E2] |                   | 2.7×10 <sup>-4</sup> 13<br>9.5×10 <sup>-5</sup> 28<br>1.836×10 <sup>-5</sup> 26<br>0.0002243 31<br>0.000254 4<br>0.00090 7 | E <sub>γ</sub> : from <sup>208</sup> Pb( <sup>36</sup> S, <sup>35</sup> Sγ).<br>E <sub>γ</sub> ,I <sub>γ</sub> : from (n,γ) E=thermal.<br>E <sub>γ</sub> : from (n,γ) E=thermal. Other: 1211 2 from<br><sup>208</sup> Pb( <sup>36</sup> S, <sup>35</sup> Sγ).<br>I <sub>γ</sub> : weighted average of 54 6 from (n,γ) E=thermal<br>and 54 6 from (d,αγ). Other: 18 7 from<br><sup>208</sup> Pb( <sup>36</sup> S, <sup>35</sup> Sγ).<br>E <sub>γ</sub> : from (n,γ) E=thermal. Other: 1567 2 from<br><sup>208</sup> Pb( <sup>36</sup> S, <sup>35</sup> Sγ).<br>I <sub>γ</sub> : from (d,αγ). Others: 100 19 from (n,γ)<br>E=thermal and 100 10 from <sup>208</sup> Pb( <sup>36</sup> S, <sup>35</sup> Sγ).<br>E <sub>γ</sub> ,I <sub>γ</sub> : from <sup>208</sup> Pb( <sup>36</sup> S, <sup>35</sup> Sγ).<br>E <sub>γ</sub> ,I <sub>γ</sub> : from (n,γ) E=thermal. |
| 3558.080                            | (5/2 <sup>+</sup> ) | 619.23 19<br>1210.28 4                                 | 8.9 17<br>54 6                     | 2938.63<br>2347.779                                          | 3/2 <sup>+</sup><br>3/2 <sup>-</sup>                                                                                 |                                                       |                   |                                                                                                                            | E <sub>γ</sub> ,I <sub>γ</sub> : from ( <sup>14</sup> N,3pγ). Other: I <sub>γ</sub> <11 from<br><sup>34</sup> S(d,pγ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                     |                     | 1566.7 3                                               | 100 6                              | 1991.27                                                      | 7/2 <sup>-</sup>                                                                                                     |                                                       |                   |                                                                                                                            | E <sub>γ</sub> : weighted average of 3594.7 7 from ( <sup>14</sup> N,3pγ),<br>3594.4 11 from ( <sup>18</sup> O,2αγ), and 3593 2 from                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 3594.62                             | (7/2 <sup>+</sup> ) | 1986 & 2<br>3558.1 5<br>655.98<br>877.6 5              | 39 6<br>18 4<br><10<br>9.3 23      | 1572.369<br>0.0<br>2938.63<br>2717.06                        | 1/2 <sup>+</sup><br>3/2 <sup>+</sup><br>3/2 <sup>+</sup><br>5/2 <sup>+</sup>                                         |                                                       |                   |                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                     |                     | 1246.82<br>1603.31<br>3594.5 7                         | <15<br><17<br>100 9                | 2347.779<br>1991.27<br>0.0                                   | 3/2 <sup>-</sup><br>7/2 <sup>-</sup><br>3/2 <sup>+</sup>                                                             | Q                                                     |                   |                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

Adopted Levels, Gammas (continued)

| <u><math>\gamma(^{35}\text{S})</math> (continued)</u> |                             |                                      |                                      |                         |                             |                          |                                     |                               |                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------------------------------|-----------------------------|--------------------------------------|--------------------------------------|-------------------------|-----------------------------|--------------------------|-------------------------------------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <u><math>E_i(\text{level})</math></u>                 | <u><math>J_i^\pi</math></u> | <u><math>E_\gamma^\dagger</math></u> | <u><math>I_\gamma^\dagger</math></u> | <u><math>E_f</math></u> | <u><math>J_f^\pi</math></u> | <u>Mult.<sup>‡</sup></u> | <u><math>\delta^\ddagger</math></u> | <u><math>\alpha^\#</math></u> | <u>Comments</u>                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 3801.952                                              | 3/2 <sup>-</sup>            | 863.28 28                            | 1.3 4                                | 2938.63                 | 3/2 <sup>+</sup>            | [E1]                     |                                     | 2.82×10 <sup>-5</sup> 4       | <sup>208</sup> Pb( <sup>36</sup> S, <sup>35</sup> S $\gamma$ ).<br>I $\gamma$ : from ( <sup>14</sup> N,3p $\gamma$ ).<br>Mult.: from $\gamma\gamma(\theta)$ (ADO) in ( <sup>14</sup> N,3p $\gamma$ ).<br>B(E1)(W.u.)=2.4×10 <sup>-4</sup> +30-12<br>E $\gamma$ , I $\gamma$ : from (n, $\gamma$ ) E=thermal. Other: I $\gamma$ <8.9 from <sup>34</sup> S(d,p $\gamma$ ).                                                         |
|                                                       |                             | 1084.79 15                           | 2.2 7                                | 2717.06                 | 5/2 <sup>+</sup>            | [E1]                     |                                     | 1.801×10 <sup>-5</sup> 25     | B(E1)(W.u.)=2.0×10 <sup>-4</sup> +28-11<br>E $\gamma$ : from (n, $\gamma$ ) E=thermal.<br>I $\gamma$ : weighted average of 2.1 4 from (n, $\gamma$ ) E=thermal and 11 5 from <sup>34</sup> S(d,p $\gamma$ ).                                                                                                                                                                                                                     |
|                                                       |                             | 1454.09 4                            | 18 5                                 | 2347.779                | 3/2 <sup>-</sup>            | [M1,E2]                  |                                     | 8.0×10 <sup>-5</sup> 12       | E $\gamma$ : from (n, $\gamma$ ) E=thermal.<br>I $\gamma$ : weighted average of 18 4 from (n, $\gamma$ ) E=thermal and 18 7 from <sup>34</sup> S(d,p $\gamma$ ).                                                                                                                                                                                                                                                                 |
|                                                       |                             | 1810.63                              | <8.9                                 | 1991.27                 | 7/2 <sup>-</sup>            | [E2]                     |                                     | 0.0002364 33                  | B(M1)(W.u.)=0.024 +32-12 if M1,<br>B(E2)(W.u.)=4×10 <sup>1</sup> +6-2 if E2.                                                                                                                                                                                                                                                                                                                                                     |
|                                                       |                             | 2229.510 16                          | 88 15                                | 1572.369                | 1/2 <sup>+</sup>            | [E1]                     |                                     | 0.000804 11                   | B(E1)(W.u.)=9×10 <sup>-4</sup> +11-4<br>E $\gamma$ : from (n, $\gamma$ ) E=thermal.<br>I $\gamma$ : unweighted average of 110 11 from (n, $\gamma$ ) E=thermal, 93 9 from <sup>34</sup> S(d,p $\gamma$ ), and 61 5 from (d, $\alpha\gamma$ ).                                                                                                                                                                                    |
|                                                       |                             | 3801.73 4                            | 100 5                                | 0.0                     | 3/2 <sup>+</sup>            | [E1]                     |                                     | 1.60×10 <sup>-3</sup> 2       | B(E1)(W.u.)=2.2×10 <sup>-4</sup> +27-10<br>E $\gamma$ : from (n, $\gamma$ ) E=thermal.<br>I $\gamma$ : from (d, $\alpha\gamma$ ). Other: 100 9 from <sup>34</sup> S(d,p $\gamma$ ), 100 9 from (n, $\gamma$ ) E=thermal.                                                                                                                                                                                                         |
| 3816.01                                               | (9/2) <sup>-</sup>          | 1824.7 3                             | 100                                  | 1991.27                 | 7/2 <sup>-</sup>            | M1+E2                    | +0.55 9                             | 0.000200 4                    | B(M1)(W.u.)=0.0099 +14-12; B(E2)(W.u.)=3.4 +10-9<br>E $\gamma$ : weighted average of 1824.6 2 from ( <sup>14</sup> N,3p $\gamma$ ), 1824.4 6 from ( <sup>18</sup> O,2 $\alpha\gamma$ ), and 1827 1 from <sup>208</sup> Pb( <sup>36</sup> S, <sup>35</sup> S $\gamma$ ).<br>Mult., $\delta$ : D+Q and $\delta$ from $\gamma(\theta)$ and $\gamma\gamma(\theta)$ (ADO) in ( <sup>14</sup> N,3p $\gamma$ ); E1+M2 ruled out by RUL. |
| 3886.28                                               | (5/2)                       | 465                                  | <30                                  | 3421.2                  | 5/2 <sup>+</sup>            |                          |                                     |                               |                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                       |                             | 947.64                               | <12                                  | 2938.63                 | 3/2 <sup>+</sup>            |                          |                                     |                               |                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                       |                             | 1169.20                              | <15                                  | 2717.06                 | 5/2 <sup>+</sup>            |                          |                                     |                               |                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                       |                             | 1538.4 8                             | 57 33                                | 2347.779                | 3/2 <sup>-</sup>            |                          |                                     |                               | E $\gamma$ : from ( <sup>14</sup> N,3p $\gamma$ ).<br>I $\gamma$ : unweighted average of 24 10 from ( <sup>14</sup> N,3p $\gamma$ ) and 89 9 from (d, $\alpha\gamma$ ). Other: <18 from <sup>34</sup> S(d,p $\gamma$ ).<br>E $\gamma$ , Mult.: from ( <sup>14</sup> N,3p $\gamma$ ).                                                                                                                                             |
|                                                       |                             | 1894.9 4                             | 100 11                               | 1991.27                 | 7/2 <sup>-</sup>            | D                        |                                     |                               |                                                                                                                                                                                                                                                                                                                                                                                                                                  |

Adopted Levels, Gammas (continued)

| $\gamma(^{35}\text{S})$ (continued) |                                       |                                                                              |                                                              |                                                                                    |                                                                                                                                                                 |                    |                   |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------|---------------------------------------|------------------------------------------------------------------------------|--------------------------------------------------------------|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $E_i(\text{level})$                 | $J_i^\pi$                             | $E_\gamma^\dagger$                                                           | $I_\gamma^\dagger$                                           | $E_f$                                                                              | $J_f^\pi$                                                                                                                                                       | Mult. <sup>‡</sup> | $\delta^\ddagger$ | $\alpha^\#$  | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 3886.28                             | (5/2)                                 | 2313.83                                                                      | <16                                                          | 1572.369                                                                           | 1/2 <sup>+</sup>                                                                                                                                                |                    |                   |              | $I_\gamma$ : from (d, $\alpha\gamma$ ). Other: 100 14 from ( <sup>14</sup> N,3p $\gamma$ ) and 100 from <sup>34</sup> S(d,p $\gamma$ ).                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 4023.24                             | (11/2) <sup>-</sup>                   | 3886<br>207.0 5                                                              | 33 7<br>0.59 30                                              | 0.0<br>3816.01                                                                     | 3/2 <sup>+</sup><br>(9/2) <sup>-</sup>                                                                                                                          | [M1+E2]            | <0.25             | 0.00124 24   | $I_\gamma$ : from (d, $\alpha\gamma$ ). Other: <16 from <sup>34</sup> S(d,p $\gamma$ ).<br>B(M1)(W.u.)=0.05 +4-3 if M1.<br>$\delta$ : calculated by evaluators from the measured half-life of the level and RUL=100 for B(E2)(W.u.).<br>B(E2)(W.u.)=7.5 +8-7<br>$E_\gamma$ : weighted average of 2031.8 3 from ( <sup>14</sup> N,3p $\gamma$ ) and 2032.4 4 from ( <sup>18</sup> O,2 $\alpha\gamma$ ). Other: 2032 2 from <sup>208</sup> Pb( <sup>36</sup> S, <sup>35</sup> S $\gamma$ ).<br>Mult.: Q from $\gamma\gamma(\theta)$ (ADO) in ( <sup>14</sup> N,3p $\gamma$ ) and M2 ruled out by RUL. |
|                                     |                                       | 2032.0 3                                                                     | 100 6                                                        | 1991.27                                                                            | 7/2 <sup>-</sup>                                                                                                                                                | E2                 |                   | 0.000341 5   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 4027.2                              | (3/2 <sup>+</sup> ,5/2 <sup>+</sup> ) | 469.1<br>606.0<br>1088.6                                                     | <14<br><28<br>95 12                                          | 3558.080<br>3421.2<br>2938.63                                                      | (5/2 <sup>+</sup> )<br>5/2 <sup>+</sup><br>3/2 <sup>+</sup>                                                                                                     |                    |                   |              | $I_\gamma$ : weighted average of 86 22 from <sup>34</sup> S(d,p $\gamma$ ) and 97 12 from (d, $\alpha\gamma$ ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                     |                                       | 1310.1<br>1679.4<br>2035.9<br>2454.7<br>4027.0                               | <22<br>97 12<br><28<br>100 18<br><36                         | 2717.06<br>2347.779<br>1991.27<br>1572.369<br>0.0                                  | 5/2 <sup>+</sup><br>3/2 <sup>-</sup><br>7/2 <sup>-</sup><br>1/2 <sup>+</sup><br>3/2 <sup>+</sup>                                                                |                    |                   |              | $I_\gamma$ : from (d, $\alpha\gamma$ ). Other: 92 33 from <sup>34</sup> S(d,p $\gamma$ ).<br>$I_\gamma$ : from (d, $\alpha\gamma$ ). Other: 100 33 from <sup>34</sup> S(d,p $\gamma$ ).                                                                                                                                                                                                                                                                                                                                                                                                             |
| 4105.6                              | (5/2 <sup>+</sup> )                   | 547.5<br>684.4<br>1167.0<br>1388.5<br>1757.8<br>2114.3<br>2533.1<br>4105.3 8 | <4.6<br><4.6<br>15 7<br><12<br><5.8<br><8.1<br><5.8<br>100 7 | 3558.080<br>3421.2<br>2938.63<br>2717.06<br>2347.779<br>1991.27<br>1572.369<br>0.0 | (5/2 <sup>+</sup> )<br>5/2 <sup>+</sup><br>3/2 <sup>+</sup><br>5/2 <sup>+</sup><br>3/2 <sup>-</sup><br>7/2 <sup>-</sup><br>1/2 <sup>+</sup><br>3/2 <sup>+</sup> |                    |                   |              | $E_\gamma$ : from (n, $\gamma$ ) E=thermal.<br>$I_\gamma$ : from (d, $\alpha\gamma$ ).<br>$I_\gamma$ : from (d, $\alpha\gamma$ ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 4180                                | (1/2,3/2,5/2 <sup>+</sup> )           | 2608<br>4180                                                                 | 22 6<br>100 6                                                | 1572.369<br>0.0                                                                    | 1/2 <sup>+</sup><br>3/2 <sup>+</sup>                                                                                                                            |                    |                   |              | $E_\gamma, I_\gamma$ : from (n, $\gamma$ ) E=thermal. Other: $I_\gamma$ <7.7 from <sup>34</sup> S(d,p $\gamma$ ).<br>$E_\gamma, I_\gamma$ : from (n, $\gamma$ ) E=thermal. Other: $I_\gamma$ <9.6 from <sup>34</sup> S(d,p $\gamma$ ).<br>$E_\gamma$ : from (n, $\gamma$ ) E=thermal.<br>$I_\gamma$ : weighted average of 87 16 from (n, $\gamma$ ) E=thermal and 92 10 from <sup>34</sup> S(d,p $\gamma$ ).                                                                                                                                                                                        |
| 4189.240                            | 1/2 <sup>-</sup>                      | 631.32 @ 24                                                                  | <2.8                                                         | 3558.080                                                                           | (5/2 <sup>+</sup> )                                                                                                                                             |                    |                   |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                     |                                       | 1250.61 5                                                                    | 8.4 10                                                       | 2938.63                                                                            | 3/2 <sup>+</sup>                                                                                                                                                | [E1]               |                   | 0.0001047 15 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                     |                                       | 1840.2 12                                                                    | 91 10                                                        | 2347.779                                                                           | 3/2 <sup>-</sup>                                                                                                                                                | [M1,E2]            |                   | 0.000221 29  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |



Adopted Levels, Gammas (continued)

| $\gamma(^{35}\text{S})$ (continued) |                                        |                    |                    |          |                     |                    |                         |                                                                                                                                                                                                                                                                                                                                              |
|-------------------------------------|----------------------------------------|--------------------|--------------------|----------|---------------------|--------------------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $E_i(\text{level})$                 | $J_i^\pi$                              | $E_\gamma^\dagger$ | $I_\gamma^\dagger$ | $E_f$    | $J_f^\pi$           | Mult. <sup>‡</sup> | $\alpha^\#$             | Comments                                                                                                                                                                                                                                                                                                                                     |
| 4189.240                            | 1/2 <sup>-</sup>                       | 2616.8 13          | 9.5 28             | 1572.369 | 1/2 <sup>+</sup>    | [E1]               | 1.04×10 <sup>-3</sup> 2 | $E_\gamma, I_\gamma$ : from (n, $\gamma$ ) E=thermal. Other: $I_\gamma < 12$ from $^{34}\text{S}(\text{d},\text{p}\gamma)$ .                                                                                                                                                                                                                 |
|                                     |                                        | 4188.95 5          | 100 10             | 0.0      | 3/2 <sup>+</sup>    | [E1]               | 1.75×10 <sup>-3</sup> 3 | $E_\gamma$ : from (n, $\gamma$ ) E=thermal.<br>$I_\gamma$ : other: 100 11 from (n, $\gamma$ ) E=thermal.                                                                                                                                                                                                                                     |
| 4302.6                              | (1/2 <sup>-</sup> , 3/2 <sup>-</sup> ) | 1954.8             | 100 9              | 2347.779 | 3/2 <sup>-</sup>    |                    |                         | $I_\gamma$ : from (d, $\alpha\gamma$ ).                                                                                                                                                                                                                                                                                                      |
|                                     |                                        | 4302.3             | 70 9               | 0.0      | 3/2 <sup>+</sup>    |                    |                         | $I_\gamma$ : from (d, $\alpha\gamma$ ).                                                                                                                                                                                                                                                                                                      |
| 4477.63                             | (5/2 <sup>+</sup> )                    | 591.35             | <8.2               | 3886.28  | (5/2)               |                    | 1.5×10 <sup>-4</sup> 6  |                                                                                                                                                                                                                                                                                                                                              |
|                                     |                                        | 675.67             | <8.2               | 3801.952 | 3/2 <sup>-</sup>    | [E1]               | 4.82×10 <sup>-5</sup> 7 |                                                                                                                                                                                                                                                                                                                                              |
|                                     |                                        | 883.00             | <8.2               | 3594.62  | (7/2 <sup>+</sup> ) | [M1,E2]            | 5.4×10 <sup>-5</sup> 12 |                                                                                                                                                                                                                                                                                                                                              |
|                                     |                                        | 919.54             | <8.2               | 3558.080 | (5/2 <sup>+</sup> ) | [M1,E2]            | 4.9×10 <sup>-5</sup> 10 |                                                                                                                                                                                                                                                                                                                                              |
|                                     |                                        | 1056.4             | <16                | 3421.2   | 5/2 <sup>+</sup>    | [M1,E2]            | 3.6×10 <sup>-5</sup> 6  |                                                                                                                                                                                                                                                                                                                                              |
|                                     |                                        | 1538.96            | <12                | 2938.63  | 3/2 <sup>+</sup>    | [M1,E2]            | 0.000106 16             |                                                                                                                                                                                                                                                                                                                                              |
|                                     |                                        | 1760.55 11         | 100 12             | 2717.06  | 5/2 <sup>+</sup>    | [M1,E2]            | 0.000189 25             | $E_\gamma$ : from (n, $\gamma$ ) E=thermal.<br>$I_\gamma$ : others: 100 15 from (n, $\gamma$ ) E=thermal and 100 11 from (d, $\alpha\gamma$ ).                                                                                                                                                                                               |
|                                     |                                        | 2129.78            | <13                | 2347.779 | 3/2 <sup>-</sup>    | [E1]               | 0.000738 10             |                                                                                                                                                                                                                                                                                                                                              |
|                                     |                                        | 2486.27            | <16                | 1991.27  | 7/2 <sup>-</sup>    | [E1]               | 0.000962 13             | $I_\gamma$ : other: 42 11 from (d, $\alpha\gamma$ ).                                                                                                                                                                                                                                                                                         |
|                                     |                                        | 2905.1 4           | 90 14              | 1572.369 | 1/2 <sup>+</sup>    | [E2]               | 0.000749 10             | $E_\gamma$ : from (n, $\gamma$ ) E=thermal.<br>$I_\gamma$ : unweighted average of 96 28 from (n, $\gamma$ ) E=thermal, 64 12 from $^{34}\text{S}(\text{d},\text{p}\gamma)$ , and 109 11 from (d, $\alpha\gamma$ ).                                                                                                                           |
| 4822.79                             | (9/2 <sup>+</sup> )                    | 4477               | <20                | 0.0      | 3/2 <sup>+</sup>    | [M1,E2]            | 0.00126 8               | $I_\gamma$ : other: 28 6 from (d, $\alpha\gamma$ ).                                                                                                                                                                                                                                                                                          |
|                                     |                                        | 1006.8 2           | 66 15              | 3816.01  | (9/2) <sup>-</sup>  | D                  |                         | $E_\gamma$ : from ( $^{14}\text{N}, 3\text{p}\gamma$ ). Other: 1007.2 11 from ( $^{18}\text{O}, 2\alpha\text{n}\gamma$ ).<br>$I_\gamma$ : weighted average of 73 9 from ( $^{14}\text{N}, 3\text{p}\gamma$ ) and 36 18 from ( $^{18}\text{O}, 2\alpha\text{n}\gamma$ ).                                                                      |
|                                     |                                        | 1228.1 3           | 100 9              | 3594.62  | (7/2 <sup>+</sup> ) | D                  |                         | Mult.: from $\gamma\gamma(\theta)(\text{ADO})$ in ( $^{14}\text{N}, 3\text{p}\gamma$ ) and ( $^{18}\text{O}, 2\alpha\text{n}\gamma$ ), consistent with $\Delta J=0$ .<br>$E_\gamma, I_\gamma$ : other: 1228.3 6 with $I_\gamma=100$ 18 from ( $^{18}\text{O}, 2\alpha\text{n}\gamma$ ).<br>Mult.: from ( $^{14}\text{N}, 3\text{p}\gamma$ ). |
| 4899.82                             | (9/2 <sup>+</sup> )                    | 2831.4 9           | 36 9               | 1991.27  | 7/2 <sup>-</sup>    |                    |                         |                                                                                                                                                                                                                                                                                                                                              |
|                                     |                                        | 1305.3 4           | 55 18              | 3594.62  | (7/2 <sup>+</sup> ) |                    |                         | $E_\gamma$ : from ( $^{18}\text{O}, 2\alpha\text{n}\gamma$ ). Other: 1304.8 12 from ( $^{14}\text{N}, 3\text{p}\gamma$ ).                                                                                                                                                                                                                    |
|                                     |                                        | 2908.3 5           | 100 18             | 1991.27  | 7/2 <sup>-</sup>    | D                  |                         | Mult.: from $\gamma\gamma(\theta)(\text{ADO})$ in ( $^{14}\text{N}, 3\text{p}\gamma$ ).                                                                                                                                                                                                                                                      |
| 4903.371                            | 1/2 <sup>+</sup>                       | 1101.92 31         | 0.43 11            | 3801.952 | 3/2 <sup>-</sup>    |                    |                         |                                                                                                                                                                                                                                                                                                                                              |
|                                     |                                        | 1964.8 2           | 4.9 13             | 2938.63  | 3/2 <sup>+</sup>    |                    |                         |                                                                                                                                                                                                                                                                                                                                              |
|                                     |                                        | 2555.492 14        | 41 4               | 2347.779 | 3/2 <sup>-</sup>    |                    |                         |                                                                                                                                                                                                                                                                                                                                              |
|                                     |                                        | 3330.84 13         | 100 9              | 1572.369 | 1/2 <sup>+</sup>    |                    |                         |                                                                                                                                                                                                                                                                                                                                              |
|                                     |                                        | 4902.99 5          | 53 5               | 0.0      | 3/2 <sup>+</sup>    |                    |                         |                                                                                                                                                                                                                                                                                                                                              |
| 4963.082                            | 5/2 <sup>+</sup>                       | 1161.05 20         | 0.98 14            | 3801.952 | 3/2 <sup>-</sup>    |                    |                         |                                                                                                                                                                                                                                                                                                                                              |
|                                     |                                        | 1404.967 24        | 10.7 11            | 3558.080 | (5/2 <sup>+</sup> ) |                    |                         |                                                                                                                                                                                                                                                                                                                                              |
|                                     |                                        | 2615.2 2           | 19.5 18            | 2347.779 | 3/2 <sup>-</sup>    |                    |                         |                                                                                                                                                                                                                                                                                                                                              |
|                                     |                                        | 2972.0 4           | 3.2 11             | 1991.27  | 7/2 <sup>-</sup>    |                    |                         |                                                                                                                                                                                                                                                                                                                                              |

Adopted Levels, Gammas (continued)

| $\gamma(^{35}\text{S})$ (continued) |                                        |                         |                    |                    |                                      |                    |                         |                                                                                                                                                                                                                                                                                    |
|-------------------------------------|----------------------------------------|-------------------------|--------------------|--------------------|--------------------------------------|--------------------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $E_i(\text{level})$                 | $J_i^\pi$                              | $E_\gamma^\dagger$      | $I_\gamma^\dagger$ | $E_f$              | $J_f^\pi$                            | Mult. <sup>‡</sup> | $\alpha^\#$             | Comments                                                                                                                                                                                                                                                                           |
| 4963.082                            | 5/2 <sup>+</sup>                       | 3390.62 7<br>4963.06 22 | 100 9<br>47 5      | 1572.369<br>0.0    | 1/2 <sup>+</sup><br>3/2 <sup>+</sup> |                    |                         |                                                                                                                                                                                                                                                                                    |
| 5010.05                             | (11/2) <sup>-</sup>                    | 986.8 3                 | 100 5              | 4023.24            | (11/2) <sup>-</sup>                  | (M1)               | 3.45×10 <sup>-5</sup> 5 | B(M1)(W.u.)=0.036 +8-6<br>E <sub>γ</sub> : weighted average of 986.9 3 from ( <sup>14</sup> N,3pγ) and 986.6 3 from ( <sup>18</sup> O,2αnγ).<br>Mult.: D from ( <sup>18</sup> O,2αnγ) and ( <sup>14</sup> N,3pγ); Δπ=no from level scheme.                                         |
|                                     |                                        | 3018.5 7                | 43 5               | 1991.27            | 7/2 <sup>-</sup>                     | E2                 | 0.000798 11             | B(E2)(W.u.)=0.22 +5-4<br>Mult.: Q from γγ(θ)(ADO) and M2 ruled out by RUL in ( <sup>14</sup> N,3pγ).                                                                                                                                                                               |
| 5343                                | (-)                                    | 5343                    |                    | 0.0                | 3/2 <sup>+</sup>                     |                    |                         |                                                                                                                                                                                                                                                                                    |
| 5412.28                             | (9/2 <sup>+</sup> )                    | 1389.0 4                | 100 10             | 4023.24            | (11/2) <sup>-</sup>                  | D                  |                         | E <sub>γ</sub> : other: E <sub>γ</sub> =1389.0 8 from ( <sup>18</sup> O,2αnγ).<br>Mult.: from γγ(θ)(ADO) in ( <sup>14</sup> N,3pγ).                                                                                                                                                |
|                                     |                                        | 1525.7 8<br>3420.6 9    | 15.4 26<br>28.2 26 | 3886.28<br>1991.27 | (5/2)<br>7/2 <sup>-</sup>            |                    |                         |                                                                                                                                                                                                                                                                                    |
| 5752.5                              | 5/2 <sup>-</sup> , 7/2 <sup>-</sup>    | 5752.0 8                | 100                | 0.0                | 3/2 <sup>+</sup>                     |                    |                         |                                                                                                                                                                                                                                                                                    |
| 5767                                | 5/2 <sup>+</sup>                       | 5767                    |                    | 0.0                | 3/2 <sup>+</sup>                     |                    |                         |                                                                                                                                                                                                                                                                                    |
| 5841.482                            | (3/2 <sup>+</sup> , 5/2 <sup>+</sup> ) | 4268.65 14              | 100                | 1572.369           | 1/2 <sup>+</sup>                     |                    |                         | E <sub>γ</sub> , I <sub>γ</sub> : from (n,γ) E=thermal.                                                                                                                                                                                                                            |
| 5877.70                             | (11/2 <sup>+</sup> )                   | 465.4 2                 | 38 6               | 5412.28            | (9/2 <sup>+</sup> )                  | D                  |                         | E <sub>γ</sub> : weighted average of 465.3 3 from ( <sup>14</sup> N,3pγ) and 465.5 2 from ( <sup>18</sup> O,2αnγ).<br>I <sub>γ</sub> : unweighted average of 31.6 26 from ( <sup>14</sup> N,3pγ) and 43.8 32 from ( <sup>18</sup> O,2αnγ).<br>Mult.: from ( <sup>18</sup> O,2αnγ). |
|                                     |                                        | 867.6 4                 | 50.0 26            | 5010.05            | (11/2) <sup>-</sup>                  | D                  |                         | E <sub>γ</sub> : weighted average of 867.3 3 from ( <sup>14</sup> N,3pγ) and 868.2 4 from ( <sup>18</sup> O,2αnγ).<br>I <sub>γ</sub> : other: 62 7 from ( <sup>18</sup> O,2αnγ).<br>Mult.: from ( <sup>18</sup> O,2αnγ).                                                           |
|                                     |                                        | 977.9 3                 | 43 5               | 4899.82            | (9/2 <sup>+</sup> )                  | D+Q                |                         | E <sub>γ</sub> : weighted average of 977.8 3 from ( <sup>14</sup> N,3pγ) and 978.4 10 from ( <sup>18</sup> O,2αnγ).<br>I <sub>γ</sub> : weighted average of 42.1 26 from ( <sup>14</sup> N,3pγ) and 82 22 from ( <sup>18</sup> O,2αnγ).<br>Mult.: from ( <sup>14</sup> N,3pγ).     |
|                                     |                                        | 1055.1 6                | 80 7               | 4822.79            | (9/2 <sup>+</sup> )                  | D                  |                         | E <sub>γ</sub> : unweighted average of 1054.5 3 from ( <sup>14</sup> N,3pγ) and 1055.6 4 from ( <sup>18</sup> O,2αnγ).<br>I <sub>γ</sub> : weighted average of 79 5 from ( <sup>14</sup> N,3pγ) and 122 32 from ( <sup>18</sup> O,2αnγ).<br>Mult.: from ( <sup>18</sup> O,2αnγ).   |
|                                     |                                        | 1854.4 4                | 44.7 26            | 4023.24            | (11/2) <sup>-</sup>                  |                    |                         | E <sub>γ</sub> , I <sub>γ</sub> : other: 1854.5 42 with I <sub>γ</sub> =88 44 from ( <sup>18</sup> O,2αnγ).                                                                                                                                                                        |

## Adopted Levels, Gammas (continued)

| $\gamma(^{35}\text{S})$ (continued)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                     |                    |                    |          |                                        |                    |                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|--------------------|--------------------|----------|----------------------------------------|--------------------|-------------------------|
| $E_i(\text{level})$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $J_i^\pi$                           | $E_\gamma^\dagger$ | $I_\gamma^\dagger$ | $E_f$    | $J_f^\pi$                              | Mult. <sup>‡</sup> | $\alpha^\#$             |
| 5877.70                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | (11/2 <sup>+</sup> )                | 2061.6 4           | 100 5              | 3816.01  | (9/2) <sup>-</sup>                     | D                  |                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                     | 2283.0 4           | 56 12              | 3594.62  | (7/2 <sup>+</sup> )                    |                    |                         |
| 6018.8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                     | 3886.0 13          | 7.9 26             | 1991.27  | 7/2 <sup>-</sup>                       |                    |                         |
| 6078.475                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1/2 <sup>-</sup> , 3/2 <sup>-</sup> | 6018.2 6           | 100                | 0.0      | 3/2 <sup>+</sup>                       |                    |                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                     | 3139.9 5           | 22 7               | 2938.63  | 3/2 <sup>+</sup>                       |                    |                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                     | 6077.87 11         | 100 12             | 0.0      | 3/2 <sup>+</sup>                       |                    |                         |
| 6293.93                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                     | 6293.2 4           | 100                | 0.0      | 3/2 <sup>+</sup>                       |                    |                         |
| 6299.4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | (11/2)                              | 887.0 7            | 40 10              | 5412.28  | (9/2 <sup>+</sup> )                    | D                  |                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                     | 2275.9 6           | 100 20             | 4023.24  | (11/2) <sup>-</sup>                    |                    |                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                     | 2483.6 8           | 90 20              | 3816.01  | (9/2) <sup>-</sup>                     |                    |                         |
| 6352.2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | (13/2) <sup>-</sup>                 | 2329.5 9           | 100 20             | 4023.24  | (11/2) <sup>-</sup>                    | (M1)               | 0.000392 5              |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                     | 2535.8 11          | 50 6               | 3816.01  | (9/2) <sup>-</sup>                     | E2                 | 0.000581 8              |
| 6354.89                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                     | 6355.0 6           | 100                | 0.0      | 3/2 <sup>+</sup>                       |                    |                         |
| 6419.9                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                     | 6419.3 11          | 100                | 0.0      | 3/2 <sup>+</sup>                       |                    |                         |
| 6629.43                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                     | 6628.5 6           | 100                | 0.0      | 3/2 <sup>+</sup>                       |                    |                         |
| 6761.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                     | 6760.3 12          | 100                | 0.0      | 3/2 <sup>+</sup>                       |                    |                         |
| (6986.097)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 1/2 <sup>+</sup>                    | 356.66 @ 9         | <0.073             | 6629.43  |                                        |                    |                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                     | 631.32 @ 24        | <0.123             | 6354.89  |                                        |                    |                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                     | 692.16 5           | 0.238 31           | 6293.93  |                                        |                    |                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                     | 907.608 23         | 1.00 18            | 6078.475 | 1/2 <sup>-</sup> , 3/2 <sup>-</sup>    |                    |                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                     | 1144.591 20        | 0.98 11            | 5841.482 | (3/2 <sup>+</sup> , 5/2 <sup>+</sup> ) |                    |                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                     | 2022.954 9         | 19.6 18            | 4963.082 | 5/2 <sup>+</sup>                       |                    |                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                     | 2082.83 17         | 28.4 27            | 4903.371 | 1/2 <sup>+</sup>                       |                    |                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                     | 2508.39 8          | 0.68 7             | 4477.63  | (5/2 <sup>+</sup> )                    |                    |                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                     | 2796.74 4          | 8.8 7              | 4189.240 | 1/2 <sup>-</sup>                       |                    |                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                     | 3183.95 4          | 10.9 11            | 3801.952 | 3/2 <sup>-</sup>                       |                    |                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                     | 4638.14 14         | 100 9              | 2347.779 | 3/2 <sup>-</sup>                       |                    |                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                     | 6985.7 10          | 0.064 14           | 0.0      | 3/2 <sup>+</sup>                       |                    |                         |
| 7179.74                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | (15/2 <sup>+</sup> )                | 827.8 6            | 2.2 8              | 6352.2   | (13/2) <sup>-</sup>                    | [E1]               | 3.08×10 <sup>-5</sup> 4 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                     | 880.2 4            | 5.9 8              | 6299.4   | (11/2)                                 |                    |                         |
| Comments:<br>E <sub>γ</sub> : weighted average of 2061.5 3 from ( <sup>14</sup> N,3pγ) and 2063.3 14 from ( <sup>18</sup> O,2αnγ).<br>I <sub>γ</sub> : other: 100 20 from ( <sup>18</sup> O,2αnγ).<br>Mult.: from ( <sup>18</sup> O,2αnγ).<br>E <sub>γ</sub> : weighted average of 2282.9 4 from ( <sup>14</sup> N,3pγ) and 2283.5 10 from ( <sup>18</sup> O,2αnγ).<br>I <sub>γ</sub> : unweighted average of 68.4 26 from ( <sup>14</sup> N,3pγ) and 44 10 from ( <sup>18</sup> O,2αnγ).<br>E <sub>γ</sub> , I <sub>γ</sub> : from (n,γ) E=thermal.<br>E <sub>γ</sub> , I <sub>γ</sub> : from (n,γ) E=thermal.<br>B(M1)(W.u.)=0.023 +6-5<br>Mult.: D from γγ(θ)(ADO) in ( <sup>14</sup> N,3pγ); Δπ=no from level scheme.<br>B(E2)(W.u.)=5.3 +17-11<br>Mult.: Q from γγ(θ)(ADO) in ( <sup>14</sup> N,3pγ) and M2 ruled out by RUL.<br>E <sub>γ</sub> , I <sub>γ</sub> : from (n,γ) E=thermal.<br>E <sub>γ</sub> , I <sub>γ</sub> : from (n,γ) E=thermal.<br>E <sub>γ</sub> , I <sub>γ</sub> : from (n,γ) E=thermal.<br>E <sub>γ</sub> , I <sub>γ</sub> : from (n,γ) E=thermal.<br>B(E1)(W.u.)=7×10 <sup>-6</sup> +6-3 |                                     |                    |                    |          |                                        |                    |                         |

**Adopted Levels, Gammas (continued)**

$\gamma(^{35}\text{S})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$            | $E_\gamma^\dagger$ | $I_\gamma^\dagger$ | $E_f$   | $J_f^\pi$            | Mult. <sup>‡</sup> | $\alpha^\#$             | Comments                                                                                                                                                                                                                      |
|---------------------|----------------------|--------------------|--------------------|---------|----------------------|--------------------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7179.74             | (15/2 <sup>+</sup> ) | 1302.0 3           | 100 6              | 5877.70 | (11/2 <sup>+</sup> ) | E2                 | $5.56 \times 10^{-5}$ 8 | B(E2)(W.u.)=6.6 +41-19<br>E <sub>γ</sub> : weighted average of 1302.2 2 from ( <sup>14</sup> N,3pγ) and 1301.6 3 from ( <sup>18</sup> O,2αnγ).<br>Mult.: Q from γγ(θ)(ADO) in ( <sup>14</sup> N,3pγ) and M2 ruled out by RUL. |
| 8024.0              | (17/2 <sup>+</sup> ) | 844.2 6            | 100                | 7179.74 | (15/2 <sup>+</sup> ) | (M1)               | $4.66 \times 10^{-5}$ 7 | B(M1)(W.u.)=0.24 +9-5<br>E <sub>γ</sub> : weighted average of 843.9 8 from ( <sup>14</sup> N,3pγ) and 844.3 6 from ( <sup>18</sup> O,2αnγ).<br>Mult.: D from γγ(ADO) in ( <sup>18</sup> O,2αnγ); Δπ=(no) from level scheme.   |
| 8755.9              | (17/2)               | 731.6 15           | 57 22              | 8024.0  | (17/2 <sup>+</sup> ) |                    |                         | E <sub>γ</sub> ,I <sub>γ</sub> : from ( <sup>18</sup> O,2αnγ).                                                                                                                                                                |
|                     |                      | 1576.3 14          | 100 22             | 7179.74 | (15/2 <sup>+</sup> ) | D                  |                         | E <sub>γ</sub> ,I <sub>γ</sub> : from ( <sup>18</sup> O,2αnγ).                                                                                                                                                                |
| 10048.6             | (19/2 <sup>+</sup> ) | 2868.7 8           | 100                | 7179.74 | (15/2 <sup>+</sup> ) | Q                  |                         | E <sub>γ</sub> ,I <sub>γ</sub> : from ( <sup>18</sup> O,2αnγ).                                                                                                                                                                |
| 12469.3             | (21/2)               | 2420.6 11          | 100                | 10048.6 | (19/2 <sup>+</sup> ) | (D)                |                         | E <sub>γ</sub> ,I <sub>γ</sub> : from ( <sup>18</sup> O,2αnγ).                                                                                                                                                                |

<sup>†</sup> For γ rays from low-spin levels (J<9/2), values are from <sup>34</sup>S(d,pγ) up to 4478 level and from (n,γ) E=thermal above that, for γ rays from high-spin level (J≥9/2), values are from <sup>24</sup>Mg(<sup>14</sup>N,3pγ), unless otherwise noted. E<sub>γ</sub> values without uncertainties are deduced from level-energy differences.

<sup>‡</sup> Mult and δ are primarily from γ(θ) and γγ(ADO) in <sup>34</sup>S(d,pγ), <sup>24</sup>Mg(<sup>14</sup>N,3pγ), and <sup>26</sup>Mg(<sup>18</sup>O,2αnγ). The magnetic or electric nature is determined based on RUL and measured T<sub>1/2</sub> where available, unless otherwise noted.

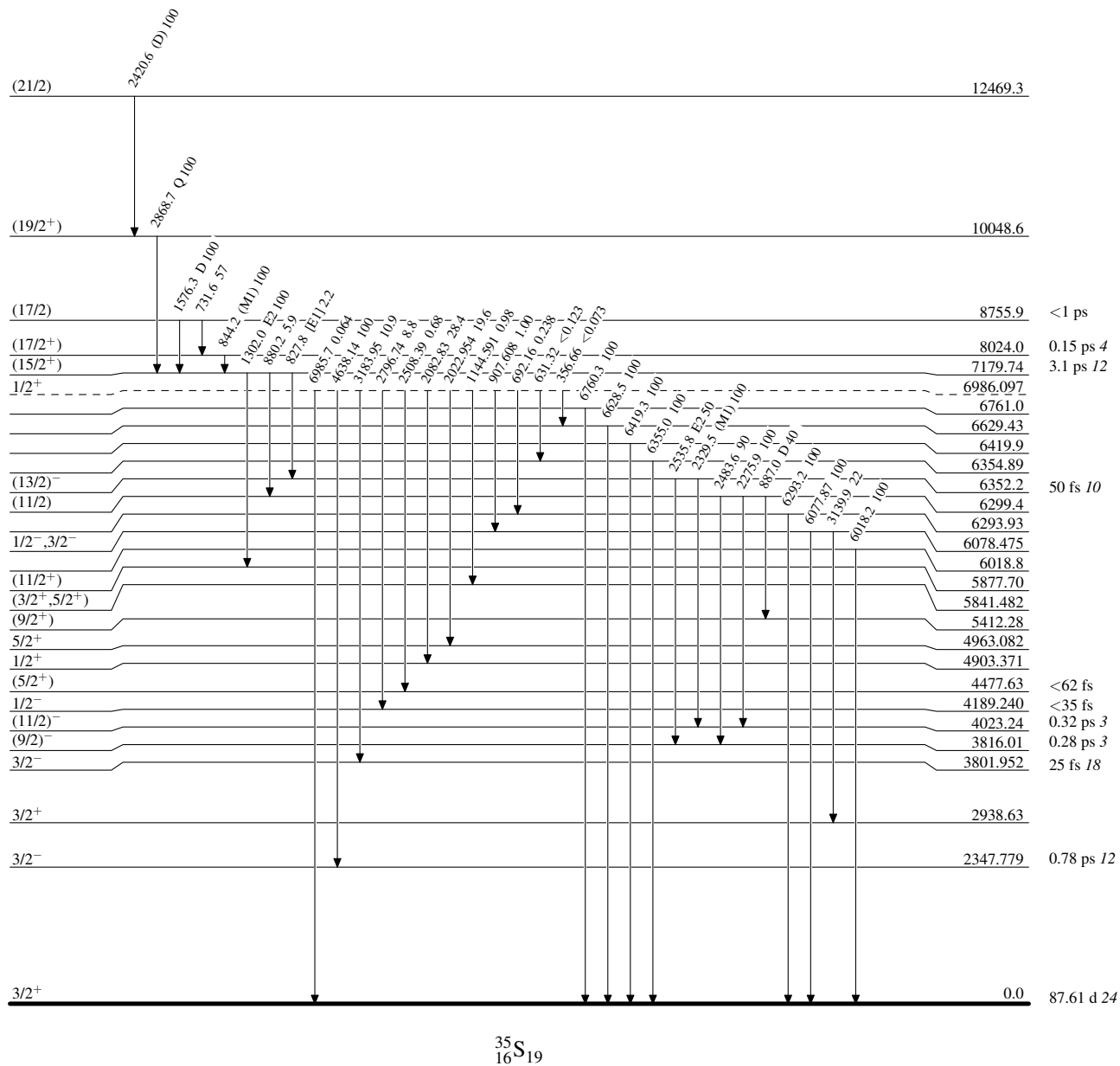
<sup>#</sup> Total theoretical internal conversion coefficients, calculated using the BrIcc code (2008Ki07) with “Frozen Orbitals” approximation based on γ-ray energies, assigned multipolarities, and mixing ratios, unless otherwise specified.

@ Multiply placed.

& Placement of transition in the level scheme is uncertain.

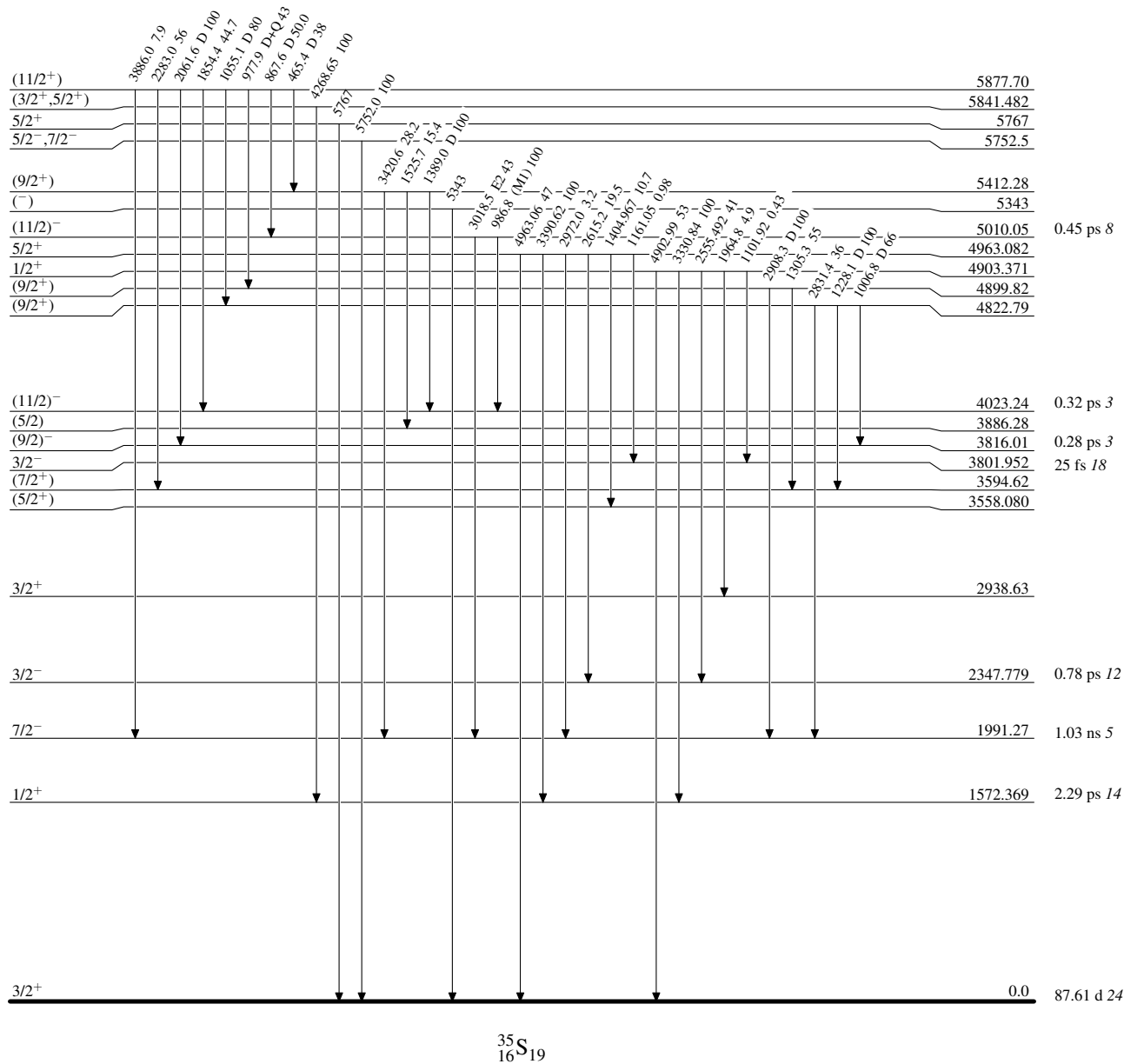
Adopted Levels, GammasLevel Scheme

Intensities: Relative photon branching from each level

 $^{35}_{16}\text{S}_{19}$

Adopted Levels, GammasLevel Scheme (continued)

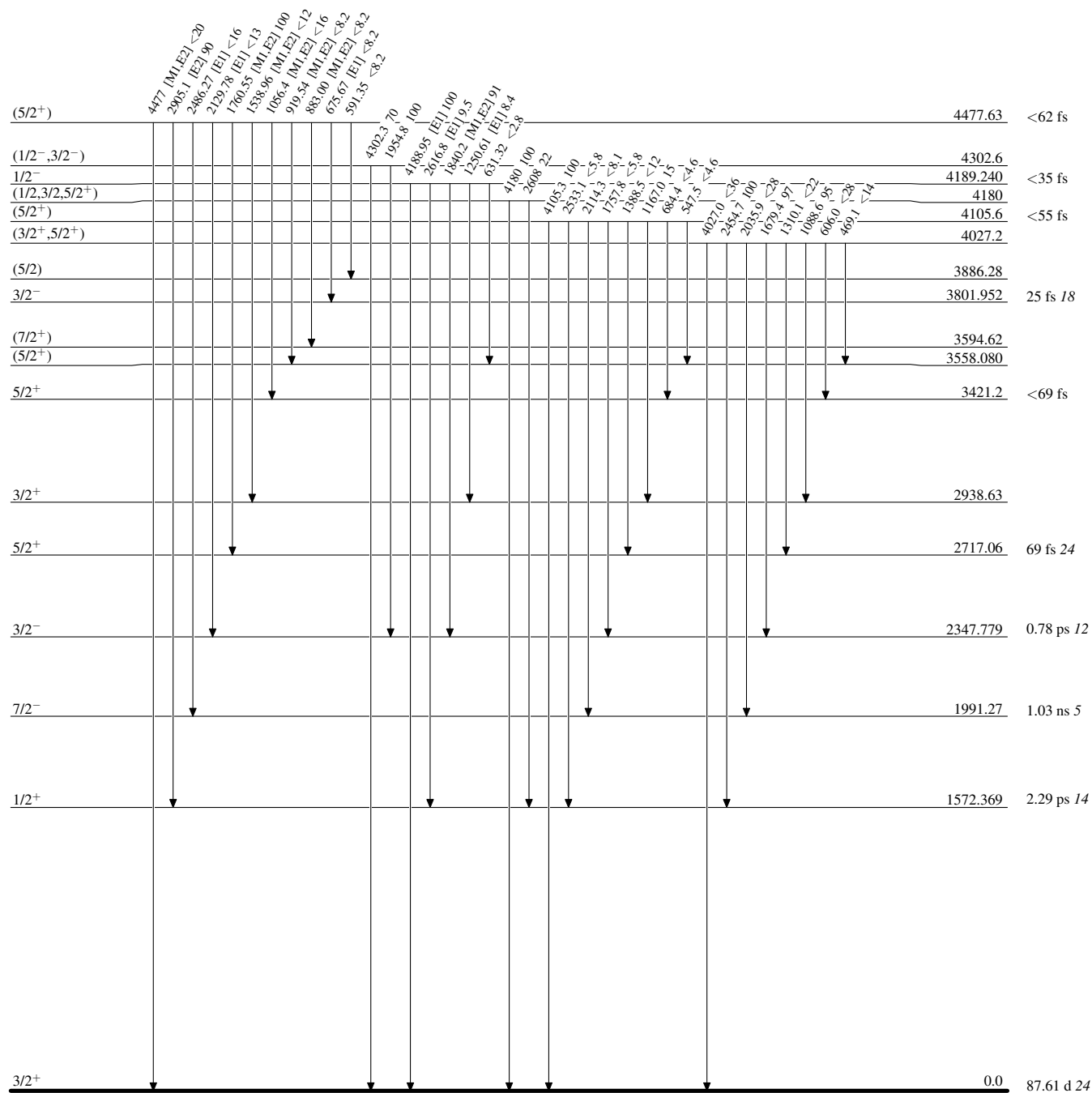
Intensities: Relative photon branching from each level



Adopted Levels, Gammas

## Level Scheme (continued)

Intensities: Relative photon branching from each level



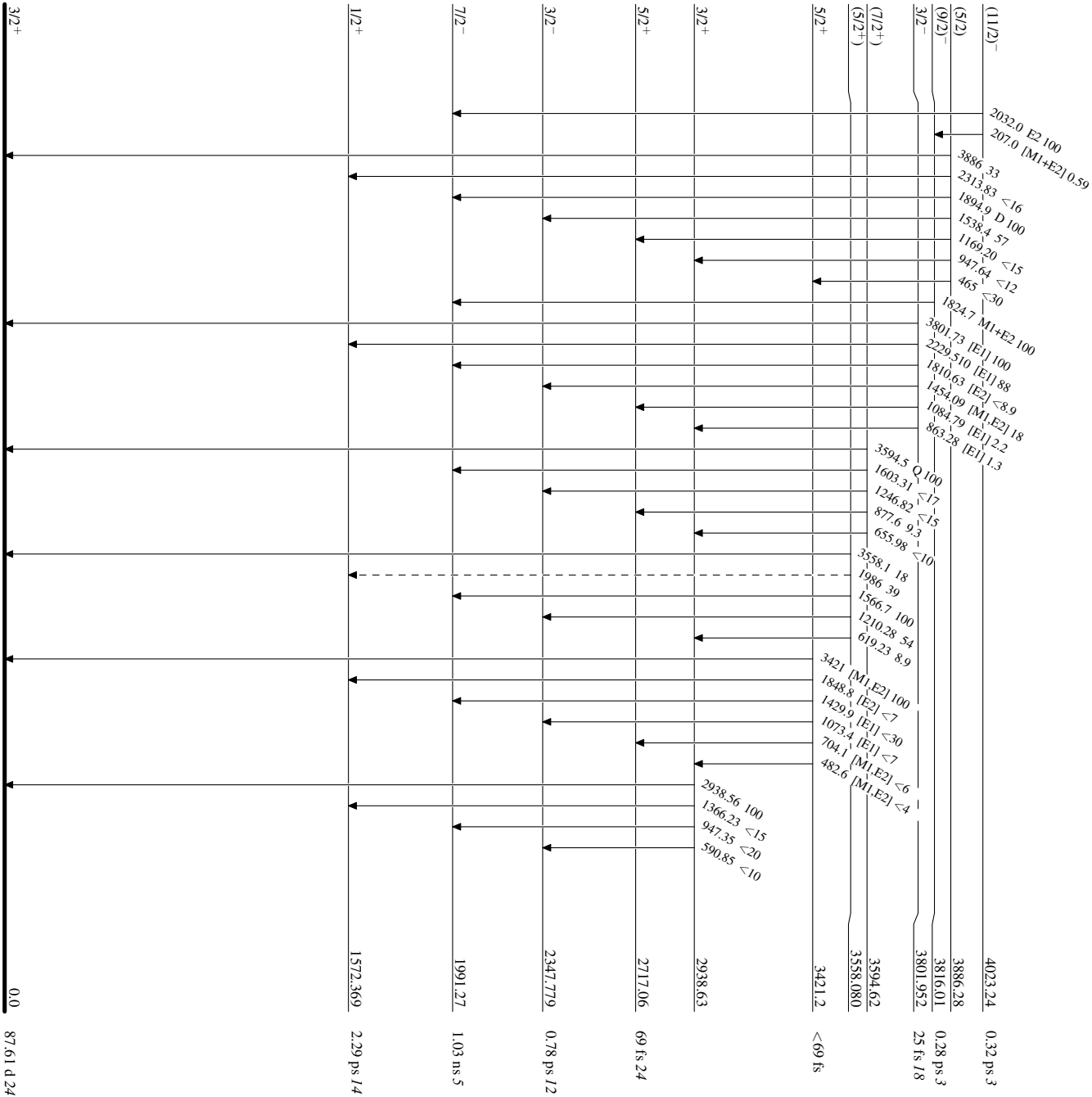
Adopted Levels, Gammas

Legend

Level Scheme (continued)

Intensities: Relative photon branching from each level

-----▶  $\gamma$  Decay (Uncertain)

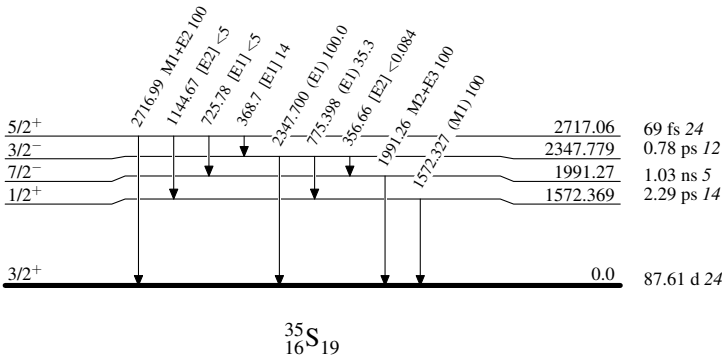




Adopted Levels, Gammas

Level Scheme (continued)

Intensities: Relative photon branching from each level



$^{35}\text{P}$   $\beta^-$  decay (47.3 s) 1986Wa22

Parent:  $^{35}\text{P}$ :  $E=0$ ;  $J^\pi=1/2^+$ ;  $T_{1/2}=47.3$  s 8;  $Q(\beta^-)=3988.4$  19;  $\% \beta^-$  decay=100

$^{35}\text{P}$ - $J^\pi, T_{1/2}$ : From Adopted Levels of  $^{35}\text{P}$ .

$^{35}\text{P}$ - $Q(\beta^-)$ : From 2021Wa16.

1986Wa22:  $^{35}\text{P}$  was produced via the  $^{36}\text{S}(t, \alpha)^{35}\text{P}$  reaction. 81% enriched  $^{36}\text{S}$  targets were bombarded with 3.4-MeV tritons from the Brookhaven National Laboratory (BNL) tandem van de Graaff facility.  $\gamma$  rays were detected using an intrinsic coaxial Ge detector. Measured  $E_\gamma$ ,  $I_\gamma$ . Deduced levels, decay branching ratios, and  $\log ft$ . Compared with shell model calculations.

1972Ap01:  $^{35}\text{P}$  was produced via the  $^{37}\text{Cl}(t, \alpha p)^{35}\text{P}$  reaction. LiCl and NaCl targets were bombarded with 16-MeV tritons from the Los Alamos tandem Van de Graaff facility.  $\gamma$  rays were detected using an 85 cm<sup>3</sup> Ge(Li) detector.  $\beta$  particles were detected using an end-window gas-flow proportional counter. Measured  $E_\gamma$ ,  $I_\gamma$ , and  $I_\beta$ . Deduced  $^{35}\text{P}$   $T_{1/2}$ , decay branching ratios, and  $\log ft$ . Compared with shell model calculations.

1972Go31:  $^{35}\text{P}$  was produced via the  $^{18}\text{O}(^{19}\text{F}, 2p)^{35}\text{P}$  and  $^{36}\text{S}(t, \alpha)^{35}\text{P}$  reactions at the second tandem of the Brookhaven National Laboratory (BNL) tandem van de Graaff facility.  $^{35}\text{P}$   $\beta$ -delayed  $\gamma$  rays were detected using a 60 cm<sup>3</sup> Ge(Li) detector.  $\beta$  particles were detected in coincidence with the 1572 $\gamma$  using a NE102 scintillator. Measured  $E_\gamma$ ,  $I_\gamma$ , and  $E_\beta$ . Deduced  $^{35}\text{P}$   $T_{1/2}$ ,  $\beta$  end-point energy, mass, decay branching ratios, and  $\log ft$ . Compared with shell model calculations.

1971Gr53:  $^{35}\text{P}$  was produced via the  $^{37}\text{Cl}(\gamma, 2p)^{35}\text{P}$  reaction. 6-g  $\text{NH}_4\text{Cl}(\text{nat})$  samples were irradiated with  $\gamma$  rays from a bremsstrahlung spectrum at the Mainz electron linear accelerator.  $\gamma$  rays were detected using NaI(Tl) and Ge(Li) detectors. Measured  $E_\gamma$ . Deduced  $T_{1/2}$  of  $^{35}\text{P}$  and levels in  $^{35}\text{S}$ .

There is a 1-MeV gap between the highest observed level at  $E=2938.4$  and  $Q(\beta^-)$  value=3988.4 19 (2021Wa16). There may be missing weak transitions from unobserved levels in the gap. Upper limits for the unobserved  $\beta$  feedings:

1991: <0.08 (1986Wa22), <2.5 (1972Go31), <0.09 (1972Ap01);

2348: <0.08 (1986Wa22), <5 (1972Go31), <0.24 (1972Ap01);

2717: <0.03 (1986Wa22), <7 (1972Go31), <0.38 (1972Ap01);

3421: <0.02 (1986Wa22);

3558: <0.11 (1986Wa22);

3597: <0.02 (1986Wa22);

3675: <0.02 (1986Wa22);

3802: <0.03 (1986Wa22).

 $^{35}\text{S}$  Levels

| $E(\text{level})^\dagger$ | $J^\pi^\ddagger$ | $T_{1/2}^\ddagger$ |
|---------------------------|------------------|--------------------|
| 0                         | $3/2^+$          | 87.61 d 24         |
| 1572.290 29               | $1/2^+$          | 2.29 ps 14         |
| 2938.4 4                  | $3/2^+$          |                    |

<sup>†</sup> From a least-squares fit to  $\gamma$ -ray energies.

<sup>‡</sup> From the Adopted Levels.

 $\beta^-$  radiations

| $E(\text{decay})$        | $E(\text{level})$ | $I\beta^-^\ddagger$ | $\log ft$ | Comments                                                                                                                              |
|--------------------------|-------------------|---------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------|
| (1050.0 22)              | 2938.4            | 0.47                | 5.0       | $I\beta^-$ : others: <8 (1972Go31), <0.45 (1972Ap01).                                                                                 |
| (2416.1 22)              | 1572.290          | 99.2                | 4.1       | $I\beta^-$ : other: 99 +1-7 from measured $\gamma/\beta$ ratio (1972Ap01), 100 (1972Go31).                                            |
| (3988.4 <sup>#</sup> 24) | 0                 | <0.73               | >7.2      | $I\beta^-$ : converted by evaluators from the shell-model calculated 0.73 in 1986Wa22. Others: <9 in 1972Ap01; not given in 1972Go31. |

<sup>†</sup> From  $I_\gamma$  intensity balance for excited levels.

<sup>‡</sup> Absolute intensity per 100 decays.

<sup>#</sup> Existence of this branch is questionable.

$^{35}\text{P} \beta^-$  decay (47.3 s) 1986Wa22 (continued) $\gamma(^{35}\text{S})$ 

$I_\gamma$  normalization: from  $\Sigma\%I(\gamma \text{ to g.s.})=99.635 \ 365$  deduced from  $100-\%I\beta^-(\text{g.s.})$ , where  $\%I\beta^-(\text{g.s.})=0.365 \ 365$  is converted by evaluators from the shell-model calculated  $\%I\beta^-(\text{g.s.})=0.73$  in 1986Wa22. Others:  $\%I\beta^-(\text{g.s.})<9$  from 1972Ap01;  $\%I\beta^-(\text{g.s.})$  is not given in 1972Go31. Given that the ground state  $\beta^-$ -feeding has not been experimentally observed, the evaluators opt to convert the shell-model calculated value to an upper limit when deducing the normalization factor.

| $E_\gamma^\dagger$ | $I_\gamma^{\dagger\#}$ | $E_i(\text{level})$ | $J_i^\pi$        | $E_f$ | $J_f^\pi$        | Mult. $^\ddagger$ | $\alpha^\text{@}$ | Comments                                                                                                                                                                  |
|--------------------|------------------------|---------------------|------------------|-------|------------------|-------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1572.252 29        | 100                    | 1572.290            | 1/2 <sup>+</sup> | 0     | 3/2 <sup>+</sup> | (M1)              | 0.0001007 14      | $\%I_\gamma=99.2$<br>$E_\gamma$ : weighted average of 1571.8 2 (1971Gr53), 1572.2 4 (1972Ap01), 1572.36 24 (1972Go31), 1572.47 25 (1972Go31), and 1572.256 24 (1986Wa22). |
| 2938.29 40         | 0.476 30               | 2938.4              | 3/2 <sup>+</sup> | 0     | 3/2 <sup>+</sup> |                   |                   | $\%I_\gamma=0.47$                                                                                                                                                         |

$^\dagger$  From 1986Wa22, unless otherwise noted.

$^\ddagger$  From the Adopted Gammas.

$^\#$  For absolute intensity per 100 decays, multiply by 0.992.

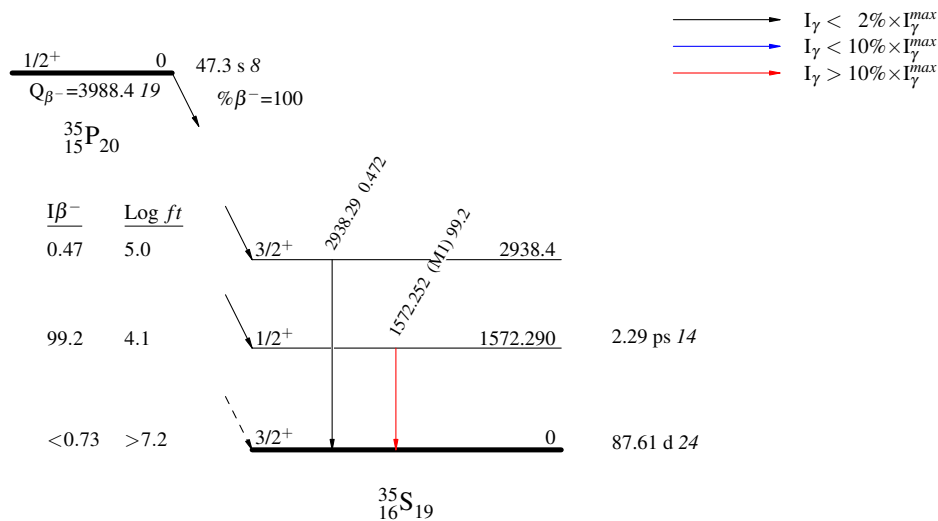
$^\text{@}$  Total theoretical internal conversion coefficients, calculated using the BrIcc code (2008Ki07) with “Frozen Orbitals” approximation based on  $\gamma$ -ray energies, assigned multiplicities, and mixing ratios, unless otherwise specified.

 $^{35}\text{P} \beta^-$  decay (47.3 s) 1986Wa22

## Decay Scheme

Intensities:  $I_\gamma$  per 100 parent decays

Legend



$^2\text{H}(^{34}\text{S},\text{p}\gamma)$  1973Wa10

$^{34}\text{S}(\text{d},\text{p})^{35}\text{S}$  in inverse kinematics.

1973Wa10: A 59.6-MeV  $^{34}\text{S}$  beam was produced from the BNL MP-tandem Van de Graaff facility. Targets were 200- $\mu\text{g}/\text{cm}^2$  TiD prepared by evaporating titanium onto the Cu, Al, and Mg target backings in a deuterium atmosphere.  $\gamma$  rays were detected using a 35-cm<sup>3</sup> Ge(Li) detector at 0° with FWHM=2 keV at 656 keV. Measured  $E_\gamma$ . Deduced  $T_{1/2}$  for two  $^{35}\text{S}$  levels using Doppler Shift Attenuation lineshape analysis.

 $^{35}\text{S}$  Levels

| <u><math>E(\text{level})^\dagger</math></u> | <u><math>J^\pi^\ddagger</math></u> | <u><math>T_{1/2}^\#</math></u> | <u>Comments</u>                              |
|---------------------------------------------|------------------------------------|--------------------------------|----------------------------------------------|
| 0                                           | $3/2^+$                            |                                |                                              |
| 1574                                        | $1/2^+$                            | 2.29 ps 35                     | $T_{1/2}$ : Lifetime=3.3 ps 5 from 1973Wa10. |
| 2350                                        | $3/2^-$                            | 0.90 ps 14                     | $T_{1/2}$ : Lifetime=1.3 ps 2 from 1973Wa10. |

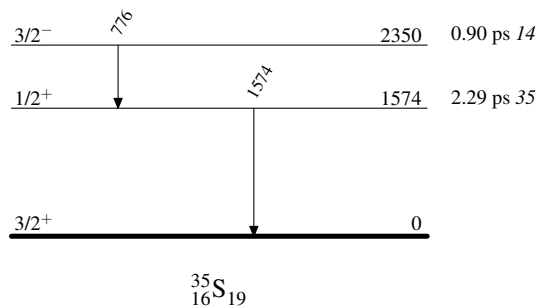
<sup>†</sup> From 1973Wa10.

<sup>‡</sup> From the Adopted Levels.

<sup>#</sup> From Doppler Shift Attenuation Method (DSAM).

 $\gamma(^{35}\text{S})$ 

| <u><math>E_\gamma</math></u> | <u><math>E_i(\text{level})</math></u> | <u><math>J_i^\pi</math></u> | <u><math>E_f</math></u> | <u><math>J_f^\pi</math></u> |
|------------------------------|---------------------------------------|-----------------------------|-------------------------|-----------------------------|
| 776                          | 2350                                  | $3/2^-$                     | 1574                    | $1/2^+$                     |
| 1574                         | 1574                                  | $1/2^+$                     | 0                       | $3/2^+$                     |

 $^2\text{H}(^{34}\text{S},\text{p}\gamma)$  1973Wa10Level Scheme

$^9\text{Be}(^{36}\text{S}, ^{35}\text{S}\gamma)$  2015MuZY, 2021JoZZ

$J^\pi=0^+$  for  $^{36}\text{S}$  ground state.

**2015MuZY**: a secondary beam of  $^{36}\text{S}$  was produced via the projectile fragmentation of a 140-MeV/nucleon  $^{48}\text{Ca}$  primary beam impinging on a 846-mg/cm<sup>2</sup>  $^9\text{Be}$  target at the coupled cyclotron facility at NSCL, MSU. The  $^{36}\text{S}$  nuclei were selected using the A1900 separator with an intensity of  $8.1 \times 10^5$  pps and a purity of 89.7%. The excited states of  $^{35}\text{S}$  were populated by the one-neutron knockout reaction from the  $^{36}\text{S}$  beam at 88 MeV/nucleon (midtarget energy) on a 100-mg/cm<sup>2</sup>  $^9\text{Be}$  secondary target located at the reaction target position of the S800 spectrometer. The projectile-like residues were identified from their energy loss measured by an ionization chamber located at the focal plane of the S800 spectrometer and from their ToF measured between two scintillators situated at the object position and at the focal plane of the S800 spectrometer. Prompt  $\gamma$  rays from the deexcitation of the  $^{35}\text{S}$  residues were detected by seven modules of the GRETINA Ge array. Measured  $E_\gamma$ ,  $I_\gamma$ , ( $^{35}\text{S}$ ) $\gamma$ -coin,  $\gamma\gamma$ -coin, the inclusive knockout cross section for producing  $^{35}\text{S}$  from  $^{36}\text{S}$ , the fractional populations and parallel momentum distributions of 11 populated states in  $^{35}\text{S}$  residues. Deduced levels,  $J$ ,  $\pi$ , L-transfers, partial knockout cross sections and spectroscopic factors.

Calculated single-particle cross sections ( $\sigma_{\text{sp}}$ ) for neutron removal and parallel momentum distributions using the eikonal model.

**2021JoZZ**: a more careful reanalysis of **2015MuZY** data.

 $^{35}\text{S}$  Levels

$b_f$  values under comments are population fraction in the knock-out reaction,  $\sigma_{\text{f}}^{\text{sp}}$  values are for calculated single-particle cross sections, and  $R_s$  values are quenching factors from a systematics study in **2014To14**. The partial cross sections to each level can be deduced from  $b_f \times \sigma_{\text{inc}}^{\text{exp}}$ .

The inclusive cross section  $\sigma_{\text{inc}}^{\text{exp}}$  for producing  $^{35}\text{S}$  from  $^{36}\text{S}$  is measured to be 63.7 mb 6 (**2015MuZY**) and 67 mb 4 (**2021JoZZ**).

| $E(\text{level})^\dagger$ | $L^\ddagger$ | $C^2S^\#$ | Comments                                                                                                                                                                                                                              |
|---------------------------|--------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0                         | 2            | 3.33 76   | $C^2S$ : 3.2 7 ( <b>2015MuZY</b> ).<br>$b_f=44.5\%$ 37 ( <b>2021JoZZ</b> ), $b_f=43.3\%$ 47, $\sigma_{\text{f}}^{\text{sp}}=13.1$ mb, $R_s=0.66$ 13 ( <b>2015MuZY</b> ).                                                              |
| 1572 1                    | 0            | 1.26 25   | $C^2S$ : 1.1 3 ( <b>2015MuZY</b> ).<br>$b_f=19.9\%$ 8 ( <b>2021JoZZ</b> ), $b_f=18.6\%$ 31, $\sigma_{\text{f}}^{\text{sp}}=16.1$ mb, $R_s=0.64$ 13 ( <b>2015MuZY</b> ).                                                               |
| 1990                      | 3            | 0.11 20   | $C^2S$ : <0.2 ( <b>2015MuZY</b> ).<br>$b_f=1.41\%$ 10 ( <b>2021JoZZ</b> ), $b_f<2.9\%$ , $\sigma_{\text{f}}^{\text{sp}}=12.7$ mb, $R_s=0.69$ 14 ( <b>2015MuZY</b> ).                                                                  |
| 2348 1                    | 1            | 0.34 1    | $C^2S$ : 0.4 1 ( <b>2015MuZY</b> ).<br>$b_f=4.33\%$ 8 ( <b>2021JoZZ</b> ), $b_f=5.3\%$ 13, $\sigma_{\text{f}}^{\text{sp}}=13.1$ mb, $R_s=0.70$ 14 ( <b>2015MuZY</b> ).                                                                |
| 2717 2                    | 2            | 0.56 12   | $C^2S$ : 0.6 2 ( <b>2015MuZY</b> ).<br>$b_f=6.14\%$ 7 ( <b>2021JoZZ</b> ), $b_f=6.4\%$ 16, $\sigma_{\text{f}}^{\text{sp}}=11.4$ mb, $R_s=0.62$ 12 ( <b>2015MuZY</b> ).                                                                |
| 2939 2                    | (2)          | 0.31 6    | L: <b>2015MuZY</b> reports $L=0$ .<br>$C^2S$ : 0.4 2 ( <b>2015MuZY</b> ).<br>$b_f=3.99\%$ 3 ( <b>2021JoZZ</b> ), $b_f=5.7\%$ 16, $\sigma_{\text{f}}^{\text{sp}}=13.5$ mb, $R_s=0.61$ 12 ( <b>2015MuZY</b> ).                          |
| 3421 2                    | 2            | 0.49 11   | $C^2S$ : 0.5 1 ( <b>2015MuZY</b> ).<br>$b_f=5.08\%$ 5 ( <b>2021JoZZ</b> ), $b_f=5.1\%$ 10, $\sigma_{\text{f}}^{\text{sp}}=11.0$ mb, $R_s=0.61$ 12 ( <b>2015MuZY</b> ).                                                                |
| 3595 5                    | (2)          | 0.21 6    | $C^2S$ : 0.3 1 ( <b>2015MuZY</b> ).<br>$b_f=2.13\%$ 5 ( <b>2021JoZZ</b> ), $b_f=3.0\%$ 7, $\sigma_{\text{f}}^{\text{sp}}=11.0$ mb, $R_s=0.60$ 12 ( <b>2015MuZY</b> ).                                                                 |
| 3818 5                    | (3)          | 0.25 7    | $C^2S$ : 0.3 1 ( <b>2015MuZY</b> ).<br>$b_f=0.49\%$ 4 ( <b>2021JoZZ</b> ), $b_f=4.0\%$ 9, $\sigma_{\text{f}}^{\text{sp}}=11.8$ mb, $R_s=0.72$ 14 ( <b>2015MuZY</b> ).                                                                 |
| 3890 4                    | (0)          | 0.29 9    | $C^2S$ : 0.15 4 ( <b>2015MuZY</b> ).<br>$b_f=3.09\%$ 6 ( <b>2021JoZZ</b> ), $b_f=2.3\%$ 5, $\sigma_{\text{f}}^{\text{sp}}=13.5$ mb, $R_s=0.72$ 14 ( <b>2015MuZY</b> ).<br>L: inconsistent with $J^\pi=(5/2^+)$ in the Adopted Levels. |
| 4025 4                    | (2)          | 0.34 9    | $C^2S$ : 0.6 1 ( <b>2015MuZY</b> ).<br>$b_f=3.98\%$ 5 ( <b>2021JoZZ</b> ), $b_f=6.4\%$ 10, $\sigma_{\text{f}}^{\text{sp}}=12.1$ mb, $R_s=0.60$ 12 ( <b>2015MuZY</b> ).                                                                |
| 4948 5                    |              |           |                                                                                                                                                                                                                                       |
| 5349 8                    |              |           |                                                                                                                                                                                                                                       |
| 5775 4                    |              |           |                                                                                                                                                                                                                                       |

<sup>†</sup> From **2021JoZZ** based on measured  $E_\gamma$ .

$^9\text{Be}(^{36}\text{S}, ^{35}\text{S}\gamma)$  2015MuZY,2021JoZZ (continued) $^{35}\text{S}$  Levels (continued)

$^{\ddagger}$  From 2021JoZZ. L-transfers are deduced by comparing the measured and eikonal-calculated parallel momentum distributions of residuals. Parentheses around L values are added by evaluators due to poor fits.

$^{\#}$  From 2021JoZZ. The spectroscopic factor  $C^2S = b_f \sigma_{\text{inc}}^{\text{exp}} / R_s \sigma_f^{\text{sp}}$ , where  $b_f$  is the population fraction,  $\sigma_{\text{inc}}^{\text{exp}}$  is the measured total inclusive cross section,  $R_s$  is the quenching factor from a systematics study 2014To14, and  $\sigma_f^{\text{sp}}$  is the eikonal model calculated single-particle cross section.

 $\gamma(^{35}\text{S})$ 

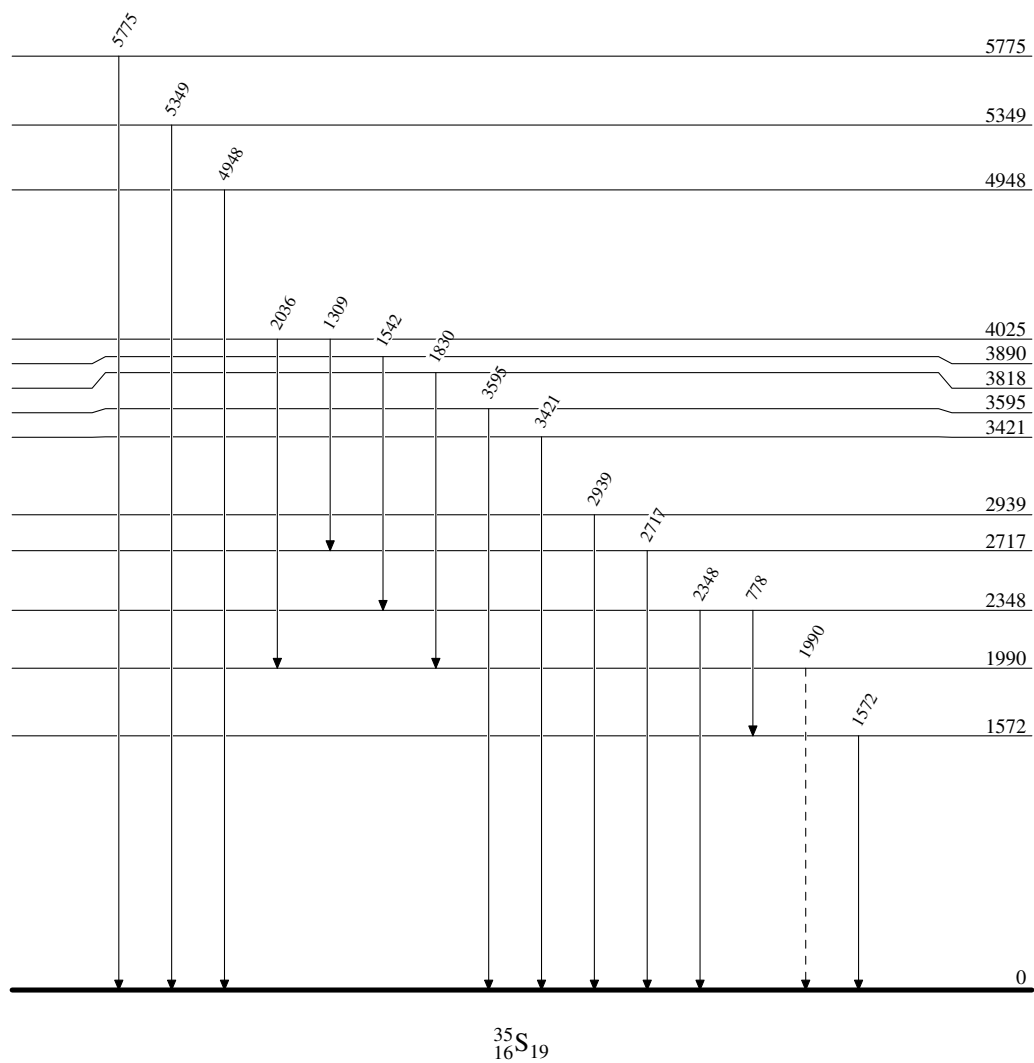
| $E_\gamma^{\dagger}$ | $E_i(\text{level})$ | $E_f$ | $E_\gamma^{\dagger}$ | $E_i(\text{level})$ | $E_f$ | $E_\gamma^{\dagger}$ | $E_i(\text{level})$ | $E_f$ | $E_\gamma^{\dagger}$ | $E_i(\text{level})$ | $E_f$ |
|----------------------|---------------------|-------|----------------------|---------------------|-------|----------------------|---------------------|-------|----------------------|---------------------|-------|
| 778                  | 2348                | 1572  | 1830                 | 3818                | 1990  | 2717                 | 2717                | 0     | 4948                 | 4948                | 0     |
| 1309                 | 4025                | 2717  | 1990 $^{\ddagger}$   | 1990                | 0     | 2939                 | 2939                | 0     | 5349                 | 5349                | 0     |
| 1542                 | 3890                | 2348  | 2036                 | 4025                | 1990  | 3421                 | 3421                | 0     | 5775                 | 5775                | 0     |
| 1572                 | 1572                | 0     | 2348                 | 2348                | 0     | 3595                 | 3595                | 0     |                      |                     |       |

$^{\dagger}$  From 2021JoZZ level scheme.  $\gamma$ -ray energy uncertainties are not given.

$^{\ddagger}$  Placement of transition in the level scheme is uncertain.

$^9\text{Be}(^{36}\text{S}, ^{35}\text{S}\gamma)$  2015MuZY,2021JoZZ

Legend

Level Scheme-----►  $\gamma$  Decay (Uncertain)

$^{24}\text{Mg}(^{14}\text{N},3\text{p}\gamma)$  2014Ay01

**2014Ay01:**  $^{24}\text{Mg}(^{14}\text{N},3\text{p})^{35}\text{S}$  fusion-evaporation reaction. A 40-MeV, 5-pnA  $^{14}\text{N}$  beam was delivered from the LNL

XTU-Tandem accelerator and impinged on 1-mg/cm<sup>2</sup> thick, 99.7% isotopically enriched  $^{24}\text{Mg}$  evaporated on an 8-mg/cm<sup>2</sup> gold layer.  $\gamma$  rays were detected using 4 $\pi$ -GASP array composed of 40 Compton-suppressed large volume HPGe detectors arranged in seven rings at different angles with respect to beam axis. Events were collected when at least two germanium detectors fired in coincidence. Measured  $E_\gamma$ ,  $I_\gamma$ ,  $\gamma\gamma$ -coin,  $\gamma(\theta)$ ,  $\gamma\gamma(\theta)$ (ADO). Deduced levels,  $J$ ,  $\pi$ , level lifetimes via the Doppler shift attenuation method (DSAM). Compared with large-scale shell model calculations.

 $^{35}\text{S}$  Levels

| <u><math>E(\text{level})^\dagger</math></u> | <u><math>J^\ddagger</math></u> | <u><math>E(\text{level})^\dagger</math></u> | <u><math>J^\ddagger</math></u> | <u><math>T_{1/2}^\#</math></u> | <u><math>E(\text{level})^\dagger</math></u> | <u><math>J^\ddagger</math></u> | <u><math>T_{1/2}^\#</math></u> |
|---------------------------------------------|--------------------------------|---------------------------------------------|--------------------------------|--------------------------------|---------------------------------------------|--------------------------------|--------------------------------|
| 0.0 <sup>&amp;</sup>                        | 3/2 <sup>+</sup>               | 3816.0 <sup>@</sup> 3                       | 9/2 <sup>-</sup>               | 0.28 ps 3                      | 5412.2 3                                    | 9/2 <sup>+</sup>               |                                |
| 1572.5 3                                    | 1/2 <sup>+</sup>               | 3886.4 4                                    | 5/2 <sup>+</sup>               |                                | 5877.6 <sup>&amp;</sup> 3                   | 11/2 <sup>+</sup>              |                                |
| 1991.4 <sup>@</sup> 2                       | 7/2 <sup>-</sup>               | 4023.2 <sup>@</sup> 3                       | 11/2 <sup>-</sup>              | 0.32 ps 3                      | 6299.4 4                                    | 11/2 <sup>+</sup>              |                                |
| 2347.9 3                                    | 3/2 <sup>-</sup>               | 4822.9 3                                    | 9/2 <sup>+</sup>               |                                | 6352.2 <sup>@</sup> 5                       | 13/2 <sup>-</sup>              | 50 fs 10                       |
| 2717.1 3                                    | 5/2 <sup>+</sup>               | 4899.7 4                                    | 9/2 <sup>+</sup>               |                                | 7179.8 <sup>&amp;</sup> 3                   | 15/2 <sup>+</sup>              | 3.1 ps 12                      |
| 3594.7 <sup>&amp;</sup> 3                   | 7/2 <sup>+</sup>               | 5010.2 3                                    | 11/2 <sup>-</sup>              | 0.45 ps 8                      | 8023.7 <sup>&amp;</sup> 9                   | 17/2 <sup>+</sup>              | 0.15 ps 4                      |

<sup>†</sup> From a least-squares fit to  $\gamma$ -ray energies.

<sup>‡</sup> As given in 2014Ay01, which quotes the known  $J^\pi$  assignments for four low-lying levels at 1572, 1991, 2347, and 2717 from 2011Ch48. 2014Ay01 assigns  $J^\pi$  for levels above 3 MeV based on the measured  $\gamma\gamma(\theta)$ (ADO) ratios. When considered in the Adopted Levels, the firm assignments here are placed within parentheses if there are no other strong arguments to support these firm assignments.

<sup>#</sup> From DSAM line-shape analyses (2014Ay01).

<sup>@</sup> Seq.(A):  $\Delta J=1$  sequence based on 7/2<sup>-</sup>.

<sup>&</sup> Seq.(B): Sequence based on 3/2<sup>+</sup>.

 $\gamma(^{35}\text{S})$ 

$R_{\text{ADO}}=[I_\gamma(34^\circ)+I_\gamma(146^\circ)]/2I_\gamma(90^\circ)$ . Expected values are 0.8 for stretched dipole ( $\Delta J=1$ ) and 1.4 for stretched quadrupole ( $\Delta J=2$ ) or unstretched dipole ( $\Delta J=0$ ) transitions.

| <u><math>E_\gamma</math></u> | <u><math>I_\gamma</math></u> | <u><math>E_i(\text{level})</math></u> | <u><math>J_i^\pi</math></u> | <u><math>E_f</math></u> | <u><math>J_f^\pi</math></u> | <u>Mult.<sup>†</sup></u> | <u>Comments</u>                                                                                                                   |
|------------------------------|------------------------------|---------------------------------------|-----------------------------|-------------------------|-----------------------------|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| 207.0 5                      | 0.4 2                        | 4023.2                                | 11/2 <sup>-</sup>           | 3816.0                  | 9/2 <sup>-</sup>            |                          |                                                                                                                                   |
| 465.3 3                      | 1.2 1                        | 5877.6                                | 11/2 <sup>+</sup>           | 5412.2                  | 9/2 <sup>+</sup>            | D                        | $R_{\text{ADO}}=0.71$ 9.                                                                                                          |
| 775.4 4                      | 1.4 2                        | 2347.9                                | 3/2 <sup>-</sup>            | 1572.5                  | 1/2 <sup>+</sup>            | D                        | $R_{\text{ADO}}=0.68$ 14.                                                                                                         |
| 827.8 6                      | 0.3 1                        | 7179.8                                | 15/2 <sup>+</sup>           | 6352.2                  | 13/2 <sup>-</sup>           |                          |                                                                                                                                   |
| 843.9 8                      | 1.6 2                        | 8023.7                                | 17/2 <sup>+</sup>           | 7179.8                  | 15/2 <sup>+</sup>           | D                        | $R_{\text{ADO}}=0.8$ 3.                                                                                                           |
| 867.3 3                      | 1.9 1                        | 5877.6                                | 11/2 <sup>+</sup>           | 5010.2                  | 11/2 <sup>-</sup>           | D                        | $R_{\text{ADO}}=1.36$ 20, consistent with $\Delta J=0$ .                                                                          |
| 877.6 5                      | 0.4 1                        | 3594.7                                | 7/2 <sup>+</sup>            | 2717.1                  | 5/2 <sup>+</sup>            |                          |                                                                                                                                   |
| 880.2 4                      | 0.8 1                        | 7179.8                                | 15/2 <sup>+</sup>           | 6299.4                  | 11/2 <sup>+</sup>           |                          |                                                                                                                                   |
| 887.0 7                      | 0.4 1                        | 6299.4                                | 11/2 <sup>+</sup>           | 5412.2                  | 9/2 <sup>+</sup>            | D                        | $R_{\text{ADO}}=0.6$ 3.                                                                                                           |
| 977.8 3                      | 1.6 1                        | 5877.6                                | 11/2 <sup>+</sup>           | 4899.7                  | 9/2 <sup>+</sup>            | D+Q                      | $R_{\text{ADO}}=1.43$ 16.                                                                                                         |
| 986.9 3                      | 4.4 2                        | 5010.2                                | 11/2 <sup>-</sup>           | 4023.2                  | 11/2 <sup>-</sup>           | D                        | $B(E1)(\text{W.u.})=0.00102 +22-16$ ; $B(M1)(\text{W.u.})=0.036 +8-6$<br>$R_{\text{ADO}}=1.30$ 13, consistent with $\Delta J=0$ . |
| 1006.8 2                     | 0.8 1                        | 4822.9                                | 9/2 <sup>+</sup>            | 3816.0                  | 9/2 <sup>-</sup>            | D                        | $R_{\text{ADO}}=1.37$ 11, consistent with $\Delta J=0$ .                                                                          |
| 1054.5 3                     | 3.0 2                        | 5877.6                                | 11/2 <sup>+</sup>           | 4822.9                  | 9/2 <sup>+</sup>            | D                        | $R_{\text{ADO}}=0.72$ 11.                                                                                                         |
| 1228.1 3                     | 1.1 1                        | 4822.9                                | 9/2 <sup>+</sup>            | 3594.7                  | 7/2 <sup>+</sup>            | D                        | $R_{\text{ADO}}=0.78$ 9.                                                                                                          |
| 1302.2 2                     | 13.5 8                       | 7179.8                                | 15/2 <sup>+</sup>           | 5877.6                  | 11/2 <sup>+</sup>           | E2                       | $B(E2)(\text{W.u.})=6.6 +42-19$<br>$R_{\text{ADO}}=1.40$ 5, $A_2=+0.27$ 5, $A_4=-0.07$ 5.                                         |
| 1304.8 12                    | 0.6 2                        | 4899.7                                | 9/2 <sup>+</sup>            | 3594.7                  | 7/2 <sup>+</sup>            |                          |                                                                                                                                   |
| 1389.0 4                     | 3.9 4                        | 5412.2                                | 9/2 <sup>+</sup>            | 4023.2                  | 11/2 <sup>-</sup>           | D                        | $R_{\text{ADO}}=0.79$ 11.                                                                                                         |

Continued on next page (footnotes at end of table)



$^{24}\text{Mg}(^{14}\text{N}, 3\text{p}\gamma)$  **2014Ay01** (continued) $\gamma(^{35}\text{S})$  (continued)

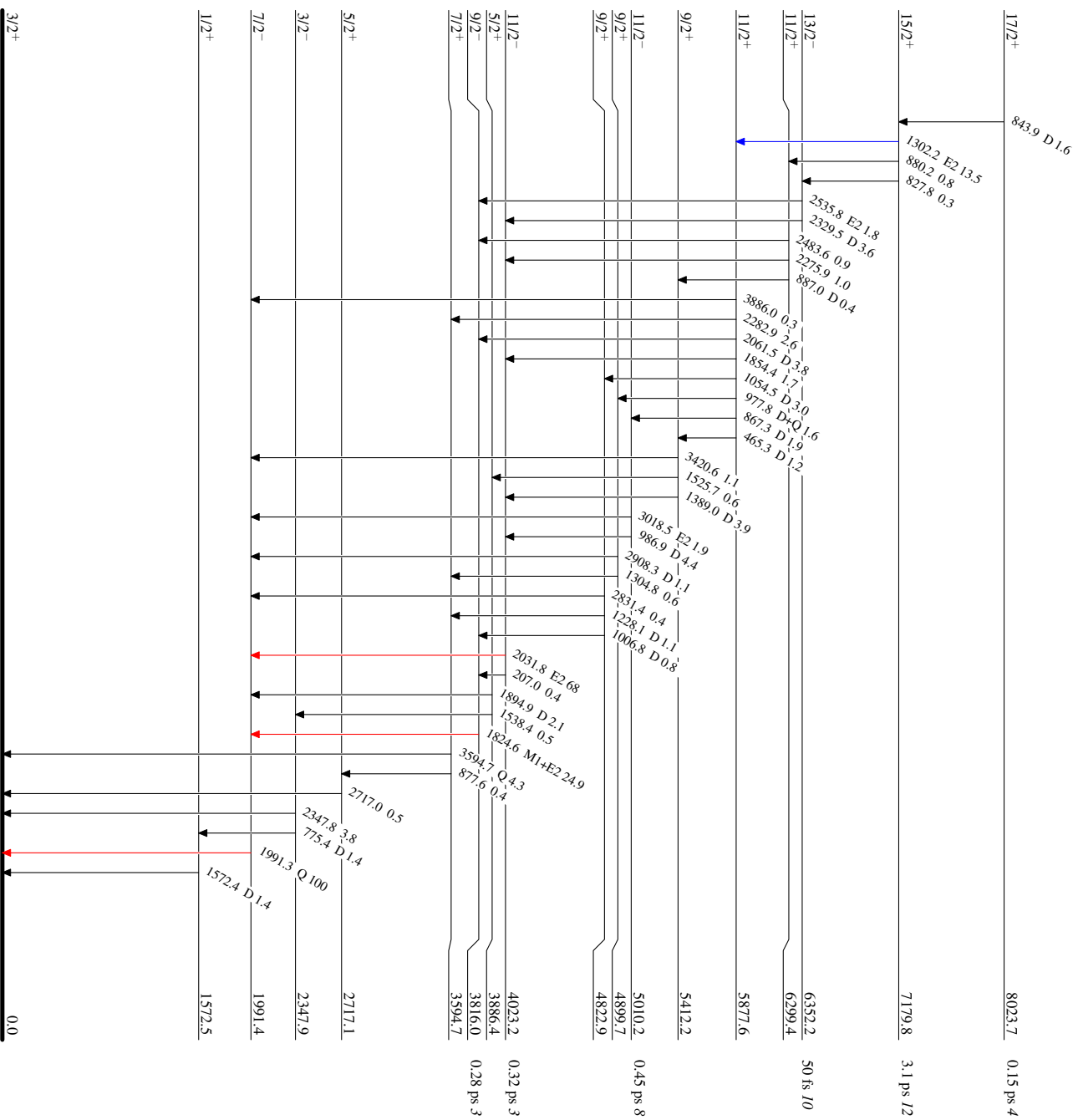
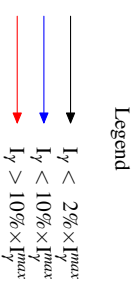
| $E_\gamma$ | $I_\gamma$ | $E_i(\text{level})$ | $J_i^\pi$         | $E_f$  | $J_f^\pi$         | Mult. <sup>†</sup> | $\delta^\dagger$ | Comments                                                                                                                                                                                                                                                                                                                                                      |
|------------|------------|---------------------|-------------------|--------|-------------------|--------------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1525.7 8   | 0.6 1      | 5412.2              | 9/2 <sup>+</sup>  | 3886.4 | 5/2 <sup>+</sup>  |                    |                  |                                                                                                                                                                                                                                                                                                                                                               |
| 1538.4 8   | 0.5 2      | 3886.4              | 5/2 <sup>+</sup>  | 2347.9 | 3/2 <sup>-</sup>  |                    |                  |                                                                                                                                                                                                                                                                                                                                                               |
| 1572.4 3   | 1.4 4      | 1572.5              | 1/2 <sup>+</sup>  | 0.0    | 3/2 <sup>+</sup>  | D                  |                  | $R_{\text{ADO}}=0.88$ 10.                                                                                                                                                                                                                                                                                                                                     |
| 1824.6 2   | 24.9 15    | 3816.0              | 9/2 <sup>-</sup>  | 1991.4 | 7/2 <sup>-</sup>  | M1+E2              | +0.55 9          | $B(\text{M1})(\text{W.u.})=0.0099$ +14-12; $B(\text{E2})(\text{W.u.})=3.4$ +10-9<br>Mult., $\delta$ : D+Q and $\delta$ from $\gamma(\theta)$ in <a href="#">2014Ay01</a> ;<br>E1+M2 ruled out by RUL.<br>$R_{\text{ADO}}=1.58$ 5 indicating $\Delta J=2$ or 0 seems questionable considering $\Delta J=1$ from level scheme.<br>$A_2=+0.48$ 4, $A_4=+0.08$ 6. |
| 1854.4 4   | 1.7 1      | 5877.6              | 11/2 <sup>+</sup> | 4023.2 | 11/2 <sup>-</sup> |                    |                  |                                                                                                                                                                                                                                                                                                                                                               |
| 1894.9 4   | 2.1 3      | 3886.4              | 5/2 <sup>+</sup>  | 1991.4 | 7/2 <sup>-</sup>  | D                  |                  | $R_{\text{ADO}}=0.69$ 10.                                                                                                                                                                                                                                                                                                                                     |
| 1991.3 2   | 100 6      | 1991.4              | 7/2 <sup>-</sup>  | 0.0    | 3/2 <sup>+</sup>  | Q                  |                  | $R_{\text{ADO}}=1.45$ 15.                                                                                                                                                                                                                                                                                                                                     |
| 2031.8 3   | 68 4       | 4023.2              | 11/2 <sup>-</sup> | 1991.4 | 7/2 <sup>-</sup>  | E2                 |                  | $B(\text{E2})(\text{W.u.})=7.5$ +8-7<br>$R_{\text{ADO}}=1.41$ 10, $A_2=+0.29$ 4, $A_4=-0.10$ 6.<br>$R_{\text{ADO}}=0.65$ 9.                                                                                                                                                                                                                                   |
| 2061.5 3   | 3.8 2      | 5877.6              | 11/2 <sup>+</sup> | 3816.0 | 9/2 <sup>-</sup>  | D                  |                  |                                                                                                                                                                                                                                                                                                                                                               |
| 2275.9 6   | 1.0 2      | 6299.4              | 11/2 <sup>+</sup> | 4023.2 | 11/2 <sup>-</sup> |                    |                  |                                                                                                                                                                                                                                                                                                                                                               |
| 2282.9 4   | 2.6 1      | 5877.6              | 11/2 <sup>+</sup> | 3594.7 | 7/2 <sup>+</sup>  |                    |                  |                                                                                                                                                                                                                                                                                                                                                               |
| 2329.5 9   | 3.6 7      | 6352.2              | 13/2 <sup>-</sup> | 4023.2 | 11/2 <sup>-</sup> | D                  |                  | $B(\text{E1})(\text{W.u.})=6.7 \times 10^{-4}$ +17-13;<br>$B(\text{M1})(\text{W.u.})=0.023$ +6-5<br>$R_{\text{ADO}}=0.8$ 3.                                                                                                                                                                                                                                   |
| 2347.8 3   | 3.8 3      | 2347.9              | 3/2 <sup>-</sup>  | 0.0    | 3/2 <sup>+</sup>  |                    |                  |                                                                                                                                                                                                                                                                                                                                                               |
| 2483.6 8   | 0.9 2      | 6299.4              | 11/2 <sup>+</sup> | 3816.0 | 9/2 <sup>-</sup>  |                    |                  |                                                                                                                                                                                                                                                                                                                                                               |
| 2535.8 11  | 1.8 2      | 6352.2              | 13/2 <sup>-</sup> | 3816.0 | 9/2 <sup>-</sup>  | E2                 |                  | $B(\text{E2})(\text{W.u.})=5.3$ +17-11<br>$R_{\text{ADO}}=1.5$ 5.                                                                                                                                                                                                                                                                                             |
| 2717.0 4   | 0.5 3      | 2717.1              | 5/2 <sup>+</sup>  | 0.0    | 3/2 <sup>+</sup>  |                    |                  |                                                                                                                                                                                                                                                                                                                                                               |
| 2831.4 9   | 0.4 1      | 4822.9              | 9/2 <sup>+</sup>  | 1991.4 | 7/2 <sup>-</sup>  |                    |                  |                                                                                                                                                                                                                                                                                                                                                               |
| 2908.3 5   | 1.1 2      | 4899.7              | 9/2 <sup>+</sup>  | 1991.4 | 7/2 <sup>-</sup>  | D                  |                  | $R_{\text{ADO}}=0.74$ 14.                                                                                                                                                                                                                                                                                                                                     |
| 3018.5 7   | 1.9 2      | 5010.2              | 11/2 <sup>-</sup> | 1991.4 | 7/2 <sup>-</sup>  | E2                 |                  | $B(\text{E2})(\text{W.u.})=0.223$ +49-39<br>$R_{\text{ADO}}=1.35$ 34.                                                                                                                                                                                                                                                                                         |
| 3420.6 9   | 1.1 1      | 5412.2              | 9/2 <sup>+</sup>  | 1991.4 | 7/2 <sup>-</sup>  |                    |                  |                                                                                                                                                                                                                                                                                                                                                               |
| 3594.7 7   | 4.3 4      | 3594.7              | 7/2 <sup>+</sup>  | 0.0    | 3/2 <sup>+</sup>  | Q                  |                  | $R_{\text{ADO}}=1.36$ 16.                                                                                                                                                                                                                                                                                                                                     |
| 3886.0 13  | 0.3 1      | 5877.6              | 11/2 <sup>+</sup> | 1991.4 | 7/2 <sup>-</sup>  |                    |                  |                                                                                                                                                                                                                                                                                                                                                               |

<sup>†</sup> From  $\gamma(\theta)$ ,  $\gamma\gamma(\theta)$ (ADO) data, and RUL when  $T_{1/2}$  is available in [2014Ay01](#).

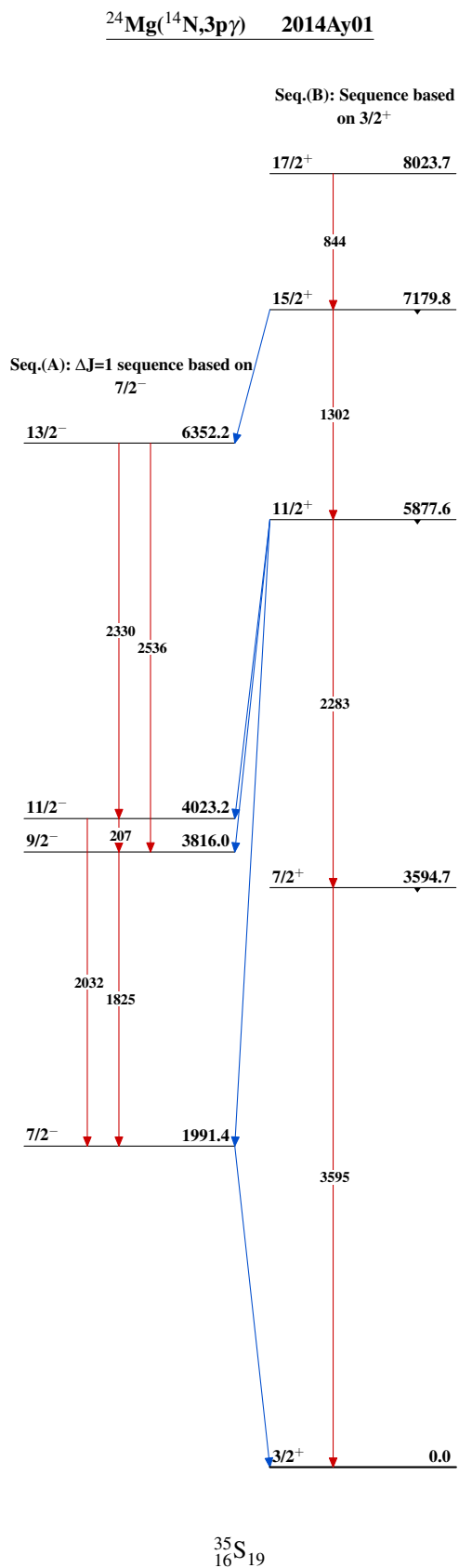
<sup>24</sup>Mg(<sup>14</sup>N,3pγ) 2014Ay01

Level Scheme

Intensities: Relative I<sub>γ</sub>



<sup>35</sup>S<sub>16</sub>→<sup>19</sup>



$^{26}\text{Mg}(^{18}\text{O}, 2\alpha n\gamma)$  **2021Go09**

**2021Go09:**  $^{26}\text{Mg}(^{18}\text{O}, 2\alpha n)^{35}\text{S}$  fusion-evaporation reaction. Exp 1 determined the beam energy for producing high-spin states of  $^{35}\text{S}$ : an 80-MeV  $^{18}\text{O}$  beam was delivered from the JAEA tandem accelerator facility and impinged on a 0.5-mg/cm<sup>2</sup> thick, self-supporting metallic enriched foil of  $^{26}\text{Mg}$ .  $\gamma$  rays were detected using the GEMINI-II array of 14 Ge detectors with BGO Compton-suppressor shields. The reaction channel was selected by detecting the evaporated charged particles with a 4 $\pi$  silicon detector. Measured  $E_\gamma$ ,  $I_\gamma$ ,  $\alpha\gamma\gamma$ -coin,  $\gamma\gamma(\theta)$ (ADO). Exp 2 used the same  $^{18}\text{O}$  beam energy, the same target, and the same charged-particle detectors as Exp 1. The beam was delivered from the ALTO-Tandem accelerator facility at IPN Orsay.  $\gamma$  rays were detected using the ORGAM array of 13 Ge detectors based on the EUROGAM array with the BGO scintillator for the anti-Compton suppressor. Compared with large-scale shell-model calculations with the SDPF-MSD4 interaction. Also see [2015GoZY](#), [2014GoZZ](#), [2013GoZW](#), and [2012IdZZ](#).

 $^{35}\text{S}$  Levels

| $E(\text{level})^\dagger$ | $J^\pi^\ddagger$     | $T_{1/2}$ | Comments                                                                                                                                     |
|---------------------------|----------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 0.0 <sup>#</sup>          | 3/2 <sup>+</sup>     |           |                                                                                                                                              |
| 1990.6 <sup>#</sup> 4     | 7/2 <sup>-</sup>     |           |                                                                                                                                              |
| 3594.2 <sup>#</sup> 7     | (7/2 <sup>+</sup> )  |           |                                                                                                                                              |
| 3815.0 6                  | (9/2 <sup>-</sup> )  |           |                                                                                                                                              |
| 4023.1 5                  | (11/2 <sup>-</sup> ) |           |                                                                                                                                              |
| 4822.4 6                  | (9/2 <sup>+</sup> )  |           |                                                                                                                                              |
| 4899.6 7                  | (9/2 <sup>+</sup> )  |           |                                                                                                                                              |
| 5009.8 6                  | (11/2 <sup>-</sup> ) |           |                                                                                                                                              |
| 5412.4 6                  | (9/2 <sup>+</sup> )  |           |                                                                                                                                              |
| 5878.0 <sup>#</sup> 6     | (11/2 <sup>+</sup> ) |           |                                                                                                                                              |
| 7179.6 <sup>#</sup> 7     | (15/2 <sup>+</sup> ) |           |                                                                                                                                              |
| 8023.9 <sup>#</sup> 9     | (17/2 <sup>+</sup> ) |           |                                                                                                                                              |
| 8755.7 <sup>#</sup> 13    | (17/2)               | <1 ps     | $T_{1/2}$ : estimated by <a href="#">2021Go09</a> from residual Doppler energy shifts observed for the 732 $\gamma$ and 1576 $\gamma$ peaks. |
| 10048.4 <sup>#</sup> 10   | (19/2)               |           |                                                                                                                                              |
| 12469.1 <sup>#</sup> 15   | (21/2)               |           |                                                                                                                                              |

<sup>†</sup> From a least-squares fit to  $\gamma$ -ray energies.

<sup>‡</sup> As given in [2021Go09](#), which quotes the  $J^\pi$  assignments for levels below 8.5 MeV from [2014Ay01](#) but adds parentheses. [2021Go09](#) assigns  $J^\pi$  for levels above 8.5 MeV based on the measured  $\gamma\gamma(\theta)$ (ADO) ratios.

<sup>#</sup> Seq.(A):  $\gamma$  cascade based on the g.s.

 $\gamma(^{35}\text{S})$ 

$R_{\text{ADO}}^{\gamma 1} = I_{\gamma 1}(47^\circ \text{ gated by } \gamma_2 \text{ at all angles}) / I_{\gamma 1}(86^\circ \text{ gated by } \gamma_2 \text{ at all angles})$ . Expected values are  $\approx 0.8$  for dipole ( $\Delta J=1$ ) and  $\approx 1.0$  for stretched quadrupole ( $\Delta J=2$ ) transitions.

| $E_\gamma^\dagger$ | $I_\gamma^\dagger$ | $E_i(\text{level})$ | $J_i^\pi$            | $E_f$  | $J_f^\pi$            | Mult. <sup>‡</sup> | Comments                                                                                                                        |
|--------------------|--------------------|---------------------|----------------------|--------|----------------------|--------------------|---------------------------------------------------------------------------------------------------------------------------------|
| 465.5 2            | 14 1               | 5878.0              | (11/2 <sup>+</sup> ) | 5412.4 | (9/2 <sup>+</sup> )  | D                  | $R_{\text{ADO}}=0.67$ 11.                                                                                                       |
| 731.6 15           | 8 3                | 8755.7              | (17/2)               | 8023.9 | (17/2 <sup>+</sup> ) |                    |                                                                                                                                 |
| 844.3 6            | 20 3               | 8023.9              | (17/2 <sup>+</sup> ) | 7179.6 | (15/2 <sup>+</sup> ) | D                  | $E_\gamma$ : 8443(6) in <a href="#">2021Go09</a> is likely a misprint.<br>$R_{\text{ADO}}=0.51$ 8.<br>$R_{\text{ADO}}=0.54$ 21. |
| 868.2 4            | 20 2               | 5878.0              | (11/2 <sup>+</sup> ) | 5009.8 | (11/2 <sup>-</sup> ) | D                  |                                                                                                                                 |
| 978.4 10           | 26 7               | 5878.0              | (11/2 <sup>+</sup> ) | 4899.6 | (9/2 <sup>+</sup> )  |                    |                                                                                                                                 |
| 986.6 3            | 26 1               | 5009.8              | (11/2 <sup>-</sup> ) | 4023.1 | (11/2 <sup>-</sup> ) |                    | $R_{\text{ADO}}=0.91$ 13 consistent with $\Delta J=(0)$ .                                                                       |
| 1007.2 11          | 4 2                | 4822.4              | (9/2 <sup>+</sup> )  | 3815.0 | (9/2 <sup>-</sup> )  | (D)                | $R_{\text{ADO}}=0.87$ 15.                                                                                                       |
| 1055.6 4           | 39 10              | 5878.0              | (11/2 <sup>+</sup> ) | 4822.4 | (9/2 <sup>+</sup> )  | D                  | $R_{\text{ADO}}=0.67$ 18.                                                                                                       |

Continued on next page (footnotes at end of table)

$^{26}\text{Mg}(^{18}\text{O}, 2\alpha n\gamma)$  2021Go09 (continued) $\gamma(^{35}\text{S})$  (continued)

| $E_\gamma^\dagger$ | $I_\gamma^\dagger$ | $E_i(\text{level})$ | $J_i^\pi$            | $E_f$   | $J_f^\pi$            | Mult. $^\ddagger$ | Comments                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------|--------------------|---------------------|----------------------|---------|----------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1228.3 6           | 11 2               | 4822.4              | (9/2 <sup>+</sup> )  | 3594.2  | (7/2 <sup>+</sup> )  |                   |                                                                                                                                                                                                                                                                                                                                                                                                     |
| 1301.6 3           | 56 11              | 7179.6              | (15/2 <sup>+</sup> ) | 5878.0  | (11/2 <sup>+</sup> ) | Q                 | $R_{\text{ADO}}=1.40$ 17.                                                                                                                                                                                                                                                                                                                                                                           |
| 1305.3 4           | 5 3                | 4899.6              | (9/2 <sup>+</sup> )  | 3594.2  | (7/2 <sup>+</sup> )  |                   |                                                                                                                                                                                                                                                                                                                                                                                                     |
| 1389.0 8           | 9 2                | 5412.4              | (9/2 <sup>+</sup> )  | 4023.1  | (11/2 <sup>-</sup> ) |                   |                                                                                                                                                                                                                                                                                                                                                                                                     |
| 1576.3 14          | 14 3               | 8755.7              | (17/2)               | 7179.6  | (15/2 <sup>+</sup> ) | D                 | $R_{\text{ADO}}=0.43$ 9. 2021Go09 Table I lists this $\Delta J=1$ ADO ratio for the 731.6 $\gamma$ , which however is a (17/2) to (17/2 <sup>+</sup> ) transition with $\Delta J=0$ . 2021Go09 repeatedly states the 8756 level has $J=(17/2)$ and therefore Evaluators reassign this $\Delta J=1$ ADO ratio to the 1576.3 $\gamma$ , corresponding to a (17/2) to (15/2 <sup>+</sup> ) transition. |
| 1824.4 6           | 48 2               | 3815.0              | (9/2 <sup>-</sup> )  | 1990.6  | 7/2 <sup>-</sup>     |                   |                                                                                                                                                                                                                                                                                                                                                                                                     |
| 1854.5 42          | 28 14              | 5878.0              | (11/2 <sup>+</sup> ) | 4023.1  | (11/2 <sup>-</sup> ) |                   |                                                                                                                                                                                                                                                                                                                                                                                                     |
| 1990.5 4           | 100                | 1990.6              | 7/2 <sup>-</sup>     | 0.0     | 3/2 <sup>+</sup>     | Q                 | $R_{\text{ADO}}=1.30$ 8.                                                                                                                                                                                                                                                                                                                                                                            |
| 2032.4 4           | 70 16              | 4023.1              | (11/2 <sup>-</sup> ) | 1990.6  | 7/2 <sup>-</sup>     |                   |                                                                                                                                                                                                                                                                                                                                                                                                     |
| 2063.3 14          | 32 6               | 5878.0              | (11/2 <sup>+</sup> ) | 3815.0  | (9/2 <sup>-</sup> )  | D                 | $R_{\text{ADO}}=0.58$ 41.                                                                                                                                                                                                                                                                                                                                                                           |
| 2283.5 10          | 14 3               | 5878.0              | (11/2 <sup>+</sup> ) | 3594.2  | (7/2 <sup>+</sup> )  |                   |                                                                                                                                                                                                                                                                                                                                                                                                     |
| 2420.6 11          | 21 6               | 12469.1             | (21/2)               | 10048.4 | (19/2)               | (D)               | $R_{\text{ADO}}=0.50$ 42.                                                                                                                                                                                                                                                                                                                                                                           |
| 2868.7 8           | 26 2               | 10048.4             | (19/2)               | 7179.6  | (15/2 <sup>+</sup> ) | Q                 | $R_{\text{ADO}}=0.98$ 13.                                                                                                                                                                                                                                                                                                                                                                           |
| 3594.4 11          | 35 2               | 3594.2              | (7/2 <sup>+</sup> )  | 0.0     | 3/2 <sup>+</sup>     | (Q)               | $R_{\text{ADO}}=0.93$ 20.                                                                                                                                                                                                                                                                                                                                                                           |

 $^\dagger$  From 2021Go09. $^\ddagger$  From  $\gamma\gamma(\theta)(\text{ADO})$  data in 2021Go09.

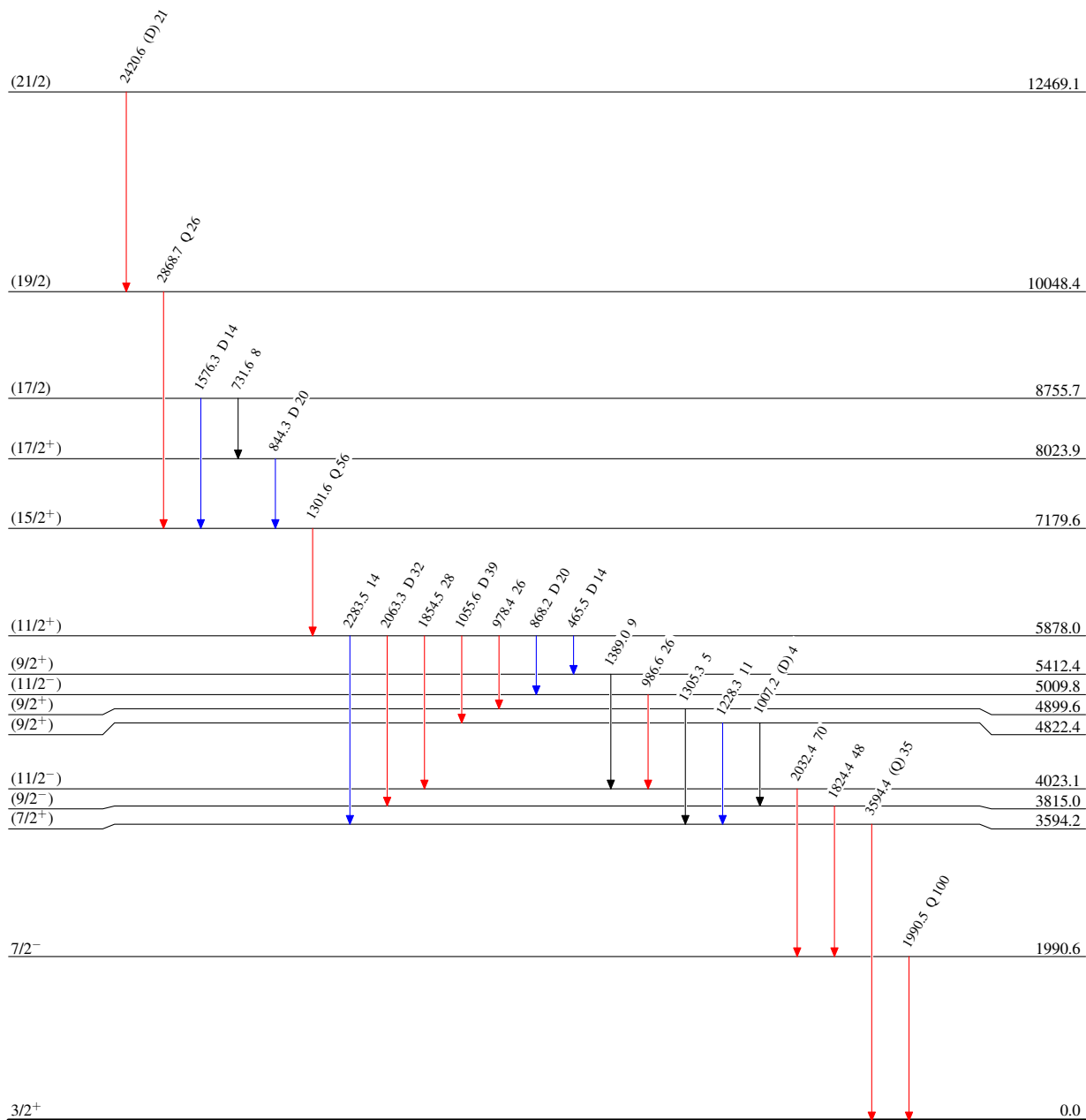
$^{26}\text{Mg}(^{18}\text{O}, 2\alpha n\gamma)$  2021Go09

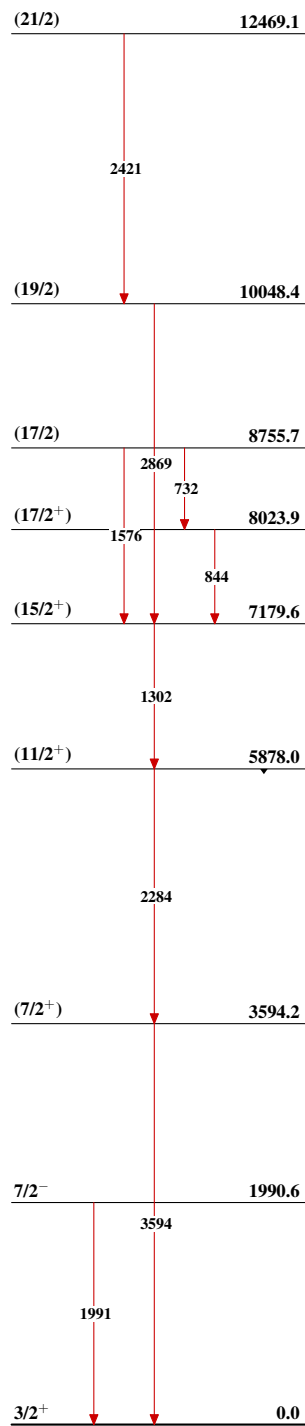
## Level Scheme

Intensities: Relative  $I_\gamma$ 

## Legend

- $\longrightarrow$   $I_\gamma < 2\% \times I_\gamma^{\max}$   
 $\longrightarrow$   $I_\gamma < 10\% \times I_\gamma^{\max}$   
 $\longrightarrow$   $I_\gamma > 10\% \times I_\gamma^{\max}$

 $^{35}_{16}\text{S}_{19}$

$^{26}\text{Mg}(^{18}\text{O},2\alpha n\gamma)$  2021Go09Seq.(A):  $\gamma$  cascade based on the  
g.s $^{35}_{16}\text{S}_{19}$

$^{34}\text{S}(\text{n},\gamma)$  E=thermal 1985Ra15

- 1985Ra15:**  $6 \times 10^{11}$  n/cm<sup>2</sup>/s thermal neutrons at the target position were produced from the Los Alamos Omega West Reactor at a nominal 8-MW reactor power.  $\gamma$  rays following thermal neutron capture by a 1.09-g, 77.7% enriched  $^{34}\text{S}$  target were detected using a 26 cm<sup>3</sup> coaxial Ge(Li) detector inside a 20-cm-diam by 30-cm-long NaI(Tl) annulus with FWHM=2.3 keV at 1 MeV, 5.5 keV at 6 MeV and 8.8 keV at 11 MeV. Measured  $\sigma_\gamma$ ,  $E_\gamma$ , and  $I_\gamma$  for 61  $\gamma$ -ray transitions. Deduced 22 levels and  $\gamma$ -ray branching ratios.
- 1972Dz13:** Thermal neutrons were produced from the thermal column of the IRT reactor of the Baghdad Nuclear Research Institute.  $\gamma$  rays following thermal neutron capture by a 1.5-g, 97%-enriched  $^{34}\text{S}$  target were detected using a 10-cm<sup>3</sup> Ge(Li) detector with FWHM=3.5 keV at 1 MeV and 8 keV at 7 MeV. Measured  $E_\gamma$  and  $I_\gamma$  for 22  $\gamma$ -ray transitions. Deduced 11 levels.
- 1985Ke08:**  $5 \times 10^{12}$  n/cm<sup>2</sup>/s thermal neutrons were produced from the McMaster University Reactor.  $\gamma$  rays following thermal neutron capture by a natural sulfur target were detected using a 10% efficient Ge centered within a NaI(Tl) annulus with FWHM=3.2-4.4 keV at 4-7 MeV. Measured  $E_\gamma$  and  $I_\gamma$  for 13  $\gamma$ -ray transitions. Deduced 9 levels.
- 1997Be42:** Thermal neutrons were provided from the reactor BR1 of the Studiecentrum voor kernenergie in Mol, Belgium.  $\gamma$  rays following thermal neutron capture by a 313.6-mg, 93.26%-enriched  $^{34}\text{S}$  and 5.933%-enriched  $^{36}\text{S}$  target were detected using a Ge detector. Measured  $\sigma_\gamma$ ,  $E_\gamma$ , and  $I_\gamma$  for 11  $\gamma$ -ray transitions. Deduced 8 levels. Compared with direct-capture (DC) model calculations.
- All 55  $E_\gamma$  values from **2007ChZX**: database of prompt gamma activation analysis (PGAA) from neutron capture are in agreement with the data below. Others: **2000Re01**, **2001ZhZZ**.

 $^{35}\text{S}$  Levels

| E(level) <sup>†</sup> | Comments                                                                                                                                                                                                                    |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0                     |                                                                                                                                                                                                                             |
| 1572.373 7            |                                                                                                                                                                                                                             |
| 1991.28 4             |                                                                                                                                                                                                                             |
| 2347.782 8            | C <sup>2</sup> S=0.48 for 3/2 <sup>-</sup> from <b>1997Be42</b> .                                                                                                                                                           |
| 2717.05 9             |                                                                                                                                                                                                                             |
| 2938.63 5             |                                                                                                                                                                                                                             |
| 3558.082 27           |                                                                                                                                                                                                                             |
| 3801.955 15           | C <sup>2</sup> S=0.08 for 3/2 <sup>-</sup> from <b>1997Be42</b> .                                                                                                                                                           |
| 4105.6 8              |                                                                                                                                                                                                                             |
| 4189.243 33           | C <sup>2</sup> S=0.15 for 1/2 <sup>-</sup> from <b>1997Be42</b> .                                                                                                                                                           |
| 4477.62 7             |                                                                                                                                                                                                                             |
| 4903.374 15           | C <sup>2</sup> S=0.49 for 1/2 <sup>-</sup> from <b>1997Be42</b> .                                                                                                                                                           |
| 4963.084 25           | C <sup>2</sup> S=0.19 for 3/2 <sup>-</sup> from <b>1997Be42</b> .                                                                                                                                                           |
| 5752.5 8              |                                                                                                                                                                                                                             |
| 5841.485 31           | This level is introduced in S. Raman to P.M. Endt private communication (mentioned in the Adopted Levels of <b>2011Ch48</b> ). Populated by 1144.6 $\gamma$ from 6986 level and deexcites by 4268.6 $\gamma$ to 1572 level. |
| 6018.8 6              |                                                                                                                                                                                                                             |
| 6078.477 32           |                                                                                                                                                                                                                             |
| 6293.93 6             |                                                                                                                                                                                                                             |
| 6354.89 23            |                                                                                                                                                                                                                             |
| 6419.9 11             |                                                                                                                                                                                                                             |
| 6629.43 9             |                                                                                                                                                                                                                             |
| 6761.0 12             |                                                                                                                                                                                                                             |
| (6986.099 24)         |                                                                                                                                                                                                                             |

<sup>†</sup> From a least-squares fit to  $\gamma$ -ray energies.



$^{34}\text{S}(\text{n},\gamma)$  E=thermal **1985Ra15 (continued)** $\gamma(^{35}\text{S})$ 

Photons per 100 thermal neutron captures are reported in [1972Dz13](#) and [1985Ke08](#).  $\gamma$ -ray cross sections in mb are reported in [1985Ra15](#), multiplying these by 0.340 yields the number of photons per 100 thermal neutron captures.  $\sigma(4638)=163$  mb *15*.  
 $I_\gamma(4638)=\sigma(4638)\times 0.340=55$  *5*.

$\gamma$ -ray intensities per incident neutron for all transitions and  $\gamma$ -ray cross sections in mb for primary transitions are reported in [1997Be42](#). The strongest 4638 $\gamma$  is used for normalization.  $\sigma(4638)=148.6$  mb *55* and  
 $I_\gamma(4638)=\sigma(1997\text{Be42})/\sigma(1985\text{Ra15})\times I_\gamma(1985\text{Ra15})=50$  *7*.

| $E_\gamma$ <sup>†</sup>              | $I_\gamma$ <sup>†#</sup> | $E_i(\text{level})$ | $E_f$    | Comments                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------------|--------------------------|---------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 356.66 <sup>@</sup> 9                | 0.037 <sup>@</sup> 4     | 2347.782            | 1991.28  |                                                                                                                                                                                                                                                                                                                                                |
| 356.66 <sup>@</sup> 9                | 0.037 <sup>@</sup> 4     | (6986.099)          | 6629.43  |                                                                                                                                                                                                                                                                                                                                                |
| 368.5 <i>4</i>                       | 0.020 <i>7</i>           | 2717.05             | 2347.782 |                                                                                                                                                                                                                                                                                                                                                |
| 619.23 <i>19</i>                     | 0.042 <i>8</i>           | 3558.082            | 2938.63  |                                                                                                                                                                                                                                                                                                                                                |
| 631.32 <sup>@</sup> 24               | 0.060 <sup>@</sup> 9     | 4189.243            | 3558.082 |                                                                                                                                                                                                                                                                                                                                                |
| 631.32 <sup>@</sup> 24               | 0.060 <sup>@</sup> 9     | (6986.099)          | 6354.89  |                                                                                                                                                                                                                                                                                                                                                |
| <sup>x</sup> 646.9 <sup>‡</sup> 5    | 0.7 <i>1</i>             |                     |          | $E_\gamma, I_\gamma$ : from <a href="#">1972Dz13</a> . <a href="#">1972Dz13</a> assigned this $\gamma$ ray as the 6986 to 6339 transition. This $\gamma$ ray and its final level 6339 have not been observed in any other $^{34}\text{S}(\text{n},\gamma)$ measurements. Evaluators consider this $\gamma$ ray questionable.                   |
| <sup>x</sup> 663.41 <sup>‡</sup> 7   | 0.09 <i>1</i>            |                     |          |                                                                                                                                                                                                                                                                                                                                                |
| 692.16 <i>5</i>                      | 0.133 <i>17</i>          | (6986.099)          | 6293.93  |                                                                                                                                                                                                                                                                                                                                                |
| 775.398 <i>6</i>                     | 17.2 <i>19</i>           | 2347.782            | 1572.373 | $E_\gamma$ : other: 775.50 <i>15</i> ( <a href="#">1972Dz13</a> ).<br>$I_\gamma$ : weighted average of 19.5 <i>20</i> ( <a href="#">1972Dz13</a> ) and 15.6 <i>17</i> ( <a href="#">1985Ra15</a> ).                                                                                                                                            |
| <sup>x</sup> 803.81 <sup>‡</sup> 9   | 0.092 <i>14</i>          |                     |          |                                                                                                                                                                                                                                                                                                                                                |
| 863.28 <i>28</i>                     | 0.034 <i>10</i>          | 3801.955            | 2938.63  |                                                                                                                                                                                                                                                                                                                                                |
| 907.608 <sup>‡</sup> 23              | 0.56 <i>10</i>           | (6986.099)          | 6078.477 | $E_\gamma$ : weighted average of 908.1 <i>3</i> ( <a href="#">1972Dz13</a> ) and 907.607 <i>14</i> ( <a href="#">1985Ra15</a> ).<br>$I_\gamma$ : weighted average of 0.5 <i>1</i> ( <a href="#">1972Dz13</a> ) and 0.61 <i>10</i> ( <a href="#">1985Ra15</a> ).                                                                                |
| 1084.79 <i>15</i>                    | 0.054 <i>10</i>          | 3801.955            | 2717.05  |                                                                                                                                                                                                                                                                                                                                                |
| 1101.92 <i>31</i>                    | 0.033 <i>8</i>           | 4903.374            | 3801.955 |                                                                                                                                                                                                                                                                                                                                                |
| 1144.591 <i>20</i>                   | 0.55 <i>6</i>            | (6986.099)          | 5841.485 | The placement of this $\gamma$ ray is changed from 2717 to 1572 ( <a href="#">1985Ra15</a> ) to 6986 to 5841 (S. Raman to P.M. Endt private communication).                                                                                                                                                                                    |
| 1161.05 <i>20</i>                    | 0.055 <i>8</i>           | 4963.084            | 3801.955 |                                                                                                                                                                                                                                                                                                                                                |
| 1210.28 <i>4</i>                     | 0.252 <i>27</i>          | 3558.082            | 2347.782 |                                                                                                                                                                                                                                                                                                                                                |
| 1250.61 <i>5</i>                     | 0.214 <i>24</i>          | 4189.243            | 2938.63  |                                                                                                                                                                                                                                                                                                                                                |
| <sup>x</sup> 1381.67 <sup>‡</sup> 24 | 0.024 <i>8</i>           |                     |          |                                                                                                                                                                                                                                                                                                                                                |
| 1404.967 <i>24</i>                   | 0.60 <i>6</i>            | 4963.084            | 3558.082 |                                                                                                                                                                                                                                                                                                                                                |
| 1454.09 <i>4</i>                     | 0.45 <i>11</i>           | 3801.955            | 2347.782 |                                                                                                                                                                                                                                                                                                                                                |
| 1566.7 <i>3</i>                      | 0.47 <i>9</i>            | 3558.082            | 1991.28  |                                                                                                                                                                                                                                                                                                                                                |
| 1572.333 <i>8</i>                    | 35.0 <i>34</i>           | 1572.373            | 0        | $E_\gamma$ : weighted average of 1572.15 <i>16</i> ( <a href="#">1972Dz13</a> ) and 1572.333 <i>7</i> ( <a href="#">1985Ra15</a> ).<br>$I_\gamma$ : others: 39 <i>4</i> ( <a href="#">1972Dz13</a> ) and 32 <i>4</i> ( <a href="#">1997Be42</a> ).                                                                                             |
| 1760.55 <i>11</i>                    | 0.146 <i>21</i>          | 4477.62             | 2717.05  |                                                                                                                                                                                                                                                                                                                                                |
| 1840.2 <i>12</i>                     | 2.2 <i>4</i>             | 4189.243            | 2347.782 | $E_\gamma$ : unweighted average of 1839 <i>1</i> ( <a href="#">1972Dz13</a> ) and 1841.426 <i>15</i> ( <a href="#">1985Ra15</a> ).<br>$I_\gamma$ : weighted average of 3.6 <i>8</i> ( <a href="#">1972Dz13</a> ) and 2.14 <i>21</i> ( <a href="#">1985Ra15</a> ).                                                                              |
| 1964.8 <i>2</i>                      | 0.37 <i>10</i>           | 4903.374            | 2938.63  |                                                                                                                                                                                                                                                                                                                                                |
| 1991.27 <i>5</i>                     | 0.54 <i>6</i>            | 1991.28             | 0        |                                                                                                                                                                                                                                                                                                                                                |
| 2022.954 <i>9</i>                    | 11 <i>1</i>              | (6986.099)          | 4963.084 | $E_\gamma$ : others: 2022.80 <i>20</i> ( <a href="#">1972Dz13</a> ) and 2023.0 <i>10</i> ( <a href="#">1985Ke08</a> ).<br>$I_\gamma$ : weighted average of 12.0 <i>15</i> ( <a href="#">1972Dz13</a> ), 11 <i>1</i> ( <a href="#">1985Ke08</a> ), 11.4 <i>10</i> ( <a href="#">1985Ra15</a> ), and 9.1 <i>12</i> ( <a href="#">1997Be42</a> ). |
| 2082.83 <i>17</i>                    | 15.9 <i>15</i>           | (6986.099)          | 4903.374 | $E_\gamma$ : unweighted average of 2082.65 <i>16</i> ( <a href="#">1972Dz13</a> ), 2083.17 <i>7</i> ( <a href="#">1985Ke08</a> ), and 2082.681 <i>12</i> ( <a href="#">1985Ra15</a> ).                                                                                                                                                         |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{n},\gamma)$  E=thermal **1985Ra15** (continued) $\gamma(^{35}\text{S})$  (continued)

| $E_\gamma$ †                                                | $I_\gamma$ ‡#                                   | $E_i(\text{level})$                                      | $E_f$                                           | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------------------------------|-------------------------------------------------|----------------------------------------------------------|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $^{*}2184.16^{+}_{-19}$                                     | 4.7 5                                           |                                                          |                                                 | <p><math>I_\gamma</math>: weighted average of 18.0 15 (1972Dz13), 17 2 (1985Ke08), 15.6 17 (1985Ra15), and 12.6 17 (1997Be42).</p> <p><math>E_\gamma, I_\gamma</math>: from 1985Ke08. 1985Ke08 assigned this <math>\gamma</math> ray as a 4903 to 2719 transition. The final level is closest to the known 2717 level. This <math>\gamma</math> ray has not been observed in any other <math>^{34}\text{S}(\text{n},\gamma)</math> measurements. Including this <math>\gamma</math> ray in the level scheme would result in a significant intensity imbalance at its initial level 4903 and its final level 2717 and also lead to poor fitting in the level scheme. Evaluators therefore consider the placement of this <math>\gamma</math> ray unlikely.</p> |
| 2229.510 16<br>2347.700 15                                  | 2.86 27<br>49 4                                 | 3801.955<br>2347.782                                     | 1572.373<br>0                                   | <p><math>E_\gamma</math>: weighted average of 2347.50 10 (1972Dz13), 2347.71 3 (1985Ke08), and 2347.701 11 (1985Ra15).</p> <p><math>I_\gamma</math>: weighted average of 54 5 (1972Dz13), 49 4 (1985Ke08), 50 5 (1985Ra15), and 40 6 (1997Be42).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 2508.39 8<br>2555.492 14                                    | 0.38 4<br>3.14 31                               | (6986.099)<br>4903.374                                   | 4477.62<br>2347.782                             | <p><math>E_\gamma</math>: other: 2555.9 4 (1972Dz13).</p> <p><math>I_\gamma</math>: weighted average of 3.4 7 (1972Dz13) and 3.09 31 (1985Ra15).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 2615.2 2                                                    | 1.09 10                                         | 4963.084                                                 | 2347.782                                        | <p><math>E_\gamma</math>: other: 2614.3 3 doublet (1972Dz13).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 2616.8 13                                                   | 0.24 7                                          | 4189.243                                                 | 1572.373                                        | <p><math>I_\gamma</math>: other: &lt;1.3 4 (1972Dz13).</p> <p><math>E_\gamma</math>: other: 2614.3 3 doublet (1972Dz13).</p> <p><math>I_\gamma</math>: other: &lt;1.3 4 (1972Dz13).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 2716.99 16<br>2796.74 4                                     | 0.30 5<br>4.9 4                                 | 2717.05<br>(6986.099)                                    | 0<br>4189.243                                   | <p><math>E_\gamma</math>: weighted average of 2796.83 13 (1985Ke08) and 2796.73 4 (1985Ra15). Other: 2796.6 5 (1972Dz13).</p> <p><math>I_\gamma</math>: weighted average of 6.1 8 (1972Dz13), 4.3 4 (1985Ke08), 5.4 5 (1985Ra15), and 4.9 7 (1997Be42).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 2905.1 4<br>2938.58 11<br>2972.0 4<br>3139.9 5<br>3183.95 4 | 0.14 4<br>0.89 9<br>0.18 6<br>0.078 24<br>6.1 6 | 4477.62<br>2938.63<br>4963.084<br>6078.477<br>(6986.099) | 1572.373<br>0<br>1991.28<br>2938.63<br>3801.955 | <p><math>E_\gamma</math>: weighted average of 3184.30 20 (1972Dz13), 3183.94 7 (1985Ke08), and 3183.94 4 (1985Ra15).</p> <p><math>I_\gamma</math>: weighted average of 7.0 9 (1972Dz13), 5.7 6 (1985Ke08), 6.2 6 (1985Ra15), and 5.7 8 (1997Be42).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 3330.84 13                                                  | 7.6 7                                           | 4903.374                                                 | 1572.373                                        | <p><math>E_\gamma</math>: unweighted average of 3331.08 15 (1972Dz13), 3330.64 6 (1985Ke08), and 3330.80 4 (1985Ra15).</p> <p><math>I_\gamma</math>: weighted average of 7.8 10 (1972Dz13), 7.7 8 (1985Ke08), and 7.4 7 (1985Ra15).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 3390.62 7                                                   | 5.6 5                                           | 4963.084                                                 | 1572.373                                        | <p><math>E_\gamma</math>: weighted average of 3390.86 20 (1972Dz13), 3390.73 7 (1985Ke08), and 3390.55 5 (1985Ra15).</p> <p><math>I_\gamma</math>: weighted average of 5.5 8 (1972Dz13), 5.7 6 (1985Ke08), and 5.5 5 (1985Ra15).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 3558.1 5<br>3801.73 4                                       | 0.085 17<br>2.59 23                             | 3558.082<br>3801.955                                     | 0<br>0                                          | <p><math>E_\gamma</math>: weighted average of 3802.0 3 (1972Dz13), 3801.49 14 (1985Ke08), and 3801.74 3 (1985Ra15).</p> <p><math>I_\gamma</math>: weighted average of 3.6 7 (1972Dz13), 2.3 2 (1985Ke08), 3.09 31 (1985Ra15), and 2.6 4 (1997Be42).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 4105.3 8<br>4188.95 5                                       | 0.048 17<br>2.54 27                             | 4105.6<br>4189.243                                       | 0<br>0                                          | <p><math>E_\gamma</math>: weighted average of 4188.80 13 (1985Ke08) and 4188.96 4 (1985Ra15). Other: 4189.6 3 (1972Dz13).</p> <p><math>I_\gamma</math>: weighted average of 1.9 4 (1972Dz13), 2.72 27 (1985Ra15), and 2.8 4 (1997Be42). Other: 24 2 (1985Ke08).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 4268.65 14                                                  | 0.34 4                                          | 5841.485                                                 | 1572.373                                        | <p><math>E_\gamma</math>: other: 4268.4 7 (assigned as 6986 to 2717 in 1972Dz13). The placement of 4268<math>\gamma</math> is changed from 6986 to 2717 (1985Ra15) to 5841 to 1572 (S. Raman to P.M. Endt, private communication, mentioned in the</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{n},\gamma)$  E=thermal **1985Ra15** (continued) $\gamma(^{35}\text{S})$  (continued)

| $E_\gamma^\dagger$                  | $I_\gamma^{\ddagger\#}$ | $E_i(\text{level})$ | $E_f$    | Comments                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------|-------------------------|---------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Adopted Levels of <b>2011Ch48</b> . |                         |                     |          |                                                                                                                                                                                                                                                                                                                                      |
| 4638.14 14                          | 56 5                    | (6986.099)          | 2347.782 | $I_\gamma$ : other: 0.4 2 ( <b>1972Dz13</b> ).<br>$E_\gamma$ : unweighted average of 4638.4 2 ( <b>1972Dz13</b> ), 4638.10 3 ( <b>1985Ke08</b> ), and 4637.91 4 ( <b>1985Ra15</b> ).<br>$I_\gamma$ : weighted average of 56 6 ( <b>1972Dz13</b> ), 59 5 ( <b>1985Ke08</b> ), 55 5 ( <b>1985Ra15</b> ), and 50 7 ( <b>1997Be42</b> ). |
| 4902.99 5                           | 4.0 4                   | 4903.374            | 0        | $E_\gamma$ : weighted average of 4903.07 7 ( <b>1985Ke08</b> ) and 4902.96 4 ( <b>1985Ra15</b> ).<br>Other: 4903.4 3 ( <b>1972Dz13</b> ).<br>$I_\gamma$ : weighted average of 3.2 8 ( <b>1972Dz13</b> ), 4.3 4 ( <b>1985Ke08</b> ), 3.7 4 ( <b>1985Ra15</b> ), and 4.2 6 ( <b>1997Be42</b> ).                                        |
| 4963.06 22                          | 2.62 27                 | 4963.084            | 0        | $E_\gamma$ : unweighted average of 4963.2 5 ( <b>1972Dz13</b> ), 4963.34 10 ( <b>1985Ke08</b> ), and 4962.63 5 ( <b>1985Ra15</b> ).<br>$I_\gamma$ : weighted average of 2.6 8 ( <b>1972Dz13</b> ), 2.92 27 ( <b>1985Ra15</b> ), and 2.18 33 ( <b>1997Be42</b> ). Other: 29 3 ( <b>1985Ke08</b> ).                                    |
| 5752.0 8                            | 0.018 5                 | 5752.5              | 0        | $E_\gamma$ : other: 6078.2 10 ( <b>1972Dz13</b> ).<br>$I_\gamma$ : weighted average of 0.3 1 ( <b>1972Dz13</b> ) and 0.36 4 ( <b>1985Ra15</b> ).                                                                                                                                                                                     |
| 6018.2 6                            | 0.020 7                 | 6018.8              | 0        |                                                                                                                                                                                                                                                                                                                                      |
| 6077.87 11                          | 0.35 4                  | 6078.477            | 0        |                                                                                                                                                                                                                                                                                                                                      |
| 6293.2 4                            | 0.065 14                | 6293.93             | 0        |                                                                                                                                                                                                                                                                                                                                      |
| 6355.0 6                            | 0.046 7                 | 6354.89             | 0        |                                                                                                                                                                                                                                                                                                                                      |
| 6419.3 11                           | 0.016 5                 | 6419.9              | 0        |                                                                                                                                                                                                                                                                                                                                      |
| 6628.5 6                            | 0.030 6                 | 6629.43             | 0        |                                                                                                                                                                                                                                                                                                                                      |
| 6760.3 12                           | 0.019 8                 | 6761.0              | 0        |                                                                                                                                                                                                                                                                                                                                      |
| 6985.7 10                           | 0.036 8                 | (6986.099)          | 0        |                                                                                                                                                                                                                                                                                                                                      |

 $^\dagger$  From **1985Ra15**, unless otherwise noted. $^\ddagger$  Not observed in **2007ChZX** (PGAA).

# Intensity per 100 neutron captures.

@ Multiply placed with undivided intensity.

 $^x$   $\gamma$  ray not placed in level scheme.

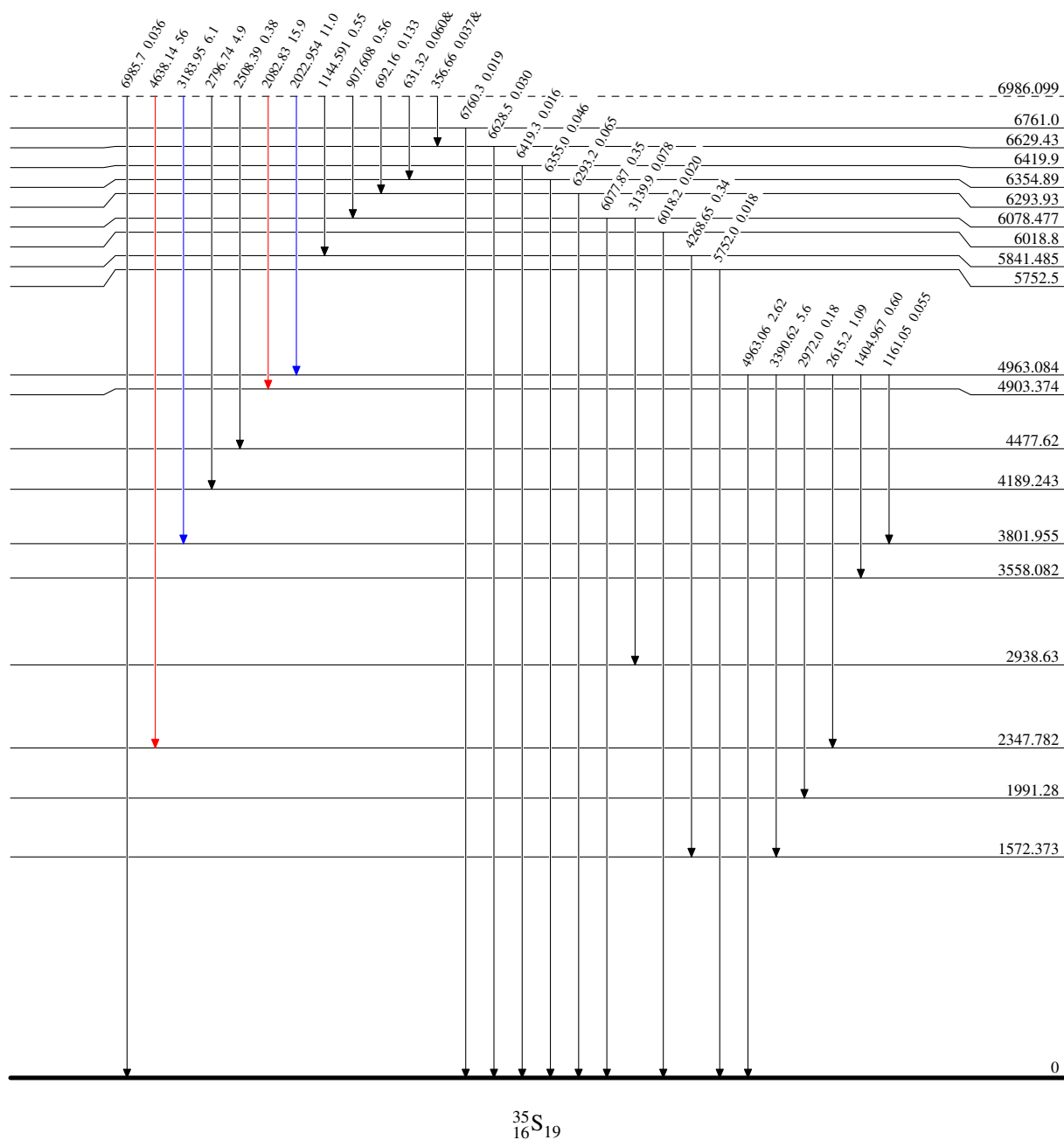
$^{34}\text{S}(n,\gamma) \text{E=thermal}$  1985Ra15

## Level Scheme

Intensities:  $I_\gamma$  per 100 neutron captures  
& Multiply placed: undivided intensity given

## Legend

—→  $I_\gamma < 2\% \times I_\gamma^{\max}$   
—→  $I_\gamma < 10\% \times I_\gamma^{\max}$   
—→  $I_\gamma > 10\% \times I_\gamma^{\max}$



<sup>34</sup>S(n,γ) E=thermal 1985Ra15

Level Scheme (continued)

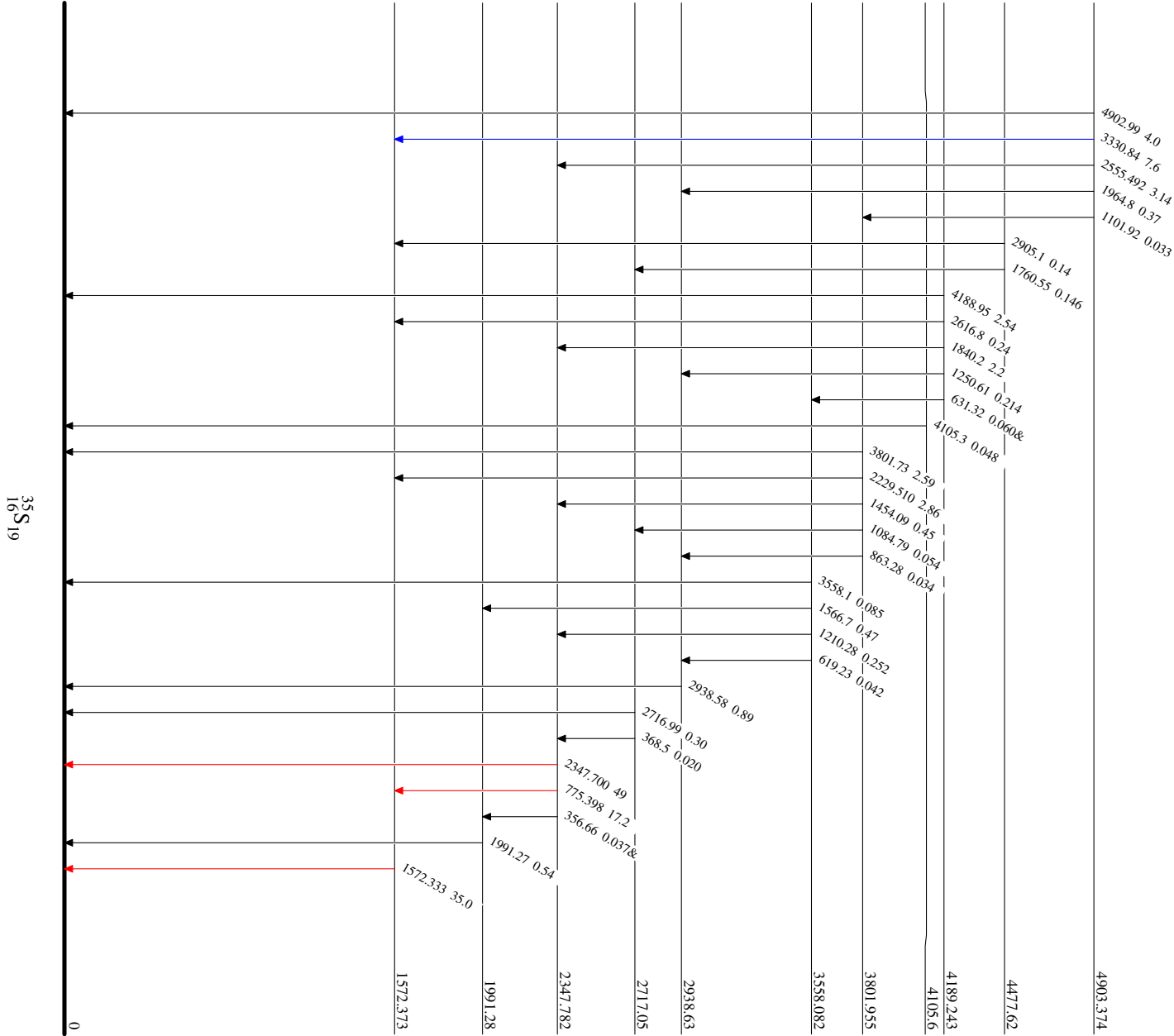
Intensities: I<sub>γ</sub> per 100 neutron captures  
& Multiply placed: undivided intensity given

Legend

I<sub>γ</sub> < 2% × I<sub>γ</sub><sup>max</sup>

I<sub>γ</sub> < 10% × I<sub>γ</sub><sup>max</sup>

I<sub>γ</sub> > 10% × I<sub>γ</sub><sup>max</sup>



$^{34}\text{S}(\text{n},\text{n}),(\text{n},\gamma):\text{resonances}$  1984Ca14

**1984Ca14:** 30-1500-keV neutrons were produced from the Oak Ridge Electron Linear Accelerator (ORELA). The target was 94.33%-enriched  $^{34}\text{S}$  with a 5.54% impurity of  $^{32}\text{S}$ . Neutrons were detected using a 2-cm-thick and 7.5-cm in diameter NE110 proton-recoil detector. Prompt  $\gamma$  rays from neutron capture were detected using two nonhydrogenous fluorocarbon scintillators (NE-226). Measured  $\sigma(E_n)$ . Deduced 81 resonances, J,  $\pi$ , L, widths and spectroscopic factors by comparing the R-matrix predictions of the cross section with the data.

**2018MuZY:** Atlas of Neutron Resonances. Evaluation of neutron resonance energies, spins, width parameters, and resonance strengths for nuclei of Z=1-60.

 $^{35}\text{S}$  Levels

J $^\pi$ , L, and resonance parameters  $E_n$ ,  $g\Gamma_n$ ,  $\Gamma_\gamma$ , and  $g\Gamma_n\Gamma_\gamma/\Gamma$  are from **1984Ca14** due to significant discrepancies, including orders of magnitude inconsistencies, between the **1984Ca14** original data and **2018MuZY** evaluation.  $g\Gamma_n=(2J+1)\Gamma_n/2$ .

| <u>E(level)<sup>†</sup></u> | <u>J<math>^\pi</math></u> | <u><math>g\Gamma_n\Gamma_\gamma/\Gamma</math></u> | <u>L</u> | <u><math>E_n(\text{lab})</math> (keV)</u> | <u>Comments</u>                                                                                                      |
|-----------------------------|---------------------------|---------------------------------------------------|----------|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| 6985.54?                    |                           |                                                   | 0        | -0.305                                    | E(level): fictitious level from analysis of resonance spectrum below the S(n) threshold.<br>$\Gamma_\gamma=0.38$ eV. |
| 7018.89 4                   |                           | 0.025 eV 1                                        | 2        | 34.03 1                                   |                                                                                                                      |
| 7072.66 5                   |                           | 0.14 eV 4                                         |          | 89.40 3                                   |                                                                                                                      |
| 7074.33 6                   |                           | 0.084 eV 3                                        |          | 91.12 4                                   |                                                                                                                      |
| 7097.72 30                  |                           | 0.009 eV 3                                        | 2        | 115.2 3                                   |                                                                                                                      |
| 7099.85 20                  |                           | 0.033 eV 5                                        | 2        | 117.4 2                                   | $g\Gamma_n<1$ eV.                                                                                                    |
| 7100.73 4                   | 3/2 <sup>-</sup>          |                                                   |          | 118.30 1                                  | $g\Gamma_n=372$ eV 3; $\Gamma_\gamma=0.70$ eV 4.                                                                     |
| 7143.27 11                  |                           | 0.42 eV 1                                         |          | 162.1 1                                   | $g\Gamma_n<1$ eV.                                                                                                    |
| 7210.42 5                   |                           |                                                   | 1,2      | 231.25 2                                  | $g\Gamma_n=62$ eV 4; $\Gamma_\gamma=0.30$ eV 2.                                                                      |
| 7218.0 4                    |                           | 0.33 eV 2                                         |          | 239.0 4                                   | $g\Gamma_n<4$ eV.                                                                                                    |
| 7234.0 5                    |                           | 0.19 eV 1                                         |          | 255.5 5                                   | $g\Gamma_n<4$ eV.                                                                                                    |
| 7239.8 6                    |                           | 0.06 eV 2                                         |          | 261.5 6                                   | $g\Gamma_n<2$ eV.                                                                                                    |
| 7253.2 6                    |                           | 0.35 eV 2                                         |          | 275.3 6                                   | $g\Gamma_n<1$ eV.                                                                                                    |
| 7275.93 5                   | 1/2 <sup>+</sup>          |                                                   |          | 298.70 3                                  | $g\Gamma_n=9600$ eV 150; $\Gamma_\gamma=2.82$ eV 67.                                                                 |
| 7279.13?                    |                           | 0.10 eV 3                                         |          | 302                                       |                                                                                                                      |
| 7289.8 8                    |                           | 0.31 eV 3                                         |          | 313.0 8                                   | $g\Gamma_n<6$ eV.                                                                                                    |
| 7294.19 6                   | 3/2 <sup>-</sup>          |                                                   |          | 317.51 5                                  | $g\Gamma_n=2500$ eV 25; $\Gamma_\gamma=0.23$ eV 8.                                                                   |
| 7331.00 5                   | 1/2 <sup>+</sup>          |                                                   |          | 355.41 2                                  | $g\Gamma_n=4810$ eV 80; $\Gamma_\gamma=0.30$ eV 30.                                                                  |
| 7338.18 20                  |                           |                                                   | 1,2      | 362.8 2                                   | $g\Gamma_n=230$ eV 10; $\Gamma_\gamma=1.06$ eV 7.                                                                    |
| 7343.91 20                  |                           | 0.72 eV 3                                         |          | 368.7 2                                   | $g\Gamma_n<2$ eV.                                                                                                    |
| 7347.2 5                    |                           | 0.44 eV 6                                         |          | 372.1 5                                   | $g\Gamma_n<2$ eV.                                                                                                    |
| 7347.9 5                    |                           | 0.30 eV 5                                         |          | 372.8 5                                   | $g\Gamma_n<2$ eV.                                                                                                    |
| 7353.9 5                    |                           | 0.44 eV 4                                         |          | 379.0 5                                   | $g\Gamma_n<10$ eV.                                                                                                   |
| 7357.6 5                    |                           | 0.16 eV 4                                         |          | 382.8 5                                   | $g\Gamma_n<6$ eV.                                                                                                    |
| 7370.56 6                   | 1/2 <sup>-</sup>          |                                                   |          | 396.14 4                                  | $g\Gamma_n=6430$ eV 150; $\Gamma_\gamma=3.08$ eV 20.                                                                 |
| 7371.97?                    |                           | 0.12 eV                                           |          | 397.6                                     |                                                                                                                      |
| 7396.0 6                    |                           | 0.44 eV 33                                        | 2        | 422.3 6                                   |                                                                                                                      |
| 7404.6 6                    |                           | 0.21 eV 5                                         |          | 431.2 6                                   | $g\Gamma_n<1$ eV.                                                                                                    |
| 7408.68 11                  | 3/2 <sup>-</sup>          |                                                   |          | 435.4 1                                   | $g\Gamma_n=1570$ eV 40; $\Gamma_\gamma=0.91$ eV 8.                                                                   |
| 7411.55 6                   |                           |                                                   |          | 438.35 5                                  | $g\Gamma_n=49$ eV 5.                                                                                                 |
| 7416.50 4                   |                           |                                                   |          | 443.45 1                                  | $g\Gamma_n=80$ eV 8; $\Gamma_\gamma=0.51$ eV 5.                                                                      |
| 7429.1 6                    |                           | 0.12 eV 3                                         |          | 456.4 6                                   | $g\Gamma_n<6$ eV; $\Gamma_\gamma=0.12$ eV 3.                                                                         |
| 7433.56 5                   |                           |                                                   |          | 461.01 2                                  | $g\Gamma_n=50$ eV 5; $\Gamma_\gamma=0.035$ eV 20.                                                                    |
| 7436.4 4                    |                           |                                                   | 1,2      | 463.9 4                                   | $g\Gamma_n=296$ eV 30; $\Gamma_\gamma=0.28$ eV 3.                                                                    |
| 7442.14 5                   | 1/2 <sup>+</sup>          |                                                   |          | 469.85 2                                  | $g\Gamma_n=1050$ eV 35; $\Gamma_\gamma=0.37$ eV 8.                                                                   |
| 7462.2 6                    |                           | 0.33 eV 4                                         |          | 490.5 6                                   | $g\Gamma_n=50$ eV 10.                                                                                                |
| 7481.15 6                   | 1/2 <sup>-</sup>          |                                                   |          | 510.02 4                                  | $g\Gamma_n=375$ eV 25; $\Gamma_\gamma=2.1$ eV 1.                                                                     |
| 7494.5 6                    |                           | 0.17 eV 4                                         |          | 523.8 6                                   | $g\Gamma_n<4$ eV.                                                                                                    |
| 7542.3 6                    |                           | 0.21 eV 3                                         |          | 573.0 6                                   | $g\Gamma_n<1$ eV.                                                                                                    |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{n,n}),(\text{n},\gamma)$ :resonances **1984Ca14** (continued) $^{35}\text{S}$  Levels (continued)

| $E(\text{level})^\dagger$ | $J^\pi$   | $g\Gamma_n\Gamma_\gamma/\Gamma$ | $L$ | $E_n(\text{lab})$ (keV) | Comments                                                  |
|---------------------------|-----------|---------------------------------|-----|-------------------------|-----------------------------------------------------------|
| 7604.0 6                  |           | 0.90 eV 8                       |     | 636.5 6                 | $g\Gamma_n < 10$ eV.                                      |
| 7609.26 5                 | $3/2^-$   |                                 |     | 641.93 3                | $g\Gamma_n = 2520$ eV 60; $\Gamma_\gamma = 0.51$ eV 6.    |
| 7648.9 6                  |           | 0.21 eV 3                       |     | 682.7 6                 | $g\Gamma_n < 30$ eV.                                      |
| 7655.7 6                  |           | 0.80 eV 5                       |     | 689.7 6                 | $g\Gamma_n < 10$ eV.                                      |
| 7663.91 11                |           |                                 | 1,2 | 698.2 1                 | $g\Gamma_n = 305$ eV 50; $\Gamma_\gamma = 0.80$ eV 8.     |
| 7678.3 7                  |           | 0.45 eV 7                       |     | 713.0 7                 | $g\Gamma_n < 4$ eV.                                       |
| 7731.0 7                  |           | 0.31 eV 13                      |     | 767.3 7                 | $g\Gamma_n < 4$ eV.                                       |
| 7749.7 4                  |           |                                 |     | 786.5 4                 | $g\Gamma_n = 18$ eV 6; $\Gamma_\gamma = 2.65$ eV 18.      |
| 7761.46 7                 | $3/2^-$   |                                 |     | 798.65 6                | $g\Gamma_n = 25350$ eV 300; $\Gamma_\gamma = 1.52$ eV 50. |
| 7776.17 11                | $1/2^-$   |                                 |     | 813.8 1                 | $g\Gamma_n = 860$ eV 60; $\Gamma_\gamma = 0.19$ eV 13.    |
| 7797.99 9                 | $1/2^+$   |                                 |     | 836.27 8                | $g\Gamma_n = 3600$ eV 160; $\Gamma_\gamma = 0.39$ eV 18.  |
| 7812.24 7                 | $3/2^+$   |                                 |     | 850.94 6                | $g\Gamma_n = 2650$ eV 110; $\Gamma_\gamma = 0.24$ eV 19.  |
| 7853.25 7                 | $3/2^-$   |                                 |     | 893.17 6                | $g\Gamma_n = 4950$ eV 160; $\Gamma_\gamma = 0.38$ eV 16.  |
| 7861.8 12                 |           | 1.13 eV 19                      |     | 902.0 12                | $g\Gamma_n < 3$ eV.                                       |
| 7880.8 12                 |           | 0.67 eV 17                      |     | 921.5 12                | $g\Gamma_n < 20$ eV.                                      |
| 7889.0 12                 |           | 0.57 eV 25                      |     | 930.0 12                | $g\Gamma_n < 20$ eV.                                      |
| 7894.63 7                 | $3/2^-$   |                                 |     | 935.78 6                | $g\Gamma_n = 4100$ eV 160; $\Gamma_\gamma = 3.48$ eV 23.  |
| 7899.70 30                |           |                                 |     | 941.0 3                 | $g\Gamma_n = 176$ eV 16; $\Gamma_\gamma = 0.5$ eV 3.      |
| 7934.1 15                 |           | 0.5 eV 3                        |     | 976.4 15                | $g\Gamma_n < 2$ eV.                                       |
| 7939.5 15                 |           | 0.89 eV 4                       |     | 982.0 15                | $g\Gamma_n < 8$ eV.                                       |
| 7954.96 11                | $3/2^-$   |                                 |     | 997.9 1                 | $g\Gamma_n = 930$ eV 80; $\Gamma_\gamma = 3.51$ eV 19.    |
| 7974.19 11                | $3/2^-$   |                                 |     | 1017.7 1                | $g\Gamma_n = 4020$ eV 180; $\Gamma_\gamma = 1.22$ eV 17.  |
| 8019.5 2                  |           |                                 | 2   | 1064.4 2                | $g\Gamma_n = 1032$ eV 90.                                 |
| 8041.20 20                |           |                                 | 2   | 1086.7 2                | $g\Gamma_n = 2650$ eV 160.                                |
| 8076.46 20                |           |                                 | 2   | 1123.0 2                | $g\Gamma_n = 5480$ eV 388.                                |
| 8077.91 30                |           |                                 | 2   | 1124.5 3                | $g\Gamma_n = 4730$ eV 320.                                |
| 8093.0 5                  | $1/2^-$   |                                 |     | 1140.0 5                | $g\Gamma_n = 34880$ eV 830.                               |
| 8143.66 20                |           |                                 | 2   | 1192.2 2                | $g\Gamma_n = 5510$ eV 290.                                |
| 8187.17 30                | $1/2^-$   |                                 |     | 1237.0 3                | $g\Gamma_n = 2945$ eV 230.                                |
| 8211.25 30                |           |                                 |     | 1261.8 3                | $g\Gamma_n = 975$ eV 86.                                  |
| 8224.2 4                  | $1/2^-$   |                                 |     | 1275.1 4                | $g\Gamma_n = 2170$ eV 180.                                |
| 8228.3 4                  |           |                                 | 2   | 1279.4 4                | $g\Gamma_n = 1140$ eV 100.                                |
| 8243.98 30                |           |                                 | 2   | 1295.5 3                | $g\Gamma_n = 1980$ eV 150.                                |
| 8256.31 20                | $3/2^-$   |                                 |     | 1308.2 2                | $g\Gamma_n = 12475$ eV 770.                               |
| 8262.4 4                  | $5/2^+$   |                                 |     | 1314.5 4                | $g\Gamma_n = 6240$ eV 500.                                |
| 8273.3 4                  | $(1/2^-)$ |                                 |     | 1325.7 4                | $g\Gamma_n = 1635$ eV 145.                                |
| 8298.27 30                |           |                                 | 2   | 1351.4 3                | $g\Gamma_n = 14470$ eV 650.                               |
| 8333.8 4                  | $3/2^-$   |                                 |     | 1388.0 4                | $g\Gamma_n = 28845$ eV 1057.                              |
| 8335.9 4                  |           |                                 | 2   | 1390.1 4                | $g\Gamma_n = 1955$ eV 170.                                |
| 8391.3 4                  | $3/2^-$   |                                 |     | 1447.2 4                | $g\Gamma_n = 27100$ eV 1390.                              |
| 8393.64 30                |           |                                 | 2   | 1449.6 3                | $g\Gamma_n = 3025$ eV 260.                                |
| 8406.55 30                |           |                                 | 2   | 1462.9 3                | $g\Gamma_n = 3885$ eV 345.                                |
| 8418.8 4                  |           |                                 | 2   | 1475.5 4                | $g\Gamma_n = 3795$ eV 340.                                |

 $^\dagger$  From  $E_{\text{c.m.}} + S(\text{n})(^{35}\text{S})$  where  $S(\text{n})(^{35}\text{S}) = 6985.84$  4 (2021Wa16) and  $E_{\text{c.m.}} = E_n(\text{lab}) \times m(^{34}\text{S}) / [m_n + m(^{34}\text{S})]$ .

$^{34}\text{S}(\text{pol d,p})$  1977Ab07 $J^\pi=0^+$  for  $^{34}\text{S}$  g.s.

1977Ab07: A vector-polarized 11.8-MeV deuteron beam was produced from the University of Wisconsin Lamb-shift polarized ion source and tandem accelerator. The target was a 90% enriched  $^{34}\text{S}$ , 610- $\mu\text{g}/\text{cm}^2$  thick, evaporated onto a 100- $\mu\text{g}/\text{cm}^2$  gold foil and covered by a 50- $\mu\text{g}/\text{cm}^2$  gold layer. The reaction products were detected using four  $\Delta E$ -E counter telescopes of freon-cooled surface-barrier and Si(Li) detectors with FWHM $\approx$ 60 keV. Measured angular distributions of cross sections  $\sigma(E_p, \theta)$  and angular distributions of analyzing powers  $iT_{11}(E_p, \theta)$ . Deduced levels, J,  $\pi$ , L-transfers, and spectroscopic factors from finite-range, non-locality corrected DWUCK-DWBA analysis of  $\sigma(E_p, \theta)$  and  $iT_{11}(E_p, \theta)$ . Also see 1976AbZX.

 $^{35}\text{S}$  Levels

Spectroscopic factor  $C^2S = \sigma(\theta)_{\text{exp}} / \sigma(\theta)_{\text{DWBA}} / N$ , where  $N=1.53$  is a normalization factor (1977Ab07).

| E(level) <sup>†</sup> | $J^\pi$          | L <sup>@</sup> | $C^2S$ <sup>@</sup>   | Comments                                                                                                                                                                                                                                                                                       |
|-----------------------|------------------|----------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0                     | $3/2^+ \ddagger$ | 2              | 0.56, 0.48 $\ddagger$ |                                                                                                                                                                                                                                                                                                |
| 1572                  | $1/2^+ \#$       | 0              | 0.27, 0.18 $\#$       |                                                                                                                                                                                                                                                                                                |
| 1992                  | $7/2^- \#$       | 3              | 1.16, 0.73 $\#$       |                                                                                                                                                                                                                                                                                                |
| 2348                  | $3/2^- \#$       | 1              | 0.56, 0.46 $\#$       |                                                                                                                                                                                                                                                                                                |
| 2718                  | $5/2^+ \#$       | 2              | 0.03, 0.02 $\#$       |                                                                                                                                                                                                                                                                                                |
| 2939                  | $3/2^+, 5/2^+$   | 2              | 0.10, 0.06            | $J^\pi$ : 1977Ab07 states that the analyzing power seems to favor $5/2^+$ but they do not think that the evidence is strong enough to support an unambiguous spin-parity assignment.<br>$C^2S$ : extracted from an incoherent superposition of direct and compound cross sections in 1977Ab07. |
| 3421                  | $5/2^+ \#$       | 2              | 0.04, 0.03 $\#$       |                                                                                                                                                                                                                                                                                                |
| 3563                  |                  |                |                       |                                                                                                                                                                                                                                                                                                |
| 3802                  | $3/2^- \#$       | 1              | 0.10, 0.08 $\#$       |                                                                                                                                                                                                                                                                                                |
| 3818                  |                  |                |                       |                                                                                                                                                                                                                                                                                                |
| 4190                  | $1/2^- \ddagger$ | 1              | 0.15, 0.13 $\ddagger$ |                                                                                                                                                                                                                                                                                                |
| 4482                  | $7/2^- \#$       | 3              | 0.06, 0.05 $\#$       |                                                                                                                                                                                                                                                                                                |
| 4575                  |                  |                |                       |                                                                                                                                                                                                                                                                                                |
| 4837                  |                  |                |                       |                                                                                                                                                                                                                                                                                                |
| 4904                  | $1/2^- \ddagger$ | 1              | 0.48, 0.41 $\ddagger$ |                                                                                                                                                                                                                                                                                                |
| 4963                  | $3/2^- \#$       | 1              | 0.19, 0.17 $\#$       |                                                                                                                                                                                                                                                                                                |
| 5058                  | $7/2^- \#$       | 3              | 0.03, 0.03 $\#$       |                                                                                                                                                                                                                                                                                                |
| 5126                  |                  |                |                       |                                                                                                                                                                                                                                                                                                |

<sup>†</sup> As given in 1977Ab07, originally from 1973EnVA rounded to the nearest integer.

$\ddagger$  L-1/2 transfer from analyzing power measurements.

$\#$  L+1/2 transfer from analyzing power measurements.

<sup>@</sup> From DWUCK-DWBA analysis of the measured  $\sigma(\theta)$  in 1977Ab07.  $C^2S$  calculations use two sets of optical model potentials. Set 1: the deuteron elastic potential and the Becchetti-Greenlees proton parameters; Set 2: Adiabatic model parameters and the Becchetti-Greenlees proton parameters.



$^{34}\text{S}(\text{d},\text{p})$  2024Ku24,1971Va18,1969Mo12

$J^\pi=0^+$  for  $^{34}\text{S}$  g.s.

**2024Ku24:** An 8-MeV/nucleon deuteron beam was produced from the 9-MV Super-FN tandem Van de Graaff accelerator at FSU.

The target was carbon-backed sulfur enriched in  $^{34}\text{S}$ . Protons were spatially dispersed by the Super-Enge Split-Pole Spectrograph and detected using a position-sensitive ionization chamber ( $\Delta E$ ) and a plastic scintillator (E) at the focal plane with FWHM $\approx$ 45 keV. An additional  $^{34}\text{S}(\text{d},\text{p}\gamma)^{35}\text{S}$  measurement using the CeBr<sub>3</sub> Array (CeBrA) demonstrator provides information on some overlapping peaks produced from the  $^{32}\text{S}$  contaminants in the target. Measured  $\sigma(E_{\text{p}},\theta)$ . Deduced 23 levels, single-neutron spectroscopic factors and strengths from finite-range PTOLEMY-DWBA analysis.

**1971Va18:** A 10-MeV deuteron beam was produced from the Utrecht 6-MV tandem accelerator. Targets were 100  $\mu\text{g}/\text{cm}^2$  PbS on carbon plus formvar foils and 5  $\mu\text{g}/\text{cm}^2$  pure  $^{34}\text{S}$  embedded in aluminum foil. Protons were detected using one 15-mm long and seven 30-mm long, 0.6-mm thick position-sensitive detectors (PSD) in the focal plane of the Utrecht split-pole magnetic spectrograph. Measured  $\sigma(E_{\text{p}},\theta)$ . Deduced 17 levels, L-transfers, J,  $\pi$ , spectroscopic factors from DWUCK-DWBA analysis for the ground state and seven excited states of  $^{35}\text{S}$ .

**1969Mo12:** A 6.495-MeV deuteron beam was produced from the ONR-CIT tandem accelerator. The target was an enriched target of 450- $\mu\text{g}/\text{cm}^2$  25 CdS (85%  $^{34}\text{S}$ ) evaporated onto a 301- $\mu\text{g}/\text{cm}^2$  15 gold foil and a natural target of 289- $\mu\text{g}/\text{cm}^2$  30 Sb<sup>2</sup>S<sup>3</sup> evaporated onto a 289- $\mu\text{g}/\text{cm}^2$  30 gold foil. Protons were detected using an array of 16 Au-Si surface barrier detectors with FWHM=30 keV. Measured  $\sigma(E_{\text{p}},\theta)$ . Deduced 37 levels.

**1984Pi03:** A 12.3-MeV deuteron beam was produced from a cyclotron. The target was CdS (61.9% in  $^{36}\text{S}$ , 37.7% in  $^{34}\text{S}$ ), 100  $\mu\text{g}/\text{cm}^2$  on a 20  $\mu\text{g}/\text{cm}^2$  carbon backing. Protons were analyzed by a multi-angle magnetic spectrograph and detected using 700-mm long nuclear emulsion plates Ilford L4. Measured  $\sigma(E_{\text{p}},\theta)$ . Deduced 13 levels, L-transfers, J,  $\pi$ , and spectroscopic factors from DWUCK-DWBA analysis for six excited states of  $^{35}\text{S}$ .

**1979So01:** A 3.55-MeV deuteron beam was produced from a Van de Graaff electrostatic generator. The target was 18- $\mu\text{g}/\text{cm}^2$  GeS (80%  $^{36}\text{S}$ , 20%  $^{34}\text{S}$ ) on a 25- $\mu\text{g}/\text{cm}^2$  carbon backing. Protons were analyzed by a magnetic spectrograph and recorded by a spark chamber with FWHM=10-15 keV. Measured  $\sigma(E_{\text{p}},\theta)$ . Deduced levels.

**1971Ko33:** A 6.6-MeV deuteron beam was produced from a cyclotron. Targets were GeS films of 70-80  $\mu\text{g}/\text{cm}^2$  (98% enrichment in  $^{34}\text{S}$ ) evaporated onto a 20  $\mu\text{g}/\text{cm}^2$  carbon substrate. Protons were detected using an 800- $\mu\text{m}$ -thick Si(Li). Measured  $\sigma(E_{\text{p}},\theta)$ . Deduced levels, J,  $\pi$ , L-transfers, and spectroscopic factors from distorted-wave analysis for the ground state and five excited states of  $^{35}\text{S}$ .

**1971Me12:** An 18-MeV deuteron beam was produced from the Yale MP tandem Van de Graaff accelerator. The target was H<sup>2</sup>S gas of natural isotopic composition (95.06%  $^{32}\text{S}$ , 4.18%  $^{34}\text{S}$ ). Protons were detected using a  $\Delta E$ -E telescope of silicon surface-barrier detectors. Measured  $\sigma(E_{\text{p}},\theta)$ . Deduced levels, J,  $\pi$ , L-transfers, and spectroscopic factors from local, zero-range JULIE-DWBA analysis of  $\sigma(E_{\text{p}},\theta)$  for the ground state and two excited states of  $^{35}\text{S}$ .

**1966Sc09:** 9- and 12-MeV deuteron beams were produced from the Argonne tandem accelerator. The target was PbS. Protons were detected using silicon surface barrier detectors. Measured  $\sigma(E_{\text{p}},\theta)$  for the ground state and one excited state of  $^{35}\text{S}$ .

**1958En51:** 6.006- and 6.542-MeV deuteron beams were produced from the MIT-ONR electrostatic generator. Targets were Sb<sup>2</sup>S<sup>3</sup> of natural isotopic constitution (4.2%  $^{34}\text{S}$ ). Energies of charged reaction products emitted from the target at angles of 50, 90, and 130° were measured by a broad-range magnetic spectrograph and nuclear emulsions. Measured  $\sigma(E_{\text{p}},\theta)$ . Deduced 5 levels.

Other measurements: **1976We29:**  $^{34}\text{S}(\text{d},\text{p})^{35}\text{S}$  angular distributions for the ground state and the first excited state of  $^{35}\text{S}$ .

**1973Co25:**  $^{34}\text{S}(\text{d},\text{p})^{35}\text{S}$  excitation functions for the two lowest excited states of  $^{35}\text{S}$ .

Theoretical calculations: **1977Os07**, **1974Os02**.

 $^{35}\text{S}$  Levels

C<sup>2</sup>S from **2024Ku24**. The normalization was determined from the L=3 strengths,  $G_{L=3}$ , summed up through  $E_x=5.1$  MeV and divided by the expected 1f<sub>7/2</sub> vacancy, i.e.,  $\Sigma G_{L=3}/(2j+1)$ , where  $j=7/2$ . This is equivalent to  $\Sigma C^2S_{L=3}=1$  up to 5.1 MeV. For C<sup>2</sup>S given under comments,  $C^2S=\sigma(\theta)_{\text{exp}}/\sigma(\theta)_{\text{DWBA}}/N/(2J_f+1)\times(2J_i+1)$ , where the normalization factor  $N=1.53$ . The C<sup>2</sup>S from **1971Me12** based on  $N=1.58$  has been renormalized by evaluators using  $N=1.53$ .

| E(level)                | $J^\pi$          | L <sup>a</sup> | C <sup>2</sup> S <sup>a</sup> | Comments                                                                                                                                                                                   |
|-------------------------|------------------|----------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0                       | 3/2 <sup>+</sup> | 2              | 0.54 12                       | L: 2 also from <b>1971Va18</b> , <b>1971Ko33</b> , and <b>1971Me12</b> .<br>C <sup>2</sup> S: others: 0.43 11 ( <b>1971Va18</b> ), 0.33 ( <b>1971Ko33</b> ), and 0.48 ( <b>1971Me12</b> ). |
| 1571.92 <sup>#</sup> 19 | 1/2 <sup>+</sup> | 0              | 0.14 7                        | L: 0 also from <b>1971Va18</b> , <b>1971Ko33</b> , and <b>1984Pi03</b> .                                                                                                                   |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{d,p})$  [2024Ku24,1971Va18,1969Mo12](#) (continued) $^{35}\text{S}$  Levels (continued)

| E(level)                | $J^\pi$                                | $L^a$ | $C^2S^a$   | Comments                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------|----------------------------------------|-------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1991.08 <sup>#</sup> 16 | 7/2 <sup>-</sup>                       | 3     | 0.87 9     | C <sup>2</sup> S: others: 0.17 4 ( <a href="#">1971Va18</a> ), 0.25 ( <a href="#">1971Ko33</a> ), and 0.154 ( <a href="#">1984Pi03</a> ).<br>C <sup>2</sup> S: others: 0.68 17 ( <a href="#">1971Va18</a> ), 0.38 8 ( <a href="#">1971Ko33</a> ), 0.65 10 ( <a href="#">1971Me12</a> ), and 0.841 12 ( <a href="#">1984Pi03</a> ).                         |
| 2347.59 <sup>#</sup> 15 | 3/2 <sup>-</sup>                       | 1     | 0.62 7     | E(level): others: 2347 8 ( <a href="#">1969Mo12</a> ) and 2336 10 ( <a href="#">1971Va18</a> ).<br>L: 1 also from <a href="#">1971Va18</a> and <a href="#">1984Pi03</a> .<br>C <sup>2</sup> S: others: 0.52 13 ( <a href="#">1971Va18</a> ), 0.52 ( <a href="#">1971Me12</a> ), 0.33 ( <a href="#">1971Ko33</a> ), and 0.505 ( <a href="#">1984Pi03</a> ). |
| 2724 8                  | 5/2 <sup>+</sup>                       | 2     | 0.02 1     | E(level): weighted average of 2722 8 ( <a href="#">1969Mo12</a> ) and 2726 8 ( <a href="#">1971Va18</a> ).<br>L: other: (2,3) from <a href="#">1971Va18</a> .                                                                                                                                                                                              |
| 2941 10                 | 3/2 <sup>+</sup>                       | 2     | <0.02      | E(level): weighted average of 2943 10 ( <a href="#">1969Mo12</a> ) and 2939 10 ( <a href="#">1971Va18</a> ).                                                                                                                                                                                                                                               |
| 3420 8                  | 5/2 <sup>+</sup>                       | 2     | 0.02 1     | E(level): weighted average of 3422 8 ( <a href="#">1969Mo12</a> ) and 3415 12 ( <a href="#">1971Va18</a> ).                                                                                                                                                                                                                                                |
| 3560 8                  |                                        |       |            | E(level): weighted average of 3563 8 ( <a href="#">1969Mo12</a> ) and 3555 9 ( <a href="#">1971Va18</a> ).                                                                                                                                                                                                                                                 |
| 3596 <sup>†</sup> 8     |                                        |       |            | E(level): other: 3595 9 ( <a href="#">1971Va18</a> ).                                                                                                                                                                                                                                                                                                      |
| 3675 <sup>‡</sup> 10    | (1/2 <sup>-</sup> , 3/2 <sup>-</sup> ) | (1)   |            | $J^\pi$ : assigned from L=1 and also adopted in the Adopted Levels.<br>L: from <a href="#">1971Va18</a> .<br>C <sup>2</sup> S: <2.5×10 <sup>-3</sup> for $J^\pi=1/2^-$ and <1.25×10 <sup>-3</sup> for $J^\pi=3/2^-$ ( <a href="#">1971Va18</a> ).                                                                                                          |
| 3801.90 <sup>#</sup> 30 | 3/2 <sup>-</sup>                       | 1     | 0.07 4     | E(level): others: 3804 8 ( <a href="#">1969Mo12</a> ) and 3795 10 ( <a href="#">1971Va18</a> ).<br>L: 1 also from <a href="#">1971Va18</a> .<br>C <sup>2</sup> S: 0.09 3 ( <a href="#">1971Va18</a> ).                                                                                                                                                     |
| 3885 <sup>†</sup> 10    |                                        |       |            | E(level): other: tentative 3866 10 ( <a href="#">1971Va18</a> ) and 3907 10 ( <a href="#">1971Va18</a> ).                                                                                                                                                                                                                                                  |
| 4025 10                 |                                        |       |            | E(level): weighted average of 4025 10 ( <a href="#">1969Mo12</a> ) and 4025 10 ( <a href="#">1971Va18</a> ).                                                                                                                                                                                                                                               |
| 4109 10                 |                                        |       |            | E(level): weighted average of 4112 10 ( <a href="#">1969Mo12</a> ) and 4105 10 ( <a href="#">1971Va18</a> ).                                                                                                                                                                                                                                               |
| 4189.87 <sup>#</sup> 26 | 1/2 <sup>-</sup>                       | 1     | 0.17 2     | E(level): others: 4190 8 ( <a href="#">1969Mo12</a> ) and 4196 12 ( <a href="#">1971Va18</a> ).<br>C <sup>2</sup> S: other: 0.12 3 ( <a href="#">1971Va18</a> ).                                                                                                                                                                                           |
| 4305 8                  |                                        |       |            | E(level): weighted average of 4302 8 ( <a href="#">1969Mo12</a> ) and 4312 12 ( <a href="#">1971Va18</a> ).                                                                                                                                                                                                                                                |
| 4481 <sup>†</sup> 8     | 3/2 <sup>+</sup> , 7/2 <sup>-</sup>    | 2+3   | 0.07, 0.05 | E(level): other: 4477 doublet ( <a href="#">2024Ku24</a> ).<br>$J^\pi$ : for two components, as given in <a href="#">2024Ku24</a> .<br>L: 2,3 doublet ( <a href="#">2024Ku24</a> ).<br>C <sup>2</sup> S: others: 0.07 2 for 3/2 <sup>+</sup> component and 0.05 2 for 7/2 <sup>-</sup> component ( <a href="#">2024Ku24</a> ).                             |
| 4575 <sup>†</sup> 8     |                                        |       |            |                                                                                                                                                                                                                                                                                                                                                            |
| 4837 <sup>†</sup> 8     |                                        |       |            |                                                                                                                                                                                                                                                                                                                                                            |
| 4903.28 <sup>#</sup> 16 | 1/2 <sup>-</sup>                       | 1     | 0.78 8     | E(level): other: 4903 8 ( <a href="#">1969Mo12</a> ).<br>L: 1 also from <a href="#">1984Pi03</a> .<br>C <sup>2</sup> S: other: 0.776 ( <a href="#">1984Pi03</a> ).                                                                                                                                                                                         |
| 4963.12 <sup>#</sup> 16 | 3/2 <sup>-</sup>                       | 1     | 0.32 3     | E(level): other: 4965 8 ( <a href="#">1969Mo12</a> ).<br>L: 1 also from <a href="#">1984Pi03</a> .<br>C <sup>2</sup> S: other: 0.218 ( <a href="#">1984Pi03</a> ).                                                                                                                                                                                         |
| 5058 <sup>†</sup> 8     | 7/2 <sup>-</sup>                       | 3     | 0.08 1     |                                                                                                                                                                                                                                                                                                                                                            |
| 5126 <sup>†</sup> 11    |                                        |       |            |                                                                                                                                                                                                                                                                                                                                                            |
| 5342 <sup>†</sup> 8     |                                        |       |            |                                                                                                                                                                                                                                                                                                                                                            |
| 5475 <sup>†</sup> 10    |                                        |       |            | Only observed in <a href="#">1969Mo12</a> and not included in the Adopted Levels.                                                                                                                                                                                                                                                                          |
| 5542 <sup>†</sup> 8     |                                        |       |            |                                                                                                                                                                                                                                                                                                                                                            |
| 5740 <sup>@</sup> 20    | 5/2 <sup>-</sup> , 7/2 <sup>-</sup>    | 3     | 0.03 1     | $J^\pi$ : assigned from L=3 and also adopted in the Adopted Levels. <a href="#">2024Ku24</a> assigns (5/2 <sup>-</sup> ) and states that the $J^\pi$ assignment is purely speculative.                                                                                                                                                                     |
| 5890 <sup>@</sup> 20    | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>    | 2     | 0.11 2     | $J^\pi$ : assigned from L=2 and also adopted in the Adopted Levels. <a href="#">2024Ku24</a> assigns (3/2 <sup>+</sup> ) and states that the $J^\pi$ assignment is purely speculative.                                                                                                                                                                     |
| 5980 <sup>†</sup> 10    |                                        |       |            |                                                                                                                                                                                                                                                                                                                                                            |
| 6078.6 <sup>#</sup> 13  | 1/2 <sup>-</sup> , 3/2 <sup>-</sup>    | 1     |            | E(level): other: 6080 8 ( <a href="#">1969Mo12</a> ).                                                                                                                                                                                                                                                                                                      |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{d,p})$  [2024Ku24](#), [1971Va18](#), [1969Mo12](#) (continued) $^{35}\text{S}$  Levels (continued)

| E(level)                                                                                                                                                                                                                      | $J^\pi$ &                   | L <sup>a</sup> | Comments                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|----------------|--------------------------------------------------------------------------------------------------------------|
| J <sup>π</sup> : assigned from L=1 and also adopted in the Adopted Levels.<br>L: from <a href="#">1984Pi03</a> . C <sup>2</sup> S=0.086 for J <sup>π</sup> =1/2 <sup>-</sup> and 0.042 for J <sup>π</sup> =3/2 <sup>-</sup> . |                             |                |                                                                                                              |
| 6292 <sup>†</sup> 8                                                                                                                                                                                                           |                             |                |                                                                                                              |
| 6334 <sup>†</sup> 8                                                                                                                                                                                                           |                             |                |                                                                                                              |
| 6344 <sup>†</sup> 8                                                                                                                                                                                                           |                             |                |                                                                                                              |
| 6446 <sup>†</sup> 8                                                                                                                                                                                                           |                             |                |                                                                                                              |
| 6496 <sup>†</sup> 8                                                                                                                                                                                                           |                             |                |                                                                                                              |
| 6537.7 <sup>#</sup> 14                                                                                                                                                                                                        |                             |                |                                                                                                              |
| 6545.1 <sup>#</sup> 13                                                                                                                                                                                                        |                             |                | E(level): other: 6543 8 ( <a href="#">1969Mo12</a> ).                                                        |
| 6584 <sup>†</sup> 10                                                                                                                                                                                                          |                             |                |                                                                                                              |
| 6635.2 <sup>#</sup> 13                                                                                                                                                                                                        |                             | (2,3)          | E(level): others: 6634 8 ( <a href="#">1969Mo12</a> ) and 6625 30 ( <a href="#">2024Ku24</a> ).              |
| 6677 <sup>†</sup> 8                                                                                                                                                                                                           |                             |                |                                                                                                              |
| 6891.3 <sup>#</sup> 14                                                                                                                                                                                                        | (1/2 to 7/2) <sup>(-)</sup> | (1,3)          | E(level): others: 6892 10 ( <a href="#">1969Mo12</a> ) and 6915 30 ( <a href="#">2024Ku24</a> ).             |
| 7021 10                                                                                                                                                                                                                       |                             | (2,3)          | E(level): weighted average of 7022 10 ( <a href="#">1969Mo12</a> ) and 7005 35 ( <a href="#">2024Ku24</a> ). |
| 7150 <sup>@</sup> 35                                                                                                                                                                                                          |                             | (0,3)          |                                                                                                              |
| 7205 <sup>@</sup> 35                                                                                                                                                                                                          |                             | (1,2)          |                                                                                                              |
| 7482.7 <sup>#</sup> 13                                                                                                                                                                                                        |                             | (1,2)          | E(level): other: 7495 40 ( <a href="#">2024Ku24</a> ).                                                       |
| 7570 <sup>@</sup> 40                                                                                                                                                                                                          |                             | (2,3)          |                                                                                                              |
| 7640 <sup>@</sup> 40                                                                                                                                                                                                          |                             | (2,3)          |                                                                                                              |

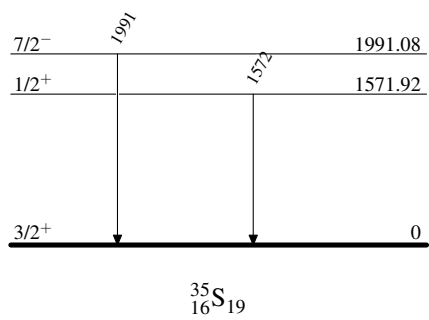
<sup>†</sup> From [1969Mo12](#).<sup>‡</sup> From [1971Va18](#).<sup>#</sup> From [1984Pi03](#).<sup>@</sup> From [2024Ku24](#).& Assumed for extracting C<sup>2</sup>S, unless otherwise noted.<sup>a</sup> From PTOLEMY-DWBA analysis of the measured  $\sigma(\theta)$  in [2024Ku24](#), unless otherwise noted. $\gamma(^{35}\text{S})$ 

| $E_\gamma$ | $E_i(\text{level})$ | $J_i^\pi$        | $E_f$ | $J_f^\pi$        | Comments                                     |
|------------|---------------------|------------------|-------|------------------|----------------------------------------------|
| 1572       | 1571.92             | 1/2 <sup>+</sup> | 0     | 3/2 <sup>+</sup> | $E_\gamma$ : from <a href="#">2024Ku24</a> . |
| 1991       | 1991.08             | 7/2 <sup>-</sup> | 0     | 3/2 <sup>+</sup> | $E_\gamma$ : from <a href="#">2024Ku24</a> . |

---

 $^{34}\text{S}(\text{d,p})$  2024Ku24,1971Va18,1969Mo12

---

Level Scheme

**$^{34}\text{S}(\text{d},\text{p}\gamma)$  1972Fr11,1975VaYG,1970Bu18**

**1972Fr11:** 3.50- and 4.00-MeV deuteron beams were produced from the 4-MV Van de Graaff accelerator at the Nuclear Research Center of Strasbourg, France. Targets were antimony sulphide enriched to 85.6%,  $550\text{ }\mu\text{g}/\text{cm}^2$  on a 0.03 mm molybdenum layer and  $150\text{ }\mu\text{g}/\text{cm}^2$  on a 0.05 mm silver layer. Protons were detected using a surface-barrier annular silicon detector at  $180^\circ$ .  $\gamma$  rays were detected using a  $40\text{ cm}^3$  Ge(Li) detector at  $0^\circ$ ,  $55^\circ$  and  $90^\circ$  with FWHM=3.2 keV at  $E_\gamma=1.332\text{ MeV}$  in the first experiment and a NaI crystal at  $0^\circ$ ,  $30^\circ$ ,  $45^\circ$ ,  $60^\circ$  and  $90^\circ$  in the second experiment. Measured  $E_p$ ,  $E_\gamma$ ,  $I_\gamma$ , and  $\text{p}\gamma(\theta)$ -coin. Deduced 12 levels, J,  $\gamma$ -ray branching ratios, and mixing ratios. Deduced  $T_{1/2}$  for 7 levels using the Doppler-Shift Attenuation Method (DSAM).

**1975VaYG,1972Va07:** A 4.72-MeV deuteron beam of 175 nA was provided from the Groningen 5 MV Van de Graaff generator. Targets were  $80\text{-}\mu\text{g}/\text{cm}^2$   $\text{Zn}^{34}\text{S}$  enriched to 89% evaporated onto  $10\text{-}\mu\text{g}/\text{cm}^2$  Formvar plus  $10\text{-}\mu\text{g}/\text{cm}^2$  carbon. Protons were detected using a 2-mm annular silicon detector.  $\gamma$  rays were detected using a  $7.6\text{-cm}\times 7.6\text{-cm}$  NaI(Tl) detector. Measured  $E_p$ ,  $E_\gamma$ ,  $I_\gamma$ , and  $\text{p}\gamma(\theta)$ -coin. Deduced levels, J,  $\gamma$ -ray branching ratios, and mixing ratios. Also see **1971VaZG**.

**1970Bu18:** 2-3.25 MeV deuteron beams were produced from the University of Arizona 5.5 MV Van de Graaff accelerator. Targets were  $480\text{-}\mu\text{g}/\text{cm}^2$   $^{120}\text{Ag}_2\text{S}$  made by heating 67% enriched  $^{34}\text{S}$  powder and silver foil. Protons were detected using a 1-mm annular silicon surface barrier detector and  $\gamma$  rays were detected using a  $7.62\text{ cm}\times 7.62\text{ cm}$  NaI(Tl) detector in the angular correlation experiment and a Ge(Li) detector in the lifetime experiment. Measured  $E_p$ ,  $E_\gamma$ ,  $I_\gamma$ , and  $\text{p}\gamma(\theta)$ -coin. Deduced 3 levels, J,  $\gamma$ -ray branching ratios, and mixing ratios. Deduced  $T_{1/2}$  for 3 levels using the Doppler-shift attenuation method (DSAM).

**2024Co04:** A 16-MeV deuteron beam from the John D. Fox Superconducting Linear Accelerator, FSU. Particles were momentum-analyzed by the Super-Enge Split-Pole Spectrograph and identified using a position-sensitive ionization chamber with two anode wires in the isobutane gas ( $\Delta E$ ) and a large plastic scintillator (E) at the focal plane.  $\gamma$  rays were detected using a  $\text{CeBr}_3$  Array of five  $\text{CeBr}_3$  detectors. Measured  $\text{p}\gamma$  delayed coincidence time spectra and deduced  $T_{1/2}$  for the 1991 level of  $^{35}\text{S}$ .

**1971Pr11:** A 4.48-MeV deuteron beam was produced from the insulated-core transformer tandem accelerator at the Aerospace Research Laboratories (ARL). The target was  $\text{Ag}_2\text{S}$  enriched to 85.61% in  $^{34}\text{S}$ . Protons were detected using a  $500\text{ }\mu\text{m}$ -thick,  $50\text{-mm}^2$  silicon detector and  $\gamma$  rays were detected using a 1-in.-thick, 1.5-in. diameter NE111 scintillator. Measured  $\text{p}\gamma$ -delayed coincidence time spectrum and deduced  $T_{1/2}$  for the 1.99-MeV level.

**1972Go31:** A 3.3-MeV deuteron beam was produced from the BNL 3.5-MeV van de Graaff facility. The target was  $\text{Sb}_2\text{S}_3$  enriched to 86% in  $^{34}\text{S}$ .  $\gamma$  rays were detected using a Ge(Li) detector. Measured  $E_\gamma=1572.24\text{ }17$  deexciting the first excited state.

 $^{35}\text{S}$  Levels

| $E(\text{level})^\dagger$ | $J^\pi^\ddagger$ | $T_{1/2}$            | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------|------------------|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0                         | $3/2^+$          |                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 1572.33 20                | $1/2^+$          | $>1.3\text{ ps}$     | E(level): weighted average of 1574.2 10 ( <b>1970Bu18</b> ), 1572.3 3 ( <b>1972Fr11</b> ), and 1572.28 17 ( <b>1972Go31</b> ).<br>$J^\pi$ : spin= $1/2$ from isotropic $\gamma(\theta)$ in <b>1972Fr11</b> and <b>1972Va07</b> .<br>$T_{1/2}$ : from $\tau>1.8\text{ ps}$ in <b>1970Bu18</b> by DSAM. Other: $>1.1\text{ ps}$ ( <b>1972Fr11</b> , $\tau>1.6\text{ ps}$ ).<br>E(level): unweighted average of 1994.6 10 ( <b>1970Bu18</b> ) and 1991.2 3 ( <b>1972Fr11</b> ).<br>$J^\pi$ : spin= $5/2$ , $7/2$ from $\gamma(\theta)$ in <b>1972Fr11</b> , <b>1972Va07</b> , and <b>1975VaYG</b> .<br>$T_{1/2}$ : from $\tau=1.48\text{ ns}$ 7 from weighted average of 1.7 ns 3 ( <b>2024Co04</b> ,) and 1.47 ns 7 ( <b>1971Pr11</b> ), both from $\text{p}\gamma(\text{t})$ . Other: $\tau>4.5\text{ ps}$ using DSAM ( <b>1970Bu18</b> ). |
| 1992.9 17                 | $7/2^-$          | $1.03\text{ ns } 5$  | E(level): unweighted average of 2350.0 10 ( <b>1970Bu18</b> ) and 2347.5 2 ( <b>1972Fr11</b> ).<br>$J^\pi$ : spin from $\gamma(\theta)$ in <b>1972Va07</b> , <b>1975VaYG</b> , and <b>1972Fr11</b> .<br>$T_{1/2}$ : from $\tau=1.03\text{ ns}$ 21 from weighted average of 0.8 ps 2 ( <b>1970Bu18</b> ) and 1.22 ps 18 ( <b>1972Fr11</b> ), both by DSAM.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 2348.8 13                 | $3/2^-$          | $0.71\text{ ps } 15$ | $J^\pi$ : spin= $3/2$ , $5/2$ , $7/2$ from $\gamma(\theta)$ in <b>1972Va07</b> and <b>1972Fr11</b> . 2717.9 $\gamma$ M1+E2 to $3/2^+$ .<br>$T_{1/2}$ : from $\tau=100\text{ fs}$ 35 in <b>1972Fr11</b> by DSAM.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 2718.0 10                 | $5/2^+$          | $69\text{ fs } 24$   | $J^\pi$ : spin= $3/2$ , $5/2$ from $\gamma(\theta)$ in <b>1972Va07</b> and <b>1975VaYG</b> .<br>$T_{1/2}$ : from $\tau=100\text{ fs}$ in <b>1972Fr11</b> by DSAM.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 2935 3                    | $3/2^+$          |                      | $J^\pi$ : spin= $3/2$ , $5/2$ from $\gamma(\theta)$ in <b>1972Va07</b> and <b>1975VaYG</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 3421.0 10                 | $5/2^+$          | $<69\text{ fs}$      | $J^\pi$ : spin= $3/2$ , $5/2$ , $7/2$ from $\gamma(\theta)$ in <b>1972Fr11</b> .<br>$T_{1/2}$ : from $\tau<100\text{ fs}$ in <b>1972Fr11</b> by DSAM.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 3558                      | $(5/2^+)$        |                      | E(level): rounded to the nearest integer based on the Adopted Levels. <b>1972Fr11</b> quotes 3563 8 from <b>1969Mo12</b> without reporting their own E(level).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 3592 5                    | $(7/2^+)$        |                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 3802.0 10                 | $3/2^-$          | $25\text{ fs } 18$   | $J^\pi$ : spin= $3/2$ from $\gamma(\theta)$ in <b>1972Fr11</b> .<br>$T_{1/2}$ : from $\tau=36\text{ fs}$ 26 in <b>1972Fr11</b> by DSAM.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 3815 3                    | $(9/2)^-$        |                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 3885.0 10                 | $(5/2)$          |                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{d},\text{p}\gamma)$  **1972Fr11,1975VaYG,1970Bu18** (continued) $^{35}\text{S}$  Levels (continued)

| $E(\text{level})^\dagger$ | $J^\pi^\ddagger$                      | $T_{1/2}$ | Comments                                             |
|---------------------------|---------------------------------------|-----------|------------------------------------------------------|
| 4025.0 10                 | (3/2 <sup>+</sup> ,5/2 <sup>+</sup> ) |           |                                                      |
| 4107 2                    | (5/2 <sup>+</sup> )                   | <55 fs    | $T_{1/2}$ : from $\tau < 80$ fs in 1972Fr11 by DSAM. |
| 4189 2                    | 1/2 <sup>-</sup>                      | <35 fs    | $T_{1/2}$ : from $\tau < 50$ fs in 1972Fr11 by DSAM. |
| 4480 2                    | (5/2 <sup>+</sup> )                   | <62 fs    | $T_{1/2}$ : from $\tau < 90$ fs in 1972Fr11 by DSAM. |

<sup>†</sup> From the average of available measurements for the lowest three excited states, and from 1972Fr11 for higher-lying states.

<sup>‡</sup> From the Adopted Levels. Supporting arguments from this dataset are given under comments if available.

 $\gamma(^{35}\text{S})$ 

| $E_i(\text{level})$ | $J_i^\pi$        | $E_\gamma^\dagger$                 | $I_\gamma^\ddagger$   | $E_f$                            | $J_f^\pi$                                                                    | Mult. | $\delta$ | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------|------------------|------------------------------------|-----------------------|----------------------------------|------------------------------------------------------------------------------|-------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1572.33             | 1/2 <sup>+</sup> | 1572.24 17                         | 100                   | 0                                | 3/2 <sup>+</sup>                                                             |       |          | $E_\gamma$ : from 1972Go31.<br>$A_2 = -0.09$ 5, $A_4 = +0.07$ 6 (1972Fr11); $A_2 = -0.007$ 10, $A_4 = +0.004$ 20 (1975VaYG).                                                                                                                                                                                                                                                                                                                                                                                           |
| 1992.9              | 7/2 <sup>-</sup> | 1992.8                             | 100                   | 0                                | 3/2 <sup>+</sup>                                                             | M2+E3 | +0.17 6  | Mult.: D+Q or Q+O from $\gamma(\theta)$ in 1970Bu18, 1972Fr11, 1972Va07, and 1975VaYG; E2+M3 is ruled out by RUL. D+Q is inconsistent with the $J^\pi = 7/2^-$ .<br>$\delta$ : unweighted average of $\delta(\text{E3/M2}) = 0.11$ 4 (1970Bu18), +0.11 4 (1972Fr11), and +0.28 3 (1972Va07,1975VaYG) for $J_i = 7/2$ . Others: $\delta(\text{Q/D}) = -1.9$ 7 (1970Bu18) and -0.43 3 (1972Va07,1975VaYG) for $J_i = 5/2$ .<br>$A_2 = +0.35$ 4, $A_4 = -0.39$ 4 (1972Fr11); $A_2 = +0.16$ 3, $A_4 = -0.10$ 4 (1975VaYG). |
| 2348.8              | 3/2 <sup>-</sup> | 776.5                              | 27 2                  | 1572.33                          | 1/2 <sup>+</sup>                                                             | (E1)  |          | Mult., $\delta$ : D(+Q) from $\gamma(\theta)$ , $\Delta\pi = \text{yes}$ from level scheme. M2 component is less likely. $\delta < 0.04$ from $\gamma(\theta)$ and RUL in 1975VaYG, superseding $0 < \delta < +1.7$ from $\gamma(\theta)$ in 1972Va07.<br>$A_2 = -0.37$ 3, $A_4 = -0.03$ 3 (1972Fr11); $A_2 = -0.38$ 2, $A_4 = +0.01$ 4 (1975VaYG).                                                                                                                                                                    |
|                     |                  | 2348.7                             | 73 2                  | 0                                | 3/2 <sup>+</sup>                                                             | (E1)  |          | Mult., $\delta$ : D(+Q) from $\gamma(\theta)$ , $\Delta\pi = \text{yes}$ from level scheme. M2 component is less likely. $\delta < 0.04$ from $\gamma(\theta)$ and RUL in 1975VaYG, superseding -0.27 12 from $\gamma(\theta)$ in 1972Va07; +0.01 4 from $\gamma(\theta)$ in 1972Fr11 assuming pure E1 for 776.5 $\gamma$ .<br>$A_2 = +0.33$ 2, $A_4 = -0.05$ 2 (1972Fr11); $A_2 = +0.30$ 2, $A_4 = -0.02$ 4 (1975VaYG).                                                                                               |
| 2718.0              | 5/2 <sup>+</sup> | 369.2<br>725.1<br>1145.7<br>2717.9 | <5<br><5<br><5<br>100 | 2348.8<br>1992.9<br>1572.33<br>0 | 3/2 <sup>-</sup><br>7/2 <sup>-</sup><br>1/2 <sup>+</sup><br>3/2 <sup>+</sup> | M1+E2 | -0.39 5  | Mult., $\delta$ : D+Q from $\gamma(\theta)$ in 1975VaYG with $-6.9 < \delta < -0.11$ or $+0.42 < \delta < +6.8$ for $J_i = 3/2$ , -0.39 5 or -15.0 34 for $J_i = 5/2$ , 0.18 4 or 3.8 7 for $J_i = 7/2$ . E1+M2 is ruled out by RUL. $\delta = -0.39$ 5 is adopted here based on $J^\pi = 5/2^+$ in the Adopted Levels. -15.0 34 is unlikely based on RUL.<br>$A_2 = +0.47$ 6, $A_4 = -0.08$ 7 (1972Fr11); $A_2 = +0.22$ 2, $A_4 = -0.02$ 3 (1975VaYG).                                                                |
| 2935                | 3/2 <sup>+</sup> | 586<br>942                         | <10<br><20            | 2348.8<br>1992.9                 | 3/2 <sup>-</sup><br>7/2 <sup>-</sup>                                         |       |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{d},\text{p}\gamma)$  **1972Fr11,1975VaYG,1970Bu18** (continued) $\gamma(^{35}\text{S})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$        | $E_\gamma^\dagger$ | $I_\gamma^\ddagger$ | $E_f$   | $J_f^\pi$ | Comments                                 |
|---------------------|------------------|--------------------|---------------------|---------|-----------|------------------------------------------|
| 2935                | $3/2^+$          | 1363               | <15                 | 1572.33 | $1/2^+$   |                                          |
|                     |                  | 2935               | 100                 | 0       | $3/2^+$   | $A_2=-0.06\ 3, A_4=-0.01\ 6$ (1975VaYG). |
| 3421.0              | $5/2^+$          | 486                | <4                  | 2935    | $3/2^+$   |                                          |
|                     |                  | 703.0              | <6                  | 2718.0  | $5/2^+$   |                                          |
|                     |                  | 1072.2             | <7                  | 2348.8  | $3/2^-$   |                                          |
|                     |                  | 1428.1             | <30                 | 1992.9  | $7/2^-$   |                                          |
|                     |                  | 1848.6             | <7                  | 1572.33 | $1/2^+$   |                                          |
|                     |                  | 3420.8             | 100                 | 0       | $3/2^+$   | $A_2=-0.54\ 8, A_4=-0.11\ 7$ (1972Fr11). |
| 3592                | $(7/2^+)$        | 657                | <10                 | 2935    | $3/2^+$   |                                          |
|                     |                  | 874                | <11                 | 2718.0  | $5/2^+$   |                                          |
|                     |                  | 1243               | <15                 | 2348.8  | $3/2^-$   |                                          |
|                     |                  | 1599               | <17                 | 1992.9  | $7/2^-$   |                                          |
|                     |                  | 2020 <sup>#@</sup> | <17                 | 1572.33 | $1/2^+$   |                                          |
|                     |                  | 3592               | 100                 | 0       | $3/2^+$   |                                          |
| 3802.0              | $3/2^-$          | 867                | <4                  | 2935    | $3/2^+$   |                                          |
|                     |                  | 1084.0             | 5 2                 | 2718.0  | $5/2^+$   |                                          |
|                     |                  | 1453.2             | 8 3                 | 2348.8  | $3/2^-$   |                                          |
|                     |                  | 1809.1             | <4                  | 1992.9  | $7/2^-$   |                                          |
|                     |                  | 2229.6             | 42 4                | 1572.33 | $1/2^+$   | $A_2=-0.56\ 15, A_4=0.0\ 1$ (1972Fr11).  |
|                     |                  | 3801.8             | 45 4                | 0       | $3/2^+$   | $A_2=+0.19\ 12, A_4=0.0\ 1$ (1972Fr11).  |
| 3815                | $(9/2)^-$        | 1822               | 100                 | 1992.9  | $7/2^-$   |                                          |
| 3885.0              | $(5/2)$          | 464.0              | <30                 | 3421.0  | $5/2^+$   |                                          |
|                     |                  | 950                | <12                 | 2935    | $3/2^+$   |                                          |
|                     |                  | 1167.0             | <15                 | 2718.0  | $5/2^+$   |                                          |
|                     |                  | 1536.2             | <18                 | 2348.8  | $3/2^-$   |                                          |
|                     |                  | 1892.1             | 100                 | 1992.9  | $7/2^-$   |                                          |
|                     |                  | 2312.6             | <16                 | 1572.33 | $1/2^+$   |                                          |
|                     |                  | 3884.8             | <16                 | 0       | $3/2^+$   |                                          |
| 4025.0              | $(3/2^+, 5/2^+)$ | 467                | <5                  | 3558    | $(5/2^+)$ |                                          |
|                     |                  | 604.0              | <10                 | 3421.0  | $5/2^+$   |                                          |
|                     |                  | 1090               | 31 8                | 2935    | $3/2^+$   |                                          |
|                     |                  | 1307.0             | <8                  | 2718.0  | $5/2^+$   |                                          |
|                     |                  | 1676.2             | 33 12               | 2348.8  | $3/2^-$   |                                          |
|                     |                  | 2032.0             | <10                 | 1992.9  | $7/2^-$   |                                          |
|                     |                  | 2452.6             | 36 12               | 1572.33 | $1/2^+$   |                                          |
|                     |                  | 4024.8             | <13                 | 0       | $3/2^+$   |                                          |
| 4107                | $(5/2^+)$        | 549                | <4                  | 3558    | $(5/2^+)$ |                                          |
|                     |                  | 686                | <4                  | 3421.0  | $5/2^+$   |                                          |
|                     |                  | 1172               | 13 6                | 2935    | $3/2^+$   |                                          |
|                     |                  | 1389               | <10                 | 2718.0  | $5/2^+$   |                                          |
|                     |                  | 1758               | <5                  | 2348.8  | $3/2^-$   |                                          |
|                     |                  | 2114               | <7                  | 1992.9  | $7/2^-$   |                                          |
|                     |                  | 2535               | <5                  | 1572.33 | $1/2^+$   |                                          |
|                     |                  | 4107               | 87 6                | 0       | $3/2^+$   |                                          |
| 4189                | $1/2^-$          | 387 <sup>#@</sup>  | <4                  | 3802.0  | $3/2^-$   |                                          |
|                     |                  | 597 <sup>#@</sup>  | <4                  | 3592    | $(7/2^+)$ |                                          |
|                     |                  | 631                | <4                  | 3558    | $(5/2^+)$ |                                          |
|                     |                  | 768 <sup>#@</sup>  | <4                  | 3421.0  | $5/2^+$   |                                          |
|                     |                  | 1254               | <5                  | 2935    | $3/2^+$   |                                          |
|                     |                  | 1471 <sup>#@</sup> | <5                  | 2718.0  | $5/2^+$   |                                          |
|                     |                  | 1840               | 48 5                | 2348.8  | $3/2^-$   |                                          |
|                     |                  | 2196 <sup>#@</sup> | <8                  | 1992.9  | $7/2^-$   |                                          |
|                     |                  | 2617               | <6                  | 1572.33 | $1/2^+$   |                                          |
|                     |                  | 4189               | 52 5                | 0       | $3/2^+$   |                                          |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{d},\text{p}\gamma)$  **1972Fr11,1975VaYG,1970Bu18** (continued) $\gamma(^{35}\text{S})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$           | $E_\gamma^\dagger$ | $I_\gamma^\ddagger$ | $E_f$  | $J_f^\pi$           | $E_i(\text{level})$ | $E_\gamma^\dagger$ | $I_\gamma^\ddagger$ | $E_f$   | $J_f^\pi$        |
|---------------------|---------------------|--------------------|---------------------|--------|---------------------|---------------------|--------------------|---------------------|---------|------------------|
| 4480                | (5/2 <sup>+</sup> ) | 595                | <5                  | 3885.0 | (5/2)               | 4480                | 1762               | 61 7                | 2718.0  | 5/2 <sup>+</sup> |
|                     |                     | 678                | <5                  | 3802.0 | 3/2 <sup>-</sup>    |                     | 2131               | <8                  | 2348.8  | 3/2 <sup>-</sup> |
|                     |                     | 888                | <5                  | 3592   | (7/2 <sup>+</sup> ) |                     | 2487               | <10                 | 1992.9  | 7/2 <sup>-</sup> |
|                     |                     | 922                | <5                  | 3558   | (5/2 <sup>+</sup> ) |                     | 2908               | 39 7                | 1572.33 | 1/2 <sup>+</sup> |
|                     |                     | 1059               | <10                 | 3421.0 | 5/2 <sup>+</sup>    |                     | 4480               | <12                 | 0       | 3/2 <sup>+</sup> |
|                     |                     | 1545               | <7                  | 2935   | 3/2 <sup>+</sup>    |                     |                    |                     |         |                  |

<sup>†</sup> Deduced by evaluators from level-energy difference, unless otherwise noted.

<sup>‡</sup> From **1972Fr11**, unless otherwise noted.

<sup>#</sup>  $\gamma$  transitions are considered questionable and not included in the Adopted Levels. The transition strengths estimated by evaluators exceed RUL.  $^{34}\text{S}(\text{n},\gamma)$  E=thermal observed other  $\gamma$  transitions from the same level but did not observe these questionable transitions.

<sup>@</sup> Placement of transition in the level scheme is uncertain.

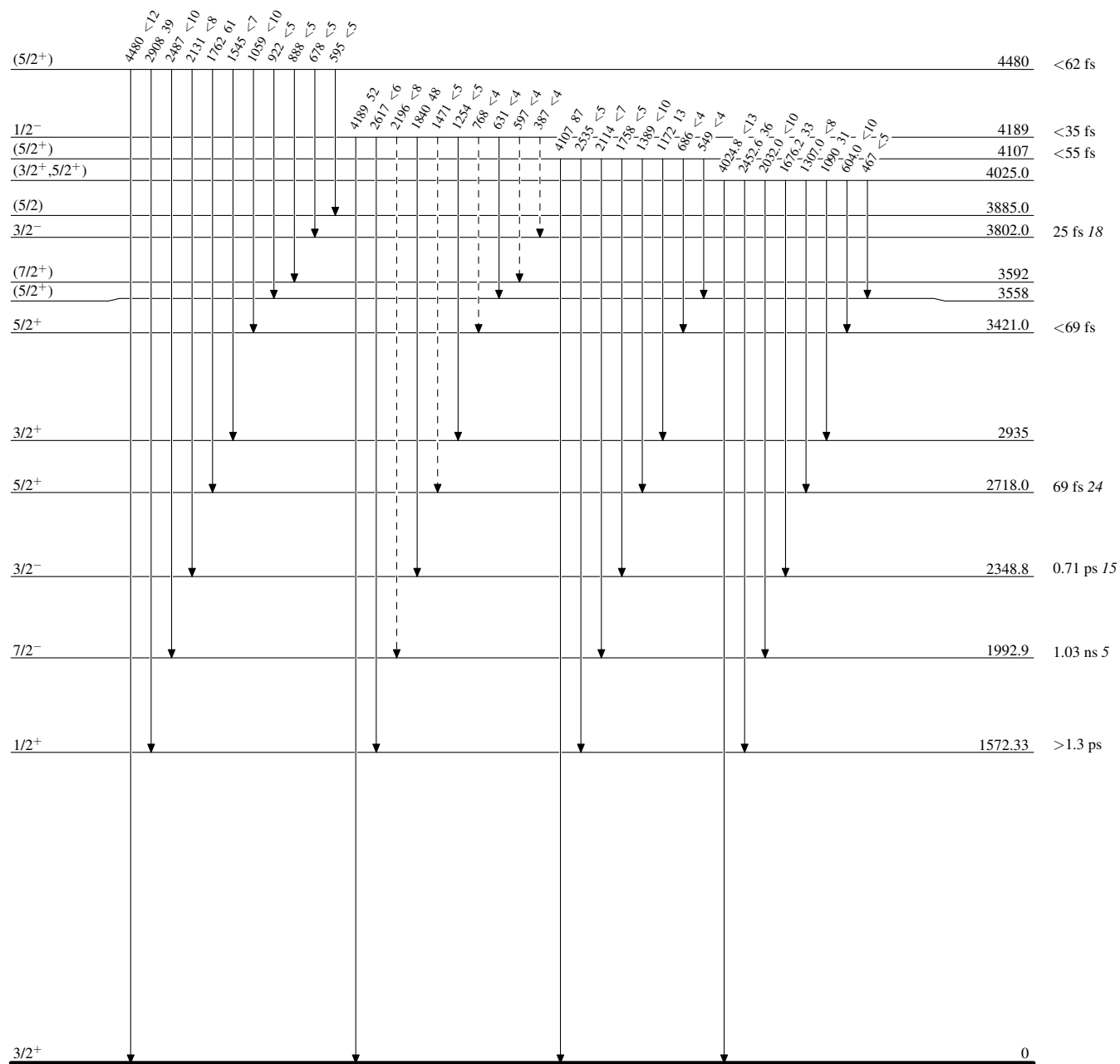


$^{34}\text{S}(\text{d},\text{p}\gamma)$  1972Fr11,1975VaYG,1970Bu18

Legend

## Level Scheme

Intensities: % photon branching from each level

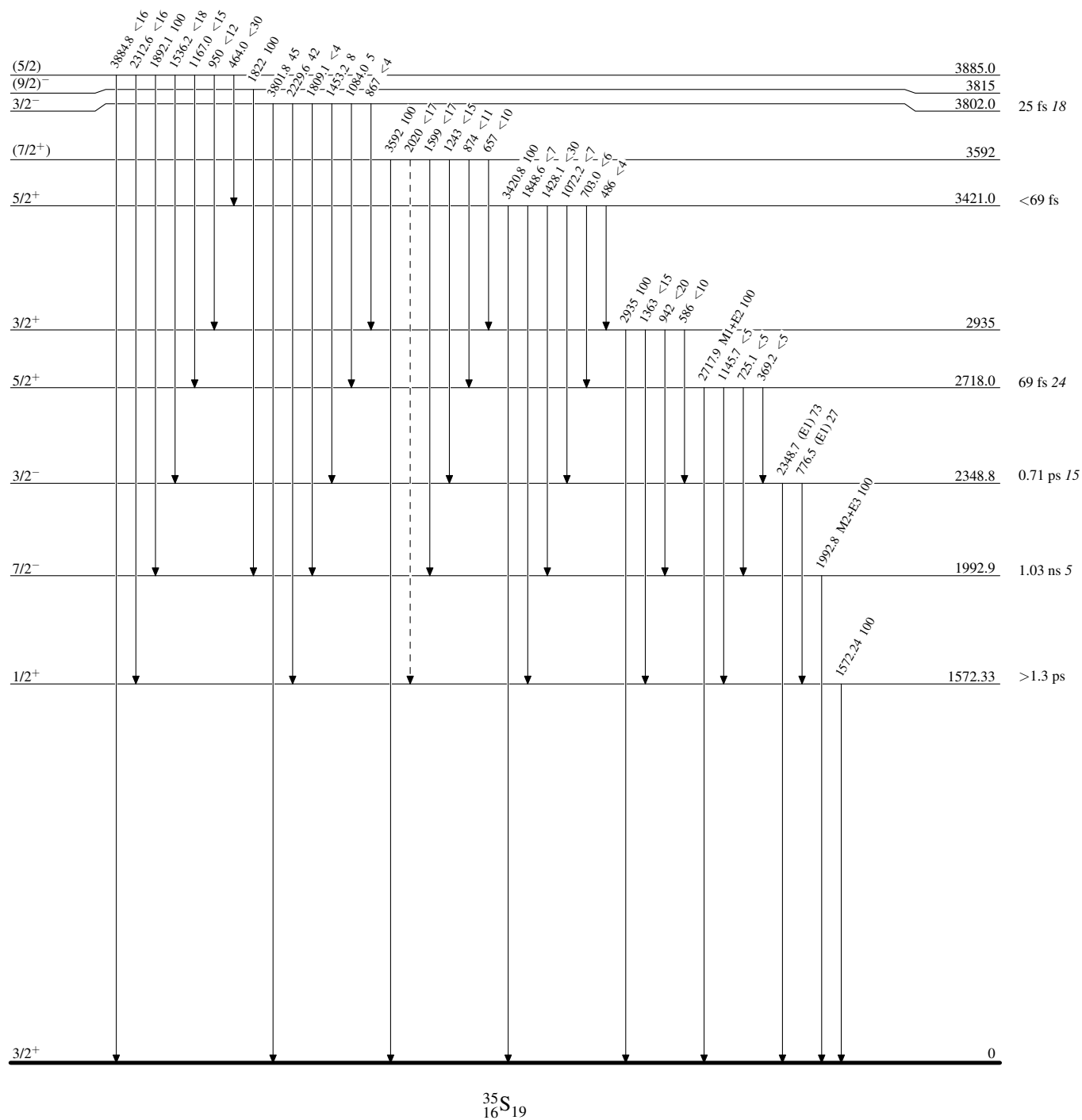
-----►  $\gamma$  Decay (Uncertain) $^{35}_{16}\text{S}_{19}$

$^{34}\text{S}(\text{d},\text{p}\gamma)$  1972Fr11,1975VaYG,1970Bu18

Legend

## Level Scheme (continued)

Intensities: % photon branching from each level

-----►  $\gamma$  Decay (Uncertain)

$^{36}\text{S}(\text{p,d})$  2026Jo01

$J^\pi=0^+$ , T=2 for  $^{36}\text{S}$  g.s.

**2026Jo01:** E=66 MeV dispersion-matched proton beam was produced from the Separated Sector Cyclotron facility of iThemba LABS, South Africa. The target was 1-mg/cm<sup>2</sup>, 99.24%-enriched  $^{36}\text{S}$  powder encapsulated between two 0.5- $\mu\text{m}$  Mylar foils. The outgoing deuteron and triton reaction products were momentum-analyzed by the K=600 quadrupole-dipole-dipole magnetic spectrometer and detected by the focal-plane detectors consisting of two multiwire drift chambers (MWDC) and a downstream plastic scintillator. Measured  $\sigma(E_d, \theta)$ . Deduced 98 levels, L-transfers, and single-neutron spectroscopic factors for 81 levels from zero-range adiabatic distorted-wave approximation (ADWA) calculations using DWUCK4. Deduced J-dependence of L=2 angular distributions and reproduced with finite-range ADWA calculations. Comparisons with large-scale shell-model and ab initio calculations. Confirmed the N=20 shell closure in  $^{36}\text{S}$ .

Also see Ph.D. Thesis [2021JoZZ](#).

 $^{35}\text{S}$  Levels

T: T=3/2 or 5/2 for  $^{35}\text{S}$  levels populated in the  $^{36}\text{S}(\text{p,d})$  reaction based on T=2 for the  $0^+$  g.s. of  $^{36}\text{S}$ .  $^{35}\text{S}$  levels below 9 MeV are likely T=3/2 and those above 9 MeV are likely T=5/2 considering the binding, Coulomb, and proton-neutron mass differences suggest the IAS of  $^{35}\text{P}$  ground state (T=5/2) should be around 9017 keV in  $^{35}\text{S}$ .  $\text{C}^2\text{S}$  extracted for these  $^{35}\text{S}$  levels show excellent agreement with proton-removal data from  $^{35}\text{P}$  when scaled by expected Clebsch-Gordan ratio of 0.167 ([2026Jo01](#)).

| E(level) | $J^\pi \uparrow \ddagger$ | $L \uparrow$ | $\text{C}^2\text{S} \uparrow \#$ | Comments                                    |
|----------|---------------------------|--------------|----------------------------------|---------------------------------------------|
| 0 5      | $3/2^+$                   | 2            | 3.5 4                            | $J^\pi$ : L-1/2 transfer from J-dependence. |
| 1569 5   | $1/2^+$                   | 0            | 1.28 14                          |                                             |
| 1990 5   | $7/2^-$                   | 3            | 0.170 19                         |                                             |
| 2348 5   | $3/2^-$                   | 1            | 0.033 4                          |                                             |
| 2718 5   | $5/2^+$                   | 2            | 0.53 6                           | $J^\pi$ : L+1/2 transfer from J-dependence. |
| 2937 5   | $3/2^+$                   | 2            | 0.034 4                          | $J^\pi$ : L-1/2 transfer from J-dependence. |
| 3420 5   | $5/2^+$                   | 2            | 0.50 6                           | $J^\pi$ : L+1/2 transfer from J-dependence. |
| 3558 5   | ( $5/2^+$ )               | (2)          | 0.0300 34                        |                                             |
| 3592 5   | ( $5/2^+$ )               | (2)          | 0.0040 5                         |                                             |
| 3800 5   | $3/2^-$                   | 1            | 0.0200 23                        |                                             |
| 3883 5   | $7/2^-$                   | 3            | 0.0180 20                        |                                             |
| 4023 5   | ( $3/2^+$ )               | (2)          | 0.0056 6                         |                                             |
| 4107 7   | ( $3/2^+$ )               | (2)          | 0.00230 26                       |                                             |
| 4182 5   | $1/2^-$                   | 1            | 0.00100 11                       |                                             |
| 4300 7   | ( $1/2^-, 3/2^-$ )        | (1)          | $3.00 \times 10^{-4}$ 34         |                                             |
| 4486 7   | $7/2^-$                   | 3            | 0.045 5                          |                                             |
| 4574 5   | $3/2^+$                   | 2            | 0.080 9                          | $J^\pi$ : L-1/2 transfer from J-dependence. |
| 4614 5   | $5/2^+$                   | 2            | 0.110 12                         |                                             |
| 4838 5   | ( $1/2^+$ )               | (0)          | 0.0035 4                         |                                             |
| 4907 5   | $1/2^+$                   | 0            | 0.0110 12                        |                                             |
| 4955 5   | $5/2^+$                   | 2            | 0.250 28                         | $J^\pi$ : L+1/2 transfer from J-dependence. |
| 5121 5   | $1/2^+$                   | 0            | 0.0092 10                        |                                             |
| 5286 5   | ( $1/2^-, 3/2^-$ )        | (1)          | $6.5 \times 10^{-4}$ 7           |                                             |
| 5476 5   | $3/2^+$                   | 2            | 0.0090 10                        |                                             |
| 5630 5   | $5/2^+$                   | 2            | 0.110 12                         |                                             |
| 5766 5   | $5/2^+$                   | 2            | 0.76 9                           | $J^\pi$ : L+1/2 transfer from J-dependence. |
| 5844 5   | ( $5/2^+$ )               | (2)          | 0.080 9                          |                                             |
| 6121 5   | ( $3/2^+$ )               | (2)          | 0.035 4                          |                                             |
| 6218 5   | ( $7/2^-$ )               | (3)          | 0.00300 34                       |                                             |
| 6338 5   | $1/2^+$                   | 0            | 0.044 5                          |                                             |
| 6439 5   | ( $1/2^+$ )               | (0)          | 0.00260 29                       |                                             |
| 6550 5   | $3/2^+$                   | 2            | 0.0035 4                         |                                             |
| 6639 5   | $5/2^+$                   | 2            | 0.250 28                         |                                             |
| 6682 5   | $1/2^+$                   | 0            | 0.056 6                          |                                             |
| 6790 5   | ( $5/2^+$ )               | (2)          | 0.037 4                          |                                             |
| 6962 5   | ( $1/2^-, 3/2^-$ )        | (1)          | 0.0050 6                         |                                             |

Continued on next page (footnotes at end of table)

$^{36}\text{S}(\text{p,d})$  [2026Jo01](#) (continued) $^{35}\text{S}$  Levels (continued)

| E(level)                  | $J^\pi$                               | $L^\dagger$ | $C^2S^\ddagger$ | Comments                                                                                                                       |
|---------------------------|---------------------------------------|-------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------|
| 7019 5                    | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | (1)         | 0.00150 17      |                                                                                                                                |
| 7102 5                    | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | (1)         | 0.00200 23      |                                                                                                                                |
| 7151 5                    | (7/2 <sup>-</sup> )                   | (3)         | 0.0090 10       |                                                                                                                                |
| 7215 5                    | 3/2 <sup>+</sup>                      | 2           | 0.0120 14       |                                                                                                                                |
| 7249 5                    | (3/2 <sup>+</sup> )                   | (2)         | 0.0120 14       |                                                                                                                                |
| 7279 5                    | 1/2 <sup>+</sup>                      | 0           | 0.0068 8        |                                                                                                                                |
| 7326 5                    | 1/2 <sup>+</sup>                      | 0           | 0.0073 8        |                                                                                                                                |
| 7349 5                    | (1/2 <sup>+</sup> )                   | (0)         | 0.0095 11       |                                                                                                                                |
| 7441 5                    | 1/2 <sup>+</sup>                      | 0           | 0.0094 11       |                                                                                                                                |
| 7490 5                    | 1/2 <sup>-</sup>                      | 1           | 0.0060 7        |                                                                                                                                |
| 7753 5                    | 5/2 <sup>+</sup>                      | 2           | 0.200 23        |                                                                                                                                |
| 7885 5                    | (5/2 <sup>+</sup> )                   | (2)         | 0.0300 34       |                                                                                                                                |
| 7980 7                    | 3/2 <sup>-</sup>                      | 1           | 0.00200 23      |                                                                                                                                |
| 8073 7                    | (5/2 <sup>+</sup> )                   | (2)         | 0.0180 20       |                                                                                                                                |
| 8103 5                    | 1/2 <sup>-</sup>                      | 1           | 0.00170 19      |                                                                                                                                |
| 8221 5                    | 1/2 <sup>-</sup>                      | 1           | 0.0040 5        |                                                                                                                                |
| 8270 5                    | 1/2 <sup>-</sup>                      | 1           | 0.0090 10       |                                                                                                                                |
| 8410 7                    | (5/2 <sup>+</sup> )                   | (2)         | 0.0100 11       |                                                                                                                                |
| 8465 7                    | (5/2 <sup>+</sup> )                   | (2)         | 0.0100 11       |                                                                                                                                |
| 8509 5                    | 5/2 <sup>+</sup>                      | 2           | 0.0120 14       |                                                                                                                                |
| 8557 7                    | (7/2 <sup>-</sup> )                   | (3)         | 0.00300 34      |                                                                                                                                |
| 8602 5                    | (7/2 <sup>-</sup> )                   | (3)         | 0.0100 11       |                                                                                                                                |
| 8649 7                    | (7/2 <sup>-</sup> )                   | (3)         | 0.0065 7        |                                                                                                                                |
| 8707 5                    | (5/2 <sup>+</sup> )                   | (2)         | 0.0220 25       |                                                                                                                                |
| 8761 7                    | (5/2 <sup>+</sup> )                   | (2)         | 0.0110 12       |                                                                                                                                |
| 8822 5                    | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | (1)         | 0.0060 7        |                                                                                                                                |
| 8876 5                    | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | (1)         | 0.00250 28      |                                                                                                                                |
| 8947 7                    | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | (1)         | 0.00100 11      |                                                                                                                                |
| 9032 15                   | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | (1)         | 0.0070 8        |                                                                                                                                |
| 9065 15                   | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | (1)         | 0.0070 8        |                                                                                                                                |
| 9126 7                    | 1/2 <sup>+</sup>                      | 0           | 0.35 4          | T=5/2<br>$J^\pi, T$ : isobaric analog state of 1/2 <sup>+</sup> , T=5/2, g.s. of $^{35}\text{P}$ ( <a href="#">2026Jo01</a> ). |
| 9180 5                    | 1/2 <sup>+</sup>                      | 0           | 0.060 7         |                                                                                                                                |
| 9219 15                   | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | (1)         | 0.00300 34      |                                                                                                                                |
| 9294 8                    | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | (1)         | 0.0100 11       |                                                                                                                                |
| 9339 8                    | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | (1)         | 0.00200 23      |                                                                                                                                |
| 9395 5                    | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | (1)         | 0.0045 5        |                                                                                                                                |
| 9446 7                    | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | (1)         | 0.00180 20      |                                                                                                                                |
| 9510 10                   | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | (1)         | 0.0035 4        |                                                                                                                                |
| 9565 20                   | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | (1)         | 0.0033 4        |                                                                                                                                |
| 9615 10                   | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | (1)         | 0.00280 32      |                                                                                                                                |
| 9666 10                   | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | (1)         |                 |                                                                                                                                |
| 9716 10                   | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | (1)         |                 |                                                                                                                                |
| 9774 10                   | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | (1)         |                 |                                                                                                                                |
| 9849 10                   | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | (1)         |                 |                                                                                                                                |
| 9911 10                   | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | (1)         |                 |                                                                                                                                |
| 9989 10                   | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | (1)         |                 |                                                                                                                                |
| 1.0049×10 <sup>4</sup> 10 | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | (1)         |                 |                                                                                                                                |
| 1.0166×10 <sup>4</sup> 10 | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | (1)         |                 |                                                                                                                                |
| 1.0225×10 <sup>4</sup> 10 | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | (1)         |                 |                                                                                                                                |
| 1.0277×10 <sup>4</sup> 10 | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | (1)         |                 |                                                                                                                                |
| 1.0350×10 <sup>4</sup> 10 | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | (1)         |                 |                                                                                                                                |
| 1.0421×10 <sup>4</sup> 10 | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> ) | (1)         |                 |                                                                                                                                |
| 1.0493×10 <sup>4</sup> 10 | (7/2 <sup>-</sup> )                   | (3)         | 0.0200 23       |                                                                                                                                |

Continued on next page (footnotes at end of table)

$^{36}\text{S}(\text{p,d})$  [2026Jo01](#) (continued) $^{35}\text{S}$  Levels (continued)

| E(level)                | $J^\pi$ <sup>†‡</sup>                  | $L$ <sup>†</sup> | $C^2S$ <sup>†#</sup> | Comments                                                                                                           |
|-------------------------|----------------------------------------|------------------|----------------------|--------------------------------------------------------------------------------------------------------------------|
| $1.0563 \times 10^4$ 10 | (1/2 <sup>-</sup> , 3/2 <sup>-</sup> ) | (1)              |                      |                                                                                                                    |
| $1.0663 \times 10^4$ 10 | (1/2 <sup>-</sup> , 3/2 <sup>-</sup> ) | (1)              |                      |                                                                                                                    |
| $1.0746 \times 10^4$ 10 | (1/2 <sup>-</sup> , 3/2 <sup>-</sup> ) | (1)              |                      |                                                                                                                    |
| $1.0810 \times 10^4$ 10 | (1/2 <sup>-</sup> , 3/2 <sup>-</sup> ) | (1)              |                      |                                                                                                                    |
| $1.147 \times 10^4$ 2   | 3/2 <sup>+</sup>                       | 2                | 0.052 6              | T=5/2<br>$J^\pi, T$ : IAS of 3/2 <sup>+</sup> , T=5/2, 2387 level in $^{35}\text{P}$ ( <a href="#">2026Jo01</a> ). |
| $1.288 \times 10^4$ 10  | 5/2 <sup>+</sup>                       | 2                | 0.49 6               | T=5/2<br>$J^\pi, T$ : IAS of 5/2 <sup>+</sup> , T=5/2, 3860 level in $^{35}\text{P}$ ( <a href="#">2026Jo01</a> ). |
| $1.368 \times 10^4$ 10  | 5/2 <sup>+</sup>                       | 2                | 0.156 18             | T=5/2<br>$J^\pi, T$ : IAS of 5/2 <sup>+</sup> , T=5/2, 4666 level in $^{35}\text{P}$ ( <a href="#">2026Jo01</a> ). |
| $1.422 \times 10^4$ 10  | 5/2 <sup>+</sup>                       | 2                | 0.247 28             | T=5/2<br>$J^\pi, T$ : IAS of 5/2 <sup>+</sup> , T=5/2, 5199 level in $^{35}\text{P}$ ( <a href="#">2026Jo01</a> ). |
| $1.470 \times 10^4$ 10  |                                        |                  |                      |                                                                                                                    |

<sup>†</sup> From ADWA analysis of the measured  $\sigma(\theta)$  in [2026Jo01](#). L-transfers are not specifically given but can be inferred from the reported  $J^\pi$ .

<sup>‡</sup> As given in [2026Jo01](#), which selected some  $J=L-1/2$  or  $L+1/2$  based on known  $J^\pi$  values ([2011Ch48](#)). When considered in the Adopted Levels,  $J=L-1/2$  and  $L+1/2$  are both possible unless other constraints are available.

<sup>#</sup>  $C^2S$  from [2026Jo01](#). For levels with  $J^\pi=(1/2^-, 3/2^-)$ , the  $C^2S$  values are given for 3/2<sup>-</sup>.  $C^2S$  is normalized to a  $^{40}\text{Ca}$  benchmark where Woods-Saxon parameters ( $r_0=1.27$  fm,  $a_0=0.70$  fm) were adjusted to ensure the summed strength of the  $1d_{3/2}$ ,  $2s_{1/2}$ , and  $1f_{7/2}$  orbitals equals 6. A systematic uncertainty of 11.2% is associated with all spectroscopic factors due to the effective target-thickness variation (10%) and the nominal target thickness uncertainty (5%) ([2026Jo01](#)). This uncertainty is included by the evaluators.

$^{37}\text{Cl}(\text{p}, ^3\text{He})$  1975Gu15

$^{37}\text{Cl} \rightarrow ^{35}\text{S} + \text{p} + \text{n}$  from  $J^\pi = 3/2^+$   $^{37}\text{Cl}$  ground state.

**1975Gu15:** A 40.2-MeV proton beam was produced from the Michigan State University cyclotron. The target was  $55 \mu\text{g}/\text{cm}^2$  NaCl with 97% enriched  $^{37}\text{Cl}$  made by evaporation of the salt onto a  $30\text{-}\mu\text{g}/\text{cm}^2$  carbon backing. The reaction products were detected using a wire-counter plastic-scintillator in the focal plane of an Enge split-pole spectrograph with FWHM=30 keV. Measured  $\sigma(\text{E}(^3\text{He}), \theta)$ . Deduced levels, L from zero-range DWUCK-DWBA analysis of the measured  $\sigma(\theta)$ .

**1971Vi02:** A 40-MeV proton beam was produced from the Grenoble variable energy cyclotron at intensities of 20-200 nA depending on the scattering angle with a 90-keV energy resolution. The target was natural purified chlorine gas 100 mm in diameter and 25 mm in height. The reaction particles were detected using two separate counter telescopes with each consisting of a 200- $\mu\text{m}$  phosphorous-drifted silicon  $\Delta\text{E}$  detector, a 2-mm lithium-drifted silicon E detector and a 3-mm lithium-drifted silicon E-reject detector with FWHM=180 keV for  $^3\text{He}$ . Measured  $\sigma(\text{E}(^3\text{He}), \theta)$ . Deduced L for  $^{35}\text{S}$  ground state from zero-range Nelson spin-dependent-DWBA analysis of the measured  $\sigma(\theta)$ .

 $^{35}\text{S}$  Levels

$\sigma_{\text{max}}$  under comments are from 1975Gu15.

| $\text{E}(\text{level})^\dagger$ | $\text{L}^\ddagger$ | Comments                                             |
|----------------------------------|---------------------|------------------------------------------------------|
| 0                                | 0+2+4               | $\sigma_{\text{max}} = 26.5 \mu\text{b}/\text{sr}$ . |
| 1575 10                          | 0+2                 | $\sigma_{\text{max}} = 19 \mu\text{b}/\text{sr}$ .   |
| 1992 10                          | 3+5                 | $\sigma_{\text{max}} = 1.5 \mu\text{b}/\text{sr}$ .  |
| 2717 10                          | 0+2+4               | $\sigma_{\text{max}} = 53.9 \mu\text{b}/\text{sr}$ . |
| 2938 10                          | 0+2+4               | $\sigma_{\text{max}} = 29.2 \mu\text{b}/\text{sr}$ . |
| 3421 10                          | 0+2+4               | $\sigma_{\text{max}} = 81.5 \mu\text{b}/\text{sr}$ . |
| 3598 10                          | 2+4                 | $\sigma_{\text{max}} = 27.8 \mu\text{b}/\text{sr}$ . |
| 3811 10                          | 3                   | $\sigma_{\text{max}} = 2.6 \mu\text{b}/\text{sr}$ .  |
| 4027 10                          | 2                   | $\sigma_{\text{max}} = 2.9 \mu\text{b}/\text{sr}$ .  |
| 4114 10                          | 0+2                 | $\sigma_{\text{max}} = 8.8 \mu\text{b}/\text{sr}$ .  |
| 4186 10                          | (2,3)               | $\sigma_{\text{max}} = 5.6 \mu\text{b}/\text{sr}$ .  |
| 4290 10                          | (2)                 | $\sigma_{\text{max}} = 2.8 \mu\text{b}/\text{sr}$ .  |
| 4489 10                          | 2                   | $\sigma_{\text{max}} = 2.7 \mu\text{b}/\text{sr}$ .  |
| 4577 10                          | 0+2                 | $\sigma_{\text{max}} = 10.7 \mu\text{b}/\text{sr}$ . |
| 4617 10                          | (1,2)               | $\sigma_{\text{max}} = 13.2 \mu\text{b}/\text{sr}$ . |
| 4843 10                          | 2                   | $\sigma_{\text{max}} = 15.9 \mu\text{b}/\text{sr}$ . |
| 4963 10                          | (0+2)               | $\sigma_{\text{max}} = 17.3 \mu\text{b}/\text{sr}$ . |
| 4990 10                          | 0+2                 | $\sigma_{\text{max}} = 19.5 \mu\text{b}/\text{sr}$ . |
| 5127 10                          | 2                   | $\sigma_{\text{max}} = 3.5 \mu\text{b}/\text{sr}$ .  |
| 5345 10                          | 3                   | $\sigma_{\text{max}} = 2.2 \mu\text{b}/\text{sr}$ .  |
| 5550 10                          | (3)                 | $\sigma_{\text{max}} = 6.2 \mu\text{b}/\text{sr}$ .  |
| 5771 10                          | 2                   | $\sigma_{\text{max}} = 10.1 \mu\text{b}/\text{sr}$ . |
| 5915? 10                         | (2,3)               | $\sigma_{\text{max}} = 3.4 \mu\text{b}/\text{sr}$ .  |
| 6129 10                          | 0+2                 | $\sigma_{\text{max}} = 7.8 \mu\text{b}/\text{sr}$ .  |
| 6347 10                          | (2)                 | $\sigma_{\text{max}} = 4.8 \mu\text{b}/\text{sr}$ .  |
| 6654 10                          | (3)                 | $\sigma_{\text{max}} = 7.7 \mu\text{b}/\text{sr}$ .  |
| 6696 10                          | 2                   | $\sigma_{\text{max}} = 6.7 \mu\text{b}/\text{sr}$ .  |
| 7151? 10                         | (4)                 | $\sigma_{\text{max}} = 3.3 \mu\text{b}/\text{sr}$ .  |
| 7375? 10                         |                     |                                                      |
| 7712? 10                         | 4                   | $\sigma_{\text{max}} = 4.6 \mu\text{b}/\text{sr}$ .  |
| 7770 10                          |                     | $\sigma_{\text{max}} = 5.5 \mu\text{b}/\text{sr}$ .  |
| 8103 10                          | (1+3)               | $\sigma_{\text{max}} = 10.6 \mu\text{b}/\text{sr}$ . |
| 8160 10                          | (1)                 | $\sigma_{\text{max}} = 8.4 \mu\text{b}/\text{sr}$ .  |
| 8430 10                          | 2                   | $\sigma_{\text{max}} = 5.7 \mu\text{b}/\text{sr}$ .  |
| 9155 10                          | 2                   | $\sigma_{\text{max}} = 10 \mu\text{b}/\text{sr}$ .   |

Continued on next page (footnotes at end of table)

---

$^{37}\text{Cl}(\text{p},^3\text{He})$     **1975Gu15** (continued)

---

$^{35}\text{S}$  Levels (continued)

<sup>†</sup> From **1975Gu15**.

<sup>‡</sup> From DWBA analysis of the measured  $\sigma(\theta)$  in **1975Gu15**. L is not considered firm by the evaluators when assigning  $J^\pi$  in the Adopted Levels because the theory of  $(\text{p},^3\text{He})$  reactions is not robust.

$^{37}\text{Cl}(\text{d},\alpha\gamma)$  1975VaYG,1972Va07

**1975VaYG,1972Va07:** A 4.25-MeV deuteron beam of 55 nA was provided from the Groningen 5 MV Van de Graaff generator. The target was 100  $\mu\text{g}/\text{cm}^2$   $\text{Co}^{37}\text{Cl}$  enriched to 98% evaporated onto 10- $\mu\text{g}/\text{cm}^2$  Formvar plus 10- $\mu\text{g}/\text{cm}^2$  carbon.  $\alpha$  particles were detected using a 60- $\mu\text{m}$  annular silicon detector.  $\gamma$  rays were detected using a 120  $\text{cm}^3$  Ge(Li) detector at 90°. Measured  $\sigma(E_\alpha)$ ,  $E_\gamma$ ,  $I_\gamma$ , and  $\alpha\gamma$ -coin. Deduced levels,  $\gamma$ -ray branching ratios. Also see [1971VaZG](#).

**1968Te06:** 3.1-4.6-MeV deuteron beams of 50 nA were provided from the Groningen 5-MV Van de Graaff generator. The target was a 100- $\mu\text{g}/\text{cm}^2$   $\text{Co}^{37}\text{Cl}$  both of natural  $^{37}\text{Cl}$  abundance and enriched to 93% evaporated onto 10- $\mu\text{g}/\text{cm}^2$  Formvar plus 10- $\mu\text{g}/\text{cm}^2$  carbon.  $\alpha$  particles were detected using an annular solid-state detector at 168-173°.  $\gamma$  rays were detected using a 3 in. by 3 in. NaI(Tl) scintillator at 55°. Measured  $\sigma(E_\alpha)$ ,  $E_\gamma$ , and  $\alpha\gamma$ -coin. Deduced levels.

**1955Pa54:**  $^{37}\text{Cl}(\text{d},\alpha)^{35}\text{S}$  with 3.0, 5.6, 6.0, 7.0, and 7.5-MeV deuteron beams from the MIT-ONR electrostatic generator. Targets were 80 and 300- $\mu\text{g}/\text{cm}^2$  barium chloride (75.4%  $^{35}\text{Cl}$ , 24.6%  $^{37}\text{Cl}$ ) evaporated onto Formvar films on a gold layer. Charged reaction products emitted at 90° were magnetically analyzed by a broad-range spectrograph. Measured  $\sigma(E_\alpha)$ . Deduced levels.

 $^{35}\text{S}$  Levels

| $E(\text{level})^\dagger$ | Comments                                                 |
|---------------------------|----------------------------------------------------------|
| 0                         |                                                          |
| 1572.2 12                 |                                                          |
| 1990.0 10                 | E(level): Other: 1992 10 from <a href="#">1955Pa54</a> . |
| 2348.2 10                 | E(level): Other: 2348 10 from <a href="#">1955Pa54</a> . |
| 2716.7 10                 | E(level): Other: 2714 10 from <a href="#">1955Pa54</a> . |
| 2939.2 13                 |                                                          |
| 3423 5                    |                                                          |
| 3560.8 19                 |                                                          |
| 3598.4 21                 |                                                          |
| 3803.6 19                 |                                                          |
| 3818.1 11                 |                                                          |
| 3889.0 19                 |                                                          |
| 4022.2 22                 |                                                          |
| 4027.7 10                 | E(level): Other: 4025 10 from <a href="#">1955Pa54</a> . |
| 4108 3                    |                                                          |
| 4180 3                    |                                                          |
| 4187 3                    |                                                          |
| 4302 4                    |                                                          |
| 4480.0 16                 |                                                          |

<sup>†</sup> From [1975VaYG](#) based on measured  $E_\gamma$ , but the  $E_\gamma$  values are not provided in [1975VaYG](#).

 $\gamma(^{35}\text{S})$ 

| $E_i(\text{level})$ | $E_\gamma^\dagger$ | $I_\gamma^\ddagger$ | $E_f$  | Comments                                       |
|---------------------|--------------------|---------------------|--------|------------------------------------------------|
| 1572.2              | 1572.2             | 100                 | 0      |                                                |
| 1990.0              | 1989.9             | 100                 | 0      |                                                |
| 2348.2              | 776.0              | 25 2                | 1572.2 |                                                |
|                     | 2348.1             | 75 2                | 0      |                                                |
| 2716.7              | 2716.6             | 100                 | 0      |                                                |
| 2939.2              | 2939.1             | 100                 | 0      |                                                |
| 3423                | 3423               | 100                 | 0      |                                                |
| 3560.8              | 1212.6             | 35 4                | 2348.2 |                                                |
|                     | 1570.8             | 65 4                | 1990.0 |                                                |
| 3598.4              | 3598.2             | 100                 | 0      | $I_\gamma$ : >95 in <a href="#">1975VaYG</a> . |
| 3803.6              | 2231.3             | 38 3                | 1572.2 |                                                |
|                     | 3803.4             | 62 3                | 0      |                                                |
| 3818.1              | 1828.1             | 100                 | 1990.0 |                                                |
| 3889.0              | 1540.8             | 40 4                | 2348.2 |                                                |
|                     | 1898.9             | 45 5                | 1990.0 |                                                |

Continued on next page (footnotes at end of table)



$^{37}\text{Cl}(\text{d},\alpha\gamma)$  [1975VaYG,1972Va07](#) (continued) $\gamma(^{35}\text{S})$  (continued)

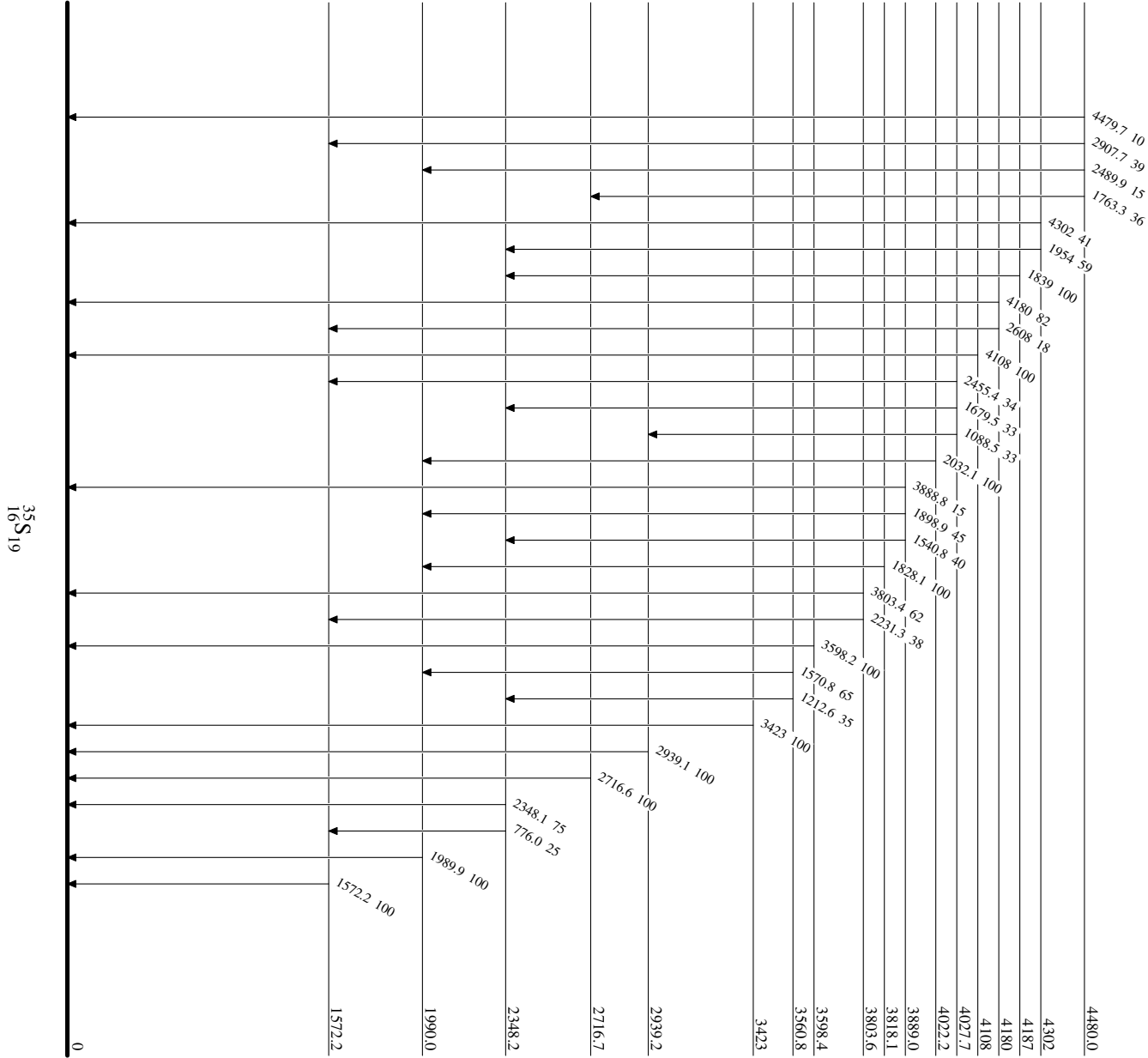
| $E_i(\text{level})$ | $E_\gamma^\dagger$ | $I_\gamma^\ddagger$ | $E_f$  | Comments                                       |
|---------------------|--------------------|---------------------|--------|------------------------------------------------|
| 3889.0              | 3888.8             | 15 3                | 0      |                                                |
| 4022.2              | 2032.1             | 100                 | 1990.0 |                                                |
| 4027.7              | 1088.5             | 33 4                | 2939.2 |                                                |
|                     | 1679.5             | 33 4                | 2348.2 |                                                |
|                     | 2455.4             | 34 6                | 1572.2 |                                                |
| 4108                | 4108               | 100                 | 0      | $I_\gamma$ : >95 in <a href="#">1975VaYG</a> . |
| 4180                | 2608               | 18 5                | 1572.2 |                                                |
|                     | 4180               | 82 5                | 0      |                                                |
| 4187                | 1839               | 100                 | 2348.2 | $I_\gamma$ : >95 in <a href="#">1975VaYG</a> . |
| 4302                | 1954               | 59 5                | 2348.2 |                                                |
|                     | 4302               | 41 5                | 0      |                                                |
| 4480.0              | 1763.3             | 36 4                | 2716.7 |                                                |
|                     | 2489.9             | 15 4                | 1990.0 |                                                |
|                     | 2907.7             | 39 4                | 1572.2 |                                                |
|                     | 4479.7             | 10 2                | 0      |                                                |

<sup>†</sup> Deduced by evaluators from level-energy difference in [1975VaYG](#).<sup>‡</sup> From [1975VaYG](#).

<sup>37</sup>Cl(d,αγ) <sup>197</sup>5VaYG,1972Va07

Level Scheme

Intensities: % photon branching from each level



$^{160}\text{Gd}(^{34}\text{S},\text{X}\gamma),(^{37}\text{Cl},\text{X}\gamma)$  1994Fo04

**1994Fo04:** 159-MeV  $^{34}\text{S}$  and 167-MeV  $^{37}\text{Cl}$  beams were produced from the Argonne Tandem Linac Accelerator System (ATLAS). Targets were 1 mg/cm<sup>2</sup> 98.1% enriched  $^{160}\text{Gd}$  backed by 15 mg/cm<sup>2</sup> gold.  $\gamma$  rays were detected using the Argonne-Notre Dame BGO  $\gamma$ -ray facility consisting of 12 Compton-suppressed Ge detectors and a 50-element bismuth germanate (BGO) array. Measured  $E\gamma$ ,  $I\gamma$ ,  $\gamma\gamma$ -coin. Deduced levels.  $\gamma\gamma$ -coin gates were placed on known  $\gamma$  rays in specific  $A\approx 160$  products to select individual reaction channels and identify coincident  $\gamma$  rays in light product partners.  $\gamma\gamma$ -correlation matrices produced with the fold condition  $4 \leq k \leq 9$  were found to include most of the events associated with deep inelastic processes.

 $^{35}\text{S}$  Levels

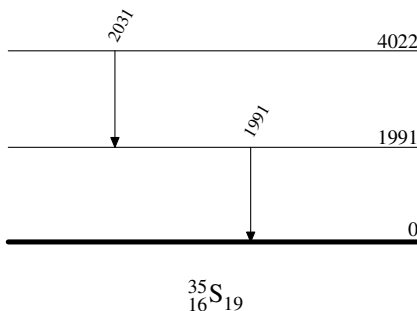
| <u><math>E(\text{level})^\dagger</math></u> | <u>Comments</u>                                                                                                                                              |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0                                           |                                                                                                                                                              |
| 1991                                        |                                                                                                                                                              |
| 4022                                        | $J^\pi$ : 1994Fo04 suggests this level is a likely candidate to be the expected $9/2^-$ yrast excitation, while $(11/2)^-$ is adopted in the Adopted Levels. |

$^\dagger$  From  $E\gamma$  data in 1994Fo04.

 $\gamma(^{35}\text{S})$ 

| <u><math>E_\gamma^\dagger</math></u> | <u><math>E_i(\text{level})</math></u> | <u><math>E_f</math></u> |
|--------------------------------------|---------------------------------------|-------------------------|
| 1991                                 | 1991                                  | 0                       |
| 2031                                 | 4022                                  | 1991                    |

$^\dagger$  From 1994Fo04.

 $^{160}\text{Gd}(^{34}\text{S},\text{X}\gamma),(^{37}\text{Cl},\text{X}\gamma)$  1994Fo04Level Scheme

$^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{S}\gamma)$  2010WaZT,2022Gr07

**2022Gr07:** A 215-MeV  $^{36}\text{S}$  beam was produced from the XTU-Tandem ALPI-superconducting linear accelerator complex at the INFN Legnaro National Laboratory, Italy. The target was 1 mg/cm<sup>2</sup> 99.7% enriched  $^{208}\text{Pb}$  with 1 mg/cm<sup>2</sup> Nb backing and mounted onto the Cologne differential plunger. Projectile-like fragments produced in binary grazing reactions were separated and identified by the PRISMA spectrometer.  $\gamma$  rays were detected using the AGATA demonstrator array of five triple cluster modules of 36-fold segmented Ge crystals covering backward angles from 135° to 175°. Measured  $E_\gamma$  with Doppler corrections, ( $^{35}\text{S}$ ) $\gamma$ -coin, and level lifetimes using the differential recoil-distance method (DRDM). Compared with shell-model calculations. Measured  $E_\gamma$ ,  $I_\gamma$ , fragment- $\gamma$ -coin, recoil distance. Deduced levels, lifetimes. Compared with shell-model calculations with PSDPF, SDPF-U, and FSU effective interactions.

**2010WaZT,2010Wa12:** A 215-MeV  $^{36}\text{S}^9$  beam was produced from the XTU-Tandem ALPI-superconducting linear accelerator complex at the INFN Legnaro National Laboratory, Italy. The target was 300  $\mu\text{g}/\text{cm}^2$  99.7% enriched  $^{208}\text{Pb}$  with 20  $\mu\text{g}/\text{cm}^2$  carbon backing. Projectile-like fragments produced in binary grazing reactions were separated and identified by the PRISMA spectrometer.  $\gamma$  rays were detected using the CLARA array of 22 escape-suppressed Ge clover detectors covering the azimuthal angles from 98° to 180°. Measured  $E_\gamma$  with Doppler corrections,  $I_\gamma$ , and ( $^{35}\text{S}$ ) $\gamma$ -coin. Deduced levels. Compared with shell-model calculations.

**2010WaZT** states that the observed  $\gamma$ -ray transitions are consistent with previous published work and no attempt has been made here to construct an independent level scheme.

 $^{35}\text{S}$  Levels

| $E(\text{level})^\dagger$ | $J^\pi^\ddagger$ | $T_{1/2}$  | Comments                                                                                                                                                                                                                                                     |
|---------------------------|------------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0                         | $3/2^+$          |            | <b>2022Gr07</b> shell-model calculated configuration: $\pi(1d_{5/2})^6(2s_{1/2})^2 \otimes \nu(1d_{5/2})^6(2s_{1/2})^2(1d_{3/2})^3$ (77%).                                                                                                                   |
| 1572.0 9                  | $1/2^+$          | 2.29 ps 14 | <b>2022Gr07</b> shell-model calculated configuration: $\pi(1d_{5/2})^6(2s_{1/2})^2 \otimes \nu(1d_{5/2})^6(2s_{1/2})^1(1d_{3/2})^4$ (44%).<br>$T_{1/2}$ : Lifetime=3.3 ps 2 from <b>2022Gr07</b> using the recoil-distance method with decay-curve analysis. |
| 1991.1 9                  | $7/2^-$          |            |                                                                                                                                                                                                                                                              |
| 2347.1 13                 | $3/2^-$          |            |                                                                                                                                                                                                                                                              |
| 2717.1 14                 | $5/2^+$          |            |                                                                                                                                                                                                                                                              |
| 3421 2                    | $5/2^+$          |            |                                                                                                                                                                                                                                                              |
| 3558.1 14                 | $(5/2^+)$        |            |                                                                                                                                                                                                                                                              |
| 3593 2                    | $(7/2^+)$        |            |                                                                                                                                                                                                                                                              |
| 3818.1 14                 | $(9/2)^-$        |            |                                                                                                                                                                                                                                                              |
| 4023.1 22                 | $(11/2)^-$       |            |                                                                                                                                                                                                                                                              |

<sup>†</sup> From a least-squares fit to  $\gamma$ -ray energies.

<sup>‡</sup> From the Adopted Levels.

 $\gamma(^{35}\text{S})$ 

| $E_\gamma^\dagger$  | $I_\gamma^\dagger$ | $E_i(\text{level})$ | $J_i^\pi$  | $E_f$  | $J_f^\pi$ | Comments                                                                                                                                                                                                                                                              |
|---------------------|--------------------|---------------------|------------|--------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 370 1               | 3.2 6              | 2717.1              | $5/2^+$    | 2347.1 | $3/2^-$   |                                                                                                                                                                                                                                                                       |
| 1211 $\frac{2}{2}$  | 1.7 7              | 3558.1              | $(5/2^+)$  | 2347.1 | $3/2^-$   |                                                                                                                                                                                                                                                                       |
| <sup>x</sup> 1227 1 | 5.2 9              |                     |            |        |           |                                                                                                                                                                                                                                                                       |
| 1567 $\frac{2}{2}$  | 9.6 9              | 3558.1              | $(5/2^+)$  | 1991.1 | $7/2^-$   |                                                                                                                                                                                                                                                                       |
| 1572 1              | 100.0 24           | 1572.0              | $1/2^+$    | 0      | $3/2^+$   | <b>2022Gr07</b> shell-model calculated $B(E2; 1/2^+ \text{ to } 3/2^+) = 35.2 \text{ e}^2 \text{ fm}^4$ (5.2 W.u.), $B(M1; 1/2^+ \text{ to } 3/2^+) = 0.0204 \mu_N^2$ ( $1.14 \times 10^{-2}$ W.u.), and mixing ratio $[\delta^2 = \lambda(E2)/\lambda(M1)] = 0.30$ . |
| 1827 1              | 12.9 11            | 3818.1              | $(9/2)^-$  | 1991.1 | $7/2^-$   |                                                                                                                                                                                                                                                                       |
| 1986 $\frac{2}{2}$  | 3.7 6              | 3558.1              | $(5/2^+)$  | 1572.0 | $1/2^+$   |                                                                                                                                                                                                                                                                       |
| 1991 1              | 37.2 18            | 1991.1              | $7/2^-$    | 0      | $3/2^+$   |                                                                                                                                                                                                                                                                       |
| 2032 2              | 13.1 12            | 4023.1              | $(11/2)^-$ | 1991.1 | $7/2^-$   |                                                                                                                                                                                                                                                                       |

Continued on next page (footnotes at end of table)

<sup>208</sup>Pb(<sup>36</sup>S, <sup>35</sup>Sγ)    [2010WaZT,2022Gr07](#) (continued)

γ(<sup>35</sup>S) (continued)

| <u>E<sub>γ</sub><sup>†</sup></u> | <u>I<sub>γ</sub><sup>†</sup></u> | <u>E<sub>i</sub>(level)</u> | <u>J<sub>i</sub><sup>π</sup></u> | <u>E<sub>f</sub></u> | <u>J<sub>f</sub><sup>π</sup></u> |
|----------------------------------|----------------------------------|-----------------------------|----------------------------------|----------------------|----------------------------------|
| 2347 2                           | 6.5 9                            | 2347.1                      | 3/2 <sup>-</sup>                 | 0                    | 3/2 <sup>+</sup>                 |
| 2717 2                           | 14.5 15                          | 2717.1                      | 5/2 <sup>+</sup>                 | 0                    | 3/2 <sup>+</sup>                 |
| <sup>x</sup> 3034 3              | 7.2 9                            |                             |                                  |                      |                                  |
| 3421 2                           | 8.6 10                           | 3421                        | 5/2 <sup>+</sup>                 | 0                    | 3/2 <sup>+</sup>                 |
| 3593 2                           | 8.8 11                           | 3593                        | (7/2 <sup>+</sup> )              | 0                    | 3/2 <sup>+</sup>                 |

<sup>†</sup> From [2010WaZT](#).

<sup>‡</sup> Placement of transition in the level scheme is uncertain.

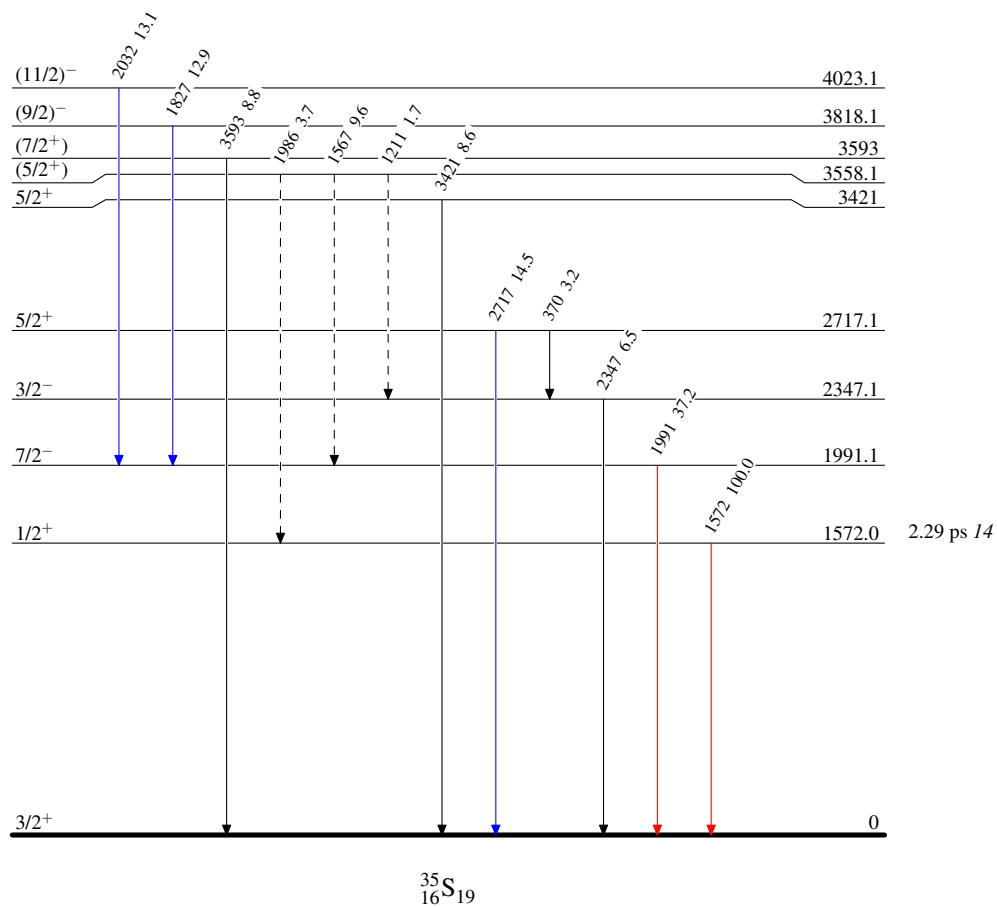
<sup>x</sup> γ ray not placed in level scheme.

$^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{S}\gamma)$  2010WaZT,2022Gr07

Legend

Level Scheme  
Intensities: Relative  $I_\gamma$

- $\longrightarrow$   $I_\gamma < 2\% \times I_\gamma^{\max}$
- $\longrightarrow$   $I_\gamma < 10\% \times I_\gamma^{\max}$
- $\longrightarrow$   $I_\gamma > 10\% \times I_\gamma^{\max}$
- $\cdots\cdots\cdots\longrightarrow$   $\gamma$  Decay (Uncertain)



Adopted Levels, Gammas

$Q(\beta^-) = -5966.2$  7;  $S(n) = 12644.76$  5;  $S(p) = 6370.81$  4;  $Q(\alpha) = -6997.90$  4    [2021Wa16](#)

$S(2n) = 24152.8$  4,  $S(2p) = 17254.1$  11 ([2021Wa16](#)).

Isotope discovery ([2012Th10](#)): positive-ray mass spectrograph at Cambridge, UK ([1919As01](#)).

 $^{35}\text{Cl}$  LevelsCross Reference (XREF) Flags

|          |                                                              |          |                                                   |           |                                                       |
|----------|--------------------------------------------------------------|----------|---------------------------------------------------|-----------|-------------------------------------------------------|
| <b>A</b> | $^{35}\text{S}$ $\beta^-$ decay (87.61 d)                    | <b>K</b> | $^{31}\text{P}$ ( $^6\text{Li}$ ,d)               | <b>U</b>  | $^{35}\text{Cl}$ (p,p' $\gamma$ )                     |
| <b>B</b> | $^{35}\text{Ar}$ $\varepsilon$ decay (1.7755 s)              | <b>L</b> | $^{32}\text{S}$ ( $\alpha$ ,p)                    | <b>V</b>  | $^{36}\text{Ar}$ (n,d)                                |
| <b>C</b> | $^{36}\text{K}$ $\varepsilon$ p decay (341 ms)               | <b>M</b> | $^{32}\text{S}$ ( $\alpha$ ,p $\gamma$ )          | <b>W</b>  | $^{36}\text{Ar}$ (d, $^3\text{He}$ )                  |
| <b>D</b> | $^1\text{H}$ ( $^{34}\text{S}$ , $^{35}\text{Cl}$ $\gamma$ ) | <b>N</b> | $^{33}\text{S}$ ( $\alpha$ ,d)                    | <b>X</b>  | $^{36}\text{Ar}$ (pol d, $^3\text{He}$ )              |
| <b>E</b> | $^2\text{H}$ ( $^{34}\text{S}$ ,n $\gamma$ )                 | <b>O</b> | $^{34}\text{S}$ (p, $\gamma$ )                    | <b>Y</b>  | $^{37}\text{Cl}$ (p,t)                                |
| <b>F</b> | $^{12}\text{C}$ ( $^{28}\text{Si}$ , $\alpha$ p $\gamma$ )   | <b>P</b> | $^{34}\text{S}$ (p,p),(p,p' $\gamma$ ):resonances | <b>Z</b>  | $^{39}\text{K}$ ( $\gamma$ , $\alpha$ )               |
| <b>G</b> | $^{16}\text{O}$ ( $^{24}\text{Mg}$ , $\alpha$ p $\gamma$ )   | <b>Q</b> | $^{34}\text{S}$ (d,n)                             | Others:   |                                                       |
| <b>H</b> | $^{24}\text{Mg}$ ( $^{16}\text{O}$ , $\alpha$ p $\gamma$ )   | <b>R</b> | $^{34}\text{S}$ ( $^3\text{He}$ ,d)               | <b>AA</b> | $^{40}\text{Ca}$ ( $\mu^-$ , $\nu\alpha$ n $\gamma$ ) |
| <b>I</b> | $^{27}\text{Al}$ ( $^{14}\text{N}$ , $\alpha$ p $\gamma$ )   | <b>S</b> | $^{35}\text{Cl}$ ( $\gamma$ , $\gamma'$ )         | <b>AB</b> | Coulomb excitation                                    |
| <b>J</b> | $^{31}\text{P}$ ( $\alpha$ ,p),( $\alpha$ ,n):resonances     | <b>T</b> | $^{35}\text{Cl}$ (n,n' $\gamma$ )                 |           |                                                       |

| E(level) <sup>†</sup> | J <sup>π</sup> <sup>‡</sup> | T <sub>1/2</sub> or Γ <sup>&amp;</sup> | XREF                                                                                                                                    | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------|-----------------------------|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0.0                   | 3/2 <sup>+</sup>            | stable                                 | <a href="#">ABC</a> <a href="#">EFGHI</a> <a href="#">KLM</a> <a href="#">O</a> <a href="#">QRSTU</a> <a href="#">VWXYZ</a>             | XREF: Others: <a href="#">AA</a> , <a href="#">AB</a><br>$\mu = +0.821870$ 2 ( <a href="#">2013Ja17</a> , <a href="#">2019StZV</a> )<br>$Q = -0.0817$ 8 ( <a href="#">2008Py02</a> , <a href="#">2021StZZ</a> )<br>J <sup>π</sup> : L=2 from 0 <sup>+</sup> in (d,n), (n,d), ( $^3\text{He}$ ,d), (d, $^3\text{He}$ ), and (pol d, $^3\text{He}$ ) and L-1/2 transfer from analyzing power. Spin=3/2 from microwave spectroscopy and atomic beam methods ( <a href="#">1948To10</a> , <a href="#">1949Da14</a> ).<br>$\mu$ : Nuclear Magnetic Resonance ( <a href="#">2013Ja17</a> ).<br>Q: Atomic beam magnetic resonance ( <a href="#">2008Py02</a> ). Others: $-0.08249$ 2 ( <a href="#">1972St38</a> ), $-0.076$ 5 ( <a href="#">1986El09</a> ), $-0.817$ 8 ( <a href="#">1993Su36</a> ), $0.0819$ 11 ( <a href="#">2000Ha64</a> ), $0.0850$ 11 ( <a href="#">2004Al08</a> ).<br>Evaluated rms nuclear charge radius $R = 3.365$ fm 19 ( <a href="#">2013An02</a> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 1219.38 7             | 1/2 <sup>+</sup>            | 0.119 ps 14                            | <a href="#">B</a> <a href="#">E</a> <a href="#">H</a> <a href="#">KLM</a> <a href="#">O</a> <a href="#">QRSTU</a> <a href="#">VWXYZ</a> | XREF: Others: <a href="#">AA</a> , <a href="#">AB</a><br>B(E2) <sup>†</sup> =0.00076 7<br>B(E2) <sup>†</sup> : from Coulomb excitation.<br>E(level): weighted average of 1219.2 2 from $^{35}\text{Ar}$ $\varepsilon$ decay, 1219.39 7 from (p, $\gamma$ ), and 1219.42 10 from (p,p' $\gamma$ ), 1219.0 10 from ( $^{16}\text{O}$ , $\alpha$ p $\gamma$ ), and 1219.4 6 from ( $\alpha$ ,p $\gamma$ ).<br>J <sup>π</sup> : L=0 from 0 <sup>+</sup> in (d,n), (n,d), ( $^3\text{He}$ ,d), (d, $^3\text{He}$ ), and (pol d, $^3\text{He}$ ).<br>T <sub>1/2</sub> : from lifetime $\tau = 0.172$ ps 20: weighted average of 0.29 ps 4 from ( $^{34}\text{S}$ ,n $\gamma$ ) ( <a href="#">1973Wa10</a> , DSAM), 0.21 ps +10-8 from ( $\alpha$ ,p $\gamma$ ) ( <a href="#">1969In04</a> , DSAM), 80 fs 40 ( <a href="#">1971Wi13</a> , DSAM), 175 fs 20 ( <a href="#">1972Hu11</a> , DSAM), 150 fs 20 ( <a href="#">1973Fa07</a> , DSAM), 145 fs 30 ( <a href="#">1976Me12</a> , DSAM) from (p, $\gamma$ ), 0.13 ps +3-2 from ( $\gamma$ , $\gamma'$ ) ( <a href="#">1966Ho02</a> , resonance fluorescence), 200 fs 80 from (n,n' $\gamma$ ) ( <a href="#">1989Ge09</a> , DSAM), 0.215 ps 75 ( <a href="#">1972Va06</a> , DSAM), 0.21 ps +6-4 ( <a href="#">1972Br45</a> , DSAM), 0.27 ps +19-10 ( <a href="#">1969Du08</a> , DSAM) from (p,p' $\gamma$ ), 0.16 ps 5 ( <a href="#">1969Ha17</a> , DSAM), and 0.230 ps 25 ( <a href="#">1977Sc36</a> , DSAM) from Coulomb excitation. |
| 1763.15 4             | 5/2 <sup>+</sup>            | 0.36 ps 3                              | <a href="#">B</a> <a href="#">EFGHI</a> <a href="#">KLM</a> <a href="#">O</a> <a href="#">QRSTU</a> <a href="#">VWXYZ</a>               | XREF: Others: <a href="#">AA</a> , <a href="#">AB</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued) $^{35}\text{Cl}$  Levels (continued)

| <u>E(level)<sup>†</sup></u> | <u>J<sup>π</sup><sup>‡</sup></u> | <u>T<sub>1/2</sub> or Γ<sup>&amp;</sup></u> | <u>XREF</u>         | <u>Comments</u>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------|----------------------------------|---------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             |                                  |                                             |                     | <p>T=1/2<br/> B(E2)↑=0.0106 7<br/> XREF: X(1785)Y(1750)<br/> B(E2)↑: from Coulomb excitation.<br/> E(level): weighted average of 1763.1 2 from <math>^{35}\text{Ar}</math> <math>\varepsilon</math> decay, 1763.0 6 from (<math>^{28}\text{Si},\alpha\text{p}\gamma</math>), 1762.95 17 from (<math>^{24}\text{Mg},\alpha\text{p}\gamma</math>), 1763.3 2 from (<math>^{16}\text{O},\alpha\text{p}\gamma</math>), 1763.19 10 from (<math>^{14}\text{N},\alpha\text{p}\text{n}\gamma</math>), 1763.14 4 from (p,<math>\gamma</math>), 1763.26 10 from (p,p'<math>\gamma</math>).<br/> J<sup>π</sup>: L=2 from 0<sup>+</sup> in (d,n); spin=5/2 from p<math>\gamma</math>(<math>\theta</math>) or <math>\gamma</math>(<math>\theta</math>) in (<math>\alpha,\text{p}\gamma</math>), (p,<math>\gamma</math>), and (p,p'<math>\gamma</math>).<br/> T<sub>1/2</sub>: lifetime <math>\tau</math>=0.52 ps 5: weighted average of 0.54 ps 7 from (<math>^{34}\text{S},\text{n}\gamma</math>) (1973Wa10, DSAM), 0.55 ps 15 from (<math>\alpha,\text{p}\gamma</math>) (1969In04, DSAM), 0.50 ps 10 (1972Hu11, DSAM), 0.40 ps 14 (1973Fa07, DSAM), 0.53 ps 9 (1976Me12, DSAM) from (p,<math>\gamma</math>), 0.60 ps 21 from (<math>\gamma,\gamma'</math>) (1962Bo17, resonance fluorescence), 0.205 ps 75 (1972Va06, DSAM), 0.49 ps +9-6 (1972Br45, DSAM), 1.15 ps +45-63 (1969Du08, DSAM) from (p,p'<math>\gamma</math>), 0.57 ps 7 (1969Ha17, DSAM), and 0.63 ps 5 (1977Sc36, DSAM) from Coulomb excitation.</p>                     |
| 2645.67 8                   | 7/2 <sup>+</sup>                 | 0.159 ps 21                                 | B FGHI LM O Qr TU y | <p>XREF: Others: AA, AB<br/> XREF: r(2645)y(2650)<br/> E(level): weighted average of 2645.81 18 from (<math>^{24}\text{Mg},\alpha\text{p}\gamma</math>), 2645.9 3 from (<math>^{16}\text{O},\alpha\text{p}\gamma</math>), 2645.45 12 from (<math>^{14}\text{N},\alpha\text{p}\text{n}\gamma</math>), 2645.70 8 from (p,<math>\gamma</math>), and 2645.74 17 from (p,p'<math>\gamma</math>). Others: 2645.1 10 from <math>^{35}\text{Ar}</math> <math>\varepsilon</math> decay, 2645.8 6 from (<math>^{28}\text{Si},\alpha\text{p}\gamma</math>), and 2645.8 8 from (<math>\alpha,\text{p}\gamma</math>).<br/> J<sup>π</sup>: spin=7/2 from p<math>\gamma</math>(<math>\theta</math>) in (<math>\alpha,\text{p}\gamma</math>) and <math>\gamma</math>(<math>\theta</math>) in (p,<math>\gamma</math>); parity from 882<math>\gamma</math>, M1+E2 to 5/2<sup>+</sup>, 1763 level in (<math>\alpha,\text{p}\gamma</math>).<br/> T<sub>1/2</sub>: lifetime <math>\tau</math>=0.229 ps 30: weighted average of 0.30 ps 9 from (<math>\alpha,\text{p}\gamma</math>) (1969In04, DSAM), 0.255 ps 65 (1972Hu11, DSAM), 0.200 ps 30 (1976Me12, DSAM) in (p,<math>\gamma</math>), 0.27 ps 9 in (n,n'<math>\gamma</math>) (1989Ge09, DSAM), 0.185 ps 50 (1972Va06, DSAM), 0.26 ps 6 (1972Br45, DSAM), 0.35 ps +10-7 (1969Du08, DSAM) in (p,p'<math>\gamma</math>), and 0.26 ps 9 in Coulomb excitation (1977Sc36, DSAM). Other: <math>\tau</math>&lt;1.3 ps in (<math>^{28}\text{Si},\alpha\text{p}\gamma</math>) (2007Ks01, DSAM).</p> |
| 2694.00 20                  | 3/2 <sup>+</sup>                 | 32 fs 6                                     | B LM O QRSTU Wxy    | <p>XREF: Others: AA<br/> XREF: L(2693?)X(2676)y(2650)<br/> E(level): weighted average of 2693.8 2 from <math>^{35}\text{Ar}</math> <math>\varepsilon</math> decay, 2693.6 13 from (<math>\alpha,\text{p}\gamma</math>), 2693.9 3 from (p,<math>\gamma</math>), 2694.50 28 from (p,p'<math>\gamma</math>).<br/> J<sup>π</sup>: spin=3/2 from <math>\gamma</math>(<math>\theta</math>) in (p,p'<math>\gamma</math>); L=2 from 0<sup>+</sup> in (<math>^3\text{He},\text{d}</math>) and (d,<math>^3\text{He}</math>).<br/> T<sub>1/2</sub>: lifetime <math>\tau</math>=46 fs 9: unweighted average of 70 fs 30 in (<math>\alpha,\text{p}\gamma</math>) from 1972Br33 with DSAM, 21 fs 3 in (<math>\alpha,\text{p}\gamma</math>) from 1972Hu11 with DSAM, 19 fs 4 in (<math>\alpha,\text{p}\gamma</math>) from 1973Fa07 with DSAM, 20 fs 4 in (p,<math>\gamma</math>) from 1976Me12 with DSAM, 65 fs 20 in (p,<math>\gamma</math>) from 1972Va06 with DSAM, 62 fs 16 in (p,p'<math>\gamma</math>) from 1969Du08 with DSAM, and 62 fs 8 in (n,n'<math>\gamma</math>) from 1989Ge09 with DSAM. Others: 115 fs +95-59 in</p>                                                                                                                                                                                                                                                                                                                                                                                                       |

Continued on next page (footnotes at end of table)



**Adopted Levels, Gammas (continued)** $^{35}\text{Cl}$  Levels (continued)

| <u>E(level)<sup>†</sup></u> | <u>J<sup>π</sup><sup>‡</sup></u> | <u>T<sub>1/2</sub> or Γ<sup>&amp;</sup></u> | <u>XREF</u>           | <u>Comments</u>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------|----------------------------------|---------------------------------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3002.64 11                  | 5/2 <sup>+</sup>                 | 22 fs 6                                     | B F HI LM O QRSTU WXY | (p,p'γ) from 1969Du08 (using 931γ) is considered less reliable by the authors, and 11 fs 7 in (p,p'γ) from 1971Ca40 with DSAM is considered an outlier.<br>E(level): weighted average of 3002.3 3 from $^{35}\text{Ar}$ ε decay, 3003.1 8 from ( $^{28}\text{Si},\alpha\text{p}\gamma$ ), 3002.0 10 from ( $^{16}\text{O},\alpha\text{p}\gamma$ ), 3001.98 22 from ( $^{14}\text{N},\alpha\text{p}\text{n}\gamma$ ), 3002.6 15 from ( $\alpha,\text{p}\gamma$ ), 3002.75 8 from (p,γ), 3002.6 5 from (p,p'γ).<br>J <sup>π</sup> : L=2 from 0 <sup>+</sup> in ( $^3\text{He},\text{d}$ ), ( $\text{d},^3\text{He}$ ), and (pol d, $^3\text{He}$ ) with L+1/2 transfer from analyzing power. Discrepancy: L=3 from 0 <sup>+</sup> in (d,n) and the angular distribution fit seems good.<br>T <sub>1/2</sub> : lifetime τ=32 fs 8: unweighted average of 48 fs 9 in (α,pγ) from 1972Br33 with DSAM, 22 fs 3 in (α,pγ) from 1972Hu11 with DSAM, 14 fs 2 in (α,pγ) from 1973Fa07 with DSAM, 16 fs 4 in (p,γ) from 1976Me12 with DSAM, 72 fs 12 in (n,n'γ) from 1989Ge09 with DSAM, 18 fs +26-15 in (p,p'γ) from 1972Va06 with DSAM, and 31 fs 13 in (p,p'γ) from 1969Du08 with DSAM. |
| 3162.87 9                   | 7/2 <sup>-</sup>                 | 29.2 ps 12                                  | FGHI KLM O QR TU WX   | T=1/2<br>E(level): weighted average of 3163.1 6 from ( $^{28}\text{Si},\alpha\text{p}\gamma$ ), 3163.43 19 from ( $^{24}\text{Mg},\alpha\text{p}\gamma$ ), 3162.9 3 from ( $^{16}\text{O},\alpha\text{p}\gamma$ ), 3162.64 9 from ( $^{14}\text{N},\alpha\text{p}\text{n}\gamma$ ), 3163.1 11 from (α,pγ), 3162.99 7 from (p,γ), 3162.75 10 from (p,p'γ).<br>J <sup>π</sup> : spin=7/2 from pγ(θ) in (α,pγ) and γ(θ) in (p,γ) and (p,p'γ); L=3 from 0 <sup>+</sup> in (d,n), ( $^3\text{He},\text{d}$ ), ( $\text{d},^3\text{He}$ ), and (pol d, $^3\text{He}$ ).<br>T <sub>1/2</sub> : lifetime τ=42.1 ps 17: weighted average of 42 ps 2 (1974Va13, RDM), 41.7 ps 17 (1976Ke02, RDM), 45 ps 6 (1991Ja11, RDM) from $^{24}\text{Mg}(\text{}^{16}\text{O},\alpha\text{p}\gamma)$ , 41.8 ps 18 from $^{27}\text{Al}(\text{}^{14}\text{N},\alpha\text{p}\text{n}\gamma)$ (1976Wa11, RDM), 60 ps 7 from (α,pγ) (1969In04, DSAM), 37 ps 4 from (p,γ), (1971Ba23, delayed coincidence), and 46 ps 6 from (p,p'γ) (1974Hu09, RDM). Others: τ=140 ps 40 (1968Az02, delayed coincidence; obtained using NaI(Tl) detectors with broad time resolution) from (p,γ).                       |
| 3918.47 12                  | 3/2 <sup>+</sup>                 | 4.9 fs 14                                   | B M O RS u x          | XREF: R(3908)u(3930)x(3948)<br>E(level): weighted average of 3918.4 5 from $^{35}\text{Ar}$ ε decay, 3915 2 from (α,pγ), 3918.48 10 from (p,γ).<br>J <sup>π</sup> : spin=3/2 determined by the feeding 5162γ(θ) from 5/2 <sup>+</sup> , 9081 level in (p,γ); parity=+ determined by 3918γ, M1+E2 to 3/2 <sup>+</sup> ground state. L=2 from 0 <sup>+</sup> in (pol d, $^3\text{He}$ ).<br>T <sub>1/2</sub> : lifetime τ=7 fs 2 from (p,γ), (1972Hu11, DSAM).<br>Others: <22 fs from (α,pγ) (1972Br33, DSAM).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 3942.9 2                    | 9/2 <sup>+</sup>                 | 0.18 ps 3                                   | F HI LM O             | E(level): weighted average of 3942.0 7 from ( $^{28}\text{Si},\alpha\text{p}\gamma$ ), 3943.4 10 from ( $^{16}\text{O},\alpha\text{p}\gamma$ ), 3942.3 10 from ( $^{14}\text{N},\alpha\text{p}\text{n}\gamma$ ), 3943.4 12 from (α,pγ), and 3942.9 2 from (p,γ).<br>J <sup>π</sup> : spin=9/2 from pγ(θ) in (α,pγ); 2180.1γ E2 to 5/2 <sup>+</sup> .<br>T <sub>1/2</sub> : lifetime τ=0.26 ps 5: weighted average of 0.33 ps 5 in (α,pγ) (1972Br33, DSAM), 0.39 ps 11 (1972Hu11,                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

Continued on next page (footnotes at end of table)

**Adopted Levels, Gammas (continued)** $^{35}\text{Cl}$  Levels (continued)

| <u>E(level)<sup>†</sup></u> | <u>J<sup>π</sup><sup>‡</sup></u> | <u>T<sub>1/2</sub> or Γ<sup>&amp;</sup></u> | <u>XREF</u> | <u>Comments</u>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------|----------------------------------|---------------------------------------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3967.6 4                    | 1/2 <sup>+</sup>                 | 11.8 fs 28                                  | B M O R     | DSAM), 0.145 ps 50 (1973Fa07, DSAM), and 0.29 ps 5 (1976Me12, DSAM) from (p,γ). Other: τ<0.5 ps in ( <sup>28</sup> Si,αpγ) (2007Ks01, DSAM).<br>E(level): weighted average of 3967.7 4 from <sup>35</sup> Ar ε decay, 3968 2 from (α,pγ), and 3967.3 6 from (p,γ).<br>J <sup>π</sup> : L=0 from 0 <sup>+</sup> in ( <sup>3</sup> He,d).<br>T <sub>1/2</sub> : lifetime τ=17 fs 4: weighted average of 13 fs 4 (1972Hu11, DSAM), 20 fs 5 (1973Fa07, DSAM), and 20 fs 4 (1976Me12, DSAM) in (p,γ). Other: τ<54 fs from (α,pγ) (1972Br33, DSAM).                                                                                                                                  |
| 4059.1 4                    | (3/2) <sup>-</sup>               | 13.9 fs 21                                  | LM O QR     | E(level): weighted average of 4056.9 27 from (α,p), 4058 3 from (α,pγ), 4059.2 4 from (p,γ).<br>J <sup>π</sup> : L=1 from 0 <sup>+</sup> in ( <sup>3</sup> He,d); (3/2 <sup>-</sup> ) from FRESCO-DWBA analysis of measured σ(θ) of (α,p) (2019Se09).<br>T <sub>1/2</sub> : lifetime τ=20 fs 3: weighted average of 31 fs 13 from (α,pγ) (1972Br33, DSAM), 22 fs 3 (1972Hu11, DSAM), 20 fs 4 (1973Fa07, DSAM), and 18 fs 3 (1976Me12, DSAM) from (p,γ).                                                                                                                                                                                                                        |
| 4112.4 16                   | 7/2 <sup>+</sup>                 | 49 fs 11                                    | LM O Q      | E(level): weighted average of 4111.1 29 from (α,p), 4108 2 from (α,pγ), and 4113.7 10 from (p,γ).<br>J <sup>π</sup> : 4113.4γ E2(+M3) to 3/2 <sup>+</sup> .<br>T <sub>1/2</sub> : lifetime τ=70 fs 16 from (α,pγ) (1972Br33, DSAM).                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 4173.43 23                  | (5/2 <sup>+</sup> )              | 35 fs 6                                     | M O q       | T=1/2<br>XREF: q(4170)<br>E(level): weighted average of 4171 2 from (α,pγ) and 4173.45 19 from (p,γ).<br>J <sup>π</sup> : based on primary transitions from proton resonances in (p,γ): 3827.3γ from 8001.0, (7/2 <sup>+</sup> ), 4456.4γ from 8630.1, 7/2 <sup>-</sup> level, 4243.0γ from 8416.7, 1/2 <sup>+</sup> level. Note that primary 3099.1γ from 7272.7, 1/2 <sup>-</sup> resonance disfavors 5/2 <sup>+</sup> assignment, but this γ branch is relatively weak.<br>T <sub>1/2</sub> : lifetime τ=51 fs 8: weighted average of 68 fs 24 in (α,pγ) (1972Br33, DSAM), 105 fs +35-30 (1971Wi13, DSAM), 55 fs 6 (1972Hu11, DSAM), and 34 fs 9 (1976Me12, DSAM) in (p,γ). |
| 4177.9 2                    | 3/2 <sup>-</sup>                 | 27.0 fs 28                                  | L O qR      | XREF: q(4170)R(4167)<br>J <sup>π</sup> : spin=3/2 from γ(θ) in (p,γ); L=1 from 0 <sup>+</sup> in ( <sup>3</sup> He,d).<br>T <sub>1/2</sub> : lifetime τ=39 fs 4: weighted average of 32 fs 5 (1972Hu11, DSAM), 42 fs 4 (1973Fa07, DSAM), and 47 fs 10 (1976Me12, DSAM) from (p,γ).                                                                                                                                                                                                                                                                                                                                                                                             |
| 4347.76 7                   | 9/2 <sup>-</sup>                 | 1.0 ps 6                                    | FGHI LM O   | E(level): weighted average of 4347.8 6 from ( <sup>28</sup> Si,αpγ), 4348.29 24 from ( <sup>24</sup> Mg,αpγ), 4347.8 3 from ( <sup>16</sup> O,αpγ), 4347.43 17 from ( <sup>14</sup> N,αpnγ), 4347.8 12 from (α,pγ), and 4347.76 6 from (p,γ).<br>J <sup>π</sup> : spin from γ(θ) in (p,γ); parity from 1184.6γ, M1+E2 to 7/2 <sup>-</sup> , 3163 level in (p,γ) (1974Lo17).<br>T <sub>1/2</sub> : lifetime τ=1.5 ps 8: unweighted average of 1.31 ps +30-23 from ( <sup>28</sup> Si,αpγ) (2007Ks01, DSAM), 2.9 ps +14-7 (1972Br33, DSAM), and 0.33 ps 6 (1976Me12, DSAM) from (p,γ).                                                                                           |
| 4624.3 3                    | 3/2,5/2                          | 40 fs 17                                    | B LM O      | XREF: B(?)<br>E(level): other: 4625.3 28 from (α,p); 4618 2 from                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

Continued on next page (footnotes at end of table)

**Adopted Levels, Gammas (continued)** $^{35}\text{Cl}$  Levels (continued)

| <u>E(level)<sup>†</sup></u> | <u>J<sup>π</sup></u>                | <u>T<sub>1/2</sub> or Γ<sup>&amp;</sup></u> | <u>XREF</u> | <u>Comments</u>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------|-------------------------------------|---------------------------------------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             |                                     |                                             |             | ( $\alpha, \text{p}\gamma$ ) is discrepant.<br>J <sup>π</sup> : from $\text{p}\gamma(\theta)$ in ( $\alpha, \text{p}\gamma$ ).<br>T <sub>1/2</sub> : lifetime $\tau=58$ fs 24 from ( $\alpha, \text{p}\gamma$ ) in <a href="#">1972Br33</a> with DSAM.<br>E(level): others: 4770.4 26 from ( $\alpha, \text{p}$ ), 4770 15 from ( $\alpha, \text{p}\gamma$ ).<br>J <sup>π</sup> : from primary transitions in ( $\text{p}, \gamma$ ): 4059.2 $\gamma$ from 8829.3, 1/2 <sup>-</sup> level, 2778.9 $\gamma$ from 7548.9, 7/2 <sup>-</sup> level, 3231.0 $\gamma$ from 8001.0, (7/2 <sup>+</sup> ) level. Other: 7/2, 9/2 from $\text{p}\gamma(\theta)$ for a group at E=4770 15 in ( $\alpha, \text{p}\gamma$ ) is inconsistent and disfavored by $\gamma$ rays to 1/2 <sup>+</sup> and 3/2 <sup>+</sup> .<br>T <sub>1/2</sub> : lifetime $\tau=110$ fs 29: weighted average of 270 fs 95 ( <a href="#">1972Hu11</a> , DSAM) and 105 fs 17 ( <a href="#">1976Me12</a> , DSAM) in ( $\text{p}, \gamma$ ).<br>XREF: x(4839) |
| 4769.86 23                  | (5/2 <sup>-</sup> )                 | 76 fs 20                                    | LM 0        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 4839.10 11                  | (1/2 <sup>+</sup> , 3/2)            | 9.7 fs 35                                   | 0 x         | J <sup>π</sup> : from primary $\gamma$ transitions in ( $\text{p}, \gamma$ ): 2339.4 $\gamma$ from 7178.6, 1/2 <sup>(+)</sup> level, 2355.4 $\gamma$ from 7194.6, 1/2 <sup>-</sup> level, 2867.2 $\gamma$ from 7706.4, 5/2 <sup>+</sup> level.<br>T <sub>1/2</sub> : lifetime $\tau=14$ fs 5 from ( $\text{p}, \gamma$ ) ( <a href="#">1976Me12</a> , DSAM).<br>XREF: x(4839)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 4854.4 19                   | (1/2, 3/2)                          | 4.9 fs 14                                   | L 0 x       | E(level): other: 4859.2 28 from ( $\alpha, \text{p}$ ).<br>J <sup>π</sup> : from primary $\gamma$ transitions in ( $\text{p}, \gamma$ ): 2340.1 $\gamma$ from 7194.6, 1/2 <sup>-</sup> level, 3183.9 $\gamma$ from 8038.4, 1/2 <sup>+</sup> , 3/2 <sup>+</sup> level; decay 3634.8 $\gamma$ to 1219.38, 1/2 <sup>+</sup> .<br>T <sub>1/2</sub> : lifetime $\tau=7$ fs 2 from ( $\text{p}, \gamma$ ) ( <a href="#">1976Me12</a> , DSAM).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 4880.96 15                  | 7/2 <sup>(+)</sup>                  | 5.6 fs 14                                   | LM 0        | E(level): others: 4881.8 29 from ( $\alpha, \text{p}$ ), 4880.8 21 from ( $\alpha, \text{p}\gamma$ ).<br>J <sup>π</sup> : spin=7/2 from $\text{p}\gamma(\theta)$ in ( $\alpha, \text{p}\gamma$ ); 3417.1 primary $\gamma$ from 8298.2, 3/2 <sup>+</sup> resonance in ( $\text{p}, \gamma$ ).<br>T <sub>1/2</sub> : lifetime $\tau=8$ fs 2: weighted average of 7 fs 2 ( <a href="#">1972Hu11</a> , DSAM), 7 fs 3 ( <a href="#">1973Fa07</a> , DSAM), and 10 fs 3 ( <a href="#">1976Me12</a> , DSAM) from ( $\text{p}, \gamma$ ). Other: $\tau=0.28$ ps 6 from ( $\alpha, \text{p}\gamma$ ) in <a href="#">1972Br33</a> with DSAM.                                                                                                                                                                                                                                                                                                                                                                                        |
| 5009.1 18                   | (1/2, 3/2)                          | 7.6 fs 21                                   | LM 0 R      | E(level): weighted average of 5006.9 27 from ( $\alpha, \text{p}$ ) and 5010.1 18 from ( $\text{p}, \gamma$ ). Others: 5015 20 from ( $\alpha, \text{p}\gamma$ ) and 5010 15 from ( $^3\text{He}, \text{d}$ ).<br>J <sup>π</sup> : primary 2185.4 $\gamma$ from 7194.6, 1/2 <sup>-</sup> level and 3407.4 $\gamma$ from 8416.7, 1/2 <sup>+</sup> level.<br>T <sub>1/2</sub> : lifetime $\tau=11$ fs 3 from ( $\text{p}, \gamma$ ) ( <a href="#">1976Me12</a> , DSAM).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 5158.6 29                   | 3/2 <sup>+</sup> , 5/2 <sup>+</sup> |                                             | Lm R WX     | XREF: m(5175)<br>E(level): weighted average of 5158.9 29 from ( $\alpha, \text{p}$ ), 5163 15 from ( $^3\text{He}, \text{d}$ ), 5150 20 from $^{36}\text{Ar}(\text{d}, ^3\text{He})$ , and 5155 11 from $^{36}\text{Ar}(\text{pol d}, ^3\text{He})$ .<br>J <sup>π</sup> : L=2 from 0 <sup>+</sup> in (pol d, $^3\text{He}$ ) and (d, $^3\text{He}$ ).<br>XREF: m(5175)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5163.34 22                  | 7/2 <sup>-</sup>                    |                                             | m 0         | J <sup>π</sup> : spin=7/2 from the feeding 3466 $\gamma(\theta)$ from 8630, 7/2 <sup>-</sup> level and RUL in ( $\text{p}, \gamma$ ); parity=- from 3466 $\gamma$ , M1+E2 from 8630, 7/2 <sup>-</sup> level.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 5214.5 20                   | (5/2 <sup>+</sup> ) <sup>#</sup>    | <4.85 fs                                    | LM 0        | XREF: M(5230)<br>E(level): weighted average of 5209.5 28 from ( $\alpha, \text{p}$ ),                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

Continued on next page (footnotes at end of table)

**Adopted Levels, Gammas (continued)**

| <sup>35</sup> Cl Levels (continued) |                                    |                                        |      |         |                                                                                                                                                                                                                        |
|-------------------------------------|------------------------------------|----------------------------------------|------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E(level) <sup>†</sup>               | J <sup>π</sup> <sup>‡</sup>        | T <sub>1/2</sub> or Γ <sup>&amp;</sup> | XREF |         | Comments                                                                                                                                                                                                               |
|                                     |                                    |                                        |      |         | 5230 20 from (α,pγ), and 5215.8 15 from (p,γ).                                                                                                                                                                         |
|                                     |                                    |                                        |      |         | J <sup>π</sup> : other: 3/2,5/2 from pγ(θ) in (α,pγ).                                                                                                                                                                  |
|                                     |                                    |                                        |      |         | T <sub>1/2</sub> : lifetime τ<7 fs from (p,γ) (1976Me12, DSAM).                                                                                                                                                        |
| 5403.3 10                           | 3/2 <sup>-</sup>                   | 11.8 fs 28                             | L    | O R     | E(level): weighted average of 5402.0 29 from (α,p) and 5403.5 10 from (p,γ). Other: 5409 12 from ( <sup>3</sup> He,d).                                                                                                 |
|                                     |                                    |                                        |      |         | J <sup>π</sup> : L=1 from 0 <sup>+</sup> in ( <sup>3</sup> He,d); 2400.6γ to 5/2 <sup>+</sup> rules out 1/2 <sup>-</sup> based on RUL, since it would require a large B(M2)(W.u.) exceeding RUL.                       |
|                                     |                                    |                                        |      |         | T <sub>1/2</sub> : lifetime τ=17 fs 4 from (p,γ) (1976Me12, DSAM).                                                                                                                                                     |
|                                     |                                    |                                        |      |         | The contribution of Iγ to unknown levels adds up to 36% of Iγ(4183)=100 18 from (p,γ).                                                                                                                                 |
| 5407.07 <sup>a</sup> 21             | 11/2 <sup>-</sup>                  | 0.28 ps 7                              | FGHI | LM      | E(level): weighted average of 5407.4 8 from ( <sup>28</sup> Si,αpγ), 5407.64 27 from ( <sup>24</sup> Mg,αpγ), 5407.3 3 from ( <sup>16</sup> O,αpγ), 5406.68 19 from ( <sup>14</sup> N,αpγ), and 5407.1 15 from (α,pγ). |
|                                     |                                    |                                        |      |         | J <sup>π</sup> : spin=11/2 from pγ(θ) in (p,γ); parity from 1059γ, M1+E2 to 9/2 <sup>-</sup> , 4347 level.                                                                                                             |
|                                     |                                    |                                        |      |         | T <sub>1/2</sub> : lifetime τ=0.4 ps 1 from (α,pγ) (1974Lo17, DSAM). Other: τ<1.1 ps from ( <sup>28</sup> Si,αpγ) (2007Ks01, DSAM).                                                                                    |
| 5531 5                              |                                    |                                        | LM   | O       | E(level): weighted average of 5531 5 from (α,p) and 5535 20 from (α,pγ).                                                                                                                                               |
| 5586.0 20                           | 5/2 <sup>+</sup>                   |                                        | Lm   | O wx    | XREF: m(5600)w(5600)x(5583)                                                                                                                                                                                            |
|                                     |                                    |                                        |      |         | E(level): weighted average of 5586.0 26 from (α,p) and 5586 2 from (p,γ).                                                                                                                                              |
|                                     |                                    |                                        |      |         | J <sup>π</sup> : L=2 from 0 <sup>+</sup> in (pol d, <sup>3</sup> He) and L+1/2 transfer from analyzing power.                                                                                                          |
| 5598.4 22                           | 3/2 <sup>+</sup> ,5/2 <sup>+</sup> | 2.1 fs 7                               | Lm   | O wx    | XREF: m(5600)w(5600)x(5583)                                                                                                                                                                                            |
|                                     |                                    |                                        |      |         | E(level): weighted average of 5594.6 26 from (α,p) and 5599.7 15 from (p,γ).                                                                                                                                           |
|                                     |                                    |                                        |      |         | J <sup>π</sup> : L(d, <sup>3</sup> He)=2 from 0 <sup>+</sup> .                                                                                                                                                         |
|                                     |                                    |                                        |      |         | T <sub>1/2</sub> : lifetime τ=3 fs 1 from (p,γ) (1976Me12, DSAM).                                                                                                                                                      |
| 5633.6 30                           |                                    |                                        | Lm   | r w y   | XREF: m(5650)r(5660)w(5600)y(5650)                                                                                                                                                                                     |
|                                     |                                    |                                        |      |         | E(level): from (α,p).                                                                                                                                                                                                  |
| 5646 2                              | (5/2,7/2,9/2 <sup>+</sup> )        | 2.8 fs 7                               | Lm   | O r y   | XREF: m(5650)r(5660)y(5650)                                                                                                                                                                                            |
|                                     |                                    |                                        |      |         | E(level): weighted average of 5645 4 from (α,p) and 5646 2 from (p,γ).                                                                                                                                                 |
|                                     |                                    |                                        |      |         | J <sup>π</sup> : from primary transitions in (p,γ): 1903γ from 7548.9, 7/2 <sup>-</sup> level, 2355γ from 8001.0, (7/2 <sup>+</sup> ) level, 3261γ from 8906.8, 5/2 <sup>+</sup> level.                                |
|                                     |                                    |                                        |      |         | T <sub>1/2</sub> : lifetime τ=4 fs 1 from (p,γ) in 1976Me12 with DSAM.                                                                                                                                                 |
| 5655 2                              | (3/2) <sup>+</sup>                 | 13.9 fs 35                             | Lm   | O r w y | T=3/2                                                                                                                                                                                                                  |
|                                     |                                    |                                        |      |         | XREF: m(5650)r(5660)w(5600)y(5650)                                                                                                                                                                                     |
|                                     |                                    |                                        |      |         | E(level): weighted average of 5653 4 from (α,p) and 5655 2 from (p,γ).                                                                                                                                                 |
|                                     |                                    |                                        |      |         | J <sup>π</sup> : L=2 from 0 <sup>+</sup> in ( <sup>3</sup> He,d); primary 2142γ from 7797.2, 1/2 <sup>-</sup> level in (p,γ). Note that a weak primary 1903γ from 7548.9, 7/2 <sup>-</sup> level in                    |

Continued on next page (footnotes at end of table)

**Adopted Levels, Gammas (continued)**

| $^{35}\text{Cl}$ Levels (continued) |                                    |                                        |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------|------------------------------------|----------------------------------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E(level) <sup>†</sup>               | J <sup>π</sup> <sup>‡</sup>        | T <sub>1/2</sub> or Γ <sup>&amp;</sup> | XREF    | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                     |                                    |                                        |         | (p,γ) favors 5/2 <sup>+</sup> .<br>T <sub>1/2</sub> : lifetime τ=20 fs 5 from (p,γ) (1976Me12, DSAM).<br>The contribution of I <sub>γ</sub> to unknown levels adds up to 18% of I <sub>γ</sub> (2652)=100 6 from (p,γ).<br>XREF: m(5650)<br>E(level): weighted average of 5679.7 34 from (α,p), 5683 2 from (p,γ), and 5684 12 from ( <sup>3</sup> He,d).<br>J <sup>π</sup> : L=1 from 0 <sup>+</sup> in ( <sup>3</sup> He,d).<br>The contribution of I <sub>γ</sub> to unknown levels adds up to 18% of I <sub>γ</sub> (1504)=100 from (p,γ).<br>E(level): others: 5731.2 25 from (α,p), 5720 20 from <sup>36</sup> Ar(d, <sup>3</sup> He), and 5720 11 from <sup>36</sup> Ar(pol d, <sup>3</sup> He).<br>J <sup>π</sup> : L=2 from 0 <sup>+</sup> in (pol d, <sup>3</sup> He) and (d, <sup>3</sup> He) and L+1/2 transfer from analyzing power.<br>The contribution of I <sub>γ</sub> to unknown levels adds up to 100% of I <sub>γ</sub> (3960)=100 11 from (p,γ).<br>E(level): weighted average of 5757 4 from (α,p), 5758 3 from (p,γ), and 5760 12 from ( <sup>3</sup> He,d).<br>J <sup>π</sup> : L( <sup>3</sup> He,d)=(0,1) from 0 <sup>+</sup> .<br>The contribution of I <sub>γ</sub> to unknown levels adds up to 56% of I <sub>γ</sub> (5758)=100 22 from (p,γ).<br>E(level): weighted average of 5808.1 31 from (α,p) and 5806 2 from (p,γ).<br>J <sup>π</sup> : primary 2890.2γ from 8696.9, 3/2 <sup>-</sup> level; 5806.1γ to 3/2 <sup>+</sup> g.s.<br>T <sub>1/2</sub> : lifetime τ=5 fs 1 from (p,γ) (1976Me12, DSAM).<br>XREF: M(5850)<br>E(level): weighted average of 5829 4 from (α,p) and 5850 25 from (α,pγ).<br>J <sup>π</sup> : other: spin=5/2,9/2 from pγ(θ) in (α,pγ).<br>E(level): weighted average of 5927.1 7 from ( <sup>28</sup> Si,αpγ), 5926.9 5 from ( <sup>16</sup> O,αpγ), 5926.47 29 from ( <sup>14</sup> N,αpγ), 5928 3 from (α,p), and 5927.1 19 from (α,pγ).<br>J <sup>π</sup> : 1579γ M1+E2, ΔJ=1 to 4348, 9/2 <sup>-</sup> in <sup>12</sup> C( <sup>28</sup> Si,αpγ), also assuming spins increase with excitation energies in fusion-evaporation reactions. Spin=7/2,11/2 from pγ(θ) in (α,pγ).<br>T <sub>1/2</sub> : lifetime τ<0.4 ps from ( <sup>28</sup> Si,αpγ) (2007Ks01, DSAM).<br>E(level): weighted average of 6087.6 8 from ( <sup>28</sup> Si,αpγ), 6088.05 29 from ( <sup>24</sup> Mg,αpγ), 6087.2 4 from ( <sup>16</sup> O,αpγ), 6086.89 23 from ( <sup>14</sup> N,αpγ), 6084.2 29 from (α,p), and 6087.0 18 from (α,pγ).<br>J <sup>π</sup> : spin=13/2 from pγ(θ) in (α,pγ); parity from 680γ, M1+E2 to 11/2 <sup>-</sup> , 5406 level in (α,pγ).<br>T <sub>1/2</sub> : lifetime τ=8.9 ps 8: weighted average of 9.3 ps 8 from (α,pγ) in 1974Lo17 with RDM and 7.7 ps 14 from ( <sup>14</sup> N,αpγ) in 1976Wa11.<br>XREF: l(6104)w(6140)x(6136)<br>E(level): weighted average of 6104 4 from (α,p) |
| 5682.2 20                           | 1/2 <sup>-</sup> ,3/2 <sup>-</sup> |                                        | Lm O R  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5723.52 18                          | 5/2 <sup>+</sup>                   |                                        | L O WX  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5758 3                              | (1/2,3/2 <sup>-</sup> )            |                                        | L O R   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5806.6 20                           | (1/2,3/2,5/2)                      | 3.5 fs 7                               | L O     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5830 4                              | (5/2 <sup>-</sup> ) <sup>#</sup>   |                                        | LM      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5926.65 29                          | (11/2) <sup>-</sup>                | <0.28 ps                               | F HI LM |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 6087.31 24                          | 13/2 <sup>-</sup>                  | 6.2 ps 6                               | FGHI LM |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 6106.2 21                           | (3/2,5/2 <sup>+</sup> )            | 8.3 fs 21                              | 1 O wx  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

Continued on next page (footnotes at end of table)

**Adopted Levels, Gammas (continued)** $^{35}\text{Cl}$  Levels (continued)

| <u>E(level)<sup>†</sup></u> | <u>J<sup>π</sup></u>                    | <u>XREF</u> |      | <u>Comments</u>                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------|-----------------------------------------|-------------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             |                                         |             |      | and 6106.2 21 from (p,γ).<br>J <sup>π</sup> : 4886.5γ to 1/2 <sup>+</sup> , 4342.8γ to 5/2 <sup>+</sup> ; primary 1675.5γ from 7781.7, (5/2 <sup>-</sup> ) level.<br>T <sub>1/2</sub> : lifetime τ=12 fs 3 from (p,γ) in 1976Me12 with DSAM.<br>XREF: w(6140)x(6136)<br>J <sup>π</sup> : L=2 from 0 <sup>+</sup> in (d, <sup>3</sup> He) and (pol d, <sup>3</sup> He), and L+1/2 transfer from analyzing power.                                |
| 6141 4                      | 5/2 <sup>+</sup>                        | L           | WX   | XREF: n(6200)<br>E(level): other: 6180 3 from (α,p).<br>J <sup>π</sup> : 6180γ to 3/2 <sup>+</sup> g.s.; primary 1438γ from 7618.7, 5/2 <sup>+</sup> level in (p,γ).<br>Apart from the γ branch I <sub>γ</sub> (6180)=100 observed in (p,γ), the contribution of I <sub>γ</sub> to unknown levels adds up to 122% of I <sub>γ</sub> (6180).                                                                                                    |
| 6181 3                      | (1/2 <sup>+</sup> to 7/2 <sup>+</sup> ) | L n0        |      | XREF: n(6200)<br>E(level): other: 6180 3 from (α,p).<br>J <sup>π</sup> : 6180γ to 3/2 <sup>+</sup> g.s.; primary 1438γ from 7618.7, 5/2 <sup>+</sup> level in (p,γ).<br>Apart from the γ branch I <sub>γ</sub> (6180)=100 observed in (p,γ), the contribution of I <sub>γ</sub> to unknown levels adds up to 122% of I <sub>γ</sub> (6180).                                                                                                    |
| 6224.4 30                   |                                         | L n         |      | XREF: n(6200)<br>J <sup>π</sup> : L(α,d)=6 for a group at 6200 10.                                                                                                                                                                                                                                                                                                                                                                             |
| 6283 3                      | (5/2 <sup>+</sup> ) <sup>#</sup>        | L           | R    | E(level): weighted average of 6283 3 from (α,p) and 6284 4 from ( <sup>3</sup> He,d).<br>J <sup>π</sup> : other: L=(2) from 0 <sup>+</sup> in ( <sup>3</sup> He,d).<br>J <sup>π</sup> : L=(0,1) from 0 <sup>+</sup> in ( <sup>3</sup> He,d).                                                                                                                                                                                                   |
| 6329 4                      | (1/2,3/2 <sup>-</sup> )                 |             | R    | XREF: R(6377)<br>E(level): weighted average of 6380.8 8 from ( <sup>16</sup> O,αpγ) and 6379.3 25 from (α,p).<br>J <sup>π</sup> : other: L=(2,3) from 0 <sup>+</sup> in ( <sup>3</sup> He,d) for a group at E=6377 2 based on very few data points in a very narrow range of angles in the DWBA fit.                                                                                                                                           |
| 6380.2 9                    | (9/2 <sup>-</sup> ) <sup>#</sup>        | H L         | R    |                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 6401.2 22                   | (1/2 <sup>-</sup> ) <sup>#</sup>        | LM          |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 6428 2                      | (1/2 <sup>+</sup> ) <sup>#</sup>        | L           | R    | E(level): weighted average of 6429 3 from (α,p) and 6427 2 from ( <sup>3</sup> He,d).<br>J <sup>π</sup> : other: L=(3) from 0 <sup>+</sup> in ( <sup>3</sup> He,d) is inconsistent.                                                                                                                                                                                                                                                            |
| 6469 3                      | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> )   | L           | R    | E(level): weighted average of 6475 4 from (α,p) and 6468 2 from ( <sup>3</sup> He,d).<br>J <sup>π</sup> : L=(1) from 0 <sup>+</sup> in ( <sup>3</sup> He,d).                                                                                                                                                                                                                                                                                   |
| 6492 2                      | (3/2 <sup>+</sup> ) <sup>#</sup>        | L O R       |      | E(level): weighted average of 6492 3 from (α,p), 6492 3 from (p,γ), and 6491 2 from ( <sup>3</sup> He,d).<br>J <sup>π</sup> : other: L=(2) from 0 <sup>+</sup> in ( <sup>3</sup> He,d).                                                                                                                                                                                                                                                        |
| 6546 2                      | (1/2 <sup>+</sup> ) <sup>#</sup>        | L           | R    | E(level): weighted average of 6549 3 from (α,p) and 6545 2 from ( <sup>3</sup> He,d).<br>J <sup>π</sup> : other: L=(0,1) from 0 <sup>+</sup> in ( <sup>3</sup> He,d), indicating (1/2,3/2 <sup>-</sup> ).<br>E(level): from ( <sup>3</sup> He,d).<br>J <sup>π</sup> : from L=(1) from 0 <sup>+</sup> in ( <sup>3</sup> He,d).                                                                                                                  |
| 6643 2                      | (1/2 <sup>-</sup> ,3/2 <sup>-</sup> )   | D           | R    |                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 6659.1 30                   | (7/2 <sup>+</sup> ) <sup>#</sup>        | L           | y    | XREF: y(6677)<br>J <sup>π</sup> : 2312.4γ to 9/2 <sup>-</sup> , 3497.1γ to 7/2 <sup>-</sup> in ( <sup>28</sup> Si,αpγ), assuming spin increasing as excitation energy increases.                                                                                                                                                                                                                                                               |
| 6660.2 8                    | (11/2 <sup>-</sup> )                    | F           |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 6675.6 25                   |                                         | D L         | R y  | XREF: y(6677)<br>E(level): weighted average of 6679.4 31 from (α,p) and 6674 2 from ( <sup>3</sup> He,d).<br>XREF: X(6745)<br>E(level): weighted average of 6761 2 from ( <sup>3</sup> He,d), 6750 20 from <sup>36</sup> Ar(d, <sup>3</sup> He), and 6745 12 from <sup>36</sup> Ar(pol d, <sup>3</sup> He).<br>J <sup>π</sup> : L(d, <sup>3</sup> He)=2 from 0 <sup>+</sup> .<br>E(level): weighted average of 6781.3 30 from (α,p) and 6778 2 |
| 6761 2                      | 3/2 <sup>+</sup> ,5/2 <sup>+</sup>      | D           | R WX |                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 6779.0 20                   |                                         | D L         | R    |                                                                                                                                                                                                                                                                                                                                                                                                                                                |

Continued on next page (footnotes at end of table)

**Adopted Levels, Gammas (continued)** $^{35}\text{Cl}$  Levels (continued)

| E(level) <sup>†</sup> | J <sup>π</sup> <sup>‡</sup>      | T <sub>1/2</sub> or Γ <sup>&amp;</sup> | XREF |    |       | Comments                                                                                                                                                                                                                                                                                  |
|-----------------------|----------------------------------|----------------------------------------|------|----|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6800 4                |                                  |                                        | D    | L  |       | from ( $^3\text{He},d$ ).<br>XREF: D(6802)<br>E(level): from ( $\alpha,p$ ).                                                                                                                                                                                                              |
| 6823 2                |                                  |                                        | D    |    | R     | E(level): from ( $^3\text{He},d$ ).                                                                                                                                                                                                                                                       |
| 6842 2                |                                  |                                        | D    | L  | R     | E(level): weighted average of 6842 4 from ( $\alpha,p$ )<br>and 6842 2 from ( $^3\text{He},d$ ).                                                                                                                                                                                          |
| 6863 3                | (9/2 <sup>+</sup> ) <sup>#</sup> |                                        |      | L  |       |                                                                                                                                                                                                                                                                                           |
| 6866.7 6              | (1/2,3/2 <sup>-</sup> )          |                                        | D    | O  | R     | E(level): other: 6866 2 from ( $^3\text{He},d$ ).<br>J <sup>π</sup> : L( $^3\text{He},d$ )=(0,1) from 0 <sup>+</sup> .                                                                                                                                                                    |
| 6891.6 30             | (9/2 <sup>+</sup> ) <sup>#</sup> |                                        |      | L  |       |                                                                                                                                                                                                                                                                                           |
| 6948.1 34             | 5/2 <sup>+</sup>                 |                                        |      | L  | WX    | XREF: W(6930)<br>E(level): weighted average of 6948.6 34 from ( $\alpha,p$ )<br>and 6930 20 from $^{36}\text{Ar}(d,^3\text{He})$ . Other: 6953 43<br>from $^{36}\text{Ar}(\text{pol } d,^3\text{He})$ .<br>J <sup>π</sup> : L(pol d, $^3\text{He}$ )=2 and L+1/2 from analyzing<br>power. |
| 6986.3 34             | (9/2 <sup>+</sup> ) <sup>#</sup> |                                        |      | L  |       |                                                                                                                                                                                                                                                                                           |
| 7066.2 7              | (5/2 <sup>+</sup> ) <sup>#</sup> |                                        |      | L  | O R   | E(level): weighted average of 7066.2 7 from (p, $\gamma$ ),<br>and 7066 2 from ( $^3\text{He},d$ ).                                                                                                                                                                                       |
| 7103.4 7              | 3/2 <sup>(-)</sup>               |                                        |      | L  | O R   | XREF: L(7105?)<br>E(level): weighted average of 7103.4 7 from (p, $\gamma$ ),<br>and 7103 2 from ( $^3\text{He},d$ ).<br>J <sup>π</sup> : spin=3/2 from $\gamma(\theta)$ in (p, $\gamma$ ); parity from<br>L( $^3\text{He},d$ )=(1,3) from 0 <sup>+</sup> .                               |
| 7121.7 24             |                                  |                                        |      | L  |       |                                                                                                                                                                                                                                                                                           |
| 7178.6 8              | 1/2 <sup>(+)</sup>               |                                        |      |    | O     | T=3/2<br>J <sup>π</sup> : spin=1/2 from $\gamma(\theta)$ in (p, $\gamma$ ); parity from IAS<br>in $^{35}\text{S}$ .                                                                                                                                                                       |
| 7179 4                | (7/2 <sup>+</sup> ) <sup>#</sup> |                                        |      | L  | N     | E(level): weighted average of 7180 4 from $^{32}\text{S}(\alpha,p)$<br>and 7170 10 from $^{33}\text{S}(\alpha,d)$ .<br>J <sup>π</sup> : also L=6 from 3/2 <sup>+</sup> in $^{33}\text{S}(\alpha,d)$ , indicating<br>(7/2;17/2) <sup>+</sup> .                                             |
| 7185                  | 5/2 <sup>+</sup>                 |                                        |      |    | O R X | XREF: R(7178)X(7181)<br>J <sup>π</sup> : L(pol d, $^3\text{He}$ )=2 and L+1/2 from analyzing<br>power for a group at E=7181 60. Other:<br>L( $^3\text{He},d$ )=(2) from 0 <sup>+</sup> for a group at E=7178 2<br>considered the same level.                                              |
| 7194.6 7              | 1/2 <sup>-</sup>                 | 0.027 keV 8                            |      | OP | R     | J <sup>π</sup> : from resonance analysis of measured $\sigma(\theta)$<br>(1971BrXT) in $^{34}\text{S}(p,p),(p,p'\gamma)$ :resonances.<br>Other: 1/2,3/2 from $\gamma(\theta)$ in (p, $\gamma$ ).                                                                                          |
| 7213.7 25             |                                  |                                        |      | L  |       | y XREF: y(7250)                                                                                                                                                                                                                                                                           |
| 7225.5 8              | 5/2                              |                                        |      | O  |       | y XREF: y(7250)                                                                                                                                                                                                                                                                           |
| 7228.2 20             | (3/2 <sup>+</sup> ) <sup>#</sup> |                                        |      | L  | R y   | J <sup>π</sup> : from $\gamma(\theta)$ in (p, $\gamma$ ).<br>XREF: R(7227)y(7250)                                                                                                                                                                                                         |
| 7234.0 8              | 5/2 <sup>(+)</sup>               |                                        |      | O  |       | E(level): weighted average of 7230.9 30 from<br>$^{32}\text{S}(\alpha,p)$ , 7227 2 from $^{34}\text{S}(\alpha,d)$ .<br>J <sup>π</sup> : other: L=(0,1) from 0 <sup>+</sup> in ( $^3\text{He},d$ ) is<br>inconsistent.                                                                     |
| 7269.2 1              |                                  |                                        |      | O  |       | J <sup>π</sup> : spin=5/2 from $\gamma(\theta)$ in (p, $\gamma$ ); primary 6014.1 $\gamma$<br>to 1/2 <sup>+</sup> .                                                                                                                                                                       |
| 7272.7 9              | 1/2 <sup>-</sup>                 | 0.014 keV 4                            |      | OP | R y   | XREF: P(7274)R(7273)y(7250)                                                                                                                                                                                                                                                               |

Continued on next page (footnotes at end of table)



**Adopted Levels, Gammas (continued)**

| $^{35}\text{Cl}$ Levels (continued) |                                             |                                        |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------|---------------------------------------------|----------------------------------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E(level) <sup>†</sup>               | J <sup>π</sup> <sup>‡</sup>                 | T <sub>1/2</sub> or Γ <sup>&amp;</sup> | XREF  | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                     |                                             |                                        |       | J <sup>π</sup> : from resonance analysis of measured $\sigma(\theta)$ (1971BrXT) in $^{34}\text{S}(\text{p},\text{p}), (\text{p},\text{p}'\gamma)$ : resonances. Other: (1/2,3/2) from $\gamma(\theta)$ in (p, $\gamma$ ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 7348.3                              | (7/2) <sup>#</sup>                          |                                        | L     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 7361.9 8                            | 3/2 <sup>(+)</sup>                          |                                        | L 0 R | E(level): weighted average of 7362.3 from ( $\alpha$ ,p), 7362.0 8 from (p, $\gamma$ ), and 7361.2 from ( $^3\text{He}$ ,d). J <sup>π</sup> : spin=3/2 from $\gamma(\theta)$ in (p, $\gamma$ ); 6141.9 $\gamma$ (M1+E2) to 1/2 <sup>+</sup> .                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 7396.4 10                           | 7/2 <sup>(-)</sup>                          |                                        | 0 R   | E(level): weighted average of 7396.0 10 from (p, $\gamma$ ), and 7398.2 from ( $^3\text{He}$ ,d). J <sup>π</sup> : spin=7/2 from $\gamma(\theta)$ in (p, $\gamma$ ); 4233.3 $\gamma$ (M1+E2) to 7/2 <sup>-</sup> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 7413.8 30                           |                                             |                                        | L     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 7450.9 8                            | 3/2                                         |                                        | L 0   | E(level): weighted average of 7447.3 from ( $\alpha$ ,p) and 7451.1 6 from (p, $\gamma$ ). J <sup>π</sup> : from $\gamma(\theta)$ in (p, $\gamma$ ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 7496                                |                                             |                                        | 0     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 7501.1 8                            | (5/2 <sup>+</sup> , 7/2, 9/2 <sup>+</sup> ) |                                        | 1 0   | XREF: l(7503)<br>J <sup>π</sup> : primary 3558.0 $\gamma$ to 9/2 <sup>+</sup> , 4498.2 $\gamma$ to 5/2 <sup>+</sup> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 7502.9 7                            |                                             |                                        | 1 0   | XREF: l(7503)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 7518.8 7                            | 7/2                                         |                                        | 0     | J <sup>π</sup> : from $\gamma(\theta)$ in (p, $\gamma$ ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 7548.9 6                            | 7/2 <sup>-</sup>                            | <0.69 fs                               | 0P    | T=3/2<br>XREF: P(7550)<br>J <sup>π</sup> : from resonance analysis of measured $\sigma(\theta)$ in $^{34}\text{S}(\text{p},\text{p}), (\text{p},\text{p}'\gamma)$ : resonances. Spin=7/2 also from $\gamma(\theta)$ in (p, $\gamma$ ).<br>T <sub>1/2</sub> : lifetime $\tau < 1$ fs from (p, $\gamma$ ) (1973Fa07, DSAM). Other: $\Gamma_p = 4.6$ eV 10 and $\Gamma = 5$ eV +4-2 from $^{34}\text{S}(\text{p},\text{p})$ elastic scattering excitation function analysis (1971BrXT). Other $\Gamma_p$ , $\Gamma_\gamma$ , $\Gamma$ values/ratios are mostly deduced from (p, $\gamma$ ) resonance strengths, which have large discrepancies. See comments in the (p, $\gamma$ ) dataset for details. |
| 7561.3 6                            | 1/2, 3/2                                    |                                        | L 0   | E(level): other: 7566.5 35 from ( $\alpha$ ,p). J <sup>π</sup> : from $\gamma(\theta)$ in (p, $\gamma$ ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 7565.20                             | 5/2 <sup>+</sup>                            |                                        |       | X J <sup>π</sup> : L=2 from 0 <sup>+</sup> in (pol d, $^3\text{He}$ ) and L+1/2 transfer from analyzing power.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 7590.4                              |                                             |                                        | L     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 7600.8 8                            | 5/2 <sup>+</sup>                            | <13.9 fs                               | 0     | J <sup>π</sup> : spin=5/2 from $\gamma(\theta)$ in (p, $\gamma$ ); 7599.9 $\gamma$ M1+E2 to 3/2 <sup>+</sup> .<br>T <sub>1/2</sub> : lifetime $\tau < 20$ fs from (p, $\gamma$ ) (1971Wi13, DSAM), which is equivalent to $\Gamma > 0.033$ eV.                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 7618.7 5                            | 5/2 <sup>+</sup>                            |                                        | 0     | J <sup>π</sup> : spin=5/2 from $\gamma(\theta)$ in (p, $\gamma$ ); 7617.8 $\gamma$ M1+E2 to 3/2 <sup>+</sup> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 7648.1 30                           |                                             |                                        | L     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 7656.6 8                            | (1/2, 3/2, 5/2 <sup>+</sup> )               |                                        | 0     | J <sup>π</sup> : primary 3688.8 $\gamma$ to 1/2 <sup>+</sup> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 7670.10                             | (7/2 to 17/2) <sup>+</sup>                  |                                        | N     | J <sup>π</sup> : L( $\alpha$ ,d)=6 from 3/2 <sup>+</sup> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 7671.9 8                            | 5/2 <sup>+</sup>                            |                                        | 1 0p  | XREF: p(7686)<br>J <sup>π</sup> : spin=5/2 from $\gamma(\theta)$ in (p, $\gamma$ ); 4668.8 $\gamma$ , M1+E2 to 5/2 <sup>+</sup> ; 5908.2 $\gamma$ , M1+E2 to 5/2 <sup>+</sup> ; other: 3324.0 $\gamma$ to 9/2 <sup>-</sup> seems to favor 5/2 <sup>-</sup> .                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 7685.5 8                            | 3/2 <sup>-</sup>                            |                                        | 0P    | J <sup>π</sup> : spin=3/2 from $\gamma(\theta)$ in (p, $\gamma$ ) and 3/2 <sup>-</sup> from R-Matrix analysis in (p,p): resonances. $\Gamma_p = 446$ eV 15 from (p,p): resonances.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

Continued on next page (footnotes at end of table)



**Adopted Levels, Gammas (continued)**

| $^{35}\text{Cl}$ Levels (continued) |                                             |                                        |         |                                                                                                                                                                                                                                                                                                  |
|-------------------------------------|---------------------------------------------|----------------------------------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E(level) <sup>†</sup>               | J <sup>π</sup> <sup>‡</sup>                 | T <sub>1/2</sub> or Γ <sup>&amp;</sup> | XREF    | Comments                                                                                                                                                                                                                                                                                         |
| 7693.9 8                            | (1/2 <sup>+</sup> , 3/2, 5/2)               |                                        | 0       | J <sup>π</sup> : primary 7693.0γ to 3/2 <sup>+</sup> , 3634.6γ to (3/2) <sup>-</sup> , 4690.9γ to 5/2 <sup>+</sup> .                                                                                                                                                                             |
| 7706.4 8                            | 5/2 <sup>+</sup>                            |                                        | L OP    | E(level): other: 7709.9 25 from (α,p).<br>J <sup>π</sup> : spin=5/2 from γ(θ) in (p,γ) and 3/2 <sup>+</sup> , 5/2 <sup>+</sup> from R-Matrix analysis in (p,p):resonances.                                                                                                                       |
| 7744.8 9                            | 7/2 <sup>-</sup>                            |                                        | L 0     | Γ <sub>p</sub> =4 eV 1 in (p,p).<br>E(level): other: 7744.1 30 from (α,p).<br>J <sup>π</sup> : spin=7/2 from γ(θ) in (p,γ); 4581.6γ M1+E2 to 7/2 <sup>-</sup> .                                                                                                                                  |
| 7748                                | (7/2 to 17/2) <sup>+</sup>                  |                                        | NO      | XREF: N(7750)<br>E(level): from (p,γ).<br>J <sup>π</sup> : L(α,d)=6 from 3/2 <sup>+</sup> .                                                                                                                                                                                                      |
| 7777.1 9                            | (5/2 <sup>+</sup> )                         |                                        | L 0     | E(level): from (p,γ). Other: 7770.3 33 from (α,p).<br>J <sup>π</sup> : based on primary transitions: 4613.9γ to 7/2 <sup>-</sup> , 5131.0γ to 7/2 <sup>+</sup> , 6557.1γ to 1/2 <sup>+</sup> .                                                                                                   |
| 7781.7 11                           | (5/2 <sup>-</sup> )                         |                                        | 0       | J <sup>π</sup> : primary 3433.8γ to 9/2 <sup>-</sup> ; relatively weak 6561.7γ to 1/2 <sup>+</sup> .                                                                                                                                                                                             |
| 7797.2 9                            | 1/2 <sup>-</sup>                            | 0.031 keV 10                           | L OP    | E(level): weighted average of 7799.8 30 from (α,p), 7797.1 9 from (p,γ), and 7796 3 from <sup>34</sup> S(p,p),(p,p'γ):resonances.<br>J <sup>π</sup> : spin=1/2 from γ(θ) in (p,γ); (1/2) <sup>-</sup> from R-Matrix analysis of in (p,p):resonances.                                             |
| 7837.2 10                           | 3/2 <sup>-</sup>                            | 3.7 keV 4                              | OP      | T=3/2<br>E(level): weighted average of 7837.2 10 from (p,γ), and 7837 3 from <sup>34</sup> S(p,p),(p,p'γ):resonances.<br>J <sup>π</sup> : spin=3/2 from γ(θ) in (p,γ) and 3/2 <sup>-</sup> from R-Matrix analysis in (p,p):resonances.<br>Γ: other: lifetime τ<5 fs from (p,γ) (1971Wi13, DSAM). |
| 7839.0 9                            | (5/2 <sup>+</sup> )                         |                                        | 0       | J <sup>π</sup> : primary 6619.0γ to 1/2 <sup>+</sup> , 3895.9γ to 9/2 <sup>+</sup> .                                                                                                                                                                                                             |
| 7868.7 10                           | (1/2 <sup>+</sup> , 3/2, 5/2 <sup>+</sup> ) |                                        | L 0     | E(level): other: 7868.5 35 from (α,p).<br>J <sup>π</sup> : primary 6648.6γ to 1/2 <sup>+</sup> , 2654.1γ to (5/2 <sup>+</sup> ).                                                                                                                                                                 |
| 7873.03 <sup>b</sup> 30             | 13/2 <sup>+</sup>                           |                                        | F HI NO | XREF: N(7870)<br>E(level): weighted average of 7873.3 8 from ( <sup>28</sup> Si,αpγ), 7873.4 4 from ( <sup>16</sup> O,αpγ), and 7872.78 30 from ( <sup>14</sup> N,αpnγ).<br>J <sup>π</sup> : 1946.35γ E1+M2, ΔJ=1 to 5927, (11/2) <sup>-</sup> ; 1786.2γ D(+Q) to 13/2 <sup>-</sup> ; band head. |
| 7880.8 9                            | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>         | 0.008 keV 4                            | OP      | J <sup>π</sup> : from γ(θ) in (p,γ) and R-Matrix analysis in (p,p):resonances.                                                                                                                                                                                                                   |
| 7885                                |                                             |                                        | 0       |                                                                                                                                                                                                                                                                                                  |
| 7889.1 15                           |                                             |                                        | 0       |                                                                                                                                                                                                                                                                                                  |
| 7899.2 7                            | (5/2)                                       |                                        | 1 0     | XREF: l(7901)<br>J <sup>π</sup> : primary 2735.8γ to 7/2 <sup>-</sup> , 3018.1γ to                                                                                                                                                                                                               |

Continued on next page (footnotes at end of table)

**Adopted Levels, Gammas (continued)**

| $^{35}\text{Cl}$ Levels (continued) |                                   |                                        |      |    |                                                                                                                                                                                                                                                                                                    |
|-------------------------------------|-----------------------------------|----------------------------------------|------|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E(level) <sup>†</sup>               | $J^{\pi\ddagger}$                 | $T_{1/2}$ or $\Gamma$ <sup>&amp;</sup> | XREF |    | Comments                                                                                                                                                                                                                                                                                           |
| 7903                                |                                   |                                        | L    | 0  | $7/2^{(+)}$ , 3721.1 $\gamma$ to $3/2^{-}$ , 5204.9 $\gamma$ to $3/2^{+}$ .<br>XREF: I(7901)                                                                                                                                                                                                       |
| 7923.3 9                            | ( $3/2^{+}$ , $5/2^{+}$ )         |                                        |      | 0  | $J^{\pi}$ : primary 5277.2 $\gamma$ to $7/2^{+}$ , 6703.2 $\gamma$ to $1/2^{+}$ .                                                                                                                                                                                                                  |
| 7930                                |                                   |                                        |      | 0  |                                                                                                                                                                                                                                                                                                    |
| 7949                                |                                   |                                        |      | 0  |                                                                                                                                                                                                                                                                                                    |
| 7970.3 10                           | ( $5/2^{-}$ )                     |                                        |      | 0  | $J^{\pi}$ : primary 3622.3 $\gamma$ to $9/2^{-}$ , weak 6750.2 $\gamma$ to $1/2^{+}$ .                                                                                                                                                                                                             |
| 7978.6 35                           |                                   |                                        | L    |    |                                                                                                                                                                                                                                                                                                    |
| 7987.8 10                           | $3/2^{+}$                         |                                        |      | 0  | $J^{\pi}$ : spin= $3/2$ from $\gamma(\theta)$ in (p, $\gamma$ ); 7986.8 $\gamma$ M1+E2 to $3/2^{+}$ .                                                                                                                                                                                              |
| 7995.6 10                           | ( $5/2$ )                         |                                        |      | 0  | $J^{\pi}$ : primary 2592.2 $\gamma$ to $3/2^{-}$ , 7994.6 $\gamma$ to $3/2^{+}$ , 2832.1 $\gamma$ to $7/2^{-}$ , 5349.5 $\gamma$ to $7/2^{+}$ .                                                                                                                                                    |
| 8001.0 9                            | ( $7/2^{+}$ )                     |                                        | L    | 0  | E(level): other: 8000.7 41 from ( $\alpha$ ,p).<br>$J^{\pi}$ : from primary transitions in (p, $\gamma$ ):<br>3653.0 $\gamma$ to $9/2^{-}$ , 4057.9 to $9/2^{+}$ , 3827.3 $\gamma$ to $5/2^{-}$ , 6237.4 $\gamma$ to $5/2^{+}$ , 8000.0 $\gamma$ to $3/2^{+}$ .<br>XREF: P(8004)W(8010)X(8007)     |
| 8005.0 10                           | $5/2^{+}$                         | 0.011 keV 5                            | OP   | WX | $J^{\pi}$ : L(d, $^3\text{He}$ )=2 from $0^{+}$ ; spin= $5/2$ from $\gamma(\theta)$ in (p, $\gamma$ ).<br>$J^{\pi}$ : L( $\alpha$ ,d)=6 from $3/2^{+}$ .                                                                                                                                           |
| 8010 10                             | ( $7/2$ to $17/2$ ) <sup>+</sup>  |                                        | N    |    |                                                                                                                                                                                                                                                                                                    |
| 8019.2 35                           |                                   |                                        | L    |    |                                                                                                                                                                                                                                                                                                    |
| 8035.7 10                           | $1/2^{-}$ , $3/2^{-}$             | 0.026 keV 7                            |      | OP | E(level): other: 8034 3 from $^{34}\text{S}(\text{p,p}), (\text{p,p}'\gamma)$ :resonances.                                                                                                                                                                                                         |
| 8038.4 11                           | $1/2^{+}$ , $3/2^{+}$             | 0.30 keV 2                             |      | OP | E(level): weighted average of 8038.5 11 from (p, $\gamma$ ), and 8038 3 from $^{34}\text{S}(\text{p,p}), (\text{p,p}'\gamma)$ :resonances.<br>$J^{\pi}$ : spin= $3/2$ from $\gamma(\theta)$ in (p, $\gamma$ ), but $1/2^{+}$ from R-Matrix analysis in (p,p); 8037.5 $\gamma$ M1+E2 to $3/2^{+}$ . |
| 8075.9 10                           | ( $5/2$ )                         |                                        | L    | 0  | E(level): weighted average of 8075.1 46 from ( $\alpha$ ,p) and 8075.9 10 from (p, $\gamma$ ).<br>$J^{\pi}$ : primary 3897.8 $\gamma$ to $3/2^{-}$ , 8074.9 $\gamma$ to $3/2^{+}$ ; 2912.4 $\gamma$ to $7/2^{-}$ , 3962.0 $\gamma$ to $7/2^{+}$ .                                                  |
| 8080                                |                                   |                                        |      | 0  |                                                                                                                                                                                                                                                                                                    |
| 8096.1 11                           | ( $5/2$ )                         |                                        |      | 0  | $J^{\pi}$ : primary 2932.6 $\gamma$ to $7/2^{-}$ , 5450.0 $\gamma$ to $7/2^{+}$ ; 3918.0 $\gamma$ to $3/2^{-}$ , 4177.4 $\gamma$ to $3/2^{+}$ .                                                                                                                                                    |
| 8100 10                             | ( $7/2$ to $17/2$ ) <sup>+</sup>  |                                        | N    |    | $J^{\pi}$ : L( $\alpha$ ,d)=6 from $3/2^{+}$ .                                                                                                                                                                                                                                                     |
| 8106.4 11                           | $3/2^{+}$                         |                                        | L    | 0  | E(level): other: 8103.8 42 from ( $\alpha$ ,p).<br>$J^{\pi}$ : spin= $3/2$ from $\gamma(\theta)$ in (p, $\gamma$ ); 8105.4 $\gamma$ M1+E2 to $3/2^{+}$ .                                                                                                                                           |
| 8113.3 11                           | ( $1/2^{+}$ , $3/2$ , $5/2^{+}$ ) |                                        | L    | 0  | XREF: L(8121?)<br>$J^{\pi}$ : primary 2897.4 $\gamma$ ( $5/2^{+}$ ), 4145.4 $\gamma$ to $1/2^{+}$ .                                                                                                                                                                                                |
| 8147.2 11                           | $3/2^{-}$                         | 0.56 keV 6                             |      | OP | XREF: P(8150)<br>$J^{\pi}$ : from R-Matrix analysis in (p,p); also proposed by 1976Me12 in (p, $\gamma$ ) based on strengths of primary $\gamma$ rays.<br>$\Gamma$ : from $^{34}\text{S}(\text{p,p}), (\text{p,p}'\gamma)$ :resonances in 1977Ou01.                                                |
| 8148 3                              | $1/2^{-}$                         | 2.66 keV 27                            |      | P  |                                                                                                                                                                                                                                                                                                    |
| 8150 3                              | $3/2^{-}$                         | 0.56 keV 6                             |      | P  |                                                                                                                                                                                                                                                                                                    |
| 8156.8 11                           | ( $5/2^{+}$ )                     |                                        |      | 0  | $J^{\pi}$ : primary 3978.7 $\gamma$ to $3/2^{-}$ , 8155.8 $\gamma$ to $3/2^{+}$ , 4213.6 $\gamma$ to $9/2^{+}$ .                                                                                                                                                                                   |
| 8171.7 42                           |                                   |                                        | L    |    |                                                                                                                                                                                                                                                                                                    |

Continued on next page (footnotes at end of table)

**Adopted Levels, Gammas (continued)**

| <sup>35</sup> Cl Levels (continued) |                                           |                                        |      |    |                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------|-------------------------------------------|----------------------------------------|------|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E(level) <sup>†</sup>               | J <sup>π</sup> <sup>‡</sup>               | T <sub>1/2</sub> or Γ <sup>&amp;</sup> | XREF |    | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 8179.4 12                           | (3/2 <sup>-</sup> ,5/2,7/2 <sup>+</sup> ) |                                        | 0    |    | J <sup>π</sup> : primary 5016.1γ to 7/2 <sup>-</sup> , 5484.9γ to 3/2 <sup>+</sup> .                                                                                                                                                                                                                                                                                                                                                    |
| 8208.2 10                           | 5/2 <sup>+</sup>                          | 0.033 keV 10                           | OP   | WX | T=3/2<br>XREF: W(8210)X(8204)<br>E(level): other: 8209 3 from <sup>34</sup> S(p,p),(p,p'γ):resonances.<br>J <sup>π</sup> : L(pol d, <sup>3</sup> He)=2 and L+1/2 from analyzing power.                                                                                                                                                                                                                                                  |
| 8211 3                              | 1/2 <sup>+</sup>                          | 0.094 keV 15                           | L    | OP | XREF: O(8209)<br>E(level): from (p,p):resonances. Other: 8210.2 46 from (α,p).                                                                                                                                                                                                                                                                                                                                                          |
| 8216.3 10                           | 5/2 <sup>+</sup>                          | 0.014 keV 3                            | 1    | OP | wx<br>XREF: w(8210)x(8204)<br>J <sup>π</sup> : from γ(θ) in (p,γ) and R-Matrix analysis in (p,p):resonances.                                                                                                                                                                                                                                                                                                                            |
| 8242.4 12                           | 3/2 <sup>-</sup>                          | 0.140 keV 15                           | 1    | OP | XREF: l(8246.0)P(8243)<br>J <sup>π</sup> : from γ(θ) in (p,γ) and R-Matrix analysis in (p,p):resonances.                                                                                                                                                                                                                                                                                                                                |
| 8251 5                              | (3/2 <sup>+</sup> ,5/2 <sup>+</sup> )     |                                        | 1    | 0  | XREF: l(8246.0)                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 8269.0 5                            | 5/2 <sup>+</sup>                          | 0.005 keV 3                            | L    | OP | J <sup>π</sup> : primary 4138γ to 7/2 <sup>+</sup> , 7031γ to 1/2 <sup>+</sup> .<br>XREF: L(8270.6)<br>E(level): other: 8271 3 from <sup>34</sup> S(p,p),(p,p'γ):resonances.                                                                                                                                                                                                                                                            |
| 8277.2 5                            | (5/2 <sup>+</sup> )                       | 0.006 keV 3                            |      | OP | J <sup>π</sup> : 3/2 <sup>+</sup> ,5/2 <sup>+</sup> from R-Matrix analysis in (p,p):resonances; primary 5105.5γ to 7/2 <sup>-</sup> rules out 3/2 <sup>+</sup> , since it would require Mult=M2 which is unlikely based on RUL and Γ=0.005 keV.                                                                                                                                                                                         |
| 8282.4 5                            | (3/2 <sup>-</sup> ,5/2)                   |                                        |      | 0  | XREF: P(8279)<br>J <sup>π</sup> : 3/2 <sup>+</sup> ,5/2 <sup>+</sup> from R-Matrix analysis in (p,p):resonances; primary 3939.2γ to 9/2 <sup>-</sup> disfavors both 3/2 <sup>+</sup> and 5/2 <sup>+</sup> , but 3/2 <sup>+</sup> is much less likely.                                                                                                                                                                                   |
| 8287.82 30                          | 1/2 <sup>-</sup>                          | 0.04 keV 1                             | L    | OP | J <sup>π</sup> : primary 3118.9γ to 7/2 <sup>-</sup> , 4104.2γ to 3/2 <sup>-</sup> , 8281.4γ to 3/2 <sup>+</sup> .<br>XREF: L(8292)P(8290)                                                                                                                                                                                                                                                                                              |
| 8297.2 9                            | 3/2 <sup>-</sup>                          | 0.073 keV 15                           |      | OP | J <sup>π</sup> : from γ(θ) in (p,γ) and R-Matrix analysis in (p,p):resonances.                                                                                                                                                                                                                                                                                                                                                          |
| 8298.2 5                            | 3/2 <sup>+</sup>                          |                                        | 1    | 0  | XREF: P(8300)<br>XREF: l(8292)                                                                                                                                                                                                                                                                                                                                                                                                          |
| 8318.1 3                            | (5/2 <sup>+</sup> )                       |                                        | L    | 0  | J <sup>π</sup> : spin=3/2 from γ(θ) in (p,γ); 8297.1γ M1+E2 to 3/2 <sup>+</sup> .<br>XREF: L(8321.0)                                                                                                                                                                                                                                                                                                                                    |
| 8319.6 <sup>a</sup> 5               | 15/2 <sup>-</sup>                         | 12.5 fs 7                              | F    | H  | J <sup>π</sup> : primary 4350.2γ to 1/2 <sup>+</sup> , 5154.8γ to 7/2 <sup>-</sup> , 5671.9γ to 7/2 <sup>+</sup> .<br>E(level): weighted average of 8319.4 9 from ( <sup>28</sup> Si,αpγ) and 8319.7 5 from ( <sup>16</sup> O,αpγ).<br>J <sup>π</sup> : 2911.9γ E2, ΔI=2 to 11/2 <sup>-</sup> , 2232.7γ to 13/2 <sup>-</sup> ; band assignment.<br>T <sub>1/2</sub> : lifetime τ=18 fs 1 from ( <sup>28</sup> Si,αpγ) (2013Bi10, DSAM). |
| 8323.1 13                           | (3/2 <sup>+</sup> ,5/2 <sup>+</sup> )     |                                        | L    | 0  | T <sub>1/2</sub> : from (HI,xnγ).<br>XREF: L(8321.0)                                                                                                                                                                                                                                                                                                                                                                                    |
| 8346 5                              |                                           |                                        |      | 0  | J <sup>π</sup> : primary 4210.4γ to 7/2 <sup>+</sup> , 7103.0γ to 1/2 <sup>+</sup> .                                                                                                                                                                                                                                                                                                                                                    |

Continued on next page (footnotes at end of table)

**Adopted Levels, Gammas (continued)**

| <sup>35</sup> Cl Levels (continued) |                                                                   |                                        |      |              |                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------|-------------------------------------------------------------------|----------------------------------------|------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E(level) <sup>†</sup>               | J <sup>π</sup> <sup>‡</sup>                                       | T <sub>1/2</sub> or Γ <sup>&amp;</sup> | XREF |              | Comments                                                                                                                                                                                                                                                                                                                                                                  |
| 8381.8 8                            | 5/2 <sup>+</sup>                                                  | 0.023 keV 7                            |      | OP           | XREF: P(8383)<br>J <sup>π</sup> : from γ(θ) in (p,γ) and R-Matrix analysis in (p,p):resonances.                                                                                                                                                                                                                                                                           |
| 8389.4 16<br>8402.9 5               | 5/2 <sup>-</sup>                                                  | 0.002 keV 1                            | L    | 0<br>OP      | XREF: L(8398.8)P(8404)<br>E(level): others: 8398.8 38 from (α,p), and 8404 3 from <sup>34</sup> S(p,p),(p,p'γ):resonances.<br>J <sup>π</sup> : 5/2 <sup>-</sup> , 7/2 <sup>-</sup> from R-Matrix analysis in (p,p):resonances; its primary 5708.5γ to 3/2 <sup>+</sup> , 2694 level rules out 7/2 <sup>-</sup> based on RUL and Γ=0.002 keV.                              |
| 8405.8 14                           | (5/2 <sup>-</sup> )                                               | 0.001 keV 1                            |      | OP           | E(level): weighted average of 8405.7 14 from (p,γ), and 8406 3 from <sup>34</sup> S(p,p),(p,p'γ):resonances.<br>J <sup>π</sup> : (5/2 <sup>-</sup> , 7/2 <sup>-</sup> ) from R-Matrix analysis in (p,p):resonances; primary 8404.6γ to 3/2 <sup>+</sup> .                                                                                                                 |
| 8409.4 18                           | 1/2 <sup>-</sup>                                                  | 0.125 keV 15                           |      | OP           | E(level): weighted average of 8409.6 18 from <sup>34</sup> S(p,γ) and 8409 3 from <sup>34</sup> S(p,p),(p,p'γ):resonances.                                                                                                                                                                                                                                                |
| 8416.7 5                            | 1/2 <sup>+</sup>                                                  | 0.026 keV 7                            |      | OP           | E(level): weighted average of 8416.7 5 from (p,γ), and 8418 3 from <sup>34</sup> S(p,p),(p,p'γ):resonances.<br>J <sup>π</sup> : 1/2 <sup>+</sup> from R-Matrix analysis in (p,p):resonances; but primary 4473.5γ to 9/2 <sup>+</sup> and 7196.5γ to 1/2 <sup>+</sup> in (p,γ) gives (5/2 <sup>+</sup> ), which may indicate two different resonances with close energies. |
| 8430.3 5<br>8434.8 5<br>8464.3 5    | (3/2) <sup>+</sup><br>(1/2 <sup>+</sup> , 3/2, 5/2 <sup>+</sup> ) | 0.090 keV 15                           | L    | 0<br>OP<br>0 | E(level): other: 8429.2 40 from (α,p).<br>XREF: P(8435)<br>J <sup>π</sup> : primary 4496.4γ to 1/2 <sup>+</sup> , 3249.6γ to (5/2 <sup>+</sup> ).                                                                                                                                                                                                                         |
| 8484.4 5                            | (3/2) <sup>+</sup>                                                | 0.012 keV 5                            | 1    | OP           | E(level): weighted average of 8484.4 5 from (p,γ), and 8486 3 from <sup>34</sup> S(p,p),(p,p'γ):resonances.<br>J <sup>π</sup> : 3/2 <sup>+</sup> , 5/2 <sup>+</sup> from R-Matrix analysis in (p,p):resonances; 3/2 <sup>+</sup> proposed by 2001Vo24 in (p,γ) based on proposed isobaric analog to 2940, 3/2 <sup>+</sup> level in <sup>35</sup> S.                      |
| 8486.2 5                            | 3/2 <sup>-</sup>                                                  | 0.150 keV 15                           | 1    | OP           | E(level): weighted average of 8486.1 5 from (p,γ), and 8488 3 from <sup>34</sup> S(p,p),(p,p'γ):resonances.                                                                                                                                                                                                                                                               |
| 8487.5 5                            | 15/2 <sup>-</sup>                                                 | <69 fs                                 | F H  |              | E(level): weighted average of 8487.8 10 from ( <sup>28</sup> Si,αpγ) and 8487.4 5 from ( <sup>16</sup> O,αpγ).<br>J <sup>π</sup> : 3080.0γ E2, ΔJ=2 to 11/2 <sup>-</sup> , 2399.8γ D(+Q) to 13/2 <sup>-</sup> .<br>T <sub>1/2</sub> : lifetime τ<0.1 ps from ( <sup>28</sup> Si,αpγ) (2007Ks01, DSAM).                                                                    |
| 8506.5 5<br>8514.3 5<br>8533.9 5    | 1/2 <sup>-</sup>                                                  | 0.150 keV 15                           | L    | 0<br>OP<br>0 | E(level): other: 8505.1 47 from (α,p).<br>XREF: P(8515)<br>E(level): weighted average of 8532.4 43 from (α,p) and 8533.9 5 from (p,γ).                                                                                                                                                                                                                                    |
| 8558.2? 13                          | (13/2 <sup>-</sup> )                                              | <0.14 ps                               | F    |              | XREF: F(?)<br>J <sup>π</sup> : 2631γ D(+Q), ΔJ=1 to 11/2 <sup>-</sup> in                                                                                                                                                                                                                                                                                                  |

Continued on next page (footnotes at end of table)

**Adopted Levels, Gammas (continued)**

| <sup>35</sup> Cl Levels (continued) |                                    |                                        |      |    |   |                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------------------|------------------------------------|----------------------------------------|------|----|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E(level) <sup>†</sup>               | J <sup>π</sup> <sup>‡</sup>        | T <sub>1/2</sub> or Γ <sup>&amp;</sup> | XREF |    |   | Comments                                                                                                                                                                                                                                                                                                                                                                   |
| 8571.9 5                            | (5/2) <sup>+</sup>                 | 0.08 keV 1                             | L    | OP | X | ( <sup>28</sup> Si,αpγ).<br>T <sub>1/2</sub> : lifetime τ<0.2 ps from ( <sup>28</sup> Si,αpγ) (2007Ks01, DSAM).<br>T=(3/2)<br>XREF: P(8573)X(8580)<br>E(level): weighted average of 8571.9 5 from (p,γ), and 8573 3 from <sup>34</sup> S(p,p),(p,p'γ):resonances. Other: 8567.9 37 from (α,p).<br>XREF: P(8582)                                                            |
| 8580.5 5                            | 1/2 <sup>+</sup>                   | 0.75 keV 8                             |      | OP |   | XREF: P(8582)                                                                                                                                                                                                                                                                                                                                                              |
| 8585.8 5                            |                                    |                                        |      | O  |   |                                                                                                                                                                                                                                                                                                                                                                            |
| 8590.1 5                            | 5/2 <sup>+</sup>                   | 0.003 keV 2                            |      | OP |   | XREF: P(8592)<br>E(level): weighted average of 8590.0 5 from (p,γ), and 8592 3 from <sup>34</sup> S(p,p),(p,p'γ):resonances.<br>J <sup>π</sup> : 3/2 <sup>+</sup> ,5/2 <sup>+</sup> from R-Matrix analysis in (p,p):resonances; primary 5426.6γ to 7/2 <sup>-</sup> ruled out 3/2 <sup>+</sup> based on RUL and Γ=0.003 keV.                                               |
| 8612.0 5                            | 3/2 <sup>+</sup> ,5/2 <sup>+</sup> |                                        | 1    | O  | W | XREF: W(8610)<br>J <sup>π</sup> : L(d, <sup>3</sup> He)=2 from 0 <sup>+</sup> .                                                                                                                                                                                                                                                                                            |
| 8613.7 5                            | 5/2 <sup>+</sup>                   | 0.175 keV 20                           | 1    | OP |   | XREF: P(8616)                                                                                                                                                                                                                                                                                                                                                              |
| 8618.2 5                            | 3/2 <sup>+</sup> ,5/2 <sup>+</sup> | 0.002 keV 1                            | 1    | OP |   | XREF: P(8620)<br>E(level): weighted average of 8618.1 5 from (p,γ), and 8620 3 from <sup>34</sup> S(p,p),(p,p'γ):resonances.<br>J <sup>π</sup> : 4282.1γ M1+E2 to 9/2 <sup>-</sup> , 6866.2γ D+Q to 5/2 <sup>+</sup> , spin=3/2,7/2 from γ(θ) in (p,γ).                                                                                                                    |
| 8630.1 3                            | 7/2 <sup>-</sup>                   | 0.001 keV 1                            |      | O  |   | J <sup>π</sup> : spin=(5/2) from γ(θ) in (p,γ); primary 4697γ to 9/2 <sup>+</sup> .<br>XREF: P(8643)                                                                                                                                                                                                                                                                       |
| 8633 3                              | (3/2,5/2 <sup>+</sup> )            | 0.001 keV 1                            |      | P  |   |                                                                                                                                                                                                                                                                                                                                                                            |
| 8640                                | (5/2 <sup>+</sup> )                |                                        |      | O  |   |                                                                                                                                                                                                                                                                                                                                                                            |
| 8641.4 5                            | 3/2 <sup>+</sup> ,5/2 <sup>+</sup> | 0.003 keV 2                            |      | OP |   |                                                                                                                                                                                                                                                                                                                                                                            |
| 8654 4                              |                                    |                                        | L    |    |   |                                                                                                                                                                                                                                                                                                                                                                            |
| 8686.2 5                            | 5/2 <sup>-</sup>                   | 0.001 keV 1                            |      | OP |   | E(level): weighted average of 8686.2 5 from (p,γ), and 8687 3 from <sup>34</sup> S(p,p),(p,p'γ):resonances.<br>J <sup>π</sup> : 5/2 <sup>-</sup> ,7/2 <sup>-</sup> from R-Matrix analysis in (p,p):resonances; primary 7446.0γ to 1219, 1/2 <sup>+</sup> disfavors both 5/2 <sup>-</sup> and 7/2 <sup>-</sup> , but 7/2 <sup>-</sup> is much less likely.<br>XREF: P(8689) |
| 8688.3 5                            | 1/2 <sup>+</sup>                   | 0.20 keV 2                             |      | OP |   |                                                                                                                                                                                                                                                                                                                                                                            |
| 8691 3                              | 1/2 <sup>-</sup>                   | 6.4 keV 7                              |      | P  |   |                                                                                                                                                                                                                                                                                                                                                                            |
| 8696.9 5                            | 3/2 <sup>-</sup>                   | 0.8 keV 1                              | L    | OP |   | XREF: L(8698)P(8698)                                                                                                                                                                                                                                                                                                                                                       |
| 8700 10                             | (7/2 to 17/2) <sup>+</sup>         |                                        |      | N  |   | J <sup>π</sup> : L(α,d)=6 from 3/2 <sup>+</sup> .                                                                                                                                                                                                                                                                                                                          |
| 8706.3 5                            |                                    |                                        |      | O  |   |                                                                                                                                                                                                                                                                                                                                                                            |
| 8717.8 5                            |                                    |                                        | L    | O  |   | E(level): weighted average of 8720.9 37 from (α,p) and 8717.7 5 from (p,γ).                                                                                                                                                                                                                                                                                                |
| 8750.9 5                            | 3/2 <sup>-</sup>                   | 0.30 keV 3                             |      | OP |   | E(level): weighted average of 8750.9 5 from (p,γ), and 8750 3 from <sup>34</sup> S(p,p),(p,p'γ):resonances.                                                                                                                                                                                                                                                                |
| 8767.0 5                            |                                    |                                        |      | O  |   |                                                                                                                                                                                                                                                                                                                                                                            |
| 8772.9 5                            | 1/2 <sup>-</sup>                   | 0.57 keV 6                             |      | OP |   | XREF: P(8775)                                                                                                                                                                                                                                                                                                                                                              |
| 8779.8 5                            | 3/2 <sup>-</sup>                   | 0.214 keV 25                           |      | OP |   | XREF: P(8781)<br>E(level): weighted average of 8779.8 5 from (p,γ), and 8781 3 from <sup>34</sup> S(p,p),(p,p'γ):resonances.                                                                                                                                                                                                                                               |

Continued on next page (footnotes at end of table)

**Adopted Levels, Gammas (continued)** $^{35}\text{Cl}$  Levels (continued)

| <u>E(level)<sup>†</sup></u> | <u>J<sup>π</sup><sub>‡</sub></u>        | <u>T<sub>1/2</sub> or Γ<sup>&amp;</sup></u> | <u>XREF</u> | <u>Comments</u>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------|-----------------------------------------|---------------------------------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8787.2 5                    | (5/2 <sup>+</sup> )                     | 0.001 keV 1                                 | L OP        | E(level): others: 8786.3 38 from (α,p), and 8788 3 from $^{34}\text{S}(p,p),(p,p'\gamma)$ :resonances. J <sup>π</sup> : (3/2 <sup>+</sup> , 5/2 <sup>+</sup> ) from R-Matrix analysis in (p,p):resonances; primary 3623.7γ and 5623.8γ to 7/2 <sup>-</sup> .                                                                                                                                                                                                                                              |
| 8788.3 <sup>b</sup> 6       | (15/2 <sup>+</sup> )                    | <0.28 ps                                    | F H         | XREF: F(?)<br>J <sup>π</sup> : 915.4γ D, ΔJ=1 to 13/2 <sup>+</sup> ; band assignment.<br>T <sub>1/2</sub> : lifetime τ<0.4 ps from ( $^{28}\text{Si},\alpha p\gamma$ ) (2007Ks01, DSAM).                                                                                                                                                                                                                                                                                                                  |
| 8798.4 5                    | (3/2 <sup>+</sup> , 5/2 <sup>+</sup> )  | 0.001 keV 1                                 | OP          | E(level): weighted average of 8798.4 5 from (p,γ), and 8799 3 from $^{34}\text{S}(p,p),(p,p'\gamma)$ :resonances.                                                                                                                                                                                                                                                                                                                                                                                         |
| 8820.9 5                    | (1/2 <sup>+</sup> to 7/2 <sup>+</sup> ) |                                             | 0           | J <sup>π</sup> : primary 5817.6γ and 7057.0γ to 5/2 <sup>+</sup> ; 6126.3γ and 8819.7γ to 3/2 <sup>+</sup> , which yields (1/2 <sup>+</sup> , 3/2 <sup>±</sup> , 5/2 <sup>±</sup> , 7/2 <sup>+</sup> ).                                                                                                                                                                                                                                                                                                   |
| 8824.2 5                    | 1/2 <sup>+</sup>                        | 1.70 keV 17                                 | OP          | XREF: P(8825)<br>E(level): weighted average of 8824.2 5 from (p,γ), and 8825 3 from $^{34}\text{S}(p,p),(p,p'\gamma)$ :resonances.                                                                                                                                                                                                                                                                                                                                                                        |
| 8829.3 5                    | 1/2 <sup>-</sup>                        | 12.2 keV 12                                 | OP          | E(level): weighted average of 8829.3 5 from (p,γ), and 8829 3 from $^{34}\text{S}(p,p),(p,p'\gamma)$ :resonances.<br>J <sup>π</sup> : 1/2 <sup>-</sup> from R-Matrix analysis in (p,p):resonances; but primary 6183.0γ to 7/2 <sup>+</sup> gives J≥3/2, which may indicate a different level.                                                                                                                                                                                                             |
| 8830 3                      | 1/2 <sup>+</sup>                        | 0.080 keV 15                                | P           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 8833.1 5                    | (5/2 <sup>-</sup> , 7/2)                |                                             | 0           | J <sup>π</sup> : primary 3618.4γ to (5/2 <sup>+</sup> ), 4063.0γ to (5/2 <sup>-</sup> ); 4720.4γ to 7/2 <sup>+</sup> , 5669.6γ to 7/2 <sup>-</sup> .                                                                                                                                                                                                                                                                                                                                                      |
| 8838.1 5                    | 5/2 <sup>-</sup> , 7/2 <sup>-</sup>     | 0.001 keV 1                                 | 1 OP        | XREF: P(8839)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 8844.4 <sup>b</sup> 4       | 17/2 <sup>+</sup>                       | 5.9 ps 11                                   | F HI N      | XREF: N(8840)<br>E(level): weighted average of 8845.1 9 from ( $^{28}\text{Si},\alpha p\gamma$ ), 8844.6 5 from ( $^{16}\text{O},\alpha p\gamma$ ), and 8844.2 4 from ( $^{14}\text{N},\alpha p\gamma$ ).<br>J <sup>π</sup> : 971.38γ E2, ΔJ=2 to 13/2 <sup>+</sup> , 524.8γ to 15/2 <sup>-</sup> ; band assignment.<br>T <sub>1/2</sub> : weighted average of 6.1 ps 11 from ( $^{16}\text{O},\alpha p\gamma$ ) (1991Ja11, DSAM) and 5.5 ps 14 from ( $^{14}\text{N},\alpha p\gamma$ ) (1976Wa11, DSAM). |
| 8856.2 5                    | 5/2 <sup>+</sup>                        | 0.010 keV 5                                 | OP          | E(level): weighted average of 8856.1 5 from (p,γ), and 8858 3 from $^{34}\text{S}(p,p),(p,p'\gamma)$ :resonances.<br>J <sup>π</sup> : 3/2 <sup>+</sup> , 5/2 <sup>+</sup> from R-Matrix analysis in (p,p):resonances; primary 5692.6γ to 7/2 <sup>-</sup> rules out 3/2 <sup>+</sup> based on RUL and Γ=0.010 keV.                                                                                                                                                                                        |
| 8868.6 5                    | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>     | 0.027 keV 10                                | OP          | E(level): weighted average of 8868.6 5 from (p,γ), and 8870 3 from $^{34}\text{S}(p,p),(p,p'\gamma)$ :resonances.                                                                                                                                                                                                                                                                                                                                                                                         |
| 8884.1 5                    |                                         |                                             | 0           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 8886.1 5                    | (5/2)                                   |                                             | L 0         | E(level): other: 8887.5 41 from (α,p).<br>J <sup>π</sup> : primary 3722.6γ to 7/2 <sup>-</sup> , 6239.8γ to 7/2 <sup>+</sup> ; 4707.9γ to 3/2 <sup>-</sup> , 4967.2γ to 3/2 <sup>+</sup> .                                                                                                                                                                                                                                                                                                                |

Continued on next page (footnotes at end of table)

**Adopted Levels, Gammas (continued)**

| $^{35}\text{Cl}$ Levels (continued) |                                             |                                        |      |    |                                                                                                                                                                                                                                                          |
|-------------------------------------|---------------------------------------------|----------------------------------------|------|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E(level) <sup>†</sup>               | J <sup>π</sup> <sup>‡</sup>                 | T <sub>1/2</sub> or Γ <sup>&amp;</sup> | XREF |    | Comments                                                                                                                                                                                                                                                 |
| 8893.3 5                            | (5/2 <sup>+</sup> , 7/2 <sup>+</sup> )      |                                        | 0    |    | J <sup>π</sup> : primary 4950.0γ to 9/2 <sup>+</sup> , 6198.7γ to 3/2 <sup>+</sup> .                                                                                                                                                                     |
| 8904.8 5                            | (1/2, 3/2, 5/2 <sup>+</sup> )               |                                        | 0    |    | J <sup>π</sup> : primary 4726.6γ to 3/2 <sup>-</sup> , 6210.2γ to 3/2 <sup>+</sup> , 7684.5γ to 1/2 <sup>+</sup> .                                                                                                                                       |
| 8906.8 5                            | 5/2 <sup>+</sup>                            | 0.002 keV 1                            | OP   | x  | XREF: P(8909)x(8921)<br>J <sup>π</sup> : 3/2 <sup>+</sup> , 5/2 <sup>+</sup> from R-Matrix analysis in (p,p):resonances; primary 4963.5γ to 9/2 <sup>+</sup> rules out 3/2 <sup>+</sup> based on RUL and Γ=0.002 keV.                                    |
| 8919.8 5                            | (5/2 <sup>-</sup> , 7/2, 9/2 <sup>+</sup> ) |                                        | 0    |    | XREF: x(8921)<br>J <sup>π</sup> : primary 4571.7γ to 9/2 <sup>-</sup> , 7155.9γ to 5/2 <sup>+</sup> .                                                                                                                                                    |
| 8933.3 5                            |                                             |                                        | 0    |    |                                                                                                                                                                                                                                                          |
| 8953.1 5                            | 3/2 <sup>+</sup>                            | 0.075 keV 15                           | OP   |    | XREF: P(8955)<br>E(level): weighted average of 8953.0 5 from (p,γ), and 8955 3 from <sup>34</sup> S(p,p),(p,p'γ):resonances.<br>J <sup>π</sup> : spin=3/2 from γ(θ) in (p,γ) and (3/2) <sup>+</sup> from R-Matrix analysis in (p,p):resonances.          |
| 8957.9 5                            | (1/2 <sup>-</sup> , 3/2 <sup>-</sup> )      | 0.04 keV 1                             | L    | OP | XREF: P(8959)<br>E(level): others: 8957.7 44 from (α,p) and 8959 3 from (p,p):resonances.                                                                                                                                                                |
| 8981.9 5                            | 5/2 <sup>-</sup> , 7/2 <sup>-</sup>         | 0.003 keV 2                            | OP   |    | XREF: P(8983)<br>E(level): weighted average of 8981.9 5 from (p,γ), and 8983 3 from <sup>34</sup> S(p,p),(p,p'γ):resonances.                                                                                                                             |
| 8984.2 5                            | 5/2 <sup>+</sup>                            | 0.025 keV 10                           | OP   |    | XREF: P(8985)<br>J <sup>π</sup> : 3/2 <sup>+</sup> , 5/2 <sup>+</sup> from R-Matrix analysis in (p,p):resonances; primary 5820.7γ to 7/2 <sup>-</sup> rules out 3/2 <sup>+</sup> based on RUL and Γ=0.025 keV.                                           |
| 8988.4 5                            |                                             |                                        | 0    |    |                                                                                                                                                                                                                                                          |
| 8992.4 5                            |                                             |                                        | 0    |    |                                                                                                                                                                                                                                                          |
| 8996.7 5                            | 5/2 <sup>-</sup> , 7/2 <sup>-</sup>         | 0.002 keV 1                            | L    | OP | XREF: L(8996.9)P(8998)                                                                                                                                                                                                                                   |
| 9001.1 5                            |                                             |                                        | 0    |    |                                                                                                                                                                                                                                                          |
| 9019.3 5                            | 3/2 <sup>-</sup>                            | 3.27 keV 30                            | OP   |    | XREF: P(9021)<br>Γ: weighted average of 3.1 keV 3 from (p,γ) and 3.50 keV 35 from (p,p):resonances.                                                                                                                                                      |
| 9024.5 5                            |                                             |                                        | 1    | 0  |                                                                                                                                                                                                                                                          |
| 9029.9 5                            | (5/2) <sup>+</sup>                          | 0.04 keV 1                             | 1    | OP | XREF: P(9031)                                                                                                                                                                                                                                            |
| 9033.1 5                            |                                             |                                        | 0    |    |                                                                                                                                                                                                                                                          |
| 9038.4 5                            | 1/2 <sup>-</sup>                            | 0.292 keV 30                           | OP   |    | XREF: P(9040)<br>E(level): weighted average of 9038.3 5 from (p,γ), and 9040 3 from <sup>34</sup> S(p,p),(p,p'γ):resonances.                                                                                                                             |
| 9048.4 5                            | 5/2 <sup>-</sup> , 7/2 <sup>-</sup>         | 0.001 keV 1                            | OP   |    | XREF: P(9048)                                                                                                                                                                                                                                            |
| 9050 3                              | (5/2) <sup>+</sup>                          | 0.095 keV 15                           | P    |    |                                                                                                                                                                                                                                                          |
| 9081.5 4                            | 5/2 <sup>+</sup>                            | 59 eV 10                               | L    | OP | XREF: L(9085.7)P(9084)<br>E(level): weighted average of 9081.4 4 from (p,γ), and 9084 3 from <sup>34</sup> S(p,p),(p,p'γ):resonances.<br>J <sup>π</sup> : spin=5/2 from γ(θ) in (p,γ) and (5/2) <sup>+</sup> from R-Matrix analysis in (p,p):resonances. |
|                                     |                                             |                                        |      |    | Γ: weighted average of 65 eV 20 from (p,γ) and 57 eV 10 from (p,p):resonances.                                                                                                                                                                           |
| 9088.5 5                            |                                             |                                        | L    | OP | XREF: L(9085.7)P(9089)<br>E(level): weighted average of 9088.5 5 from                                                                                                                                                                                    |

Continued on next page (footnotes at end of table)

**Adopted Levels, Gammas (continued)**

| $^{35}\text{Cl}$ Levels (continued) |                             |                                        |      |                                                                                                                                                                  |
|-------------------------------------|-----------------------------|----------------------------------------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E(level) <sup>†</sup>               | J <sup>π</sup> <sup>‡</sup> | T <sub>1/2</sub> or Γ <sup>&amp;</sup> | XREF | Comments                                                                                                                                                         |
|                                     |                             |                                        |      | (p,γ), and 9089 4 from<br>$^{34}\text{S}(\text{p,p}), (\text{p,p}'\gamma)$ :resonances.                                                                          |
| 9099.3 5                            |                             |                                        | O    |                                                                                                                                                                  |
| 9100.4 5                            | 3/2 <sup>-</sup>            | 0.20 keV 2                             | OP   | XREF: P(9101)                                                                                                                                                    |
| 9107.4 5                            |                             |                                        | L OP | XREF: L(9109.3)P(9108)<br>E(level): weighted average of 9107.4 5 from<br>(p,γ), and 9108 4 from<br>$^{34}\text{S}(\text{p,p}), (\text{p,p}'\gamma)$ :resonances. |
| 9110.1 5                            |                             |                                        | L OP | XREF: L(9109.3)P(9114)<br>E(level): weighted average of 9110.0 5 from<br>(p,γ), and 9114 4 from<br>$^{34}\text{S}(\text{p,p}), (\text{p,p}'\gamma)$ :resonances. |
| 9124.0 5                            |                             |                                        | OP   | XREF: P(9120)                                                                                                                                                    |
| 9127 4                              | 5/2 <sup>@</sup>            |                                        | J P  | E(level): from (p,p):resonances. Other: 9127 9<br>from $^{31}\text{P}(\alpha, \text{p})$ :resonances.                                                            |
| 9135.1 5                            |                             | 2.0 keV 4                              | Op   | XREF: p(9136)<br>Γ: from (p,γ).                                                                                                                                  |
| 9138.3 5                            |                             |                                        | Op   | XREF: p(9136)                                                                                                                                                    |
| 9147 4                              | (7/2 to 17/2) <sup>+</sup>  |                                        | N P  | XREF: N(9150)P(9147)<br>J <sup>π</sup> : L(α,d)=6 from 3/2 <sup>+</sup> .                                                                                        |
| 9155.7 5                            |                             |                                        | O    |                                                                                                                                                                  |
| 9157.1 5                            | 5/2 <sup>-</sup>            |                                        | OP   | XREF: P(9159)<br>J <sup>π</sup> : spin=5/2 from γ(θ) in (p,γ); 5993.7γ<br>M1+E2 to 7/2 <sup>-</sup> .                                                            |
| 9161 4                              | 1/2 <sup>@</sup>            |                                        | J 1  | XREF: l(9160.0)                                                                                                                                                  |
| 9163.3 5                            | (3/2 <sup>-</sup> )         |                                        | 1 OP | XREF: l(9160.0)P(9168)<br>E(level): other: 9168 4 from (p,p):resonances.                                                                                         |
| 9184.2 5                            |                             |                                        | Op   | XREF: p(9185)                                                                                                                                                    |
| 9188.8 5                            |                             |                                        | Op   | XREF: p(9185)                                                                                                                                                    |
| 9194.1 5                            |                             |                                        | L OP | E(level): others: 9194.0 39 from (α,p), and<br>9196 4 from (p,p):resonances.                                                                                     |
| 9207 4                              | 9/2 <sup>+</sup>            |                                        | P    |                                                                                                                                                                  |
| 9222 4                              | 5/2 <sup>+</sup>            |                                        | P    | X XREF: X(9220)<br>J <sup>π</sup> : L(pol d, $^3\text{He}$ )=2 and L+1/2 from analyzing<br>power for a group at E=9220 120.                                      |
| 9227 4                              | 5/2 <sup>-</sup>            |                                        | P    |                                                                                                                                                                  |
| 9241 4                              | 7/2 <sup>+</sup>            |                                        | P    |                                                                                                                                                                  |
| 9255 4                              | 1/2 <sup>@</sup>            |                                        | J P  | E(level): from (p,p):resonances. Other: 9256 4<br>from $^{31}\text{P}(\alpha, \text{p})$ :resonances.                                                            |
| 9264.9 39                           |                             |                                        | L    |                                                                                                                                                                  |
| 9271 4                              |                             |                                        | P    |                                                                                                                                                                  |
| 9276 4                              |                             |                                        | P    |                                                                                                                                                                  |
| 9284 4                              | 5/2 <sup>-</sup>            |                                        | P    |                                                                                                                                                                  |
| 9294 4                              |                             |                                        | P    |                                                                                                                                                                  |
| 9299 4                              |                             |                                        | P    |                                                                                                                                                                  |
| 9316.2 39                           |                             |                                        | L    |                                                                                                                                                                  |
| 9325 4                              | 5/2 <sup>+</sup>            |                                        | P    |                                                                                                                                                                  |
| 9335 4                              | 7/2 <sup>+</sup>            |                                        | L P  | E(level): weighted average of 9336 4 from<br>$^{34}\text{S}(\text{p,p}), (\text{p,p}'\gamma)$ :resonances and 9334 5 from<br>$^{32}\text{S}(\alpha, \text{p})$ . |
| 9345 4                              |                             |                                        | P    |                                                                                                                                                                  |
| 9349 4                              |                             |                                        | P    |                                                                                                                                                                  |
| 9355 4                              |                             |                                        | P    |                                                                                                                                                                  |
| 9376 4                              |                             |                                        | L    |                                                                                                                                                                  |

Continued on next page (footnotes at end of table)



**Adopted Levels, Gammas (continued)**

| $^{35}\text{Cl}$ Levels (continued) |                                                     |                                        |      |                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------|-----------------------------------------------------|----------------------------------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E(level) <sup>†</sup>               | J <sup>π</sup> <sup>‡</sup>                         | T <sub>1/2</sub> or Γ <sup>&amp;</sup> | XREF | Comments                                                                                                                                                                                                                                                                                                                        |
| 9400 9                              | 1/2 <sup>@</sup>                                    |                                        | J    |                                                                                                                                                                                                                                                                                                                                 |
| 9450 10                             | (7/2 to 17/2) <sup>+</sup>                          |                                        | N    | J <sup>π</sup> : L(α,d)=6 from 3/2 <sup>+</sup> .                                                                                                                                                                                                                                                                               |
| 9456 4                              | 3/2 <sup>@</sup>                                    |                                        | J    |                                                                                                                                                                                                                                                                                                                                 |
| 9476 4                              |                                                     |                                        | L    |                                                                                                                                                                                                                                                                                                                                 |
| 9481.0 18                           | 3/2 <sup>@</sup>                                    |                                        | J    |                                                                                                                                                                                                                                                                                                                                 |
| 9508 4                              | (1/2 <sup>-</sup> , 3/2, 5/2 <sup>+</sup> )         |                                        | L X  | XREF: X(9514)<br>J <sup>π</sup> : L(pol d, <sup>3</sup> He)=(1,2) from 0 <sup>+</sup> .                                                                                                                                                                                                                                         |
| 9551 6                              | 5/2 <sup>@</sup>                                    |                                        | J    |                                                                                                                                                                                                                                                                                                                                 |
| 9673 6                              | 1/2 <sup>@</sup>                                    |                                        | J    |                                                                                                                                                                                                                                                                                                                                 |
| 9712.7 27                           | 1/2 <sup>@</sup>                                    |                                        | J L  | E(level): weighted average of 9713.0 27 from <sup>31</sup> P(α,p),(α,n):resonances and 9711.8 44 from (α,p).                                                                                                                                                                                                                    |
| 9740 4                              |                                                     |                                        | L    |                                                                                                                                                                                                                                                                                                                                 |
| 9751.1 27                           | 7/2 <sup>@</sup>                                    |                                        | J    |                                                                                                                                                                                                                                                                                                                                 |
| 9799 5                              |                                                     |                                        | L    |                                                                                                                                                                                                                                                                                                                                 |
| 9814.0 27                           | 5/2 <sup>(+)</sup>                                  |                                        | J X  | XREF: X(9820)<br>J <sup>π</sup> : spin=5/2 from angular distributions in <sup>31</sup> P(α,p):resonances; L=(1,2) from 0 <sup>+</sup> in (pol d, <sup>3</sup> He) gives (1/2 <sup>-</sup> , 3/2, 5/2 <sup>+</sup> ).                                                                                                            |
| 9836 5                              |                                                     |                                        | L    |                                                                                                                                                                                                                                                                                                                                 |
| 9869.8 27                           | 1/2 <sup>@</sup>                                    |                                        | J L  | XREF: L(9870)                                                                                                                                                                                                                                                                                                                   |
| 9900.8 27                           | 3/2 <sup>@</sup>                                    |                                        | J    |                                                                                                                                                                                                                                                                                                                                 |
| 9923 4                              | 3/2 <sup>-@</sup>                                   |                                        | J    |                                                                                                                                                                                                                                                                                                                                 |
| 9958 6                              | 3/2 <sup>@</sup>                                    |                                        | J    |                                                                                                                                                                                                                                                                                                                                 |
| 9969 5                              | (1/2, 3/2) <sup>@</sup>                             |                                        | J L  | XREF: J(9968)<br>E(level): from (α,p) and (α,n):resonances.                                                                                                                                                                                                                                                                     |
| 10031 5                             | (1/2 <sup>+</sup> , 3/2 <sup>-</sup> ) <sup>@</sup> |                                        | J    |                                                                                                                                                                                                                                                                                                                                 |
| 10058 5                             | 3/2 <sup>@</sup>                                    |                                        | J    |                                                                                                                                                                                                                                                                                                                                 |
| 10075 5                             | (1/2) <sup>@</sup>                                  |                                        | J L  | E(level): from (α,p) and (α,n):resonances.                                                                                                                                                                                                                                                                                      |
| 10090 5                             | 5/2 <sup>+</sup> <sup>@</sup>                       |                                        | J    |                                                                                                                                                                                                                                                                                                                                 |
| 10132 5                             | 3/2 <sup>-@</sup>                                   |                                        | J    |                                                                                                                                                                                                                                                                                                                                 |
| 10163 5                             | <sup>@</sup>                                        |                                        | L    |                                                                                                                                                                                                                                                                                                                                 |
| 10172 5                             | 3/2 <sup>-</sup> , 5/2 <sup>-@</sup>                |                                        | J    |                                                                                                                                                                                                                                                                                                                                 |
| 10179 5                             |                                                     |                                        | L    |                                                                                                                                                                                                                                                                                                                                 |
| 10181.1 <sup>a</sup> 5              | 19/2 <sup>-</sup>                                   | 42.3 fs 28                             | F H  | E(level): weighted average of 10181.1 10 from ( <sup>28</sup> Si, αpγ) and 10181.1 5 from ( <sup>16</sup> O, αpγ).<br>J <sup>π</sup> : 1693.5γ E2, ΔJ=2 to 15/2 <sup>-</sup> ; 1336.3γ D, ΔJ=1 to 17/2 <sup>+</sup> ; band assignment.<br>T <sub>1/2</sub> : lifetime τ=61 fs 4 from ( <sup>28</sup> Si, αpγ) (2013Bi10, DSAM). |
| 10215 5                             | 3/2 <sup>@</sup>                                    |                                        | J L  | XREF: L(10218.0)                                                                                                                                                                                                                                                                                                                |
| 10221.9 11                          | (17/2 <sup>-</sup> )                                | <0.28 ps                               | F H  | XREF: F(?)<br>E(level): from ( <sup>16</sup> O, αpγ).<br>J <sup>π</sup> : 1377.8γ to 17/2 <sup>+</sup> consistent with ΔJ=0.<br>T <sub>1/2</sub> : lifetime τ<0.4 ps from ( <sup>28</sup> Si, αpγ) (2007Ks01, DSAM).                                                                                                            |
| 10236 5                             | 3/2 <sup>-@</sup>                                   |                                        | J    |                                                                                                                                                                                                                                                                                                                                 |
| 10278 5                             | 1/2 <sup>@</sup>                                    |                                        | J    |                                                                                                                                                                                                                                                                                                                                 |

Continued on next page (footnotes at end of table)

**Adopted Levels, Gammas (continued)** $^{35}\text{Cl}$  Levels (continued)

| E(level) <sup>†</sup>  | J <sup>π</sup> <sup>‡</sup>           | T <sub>1/2</sub> or Γ <sup>&amp;</sup> | XREF | Comments                                                                                                                                                                                                                             |
|------------------------|---------------------------------------|----------------------------------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10295 5                |                                       |                                        | J    |                                                                                                                                                                                                                                      |
| 10322 5                | 5/2 <sup>+</sup> @                    |                                        | J    |                                                                                                                                                                                                                                      |
| 10340 5                | (3/2,5/2) @                           |                                        | J    |                                                                                                                                                                                                                                      |
| 10359 5                | 3/2 @                                 |                                        | J    |                                                                                                                                                                                                                                      |
| 10385 5                |                                       |                                        | J    |                                                                                                                                                                                                                                      |
| 10395 5                |                                       |                                        | L    |                                                                                                                                                                                                                                      |
| 10399 5                | 3/2 <sup>+</sup> , 7/2 <sup>+</sup> @ |                                        | J    |                                                                                                                                                                                                                                      |
| 10431 5                | 5/2 <sup>+</sup> @                    |                                        | J    |                                                                                                                                                                                                                                      |
| 10465 5                | 5/2 <sup>+</sup> @                    |                                        | J L  | E(level): weighted average of 10467 5 from $^{31}\text{P}(\alpha, \text{p}), (\alpha, \text{n})$ :resonances and 10463.2 49 from $(\alpha, \text{p})$ .                                                                              |
| 10486 5                | 5/2 <sup>+</sup> @                    |                                        | J    |                                                                                                                                                                                                                                      |
| 10517 6                |                                       |                                        | L    |                                                                                                                                                                                                                                      |
| 10534 5                | 1/2 <sup>+</sup> , 5/2 <sup>+</sup> @ |                                        | J    |                                                                                                                                                                                                                                      |
| 10548 6                |                                       |                                        | L    |                                                                                                                                                                                                                                      |
| 10584 5                | 3/2 <sup>+</sup> @                    |                                        | J L  | E(level): weighted average of 10587 5 from $^{31}\text{P}(\alpha, \text{p}), (\alpha, \text{n})$ :resonances and 10579.5 54 from $(\alpha, \text{p})$ .                                                                              |
| 10642 5                | 5/2 <sup>+</sup> @                    |                                        | J L  | E(level): weighted average of 10641 5 from $^{31}\text{P}(\alpha, \text{p}), (\alpha, \text{n})$ :resonances and 10643.0 50 from $(\alpha, \text{p})$ .                                                                              |
| 10678 5                | 3/2 <sup>+</sup> , 5/2 <sup>+</sup> @ |                                        | J    |                                                                                                                                                                                                                                      |
| 10696 5                | 7/2 <sup>-</sup> @                    |                                        | J    |                                                                                                                                                                                                                                      |
| 10733 5                | 3/2 <sup>+</sup> @                    |                                        | J L  | E(level): weighted average of 10734 5 from $^{31}\text{P}(\alpha, \text{p}), (\alpha, \text{n})$ :resonances and 10732.1 50 from $(\alpha, \text{p})$ .                                                                              |
| 10760 5                | 3/2 <sup>+</sup> , 7/2 <sup>-</sup> @ |                                        | J L  | E(level): weighted average of 10760 5 from $^{31}\text{P}(\alpha, \text{p}), (\alpha, \text{n})$ :resonances and 10759.1 55 from $(\alpha, \text{p})$ .                                                                              |
| 10800 5                | 5/2 <sup>+</sup> @                    |                                        | J    |                                                                                                                                                                                                                                      |
| 10816 5                | 1/2 <sup>+</sup> , 3/2 <sup>-</sup> @ |                                        | J    |                                                                                                                                                                                                                                      |
| 10844 5                | 3/2 <sup>-</sup> @                    |                                        | J    |                                                                                                                                                                                                                                      |
| 10858.5 <sup>b</sup> 8 | 19/2 <sup>+</sup>                     | <0.28 ps                               | F H  | XREF: F(?)<br>J <sup>π</sup> : 2069.8γ E2, ΔJ=2 to 15/2 <sup>+</sup> , 2014.7γ D, ΔJ=1 to 17/2 <sup>+</sup> ; band assignment.<br>T <sub>1/2</sub> : lifetime τ<0.4 ps from ( $^{28}\text{Si}, \alpha \text{py}$ ) (2007Ks01, DSAM). |
| 10870 5                | (5/2, 7/2) @                          |                                        | J    |                                                                                                                                                                                                                                      |
| 10886 5                |                                       |                                        | J    |                                                                                                                                                                                                                                      |
| 10909 5                | 3/2 <sup>+</sup> , 5/2 <sup>-</sup> @ |                                        | J    |                                                                                                                                                                                                                                      |
| 10928 5                | 7/2 @                                 |                                        | J    |                                                                                                                                                                                                                                      |
| 10942 5                |                                       |                                        | J    |                                                                                                                                                                                                                                      |
| 10964 5                | 5/2 <sup>+</sup> @                    |                                        | J    |                                                                                                                                                                                                                                      |
| 10973 5                |                                       |                                        | J    |                                                                                                                                                                                                                                      |
| 10998 5                | 3/2 <sup>+</sup> , 7/2 <sup>-</sup> @ |                                        | J    |                                                                                                                                                                                                                                      |
| 11034 5                | 3/2 <sup>+</sup> @                    |                                        | J    |                                                                                                                                                                                                                                      |
| 11063 5                | 1/2 <sup>+</sup> , 9/2 <sup>+</sup> @ |                                        | J    |                                                                                                                                                                                                                                      |
| 11082 5                | 5/2 <sup>+</sup> , 9/2 <sup>+</sup> @ |                                        | J    |                                                                                                                                                                                                                                      |

Continued on next page (footnotes at end of table)

**Adopted Levels, Gammas (continued)** $^{35}\text{Cl}$  Levels (continued)

| E(level) <sup>†</sup>  | J <sup>π</sup> <sup>‡</sup>                      | T <sub>1/2</sub> or Γ <sup>&amp;</sup> | XREF  | Comments                                                                                                                                                                                                                                                                                                                                         |
|------------------------|--------------------------------------------------|----------------------------------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 11105 5                |                                                  |                                        | J     |                                                                                                                                                                                                                                                                                                                                                  |
| 11123 5                | 3/2 <sup>@</sup>                                 |                                        | J     |                                                                                                                                                                                                                                                                                                                                                  |
| 11142 5                | 5/2 <sup>+</sup> <sup>@</sup>                    |                                        | J     |                                                                                                                                                                                                                                                                                                                                                  |
| 11154 5                |                                                  |                                        | J     |                                                                                                                                                                                                                                                                                                                                                  |
| 11169 5                |                                                  |                                        | J     |                                                                                                                                                                                                                                                                                                                                                  |
| 11179 5                | 3/2 <sup>-</sup> , 5/2 <sup>+</sup> <sup>@</sup> |                                        | J     |                                                                                                                                                                                                                                                                                                                                                  |
| 11185 5                |                                                  |                                        | J     |                                                                                                                                                                                                                                                                                                                                                  |
| 11198 5                | 3/2 <sup>@</sup>                                 |                                        | J     |                                                                                                                                                                                                                                                                                                                                                  |
| 11230 5                |                                                  |                                        | J     |                                                                                                                                                                                                                                                                                                                                                  |
| 11246 5                | 3/2 <sup>@</sup>                                 |                                        | J     |                                                                                                                                                                                                                                                                                                                                                  |
| 11265 5                | 3/2 <sup>-</sup> <sup>@</sup>                    |                                        | J     |                                                                                                                                                                                                                                                                                                                                                  |
| 11287 5                |                                                  |                                        | J     |                                                                                                                                                                                                                                                                                                                                                  |
| 11307 5                |                                                  |                                        | J     |                                                                                                                                                                                                                                                                                                                                                  |
| 11314 5                | 1/2 <sup>-</sup> , 5/2 <sup>+</sup> <sup>@</sup> |                                        | J     |                                                                                                                                                                                                                                                                                                                                                  |
| 11327 5                | 1/2 <sup>-</sup> , 5/2 <sup>+</sup> <sup>@</sup> |                                        | J     |                                                                                                                                                                                                                                                                                                                                                  |
| 11357 5                |                                                  |                                        | J     |                                                                                                                                                                                                                                                                                                                                                  |
| 11375 5                |                                                  |                                        | J     |                                                                                                                                                                                                                                                                                                                                                  |
| 11390 5                |                                                  |                                        | J     |                                                                                                                                                                                                                                                                                                                                                  |
| 11410 5                |                                                  |                                        | J     |                                                                                                                                                                                                                                                                                                                                                  |
| 11421 5                | 3/2 <sup>-</sup> <sup>@</sup>                    |                                        | J     |                                                                                                                                                                                                                                                                                                                                                  |
| 11434 5                |                                                  |                                        | J     |                                                                                                                                                                                                                                                                                                                                                  |
| 11459.1 <sup>b</sup> 7 | 21/2 <sup>+</sup>                                | 43.0 fs 21                             | F H J | E(level): weighted average of 11459.2 12 from ( <sup>28</sup> Si,αγ) and 11459.0 7 from ( <sup>16</sup> O,αγ).<br>Other: 11460 5 from <sup>31</sup> P(α,p),(α,n):resonances.<br>J <sup>π</sup> : 2614.5γ E2, ΔJ=2 to 17/2 <sup>+</sup> ; band assignment.<br>T <sub>1/2</sub> : lifetime τ=62 fs 3 from ( <sup>28</sup> Si,αγ) (2013Bi10, DSAM). |
| 11473 5                | 1/2 <sup>-</sup> , 5/2 <sup>-</sup> <sup>@</sup> |                                        | J     |                                                                                                                                                                                                                                                                                                                                                  |
| 11492 5                |                                                  |                                        | J     |                                                                                                                                                                                                                                                                                                                                                  |
| 11505 5                | 3/2 <sup>-</sup> , 5/2 <sup>-</sup> <sup>@</sup> |                                        | J     |                                                                                                                                                                                                                                                                                                                                                  |
| 11525 5                |                                                  |                                        | J     |                                                                                                                                                                                                                                                                                                                                                  |
| 11540 5                |                                                  |                                        | J     |                                                                                                                                                                                                                                                                                                                                                  |
| 11551 5                | 3/2 <sup>-</sup> , 5/2 <sup>+</sup> <sup>@</sup> |                                        | J     |                                                                                                                                                                                                                                                                                                                                                  |
| 11565 5                |                                                  |                                        | J     |                                                                                                                                                                                                                                                                                                                                                  |
| 11589 5                |                                                  |                                        | J     |                                                                                                                                                                                                                                                                                                                                                  |
| 11609 5                | 5/2 <sup>+</sup> , 7/2 <sup>-</sup> <sup>@</sup> |                                        | J     |                                                                                                                                                                                                                                                                                                                                                  |
| 11627 5                | 1/2 <sup>+</sup> , 5/2 <sup>+</sup> <sup>@</sup> |                                        | J     |                                                                                                                                                                                                                                                                                                                                                  |
| 11638 5                |                                                  |                                        | J     |                                                                                                                                                                                                                                                                                                                                                  |
| 11651 5                |                                                  |                                        | J     |                                                                                                                                                                                                                                                                                                                                                  |
| 11684 5                | 1/2 <sup>+</sup> , 5/2 <sup>+</sup> <sup>@</sup> |                                        | J     |                                                                                                                                                                                                                                                                                                                                                  |
| 11773 5                | 3/2 <sup>-</sup> , 7/2 <sup>-</sup> <sup>@</sup> |                                        | J     |                                                                                                                                                                                                                                                                                                                                                  |
| 11783 5                | 3/2 <sup>-</sup> <sup>@</sup>                    |                                        | J     |                                                                                                                                                                                                                                                                                                                                                  |
| 1.25×10 <sup>4</sup> 5 |                                                  |                                        | J     | X T=3/2<br>XREF: J(12900)<br>E(level): from (pol d, <sup>3</sup> He).<br>E(level): weighted average of 12572.2 12 from ( <sup>28</sup> Si,αγ) and 12572.2 6 from ( <sup>16</sup> O,αγ).<br>J <sup>π</sup> : 2390.8γ E2, ΔJ=2 to 19/2 <sup>-</sup> , 1113.3γ D, ΔJ=1 to 21/2 <sup>+</sup> ; band assignment.                                      |
| 12572.2 <sup>d</sup> 6 | 23/2 <sup>-</sup>                                | 33 fs 4                                | F H   |                                                                                                                                                                                                                                                                                                                                                  |

Continued on next page (footnotes at end of table)

**Adopted Levels, Gammas (continued)** $^{35}\text{Cl}$  Levels (continued)

| <u>E(level)<sup>†</sup></u> | <u>J<sup>π</sup><sup>‡</sup></u> | <u>T<sub>1/2</sub> or Γ<sup>&amp;</sup></u> | <u>XREF</u> | <u>Comments</u>                                                                                                                                                                        |
|-----------------------------|----------------------------------|---------------------------------------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 13900                       |                                  |                                             |             | T <sub>1/2</sub> : lifetime $\tau=48$ fs 6 from ( $^{28}\text{Si},\alpha p\gamma$ ) (2013Bi10, DSAM).                                                                                  |
| 16306.4 <sup>a</sup> 16     | (27/2 <sup>-</sup> )             | <9.0 fs                                     | F J         | T=3/2<br>J <sup>π</sup> : 3734 $\gamma$ to 23/2 <sup>-</sup> ; band assignment.<br>T <sub>1/2</sub> : lifetime $\tau<13$ fs from ( $^{28}\text{Si},\alpha p\gamma$ ) (2013Bi10, DSAM). |

<sup>†</sup> Most of Adopted Levels are reported in (p, $\gamma$ ) dataset with precise E(level) values, deduced based on measured  $\gamma$ -ray energies which however are not reported in references in (p, $\gamma$ ) dataset, while E(level) values from E $\gamma$  values with uncertainties in other datasets are less precise and the numbers of  $\gamma$  rays are much less than that of (p, $\gamma$ ) datasets. Therefore, a least-squares fit to available E $\gamma$  values with uncertainties would not produce E(level) values more precise and complete than those already reported in (p, $\gamma$ ) and the evaluators have adopted E(level) values from (p, $\gamma$ ) or averages of values based on E $\gamma$  data from all available datasets where applicable, as noted.

<sup>‡</sup> For proton-resonance levels populated in (p,p):resonances, J<sup>π</sup> assignments are from R-Matrix analysis, unless otherwise noted.

<sup>#</sup> From FRESKO-DWBA analysis of measured  $\sigma(\theta)$  of ( $\alpha$ ,p) (2019Se09).

<sup>@</sup> From angular distributions in  $^{31}\text{P}(\alpha,p)$ :resonances.

<sup>&</sup> Sources of T<sub>1/2</sub> values are listed under comments;  $\Gamma$  from  $^{34}\text{S}(p,p),(p,p'\gamma)$ :resonances, unless otherwise noted.

<sup>a</sup> Band(A): Band based on f<sub>7/2</sub> orbital.

<sup>b</sup> Band(B): Band based on 13/2<sup>+</sup>.

Adopted Levels, Gammas (continued)

| $\gamma(^{35}\text{Cl})$ |                  |                       |                     |                |                                      |                    |             |                                        | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------|------------------|-----------------------|---------------------|----------------|--------------------------------------|--------------------|-------------|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $E_i(\text{level})$      | $J_i^\pi$        | $E_\gamma^\dagger$    | $I_\gamma^\ddagger$ | $E_f$          | $J_f^\pi$                            | Mult. <sup>#</sup> | $\delta^\#$ | $\alpha^a$                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 1219.38                  | 1/2 <sup>+</sup> | 1219.4 1              | 100                 | 0.0            | 3/2 <sup>+</sup>                     | M1+E2              | 0.093 10    | 3.66×10 <sup>-5</sup> 5                | B(M1)(W.u.)=0.101 +14-11; B(E2)(W.u.)=2.2 +6-5<br>E <sub>γ</sub> : weighted average of 1219.3 2 from <sup>35</sup> Ar ε decay and 1219.4 1 from (p,p'γ), and 1219.4 6 from (α,pγ).<br>Mult.,δ: ΔJ=1 from γ(θ) in (p,p'γ), E1+M2 ruled out by RUL. δ from adopted B(E2)†=0.00076 7 measured in Coulomb excitation and adopted lifetime τ=0.172 ps 20.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 1763.15                  | 5/2 <sup>+</sup> | 543.77<br>1763.16 10  | <0.2<br>100         | 1219.38<br>0.0 | 1/2 <sup>+</sup><br>3/2 <sup>+</sup> | [E2]<br>M1+E2      | -2.85 10    | 0.000323 5<br>0.0002124 30             | I <sub>γ</sub> : from (p,γ). Other: <0.5 from (α,pγ).<br>B(M1)(W.u.)=0.00122 +14-12; B(E2)(W.u.)=12.1 +11-9<br>E <sub>γ</sub> : weighted average of 1763.0 2 from <sup>35</sup> Ar ε decay, 1763.1 2 from ( <sup>24</sup> Mg,αpγ), 1763.3 2 from ( <sup>16</sup> O,αpγ), 1763.15 10 from ( <sup>14</sup> N,αpnγ), and 1763.2 1 from (p,p'γ). Other: 1763.2 6 from (α,pγ).<br>Mult.: from γγ(ADO) in ( <sup>16</sup> O,αpγ) with ΔJ=1 and γ(lin pol) in ( <sup>28</sup> Si,αpγ).<br>δ: weighted average of -3.0 1 from (α,pγ), -2.66 12 from (p,p'γ), -2.95 45 from Coulomb excitation, -2.6 4 from ( <sup>16</sup> O,αpγ), all from γ(θ) or pγ(θ). Other: -0.42 17 from γγ(θ) in (p,γ); 1.8 +10-4; deduced by the evaluators from adopted B(E2)†=0.0106 7 and T <sub>1/2</sub> =0.36 ps 3.                                                      |
| 2645.67                  | 7/2 <sup>+</sup> | 882.80 11             | 11.4 10             | 1763.15        | 5/2 <sup>+</sup>                     | M1+E2              | +0.21 5     | 5.29×10 <sup>-5</sup> 9                | B(M1)(W.u.)=0.0196 +35-27; B(E2)(W.u.)=4.2 +23-18<br>E <sub>γ</sub> : weighted average of 882.9 1 from ( <sup>24</sup> Mg,αpγ), 882.7 5 from ( <sup>16</sup> O,αpγ), 882.32 35 from ( <sup>14</sup> N,αpnγ), 882.4 10 from (α,pγ), and 882.3 3 from (p,p'γ).<br>I <sub>γ</sub> : unweighted average of 8.7 6 from ( <sup>28</sup> Si,αpγ), 15.2 13 from ( <sup>24</sup> Mg,αpγ), 11.08 28 from ( <sup>16</sup> O,αpγ), 11.1 22 from (α,pγ), 9.5 11 from (p,γ), and 13 5 from (p,p'γ).<br>Mult.: D+Q from pγ(θ) in (α,pγ); E1+M2 ruled out by RUL.<br>δ: weighted average of +0.25 5 from (α,pγ) and +0.17 5 from (p,p'γ).<br>I <sub>γ</sub> : from (α,pγ). Other: <2.2 from (p,γ).<br>B(E2)(W.u.)=3.6 +6-4<br>E <sub>γ</sub> : weighted average of 2645.5 4 from ( <sup>24</sup> Mg,αpγ), 2645.8 5 from ( <sup>16</sup> O,αpγ), 2645.79 30 from |
|                          |                  | 1426.26<br>2645.70 20 | <1.2<br>100.0 11    | 1219.38<br>0.0 | 1/2 <sup>+</sup><br>3/2 <sup>+</sup> | [M3]<br>E2         |             | 7.52×10 <sup>-5</sup> 11<br>0.000633 9 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

Adopted Levels, Gammas (continued)

| $\gamma(^{35}\text{Cl})$ (continued) |                  |                    |                     |         |                  |         |             |                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------------|------------------|--------------------|---------------------|---------|------------------|---------|-------------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $E_i(\text{level})$                  | $J_i^\pi$        | $E_\gamma^\dagger$ | $I_\gamma^\ddagger$ | $E_f$   | $J_f^\pi$        | Mult.#  | $\delta^\#$ | $\alpha^a$              | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                      |                  |                    |                     |         |                  |         |             |                         | $(^{14}\text{N},\alpha\text{p}\gamma)$ , and 2645.7 2 from $(\text{p},\text{p}'\gamma)$ . Other: 2646.2 15 from $(\alpha,\text{p}\gamma)$ .<br>$I_\gamma$ : others: 100.0 31 from $(^{24}\text{Mg},\alpha\text{p}\gamma)$ , 100.0 22 from $(^{16}\text{O},\alpha\text{p}\gamma)$ , 100.0 22 from $(\alpha,\text{p}\gamma)$ , and 100.0 20 from $(\text{p},\text{p}'\gamma)$ . Other: 100 6 from $(^{28}\text{Si},\alpha\text{p}\gamma)$ .<br>Mult.: from $\gamma\gamma(\theta)(\text{DCO})$ and $\gamma\gamma(\text{lin pol})$ in $(^{28}\text{Si},\alpha\text{p}\gamma)$ ; $\gamma(\theta)$ and $\gamma(\text{lin pol})$ in $(\text{p},\text{p}'\gamma)$ .<br>B(M1)(W.u.)=0.110 +26−19; B(E2)(W.u.)=3.9 +33−22<br>$E_\gamma$ : weighted average of 930.7 2 from $^{35}\text{Ar}$ $\varepsilon$ decay, 930.5 15 from $(\alpha,\text{p}\gamma)$ , and 931.3 3 from $(\text{p},\text{p}'\gamma)$ .<br>$I_\gamma$ : weighted average of 14.3 18 from $^{35}\text{Ar}$ $\varepsilon$ decay, 18.2 26 from $(\alpha,\text{p}\gamma)$ , 16.3 13 from $(\text{p},\gamma)$ , and 18.6 25 from $(\text{p},\text{p}'\gamma)$ .<br>Mult., $\delta$ : from $\gamma(\theta)$ and $\gamma(\text{lin pol})$ in $(\text{p},\text{p}'\gamma)$ .<br>B(M1)(W.u.)=0.0116 +37−29; B(E2)(W.u.)=8.0 +45−39<br>$E_\gamma$ : weighted average of 1474.8 3 from $^{35}\text{Ar}$ $\varepsilon$ decay and 1475 2 from $(\text{p},\text{p}'\gamma)$ .<br>$I_\gamma$ : weighted average of 9.3 10 from $^{35}\text{Ar}$ $\varepsilon$ decay, 11.7 26 from $(\alpha,\text{p}\gamma)$ , 9.5 13 from $(\text{p},\gamma)$ , and 8.3 25 from $(\text{p},\text{p}'\gamma)$ .<br>Mult., $\delta$ : from $\gamma(\theta)$ and $\gamma(\text{lin pol})$ in $(\text{p},\text{p}'\gamma)$ .<br>B(M1)(W.u.)=0.026 +6−4; B(E2)(W.u.)=0.92 +26−19<br>$E_\gamma$ : weighted average of 2693.7 2 from $^{35}\text{Ar}$ $\varepsilon$ decay and 2694.1 6 from $(\text{p},\text{p}'\gamma)$ . Other: 2693.2 20 from $(\alpha,\text{p}\gamma)$ .<br>$I_\gamma$ : weighted average of 100.0 17 from $^{35}\text{Ar}$ $\varepsilon$ decay, 100 4 from $(\alpha,\text{p}\gamma)$ , 100.0 25 from $(\text{p},\gamma)$ , and 100.0 30 from $(\text{p},\text{p}'\gamma)$ .<br>Mult.: from $\gamma(\theta)$ and $\gamma(\text{lin pol})$ in $(\text{p},\text{p}'\gamma)$ .<br>$\delta$ : weighted average of +0.17 8 from $(\alpha,\text{p}\gamma)$ and +0.26 2 from $(\text{p},\text{p}'\gamma)$ .<br>$I_\gamma$ : from $(\alpha,\text{p}\gamma)$ .<br>$I_\gamma$ : from $(\alpha,\text{p}\gamma)$ .<br>$I_\gamma$ : from $(\text{p},\text{p}'\gamma)$ .<br>B(M1)(W.u.)=0.011 +8−5 if M1, B(E2)(W.u.)=26 +19−12 if E2.<br>B(E2)(W.u.)=6.4 +35−22<br>$I_\gamma$ : from $(\text{p},\text{p}'\gamma)$ .<br>B(M1)(W.u.)=0.034 +14−7; B(E2)(W.u.)=0.09 +7−4<br>$E_\gamma$ : weighted average of 3002.1 4 from $^{35}\text{Ar}$ $\varepsilon$ decay, |
| 2694.00                              | 3/2 <sup>+</sup> | 930.9 2            | 16.3 13             | 1763.15 | 5/2 <sup>+</sup> | M1+E2   | +0.09 3     | 4.68×10 <sup>−5</sup> 7 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                      |                  | 1474.8 3           | 9.5 10              | 1219.38 | 1/2 <sup>+</sup> | M1+E2   | −0.63 21    | 8.4×10 <sup>−5</sup> 4  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                      |                  | 2693.7 2           | 100.0 17            | 0.0     | 3/2 <sup>+</sup> | M1+E2   | +0.26 2     | 0.000553 8              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 3002.64                              | 5/2 <sup>+</sup> | 308.64             | <5                  | 2694.00 | 3/2 <sup>+</sup> | [M1,E2] |             | 0.0014 9                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                      |                  | 356.97             | <1                  | 2645.67 | 7/2 <sup>+</sup> | [M1,E2] |             | 9                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                      |                  | 1239.47            | 2.2 11              | 1763.15 | 5/2 <sup>+</sup> | [M1,E2] |             | 4.4×10 <sup>−5</sup> 6  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                      |                  | 1783.21            | 3.3 11              | 1219.38 | 1/2 <sup>+</sup> | [E2]    |             | 0.0002268 32            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                      |                  | 3002.3 4           | 100.0 10            | 0.0     | 3/2 <sup>+</sup> | M1+E2   | +0.08 2     | 0.000671 9              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

Adopted Levels, Gammas (continued)

| $\gamma(^{35}\text{Cl})$ (continued) |                  |                    |                     |         |                  |                    |             |              | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------------|------------------|--------------------|---------------------|---------|------------------|--------------------|-------------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $E_i(\text{level})$                  | $J_i^\pi$        | $E_\gamma^\dagger$ | $I_\gamma^\ddagger$ | $E_f$   | $J_f^\pi$        | Mult. <sup>#</sup> | $\delta^\#$ | $\alpha^a$   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 3162.87                              | 7/2 <sup>-</sup> | 160.67 20          | 1.5 4               | 3002.64 | 5/2 <sup>+</sup> | [E1]               |             | 0.00330 5    | 3002.5 15 from ( $\alpha$ ,p $\gamma$ ), and 3002.5 5 from (p,p' $\gamma$ ).<br>I $_\gamma$ : weighted average of 100.0 19 from <sup>35</sup> Ar $\epsilon$ decay and 100.0 10 from (p,p' $\gamma$ ).<br>Mult.: from $\gamma(\theta)$ and $\gamma(\text{lin pol})$ in (p,p' $\gamma$ ).<br>$\delta$ : weighted average of +0.09 3 from ( $\alpha$ ,p $\gamma$ ) and +0.07 2 from (p,p' $\gamma$ ). Other: -0.22 1 from (p, $\gamma$ ).<br>B(E1)(W.u.)=6.6 $\times$ 10 <sup>-5</sup> 17<br>E $_\gamma$ : weighted average of 161 1 from ( <sup>16</sup> O, $\alpha$ p $\gamma$ ) and 160.66 20 from ( <sup>14</sup> N, $\alpha$ p $\gamma$ ).<br>I $_\gamma$ : unweighted average of 1.10 30 from ( <sup>28</sup> Si, $\alpha$ p $\gamma$ ) and 1.89 22 from (p, $\gamma$ ).<br>B(E1)(W.u.)=1.61 $\times$ 10 <sup>-5</sup> 20<br>E $_\gamma$ : weighted average of 517.7 1 from ( <sup>24</sup> Mg, $\alpha$ p $\gamma$ ), 517.3 5 from ( <sup>16</sup> O, $\alpha$ p $\gamma$ ), 517.26 10 from ( <sup>14</sup> N, $\alpha$ p $\gamma$ ), 517.4 15 from ( $\alpha$ ,p $\gamma$ ), and 517.0 8 from (p,p' $\gamma$ ).<br>I $_\gamma$ : unweighted average of 9.2 5 from ( <sup>28</sup> Si, $\alpha$ p $\gamma$ ), 14.3 8 from ( <sup>24</sup> Mg, $\alpha$ p $\gamma$ ), 11.1 13 from ( <sup>16</sup> O, $\alpha$ p $\gamma$ ), 11.1 33 from ( $\alpha$ ,p $\gamma$ ), 8.9 11 from (p, $\gamma$ ), and 19 8 from (p,p' $\gamma$ ).<br>Mult.: D from $\gamma\gamma(\theta)$ (DCO) in ( <sup>28</sup> Si, $\alpha$ p $\gamma$ ); $\Delta\pi$ =yes from level scheme.<br>B(E1)(W.u.)=1.88 $\times$ 10 <sup>-8</sup> +31-34; B(M2)(W.u.)=0.0085 +43-38<br>I $_\gamma$ : weighted average of 0.50 20 from ( <sup>16</sup> O, $\alpha$ p $\gamma$ ) and 0.33 5 from (p, $\gamma$ ).<br>Mult., $\delta$ : D+Q from $\gamma(\theta)$ in (p, $\gamma$ ) (1971Pr11); $\Delta\pi$ =yes from level scheme. |
|                                      |                  | 517.47 11          | 12.3 16             | 2645.67 | 7/2 <sup>+</sup> | (E1)               |             | 0.0001097 15 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                      |                  | 1399.9 7           | 0.34 5              | 1763.15 | 5/2 <sup>+</sup> | (E1+M2)            | +0.44 12    | 0.000181 12  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                      |                  | 1943.43            | <0.22               | 1219.38 | 1/2 <sup>+</sup> | [E3]               |             | 0.0001742 24 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                      |                  | 3162.53 10         | 100.0 11            | 0.0     | 3/2 <sup>+</sup> | M2+E3              | +0.17 2     | 0.000523 7   | B(M2)(W.u.)=0.255 9; B(E3)(W.u.)=4.2 +11-9<br>E $_\gamma$ : weighted average of 3162.7 4 from ( <sup>24</sup> Mg, $\alpha$ p $\gamma$ ), 3162.6 3 from ( <sup>16</sup> O, $\alpha$ p $\gamma$ ), 3162.43 10 from ( <sup>14</sup> N, $\alpha$ p $\gamma$ ), and 3162.6 1 from (p,p' $\gamma$ ), and 3162.6 15 from ( $\alpha$ ,p $\gamma$ ).<br>I $_\gamma$ : weighted average of 100 5 from ( <sup>28</sup> Si, $\alpha$ p $\gamma$ ), 100.0 30 from ( <sup>24</sup> Mg, $\alpha$ p $\gamma$ ), 100.0 20 from ( <sup>16</sup> O, $\alpha$ p $\gamma$ ), 100.0 33 from ( $\alpha$ ,p $\gamma$ ), 100.0 11 from (p, $\gamma$ ), and 100.0 20 from (p,p' $\gamma$ ).<br>Mult.: from $\gamma(\theta)$ and $\gamma(\text{lin pol})$ in (p, $\gamma$ ).<br>$\delta$ : weighted average of +0.16 1 from ( <sup>28</sup> Si, $\alpha$ p $\gamma$ ), +0.18 7 from ( <sup>16</sup> O, $\alpha$ p $\gamma$ ), and +0.25 4 from (p, $\gamma$ ). Others: -0.14 2 from ( $\alpha$ ,p $\gamma$ ) and -0.22 5 from (p,p' $\gamma$ ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

## Adopted Levels, Gammas (continued)

| $\gamma(^{35}\text{Cl})$ (continued) |                    |                    |                     |         |                  |                    |                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------------|--------------------|--------------------|---------------------|---------|------------------|--------------------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $E_i(\text{level})$                  | $J_i^\pi$          | $E_\gamma^\dagger$ | $I_\gamma^\ddagger$ | $E_f$   | $J_f^\pi$        | Mult. <sup>#</sup> | $\alpha^a$              | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 3918.47                              | 3/2 <sup>+</sup>   | 1272.78            | <3.7                | 2645.67 | 7/2 <sup>+</sup> | [E2]               | 5.56×10 <sup>-5</sup> 8 | $E_\gamma$ : from $^{35}\text{Ar}$ $\varepsilon$ decay (1.7755 s).<br>$I_\gamma$ : weighted average of 29 12 from $^{35}\text{Ar}$ $\varepsilon$ decay and 23.9 28 from (p, $\gamma$ ).<br>B(M1)(W.u.)=0.075 +32-18 if M1, B(E2)(W.u.)=61 +26-14 if E2.<br>$I_\gamma$ : from ( $\alpha$ ,p $\gamma$ ).<br>B(M1)(W.u.)=0.030 +15-9 if M1, B(E2)(W.u.)=16 +8-5 if E2.<br>$I_\gamma$ : weighted average of 100 6 from $^{35}\text{Ar}$ $\varepsilon$ decay, 100 5 from ( $\alpha$ ,p $\gamma$ ), and 100.0 19 from (p, $\gamma$ ).<br>$\delta$ : -0.21 2 or +20 +10-5 from $\gamma(\theta)$ in (p, $\gamma$ ).<br>B(M1)(W.u.)=0.052 +22-11 if M1, B(E2)(W.u.)=13 +6-3 if E2.<br>$E_\gamma$ : from ( $\alpha$ ,p $\gamma$ ).<br>$I_\gamma$ : weighted average of 6.2 31 from ( $^{28}\text{Si}$ , $\alpha$ p $\gamma$ ), 8.1 21 from ( $\alpha$ ,p $\gamma$ ), and 8.7 22 from (p, $\gamma$ ).<br>B(M1)(W.u.)=0.0042 +13-12 if M1, B(E2)(W.u.)=9.3 +29-27 if E2.<br>B(E2)(W.u.)=8.7 +17-13<br>$E_\gamma$ : weighted average of 2180 1 from ( $^{16}\text{O}$ , $\alpha$ p $\gamma$ ) and 2180.2 15 from ( $\alpha$ ,p $\gamma$ ).<br>$I_\gamma$ : weighted average of 100.0 21 from ( $\alpha$ ,p $\gamma$ ) and 100.0 22 from (p, $\gamma$ ). Other: 100 11 from ( $^{28}\text{Si}$ , $\alpha$ p $\gamma$ ).<br>B(M1)(W.u.)=0.019 +8-6 if M1, B(E2)(W.u.)=45 +18-13 if E2.<br>B(M1)(W.u.)=0.071 +23-14<br>$E_\gamma$ : from $^{35}\text{Ar}$ $\varepsilon$ decay (1.7755 s).<br>$I_\gamma$ : other: 100 7 from $^{35}\text{Ar}$ $\varepsilon$ decay.<br>$I_\gamma$ : weighted average of 22 5 from $^{35}\text{Ar}$ $\varepsilon$ decay and 24 4 from (p, $\gamma$ ).<br>B(M1)(W.u.)=0.0054 +19-13 if M1, B(E2)(W.u.)=1.31 +46-30 if E2.<br>B(E1)(W.u.)=2.1×10 <sup>-4</sup> +8-7<br>B(E1)(W.u.)=1.9×10 <sup>-4</sup> +7-6<br>$I_\gamma$ : weighted average of 8 4 from ( $\alpha$ ,p $\gamma$ ) and 5.1 16 from (p, $\gamma$ ).<br>B(E1)(W.u.)=0.00179 +34-22<br>$I_\gamma$ : weighted average of 100 4 from ( $\alpha$ ,p $\gamma$ ) and 100.0 21 from (p, $\gamma$ ). |
|                                      |                    | 2155.1 15          | 24.2 28             | 1763.15 | 5/2 <sup>+</sup> | [M1,E2]            | 0.00036 4               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                      |                    | 2698.98            | 19 5                | 1219.38 | 1/2 <sup>+</sup> | [M1,E2]            | 0.00060 6               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                      |                    | 3918.24            | 100.0 19            | 0.0     | 3/2 <sup>+</sup> | M1+E2              | 0.00108 7               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 3942.9                               | 9/2 <sup>+</sup>   | 1297.4 15          | 8.0 21              | 2645.67 | 7/2 <sup>+</sup> | [M1,E2]            | 5.2×10 <sup>-5</sup> 8  | $E_\gamma$ : from $^{35}\text{Ar}$ $\varepsilon$ decay (1.7755 s).<br>$I_\gamma$ : weighted average of 29 12 from $^{35}\text{Ar}$ $\varepsilon$ decay and 23.9 28 from (p, $\gamma$ ).<br>B(M1)(W.u.)=0.075 +32-18 if M1, B(E2)(W.u.)=61 +26-14 if E2.<br>$I_\gamma$ : from ( $\alpha$ ,p $\gamma$ ).<br>B(M1)(W.u.)=0.030 +15-9 if M1, B(E2)(W.u.)=16 +8-5 if E2.<br>$I_\gamma$ : weighted average of 100 6 from $^{35}\text{Ar}$ $\varepsilon$ decay, 100 5 from ( $\alpha$ ,p $\gamma$ ), and 100.0 19 from (p, $\gamma$ ).<br>$\delta$ : -0.21 2 or +20 +10-5 from $\gamma(\theta)$ in (p, $\gamma$ ).<br>B(M1)(W.u.)=0.052 +22-11 if M1, B(E2)(W.u.)=13 +6-3 if E2.<br>$E_\gamma$ : from ( $\alpha$ ,p $\gamma$ ).<br>$I_\gamma$ : weighted average of 6.2 31 from ( $^{28}\text{Si}$ , $\alpha$ p $\gamma$ ), 8.1 21 from ( $\alpha$ ,p $\gamma$ ), and 8.7 22 from (p, $\gamma$ ).<br>B(M1)(W.u.)=0.0042 +13-12 if M1, B(E2)(W.u.)=9.3 +29-27 if E2.<br>B(E2)(W.u.)=8.7 +17-13<br>$E_\gamma$ : weighted average of 2180 1 from ( $^{16}\text{O}$ , $\alpha$ p $\gamma$ ) and 2180.2 15 from ( $\alpha$ ,p $\gamma$ ).<br>$I_\gamma$ : weighted average of 100.0 21 from ( $\alpha$ ,p $\gamma$ ) and 100.0 22 from (p, $\gamma$ ). Other: 100 11 from ( $^{28}\text{Si}$ , $\alpha$ p $\gamma$ ).<br>B(M1)(W.u.)=0.019 +8-6 if M1, B(E2)(W.u.)=45 +18-13 if E2.<br>B(M1)(W.u.)=0.071 +23-14<br>$E_\gamma$ : from $^{35}\text{Ar}$ $\varepsilon$ decay (1.7755 s).<br>$I_\gamma$ : other: 100 7 from $^{35}\text{Ar}$ $\varepsilon$ decay.<br>$I_\gamma$ : weighted average of 22 5 from $^{35}\text{Ar}$ $\varepsilon$ decay and 24 4 from (p, $\gamma$ ).<br>B(M1)(W.u.)=0.0054 +19-13 if M1, B(E2)(W.u.)=1.31 +46-30 if E2.<br>B(E1)(W.u.)=2.1×10 <sup>-4</sup> +8-7<br>B(E1)(W.u.)=1.9×10 <sup>-4</sup> +7-6<br>$I_\gamma$ : weighted average of 8 4 from ( $\alpha$ ,p $\gamma$ ) and 5.1 16 from (p, $\gamma$ ).<br>B(E1)(W.u.)=0.00179 +34-22<br>$I_\gamma$ : weighted average of 100 4 from ( $\alpha$ ,p $\gamma$ ) and 100.0 21 from (p, $\gamma$ ). |
|                                      |                    | 2180.1 10          | 100.0 21            | 1763.15 | 5/2 <sup>+</sup> | E2                 | 0.000414 6              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 3967.6                               | 1/2 <sup>+</sup>   | 1273.6             | 2.7 6               | 2694.00 | 3/2 <sup>+</sup> | [M1,E2]            | 4.9×10 <sup>-5</sup> 7  | $E_\gamma$ : from $^{35}\text{Ar}$ $\varepsilon$ decay (1.7755 s).<br>$I_\gamma$ : other: 100 7 from $^{35}\text{Ar}$ $\varepsilon$ decay.<br>$I_\gamma$ : weighted average of 22 5 from $^{35}\text{Ar}$ $\varepsilon$ decay and 24 4 from (p, $\gamma$ ).<br>B(M1)(W.u.)=0.0054 +19-13 if M1, B(E2)(W.u.)=1.31 +46-30 if E2.<br>B(E1)(W.u.)=2.1×10 <sup>-4</sup> +8-7<br>B(E1)(W.u.)=1.9×10 <sup>-4</sup> +7-6<br>$I_\gamma$ : weighted average of 8 4 from ( $\alpha$ ,p $\gamma$ ) and 5.1 16 from (p, $\gamma$ ).<br>B(E1)(W.u.)=0.00179 +34-22<br>$I_\gamma$ : weighted average of 100 4 from ( $\alpha$ ,p $\gamma$ ) and 100.0 21 from (p, $\gamma$ ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                      |                    | 2204.4             | ≤1.3                | 1763.15 | 5/2 <sup>+</sup> | [E2]               | 0.000426 6              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                      |                    | 2748.5 6           | 100 6               | 1219.38 | 1/2 <sup>+</sup> | [M1]               | 0.000569 8              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                      |                    | 3967.4             | 23 4                | 0.0     | 3/2 <sup>+</sup> | [M1,E2]            | 0.00109 7               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 4059.1                               | (3/2) <sup>-</sup> | 1365.1             | 1.3 4               | 2694.00 | 3/2 <sup>+</sup> | [E1]               | 0.0001824 26            | $E_\gamma$ : from $^{35}\text{Ar}$ $\varepsilon$ decay (1.7755 s).<br>$I_\gamma$ : other: 100 7 from $^{35}\text{Ar}$ $\varepsilon$ decay.<br>$I_\gamma$ : weighted average of 22 5 from $^{35}\text{Ar}$ $\varepsilon$ decay and 24 4 from (p, $\gamma$ ).<br>B(M1)(W.u.)=0.0054 +19-13 if M1, B(E2)(W.u.)=1.31 +46-30 if E2.<br>B(E1)(W.u.)=2.1×10 <sup>-4</sup> +8-7<br>B(E1)(W.u.)=1.9×10 <sup>-4</sup> +7-6<br>$I_\gamma$ : weighted average of 8 4 from ( $\alpha$ ,p $\gamma$ ) and 5.1 16 from (p, $\gamma$ ).<br>B(E1)(W.u.)=0.00179 +34-22<br>$I_\gamma$ : weighted average of 100 4 from ( $\alpha$ ,p $\gamma$ ) and 100.0 21 from (p, $\gamma$ ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                      |                    | 2295.9             | 5.5 16              | 1763.15 | 5/2 <sup>+</sup> | [E1]               | 0.000848 12             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                      |                    | 2839.6             | 100.0 21            | 1219.38 | 1/2 <sup>+</sup> | [E1]               | 1.17×10 <sup>-3</sup> 2 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                      |                    | 4058.9             | <2.1                | 0.0     | 3/2 <sup>+</sup> | [E1]               | 1.70×10 <sup>-3</sup> 2 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |



**Adopted Levels, Gammas (continued)**

$\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$           | $E_\gamma^\dagger$ | $I_\gamma^\ddagger$ | $E_f$   | $J_f^\pi$        | Mult.#  | $\delta^\#$ | $\alpha^a$              | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------|---------------------|--------------------|---------------------|---------|------------------|---------|-------------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4112.4              | 7/2 <sup>+</sup>    | 2349.2             | 89 6                | 1763.15 | 5/2 <sup>+</sup> | M1+E2   | -0.16 2     | 0.000405 6              | B(M1)(W.u.)=0.0159 +48-31; B(E2)(W.u.)=0.28 +12-8<br>I <sub>γ</sub> : other: 16 5 from (α,pγ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                     |                     | 4112.1             | 100 6               | 0.0     | 3/2 <sup>+</sup> | E2(+M3) | 0.00 10     | 1.21×10 <sup>-3</sup> 2 | B(E2)(W.u.)=0.76 +22-15<br>I <sub>γ</sub> : other: 100 5 from (α,pγ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 4173.43             | (5/2 <sup>+</sup> ) | 1479.40            | 45 14               | 2694.00 | 3/2 <sup>+</sup> | [M1,E2] |             | 9.1×10 <sup>-5</sup> 13 | B(M1)(W.u.)=0.048 +20-12 if M1, B(E2)(W.u.)=82<br>+34-21 if E2.<br>B(E2)(W.u.)=82 +34-21 upper bound exceeds<br>RUL=100 if E2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                     |                     | 1527.72            | <17                 | 2645.67 | 7/2 <sup>+</sup> | [M1,E2] |             | 0.000106 15             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                     |                     | 2410.19            | 28 9                | 1763.15 | 5/2 <sup>+</sup> | [M1,E2] |             | 0.00048 5               | B(M1)(W.u.)=0.0068 +33-19 if M1, B(E2)(W.u.)=4.5<br>+21-13 if E2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                     |                     | 2953.92            | ≤5.2                | 1219.38 | 1/2 <sup>+</sup> | [E2]    |             | 0.000771 11             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                     |                     | 4173.16            | 100 17              | 0.0     | 3/2 <sup>+</sup> | [M1,E2] |             | 0.00116 7               | B(M1)(W.u.)=0.0047 +15-7 if M1, B(E2)(W.u.)=1.02<br>+33-14 if E2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 4177.9              | 3/2 <sup>-</sup>    | 1175.2             | <0.82               | 3002.64 | 5/2 <sup>+</sup> | [E1]    |             | 6.06×10 <sup>-5</sup> 8 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                     |                     | 1483.9             | 13.1 33             | 2694.00 | 3/2 <sup>+</sup> | [E1]    |             | 0.000265 4              | B(E1)(W.u.)=5.7×10 <sup>-4</sup> +16-14                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                     |                     | 1532.2             | <0.82               | 2645.67 | 7/2 <sup>+</sup> | [M2]    |             | 6.18×10 <sup>-5</sup> 9 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                     |                     | 2414.7             | <1.6                | 1763.15 | 5/2 <sup>+</sup> | [E1]    |             | 0.000921 13             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                     |                     | 2958.4             | 51 8                | 1219.38 | 1/2 <sup>+</sup> | (E1+M2) | +0.11 4     | 1.22×10 <sup>-3</sup> 2 | B(E1)(W.u.)=2.8×10 <sup>-4</sup> +5-4; B(M2)(W.u.)=1.7 +16-11<br>Mult.,δ: D+Q from γ(θ) in (p,γ); Δπ=yes from level<br>scheme.<br>B(M2)(W.u.)=1.7 +16-11 upper bound exceeds RUL=3.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                     |                     | 4177.6             | 100 8               | 0.0     | 3/2 <sup>+</sup> | (E1+M2) | +0.06 3     | 1.74×10 <sup>-3</sup> 3 | B(E1)(W.u.)=1.93×10 <sup>-4</sup> +28-20; B(M2)(W.u.)=0.18<br>+24-14<br>Mult.,δ: D+Q from γ(θ) in (p,γ); Δπ=yes from level<br>scheme.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 4347.76             | 9/2 <sup>-</sup>    | 1184.90 20         | 100.0 29            | 3162.87 | 7/2 <sup>-</sup> | M1+E2   | -0.36 3     | 3.57×10 <sup>-5</sup> 5 | B(M1)(W.u.)=0.007 +7-3; B(E2)(W.u.)=2.5 +27-10<br>E <sub>γ</sub> : weighted average of 1185.0 2 from ( <sup>24</sup> Mg,αpγ),<br>1184.9 3 from ( <sup>16</sup> O,αpγ), and 1184.80 20 from<br>( <sup>14</sup> N,αpnγ). Other: 1184.6 10 from (α,pγ).<br>I <sub>γ</sub> : weighted average of 100 4 from ( <sup>28</sup> Si,αpγ), 100.0<br>31 from ( <sup>24</sup> Mg,αpγ), 100 4 from ( <sup>16</sup> O,αpγ), 100.0<br>31 from (α,pγ), and 100.0 29 from (p,γ).<br>Mult.: from γγ(DCO) and γγ(lin pol) in ( <sup>28</sup> Si,αpγ),<br>ΔJ=1.<br>δ: weighted average of -0.36 3 from (α,pγ), -0.3 1<br>from ( <sup>28</sup> Si,αpγ), and -0.65 30 from ( <sup>16</sup> O,αpγ). |
|                     |                     | 1701.88 25         | 46.8 24             | 2645.67 | 7/2 <sup>+</sup> | E1+M2   | -0.018 12   | 0.000434 6              | B(E1)(W.u.)=3.7×10 <sup>-5</sup> +39-14; B(M2)(W.u.)=0.019                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

## Adopted Levels, Gammas (continued)

| $\gamma(^{35}\text{Cl})$ (continued) |                          |                                                                             |                                             |                                                            |                                                                                                                      |                                          |                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------------------|--------------------------|-----------------------------------------------------------------------------|---------------------------------------------|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $E_i(\text{level})$                  | $J_i^\pi$                | $E_\gamma^\dagger$                                                          | $I_\gamma^\ddagger$                         | $E_f$                                                      | $J_f^\pi$                                                                                                            | Mult. <sup>#</sup>                       | $\alpha^a$                                                                        | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                      |                          |                                                                             |                                             |                                                            |                                                                                                                      |                                          |                                                                                   | +47-17<br>$E_\gamma$ : weighted average of 1702.0 3 from ( $^{24}\text{Mg}, \alpha\text{py}$ ), 1701.7 5 from ( $^{16}\text{O}, \alpha\text{py}$ ), and 1701.85 25 from ( $^{14}\text{N}, \alpha\text{pny}$ ). Other: 1702.1 15 from ( $\alpha, \text{py}$ ).<br>$I_\gamma$ : unweighted average of 45.6 24 from ( $^{28}\text{Si}, \alpha\text{py}$ ), 39.5 24 from ( $^{24}\text{Mg}, \alpha\text{py}$ ), 52.0 14 from ( $^{16}\text{O}, \alpha\text{py}$ ), 52 7 from ( $\alpha, \text{py}$ ), and 44.9 29 from ( $\text{p}, \gamma$ ).<br>Mult.: from $\gamma\gamma(\text{DCO})$ and $\gamma\gamma(\text{lin pol})$ in ( $^{28}\text{Si}, \alpha\text{py}$ ) with $\Delta J=1$ and $\gamma(\theta)$ and $\gamma(\text{lin pol})$ in ( $\alpha, \text{py}$ ).<br>$\delta$ : from ( $\text{p}, \gamma$ ). Others: +0.5 +5-3 from ( $^{28}\text{Si}, \alpha\text{py}$ ), +0.5 4 from ( $^{16}\text{O}, \alpha\text{py}$ ). |
| 4347.76                              | 9/2 <sup>-</sup>         | 2584.51<br>4347.8 8                                                         | $\leq 4.4$<br>12.4 11                       | 1763.15<br>0.0                                             | 5/2 <sup>+</sup><br>3/2 <sup>+</sup>                                                                                 | [M2]<br>[E3]                             | 0.000329 5<br>0.001000 14                                                         | B(E3)(W.u.)=44 +46-17<br>$E_\gamma, I_\gamma$ : from ( $^{16}\text{O}, \alpha\text{py}$ ).<br>$E_\gamma$ : other: 4618 2 from ( $\alpha, \text{py}$ ).<br>Mult., $\delta$ : from $\text{py}(\theta)$ with -0.08 14 for J=3/2; +0.36 15 for J=5/2 in ( $\alpha, \text{py}$ ). If J(4624=5/2), Mult would be M1+E2 based on RUL.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 4624.3                               | 3/2, 5/2                 | 4624.0                                                                      | 100                                         | 0.0                                                        | 3/2 <sup>+</sup>                                                                                                     | D+Q                                      |                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 4769.86                              | (5/2 <sup>-</sup> )      | 1606.95<br>1767.17<br>2075.79<br>3550.29<br>4769.51                         | 100 16<br>54 16<br><15<br><15<br><15        | 3162.87<br>3002.64<br>2694.00<br>1219.38<br>0.0            | 7/2 <sup>-</sup><br>5/2 <sup>+</sup><br>3/2 <sup>+</sup><br>1/2 <sup>+</sup><br>3/2 <sup>+</sup>                     | [M1, E2]<br>[E1]<br>[E1]<br>[M2]<br>[E1] | 0.000133 19<br>0.000483 7<br>0.000702 10<br>0.000645 9<br>1.95 $\times 10^{-3}$ 3 | B(M1)(W.u.)=0.040 +23-5 if M1, B(E2)(W.u.)=58 +34-7 if E2.<br>B(E1)(W.u.)=4.6 $\times 10^{-4}$ +30-9                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 4839.10                              | (1/2 <sup>+</sup> , 3/2) | 1836.41<br>2145.03<br>2193.36 <sup>b</sup><br>3075.81<br>3619.52<br>4838.74 | <12<br><6.8<br><5.1<br>100 9<br><12<br>70 9 | 3002.64<br>2694.00<br>2645.67<br>1763.15<br>1219.38<br>0.0 | 5/2 <sup>+</sup><br>3/2 <sup>+</sup><br>7/2 <sup>+</sup><br>5/2 <sup>+</sup><br>1/2 <sup>+</sup><br>3/2 <sup>+</sup> |                                          |                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 4854.4                               | (1/2, 3/2)               | 3634.8<br>4854.0                                                            | 100 7<br>33 7                               | 1219.38<br>0.0                                             | 1/2 <sup>+</sup><br>3/2 <sup>+</sup>                                                                                 |                                          |                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 4880.96                              | 7/2 <sup>(+)</sup>       | 1878.27<br>2186.89<br>2235.21                                               | 15 5<br><16<br>47 8                         | 3002.64<br>2694.00<br>2645.67                              | 5/2 <sup>+</sup><br>3/2 <sup>+</sup><br>7/2 <sup>+</sup>                                                             | [M1, E2]<br>[E2]<br>[M1, E2]             | 0.000240 30<br>0.000417 6<br>0.00040 4                                            | B(M1)(W.u.)=0.051 +30-16 if M1, B(E2)(W.u.)=55 +32-17 if E2.<br>$I_\gamma$ : other: 52 9 from ( $\alpha, \text{py}$ ) for a doublet.<br>B(M1)(W.u.)=0.095 +43-18 if M1, B(E2)(W.u.)=72 +33-14 if E2.<br>B(E2)(W.u.)=72 +33-14 upper bound exceeds RUL=100 if E2.<br>$E_\gamma$ : from ( $\alpha, \text{py}$ ).<br>$I_\gamma$ : weighted average of 100 9 from ( $\alpha, \text{py}$ ) and 100 8 from ( $\text{p}, \gamma$ ).<br>B(M1)(W.u.)=0.075 +33-12 if M1, B(E2)(W.u.)=29 +13-5 if E2.                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                      |                          | 3117.4 20                                                                   | 100 8                                       | 1763.15                                                    | 5/2 <sup>+</sup>                                                                                                     | [M1, E2]                                 | 0.00078 6                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                      |                          | 4880.60                                                                     | <8.1                                        | 0.0                                                        | 3/2 <sup>+</sup>                                                                                                     | [E2]                                     | 1.46 $\times 10^{-3}$ 2                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

Adopted Levels, Gammas (continued)

| $\gamma(^{35}\text{Cl})$ (continued) |                     |                      |                     |         |                     |                    |             |                         |                                                                                                                                                                                                                                                   |
|--------------------------------------|---------------------|----------------------|---------------------|---------|---------------------|--------------------|-------------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $E_i(\text{level})$                  | $J_i^\pi$           | $E_\gamma^\dagger$   | $I_\gamma^\ddagger$ | $E_f$   | $J_f^\pi$           | Mult. <sup>#</sup> | $\delta^\#$ | $\alpha^a$              | Comments                                                                                                                                                                                                                                          |
| 5009.1                               | (1/2,3/2)           | 835.7                | <25                 | 4173.43 | (5/2 <sup>+</sup> ) |                    |             |                         |                                                                                                                                                                                                                                                   |
|                                      |                     | 950.0                | <10                 | 4059.1  | (3/2) <sup>-</sup>  |                    |             |                         |                                                                                                                                                                                                                                                   |
|                                      |                     | 1846.2 <sup>b</sup>  | <50                 | 3162.87 | 7/2 <sup>-</sup>    |                    |             |                         |                                                                                                                                                                                                                                                   |
|                                      |                     | 2006.4               | <20                 | 3002.64 | 5/2 <sup>+</sup>    |                    |             |                         |                                                                                                                                                                                                                                                   |
|                                      |                     | 2315.0               | <25                 | 2694.00 | 3/2 <sup>+</sup>    |                    |             |                         |                                                                                                                                                                                                                                                   |
|                                      |                     | 2363.3 <sup>b</sup>  | <5                  | 2645.67 | 7/2 <sup>+</sup>    |                    |             |                         |                                                                                                                                                                                                                                                   |
|                                      |                     | 3245.8               | <10                 | 1763.15 | 5/2 <sup>+</sup>    |                    |             |                         |                                                                                                                                                                                                                                                   |
|                                      |                     | 3789.5               | <7                  | 1219.38 | 1/2 <sup>+</sup>    |                    |             |                         |                                                                                                                                                                                                                                                   |
|                                      |                     | 5008.7               | 100                 | 0.0     | 3/2 <sup>+</sup>    |                    |             |                         |                                                                                                                                                                                                                                                   |
| 5163.34                              | 7/2 <sup>-</sup>    | 2000.41              | 93 12               | 3162.87 | 7/2 <sup>-</sup>    | D+Q                | +0.44 20    |                         |                                                                                                                                                                                                                                                   |
|                                      |                     | 2160.63              | 23 9                | 3002.64 | 5/2 <sup>+</sup>    |                    |             |                         |                                                                                                                                                                                                                                                   |
|                                      |                     | 3400.01              | 100 12              | 1763.15 | 5/2 <sup>+</sup>    | D(+Q)              | 0.00 3      |                         |                                                                                                                                                                                                                                                   |
|                                      |                     | 3943.72 <sup>b</sup> | <23                 | 1219.38 | 1/2 <sup>+</sup>    |                    |             |                         |                                                                                                                                                                                                                                                   |
|                                      |                     | 5162.93              | <23                 | 0.0     | 3/2 <sup>+</sup>    |                    |             |                         |                                                                                                                                                                                                                                                   |
| 5214.5                               | (5/2 <sup>+</sup> ) | 205.4                | <23                 | 5009.1  | (1/2,3/2)           |                    |             |                         |                                                                                                                                                                                                                                                   |
|                                      |                     | 1041.1               | <60                 | 4173.43 | (5/2 <sup>+</sup> ) |                    |             |                         |                                                                                                                                                                                                                                                   |
|                                      |                     | 1155.4               | <10                 | 4059.1  | (3/2) <sup>-</sup>  |                    |             |                         |                                                                                                                                                                                                                                                   |
|                                      |                     | 2051.6               | <15                 | 3162.87 | 7/2 <sup>-</sup>    |                    |             |                         |                                                                                                                                                                                                                                                   |
|                                      |                     | 2211.8               | <25                 | 3002.64 | 5/2 <sup>+</sup>    |                    |             |                         |                                                                                                                                                                                                                                                   |
|                                      |                     | 2520.4               | <15                 | 2694.00 | 3/2 <sup>+</sup>    |                    |             |                         |                                                                                                                                                                                                                                                   |
|                                      |                     | 2568.7               | <15                 | 2645.67 | 7/2 <sup>+</sup>    |                    |             |                         |                                                                                                                                                                                                                                                   |
|                                      |                     | 3451.2               | <6                  | 1763.15 | 5/2 <sup>+</sup>    |                    |             |                         |                                                                                                                                                                                                                                                   |
|                                      |                     | 3994.9               | <4                  | 1219.38 | 1/2 <sup>+</sup>    |                    |             |                         |                                                                                                                                                                                                                                                   |
|                                      |                     | 5214.1               | 100                 | 0.0     | 3/2 <sup>+</sup>    | [M1,E2]            |             | 0.00147 8               |                                                                                                                                                                                                                                                   |
|                                      |                     | 2400.6               | 46 9                | 3002.64 | 5/2 <sup>+</sup>    | [E1]               |             | 0.000913 13             | B(E1)(W.u.)=0.0011 +6-2                                                                                                                                                                                                                           |
| 5403.3                               | 3/2 <sup>-</sup>    | 3640.0               | <18                 | 1763.15 | 5/2 <sup>+</sup>    | [E1]               |             | 1.54×10 <sup>-3</sup> 2 |                                                                                                                                                                                                                                                   |
|                                      |                     | 4183.7               | 100 18              | 1219.38 | 1/2 <sup>+</sup>    | [E1]               |             | 1.75×10 <sup>-3</sup> 2 | B(E1)(W.u.)=4.5×10 <sup>-4</sup> +20-6                                                                                                                                                                                                            |
|                                      |                     | 5402.9               | <13                 | 0.0     | 3/2 <sup>+</sup>    | [E1]               |             | 2.14×10 <sup>-3</sup> 3 |                                                                                                                                                                                                                                                   |
|                                      |                     |                      |                     |         |                     |                    |             |                         |                                                                                                                                                                                                                                                   |
| 5407.07                              | 11/2 <sup>-</sup>   | 1059.32 20           | 14.5 22             | 4347.76 | 9/2 <sup>-</sup>    | M1+E2              | +0.14 3     | 3.67×10 <sup>-5</sup> 5 | B(M1)(W.u.)=0.0082 +30-20; B(E2)(W.u.)=0.54 +35-23                                                                                                                                                                                                |
|                                      |                     |                      |                     |         |                     |                    |             |                         | $E_\gamma$ : weighted average of 1059.3 2 from ( <sup>24</sup> Mg, $\alpha\gamma$ ), 1059.5 2 from ( <sup>16</sup> O, $\alpha\gamma$ ), and 1059.17 20 from ( <sup>14</sup> N, $\alpha p n\gamma$ ). Other: 1059.3 10 from ( $\alpha, p\gamma$ ). |
|                                      |                     |                      |                     |         |                     |                    |             |                         | $I_\gamma$ : unweighted average of 12.0 12 from ( <sup>28</sup> Si, $\alpha\gamma$ ), 16.9 17 from ( <sup>24</sup> Mg, $\alpha\gamma$ ), 19.2 5 from ( <sup>16</sup> O, $\alpha\gamma$ ), and 9.9 22 from ( $\alpha, p\gamma$ ).                  |
|                                      |                     |                      |                     |         |                     |                    |             |                         | Mult.: from $\gamma\gamma$ (DCO) and $\gamma\gamma$ (lin pol) in ( <sup>28</sup> Si, $\alpha\gamma$ ), $\Delta J=1$ .                                                                                                                             |
|                                      |                     |                      |                     |         |                     |                    |             |                         | $\delta$ : weighted average of +0.12 3 from ( <sup>16</sup> O, $\alpha\gamma$ ),                                                                                                                                                                  |

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$                           | $E_\gamma^\dagger$                   | $I_\gamma^\ddagger$           | $E_f$                                                                                                    | $J_f^\pi$        | Mult. <sup>#</sup> | $\delta^\#$ | $\alpha^a$   | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------|-------------------------------------|--------------------------------------|-------------------------------|----------------------------------------------------------------------------------------------------------|------------------|--------------------|-------------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5407.07             | 11/2 <sup>-</sup>                   | 2243.71 23                           | 100.0 22                      | 3162.87                                                                                                  | 7/2 <sup>-</sup> | E2                 |             | 0.000444 6   | +0.2 1 from ( <sup>28</sup> Si, $\alpha$ py), and +0.25 8 from ( $\alpha$ ,py).<br>B(E2)(W.u.)=4.6 +15-9<br>E <sub>γ</sub> : weighted average of 2244.2 3 from ( <sup>24</sup> Mg, $\alpha$ py),<br>2243.3 2 from ( <sup>16</sup> O, $\alpha$ py), 2244.00 25 from<br>( <sup>14</sup> N, $\alpha$ pny), and 2244 3 from ( $\alpha$ ,py).<br>I <sub>γ</sub> : weighted average of 100.0 33 from ( <sup>28</sup> Si, $\alpha$ py), 100.0<br>32 from ( <sup>24</sup> Mg, $\alpha$ py), 100.0 23 from ( <sup>16</sup> O, $\alpha$ py), and<br>100.0 22 from ( $\alpha$ ,py).<br>Mult.: from $\gamma\gamma$ (DCO) and $\gamma\gamma$ (lin pol) in ( <sup>28</sup> Si, $\alpha$ py),<br>$\Delta J=2$ . |
| 5586.0              | 5/2 <sup>+</sup>                    | 2940.2<br>3822.6<br>4366.3<br>5585.5 | 67 17<br>100 17<br><17<br><17 | 2645.67 7/2 <sup>+</sup><br>1763.15 5/2 <sup>+</sup><br>1219.38 1/2 <sup>+</sup><br>0.0 3/2 <sup>+</sup> |                  |                    |             |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 5598.4              | 3/2 <sup>+</sup> , 5/2 <sup>+</sup> | 3835.0                               | 37 14                         | 1763.15                                                                                                  | 5/2 <sup>+</sup> | [M1,E2]            |             | 0.00105 7    | B(M1)(W.u.)=0.050 +29-19 if M1, B(E2)(W.u.)=13 +8-5<br>if E2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                     |                                     | 5597.9                               | 100 14                        | 0.0 3/2 <sup>+</sup>                                                                                     |                  | [M1,E2]            |             | 0.00157 8    | B(M1)(W.u.)=0.044 +23-12 if M1, B(E2)(W.u.)=5.3<br>+28-14 if E2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5646                | (5/2, 7/2, 9/2 <sup>+</sup> )       | 2483<br>2643<br>3883                 | 100                           | 3162.87 7/2 <sup>-</sup><br>3002.64 5/2 <sup>+</sup><br>1763.15 5/2 <sup>+</sup>                         |                  |                    |             |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 5655                | (3/2) <sup>+</sup>                  | 5646<br>2652                         | <8<br>100 6                   | 0.0 3/2 <sup>+</sup><br>3002.64 5/2 <sup>+</sup>                                                         |                  | [M1,E2]            |             | 0.00058 5    | B(M1)(W.u.)=0.076 +30-14 if M1, B(E2)(W.u.)=41<br>+16-7 if E2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                     |                                     | 3892                                 | 7.5 25                        | 1763.15 5/2 <sup>+</sup>                                                                                 |                  | [M1,E2]            |             | 0.00107 7    | B(M1)(W.u.)=0.0018 +10-6 if M1, B(E2)(W.u.)=0.45<br>+25-15 if E2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 5682.2              | 1/2 <sup>-</sup> , 3/2 <sup>-</sup> | 4435<br>1504.3                       | <7.5<br>100                   | 1219.38 1/2 <sup>+</sup><br>4177.9 3/2 <sup>-</sup>                                                      |                  | [M1,E2]            |             | 0.00125 8    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 5723.52             | 5/2 <sup>+</sup>                    | 3029.38<br>3960.13                   | 63 8<br>100 11                | 2694.00 3/2 <sup>+</sup><br>1763.15 5/2 <sup>+</sup>                                                     |                  |                    |             |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 5758                | (1/2, 3/2 <sup>-</sup> )            | 1580<br>4538<br>5758                 | 67 22<br><13<br>100 22        | 4177.9 3/2 <sup>-</sup><br>1219.38 1/2 <sup>+</sup><br>0.0 3/2 <sup>+</sup>                              |                  |                    |             |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 5806.6              | (1/2, 3/2, 5/2)                     | 5806.1                               | 100                           | 0.0 3/2 <sup>+</sup>                                                                                     |                  |                    |             |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 5830                | (5/2 <sup>-</sup> )                 | 2667                                 | 100                           | 3162.87 7/2 <sup>-</sup>                                                                                 |                  | D+Q                |             |              | I <sub>γ</sub> : from ( $\alpha$ ,py).<br>$\delta$ : -2.2 5 or -0.6 1 for J=5/2 from $\gamma(\theta)$ in ( $\alpha$ ,py).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 5926.65             | (11/2) <sup>-</sup>                 | 1579.09 30                           | 100 11                        | 4347.76 9/2 <sup>-</sup>                                                                                 |                  | M1+E2              | -0.2 1      | 0.0001074 22 | E <sub>γ</sub> : weighted average of 1578.9 5 from ( <sup>16</sup> O, $\alpha$ py),<br>1579.15 30 from ( <sup>14</sup> N, $\alpha$ pny), and 1579.3 15 from<br>( $\alpha$ ,py).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$                               | $E_\gamma^\dagger$ | $I_\gamma^\ddagger$ | $E_f$   | $J_f^\pi$           | Mult. # | $\delta^\#$ | $\alpha^a$               | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------|-----------------------------------------|--------------------|---------------------|---------|---------------------|---------|-------------|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     |                                         |                    |                     |         |                     |         |             |                          | Mult., $\delta$ : D+Q and $\delta$ from $\gamma\gamma(\theta)$ (DCO) in $(^{28}\text{Si}, \alpha\gamma)$ with $\Delta J=1$ ; E1+M2 ruled out by RUL. Other: $-0.8$ 4 from $\gamma(\theta)$ in $(\alpha, \text{py})$ .<br>$I_\gamma$ : from $(^{28}\text{Si}, \alpha\gamma)$ .                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5926.65             | (11/2) <sup>-</sup>                     | 1983.7             | 11.1 22             | 3942.9  | 9/2 <sup>+</sup>    | [E1]    |             | 0.000639 9               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                     |                                         | 2763.66            |                     | 3162.87 | 7/2 <sup>-</sup>    |         |             |                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 6087.31             | 13/2 <sup>-</sup>                       | 680.35 10          | 100.0 21            | 5407.07 | 11/2 <sup>-</sup>   | M1+E2   | +0.020 15   | 8.70×10 <sup>-5</sup> 12 | B(M1)(W.u.)=0.0107 10<br>$E_\gamma$ : weighted average of 680.4 1 from $(^{24}\text{Mg}, \alpha\text{py})$ , 680.5 5 from $(^{16}\text{O}, \alpha\text{py})$ , and 680.22 15 from $(^{14}\text{N}, \alpha\text{pn}\gamma)$ . Other: 679.9 10 from $(\alpha, \text{py})$ .<br>$I_\gamma$ : weighted average of 100.0 31 from $(^{28}\text{Si}, \alpha\text{py})$ , 100.0 21 from $(^{16}\text{O}, \alpha\text{py})$ , and 100 6 from $(\alpha, \text{py})$ .<br>Mult., $\delta$ : from $\gamma(\theta)$ and $\gamma(\text{lin pol})$ in $(\alpha, \text{py})$ . Other: $\delta=+0.1$ 1 from $\gamma\gamma(\text{DCO})$ in $(^{28}\text{Si}, \alpha\text{py})$ , $\Delta J=1$ . |
|                     |                                         | 1739.4 4           | 6.9 9               | 4347.76 | 9/2 <sup>-</sup>    | E2      |             | 0.0002076 29             | B(E2)(W.u.)=0.054 +9-8<br>$E_\gamma$ : weighted average of 1739.6 5 from $(^{16}\text{O}, \alpha\text{py})$ and 1739.3 4 from $(^{14}\text{N}, \alpha\text{pn}\gamma)$ .<br>$I_\gamma$ : weighted average of 6.3 6 from $(^{28}\text{Si}, \alpha\text{py})$ , 7.9 9 from $(^{16}\text{O}, \alpha\text{py})$ , and 18 6 from $(\alpha, \text{py})$ .<br>Mult.: from $\gamma\gamma(\theta)$ (DCO) and $\gamma\gamma(\text{lin pol})$ in $(^{28}\text{Si}, \alpha\text{py})$ , $\Delta J=2$ .                                                                                                                                                                                    |
| 6106.2              | (3/2, 5/2 <sup>+</sup> )                | 4342.8             | 10.3 35             | 1763.15 | 5/2 <sup>+</sup>    |         |             |                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                     |                                         | 4886.5             | 62 17               | 1219.38 | 1/2 <sup>+</sup>    |         |             |                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                     |                                         | 6105.6             | 100 17              | 0.0     | 3/2 <sup>+</sup>    |         |             |                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 6181                | (1/2 <sup>+</sup> to 7/2 <sup>+</sup> ) | 6180               | 100                 | 0.0     | 3/2 <sup>+</sup>    |         |             |                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 6380.2              | (9/2 <sup>-</sup> )                     | 296 1              |                     | 6087.31 | 13/2 <sup>-</sup>   |         |             |                          | $I_\gamma$ : from $(^{16}\text{O}, \alpha\text{py})$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                     |                                         | 971 1              |                     | 5407.07 | 11/2 <sup>-</sup>   |         |             |                          | $I_\gamma$ : from $(^{16}\text{O}, \alpha\text{py})$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 6492                | (3/2 <sup>+</sup> )                     | 6491               | 100                 | 0.0     | 3/2 <sup>+</sup>    |         |             |                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 6660.2              | (11/2 <sup>-</sup> )                    | 2312.4             |                     | 4347.76 | 9/2 <sup>-</sup>    |         |             |                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                     |                                         | 3497.1             |                     | 3162.87 | 7/2 <sup>-</sup>    |         |             |                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 7066.2              | (5/2 <sup>+</sup> )                     | 1902.8             | 1.8                 | 5163.34 | 7/2 <sup>-</sup>    |         |             |                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                     |                                         | 2892.6             | 1.0                 | 4173.43 | (5/2 <sup>+</sup> ) |         |             |                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                     |                                         | 2953.7             | 1.2                 | 4112.4  | 7/2 <sup>+</sup>    |         |             |                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                     |                                         | 3007.0             | 2.9                 | 4059.1  | (3/2) <sup>-</sup>  |         |             |                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                     |                                         | 3903.1             | 12                  | 3162.87 | 7/2 <sup>-</sup>    |         |             |                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                     |                                         | 4063.3             | 6.1                 | 3002.64 | 5/2 <sup>+</sup>    |         |             |                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$  (continued)

| <u>E<sub>i</sub>(level)</u> | <u>J<sup><math>\pi</math></sup><sub>i</sub></u> | <u>E<sub><math>\gamma</math></sub><sup><math>\dagger</math></sup></u> | <u>I<sub><math>\gamma</math></sub><sup><math>\ddagger</math></sup></u> | <u>E<sub>f</sub></u> | <u>J<sup><math>\pi</math></sup><sub>f</sub></u> | <u>Mult.<sup>#</sup></u> | <u><math>\delta^{\#}</math></u> |
|-----------------------------|-------------------------------------------------|-----------------------------------------------------------------------|------------------------------------------------------------------------|----------------------|-------------------------------------------------|--------------------------|---------------------------------|
| 7066.2                      | (5/2 <sup>+</sup> )                             | 4371.9                                                                | 10                                                                     | 2694.00              | 3/2 <sup>+</sup>                                |                          |                                 |
|                             |                                                 | 4420.2                                                                | 3.3                                                                    | 2645.67              | 7/2 <sup>+</sup>                                |                          |                                 |
|                             |                                                 | 5302.6                                                                | 29                                                                     | 1763.15              | 5/2 <sup>+</sup>                                |                          |                                 |
|                             |                                                 | 5846.3                                                                | 37                                                                     | 1219.38              | 1/2 <sup>+</sup>                                |                          |                                 |
|                             |                                                 | 7065.4                                                                | 100                                                                    | 0.0                  | 3/2 <sup>+</sup>                                |                          |                                 |
| 7103.4                      | 3/2 <sup>(-)</sup>                              | 2925.4                                                                | 2.3                                                                    | 4177.9               | 3/2 <sup>-</sup>                                |                          |                                 |
|                             |                                                 | 3184.8                                                                | 2.3                                                                    | 3918.47              | 3/2 <sup>+</sup>                                |                          |                                 |
|                             |                                                 | 4100.5                                                                | 4.7                                                                    | 3002.64              | 5/2 <sup>+</sup>                                |                          |                                 |
|                             |                                                 | 4409.1                                                                | 17                                                                     | 2694.00              | 3/2 <sup>+</sup>                                | D(+Q)                    | 0.00 3                          |
|                             |                                                 | 5339.8                                                                | 6.3                                                                    | 1763.15              | 5/2 <sup>+</sup>                                |                          |                                 |
|                             |                                                 | 5883.5                                                                | 100                                                                    | 1219.38              | 1/2 <sup>+</sup>                                | D(+Q)                    | 0.00 3                          |
|                             |                                                 | 7102.6                                                                | 23                                                                     | 0.0                  | 3/2 <sup>+</sup>                                |                          |                                 |
| 7178.6                      | 1/2 <sup>(+)</sup>                              | 1072.4                                                                | 7.9                                                                    | 6106.2               | (3/2,5/2 <sup>+</sup> )                         |                          |                                 |
|                             |                                                 | 1775.3                                                                | 4.5                                                                    | 5403.3               | 3/2 <sup>-</sup>                                |                          |                                 |
|                             |                                                 | 2339.4                                                                | 4                                                                      | 4839.10              | (1/2 <sup>+</sup> ,3/2)                         |                          |                                 |
|                             |                                                 | 3005.0                                                                | 2.1                                                                    | 4173.43              | (5/2 <sup>+</sup> )                             |                          |                                 |
|                             |                                                 | 3119.4                                                                | 58                                                                     | 4059.1               | (3/2) <sup>-</sup>                              |                          |                                 |
|                             |                                                 | 3210.8                                                                | 24                                                                     | 3967.6               | 1/2 <sup>+</sup>                                |                          |                                 |
|                             |                                                 | 3260.0                                                                | 7.9                                                                    | 3918.47              | 3/2 <sup>+</sup>                                |                          |                                 |
|                             |                                                 | 4484.3                                                                | 11                                                                     | 2694.00              | 3/2 <sup>+</sup>                                |                          |                                 |
|                             |                                                 | 5958.7                                                                | 45                                                                     | 1219.38              | 1/2 <sup>+</sup>                                |                          |                                 |
|                             |                                                 | 7177.8                                                                | 100                                                                    | 0.0                  | 3/2 <sup>+</sup>                                |                          |                                 |
|                             |                                                 | 1970                                                                  | <1.1                                                                   | 5214.5               | (5/2 <sup>+</sup> )                             |                          |                                 |
|                             |                                                 | 2176                                                                  | <1.1                                                                   | 5009.1               | (1/2,3/2)                                       |                          |                                 |
| 7185                        | 5/2 <sup>+</sup>                                | 3011                                                                  | <11                                                                    | 4173.43              | (5/2 <sup>+</sup> )                             |                          |                                 |
|                             |                                                 | 3126                                                                  | 57 13                                                                  | 4059.1               | (3/2) <sup>-</sup>                              |                          |                                 |
|                             |                                                 | 4022                                                                  | <6.5                                                                   | 3162.87              | 7/2 <sup>-</sup>                                |                          |                                 |
|                             |                                                 | 4182                                                                  | <6.5                                                                   | 3002.64              | 5/2 <sup>+</sup>                                |                          |                                 |
|                             |                                                 | 4491                                                                  | 4.4 22                                                                 | 2694.00              | 3/2 <sup>+</sup>                                |                          |                                 |
|                             |                                                 | 4539                                                                  | 13 5                                                                   | 2645.67              | 7/2 <sup>+</sup>                                |                          |                                 |
|                             |                                                 | 5421                                                                  | 9 7                                                                    | 1763.15              | 5/2 <sup>+</sup>                                |                          |                                 |
|                             |                                                 | 5965                                                                  | 35 15                                                                  | 1219.38              | 1/2 <sup>+</sup>                                |                          |                                 |
|                             |                                                 | 7184                                                                  | 100 22                                                                 | 0.0                  | 3/2 <sup>+</sup>                                |                          |                                 |
|                             |                                                 | 2185.4                                                                | 1.2                                                                    | 5009.1               | (1/2,3/2)                                       |                          |                                 |
|                             |                                                 | 2340.1                                                                | 3.0                                                                    | 4854.4               | (1/2,3/2)                                       |                          |                                 |
|                             |                                                 | 2355.4                                                                | 2.7                                                                    | 4839.10              | (1/2 <sup>+</sup> ,3/2)                         |                          |                                 |
| 7194.6                      | 1/2 <sup>-</sup>                                | 3016.6                                                                | 24                                                                     | 4177.9               | 3/2 <sup>-</sup>                                |                          |                                 |
|                             |                                                 | 3135.4                                                                | 4.5                                                                    | 4059.1               | (3/2) <sup>-</sup>                              |                          |                                 |
|                             |                                                 | 3226.8                                                                | 2.7                                                                    | 3967.6               | 1/2 <sup>+</sup>                                |                          |                                 |
|                             |                                                 | 3276.0                                                                | 6                                                                      | 3918.47              | 3/2 <sup>+</sup>                                |                          |                                 |
|                             |                                                 | 4500.3                                                                | 1.6                                                                    | 2694.00              | 3/2 <sup>+</sup>                                |                          |                                 |
|                             |                                                 | 5431.0                                                                | 0.75                                                                   | 1763.15              | 5/2 <sup>+</sup>                                |                          |                                 |
|                             |                                                 |                                                                       |                                                                        |                      |                                                 |                          |                                 |
|                             |                                                 |                                                                       |                                                                        |                      |                                                 |                          |                                 |

Adopted Levels, Gammas (continued)

| $\gamma(^{35}\text{Cl})$ (continued) |                    |                    |                     |         |                                    |            |                                                                       |          |
|--------------------------------------|--------------------|--------------------|---------------------|---------|------------------------------------|------------|-----------------------------------------------------------------------|----------|
| $E_i(\text{level})$                  | $J_i^\pi$          | $E_\gamma^\dagger$ | $I_\gamma^\ddagger$ | $E_f$   | $J_f^\pi$                          | Mult. $\#$ | $\delta^\#$                                                           | Comments |
| 7194.6                               | 1/2 <sup>-</sup>   | 5974.7             | 100                 | 1219.38 | 1/2 <sup>+</sup>                   |            |                                                                       |          |
|                                      |                    | 7193.8             | 3.0                 | 0.0     | 3/2 <sup>+</sup>                   |            |                                                                       |          |
| 7225.5                               | 5/2                | 1119.3             | 3.1                 | 6106.2  | (3/2,5/2 <sup>+</sup> )            |            |                                                                       |          |
|                                      |                    | 1418.9             |                     | 5806.6  | (1/2,3/2,5/2)                      |            |                                                                       |          |
|                                      |                    | 1543.3             | 3.4                 | 5682.2  | 1/2 <sup>-</sup> ,3/2 <sup>-</sup> |            |                                                                       |          |
|                                      |                    | 1571               | 1.4                 | 5655    | (3/2) <sup>+</sup>                 |            |                                                                       |          |
|                                      |                    | 1822.2             | 1.5                 | 5403.3  | 3/2 <sup>-</sup>                   |            |                                                                       |          |
|                                      |                    | 2010.9             | 12                  | 5214.5  | (5/2 <sup>+</sup> )                |            |                                                                       |          |
|                                      |                    | 3047.5             | 14                  | 4177.9  | 3/2 <sup>-</sup>                   |            |                                                                       |          |
|                                      |                    | 3166.3             | 3.4                 | 4059.1  | (3/2) <sup>-</sup>                 |            |                                                                       |          |
|                                      |                    | 3306.9             | 2.5                 | 3918.47 | 3/2 <sup>+</sup>                   |            |                                                                       |          |
|                                      |                    | 4222.6             |                     | 3002.64 | 5/2 <sup>+</sup>                   |            |                                                                       |          |
|                                      |                    | 4531.2             | 17                  | 2694.00 | 3/2 <sup>+</sup>                   | D+Q        | +0.36 4                                                               |          |
|                                      |                    | 5461.9             | 12                  | 1763.15 | 5/2 <sup>+</sup>                   | D+Q        | -0.36 7                                                               |          |
|                                      |                    | 7224.7             | 100                 | 0.0     | 3/2 <sup>+</sup>                   | D+Q        | +0.36 4                                                               |          |
| 7234.0                               | 5/2 <sup>(+)</sup> | 1579               | 0.22                | 5655    | (3/2) <sup>+</sup>                 |            |                                                                       |          |
|                                      |                    | 3056.0             |                     | 4177.9  | 3/2 <sup>-</sup>                   |            |                                                                       |          |
|                                      |                    | 4539.7             | 1.9                 | 2694.00 | 3/2 <sup>+</sup>                   |            |                                                                       |          |
|                                      |                    | 4588.0             | 3.2                 | 2645.67 | 7/2 <sup>+</sup>                   |            |                                                                       |          |
|                                      |                    | 5470.4             | 0.22                | 1763.15 | 5/2 <sup>+</sup>                   |            |                                                                       |          |
|                                      |                    | 6014.1             | 1.9                 | 1219.38 | 1/2 <sup>+</sup>                   |            |                                                                       |          |
|                                      |                    | 7233.2             | 100                 | 0.0     | 3/2 <sup>+</sup>                   | D+Q        | -0.10                                                                 |          |
| 7272.7                               | 1/2 <sup>-</sup>   | 1515               | 1.7                 | 5758    | (1/2,3/2 <sup>-</sup> )            |            |                                                                       |          |
|                                      |                    | 1869.4             | 0.29                | 5403.3  | 3/2 <sup>-</sup>                   |            |                                                                       |          |
|                                      |                    | 2263.5             | 2.0                 | 5009.1  | (1/2,3/2)                          |            |                                                                       |          |
|                                      |                    | 2433.5             | 1.5                 | 4839.10 | (1/2 <sup>+</sup> ,3/2)            |            |                                                                       |          |
|                                      |                    | 3099.1             | 0.58                | 4173.43 | (5/2 <sup>+</sup> )                |            |                                                                       |          |
|                                      |                    | 3213.4             | 1.3                 | 4059.1  | (3/2) <sup>-</sup>                 |            |                                                                       |          |
|                                      |                    | 3304.9             | 1.2                 | 3967.6  | 1/2 <sup>+</sup>                   |            |                                                                       |          |
|                                      |                    | 4578.4             | 1.6                 | 2694.00 | 3/2 <sup>+</sup>                   |            |                                                                       |          |
|                                      |                    | 6052.8             | 35                  | 1219.38 | 1/2 <sup>+</sup>                   |            |                                                                       |          |
|                                      |                    | 7271.9             | 100                 | 0.0     | 3/2 <sup>+</sup>                   |            |                                                                       |          |
| 7361.9                               | 3/2 <sup>(+)</sup> | 1707               | 0.41                | 5655    | (3/2) <sup>+</sup>                 |            |                                                                       |          |
|                                      |                    | 2352.7             | 0.82                | 5009.1  | (1/2,3/2)                          |            |                                                                       |          |
|                                      |                    | 2522.7             | 1.4                 | 4839.10 | (1/2 <sup>+</sup> ,3/2)            |            |                                                                       |          |
|                                      |                    | 3183.8             | 0.69                | 4177.9  | 3/2 <sup>-</sup>                   |            |                                                                       |          |
|                                      |                    | 3302.6             | 2.7                 | 4059.1  | (3/2) <sup>-</sup>                 |            |                                                                       |          |
|                                      |                    | 3394.1             | 4.1                 | 3967.6  | 1/2 <sup>+</sup>                   |            |                                                                       |          |
|                                      |                    | 3443.3             | 1.1                 | 3918.47 | 3/2 <sup>+</sup>                   |            |                                                                       |          |
|                                      |                    | 4359.0             | 0.41                | 3002.64 | 5/2 <sup>+</sup>                   |            |                                                                       |          |
|                                      |                    | 4667.6             | 0.69                | 2694.00 | 3/2 <sup>+</sup>                   |            |                                                                       |          |
|                                      |                    | 5598.3             | 11                  | 1763.15 | 5/2 <sup>+</sup>                   | D+Q        | $\delta$ : >+2.5 or -0.01 10 from $\gamma(\theta)$ in (p, $\gamma$ ). |          |

Adopted Levels, Gammas (continued)

| $\gamma(^{35}\text{Cl})$ (continued) |                                             |                    |                     |         |                                     |                    |             |                         |                                                                                                                                                                                                                                                                  |
|--------------------------------------|---------------------------------------------|--------------------|---------------------|---------|-------------------------------------|--------------------|-------------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $E_i(\text{level})$                  | $J_i^\pi$                                   | $E_\gamma^\dagger$ | $I_\gamma^\ddagger$ | $E_f$   | $J_f^\pi$                           | Mult. <sup>#</sup> | $\delta^\#$ | $\alpha^a$              | Comments                                                                                                                                                                                                                                                         |
| 7361.9                               | 3/2 <sup>(+)</sup>                          | 6141.9             | 100                 | 1219.38 | 1/2 <sup>+</sup>                    | (M1+E2)            |             |                         | Mult.: D+Q from $\gamma(\theta)$ in (p, $\gamma$ ); E1+M2 is less likely for this strong primary transition.<br>$\delta$ : $-0.05$ 3 or $+2.5$ 3 in (p, $\gamma$ ).<br>$\delta$ : $-0.16 < \delta(Q/D) < -0.12$ or $+7.5 < \delta(Q/D) < +10$ in (p, $\gamma$ ). |
|                                      |                                             | 7361.1             | 14                  | 0.0     | 3/2 <sup>+</sup>                    | D+Q                |             |                         |                                                                                                                                                                                                                                                                  |
| 7396.4                               | 7/2 <sup>(-)</sup>                          | 3048.5             | 23                  | 4347.76 | 9/2 <sup>-</sup>                    |                    |             |                         | Mult.: D+Q from $\gamma(\theta)$ in (p, $\gamma$ ); E1+M2 is less likely for this strong primary transition.                                                                                                                                                     |
|                                      |                                             | 3283.8             | 17                  | 4112.4  | 7/2 <sup>+</sup>                    |                    |             |                         |                                                                                                                                                                                                                                                                  |
|                                      |                                             | 3453.3             | 11                  | 3942.9  | 9/2 <sup>+</sup>                    |                    |             |                         |                                                                                                                                                                                                                                                                  |
|                                      |                                             | 4233.3             | 100                 | 3162.87 | 7/2 <sup>-</sup>                    | (M1+E2)            | +0.28 25    | $1.12 \times 10^{-3}$ 3 |                                                                                                                                                                                                                                                                  |
|                                      |                                             | 4393.5             | 30                  | 3002.64 | 5/2 <sup>+</sup>                    | D(+Q)              | +0.011 18   |                         |                                                                                                                                                                                                                                                                  |
|                                      |                                             | 4750.4             | 28                  | 2645.67 | 7/2 <sup>+</sup>                    |                    |             |                         |                                                                                                                                                                                                                                                                  |
|                                      |                                             | 5632.8             | 4.3                 | 1763.15 | 5/2 <sup>+</sup>                    |                    |             |                         |                                                                                                                                                                                                                                                                  |
| 7450.9                               | 3/2                                         | 3532.2             | 6.9 34              | 3918.47 | 3/2 <sup>+</sup>                    |                    |             |                         |                                                                                                                                                                                                                                                                  |
|                                      |                                             | 4448.0             | 13.7 14             | 3002.64 | 5/2 <sup>+</sup>                    |                    |             |                         |                                                                                                                                                                                                                                                                  |
|                                      |                                             | 4756.6             | 11.0 11             | 2694.00 | 3/2 <sup>+</sup>                    |                    |             |                         |                                                                                                                                                                                                                                                                  |
|                                      |                                             | 5687.3             | 5.5 28              | 1763.15 | 5/2 <sup>+</sup>                    |                    |             |                         |                                                                                                                                                                                                                                                                  |
|                                      |                                             | 6230.9             | 100 10              | 1219.38 | 1/2 <sup>+</sup>                    |                    |             |                         |                                                                                                                                                                                                                                                                  |
| 7501.1                               | (5/2 <sup>+</sup> , 7/2, 9/2 <sup>+</sup> ) | 3558.0             | 46 6                | 3942.9  | 9/2 <sup>+</sup>                    |                    |             |                         |                                                                                                                                                                                                                                                                  |
|                                      |                                             | 4498.2             | 23.6 24             | 3002.64 | 5/2 <sup>+</sup>                    |                    |             |                         |                                                                                                                                                                                                                                                                  |
|                                      |                                             | 4855.1             | 100 11              | 2645.67 | 7/2 <sup>+</sup>                    |                    |             |                         |                                                                                                                                                                                                                                                                  |
|                                      |                                             | 5737.5             | 12.7 13             | 1763.15 | 5/2 <sup>+</sup>                    |                    |             |                         |                                                                                                                                                                                                                                                                  |
| 7502.9                               |                                             | 1396.7 @           | 14 @                | 6106.2  | (3/2, 5/2 <sup>+</sup> )            |                    |             |                         |                                                                                                                                                                                                                                                                  |
|                                      |                                             | 1904.4 @           | 4 @                 | 5598.4  | 3/2 <sup>+</sup> , 5/2 <sup>+</sup> |                    |             |                         |                                                                                                                                                                                                                                                                  |
|                                      |                                             | 2493.7 @           | 4 @                 | 5009.1  | (1/2, 3/2)                          |                    |             |                         |                                                                                                                                                                                                                                                                  |
|                                      |                                             | 3324.8 @           | 6 @                 | 4177.9  | 3/2 <sup>-</sup>                    |                    |             |                         |                                                                                                                                                                                                                                                                  |
|                                      |                                             | 3443.6 @           | 100 @               | 4059.1  | (3/2) <sup>-</sup>                  |                    |             |                         |                                                                                                                                                                                                                                                                  |
|                                      |                                             | 3559.8 @           | 4 @                 | 3942.9  | 9/2 <sup>+</sup>                    |                    |             |                         |                                                                                                                                                                                                                                                                  |
|                                      |                                             | 4339.7 @           | 8 @                 | 3162.87 | 7/2 <sup>-</sup>                    |                    |             |                         |                                                                                                                                                                                                                                                                  |
|                                      |                                             | 4500.0 @           | 8 @                 | 3002.64 | 5/2 <sup>+</sup>                    |                    |             |                         |                                                                                                                                                                                                                                                                  |
|                                      |                                             | 4856.9 @           | 14 @                | 2645.67 | 7/2 <sup>+</sup>                    |                    |             |                         |                                                                                                                                                                                                                                                                  |
|                                      |                                             | 5739.2 @           | 1.4 @               | 1763.15 | 5/2 <sup>+</sup>                    |                    |             |                         |                                                                                                                                                                                                                                                                  |
|                                      |                                             | 6282.9 @           | 36 @                | 1219.38 | 1/2 <sup>+</sup>                    |                    |             |                         |                                                                                                                                                                                                                                                                  |
|                                      |                                             | 7502.0 @           | 0.6 @               | 0.0     | 3/2 <sup>+</sup>                    |                    |             |                         |                                                                                                                                                                                                                                                                  |
| 7518.8                               | 7/2                                         | 1920.3             | 2.9 15              | 5598.4  | 3/2 <sup>+</sup> , 5/2 <sup>+</sup> |                    |             |                         |                                                                                                                                                                                                                                                                  |
|                                      |                                             | 1932.7             | 4.4 22              | 5586.0  | 5/2 <sup>+</sup>                    |                    |             |                         |                                                                                                                                                                                                                                                                  |
|                                      |                                             | 3170.9             | 2.9 15              | 4347.76 | 9/2 <sup>-</sup>                    |                    |             |                         |                                                                                                                                                                                                                                                                  |
|                                      |                                             | 3575.7             | 2.9 15              | 3942.9  | 9/2 <sup>+</sup>                    |                    |             |                         |                                                                                                                                                                                                                                                                  |



Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$        | $E_\gamma^\dagger$ | $I_\gamma^\ddagger$ | $E_f$   | $J_f^\pi$                               | Mult.#  | $\delta^\#$ | $\alpha^a$              | Comments                                                                                                                                                                  |
|---------------------|------------------|--------------------|---------------------|---------|-----------------------------------------|---------|-------------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7518.8              | 7/2              | 4355.6             | 100 10              | 3162.87 | 7/2 <sup>-</sup>                        | D(+Q)   | +0.1 2      |                         |                                                                                                                                                                           |
|                     |                  | 4872.8             | 4.4 22              | 2645.67 | 7/2 <sup>+</sup>                        |         |             |                         |                                                                                                                                                                           |
|                     |                  | 5755.1             | 29.4 30             | 1763.15 | 5/2 <sup>+</sup>                        | D+Q     | +0.098 22   |                         |                                                                                                                                                                           |
| 7548.9              | 7/2 <sup>-</sup> | 1903               | 0.59 7              | 5646    | (5/2,7/2,9/2 <sup>+</sup> )             |         |             |                         |                                                                                                                                                                           |
|                     |                  | 1962.8             | 0.11                | 5586.0  | 5/2 <sup>+</sup>                        | [E1]    |             | 0.000624 9              |                                                                                                                                                                           |
|                     |                  | 2778.9             | 1.49 11             | 4769.86 | (5/2 <sup>-</sup> )                     | [M1,E2] |             | 0.00064 6               |                                                                                                                                                                           |
|                     |                  | 4385.7             | 100.0 11            | 3162.87 | 7/2 <sup>-</sup>                        | M1+E2   | -0.07 2     | 1.16×10 <sup>-3</sup> 2 |                                                                                                                                                                           |
|                     |                  | 4545.9             | 2.13 11             | 3002.64 | 5/2 <sup>+</sup>                        | (E1+M2) | +0.6 4      | 0.00163 22              | Mult., $\delta$ : D+Q and $\delta$ from $\gamma(\theta)$ in (p, $\gamma$ );<br>$\Delta\pi$ =yes from level scheme.                                                        |
|                     |                  | 4902.9             | 0.72 8              | 2645.67 | 7/2 <sup>+</sup>                        | [E1]    |             | 1.99×10 <sup>-3</sup> 3 |                                                                                                                                                                           |
|                     |                  | 5785.2             | 0.35 4              | 1763.15 | 5/2 <sup>+</sup>                        | [E1]    |             | 2.24×10 <sup>-3</sup> 3 |                                                                                                                                                                           |
|                     |                  | 7548.0             | 0.287 32            | 0.0     | 3/2 <sup>+</sup>                        | (M2)    |             |                         | Mult.: Q(+O) with $\delta(\text{O/Q})=0.1$ 2 from $\gamma(\theta)$<br>in (p, $\gamma$ ); $\Delta\pi$ =yes from level scheme; E3<br>component is less likely based on RUL. |
| 7561.3              | 1/2,3/2          | 3383.2             | 2.6                 | 4177.9  | 3/2 <sup>-</sup>                        |         |             |                         |                                                                                                                                                                           |
|                     |                  | 3502.0             | 17                  | 4059.1  | (3/2 <sup>-</sup> )                     |         |             |                         |                                                                                                                                                                           |
|                     |                  | 3593.5             | 1.4                 | 3967.6  | 1/2 <sup>+</sup>                        |         |             |                         |                                                                                                                                                                           |
|                     |                  | 3642.6             | 4.6                 | 3918.47 | 3/2 <sup>+</sup>                        |         |             |                         |                                                                                                                                                                           |
|                     |                  | 4866.9             | 63                  | 2694.00 | 3/2 <sup>+</sup>                        |         |             |                         |                                                                                                                                                                           |
|                     |                  | 6341.3             | 100                 | 1219.38 | 1/2 <sup>+</sup>                        |         |             |                         |                                                                                                                                                                           |
|                     |                  | 7560.4             | 97                  | 0.0     | 3/2 <sup>+</sup>                        |         |             |                         |                                                                                                                                                                           |
| 7600.8              | 5/2 <sup>+</sup> | 2437.4             | 5.8                 | 5163.34 | 7/2 <sup>-</sup>                        | [E1]    |             | 0.000934 13             |                                                                                                                                                                           |
|                     |                  | 2719.7             | 6.1                 | 4880.96 | 7/2 <sup>(+)</sup>                      |         |             |                         |                                                                                                                                                                           |
|                     |                  | 2761.6             | 6.1                 | 4839.10 | (1/2 <sup>+</sup> ,3/2)                 |         |             |                         |                                                                                                                                                                           |
|                     |                  | 2830.8             | 3.9                 | 4769.86 | (5/2 <sup>-</sup> )                     |         |             |                         |                                                                                                                                                                           |
|                     |                  | 2976.4             | 5.5                 | 4624.3  | 3/2,5/2                                 |         |             |                         |                                                                                                                                                                           |
|                     |                  | 3422.7             | 9.1                 | 4177.9  | 3/2 <sup>-</sup>                        | [E1]    |             | 1.44×10 <sup>-3</sup> 2 |                                                                                                                                                                           |
|                     |                  | 3427.2             |                     | 4173.43 | (5/2 <sup>+</sup> )                     |         |             |                         |                                                                                                                                                                           |
|                     |                  | 3541.5             | 6.1                 | 4059.1  | (3/2 <sup>-</sup> )                     | [E1]    |             | 1.50×10 <sup>-3</sup> 2 |                                                                                                                                                                           |
|                     |                  | 4437.6             | 15                  | 3162.87 | 7/2 <sup>-</sup>                        | [E1]    |             | 1.85×10 <sup>-3</sup> 3 |                                                                                                                                                                           |
|                     |                  | 4597.8             | 24                  | 3002.64 | 5/2 <sup>+</sup>                        | M1+E2   | +0.50 22    | 1.25×10 <sup>-3</sup> 3 | Mult.: D+Q from $\gamma(\theta)$ in (p, $\gamma$ ); E1+M2 is<br>less likely for this strong primary transition.                                                           |
|                     |                  | 4906.4             | 55                  | 2694.00 | 3/2 <sup>+</sup>                        | (M1+E2) |             | 0.00139 8               | Mult.: D+Q from $\gamma(\theta)$ in (p, $\gamma$ ); $\Delta\pi$ =no from<br>level scheme.                                                                                 |
|                     |                  | 4954.8             | 5.8                 | 2645.67 | 7/2 <sup>+</sup>                        | [M1,E2] |             | 0.00140 8               |                                                                                                                                                                           |
|                     |                  | 5837.1             | 58                  | 1763.15 | 5/2 <sup>+</sup>                        | M1+E2   | +0.31 7     | 1.57×10 <sup>-3</sup> 2 |                                                                                                                                                                           |
|                     |                  | 6380.8             | 3.3                 | 1219.38 | 1/2 <sup>+</sup>                        | [E2]    |             |                         |                                                                                                                                                                           |
|                     |                  | 7599.9             | 100                 | 0.0     | 3/2 <sup>+</sup>                        | M1+E2   | -0.19 4     |                         |                                                                                                                                                                           |
| 7618.7              | 5/2 <sup>+</sup> | 1438               | 0.77                | 6181    | (1/2 <sup>+</sup> to 7/2 <sup>+</sup> ) |         |             |                         |                                                                                                                                                                           |
|                     |                  | 2404.1             | 2.6                 | 5214.5  | (5/2 <sup>+</sup> )                     |         |             |                         |                                                                                                                                                                           |
|                     |                  | 2455.3             | 0.9                 | 5163.34 | 7/2 <sup>-</sup>                        |         |             |                         |                                                                                                                                                                           |

## Adopted Levels, Gammas (continued)

| $\gamma(^{35}\text{Cl})$ (continued) |                     |                    |                     |         |                |                    |                |                         |          |  |
|--------------------------------------|---------------------|--------------------|---------------------|---------|----------------|--------------------|----------------|-------------------------|----------|--|
| $E_i(\text{level})$                  | $J_i^\pi$           | $E_\gamma^\dagger$ | $I_\gamma^\ddagger$ | $E_f$   | $J_f^\pi$      | Mult. <sup>#</sup> | $\delta^\#$    | $\alpha^a$              | Comments |  |
| 7618.7                               | $5/2^+$             | 2779.5             | 0.9                 | 4839.10 | $(1/2^+, 3/2)$ |                    |                |                         |          |  |
|                                      |                     | 3440.6             | 3.9                 | 4177.9  | $3/2^-$        |                    |                |                         |          |  |
|                                      |                     | 3559.4             | 2.6                 | 4059.1  | $(3/2)^-$      |                    |                |                         |          |  |
|                                      |                     | 4455.5             | 3.9                 | 3162.87 | $7/2^-$        |                    |                |                         |          |  |
|                                      |                     | 5855.0             | 13                  | 1763.15 | $5/2^+$        | M1+E2              | +0.48 11       | $1.59 \times 10^{-3}$ 3 |          |  |
|                                      |                     | 7617.8             | 100                 | 0.0     | $3/2^+$        | M1+E2              | +2.2 3         |                         |          |  |
| 7656.6                               | $(1/2, 3/2, 5/2^+)$ | 1550.4             | 5.3 26              | 6106.2  | $(3/2, 5/2^+)$ |                    |                |                         |          |  |
|                                      |                     | 1974.3             | 1.8 9               | 5682.2  | $1/2^-, 3/2^-$ |                    |                |                         |          |  |
|                                      |                     | 2002               | 1.8 9               | 5655    | $(3/2)^+$      |                    |                |                         |          |  |
|                                      |                     | 2817.4             | 5.3 26              | 4839.10 | $(1/2^+, 3/2)$ |                    |                |                         |          |  |
|                                      |                     | 3478.5             | 1.8 9               | 4177.9  | $3/2^-$        |                    |                |                         |          |  |
|                                      |                     | 3597.3             | 56 6                | 4059.1  | $(3/2)^-$      |                    |                |                         |          |  |
|                                      |                     | 3688.8             | 3.5 18              | 3967.6  | $1/2^+$        |                    |                |                         |          |  |
|                                      |                     | 7655.7             | 100 11              | 0.0     | $3/2^+$        |                    |                |                         |          |  |
|                                      |                     | 7671.9             | $5/2^+$             | 2073.4  | 2.1            | 5598.4             | $3/2^+, 5/2^+$ |                         |          |  |
| 2508.5                               | 7                   |                    |                     | 5163.34 | $7/2^-$        |                    |                |                         |          |  |
| 2790.8                               | 2.1                 |                    |                     | 4880.96 | $7/2^{(+)}$    |                    |                |                         |          |  |
| 2901.9                               | 1.8                 |                    |                     | 4769.86 | $(5/2^-)$      |                    |                |                         |          |  |
| 3324.0                               | 11                  |                    |                     | 4347.76 | $9/2^-$        |                    |                |                         |          |  |
| 3498.3                               | 8.8                 |                    |                     | 4173.43 | $(5/2^+)$      |                    |                |                         |          |  |
| 4508.7                               | 3.2                 |                    |                     | 3162.87 | $7/2^-$        |                    |                |                         |          |  |
| 4668.9                               | 16                  |                    |                     | 3002.64 | $5/2^+$        | M1+E2              | -1.4 5         | 0.00134 4               |          |  |
| 5025.8                               | 23                  |                    |                     | 2645.67 | $7/2^+$        | D(+Q)              | +0.3 22        |                         |          |  |
| 5908.2                               | 100                 |                    |                     | 1763.15 | $5/2^+$        | M1+E2              | +0.86 14       | $1.64 \times 10^{-3}$ 3 |          |  |
| 7671.0                               | 1.4                 |                    |                     | 0.0     | $3/2^+$        |                    |                |                         |          |  |
| 7685.5                               | $3/2^-$             |                    |                     | 1927    |                | 5758               | $(1/2, 3/2^-)$ |                         |          |  |
|                                      |                     | 2087.0             | 2.9                 | 5598.4  | $3/2^+, 5/2^+$ |                    |                |                         |          |  |
|                                      |                     | 2282.1             | 2.9                 | 5403.3  | $3/2^-$        |                    |                |                         |          |  |
|                                      |                     | 2470.9             | 1                   | 5214.5  | $(5/2^+)$      |                    |                |                         |          |  |
|                                      |                     | 2676.3             | 0.86                | 5009.1  | $(1/2, 3/2)$   |                    |                |                         |          |  |
|                                      |                     | 3507.4             | 4.3                 | 4177.9  | $3/2^-$        |                    |                |                         |          |  |
|                                      |                     | 3511.9             | 5.7                 | 4173.43 | $(5/2^+)$      |                    |                |                         |          |  |
|                                      |                     | 3626.2             | 8.6                 | 4059.1  | $(3/2)^-$      |                    |                |                         |          |  |
|                                      |                     | 4682.5             | 14                  | 3002.64 | $5/2^+$        | D(+Q)              | -0.01 5        |                         |          |  |
|                                      |                     | 5921.8             | 1                   | 1763.15 | $5/2^+$        |                    |                |                         |          |  |
|                                      |                     | 6465.5             | 1.4                 | 1219.38 | $1/2^+$        |                    |                |                         |          |  |
|                                      |                     | 7684.6             | 100                 | 0.0     | $3/2^+$        | D+Q                | +6.0 7         |                         |          |  |
|                                      |                     |                    |                     |         |                |                    |                |                         |          |  |
|                                      |                     |                    |                     |         |                |                    |                |                         |          |  |

Mult.: D+Q with  $\delta=+6.0$  7 from  $\gamma(\theta)$  in (p, $\gamma$ );

Mult.: D+Q with  $\delta=+6.0$  7 from  $\gamma(\theta)$  in (p, $\gamma$ );  
 $\Delta\pi$ =yes from level scheme; but E1+M2 with such a large  $\delta$ (M2/E1) is unlikely for this strong primary transition based on RUL, which might suggest a different level in (p, $\gamma$ ).

| Adopted Levels, Gammas (continued)   |                               |                     |                     |         |                                     |        |             |                         |                                                                                                                                                                          |
|--------------------------------------|-------------------------------|---------------------|---------------------|---------|-------------------------------------|--------|-------------|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\gamma(^{35}\text{Cl})$ (continued) |                               |                     |                     |         |                                     |        |             |                         |                                                                                                                                                                          |
| $E_i(\text{level})$                  | $J_i^\pi$                     | $E_\gamma^\dagger$  | $I_\gamma^\ddagger$ | $E_f$   | $J_f^\pi$                           | Mult.# | $\delta^\#$ | $\alpha^a$              | Comments                                                                                                                                                                 |
| 7693.9                               | (1/2 <sup>+</sup> , 3/2, 5/2) | 2095.4              | 1.0 5               | 5598.4  | 3/2 <sup>+</sup> , 5/2 <sup>+</sup> |        |             |                         |                                                                                                                                                                          |
|                                      |                               | 3634.6              | 1.0 5               | 4059.1  | (3/2) <sup>-</sup>                  |        |             |                         |                                                                                                                                                                          |
|                                      |                               | 4690.9              | 2.1 11              | 3002.64 | 5/2 <sup>+</sup>                    |        |             |                         |                                                                                                                                                                          |
|                                      |                               | 7693.0              | 100 11              | 0.0     | 3/2 <sup>+</sup>                    |        |             |                         |                                                                                                                                                                          |
| 7706.4                               | 5/2 <sup>+</sup>              | 2051                | 4.8                 | 5655    | (3/2) <sup>+</sup>                  | M1+E2  |             | 0.00032 4               | $\delta$ : <-2 or $\geq 13$ from $\gamma(\theta)$ in (p, $\gamma$ ).                                                                                                     |
|                                      |                               | 2825.3              | 0.48                | 4880.96 | 7/2 <sup>(+)</sup>                  |        |             |                         |                                                                                                                                                                          |
|                                      |                               | 2867.2              | 1.6                 | 4839.10 | (1/2 <sup>+</sup> , 3/2)            |        |             |                         |                                                                                                                                                                          |
|                                      |                               | 3082.0              | 0.96                | 4624.3  | 3/2, 5/2                            |        |             |                         |                                                                                                                                                                          |
|                                      |                               | 3528.3              | 2.4                 | 4177.9  | 3/2 <sup>-</sup>                    |        |             |                         |                                                                                                                                                                          |
|                                      |                               | 3593.8              | 2.4                 | 4112.4  | 7/2 <sup>+</sup>                    |        |             |                         |                                                                                                                                                                          |
|                                      |                               | 3787.7              | 0.72                | 3918.47 | 3/2 <sup>+</sup>                    |        |             |                         |                                                                                                                                                                          |
|                                      |                               | 4703.4              | 1.2                 | 3002.64 | 5/2 <sup>+</sup>                    |        |             |                         |                                                                                                                                                                          |
|                                      |                               | 5060.3              | 4.8                 | 2645.67 | 7/2 <sup>+</sup>                    | M1+E2  |             | 0.00143 8               | $\delta$ : -1.01 10 or $\geq 13$ from (p, $\gamma$ ).                                                                                                                    |
|                                      |                               | 5942.7              | 0.48                | 1763.15 | 5/2 <sup>+</sup>                    |        |             |                         |                                                                                                                                                                          |
|                                      |                               | 6486.4              | 0.6                 | 1219.38 | 1/2 <sup>+</sup>                    |        |             |                         |                                                                                                                                                                          |
|                                      |                               | 7705.5              | 100                 | 0.0     | 3/2 <sup>+</sup>                    | D(+Q)  | +0.1 1      |                         |                                                                                                                                                                          |
|                                      |                               | 2863.7              | 47                  | 4880.96 | 7/2 <sup>(+)</sup>                  |        |             |                         |                                                                                                                                                                          |
|                                      |                               | 2974.8              | 4.7                 | 4769.86 | (5/2) <sup>-</sup>                  |        |             |                         |                                                                                                                                                                          |
| 7744.8                               | 7/2 <sup>-</sup>              | 3632.2              | 14                  | 4112.4  | 7/2 <sup>+</sup>                    |        |             |                         |                                                                                                                                                                          |
|                                      |                               | 3801.7              | 23                  | 3942.9  | 9/2 <sup>+</sup>                    | (E1)   |             | 1.60 $\times 10^{-3}$ 2 | Mult.: D(+Q) with $\delta=-0.16$ 17 from $\gamma(\theta)$ in (p, $\gamma$ ); $\Delta\pi$ =yes from level scheme; M2 component is less likely for the primary transition. |
|                                      |                               | 4581.6              | 37                  | 3162.87 | 7/2 <sup>-</sup>                    | M1+E2  | +0.18 8     | 1.23 $\times 10^{-3}$ 2 |                                                                                                                                                                          |
|                                      |                               | 5098.7              | 100                 | 2645.67 | 7/2 <sup>+</sup>                    | (E1)   |             | 2.05 $\times 10^{-3}$ 3 | Mult.: D(+Q) with $\delta=+0.1$ 30 from $\gamma(\theta)$ in (p, $\gamma$ ); $\Delta\pi$ =yes from level scheme; M2 component is less likely for the primary transition.  |
|                                      |                               | 5981.1              | 5.6                 | 1763.15 | 5/2 <sup>+</sup>                    |        |             |                         |                                                                                                                                                                          |
|                                      |                               | 6524.8 <sup>b</sup> | 1.4                 | 1219.38 | 1/2 <sup>+</sup>                    | [E3]   |             |                         | This transition is considered questionable by the evaluators since it would require Mult=E3 which is unlikely for this primary transition based on RUL.                  |
|                                      |                               |                     |                     |         |                                     |        |             |                         |                                                                                                                                                                          |
|                                      |                               |                     |                     |         |                                     |        |             |                         |                                                                                                                                                                          |
|                                      |                               |                     |                     |         |                                     |        |             |                         |                                                                                                                                                                          |
|                                      |                               |                     |                     |         |                                     |        |             |                         |                                                                                                                                                                          |
| 7777.1                               | (5/2 <sup>+</sup> )           | 2122                | 1.3                 | 5655    | (3/2) <sup>+</sup>                  |        |             |                         |                                                                                                                                                                          |
|                                      |                               | 2613.7              | 2.8                 | 5163.34 | 7/2 <sup>-</sup>                    |        |             |                         |                                                                                                                                                                          |
|                                      |                               | 2937.9              | 5                   | 4839.10 | (1/2 <sup>+</sup> , 3/2)            |        |             |                         |                                                                                                                                                                          |
|                                      |                               | 3007.1              | 7.5                 | 4769.86 | (5/2) <sup>-</sup>                  |        |             |                         |                                                                                                                                                                          |
|                                      |                               | 3599.0              | 18                  | 4177.9  | 3/2 <sup>-</sup>                    |        |             |                         |                                                                                                                                                                          |
|                                      |                               | 3717.8              | 2.8                 | 4059.1  | (3/2) <sup>-</sup>                  |        |             |                         |                                                                                                                                                                          |
|                                      |                               | 4613.9              | 5                   | 3162.87 | 7/2 <sup>-</sup>                    |        |             |                         |                                                                                                                                                                          |
|                                      |                               | 4774.1              | 10                  | 3002.64 | 5/2 <sup>+</sup>                    |        |             |                         |                                                                                                                                                                          |
|                                      |                               | 5082.7              | 23                  | 2694.00 | 3/2 <sup>+</sup>                    |        |             |                         |                                                                                                                                                                          |
|                                      |                               | 5131.0              | 3.3                 | 2645.67 | 7/2 <sup>+</sup>                    |        |             |                         |                                                                                                                                                                          |

## Adopted Levels, Gammas (continued)

| $\gamma(^{35}\text{Cl})$ (continued) |                     |                          |                       |         |                                         |                    |             |             |                                                                                                                                                                                                                             |
|--------------------------------------|---------------------|--------------------------|-----------------------|---------|-----------------------------------------|--------------------|-------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $E_i(\text{level})$                  | $J_i^\pi$           | $E_\gamma^\dagger$       | $I_\gamma^\ddagger$   | $E_f$   | $J_f^\pi$                               | Mult. <sup>#</sup> | $\delta^\#$ | $\alpha^a$  | Comments                                                                                                                                                                                                                    |
| 7777.1                               | (5/2 <sup>+</sup> ) | 6013.4                   | 100                   | 1763.15 | 5/2 <sup>+</sup>                        |                    |             |             |                                                                                                                                                                                                                             |
|                                      |                     | 6557.1                   | 28                    | 1219.38 | 1/2 <sup>+</sup>                        |                    |             |             |                                                                                                                                                                                                                             |
|                                      |                     | 7776.2                   | 45                    | 0.0     | 3/2 <sup>+</sup>                        |                    |             |             |                                                                                                                                                                                                                             |
| 7781.7                               | (5/2 <sup>-</sup> ) | 1675.5                   | 3.9                   | 6106.2  | (3/2, 5/2 <sup>+</sup> )                |                    |             |             |                                                                                                                                                                                                                             |
|                                      |                     | 2183.2                   |                       | 5598.4  | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>     |                    |             |             |                                                                                                                                                                                                                             |
|                                      |                     | 2378.3                   |                       | 5403.3  | 3/2 <sup>-</sup>                        |                    |             |             |                                                                                                                                                                                                                             |
|                                      |                     | 2567.1                   | 4.2                   | 5214.5  | (5/2 <sup>+</sup> )                     |                    |             |             |                                                                                                                                                                                                                             |
|                                      |                     | 3011.7                   | 19                    | 4769.86 | (5/2 <sup>-</sup> )                     |                    |             |             |                                                                                                                                                                                                                             |
|                                      |                     | 3433.8                   | 5.6                   | 4347.76 | 9/2 <sup>-</sup>                        |                    |             |             |                                                                                                                                                                                                                             |
|                                      |                     | 3608.1                   | 100                   | 4173.43 | (5/2 <sup>+</sup> )                     |                    |             |             |                                                                                                                                                                                                                             |
|                                      |                     | 3669.1                   | 17                    | 4112.4  | 7/2 <sup>+</sup>                        |                    |             |             |                                                                                                                                                                                                                             |
|                                      |                     | 3722.4                   | 3.3                   | 4059.1  | (3/2) <sup>-</sup>                      |                    |             |             |                                                                                                                                                                                                                             |
|                                      |                     | 3863.0                   | 31                    | 3918.47 | 3/2 <sup>+</sup>                        |                    |             |             |                                                                                                                                                                                                                             |
|                                      |                     | 4618.5                   | 14                    | 3162.87 | 7/2 <sup>-</sup>                        |                    |             |             |                                                                                                                                                                                                                             |
|                                      |                     | 5135.6                   | 22                    | 2645.67 | 7/2 <sup>+</sup>                        |                    |             |             |                                                                                                                                                                                                                             |
|                                      |                     | 6018.0                   | 50                    | 1763.15 | 5/2 <sup>+</sup>                        |                    |             |             |                                                                                                                                                                                                                             |
|                                      |                     | 6561.7                   | 3.3                   | 1219.38 | 1/2 <sup>+</sup>                        |                    |             |             |                                                                                                                                                                                                                             |
|                                      |                     | 7780.8                   | 4.7                   | 0.0     | 3/2 <sup>+</sup>                        |                    |             |             |                                                                                                                                                                                                                             |
| 7797.2                               | 1/2 <sup>-</sup>    | 2039                     |                       | 5758    | (1/2, 3/2 <sup>-</sup> )                |                    |             |             |                                                                                                                                                                                                                             |
|                                      |                     | 2142                     | 1.6                   | 5655    | (3/2) <sup>+</sup>                      |                    |             |             |                                                                                                                                                                                                                             |
|                                      |                     | 2393.8                   | 2.4                   | 5403.3  | 3/2 <sup>-</sup>                        |                    |             |             |                                                                                                                                                                                                                             |
|                                      |                     | 2788.0                   | 2.4                   | 5009.1  | (1/2, 3/2)                              |                    |             |             |                                                                                                                                                                                                                             |
|                                      |                     | 2942.7                   | 1.6                   | 4854.4  | (1/2, 3/2)                              |                    |             |             |                                                                                                                                                                                                                             |
|                                      |                     | 2958.0                   | 1.3                   | 4839.10 | (1/2 <sup>+</sup> , 3/2)                |                    |             |             |                                                                                                                                                                                                                             |
|                                      |                     | 3737.9                   | 0.36                  | 4059.1  | (3/2) <sup>-</sup>                      |                    |             |             |                                                                                                                                                                                                                             |
|                                      |                     | 6577.2                   | 11                    | 1219.38 | 1/2 <sup>+</sup>                        |                    |             |             |                                                                                                                                                                                                                             |
|                                      |                     | 7796.3                   | 100                   | 0.0     | 3/2 <sup>+</sup>                        |                    |             |             |                                                                                                                                                                                                                             |
| 7837.2                               | 3/2 <sup>-</sup>    | 1656 <sup>&amp;</sup>    | 1.4 <sup>&amp;</sup>  | 6181    | (1/2 <sup>+</sup> to 7/2 <sup>+</sup> ) |                    |             |             |                                                                                                                                                                                                                             |
|                                      |                     | 2238.7 <sup>&amp;</sup>  | 1.4 <sup>&amp;</sup>  | 5598.4  | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>     |                    |             |             |                                                                                                                                                                                                                             |
|                                      |                     | 2622.6 <sup>&amp;</sup>  | 3.8 <sup>&amp;</sup>  | 5214.5  | (5/2 <sup>+</sup> )                     |                    |             |             |                                                                                                                                                                                                                             |
|                                      |                     | 3659.1 <sup>&amp;</sup>  | 76 <sup>&amp;</sup>   | 4177.9  | 3/2 <sup>-</sup>                        | (M1+E2)            | -0.05 3     | 0.000917 13 | Mult.: D+Q from (p, $\gamma$ ); $\Delta\pi$ =no from level scheme.<br>B(M1)(W.u.)=1.03 $\times 10^3$ 19 exceeds RUL=10.<br>B(E2)(W.u.)=7 $\times 10^2$ 11-6 exceeds RUL=100.                                                |
|                                      |                     | 3663.6 <sup>&amp;</sup>  | 8.1 <sup>&amp;</sup>  | 4173.43 | (5/2 <sup>+</sup> )                     |                    |             |             |                                                                                                                                                                                                                             |
|                                      |                     | 3777.9 <sup>&amp;</sup>  | 5.4 <sup>&amp;</sup>  | 4059.1  | (3/2) <sup>-</sup>                      |                    |             |             |                                                                                                                                                                                                                             |
|                                      |                     | 3894.1 <sup>&amp;b</sup> | 1.4 <sup>&amp;</sup>  | 3942.9  | 9/2 <sup>+</sup>                        |                    |             |             | This $\gamma$ is a primary $\gamma$ from the unresolved 7837+7839<br>doublet goes to 9/2 <sup>+</sup> , which is unlikely from a 3/2 <sup>-</sup> ,<br>7837 level. It is likely from the (5/2 <sup>+</sup> ), 7839.0 level. |
|                                      |                     | 3918.5 <sup>&amp;</sup>  | 0.54 <sup>&amp;</sup> | 3918.47 | 3/2 <sup>+</sup>                        |                    |             |             |                                                                                                                                                                                                                             |

Adopted Levels, Gammas (continued) $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$             | $E_\gamma^\dagger$ | $I_\gamma^\ddagger$ | $E_f$   | $J_f^\pi$                   | Mult. <sup>#</sup> | $\delta^\#$ | $\alpha^a$  | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------|-----------------------|--------------------|---------------------|---------|-----------------------------|--------------------|-------------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7837.2              | $3/2^-$               | 4834.2 &           | 11 &                | 3002.64 | $5/2^+$                     |                    |             |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                     |                       | 6073.5 &           | 5.1 &               | 1763.15 | $5/2^+$                     |                    |             |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                     |                       | 6617.2 &           | 100 &               | 1219.38 | $1/2^+$                     | (E1)               |             |             | Mult.: D(+Q) with $\delta=+0.06$ 3 from (p, $\gamma$ );<br>$\Delta\pi$ =yes from level scheme; M2 component is less likely for this strong primary transition.<br>B(E1)(W.u.)=6.5 +10-8 exceeds RUL=0.1.                                                                                                                                                                                                                                                                                                                                       |
|                     |                       | 7836.3 &           | 57 &                | 0.0     | $3/2^+$                     | (E1)               |             |             | Mult.: D(+Q) with $\delta=+0.02$ 3 from (p, $\gamma$ );<br>$\Delta\pi$ =yes from level scheme; M2 component is less likely for this strong primary transition.<br>B(E1)(W.u.)=2.24 +47-44 exceeds RUL=0.1.                                                                                                                                                                                                                                                                                                                                     |
| 7839.0              | $(5/2^+)$             | 1658 &             | 1.4 &               | 6181    | $(1/2^+ \text{ to } 7/2^+)$ |                    |             |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                     |                       | 2240.5 &           | 1.4 &               | 5598.4  | $3/2^+, 5/2^+$              |                    |             |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                     |                       | 2624.4 &           | 3.8 &               | 5214.5  | $(5/2^+)$                   |                    |             |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                     |                       | 3660.9 &           | 76 &                | 4177.9  | $3/2^-$                     |                    |             |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                     |                       | 3665.4 &           | 8.1 &               | 4173.43 | $(5/2^+)$                   |                    |             |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                     |                       | 3779.7 &           | 5.4 &               | 4059.1  | $(3/2)^-$                   |                    |             |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                     |                       | 3895.9 &           | 1.4 &               | 3942.9  | $9/2^+$                     |                    |             |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                     |                       | 3920.3 &           | 0.54 &              | 3918.47 | $3/2^+$                     |                    |             |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                     |                       | 4836.0 &           | 11 &                | 3002.64 | $5/2^+$                     |                    |             |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                     |                       | 6075.3 &           | 5.1 &               | 1763.15 | $5/2^+$                     |                    |             |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                     |                       | 6619.0 &           | 100 &               | 1219.38 | $1/2^+$                     |                    |             |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                     |                       | 7838.1 &           | 57 &                | 0.0     | $3/2^+$                     |                    |             |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                     |                       | 2654.1             | 1.6                 | 5214.5  | $(5/2^+)$                   |                    |             |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                     |                       | 2859.5             | 4.9                 | 5009.1  | $(1/2, 3/2)$                |                    |             |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 7868.7              | $(1/2^+, 3/2, 5/2^+)$ | 3695.1             | 8.2                 | 4173.43 | $(5/2^+)$                   |                    |             |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                     |                       | 6105.0             | 28                  | 1763.15 | $5/2^+$                     |                    |             |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                     |                       | 6648.6             | 21                  | 1219.38 | $1/2^+$                     |                    |             |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                     |                       | 7867.8             | 100                 | 0.0     | $3/2^+$                     |                    |             |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                     |                       | 1212.8             |                     | 6660.2  | $(11/2^-)$                  |                    |             |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                     |                       | 1786.2 5           | 9.6 4               | 6087.31 | $13/2^-$                    | D(+Q)              | -0.6 6      |             | $E_\gamma$ : from ( $^{28}\text{Si}, \alpha p \gamma$ ) only.<br>$I_\gamma$ : weighted average of 8.2 18 from ( $^{28}\text{Si}, \alpha p \gamma$ ) and 9.7 4 from ( $^{16}\text{O}, \alpha p \gamma$ ).<br>Mult., $\delta$ : from $\gamma\gamma(\text{DCO})$ in ( $^{28}\text{Si}, \alpha p \gamma$ ).<br>$E_\gamma$ : weighted average of 1946.2 5 from ( $^{16}\text{O}, \alpha p \gamma$ ) and 1946.40 30 from ( $^{14}\text{N}, \alpha p n \gamma$ ).<br>$I_\gamma$ : weighted average of 46 5 from ( $^{28}\text{Si}, \alpha p \gamma$ ) |
| 7873.03             | $13/2^+$              | 1946.35 30         | 48.0 26             | 5926.65 | $(11/2)^-$                  | E1+M2              | +0.2 1      | 0.000594 22 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

Adopted Levels, Gammas (continued)

| $\gamma(^{35}\text{Cl})$ (continued) |                                       |                    |                     |         |                             |                    |             |            | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------------------|---------------------------------------|--------------------|---------------------|---------|-----------------------------|--------------------|-------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $E_i(\text{level})$                  | $J_i^\pi$                             | $E_\gamma^\dagger$ | $I_\gamma^\ddagger$ | $E_f$   | $J_f^\pi$                   | Mult. <sup>#</sup> | $\delta^\#$ | $\alpha^a$ |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 7873.03                              | 13/2 <sup>+</sup>                     | 2465.94 30         | 100.0 20            | 5407.07 | 11/2 <sup>-</sup>           | E1+M2              | -0.25 10    | 0.00091 4  | and 48.5 26 from ( $^{16}\text{O},\alpha p\gamma$ ).<br>Mult., $\delta$ : from $\gamma\gamma(\text{DCO})$ and $\gamma\gamma(\text{lin pol})$ in ( $^{28}\text{Si},\alpha p\gamma$ ), $\Delta J=1$ .<br>$E_\gamma$ : weighted average of 2466.2 5 from ( $^{16}\text{O},\alpha p\gamma$ ) and 2465.85 30 from ( $^{14}\text{N},\alpha p n\gamma$ ).<br>$I_\gamma$ : weighted average of 100 9 from ( $^{28}\text{Si},\alpha p\gamma$ ) and 100.0 20 from ( $^{16}\text{O},\alpha p\gamma$ ).<br>Mult., $\delta$ : from $\gamma\gamma(\text{DCO})$ and $\gamma\gamma(\text{lin pol})$ in ( $^{28}\text{Si},\alpha p\gamma$ ), $\Delta J=1$ . |
| 7880.8                               | 3/2 <sup>+</sup> ,5/2 <sup>+</sup>    | 2226               | 40                  | 5655    | (3/2) <sup>+</sup>          |                    |             |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                      |                                       | 3041.6             | 5.7                 | 4839.10 | (1/2 <sup>+</sup> ,3/2)     |                    |             |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                      |                                       | 3256.3             | 6.7                 | 4624.3  | 3/2,5/2                     |                    |             |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                      |                                       | 3702.7             | 10                  | 4177.9  | 3/2 <sup>-</sup>            |                    |             |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                      |                                       | 3768.2             | 4.3                 | 4112.4  | 7/2 <sup>+</sup>            |                    |             |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                      |                                       | 3821.5             | 6.7                 | 4059.1  | (3/2) <sup>-</sup>          |                    |             |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                      |                                       | 3962.1             | 6.7                 | 3918.47 | 3/2 <sup>+</sup>            |                    |             |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                      |                                       | 4877.8             | 37                  | 3002.64 | 5/2 <sup>+</sup>            |                    |             |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                      |                                       | 5186.4             | 83                  | 2694.00 | 3/2 <sup>+</sup>            |                    |             |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                      |                                       | 6117.1             | 100                 | 1763.15 | 5/2 <sup>+</sup>            |                    |             |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                      |                                       | 7879.9             | 33                  | 0.0     | 3/2 <sup>+</sup>            |                    |             |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 7899.2                               | (5/2)                                 | 2253               | 2.2                 | 5646    | (5/2,7/2,9/2 <sup>+</sup> ) |                    |             |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                      |                                       | 2735.8             | 1.3                 | 5163.34 | 7/2 <sup>-</sup>            |                    |             |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                      |                                       | 3018.1             | 1.2                 | 4880.96 | 7/2 <sup>(+)</sup>          |                    |             |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                      |                                       | 3274.7             | 0.74                | 4624.3  | 3/2,5/2                     |                    |             |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                      |                                       | 3721.1             | 12                  | 4177.9  | 3/2 <sup>-</sup>            |                    |             |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                      |                                       | 3725.6             | 12                  | 4173.43 | (5/2 <sup>+</sup> )         |                    |             |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                      |                                       | 3839.9             | 4.4                 | 4059.1  | (3/2) <sup>-</sup>          |                    |             |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                      |                                       | 4736.0             | 5.9                 | 3162.87 | 7/2 <sup>-</sup>            |                    |             |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                      |                                       | 5204.8             | 1.9                 | 2694.00 | 3/2 <sup>+</sup>            |                    |             |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                      |                                       | 6135.5             | 100                 | 1763.15 | 5/2 <sup>+</sup>            |                    |             |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                      |                                       | 7898.2             | 5.9                 | 0.0     | 3/2 <sup>+</sup>            |                    |             |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 7923.3                               | (3/2 <sup>+</sup> ,5/2 <sup>+</sup> ) | 3084.1             | 3.4                 | 4839.10 | (1/2 <sup>+</sup> ,3/2)     |                    |             |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                      |                                       | 3153.3             | 4.9                 | 4769.86 | (5/2 <sup>-</sup> )         |                    |             |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                      |                                       | 3298.8             | 4.6                 | 4624.3  | 3/2,5/2                     |                    |             |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                      |                                       | 3745.2             | 7.3                 | 4177.9  | 3/2 <sup>-</sup>            |                    |             |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                      |                                       | 3749.7             | 12                  | 4173.43 | (5/2 <sup>+</sup> )         |                    |             |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                      |                                       | 3864.0             | 4.2                 | 4059.1  | (3/2) <sup>-</sup>          |                    |             |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                      |                                       | 3955.5             | 3.7                 | 3967.6  | 1/2 <sup>+</sup>            |                    |             |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                      |                                       | 4004.6             | 2.9                 | 3918.47 | 3/2 <sup>+</sup>            |                    |             |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                      |                                       | 4920.3             | 2.2                 | 3002.64 | 5/2 <sup>+</sup>            |                    |             |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                      |                                       | 5228.9             | 27                  | 2694.00 | 3/2 <sup>+</sup>            |                    |             |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

Adopted Levels, Gammas (continued)

| $\gamma(^{35}\text{Cl})$ (continued) |                                        |                    |                     |         |                                     |                                                         |
|--------------------------------------|----------------------------------------|--------------------|---------------------|---------|-------------------------------------|---------------------------------------------------------|
| $E_i(\text{level})$                  | $J_i^\pi$                              | $E_\gamma^\dagger$ | $I_\gamma^\ddagger$ | $E_f$   | $J_f^\pi$                           | Mult. <sup>#</sup>                                      |
| 7923.3                               | (3/2 <sup>+</sup> , 5/2 <sup>+</sup> ) | 5277.2             | 2.2                 | 2645.67 | 7/2 <sup>+</sup>                    |                                                         |
|                                      |                                        | 6159.6             | 3.7                 | 1763.15 | 5/2 <sup>+</sup>                    |                                                         |
|                                      |                                        | 6703.2             | 66                  | 1219.38 | 1/2 <sup>+</sup>                    |                                                         |
|                                      |                                        | 7922.3             | 100                 | 0.0     | 3/2 <sup>+</sup>                    |                                                         |
| 7970.3                               | (5/2 <sup>-</sup> )                    | 2324               | 15                  | 5646    | (5/2, 7/2, 9/2 <sup>+</sup> )       |                                                         |
|                                      |                                        | 2384.2             | 15                  | 5586.0  | 5/2 <sup>+</sup>                    |                                                         |
|                                      |                                        | 2755.7             | 15                  | 5214.5  | (5/2 <sup>+</sup> )                 |                                                         |
|                                      |                                        | 3089.2             | 50                  | 4880.96 | 7/2 <sup>(+)</sup>                  |                                                         |
|                                      |                                        | 3200.3             | 25                  | 4769.86 | (5/2 <sup>-</sup> )                 |                                                         |
|                                      |                                        | 3622.3             | 40                  | 4347.76 | 9/2 <sup>-</sup>                    |                                                         |
|                                      |                                        | 3792.2             | 30                  | 4177.9  | 3/2 <sup>-</sup>                    |                                                         |
|                                      |                                        | 3857.7             | 30                  | 4112.4  | 7/2 <sup>+</sup>                    |                                                         |
|                                      |                                        | 4807.1             | 50                  | 3162.87 | 7/2 <sup>-</sup>                    |                                                         |
|                                      |                                        | 4967.3             | 25                  | 3002.64 | 5/2 <sup>+</sup>                    |                                                         |
|                                      |                                        | 5324.2             | 100                 | 2645.67 | 7/2 <sup>+</sup>                    |                                                         |
|                                      |                                        | 6206.6             | 100                 | 1763.15 | 5/2 <sup>+</sup>                    |                                                         |
|                                      |                                        | 6750.2             | 5                   | 1219.38 | 1/2 <sup>+</sup>                    |                                                         |
|                                      |                                        | 7987.8             | 3/2 <sup>+</sup>    | 1881.6  | 0.83                                | 6106.2 (3/2, 5/2 <sup>+</sup> )                         |
|                                      |                                        | 2333               | 2.8                 | 5655    | (3/2 <sup>+</sup> )                 |                                                         |
|                                      |                                        | 3809.7             | <1.3                | 4177.9  | 3/2 <sup>-</sup>                    |                                                         |
|                                      |                                        | 3814.2             | <1.3                | 4173.43 | (5/2 <sup>+</sup> )                 |                                                         |
|                                      |                                        | 3928.5             | 1.7                 | 4059.1  | (3/2 <sup>-</sup> )                 |                                                         |
|                                      |                                        | 6767.7             | 100                 | 1219.38 | 1/2 <sup>+</sup>                    | M1+E2 $\delta$ : +0.21 4 or -3.1 4 from (p, $\gamma$ ). |
|                                      |                                        | 7986.8             | 60                  | 0.0     | 3/2 <sup>+</sup>                    | M1+E2 $\delta$ : -0.36 2 or -11 3 from (p, $\gamma$ ).  |
| 7995.6                               | (5/2)                                  | 2188.9             | 0.77                | 5806.6  | (1/2, 3/2, 5/2)                     |                                                         |
|                                      |                                        | 2397.1             | 1.4                 | 5598.4  | 3/2 <sup>+</sup> , 5/2 <sup>+</sup> |                                                         |
|                                      |                                        | 2592.2             | 3.9                 | 5403.3  | 3/2 <sup>-</sup>                    |                                                         |
|                                      |                                        | 2832.1             | 1.9                 | 5163.34 | 7/2 <sup>-</sup>                    |                                                         |
|                                      |                                        | 3817.5             | 5.1                 | 4177.9  | 3/2 <sup>-</sup>                    |                                                         |
|                                      |                                        | 3936.3             | 1.4                 | 4059.1  | (3/2 <sup>-</sup> )                 |                                                         |
|                                      |                                        | 4832.4             | 9                   | 3162.87 | 7/2 <sup>-</sup>                    |                                                         |
|                                      |                                        | 4992.6             | 2.6                 | 3002.64 | 5/2 <sup>+</sup>                    |                                                         |
|                                      |                                        | 5301.2             | 0.26                | 2694.00 | 3/2 <sup>+</sup>                    |                                                         |
|                                      |                                        | 5349.5             | 0.77                | 2645.67 | 7/2 <sup>+</sup>                    |                                                         |
|                                      |                                        | 6231.9             | 1.2                 | 1763.15 | 5/2 <sup>+</sup>                    |                                                         |
|                                      |                                        | 7994.6             | 100                 | 0.0     | 3/2 <sup>+</sup>                    |                                                         |
|                                      |                                        | 8001.0             | (7/2 <sup>+</sup> ) | 2355    | 2.6                                 | 5646 (5/2, 7/2, 9/2 <sup>+</sup> )                      |
|                                      |                                        | 2786.4             | 1.1                 | 5214.5  | (5/2 <sup>+</sup> )                 |                                                         |
|                                      |                                        | 3119.9             | 1                   | 4880.96 | 7/2 <sup>(+)</sup>                  |                                                         |
|                                      |                                        | 3231.0             | 11                  | 4769.86 | (5/2 <sup>-</sup> )                 |                                                         |
|                                      |                                        | 3653.0             | 2.7                 | 4347.76 | 9/2 <sup>-</sup>                    |                                                         |
|                                      |                                        | 3827.4             | 14                  | 4173.43 | (5/2 <sup>+</sup> )                 |                                                         |

$I_\gamma$ : total  $I_\gamma=1.3$  for  $\gamma$  decays to 4173 and 4178 doublet final levels.  
 $I_\gamma$ : total  $I_\gamma=1.3$  for  $\gamma$  decays to 4173 and 4178 doublet final levels.

## Adopted Levels, Gammas (continued)

| $\gamma(^{35}\text{Cl})$ (continued) |                                     |                    |                                     |         |                          |                    |                  |                         |                                                                                                                                                                                                                                         |  |  |
|--------------------------------------|-------------------------------------|--------------------|-------------------------------------|---------|--------------------------|--------------------|------------------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
| $E_i(\text{level})$                  | $J_i^\pi$                           | $E_\gamma^\dagger$ | $I_\gamma^\ddagger$                 | $E_f$   | $J_f^\pi$                | Mult. <sup>#</sup> | $\delta^\#$      | $\alpha^a$              | Comments                                                                                                                                                                                                                                |  |  |
| 8001.0                               | (7/2 <sup>+</sup> )                 | 4057.9             | 4.3                                 | 3942.9  | 9/2 <sup>+</sup>         |                    |                  |                         |                                                                                                                                                                                                                                         |  |  |
|                                      |                                     | 4837.8             | 1.1                                 | 3162.87 | 7/2 <sup>-</sup>         |                    |                  |                         |                                                                                                                                                                                                                                         |  |  |
|                                      |                                     | 5354.9             | 2.3                                 | 2645.67 | 7/2 <sup>+</sup>         |                    |                  |                         |                                                                                                                                                                                                                                         |  |  |
|                                      |                                     | 6237.3             | 100                                 | 1763.15 | 5/2 <sup>+</sup>         |                    |                  |                         |                                                                                                                                                                                                                                         |  |  |
|                                      |                                     | 8000.0             | 2                                   | 0.0     | 3/2 <sup>+</sup>         |                    |                  |                         |                                                                                                                                                                                                                                         |  |  |
| 8005.0                               | 5/2 <sup>+</sup>                    | 2350               | 3.3                                 | 5655    | (3/2) <sup>+</sup>       |                    |                  |                         |                                                                                                                                                                                                                                         |  |  |
|                                      |                                     | 3123.9             | 1.1                                 | 4880.96 | 7/2 <sup>(+)</sup>       |                    |                  |                         |                                                                                                                                                                                                                                         |  |  |
|                                      |                                     | 3165.8             | 5.5                                 | 4839.10 | (1/2 <sup>+</sup> , 3/2) |                    |                  |                         |                                                                                                                                                                                                                                         |  |  |
|                                      |                                     | 3826.9             | <2.4                                | 4177.9  | 3/2 <sup>-</sup>         |                    |                  |                         | $I_\gamma$ : total $I_\gamma=2.4$ for $\gamma$ decays to 4173 and 4178 doublet final levels.                                                                                                                                            |  |  |
|                                      |                                     | 3831.3             | <2.4                                | 4173.43 | (5/2 <sup>+</sup> )      |                    |                  |                         | $I_\gamma$ : total $I_\gamma=2.4$ for $\gamma$ decays to 4173 and 4178 doublet final levels.                                                                                                                                            |  |  |
|                                      |                                     | 4841.8             | 27                                  | 3162.87 | 7/2 <sup>-</sup>         |                    |                  |                         |                                                                                                                                                                                                                                         |  |  |
|                                      |                                     | 5002.0             | 3.6                                 | 3002.64 | 5/2 <sup>+</sup>         |                    |                  |                         |                                                                                                                                                                                                                                         |  |  |
|                                      |                                     | 5310.6             | 3.1                                 | 2694.00 | 3/2 <sup>+</sup>         |                    |                  |                         |                                                                                                                                                                                                                                         |  |  |
|                                      |                                     | 5358.9             | 7.3                                 | 2645.67 | 7/2 <sup>+</sup>         |                    |                  |                         |                                                                                                                                                                                                                                         |  |  |
|                                      |                                     | 6241.3             | 27                                  | 1763.15 | 5/2 <sup>+</sup>         |                    |                  |                         |                                                                                                                                                                                                                                         |  |  |
|                                      |                                     | 6784.9             | 1.1                                 | 1219.38 | 1/2 <sup>+</sup>         |                    |                  |                         |                                                                                                                                                                                                                                         |  |  |
|                                      |                                     | 8004.0             | 100                                 | 0.0     | 3/2 <sup>+</sup>         |                    |                  |                         |                                                                                                                                                                                                                                         |  |  |
|                                      |                                     | 8035.7             | 1/2 <sup>-</sup> , 3/2 <sup>-</sup> | 2229.0  | 10.7 11                  | 5806.6             | (1/2, 3/2, 5/2)  |                         |                                                                                                                                                                                                                                         |  |  |
|                                      |                                     |                    |                                     | 3857.6  | 26.8 27                  | 4177.9             | 3/2 <sup>-</sup> |                         |                                                                                                                                                                                                                                         |  |  |
| 3976.4                               | 19.6 20                             |                    |                                     | 4059.1  | (3/2) <sup>-</sup>       |                    |                  |                         |                                                                                                                                                                                                                                         |  |  |
| 5341.3                               | 9 5                                 |                    |                                     | 2694.00 | 3/2 <sup>+</sup>         |                    |                  |                         |                                                                                                                                                                                                                                         |  |  |
| 6272.0                               | 9 5                                 |                    |                                     | 1763.15 | 5/2 <sup>+</sup>         |                    |                  |                         |                                                                                                                                                                                                                                         |  |  |
| 6815.6                               | 100 11                              |                    |                                     | 1219.38 | 1/2 <sup>+</sup>         |                    |                  |                         |                                                                                                                                                                                                                                         |  |  |
| 8034.7                               | 3.6 18                              |                    |                                     | 0.0     | 3/2 <sup>+</sup>         |                    |                  |                         |                                                                                                                                                                                                                                         |  |  |
| 3183.8                               | 1.1                                 |                    |                                     | 4854.4  | (1/2, 3/2)               |                    |                  |                         |                                                                                                                                                                                                                                         |  |  |
| 8038.4                               | 1/2 <sup>+</sup> , 3/2 <sup>+</sup> | 3199.1             | 2.8                                 | 4839.10 | (1/2 <sup>+</sup> , 3/2) |                    |                  |                         |                                                                                                                                                                                                                                         |  |  |
|                                      |                                     | 3413.9             | 5.3                                 | 4624.3  | 3/2, 5/2                 |                    |                  |                         |                                                                                                                                                                                                                                         |  |  |
|                                      |                                     | 3860.3             | 2.8                                 | 4177.9  | 3/2 <sup>-</sup>         |                    |                  |                         |                                                                                                                                                                                                                                         |  |  |
|                                      |                                     | 3979.1             | 5.3                                 | 4059.1  | (3/2) <sup>-</sup>       |                    |                  |                         |                                                                                                                                                                                                                                         |  |  |
|                                      |                                     | 4070.6             | 1.4                                 | 3967.6  | 1/2 <sup>+</sup>         |                    |                  |                         |                                                                                                                                                                                                                                         |  |  |
|                                      |                                     | 4119.7             | 2.5                                 | 3918.47 | 3/2 <sup>+</sup>         |                    |                  |                         |                                                                                                                                                                                                                                         |  |  |
|                                      |                                     | 5035.4             | 7                                   | 3002.64 | 5/2 <sup>+</sup>         |                    |                  |                         |                                                                                                                                                                                                                                         |  |  |
|                                      |                                     | 5344.0             | 100                                 | 2694.00 | 3/2 <sup>+</sup>         | M1+E2              | -0.20 2          | 1.44×10 <sup>-3</sup> 2 | Mult., $\delta$ : D+Q with $\delta=-0.20$ 2 or +19 8 from (p, $\gamma$ ); E1+M2 and the larger $\delta(E2/M1)$ is ruled out by RUL.<br>B(M1)(W.u.)=52.1 +43-37 exceeds RUL=10.<br>B(E2)(W.u.)=2.8×10 <sup>2</sup> +6-5 exceeds RUL=100. |  |  |
|                                      |                                     | 6274.7             | 26                                  | 1763.15 | 5/2 <sup>+</sup>         |                    |                  |                         |                                                                                                                                                                                                                                         |  |  |
|                                      |                                     | 6818.3             | 7                                   | 1219.38 | 1/2 <sup>+</sup>         |                    |                  |                         |                                                                                                                                                                                                                                         |  |  |
|                                      |                                     | 8037.4             | 14                                  | 0.0     | 3/2 <sup>+</sup>         | M1+E2              | -0.13 4          |                         | B(M1)(W.u.)=2.20 44; B(E2)(W.u.)=2.2 +16-12                                                                                                                                                                                             |  |  |
|                                      |                                     |                    |                                     |         |                          |                    |                  |                         |                                                                                                                                                                                                                                         |  |  |
|                                      |                                     |                    |                                     |         |                          |                    |                  |                         |                                                                                                                                                                                                                                         |  |  |
|                                      |                                     |                    |                                     |         |                          |                    |                  |                         |                                                                                                                                                                                                                                         |  |  |
|                                      |                                     |                    |                                     |         |                          |                    |                  |                         |                                                                                                                                                                                                                                         |  |  |



Adopted Levels, Gammas (continued)

| $\gamma(^{35}\text{Cl})$ (continued)                                                                                                             |                                             |                    |                     |         |                                     |         |                                          |
|--------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|--------------------|---------------------|---------|-------------------------------------|---------|------------------------------------------|
| $E_i(\text{level})$                                                                                                                              | $J_i^\pi$                                   | $E_\gamma^\dagger$ | $I_\gamma^\ddagger$ | $E_f$   | $J_f^\pi$                           | Mult. # | $\alpha^a$                               |
| Comments                                                                                                                                         |                                             |                    |                     |         |                                     |         |                                          |
| Mult., $\delta$ : D+Q with $\delta = -0.13$ 4 or $+8.1$ 26 from (p, $\gamma$ ); E1+M2 and the larger $\delta(\text{E2/M1})$ is ruled out by RUL. |                                             |                    |                     |         |                                     |         |                                          |
| 8075.9                                                                                                                                           | (5/2)                                       | 2430               | 20                  | 5646    | (5/2,7/2,9/2 <sup>+</sup> )         |         |                                          |
|                                                                                                                                                  |                                             | 2489.8             | 20                  | 5586.0  | 5/2 <sup>+</sup>                    |         |                                          |
|                                                                                                                                                  |                                             | 2861.3             | 14                  | 5214.5  | (5/2 <sup>+</sup> )                 |         |                                          |
|                                                                                                                                                  |                                             | 2912.4             |                     | 5163.34 | 7/2 <sup>-</sup>                    |         |                                          |
|                                                                                                                                                  |                                             | 3194.8             |                     | 4880.96 | 7/2 <sup>(+)</sup>                  |         |                                          |
|                                                                                                                                                  |                                             | 3305.9             | 7.1                 | 4769.86 | (5/2 <sup>-</sup> )                 |         |                                          |
|                                                                                                                                                  |                                             | 3897.8             | 14                  | 4177.9  | 3/2 <sup>-</sup>                    |         |                                          |
|                                                                                                                                                  |                                             | 3963.3             | 3.6                 | 4112.4  | 7/2 <sup>+</sup>                    |         |                                          |
|                                                                                                                                                  |                                             | 5072.9             | 3.6                 | 3002.64 | 5/2 <sup>+</sup>                    |         |                                          |
|                                                                                                                                                  |                                             | 6312.1             | 100                 | 1763.15 | 5/2 <sup>+</sup>                    |         |                                          |
|                                                                                                                                                  |                                             | 8074.9             |                     | 0.0     | 3/2 <sup>+</sup>                    |         |                                          |
| 8096.1                                                                                                                                           | (5/2)                                       | 2289.4             | 5.1                 | 5806.6  | (1/2,3/2,5/2)                       |         |                                          |
|                                                                                                                                                  |                                             | 2450               | 13                  | 5646    | (5/2,7/2,9/2 <sup>+</sup> )         |         |                                          |
|                                                                                                                                                  |                                             | 2497.6             | 13                  | 5598.4  | 3/2 <sup>+</sup> , 5/2 <sup>+</sup> |         |                                          |
|                                                                                                                                                  |                                             | 2881.5             | 3.3                 | 5214.5  | (5/2 <sup>+</sup> )                 |         |                                          |
|                                                                                                                                                  |                                             | 2932.6             | 5.1                 | 5163.34 | 7/2 <sup>-</sup>                    |         |                                          |
|                                                                                                                                                  |                                             | 3256.8             | 7.7                 | 4839.10 | (1/2 <sup>+</sup> , 3/2)            |         |                                          |
|                                                                                                                                                  |                                             | 3326.1             | 5.1                 | 4769.86 | (5/2 <sup>-</sup> )                 |         |                                          |
|                                                                                                                                                  |                                             | 3918.0             | 4.9                 | 4177.9  | 3/2 <sup>-</sup>                    |         |                                          |
|                                                                                                                                                  |                                             | 4036.8             | 5.1                 | 4059.1  | (3/2) <sup>-</sup>                  |         |                                          |
|                                                                                                                                                  |                                             | 4177.4             | 4.6                 | 3918.47 | 3/2 <sup>+</sup>                    |         |                                          |
|                                                                                                                                                  |                                             | 4932.9             | 13                  | 3162.87 | 7/2 <sup>-</sup>                    |         |                                          |
|                                                                                                                                                  |                                             | 5401.7             | 13                  | 2694.00 | 3/2 <sup>+</sup>                    |         |                                          |
|                                                                                                                                                  |                                             | 5450.0             | 44                  | 2645.67 | 7/2 <sup>+</sup>                    |         |                                          |
|                                                                                                                                                  |                                             | 6332.3             | 21                  | 1763.15 | 5/2 <sup>+</sup>                    |         |                                          |
| 8106.4                                                                                                                                           | 3/2 <sup>+</sup>                            | 8095.1             | 100                 | 0.0     | 3/2 <sup>+</sup>                    |         |                                          |
|                                                                                                                                                  |                                             | 2507.9             | 1.4                 | 5598.4  | 3/2 <sup>+</sup> , 5/2 <sup>+</sup> |         |                                          |
|                                                                                                                                                  |                                             | 3251.8             | 1.6                 | 4854.4  | (1/2,3/2)                           |         |                                          |
|                                                                                                                                                  |                                             | 3267.1             | 1.4                 | 4839.10 | (1/2 <sup>+</sup> , 3/2)            |         |                                          |
|                                                                                                                                                  |                                             | 3928.3             | 4.3                 | 4177.9  | 3/2 <sup>-</sup>                    |         |                                          |
|                                                                                                                                                  |                                             | 4047.1             | 3.5                 | 4059.1  | (3/2) <sup>-</sup>                  |         |                                          |
|                                                                                                                                                  |                                             | 4138.5             | 8.1                 | 3967.6  | 1/2 <sup>+</sup>                    |         |                                          |
|                                                                                                                                                  |                                             | 5103.4             | 24                  | 3002.64 | 5/2 <sup>+</sup>                    |         |                                          |
|                                                                                                                                                  |                                             | 5412.0             | 24                  | 2694.00 | 3/2 <sup>+</sup>                    | M1+E2   | 0.00153 8 $\delta$ : +4.9 10 or -0.05 4. |
|                                                                                                                                                  |                                             | 6342.6             | 8.1                 | 1763.15 | 5/2 <sup>+</sup>                    |         |                                          |
|                                                                                                                                                  |                                             | 6886.3             | 95                  | 1219.38 | 1/2 <sup>+</sup>                    |         |                                          |
|                                                                                                                                                  |                                             | 8105.4             | 100                 | 0.0     | 3/2 <sup>+</sup>                    | M1+E2   | $\delta$ : +1.2 1 or -0.46 4.            |
| 8113.3                                                                                                                                           | (1/2 <sup>+</sup> , 3/2, 5/2 <sup>+</sup> ) | 1621               | 19                  | 6492    | (3/2 <sup>+</sup> )                 |         |                                          |
|                                                                                                                                                  |                                             | 2007.0             | 19                  | 6106.2  | (3/2, 5/2 <sup>+</sup> )            |         |                                          |
|                                                                                                                                                  |                                             | 2306.6             | 3.4                 | 5806.6  | (1/2, 3/2, 5/2)                     |         |                                          |

Adopted Levels, Gammas (continued) $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$             | $E_\gamma^\dagger$ | $I_\gamma^\ddagger$ | $E_f$   | $J_f^\pi$           |
|---------------------|-----------------------|--------------------|---------------------|---------|---------------------|
| 8113.3              | $(1/2^+, 3/2, 5/2^+)$ | 2458               | 2.8                 | 5655    | $(3/2)^+$           |
|                     |                       | 2514.8             |                     | 5598.4  | $3/2^+, 5/2^+$      |
|                     |                       | 2898.7             | 5.6                 | 5214.5  | $(5/2^+)$           |
|                     |                       | 3939.6             | 22                  | 4173.43 | $(5/2^+)$           |
|                     |                       | 4054.0             | 4.1                 | 4059.1  | $(3/2)^-$           |
|                     |                       | 4145.4             | 16                  | 3967.6  | $1/2^+$             |
|                     |                       | 5110.3             | 5.9                 | 3002.64 | $5/2^+$             |
|                     |                       | 5418.9             | 41                  | 2694.00 | $3/2^+$             |
|                     |                       | 6349.5             | 16                  | 1763.15 | $5/2^+$             |
|                     |                       | 6893.2             | 100                 | 1219.38 | $1/2^+$             |
|                     |                       | 8112.3             | 78                  | 0.0     | $3/2^+$             |
|                     |                       | 2464.9             | 1.6                 | 5682.2  | $1/2^-, 3/2^-$      |
|                     |                       | 3138.0             | 4.8                 | 5009.1  | $(1/2, 3/2)$        |
|                     |                       | 3969.1             | 13                  | 4177.9  | $3/2^-$             |
| 8147.2              | $3/2^-$               | 4087.8             | 4.8                 | 4059.1  | $(3/2)^-$           |
|                     |                       | 4179.3             | 3.2                 | 3967.6  | $1/2^+$             |
|                     |                       | 4228.5             | 4.8                 | 3918.47 | $3/2^+$             |
|                     |                       | 5452.7             | 4.8                 | 2694.00 | $3/2^+$             |
|                     |                       | 6927.1             | 24                  | 1219.38 | $1/2^+$             |
|                     |                       | 8146.2             | 100                 | 0.0     | $3/2^+$             |
|                     |                       | 2511               | 7.8                 | 5646    | $(5/2, 7/2, 9/2^+)$ |
|                     |                       | 2570.7             | 16                  | 5586.0  | $5/2^+$             |
|                     |                       | 2942.2             | 12                  | 5214.5  | $(5/2^+)$           |
|                     |                       | 3386.8             | 2.0                 | 4769.86 | $(5/2^-)$           |
|                     |                       | 3978.7             | 9.8                 | 4177.9  | $3/2^-$             |
|                     |                       | 4044.2             | 14                  | 4112.4  | $7/2^+$             |
|                     |                       | 4097.4             | 2.0                 | 4059.1  | $(3/2)^-$           |
|                     |                       | 4213.6             | 3.9                 | 3942.9  | $9/2^+$             |
| 8156.8              | $(5/2^+)$             | 4993.6             | 9.8                 | 3162.87 | $7/2^-$             |
|                     |                       | 5153.8             | 12                  | 3002.64 | $5/2^+$             |
|                     |                       | 5510.7             | 100                 | 2645.67 | $7/2^+$             |
|                     |                       | 6393.0             | 2.0                 | 1763.15 | $5/2^+$             |
|                     |                       | 8155.8             | 5.9                 | 0.0     | $3/2^+$             |
|                     |                       | 3554.9             | 13.2 <i>I3</i>      | 4624.3  | $3/2, 5/2$          |
|                     |                       | 5016.1             | 11.8 <i>I2</i>      | 3162.87 | $7/2^-$             |
|                     |                       | 5176.4             | 5.9 <i>30</i>       | 3002.64 | $5/2^+$             |
|                     |                       | 5484.9             | 4.4 <i>22</i>       | 2694.00 | $3/2^+$             |
|                     |                       | 6415.6             | 100 <i>I0</i>       | 1763.15 | $5/2^+$             |
|                     |                       | 8178.4             | 11.8 <i>I2</i>      | 0.0     | $3/2^+$             |
|                     |                       | 2553               | 0.9                 | 5655    | $(3/2)^+$           |
|                     |                       | 3327.1             | 1.0                 | 4880.96 | $7/2^{(+)}$         |
|                     |                       | 3368.9             | 0.51                | 4839.10 | $(1/2^+, 3/2)$      |
| 8208.2              | $5/2^+$               |                    |                     |         |                     |
|                     |                       |                    |                     |         |                     |
|                     |                       |                    |                     |         |                     |

## Adopted Levels, Gammas (continued)

| $\gamma(^{35}\text{Cl})$ (continued) |                  |                    |                     |         |                |                    |             |                         |                                                                                                                                                                                                                  |
|--------------------------------------|------------------|--------------------|---------------------|---------|----------------|--------------------|-------------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $E_i(\text{level})$                  | $J_i^\pi$        | $E_\gamma^\dagger$ | $I_\gamma^\ddagger$ | $E_f$   | $J_f^\pi$      | Mult. <sup>#</sup> | $\delta^\#$ | $\alpha^a$              | Comments                                                                                                                                                                                                         |
| 8208.2                               | $5/2^+$          | 4148.8             | 1.9                 | 4059.1  | $(3/2)^-$      |                    |             |                         |                                                                                                                                                                                                                  |
|                                      |                  | 4289.5             |                     | 3918.47 | $3/2^+$        |                    |             |                         |                                                                                                                                                                                                                  |
|                                      |                  | 5205.1             | 1.4                 | 3002.64 | $5/2^+$        |                    |             |                         |                                                                                                                                                                                                                  |
|                                      |                  | 5513.7             | 0.64                | 2694.00 | $3/2^+$        |                    |             |                         |                                                                                                                                                                                                                  |
|                                      |                  | 6444.4             | 18                  | 1763.15 | $5/2^+$        | M1+E2              | +0.33 4     |                         | B(M1)(W.u.)=0.75 +35-22; B(E2)(W.u.)=7.5 +41-25                                                                                                                                                                  |
|                                      |                  | 6988.1             | 3.9                 | 1219.38 | $1/2^+$        |                    |             |                         |                                                                                                                                                                                                                  |
|                                      |                  | 8207.2             | 100                 | 0.0     | $3/2^+$        | M1+E2              | -0.36 4     |                         | B(M1)(W.u.)=2.0 +9-5; B(E2)(W.u.)=15 +7-4                                                                                                                                                                        |
| 8216.3                               | $5/2^+$          | 2561               | 1.8                 | 5655    | $(3/2)^+$      |                    |             |                         |                                                                                                                                                                                                                  |
|                                      |                  | 3446.3             | 1.8                 | 4769.86 | $(5/2)^-$      |                    |             |                         |                                                                                                                                                                                                                  |
|                                      |                  | 4038.2             | <6.7                | 4177.9  | $3/2^-$        |                    |             |                         |                                                                                                                                                                                                                  |
|                                      |                  | 4042.6             | <6.7                | 4173.43 | $(5/2^+)$      |                    |             |                         |                                                                                                                                                                                                                  |
|                                      |                  | 5053.0             | 91                  | 3162.87 | $7/2^-$        | (E1)               |             | $2.04 \times 10^{-3}$ 3 | B(E1)(W.u.)=0.062 +21-12<br>Mult.: D+Q with $\delta=+0.11$ 2 from $\gamma(\theta)$ in (p, $\gamma$ ); $\Delta\pi=\text{yes}$ from level scheme; but M2 component is unlikely for this strong primary transition. |
|                                      |                  | 5213.2             | 11                  | 3002.64 | $5/2^+$        |                    |             |                         |                                                                                                                                                                                                                  |
|                                      |                  | 5521.8             | 6.7                 | 2694.00 | $3/2^+$        |                    |             |                         |                                                                                                                                                                                                                  |
| 8242.4                               | $3/2^-$          | 6452.5             | 3.1                 | 1763.15 | $5/2^+$        |                    |             |                         |                                                                                                                                                                                                                  |
|                                      |                  | 8215.3             | 100                 | 0.0     | $3/2^+$        | M1+E2              | +0.06 3     |                         | B(M1)(W.u.)=0.55 +18-9; B(E2)(W.u.)=0.11 +16-8                                                                                                                                                                   |
|                                      |                  | 4068.7             | 7.7                 | 4173.43 | $(5/2^+)$      |                    |             |                         |                                                                                                                                                                                                                  |
|                                      |                  | 5239.3             | 2.2                 | 3002.64 | $5/2^+$        |                    |             |                         |                                                                                                                                                                                                                  |
|                                      |                  | 7022.3             | 100                 | 1219.38 | $1/2^+$        |                    |             |                         |                                                                                                                                                                                                                  |
| 8251                                 | $(3/2^+, 5/2^+)$ | 4138               | 4.9                 | 4112.4  | $7/2^+$        |                    |             |                         |                                                                                                                                                                                                                  |
|                                      |                  | 5248               | 7.3                 | 3002.64 | $5/2^+$        |                    |             |                         |                                                                                                                                                                                                                  |
|                                      |                  | 7031               | 100                 | 1219.38 | $1/2^+$        |                    |             |                         |                                                                                                                                                                                                                  |
|                                      |                  | 8250               | 9.8                 | 0.0     | $3/2^+$        |                    |             |                         |                                                                                                                                                                                                                  |
| 8269.0                               | $5/2^+$          | 3499.0             | 4.4                 | 4769.86 | $(5/2)^-$      |                    |             |                         |                                                                                                                                                                                                                  |
|                                      |                  | 4090.8             | 27                  | 4177.9  | $3/2^-$        |                    |             |                         |                                                                                                                                                                                                                  |
|                                      |                  | 4350.2             | 6.7                 | 3918.47 | $3/2^+$        |                    |             |                         |                                                                                                                                                                                                                  |
|                                      |                  | 5105.7             | 16                  | 3162.87 | $7/2^-$        |                    |             |                         |                                                                                                                                                                                                                  |
|                                      |                  | 5265.9             | 29                  | 3002.64 | $5/2^+$        |                    |             |                         |                                                                                                                                                                                                                  |
|                                      |                  | 5574.5             | 13                  | 2694.00 | $3/2^+$        |                    |             |                         |                                                                                                                                                                                                                  |
|                                      |                  | 5622.9             | 8.9                 | 2645.67 | $7/2^+$        |                    |             |                         |                                                                                                                                                                                                                  |
|                                      |                  | 6505.2             | 18                  | 1763.15 | $5/2^+$        |                    |             |                         |                                                                                                                                                                                                                  |
|                                      |                  | 8268.0             | 100                 | 0.0     | $3/2^+$        |                    |             |                         |                                                                                                                                                                                                                  |
|                                      |                  | 2622               | 5.2                 | 5655    | $(3/2)^+$      |                    |             |                         |                                                                                                                                                                                                                  |
| 8277.2                               | $(5/2)^+$        | 3396.1             | 17                  | 4880.96 | $7/2^{(+)}$    |                    |             |                         |                                                                                                                                                                                                                  |
|                                      |                  | 3437.9             | 38                  | 4839.10 | $(1/2^+, 3/2)$ |                    |             |                         |                                                                                                                                                                                                                  |
|                                      |                  | 3507.2             | 4.1                 | 4769.86 | $(5/2)^-$      |                    |             |                         |                                                                                                                                                                                                                  |
|                                      |                  | 3652.7             | 10                  | 4624.3  | $3/2, 5/2$     |                    |             |                         |                                                                                                                                                                                                                  |
|                                      |                  | 3929.2             | 6.2                 | 4347.76 | $9/2^-$        |                    |             |                         |                                                                                                                                                                                                                  |
|                                      |                  | 4103.5             | 6.9                 | 4173.43 | $(5/2^+)$      |                    |             |                         |                                                                                                                                                                                                                  |

## Adopted Levels, Gammas (continued)

| $\gamma(^{35}\text{Cl})$ (continued) |                |                    |                     |         |                   |                    |                                                                                                                                                                                                                      |
|--------------------------------------|----------------|--------------------|---------------------|---------|-------------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $E_i(\text{level})$                  | $J_i^\pi$      | $E_\gamma^\dagger$ | $I_\gamma^\ddagger$ | $E_f$   | $J_f^\pi$         | Mult. <sup>#</sup> | Comments                                                                                                                                                                                                             |
| 8277.2                               | $(5/2)^+$      | 4217.8             | 5.2                 | 4059.1  | $(3/2)^-$         |                    |                                                                                                                                                                                                                      |
|                                      |                | 4358.4             | 17                  | 3918.47 | $3/2^+$           |                    |                                                                                                                                                                                                                      |
|                                      |                | 5274.1             | 35                  | 3002.64 | $5/2^+$           |                    |                                                                                                                                                                                                                      |
|                                      |                | 5582.7             | 10                  | 2694.00 | $3/2^+$           |                    |                                                                                                                                                                                                                      |
|                                      |                | 5631.0             | 10                  | 2645.67 | $7/2^+$           |                    |                                                                                                                                                                                                                      |
|                                      |                | 6513.4             | 79                  | 1763.15 | $5/2^+$           |                    |                                                                                                                                                                                                                      |
|                                      |                | 8276.2             | 100                 | 0.0     | $3/2^+$           |                    |                                                                                                                                                                                                                      |
| 8282.4                               | $(3/2^-, 5/2)$ | 2696.3             | 29                  | 5586.0  | $5/2^+$           |                    |                                                                                                                                                                                                                      |
|                                      |                | 3067.8             | 29                  | 5214.5  | $(5/2^+)$         |                    |                                                                                                                                                                                                                      |
|                                      |                | 3118.9             | 11                  | 5163.34 | $7/2^-$           |                    |                                                                                                                                                                                                                      |
|                                      |                | 4104.2             | 50                  | 4177.9  | $3/2^-$           |                    |                                                                                                                                                                                                                      |
|                                      |                | 4223.0             | 21                  | 4059.1  | $(3/2)^-$         |                    |                                                                                                                                                                                                                      |
|                                      |                | 5119.1             | 93                  | 3162.87 | $7/2^-$           |                    |                                                                                                                                                                                                                      |
|                                      |                | 6518.6             | 25                  | 1763.15 | $5/2^+$           |                    |                                                                                                                                                                                                                      |
| 8287.82                              | $1/2^-$        | 8281.4             | 100                 | 0.0     | $3/2^+$           |                    |                                                                                                                                                                                                                      |
|                                      |                | 2633               | 4.6                 | 5655    | $(3/2)^+$         |                    |                                                                                                                                                                                                                      |
|                                      |                | 3278.6             | 6.8                 | 5009.1  | $(1/2, 3/2)$      |                    |                                                                                                                                                                                                                      |
|                                      |                | 4109.7             | 21                  | 4177.9  | $3/2^-$           |                    |                                                                                                                                                                                                                      |
|                                      |                | 4228.5             | 4.6                 | 4059.1  | $(3/2)^-$         |                    |                                                                                                                                                                                                                      |
|                                      |                | 4319.9             | 6.8                 | 3967.6  | $1/2^+$           |                    |                                                                                                                                                                                                                      |
|                                      |                | 4369.06            | 2.3                 | 3918.47 | $3/2^+$           |                    |                                                                                                                                                                                                                      |
|                                      |                | 5593.3             | 52                  | 2694.00 | $3/2^+$           |                    |                                                                                                                                                                                                                      |
|                                      |                | 7067.67            | 100                 | 1219.38 | $1/2^+$           | (E1)               | B(E1)(W.u.)=0.069 +24-14<br>Mult.: D+Q with $\delta(Q/D)=+0.21$ 2 or $-3.2$ 2 in (p, $\gamma$ ); but M2 component is unlikely for this primary transition based on resonance $\Gamma=0.04$ keV.                      |
|                                      |                | 8298.2             | $3/2^+$             | 8286.77 | 30                | 0.0                | $3/2^+$                                                                                                                                                                                                              |
| 3288.9                               | 13             |                    |                     | 5009.1  | $(1/2, 3/2)$      |                    |                                                                                                                                                                                                                      |
| 3417.1                               | 17             |                    |                     | 4880.96 | $7/2^{(+)}$       |                    |                                                                                                                                                                                                                      |
| 3443.6                               | 21             |                    |                     | 4854.4  | $(1/2, 3/2)$      |                    |                                                                                                                                                                                                                      |
| 4124.5                               | 25             |                    |                     | 4173.43 | $(5/2^+)$         |                    |                                                                                                                                                                                                                      |
| 4238.8                               | 42             |                    |                     | 4059.1  | $(3/2)^-$         |                    |                                                                                                                                                                                                                      |
| 5134.9                               | 4.2            |                    |                     | 3162.87 | $7/2^-$           |                    | This $\gamma$ is questionable if $J^\pi(8298)=3/2^+$ , since it would require Mult=M2 which is probably unlikely; however $J^\pi(8298)=3/2^-$ would make the strong 7078.0 $\gamma$ and 8297.1 $\gamma$ less likely. |
| 5295.1                               | 4.2            |                    |                     | 3002.64 | $5/2^+$           |                    |                                                                                                                                                                                                                      |
| 5603.7                               | 42             |                    |                     | 2694.00 | $3/2^+$           |                    |                                                                                                                                                                                                                      |
| 5652.0                               | 25             |                    |                     | 2645.67 | $7/2^+$           |                    |                                                                                                                                                                                                                      |
| 8318.1                               | $(5/2^+)$      | 6534.4             | 33                  | 1763.15 | $5/2^+$           |                    |                                                                                                                                                                                                                      |
|                                      |                | 7078.1             | 100                 | 1219.38 | $1/2^+$           | M1+E2              | $\delta$ : $-1.7$ 1 or $-0.02$ 2 from $\gamma(\theta)$ in (p, $\gamma$ ).                                                                                                                                            |
|                                      |                | 8297.1             | 54                  | 0.0     | $3/2^+$           | M1+E2              | $\delta$ : $-0.15$ 4 or $+9.5$ 36 from $\gamma(\theta)$ in (p, $\gamma$ ).                                                                                                                                           |
|                                      |                | 2511.4             | 0.9                 | 5806.6  | $(1/2, 3/2, 5/2)$ |                    |                                                                                                                                                                                                                      |

## Adopted Levels, Gammas (continued)

| $\gamma(^{35}\text{Cl})$ (continued) |                  |                    |                     |         |                     |                    |             |                         |                                                                                                                                                                                    |
|--------------------------------------|------------------|--------------------|---------------------|---------|---------------------|--------------------|-------------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $E_i(\text{level})$                  | $J_i^\pi$        | $E_\gamma^\dagger$ | $I_\gamma^\ddagger$ | $E_f$   | $J_f^\pi$           | Mult. <sup>#</sup> | $\delta^\#$ | $\alpha^a$              | Comments                                                                                                                                                                           |
| 8318.1                               | $(5/2^+)$        | 2672               | 0.9                 | 5646    | $(5/2, 7/2, 9/2^+)$ |                    |             |                         |                                                                                                                                                                                    |
|                                      |                  | 2719.6             | 1.9                 | 5598.4  | $3/2^+, 5/2^+$      |                    |             |                         |                                                                                                                                                                                    |
|                                      |                  | 3548.1             | 0.39                | 4769.86 | $(5/2^-)$           |                    |             |                         |                                                                                                                                                                                    |
|                                      |                  | 3693.6             | 2.3                 | 4624.3  | $3/2, 5/2$          |                    |             |                         |                                                                                                                                                                                    |
|                                      |                  | 4139.9             | <3.9                | 4177.9  | $3/2^-$             |                    |             |                         |                                                                                                                                                                                    |
|                                      |                  | 4144.4             | <3.9                | 4173.43 | $(5/2^+)$           |                    |             |                         |                                                                                                                                                                                    |
|                                      |                  | 4258.7             | 1.5                 | 4059.1  | $(3/2)^-$           |                    |             |                         |                                                                                                                                                                                    |
|                                      |                  | 4350.2             | 3.9                 | 3967.6  | $1/2^+$             |                    |             |                         |                                                                                                                                                                                    |
|                                      |                  | 5154.8             | 0.64                | 3162.87 | $7/2^-$             |                    |             |                         |                                                                                                                                                                                    |
|                                      |                  | 5315.0             | 3.9                 | 3002.64 | $5/2^+$             |                    |             |                         |                                                                                                                                                                                    |
|                                      |                  | 5623.6             | 2.1                 | 2694.00 | $3/2^+$             |                    |             |                         |                                                                                                                                                                                    |
|                                      |                  | 5671.9             | 1.2                 | 2645.67 | $7/2^+$             |                    |             |                         |                                                                                                                                                                                    |
|                                      |                  | 6554.3             | 2.3                 | 1763.15 | $5/2^+$             |                    |             |                         |                                                                                                                                                                                    |
|                                      |                  | 7098.0             | 2.6                 | 1219.38 | $1/2^+$             |                    |             |                         |                                                                                                                                                                                    |
|                                      |                  | 8317.0             | 100                 | 0.0     | $3/2^+$             |                    |             |                         |                                                                                                                                                                                    |
| 8319.6                               | $15/2^-$         | 2232.7 6           | 57.2 28             | 6087.31 | $13/2^-$            | [M1,E2]            |             | 0.00040 4               | $E_\gamma, I_\gamma$ : from ( $^{16}\text{O}, \alpha\gamma$ ).<br>B(M1)(W.u.)=0.0577 4I if M1,<br>B(E2)(W.u.)=43.8 3I if E2.                                                       |
|                                      |                  | 2911.9 8           | 100 5               | 5407.07 | $11/2^-$            | E2                 |             | 0.000753 1I             | B(E2)(W.u.)=20.3 12<br>$E_\gamma, I_\gamma, \text{Mult.}$ : from ( $^{16}\text{O}, \alpha\gamma$ ). Mult=Q, $\Delta J=2$<br>from $\gamma\gamma(\text{DCO})$ ; M2 ruled out by RUL. |
| 8323.1                               | $(3/2^+, 5/2^+)$ | 4210.4             | 1.2                 | 4112.4  | $7/2^+$             |                    |             |                         |                                                                                                                                                                                    |
|                                      |                  | 5320.0             | 3.7                 | 3002.64 | $5/2^+$             |                    |             |                         |                                                                                                                                                                                    |
|                                      |                  | 5676.9             | 4.9                 | 2645.67 | $7/2^+$             |                    |             |                         |                                                                                                                                                                                    |
|                                      |                  | 6559.3             | <2.4                | 1763.15 | $5/2^+$             |                    |             |                         |                                                                                                                                                                                    |
|                                      |                  | 7103.0             | 9.8                 | 1219.38 | $1/2^+$             |                    |             |                         |                                                                                                                                                                                    |
|                                      |                  | 8322.0             | 100                 | 0.0     | $3/2^+$             |                    |             |                         |                                                                                                                                                                                    |
| 8381.8                               | $5/2^+$          | 3611.7             |                     | 4769.86 | $(5/2^-)$           |                    |             |                         |                                                                                                                                                                                    |
|                                      |                  | 3757.3             |                     | 4624.3  | $3/2, 5/2$          |                    |             |                         |                                                                                                                                                                                    |
|                                      |                  | 4269.1             | 23.5 24             | 4112.4  | $7/2^+$             |                    |             |                         |                                                                                                                                                                                    |
|                                      |                  | 4463.0             | 71 7                | 3918.47 | $3/2^+$             | M1+E2              | +0.15 3     | $1.19 \times 10^{-3}$ 2 | B(M1)(W.u.)=2.9 +13-7; B(E2)(W.u.)=13<br>+8-5                                                                                                                                      |
|                                      |                  | 5378.7             | 74 8                | 3002.64 | $5/2^+$             | M1+E2              | +0.02 3     | $1.44 \times 10^{-3}$ 2 | B(M1)(W.u.)=1.8 +14-7                                                                                                                                                              |
|                                      |                  | 5687.3             | 2.9 15              | 2694.00 | $3/2^+$             | M1+E2              | -0.72 3I    | 0.00157 4               | B(M1)(W.u.)=0.039 +32-2I;<br>B(E2)(W.u.)=2.4 +23-18                                                                                                                                |
|                                      |                  | 5735.6             | 15 8                | 2645.67 | $7/2^+$             |                    |             |                         |                                                                                                                                                                                    |
|                                      |                  | 6618.0             | 100 10              | 1763.15 | $5/2^+$             | M1+E2              | +0.06 3     |                         | B(M1)(W.u.)=1.3 +6-3; B(E2)(W.u.)=0.4<br>+6-3                                                                                                                                      |
|                                      |                  | 8380.7             | 9 5                 | 0.0     | $3/2^+$             | D+Q                | 1.65        |                         |                                                                                                                                                                                    |
|                                      |                  | 3632.8             | 16 8                | 4769.86 | $(5/2^-)$           |                    |             |                         |                                                                                                                                                                                    |
| 8402.9                               | $5/2^-$          | 4224.7             | 29 16               | 4177.9  | $3/2^-$             |                    |             |                         |                                                                                                                                                                                    |

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$             | $E_\gamma^\dagger$ | $I_\gamma^\ddagger$ | $E_f$   | $J_f^\pi$    |
|---------------------|-----------------------|--------------------|---------------------|---------|--------------|
| 8402.9              | $5/2^-$               | 4343.5             | 74 20               | 4059.1  | $(3/2)^-$    |
|                     |                       | 5708.4             | 32 16               | 2694.00 | $3/2^+$      |
|                     |                       | 5756.7             | 23 13               | 2645.67 | $7/2^+$      |
|                     |                       | 6639.1             | 48 13               | 1763.15 | $5/2^+$      |
|                     |                       | 8401.8             | 100 26              | 0.0     | $3/2^+$      |
| 8405.8              | $(5/2^-)$             | 3524.7             | 35 9                | 4880.96 | $7/2^{(+)}$  |
|                     |                       | 4232.1             | 3.7 19              | 4173.43 | $(5/2^+)$    |
|                     |                       | 4293.1             | 3.3 16              | 4112.4  | $7/2^+$      |
|                     |                       | 5402.7             | 35 9                | 3002.64 | $5/2^+$      |
|                     |                       | 6642.0             | 100 26              | 1763.15 | $5/2^+$      |
|                     |                       | 8404.7             | 56 14               | 0.0     | $3/2^+$      |
|                     |                       | 3407.4             | 14                  | 5009.1  | $(1/2,3/2)$  |
| 8416.7              | $1/2^+$               | 4243.0             | 31                  | 4173.43 | $(5/2^+)$    |
|                     |                       | 4473.5             | 8.6                 | 3942.9  | $9/2^+$      |
|                     |                       | 5413.6             | 20                  | 3002.64 | $5/2^+$      |
|                     |                       | 5770.5             | 23                  | 2645.67 | $7/2^+$      |
|                     |                       | 6652.9             | 63                  | 1763.15 | $5/2^+$      |
|                     |                       | 7196.5             | 26                  | 1219.38 | $1/2^+$      |
|                     |                       | 8415.6             | 100                 | 0.0     | $3/2^+$      |
|                     |                       | 2809               | 47 24               | 5655    | $(3/2)^+$    |
|                     |                       | 3249.6             | 34 17               | 5214.5  | $(5/2^+)$    |
| 8464.3              | $(1/2^+, 3/2, 5/2^+)$ | 3455.0             | 47 24               | 5009.1  | $(1/2, 3/2)$ |
|                     |                       | 3609.7             | 19 10               | 4854.4  | $(1/2, 3/2)$ |
|                     |                       | 3839.8             | 25 13               | 4624.3  | $3/2, 5/2$   |
|                     |                       | 4286.1             | 31 15               | 4177.9  | $3/2^-$      |
|                     |                       | 4290.6             | 32 17               | 4173.43 | $(5/2^+)$    |
|                     |                       | 4496.4             | 31 15               | 3967.6  | $1/2^+$      |
|                     |                       | 4545.5             | 15 8                | 3918.47 | $3/2^+$      |
|                     |                       | 5461.2             | 100 24              | 3002.64 | $5/2^+$      |
|                     |                       | 5769.8             | 94 24               | 2694.00 | $3/2^+$      |
|                     |                       | 7244.1             | 53 24               | 1219.38 | $1/2^+$      |
|                     |                       | 8463.2             | 59 18               | 0.0     | $3/2^+$      |
|                     |                       | 2829               | 12 6                | 5655    | $(3/2)^+$    |
|                     |                       | 3603.2             | 15 8                | 4880.96 | $7/2^{(+)}$  |
| 8484.4              | $(3/2)^+$             | 3859.9             | 2.2 11              | 4624.3  | $3/2, 5/2$   |
|                     |                       | 4516.5             | 2.6 13              | 3967.6  | $1/2^+$      |
|                     |                       | 4565.6             | 11 6                | 3918.47 | $3/2^+$      |
|                     |                       | 5481.3             | 15 9                | 3002.64 | $5/2^+$      |
|                     |                       | 5789.9             | 44 11               | 2694.00 | $3/2^+$      |
|                     |                       | 5838.2             | 6.1 31              | 2645.67 | $7/2^+$      |
|                     |                       | 6720.6             | 100 26              | 1763.15 | $5/2^+$      |
|                     |                       | 8483.3             | 9 5                 | 0.0     | $3/2^+$      |

**Adopted Levels, Gammas (continued)**

$\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$            | $E_\gamma^\dagger$  | $I_\gamma^\ddagger$ | $E_f$   | $J_f^\pi$                          | Mult. <sup>#</sup> | $\delta^\#$ | $\alpha^a$  | Comments                                                                                                                                                                |
|---------------------|----------------------|---------------------|---------------------|---------|------------------------------------|--------------------|-------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8486.2              | 3/2 <sup>-</sup>     | 2728                | 2.4 13              | 5758    | (1/2,3/2 <sup>-</sup> )            |                    |             |             |                                                                                                                                                                         |
|                     |                      | 3861.7              | 3.9 20              | 4624.3  | 3/2,5/2                            |                    |             |             |                                                                                                                                                                         |
|                     |                      | 4308.0              | <15                 | 4177.9  | 3/2 <sup>-</sup>                   |                    |             |             | $I_\gamma$ : total $I_\gamma=10$ 5 for $\gamma$ decays to 4173 and 4178 doublet levels ( <a href="#">1976Sp08</a> ).                                                    |
|                     |                      | 4312.5              | <15                 | 4173.43 | (5/2 <sup>+</sup> )                |                    |             |             | $I_\gamma$ : total $I_\gamma=10$ 5 for $\gamma$ decays to 4173 and 4178 doublet levels ( <a href="#">1976Sp08</a> ).                                                    |
|                     |                      | 4426.8              | 4.1 22              | 4059.1  | (3/2) <sup>-</sup>                 |                    |             |             |                                                                                                                                                                         |
|                     |                      | 4518.3              | 15 9                | 3967.6  | 1/2 <sup>+</sup>                   |                    |             |             |                                                                                                                                                                         |
|                     |                      | 5483.1              | 17 9                | 3002.64 | 5/2 <sup>+</sup>                   |                    |             |             |                                                                                                                                                                         |
|                     |                      | 5791.7              | 41 11               | 2694.00 | 3/2 <sup>+</sup>                   |                    |             |             |                                                                                                                                                                         |
|                     |                      | 6722.4              | 8 4                 | 1763.15 | 5/2 <sup>+</sup>                   |                    |             |             |                                                                                                                                                                         |
|                     |                      | 7266.0              | 14 7                | 1219.38 | 1/2 <sup>+</sup>                   |                    |             |             |                                                                                                                                                                         |
|                     |                      | 8485.1              | 100 26              | 0.0     | 3/2 <sup>+</sup>                   |                    |             |             |                                                                                                                                                                         |
| 8487.5              | 15/2 <sup>-</sup>    | 2399.8 8            | 100.0 34            | 6087.31 | 13/2 <sup>-</sup>                  | (M1(+E2))          | +0.2 +3-2   | 0.000427 17 | Mult., $\delta$ : D+Q and $\delta$ from $\gamma\gamma(\text{DCO})$ in ( <sup>28</sup> Si, $\alpha\gamma\gamma$ ); $\Delta\pi$ =no from level scheme.                    |
|                     |                      | 3080.0 7            | 31.8 20             | 5407.07 | 11/2 <sup>-</sup>                  | E2                 |             | 0.000825 12 | $E_\gamma, I_\gamma, \text{Mult.}$ : from ( <sup>16</sup> O, $\alpha\gamma\gamma$ ), with Mult=Q, $\Delta J=2$ from $\gamma\gamma(\text{ADO})$ and M2 ruled out by RUL. |
| 8514.3              | 1/2 <sup>-</sup>     | 3659.7              | 16 8                | 4854.4  | (1/2,3/2)                          |                    |             |             |                                                                                                                                                                         |
|                     |                      | 4336.1              | 7 4                 | 4177.9  | 3/2 <sup>-</sup>                   |                    |             |             |                                                                                                                                                                         |
|                     |                      | 4454.9              | 100 26              | 4059.1  | (3/2) <sup>-</sup>                 |                    |             |             |                                                                                                                                                                         |
|                     |                      | 4546.4              | 85 22               | 3967.6  | 1/2 <sup>+</sup>                   |                    |             |             |                                                                                                                                                                         |
|                     |                      | 4595.5              | 24 12               | 3918.47 | 3/2 <sup>+</sup>                   |                    |             |             |                                                                                                                                                                         |
|                     |                      | 7294.1              | 37 11               | 1219.38 | 1/2 <sup>+</sup>                   |                    |             |             |                                                                                                                                                                         |
|                     |                      | 8513.2              | 100 26              | 0.0     | 3/2 <sup>+</sup>                   |                    |             |             |                                                                                                                                                                         |
| 8558.2?             | (13/2 <sup>-</sup> ) | 2631.4 <sup>b</sup> | 100                 | 5926.65 | (11/2) <sup>-</sup>                | D(+Q)              | -0.2 +2-3   |             | $E_\gamma, \text{Mult.}, \delta$ : from ( <sup>28</sup> Si, $\alpha\gamma\gamma$ ); $\Delta\pi$ =no from level scheme; Mult=Q, $\Delta J=1$ .                           |
| 8571.9              | (5/2) <sup>+</sup>   | 2848.3              | 3.1 16              | 5723.52 | 5/2 <sup>+</sup>                   |                    |             |             |                                                                                                                                                                         |
|                     |                      | 2917                | 1.9 10              | 5655    | (3/2) <sup>+</sup>                 |                    |             |             |                                                                                                                                                                         |
|                     |                      | 2973.4              | 1.0 6               | 5598.4  | 3/2 <sup>+</sup> ,5/2 <sup>+</sup> |                    |             |             |                                                                                                                                                                         |
|                     |                      | 3408.4              | 2.6 13              | 5163.34 | 7/2 <sup>-</sup>                   |                    |             |             |                                                                                                                                                                         |
|                     |                      | 3690.7              | 1.3 7               | 4880.96 | 7/2 <sup>(+)</sup>                 |                    |             |             |                                                                                                                                                                         |
|                     |                      | 4398.2              | 1.0 6               | 4173.43 | (5/2 <sup>+</sup> )                |                    |             |             |                                                                                                                                                                         |
|                     |                      | 4459.2              | 1.6 9               | 4112.4  | 7/2 <sup>+</sup>                   |                    |             |             |                                                                                                                                                                         |
|                     |                      | 4653.1              | 1.6 9               | 3918.47 | 3/2 <sup>+</sup>                   |                    |             |             |                                                                                                                                                                         |
|                     |                      | 5408.6              | 2.3 12              | 3162.87 | 7/2 <sup>-</sup>                   |                    |             |             |                                                                                                                                                                         |
|                     |                      | 5568.8              | 6.4 33              | 3002.64 | 5/2 <sup>+</sup>                   |                    |             |             |                                                                                                                                                                         |
|                     |                      | 5877.4              | 0.9 4               | 2694.00 | 3/2 <sup>+</sup>                   |                    |             |             |                                                                                                                                                                         |
|                     |                      | 5925.7              | 11 6                | 2645.67 | 7/2 <sup>+</sup>                   |                    |             |             |                                                                                                                                                                         |
|                     |                      | 7351.7              | 8 4                 | 1219.38 | 1/2 <sup>+</sup>                   |                    |             |             |                                                                                                                                                                         |
|                     |                      | 8570.8              | 100 10              | 0.0     | 3/2 <sup>+</sup>                   |                    |             |             |                                                                                                                                                                         |
| 8580.5              | 1/2 <sup>+</sup>     | 2088                | 8 4                 | 6492    | (3/2 <sup>+</sup> )                |                    |             |             |                                                                                                                                                                         |

**Adopted Levels, Gammas (continued)**

$\gamma(^{35}\text{Cl})$  (continued)

| <u>E<sub>i</sub>(level)</u> | <u>J<sub>i</sub><sup><math>\pi</math></sup></u> | <u>E<sub><math>\gamma</math></sub><sup><math>\dagger</math></sup></u> | <u>I<sub><math>\gamma</math></sub><sup><math>\ddagger</math></sup></u> | <u>E<sub>f</sub></u> | <u>J<sub>f</sub><sup><math>\pi</math></sup></u> |
|-----------------------------|-------------------------------------------------|-----------------------------------------------------------------------|------------------------------------------------------------------------|----------------------|-------------------------------------------------|
| 8580.5                      | 1/2 <sup>+</sup>                                | 3741.2                                                                | 2.4 13                                                                 | 4839.10              | (1/2 <sup>+</sup> ,3/2)                         |
|                             |                                                 | 4402.3                                                                | 8 4                                                                    | 4177.9               | 3/2 <sup>-</sup>                                |
|                             |                                                 | 4521.1                                                                | 3.7 19                                                                 | 4059.1               | (3/2) <sup>-</sup>                              |
|                             |                                                 | 4661.7                                                                | 4.3 22                                                                 | 3918.47              | 3/2 <sup>+</sup>                                |
|                             |                                                 | 5886.0                                                                | 10 5                                                                   | 2694.00              | 3/2 <sup>+</sup>                                |
|                             |                                                 | 7360.3                                                                | 100 11                                                                 | 1219.38              | 1/2 <sup>+</sup>                                |
|                             |                                                 | 8579.4                                                                | 48 13                                                                  | 0.0                  | 3/2 <sup>+</sup>                                |
| 8590.1                      | 5/2 <sup>+</sup>                                | 2832                                                                  | 22 11                                                                  | 5758                 | (1/2,3/2 <sup>-</sup> )                         |
|                             |                                                 | 2935                                                                  | 17 9                                                                   | 5655                 | (3/2) <sup>+</sup>                              |
|                             |                                                 | 3820.0                                                                | 73 18                                                                  | 4769.86              | (5/2 <sup>-</sup> )                             |
|                             |                                                 | 4416.4                                                                | 59 14                                                                  | 4173.43              | (5/2 <sup>+</sup> )                             |
|                             |                                                 | 4477.4                                                                | 16 8                                                                   | 4112.4               | 7/2 <sup>+</sup>                                |
|                             |                                                 | 5426.8                                                                | 36 18                                                                  | 3162.87              | 7/2 <sup>-</sup>                                |
|                             |                                                 | 5587.0                                                                | 100 27                                                                 | 3002.64              | 5/2 <sup>+</sup>                                |
|                             |                                                 | 5943.9                                                                | 25 13                                                                  | 2645.67              | 7/2 <sup>+</sup>                                |
|                             |                                                 | 6826.2                                                                | 64 18                                                                  | 1763.15              | 5/2 <sup>+</sup>                                |
|                             |                                                 | 8589.0                                                                | 46 14                                                                  | 0.0                  | 3/2 <sup>+</sup>                                |
|                             |                                                 | 6848.1                                                                | 100 10                                                                 | 1763.15              | 5/2 <sup>+</sup>                                |
| 8612.0                      | 3/2 <sup>+</sup> ,5/2 <sup>+</sup>              | 7391.8                                                                | 34 8                                                                   | 1219.38              | 1/2 <sup>+</sup>                                |
|                             |                                                 | 8610.9                                                                | 66 16                                                                  | 0.0                  | 3/2 <sup>+</sup>                                |
|                             |                                                 | 3732.5                                                                | 2.9 14                                                                 | 4880.96              | 7/2 <sup>(+)</sup>                              |
| 8613.7                      | 5/2 <sup>+</sup>                                | 4554.3                                                                | 10 5                                                                   | 4059.1               | (3/2) <sup>-</sup>                              |
|                             |                                                 | 4670.5                                                                | 7.1                                                                    | 3942.9               | 9/2 <sup>+</sup>                                |
|                             |                                                 | 4694.9                                                                | 13 7                                                                   | 3918.47              | 3/2 <sup>+</sup>                                |
|                             |                                                 | 5610.6                                                                | 4.5 23                                                                 | 3002.64              | 5/2 <sup>+</sup>                                |
|                             |                                                 | 5919.2                                                                | 8 4                                                                    | 2694.00              | 3/2 <sup>+</sup>                                |
|                             |                                                 | 5967.5                                                                | 41 11                                                                  | 2645.67              | 7/2 <sup>+</sup>                                |
|                             |                                                 | 6849.8                                                                | 16                                                                     | 1763.15              | 5/2 <sup>+</sup>                                |
|                             |                                                 | 7393.5                                                                | 3.6                                                                    | 1219.38              | 1/2 <sup>+</sup>                                |
|                             |                                                 | 8612.6                                                                | 100 11                                                                 | 0.0                  | 3/2 <sup>+</sup>                                |
|                             |                                                 | 2894.6                                                                | 6.4 33                                                                 | 5723.52              | 5/2 <sup>+</sup>                                |
|                             |                                                 | 2963                                                                  | 5.2 27                                                                 | 5655                 | (3/2) <sup>+</sup>                              |
| 8618.2                      | 3/2 <sup>+</sup> ,5/2 <sup>+</sup>              | 3763.6                                                                | 4.2 21                                                                 | 4854.4               | (1/2,3/2)                                       |
|                             |                                                 | 3993.7                                                                | 4.6 24                                                                 | 4624.3               | 3/2,5/2                                         |
|                             |                                                 | 4650.3                                                                | 9 5                                                                    | 3967.6               | 1/2 <sup>+</sup>                                |
|                             |                                                 | 4699.4                                                                | 7 4                                                                    | 3918.47              | 3/2 <sup>+</sup>                                |
|                             |                                                 | 5615.1                                                                | 49 12                                                                  | 3002.64              | 5/2 <sup>+</sup>                                |
|                             |                                                 | 5923.7                                                                | 42 12                                                                  | 2694.00              | 3/2 <sup>+</sup>                                |
|                             |                                                 | 5972.0                                                                | 5.2 27                                                                 | 2645.67              | 7/2 <sup>+</sup>                                |
|                             |                                                 | 6854.3                                                                | 70 18                                                                  | 1763.15              | 5/2 <sup>+</sup>                                |
|                             |                                                 | 8617.1                                                                | 100 24                                                                 | 0.0                  | 3/2 <sup>+</sup>                                |
|                             |                                                 | 2906.5                                                                | 0.6 4                                                                  | 5723.52              | 5/2 <sup>+</sup>                                |
| 8630.1                      | 7/2 <sup>-</sup>                                |                                                                       |                                                                        |                      |                                                 |



Adopted Levels, Gammas (continued)

| $\gamma(^{35}\text{Cl})$ (continued) |                                    |                    |                     |         |                         |                    |             |                         |                                                                                                                 |
|--------------------------------------|------------------------------------|--------------------|---------------------|---------|-------------------------|--------------------|-------------|-------------------------|-----------------------------------------------------------------------------------------------------------------|
| $E_i(\text{level})$                  | $J_i^\pi$                          | $E_\gamma^\dagger$ | $I_\gamma^\ddagger$ | $E_f$   | $J_f^\pi$               | Mult. <sup>#</sup> | $\delta^\#$ | $\alpha^a$              | Comments                                                                                                        |
| 8630.1                               | 7/2 <sup>-</sup>                   | 3466.6             | 17 9                | 5163.34 | 7/2 <sup>-</sup>        | M1+E2              | +0.60 10    | 0.000884 15             | Mult., $\delta$ : D+Q and $\delta$ from $\gamma(\theta)$ in (p, $\gamma$ ); E1+M2 ruled out by RUL.             |
|                                      |                                    | 3860.0             | 3.0 15              | 4769.86 | (5/2 <sup>-</sup> )     |                    |             |                         |                                                                                                                 |
|                                      |                                    | 4282.1             | 26 7                | 4347.76 | 9/2 <sup>-</sup>        | M1+E2              | -0.184 14   | 1.13×10 <sup>-3</sup> 2 | Mult., $\delta$ : D+Q and $\delta$ from $\gamma(\theta)$ in (p, $\gamma$ ); E1+M2 ruled out by RUL.             |
|                                      |                                    | 4456.4             | 8 4                 | 4173.43 | (5/2 <sup>+</sup> )     |                    |             |                         |                                                                                                                 |
|                                      |                                    | 4517.4             | 21 11               | 4112.4  | 7/2 <sup>+</sup>        | (E1+M2)            | -0.06 2     | 1.86×10 <sup>-3</sup> 3 | Mult., $\delta$ : D+Q and $\delta$ from $\gamma(\theta)$ in (p, $\gamma$ ); $\Delta\pi$ =yes from level scheme. |
|                                      |                                    | 4686.9             | 21 11               | 3942.9  | 9/2 <sup>+</sup>        |                    |             |                         |                                                                                                                 |
|                                      |                                    | 5466.8             | 7 4                 | 3162.87 | 7/2 <sup>-</sup>        |                    |             |                         |                                                                                                                 |
|                                      |                                    | 5627.0             | 3.0 15              | 3002.64 | 5/2 <sup>+</sup>        |                    |             |                         |                                                                                                                 |
|                                      |                                    | 5983.9             | 5.5 28              | 2645.67 | 7/2 <sup>+</sup>        |                    |             |                         |                                                                                                                 |
|                                      |                                    | 6866.2             | 100 26              | 1763.15 | 5/2 <sup>+</sup>        | (E1+M2)            | -0.011 3    |                         | Mult., $\delta$ : D+Q and $\delta$ from $\gamma(\theta)$ in (p, $\gamma$ ); $\Delta\pi$ =yes from level scheme. |
| 8640                                 | (5/2 <sup>+</sup> )                | 8629.0             | 1.1 7               | 0.0     | 3/2 <sup>+</sup>        |                    |             |                         |                                                                                                                 |
|                                      |                                    | 4697               | 61                  | 3942.9  | 9/2 <sup>+</sup>        |                    |             |                         |                                                                                                                 |
|                                      |                                    | 5477               | 53                  | 3162.87 | 7/2 <sup>-</sup>        |                    |             |                         |                                                                                                                 |
|                                      |                                    | 5637               | 16                  | 3002.64 | 5/2 <sup>+</sup>        |                    |             |                         |                                                                                                                 |
|                                      |                                    | 5946               | 34                  | 2694.00 | 3/2 <sup>+</sup>        |                    |             |                         |                                                                                                                 |
|                                      |                                    | 6876               | 100                 | 1763.15 | 5/2 <sup>+</sup>        |                    |             |                         |                                                                                                                 |
| 8641.4                               | 3/2 <sup>+</sup> ,5/2 <sup>+</sup> | 2986               | 10 6                | 5655    | (3/2 <sup>+</sup> )     |                    |             |                         |                                                                                                                 |
|                                      |                                    | 3055.3             | 9 5                 | 5586.0  | 5/2 <sup>+</sup>        |                    |             |                         |                                                                                                                 |
|                                      |                                    | 3237.9             | 17 9                | 5403.3  | 3/2 <sup>-</sup>        |                    |             |                         |                                                                                                                 |
|                                      |                                    | 3802.1             | 9 5                 | 4839.10 | (1/2 <sup>+</sup> ,3/2) |                    |             |                         |                                                                                                                 |
|                                      |                                    | 3871.3             | 86 22               | 4769.86 | (5/2 <sup>-</sup> )     |                    |             |                         |                                                                                                                 |
|                                      |                                    | 4016.9             | 7 4                 | 4624.3  | 3/2,5/2                 |                    |             |                         |                                                                                                                 |
|                                      |                                    | 4463.2             | <64                 | 4177.9  | 3/2 <sup>-</sup>        |                    |             |                         | I $_\gamma$ : total I $_\gamma$ =50 14 for $\gamma$ decays to 4173 and 4178 doublet levels (1976Sp08).          |
|                                      |                                    | 4467.7             | <64                 | 4173.43 | (5/2 <sup>+</sup> )     |                    |             |                         | I $_\gamma$ : total I $_\gamma$ =50 14 for $\gamma$ decays to 4173 and 4178 doublet levels (1976Sp08).          |
|                                      |                                    | 4582.0             | 32 18               | 4059.1  | (3/2 <sup>-</sup> )     |                    |             |                         |                                                                                                                 |
|                                      |                                    | 4722.6             | 21 11               | 3918.47 | 3/2 <sup>+</sup>        |                    |             |                         |                                                                                                                 |
|                                      |                                    | 5946.9             | 19 9                | 2694.00 | 3/2 <sup>+</sup>        |                    |             |                         |                                                                                                                 |
|                                      |                                    | 8640.3             | 100 25              | 0.0     | 3/2 <sup>+</sup>        |                    |             |                         |                                                                                                                 |
| 8686.2                               | 5/2 <sup>-</sup>                   | 3100.1             | 2.8 15              | 5586.0  | 5/2 <sup>+</sup>        |                    |             |                         |                                                                                                                 |
|                                      |                                    | 3282.7             | 4.1 21              | 5403.3  | 3/2 <sup>-</sup>        |                    |             |                         |                                                                                                                 |
|                                      |                                    | 4508.0             | 2.6 13              | 4177.9  | 3/2 <sup>-</sup>        |                    |             |                         |                                                                                                                 |
|                                      |                                    | 4626.8             | 4.6 24              | 4059.1  | (3/2 <sup>-</sup> )     |                    |             |                         |                                                                                                                 |
|                                      |                                    | 5522.9             | 4.9 25              | 3162.87 | 7/2 <sup>-</sup>        |                    |             |                         |                                                                                                                 |
|                                      |                                    | 5991.7             | 3.7 19              | 2694.00 | 3/2 <sup>+</sup>        |                    |             |                         |                                                                                                                 |
|                                      |                                    | 6922.3             | 6.3 32              | 1763.15 | 5/2 <sup>+</sup>        |                    |             |                         |                                                                                                                 |
|                                      |                                    | 7466.0             | 2.5 13              | 1219.38 | 1/2 <sup>+</sup>        |                    |             |                         |                                                                                                                 |

Adopted Levels, Gammas (continued)

| $\gamma(^{35}\text{Cl})$ (continued) |                                         |                    |                     |         |                                    |         |                         |                                                                                                                                                                       |
|--------------------------------------|-----------------------------------------|--------------------|---------------------|---------|------------------------------------|---------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $E_i(\text{level})$                  | $J_i^\pi$                               | $E_\gamma^\dagger$ | $I_\gamma^\ddagger$ | $E_f$   | $J_f^\pi$                          | Mult. # | $\alpha^a$              | Comments                                                                                                                                                              |
| 8686.2                               | 5/2 <sup>-</sup>                        | 8685.0             | 100 11              | 0.0     | 3/2 <sup>+</sup>                   |         |                         |                                                                                                                                                                       |
| 8696.9                               | 3/2 <sup>-</sup>                        | 2890.2             | 3.6 19              | 5806.6  | (1/2,3/2,5/2)                      |         |                         |                                                                                                                                                                       |
|                                      |                                         | 3098.4             | 6.0 30              | 5598.4  | 3/2 <sup>+</sup> ,5/2 <sup>+</sup> |         |                         |                                                                                                                                                                       |
|                                      |                                         | 7476.7             | 15 4                | 1219.38 | 1/2 <sup>+</sup>                   |         |                         |                                                                                                                                                                       |
|                                      |                                         | 8695.7             | 100 10              | 0.0     | 3/2 <sup>+</sup>                   |         |                         |                                                                                                                                                                       |
| 8750.9                               | 3/2 <sup>-</sup>                        | 2944.2             | 4.2 21              | 5806.6  | (1/2,3/2,5/2)                      |         |                         |                                                                                                                                                                       |
|                                      |                                         | 3347.4             | 9 5                 | 5403.3  | 3/2 <sup>-</sup>                   |         |                         |                                                                                                                                                                       |
|                                      |                                         | 3536.2             | 4.5 23              | 5214.5  | (5/2 <sup>+</sup> )                |         |                         |                                                                                                                                                                       |
|                                      |                                         | 4572.7             | 4.0 21              | 4177.9  | 3/2 <sup>-</sup>                   |         |                         |                                                                                                                                                                       |
|                                      |                                         | 4577.2             | 4.3 23              | 4173.43 | (5/2 <sup>+</sup> )                |         |                         |                                                                                                                                                                       |
|                                      |                                         | 6056.3             | 5.7 28              | 2694.00 | 3/2 <sup>+</sup>                   |         |                         |                                                                                                                                                                       |
|                                      |                                         | 6987.0             | 57 15               | 1763.15 | 5/2 <sup>+</sup>                   |         |                         |                                                                                                                                                                       |
|                                      |                                         | 7530.7             | 3.8                 | 1219.38 | 1/2 <sup>+</sup>                   |         |                         |                                                                                                                                                                       |
|                                      |                                         | 8749.7             | 100 10              | 0.0     | 3/2 <sup>+</sup>                   |         |                         |                                                                                                                                                                       |
| 8779.8                               | 3/2 <sup>-</sup>                        | 3181.2             | 16 8                | 5598.4  | 3/2 <sup>+</sup> ,5/2 <sup>+</sup> |         |                         |                                                                                                                                                                       |
|                                      |                                         | 3565.1             | 24 12               | 5214.5  | (5/2 <sup>+</sup> )                |         |                         |                                                                                                                                                                       |
|                                      |                                         | 4601.6             | <71                 | 4177.9  | 3/2 <sup>-</sup>                   |         |                         | $I_\gamma$ : total $I_\gamma=56$ 15 for $\gamma$ decays to 4173 and 4178 doublet levels (1976Sp08).                                                                   |
|                                      |                                         | 4606.0             | <71                 | 4173.43 | (5/2 <sup>+</sup> )                |         |                         | $I_\gamma$ : total $I_\gamma=56$ 15 for $\gamma$ decays to 4173 and 4178 doublet levels (1976Sp08).                                                                   |
|                                      |                                         | 4861.0             | 16 8                | 3918.47 | 3/2 <sup>+</sup>                   |         |                         |                                                                                                                                                                       |
|                                      |                                         | 5776.7             | 100 26              | 3002.64 | 5/2 <sup>+</sup>                   |         |                         |                                                                                                                                                                       |
|                                      |                                         | 6085.2             | 44 11               | 2694.00 | 3/2 <sup>+</sup>                   |         |                         |                                                                                                                                                                       |
|                                      |                                         | 6133.6             | 48 11               | 2645.67 | 7/2 <sup>+</sup>                   |         |                         |                                                                                                                                                                       |
|                                      |                                         | 7015.9             | 25 13               | 1763.15 | 5/2 <sup>+</sup>                   |         |                         |                                                                                                                                                                       |
|                                      |                                         | 8778.6             | 41 11               | 0.0     | 3/2 <sup>+</sup>                   |         |                         |                                                                                                                                                                       |
| 8787.2                               | (5/2 <sup>+</sup> )                     | 3063.5             | 27 14               | 5723.52 | 5/2 <sup>+</sup>                   |         |                         |                                                                                                                                                                       |
|                                      |                                         | 3623.7             | 76 20               | 5163.34 | 7/2 <sup>-</sup>                   |         |                         |                                                                                                                                                                       |
|                                      |                                         | 4017.1             | 27 14               | 4769.86 | (5/2 <sup>-</sup> )                |         |                         |                                                                                                                                                                       |
|                                      |                                         | 4609.0             | 36 20               | 4177.9  | 3/2 <sup>-</sup>                   |         |                         |                                                                                                                                                                       |
|                                      |                                         | 5623.8             | 23 12               | 3162.87 | 7/2 <sup>-</sup>                   |         |                         |                                                                                                                                                                       |
|                                      |                                         | 5784.1             | 80 20               | 3002.64 | 5/2 <sup>+</sup>                   |         |                         |                                                                                                                                                                       |
|                                      |                                         | 7023.3             | 100 24              | 1763.15 | 5/2 <sup>+</sup>                   |         |                         |                                                                                                                                                                       |
|                                      |                                         | 8786.0             | 32 16               | 0.0     | 3/2 <sup>+</sup>                   |         |                         |                                                                                                                                                                       |
| 8788.3                               | (15/2 <sup>+</sup> )                    | 915.4 4            | 100                 | 7873.03 | 13/2 <sup>+</sup>                  | (M1)    | $4.82 \times 10^{-5}$ 7 | $E_\gamma$ , Mult.: from ( $^{16}\text{O}, \alpha p \gamma$ ), with Mult=D, $\Delta J=1$ from $\gamma\gamma(\text{ADO})$ ; $\Delta\pi=(\text{no})$ from level scheme. |
| 8798.4                               | (3/2 <sup>+</sup> ,5/2 <sup>+</sup> )   | 4739.0             | 13 7                | 4059.1  | (3/2 <sup>-</sup> )                |         |                         |                                                                                                                                                                       |
|                                      |                                         | 5795.2             | 26 7                | 3002.64 | 5/2 <sup>+</sup>                   |         |                         |                                                                                                                                                                       |
|                                      |                                         | 7034.5             | 11 6                | 1763.15 | 5/2 <sup>+</sup>                   |         |                         |                                                                                                                                                                       |
|                                      |                                         | 7578.1             | 11 6                | 1219.38 | 1/2 <sup>+</sup>                   |         |                         |                                                                                                                                                                       |
|                                      |                                         | 8797.2             | 100 10              | 0.0     | 3/2 <sup>+</sup>                   |         |                         |                                                                                                                                                                       |
| 8820.9                               | (1/2 <sup>+</sup> to 7/2 <sup>+</sup> ) | 5817.7             | 17 10               | 3002.64 | 5/2 <sup>+</sup>                   |         |                         |                                                                                                                                                                       |

Adopted Levels, Gammas (continued)

| $\gamma(^{35}\text{Cl})$ (continued) |                                         |                    |                     |         |                          |         |                         |                                                                                                                                                |
|--------------------------------------|-----------------------------------------|--------------------|---------------------|---------|--------------------------|---------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| $E_i(\text{level})$                  | $J_i^\pi$                               | $E_\gamma^\dagger$ | $I_\gamma^\ddagger$ | $E_f$   | $J_f^\pi$                | Mult. # | $\alpha^a$              | Comments                                                                                                                                       |
| 8820.9                               | (1/2 <sup>+</sup> to 7/2 <sup>+</sup> ) | 6126.3             | 35 10               | 2694.00 | 3/2 <sup>+</sup>         |         |                         |                                                                                                                                                |
|                                      |                                         | 7057.0             | 40 10               | 1763.15 | 5/2 <sup>+</sup>         |         |                         |                                                                                                                                                |
|                                      |                                         | 8819.7             | 100 10              | 0.0     | 3/2 <sup>+</sup>         |         |                         |                                                                                                                                                |
| 8824.2                               | 1/2 <sup>+</sup>                        | 5821.0             | 100 24              | 3002.64 | 5/2 <sup>+</sup>         |         |                         |                                                                                                                                                |
|                                      |                                         | 6129.6             | 16.2 8              | 2694.00 | 3/2 <sup>+</sup>         |         |                         |                                                                                                                                                |
|                                      |                                         | 7060.3             | 43 11               | 1763.15 | 5/2 <sup>+</sup>         |         |                         |                                                                                                                                                |
|                                      |                                         | 7603.9             | 60 16               | 1219.38 | 1/2 <sup>+</sup>         |         |                         |                                                                                                                                                |
|                                      |                                         | 8823.0             | 51 14               | 0.0     | 3/2 <sup>+</sup>         |         |                         |                                                                                                                                                |
| 8829.3                               | 1/2 <sup>-</sup>                        | 3174               | 30 15               | 5655    | (3/2) <sup>+</sup>       |         |                         |                                                                                                                                                |
|                                      |                                         | 3990.0             | 3.9 21              | 4839.10 | (1/2 <sup>+</sup> , 3/2) |         |                         |                                                                                                                                                |
|                                      |                                         | 4059.2             | 4.6 24              | 4769.86 | (5/2 <sup>-</sup> )      |         |                         |                                                                                                                                                |
|                                      |                                         | 4910.5             | 14 7                | 3918.47 | 3/2 <sup>+</sup>         |         |                         |                                                                                                                                                |
|                                      |                                         | 5826.1             | 67 18               | 3002.64 | 5/2 <sup>+</sup>         |         |                         |                                                                                                                                                |
|                                      |                                         | 6134.7             | 39 9                | 2694.00 | 3/2 <sup>+</sup>         |         |                         |                                                                                                                                                |
|                                      |                                         | 6183.0             | 100 24              | 2645.67 | 7/2 <sup>+</sup>         |         |                         |                                                                                                                                                |
|                                      |                                         | 7065.4             | 27 15               | 1763.15 | 5/2 <sup>+</sup>         |         |                         |                                                                                                                                                |
|                                      |                                         | 7609.0             | 8 4                 | 1219.38 | 1/2 <sup>+</sup>         |         |                         |                                                                                                                                                |
|                                      |                                         | 8828.1             | 12 6                | 0.0     | 3/2 <sup>+</sup>         |         |                         |                                                                                                                                                |
|                                      |                                         | 3618.4             | 30 15               | 5214.5  | (5/2 <sup>+</sup> )      |         |                         |                                                                                                                                                |
|                                      |                                         | 4063.0             | 11 6                | 4769.86 | (5/2 <sup>-</sup> )      |         |                         |                                                                                                                                                |
|                                      |                                         | 4659.3             | 61 18               | 4173.43 | (5/2 <sup>+</sup> )      |         |                         |                                                                                                                                                |
|                                      |                                         | 4720.4             | 20 11               | 4112.4  | 7/2 <sup>+</sup>         |         |                         |                                                                                                                                                |
| 8833.1                               | (5/2, 7/2)                              | 4773.7             | 48 13               | 4059.1  | (3/2) <sup>-</sup>       |         |                         |                                                                                                                                                |
|                                      |                                         | 4914.3             | 48 13               | 3918.47 | 3/2 <sup>+</sup>         |         |                         |                                                                                                                                                |
|                                      |                                         | 5669.7             | 16 8                | 3162.87 | 7/2 <sup>-</sup>         |         |                         |                                                                                                                                                |
|                                      |                                         | 5829.9             | 29 15               | 3002.64 | 5/2 <sup>+</sup>         |         |                         |                                                                                                                                                |
|                                      |                                         | 6138.5             | 20 10               | 2694.00 | 3/2 <sup>+</sup>         |         |                         |                                                                                                                                                |
|                                      |                                         | 6186.8             | 100 26              | 2645.67 | 7/2 <sup>+</sup>         |         |                         |                                                                                                                                                |
|                                      |                                         | 7069.2             | 39 22               | 1763.15 | 5/2 <sup>+</sup>         |         |                         |                                                                                                                                                |
|                                      |                                         | 8831.9             | 13 7                | 0.0     | 3/2 <sup>+</sup>         |         |                         |                                                                                                                                                |
|                                      |                                         | 3674.6             |                     | 5163.34 | 7/2 <sup>-</sup>         |         |                         |                                                                                                                                                |
|                                      |                                         | 3956.9             |                     | 4880.96 | 7/2 <sup>(+)</sup>       |         |                         |                                                                                                                                                |
|                                      |                                         | 4068.0             | 13 7                | 4769.86 | (5/2 <sup>-</sup> )      |         |                         |                                                                                                                                                |
|                                      |                                         | 4490.0             |                     | 4347.76 | 9/2 <sup>-</sup>         |         |                         |                                                                                                                                                |
|                                      |                                         | 4778.7             |                     | 4059.1  | (3/2) <sup>-</sup>       |         |                         |                                                                                                                                                |
|                                      |                                         | 5674.7             | 46 11               | 3162.87 | 7/2 <sup>-</sup>         |         |                         |                                                                                                                                                |
| 8844.4                               | 17/2 <sup>+</sup>                       | 7074.2             | 100 10              | 1763.15 | 5/2 <sup>+</sup>         |         |                         |                                                                                                                                                |
|                                      |                                         | 524.8              |                     | 8319.6  | 15/2 <sup>-</sup>        |         |                         |                                                                                                                                                |
|                                      |                                         | 971.38 20          | 100.0 30            | 7873.03 | 13/2 <sup>+</sup>        | E2      | 6.16×10 <sup>-5</sup> 9 | B(E2)(W.u.)=16.3 +37-26<br>E <sub>γ</sub> : weighted average of 971.4 5 from ( <sup>16</sup> O,αpγ) and 971.38 20 from ( <sup>14</sup> N,αpγ). |

Adopted Levels, Gammas (continued)

| <u><math>\gamma(^{35}\text{Cl})</math> (continued)</u> |                                       |                                      |                                       |                         |                                         |                 |
|--------------------------------------------------------|---------------------------------------|--------------------------------------|---------------------------------------|-------------------------|-----------------------------------------|-----------------|
| <u><math>E_i(\text{level})</math></u>                  | <u><math>J_i^\pi</math></u>           | <u><math>E_\gamma^\dagger</math></u> | <u><math>I_\gamma^\ddagger</math></u> | <u><math>E_f</math></u> | <u><math>J_f^\pi</math></u>             | <u>Comments</u> |
| 8844.4                                                 | 17/2 <sup>+</sup>                     | 2757.0                               |                                       | 6087.31                 | 13/2 <sup>-</sup>                       |                 |
| 8856.2                                                 | 5/2 <sup>+</sup>                      | 3975.0                               | 6.7 33                                | 4880.96                 | 7/2 <sup>(+)</sup>                      |                 |
|                                                        |                                       | 4231.6                               | 10 5                                  | 4624.3                  | 3/2,5/2                                 |                 |
|                                                        |                                       | 4678.0                               | 7 4                                   | 4177.9                  | 3/2 <sup>-</sup>                        |                 |
|                                                        |                                       | 4937.4                               | 22 6                                  | 3918.47                 | 3/2 <sup>+</sup>                        |                 |
|                                                        |                                       | 5692.8                               | 100 10                                | 3162.87                 | 7/2 <sup>-</sup>                        |                 |
|                                                        |                                       | 6161.6                               | 5.5 28                                | 2694.00                 | 3/2 <sup>+</sup>                        |                 |
|                                                        |                                       | 6209.9                               |                                       | 2645.67                 | 7/2 <sup>+</sup>                        |                 |
|                                                        |                                       | 7092.3                               | 14 8                                  | 1763.15                 | 5/2 <sup>+</sup>                        |                 |
|                                                        |                                       | 8855.0                               | 31 8                                  | 0.0                     | 3/2 <sup>+</sup>                        |                 |
| 8868.6                                                 | 3/2 <sup>+</sup> ,5/2 <sup>+</sup>    | 4690.4                               | 5.3 27                                | 4177.9                  | 3/2 <sup>-</sup>                        |                 |
|                                                        |                                       | 5865.4                               | 33 8                                  | 3002.64                 | 5/2 <sup>+</sup>                        |                 |
|                                                        |                                       | 6174.0                               | 25 6                                  | 2694.00                 | 3/2 <sup>+</sup>                        |                 |
|                                                        |                                       | 7104.7                               | 9 5                                   | 1763.15                 | 5/2 <sup>+</sup>                        |                 |
|                                                        |                                       | 7648.3                               | 33 8                                  | 1219.38                 | 1/2 <sup>+</sup>                        |                 |
|                                                        |                                       | 8867.4                               | 100 25                                | 0.0                     | 3/2 <sup>+</sup>                        |                 |
| 8886.1                                                 | (5/2)                                 | 3231                                 | 23 12                                 | 5655                    | (3/2) <sup>+</sup>                      |                 |
|                                                        |                                       | 3287.5                               | 25 13                                 | 5598.4                  | 3/2 <sup>+</sup> ,5/2 <sup>+</sup>      |                 |
|                                                        |                                       | 3671.4                               | 9 4                                   | 5214.5                  | (5/2 <sup>+</sup> )                     |                 |
|                                                        |                                       | 3722.6                               | 35 16                                 | 5163.34                 | 7/2 <sup>-</sup>                        |                 |
|                                                        |                                       | 4116.0                               | 17 9                                  | 4769.86                 | (5/2 <sup>-</sup> )                     |                 |
|                                                        |                                       | 4707.9                               | 31 16                                 | 4177.9                  | 3/2 <sup>-</sup>                        |                 |
|                                                        |                                       | 4967.3                               | 16 8                                  | 3918.47                 | 3/2 <sup>+</sup>                        |                 |
|                                                        |                                       | 5882.9                               | 31 16                                 | 3002.64                 | 5/2 <sup>+</sup>                        |                 |
|                                                        |                                       | 6239.8                               | 100 27                                | 2645.67                 | 7/2 <sup>+</sup>                        |                 |
|                                                        |                                       | 7122.2                               | 46 12                                 | 1763.15                 | 5/2 <sup>+</sup>                        |                 |
|                                                        |                                       | 8884.9                               | 54 16                                 | 0.0                     | 3/2 <sup>+</sup>                        |                 |
| 8893.3                                                 | (5/2 <sup>+</sup> ,7/2 <sup>+</sup> ) | 3307.1                               | 4.3 22                                | 5586.0                  | 5/2 <sup>+</sup>                        |                 |
|                                                        |                                       | 4780.6                               | 24 11                                 | 4112.4                  | 7/2 <sup>+</sup>                        |                 |
|                                                        |                                       | 4950.0                               | 11 6                                  | 3942.9                  | 9/2 <sup>+</sup>                        |                 |
|                                                        |                                       | 5729.9                               | 78 19                                 | 3162.87                 | 7/2 <sup>-</sup>                        |                 |
|                                                        |                                       | 6198.7                               | 100 24                                | 2694.00                 | 3/2 <sup>+</sup>                        |                 |
|                                                        |                                       | 7129.4                               | 51 14                                 | 1763.15                 | 5/2 <sup>+</sup>                        |                 |
|                                                        |                                       | 8892.1                               | 2.7 14                                | 0.0                     | 3/2 <sup>+</sup>                        |                 |
| 8904.8                                                 | (1/2,3/2,5/2 <sup>+</sup> )           | 2724                                 | 15 8                                  | 6181                    | (1/2 <sup>+</sup> to 7/2 <sup>+</sup> ) |                 |
|                                                        |                                       | 3250                                 | 12 6                                  | 5655                    | (3/2) <sup>+</sup>                      |                 |
|                                                        |                                       | 3895.5                               | 13 7                                  | 5009.1                  | (1/2,3/2)                               |                 |
|                                                        |                                       | 4050.2                               | 21 11                                 | 4854.4                  | (1/2,3/2)                               |                 |
|                                                        |                                       | 4065.5                               | 16 8                                  | 4839.10                 | (1/2 <sup>+</sup> ,3/2)                 |                 |
|                                                        |                                       | 4726.6                               | 62 17                                 | 4177.9                  | 3/2 <sup>-</sup>                        |                 |
|                                                        |                                       | 4845.3                               | 20 10                                 | 4059.1                  | (3/2) <sup>-</sup>                      |                 |
|                                                        |                                       | 6210.2                               | 52 14                                 | 2694.00                 | 3/2 <sup>+</sup>                        |                 |

Adopted Levels, Gammas (continued)

| $E_i(\text{level})$ | $J_i^\pi$                                 | $E_\gamma^\dagger$ | $I_\gamma^\ddagger$ | $E_f$   | $J_f^\pi$                          | $\gamma(^{35}\text{Cl})$ (continued) |             | Comments                                                                                              |
|---------------------|-------------------------------------------|--------------------|---------------------|---------|------------------------------------|--------------------------------------|-------------|-------------------------------------------------------------------------------------------------------|
|                     |                                           |                    |                     |         |                                    | Mult. #                              | $\delta^\#$ |                                                                                                       |
| 8904.8              | (1/2,3/2,5/2 <sup>+</sup> )               | 7684.5             | 35 10               | 1219.38 | 1/2 <sup>+</sup>                   |                                      |             |                                                                                                       |
|                     |                                           | 8903.6             | 100 24              | 0.0     | 3/2 <sup>+</sup>                   |                                      |             |                                                                                                       |
| 8906.8              | 5/2 <sup>+</sup>                          | 3252               |                     | 5655    | (3/2) <sup>+</sup>                 |                                      |             |                                                                                                       |
|                     |                                           | 3261               | 5.4 28              | 5646    | (5/2,7/2,9/2 <sup>+</sup> )        |                                      |             |                                                                                                       |
|                     |                                           | 3308.2             | 3.2 16              | 5598.4  | 3/2 <sup>+</sup> ,5/2 <sup>+</sup> |                                      |             |                                                                                                       |
|                     |                                           | 3320.6             | 5.8                 | 5586.0  | 5/2 <sup>+</sup>                   |                                      |             |                                                                                                       |
|                     |                                           | 3376               | 3.6 19              | 5531    |                                    |                                      |             |                                                                                                       |
|                     |                                           | 3692.1             | 6.4 32              | 5214.5  | (5/2 <sup>+</sup> )                |                                      |             |                                                                                                       |
|                     |                                           | 4136.7             | 8 4                 | 4769.86 | (5/2 <sup>-</sup> )                |                                      |             |                                                                                                       |
|                     |                                           | 4963.5             | 19 5                | 3942.9  | 9/2 <sup>+</sup>                   |                                      |             |                                                                                                       |
|                     |                                           | 7142.9             | 100 10              | 1763.15 | 5/2 <sup>+</sup>                   |                                      |             |                                                                                                       |
| 8919.8              | (5/2 <sup>-</sup> ,7/2,9/2 <sup>+</sup> ) | 3274               | 10 5                | 5646    | (5/2,7/2,9/2 <sup>+</sup> )        |                                      |             |                                                                                                       |
|                     |                                           | 4571.7             | 14 7                | 4347.76 | 9/2 <sup>-</sup>                   |                                      |             |                                                                                                       |
|                     |                                           | 5756.4             | 100 11              | 3162.87 | 7/2 <sup>-</sup>                   |                                      |             |                                                                                                       |
|                     |                                           | 6273.5             | 46 12               | 2645.67 | 7/2 <sup>+</sup>                   |                                      |             |                                                                                                       |
|                     |                                           | 7155.9             | 7 4                 | 1763.15 | 5/2 <sup>+</sup>                   |                                      |             |                                                                                                       |
| 8953.1              | 3/2 <sup>+</sup>                          | 7732.8             |                     | 1219.38 | 1/2 <sup>+</sup>                   | M1+E2                                | -0.34 2     | Mult., $\delta$ : or $\delta=-5.0$ 4 from $\gamma(\theta)$ in (p, $\gamma$ ). E1+M2 ruled out by RUL. |
|                     |                                           | 8951.9             |                     | 0.0     | 3/2 <sup>+</sup>                   | M1+E2                                | -0.39 5     | Mult., $\delta$ : or $\delta=-8.2$ 7 from $\gamma(\theta)$ in (p, $\gamma$ ). E1+M2 ruled out by RUL. |
| 8984.2              | 5/2 <sup>+</sup>                          | 3260.5             | 1.3 7               | 5723.52 | 5/2 <sup>+</sup>                   |                                      |             |                                                                                                       |
|                     |                                           | 4144.8             | 2.0 10              | 4839.10 | (1/2 <sup>+</sup> ,3/2)            |                                      |             |                                                                                                       |
|                     |                                           | 4214.1             | 0.8 5               | 4769.86 | (5/2 <sup>-</sup> )                |                                      |             |                                                                                                       |
|                     |                                           | 4359.6             | 0.8 5               | 4624.3  | 3/2,5/2                            |                                      |             |                                                                                                       |
|                     |                                           | 4806.0             | 2.5 13              | 4177.9  | 3/2 <sup>-</sup>                   |                                      |             |                                                                                                       |
|                     |                                           | 4810.4             | 2.5 13              | 4173.43 | (5/2 <sup>+</sup> )                |                                      |             |                                                                                                       |
|                     |                                           | 5065.3             | 15 8                | 3918.47 | 3/2 <sup>+</sup>                   |                                      |             |                                                                                                       |
|                     |                                           | 5820.8             | 10 5                | 3162.87 | 7/2 <sup>-</sup>                   |                                      |             |                                                                                                       |
|                     |                                           | 6289.6             | 3.0 15              | 2694.00 | 3/2 <sup>+</sup>                   |                                      |             |                                                                                                       |
|                     |                                           | 6337.9             | 2.0 10              | 2645.67 | 7/2 <sup>+</sup>                   |                                      |             |                                                                                                       |
|                     |                                           | 7220.3             | 27 7                | 1763.15 | 5/2 <sup>+</sup>                   |                                      |             |                                                                                                       |
|                     |                                           | 8983.0             | 100 10              | 0.0     | 3/2 <sup>+</sup>                   |                                      |             |                                                                                                       |
| 9081.5              | 5/2 <sup>+</sup>                          | 3357.8             | 1.7                 | 5723.52 | 5/2 <sup>+</sup>                   |                                      |             |                                                                                                       |
|                     |                                           | 4200.3             | 1.7                 | 4880.96 | 7/2 <sup>(+)</sup>                 |                                      |             |                                                                                                       |
|                     |                                           | 4242.1             |                     | 4839.10 | (1/2 <sup>+</sup> ,3/2)            |                                      |             |                                                                                                       |
|                     |                                           | 4311.4             |                     | 4769.86 | (5/2 <sup>-</sup> )                |                                      |             |                                                                                                       |
|                     |                                           | 4456.9             |                     | 4624.3  | 3/2,5/2                            |                                      |             |                                                                                                       |
|                     |                                           | 4903.2             | 3.3                 | 4177.9  | 3/2 <sup>-</sup>                   |                                      |             |                                                                                                       |
|                     |                                           | 4907.7             | 3.3                 | 4173.43 | (5/2 <sup>+</sup> )                |                                      |             |                                                                                                       |
|                     |                                           | 5162.6             | 15                  | 3918.47 | 3/2 <sup>+</sup>                   | D(+Q)                                | -0.002 9    |                                                                                                       |
|                     |                                           | 5918.1             | 10                  | 3162.87 | 7/2 <sup>-</sup>                   |                                      |             |                                                                                                       |
|                     |                                           | 6386.9             | 3.3                 | 2694.00 | 3/2 <sup>+</sup>                   |                                      |             |                                                                                                       |
|                     |                                           | 6435.2             | 1.7                 | 2645.67 | 7/2 <sup>+</sup>                   |                                      |             |                                                                                                       |

## Adopted Levels, Gammas (continued)

| $\gamma(^{35}\text{Cl})$ (continued) |                      |                    |                     |         |                      |                    |             |                         |                                                                                                                                                                            |
|--------------------------------------|----------------------|--------------------|---------------------|---------|----------------------|--------------------|-------------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $E_i(\text{level})$                  | $J_i^\pi$            | $E_\gamma^\dagger$ | $I_\gamma^\ddagger$ | $E_f$   | $J_f^\pi$            | Mult. <sup>#</sup> | $\delta^\#$ | $\alpha^a$              | Comments                                                                                                                                                                   |
| 9081.5                               | 5/2 <sup>+</sup>     | 7317.5             | 27                  | 1763.15 | 5/2 <sup>+</sup>     | M1+E2              | +0.11 2     |                         | B(M1)(W.u.)=1.16 +31-25; B(E2)(W.u.)=0.99 +50-38                                                                                                                           |
|                                      |                      | 9080.2             | 100                 | 0.0     | 3/2 <sup>+</sup>     | M1+E2              | +0.089 9    |                         | B(M1)(W.u.)=2.26 +49-33; B(E2)(W.u.)=0.82 +27-19                                                                                                                           |
| 9157.1                               | 5/2 <sup>-</sup>     | 3942.4             | 17 5                | 5214.5  | (5/2 <sup>+</sup> )  |                    |             |                         |                                                                                                                                                                            |
|                                      |                      | 4386.9             | 16 4                | 4769.86 | (5/2 <sup>-</sup> )  |                    |             |                         |                                                                                                                                                                            |
|                                      |                      | 4978.8             | <37                 | 4177.9  | 3/2 <sup>-</sup>     |                    |             |                         | $I_\gamma$ : total $I_\gamma=30$ 7 for $\gamma$ decays to 4173 and 4178 doublet final levels (1976Sp08).                                                                   |
|                                      |                      | 4983.3             | <37                 | 4173.43 | (5/2 <sup>+</sup> )  |                    |             |                         | $I_\gamma$ : total $I_\gamma=30$ 7 for $\gamma$ decays to 4173 and 4178 doublet final levels (1976Sp08).                                                                   |
|                                      |                      | 5993.7             | 22 6                | 3162.87 | 7/2 <sup>-</sup>     | M1+E2              | +0.10 2     | 1.59×10 <sup>-3</sup> 2 |                                                                                                                                                                            |
|                                      |                      | 6153.9             | 20 5                | 3002.64 | 5/2 <sup>+</sup>     |                    |             |                         |                                                                                                                                                                            |
|                                      |                      | 6462.5             | 3 2                 | 2694.00 | 3/2 <sup>+</sup>     |                    |             |                         |                                                                                                                                                                            |
|                                      |                      | 6510.8             | 7 4                 | 2645.67 | 7/2 <sup>+</sup>     |                    |             |                         |                                                                                                                                                                            |
|                                      |                      | 7393.1             | 17 5                | 1763.15 | 5/2 <sup>+</sup>     |                    |             |                         |                                                                                                                                                                            |
|                                      |                      | 9155.8             | 100 9               | 0.0     | 3/2 <sup>+</sup>     | (E1)               |             |                         | Mult.: D+Q with $\delta=+0.09$ 1 from $\gamma(\theta)$ in (p, $\gamma$ ); $\Delta\pi$ =yes from level scheme; M2 component is less likely.                                 |
| 10181.1                              | 19/2 <sup>-</sup>    | 1336.3 5           | 100.0 34            | 8844.4  | 17/2 <sup>+</sup>    | (E1)               |             | 0.0001637 23            | B(E1)(W.u.)=0.00235 +21-18                                                                                                                                                 |
|                                      |                      |                    |                     |         |                      |                    |             |                         | $I_\gamma$ : weighted average of 100 17 from ( $^{28}\text{Si},\alpha\gamma$ ) and 100.0 34 from ( $^{16}\text{O},\alpha\gamma$ ).                                         |
|                                      |                      |                    |                     |         |                      |                    |             |                         | $E_\gamma$ ,Mult.: from ( $^{16}\text{O},\alpha\gamma$ ), with Mult=D from $\gamma\gamma(\text{ADO})$ with $\Delta J=1$ ; $\Delta\pi$ =(yes) from level scheme.            |
|                                      |                      | 1693.5 5           | 84 9                | 8487.5  | 15/2 <sup>-</sup>    | E2                 |             | 0.0001880 26            | B(E2)(W.u.)=44.4 46                                                                                                                                                        |
|                                      |                      |                    |                     |         |                      |                    |             |                         | $E_\gamma$ , $I_\gamma$ ,Mult.: from ( $^{16}\text{O},\alpha\gamma$ ), with Mult=Q, $\Delta J=2$ from $\gamma\gamma(\text{ADO})$ and M2 ruled out by RUL.                  |
|                                      |                      | 1861.3 5           | 83 6                | 8319.6  | 15/2 <sup>-</sup>    | E2                 |             | 0.000262 4              | B(E2)(W.u.)=27.4 +26-24                                                                                                                                                    |
|                                      |                      |                    |                     |         |                      |                    |             |                         | $E_\gamma$ , $I_\gamma$ ,Mult.: from ( $^{16}\text{O},\alpha\gamma$ ), with Mult=Q, $\Delta J=2$ from $\gamma\gamma(\text{ADO})$ and M2 ruled out by RUL.                  |
| 10221.9                              | (17/2 <sup>-</sup> ) | 1377.8 10          | 100                 | 8844.4  | 17/2 <sup>+</sup>    |                    |             |                         |                                                                                                                                                                            |
| 10858.5                              | 19/2 <sup>+</sup>    | 2014.7 9           | 100 5               | 8844.4  | 17/2 <sup>+</sup>    | (M1)               |             | 0.000265 4              | $E_\gamma$ , $I_\gamma$ ,Mult.: from ( $^{16}\text{O},\alpha\gamma$ ), with Mult=D from $\gamma\gamma(\text{ADO})$ with $\Delta J=1$ ; $\Delta\pi$ =no from level scheme.  |
|                                      |                      | 2069.8 10          | 32.5 22             | 8788.3  | (15/2 <sup>+</sup> ) | E2                 |             | 0.000361 5              | $E_\gamma$ , $I_\gamma$ ,Mult.: from ( $^{16}\text{O},\alpha\gamma$ ) with Mult=Q, $\Delta J=2$ from $\gamma\gamma(\text{ADO})$ ; M2 ruled out by RUL.                     |
| 11459.1                              | 21/2 <sup>+</sup>    | 2614.5 5           | 100                 | 8844.4  | 17/2 <sup>+</sup>    | E2                 |             | 0.000619 9              | B(E2)(W.u.)=15.9 8                                                                                                                                                         |
|                                      |                      |                    |                     |         |                      |                    |             |                         | $E_\gamma$ ,Mult.: from ( $^{16}\text{O},\alpha\gamma$ ), with Mult=Q, $\Delta J=2$ from $\gamma\gamma(\text{ADO})$ ; M2 ruled out by RUL.                                 |
| 12572.2                              | 23/2 <sup>-</sup>    | 1113.3 5           | 24.4 18             | 11459.1 | 21/2 <sup>+</sup>    | (E1)               |             | 3.60×10 <sup>-5</sup> 5 | B(E1)(W.u.)=0.00272 +41-35                                                                                                                                                 |
|                                      |                      |                    |                     |         |                      |                    |             |                         | $E_\gamma$ , $I_\gamma$ ,Mult.: from ( $^{16}\text{O},\alpha\gamma$ ), with Mult=D from $\gamma\gamma(\text{ADO})$ with $\Delta J=1$ ; $\Delta\pi$ =yes from level scheme. |

**Adopted Levels, Gammas (continued)**

$\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$            | $E_\gamma^\dagger$ | $I_\gamma^\ddagger$ | $E_f$   | $J_f^\pi$         | Mult. <sup>#</sup> | $\alpha^a$ | Comments                                                                                                                                               |
|---------------------|----------------------|--------------------|---------------------|---------|-------------------|--------------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12572.2             | 23/2 <sup>-</sup>    | 2390.8 5           | 100 4               | 10181.1 | 19/2 <sup>-</sup> | E2                 | 0.000515 7 | B(E2)(W.u.)=26.0 +36-29<br>E <sub>γ</sub> , I <sub>γ</sub> , Mult.: from ( <sup>16</sup> O, αpγ), with Mult=Q, ΔJ=2 from γγ(ADO); M2 ruled out by RUL. |
| 16306.4             | (27/2 <sup>-</sup> ) | 3734.0             |                     | 12572.2 | 23/2 <sup>-</sup> |                    |            |                                                                                                                                                        |

<sup>†</sup> Sources of values with uncertainties are listed under comments and those without uncertainties are deduced from level-energy differences, unless otherwise noted.

<sup>‡</sup> From (p,γ), unless otherwise noted.

<sup>#</sup> From γ(θ) and/or γγ(θ) in (p,γ) for γ rays originating from low-spin levels (J<9/2) and from γγ(θ)(DCO) and γ(lin pol) in (<sup>28</sup>Si, αpγ) for γ rays originating from high-spin (J≥9/2) levels, unless otherwise noted. Magnetic/electric characters are determined based on γ(lin pol) or RUL with measured T<sub>1/2</sub>, unless otherwise noted. For primary γ transitions from proton resonances where widths or resonance strengths are provided in (p,γ), a D+Q with definite δ(Q/D)>0 from γ(θ) is firmly assigned as M1+E2. This assignment is due to the fact that for these resonances, the proton width Γ<sub>p</sub> is dominant over Γ<sub>γ</sub> and thus the total width Γ is much greater than Γ<sub>γ</sub>. The total width Γ typically falls within the same scale as the resonance strength, usually a few eV in the <sup>34</sup>S(p,γ) dataset.

@ Primary transitions probably from the unresolved 7501+7503 doublet initial levels in 1976Me12 in (p,γ).

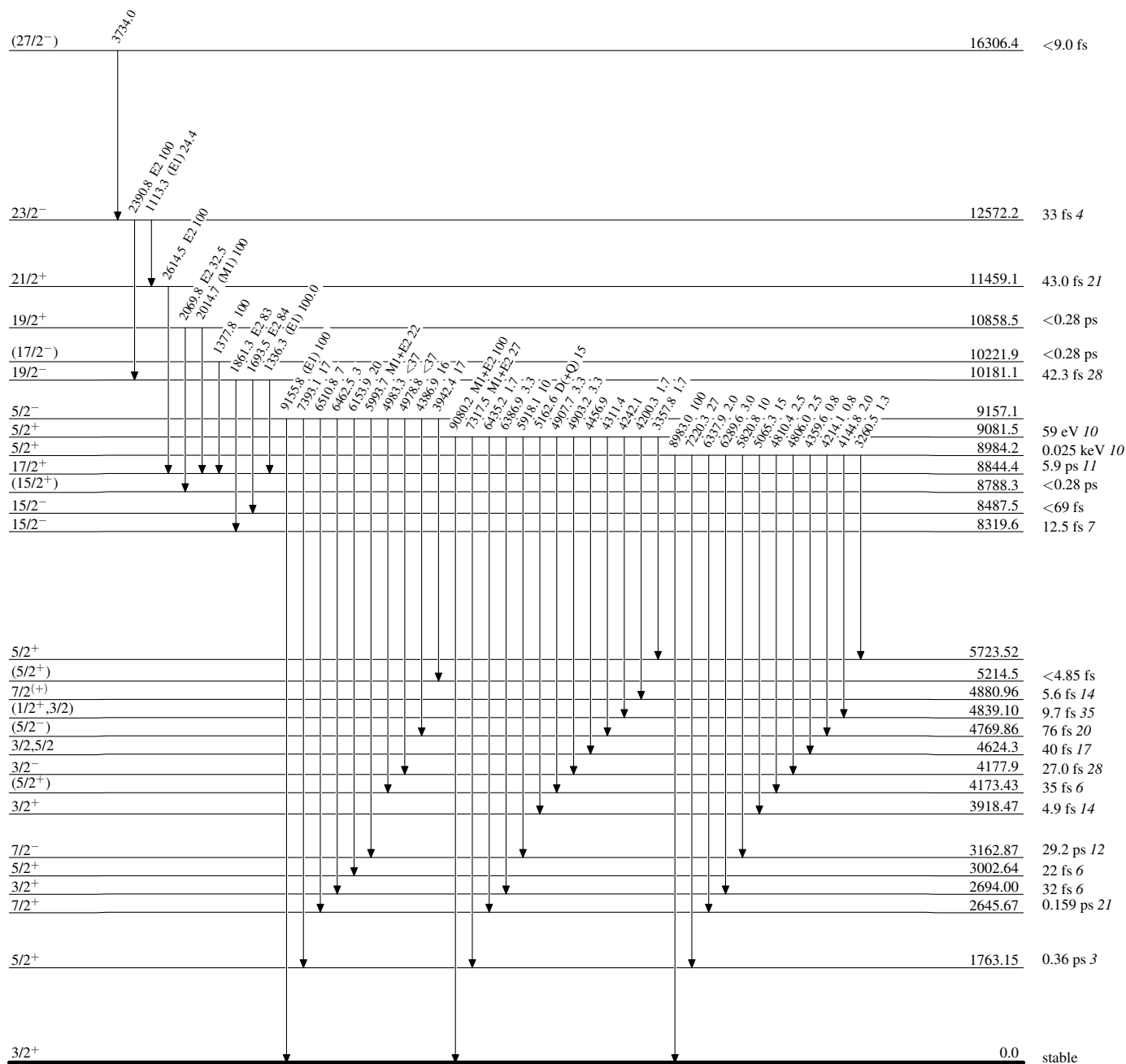
& Primary transitions probably from the unresolved 7837+7839 doublet initial levels in 1976Me12 in (p,γ).

<sup>a</sup> Total theoretical internal conversion coefficients, calculated using the BrIcc code (2008Ki07) with "Frozen Orbitals" approximation based on γ-ray energies, assigned multipolarities, and mixing ratios, unless otherwise specified.

<sup>b</sup> Placement of transition in the level scheme is uncertain.

**Adopted Levels, Gammas****Level Scheme**

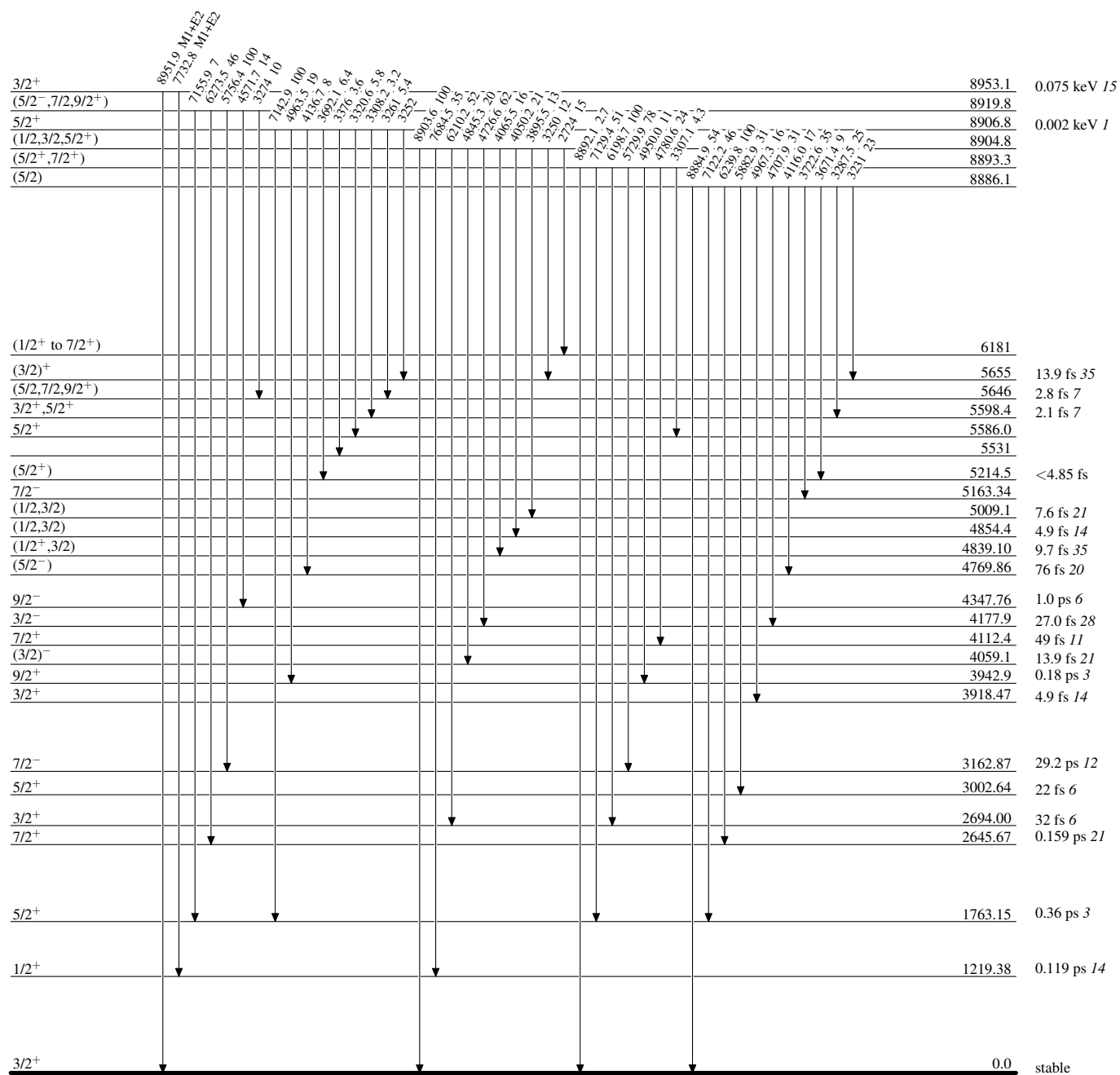
Intensities: Relative photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$



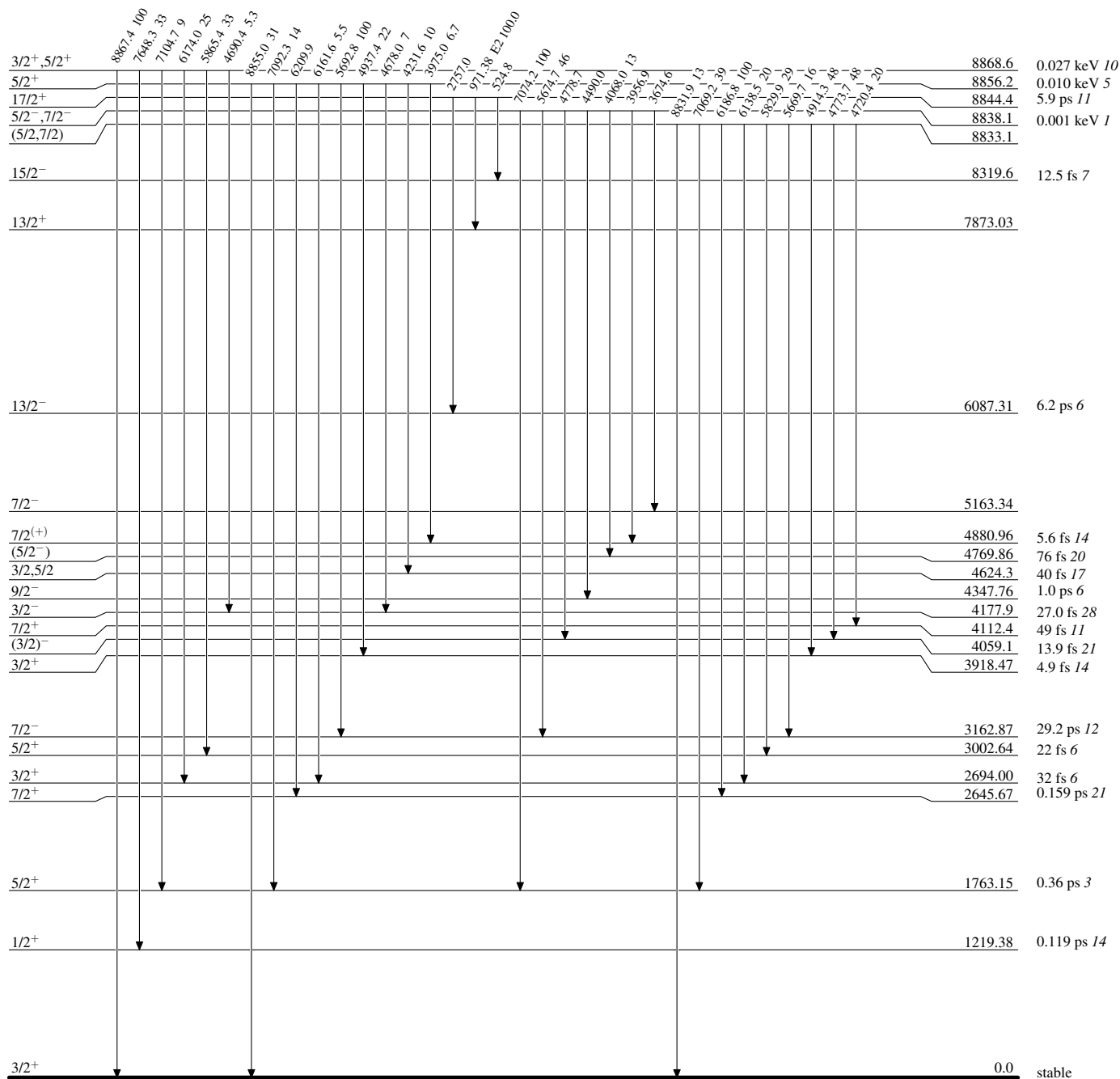
Adopted Levels, GammasLevel Scheme (continued)

Intensities: Relative photon branching from each level



**Adopted Levels, Gammas****Level Scheme (continued)**

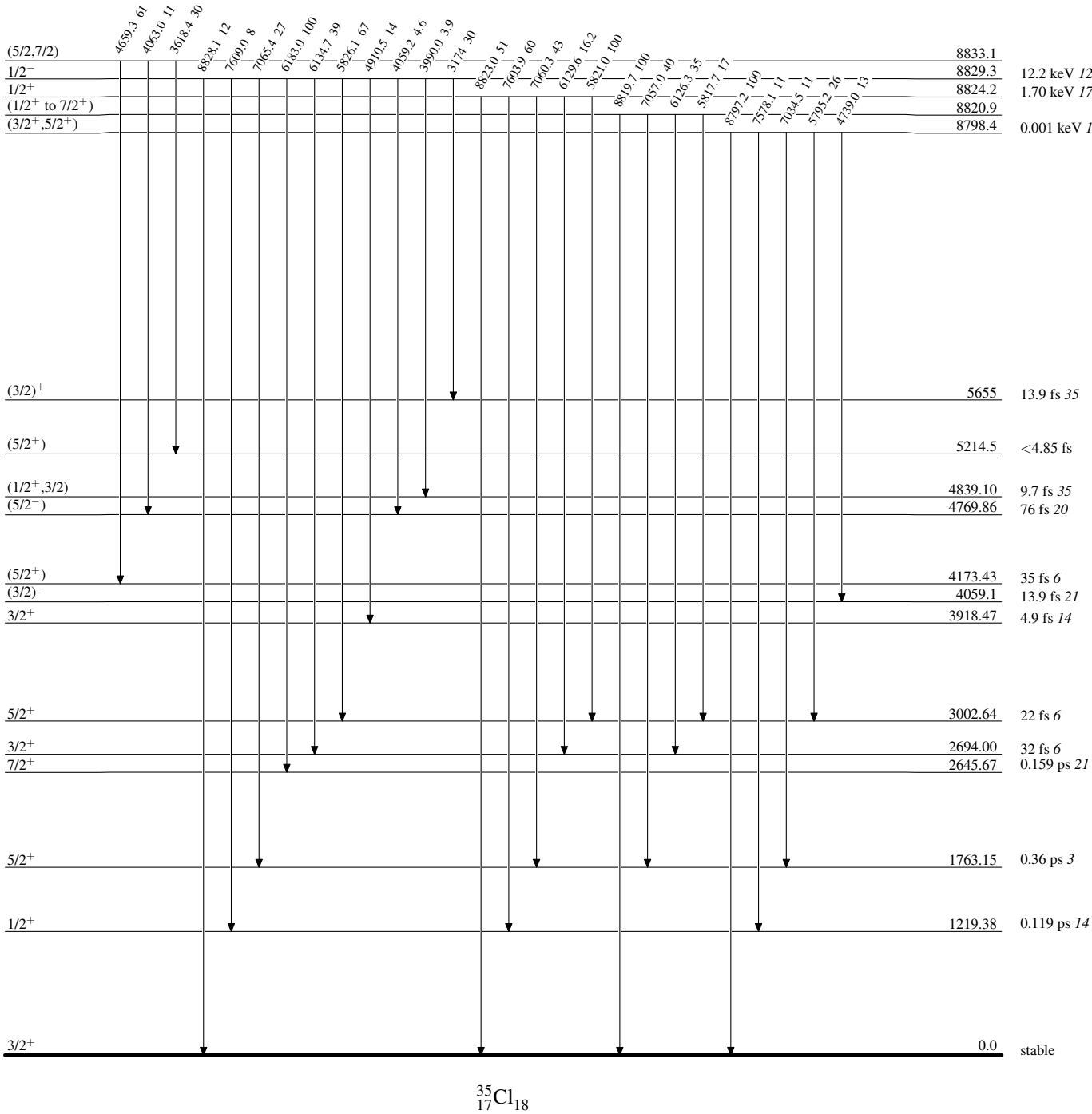
Intensities: Relative photon branching from each level



Adopted Levels, Gammas

Level Scheme (continued)

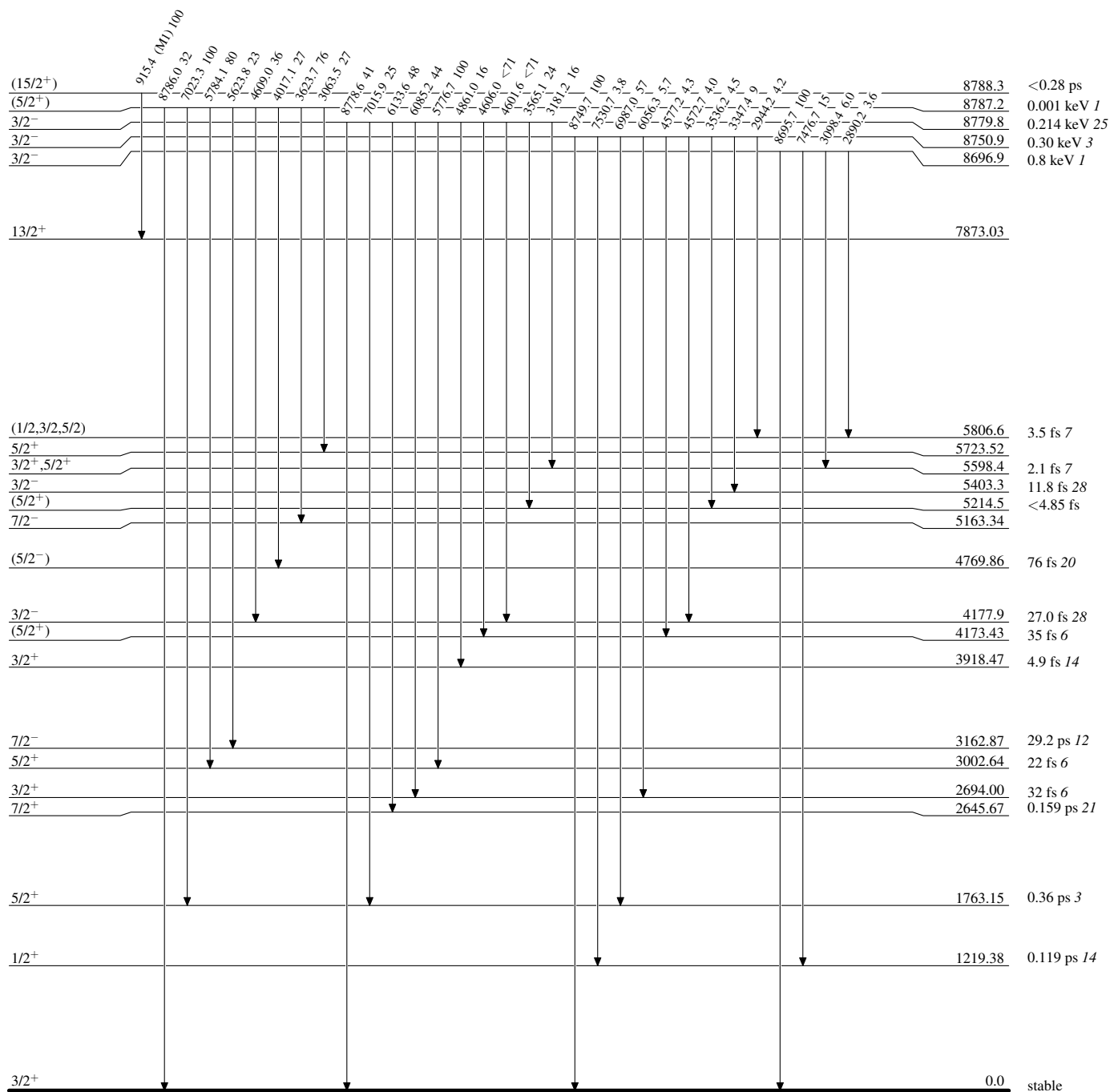
Intensities: Relative photon branching from each level



<sup>35</sup><sub>17</sub>Cl<sub>18</sub>

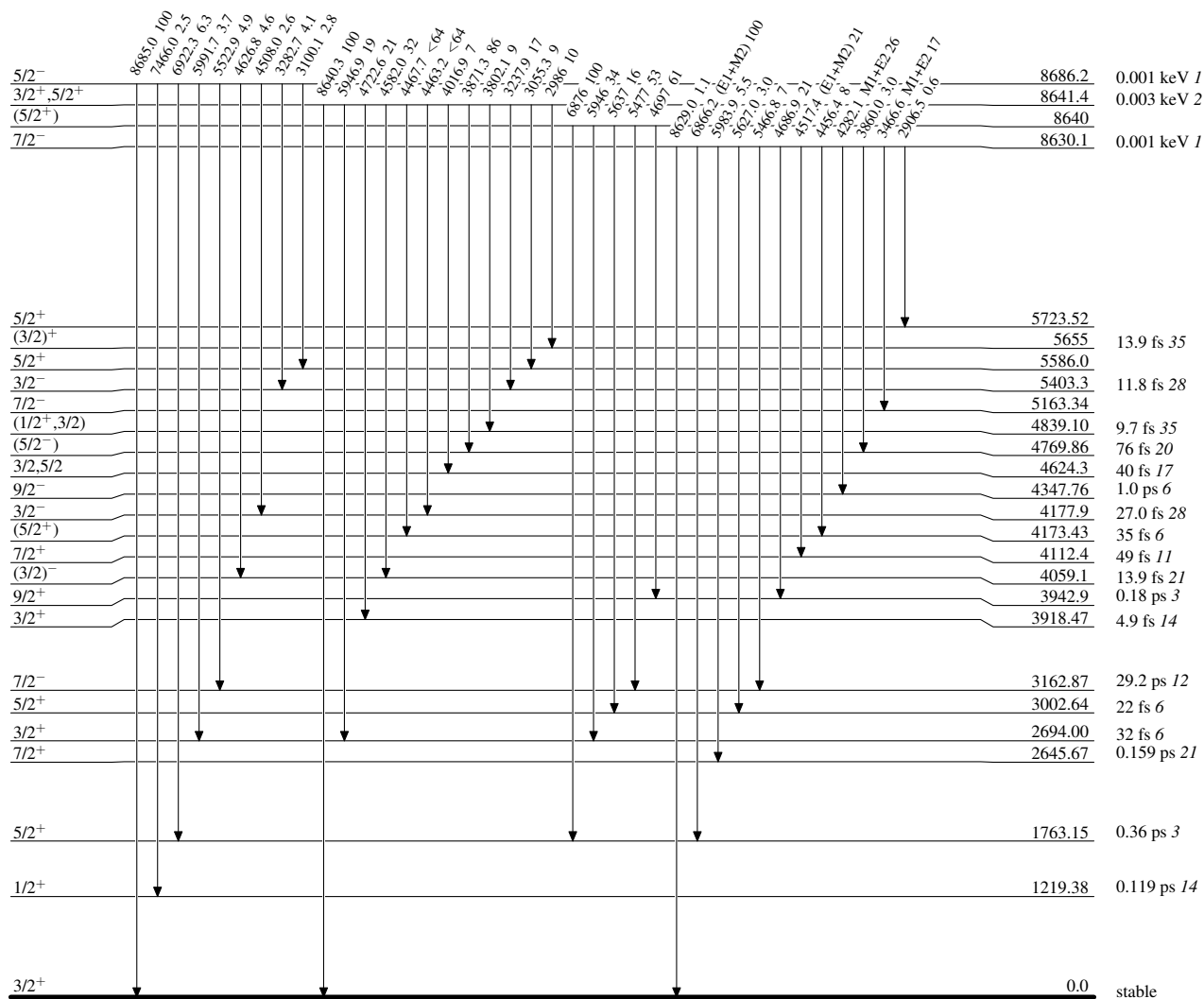
**Adopted Levels, Gammas****Level Scheme (continued)**

Intensities: Relative photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

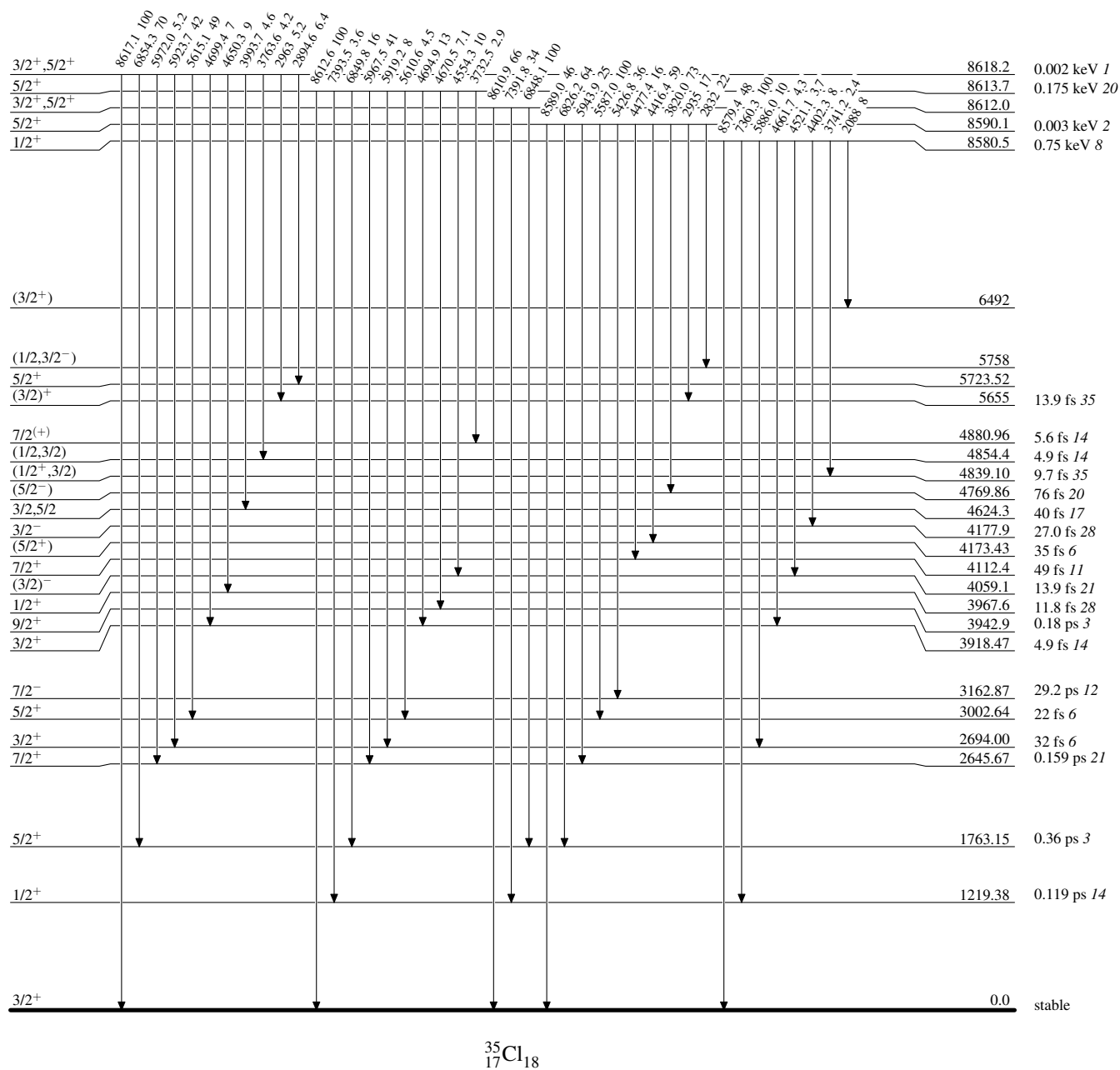
Adopted Levels, GammasLevel Scheme (continued)

Intensities: Relative photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

**Adopted Levels, Gammas****Level Scheme (continued)**

Intensities: Relative photon branching from each level

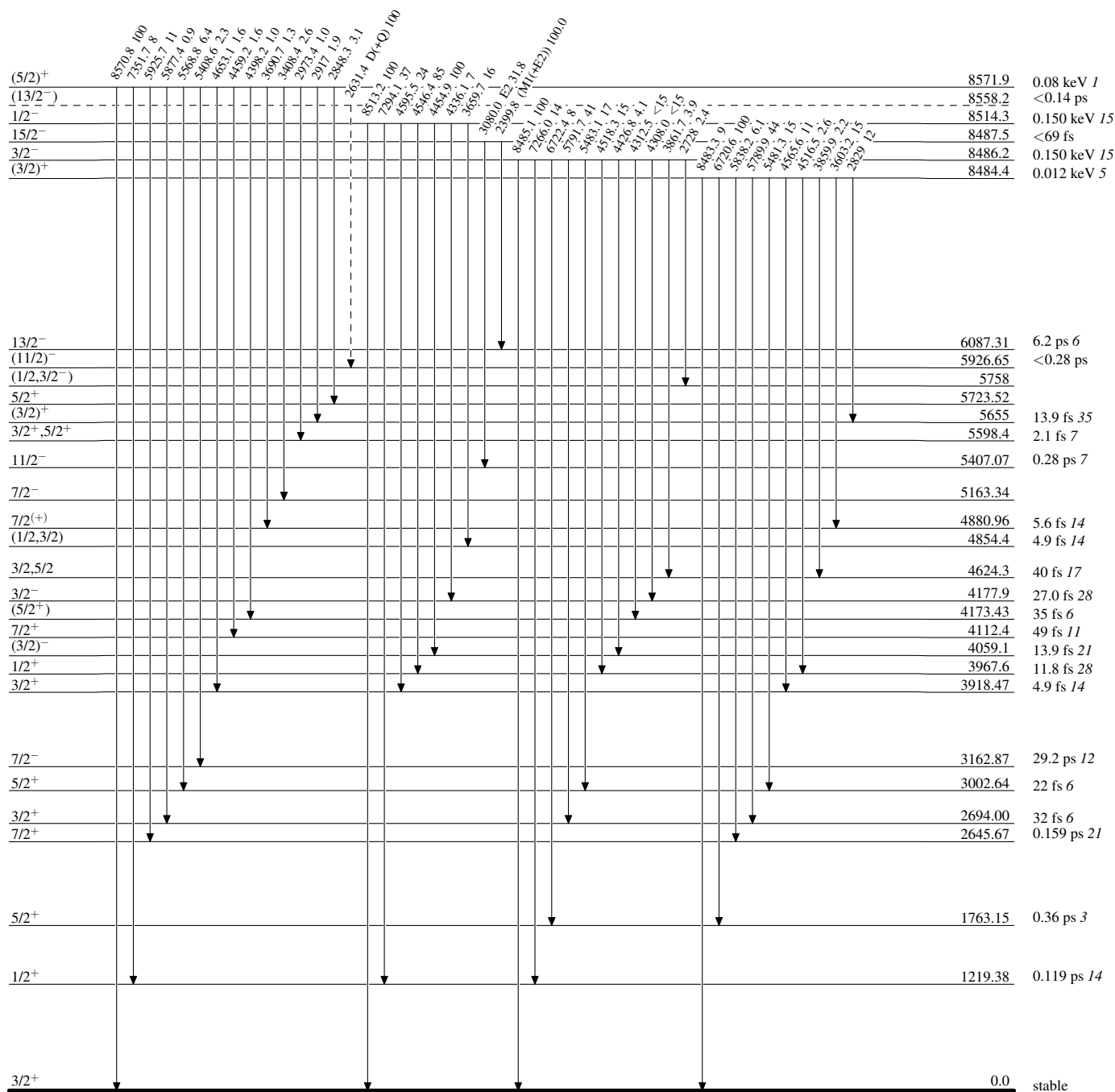
 $^{35}_{17}\text{Cl}_{18}$

## Adopted Levels, Gammas

Legend

## Level Scheme (continued)

Intensities: Relative photon branching from each level

-----►  $\gamma$  Decay (Uncertain) $^{35}_{17}\text{Cl}_{18}$

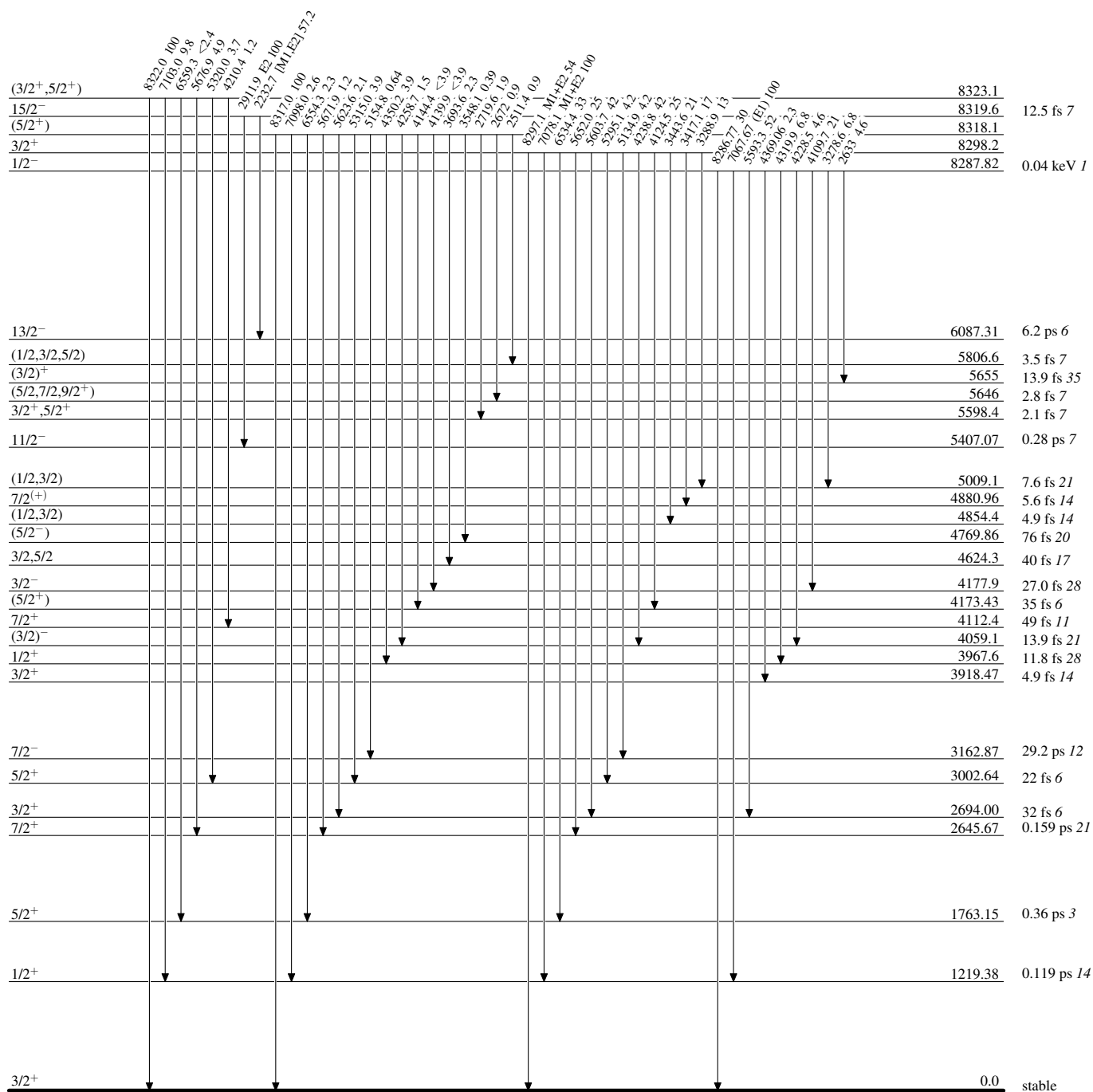




## Adopted Levels, Gammas

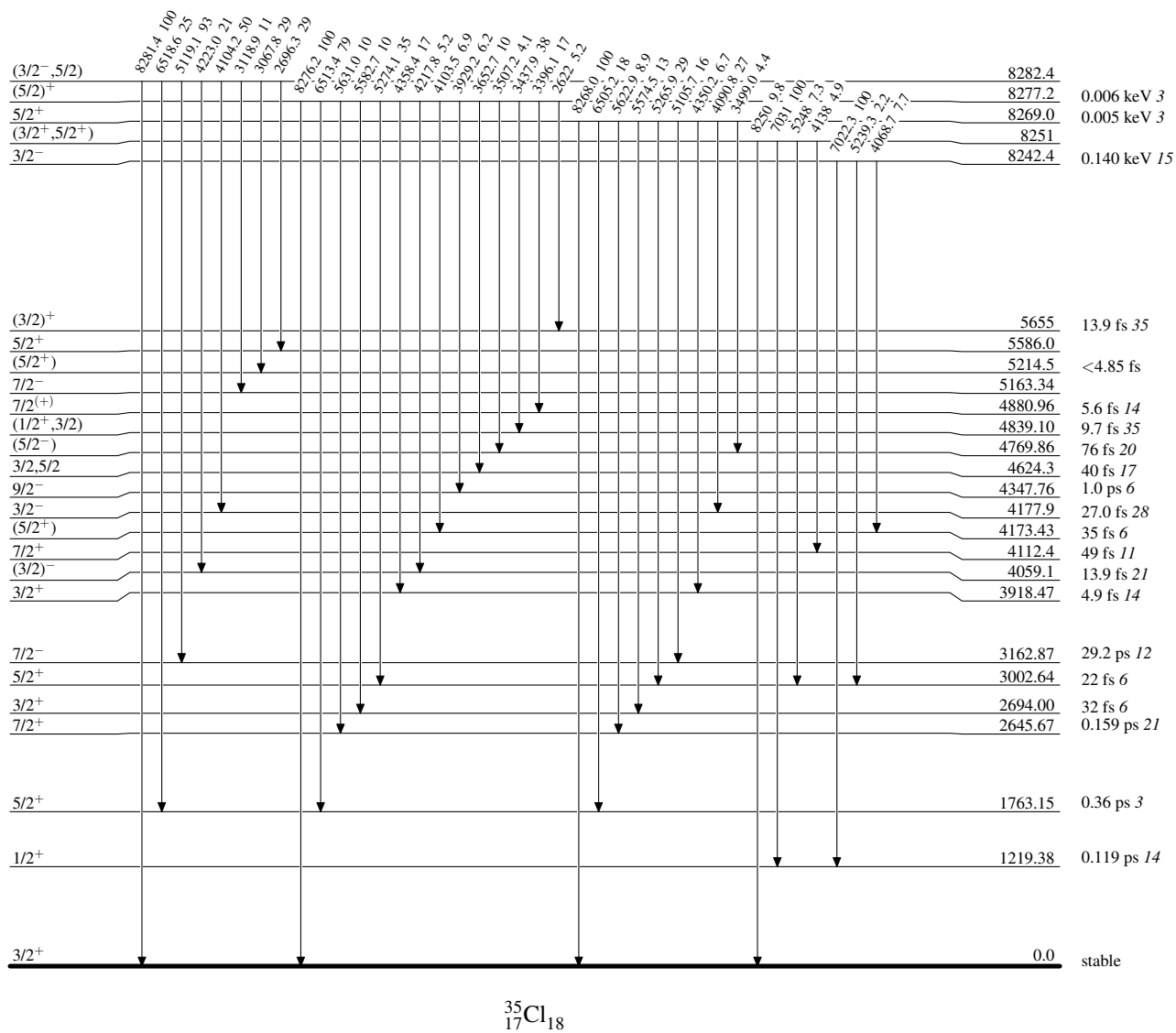
## Level Scheme (continued)

Intensities: Relative photon branching from each level



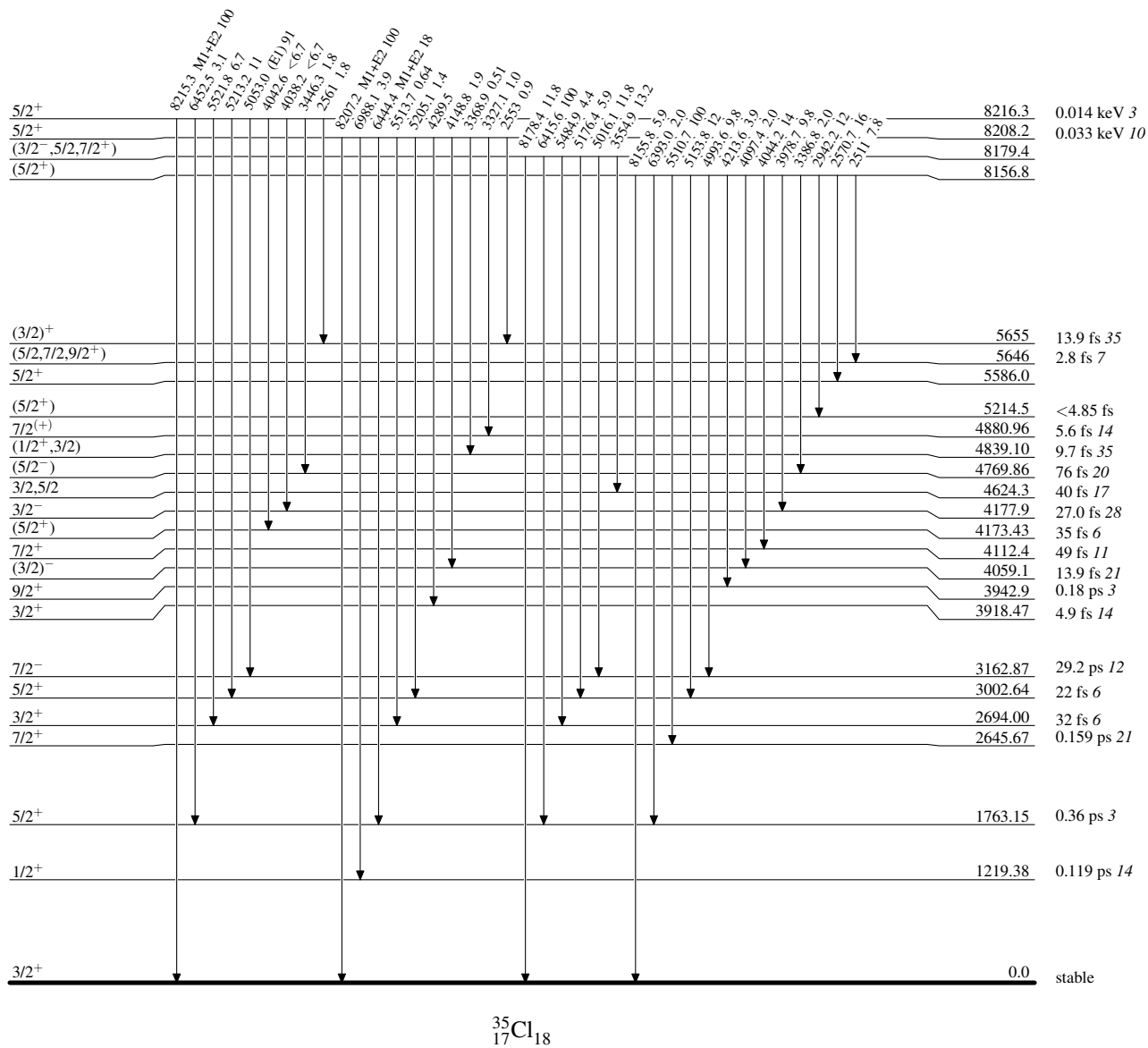
Adopted Levels, GammasLevel Scheme (continued)

Intensities: Relative photon branching from each level



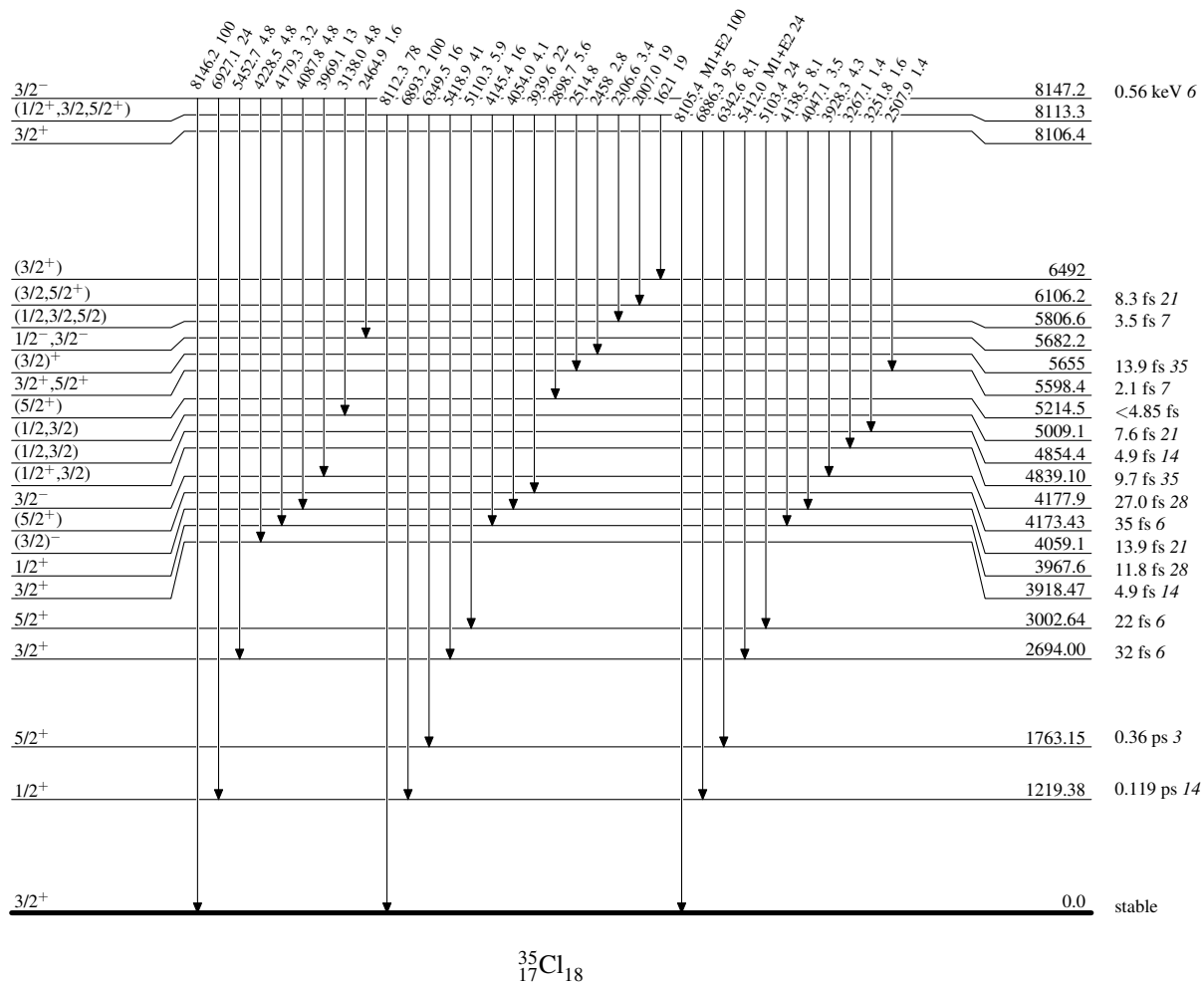
Adopted Levels, GammasLevel Scheme (continued)

Intensities: Relative photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

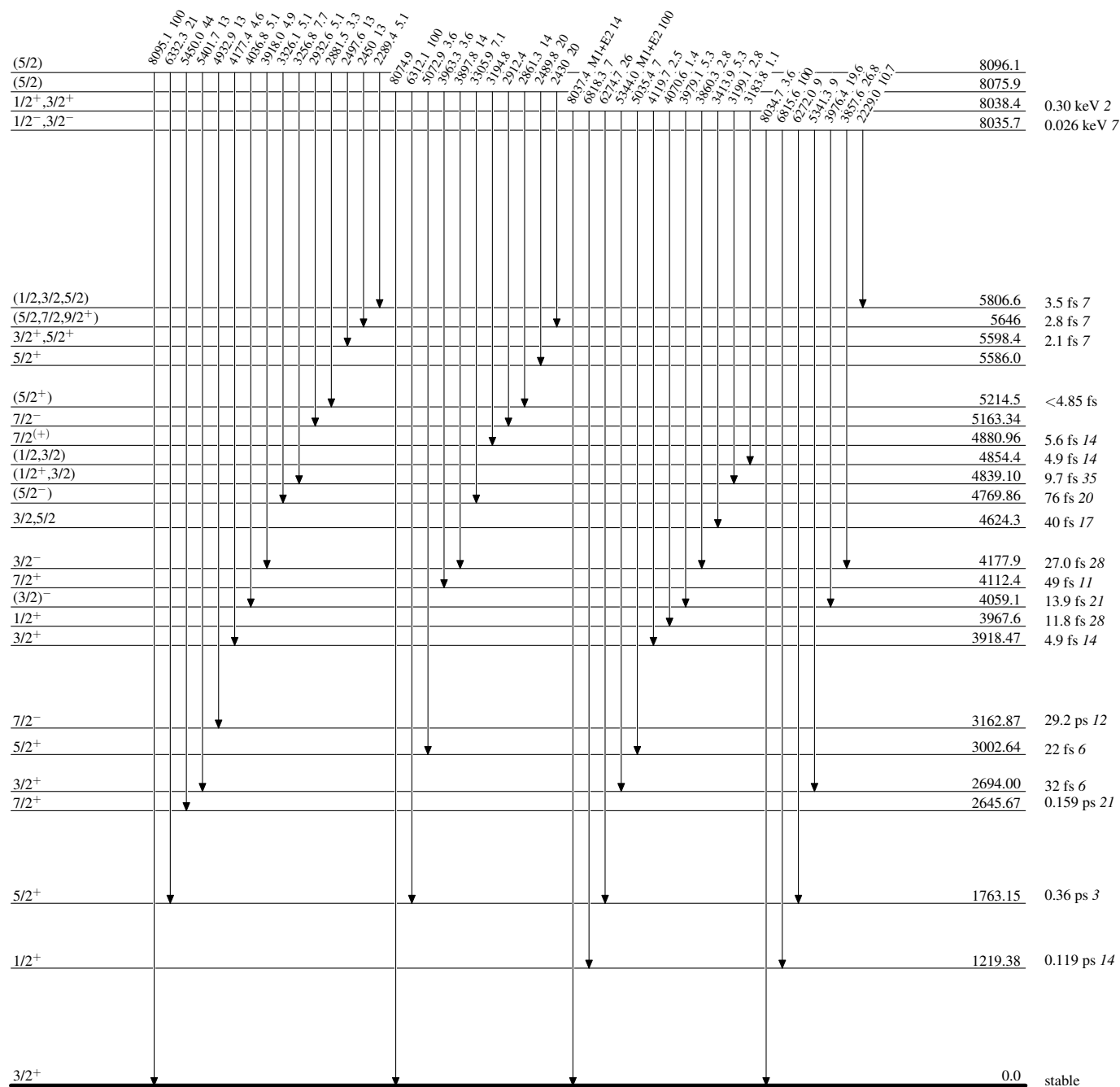
**Adopted Levels, Gammas****Level Scheme (continued)**

Intensities: Relative photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

**Adopted Levels, Gammas****Level Scheme (continued)**

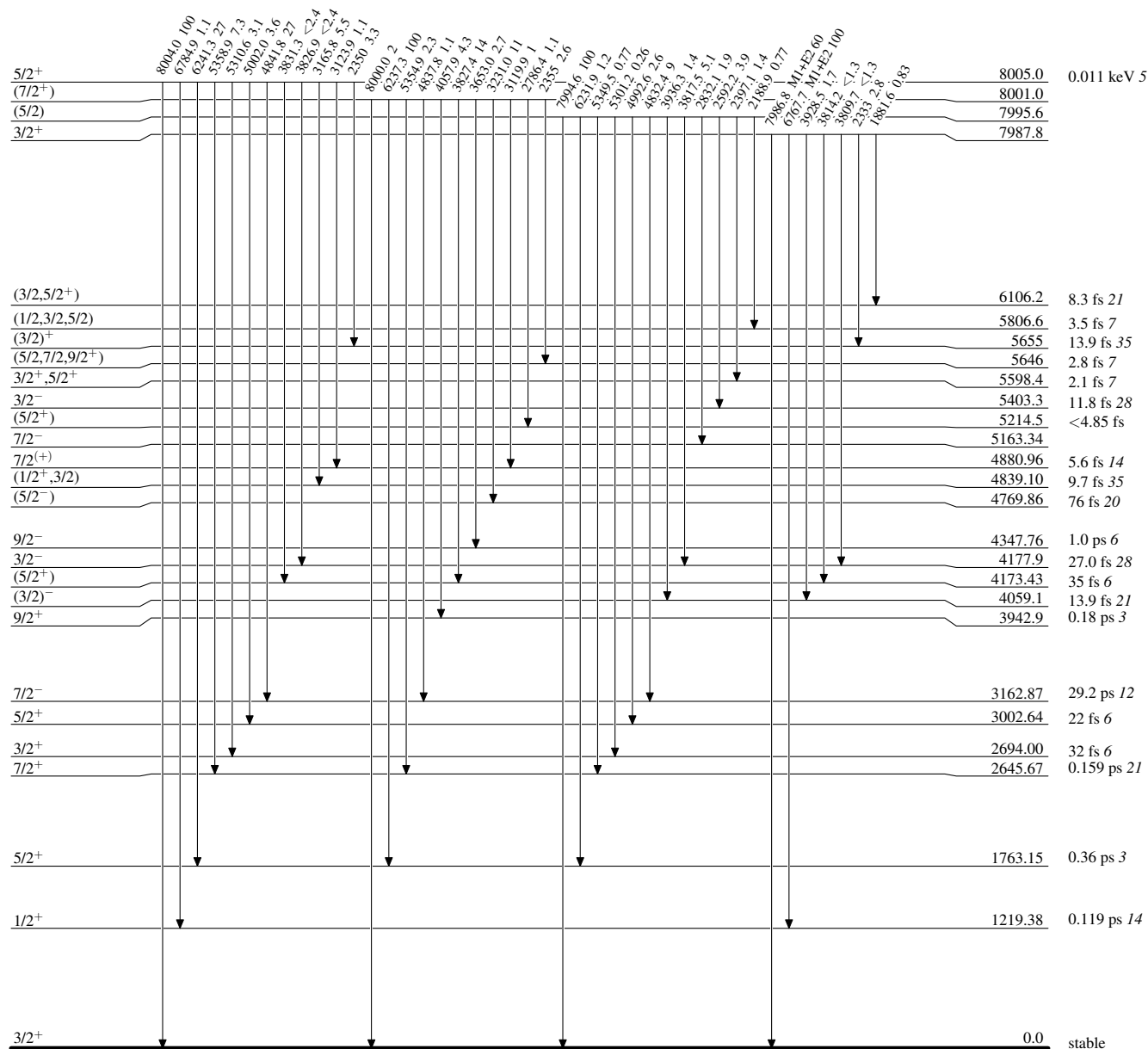
Intensities: Relative photon branching from each level



## Adopted Levels, Gammas

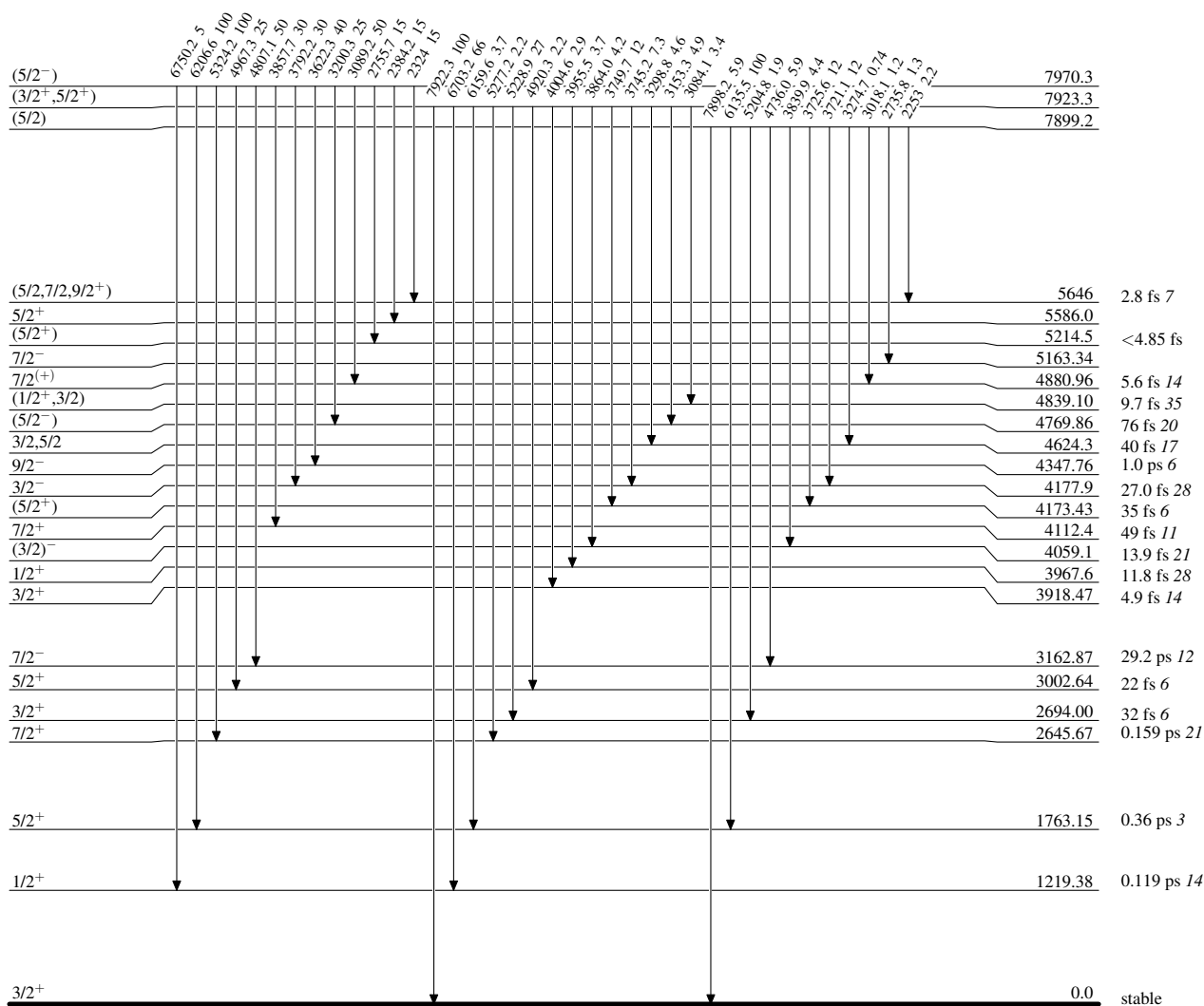
## Level Scheme (continued)

Intensities: Relative photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

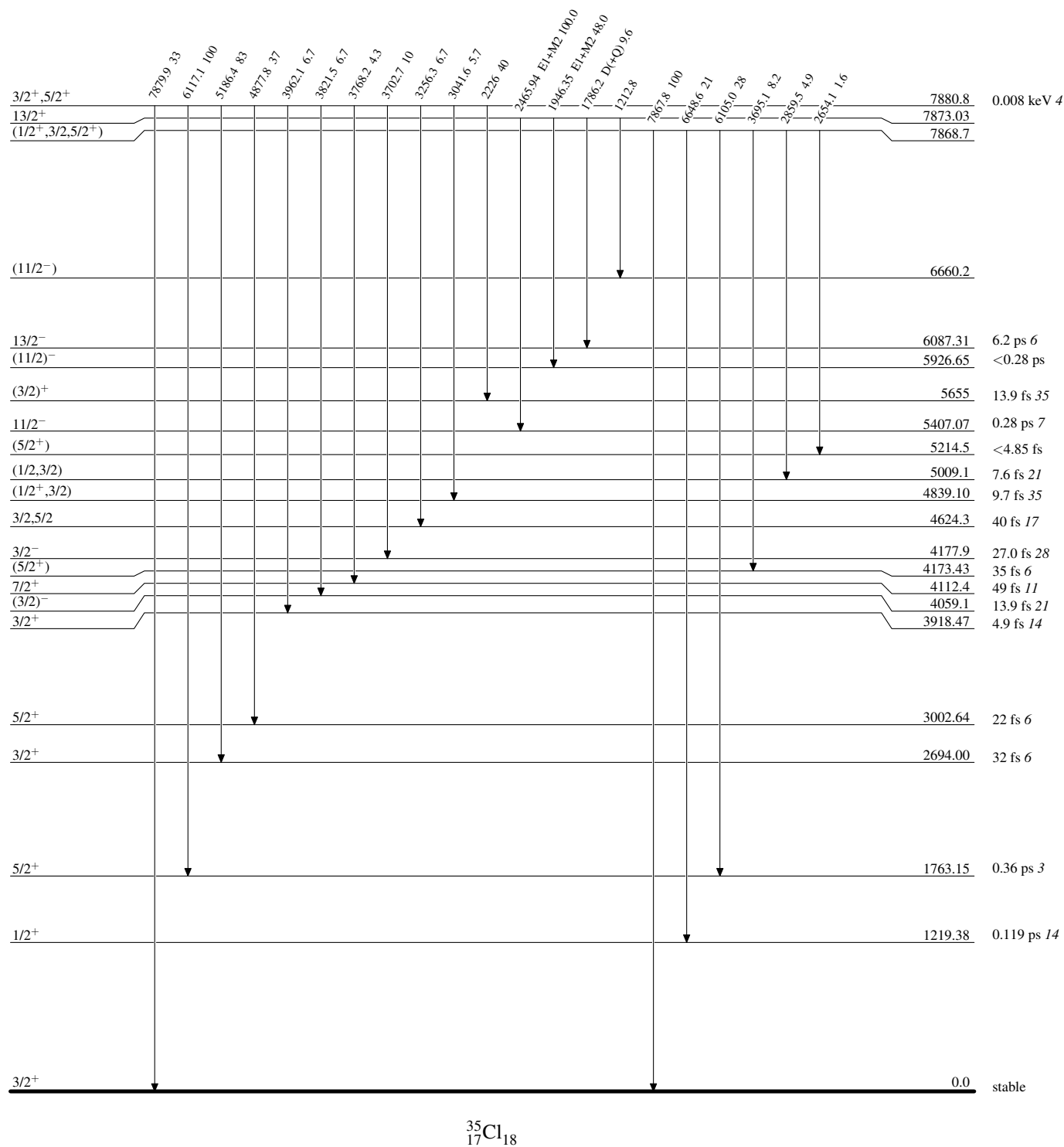
Adopted Levels, GammasLevel Scheme (continued)

Intensities: Relative photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

**Adopted Levels, Gammas****Level Scheme (continued)**

Intensities: Relative photon branching from each level



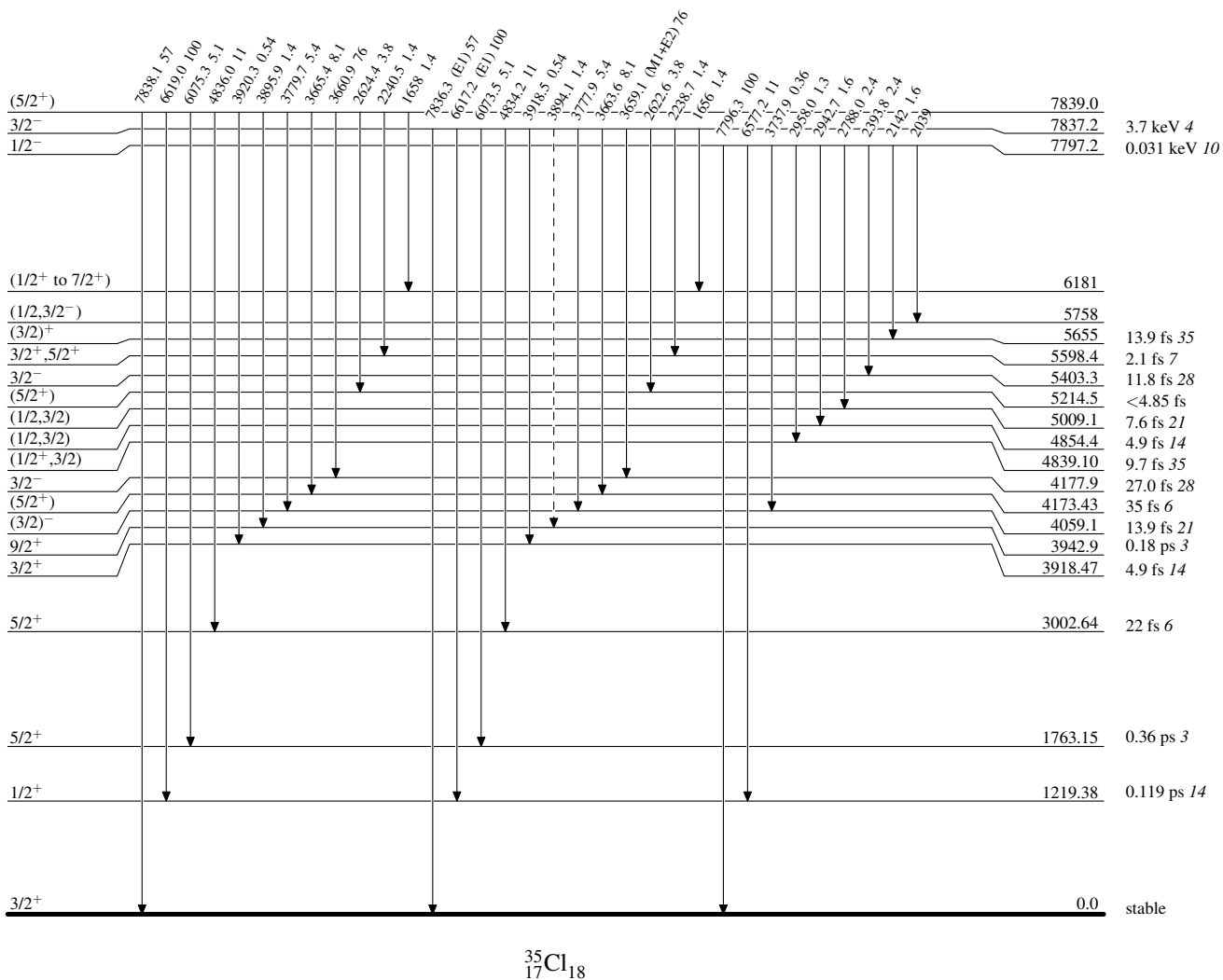


**Adopted Levels, Gammas**

Legend

**Level Scheme (continued)**

Intensities: Relative photon branching from each level

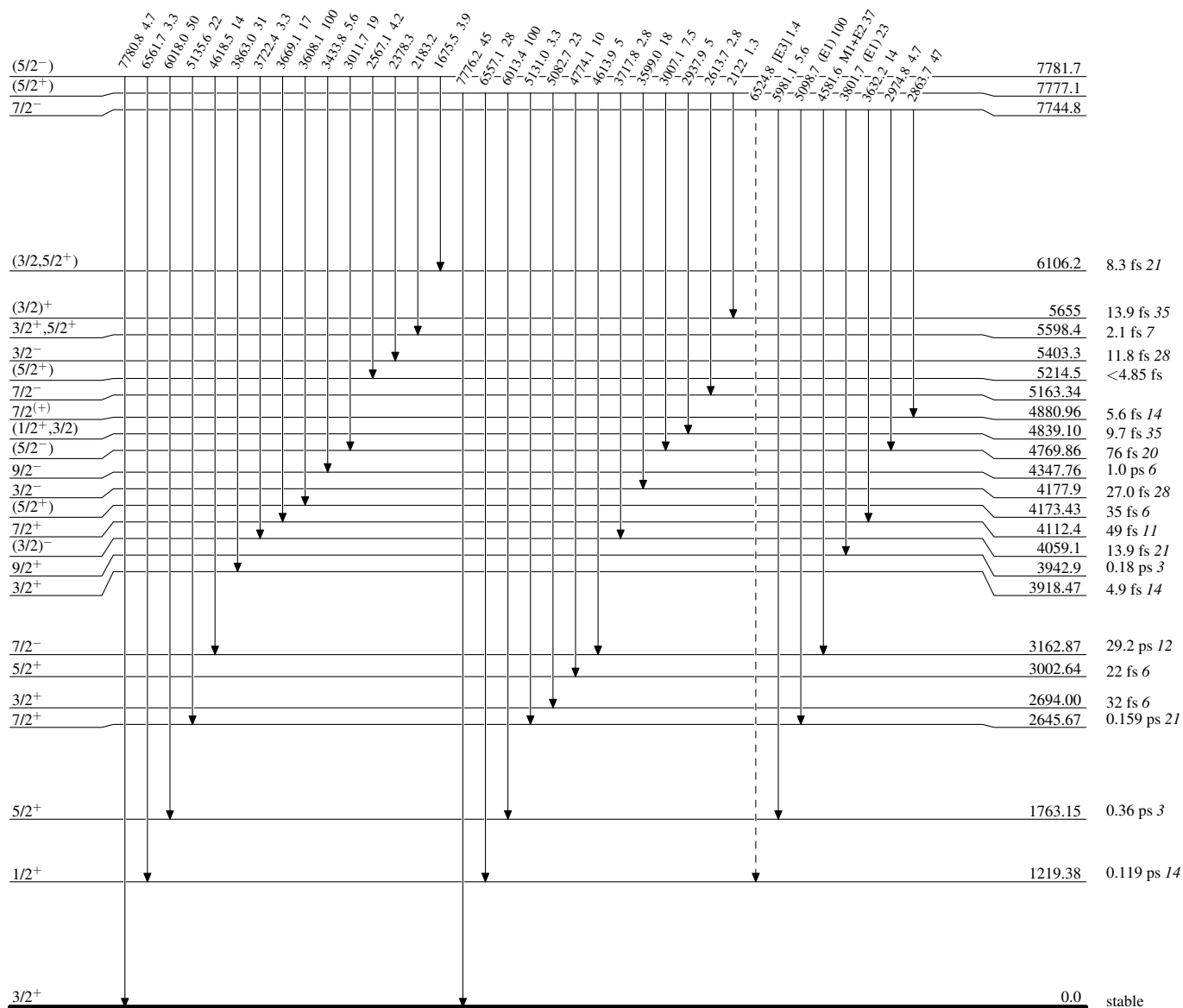
-----►  $\gamma$  Decay (Uncertain) $^{35}_{17}\text{Cl}_{18}$

Adopted Levels, Gammas

Legend

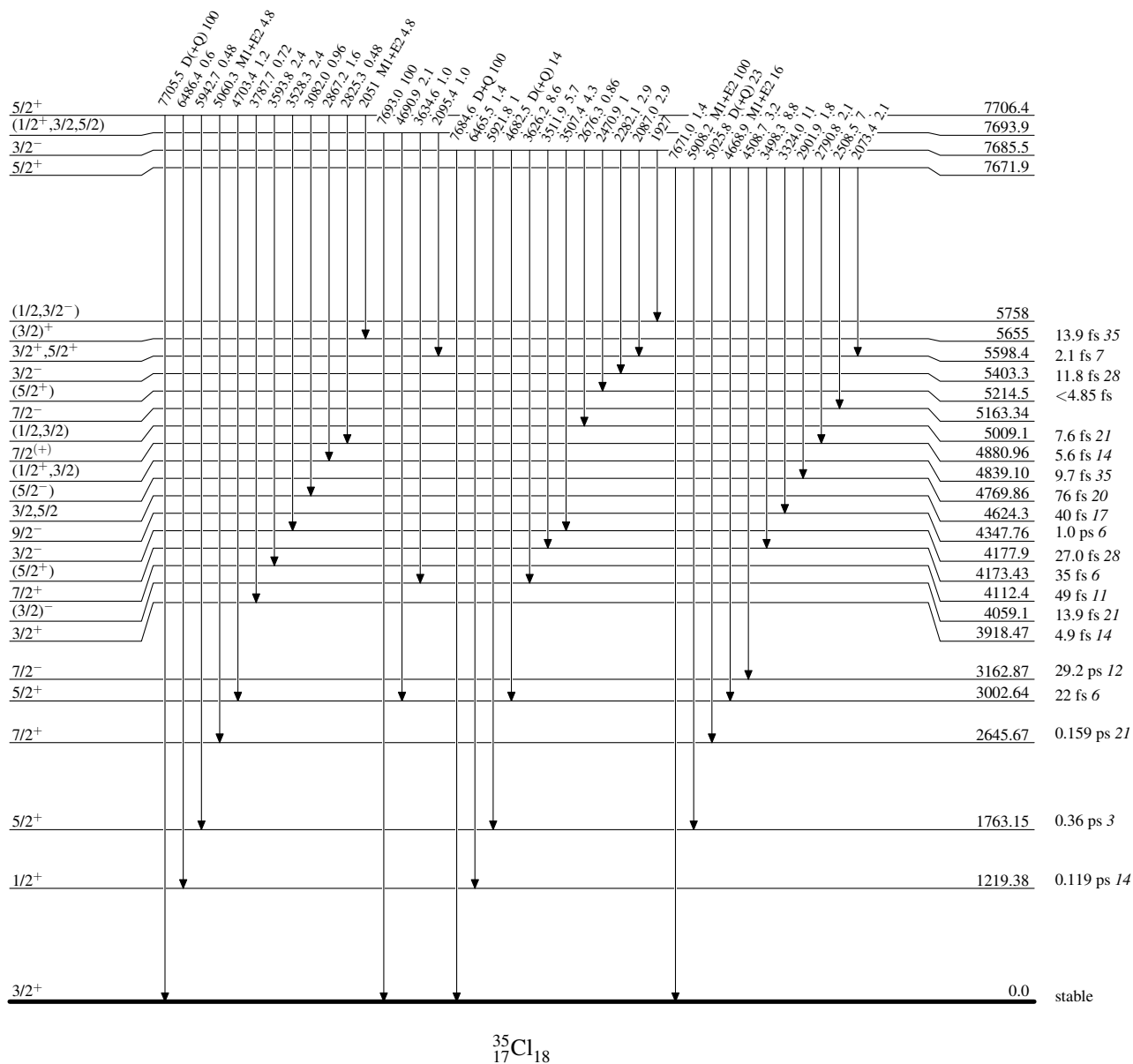
Level Scheme (continued)

Intensities: Relative photon branching from each level

-----►  $\gamma$  Decay (Uncertain) $^{35}_{17}\text{Cl}_{18}$

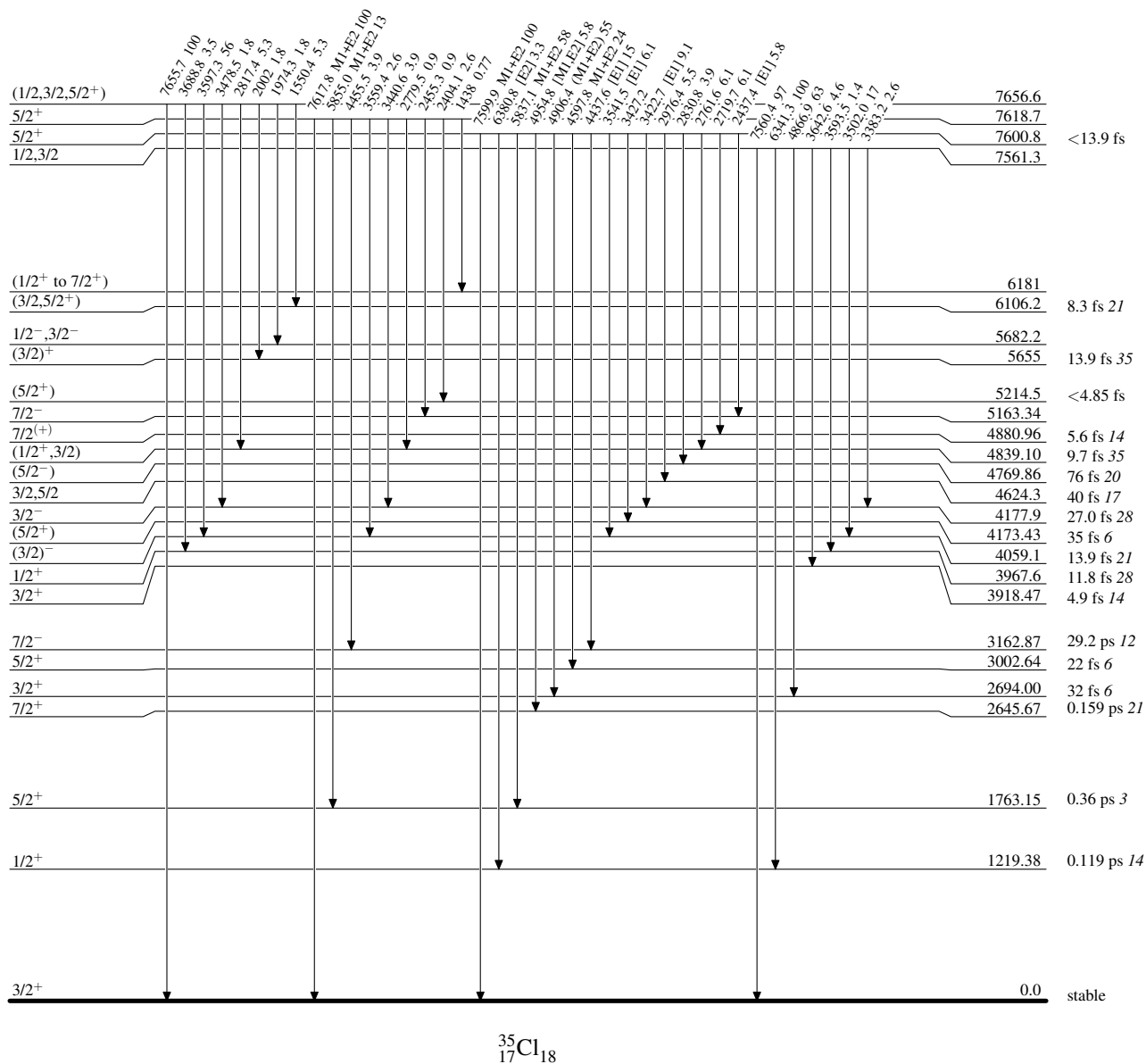
Adopted Levels, GammasLevel Scheme (continued)

Intensities: Relative photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

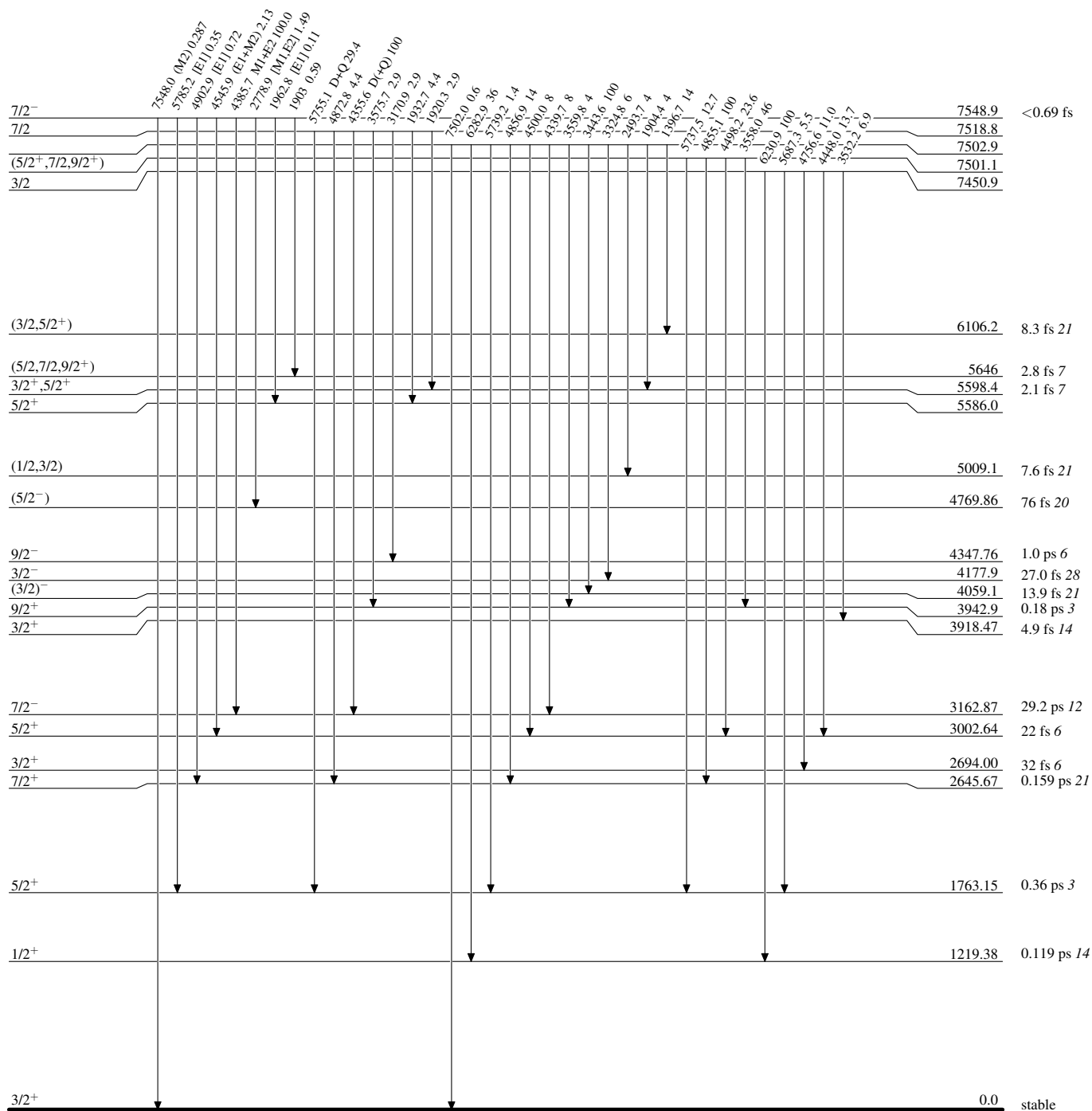
Adopted Levels, GammasLevel Scheme (continued)

Intensities: Relative photon branching from each level



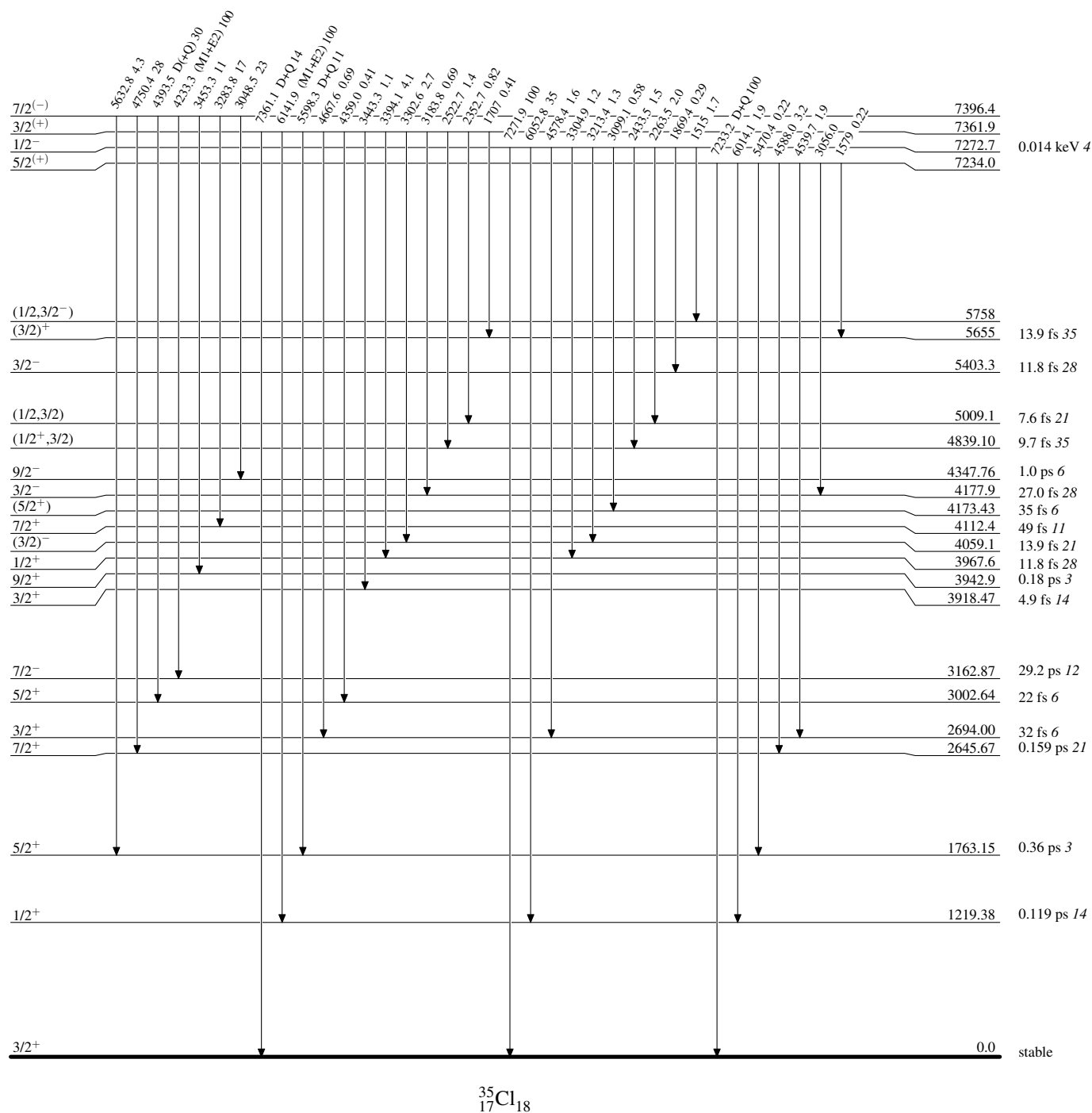
**Adopted Levels, Gammas****Level Scheme (continued)**

Intensities: Relative photon branching from each level



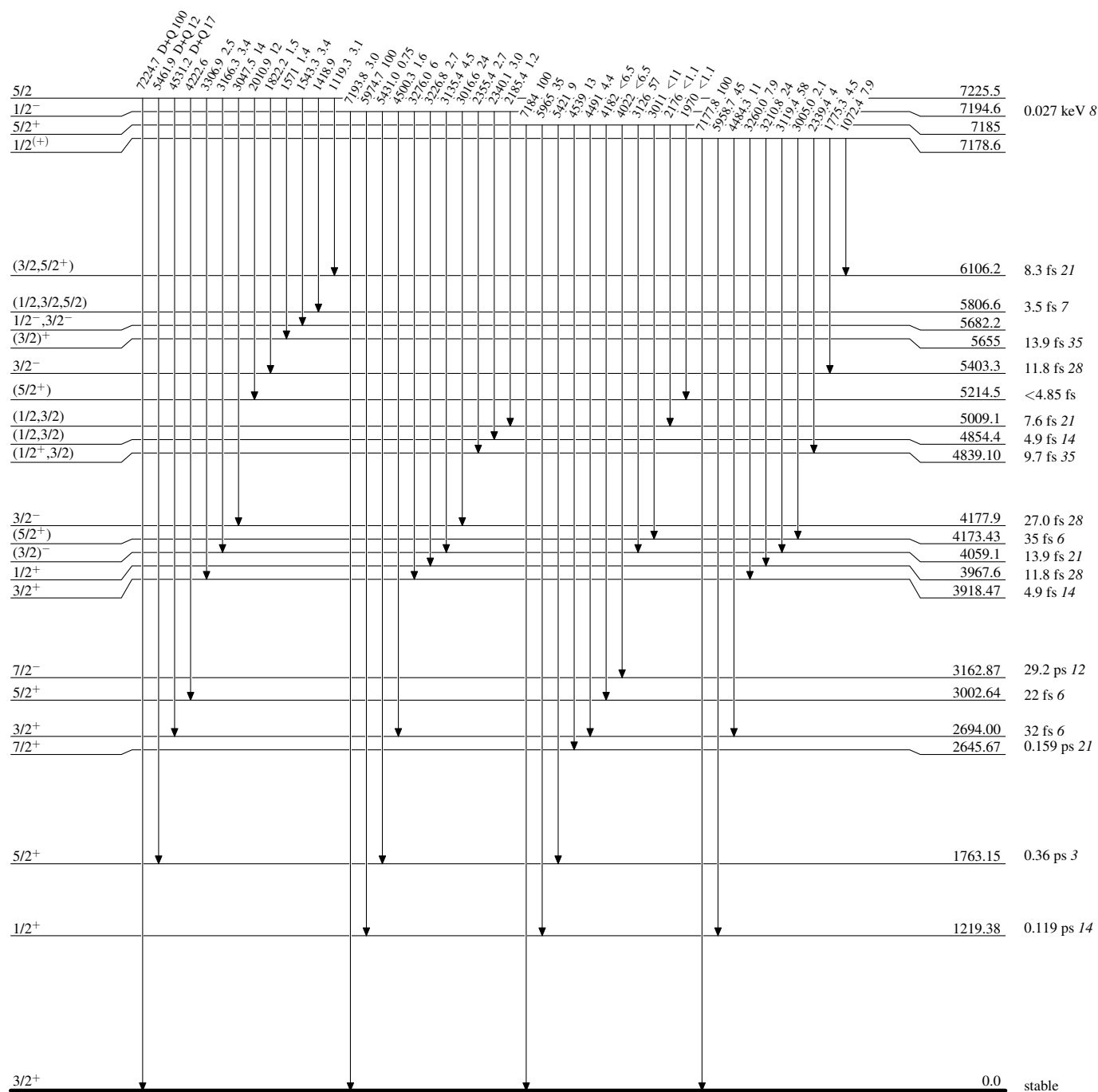
**Adopted Levels, Gammas****Level Scheme (continued)**

Intensities: Relative photon branching from each level



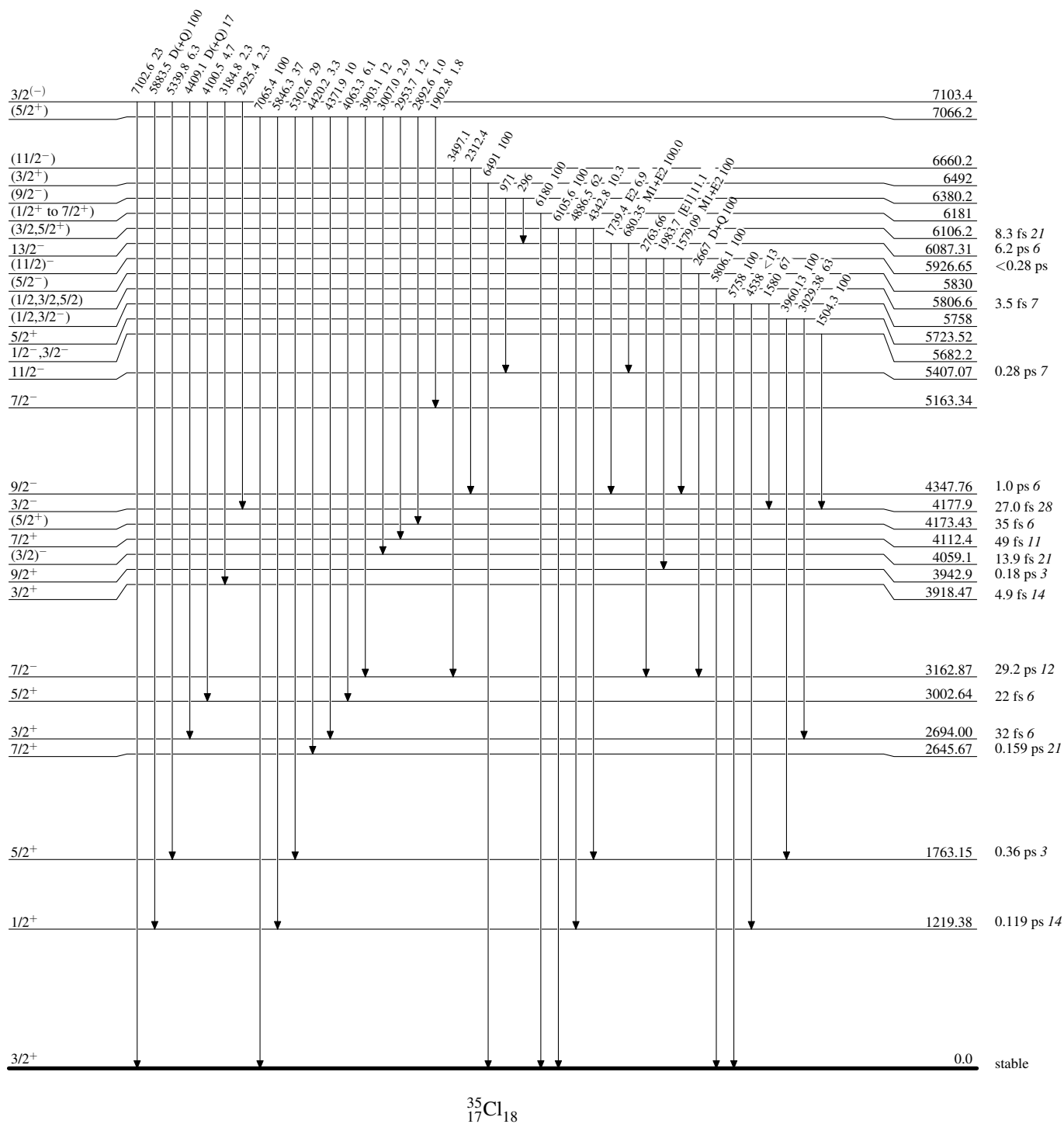
**Adopted Levels, Gammas****Level Scheme (continued)**

Intensities: Relative photon branching from each level



**Adopted Levels, Gammas****Level Scheme (continued)**

Intensities: Relative photon branching from each level



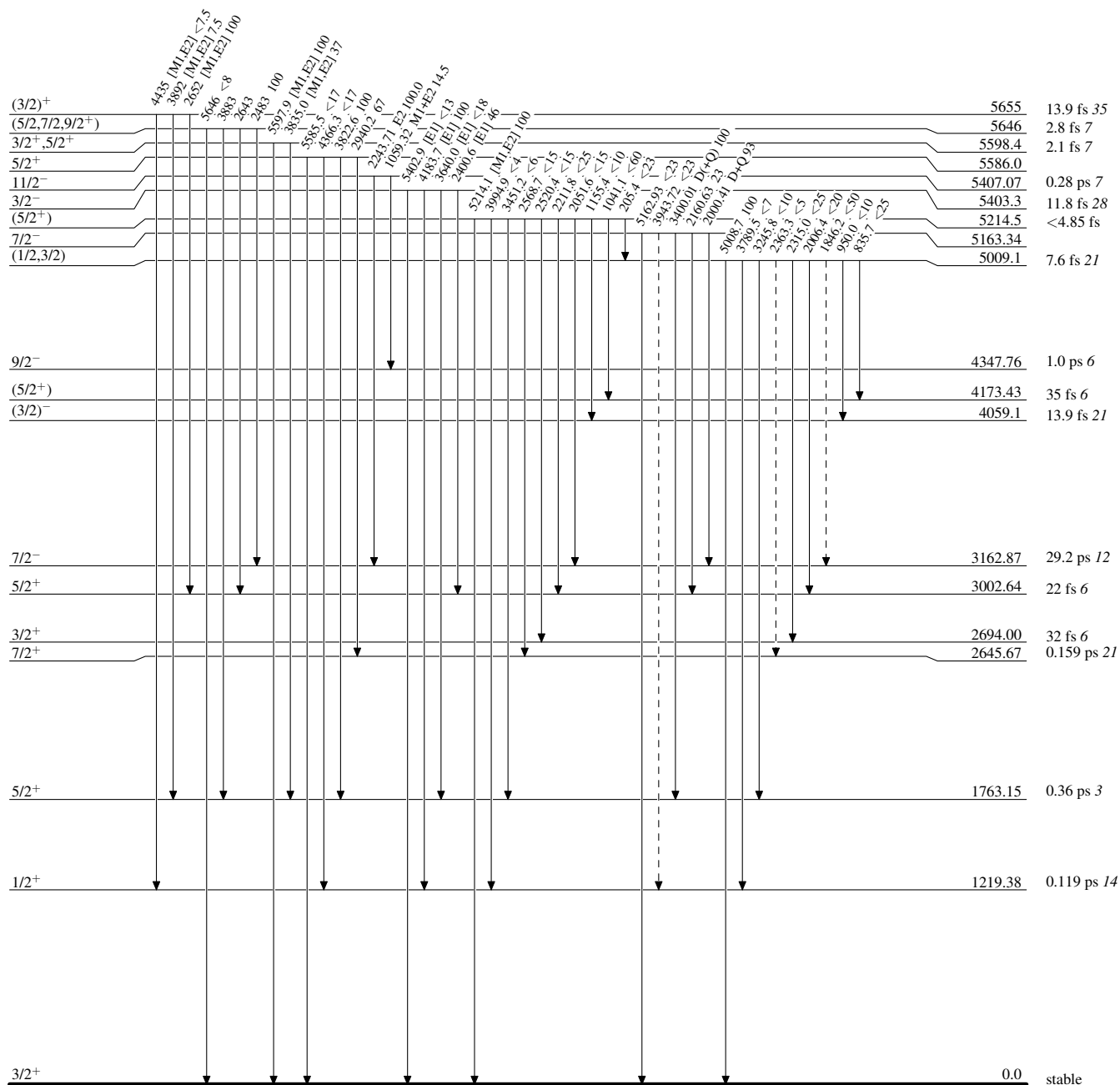


## Adopted Levels, Gammas

Legend

## Level Scheme (continued)

Intensities: Relative photon branching from each level

 -----►  $\gamma$  Decay (Uncertain)


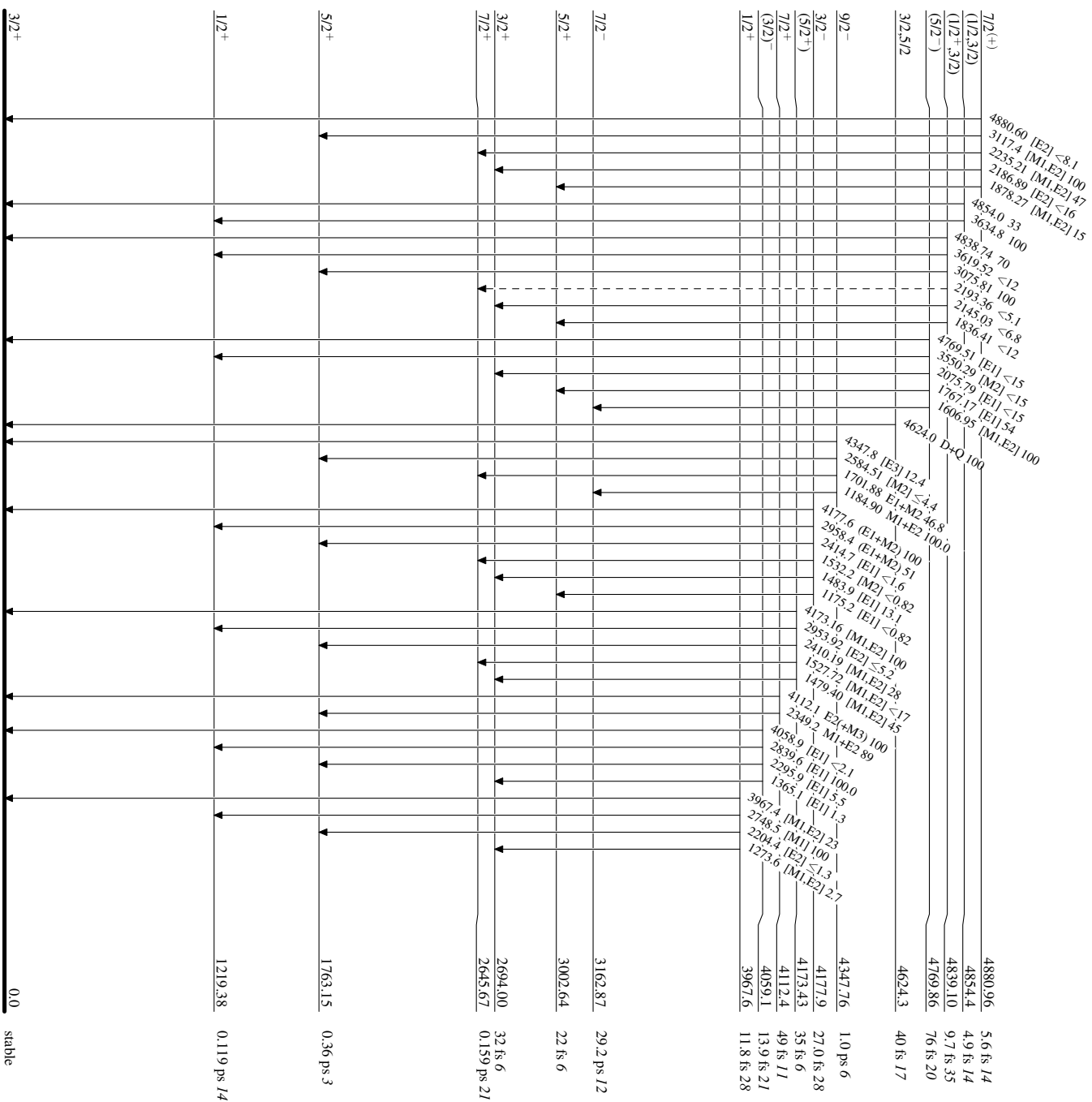
Adopted Levels, Gammas

Legend

Level Scheme (continued)

Intensities: Relative photon branching from each level

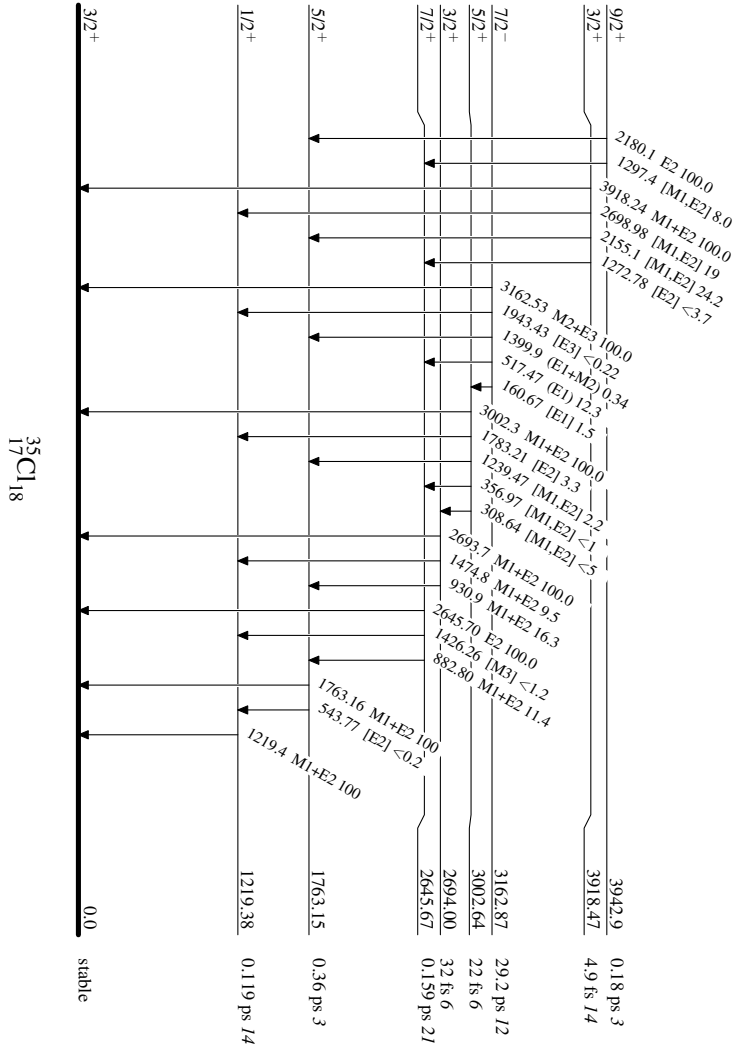
-----▶  $\gamma$  Decay (Uncertain)



Adopted Levels, Gammas

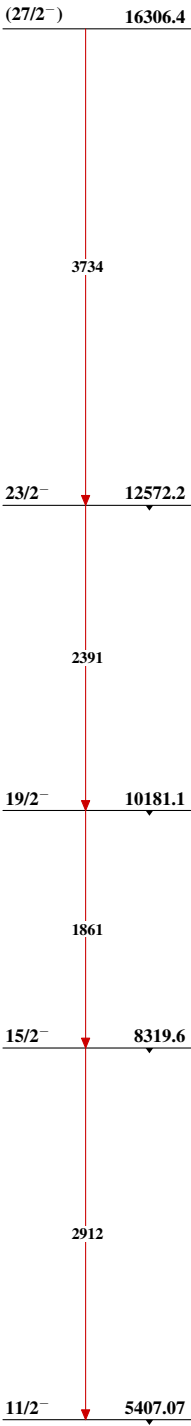
Level Scheme (continued)

Intensities: Relative photon branching from each level



Adopted Levels, Gammas

Band(A): Band based on f<sub>7/2</sub>  
orbital



<sup>35</sup><sub>17</sub>Cl<sub>18</sub>

Adopted Levels, Gammas (continued)Band(B): Band based on  $13/2^+$  $21/2^+$  11459.1 $19/2^+$  10858.5

2615

2070

2015

 $17/2^+$  8844.4  
 $(15/2^+)$  8788.3

971

915

 $13/2^+$  7873.03 $^{35}_{17}\text{Cl}_{18}$

$^{35}\text{S}$   $\beta^-$  decay (87.61 d)

Parent:  $^{35}\text{S}$ :  $E=0$ ;  $J^\pi=3/2^+$ ;  $T_{1/2}=87.61$  d 24;  $Q(\beta^-)=167.322$  26;  $\%\beta^-$  decay=100

$^{35}\text{S}$ - $Q(\beta^-)$ : From [2021Wa16](#).

$^{35}\text{S}$ - $J^\pi, T_{1/2}$ : From the Adopted Levels of  $^{35}\text{S}$ .

[1983Ra04](#):  $Q(\beta^-)=166.74$  keV 26.

[1985Ap01](#):  $Q(\beta^-)=167.29$  keV 3.

[1985Oh06](#):  $Q(\beta^-)=167.4$  keV 1.

[1985Al11](#): Upper limit for 17 keV neutrino: 0.4%.

[1985Oh06](#), [1992Hi06](#), [1993Ab11](#), [1993Be21](#), [1993Mo01](#): Deduced evidence against 17 keV neutrino.

[1994Bo33](#): Deduced spurious 17 keV neutrino signal origin.

[1995Bo43](#):  $Q(\beta^-)=167.60$  keV 5 and upper limit for 17 keV neutrino is  $1.8 \times 10^{-3}$ .

[1995Mo17](#): Proposed 17 keV neutrino hypothesis exclusion.

Others: [2008Al39](#), [2000Ho13](#), [1999Oh09](#), [1996Je06](#), [1994Gr04](#), [1994Ho35](#), [1994Mu26](#), [1993Gi08](#), [1993Gr07](#), [1992Ch27](#), [1989Ba04](#), [1989Ta08](#), [1988Ch45](#), [1987Ge04](#), [1987Na22](#), [1985Ma59](#), [1999Pa18](#).

$^{35}\text{S}$  decay leads only to the ground state of  $^{35}\text{Cl}$ .

 $^{35}\text{Cl}$  Levels

| <u>E(level)</u> | <u><math>J^\pi</math></u> | <u>Comments</u>                    |
|-----------------|---------------------------|------------------------------------|
| 0               | $3/2^+$                   | $J^\pi$ : From the Adopted Levels. |

 $\beta^-$  radiations

| <u>E(decay)</u> | <u>E(level)</u> | <u><math>I\beta^-^\dagger</math></u> | <u>Log <math>ft</math></u> |
|-----------------|-----------------|--------------------------------------|----------------------------|
| (167.3 14)      | 0               | 100                                  | 5.023 2                    |

$^\dagger$  Absolute intensity per 100 decays.

**$^{35}\text{Ar}$   $\varepsilon+\beta^+$  decay (1.7755 s) 1980Wi13,1984Ad01**

Parent:  $^{35}\text{Ar}$ :  $E=0$ ;  $J^\pi=3/2^+$ ;  $T_{1/2}=1.7755$  s 14;  $Q(\varepsilon)=5966.2$  7;  $\% \varepsilon+\% \beta^+$  decay=100

$^{35}\text{Ar}$ - $J^\pi$ ,  $T_{1/2}$ : From Adopted Levels of  $^{35}\text{Ar}$ .

$^{35}\text{Ar}$ - $Q(\varepsilon)$ : From 2021Wa16.

1980Wi13:  $^{35}\text{Ar}$  was produced via the  $^{35}\text{Cl}(p,n)^{35}\text{Ar}$  reaction with 11.0-MeV protons from the ONR-CIT tandem accelerator impinging on  $\text{LiCl}$  and  $\text{PbCl}_2$  wrapped in 25- $\mu\text{m}$  Ta foil.  $\gamma$  rays were detected using a 100- $\text{cm}^3$   $\text{Ge}(\text{Li})$  detector. Measured  $E_\gamma$  and  $I_\gamma$  for 20  $\gamma$ -ray peaks. Deduced decay branching ratios for 9 levels, and log  $ft$ . Compared with shell-model calculations.

1984Ad01:  $^{35}\text{Ar}$  was produced via the  $^{35}\text{Cl}(p,n)^{35}\text{Ar}$  reaction with 13-MeV protons from the University of Washington FN tandem accelerator impinging on a thick  $\text{KCl}$  target.  $\gamma$  rays were detected using a 28%  $\text{Ge}(\text{Li})$  detector. Measured  $E_\gamma$  and  $I_\gamma$  for 8  $\gamma$ -ray peaks. Deduced decay branching ratios for 6 levels, and log  $ft$ . Compared with shell-model calculations.

1985Da04:  $^{35}\text{Ar}$  was produced via the  $^{35}\text{Cl}(p,n)^{35}\text{Ar}$  reaction with 9-MeV protons from the Van de Graaff accelerator at the University of Pittsburgh impinging on a  $\text{NaCl}$  target.  $\gamma$  rays were detected using a  $\text{Ge}(\text{Li})$  detector with  $\text{NaI}$  anti-Compton shielding. Measured  $E_\gamma$  and  $I_\gamma$  for 4  $\gamma$ -ray peaks. Deduced 3 levels.

1979Ha04:  $^{35}\text{Ar}$  produced via the  $^{51}\text{V}(p,p11n)$  spallation reaction with 600-MeV protons from the CERN synchrocyclotron bombarding a 38-g/ $\text{cm}^2$  vanadium target in the form of VC powder.  $\gamma$  rays were detected using a  $\text{Ge}(\text{Li})$  detector. Measured  $E_\gamma$  and  $I_\gamma$  for the 1219 $\gamma$ . Deduced the decay branching ratio for the 1219-keV level.

1977Az01:  $^{35}\text{Ar}$  was produced via the  $^{35}\text{Cl}(p,n)^{35}\text{Ar}$  reaction with 10-13-MeV protons from the McGill synchrocyclotron impinging on  $\text{CCl}_4$  and  $\text{LiCl}$  targets and 8-MeV protons from the University of Montreal impinging a thick  $\text{LiCl}$  pellet. Positrons were detected using an NE102 plastic scintillator coupled to an RCA 6342A photomultiplier tube.  $\gamma$  rays were detected using an 18%  $\text{Ge}(\text{Li})$  detector. Measured  $E_\gamma$  and  $I_\gamma$  for 3  $\gamma$ -ray peaks. Deduced decay branching ratios for 3 levels and parent  $^{35}\text{Ar}$   $T_{1/2}$ . Related:  $^{35}\text{Cl}(p,n)^{35}\text{Ar}$  reaction threshold measurement in 1978Az01.

1971Ge04:  $^{35}\text{Ar}$  was produced via the  $^{35}\text{Cl}(p,n)^{35}\text{Ar}$  reaction with 10-MeV protons from the Chalk River tandem accelerator impinging on  $\text{PbCl}_2$  targets.  $\gamma$  rays were detected using two coaxial  $\text{Ge}(\text{Li})$  detectors. Measured  $E_\gamma$  and  $I_\gamma$  for 4  $\gamma$ -ray peaks. Deduced 4 levels, decay branching ratios, log  $ft$ , and parent  $^{35}\text{Ar}$   $T_{1/2}$ .

1971De05:  $^{35}\text{Ar}$  was produced via the  $^{35}\text{Cl}(p,n)^{35}\text{Ar}$  reaction on  $\text{KCl}$  targets.  $\gamma$  rays were detected using a 25  $\text{cm}^3$   $\text{Ge}(\text{Li})$  detector. Measured  $E_\gamma$  and  $I_\gamma$ . Deduced 6 levels, decay branching ratios, and log  $ft$ .

1969Wi18,1969Fr09:  $^{35}\text{Ar}$  was produced via the  $^{35}\text{Cl}(p,n)^{35}\text{Ar}$  reaction with a proton beam from the Harwell tandem accelerator impinging on 3.75-mg/ $\text{cm}^2$   $\text{KCl}$  deposited on 0.13-mm Au backings and 160-mmHg gas  $\text{CCl}_4$  targets.  $\gamma$  rays were detected using a  $\text{Ge}(\text{Li})$  detector. Measured  $E_\gamma$  and  $I_\gamma$  for 2  $\gamma$ -ray peaks. Deduced 2 levels, decay branching ratios, (p,n) reaction threshold and  $\beta$  end-point energy, log  $ft$ , and parent  $^{35}\text{Ar}$   $T_{1/2}$ .

1960Wa04:  $^{35}\text{Ar}$  was produced at Crocker Laboratory and the Lawrence Radiation Laboratory.  $\beta$  spectra were measured with a uniform-field 180°-deflection single-focusing spectrometer. Deduced Coulomb-energy differences between mirror ground-state transitions.

1956Ki29:  $^{35}\text{Ar}$  was produced via the  $^{35}\text{Cl}(p,n)^{35}\text{Ar}$  reaction with 10-MeV protons from the BNL 60-inch cyclotron impinging on a liquid  $\text{CCl}_4$  target.  $\beta$  particles were detected using a conventional thin-lens, iron-free, magnetic spectrometer.  $\gamma$  rays were detected using a  $\text{NaI}(\text{Tl})$  crystal scintillation counter. Measured  $E_\beta$  and total  $I_\beta$  for two positron decays to  $^{35}\text{Cl}$  excited states,  $E_\gamma$  and  $I_\gamma$  for two  $\gamma$ -ray peaks. Deduced two levels and decay branching ratios.

$^{35}\text{Ar}$   $\beta$  asymmetry parameter measurements: 1993Co07, 1991Na05, 1991Mi18, 1988Ga02, 1965Ca04.

 $^{35}\text{Cl}$  Levels

| $E(\text{level})^\dagger$ | $J^\pi^\ddagger$ | $T_{1/2}^\ddagger$ |
|---------------------------|------------------|--------------------|
| 0                         | $3/2^+$          | stable             |
| 1219.2 2                  | $1/2^+$          | 0.119 ps 14        |
| 1763.1 2                  | $5/2^+$          | 0.36 ps 3          |
| 2645.1 10                 | $7/2^+$          | 0.159 ps 21        |
| 2693.9 2                  | $3/2^+$          | 32 fs 6            |
| 3002.2 4                  | $5/2^+$          | 22 fs 6            |
| 3918.3 15                 | $3/2^+$          | 4.9 fs 14          |
| 3967.9 6                  | $1/2^+$          | 11.8 fs 28         |
| 4624.5? 5                 | $3/2, 5/2$       | 40 fs 17           |

$^\dagger$  From a least-squares fit to  $\gamma$ -ray energies.

$^\ddagger$  From the Adopted Levels.

$^{35}\text{Ar}$   $\varepsilon+\beta^+$  decay (1.7755 s) 1980Wi13,1984Ad01 (continued)

| $\varepsilon, \beta^+$ radiations |          |                          |                              |           |                                                | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------------|----------|--------------------------|------------------------------|-----------|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E(decay)                          | E(level) | $I\beta^+$ $^{\ddagger}$ | $I\varepsilon$ $^{\ddagger}$ | Log $ft$  | $I(\varepsilon+\beta^+)$ $^{\ddagger\ddagger}$ |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| (1341.7 $^{\#}$ 13)               | 4624.5?  | $<8.0\times 10^{-4}$     | $<0.0035$                    | $>3.8$    | $<0.0043$                                      | $I(\varepsilon+\beta^+)$ : deduced from $100-\Sigma I(\varepsilon+\beta^+)$ (excited states), where $\Sigma I(\varepsilon+\beta^+)$ (excited states)=1.76 5. Level scheme fed by $\beta^+$ decay may be complete considering the highest observed level is at 4.6 MeV. Observed $\Sigma I(\varepsilon)\approx 0.08$ . Evaluators assume $\Sigma I(\varepsilon)\leq 0.08$ for all unobserved $\varepsilon$ feedings to high-lying levels and include this limit in the uncertainty for $\Sigma I(\varepsilon+\beta^+)$ (excited states)=1.76 10 and $I(\varepsilon+\beta^+)$ (gs)=98.24 10. |
| (1998.3 14)                       | 3967.9   | 0.015 10                 | 0.0016 10                    | $4.5+5-2$ | 0.017 11                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| (2047.9 19)                       | 3918.3   | 0.0096 30                | $8.3\times 10^{-4}$ 26       | $4.8+2-1$ | 0.0104 32                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| (2964.0 13)                       | 3002.2   | 0.089 14                 | 0.0011 2                     | 5.0 1     | 0.090 14                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| (3272.3 13)                       | 2693.9   | 0.164 9                  | 0.00126 7                    | 5.00 3    | 0.165 9                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| (3321.1 16)                       | 2645.1   | $<0.00146$               | $<2.82\times 10^{-5}$        | $>8.3$    | $<0.00149$                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| (4203.1 13)                       | 1763.1   | 0.256 13                 | $7.0\times 10^{-4}$ 4        | 5.47 2    | 0.257 13                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| (4747.0 13)                       | 1219.2   | 1.22 4                   | 0.00208 7                    | 5.10 2    | 1.22 4                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| (5966.2 16)                       | 0        | 98.17 10                 | 0.07309 8                    | 3.759 1   | 98.24 10                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

$^{\dagger}$  Deduced from  $\gamma$ -ray intensity balance, relative to  $I_{\gamma}(1219\gamma)=100$ , normalized to absolute  $\%I_{\gamma}(1219\gamma)=1.26$  5.

$^{\ddagger}$  Absolute intensity per 100 decays.

$^{\#}$  Existence of this branch is questionable.



<sup>35</sup>Ar ε+β<sup>+</sup> decay (1.7755 s) [1980Wi13](#),[1984Ad01](#) (continued)

γ(<sup>35</sup>Cl)

I<sub>γ</sub> normalization: absolute %I<sub>γ</sub>(1219γ)=1.243 30 deduced from weighted average of 1.223 46 ([1969Wi18](#),[1969Fr09](#)), 1.22 20 ([1971Ge04](#)), 1.55 15 ([1977Az01](#)), 1.34 8 ([1979Ha04](#)), 1.23 7 ([1984Ad01](#)), 1.228 30 ([1980Wi13](#)).

| E <sub>γ</sub> <sup>†</sup> | I <sub>γ</sub> <sup>‡&amp;</sup> | E <sub>i</sub> (level) | J <sub>i</sub> <sup>π</sup> | E <sub>f</sub> | J <sub>f</sub> <sup>π</sup> | Mult. <sup>#</sup> | δ <sup>#</sup> | α <sup>a</sup>          | Comments                                                                                                                                                                                                                                                                                               |
|-----------------------------|----------------------------------|------------------------|-----------------------------|----------------|-----------------------------|--------------------|----------------|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 916 <sup>b</sup>            |                                  | 3918.3                 | 3/2 <sup>+</sup>            | 3002.2         | 5/2 <sup>+</sup>            |                    |                |                         | The intensity could not be determined due to being obscured by another peak ( <a href="#">1980Wi13</a> ).                                                                                                                                                                                              |
| 930.7 2                     | 1.57 20                          | 2693.9                 | 3/2 <sup>+</sup>            | 1763.1         | 5/2 <sup>+</sup>            | M1+E2              | +0.09 3        | 4.69×10 <sup>-5</sup> 7 | %I <sub>γ</sub> =0.0195 25<br>I <sub>γ</sub> : weighted average of 1.13 26 ( <a href="#">1980Wi13</a> ) and 1.66 12 ( <a href="#">1984Ad01</a> ).                                                                                                                                                      |
| 965 <sup>b</sup>            | <0.87 <sup>@</sup>               | 3967.9                 | 1/2 <sup>+</sup>            | 3002.2         | 5/2 <sup>+</sup>            |                    |                |                         | %I <sub>γ</sub> <0.0111                                                                                                                                                                                                                                                                                |
| 1219.3 2                    | 100.0 15                         | 1219.2                 | 1/2 <sup>+</sup>            | 0              | 3/2 <sup>+</sup>            | M1+E2              | 0.093 10       | 3.66×10 <sup>-5</sup> 5 | %I <sub>γ</sub> =1.24 4<br>I <sub>γ</sub> : others: 100.0 30 from <a href="#">1984Ad01</a> , 100 10 from <a href="#">1977Az01</a> , 100 from <a href="#">1969Wi18</a> , <a href="#">1971Ge04</a> , <a href="#">1979Ha04</a> , and <a href="#">1985Da04</a> .                                           |
| 1225 <sup>b</sup>           |                                  | 3918.3                 | 3/2 <sup>+</sup>            | 2693.9         | 3/2 <sup>+</sup>            |                    |                |                         | The intensity could not be determined due to being obscured by another peak ( <a href="#">1980Wi13</a> ).                                                                                                                                                                                              |
| 1239 <sup>b</sup>           | <0.30 <sup>@</sup>               | 3002.2                 | 5/2 <sup>+</sup>            | 1763.1         | 5/2 <sup>+</sup>            | [M1,E2]            |                | 4.4×10 <sup>-5</sup> 6  | %I <sub>γ</sub> <0.00382                                                                                                                                                                                                                                                                               |
| 1274 <sup>b</sup>           | <0.42 <sup>@</sup>               | 3967.9                 | 1/2 <sup>+</sup>            | 2693.9         | 3/2 <sup>+</sup>            | [M1,E2]            |                | 4.9×10 <sup>-5</sup> 7  | %I <sub>γ</sub> <0.00535                                                                                                                                                                                                                                                                               |
| 1474.8 3                    | 1.02 11                          | 2693.9                 | 3/2 <sup>+</sup>            | 1219.2         | 1/2 <sup>+</sup>            | M1+E2              | -0.63 21       | 8.4×10 <sup>-5</sup> 4  | %I <sub>γ</sub> =0.0127 14<br>I <sub>γ</sub> : weighted average of 0.83 13 ( <a href="#">1980Wi13</a> ) and 1.08 7 ( <a href="#">1984Ad01</a> ).                                                                                                                                                       |
| 1763.0 2                    | 22.7 4                           | 1763.1                 | 5/2 <sup>+</sup>            | 0              | 3/2 <sup>+</sup>            | M1+E2              | -2.85 10       | 0.0002124 30            | %I <sub>γ</sub> =0.282 9<br>I <sub>γ</sub> : weighted average of 23.4 13 ( <a href="#">1969Wi18</a> ), 20.0 27 ( <a href="#">1971Ge04</a> ), 22 6 ( <a href="#">1977Az01</a> ), 23.08 37 ( <a href="#">1980Wi13</a> ), 21.85 67 ( <a href="#">1984Ad01</a> ), and 21.7 9 ( <a href="#">1985Da04</a> ). |
| 1783 <sup>b</sup>           | <0.39 <sup>@</sup>               | 3002.2                 | 5/2 <sup>+</sup>            | 1219.2         | 1/2 <sup>+</sup>            | [E2]               |                | 0.0002267 32            | %I <sub>γ</sub> <0.00497                                                                                                                                                                                                                                                                               |
| 1931 <sup>b</sup>           | <0.21 <sup>@</sup>               | 4624.5?                | 3/2,5/2                     | 2693.9         | 3/2 <sup>+</sup>            |                    |                |                         | %I <sub>γ</sub> <0.00267                                                                                                                                                                                                                                                                               |
| 2155.1 15                   | 0.15 6                           | 3918.3                 | 3/2 <sup>+</sup>            | 1763.1         | 5/2 <sup>+</sup>            | [M1,E2]            |                | 0.00036 4               | %I <sub>γ</sub> =0.0019 8<br>I <sub>γ</sub> : from <a href="#">1984Ad01</a> . Other: <0.22 ( <a href="#">1980Wi13</a> ).                                                                                                                                                                               |
| 2204 <sup>b</sup>           | <0.27 <sup>@</sup>               | 3967.9                 | 1/2 <sup>+</sup>            | 1763.1         | 5/2 <sup>+</sup>            | [E2]               |                | 0.000425 6              | %I <sub>γ</sub> <0.00344                                                                                                                                                                                                                                                                               |
| 2645 <sup>b</sup>           | <0.12 <sup>@</sup>               | 2645.1                 | 7/2 <sup>+</sup>            | 0              | 3/2 <sup>+</sup>            | E2                 |                | 0.000633 9              | %I <sub>γ</sub> <0.00153                                                                                                                                                                                                                                                                               |
| 2693.7 2                    | 10.96 18                         | 2693.9                 | 3/2 <sup>+</sup>            | 0              | 3/2 <sup>+</sup>            | M1+E2              | +0.26 2        | 0.000553 8              | %I <sub>γ</sub> =0.136 4<br>I <sub>γ</sub> : weighted average of 10.2 15 ( <a href="#">1971Ge04</a> ), 10.96 18 ( <a href="#">1980Wi13</a> ) and 11.02 34 ( <a href="#">1984Ad01</a> ).                                                                                                                |
| 2699 <sup>b</sup>           | <0.33 <sup>@</sup>               | 3918.3                 | 3/2 <sup>+</sup>            | 1219.2         | 1/2 <sup>+</sup>            | [M1,E2]            |                | 0.00060 6               | %I <sub>γ</sub> <0.00420                                                                                                                                                                                                                                                                               |
| 2748.5 6                    | 0.477 34                         | 3967.9                 | 1/2 <sup>+</sup>            | 1219.2         | 1/2 <sup>+</sup>            | [M1]               |                | 0.000569 8              | %I <sub>γ</sub> =0.0059 5<br>I <sub>γ</sub> : weighted average of 0.48 6 ( <a href="#">1980Wi13</a> ) and 0.476 34 ( <a href="#">1984Ad01</a> ).                                                                                                                                                       |

<sup>35</sup>Ar  $\varepsilon+\beta^+$  decay (1.7755 s) [1980Wi13](#),[1984Ad01](#) (continued)

$\gamma(^{35}\text{Cl})$  (continued)

| $E_\gamma$ <sup>†</sup> | $I_\gamma$ <sup>‡&amp;</sup> | $E_i(\text{level})$ | $J_i^\pi$        | $E_f$  | $J_f^\pi$        | Mult. <sup>#</sup> | $\delta$ <sup>#</sup> | $\alpha$ <sup>a</sup> | Comments                                                                                                                                                                                                                      |
|-------------------------|------------------------------|---------------------|------------------|--------|------------------|--------------------|-----------------------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2861 <sup>b</sup>       | <0.099 <sup>@</sup>          | 4624.5?             | 3/2,5/2          | 1763.1 | 5/2 <sup>+</sup> |                    |                       |                       | %I <sub>γ</sub> <0.00126                                                                                                                                                                                                      |
| 3002.1 4                | 7.29 14                      | 3002.2              | 5/2 <sup>+</sup> | 0      | 3/2 <sup>+</sup> | M1+E2              | +0.08 2               | 0.000671 9            | %I <sub>γ</sub> =0.0906 28<br>I <sub>γ</sub> : weighted average of 9.0 15 ( <a href="#">1971Ge04</a> ), 7.24 14 ( <a href="#">1980Wi13</a> ),<br>7.45 23 ( <a href="#">1984Ad01</a> ) and 6.9 5 ( <a href="#">1985Da04</a> ). |
| 3405 <sup>b</sup>       |                              | 4624.5?             | 3/2,5/2          | 1219.2 | 1/2 <sup>+</sup> |                    |                       |                       | I <sub>γ</sub> : could not be determined due to being obscured by another peak<br>( <a href="#">1980Wi13</a> ).                                                                                                               |
| 3918                    | 0.52 3                       | 3918.3              | 3/2 <sup>+</sup> | 0      | 3/2 <sup>+</sup> | M1+E2              |                       | 0.00108 7             | %I <sub>γ</sub> =0.0065 4                                                                                                                                                                                                     |
| 3968                    | 0.106 25                     | 3967.9              | 1/2 <sup>+</sup> | 0      | 3/2 <sup>+</sup> | [M1,E2]            |                       | 0.00109 7             | %I <sub>γ</sub> =0.00132 31                                                                                                                                                                                                   |
| 4624 <sup>b</sup>       | <0.035 <sup>@</sup>          | 4624.5?             | 3/2,5/2          | 0      | 3/2 <sup>+</sup> | D+Q                |                       |                       | %I <sub>γ</sub> <4.46×10 <sup>-4</sup>                                                                                                                                                                                        |

<sup>†</sup> From [1984Ad01](#), unless otherwise noted.

<sup>‡</sup> from [1980Wi13](#), unless otherwise noted.

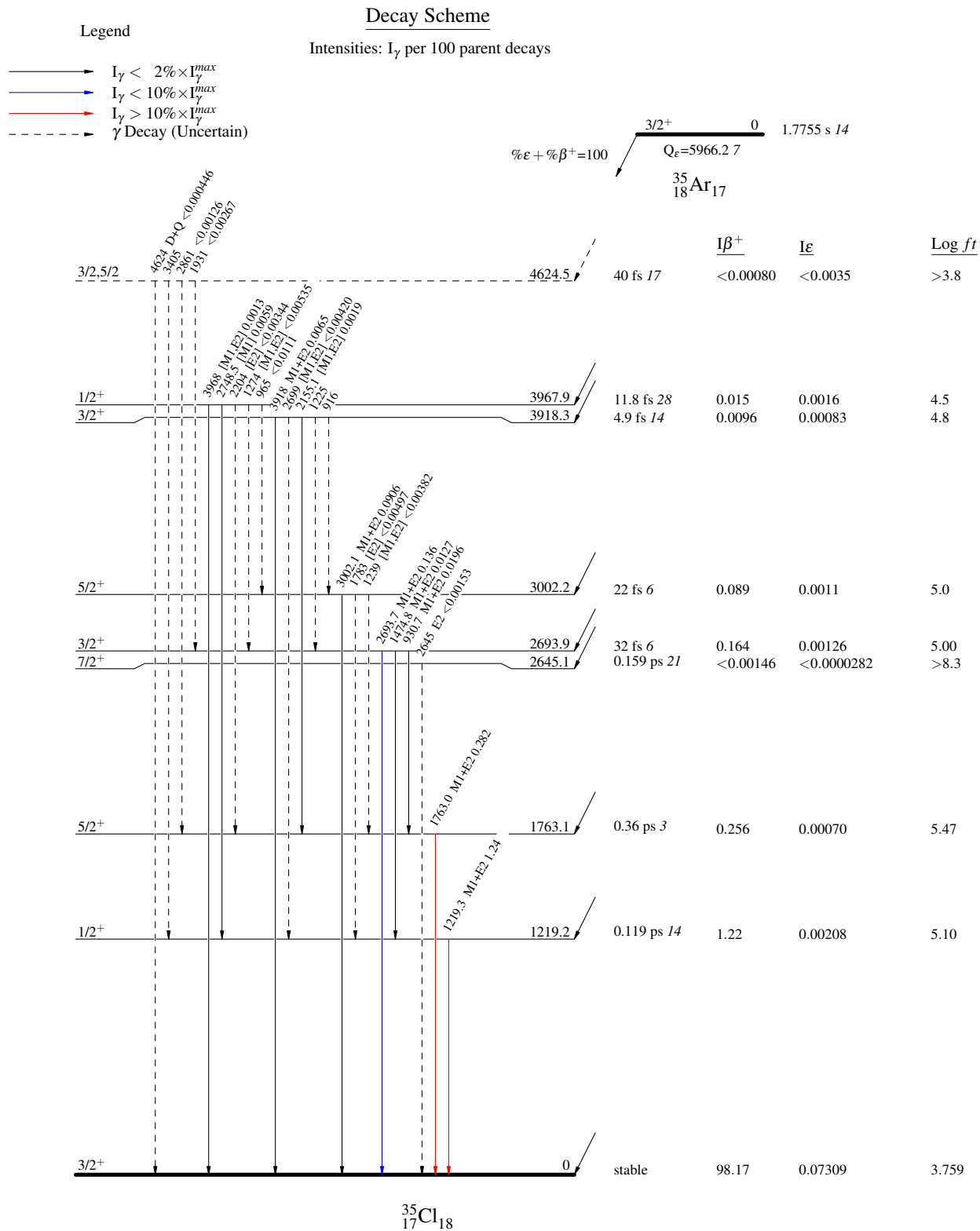
<sup>#</sup> From Adopted Gammas.

<sup>@</sup> [1980Wi13](#) gives upper limits of I<sub>γ</sub> at 98% confidence level. These peaks are not observed in the spectra of [1980Wi13](#) or in [1984Ad01](#).

<sup>&</sup> For absolute intensity per 100 decays, multiply by 0.01243 30.

<sup>a</sup> Total theoretical internal conversion coefficients, calculated using the BrIcc code ([2008Ki07](#)) with “Frozen Orbitals” approximation based on  $\gamma$ -ray energies, assigned multipolarities, and mixing ratios, unless otherwise specified.

<sup>b</sup> Placement of transition in the level scheme is uncertain.

$^{35}\text{Ar}$   $\varepsilon + \beta^+$  decay (1.7755 s) 1980Wi13,1984Ad01

### <sup>36</sup>K εp decay (341 ms) 1996II02,1980Es01

Parent: <sup>36</sup>K: E=0; J<sup>π</sup>=2<sup>+</sup>; T<sub>1/2</sub>=341 ms 3; Q(εp)=4307.4 3; %εp decay=5.1×10<sup>-2</sup>

<sup>36</sup>K-J<sup>π</sup>, T<sub>1/2</sub>: From the Adopted Levels of <sup>36</sup>K (2012Ni01).

<sup>36</sup>K-Q(εp): From 2021Wa16.

<sup>36</sup>K-%εp decay: %ε+β<sup>+</sup>p=5.1×10<sup>-4</sup> is scaled by evaluators from %ε+β<sup>+</sup>p=4.9×10<sup>-4</sup> reported in 1996II02. %ε+β<sup>+</sup>p=4.9×10<sup>-4</sup> in 1996II02 was normalized based on the strongest 970-keV ε+β<sup>+</sup>-delayed proton I<sub>p</sub>=3.3×10<sup>-4</sup> 9 measured by 1980Es01. 1980Es01 reported %ε+β<sup>+</sup>p=4.8×10<sup>-4</sup> and their I<sub>p</sub> was normalized based on the strongest 1970-keV ε+β<sup>+</sup>-delayed γ ray %I<sub>γ</sub>=79 8 measured by 1972Mi13. Evaluators adopt a more precise %I<sub>γ</sub>(1970)=81.8 16 from the evaluated <sup>36</sup>K ε decay dataset (2012Ni01,1976Fr03) as the new normalization reference, and therefore, all original I<sub>p</sub> values from 1996II02 and 1980Es01 are scaled by 1.04 11.

1996II02,1997II03: a 500-MeV proton beam was produced from the TRIUMF cyclotron and bombarded a CaO powder target. The spallation reaction products diffused out of the target and were selectively ionized at a Re surface. A=36 singly charged ions were selected by a magnetic mass analyzer and implanted into a carbon foil at 7.0×10<sup>5</sup> pps. Protons and α particles were detected using a 30-μm silicon surface barrier detector with FWHM=23 keV. <sup>32</sup>S and <sup>35</sup>Cl recoils were detected using an MCP in a back-to-back geometry for timing stop signals. Measured E<sub>p</sub>(>0.5 MeV) and I<sub>p</sub> for 15 proton peaks, E<sub>α</sub>(>1.5 MeV) and I<sub>α</sub> for 19 α peaks, p(recoil)-coin, and α(recoil)-coin. Deduced levels, J, π, decay branching ratios, log ft, partial decay widths for <sup>36</sup>Ar, and ratios of <sup>35</sup>Cl(p,α)<sup>32</sup>S to <sup>35</sup>Cl(p,γ)<sup>36</sup>Ar reaction rates.

1980Es01: a 20-MeV proton beam was produced from the University of Jyväskylä MC-20 cyclotron and bombarded a stack of four <sup>36</sup>Ar-implanted Al foils. The <sup>36</sup>Ar(p,n) reaction products were thermalized by helium, transported with NaCl-loaded helium, and deposited onto a fast-transport Al-coated mylar tape in a measurement chamber. Protons and α particles were detected using 14 and 26-μm fully depleted (FWHM=40 keV) and 100-μm partially depleted Si(Au) surface barrier detectors. A polyethylene absorber foil was placed in front of Si(Au) to cause an energy shift for distinguishing p/α. Measured E<sub>p</sub>(>0.5 MeV) and I<sub>p</sub> for 11 proton peaks, E<sub>α</sub>(>2.0 MeV) and I<sub>α</sub> for 12 α peaks. Deduced levels, J, π, decay branching ratios, log ft, and partial decay widths for <sup>36</sup>Ar.

1980Ew01: a 600-MeV proton beam was produced from the synchrocyclotron at CERN-ISOLDE and bombarded a ScC<sub>2</sub> target. The <sup>45</sup>Sc(p,7n3p) spallation reaction products diffused out of the target and reached a tungsten surface ionization source where potassium isotopes were selectively ionized. The beam was extracted from the ion source, separated by the ISOLDE analyzing magnet, and collected by a carbon foil for proton measurements. Protons and α particles were detected using 20-μm and 100-μm silicon surface barrier detectors with FWHM=35 and 23 keV. Measured E<sub>p</sub>(>0.5 MeV) and I<sub>p</sub> for 3 proton peaks, E<sub>α</sub> and I<sub>α</sub> for 4 α peaks. Deduced levels, J, π, and decay branching ratios for <sup>36</sup>Ar.

#### <sup>35</sup>Cl Levels

| E(level) | J <sup>π</sup> † |
|----------|------------------|
| 0        | 3/2 <sup>+</sup> |

† From the Adopted Levels.

#### Delayed Protons (<sup>35</sup>Cl)

| E(p)†     | E( <sup>35</sup> Cl) | I(p)†@                   | E( <sup>36</sup> Ar)# | Comments                                                                                                                             |
|-----------|----------------------|--------------------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| 501.8 11  | 0                    | 1.14×10 <sup>-6</sup> 33 | 9023                  | E(p): other: 501 10 (1980Es01).<br>I(p): other: 3.6×10 <sup>-6</sup> 13 (1980Es01).                                                  |
| 622.7 14  | 0                    | 1.6×10 <sup>-6</sup> 5   | 9147                  |                                                                                                                                      |
| 693.5 8   | 0                    | 7.9×10 <sup>-5</sup> 22  | 9220                  | E(p): other: 693 5 (1980Es01), 691 8 (1980Ew01).<br>I(p): other: 7.8×10 <sup>-5</sup> 24 (1980Es01).                                 |
| 851.7 10  | 0                    | 2.3×10 <sup>-5</sup> 7   | 9383                  | E(p): other: 849 5 (1980Es01).<br>I(p): other: 2.0×10 <sup>-5</sup> 7 (1980Es01).                                                    |
| 969.6 12  | 0                    | 3.4×10 <sup>-4</sup> 10  | 9504                  | E(p): other: 970 5 (1980Es01), 959 7 (1980Ew01).<br>I(p): from 1980Es01 and also adopted as the normalization reference by 1996II02. |
| 1168.5 14 | 0                    | 6.2×10 <sup>-7</sup> 7   | 9709                  |                                                                                                                                      |

Continued on next page (footnotes at end of table)

**$^{36}\text{K}$   $\varepsilon$ p decay (341 ms) 1996II02,1980Es01 (continued)**Delayed Protons (continued)

| $E(p)^{\dagger}$      | $E(^{35}\text{Cl})$ | $I(p)^{\dagger @}$       | $E(^{36}\text{Ar})^{\#}$ | Comments                                                                                               |
|-----------------------|---------------------|--------------------------|--------------------------|--------------------------------------------------------------------------------------------------------|
| 1198.5 11             | 0                   | $2.5 \times 10^{-6}$ 7   | 9740                     |                                                                                                        |
| 1272.0 20             | 0                   | $1.6 \times 10^{-6}$ 5   | 9815                     |                                                                                                        |
| 1334.2 7              | 0                   | $4.5 \times 10^{-5}$ 12  | 9879                     | E(p): other: 1333 5 (1980Es01), 1325 7 (1980Ew01).<br>I(p): other: $3.6 \times 10^{-5}$ 11 (1980Es01). |
| 1409.3 22             | 0                   | $1.04 \times 10^{-6}$ 33 | 9957                     |                                                                                                        |
| 1874 $^{\ddagger}$ 10 | 0                   | $1.1 \times 10^{-6}$ 5   | 10435                    |                                                                                                        |
| 1992 $^{\ddagger}$ 10 | 0                   | $5.0 \times 10^{-6}$ 20  | 10556                    |                                                                                                        |
| 2048 $^{\ddagger}$ 10 | 0                   | $4.9 \times 10^{-6}$ 19  | 10613                    |                                                                                                        |
| 2458 $^{\ddagger}$ 10 | 0                   | $3.0 \times 10^{-6}$ 13  | 11035                    |                                                                                                        |
| 2640 $^{\ddagger}$ 10 | 0                   | $2.1 \times 10^{-6}$ 10  | 11222                    |                                                                                                        |

$^{\dagger}$  From 1996II02, unless otherwise noted. E(p) is in lab frame.

$^{\ddagger}$  From 1980Es01 and also adopted by 1996II02.

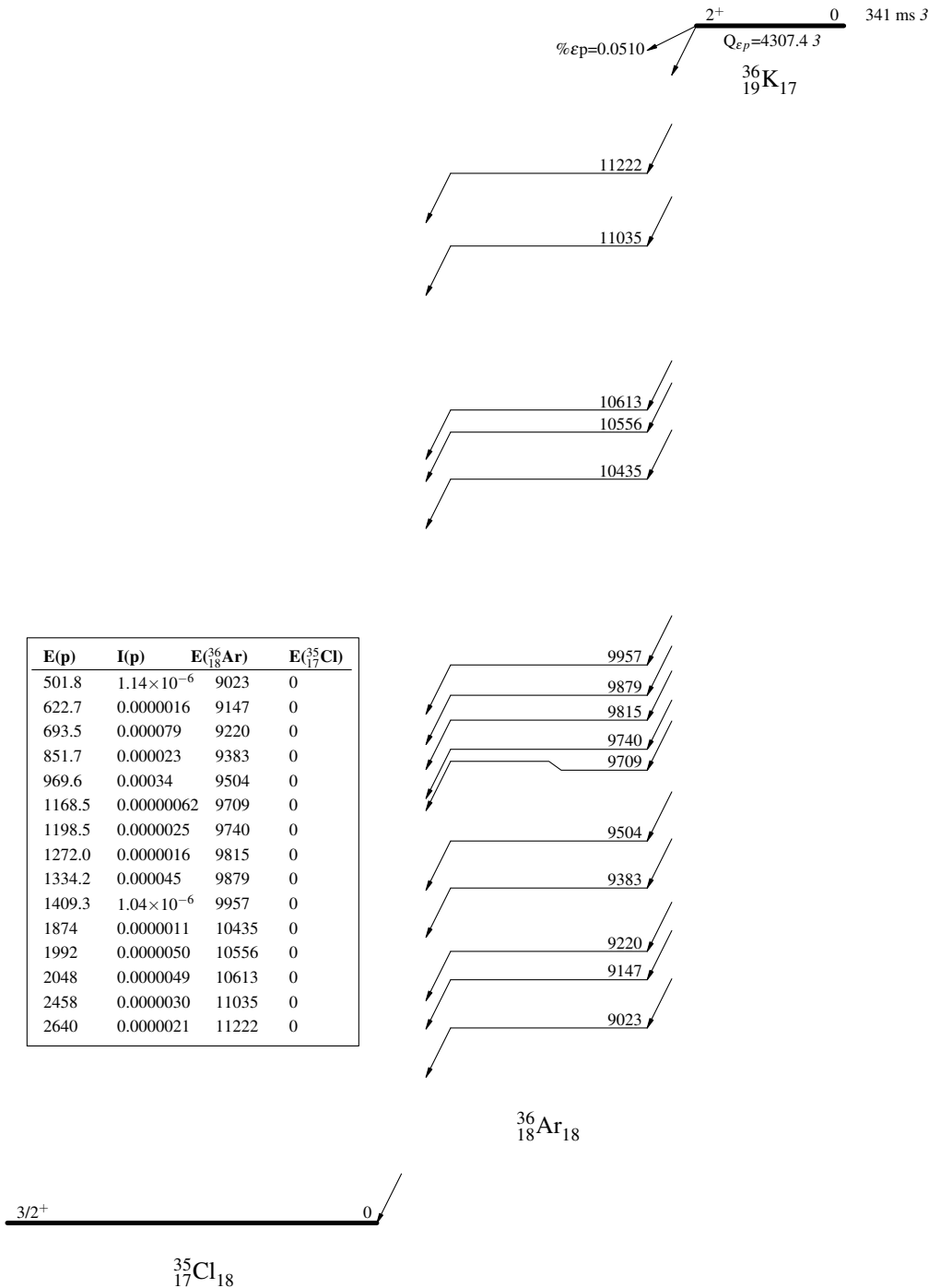
$^{\#}$   $E(\text{level})(^{36}\text{Ar}) = E(p)_{\text{lab}} \times [m(p) + m(^{35}\text{Cl})] / m(^{35}\text{Cl}) + S(p)(^{36}\text{Ar}) + E(\text{level})(^{35}\text{Cl})$ , where  $S(p)(^{36}\text{Ar}) = 8506.97$  4 (2021Wa16). Proton decay is assumed to populate only the ground state of the daughter nucleus  $^{35}\text{Cl}$  based on the arguments proposed in 1996II02: (A) unreasonably small  $\log ft$  values if populating excited states; (B) no pairs of particle groups characteristic of the g.s./first excited state energy difference; (C) compatible level matching with resonance reaction studies.

$^{\textcircled{a}}$  Absolute intensity per 100 decays.

<sup>36</sup>K  $\epsilon$ p decay (341 ms) 1996IN02,1980Es01

Decay Scheme

I(p) Intensities: I(p) per 100 parent decays



$^1\text{H}(^{34}\text{S}, ^{35}\text{Cl}\gamma)$  **2021Lo09**

$^{34}\text{S}(\text{p},\gamma)^{35}\text{Cl}$  in inverse kinematics.

**2021Lo09:**  $^{34}\text{S}$  was produced by the off-line ion source (OLIS) and accelerated by the ISAC accelerator at TRIUMF.  $10^{10}$ – $10^{11}$  pps  $^{34}\text{S}^{+7}$  ions were delivered to a windowless 2-10 Torr hydrogen gas target with an effective length of 12.3 cm. Elastically scattered protons were detected by two silicon surface barrier detectors inside the gas target. Prompt  $\gamma$  rays from the compound recoil nuclei were detected by a BGO array in coincidence with the heavy-ion recoils separated by the DRAGON separator and identified by two MCP detectors, a DSSD, and an ionization chamber at the focal plane. Deduced resonance strengths ( $\omega\gamma$ ) for 8 resonances and the  $^{34}\text{S}(\text{p},\gamma)^{35}\text{Cl}$  astrophysical reaction rate.

 $^{35}\text{Cl}$  Levels

Values of resonance strength  $\omega\gamma$  under comments are from direct measurements in **2021Lo09**.

| <u>E(level)<sup>†</sup></u> | <u>Comments</u>                                                   |
|-----------------------------|-------------------------------------------------------------------|
| 6643                        | $E_{\text{R}}=272$ keV. $\omega\gamma=1.22\times 10^{-5}$ eV 36.  |
| 6674                        | $E_{\text{R}}=303$ keV. $\omega\gamma \leq 7.2\times 10^{-6}$ eV. |
| 6761                        | $E_{\text{R}}=390$ keV. $\omega\gamma \leq 6.9\times 10^{-6}$ eV. |
| 6778                        | $E_{\text{R}}=407$ keV. $\omega\gamma=8.0\times 10^{-4}$ eV 19.   |
| 6802                        | $E_{\text{R}}=431$ keV. $\omega\gamma \leq 3.9\times 10^{-5}$ eV. |
| 6823                        | $E_{\text{R}}=452$ keV. $\omega\gamma \leq 3.1\times 10^{-5}$ eV. |
| 6842                        | $E_{\text{R}}=471$ keV. $\omega\gamma=9.4\times 10^{-5}$ eV 29.   |
| 6866                        | $E_{\text{R}}=496$ keV. $\omega\gamma=1.37\times 10^{-2}$ eV 22.  |

<sup>†</sup> Excitation energies are deduced by evaluators from resonance energies  $E_{\text{R}}+S_{\text{p}}(^{35}\text{Cl})$ , where  $S_{\text{p}}(^{35}\text{Cl})=6370.81$  4 (**2021Wa16**).

$^2\text{H}(^{34}\text{S}, \text{n}\gamma)$  1973Wa10

$^{34}\text{S}(\text{d}, \text{n})^{35}\text{Cl}$  in inverse kinematics.

1973Wa10: A 59.6-MeV  $^{34}\text{S}$  beam was produced from the BNL MP-tandem Van de Graaff facility. Targets were 200- $\mu\text{g}/\text{cm}^2$  TiD prepared by evaporating titanium onto the Cu, Al, and Mg target backings in a deuterium atmosphere.  $\gamma$  rays were detected using a 35-cm<sup>3</sup> Ge(Li) detector at 0° with FWHM=2 keV at 656 keV. Measured  $E_\gamma$ . Deduced  $T_{1/2}$  for two  $^{35}\text{Cl}$  levels using Doppler Shift Attenuation lineshape analysis.

 $^{35}\text{Cl}$  Levels

| <u><math>E(\text{level})^\dagger</math></u> | <u><math>J^\pi^\ddagger</math></u> | <u><math>T_{1/2}^\#</math></u> | Comments                                      |
|---------------------------------------------|------------------------------------|--------------------------------|-----------------------------------------------|
| 0                                           | $3/2^+$                            |                                |                                               |
| 1219                                        | $1/2^+$                            | 201 fs 28                      | $T_{1/2}$ : Lifetime=0.29 ps 4 from 1973Wa10. |
| 1762                                        | $5/2^+$                            | 374 fs 49                      | $T_{1/2}$ : Lifetime=0.54 ps 7 from 1973Wa10. |

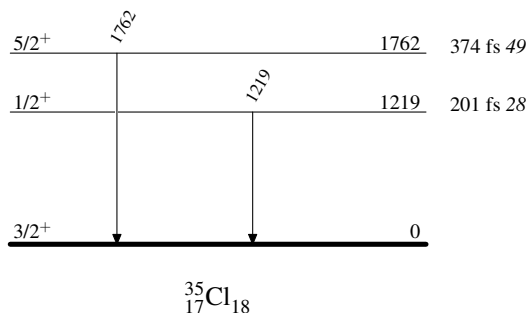
$^\dagger$  From 1973Wa10.

$^\ddagger$  From the Adopted Levels.

$^\#$  From Doppler Shift Attenuation Method (DSAM) in 1973Wa10.

 $\gamma(^{35}\text{Cl})$ 

| <u><math>E_\gamma</math></u> | <u><math>E_i(\text{level})</math></u> | <u><math>J_i^\pi</math></u> | <u><math>E_f</math></u> | <u><math>J_f^\pi</math></u> |
|------------------------------|---------------------------------------|-----------------------------|-------------------------|-----------------------------|
| 1219                         | 1219                                  | $1/2^+$                     | 0                       | $3/2^+$                     |
| 1762                         | 1762                                  | $5/2^+$                     | 0                       | $3/2^+$                     |

 $^2\text{H}(^{34}\text{S}, \text{n}\gamma)$  1973Wa10Level Scheme



$^{12}\text{C}(^{28}\text{Si}, \alpha p \gamma)$  **2007Ks01, 2013Bi10, 2020Ra14**

**2007Ks01:** two different experiments: one with  $E=70$  MeV  $^{28}\text{Si}$  beam at BARC-TIFR facility and the second with  $E=88$  MeV  $^{28}\text{Si}$  beam at NSC facility, both using a target of about  $50 \mu\text{g}/\text{cm}^2$   $^{12}\text{C}$  on a  $\approx 10.5\text{-mg}/\text{cm}^2$  gold backing.  $\gamma$  rays were detected with almost identical (INGA) arrays of eight Compton-suppressed Clover HPGe detectors. Measured  $E\gamma$ ,  $I\gamma$ ,  $\gamma\gamma$ -coin,  $\gamma\gamma(\theta)(\text{DCO})$ ,  $\gamma\gamma(\text{lin pol})$ , Doppler-shift attenuations. Deduced levels,  $J$ ,  $\pi$ ,  $T_{1/2}$ ,  $\gamma$ -ray multipolarities and mixing ratios. Comparisons with available data and large-scale shell-model calculations.

**2013Bi10:**  $E=110$  MeV  $^{28}\text{Si}$  beam was produced from the IUAC-New Delhi accelerator facility. Target was  $50 \mu\text{g}/\text{cm}^2$  thick  $^{12}\text{C}$  on  $18 \text{mg}/\text{cm}^2$  Au backing.  $\gamma$  rays were detected with the INGA array of thirteen Compton-suppressed Clover detectors. Measured  $E\gamma$ ,  $I\gamma$ ,  $\gamma\gamma$ -coin,  $\gamma\gamma(\theta)(\text{DCO})$ , Doppler-shift attenuations. Deduced levels,  $J$ ,  $\pi$ ,  $T_{1/2}$ , evidence for an SD band,  $\gamma$ -ray multipolarities. Comparison with large-scale shell-model calculations, two-level mixing calculation, and with structure of SD states in  $^{36}\text{Ar}$ .

**2020Ra14:**  $E=110$  MeV  $^{28}\text{Si}$  beam was produced from the 15UD Pelletron accelerator at the Inter-University Accelerator Centre (IUAC), New Delhi. Target was  $50 \mu\text{g}/\text{cm}^2$   $^{12}\text{C}$  evaporated onto  $18 \text{mg}/\text{cm}^2$  Au backing.  $\gamma$  rays were detected with a multi-detector array (Indian National Gamma Array) consisting of 13 Compton-suppressed clovers. Measured  $E\gamma$ ,  $I\gamma$ ,  $\gamma\gamma$ -coin,  $\gamma\gamma(\text{DCO})$ ,  $\gamma\gamma(\text{ADO})$ ,  $\gamma\gamma(\text{IPDCO})$ . Deduced levels,  $J$ ,  $\pi$ ,  $\gamma$ -ray multipolarities. Comparisons with theoretical calculations.

Other reactions:

**1982Le04:**  $^{28}\text{Si}(^{12}\text{C}, X\gamma) ^{12}\text{C}$  ( $E_{\text{cm}}=23\text{-}35$  MeV) beam produced from the University of Washington FN Tandem Van de Graaff.

A target of  $100 \mu\text{g}/\text{cm}^2$   $^{28}\text{Si}$  on a thick tantalum backing. A Ge(Li) detector. Measured  $\sigma(E)$ ,  $E\gamma$ .

**1985Kr21:**  $^{27}\text{Al}(^{12}\text{C}, X) E=85$  MeV/nucleon. Measured particle-particle angular correlations.

 $^{35}\text{Cl}$  Levels

| $E(\text{level})^\dagger$ | $J\pi^\ddagger$ | $T_{1/2}^\#$         | Comments                                                                                                                                                                                                                              |
|---------------------------|-----------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0.0                       | $3/2^+$         |                      |                                                                                                                                                                                                                                       |
| 1763.0 6                  | $5/2^+$         |                      |                                                                                                                                                                                                                                       |
| 2645.8 6                  | $7/2^+$         | $<0.90$ ps           |                                                                                                                                                                                                                                       |
| 3003.1 8                  | $5/2^+$         |                      |                                                                                                                                                                                                                                       |
| 3163.1 6                  | $7/2^-$         |                      |                                                                                                                                                                                                                                       |
| 3942.0 7                  | $9/2^+$         | $<0.35$ ps           |                                                                                                                                                                                                                                       |
| 4347.8 6                  | $9/2^-$         | $0.91$ ps $+21-16$   |                                                                                                                                                                                                                                       |
| 5407.4 8                  | $11/2^-$        | $<0.76$ ps           |                                                                                                                                                                                                                                       |
| 5927.1 7                  | $(11/2)^-$      | $<0.28$ ps           |                                                                                                                                                                                                                                       |
| 6087.6 8                  | $13/2^-$        |                      |                                                                                                                                                                                                                                       |
| 6119.5? 13                | $(11/2^-)$      |                      |                                                                                                                                                                                                                                       |
| 6660.2 8                  | $(11/2^-)$      |                      |                                                                                                                                                                                                                                       |
| 7873.3 8                  | $13/2^+$        |                      |                                                                                                                                                                                                                                       |
| 8319.4 <sup>a</sup> 9     | $15/2^-$        | $12.5\text{&}$ fs 7  | Tentative 8324, $(13/2^-)$ level in <b>2007Ks01</b> is considered the same level here by the evaluators.<br>$T_{1/2}$ : other: $<70$ fs ( <b>2007Ks01</b> ).<br>$Q(\text{transition})=0.65$ 2, $\beta_2=0.30$ ( <b>2013Bi10</b> ).    |
| 8487.8 10                 | $15/2^-$        | $<0.07$ ps           | Tentative 8481, $(15/2^-)$ level in <b>2007Ks01</b> is considered the same level here by the evaluators.                                                                                                                              |
| 8558.2? 13                | $(13/2^-)$      | $<0.14$ ps           |                                                                                                                                                                                                                                       |
| 8787.3? 13                | $(15/2^+)$      | $<0.28$ ps           |                                                                                                                                                                                                                                       |
| 8845.1 <sup>@</sup> 9     | $17/2^+$        |                      |                                                                                                                                                                                                                                       |
| 10181.1 <sup>a</sup> 10   | $19/2^-$        | $42.3\text{&}$ fs 28 | Tentative 10181, $(19/2^+)$ level in <b>2007Ks01</b> is considered the same level here by the evaluators.<br>$T_{1/2}$ : other: $<0.14$ ps ( <b>2007Ks01</b> ).<br>$Q(\text{transition})=0.82$ 6, $\beta_2=0.37$ ( <b>2013Bi10</b> ). |
| 10223.2? 14               | $(17/2^-)$      | $<0.28$ ps           |                                                                                                                                                                                                                                       |
| 10862.2? 14               | $19/2^+$        | $<0.28$ ps           |                                                                                                                                                                                                                                       |
| 11459.2 <sup>@</sup> 12   | $21/2^+$        | $43.0\text{&}$ fs 21 | $Q(\text{transition})=0.56$ 2, $\beta_2=0.27$ ( <b>2013Bi10</b> ).                                                                                                                                                                    |
| 12572.2 <sup>a</sup> 12   | $23/2^-$        | $33\text{&}$ fs 4    | $Q(\text{transition})=0.71$ 5, $\beta_2=0.32$ ( <b>2013Bi10</b> ).                                                                                                                                                                    |
| 16306.4 <sup>a</sup> 16   | $(27/2^-)$      | $<9.0\text{&}$ fs    | $Q(\text{transition})>0.5$ , $\beta_2=0.24$ ( <b>2013Bi10</b> ).                                                                                                                                                                      |

Continued on next page (footnotes at end of table)

$^{12}\text{C}(^{28}\text{Si}, \alpha p \gamma)$  **2007Ks01, 2013Bi10, 2020Ra14 (continued)** $^{35}\text{Cl}$  Levels (continued)<sup>†</sup> From a least-squares fit to  $\gamma$ -ray energies, assuming  $\Delta E_{\gamma}=1$  keV in the fitting procedure.<sup>‡</sup> From the Adopted Levels, unless otherwise noted. Assignments for tentative levels not adopted in Adopted Levels are proposed by **2007Ks01**.# From DSAM in **2007Ks01**, unless otherwise noted.@ Existence of  $17/2^{+}$  and  $21/2^{+}$  levels provides indirect evidence in favor of SD band interpretation and parity-doublet cluster structure (**2013Bi10**).& From DSAM in **2013Bi10**.<sup>a</sup> Band(A): SD band. Proposed configuration= $3p-3h$ ,  $(sd)^{16}(pf)^3$ . Negative-parity partner of SD band in  $^{36}\text{Ar}$  (**2013Bi10**). $\gamma(^{35}\text{Cl})$  $\Delta_{\text{IPDCO}}$  corresponds to integrated polarization asymmetry measurement at  $E=70$  MeV at TIFR facility (**2007Ks01**). The values are from e-mail reply of Dec 1, 2006 from one of the authors (M. Saha Sarkar) of **2007Ks01** to the previous evaluator B. Singh. $\Delta_{\text{IPDCO}}$  and DCO under comments are from **2007Ks01**, unless otherwise noted. Positive value of  $\Delta_{\text{IPDCO}}$  indicates electric character and negative value indicates magnetic character. Expected value of DCO is  $\approx 1.0$  for a stretched transition gated by a transition of same multipolarity,  $\approx 2.0$  for dipole when gated by a stretched quadrupole (DCO(Q)) and  $\approx 0.5$  for stretched quadrupole when gated by a stretched dipole (DCO(D)) (**2007Ks01**). Values deviating from above values indicate a mixed transition. DCO values are also available in **2020Ra14**, but expected values are opposite to those in **2007Ks01** for stretched dipole and quadrupole transitions when gated by a transition of different multipolarity, that is DCO(Q) $\approx 0.5$  for stretched dipole and DCO(D) $\approx 2.0$  for stretched quadrupole (**2020Ra14**).For ADO ratios under comments from **2020Ra14**, expected values are 0.6 for pure dipole transitions and 1.6 for pure quadrupole transitions or  $\Delta J=0$  transitions (**2020Ra14**).

| $E_{\gamma}^{\dagger}$ | $I_{\gamma}^{\ddagger}$ | $E_i(\text{level})$ | $J_i^{\pi}$  | $E_f$   | $J_f^{\pi}$ | Mult. <sup>#</sup> | $\delta^{\#}$ | Comments                                                                                                                                                                                                                                                                                                                                                       |
|------------------------|-------------------------|---------------------|--------------|---------|-------------|--------------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 160 <sup>@</sup>       | 1.1 3                   | 3163.1              | $7/2^{-}$    | 3003.1  | $5/2^{+}$   | D                  |               | I <sub><math>\gamma</math></sub> : other: 0.6 2 at $E=70$ MeV.<br>I <sub><math>\gamma</math></sub> : other: 10.0 2 at $E=70$ MeV.<br>DCO(Q)=0.8 1.<br>DCO(D)=1.5 2, $R_{\text{ADO}}=1.6$ 4 ( <b>2020Ra14</b> ).                                                                                                                                                |
| 518                    | 9.2 5                   | 3163.1              | $7/2^{-}$    | 2645.8  | $7/2^{+}$   |                    |               |                                                                                                                                                                                                                                                                                                                                                                |
| 526                    |                         | 8845.1              | $17/2^{+}$   | 8319.4  | $15/2^{-}$  | M1(+E2)            | +0.1 1        | I <sub><math>\gamma</math></sub> : other: 8.6 2 at $E=70$ MeV.<br>DCO(Q)=1.3 1 at $E=70$ MeV.<br>$\Delta_{\text{IPDCO}}=-0.09$ 2.                                                                                                                                                                                                                              |
| 680                    | 32 1                    | 6087.6              | $13/2^{-}$   | 5407.4  | $11/2^{-}$  |                    |               |                                                                                                                                                                                                                                                                                                                                                                |
| 712 <sup>@&amp;</sup>  | <1                      | 6119.5?             | $(11/2^{-})$ | 5407.4  | $11/2^{-}$  | D(+Q)<br>E2        | -0.2 2        | I <sub><math>\gamma</math></sub> =4.6 2 at $E=70$ MeV.<br>$\Delta_{\text{IPDCO}}=+0.01$ 5.<br>DCO(Q)=2.2 6 at $E=88$ MeV.<br>I <sub><math>\gamma</math></sub> : other: 2.0 1 at $E=70$ MeV.<br>DCO(Q)=1.0 2 at $E=70$ MeV, 0.9 1 at $E=88$ MeV.<br>$\Delta_{\text{IPDCO}}=+0.08$ 4.<br>DCO(D)=1.8 2, DCO(Q)=0.9 1, $R_{\text{ADO}}=1.6$ 2 ( <b>2020Ra14</b> ). |
| 779 <sup>@&amp;</sup>  | <1                      | 3942.0              | $9/2^{+}$    | 3163.1  | $7/2^{-}$   |                    |               |                                                                                                                                                                                                                                                                                                                                                                |
| 883                    | 3.2 2                   | 2645.8              | $7/2^{+}$    | 1763.0  | $5/2^{+}$   |                    |               |                                                                                                                                                                                                                                                                                                                                                                |
| 914 <sup>&amp;</sup>   | 2.4 5                   | 8787.3?             | $(15/2^{+})$ | 7873.3  | $13/2^{+}$  | E2                 |               | I <sub><math>\gamma</math></sub> : other: 2.0 1 at $E=70$ MeV.<br>DCO(Q)=1.0 2 at $E=70$ MeV, 0.9 1 at $E=88$ MeV.<br>$\Delta_{\text{IPDCO}}=+0.08$ 4.<br>DCO(D)=1.8 2, DCO(Q)=0.9 1, $R_{\text{ADO}}=1.6$ 2 ( <b>2020Ra14</b> ).                                                                                                                              |
| 972                    | 33 1                    | 8845.1              | $17/2^{+}$   | 7873.3  | $13/2^{+}$  |                    |               |                                                                                                                                                                                                                                                                                                                                                                |
| 1060                   | 7.3 7                   | 5407.4              | $11/2^{-}$   | 4347.8  | $9/2^{-}$   | M1+E2              | +0.2 1        | I <sub><math>\gamma</math></sub> : other: 6.9 3 at $E=70$ MeV.<br>DCO(D)=0.9 1.<br>$\Delta_{\text{IPDCO}}=-0.10$ 4.                                                                                                                                                                                                                                            |
| 1113                   |                         | 12572.2             | $23/2^{-}$   | 11459.2 | $21/2^{+}$  | M1+E2              | -0.3 1        | I <sub><math>\gamma</math></sub> : other: 25 1 at $E=70$ MeV.<br>DCO(Q)=2.0 2 at $E=70$ MeV, 2.8 3 at $E=88$ MeV.<br>$\Delta_{\text{IPDCO}}=-0.08$ 3.                                                                                                                                                                                                          |
| 1185                   | 25 1                    | 4347.8              | $9/2^{-}$    | 3163.1  | $7/2^{-}$   |                    |               |                                                                                                                                                                                                                                                                                                                                                                |

Continued on next page (footnotes at end of table)

$^{12}\text{C}(^{28}\text{Si},\alpha p\gamma)$  **2007Ks01,2013Bi10,2020Ra14** (continued) $\gamma(^{35}\text{Cl})$  (continued)

| $E_\gamma$ <sup>†</sup> | $I_\gamma$ <sup>‡</sup> | $E_i(\text{level})$ | $J_i^\pi$            | $E_f$   | $J_f^\pi$            | Mult. <sup>#</sup> | $\delta$ <sup>#</sup> | Comments                                                                                                                                                                                                                                                          |
|-------------------------|-------------------------|---------------------|----------------------|---------|----------------------|--------------------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1213                    |                         | 7873.3              | 13/2 <sup>+</sup>    | 6660.2  | (11/2 <sup>-</sup> ) |                    |                       |                                                                                                                                                                                                                                                                   |
| 1296                    | 0.4 2                   | 3942.0              | 9/2 <sup>+</sup>     | 2645.8  | 7/2 <sup>+</sup>     |                    |                       | $I_\gamma$ : other: 0.2 1 at E=70 MeV.                                                                                                                                                                                                                            |
| 1336                    | 6 1                     | 10181.1             | 19/2 <sup>-</sup>    | 8845.1  | 17/2 <sup>+</sup>    | D(+Q)              | -0.2 2                | DCO(Q)=2.0 4 at E=88 MeV.                                                                                                                                                                                                                                         |
| 1378 @&                 | 2 1                     | 10223.2?            | (17/2 <sup>-</sup> ) | 8845.1  | 17/2 <sup>+</sup>    |                    |                       |                                                                                                                                                                                                                                                                   |
| 1400                    | <1                      | 3163.1              | 7/2 <sup>-</sup>     | 1763.0  | 5/2 <sup>+</sup>     |                    |                       |                                                                                                                                                                                                                                                                   |
| 1579                    | 18 2                    | 5927.1              | (11/2 <sup>-</sup> ) | 4347.8  | 9/2 <sup>-</sup>     | (M1+E2)            | -0.2 1                | $I_\gamma$ : other: 12 1 at E=70 MeV.<br>DCO(D)=0.9 1 at E=70 MeV, 1.0 1 at E=88 MeV.<br>$\Delta\text{IPDCO}=+0.02$ 3.                                                                                                                                            |
| 1693                    |                         | 10181.1             | 19/2 <sup>-</sup>    | 8487.8  | 15/2 <sup>-</sup>    |                    |                       |                                                                                                                                                                                                                                                                   |
| 1702                    | 11.4 6                  | 4347.8              | 9/2 <sup>-</sup>     | 2645.8  | 7/2 <sup>+</sup>     | E1+M2              | +0.5 +5-3             | $I_\gamma$ : other: 10.8 5 at E=70 MeV.<br>DCO(Q)=1.0 2 from <b>2007Ks01</b> indicating $\Delta J=2$ or 0 seems inconsistent with $\Delta J=1$ from level scheme.<br>$\Delta\text{IPDCO}=+0.04$ 4.<br>DCO(Q)=0.49 6, $R_{\text{ADO}}=0.60$ 6 ( <b>2020Ra14</b> ). |
| 1740                    | 2.0 2                   | 6087.6              | 13/2 <sup>-</sup>    | 4347.8  | 9/2 <sup>-</sup>     | E2                 |                       | $I_\gamma$ : other: 2.3 2 at E=70 MeV.<br>$\Delta\text{IPDCO}=+0.12$ 7.                                                                                                                                                                                           |
| 1763                    | >9.7                    | 1763.0              | 5/2 <sup>+</sup>     | 0.0     | 3/2 <sup>+</sup>     | E2+M1              |                       | $I_\gamma$ : other: >15.7 at E=70 MeV.<br>$\Delta\text{IPDCO}=+0.13$ 5.                                                                                                                                                                                           |
| 1786                    | 1.8 4                   | 7873.3              | 13/2 <sup>+</sup>    | 6087.6  | 13/2 <sup>-</sup>    | D(+Q)              | -0.6 6                | $I_\gamma$ : other: 0.1 1 at E=70 MeV.<br>DCO(Q)=1.2 3 at E=88 MeV.                                                                                                                                                                                               |
| 1862                    |                         | 10181.1             | 19/2 <sup>-</sup>    | 8319.4  | 15/2 <sup>-</sup>    |                    |                       |                                                                                                                                                                                                                                                                   |
| 1946                    | 10 1                    | 7873.3              | 13/2 <sup>+</sup>    | 5927.1  | (11/2 <sup>-</sup> ) | E1+M2              | +0.2 1                | $I_\gamma$ : other: 0.5 1 at E=70 MeV.<br>DCO(Q)=1.3 2 at E=88 MeV.<br>$\Delta\text{IPDCO}=+0.11$ 6.<br>DCO(Q)=0.45 8, $R_{\text{ADO}}=0.59$ 8 ( <b>2020Ra14</b> ).                                                                                               |
| 1985 @                  | 2.0 4                   | 5927.1              | (11/2 <sup>-</sup> ) | 3942.0  | 9/2 <sup>+</sup>     |                    |                       | $I_\gamma$ : other: 1.6 6 at E=70 MeV.                                                                                                                                                                                                                            |
| 2017 @&                 | 7 1                     | 10862.2?            | 19/2 <sup>+</sup>    | 8845.1  | 17/2 <sup>+</sup>    |                    |                       |                                                                                                                                                                                                                                                                   |
| 2179                    | 6.5 7                   | 3942.0              | 9/2 <sup>+</sup>     | 1763.0  | 5/2 <sup>+</sup>     | E2                 |                       | $I_\gamma$ : other: 11.1 6 at E=70 MeV.<br>$\Delta\text{IPDCO}=+0.07$ 5.                                                                                                                                                                                          |
| 2232                    |                         | 8319.4              | 15/2 <sup>-</sup>    | 6087.6  | 13/2 <sup>-</sup>    |                    |                       | $E_\gamma$ : from <b>2013Bi10</b> only.                                                                                                                                                                                                                           |
| 2244                    | 61 2                    | 5407.4              | 11/2 <sup>-</sup>    | 3163.1  | 7/2 <sup>-</sup>     | E2                 |                       | $I_\gamma$ : other: 50 3 at E=70 MeV.<br>DCO(Q)=1.0 1 at E=70 MeV, 1.2 1 at E=88 MeV.<br>$\Delta\text{IPDCO}=+0.04$ 2.                                                                                                                                            |
| 2312                    |                         | 6660.2              | (11/2 <sup>-</sup> ) | 4347.8  | 9/2 <sup>-</sup>     |                    |                       |                                                                                                                                                                                                                                                                   |
| 2391                    |                         | 12572.2             | 23/2 <sup>-</sup>    | 10181.1 | 19/2 <sup>-</sup>    |                    |                       |                                                                                                                                                                                                                                                                   |
| 2400                    | <1                      | 8487.8              | 15/2 <sup>-</sup>    | 6087.6  | 13/2 <sup>-</sup>    | D(+Q)              | +0.2 +3-2             | $E_\gamma$ : other: 2394 ( <b>2007Ks01</b> ).<br>DCO(Q)=1.4 4 at E=88 MeV.                                                                                                                                                                                        |
| 2466                    | 22 2                    | 7873.3              | 13/2 <sup>+</sup>    | 5407.4  | 11/2 <sup>-</sup>    | E1+M2              | -0.25 10              | $I_\gamma$ : other: 1.2 1 at E=70 MeV.<br>DCO(Q)=2.4 3.<br>$\Delta\text{IPDCO}=+0.21$ 6.<br>DCO(Q)=0.51 8, $R_{\text{ADO}}=0.65$ 9 ( <b>2020Ra14</b> ).                                                                                                           |
| 2614                    |                         | 11459.2             | 21/2 <sup>+</sup>    | 8845.1  | 17/2 <sup>+</sup>    |                    |                       |                                                                                                                                                                                                                                                                   |
| 2631 @&                 | <1                      | 8558.2?             | (13/2 <sup>-</sup> ) | 5927.1  | (11/2 <sup>-</sup> ) | D(+Q)              | -0.2 +2-3             | DCO(Q)=2.4 7 at E=88 MeV.                                                                                                                                                                                                                                         |
| 2646                    | 37 2                    | 2645.8              | 7/2 <sup>+</sup>     | 0.0     | 3/2 <sup>+</sup>     | E2                 |                       | $I_\gamma$ : other: 47 2 at E=70 MeV.<br>$\Delta\text{IPDCO}=+0.08$ 5.<br>DCO(D)=1.7 2, $R_{\text{ADO}}=1.6$ 2 ( <b>2020Ra14</b> ).                                                                                                                               |
| 2757                    |                         | 8845.1              | 17/2 <sup>+</sup>    | 6087.6  | 13/2 <sup>-</sup>    |                    |                       |                                                                                                                                                                                                                                                                   |
| 2764                    |                         | 5927.1              | (11/2 <sup>-</sup> ) | 3163.1  | 7/2 <sup>-</sup>     |                    |                       | $E_\gamma$ : from <b>2013Bi10</b> only.                                                                                                                                                                                                                           |

Continued on next page (footnotes at end of table)

$^{12}\text{C}(^{28}\text{Si},\alpha p\gamma)$  **2007Ks01,2013Bi10,2020Ra14** (continued) $\gamma(^{35}\text{Cl})$  (continued)

| $E_\gamma$ <sup>†</sup> | $I_\gamma$ <sup>‡</sup> | $E_i(\text{level})$ | $J_i^\pi$            | $E_f$   | $J_f^\pi$         | Mult. <sup>#</sup> | $\delta$ <sup>#</sup> | Comments                                                                                                                                                     |
|-------------------------|-------------------------|---------------------|----------------------|---------|-------------------|--------------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2912                    | <1                      | 8319.4              | 15/2 <sup>-</sup>    | 5407.4  | 11/2 <sup>-</sup> |                    |                       | $E_\gamma$ : other: 2917 (2007Ks01, tentative).<br>Mult., $\delta$ : D(+Q) with $\delta=-0.1$ +3-2 if J(8319)=13/2 in 2007Ks01.<br>DCO(Q)=1.9 6 at E=88 MeV. |
| 3003 <sup>@</sup>       | >1.1                    | 3003.1              | 5/2 <sup>+</sup>     | 0.0     | 3/2 <sup>+</sup>  |                    |                       | $I_\gamma$ : other: >0.6 at E=70 MeV.                                                                                                                        |
| 3080                    | <1                      | 8487.8              | 15/2 <sup>-</sup>    | 5407.4  | 11/2 <sup>-</sup> |                    |                       | $E_\gamma$ : other: 3074 (2007Ks01).                                                                                                                         |
| 3163                    | 100 5                   | 3163.1              | 7/2 <sup>-</sup>     | 0.0     | 3/2 <sup>+</sup>  | M2+E3              | +0.16 1               | $I_\gamma$ : other: 100 2 at E=70 MeV.<br>DCO(Q)=0.9 1 at E=70 MeV, 0.7 1 at 88 MeV.<br>$\Delta\text{IPDCO}=-0.06$ 2.                                        |
| 3497                    |                         | 6660.2              | (11/2 <sup>-</sup> ) | 3163.1  | 7/2 <sup>-</sup>  |                    |                       |                                                                                                                                                              |
| 3734                    |                         | 16306.4             | (27/2 <sup>-</sup> ) | 12572.2 | 23/2 <sup>-</sup> |                    |                       |                                                                                                                                                              |
| 4347 <sup>@</sup>       | <1                      | 4347.8              | 9/2 <sup>-</sup>     | 0.0     | 3/2 <sup>+</sup>  |                    |                       | $I_\gamma$ : other: <1 at E=70 MeV.                                                                                                                          |

<sup>†</sup> From 2013Bi10, unless otherwise noted.<sup>‡</sup> From 2007Ks01, measured at E( $^{28}\text{Si}$ )=88 MeV at the NSC. Values are also available from E=70 MeV data of 2007Ks01 at TIFR facility, but less complete and are given under comments where available.<sup>#</sup> Q and D+Q with  $\delta(Q/D)$  are from  $\gamma\gamma(\theta)(\text{DCO})$  in 2007Ks01 and magnetic/electric characters are from  $\gamma\gamma(\text{lin pol})$  in 2007Ks01 where available and/or RUL where level T<sub>1/2</sub> is measured, unless otherwise noted.<sup>@</sup> From 2007Ks01 only.<sup>&</sup> Placement of transition in the level scheme is uncertain.

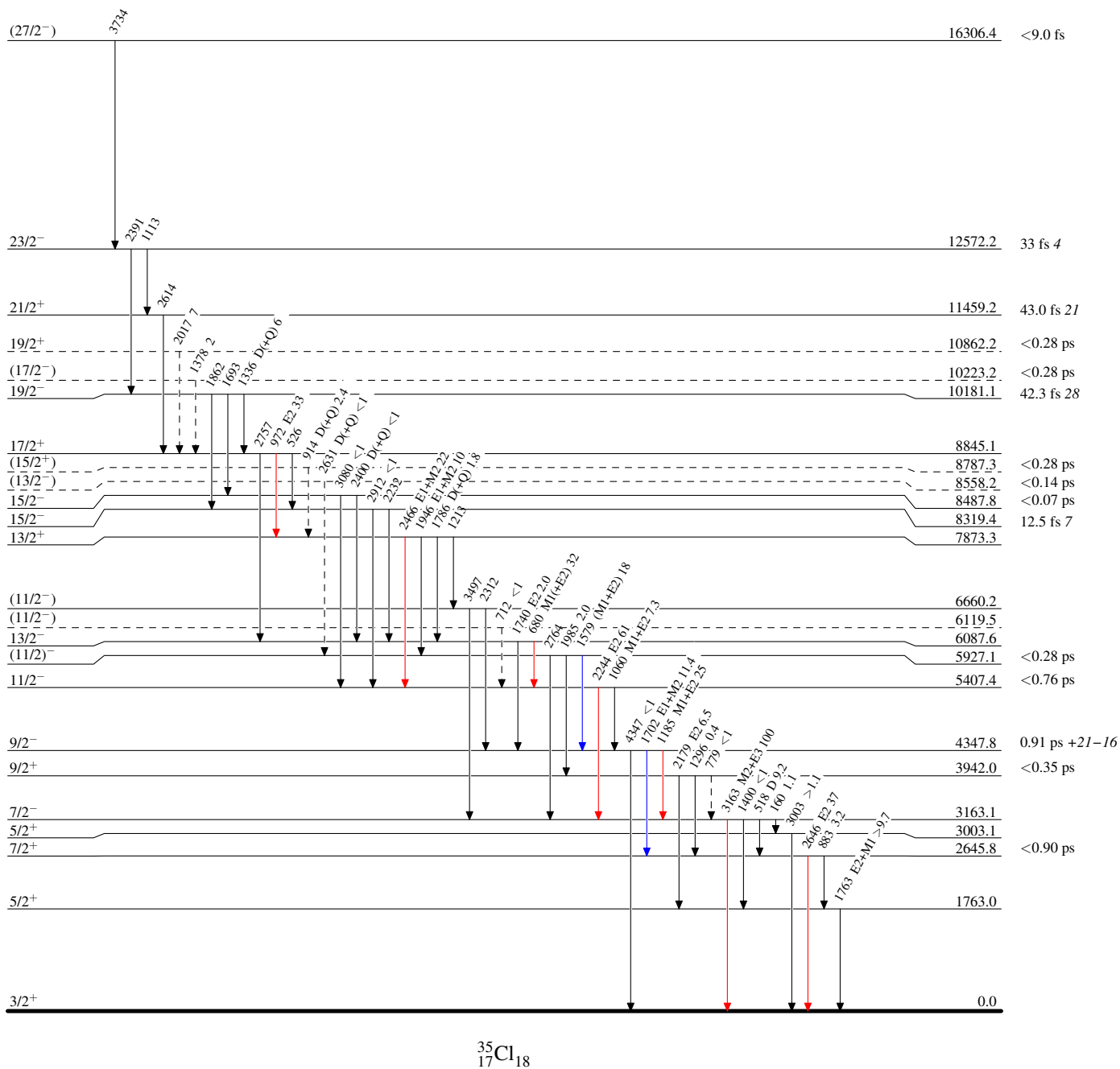
$^{12}\text{C}(^{28}\text{Si}, \alpha p \gamma)$  2007Ks01, 2013Bi10, 2020Ra14

Legend

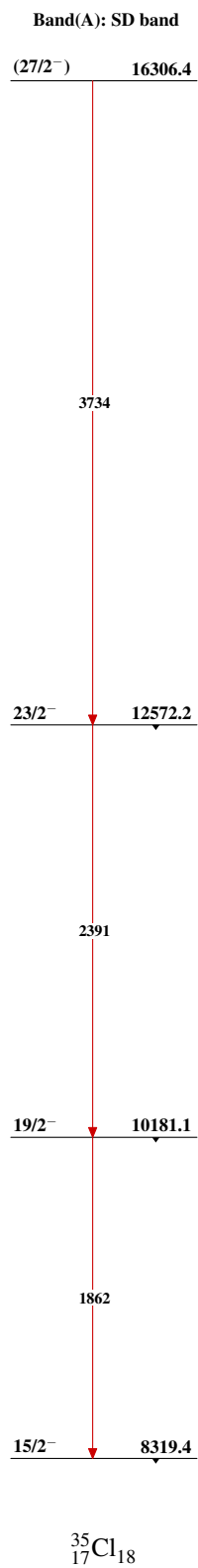
## Level Scheme

Intensities: Relative  $I_\gamma$ 

- $\longrightarrow$   $I_\gamma < 2\% \times I_\gamma^{\max}$   
 $\longrightarrow$   $I_\gamma < 10\% \times I_\gamma^{\max}$   
 $\longrightarrow$   $I_\gamma > 10\% \times I_\gamma^{\max}$   
 $\longrightarrow$   $\gamma$  Decay (Uncertain)

 $^{35}_{17}\text{Cl}_{18}$

$^{12}\text{C}(^{28}\text{Si}, \alpha p \gamma)$  2007Ks01, 2013Bi10, 2020Ra14



$^{16}\text{O}(^{24}\text{Mg}, \alpha p \gamma)$  2004Ek01, 2005Ek01

**2004Ek01, 2005Ek01:** E=60 MeV  $^{24}\text{Mg}$  beam was produced at the Legnaro National Laboratory. Target was a 0.5 mg/cm<sup>2</sup> enriched  $^{40}\text{Ca}$  target with a 7 mg/cm<sup>2</sup> tantalum backing and with oxygen inside.  $\gamma$  rays were detected with the GASP array of Ge detectors and charged particles were detected with the ISIS array of 40  $\Delta\text{E-E}$  Si telescopes. Measured  $E_\gamma$ ,  $I_\gamma$ ,  $\gamma\gamma$ -coin,  $\gamma(\theta)$ ,  $\gamma(\text{particle})$ -coin. Deduced levels, J,  $\pi$  from measured asymmetry ratios  $R(\gamma(\theta))$ .

 $^{35}\text{Cl}$  Levels

| $E(\text{level})^\dagger$ | $J^\pi^\ddagger$ |
|---------------------------|------------------|
| 0.0                       | $3/2^+$          |
| 1762.95 17                | $5/2^+$          |
| 2645.81 18                | $7/2^+$          |
| 3163.43 19                | $7/2^-$          |
| 4348.29 24                | $9/2^-$          |
| 5407.64 27                | $11/2^-$         |
| 6088.05 29                | $13/2^-$         |

<sup>†</sup> From a least-squares fit to  $\gamma$ -ray energies.

<sup>‡</sup> As given in **2004Ek01** based on measured  $\gamma(\theta)$  asymmetry and known assignments of low-lying levels. When considered in Adopted Levels, firm assignments here will be placed in parentheses if there are no strong supporting arguments.

 $\gamma(^{35}\text{Cl})$ 

$R(\gamma(\theta))$ =Asymmetry ratio measured at 35° and 81° with respect to the beam axis. Value of  $\approx 1.2$  for stretched quadrupole and  $\approx 0.7$  for stretched dipole ( $\Delta J=1$ ).

| $E_\gamma^\dagger$ | $I_\gamma^\dagger$ | $E_i(\text{level})$ | $J_i^\pi$ | $E_f$   | $J_f^\pi$ | Mult. <sup>‡</sup> | Comments                                                                                        |
|--------------------|--------------------|---------------------|-----------|---------|-----------|--------------------|-------------------------------------------------------------------------------------------------|
| 517.7 1            | 14.3 8             | 3163.43             | $7/2^-$   | 2645.81 | $7/2^+$   |                    |                                                                                                 |
| 680.4 1            | 39.1 12            | 6088.05             | $13/2^-$  | 5407.64 | $11/2^-$  | D                  | $R(\gamma(\theta))=0.65$ 3.                                                                     |
| 882.9 1            | 4.9 4              | 2645.81             | $7/2^+$   | 1762.95 | $5/2^+$   |                    |                                                                                                 |
| 1059.3 2           | 9.1 9              | 5407.64             | $11/2^-$  | 4348.29 | $9/2^-$   |                    |                                                                                                 |
| 1185.0 2           | 32.9 10            | 4348.29             | $9/2^-$   | 3163.43 | $7/2^-$   |                    |                                                                                                 |
| 1702.0 3           | 13.0 8             | 4348.29             | $9/2^-$   | 2645.81 | $7/2^+$   |                    |                                                                                                 |
| 1763.1 2           | 22.9 7             | 1762.95             | $5/2^+$   | 0.0     | $3/2^+$   |                    | $R(\gamma(\theta))=1.43$ 7 indicating Q seems inconsistent with $\Delta J=1$ from level scheme. |
| 2244.2 3           | 53.9 17            | 5407.64             | $11/2^-$  | 3163.43 | $7/2^-$   | Q                  | $R(\gamma(\theta))=1.24$ 6.                                                                     |
| 2645.5 4           | 32.3 10            | 2645.81             | $7/2^+$   | 0.0     | $3/2^+$   | Q                  | $R(\gamma(\theta))=1.21$ 6.                                                                     |
| 3162.7 4           | 100 3              | 3163.43             | $7/2^-$   | 0.0     | $3/2^+$   | Q                  | $R(\gamma(\theta))=1.43$ 7.                                                                     |

<sup>†</sup> From **2004Ek01**.

<sup>‡</sup> Deduced from  $\gamma(\theta)$  asymmetry in **2004Ek01** by the evaluators.

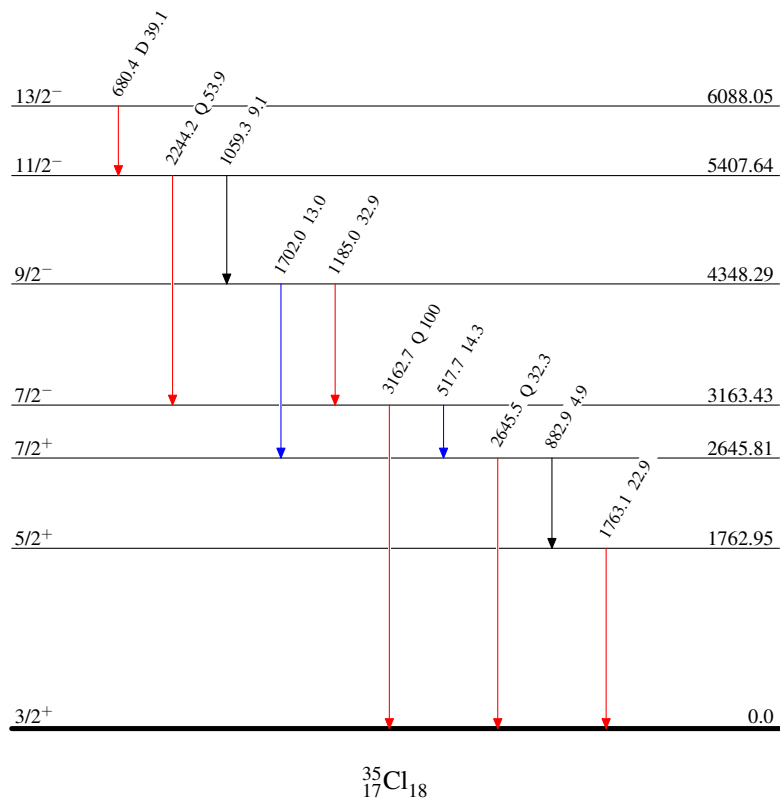
$^{16}\text{O}(^{24}\text{Mg},\alpha p\gamma)$  2004Ek01,2005Ek01

## Level Scheme

Intensities: Relative  $I_\gamma$ 

## Legend

- $\longrightarrow$   $I_\gamma < 2\% \times I_\gamma^{\max}$   
 $\longrightarrow$   $I_\gamma < 10\% \times I_\gamma^{\max}$   
 $\longrightarrow$   $I_\gamma > 10\% \times I_\gamma^{\max}$





$^{24}\text{Mg}(^{16}\text{O},\alpha p\gamma)$     **2007De14,1976Va24**

Also includes data from  $^{25}\text{Mg}(^{16}\text{O},\alpha p n\gamma)$  in [1976Ke02](#) and  $^{27}\text{Al}(^{16}\text{O},2\alpha\gamma)$  in [1991Ja11](#).

**2007De14:** E=70 MeV  $^{16}\text{O}$  beam was produced from the XTU-Tandem accelerator at Legnaro National Laboratory. Target was a  $400\text{ }\mu\text{g}/\text{cm}^2$  self-supporting target of  $^{24}\text{Mg}$ .  $\gamma$  rays were detected with the GASP spectrometer of 40 Compton-suppressed HPGe detectors and a multiplicity filter of 80 BGO scintillators and charged particles were detected with the  $4\pi$  array ISIS. Measured  $E\gamma$ ,  $I\gamma$ ,  $\gamma\gamma$ -coin,  $\gamma\gamma(\theta)$ (ADO), particle- $\gamma$ -coin. Deduced levels, J,  $\pi$ , band structures,  $\gamma$ -ray multipolarities.

**1976Va24:** E=38 and 45 MeV  $^{16}\text{O}$  beams were produced at Fysisch laboratorium in the Netherlands. Targets were  $300\text{ }\mu\text{g}/\text{cm}^2$   $^{24}\text{Mg}$  enriched to 99.94% evaporated onto  $30\text{ }\mu\text{m}$  Au backings.  $\gamma$  rays were detected with a Ge(Li)-NaI(Tl) Compton-suppression spectrometer and a three-Ge(Li) Compton polarimeter. Measured  $E\gamma$ ,  $I\gamma$ ,  $\gamma\gamma$ -coin,  $\gamma(\theta)$ ,  $\gamma(\text{lin pol})$ . Deduced levels, J,  $\pi$ ,  $\gamma$ -ray multipolarities and mixing ratios.

**1974Va13:** E=36 MeV  $^{16}\text{O}$  beam was produced at Fysisch laboratorium in the Netherlands. Targets were  $100$  and  $300\text{ }\mu\text{g}/\text{cm}^2$  Mg on  $1\text{ }\mu\text{m}$  Ni backings.  $\gamma$  rays were detected with Ge(Li) detectors. Measured  $E\gamma$ ,  $I\gamma$ ,  $\gamma\gamma$ -coin, recoil distances. Deduced the half-life for the level at 3.16 MeV.

**1982Ra25:** E=40-51 MeV  $^{16}\text{O}$  beams were produced at the Argonne National Laboratory. Target was  $50\text{ }\mu\text{g}/\text{cm}^2$  layer of  $^{24}\text{Mg}$  (>99.9%) evaporated onto a  $50\text{ }\mu\text{g}$  Au backing.  $\gamma$  rays were detected with a Ge(Li) detector. Measured  $\gamma$ -ray yields.

Other reactions:

**1991Ja11:**  $^{27}\text{Al}(^{16}\text{O},2\alpha\gamma)$  E=60 MeV  $^{16}\text{O}$  beam was produced from the 14 UD-Pelletron at TIFR. Target was  $350\text{ }\mu\text{g}/\text{cm}^2$  Al evaporated onto a  $1.8\text{ mg}/\text{cm}^2$  thick Ta backing.  $\gamma$  rays were detected with Ge(Li) detector. Measured  $E\gamma$ , recoil distances. Deduced  $T_{1/2}$  for the levels of 3162 and 8844 keV.

**1976Ke02:**  $^{25}\text{Mg}(^{16}\text{O},\alpha p n\gamma)$  E=43 and 50 MeV  $^{16}\text{O}$  beams were produced at the Institut fur Kernphysik in Germany. The target was  $150\text{ }\mu\text{g}/\text{cm}^2$  enriched  $^{25}\text{Mg}$  (99.2%) evaporated onto a Au backing.  $\gamma$  rays were detected with two true coaxial Ge(Li) detectors of  $45\text{ cm}^3$ . Measured  $E\gamma$ ,  $I\gamma$ , Doppler-shift recoil distances. Deduced  $T_{1/2}$  for the level of 3162 keV.

**1989De02:**  $^{27}\text{Al}(^{16}\text{O},X)$  E=140 MeV  $^{16}\text{O}$  from the Stony Brook LINAC facility. Measured particle-particle correlations.

**1986Br26:**  $^{27}\text{Al}(^{16}\text{O},X)$  E=65 MeV  $^{16}\text{O}$  from the Pelletron Laboratory at the Universidade de Sao Paulo, Brazil. Measured  $\sigma$ .

**1979Ta03:**  $^{27}\text{Al}(^{16}\text{O},X)$  E=50, 60, and 70 MeV  $^{16}\text{O}$  from the NBI super FN tandem. Measured particle-particle angular correlations.

**2001Pi10:**  $^{12}\text{C}(^{32}\text{S},X)$  E=19.5 MeV/nucleon  $^{32}\text{S}$  beam from the superconducting cyclotron of the Laboratori Nazionali del Sud in Catania. Measured  $\sigma$ .

 $^{35}\text{Cl}$  Levels

Band assignments are from [2007De14](#).

| E(level) <sup>†</sup>        | J $\pi$ <sup>‡</sup> | $T_{1/2}$        | Comments                                                                                                                                                                                       |
|------------------------------|----------------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0                            | $3/2^+$              |                  |                                                                                                                                                                                                |
| 1219.0 <i>10</i>             |                      |                  |                                                                                                                                                                                                |
| 1763.3 <i>2</i>              | $5/2^+$              |                  |                                                                                                                                                                                                |
| 2645.9 <i>3</i>              | $7/2^+$              |                  |                                                                                                                                                                                                |
| 3001.9 <i>10</i>             | $5/2^+$              |                  |                                                                                                                                                                                                |
| 3162.9 <sup>#</sup> <i>3</i> | $7/2^-$              | 31.0 ps <i>6</i> | $T_{1/2}$ : weighted average of 29.1 ps <i>14</i> ( <a href="#">1974Va13</a> ), 28.9 ps <i>12</i> ( <a href="#">1976Ke02</a> ), and 31.4 ps <i>4</i> ( <a href="#">1991Ja11</a> ), all by RDM. |
| 3943.4 <i>10</i>             | $9/2^+$              |                  |                                                                                                                                                                                                |
| 4347.8 <i>3</i>              | $9/2^-$              |                  |                                                                                                                                                                                                |
| 5407.3 <sup>#</sup> <i>3</i> | $11/2^-$             |                  |                                                                                                                                                                                                |
| 5926.9 <i>5</i>              | $11/2^-$             |                  |                                                                                                                                                                                                |
| 6087.2 <i>4</i>              | $13/2^-$             |                  |                                                                                                                                                                                                |
| 6380.8 <i>8</i>              |                      |                  |                                                                                                                                                                                                |
| 7873.4 <sup>@</sup> <i>4</i> | $13/2^+$             |                  |                                                                                                                                                                                                |
| 8319.7 <sup>#</sup> <i>5</i> | $15/2^-$             |                  |                                                                                                                                                                                                |
| 8487.4 <i>5</i>              | $15/2^-$             |                  |                                                                                                                                                                                                |
| 8788.9 <sup>@</sup> <i>6</i> | $15/2^+$             |                  |                                                                                                                                                                                                |
| 8844.6 <sup>@</sup> <i>5</i> | $17/2^+$             | 6.1 ps <i>11</i> | $T_{1/2}$ : from RDM in <a href="#">1991Ja11</a> .                                                                                                                                             |

Continued on next page (footnotes at end of table)

$^{24}\text{Mg}(^{16}\text{O},\alpha\text{p}\gamma)$  **2007De14,1976Va24 (continued)** $^{35}\text{Cl}$  Levels (continued)

| $E(\text{level})^\dagger$ | $J^\pi^\ddagger$  |
|---------------------------|-------------------|
| 10181.1 <sup>#</sup> 5    | 19/2 <sup>-</sup> |
| 10222.5 11                | 17/2 <sup>-</sup> |
| 10859.1 <sup>@</sup> 8    | 19/2 <sup>+</sup> |
| 11459.0 <sup>@</sup> 7    | 21/2 <sup>+</sup> |
| 12572.2 <sup>#</sup> 6    | 23/2 <sup>-</sup> |

<sup>†</sup> From a least-squares fit to  $\gamma$ -ray energies.

<sup>‡</sup> As proposed in **2007De14** based on measured  $\gamma\gamma(\theta)(\text{ADO})$  and band assignments, unless otherwise noted. When considered in Adopted Levels, firm assignments here will be placed in parentheses if there are no strong supporting arguments.

<sup>#</sup> Band(A): Band based on 3163, 7/2<sup>-</sup> level. Configuration=f<sub>7/2</sub>.

<sup>@</sup> Band(B): Band based on 7873, 13/2<sup>+</sup> level.

 $\gamma(^{35}\text{Cl})$ 

$R_{\text{ADO}}$  values under comments are from  $\gamma\gamma(\theta)(\text{ADO})$  in **2007De14**. Expected values are  $\approx 1.3$  for  $\Delta J=2$ , quadrupole (and  $\Delta J=0$ , dipole in few cases) and  $\approx 0.8$  for  $\Delta J=1$ , dipole transitions (**2007De14**).

| $E_\gamma^\dagger$ | $I_\gamma^\dagger$ | $E_i(\text{level})$ | $J_i^\pi$         | $E_f$   | $J_f^\pi$         | Mult. <sup>‡</sup> | $\delta^\#$ | Comments                                                                                        |
|--------------------|--------------------|---------------------|-------------------|---------|-------------------|--------------------|-------------|-------------------------------------------------------------------------------------------------|
| 161 <sup>@</sup> 1 |                    | 3162.9              | 7/2 <sup>-</sup>  | 3001.9  | 5/2 <sup>+</sup>  |                    |             |                                                                                                 |
| 296 <sup>@</sup> 1 |                    | 6380.8              |                   | 6087.2  | 13/2 <sup>-</sup> |                    |             |                                                                                                 |
| 517.3 5            | 11.1 13            | 3162.9              | 7/2 <sup>-</sup>  | 2645.9  | 7/2 <sup>+</sup>  |                    |             | $E_\gamma$ : weighted average of 517.4 5 ( <b>2007De14</b> ) and 517 1 ( <b>1976Va24</b> ).     |
| 680.5 5            | 33.0 7             | 6087.2              | 13/2 <sup>-</sup> | 5407.3  | 11/2 <sup>-</sup> | D                  |             | $R_{\text{ADO}}=0.84$ 2.                                                                        |
| 882.7 5            | 4.0 1              | 2645.9              | 7/2 <sup>+</sup>  | 1763.3  | 5/2 <sup>+</sup>  | D                  |             | $E_\gamma$ : weighted average of 882.9 5 ( <b>2007De14</b> ) and 882 1 ( <b>1976Va24</b> ).     |
|                    |                    |                     |                   |         |                   |                    |             | $R_{\text{ADO}}=0.75$ 3.                                                                        |
| 915.4 4            | 7.2 5              | 8788.9              | 15/2 <sup>+</sup> | 7873.4  | 13/2 <sup>+</sup> | D                  |             | $R_{\text{ADO}}=0.69$ 3.                                                                        |
| 971 <sup>@</sup> 1 |                    | 6380.8              |                   | 5407.3  | 11/2 <sup>-</sup> |                    |             |                                                                                                 |
| 971.4 5            | 75.3 19            | 8844.6              | 17/2 <sup>+</sup> | 7873.4  | 13/2 <sup>+</sup> | Q                  |             | $R_{\text{ADO}}=1.38$ 3.                                                                        |
| 1059.5 2           | 17.1 4             | 5407.3              | 11/2 <sup>-</sup> | 4347.8  | 9/2 <sup>-</sup>  | M1+E2              | +0.12 3     | $E_\gamma$ : weighted average of 1059.6 5 ( <b>2007De14</b> ) and 1059.5 2 ( <b>1976Va24</b> ). |
|                    |                    |                     |                   |         |                   |                    |             | $I_\gamma$ : other: 7.7 3 ( <b>1976Va24</b> , E=38 MeV).                                        |
|                    |                    |                     |                   |         |                   |                    |             | $R_{\text{ADO}}=1.05$ 4.                                                                        |
|                    |                    |                     |                   |         |                   |                    |             | $A_2=-0.08$ 6, $A_4=+0.049$ 46, POL=-0.11 12 ( <b>1976Va24</b> ).                               |
| 1113.3 5           | 4.1 3              | 12572.2             | 23/2 <sup>-</sup> | 11459.0 | 21/2 <sup>+</sup> | D                  |             | $R_{\text{ADO}}=0.86$ 9.                                                                        |
| 1184.9 3           | 44.4 16            | 4347.8              | 9/2 <sup>-</sup>  | 3162.9  | 7/2 <sup>-</sup>  | M1+E2              | -0.65 30    | $E_\gamma$ : weighted average of 1184.9 5 ( <b>2007De14</b> ) and 1184.9 3 ( <b>1976Va24</b> ). |
|                    |                    |                     |                   |         |                   |                    |             | $I_\gamma$ : other: 22.2 3 ( <b>1976Va24</b> , E=38 MeV).                                       |
|                    |                    |                     |                   |         |                   |                    |             | $R_{\text{ADO}}=0.54$ 2, giving $\Delta J=1$ .                                                  |
|                    |                    |                     |                   |         |                   |                    |             | $A_2=-0.663$ 25, $A_4=+0.004$ 25, POL=+0.07 7 ( <b>1976Va24</b> ).                              |
| 1336.3 5           | 23.5 8             | 10181.1             | 19/2 <sup>-</sup> | 8844.6  | 17/2 <sup>+</sup> | D                  |             | $R_{\text{ADO}}=0.75$ 2.                                                                        |
| 1377.8 10          | 5.4 4              | 10222.5             | 17/2 <sup>-</sup> | 8844.6  | 17/2 <sup>+</sup> |                    |             | $R_{\text{ADO}}=1.32$ 12 also consistent with $\Delta J=0$ .                                    |
| 1399.9 7           | 0.5 2              | 3162.9              | 7/2 <sup>-</sup>  | 1763.3  | 5/2 <sup>+</sup>  |                    |             |                                                                                                 |
| 1578.9 5           | 31.8 13            | 5926.9              | 11/2 <sup>-</sup> | 4347.8  | 9/2 <sup>-</sup>  | D                  |             | $E_\gamma$ : other: 1579 1 ( <b>1976Va24</b> ).                                                 |
|                    |                    |                     |                   |         |                   |                    |             | $R_{\text{ADO}}=0.48$ 1.                                                                        |
| 1693.5 5           | 19.8 21            | 10181.1             | 19/2 <sup>-</sup> | 8487.4  | 15/2 <sup>-</sup> | Q                  |             | $R_{\text{ADO}}=1.44$ 7.                                                                        |
| 1701.7 5           | 23.1 6             | 4347.8              | 9/2 <sup>-</sup>  | 2645.9  | 7/2 <sup>+</sup>  | E1+M2              | +0.5 4      | $E_\gamma$ : weighted average of 1701.6 5                                                       |

Continued on next page (footnotes at end of table)

$^{24}\text{Mg}(^{16}\text{O},\alpha p\gamma)$  **2007De14,1976Va24 (continued)** $\gamma(^{35}\text{Cl})$  (continued)

| $E_\gamma^\dagger$  | $I_\gamma^\dagger$ | $E_i(\text{level})$ | $J_i^\pi$         | $E_f$   | $J_f^\pi$         | Mult. $^\ddagger$ | $\delta^\#$ | Comments                                                                                                                                                                                                                                                                                                                                                  |
|---------------------|--------------------|---------------------|-------------------|---------|-------------------|-------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1739.6 5            | 2.6 3              | 6087.2              | 13/2 <sup>-</sup> | 4347.8  | 9/2 <sup>-</sup>  | Q                 |             | (2007De14) and 1702 1 (1976Va24).<br>R <sub>ADO</sub> =0.76 6, giving $\Delta J=1$ .<br>E <sub>γ</sub> : weighted average of 1739.5 5 (2007De14) and 1740 1 (1976Va24).<br>R <sub>ADO</sub> =1.38 10.                                                                                                                                                     |
| 1763.3 2            | 21.2 7             | 1763.3              | 5/2 <sup>+</sup>  | 0       | 3/2 <sup>+</sup>  | D+Q               | -2.6 4      | E <sub>γ</sub> : weighted average of 1763.1 5 (2007De14) and 1763.3 2 (1976Va24).<br>I <sub>γ</sub> : other: 18.5 3 (1976Va24, E=38 MeV).<br>A <sub>2</sub> =-0.368 18, A <sub>4</sub> =+0.13 2, POL=+0.06 20 (1976Va24).<br>R <sub>ADO</sub> =0.77 3, giving $\Delta J=1$ .<br>R <sub>ADO</sub> =1.37 7, consistent with $\Delta J=0$ from level scheme. |
| 1786.2 5            | 4.5 2              | 7873.4              | 13/2 <sup>+</sup> | 6087.2  | 13/2 <sup>-</sup> |                   |             | R <sub>ADO</sub> =1.39 7.                                                                                                                                                                                                                                                                                                                                 |
| 1861.3 5            | 19.6 13            | 10181.1             | 19/2 <sup>-</sup> | 8319.7  | 15/2 <sup>-</sup> | Q                 |             | R <sub>ADO</sub> =0.80 4.                                                                                                                                                                                                                                                                                                                                 |
| 1946.2 5            | 22.4 12            | 7873.4              | 13/2 <sup>+</sup> | 5926.9  | 11/2 <sup>-</sup> | D                 |             | R <sub>ADO</sub> =0.50 3.                                                                                                                                                                                                                                                                                                                                 |
| 2014.7 9            | 23.1 12            | 10859.1             | 19/2 <sup>+</sup> | 8844.6  | 17/2 <sup>+</sup> | D                 |             | R <sub>ADO</sub> =1.57 11.                                                                                                                                                                                                                                                                                                                                |
| 2069.8 10           | 7.5 5              | 10859.1             | 19/2 <sup>+</sup> | 8788.9  | 15/2 <sup>+</sup> | Q                 |             |                                                                                                                                                                                                                                                                                                                                                           |
| 2180 <sup>@</sup> 1 |                    | 3943.4              | 9/2 <sup>+</sup>  | 1763.3  | 5/2 <sup>+</sup>  |                   |             |                                                                                                                                                                                                                                                                                                                                                           |
| 2232.7 6            | 10.3 5             | 8319.7              | 15/2 <sup>-</sup> | 6087.2  | 13/2 <sup>-</sup> |                   |             |                                                                                                                                                                                                                                                                                                                                                           |
| 2244.3 2            | 89 2               | 5407.3              | 11/2 <sup>-</sup> | 3162.9  | 7/2 <sup>-</sup>  | E2(+M3)           | 0.000 1     | E <sub>γ</sub> : weighted average of 2243.9 5 (2007De14) and 2244.3 2 (1976Va24).<br>I <sub>γ</sub> : other: 30.3 20 (1976Va24, E=38 MeV).<br>R <sub>ADO</sub> =1.26 3, giving $\Delta J=2$ .<br>A <sub>2</sub> =+0.303 20, A <sub>4</sub> =-0.119 22, POL=+0.26 28 (1976Va24).<br>R <sub>ADO</sub> =1.34 12.                                             |
| 2390.8 5            | 16.8 7             | 12572.2             | 23/2 <sup>-</sup> | 10181.1 | 19/2 <sup>-</sup> | Q                 |             |                                                                                                                                                                                                                                                                                                                                                           |
| 2399.8 8            | 14.8 5             | 8487.4              | 15/2 <sup>-</sup> | 6087.2  | 13/2 <sup>-</sup> |                   |             | E <sub>γ</sub> : other: 2466 1 (1976Va24, unplaced).<br>R <sub>ADO</sub> =0.77 2.                                                                                                                                                                                                                                                                         |
| 2466.2 5            | 46.2 9             | 7873.4              | 13/2 <sup>+</sup> | 5407.3  | 11/2 <sup>-</sup> | D                 |             | R <sub>ADO</sub> =1.33 4.                                                                                                                                                                                                                                                                                                                                 |
| 2614.5 5            | 19.0 8             | 11459.0             | 21/2 <sup>+</sup> | 8844.6  | 17/2 <sup>+</sup> | Q                 |             | E <sub>γ</sub> : weighted average of 2645.7 5 (2007De14) and 2646 1 (1976Va24).<br>R <sub>ADO</sub> =1.35 4.                                                                                                                                                                                                                                              |
| 2645.8 5            | 36.1 8             | 2645.9              | 7/2 <sup>+</sup>  | 0       | 3/2 <sup>+</sup>  | Q                 |             | R <sub>ADO</sub> =1.35 5.                                                                                                                                                                                                                                                                                                                                 |
| 2911.9 8            | 18.0 8             | 8319.7              | 15/2 <sup>-</sup> | 5407.3  | 11/2 <sup>-</sup> | Q                 |             | R <sub>ADO</sub> =1.20 12.                                                                                                                                                                                                                                                                                                                                |
| 3080.0 7            | 4.7 3              | 8487.4              | 15/2 <sup>-</sup> | 5407.3  | 11/2 <sup>-</sup> | Q                 |             | E <sub>γ</sub> : weighted average of 3163.1 5 (2007De14) and 3162.4 3 (1976Va24).<br>I <sub>γ</sub> : other: 100.0 15 (1976Va24, E=38 MeV).<br>R <sub>ADO</sub> =1.30 2, giving $\Delta J=2$ .<br>A <sub>2</sub> =+0.416 6, A <sub>4</sub> =-0.001 6, POL=-0.57 12 (1976Va24).                                                                            |
| 3162.6 3            | 100 2              | 3162.9              | 7/2 <sup>-</sup>  | 0       | 3/2 <sup>+</sup>  | M2+E3             | +0.18 7     |                                                                                                                                                                                                                                                                                                                                                           |
| 4347.8 8            | 5.5 5              | 4347.8              | 9/2 <sup>-</sup>  | 0       | 3/2 <sup>+</sup>  |                   |             |                                                                                                                                                                                                                                                                                                                                                           |

<sup>†</sup> From 2007De14. Note that a few I<sub>γ</sub> values in 2007De14 have very small uncertainties (<1%) which is unrealistic for an in-beam measurement with Ge detectors. It is very likely the reported uncertainties are statistical only and the evaluators have added in quadrature a typical 2% systematic uncertainty from efficiency calibration.

<sup>‡</sup> D with  $\Delta J=1$  or Q with  $\Delta J=2$  are deduced by the evaluators based on measured  $\gamma\gamma(\theta)(\text{ADO})$  in 2007De14 and others are from  $\gamma(\theta)$  and  $\gamma(\text{lin pol})$  in 1976Va24. Assignments are not explicitly listed in 2007De14.

<sup>#</sup> From  $\gamma(\theta)$  in 1976Va24.

<sup>@</sup> From 1976Va24 only.

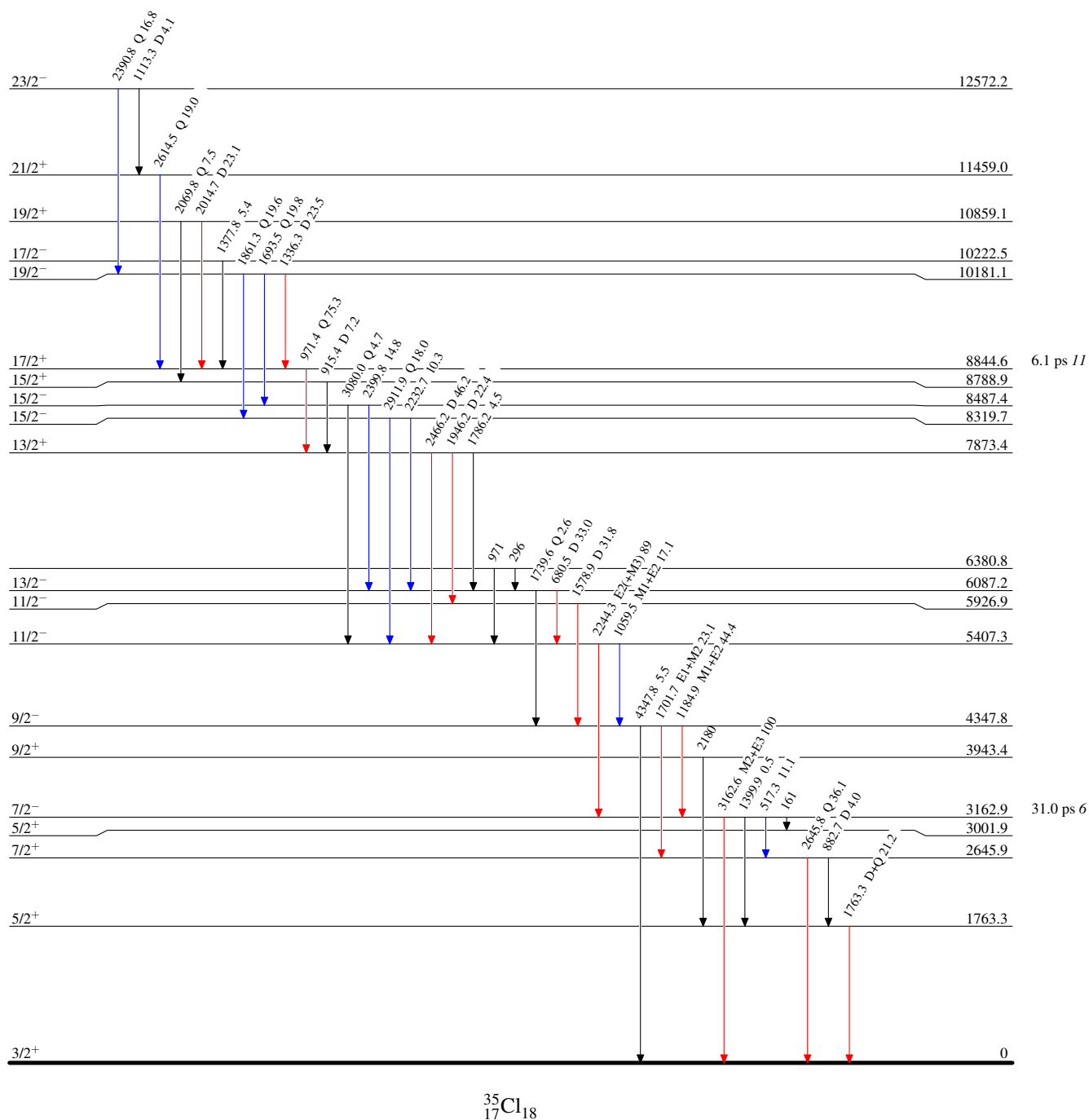
$^{24}\text{Mg}(^{16}\text{O}, \alpha p \gamma)$  2007De14, 1976Va24

## Level Scheme

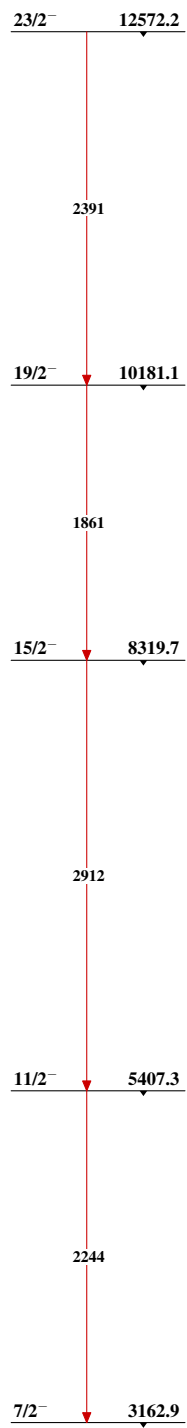
Intensities: Relative  $I_\gamma$ 

## Legend

- $\longrightarrow$   $I_\gamma < 2\% \times I_\gamma^{\max}$   
 $\longrightarrow$   $I_\gamma < 10\% \times I_\gamma^{\max}$   
 $\longrightarrow$   $I_\gamma > 10\% \times I_\gamma^{\max}$

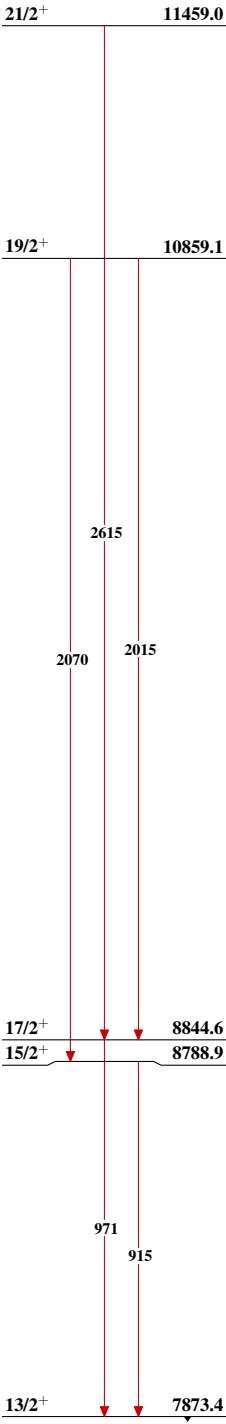


$^{24}\text{Mg}(^{16}\text{O}, \alpha p \gamma)$  2007De14,1976Va24

 Band(A): Band based on  
3163, 7/2<sup>-</sup> level

 $^{35}_{17}\text{Cl}_{18}$

<sup>24</sup>Mg(<sup>16</sup>O, $\alpha$ p $\gamma$ ) 2007De14,1976Va24 (continued)

Band(B): Band based on 7873, 13/2<sup>+</sup> level



<sup>35</sup><sub>17</sub>Cl<sub>18</sub>

$^{27}\text{Al}(^{14}\text{N},\alpha\text{pn}\gamma)$  **1976Wa11**

**1976Wa11:** E=40 MeV  $^{14}\text{N}$  beam was produced at the Brookhaven National Laboratory. Target was  $^{27}\text{Al}$ .  $\gamma$  rays were detected with Ge(Li) detectors. Measured  $E_\gamma$ ,  $I_\gamma$ ,  $\gamma(\theta)$ ,  $\gamma\gamma$ -coin, recoils distances. Deduced levels, J,  $\pi$ ,  $T_{1/2}$ ,  $\gamma$ -ray branching ratios, multipolarities.

 $^{35}\text{Cl}$  Levels

| E(level) <sup>†</sup> | J $\pi$ <sup>‡</sup> | $T_{1/2}$  | Comments                                                |
|-----------------------|----------------------|------------|---------------------------------------------------------|
| 0                     | 3/2 <sup>+</sup>     |            |                                                         |
| 1763.19 10            | 5/2 <sup>+</sup>     |            |                                                         |
| 2645.45 12            | 7/2 <sup>+</sup>     |            |                                                         |
| 3001.98 22            | 5/2 <sup>+</sup>     |            |                                                         |
| 3162.64 9             | 7/2 <sup>-</sup>     | 29.0 ps 13 | $T_{1/2}$ : from $\tau=41.8$ ps 18 ( <b>1976Wa11</b> ). |
| 3942.3 10             | 9/2 <sup>+</sup>     |            |                                                         |
| 4347.43 17            | 9/2 <sup>-</sup>     |            |                                                         |
| 5406.68 19            | 11/2 <sup>-</sup>    |            |                                                         |
| 5926.47 29            | 11/2 <sup>-</sup>    |            |                                                         |
| 6086.89 23            | 13/2 <sup>-</sup>    | 5.3 ps 10  | $T_{1/2}$ : from $\tau=7.7$ ps 14 ( <b>1976Wa11</b> ).  |
| 7872.78 30            | 13/2 <sup>+</sup>    |            | $T_{1/2}$ : 0.5 ps< $\tau$ <50 ps ( <b>1976Wa11</b> ).  |
| 8844.2 4              | 17/2 <sup>+</sup>    | 5.5 ps 14  | $T_{1/2}$ : from $\tau=8$ ps 2 ( <b>1976Wa11</b> ).     |

<sup>†</sup> From a least-squares fit to  $\gamma$ -ray energies.

<sup>‡</sup> As given in **1976Wa11** based on measured  $\gamma(\theta)$  and known assignments of low-lying levels. When considered in Adopted Levels, firm assignments here will be placed in parentheses if there are no strong supporting arguments.

 $\gamma(^{35}\text{Cl})$ 

$A_2$  and  $A_4$  under comments are from  $\gamma(\theta)$  in **1976Wa11**, with  $A_4=0$  if not given.

| $E_\gamma$ <sup>†</sup> | $I_\gamma$ <sup>†</sup> | $E_i(\text{level})$ | J $\pi_i$         | $E_f$   | J $\pi_f$         | Comments                                                                                                                     |
|-------------------------|-------------------------|---------------------|-------------------|---------|-------------------|------------------------------------------------------------------------------------------------------------------------------|
| 160.66 <sup>#</sup> 20  | 2.3                     | 3162.64             | 7/2 <sup>-</sup>  | 3001.98 | 5/2 <sup>+</sup>  | $A_2=-0.24$ 15.                                                                                                              |
| 517.26 10               | <2.2 <sup>‡</sup>       | 3162.64             | 7/2 <sup>-</sup>  | 2645.45 | 7/2 <sup>+</sup>  | $E_\gamma, I_\gamma$ : doublet with $\gamma$ rays in $^{36}\text{Cl}$ ( <b>1976Wa11</b> ).<br>$A_2=+0.47$ 25.                |
| 680.22 15               | 25                      | 6086.89             | 13/2 <sup>-</sup> | 5406.68 | 11/2 <sup>-</sup> | $A_2=-0.20$ 2.                                                                                                               |
| 882.32 35               | 2.3                     | 2645.45             | 7/2 <sup>+</sup>  | 1763.19 | 5/2 <sup>+</sup>  | $A_2=+0.31$ 40.                                                                                                              |
| 971.38 20               | 8.0                     | 8844.2              | 17/2 <sup>+</sup> | 7872.78 | 13/2 <sup>+</sup> | $A_2=+0.28$ 7, $A_4=-0.18$ 7.                                                                                                |
| 1059.17 20              | 4.6                     | 5406.68             | 11/2 <sup>-</sup> | 4347.43 | 9/2 <sup>-</sup>  | $A_2=-0.04$ 15.<br>%Branching=13 4.                                                                                          |
| 1184.80 20              | 21                      | 4347.43             | 9/2 <sup>-</sup>  | 3162.64 | 7/2 <sup>-</sup>  | $A_2=-0.66$ 4, $A_4=+0.08$ 4.<br>%Branching=69 5.                                                                            |
| 1579.15 30              | 1.9 <sup>‡</sup>        | 5926.47             | 11/2 <sup>-</sup> | 4347.43 | 9/2 <sup>-</sup>  | $A_2=-0.6$ 4.                                                                                                                |
| 1701.85 25              | 9.5                     | 4347.43             | 9/2 <sup>-</sup>  | 2645.45 | 7/2 <sup>+</sup>  | $A_2=-0.20$ 3.<br>%Branching=31 5.                                                                                           |
| 1739.3 4                | 1.7                     | 6086.89             | 13/2 <sup>-</sup> | 4347.43 | 9/2 <sup>-</sup>  |                                                                                                                              |
| 1763.15 10              | 32                      | 1763.19             | 5/2 <sup>+</sup>  | 0       | 3/2 <sup>+</sup>  | $A_2=-0.28$ 2, $A_4=+0.05$ 3.                                                                                                |
| 1786 <sup>#</sup>       | 0.63                    | 7872.78             | 13/2 <sup>+</sup> | 6086.89 | 13/2 <sup>-</sup> | $E_\gamma, I_\gamma$ : inferred from the presence of 680 $\gamma$ -971 $\gamma$ coin ( <b>1976Wa11</b> ).<br>%Branching=8 5. |
| 1946.40 30              | 1.9                     | 7872.78             | 13/2 <sup>+</sup> | 5926.47 | 11/2 <sup>-</sup> | $A_2=-0.2$ 1.<br>%Branching=22 5.                                                                                            |
| 2179                    | 5.0                     | 3942.3              | 9/2 <sup>+</sup>  | 1763.19 | 5/2 <sup>+</sup>  | $E_\gamma, I_\gamma$ : observed in $\gamma\gamma$ -coin only ( <b>1976Wa11</b> ).<br>$A_2=+0.23$ 2, $A_4=-0.08$ 2.           |
| 2244.00 25              | 32                      | 5406.68             | 11/2 <sup>-</sup> | 3162.64 | 7/2 <sup>-</sup>  | %Branching=87 4.                                                                                                             |
| 2465.85 30              | 5.9                     | 7872.78             | 13/2 <sup>+</sup> | 5406.68 | 11/2 <sup>-</sup> | $A_2=-0.21$ 6.<br>%Branching=70 5.                                                                                           |

Continued on next page (footnotes at end of table)

<sup>27</sup>Al(<sup>14</sup>N, $\alpha$ pn $\gamma$ )    **1976Wa11** (continued)

$\gamma(^{35}\text{Cl})$  (continued)

| <u>E<sub><math>\gamma</math></sub><sup>†</sup></u> | <u>I<sub><math>\gamma</math></sub><sup>†</sup></u> | <u>E<sub>i</sub>(level)</u> | <u>J<sub>i</sub><sup><math>\pi</math></sup></u> | <u>E<sub>f</sub></u> | <u>J<sub>f</sub><sup><math>\pi</math></sup></u> | <u>Comments</u>                                                                                                                                                                               |
|----------------------------------------------------|----------------------------------------------------|-----------------------------|-------------------------------------------------|----------------------|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2645.79 30                                         | <26                                                | 2645.45                     | 7/2 <sup>+</sup>                                | 0                    | 3/2 <sup>+</sup>                                | E <sub><math>\gamma</math></sub> , I <sub><math>\gamma</math></sub> : doublet with $\gamma$ rays in <sup>38</sup> K ( <b>1976Wa11</b> ).<br>A <sub>2</sub> =+0.37 3, A <sub>4</sub> =-0.04 3. |
| 3162.43 10                                         | 100                                                | 3162.64                     | 7/2 <sup>-</sup>                                | 0                    | 3/2 <sup>+</sup>                                |                                                                                                                                                                                               |

<sup>†</sup> From **1976Wa11**. Original values of relative intensities have been renormalized by the evaluators to I(3162 $\gamma$ )=100.

<sup>‡</sup> Estimated from  $\gamma\gamma$ -coin data or known branching ratios (**1976Wa11**).

# Placement of transition in the level scheme is uncertain.



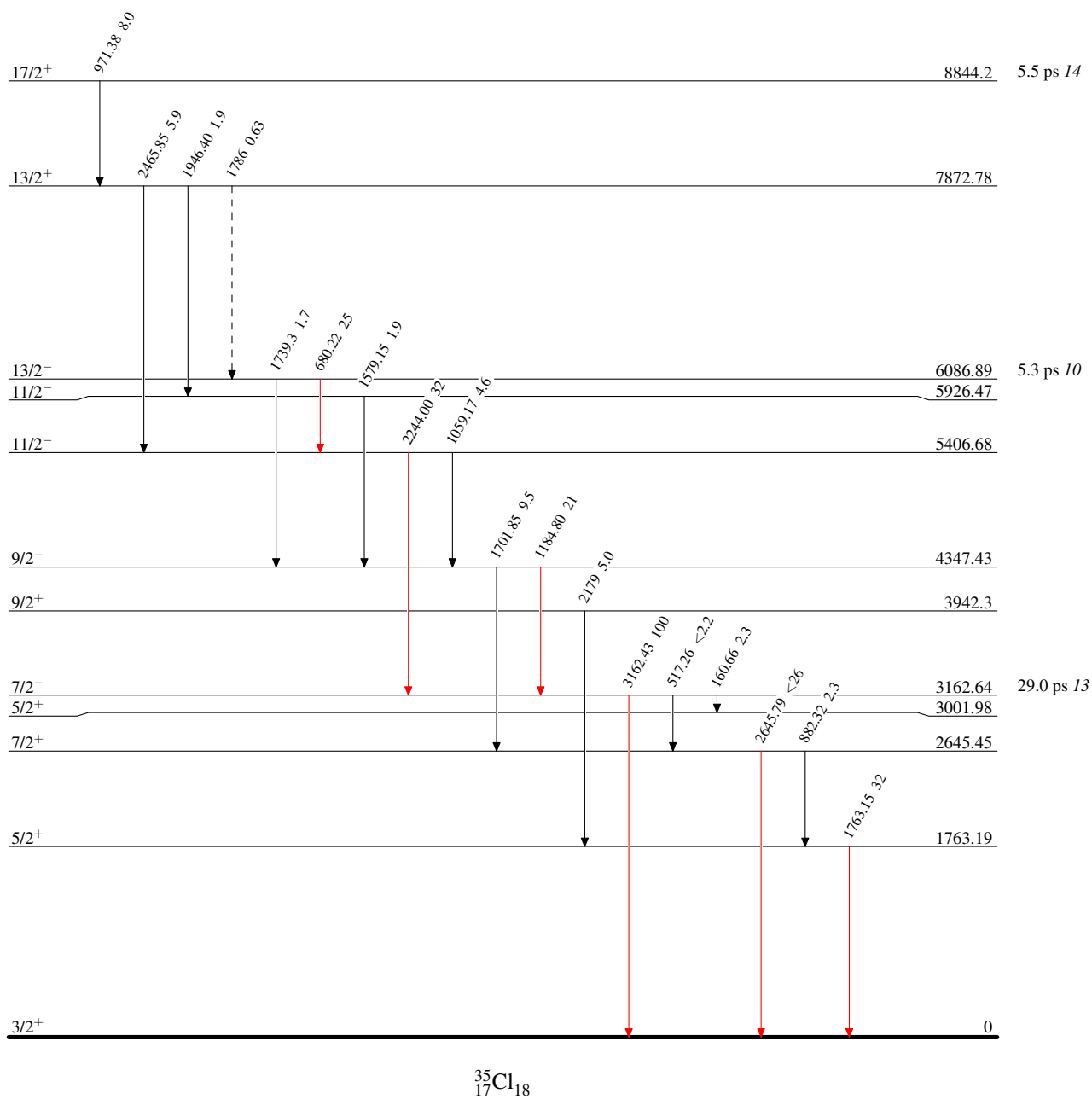
$^{27}\text{Al}(^{14}\text{N}, \alpha \text{pn} \gamma)$  1976Wa11

Legend

## Level Scheme

Intensities: Relative  $I_\gamma$ 

- $\longrightarrow$   $I_\gamma < 2\% \times I_\gamma^{\text{max}}$   
 $\longrightarrow$   $I_\gamma < 10\% \times I_\gamma^{\text{max}}$   
 $\longrightarrow$   $I_\gamma > 10\% \times I_\gamma^{\text{max}}$   
 $\longrightarrow$   $\gamma$  Decay (Uncertain)



$^{31}\text{P}(\alpha, \text{p}), (\alpha, \text{n}): \text{resonances}$     [1975Sc40](#), [1971Mc23](#), [1964Ku12](#) $J^\pi=1/2^+$   $^{31}\text{P}$  ground state.

[1975Sc40](#):  $^{31}\text{P}(\alpha, \text{p})$  with 3.25-5.55-MeV  $\alpha$  beams produced from the 5.5-MV Van de Graaff accelerator of the University of Lowell. The target was prepared by evaporating natural  $^{31}\text{P}$  onto carbon backings. Protons were detected using four silicon surface-barrier detectors. Measured proton yields and angular distributions. Deduced resonance levels at  $E_x=9.9$ -11.8 MeV, J, and  $\pi$ .

[1971Mc23](#):  $^{31}\text{P}(\alpha, \text{p})$  with 3.25-5.25-MeV  $\alpha$  beams produced from the 5.5-MeV Van de Graaff accelerator of the Southern Universities Nuclear Institute. The target was  $\text{Zn}_3\text{P}_2$  evaporated onto carbon backings. Protons were detected using four solid state detectors. Measured proton yields and angular distributions. Deduced resonance levels at  $E_x=9.9$ -11.7 MeV, J, and resonance strengths.

[1964Ku12](#):  $^{31}\text{P}(\alpha, \text{p})$  with 1.7-3.3-MeV  $\alpha$  beams produced from the Utrecht Van de Graaff accelerator. A zinc phosphide target was prepared by evaporation of 40-500-mg/cm<sup>2</sup>  $\text{Zn}_3\text{P}_2$  onto solid copper backings. Protons were detected using five silicon surface barrier counters at 87°, 120°, 135°, 150°, and 172°. Measured resonance yields and proton angular distributions. Deduced  $^{35}\text{Cl}$  resonance energies, J, and resonance strengths.

[1970Um01](#):  $^{34}\text{S}(\text{p}, \text{n})$ ,  $^{31}\text{P}(\alpha, \text{n})$ ,  $^{31}\text{P}(\alpha, \alpha_0)$  reactions with proton and  $\alpha$  beams produced from the Florida State University Tandem Van de Graaff accelerator. Targets for the  $^{34}\text{S}(\text{p}, \text{n})$  experiment were a CdS target made by evaporating  $^{34}\text{S}$  enriched CdS onto 0.12-mm gold plate and a AgS target with heating  $^{34}\text{S}$  (67% enriched) in the presence of 0.025-mm Ag foil. Targets for the  $^{31}\text{P}(\alpha, \text{n})$  experiment were prepared by evaporating natural  $\text{Ca}_3\text{P}_2$  onto 0.075-mm Ta plate and 0.025-mm Al foil. Positrons from  $^{34}\text{Cl}$  decay were detected using an NE102 plastic scintillator.  $\gamma$  rays in  $^{34}\text{S}$  following the decay were detected using a Ge(Li) detector. Measured excitation functions. Deduced level energies and isospin T=3/2 for  $^{35}\text{Cl}$  resonance states at 12.9 and 13.9 MeV.

 $^{35}\text{Cl}$  Levels

$(2J+1)\Gamma_\alpha\Gamma_p/\Gamma$ : from [1964Ku12](#) for  $E_x<9950$  and from [1971Mc23](#) for  $E_x>9950$ .

| E(level) <sup>†</sup> | $J^\pi$ <sup>‡</sup>                | $E_\alpha(\text{lab})$ (keV) <sup>&amp;</sup> | Comments                                                                                                                                                                                                                         |
|-----------------------|-------------------------------------|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9127 9                | 5/2 <sup>#</sup>                    | 2404 <sup>#</sup> 10                          | $(2J+1)\Gamma_\alpha\Gamma_p/\Gamma=0.16$ eV.                                                                                                                                                                                    |
| 9161 4                | 1/2 <sup>#</sup>                    | 2442 <sup>#</sup> 4                           | $(2J+1)\Gamma_\alpha\Gamma_p/\Gamma=1.6$ eV.                                                                                                                                                                                     |
| 9256 4                | 1/2 <sup>#</sup>                    | 2550 <sup>#</sup> 4                           | $(2J+1)\Gamma_\alpha\Gamma_p/\Gamma=2.6$ eV.                                                                                                                                                                                     |
| 9400 9                | 1/2 <sup>#</sup>                    | 2712 <sup>#</sup> 10                          | $(2J+1)\Gamma_\alpha\Gamma_p/\Gamma=0.6$ eV.                                                                                                                                                                                     |
| 9456 4                | 3/2 <sup>#</sup>                    | 2776 <sup>#</sup> 4                           | $(2J+1)\Gamma_\alpha\Gamma_p/\Gamma=2.8$ eV.                                                                                                                                                                                     |
| 9481.0 18             | 3/2 <sup>#</sup>                    | 2804 <sup>#</sup> 2                           | $(2J+1)\Gamma_\alpha\Gamma_p/\Gamma=18$ eV.                                                                                                                                                                                      |
| 9551 6                | 5/2 <sup>#</sup>                    | 2883 <sup>#</sup> 7                           | $(2J+1)\Gamma_\alpha\Gamma_p/\Gamma=1.9$ eV.                                                                                                                                                                                     |
| 9673 6                | 1/2 <sup>#</sup>                    | 3021 <sup>#</sup> 7                           | $(2J+1)\Gamma_\alpha\Gamma_p/\Gamma=1.1$ eV.                                                                                                                                                                                     |
| 9713.0 27             | 1/2 <sup>#</sup>                    | 3066 <sup>#</sup> 3                           | $(2J+1)\Gamma_\alpha\Gamma_p/\Gamma=41$ eV.                                                                                                                                                                                      |
| 9751.1 27             | 7/2 <sup>#</sup>                    | 3109 <sup>#</sup> 3                           | $(2J+1)\Gamma_\alpha\Gamma_p/\Gamma=7.6$ eV.                                                                                                                                                                                     |
| 9814.0 27             | 5/2 <sup>#</sup>                    | 3180 <sup>#</sup> 3                           | $(2J+1)\Gamma_\alpha\Gamma_p/\Gamma=20$ eV.                                                                                                                                                                                      |
| 9869.8 27             | 1/2                                 | 3243 3                                        | $E_\alpha(\text{lab})$ (keV): weighted average of 3243 3 ( <a href="#">1964Ku12</a> ) and 3242 5 ( <a href="#">1975Sc40</a> ).<br>$J^\pi$ : 1/2 from <a href="#">1964Ku12</a> .<br>$(2J+1)\Gamma_\alpha\Gamma_p/\Gamma=46$ eV.   |
| 9900.8 27             | 3/2                                 | 3278 3                                        | $E_\alpha(\text{lab})$ (keV): weighted average of 3278 3 ( <a href="#">1964Ku12</a> ) and 3278 5 ( <a href="#">1975Sc40</a> ).<br>$J^\pi$ : 3/2 from <a href="#">1964Ku12</a> .<br>$(2J+1)\Gamma_\alpha\Gamma_p/\Gamma=38$ eV.   |
| 9923 4                | 3/2 <sup>-</sup>                    | 3303 4                                        | $E_\alpha(\text{lab})$ (keV): weighted average of 3302 4 ( <a href="#">1964Ku12</a> ) and 3305 5 ( <a href="#">1975Sc40</a> ).<br>$J^\pi$ : (3/2) from <a href="#">1964Ku12</a> .<br>$(2J+1)\Gamma_\alpha\Gamma_p/\Gamma=80$ eV. |
| 9958 6                | 3/2                                 | 3342 7                                        | $E_\alpha(\text{lab})$ (keV): weighted average of 3348 5 ( <a href="#">1971Mc23</a> ) and 3335 5 ( <a href="#">1975Sc40</a> ).<br>$J^\pi$ : 3/2 from <a href="#">1971Mc23</a> .<br>$(2J+1)\Gamma_\alpha\Gamma_p/\Gamma=34$ eV.   |
| 9968 5                | (1/2, 3/2) <sup>@</sup>             | 3354 5                                        | $(2J+1)\Gamma_\alpha\Gamma_p/\Gamma=17$ eV.                                                                                                                                                                                      |
| 10031 5               | 1/2 <sup>+</sup> , 3/2 <sup>-</sup> | 3425 5                                        | $E_\alpha(\text{lab})$ (keV): weighted average of 3424 5 ( <a href="#">1971Mc23</a> ) and 3425 5 ( <a href="#">1975Sc40</a> ).<br>$J^\pi$ : (1/2) from <a href="#">1971Mc23</a> .                                                |

Continued on next page (footnotes at end of table)

$^{31}\text{P}(\alpha, \text{p}), (\alpha, \text{n})$ : resonances **1975Sc40, 1971Mc23, 1964Ku12** (continued) $^{35}\text{Cl}$  Levels (continued)

| E(level) <sup>†</sup> | J <sup>π</sup> <sup>‡</sup>         | E <sub>α</sub> (lab) (keV) <sup>&amp;</sup> | Comments                                                                                                                                                         |
|-----------------------|-------------------------------------|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       |                                     |                                             | (2J+1)Γ <sub>α</sub> Γ <sub>p</sub> /Γ=124 eV.                                                                                                                   |
| 10058 5               | 3/2 <sup>@</sup>                    | 3455 5                                      | (2J+1)Γ <sub>α</sub> Γ <sub>p</sub> /Γ=40 eV.                                                                                                                    |
| 10075 5               | (1/2) <sup>@</sup>                  | 3475 5                                      | (2J+1)Γ <sub>α</sub> Γ <sub>p</sub> /Γ=52 eV.                                                                                                                    |
| 10090 5               | 5/2 <sup>+</sup>                    | 3491 5                                      | E <sub>α</sub> (lab) (keV): weighted average of 3491 5 ( <b>1971Mc23</b> ) and 3490 5 ( <b>1975Sc40</b> ).<br>J <sup>π</sup> : 5/2 from <b>1971Mc23</b> .        |
|                       |                                     |                                             | (2J+1)Γ <sub>α</sub> Γ <sub>p</sub> /Γ=63 eV.                                                                                                                    |
| 10132 5               | 3/2 <sup>-</sup>                    | 3539 5                                      | E <sub>α</sub> (lab) (keV): weighted average of 3538 5 ( <b>1971Mc23</b> ) and 3540 5 ( <b>1975Sc40</b> ).<br>J <sup>π</sup> : 3/2 from <b>1971Mc23</b> .        |
|                       |                                     |                                             | (2J+1)Γ <sub>α</sub> Γ <sub>p</sub> /Γ=165 eV.                                                                                                                   |
| 10172 5               | 3/2 <sup>-</sup> , 5/2 <sup>-</sup> | 3584 5                                      | E <sub>α</sub> (lab) (keV): weighted average of 3587 5 ( <b>1971Mc23</b> ) and 3580 5 ( <b>1975Sc40</b> ).<br>J <sup>π</sup> : (5/2) from <b>1971Mc23</b> .      |
|                       |                                     |                                             | (2J+1)Γ <sub>α</sub> Γ <sub>p</sub> /Γ=26 eV.                                                                                                                    |
| 10215 5               | 3/2                                 | 3633 5                                      | E <sub>α</sub> (lab) (keV): weighted average of 3631 5 ( <b>1971Mc23</b> ) and 3635 5 ( <b>1975Sc40</b> ).<br>J <sup>π</sup> : 3/2 from <b>1971Mc23</b> .        |
|                       |                                     |                                             | (2J+1)Γ <sub>α</sub> Γ <sub>p</sub> /Γ=63 eV.                                                                                                                    |
| 10236 5               | 3/2 <sup>-</sup>                    | 3656 5                                      | E <sub>α</sub> (lab) (keV): weighted average of 3656 5 ( <b>1971Mc23</b> ) and 3655 5 ( <b>1975Sc40</b> ).<br>J <sup>π</sup> : 3/2 from <b>1971Mc23</b> .        |
|                       |                                     |                                             | (2J+1)Γ <sub>α</sub> Γ <sub>p</sub> /Γ=271 eV.                                                                                                                   |
| 10278 5               | 1/2 <sup>@</sup>                    | 3704 5                                      | (2J+1)Γ <sub>α</sub> Γ <sub>p</sub> /Γ=15 eV.                                                                                                                    |
| 10295 5               |                                     | 3723 5                                      |                                                                                                                                                                  |
| 10322 5               | 5/2 <sup>+</sup> , 7/2 <sup>-</sup> | 3753 5                                      | E <sub>α</sub> (lab) (keV): weighted average of 3755 5 ( <b>1971Mc23</b> ) and 3750 5 ( <b>1975Sc40</b> ).<br>J <sup>π</sup> : 5/2 from <b>1971Mc23</b> .        |
|                       |                                     |                                             | (2J+1)Γ <sub>α</sub> Γ <sub>p</sub> /Γ=63 eV.                                                                                                                    |
| 10340 5               | (3/2, 5/2) <sup>@</sup>             | 3774 5                                      | (2J+1)Γ <sub>α</sub> Γ <sub>p</sub> /Γ=50 eV.                                                                                                                    |
| 10359 5               | 3/2                                 | 3795 5                                      | E <sub>α</sub> (lab) (keV): weighted average of 3795 5 ( <b>1971Mc23</b> ) and 3795 5 ( <b>1975Sc40</b> ).<br>J <sup>π</sup> : (3/2) from <b>1971Mc23</b> .      |
|                       |                                     |                                             | (2J+1)Γ <sub>α</sub> Γ <sub>p</sub> /Γ=155 eV.                                                                                                                   |
| 10385 5               |                                     | 3825 5                                      |                                                                                                                                                                  |
| 10399 5               | 3/2 <sup>+</sup> , 7/2 <sup>+</sup> | 3841 5                                      | E <sub>α</sub> (lab) (keV): weighted average of 3842 5 ( <b>1971Mc23</b> ) and 3840 5 ( <b>1975Sc40</b> ).<br>J <sup>π</sup> : (1/2, 3/2) from <b>1971Mc23</b> . |
|                       |                                     |                                             | (2J+1)Γ <sub>α</sub> Γ <sub>p</sub> /Γ=61 eV.                                                                                                                    |
| 10431 5               | 5/2 <sup>+</sup>                    | 3877 5                                      | E <sub>α</sub> (lab) (keV): weighted average of 3876 5 ( <b>1971Mc23</b> ) and 3877 5 ( <b>1975Sc40</b> ).<br>J <sup>π</sup> : 5/2 from <b>1971Mc23</b> .        |
|                       |                                     |                                             | (2J+1)Γ <sub>α</sub> Γ <sub>p</sub> /Γ=66 eV.                                                                                                                    |
| 10467 5               | 5/2 <sup>+</sup>                    | 3917 5                                      | E <sub>α</sub> (lab) (keV): weighted average of 3918 5 ( <b>1971Mc23</b> ) and 3915 5 ( <b>1975Sc40</b> ).<br>J <sup>π</sup> : 5/2 from <b>1971Mc23</b> .        |
|                       |                                     |                                             | (2J+1)Γ <sub>α</sub> Γ <sub>p</sub> /Γ=211 eV.                                                                                                                   |
| 10486 5               | 5/2 <sup>+</sup>                    | 3939 5                                      | E <sub>α</sub> (lab) (keV): weighted average of 3939 5 ( <b>1971Mc23</b> ) and 3938 5 ( <b>1975Sc40</b> ).<br>J <sup>π</sup> : 5/2 from <b>1971Mc23</b> .        |
|                       |                                     |                                             | (2J+1)Γ <sub>α</sub> Γ <sub>p</sub> /Γ=76 eV.                                                                                                                    |
| 10534 5               | 1/2 <sup>+</sup> , 5/2 <sup>+</sup> | 3993 5                                      | E <sub>α</sub> (lab) (keV): weighted average of 3990 5 ( <b>1971Mc23</b> ) and 3995 5 ( <b>1975Sc40</b> ).<br>J <sup>π</sup> : 1/2 from <b>1971Mc23</b> .        |
|                       |                                     |                                             | (2J+1)Γ <sub>α</sub> Γ <sub>p</sub> /Γ=182 eV.                                                                                                                   |
| 10587 5               | 3/2 <sup>+</sup>                    | 4053 5                                      | E <sub>α</sub> (lab) (keV): weighted average of 4050 5 ( <b>1971Mc23</b> ) and 4055 5 ( <b>1975Sc40</b> ).<br>J <sup>π</sup> : 3/2 from <b>1971Mc23</b> .        |
|                       |                                     |                                             | (2J+1)Γ <sub>α</sub> Γ <sub>p</sub> /Γ=71 eV.                                                                                                                    |
| 10641 5               | 5/2 <sup>+</sup>                    | 4114 5                                      | E <sub>α</sub> (lab) (keV): weighted average of 4113 5 ( <b>1971Mc23</b> ) and 4114 5 ( <b>1975Sc40</b> ).<br>J <sup>π</sup> : 5/2 from <b>1971Mc23</b> .        |
|                       |                                     |                                             | (2J+1)Γ <sub>α</sub> Γ <sub>p</sub> /Γ=272 eV.                                                                                                                   |
| 10678 5               | 3/2 <sup>+</sup> , 5/2 <sup>+</sup> | 4156 5                                      | E <sub>α</sub> (lab) (keV): weighted average of 4156 5 ( <b>1971Mc23</b> ) and 4155 5 ( <b>1975Sc40</b> ).<br>J <sup>π</sup> : 3/2 from <b>1971Mc23</b> .        |
|                       |                                     |                                             | (2J+1)Γ <sub>α</sub> Γ <sub>p</sub> /Γ=132 eV.                                                                                                                   |
| 10696 5               | 7/2 <sup>-</sup>                    | 4176 5                                      | E <sub>α</sub> (lab) (keV): weighted average of 4174 5 ( <b>1971Mc23</b> ) and 4178 5 ( <b>1975Sc40</b> ).                                                       |

Continued on next page (footnotes at end of table)

$^{31}\text{P}(\alpha, \text{p}), (\alpha, \text{n})$ : resonances **1975Sc40, 1971Mc23, 1964Ku12** (continued) $^{35}\text{Cl}$  Levels (continued)

| E(level) <sup>†</sup> | J <sup>π</sup> <sup>‡</sup>         | E <sub>α</sub> (lab) (keV) <sup>&amp;</sup> | Comments                                                                                                                                                                                                                                                                                                                                         |
|-----------------------|-------------------------------------|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10734 5               | 3/2 <sup>+</sup> , 9/2 <sup>+</sup> | 4219 5                                      | J <sup>π</sup> : 7/2 from <a href="#">1971Mc23</a> .<br>(2J+1)Γ <sub>α</sub> Γ <sub>p</sub> /Γ=100 eV.<br>E <sub>α</sub> (lab) (keV): weighted average of 4218 5 ( <a href="#">1971Mc23</a> ) and 4220 5 ( <a href="#">1975Sc40</a> ).<br>J <sup>π</sup> : 3/2 from <a href="#">1971Mc23</a> .<br>(2J+1)Γ <sub>α</sub> Γ <sub>p</sub> /Γ=269 eV. |
| 10760 5               | 3/2 <sup>+</sup> , 7/2 <sup>-</sup> | 4248 5                                      | E <sub>α</sub> (lab) (keV): weighted average of 4246 5 ( <a href="#">1971Mc23</a> ) and 4250 5 ( <a href="#">1975Sc40</a> ).                                                                                                                                                                                                                     |
| 10800 5               | 5/2 <sup>+</sup>                    | 4293 5                                      | E <sub>α</sub> (lab) (keV): weighted average of 4290 5 ( <a href="#">1971Mc23</a> ) and 4295 5 ( <a href="#">1975Sc40</a> ).<br>J <sup>π</sup> : 5/2 from <a href="#">1971Mc23</a> .<br>(2J+1)Γ <sub>α</sub> Γ <sub>p</sub> /Γ=238 eV.                                                                                                           |
| 10816 5               | 1/2 <sup>+</sup> , 3/2 <sup>-</sup> | 4311 5                                      | E <sub>α</sub> (lab) (keV): weighted average of 4308 5 ( <a href="#">1971Mc23</a> ) and 4313 5 ( <a href="#">1975Sc40</a> ).<br>J <sup>π</sup> : 3/2 from <a href="#">1971Mc23</a> .<br>(2J+1)Γ <sub>α</sub> Γ <sub>p</sub> /Γ=345 eV.                                                                                                           |
| 10844 5               | 3/2 <sup>-</sup>                    | 4343 5                                      | E <sub>α</sub> (lab) (keV): weighted average of 4342 5 ( <a href="#">1971Mc23</a> ) and 4343 5 ( <a href="#">1975Sc40</a> ).<br>J <sup>π</sup> : 3/2 from <a href="#">1971Mc23</a> .<br>(2J+1)Γ <sub>α</sub> Γ <sub>p</sub> /Γ=923 eV.                                                                                                           |
| 10870 5               | (5/2, 7/2) <sup>@</sup>             | 4372 5                                      | (2J+1)Γ <sub>α</sub> Γ <sub>p</sub> /Γ=137 eV.                                                                                                                                                                                                                                                                                                   |
| 10886 5               |                                     | 4390 5                                      |                                                                                                                                                                                                                                                                                                                                                  |
| 10909 5               | 3/2 <sup>+</sup> , 5/2 <sup>-</sup> | 4416 5                                      | E <sub>α</sub> (lab) (keV): weighted average of 4413 5 ( <a href="#">1971Mc23</a> ) and 4418 5 ( <a href="#">1975Sc40</a> ).                                                                                                                                                                                                                     |
| 10928 5               | 7/2 <sup>@</sup>                    | 4438 5                                      | (2J+1)Γ <sub>α</sub> Γ <sub>p</sub> /Γ=231 eV.                                                                                                                                                                                                                                                                                                   |
| 10942 5               |                                     | 4454 5                                      |                                                                                                                                                                                                                                                                                                                                                  |
| 10964 5               | 5/2 <sup>+</sup>                    | 4478 5                                      | E <sub>α</sub> (lab) (keV): weighted average of 4478 5 ( <a href="#">1971Mc23</a> ) and 4477 5 ( <a href="#">1975Sc40</a> ).<br>J <sup>π</sup> : 5/2 from <a href="#">1971Mc23</a> .<br>(2J+1)Γ <sub>α</sub> Γ <sub>p</sub> /Γ=181 eV.                                                                                                           |
| 10973 5               |                                     | 4489 5                                      |                                                                                                                                                                                                                                                                                                                                                  |
| 10998 5               | 3/2 <sup>+</sup> , 7/2 <sup>-</sup> | 4517 5                                      | E <sub>α</sub> (lab) (keV): weighted average of 4517 5 ( <a href="#">1971Mc23</a> ) and 4517 5 ( <a href="#">1975Sc40</a> ).<br>J <sup>π</sup> : (5/2) from <a href="#">1971Mc23</a> .<br>(2J+1)Γ <sub>α</sub> Γ <sub>p</sub> /Γ=172 eV.                                                                                                         |
| 11034 5               | 3/2 <sup>+</sup>                    | 4557 5                                      | E <sub>α</sub> (lab) (keV): weighted average of 4556 5 ( <a href="#">1971Mc23</a> ) and 4558 5 ( <a href="#">1975Sc40</a> ).<br>J <sup>π</sup> : 3/2 from <a href="#">1971Mc23</a> .<br>(2J+1)Γ <sub>α</sub> Γ <sub>p</sub> /Γ=218 eV.                                                                                                           |
| 11063 5               | 1/2 <sup>+</sup> , 9/2 <sup>+</sup> | 4590 5                                      | E <sub>α</sub> (lab) (keV): weighted average of 4589 5 ( <a href="#">1971Mc23</a> ) and 4590 5 ( <a href="#">1975Sc40</a> ).                                                                                                                                                                                                                     |
| 11082 5               | 5/2 <sup>+</sup> , 9/2 <sup>+</sup> | 4612 5                                      | E <sub>α</sub> (lab) (keV): weighted average of 4611 5 ( <a href="#">1971Mc23</a> ) and 4613 5 ( <a href="#">1975Sc40</a> ).                                                                                                                                                                                                                     |
| 11105 5               |                                     | 4638 5                                      |                                                                                                                                                                                                                                                                                                                                                  |
| 11123 5               | 3/2                                 | 4658 5                                      | E <sub>α</sub> (lab) (keV): weighted average of 4657 5 ( <a href="#">1971Mc23</a> ) and 4658 5 ( <a href="#">1975Sc40</a> ).                                                                                                                                                                                                                     |
| 11142 5               | 5/2 <sup>+</sup>                    | 4679 5                                      | E <sub>α</sub> (lab) (keV): weighted average of 4678 5 ( <a href="#">1971Mc23</a> ) and 4680 5 ( <a href="#">1975Sc40</a> ).                                                                                                                                                                                                                     |
| 11154 5               |                                     | 4693 5                                      |                                                                                                                                                                                                                                                                                                                                                  |
| 11169 5               |                                     | 4710 5                                      |                                                                                                                                                                                                                                                                                                                                                  |
| 11179 5               | 3/2 <sup>-</sup> , 5/2 <sup>+</sup> | 4721 5                                      | E <sub>α</sub> (lab) (keV): weighted average of 4721 5 ( <a href="#">1971Mc23</a> ) and 4720 5 ( <a href="#">1975Sc40</a> ).                                                                                                                                                                                                                     |
| 11185 5               |                                     | 4728 5                                      |                                                                                                                                                                                                                                                                                                                                                  |
| 11198 5               | 3/2                                 | 4743 5                                      | E <sub>α</sub> (lab) (keV): weighted average of 4745 5 ( <a href="#">1971Mc23</a> ) and 4740 5 ( <a href="#">1975Sc40</a> ).                                                                                                                                                                                                                     |
| 11230 5               |                                     | 4779 5                                      |                                                                                                                                                                                                                                                                                                                                                  |
| 11246 5               | 3/2                                 | 4797 5                                      | E <sub>α</sub> (lab) (keV): weighted average of 4799 5 ( <a href="#">1971Mc23</a> ) and 4795 5 ( <a href="#">1975Sc40</a> ).                                                                                                                                                                                                                     |
| 11265 5               | 3/2 <sup>-</sup>                    | 4818 5                                      | E <sub>α</sub> (lab) (keV): weighted average of 4820 5 ( <a href="#">1971Mc23</a> ) and 4815 5 ( <a href="#">1975Sc40</a> ).                                                                                                                                                                                                                     |
| 11287 5               |                                     | 4843 5                                      |                                                                                                                                                                                                                                                                                                                                                  |
| 11307 5               |                                     | 4866 5                                      |                                                                                                                                                                                                                                                                                                                                                  |
| 11314 5               | 1/2 <sup>-</sup> , 5/2 <sup>+</sup> | 4874 5                                      | E <sub>α</sub> (lab) (keV): weighted average of 4875 5 ( <a href="#">1971Mc23</a> ) and 4873 5 ( <a href="#">1975Sc40</a> ).                                                                                                                                                                                                                     |
| 11327 5               | 1/2 <sup>-</sup> , 5/2 <sup>+</sup> | 4888 5                                      | E <sub>α</sub> (lab) (keV): weighted average of 4886 5 ( <a href="#">1971Mc23</a> ) and 4890 5 ( <a href="#">1975Sc40</a> ).                                                                                                                                                                                                                     |
| 11357 5               |                                     | 4922 5                                      |                                                                                                                                                                                                                                                                                                                                                  |
| 11375 5               |                                     | 4942 5                                      |                                                                                                                                                                                                                                                                                                                                                  |
| 11390 5               |                                     | 4960 5                                      |                                                                                                                                                                                                                                                                                                                                                  |
| 11410 5               |                                     | 4982 5                                      |                                                                                                                                                                                                                                                                                                                                                  |
| 11421 5               | 3/2 <sup>-</sup>                    | 4995 5                                      | E <sub>α</sub> (lab) (keV): weighted average of 4995 5 ( <a href="#">1971Mc23</a> ) and 4995 5 ( <a href="#">1975Sc40</a> ).                                                                                                                                                                                                                     |
| 11434 5               |                                     | 5009 5                                      |                                                                                                                                                                                                                                                                                                                                                  |
| 11460 5               |                                     | 5038 5                                      |                                                                                                                                                                                                                                                                                                                                                  |
| 11473 5               | 1/2 <sup>-</sup> , 5/2 <sup>-</sup> | 5053 5                                      | E <sub>α</sub> (lab) (keV): weighted average of 5056 5 ( <a href="#">1971Mc23</a> ) and 5050 5 ( <a href="#">1975Sc40</a> ).                                                                                                                                                                                                                     |

Continued on next page (footnotes at end of table)

$^{31}\text{P}(\alpha, \text{p}), (\alpha, \text{n})$ : resonances [1975Sc40](#), [1971Mc23](#), [1964Ku12](#) (continued) $^{35}\text{Cl}$  Levels (continued)

| E(level) <sup>†</sup> | J <sup>π</sup> <sup>‡</sup>         | E <sub>α</sub> (lab) (keV) <sup>&amp;</sup> | Comments                                                                                                                     |
|-----------------------|-------------------------------------|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| 11492 5               |                                     | 5075 5                                      |                                                                                                                              |
| 11505 5               | 3/2 <sup>-</sup> , 5/2 <sup>-</sup> | 5089 5                                      | E <sub>α</sub> (lab) (keV): weighted average of 5090 5 ( <a href="#">1971Mc23</a> ) and 5088 5 ( <a href="#">1975Sc40</a> ). |
| 11525 5               |                                     | 5112 5                                      |                                                                                                                              |
| 11540 5               |                                     | 5129 5                                      |                                                                                                                              |
| 11551 5               | 3/2 <sup>-</sup> , 5/2 <sup>+</sup> | 5141 5                                      | E <sub>α</sub> (lab) (keV): weighted average of 5142 5 ( <a href="#">1971Mc23</a> ) and 5140 5 ( <a href="#">1975Sc40</a> ). |
| 11565 5               |                                     | 5157 5                                      |                                                                                                                              |
| 11589 5               |                                     | 5184 5                                      |                                                                                                                              |
| 11609 5               | 5/2 <sup>+</sup> , 7/2 <sup>-</sup> | 5207 5                                      | E <sub>α</sub> (lab) (keV): weighted average of 5209 5 ( <a href="#">1971Mc23</a> ) and 5205 5 ( <a href="#">1975Sc40</a> ). |
| 11627 5               | 1/2 <sup>+</sup> , 5/2 <sup>+</sup> | 5227 5                                      | E <sub>α</sub> (lab) (keV): weighted average of 5224 5 ( <a href="#">1971Mc23</a> ) and 5230 5 ( <a href="#">1975Sc40</a> ). |
| 11638 5               |                                     | 5239 5                                      |                                                                                                                              |
| 11651 5               |                                     | 5254 5                                      |                                                                                                                              |
| 11684 5               | 1/2 <sup>+</sup> , 5/2 <sup>+</sup> | 5292 <sup>a</sup> 5                         |                                                                                                                              |
| 11773 5               | 3/2 <sup>-</sup> , 7/2 <sup>-</sup> | 5392 <sup>a</sup> 5                         |                                                                                                                              |
| 11783 5               | 3/2 <sup>-</sup>                    | 5403 <sup>a</sup> 5                         |                                                                                                                              |
| 12900 <sup>b</sup>    |                                     |                                             | T=3/2                                                                                                                        |
| 13900 <sup>b</sup>    |                                     |                                             | T=3/2                                                                                                                        |

<sup>†</sup> From E<sub>c.m.</sub>+S(α)( $^{35}\text{Cl}$ ), where S(α)( $^{35}\text{Cl}$ )=6997.91 4 ([2021Wa16](#)). E<sub>c.m.</sub>=E<sub>α</sub>(lab)×m( $^{31}\text{P}$ )/[m<sub>α</sub>+m( $^{31}\text{P}$ )].

<sup>‡</sup> From [1975Sc40](#) by analyzing measured angular distributions, unless otherwise noted.

# From [1964Ku12](#).

@ From [1971Mc23](#).

& From [1971Mc23](#), unless otherwise noted.

<sup>a</sup> From [1975Sc40](#).

<sup>b</sup> From [1970Um01](#). Two  $^{35}\text{Cl}$  resonances are found in  $^{34}\text{S}(\text{p}, \text{n})$  and  $^{31}\text{P}(\alpha, \text{n})$ , but not in  $^{31}\text{P}(\alpha, \alpha_0)$  reactions. T=1  $^{34}\text{S}$  g.s. and T=1/2 protons can populate either T=1/2 or T=3/2 states in  $^{35}\text{Cl}$ . T=1/2  $^{31}\text{P}$  g.s. and T=0 α particles can populate T=1/2 states in  $^{35}\text{Cl}$ . A small isospin impurity would allow formation of T=3/2 states in  $^{35}\text{Cl}$ . The lack of resonances in  $^{31}\text{P}(\alpha, \alpha_0)$  indicates that these two resonances are most likely T=3/2. A T=3/2 resonances in  $^{31}\text{P}(\alpha, \alpha_0)$  scattering is doubly isospin forbidden (both entrance and exit channels) compared with singly forbidden (entrance channel only) for  $^{31}\text{P}(\alpha, \text{n})$ .

$^{31}\text{P}(^6\text{Li},\text{d})$     [1979Es05](#)

$^{31}\text{P}+\alpha\rightarrow^{35}\text{Cl}$  from  $J^\pi=1/2^+$   $^{31}\text{P}$  ground state.

[1979Es05](#),[1978Es01](#): a 36-MeV  $^6\text{Li}^{3+}$  beam at 150-400 nA was produced from the MP tandem accelerator at the University of Rochester and bombarded a  $100\text{-}\mu\text{g}/\text{cm}^2$  cadmium phosphide  $\text{Cd}_3\text{P}_2$  target. Outgoing deuterons were momentum analyzed by an Enge split-pole magnetic spectrograph and detected using Kodak NTB-50 nuclear emulsion plates with FWHM=50 keV. Measured  $\sigma(E_{\text{d}},\theta)$ . Deduced levels, J,  $\pi$ , L-transfers, spectroscopic factors from finite-range, LOLA-DWBA analysis of the measured  $\sigma(\theta)$ . Other: [1982Ch10](#) (finite-range DWBA calculations).

 $^{35}\text{Cl}$  Levels

Spectroscopic factor  $g^*\text{S}=\sigma(\theta)_{\text{exp}}/\sigma(\theta)_{\text{DWBA}}$ , where  $g=(2J_{\text{f}}+1)/(2J_{\text{i}}+1)$ .

| <u>E(level)<sup>†</sup></u> | <u><math>J^\pi</math><sup>‡</sup></u> | <u>L<sup>#</sup></u> | <u>S<sup>#</sup></u> |
|-----------------------------|---------------------------------------|----------------------|----------------------|
| 0                           | $3/2^+$                               | 2                    | 1.00                 |
| 1220                        | $1/2^+$                               | 0                    | 1.49                 |
| 1760                        | $5/2^+$                               | 2                    | 0.24                 |
| 3160                        | $7/2^-$                               | 3                    | 0.52                 |

<sup>†</sup> From [1979Es05](#).

<sup>‡</sup> As given in [1979Es05](#) for extracting spectroscopic factors.

<sup>#</sup> From DWBA analysis of the measured  $\sigma(\theta)$  in [1979Es05](#).

$^{32}\text{S}(\alpha, \text{p})$  2019Se09, 1973Go16

$J^\pi=0^+$  for  $^{32}\text{S}$  ground state.

**2019Se09:** a 21-MeV  $\alpha$  beam was produced from the 10-MV FN tandem Van de Graaff accelerator at Triangle Universities Nuclear Laboratory (TUNL). Targets were  $15.9\text{-}\mu\text{g}/\text{cm}^2$  natS in cadmium sulfide and  $46.3\text{-}\mu\text{g}/\text{cm}^2$  natS in antimony sulfide. Reaction products were momentum-analyzed by the TUNL high-resolution Enge split-pole magnetic spectrograph and detected using a high-resolution position-sensitive focal-plane detector consisting of two position-sensitive gas avalanche counters: a gas proportionality energy loss section and a residual energy scintillator (FWHM=24 keV). Measured  $\sigma(E_p, \theta)$ , at  $\theta=10^\circ$  to  $50^\circ$ . Deduced levels, J,  $\pi$  from one-step finite range triton transfer FRESKO-DWBA analysis. Deduced the thermonuclear  $^{34}\text{S}(\text{p}, \gamma)^{35}\text{Cl}$  reaction rate via Monte Carlo methods.

**1973Go16:** 19.991-, 20.025- and 20.284-MeV  $\alpha$  beams were produced from the University of Notre Dame FN tandem Van de Graaff accelerator. Natural sulfur targets were prepared by vacuum evaporation of CdS onto  $20\text{-}\mu\text{g}/\text{cm}^2$  carbon backings. Protons were analyzed with broad-range magnetic spectrographs at  $15^\circ$ ,  $30^\circ$ ,  $70^\circ$ ,  $90^\circ$ , and  $120^\circ$  and detected using nuclear track plates at the focal planes. Measured  $\sigma(E_p, \theta)$ . Deduced 102 levels.

**1971Au07:** a 28.8-MeV  $\alpha$  beam of 0.1-0.5  $\mu\text{A}$  was produced from the Naval Research Laboratory (NRL) isochronous cyclotron. The target was made by evaporating natural PbS onto a  $30\text{-}\mu\text{g}/\text{cm}^2$  carbon backing, and the effective  $^{32}\text{S}$  thickness was  $163\text{ }\mu\text{g}/\text{cm}^2$ . Protons were detected using two counter telescopes with FWHM=250 keV. Measured  $\sigma(E_p, \theta)$ . Deduced levels, J,  $\pi$ , L-transfers, and J-dependence for the g.s. and 1763-keV level from local zero-range DWUCK-DWBA analysis of the measured  $\sigma(\theta)$ .

Other: **1996So21** (cross sections).

 $^{35}\text{Cl}$  Levels

| E(level) <sup>†</sup>  | $J^\pi$ &                 | $L^a$ | Comments                                                                                              |
|------------------------|---------------------------|-------|-------------------------------------------------------------------------------------------------------|
| 0                      |                           | 2     | $J^\pi$ : $3/2^+$ assumed in <b>1971Au07</b> for investigation of J-dependence of L=2 transfer.       |
| 1217.4 26              |                           |       |                                                                                                       |
| 1760.1 24              |                           | (2)   | $J^\pi$ : $5/2^+$ assumed in <b>1971Au07</b> for investigation of J-dependence of L=2 transfer.       |
|                        |                           |       | L: The parentheses are added by evaluators based on the DWBA fit in <b>1971Au07</b> .                 |
| 2644.9 23              |                           |       |                                                                                                       |
| 2693?                  |                           |       |                                                                                                       |
| 3001.5 31              |                           |       |                                                                                                       |
| 3158.2 25              |                           |       |                                                                                                       |
| 3945.2 23              |                           |       | E(level): weighted average of 3943.7 23 ( <b>1973Go16</b> ) and 3947.9 31 ( <b>2019Se09</b> ).        |
| 4056.9 27              | ( $3/2^-$ ) <sup>b</sup>  |       | E(level): 4059.12 quoted from <b>2011Ch48</b> and used for internal calibrations by <b>2019Se09</b> . |
| 4111.1 29              | ( $7/2^+$ ) <sup>b</sup>  |       | E(level): weighted average of 4110.2 29 ( <b>1973Go16</b> ) and 4112.0 30 ( <b>2019Se09</b> ).        |
| 4177.9 24              | ( $3/2^-$ ) <sup>b</sup>  |       | E(level): weighted average of 4177.5 24 ( <b>1973Go16</b> ) and 4179 4 ( <b>2019Se09</b> ).           |
| 4346.6 24              |                           |       | E(level): 4347.82 quoted from <b>2011Ch48</b> and used for internal calibrations by <b>2019Se09</b> . |
| 4625.3 28              |                           |       | E(level): weighted average of 4624.4 31 ( <b>1973Go16</b> ) and 4626.1 28 ( <b>2019Se09</b> ).        |
| 4770.4 26              |                           |       | E(level): weighted average of 4770.8 26 ( <b>1973Go16</b> ) and 4769.9 28 ( <b>2019Se09</b> ).        |
| 4859.2 <sup>‡</sup> 28 |                           |       |                                                                                                       |
| 4881.8 29              |                           |       | E(level): weighted average of 4883.1 29 ( <b>1973Go16</b> ) and 4878 5 ( <b>2019Se09</b> ).           |
| 5006.9 <sup>‡</sup> 27 |                           |       |                                                                                                       |
| 5158.9 29              | ( $5/2^+$ ) <sup>b</sup>  |       | E(level): weighted average of 5161.7 33 ( <b>1973Go16</b> ) and 5156.8 29 ( <b>2019Se09</b> ).        |
| 5209.5 28              | ( $5/2^+$ )               |       | E(level): weighted average of 5206.6 37 ( <b>1973Go16</b> ) and 5211.1 28 ( <b>2019Se09</b> ).        |
| 5402.0 29              |                           |       |                                                                                                       |
| 5407.2                 | ( $11/2^-$ ) <sup>b</sup> |       | E(level): quoted from <b>2011Ch48</b> and used for internal calibrations by <b>2019Se09</b> .         |
| 5531 <sup>‡</sup> 5    |                           |       |                                                                                                       |
| 5586.0 26              |                           |       | E(level): 5576 tentative level from <b>1973Go16</b> .                                                 |
| 5594.6 26              |                           |       | E(level): weighted average of 5591.8 32 ( <b>1973Go16</b> ) and 5596.5 26 ( <b>2019Se09</b> ).        |
| 5633.6 30              |                           |       | E(level): weighted average of 5633.1 32 ( <b>1973Go16</b> ) and 5634 3 ( <b>2019Se09</b> ).           |
| 5645 <sup>‡</sup> 4    |                           |       |                                                                                                       |
| 5653 <sup>‡</sup> 4    |                           |       |                                                                                                       |
| 5679.7 34              |                           |       | E(level): weighted average of 5677.7 34 ( <b>1973Go16</b> ) and 5684 5 ( <b>2019Se09</b> ).           |
| 5731.2 <sup>‡</sup> 25 |                           |       |                                                                                                       |
| 5757 <sup>‡</sup> 4    |                           |       |                                                                                                       |

Continued on next page (footnotes at end of table)

$^{32}\text{S}(\alpha, p)$  **2019Se09, 1973Go16 (continued)** $^{35}\text{Cl}$  Levels (continued)

| E(level) <sup>†</sup>  | $J^{\pi}$ &                       | Comments                                                                              |
|------------------------|-----------------------------------|---------------------------------------------------------------------------------------|
| 5808.1 31              |                                   | E(level): weighted average of 5809.2 34 (1973Go16) and 5807.1 31 (2019Se09).          |
| 5829 <sup>‡</sup> 4    | (5/2 <sup>-</sup> ) <sup>b</sup>  | E(level): 5823 tentative level from 1973Go16.                                         |
| 5928 <sup>‡</sup> 3    |                                   | E(level): other: 5927.4 35 (1973Go16).                                                |
| 6084.2 29              | (13/2 <sup>-</sup> ) <sup>b</sup> | E(level): 6087.4 quoted from 2011Ch48 and used for internal calibrations by 2019Se09. |
| 6104 <sup>‡</sup> 4    |                                   |                                                                                       |
| 6141 4                 |                                   | E(level): weighted average of 6140.2 40 (1973Go16) and 6142 4 (2019Se09).             |
| 6180 <sup>‡</sup> 3    |                                   |                                                                                       |
| 6224.4 30              |                                   | E(level): weighted average of 6224.9 36 (1973Go16) and 6224 3 (2019Se09).             |
| 6283 <sup>‡</sup> 3    | (5/2 <sup>+</sup> )               |                                                                                       |
| 6379.3 25              | (9/2 <sup>-</sup> )               | E(level): from 2019Se09. Other: 6379.0 34 (1973Go16).                                 |
| 6401.2 22              | (1/2 <sup>-</sup> )               | E(level): weighted average of 6402.4 41 (1973Go16) and 6400.9 22 (2019Se09).          |
| 6429 <sup>‡</sup> 3    | (1/2 <sup>+</sup> )               | E(level): 6427 tentative level from 1973Go16.                                         |
| 6475 <sup>‡</sup> 4    |                                   |                                                                                       |
| 6492 <sup>‡</sup> 3    | (3/2 <sup>+</sup> )               | E(level): other: 6491.9 34 (1973Go16).                                                |
| 6549 <sup>‡</sup> 3    | (1/2 <sup>+</sup> )               |                                                                                       |
| 6659.1 30              | (7/2 <sup>+</sup> )               | E(level): weighted average of 6656.0 31 (1973Go16) and 6662 3 (2019Se09).             |
| 6679.4 31              | (1/2 <sup>+</sup> )               | E(level): weighted average of 6680.8 31 (1973Go16) and 6677 4 (2019Se09).             |
| 6781.3 30              | (3/2 <sup>-</sup> )               | E(level): weighted average of 6782.8 32 (1973Go16) and 6780 3 (2019Se09).             |
| 6800 4                 |                                   | E(level): weighted average of 6802.1 42 (1973Go16) and 6795 6 (2019Se09).             |
| 6842 <sup>‡</sup> 4    | (3/2 <sup>+</sup> )               |                                                                                       |
| 6863 <sup>‡</sup> 3    | (9/2 <sup>+</sup> )               | E(level): 6867 tentative level from 1973Go16.                                         |
| 6891.6 30              | (9/2 <sup>+</sup> )               | E(level): weighted average of 6893.5 32 (1973Go16) and 6890 3 (2019Se09).             |
| 6948.6 34              | (5/2 <sup>+</sup> )               | E(level): weighted average of 6947.5 34 (1973Go16) and 6950 4 (2019Se09).             |
| 6986.3 34              | (9/2 <sup>+</sup> )               | E(level): weighted average of 6985.7 34 (1973Go16) and 6987 4 (2019Se09).             |
| 7066.2                 | (5/2 <sup>+</sup> )               | E(level): 7066.2 quoted from 2011Ch48 and used for internal calibrations by 2019Se09. |
| 7105?                  |                                   |                                                                                       |
| 7121.7 <sup>#</sup> 24 | (5/2 <sup>-</sup> )               | E(level): weighted average of 7120.8 34 (1973Go16) and 7122.1 24 (2019Se09).          |
| 7180 <sup>‡</sup> 4    | (7/2 <sup>+</sup> )               | E(level): other: 7179.0 43 (1973Go16).                                                |
| 7213.7 25              |                                   | E(level): weighted average of 7210.1 39 (1973Go16) and 7215.2 25 (2019Se09).          |
| 7230.9 30              | (3/2 <sup>+</sup> )               | E(level): weighted average of 7229.5 34 (1973Go16) and 7232 3 (2019Se09).             |
| 7301?                  |                                   |                                                                                       |
| 7348 <sup>‡</sup> 3    | (7/2)                             | E(level): other: 7348.3 44 (1973Go16).                                                |
| 7362 <sup>‡</sup> 3    |                                   |                                                                                       |
| 7413.8 30              |                                   | E(level): weighted average of 7417.5 44 (1973Go16) and 7412 3 (2019Se09).             |
| 7447 <sup>‡</sup> 3    |                                   |                                                                                       |
| 7503 4                 |                                   | E(level): weighted average of 7503.0 37 (1973Go16) and 7502 5 (2019Se09).             |
| 7566.4 35              |                                   | E(level): weighted average of 7567.6 35 (1973Go16) and 7564 5 (2019Se09).             |
| 7590 4                 |                                   | E(level): weighted average of 7587.2 40 (1973Go16) and 7595 6 (2019Se09).             |
| 7648.1 30              |                                   | E(level): weighted average of 7650.1 40 (1973Go16) and 7647 3 (2019Se09).             |
| 7674.6 <sup>‡</sup> 25 |                                   | E(level): 7671 tentative level from 1973Go16.                                         |
| 7709.9 <sup>‡</sup> 25 |                                   |                                                                                       |
| 7730?                  |                                   |                                                                                       |
| 7744.1 <sup>#</sup> 30 |                                   | E(level): weighted average of 7742.7 38 (1973Go16) and 7745 3 (2019Se09).             |
| 7770.3 33              |                                   | E(level): weighted average of 7776.3 45 (1973Go16) and 7768.5 25 (2019Se09).          |
| 7799.8 30              |                                   | E(level): weighted average of 7800.8 35 (1973Go16) and 7799 3 (2019Se09).             |
| 7868.5 35              |                                   | E(level): from 1973Go16. Other: 7867 7 (2019Se09).                                    |
| 7901 4                 |                                   | E(level): weighted average of 7900.6 38 (1973Go16) and 7899 12 (2019Se09).            |
| 7978.6 35              |                                   |                                                                                       |
| 8000.7 41              |                                   |                                                                                       |
| 8019.2 35              |                                   |                                                                                       |
| 8075.1 46              |                                   |                                                                                       |

Continued on next page (footnotes at end of table)



$^{32}\text{S}(\alpha, \text{p})$  **2019Se09,1973Go16** (continued) $^{35}\text{Cl}$  Levels (continued)

| E(level) <sup>†</sup>  | E(level) <sup>†</sup>  | E(level) <sup>†</sup>   | E(level) <sup>†</sup>   |
|------------------------|------------------------|-------------------------|-------------------------|
| 8103.8 <sup>#</sup> 42 | 8654.1 41              | 9334.0 46               | 10117?                  |
| 8121?                  | 8698?                  | 9376.0 41               | 10162.7 53              |
| 8171.7 42              | 8720.9 <sup>#</sup> 37 | 9475.5 43               | 10179.1 53              |
| 8210.2 46              | 8786.3 38              | 9508.1 <sup>#</sup> 41  | 10218.0 46              |
| 8246.0 42              | 8840.3 38              | 9525?                   | 10395.2 54              |
| 8270.6 <sup>#</sup> 42 | 8887.5 41              | 9711.8 44               | 10443?                  |
| 8292?                  | 8957.5 44              | 9740.4 40               | 10463.2 <sup>#</sup> 49 |
| 8321.0 38              | 8996.9 40              | 9799.2 47               | 10517.1 54              |
| 8398.8 38              | 9027.4 38              | 9836.1 44               | 10548.3 54              |
| 8429.2 40              | 9085.7 38              | 9870?                   | 10579.5 54              |
| 8482.3 38              | 9109.3 49              | 9968.6 <sup>@</sup> 52  | 10643.0 50              |
| 8505.1 47              | 9160.0 42              | 9992?                   | 10717?                  |
| 8532.4 43              | 9194.0 39              | 10009?                  | 10732.1 <sup>#</sup> 50 |
| 8567.9 37              | 9264.9 39              | 10075.2 <sup>@</sup> 48 | 10759.1 55              |
| 8616.9 37              | 9316.2 39              | 10098?                  |                         |

<sup>†</sup> From **1973Go16**, unless otherwise noted.

<sup>‡</sup> From **2019Se09**.

<sup>#</sup> Doublet in **1973Go16**.

<sup>@</sup> Triplet in **1973Go16**.

<sup>&</sup> From **2019Se09**, based on FRESKO-DWBA analysis of measured  $\sigma(\theta)$ , unless otherwise noted.

<sup>a</sup> From **1971Au07**.

<sup>b</sup> Original firm assignments in **2019Se09** are placed in parentheses by the evaluators when DWBA fits to the measured  $\sigma(\theta)$  were poor or when only a few data points were available for fitting.

$^{32}\text{S}(\alpha, p\gamma)$  1970Ho09, 1972Br33, 1973Al22

$J^\pi=0^+$  for  $^{32}\text{S}$  ground state.

**1970Ho09:** 10.975-, 11.960-, 12.120-MeV  $\alpha$  beams were produced from the Chalk River MP tandem accelerator. Targets were  $100\text{-}\mu\text{g}/\text{cm}^2$   $\text{Sb}_2\text{S}_3$  on a  $10\text{-}\mu\text{g}/\text{cm}^2$  carbon backing.  $\gamma$  rays were detected using 15.2-cm-long by 12.7-cm-diam NaI(Tl) detectors. Protons were detected using an annular silicon surface barrier detector close to  $180^\circ$  with FWHM=30 keV for protons and FWHM=50 keV for  $\alpha$ . Measured  $E_p$ ,  $E_\gamma$ ,  $I_\gamma$ ,  $p\gamma(\theta)$ . Deduced 25 level energies from  $E_p$ , J, branching ratios, multipole mixing ratios.

**1972Br33:** a 10-MeV  $\alpha$  beam was produced from the Chalk River MP tandem accelerator. Targets were  $340\text{-}\mu\text{g}/\text{cm}^2$   $\text{Sb}_2\text{S}_3$  on gold backings.  $\gamma$  rays were detected using a 44-cm<sup>3</sup> Ge(Li) detector inside an anticoincidence split annular NaI(Tl) detector. Measured  $E_\gamma$ ,  $I_\gamma$ . Deduced 11 level energies, 16  $\gamma$  branching ratios, 11 lifetimes using the Doppler Shift Attenuation Method (DSAM). **1970In01** from the same experiment deduced the lifetime using DSAM for a level at 3942 keV and the mixing ratio for a  $3942 \rightarrow 2646$   $\gamma$  transition.

**1973Al22:** a 14-MeV  $\alpha$  beam was produced from the 80-cm cyclotron at the Research Institute for Physics in Stockholm and 12-16-MeV  $\alpha$  beams were produced from the Uppsala tandem accelerator. Targets were prepared by depositing pulverized sulphur enriched to 99.9% in  $^{32}\text{S}$  on lead backings.  $\gamma$  rays were detected using an Ortec 60-cm<sup>3</sup> Ge(Li) detector with FWHM=2.2 keV at 1333 keV and an additional Canberra 60-cm<sup>3</sup> Ge(Li) detector for  $\gamma\gamma$ -coincidence measurements. Measured  $E_\gamma$ ,  $I_\gamma$ ,  $\gamma\gamma$ -coin,  $\gamma(\theta)$  for 4  $\gamma$  rays. Deduced 12 level energies from  $E_\gamma$ , J for 4 levels, branching ratios, and mixing ratios.

**1969In04:** 6-8-MeV  $\alpha$  beams were produced from the Chalk River MP tandem accelerator. A thick target ( $270\text{ }\mu\text{g}/\text{cm}^2$ ) of natural sulfur on a tantalum backing and a thin target ( $50\text{ }\mu\text{g}/\text{cm}^2$ ) of  $\text{Sb}_2\text{S}_3$  on a gold backing were used.  $\gamma$  rays were detected using two Ge(Li) detectors of 15 and 40 cm<sup>3</sup> volume. Measured  $E_\gamma$ ,  $I_\gamma$ . Deduced 6 level energies, branching ratios, 5 lifetimes using the Doppler Shift Attenuation Method (DSAM) and 1 lifetime for the 3163-keV level using the Recoil Distance Method (RDM), and transition strengths.

**1974Lo17:** 11.0, 12.0 and 14.7-MeV  $\alpha$  beams were produced from the Oliver Lodge Laboratory at the University of Liverpool. Targets were natural CdS. Protons were detected using an annular  $\Delta E$ -E telescope close to  $180^\circ$  with FWHM=80 keV.  $\gamma$  rays were detected using five NaI(Tl) detectors and one Ge(Li) detector. Measured  $E_\gamma$ ,  $I_\gamma$ ,  $p\gamma$ -coin,  $p\gamma(\theta)$ ,  $\gamma\gamma$ -coin,  $\gamma$ -ray linear polarization. Deduced J and  $\pi$  for 3 levels, 6  $\gamma$  branching ratios, 5 mixing ratios, and lifetimes for a 5408-keV level using DSAM and for a 6088-keV level using RDM.

**1973Br26:** a 11.2-MeV  $\alpha$  beam was produced from the Legnaro (Padova) Van de Graaff generator. Targets were natural CdS.  $\gamma$  rays were detected using a Ge(Li) detector. Deduced the lifetime using the Recoil Distance Method (RDM) for a level at 3163 keV.

**1973An01:** 5.5- and 8.6-MeV  $\alpha$  beams of 100-200 nA were produced from the Oxford Vertical Van de Graaff (Injector) or the Oxford EN Tandem. Targets were natural sulphur in AgS.  $\gamma$  rays were detected using a Ge(Li) detector at  $0^\circ$ . Deduced the lifetime using the Recoil Distance Method (RDM) for a level at 3163 keV.

 $^{35}\text{Cl}$  Levels

| $E(\text{level})^\dagger$ | $J^\pi^\ddagger$ | $T_{1/2}$    | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------|------------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0                         | $3/2^+$          |              | $J^\pi$ : from the Adopted Levels.                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 1219.4 6                  | $1/2^+$          | 0.15 ps +7-6 | $J^\pi$ : from the Adopted Levels.<br>$T_{1/2}$ : Lifetime $\tau=0.21$ ps +10-8 ( <b>1969In04</b> DSAM).                                                                                                                                                                                                                                                                                                                                                                                  |
| 1763.3 6                  | $5/2^+$          | 0.38 ps 11   | $J^\pi$ : spin=5/2 from $p\gamma(\theta)$ in <b>1970Ho09</b> ; parity=+ from 1763.2 $\gamma$ M1+E2 to $3/2^+$ .<br>$T_{1/2}$ : Lifetime $\tau=0.55$ ps 15 from <b>1969In04</b> using DSAM.                                                                                                                                                                                                                                                                                                |
| 2645.8 8                  | $7/2^+$          | 0.21 ps 6    | $J^\pi$ : spin=7/2 from $p\gamma(\theta)$ in <b>1970Ho09</b> ; parity from 882 $\gamma$ , M1+E2 to $5/2^+$ in <b>1970Ho09</b> .<br>$T_{1/2}$ : Lifetime $\tau=0.30$ ps 9 from <b>1969In04</b> using DSAM.                                                                                                                                                                                                                                                                                 |
| 2693.6 13                 | $3/2^+, 5/2^+$   | 49 fs 21     | $T_{1/2}$ : Lifetime $\tau=70$ fs 30 from <b>1972Br33</b> using DSAM. Other: $\tau < 30$ fs ( <b>1969In04</b> , DSAM).                                                                                                                                                                                                                                                                                                                                                                    |
| 3002.6 15                 | $(5/2)^+$        | 33 fs 6      | $J^\pi$ : spin=3/2, 5/2 from $p\gamma(\theta)$ in <b>1970Ho09</b> ; parity from 3002.5 $\gamma$ , M1+E2 to $3/2^+$ in <b>1970Ho09</b> ; J=3/2 is disfavored since it would give a large $\delta(E2/M1)=1.2$ 5 for 3002.5 $\gamma$ in <b>1970Ho09</b> which is less likely and also due to the absence of an expected $\gamma$ to 1219.5, $1/2^+$ level if J=3/2.<br>$T_{1/2}$ : Lifetime $\tau=48$ fs 9 from <b>1972Br33</b> using DSAM. Other: $\tau < 50$ fs ( <b>1969In04</b> , DSAM). |
| 3163.1 11                 | $7/2^-$          | 42 ps 5      | $J^\pi$ : spin=7/2 from $p\gamma(\theta)$ in <b>1970Ho09</b> ; parity from 3162.6 $\gamma$ , M2+E3 to $3/2^+$ in <b>1970Ho09</b> .                                                                                                                                                                                                                                                                                                                                                        |

Continued on next page (footnotes at end of table)

$^{32}\text{S}(\alpha, p\gamma)$  **1970Ho09, 1972Br33, 1973Al22 (continued)** $^{35}\text{Cl}$  Levels (continued)

| E(level) <sup>†</sup> | J <sup>π</sup> <sup>‡</sup> | T <sub>1/2</sub> | Comments                                                                                                                                                                                                                                                                                            |
|-----------------------|-----------------------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3915 2                |                             | <15.3 fs         | T <sub>1/2</sub> : Lifetime $\tau=60$ ps 7 from 1969In04.<br>E(level): from 1972Br33.                                                                                                                                                                                                               |
| 3943.4 12             | 9/2                         | 0.229 ps 35      | T <sub>1/2</sub> : Lifetime $\tau<22$ fs from 1972Br33 using DSAM.                                                                                                                                                                                                                                  |
| 3968 2                |                             | <37 fs           | T <sub>1/2</sub> : Lifetime $\tau=0.33$ ps 5 from 1972Br33 using DSAM.<br>E(level): from 1972Br33.                                                                                                                                                                                                  |
| 4058 3                |                             | 22 fs 9          | T <sub>1/2</sub> : Lifetime $\tau<54$ fs from 1972Br33 using DSAM.                                                                                                                                                                                                                                  |
| 4108 2                | 3/2, 7/2                    | 49 fs 11         | E(level): weighted average of 4045 10 (1970Ho09) and 4058 2 (1972Br33).<br>T <sub>1/2</sub> : Lifetime $\tau=31$ fs 13 from 1972Br33 using DSAM.                                                                                                                                                    |
| 4171 2                |                             | 47 fs 17         | E(level): weighted average of 4110 10 (1970Ho09) and 4108 2 (1972Br33).<br>T <sub>1/2</sub> : Lifetime $\tau=70$ fs 16 from 1972Br33 using DSAM.                                                                                                                                                    |
| 4347.8 12             | 9/2 <sup>-</sup>            | 2.0 ps +10-5     | E(level): weighted average of 4175 10 (1970Ho09) and 4171 2 (1972Br33).<br>T <sub>1/2</sub> : Lifetime $\tau=68$ fs 24 from 1972Br33 using DSAM.                                                                                                                                                    |
| 4618 2                | 3/2, 5/2                    | 40 fs 17         | J <sup>π</sup> : spin from $p\gamma(\theta)$ in 1974Lo17; parity from 1184.6 $\gamma$ , M1+E2 to 7/2 <sup>-</sup> in 1974Lo17. Other: J=5/2, 9/2 from $p\gamma(\theta)$ in 1970Ho09 and $p\gamma(\theta)$ in 1973Al22.<br>T <sub>1/2</sub> : Lifetime $\tau=2.9$ ps +14-7 from 1972Br33 using DSAM. |
| 4770 <sup>#</sup> 15  | 7/2, 9/2                    |                  | E(level): from 1972Br33.                                                                                                                                                                                                                                                                            |
| 4880.8 21             | 7/2                         | 0.19 ps 4        | T <sub>1/2</sub> : Lifetime $\tau=58$ fs 24 from 1972Br33 using DSAM.                                                                                                                                                                                                                               |
| 5015 <sup>#</sup> 20  |                             |                  | E(level): other: 4882 2 from 1972Br33, 4885 20 from 1970Ho09.                                                                                                                                                                                                                                       |
| 5175 <sup>#</sup> 20  |                             |                  | T <sub>1/2</sub> : Lifetime $\tau=0.28$ ps 6 from 1972Br33 using DSAM.                                                                                                                                                                                                                              |
| 5230 <sup>#</sup> 20  | 1/2, 3/2, 5/2               |                  |                                                                                                                                                                                                                                                                                                     |
| 5407.1 15             | 11/2 <sup>-</sup>           | 0.28 ps 7        | E(level): other: 5415 20 from 1970Ho09.<br>J <sup>π</sup> : spin from $p\gamma(\theta)$ in 1974Lo17 and from $\gamma(\theta)$ and yield function in 1973Al22; parity from 1059.3 $\gamma$ , M1+E2 to 9/2 <sup>-</sup> in 1974Lo17.                                                                  |
| 5535 <sup>#</sup> 20  |                             |                  | T <sub>1/2</sub> : Lifetime $\tau=0.4$ ps 1 from 1974Lo17 using DSAM.                                                                                                                                                                                                                               |
| 5600 <sup>#</sup> 20  |                             |                  |                                                                                                                                                                                                                                                                                                     |
| 5650 <sup>#</sup> 20  | 5/2, 9/2                    |                  |                                                                                                                                                                                                                                                                                                     |
| 5850 <sup>#</sup> 25  | 5/2, 9/2                    |                  |                                                                                                                                                                                                                                                                                                     |
| 5927.1 19             | 7/2, 11/2                   |                  | E(level): other: 5930 25 from 1970Ho09.                                                                                                                                                                                                                                                             |
| 6087.0 18             | 13/2 <sup>-</sup>           | 6.5 ps 6         | J <sup>π</sup> : spin from $\gamma(\theta)$ in 1973Al22.<br>J <sup>π</sup> : spin from $p\gamma(\theta)$ in 1974Lo17; parity from 679.9 $\gamma$ , M1+E2 to 11/2 <sup>-</sup> in 1974Lo17. Other: J=9/2, 13/2 from $\gamma(\theta)$ in 1973Al22.                                                    |
| 6400 <sup>#</sup> 25  |                             |                  | T <sub>1/2</sub> : Lifetime $\tau=9.3$ ps 8 from 1974Lo17 using RDM.                                                                                                                                                                                                                                |

<sup>†</sup> From a least-squares fit to E<sub>γ</sub> values with uncertainties, unless otherwise noted.<sup>‡</sup> Spin assignments are from  $p\gamma(\theta)$  in 1970Ho09, unless otherwise noted.<sup>#</sup> From 1970Ho09, derived from measured E<sub>p</sub>. $\gamma(^{35}\text{Cl})$ 

| E <sub>i</sub> (level) | J <sub>i</sub> <sup>π</sup> | E <sub>γ</sub> <sup>†</sup> | I <sub>γ</sub> <sup>‡</sup> | E <sub>f</sub> | J <sub>f</sub> <sup>π</sup> | Mult. <sup>#</sup> | δ <sup>#</sup> | Comments                                                                           |
|------------------------|-----------------------------|-----------------------------|-----------------------------|----------------|-----------------------------|--------------------|----------------|------------------------------------------------------------------------------------|
| 1219.4                 | 1/2 <sup>+</sup>            | 1219.4 6                    | 100                         | 0              | 3/2 <sup>+</sup>            |                    |                | E <sub>γ</sub> : weighted average of 1219.3 6 (1969In04) and 1219.5 10 (1973Al22). |
| 1763.3                 | 5/2 <sup>+</sup>            | 543.9                       | <0.5                        | 1219.4         | 1/2 <sup>+</sup>            |                    |                |                                                                                    |
|                        |                             | 1763.2 6                    | 100                         | 0              | 3/2 <sup>+</sup>            | M1+E2              | 3.0 1          | E <sub>γ</sub> : from 1969In04. Other: 1763.1 10 (1973Al22).                       |
| 2645.8                 | 7/2 <sup>+</sup>            | 882.4 10                    | 11.1 22                     | 1763.3         | 5/2 <sup>+</sup>            | M1+E2              | +0.25 5        | I <sub>γ</sub> : from 1970Ho09. Other: 18 (1969In04), 10 (1973Al22).               |

Continued on next page (footnotes at end of table)

$^{32}\text{S}(\alpha, p\gamma)$  **1970Ho09, 1972Br33, 1973Al22 (continued)** $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$      | $E_\gamma^\dagger$                                  | $I_\gamma^\ddagger$                           | $E_f$                                     | $J_f^\pi$                                                    | Mult.#           | $\delta^\#$       | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------|----------------|-----------------------------------------------------|-----------------------------------------------|-------------------------------------------|--------------------------------------------------------------|------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2645.8              | $7/2^+$        | 1426.4<br>2646.2 15                                 | <1.2<br>100.0 22                              | 1219.4<br>0                               | $1/2^+$<br>$3/2^+$                                           |                  |                   | $E_\gamma$ : from 1969In04. Other: 2646.0 20 (1973Al22).<br>$I_\gamma$ : from 1970Ho09. Other: 100 (1969In04), 100 (1973Al22).<br>$I_\gamma$ : from 1970Ho09.<br>$I_\gamma$ : from 1970Ho09. Other: 10 (1969In04).<br>$I_\gamma$ : from 1970Ho09. Other: 100 (1969In04), 100 (1973Al22).<br>$\delta$ : $\delta=+0.17$ 8 for $J=3/2$ ; $\delta=-0.26$ 5 for $J=5/2$ (1970Ho09).                                                                                                                                                                           |
| 2693.6              | $3/2^+, 5/2^+$ | 930.5 15<br>1474.2<br>2693.2 20                     | 18.2 26<br>11.7 26<br>100 4                   | 1763.3<br>1219.4<br>0                     | $5/2^+$<br>$1/2^+$<br>$3/2^+$                                | M1+E2            |                   | $I_\gamma$ : from 1970Ho09. Other: 100 (1969In04), 100 (1973Al22).<br>$I_\gamma$ : from 1970Ho09.<br>$I_\gamma$ : from 1970Ho09. Other: 10 (1969In04).<br>$I_\gamma$ : from 1970Ho09. Other: 100 (1969In04), 100 (1973Al22).<br>$\delta$ : $\delta=+0.17$ 8 for $J=3/2$ ; $\delta=-0.26$ 5 for $J=5/2$ (1970Ho09).                                                                                                                                                                                                                                       |
| 3002.6              | $(5/2)^+$      | 309.0<br>356.8<br>1239.3<br>1783.2<br>3002.5 15     | <5<br><1<br><2<br><2<br>100                   | 2693.6<br>2645.8<br>1763.3<br>1219.4<br>0 | $3/2^+, 5/2^+$<br>$7/2^+$<br>$5/2^+$<br>$1/2^+$<br>$3/2^+$   | M1+E2            | +0.09 3           | $I_\gamma$ : other: <4 (1972Br33).<br>$I_\gamma$ : from 1969In04, 1972Br33.<br>$E_\gamma$ : weighted average of 3003 2 (1969In04), 3003 2 (1972Br33), and 3002.0 15 (1973Al22).<br>$\delta$ : other: 1.2 5 if $J(3002.6)=3/2$ (1970Ho09).                                                                                                                                                                                                                                                                                                                |
| 3163.1              | $7/2^-$        | 160.5<br>469.5 &<br>517.4 15<br>1399.8<br>3162.6 15 | <0.54<br><0.54<br>11.1 33<br><1.1<br>100.0 33 | 3002.6<br>2693.6<br>2645.8<br>1763.3<br>0 | $(5/2)^+$<br>$3/2^+, 5/2^+$<br>$7/2^+$<br>$5/2^+$<br>$3/2^+$ | M2+E3            | -0.14 2           | $I_\gamma$ : from 1970Ho09. Other: 8.7 (1969In04), 11 (1973Al22).<br>$I_\gamma$ : from 1970Ho09. Other: 100 (1969In04), 100 (1973Al22).                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 3915                |                | 2696<br>3915                                        | 19 5<br>100 5                                 | 1219.4<br>0                               | $1/2^+$<br>$3/2^+$                                           |                  |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 3943.4              | $9/2$          | 1297.4 15<br>2180.2 15                              | 8.1 21<br>100.0 21                            | 2645.8<br>1763.3                          | $7/2^+$<br>$5/2^+$                                           |                  |                   | $I_\gamma$ : weighted average of 9.9 22 (1970Ho09) and 6.4 21 (1972Br33).<br>Other: 11 (1973Al22).<br>$I_\gamma$ : others: 100.0 22 (1970Ho09), 100 (1973Al22).                                                                                                                                                                                                                                                                                                                                                                                          |
| 3968                |                | 2749<br>3968                                        | 100<br><2                                     | 1219.4<br>0                               | $1/2^+$<br>$3/2^+$                                           |                  |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 4058                |                | 2295<br>2839<br>4058                                | 8 4<br>100 4<br><2.2                          | 1763.3<br>1219.4<br>0                     | $5/2^+$<br>$1/2^+$<br>$3/2^+$                                |                  |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 4108                | $3/2, 7/2$     | 2345<br>4108                                        | 16 5<br>100 5                                 | 1763.3<br>0                               | $5/2^+$<br>$3/2^+$                                           |                  |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 4347.8              | $9/2^-$        | 1184.6 10<br>1702.1 15                              | 100.0 31<br>52 7                              | 3163.1<br>2645.8                          | $7/2^-$<br>$7/2^+$                                           | M1+E2<br>E1(+M2) | -0.36 3<br>0.00 3 | $I_\gamma$ : from 1974Lo17. Others: 100 7 (1970Ho09) and 100 4 (1972Br33).<br>$\delta$ : from 1974Lo17. Other: -0.40 +8-9 (1973Al22); +0.42 4 (1970Ho09).<br>$A_2=-0.95$ 3, $A_4=+0.08$ 3 (1974Lo17).<br>$\text{POL}=-0.005$ 40 (1974Lo17).<br>$I_\gamma$ : unweighted average of 61 7 (1970Ho09), 39 4 (1972Br33), and 56.3 31 (1974Lo17).<br>Mult., $\delta$ : D(+Q) and $\delta$ from $p\gamma(\theta)$ and electric character from $\gamma(\text{pol})$ in 1974Lo17.<br>$A_2=-0.34$ 5, $A_4=+0.04$ 5 (1974Lo17).<br>$\text{POL}=+0.36$ 7 (1974Lo17). |

Continued on next page (footnotes at end of table)

$^{32}\text{S}(\alpha, p\gamma)$  [1970Ho09](#), [1972Br33](#), [1973Al22](#) (continued) $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$         | $E_\gamma^\dagger$ | $I_\gamma^\ddagger$ | $E_f$  | $J_f^\pi$                           | Mult. <sup>#</sup> | $\delta^\#$ | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------|-------------------|--------------------|---------------------|--------|-------------------------------------|--------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4618                | 3/2, 5/2          | 4618 2             | 100                 | 0      | 3/2 <sup>+</sup>                    |                    |             | $E_\gamma, I_\gamma$ : from <a href="#">1972Br33</a> .<br>$\delta$ : +0.08 14 for J=3/2; -0.36 15 for J=5/2 ( <a href="#">1970Ho09</a> ).                                                                                                                                                                                                                                                                                                                                                                                     |
| 4880.8              | 7/2               | 2187.1 @           | 52 @ 9              | 2693.6 | 3/2 <sup>+</sup> , 5/2 <sup>+</sup> |                    |             | $E_\gamma, I_\gamma$ : Final level is uncertain, either 2646 or 2694 ( <a href="#">1970Ho09</a> ).                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                     |                   | 2234.9 @           | 52 @ 9              | 2645.8 | 7/2 <sup>+</sup>                    |                    |             | $E_\gamma, I_\gamma$ : Final level is uncertain, either 2646 or 2694 ( <a href="#">1970Ho09</a> ).                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                     |                   | 3117.4 20          | 100 9               | 1763.3 | 5/2 <sup>+</sup>                    |                    |             | $I_\gamma$ : from <a href="#">1970Ho09</a> . Other: 100 ( <a href="#">1973Al22</a> ).                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5230                | 1/2, 3/2, 5/2     | 5230               | 100                 | 0      | 3/2 <sup>+</sup>                    |                    |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5407.1              | 11/2 <sup>-</sup> | 1059.3 10          | 9.9 22              | 4347.8 | 9/2 <sup>-</sup>                    | M1+E2              | +0.25 8     | Mult., $\delta$ : D+Q and $\delta$ from $\gamma(\theta)$ and magnetic character from $\gamma(\text{pol})$ ( <a href="#">1974Lo17</a> ).<br>$A_2=+0.19$ 11, $A_4=-0.01$ 13 ( <a href="#">1974Lo17</a> ).<br>POL=-0.40 14 ( <a href="#">1974Lo17</a> ).<br>Mult.: Q(+O) with $\delta(\text{O/Q})=0.00$ 3 from $\text{py}(\theta)$ in <a href="#">1974Lo17</a> ; M2 ruled out by RUL. Other: Q(+O) with $\delta(\text{O/Q})=+0.02$ 9 ( <a href="#">1973Al22</a> ).<br>$A_2=+0.44$ 3, $A_4=-0.19$ 3 ( <a href="#">1974Lo17</a> ). |
|                     |                   | 2244 3             | 100.0 22            | 3163.1 | 7/2 <sup>-</sup>                    | E2                 |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5850                | 5/2, 9/2          | 2687               | 100                 | 3163.1 | 7/2 <sup>-</sup>                    | D+Q                |             | Mult., $\delta$ : from $\text{py}(\theta)$ in <a href="#">1970Ho09</a> with $\delta(\text{Q/D})=-2.2$ 5 or -0.6 1 for J=5/2; +0.28 4 for J=9/2 ( <a href="#">1970Ho09</a> ).                                                                                                                                                                                                                                                                                                                                                  |
| 5927.1              | 7/2, 11/2         | 1579.3 15          | 100                 | 4347.8 | 9/2 <sup>-</sup>                    | D+Q                |             | Mult., $\delta$ : from $\gamma(\theta)$ in <a href="#">1973Al22</a> , with $\delta(\text{Q/D})=+1.1$ 5 for J=7/2; -0.8 4 for J=11/2.                                                                                                                                                                                                                                                                                                                                                                                          |
| 6087.0              | 13/2 <sup>-</sup> | 679.9 10           | 100 6               | 5407.1 | 11/2 <sup>-</sup>                   | M1+E2              | +0.020 15   | Mult., $\delta$ : D+Q and $\delta$ from $\text{py}(\theta)$ and magnetic character from $\gamma(\text{pol})$ in <a href="#">1974Lo17</a> . Other: $\delta=0.00$ 7 for $J_i=13/2$ ( <a href="#">1973Al22</a> ).<br>$A_2=-0.22$ 1, $A_4=0.005$ 15 ( <a href="#">1974Lo17</a> ).<br>POL=-0.45 5 ( <a href="#">1974Lo17</a> ).                                                                                                                                                                                                    |
|                     |                   | 1739.2             | 18 6                | 4347.8 | 9/2 <sup>-</sup>                    |                    |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

<sup>†</sup> Values with uncertainties are from [1973Al22](#) and those without uncertainties are from level-energy differences, unless otherwise noted. [1973Al22](#) is the only literature in this dataset that reported measured  $E_\gamma$  with uncertainties. [1969In04](#) and [1972Br33](#) reported  $E(\text{level})$  with uncertainties derived from their measured  $E_\gamma$  but did not report  $E_\gamma$ . Evaluators adopt the uncertainties of  $E(\text{level})$  from [1969In04](#) and [1972Br33](#) as the uncertainties of an  $E_\gamma$  if this  $\gamma$  transition populates the ground state and is also the dominant decay branch of that level.

<sup>‡</sup> From [1969In04](#) for  $\gamma$  rays from level up to 3163, from [1972Br33](#) for  $\gamma$  rays from level above 3163 up to 4618 level, and from [1974Lo17](#) for  $\gamma$  rays from level above 4618, unless otherwise noted.

<sup>#</sup> Mult=D+Q, Q or Q+O and  $\delta$  are determined from  $\text{py}(\theta)$  in [1970Ho09](#); magnetic or electric characters are determined based on RUL and measured level  $T_{1/2}$  where available, unless otherwise noted.

@ Multiply placed with undivided intensity.

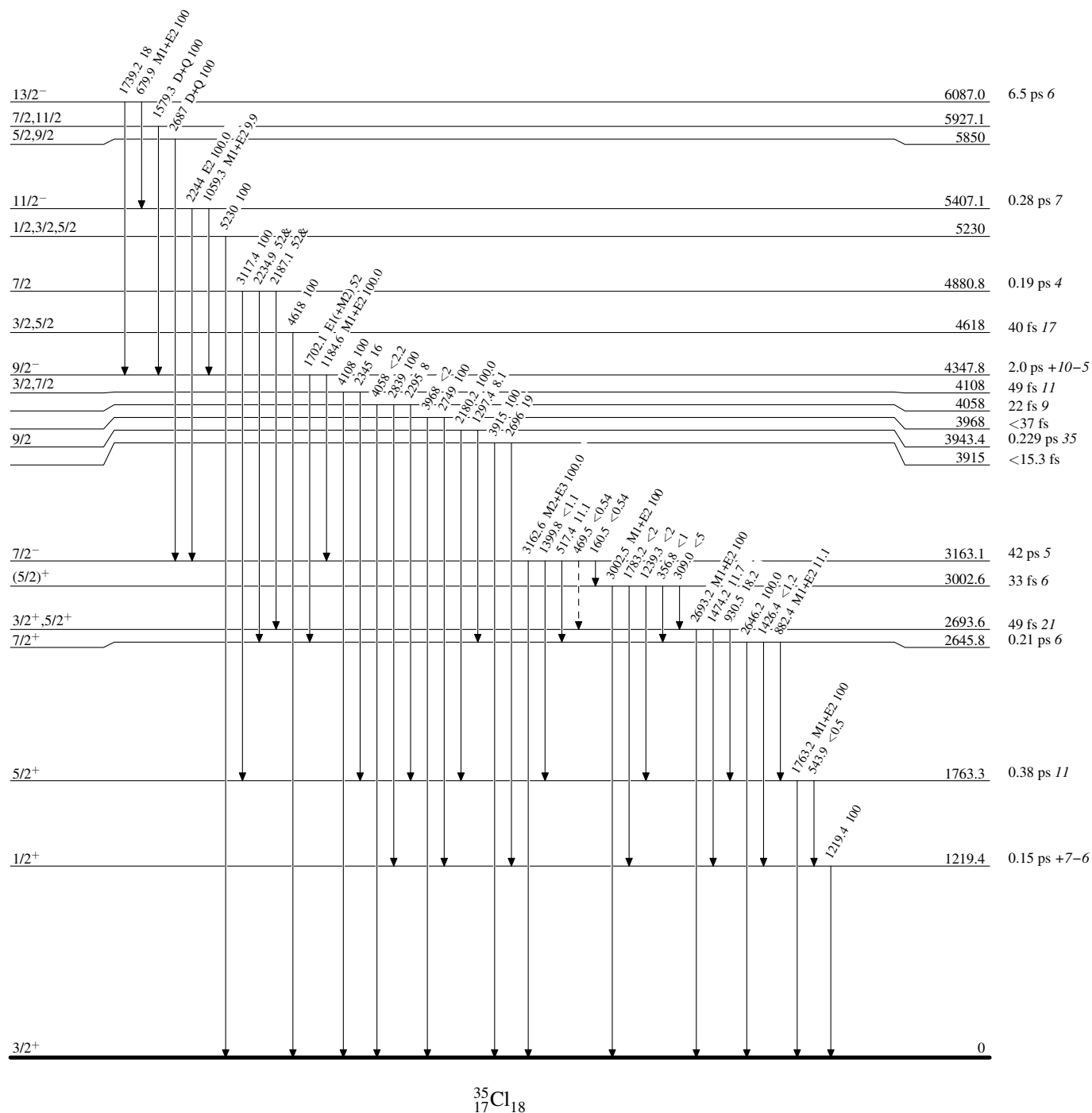
& Placement of transition in the level scheme is uncertain.

$^{32}\text{S}(\alpha, p\gamma)$  1970Ho09, 1972Br33, 1973Al22

Legend

## Level Scheme

Intensities: Relative photon branching from each level  
& Multiply placed: undivided intensity given

-----►  $\gamma$  Decay (Uncertain) $^{35}_{17}\text{Cl}_{18}$

$^{33}\text{S}(\alpha, \text{d})$     [1975Na18,1977Na10](#)

$J^\pi=3/2^+$  for  $^{33}\text{S}$  ground state.

[1975Na18,1977Na10](#): a 40-MeV  $\alpha$  beam was produced from the Michigan State University cyclotron. Targets were 76.8%-enriched  $^{33}\text{S}$  of  $15 \mu\text{g}/\text{cm}^2$  sandwiched between layers of Formvar and carbon foils. Reaction products were detected in the focal plane of an Enge split-pole magnetic spectrograph with a proportional-counter plastic-scintillator combination with FWHM=40-60 keV. Measured  $\sigma(E_d, \theta)$ . Deduced levels, J,  $\pi$ , and L-transfers.

 $^{35}\text{Cl}$  Levels

| <u>E(level)<sup>†</sup></u> | <u>L<sup>†</sup></u> | <u>Comments</u>                                                                                                                                                                                                                                                                            |
|-----------------------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6200 <i>10</i>              | 6                    |                                                                                                                                                                                                                                                                                            |
| 7170 <i>10</i>              | 6                    |                                                                                                                                                                                                                                                                                            |
| 7670 <i>10</i>              | 6                    |                                                                                                                                                                                                                                                                                            |
| 7750 <i>10</i>              | 6                    |                                                                                                                                                                                                                                                                                            |
| 7870 <i>10</i>              | 4+6                  | L: <a href="#">1977Na10</a> states that this is excited by a mixture of L=4 and 6 transfers. The L=6 admixture can be attributed to a $(f_{7/2})^2_{7,0}$ transfer whereas the L=4 component is associated with a mixture of $(f_{7/2})^2_{5,0}$ and $(f_{7/2}, p_{3/2})_{5,0}$ transfers. |
| 8010 <i>10</i>              | 6                    |                                                                                                                                                                                                                                                                                            |
| 8100 <i>10</i>              | 6                    |                                                                                                                                                                                                                                                                                            |
| 8700 <i>10</i>              | 6                    |                                                                                                                                                                                                                                                                                            |
| 8840 <i>10</i>              | 6                    |                                                                                                                                                                                                                                                                                            |
| 9150 <i>10</i>              | 6                    |                                                                                                                                                                                                                                                                                            |
| 9450 <i>10</i>              | 6                    |                                                                                                                                                                                                                                                                                            |

<sup>†</sup> From [1977Na10](#). Experimental angular distributions of L=4, 5, 6 to  $^{42}\text{Sc}$  and  $^{34}\text{Cl}$  levels with known  $J^\pi$  are compared with the  $\sigma(E_d, \theta)$  measured in [1977Na10](#) to assign L-transfers.

$^{34}\text{S}(\text{p},\gamma)$  1976Me12,1976Sp08,1972Hu10

$J^\pi=0^+$  for  $^{34}\text{S}$  ground state.

- 1976Me12:** 0.7-2.1-MeV proton beams were produced from the 1.1-MV Cockcroft-Walton generator and a 3-MV Van de Graaff accelerator. Targets were ZnS (46%  $^{34}\text{S}$ ) and CdS (37%  $^{34}\text{S}$ ) on tantalum backings.  $\gamma$  rays were detected using a 10-cm by 10-cm NaI detector and a 40-cm<sup>3</sup> Ge(Li) detector. Measured  $\sigma(E_p)$ ,  $E_\gamma$ ,  $\gamma(\theta)$ . Deduced level energies for 33 bound states and 47 resonances, J,  $\pi$ , resonance strengths,  $\gamma$  branching ratios, mixing ratios, and lifetimes for 25 bound states using DSAM.
- 1976Sp08:** 1.21, 1.95-2.91-MeV protons were produced from the Utrecht University 3-MV Van de Graaff accelerator. Targets were Ag<sup>2</sup>S (95%  $^{34}\text{S}$ ) evaporated onto a tantalum backing.  $\gamma$  rays were detected using a 10 cm by 10 cm diameter NaI scintillator and a 120- or 60-cm<sup>3</sup> Ge(Li) detector. Measured reaction yields,  $E_\gamma$ . Deduced Q value, 84 resonance energies, resonance strengths, widths, J,  $\pi$ , and  $\gamma$ -branching ratios.
- 1976Sp09:** similar setup as **1976Sp08** with 2.33-2.79-MeV protons. Measured reaction yields,  $E_\gamma$ ,  $\gamma(\theta)$ . Deduced levels, J,  $\pi$ , and  $\gamma$ -branching ratios.
- 1972Hu10:** 0.7-2.1-MeV protons were produced from the 4-MV Van de Graaff at the Centre d'Etudes Nucleaires de Bordeaux-Gradignan. Targets were 100- $\mu\text{g}/\text{cm}^2$  Ag<sub>2</sub>S (90%  $^{34}\text{S}$ ).  $\gamma$  rays were detected using a 12.7-cm by 12.7-cm NaI(Tl) detector and a 60-cm<sup>3</sup> Ge(Li) detector. Measured  $\sigma(E_p)$ ,  $E_\gamma$ ,  $I_\gamma$ . Deduced 59 resonances, 32 bound states, resonance strengths, and  $\gamma$  branching ratios.
- 1972Hu11:** similar setup as **1972Hu10**. Deduced lifetimes using DSAM.
- 1973Fa07:** 1-2-MeV protons were produced from the Helsinki 2-MV Van de Graaff accelerator. Targets were  $^{34}\text{S}$  on tantalum backings.  $\gamma$  rays were detected using a 55-cm<sup>3</sup> Ge(Li) detector. Measured reaction yields,  $E_\gamma$ ,  $I_\gamma$ . Deduced 28 resonances, resonance strengths,  $\gamma$ -branching ratios, and 9 lifetimes using DSAM.
- 2001Vo24:** 1.183-2.820-MeV protons were produced from the 4-MeV electrostatic Van de Graaff accelerator (NSC KhPEI). Enriched  $^{34}\text{S}$  target (100%). A 63 cm<sup>3</sup> Ge(Li) detector and a 15-cm-diam by 10-cm-long NaI(Tl) detector. Measured  $\sigma(E_p)$ ,  $E_\gamma$ ,  $I_\gamma$ . Deduced 12 resonances and resonance strengths.
- 2002Vo17:** 2.187-MeV protons were produced from the ESU-4 accelerator (NSC KhPEI). Enriched  $^{34}\text{S}$  target. A 63 cm<sup>3</sup> Ge(Li) detector. Measured  $E_\gamma$ ,  $I_\gamma$ . Deduced levels, resonance strengths, widths.
- 1981Bi05:** 1.4-2.8-MeV protons were produced from the Groningen 5-MV Van de Graaff accelerator. Targets were 4 and 1- $\mu\text{g}/\text{cm}^2$  Sb<sub>2</sub>S<sub>3</sub> (90% in  $^{34}\text{S}$ ) evaporated onto 10- $\mu\text{g}/\text{cm}^2$  carbon backings. Scattered protons were detected using three silicon surface barrier detectors.  $\gamma$  rays were detected using a 10-cm by 10-cm NaI(Tl) detector and a Ge(Li) detector. Measured  $\sigma(E_p, \theta)$ ,  $E_\gamma$ ,  $\gamma(\theta)$  for two  $\gamma$  rays. Deduced levels, J,  $\pi$ , widths, and resonance strengths.
- 1971Wi13:** protons were produced from the Ontario Cancer Institute. Targets were CdS.  $\gamma$  rays were detected using NaI(Tl) and Ge(Li) detectors. Measured  $\sigma(E_p)$ ,  $E_\gamma$ ,  $I_\gamma$ ,  $\gamma(\theta)$ ,  $\gamma$ -linear polarization. Deduced 1.020-, 1.214-, 1.267-, and 1.510-MeV resonances,  $J^\pi$ , branchings, mixing ratios, and half-lives using DSAM.
- 1971Pr11:** 1.211-MeV protons were produced from the 2-MeV ARL Van de Graaff at the Aerospace Research Laboratories (ARL). Targets were Ag<sub>2</sub>S (85.6%  $^{34}\text{S}$ ).  $\gamma$  rays were detected using an 80-cm<sup>3</sup> Ge(Li) detector. Measured  $\sigma(E_p)$ ,  $E_\gamma$ ,  $I_\gamma$ ,  $\gamma(\theta)$ ,  $p\gamma$ -delay. Deduced resonances, levels, J,  $\pi$ , branchings.
- 1963Ha32:** 0.7-1.2-MeV protons were produced from the Laboratorium Technische Fysika. Target was  $^{34}\text{S}$ .  $\gamma$  rays were detected using NaI(Tl) scintillation spectrometers. Measured  $\sigma(E_p)$ ,  $E_\gamma$ ,  $I_\gamma$ . Deduced levels, branchings.
- 1967Da12:** 0.8-2.5-MeV protons were produced from the 2.5 MeV Van de Graaff generator of the U.A.R. Atomic Energy Establishment. Targets were  $^{34}\text{S}$  on tantalum backings.  $\gamma$  rays were detected using two 15.2-cm-diam by 12.7-cm-thick NaI(Tl) crystals. Measured  $\sigma(E_p)$ ,  $E_\gamma$ ,  $I_\gamma$ . Deduced resonances, levels, branchings.
- 1960An06:** 0.7-1.9-MeV protons were produced from the electrostatic generator of the Physical-Technical Institute of the Ukrainian SSR Academy of Sciences. Targets were isotopic  $^{34}\text{S}$  (4.2%).  $\gamma$  rays were detected using a 2-cm-thick by 3-cm-diam CsI(Tl) crystal. Measured  $\sigma(E_p)$ ,  $E_\gamma$ ,  $\gamma(\theta)$ . Deduced levels, J, branchings.
- 1966Wa09:** 1.214 MeV proton beam. Measured  $E_\gamma$ ,  $\gamma$ -polarization. Deduced level energy and J,  $\pi$  for the level of 3163 and 7540 level.
- 1966En04:** 0.3-2.1-MeV protons were produced from the Utrecht 850-keV Cockcroft-Walton generator and the 3-MV Van de Graaff generator. Targets were natural sulphur compound on tantalum backings.  $\gamma$  rays were detected using a cylindrical 10-cm-by-10-cm NaI crystal. Measured  $\sigma(E_p)$ ,  $E_\gamma$ . Deduced the resonance strength for  $E_p=1214$  keV.
- 1966Az01:** 1.214-MeV protons were produced from the Ontario Cancer Institute 3-MeV Van de Graaff accelerator. Targets were CdS with enriched  $^{34}\text{S}$  (46%).  $\gamma$  rays were detected using a Compton polarimeter. Measured  $E_\gamma$ ,  $I_\gamma$ ,  $\gamma(\theta)$ . Deduced levels, J,  $\pi$ , branchings.
- 1967Wa22:** 1.214- and 1.512-MeV protons were produced from the Aerospace Research Laboratories. Targets were enriched CdS (40%  $^{34}\text{S}$ ) and Ag<sub>2</sub>S (86%  $^{34}\text{S}$ ).  $\gamma$  rays were detected using two large volume NaI(Tl) detectors. Measured  $\sigma(E_p)$ ,  $E_\gamma$ ,  $I_\gamma$ ,  $\gamma(\theta)$ ,  $\gamma$ -polarization. Deduced levels, J,  $\pi$ , branchings.



**$^{34}\text{S}(\text{p},\gamma)$  1976Me12,1976Sp08,1972Hu10 (continued)**

- 1967Ta08:** 1.020-, 1.213- and 1.513-MeV protons were produced from the 3-MV Van de Graaff accelerator of the Ontario Cancer Institute. Targets were CdS (46%  $^{34}\text{S}$ ) on thick metal foils.  $\gamma$  rays were detected using a 7.6-cm-long by 7.6-cm-diam NaI(Tl) detector. Measured  $\sigma(E_p)$ ,  $E_\gamma$ ,  $I_\gamma$ ,  $\gamma(\theta)$ ,  $\gamma$ -polarization. Deduced levels,  $J$ ,  $\pi$ , branchings, mixing ratios.
- 1977Ko35:** 1.055, 1.183, 1.214, and 1.416 MeV protons were produced from the 4 MV electrostatic accelerator of the Physical-Technical Institute of the Ukrainian SSR Academy of Sciences. Target of  $^{34}\text{S}$ . A 40 cm<sup>3</sup> coaxial Ge(Li) detector. Measured  $\sigma(E_p)$ ,  $E_\gamma$ ,  $\gamma(\theta)$ . Deduced levels, branchings, mixing ratios.
- 1977Ko34:** 1.0-1.4 MeV protons. Measured  $E_\gamma$ ,  $I_\gamma$ ,  $\gamma(\theta)$ . Deduced levels,  $J^\pi$ , branchings, mixing ratios.
- 1970Ko33:** 2.317-, 2.223-, and 2.336-MeV protons were produced from the 4-MV electrostatic accelerator of the Physical-Technical Institute of the Ukrainian SSR Academy of Sciences. Targets were  $^{34}\text{S}$  on tantalum backings.  $\gamma$  rays were detected using a 12-cm<sup>3</sup> coaxial Ge(Li) detector. Measured  $E_\gamma$ ,  $I_\gamma$ ,  $\gamma(\theta)$ . Deduced 3 levels,  $J$ , and branching ratios.
- 1967Ko19,1967Ko20,1967Ko23:** 2.079-MeV protons were produced from the 4 MV electrostatic accelerator of the Physical-Technical Institute of the Ukrainian SSR Academy of Sciences. Targets were made by implanting  $^{34}\text{S}$  ions in a tantalum substrate.  $\gamma$  rays were detected using a 70-mm by 50-mm and a 70-mm by 60-mm NaI(Tl) detectors. Measured  $E_\gamma$ ,  $I_\gamma$ ,  $p\gamma\gamma(\theta)$ . Deduced levels,  $J$ ,  $\pi$ , branchings, mixing ratios.
- 1968Az02:** 1.214- and 1.513-MeV protons were produced from the Ontario Cancer Institute 3-MeV Van de Graaff accelerator. Targets were CdS (37%  $^{34}\text{S}$ ) evaporated onto gold backings.  $\gamma$  rays were detected using a 5-cm-diam by 5-cm-long NaI(Tl) crystal. Measured  $E_\gamma$ ,  $\gamma\gamma$ -delay. Deduced lifetimes of the level of 3163 keV using the delayed coincidence method and DSAM.
- 1969Mi23:** 2.079-MeV protons.  $^{34}\text{S}$  target. Ge(Li) detector. Measured  $E_\gamma$ ,  $I_\gamma$ . Deduced levels, branchings.
- 1971Ko32,1971Ko46:** 0.8-2.6-MeV protons. Targets were made by evaporating  $^{34}\text{S}$  ions onto tantalum backings.  $\gamma$  rays were detected using Ge(Li) and NaI(Tl) detectors. Measured  $\sigma(E_p)$ ,  $E_\gamma$ ,  $I_\gamma$ , and  $p\gamma\gamma(\theta)$ . Deduced levels,  $J$ ,  $\pi$ , level widths, and branchings.
- 1971Ba23:** 1.214-MeV protons were produced from the 3 MeV Dynamitron of Carleton and Ottawa Universities. Target of CdS (93%  $^{34}\text{S}$ ) on a gold backing. A 28 cm<sup>3</sup> Ge(Li) detector. Measured  $E_\gamma$ ,  $I_\gamma$ . Deduced lifetime of the 3163 keV level using the delayed coincidence method.
- 1974Ai04:** 1.214-MeV protons. Targets were Ag<sub>2</sub>S, CdS, and ZnS of 300- $\mu\text{g}/\text{cm}^2$ . Ge(Li) detector. Measured  $\sigma(E_\gamma)$ . Deduced resonance strengths.
- 1975Ke11:** 1.214-MeV protons were produced from the Helsinki University 2.5-MV van de Graaff. Targets of CdS and ZnS on tantalum backings.  $\gamma$  rays were detected using a 120 cm<sup>3</sup> Ge(Li) detector with FWHM=2.9 keV at 2.6 MeV. Measured  $\sigma(E_\gamma)$ . Deduced resonance strengths.
- 1979Pa16:** 1.211-MeV protons were produced from the University of Melbourne 5U Pelletron accelerator. Natural sulphur target. NaI(Tl) and Ge(Li) detector. Measured  $\sigma(E_p)$ . Deduced resonance strength.
- 1988Va06:** 0.4-0.7 and 1.47 MeV protons were produced from the Utrecht 3 MV Van de Graaff accelerator. Targets of Ag<sub>2</sub>S (90%  $^{34}\text{S}$ ) of 10 and 34  $\mu\text{g}/\text{cm}^2$  on Ta backings. A 80-cm<sup>3</sup> coaxial Philips Ge(Li) or an  $\gamma$ -X intrinsic Ge detector, FWHM=1.85 and 1.75 keV at 1.33 MeV. Measured  $\sigma(E_p)$ ,  $E_\gamma$ ,  $I_\gamma$ ,  $\gamma(\theta)$ . Deduced levels,  $J^\pi$ , branchings, mixing ratios.
- 1988Co03:** 1.211-MeV protons were produced from the Geel 7-MV CN-type Van de Graaff. Targets of  $^{34}\text{S}$ . Ge(Li) detector. Measured  $\gamma\gamma$ -coin. Deduced weighting function.
- 1996Vo20:** 1.891, 1.900 and 2.070-MeV protons were produced by an electrostatic accelerator ESU-4 at the Physical and Technical Institute. Target made by implanting  $^{34}\text{S}$  ions in a tantalum backing. A 63 cm<sup>3</sup> Ge(Li) detector. Measured  $E_\gamma$ ,  $I_\gamma$ . Deduced levels, gamma widths.
- 1996Ka21:** 1-3 MeV protons were produced from the electrostatic accelerator of the Institute of Physica and Technology (Kharkov). Targets were 20- $\mu\text{g}/\text{cm}^2$  Ag<sub>2</sub>S.  $\gamma$  rays were detected using a 60 cm<sup>3</sup> Ge(Li) detector and a 150×100-mm<sup>2</sup> NaI(Tl) as a monitor. Measured  $E_\gamma$ ,  $I_\gamma$ ,  $\gamma(\theta)$ . Deduced  $J^\pi$  for 3 resonances, resonance strengths.
- 2000Vo21:** 1.794 and 2.275 MeV protons. Measured  $E_\gamma$ ,  $I_\gamma$ . Deduced gamma widths, transition strengths.
- 2001Ka69:** 0.8-3 MeV protons produced from the ESU-4 accelerator (NSC KhPEI). Target of  $^{34}\text{S}$ . A 15-cm-diam by 19-cm-long NaI(Tl) detector. Measured  $E_\gamma$ ,  $I_\gamma$ . Deduced levels, transition strengths.
- Others: **1962Du05**, **1963Du08**, **1964Hy01**, **1966Ko22**, **1966Ko15**, **1967Ma48**, **1969Ko26**, **1969Ah02**, **1969KrZW**, **1972LeYW**, **1974Ii01**, **1974Bi16**, **1981Va02**.
- Theoretical studies: **2020Ri06** (shell model).
- Standard resonance strength  $\omega\gamma=(2J_r+1)/[(2J_p+1)(2J_T+1)]\times\Gamma_\gamma\Gamma_p/\Gamma$ , where  $J_r$  is the spin of the resonance,  $J_p=1/2$  is the spin of the proton, and  $J_T=0$  is the spin of the target  $^{34}\text{S}$  ground state. The resonance strengths  $S$  from the original papers in this dataset are mainly obtained from resonance  $\gamma$ -ray yields and are defined without the  $(2J_p+1)$  factor. Therefore, to obtain the standard resonance strength, all the resonance strengths provided in the comments below should be divided by 2.
- $S=21$  eV at  $E_p=1213$  measured by **1966En04** has been used as a normalization standard by **1971BrXT**, **1972Hu10**, **1973Fa07**,

<sup>34</sup>S(p,γ) 1976Me12,1976Sp08,1972Hu10 (continued)

1974Bi16, and 1976Me12. 1971BrXT measured <sup>34</sup>S(p,p) elastic scattering and determined  $\Gamma_p=4.6$  eV 10 and  $\Gamma=5$  eV +4-2 from a  $Q^2$  minimization analysis of the excitation function. 1971BrXT added  $\Gamma_p\Gamma_\gamma/\Gamma=2.6$  eV 4 converted from the S measured by 1966En04 to constrain their  $Q^2$  analysis and reported a final  $\Gamma_p=5.8$  eV 7,  $\Gamma_\gamma=4.7$  eV 17, and  $\Gamma=10.5$  eV 15. 1974Bi16 measured <sup>34</sup>S(p,p) and determined  $\Gamma_p=7.2$  eV 15 from a  $Q^2$  minimization analysis of the excitation function and quoted  $S=21$  eV 3 (1966En04) to deduce  $\Gamma_\gamma=3.8$  eV 15 and  $\Gamma=11.0$  eV 15. 1974AI04 determined an absolute  $S_{1213}=9.7$  eV 9. 1975Ke11 determined an absolute  $S_{1213}=9.8$  eV 10 and quoted 1974Bi16's relative ratio of  $\Gamma_p/\Gamma_\gamma=7.2/3.9=1.9$  to deduce  $\Gamma_p=3.8$  eV 13,  $\Gamma_\gamma=2.0$  eV 13, and  $\Gamma=5.8$  eV 13. 1976Sp09 measured the resonant absorption of the  $E_p=2791$  resonance, which indicates that normalization using  $S_{1213}=9.7$  eV gives a better agreement than that using  $S_{1213}=21$  eV. 1981Bi05 determined an absolute  $S=4.1$  eV 5 for the  $E_p=1892$  resonance and quoted the relative ratio of  $S_{1213}/S_{1892}$  from 1978En02 to deduce  $S=9.5$  eV 12 for  $E_p=1213$  resonance. The  $S_{1213}/S_{1892}=2.3$  quoted by 1981Bi05 was based on two S values: first, 1978En02 quoted  $S=4.2$  eV for a 1892+1893 resonance doublet, which was scaled down by a factor of 21/9.7 from 1976Me12's original  $S_{1892}=9.0$  eV; second, 1978En02 quoted  $S_{1213}=9.7$  eV 7 from 1976Sp08, 1974AI04, and 1975Ke11. 1979Pa16 determined  $S=9.0$  eV 10 using the <sup>27</sup>Al(p,γ)<sup>28</sup>Si resonances as normalization standards. 1996Ka21 normalized all their S to  $S_{1213}=9.7$  eV 7 and cited 1976Sp08 but listed this value in the "this study" column. 2001Vo24 normalized all S to  $S_{1213}=9.7$  eV 7 and cited 1976Sp08 and 1975Ke11. However, 2001Vo24 also reported a different  $S_{1213}=9.9$  eV 7 without giving details. Although 1976Sp08 is frequently cited as the source of  $S_{1213}=9.7$  eV 7, they did not measure the absolute S value; instead, they calculated a weighted average of 1974AI04 and 1975Ke11.

In summary, the evaluators adopt  $S_{1213}=9.5$  eV 9: weighted average of all independently measured  $S_{1213}$ : 9.7 eV 9 (1974AI04), 9.8 eV 10 (1975Ke11), and 9.0 eV 10 (1979Pa16). All resonance strengths given in the comments below have been normalized to this value by the evaluators.  $\Gamma_p=4.6$  eV 10 and  $\Gamma=5$  eV +4-2 from (p,p) excitation function analysis in 1971BrXT are likely the only width data that are not influenced by the questionable  $S_{1213}=21$  eV 3.

<sup>35</sup>Cl Levels

| E(level) <sup>†</sup> | J <sup>π</sup> <sup>‡</sup> | T <sub>1/2</sub> | Comments                                                                                                                                                                                                                                                                                                                                      |
|-----------------------|-----------------------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0                     | 3/2 <sup>+</sup>            |                  |                                                                                                                                                                                                                                                                                                                                               |
| 1219.39 7             | 1/2 <sup>+</sup>            | 105 fs 14        | E(level): weighted average of 1219.1 5 (1972Hu10) and 1219.3 1 (1976Me12), and 1219.44 7 (1981Bi05).<br>T <sub>1/2</sub> : lifetime $\tau=152$ fs 20: weighted average of 80 fs 40 (1971Wi13, DSAM), 175 fs 20 (1972Hu11, DSAM), 150 fs 20 (1973Fa07, DSAM), and 145 fs 30 (1976Me12, DSAM).                                                  |
| 1763.14 4             | 5/2 <sup>+</sup>            | 0.347 ps 63      | E(level): weighted average of 1762.8 5 (1972Hu10), 1763.4 7 (1976Me12), and 1763.14 4 (1981Bi05).<br>J <sup>π</sup> : from $\gamma(\theta)$ in 1971Wi13.<br>T <sub>1/2</sub> : lifetime $\tau=0.50$ ps 9: weighted average of 0.50 ps 10 (1972Hu11, DSAM), 0.40 ps 14 (1973Fa07, DSAM), and 0.53 ps 9 (1976Me12, DSAM).                       |
| 2645.70 8             | (5/2,7/2)                   | 0.146 ps 21      | E(level): weighted average of 2645.6 5 (1972Hu10), 2644.7 13 (1976Me12), and 2645.71 8 (1981Bi05).<br>J <sup>π</sup> : from $\gamma(\theta)$ in 1971Wi13. Other: 3/2 from $\gamma(\theta)$ in 1967Ko20.<br>T <sub>1/2</sub> : lifetime $\tau=0.210$ ps 30: weighted average of 0.255 ps 65 (1972Hu11, DSAM) and 0.200 ps 30 (1976Me12, DSAM). |
| 2693.9 3              | (3/2,5/2)                   | 13.9 fs 21       | E(level): weighted average of 2694.0 3 (1972Hu10), 2694.7 12 (1976Me12), and 2693.8 3 (1981Bi05).<br>J <sup>π</sup> : from $\gamma(\theta)$ in 1971Wi13.<br>T <sub>1/2</sub> : lifetime $\tau=20$ fs 3: weighted average of 21 fs 3 (1972Hu11, DSAM), 19 fs 4 (1973Fa07, DSAM), and 20 fs 4 (1976Me12, DSAM).                                 |
| 3002.75 8             | 5/2                         | 11.1 fs 21       | E(level): weighted average of 3002.7 7 (1972Hu10), 3003.7 8 (1976Me12), and 3002.74 8 (1981Bi05).<br>J <sup>π</sup> : 5/2 from $\gamma(\theta)$ in 1967Ko20 and 1971Wi13.<br>T <sub>1/2</sub> : lifetime $\tau=16$ fs 3: weighted average of 22 fs 3 (1972Hu11, DSAM), 14 fs 2 (1973Fa07, DSAM), and 16 fs 4 (1976Me12, DSAM).                |
| 3162.99 7             | 7/2 <sup>-</sup>            | 25.7 ps 28       | T=1/2<br>E(level): weighted average of 3162.5 5 (1972Hu10), 3163.9 7 (1976Me12), and 3162.99 6 (1981Bi05).<br>J <sup>π</sup> : spin from $\gamma(\theta)$ in 1966Wa09, 1967Wa22, 1967Ta08, and 1971Wi13; parity from 3163γ, M2+E3 to 3/2 <sup>+</sup> ground state in 1966Wa09 and 1967Wa22.                                                  |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $^{35}\text{Cl}$  Levels (continued)

| E(level) <sup>†</sup> | J <sup>π</sup> <sup>‡</sup> | T <sub>1/2</sub> | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------|-----------------------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       |                             |                  | T <sub>1/2</sub> : lifetime $\tau=37$ ps 4 (1971Ba23, delayed coincidence). Others: 140 ps 40 (1968Az02, delayed coincidence), >0.6 ps (1968Az02, DSAM), >0.2 ps (1971Wi13, DSAM), >2 ps (1972Hu11, DSAM), >0.9 ps (1973Fa07, DSAM), and >1 ps (1976Me12). 1971Ba23 states that the earlier 1968Az02 value was obtained using NaI(Tl) detectors with broad time resolution and the discrepancy between 1968Az02 and 1971Ba23 is not unreasonable. |
| 3918.48 10            | 3/2 <sup>+</sup>            | 4.9 fs 14        | E(level): weighted average of 3917.9 6 (1972Hu10), 3920.7 13 (1976Me12), and 3918.48 7 (1981Bi05).<br>J <sup>π</sup> : spin=3/2 determined by the feeding 5162 $\gamma$ ( $\theta$ ) from 5/2 <sup>+</sup> , 9081 level in 1976Sp09; parity=+ determined by 3918 $\gamma$ , M1+E2 to 3/2 <sup>+</sup> ground state.                                                                                                                               |
| 3942.9 2              |                             | 0.159 ps 42      | T <sub>1/2</sub> : lifetime $\tau=7$ fs 2 (1972Hu11, DSAM). Other: <6 fs (1976Me12, DSAM).<br>E(level): weighted average of 3942.6 9 (1972Hu10), 3944.1 11 (1976Me12), and 3942.9 2 (1981Bi05).                                                                                                                                                                                                                                                   |
| 3967.3 6              |                             | 11.8 fs 28       | T <sub>1/2</sub> : lifetime $\tau=0.23$ ps 6: weighted average of 0.39 ps 11 (1972Hu11, DSAM), 0.145 ps 50 (1973Fa07, DSAM), and 0.29 ps 5 (1976Me12, DSAM).<br>E(level): from 1972Hu10. Other: 3979.0 15 (1976Me12). 1976Me12 matched their 3979 level with the 3967 level observed in other work. No other work has observed a level at 3979 keV.                                                                                               |
| 4059.2 4              |                             | 13.9 fs 21       | T <sub>1/2</sub> : lifetime $\tau=17$ fs 4: weighted average of 13 fs 4 (1972Hu11, DSAM), 20 fs 5 (1973Fa07, DSAM), and 20 fs 4 (1976Me12, DSAM).<br>E(level): weighted average of 4058.4 8 (1972Hu10), 4059.4 4 (1976Me12), and 4059.2 3 (1981Bi05).                                                                                                                                                                                             |
| 4113.7 10             | 7/2                         |                  | T <sub>1/2</sub> : lifetime $\tau=20$ fs 3: weighted average of 22 fs 3 (1972Hu11, DSAM), 20 fs 4 (1973Fa07, DSAM), and 18 fs 3 (1976Me12, DSAM).<br>E(level): weighted average of 4111 3 (1972Hu10) and 4114.0 10 (1976Me12).<br>J <sup>π</sup> : spin=7/2 from feeding 4516 $\gamma$ ( $\theta$ ) from 7/2, 8630 level in 1976Sp09.                                                                                                             |
| 4173.45 19            |                             | 34.7 fs 63       | T=1/2<br>E(level): weighted average of 4173.0 6 (1972Hu10), 4174.7 10 (1976Me12), and 4173.45 19 (1981Bi05).<br>T <sub>1/2</sub> : lifetime $\tau=50$ fs 9: weighted average of 105 fs +35–30 (1971Wi13, DSAM), 55 fs 6 (1972Hu11, DSAM), and 34 fs 9 (1976Me12, DSAM).                                                                                                                                                                           |
| 4177.9 2              | 3/2                         | 27.0 fs 28       | E(level): weighted average of 4177.2 9 (1972Hu10), 4180.1 15 (1976Me12), and 4177.9 2 (1981Bi05).<br>J <sup>π</sup> : from $\gamma$ ( $\theta$ ) in 1967Wa22.                                                                                                                                                                                                                                                                                     |
| 4347.76 6             | 5/2,9/2                     | 0.229 ps 42      | T <sub>1/2</sub> : lifetime $\tau=39$ fs 4: weighted average of 32 fs 5 (1972Hu11, DSAM), 42 fs 4 (1973Fa07, DSAM), and 47 fs 10 (1976Me12, DSAM).<br>E(level): weighted average of 4347.5 8 (1972Hu10), 4347.2 12 (1976Me12), and 4347.76 6 (1981Bi05).<br>J <sup>π</sup> : spin=5/2,9/2 based on feeding 4282 $\gamma$ ( $\theta$ ) from 7/2, 8630 level in 1976Sp09.                                                                           |
| 4624.4 3              |                             |                  | T <sub>1/2</sub> : lifetime $\tau=0.33$ ps 6 (1976Me12, DSAM). Other: >0.6 ps (1972Hu11, DSAM).<br>E(level): weighted average of 4624.2 20 (1976Me12) and 4624.4 3 (1981Bi05). Other: 4625 5 (1972Hu10).<br>15% of I $\gamma$ from this level decays to unknown final levels (1976Me12).                                                                                                                                                          |
| 4769.86 23            |                             | 76 fs 20         | E(level): weighted average of 4768.9 12 (1972Hu10), 4766.9 15 (1976Me12), and 4769.90 15 (1981Bi05).<br>T <sub>1/2</sub> : lifetime $\tau=110$ fs 29: weighted average of 270 fs 95 (1972Hu11, DSAM) and 105 fs 17 (1976Me12, DSAM).                                                                                                                                                                                                              |
| 4839.10 11            |                             | 9.7 fs 35        | E(level): weighted average of 4838 3 (1972Hu10), 4841.7 19 (1976Me12), and 4839.09 11 (1981Bi05).<br>T <sub>1/2</sub> : lifetime $\tau=14$ fs 5 (1976Me12, DSAM).                                                                                                                                                                                                                                                                                 |
| 4854.4 19             |                             | 4.9 fs 14        | E(level): weighted average of 4853 2 (1972Hu10) and 4855.7 19 (1976Me12).<br>T <sub>1/2</sub> : lifetime $\tau=7$ fs 2 (1976Me12, DSAM). Other: <6 fs (1972Hu11, DSAM).                                                                                                                                                                                                                                                                           |
| 4880.96 15            |                             | 5.6 fs 14        | E(level): weighted average of 4880.9 1 (1972Hu10), 4885 2 (1976Me12), and 4881.14 20 (1981Bi05).<br>T <sub>1/2</sub> : lifetime $\tau=8$ fs 2: weighted average of 7 fs 2 (1972Hu11, DSAM), 7 fs 3 (1973Fa07, DSAM), and 10 fs 3 (1976Me12, DSAM).                                                                                                                                                                                                |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $^{35}\text{Cl}$  Levels (continued)

| E(level) <sup>†</sup> | J <sup>π</sup> <sup>‡</sup> | T <sub>1/2</sub> | E <sub>p</sub> (lab) (keV) | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------|-----------------------------|------------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5010.1 18             |                             | 7.6 fs 21        |                            | E(level): weighted average of 5008 5 (1972Hu10) and 5010.4 18 (1976Me12).<br>T <sub>1/2</sub> : lifetime $\tau=11$ fs 3 (1976Me12, DSAM).                                                                                                                                                                                                                                                                                                                                        |
| 5163.34 22            | 7/2                         |                  |                            | E(level): weighted average of 5163 5 (1972Hu10), 5166.7 15 (1976Me12), and 5163.31 14 (1981Bi05).<br>J <sup>π</sup> : spin=3/2, 7/2 from feeding 3466 $\gamma$ ( $\theta$ ) from 7/2, 8630 level in 1976Sp09. Spin=3/2 and $\delta(\text{O}/\text{Q})=0.12$ 1 would require a large E3 or M3 component, which is ruled out by RUL.<br>10% of I $\gamma$ transitions to unknown levels (1976Me12).                                                                                |
| 5215.8 15             |                             | <4.85 fs         |                            | E(level): weighted average of 5214 3 (1972Hu10) and 5216.2 15 (1976Me12).<br>T <sub>1/2</sub> : lifetime $\tau<7$ fs (1976Me12, DSAM).                                                                                                                                                                                                                                                                                                                                           |
| 5403.5 10             |                             | 11.8 fs 28       |                            | E(level): weighted average of 5400 5 (1972Hu10) and 5403.6 10 (1976Me12).<br>T <sub>1/2</sub> : lifetime $\tau=17$ fs 4 (1976Me12, DSAM).<br>20% of I $\gamma$ from this level decays (1976Me12). 50% of I $\gamma$ from this level decays to unknown levels (1972Hu10).                                                                                                                                                                                                         |
| 5520                  |                             |                  |                            | E(level): from 1976Sp08.                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5586 2                |                             |                  |                            | E(level): weighted average of 5587 3 (1972Hu10) and 5586 2 (1976Me12).                                                                                                                                                                                                                                                                                                                                                                                                           |
| 5599.7 15             |                             | 2.08 fs 69       |                            | E(level): weighted average of 5598 3 (1972Hu10) and 5600.1 15 (1976Me12).<br>T <sub>1/2</sub> : lifetime $\tau=3$ fs 1 (1976Me12, DSAM).                                                                                                                                                                                                                                                                                                                                         |
| 5646 2                |                             | 2.77 fs 69       |                            | E(level): weighted average of 5645 3 (1972Hu10) and 5646 2 (1976Me12).<br>T <sub>1/2</sub> : lifetime $\tau=4$ fs 1 (1976Me12, DSAM).                                                                                                                                                                                                                                                                                                                                            |
| 5655 2                |                             | 13.9 fs 35       |                            | E(level): weighted average of 5651 3 (1972Hu10) and 5656 2 (1976Me12).<br>J <sup>π</sup> : Mirror level of $^{35}\text{S}$ ground state and thus 3/2 <sup>+</sup> (2001Vo24). Other: spin=5/2 based on the feeding 2051 $\gamma$ ( $\theta$ ) from the 5/2, 7706 level in 1977Ko34.<br>T <sub>1/2</sub> : lifetime $\tau=20$ fs 5 (1976Me12, DSAM).<br>$\Gamma_{\gamma}=0.033$ eV 10 (2001Vo24).<br>14% of I $\gamma$ from this level decays to unknown final levels (1976Me12). |
| 5683 2                |                             |                  |                            | E(level): weighted average of 5682 5 (1972Hu10) and 5683 2 (1976Me12).<br>40% of I $\gamma$ from this level decays to unknown final levels (1976Me12).                                                                                                                                                                                                                                                                                                                           |
| 5723.52 18            |                             |                  |                            | E(level): from 1981Bi05. 38% of I $\gamma$ from this level decays to unknown final levels (1976Sp09).                                                                                                                                                                                                                                                                                                                                                                            |
| 5758 3                |                             |                  |                            | E(level): weighted average of 5754 5 (1972Hu10) and 5759 3 (1976Me12).<br>25% of $\gamma$ from this level decays to unknown final levels and the other 75% is attributed to two $\gamma$ transitions (1976Me12). Other: 100% of $\gamma$ from this level decays to unknown final levels (1972Hu10).                                                                                                                                                                              |
| 5806 2                |                             | 3.47 fs 69       |                            | E(level): weighted average of 5802 5 (1972Hu10) and 5806 2 (1976Me12).<br>T <sub>1/2</sub> : lifetime $\tau=5$ fs 1 (1976Me12, DSAM).                                                                                                                                                                                                                                                                                                                                            |
| 6106.2 21             |                             | 8.3 fs 21        |                            | E(level): weighted average of 6102 3 (1972Hu10) and 6107.2 15 (1976Me12).<br>T <sub>1/2</sub> : lifetime $\tau=12$ fs 3 (1976Me12, DSAM).                                                                                                                                                                                                                                                                                                                                        |
| 6181 3                |                             |                  |                            | E(level): from 1976Me12.<br>55% of I $\gamma$ from this level decays to unknown final levels (1976Me12).                                                                                                                                                                                                                                                                                                                                                                         |
| 6492 3                |                             |                  |                            | E(level): weighted average of 6489 5 (1972Hu10) and 6493 3 (1976Me12).                                                                                                                                                                                                                                                                                                                                                                                                           |
| 6866.7 6              |                             |                  | 510.6 6                    | E <sub>p</sub> (lab) (keV): from 1988Va06.                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 7066.2 7              |                             |                  | 716.0 7                    | E <sub>p</sub> (lab) (keV): weighted average of 716.0 7 (1972Hu10) and 716.0 10 (1976Me12).<br>$\omega\gamma=0.09$ eV 5 (1972Hu10).<br>$\omega\gamma=0.14$ eV 4 (1976Me12).                                                                                                                                                                                                                                                                                                      |
| 7103.4 7              | 3/2                         |                  | 754.3 7                    | J <sup>π</sup> : from $\gamma$ ( $\theta$ ) in 1976Me12.<br>E <sub>p</sub> (lab) (keV): weighted average of 754.1 7 (1972Hu10) and 754.6 10 (1976Me12).                                                                                                                                                                                                                                                                                                                          |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $^{35}\text{Cl}$  Levels (continued)

| E(level) <sup>†</sup> | $J^{\pi}$ <sup>‡</sup> | $E_{\text{p}}(\text{lab})$ (keV) | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------|------------------------|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7178.6 8              | 1/2                    | 831.8 8                          | $\omega\gamma=0.04$ eV (1963Ha32).<br>$\omega\gamma=0.23$ eV 11 (1972Hu10).<br>$\omega\gamma=0.23$ eV 7 (1976Me12).<br>$J^{\pi}$ : from $\gamma(\theta)$ in 1976Me12.<br>$E_{\text{p}}(\text{lab})$ (keV): weighted average of 831.5 8 (1972Hu10), 832.2 15 (1976Me12), and 832 1 (2001Vo24).<br>$\omega\gamma=0.13$ eV 2 (2001Vo24).<br>$\Gamma_{\gamma} \geq 0.07$ eV (2001Vo24).<br>$\omega\gamma=0.18$ eV 9 (1972Hu10).<br>$\omega\gamma=0.14$ eV 4 (1976Me12). |
| 7185                  |                        | 838                              | $E_{\text{p}}(\text{lab})$ (keV): from 1963Ha32.<br>$\omega\gamma=0.056$ eV (1963Ha32).                                                                                                                                                                                                                                                                                                                                                                             |
| 7194.6 7              | 1/2,3/2                | 848.2 7                          | $J^{\pi}$ : from $\gamma(\theta)$ in 1960An06.<br>$E_{\text{p}}(\text{lab})$ (keV): weighted average of 848.2 7 (1972Hu10) and 848.1 10 (1976Me12).<br>$\omega\gamma=0.17$ eV (1963Ha32).<br>$\omega\gamma=0.50$ eV 15 (1972Hu10).<br>$\omega\gamma=0.54$ eV 16 (1976Me12).                                                                                                                                                                                         |
| 7225.5 8              | 5/2                    | 880.0 8                          | $J^{\pi}$ : from $\gamma(\theta)$ in 1976Me12.<br>$E_{\text{p}}(\text{lab})$ (keV): weighted average of 880.5 8 (1972Hu10) and 879.3 10 (1976Me12).<br>$\omega\gamma=0.073$ eV (1963Ha32).<br>$\omega\gamma=0.18$ eV 9 (1972Hu10).<br>$\omega\gamma=0.14$ eV 4 (1976Me12).                                                                                                                                                                                          |
| 7234.0 8              | 5/2 <sup>+</sup>       | 888.8 8                          | $J^{\pi}$ : spin=5/2 from $\gamma(\theta)$ in 1960An06 and 6014 $\gamma$ to 1/2 <sup>+</sup> , 1219 level in 1972Hu10 and 1976Me12.<br>$E_{\text{p}}(\text{lab})$ (keV): weighted average of 889.0 8 (1972Hu10) and 888.5 10 (1976Me12).<br>$\omega\gamma=0.23$ eV (1963Ha32).<br>$\omega\gamma=0.59$ eV 18 (1972Hu10).<br>$\omega\gamma=0.86$ eV 26 (1976Me12).                                                                                                    |
| 7269.2 1              |                        | 925.1 1                          | $E_{\text{p}}(\text{lab})$ (keV): from 1976Sp08.                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 7272.7 9              | (1/2,3/2)              | 928.6 9                          | $J^{\pi}$ : from $\gamma(\theta)$ in 1960An06.<br>$E_{\text{p}}(\text{lab})$ (keV): weighted average of 928.2 9 (1972Hu10) and 929.0 10 (1976Me12).<br>$\omega\gamma=0.31$ eV (1963Ha32).<br>$\omega\gamma=0.63$ eV 19 (1972Hu10).<br>$\omega\gamma=1.00$ eV 30 (1976Me12).                                                                                                                                                                                         |
| 7362.0 8              | 3/2                    | 1020.6 8                         | $J^{\pi}$ : from $\gamma(\theta)$ in 1971Wi13 and 1977Ko34.<br>$E_{\text{p}}(\text{lab})$ (keV): weighted average of 1020.4 8 (1972Hu10), 1020.5 11 (1973Fa07), and 1021.0 10 (1976Me12).<br>$\omega\gamma=0.48$ eV (1963Ha32).<br>$\omega\gamma=1.4$ eV 4 (1972Hu10).<br>$\omega\gamma=1.13$ eV (1973Fa07).<br>$\omega\gamma=1.4$ eV 4 (1976Me12).                                                                                                                 |
| 7396.0 10             | 7/2                    | 1055.6 10                        | $J^{\pi}$ : from $\gamma(\theta)$ in 1977Ko35.<br>$E_{\text{p}}(\text{lab})$ (keV): weighted average of 1054.9 10 (1972Hu10), 1055.8 12 (1973Fa07), and 1056.9 16 (1976Me12).<br>$\omega\gamma=0.67$ eV (1963Ha32).<br>$\omega\gamma=0.18$ eV 9 (1972Hu10).<br>$\omega\gamma=0.18$ eV (1973Fa07).<br>$\omega\gamma=0.23$ eV 7 (1976Me12).                                                                                                                           |
| 7451.1 6              | 3/2                    | 1112.3 6                         | $J^{\pi}$ : from $\gamma(\theta)$ in 1972LeYW.<br>$E_{\text{p}}(\text{lab})$ (keV): weighted average of 1112.2 10 (1972Hu10), 1112.8 13 (1973Fa07), and 1112.2 6 (1976Me12).<br>$\omega\gamma=0.14$ eV 7 (1972Hu10).<br>$\omega\gamma=0.14$ eV (1973Fa07).<br>$\omega\gamma=0.18$ eV 5 (1976Me12).                                                                                                                                                                  |
| 7496?                 |                        | 1158                             | E(level): from 1960An06.                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 7501.1 8              |                        | 1163.8 8                         | $E_{\text{p}}(\text{lab})$ (keV): from 1973Fa07. Other: 1165 (1972Hu10).                                                                                                                                                                                                                                                                                                                                                                                            |

Continued on next page (footnotes at end of table)



$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $^{35}\text{Cl}$  Levels (continued)

| E(level) <sup>†</sup> | J <sup>π</sup> <sup>‡</sup> | T <sub>1/2</sub> | E <sub>p</sub> (lab) (keV) | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------|-----------------------------|------------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7502.9 7              |                             |                  | 1165.7 7                   | $\omega\gamma=0.05$ eV (1973Fa07).<br>$\omega\gamma=0.27$ eV 14 for E <sub>p</sub> =1165+1166 doublet (1972Hu10).<br>E <sub>p</sub> (lab) (keV): from 1973Fa07. Other: 1165.4 7 (1976Me12), 1166 (1972Hu10).<br>$\omega\gamma=0.27$ eV 14 for E <sub>p</sub> =1165+1166 doublet (1972Hu10).<br>$\omega\gamma=0.23$ eV (1973Fa07).<br>$\omega\gamma=0.36$ eV 11 (1976Me12), possible doublet.<br>J <sup>π</sup> : from $\gamma(\theta)$ in 1977Ko35.<br>E <sub>p</sub> (lab) (keV): weighted average of 1183.2 11 (1972Hu10), 1182.2 12 (1973Fa07), and 1181.4 7 (1976Me12).<br>$\omega\gamma=0.09$ eV 5 (1972Hu10).<br>$\omega\gamma=0.14$ eV (1973Fa07).<br>$\omega\gamma=0.18$ eV 5 (1976Me12).<br>T=3/2<br>J <sup>π</sup> : spin from $\gamma(\theta)$ in 1966Wa09, 1967Wa22, and 1977Ko35; parity from 4386 $\gamma$ , M1+E2 to 7/2 <sup>-</sup> , 3163 level in 1966Wa09 and 1967Wa22.<br>T <sub>1/2</sub> : lifetime $\tau < 1$ fs (1973Fa07, DSAM). Others: $\tau < 13$ fs (1971Wi13, DSAM).<br>E <sub>p</sub> (lab) (keV): weighted average of 1213.7 10 (1972Hu10), 1213.6 6 (1973Fa07), 1212.4 7 (1976Me12), and 1212 1 (2001Vo24).<br>The resonance strength of this level has been intensively studied. See the general comments at the beginning of this dataset for details.<br>1971Pr11 reports a 3166 $\gamma$ with I $\gamma$ =0.37 13 to the 4382 final level.<br>J <sup>π</sup> : from $\gamma(\theta)$ in 1977Ko35.<br>E <sub>p</sub> (lab) (keV): weighted average of 1225.6 10 (1972Hu10), 1226.3 6 (1973Fa07), and 1225.3 7 (1976Me12).<br>$\omega\gamma=0.68$ eV 20 (1972Hu10).<br>$\omega\gamma=0.50$ eV (1973Fa07).<br>$\omega\gamma=0.68$ eV 20 (1976Me12).<br>$\omega\gamma=0.7$ eV 2 (2001Vo24).<br>A <sub>2</sub> =2 (2001Vo24).<br>J <sup>π</sup> : spin=5/2 from $\gamma(\theta)$ in 1971Wi13 and 1976Me12. Other: spin=3/2 from 1977Ko34.<br>T <sub>1/2</sub> : lifetime $\tau < 20$ fs (1971Wi13, DSAM).<br>E <sub>p</sub> (lab) (keV): weighted average of 1266.5 10 (1972Hu10), 1267.9 11 (1973Fa07), and 1265.7 8 (1976Me12).<br>$\omega\gamma=3.4$ eV, $\Gamma_{\gamma} \approx 0.57$ eV (1971Wi13).<br>$\omega\gamma=1.4$ eV 4 (1972Hu10).<br>$\omega\gamma=1.31$ eV (1973Fa07).<br>$\omega\gamma=1.3$ eV 4 (1976Me12).<br>J <sup>π</sup> : from $\gamma(\theta)$ in 1977Ko34. Spin=3/2,5/2 from $\gamma(\theta)$ in 1976Me12.<br>E <sub>p</sub> (lab) (keV): weighted average of 1285.5 11 (1972Hu10), 1286.1 11 (1973Fa07), and 1284.5 5 (1976Me12).<br>$\omega\gamma=0.54$ eV 16 (1972Hu10).<br>$\omega\gamma=0.50$ eV (1973Fa07).<br>$\omega\gamma=0.77$ eV 23 (1976Me12).<br>E <sub>p</sub> (lab) (keV): weighted average of 1323.7 12 (1972Hu10), 1323.5 12 (1973Fa07), and 1324.2 8 (1976Me12).<br>$\omega\gamma=0.23$ eV 11 (1972Hu10).<br>$\omega\gamma=0.23$ eV (1973Fa07).<br>$\omega\gamma=0.36$ eV 11 (1976Me12).<br>J <sup>π</sup> : spin=5/2 from $\gamma(\theta)$ in 1977Ko34. Spin=5/2,7/2 from $\gamma(\theta)$ in 1976Me12.<br>E <sub>p</sub> (lab) (keV): weighted average of 1339.8 11 (1972Hu10), 1339.4 11 (1973Fa07), and 1339.7 8 (1976Me12).<br>$\omega\gamma=0.68$ eV 20 (1972Hu10).<br>$\omega\gamma=0.63$ eV (1973Fa07). |
| 7518.8 7              | 7/2                         |                  | 1182.0 7                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 7548.9 6              | 7/2 <sup>-</sup>            | <0.69 fs         | 1213.0 6                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 7561.3 6              | 1/2,3/2                     |                  | 1225.8 6                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 7600.8 8              | 5/2                         | <13.9 fs         | 1266.5 8                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 7618.7 5              | 5/2                         |                  | 1284.9 5                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 7656.6 8              |                             |                  | 1323.9 8                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 7671.9 8              | 5/2                         |                  | 1339.7 8                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $^{35}\text{Cl}$  Levels (continued)

| E(level) <sup>†</sup> | J <sup>π</sup> <sup>‡</sup> | T <sub>1/2</sub> | E <sub>p</sub> (lab) (keV) | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------|-----------------------------|------------------|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7685.5 8              | 3/2 <sup>-</sup>            |                  | 1353.7 8                   | $\omega\gamma=0.59$ eV 18 (1976Me12).<br>J <sup>π</sup> : 3/2 <sup>-</sup> from $\sigma(E_p, \theta)$ in 1981Bi05. Spin=3/2 from $\gamma(\theta)$ in 1976Me12.<br>Spin=3/2, 5/2 from $\gamma(\theta)$ in 1977Ko34.<br>E <sub>p</sub> (lab) (keV): weighted average of 1353.7 11 (1972Hu10), 1353.4 11 (1973Fa07), and 1353.8 8 (1976Me12).<br>$\Gamma_p=445$ eV 11 (1981Bi05).<br>$\omega\gamma=1.13$ eV 34 (1972Hu10).<br>$\omega\gamma=1.09$ eV (1973Fa07).<br>$\omega\gamma=1.3$ eV 4 (1976Me12).                                                            |
| 7693.9 8              |                             |                  | 1362.3 8                   | E <sub>p</sub> (lab) (keV): weighted average of 1362.4 12 (1972Hu10) and 1362.3 8 (1976Me12).<br>$\omega\gamma=0.18$ eV 9 (1972Hu10).<br>$\omega\gamma=0.27$ eV 8 (1976Me12).                                                                                                                                                                                                                                                                                                                                                                                   |
| 7706.4 8              | 5/2 <sup>+</sup>            |                  | 1375.2 8                   | J <sup>π</sup> : 5/2 <sup>+</sup> from $\sigma(E_p, \theta)$ in 1981Bi05. Spin=5/2 from $\gamma(\theta)$ in 1977Ko34.<br>Spin=3/2, 5/2 from $\gamma(\theta)$ in 1976Me12 and 5/2 is favored.<br>E <sub>p</sub> (lab) (keV): weighted average of 1375.7 11 (1972Hu10), 1375.0 11 (1973Fa07), and 1375.1 8 (1976Me12).<br>$\omega\gamma=1.7$ eV 5 (1972Hu10).<br>$\omega\gamma=1.40$ eV (1973Fa07).<br>$\omega\gamma=2.0$ eV 6 (1976Me12).<br>$\Gamma_p=4$ eV 1 (1981Bi05).                                                                                       |
| 7744.8 9              | 7/2                         |                  | 1414.8 9                   | J <sup>π</sup> : from $\gamma(\theta)$ for J <sub>f</sub> (2645.7)=7/2, J <sub>f</sub> (3163)=7/2, J <sub>f</sub> (3942.9)=9/2 (1977Ko35).<br>E <sub>p</sub> (lab) (keV): weighted average of 1415.6 12 (1972Hu10), 1414.2 12 (1973Fa07), and 1414.7 9 (1976Me12).<br>$\omega\gamma=0.59$ eV 18 (1972Hu10).<br>$\omega\gamma=0.54$ eV (1973Fa07).<br>$\omega\gamma=0.77$ eV 23 (1976Me12).                                                                                                                                                                      |
| 7748                  |                             |                  | 1418                       | E(level): from 1960An06.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 7777.1 9              |                             |                  | 1448.0 9                   | E <sub>p</sub> (lab) (keV): weighted average of 1449.0 11 (1972Hu10), 1447.5 12 (1973Fa07), and 1447.5 9 (1976Me12).<br>$\omega\gamma=0.68$ eV 20 (1972Hu10).<br>$\omega\gamma=(0.63)$ eV (1973Fa07).<br>$\omega\gamma=0.68$ eV 20 (1976Me12).                                                                                                                                                                                                                                                                                                                  |
| 7781.7 11             | 5/2 <sup>-</sup>            |                  | 1452.8 11                  | J <sup>π</sup> : proposed by 1976Me12 based on strengths of primary transitions.<br>E <sub>p</sub> (lab) (keV): weighted average of 1453.6 11 (1972Hu10), 1451.8 15 (1973Fa07), and 1452.3 13 (1976Me12).<br>$\omega\gamma=0.36$ eV 18 (1972Hu10).<br>$\omega\gamma=(0.41)$ eV (1973Fa07).<br>$\omega\gamma=0.41$ eV 12 (1976Me12).                                                                                                                                                                                                                             |
| 7797.1 9              | 1/2                         |                  | 1468.6 9                   | J <sup>π</sup> : from $\gamma(\theta)$ in 1988Va06, which rejects spin=3/2 from $\gamma(\theta)$ in 1976Me12.<br>E <sub>p</sub> (lab) (keV): weighted average of 1469.6 12 (1972Hu10), 1468.2 12 (1973Fa07), and 1468.2 9 (1976Me12).<br>$\omega\gamma=0.68$ eV 20 (1972Hu10).<br>$\omega\gamma=0.72$ eV (1973Fa07).<br>$\omega\gamma=0.63$ eV 19 (1976Me12).                                                                                                                                                                                                   |
| 7837.2 10             | 3/2                         | <3.47 fs         | 1509.9 10                  | T=3/2<br>J <sup>π</sup> : spin from $\gamma(\theta)$ in 1967Wa22.<br>T <sub>1/2</sub> : lifetime $\tau < 5$ fs (1971Wi13, DSAM).<br>E <sub>p</sub> (lab) (keV): weighted average of 1509.1 15 (1973Fa07), 1510.2 10 (1976Me12), and 1510 1 (2001Vo24).<br>$\omega\gamma=3.4$ eV 6 (1967Wa22).<br>$\omega\gamma \approx 7.5$ eV, $\Gamma_\gamma=1.9$ eV (1971Wi13).<br>$\omega\gamma=5.0$ eV 15 for E <sub>p</sub> =1510 doublet (1972Hu10).<br>$\omega\gamma=0.59$ eV (1973Fa07).<br>$\omega\gamma=2.2$ eV 7 (1976Me12).<br>$\omega\gamma=2.1$ eV 3 (2001Vo24). |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $^{35}\text{Cl}$  Levels (continued)

| E(level) <sup>†</sup> | J <sup>π</sup> <sup>‡</sup> | E <sub>p</sub> (lab) (keV) | Comments                                                                                                                                                                                                                                                                                               |
|-----------------------|-----------------------------|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7839.0 9              |                             | 1511.8 9                   | $\Gamma_\gamma=0.55$ eV 6 (2001Vo24).<br>E <sub>p</sub> (lab) (keV): weighted average of 1512.3 15 (1973Fa07) and 1511.6 9 (1976Me12).<br>$\omega\gamma=5.0$ eV 15 for E <sub>p</sub> =1510 doublet (1972Hu10).<br>$\omega\gamma=3.94$ eV (1973Fa07).<br>$\omega\gamma=2.8$ eV 8 (1976Me12).           |
| 7868.7 10             |                             | 1542.3 10                  | E <sub>p</sub> (lab) (keV): weighted average of 1542.3 13 (1972Hu10), 1542.3 12 (1973Fa07), and 1542.2 10 (1976Me12).<br>$\omega\gamma=0.23$ eV 11 (1972Hu10).<br>$\omega\gamma=(0.23)$ eV (1973Fa07).<br>$\omega\gamma=0.45$ eV 14 (1976Me12).                                                        |
| 7873?                 |                             | 1547                       | E(level): from 1960An06.                                                                                                                                                                                                                                                                               |
| 7880.8 9              | 3/2 <sup>-</sup> ,5/2       | 1554.8 9                   | J <sup>π</sup> : from $\gamma(\theta)$ in 1976Me12.<br>E <sub>p</sub> (lab) (keV): weighted average of 1554.9 12 (1972Hu10), 1554.8 11 (1973Fa07), and 1554.7 9 (1976Me12).<br>$\omega\gamma=0.72$ eV 22 (1972Hu10).<br>$\omega\gamma=0.59$ eV (1973Fa07).<br>$\omega\gamma=1.00$ eV 30 (1976Me12).    |
| 7885?                 |                             | 1559                       | E(level): from 1960An06.                                                                                                                                                                                                                                                                               |
| 7889.1 15             |                             | 1563.3 15                  | E <sub>p</sub> (lab) (keV): from 1973Fa07.<br>$\omega\gamma=(0.05)$ eV (1973Fa07).                                                                                                                                                                                                                     |
| 7899.2 7              | (3/2 <sup>-</sup> ,5/2)     | 1573.7 7                   | E <sub>p</sub> (lab) (keV): weighted average of 1574.4 12 (1972Hu10), 1573.2 11 (1973Fa07), and 1573.7 7 (1976Me12).<br>$\omega\gamma=0.63$ eV 19 (1972Hu10).<br>$\omega\gamma=0.45$ eV (1973Fa07).<br>$\omega\gamma=1.00$ eV 30 (1976Me12).                                                           |
| 7903?                 |                             | 1578                       | E(level): from 1960An06.                                                                                                                                                                                                                                                                               |
| 7923.3 9              |                             | 1598.6 9                   | E <sub>p</sub> (lab) (keV): weighted average of 1598.9 13 (1972Hu10), 1598.7 12 (1973Fa07), and 1598.4 9 (1976Me12).<br>$\omega\gamma=0.36$ eV 18 (1972Hu10).<br>$\omega\gamma=(0.32)$ eV (1973Fa07).<br>$\omega\gamma=0.45$ eV 14 (1976Me12).                                                         |
| 7930?                 |                             | 1605                       | E(level): from 1960An06.                                                                                                                                                                                                                                                                               |
| 7949?                 |                             | 1625                       | E(level): from 1960An06.                                                                                                                                                                                                                                                                               |
| 7970.3 10             |                             | 1646.9 10                  | E <sub>p</sub> (lab) (keV): weighted average of 1647.3 12 (1972Hu10), 1645.9 13 (1973Fa07), and 1647.1 10 (1976Me12).<br>$\omega\gamma=0.54$ eV 16 (1972Hu10).<br>$\omega\gamma=(0.45)$ eV (1973Fa07).<br>$\omega\gamma=0.86$ eV 26 (1976Me12).                                                        |
| 7987.8 10             | 3/2                         | 1665.0 10                  | J <sup>π</sup> : from $\gamma(\theta)$ in 1976Me12.<br>E <sub>p</sub> (lab) (keV): weighted average of 1665.2 12 (1972Hu10), 1663.3 12 (1973Fa07), and 1666.1 10 (1976Me12).<br>$\omega\gamma=0.68$ eV 20 (1972Hu10).<br>$\omega\gamma=(0.59)$ eV (1973Fa07).<br>$\omega\gamma=0.54$ eV 16 (1976Me12). |
| 7995.6 10             | 5/2                         | 1673.0 10                  | E <sub>p</sub> (lab) (keV): weighted average of 1673.7 11 (1972Hu10), 1672.4 13 (1973Fa07), and 1672.7 10 (1976Me12).<br>$\omega\gamma=1.2$ eV 4 (1972Hu10).<br>$\omega\gamma=(1.04)$ eV (1973Fa07).<br>$\omega\gamma=0.95$ eV 28 (1976Me12).                                                          |
| 8001.0 9              | (7/2,9/2 <sup>+</sup> )     | 1678.6 9                   | E <sub>p</sub> (lab) (keV): weighted average of 1679.7 11 (1972Hu10), 1678.2 13 (1973Fa07), and 1678.1 9 (1976Me12).<br>$\omega\gamma=1.9$ eV 6 (1972Hu10).<br>$\omega\gamma=(1.18)$ eV (1973Fa07).<br>$\omega\gamma=1.3$ eV 4 (1976Me12).                                                             |
| 8005.0 10             | 5/2 <sup>+</sup>            | 1682.7 10                  | J <sup>π</sup> : from $\sigma(\text{E}_p,\theta)$ in 1981Bi05. Spin=5/2 from $\gamma(\theta)$ in 1981Bi05. Spin=3/2,5/2 from                                                                                                                                                                           |

Continued on next page (footnotes at end of table)



$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $^{35}\text{Cl}$  Levels (continued)

| E(level) <sup>†</sup> | J <sup>π</sup> <sup>‡</sup>                 | E <sub>p</sub> (lab) (keV) | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------|---------------------------------------------|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       |                                             |                            | $\gamma(\theta)$ in <a href="#">1976Me12</a> .<br>E <sub>p</sub> (lab) (keV): weighted average of 1684.3 <i>I3</i> ( <a href="#">1972Hu10</a> ), 1681.9 <i>I1</i> ( <a href="#">1973Fa07</a> ), and 1682.5 <i>I0</i> ( <a href="#">1976Me12</a> ).<br>$\omega\gamma=(1.54)$ eV ( <a href="#">1973Fa07</a> ).<br>$\omega\gamma=2.2$ eV <i>7</i> ( <a href="#">1976Me12</a> ).<br>$\omega\gamma=2.0$ eV <i>6</i> ( <a href="#">1972Hu10</a> ).<br>$\Gamma_p=21$ eV <i>3</i> ( <a href="#">1981Bi05</a> ).<br>E <sub>p</sub> (lab) (keV): weighted average of 1714.0 <i>I1</i> ( <a href="#">1972Hu10</a> ), 1713.5 <i>I4</i> ( <a href="#">1973Fa07</a> ), and 1714.9 <i>I0</i> ( <a href="#">1976Me12</a> ).<br>$\omega\gamma=0.41$ eV <i>20</i> ( <a href="#">1972Hu10</a> ).<br>$\omega\gamma=(0.41)$ eV ( <a href="#">1973Fa07</a> ).<br>$\omega\gamma=0.36$ eV <i>11</i> ( <a href="#">1976Me12</a> ).<br>J <sup>π</sup> : from $\gamma(\theta)$ in <a href="#">1976Me12</a> .<br>E <sub>p</sub> (lab) (keV): weighted average of 1717.9 <i>I1</i> ( <a href="#">1972Hu10</a> ), 1716.3 <i>I1</i> ( <a href="#">1973Fa07</a> ), and 1717.3 <i>I1</i> ( <a href="#">1976Me12</a> ).<br>$\omega\gamma=1.6$ eV <i>5</i> ( <a href="#">1972Hu10</a> ).<br>$\omega\gamma=1.36$ eV ( <a href="#">1973Fa07</a> ).<br>$\omega\gamma=1.3$ eV <i>4</i> ( <a href="#">1976Me12</a> ).<br>E <sub>p</sub> (lab) (keV): weighted average of 1756.0 <i>I0</i> ( <a href="#">1972Hu10</a> ), 1755.4 <i>I1</i> ( <a href="#">1973Fa07</a> ), and 1755.7 <i>I1</i> ( <a href="#">1976Me12</a> ).<br>$\omega\gamma=0.72$ eV <i>22</i> ( <a href="#">1972Hu10</a> ).<br>$\omega\gamma=(0.59)$ eV ( <a href="#">1973Fa07</a> ).<br>$\omega\gamma=0.72$ eV <i>22</i> ( <a href="#">1976Me12</a> ).<br>E <sub>p</sub> (lab) (keV): from <a href="#">1960An06</a> .<br>E <sub>p</sub> (lab) (keV): weighted average of 1776.7 <i>I1</i> ( <a href="#">1972Hu10</a> ), 1776.9 <i>I1</i> ( <a href="#">1973Fa07</a> ), and 1776.0 <i>I1</i> ( <a href="#">1976Me12</a> ).<br>$\omega\gamma=1.13$ eV <i>34</i> ( <a href="#">1972Hu10</a> ).<br>$\omega\gamma=1.13$ eV ( <a href="#">1973Fa07</a> ).<br>$\omega\gamma=1.1$ eV <i>3</i> ( <a href="#">1976Me12</a> ).<br>J <sup>π</sup> : from $\gamma(\theta)$ in <a href="#">1976Me12</a> .<br>E <sub>p</sub> (lab) (keV): weighted average of 1787.7 <i>I1</i> ( <a href="#">1972Hu10</a> ), 1786.6 <i>I1</i> ( <a href="#">1973Fa07</a> ), and 1786.9 <i>I1</i> ( <a href="#">1976Me12</a> ).<br>$\omega\gamma=1.9$ eV <i>6</i> ( <a href="#">1972Hu10</a> ).<br>$\omega\gamma=1.63$ eV ( <a href="#">1973Fa07</a> ).<br>$\omega\gamma=1.7$ eV <i>5</i> ( <a href="#">1976Me12</a> ).<br>E <sub>p</sub> (lab) (keV): weighted average of 1794.8 <i>I3</i> ( <a href="#">1972Hu10</a> ), 1793.3 <i>I4</i> ( <a href="#">1973Fa07</a> ), and 1794.2 <i>I1</i> ( <a href="#">1976Me12</a> ).<br>$\omega\gamma=0.54$ eV <i>16</i> ( <a href="#">1972Hu10</a> ).<br>$\omega\gamma=(0.50)$ eV ( <a href="#">1973Fa07</a> ).<br>$\omega\gamma=0.72$ eV <i>22</i> ( <a href="#">1976Me12</a> ).<br>E <sub>p</sub> (lab) (keV): weighted average of 1829.1 <i>I3</i> ( <a href="#">1972Hu10</a> ), 1828.6 <i>I2</i> ( <a href="#">1973Fa07</a> ), and 1829.5 <i>I1</i> ( <a href="#">1976Me12</a> ).<br>$\omega\gamma=1.09$ eV <i>33</i> ( <a href="#">1972Hu10</a> ).<br>$\omega\gamma=0.68$ eV ( <a href="#">1973Fa07</a> ).<br>$\omega\gamma=0.77$ eV <i>23</i> ( <a href="#">1976Me12</a> ).<br>E <sub>p</sub> (lab) (keV): weighted average of 1839.0 <i>I2</i> ( <a href="#">1972Hu10</a> ), 1838.8 <i>I2</i> ( <a href="#">1973Fa07</a> ), and 1839.1 <i>I1</i> ( <a href="#">1976Me12</a> ).<br>$\omega\gamma=0.95$ eV <i>28</i> ( <a href="#">1972Hu10</a> ).<br>$\omega\gamma=0.95$ eV ( <a href="#">1973Fa07</a> ).<br>$\omega\gamma=0.86$ eV <i>26</i> ( <a href="#">1976Me12</a> ).<br>E <sub>p</sub> (lab) (keV): weighted average of 1862.7 <i>I3</i> ( <a href="#">1972Hu10</a> ), 1862.1 <i>I3</i> ( <a href="#">1973Fa07</a> ), and 1862.1 <i>I2</i> ( <a href="#">1976Me12</a> ).<br>$\omega\gamma=0.36$ eV <i>18</i> ( <a href="#">1972Hu10</a> ).<br>$\omega\gamma=(0.32)$ eV ( <a href="#">1973Fa07</a> ).<br>$\omega\gamma=0.41$ eV <i>12</i> ( <a href="#">1976Me12</a> ).<br>J <sup>π</sup> : from $\sigma(E_p,\theta)$ in <a href="#">1981Bi05</a> . Spin=5/2 from $\gamma(\theta)$ in <a href="#">1996Ka21</a> and <a href="#">2001Vo24</a> . |
| 8035.7 <i>I0</i>      |                                             | 1714.3 <i>I0</i>           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 8038.5 <i>I1</i>      | 3/2                                         | 1717.2 <i>I1</i>           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 8075.9 <i>I0</i>      |                                             | 1755.7 <i>I0</i>           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 8080?                 |                                             | 1760                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 8096.1 <i>I1</i>      | (5/2, 7/2 <sup>+</sup> )                    | 1776.5 <i>I1</i>           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 8106.4 <i>I1</i>      | 3/2                                         | 1787.1 <i>I1</i>           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 8113.3 <i>I1</i>      | (1/2 <sup>+</sup> , 3/2, 5/2 <sup>+</sup> ) | 1794.2 <i>I1</i>           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 8147.2 <i>I1</i>      | 3/2 <sup>-</sup>                            | 1829.1 <i>I1</i>           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 8156.8 <i>I1</i>      | (5/2 <sup>+</sup> , 7/2 <sup>-</sup> )      | 1839.0 <i>I1</i>           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 8179.4 <i>I2</i>      |                                             | 1862.3 <i>I2</i>           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 8208.2 <i>I0</i>      | 5/2 <sup>+</sup>                            | 1891.9 <i>I0</i>           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $^{35}\text{Cl}$  Levels (continued)

| E(level) <sup>†</sup> | J <sup>π</sup> <sub>z</sub> | E <sub>p</sub> (lab) (keV) | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------|-----------------------------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       |                             |                            | E <sub>p</sub> (lab) (keV): weighted average of 1893.2 <i>11</i> ( <b>1972Hu10</b> ), 1891.3 <i>12</i> ( <b>1976Me12</b> ), 1892.0 <i>11</i> ( <b>1973Fa07</b> ), and 1891 <i>1</i> ( <b>2001Vo24</b> ).<br>$\omega\gamma=3.6$ eV <i>11</i> ( <b>1972Hu10</b> ).<br>$\omega\gamma=2.71$ eV ( <b>1973Fa07</b> ).<br>$\omega\gamma=4.1$ eV <i>12</i> ( <b>1976Me12</b> ).<br>$\omega\gamma=4.1$ eV 5, $\Gamma_p=39$ eV 5 ( <b>1981Bi05</b> ).<br>$\omega\gamma=2.5$ eV 7 ( <b>1996Ka21</b> ).<br>$\omega\gamma=4.0$ eV 8 ( <b>2001Vo24</b> ), sum of $\omega\gamma$ for the resonances at E <sub>p</sub> =1891 and 1893, $\Gamma_\gamma=0.70$ eV <i>14</i> ( <b>2001Vo24</b> ). |
| 8209                  | 1/2 <sup>+</sup>            | 1893                       | E(level): from <b>1974Bi16</b> ; a weak resonance in the tail of the 1892 resonance.<br>J <sup>π</sup> : from $\sigma(E_p, \theta)$ in <b>1981Bi05</b> .<br>$\Gamma_p=120$ eV <i>11</i> ( <b>1981Bi05</b> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 8216.3 <i>10</i>      | 5/2                         | 1900.3 <i>10</i>           | J <sup>π</sup> : spin=5/2 from $\gamma(\theta)$ in <b>2001Vo24</b> and <b>1996Ka21</b> .<br>E <sub>p</sub> (lab) (keV): weighted average of 1901.8 <i>11</i> ( <b>1972Hu10</b> ), 1899.7 <i>12</i> ( <b>1973Fa07</b> ), 1899.6 <i>11</i> ( <b>1976Me12</b> ), and 1900 <i>1</i> ( <b>2001Vo24</b> ).<br>$\omega\gamma=2.8$ eV 8 ( <b>1972Hu10</b> ).<br>$\omega\gamma=2.13$ eV ( <b>1973Fa07</b> ).<br>$\omega\gamma=3.2$ eV <i>10</i> ( <b>1976Me12</b> ).<br>$\omega\gamma=1.7$ eV 5 ( <b>1996Ka21</b> ).<br>$\omega\gamma=3.2$ eV 5 ( <b>2001Vo24</b> ).<br>$\Gamma_\gamma=0.55$ eV 8 ( <b>2001Vo24</b> ).                                                                 |
| 8242.4 <i>12</i>      |                             | 1927.1 <i>12</i>           | E <sub>p</sub> (lab) (keV): weighted average of 1928.2 <i>13</i> ( <b>1972Hu10</b> ), 1926.8 <i>13</i> ( <b>1973Fa07</b> ), and 1926.5 <i>12</i> ( <b>1976Me12</b> ).<br>$\omega\gamma=0.86$ eV 26 ( <b>1972Hu10</b> ).<br>$\omega\gamma=0.86$ eV ( <b>1973Fa07</b> ).<br>$\omega\gamma=1.00$ eV 30 ( <b>1976Me12</b> ).<br>$\omega\gamma=1.0$ eV 3 ( <b>2001Vo24</b> ).<br>$\Gamma_\gamma=0.25$ eV 8 ( <b>2001Vo24</b> ).<br>20% of I $\gamma$ transitions to unknown levels ( <b>1973Fa07</b> ).                                                                                                                                                                            |
| 8251? 5<br>8268.9 5   |                             | 1936 5<br>1954.4 5         | E <sub>p</sub> (lab) (keV): from <b>1967Da12</b> .<br>E <sub>p</sub> (lab) (keV): weighted average of 1955.9 <i>13</i> ( <b>1972Hu10</b> ), 1952 2 ( <b>1973Fa07</b> ), 1954.3 <i>12</i> ( <b>1976Me12</b> ), and 1954.4 5 ( <b>1976Sp08</b> ).<br>$\omega\gamma=0.81$ eV 24 ( <b>1972Hu10</b> ).<br>$\omega\gamma=(0.86)$ eV ( <b>1973Fa07</b> ).<br>$\omega\gamma=0.90$ eV 27 ( <b>1976Me12</b> ).<br>$\omega\gamma=0.6$ eV 3 ( <b>1976Sp08</b> ).                                                                                                                                                                                                                          |
| 8277.2 5              |                             | 1963.0 5                   | E <sub>p</sub> (lab) (keV): weighted average of 1964.1 <i>14</i> ( <b>1972Hu10</b> ), 1962 2 ( <b>1973Fa07</b> ), 1962.8 <i>12</i> ( <b>1976Me12</b> ), and 1962.9 5 ( <b>1976Sp08</b> ).<br>$\omega\gamma=0.72$ eV 22 ( <b>1972Hu10</b> ).<br>$\omega\gamma=(0.77)$ eV ( <b>1973Fa07</b> ).<br>$\omega\gamma=0.68$ eV 20 ( <b>1976Me12</b> ).<br>$\omega\gamma=0.7$ eV 4 ( <b>1976Sp08</b> ).                                                                                                                                                                                                                                                                                |
| 8282.4 5              | (3/2 <sup>-</sup> , 5/2)    | 1968.3 5                   | E <sub>p</sub> (lab) (keV): weighted average of 1970.0 <i>13</i> ( <b>1972Hu10</b> ), 1971 2 ( <b>1973Fa07</b> ), 1967.8 <i>12</i> ( <b>1976Me12</b> ), and 1967.9 5 ( <b>1976Sp08</b> ).<br>$\omega\gamma=1.09$ eV 33 ( <b>1972Hu10</b> ).<br>$\omega\gamma=(1.09)$ eV ( <b>1973Fa07</b> ).<br>$\omega\gamma=0.59$ eV <i>18</i> ( <b>1976Me12</b> ).<br>$\omega\gamma=0.4$ eV 2 ( <b>1976Sp08</b> ).                                                                                                                                                                                                                                                                         |
| 8287.82 30            | 1/2 <sup>-</sup>            | 1973.89 30                 | J <sup>π</sup> : from $\sigma(E_p, \theta)$ in <b>1981Bi05</b> , which also rejects spin=3/2 from $\gamma(\theta)$ in <b>1976Me12</b> .<br>E <sub>p</sub> (lab) (keV): weighted average of 1975.6 <i>13</i> ( <b>1972Hu10</b> ), 1973.8 <i>12</i> ( <b>1976Me12</b> ), and 1973.80 30 ( <b>1976Sp08</b> ).<br>$\omega\gamma=1.3$ eV 4 ( <b>1972Hu10</b> ).<br>$\omega\gamma=0.72$ eV 22 ( <b>1976Me12</b> ).<br>$\omega\gamma=0.9$ eV 5 ( <b>1976Sp08</b> ).<br>$\Gamma_p=48$ eV 6 ( <b>1981Bi05</b> ).                                                                                                                                                                       |
| 8297.2 9              |                             | 1983.5 9                   | E <sub>p</sub> (lab) (keV): weighted average of 1983.7 5 ( <b>1976Sp08</b> ) and 1980 2 ( <b>1973Fa07</b> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $^{35}\text{Cl}$  Levels (continued)

| E(level) <sup>†</sup> | J <sup>π</sup> <sup>‡</sup> | E <sub>p</sub> (lab) (keV) | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------|-----------------------------|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8298.2 5              | 3/2                         | 1984.6 5                   | $\omega\gamma=(1.36)$ eV (1973Fa07).<br>$\omega\gamma=0.6$ eV 3 (1976Sp08).<br>J <sup>π</sup> : from $\gamma(\theta)$ in 1976Me12.<br>E <sub>p</sub> (lab) (keV): weighted average of 1986.0 13 (1972Hu10), 1983.0 20 (1973Fa07), 1984.5 12 (1976Me12), and 1984.5 5 (1976Sp08).<br>$\omega\gamma=1.8$ eV 5 (1972Hu10).<br>$\omega\gamma=(0.77)$ eV (1973Fa07).<br>$\omega\gamma=1.6$ eV 5 (1976Me12).<br>$\omega\gamma=0.9$ eV 5 (1976Sp08). |
| 8318.1 3              |                             | 2005.1 3                   | E <sub>p</sub> (lab) (keV): weighted average of 2006.7 12 (1972Hu10), 2004.6 13 (1973Fa07), 2006.0 13 (1976Me12), and 2005.0 3 (1976Sp08).<br>$\omega\gamma=1.9$ eV 6 (1972Hu10).<br>$\omega\gamma=2.08$ eV (1973Fa07).<br>$\omega\gamma=1.9$ eV 6 (1976Me12).<br>$\omega\gamma=2.6$ eV 5 (1976Sp08).                                                                                                                                         |
| 8323.1 13             |                             | 2010.2 13                  | E <sub>p</sub> (lab) (keV): weighted average of 2010.5 13 (1972Hu10) and 2009.9 13 (1976Me12).<br>$\omega\gamma=0.45$ eV 14 (1972Hu10).<br>$\omega\gamma=0.72$ eV 22 (1976Me12).                                                                                                                                                                                                                                                              |
| 8346? 5               |                             | 2034 5                     | E <sub>p</sub> (lab) (keV): from 1967Da12.                                                                                                                                                                                                                                                                                                                                                                                                    |
| 8381.8 8              | 5/2 <sup>+</sup>            | 2070.7 8                   | J <sup>π</sup> : spin=5/2 from $\gamma(\theta)$ in 2001Vo24 and 1967Ko23.<br>E <sub>p</sub> (lab) (keV): weighted average of 2073.5 12 (1972Hu10), 2070.4 5 (1976Sp08), and 2070 1 (2001Vo24).<br>$\omega\gamma\approx 4$ eV (1967Ko23).<br>$\omega\gamma=6.4$ eV 19 (1972Hu10).<br>$\omega\gamma=6.0$ eV 12 (1976Sp08).<br>$\omega\gamma=5.9$ eV 6 (2001Vo24).<br>$\Gamma_\gamma=1.02$ eV 10 (2001Vo24).                                     |
| 8389.4 16             |                             | 2078.5 16                  | E <sub>p</sub> (lab) (keV): unweighted average of 2080.0 14 (1972Hu10) and 2076.9 5 (1976Sp08).<br>$\omega\gamma=0.27$ eV 14 (1972Hu10).<br>$\omega\gamma=0.4$ eV 2 (1976Sp08).                                                                                                                                                                                                                                                               |
| 8402.9 5              |                             | 2092.4 5                   | E <sub>p</sub> (lab) (keV): from 1976Sp08.<br>$\omega\gamma=0.7$ eV 4 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                             |
| 8405.7 14             |                             | 2095.3 14                  | E <sub>p</sub> (lab) (keV): unweighted average of 2096.6 12 (1972Hu10) and 2093.9 5 (1976Sp08).<br>$\omega\gamma=2.9$ eV 9 (1972Hu10).<br>$\omega\gamma=2.3$ eV 5 (1976Sp08).                                                                                                                                                                                                                                                                 |
| 8409.6 18             |                             | 2099.3 18                  | E <sub>p</sub> (lab) (keV): unweighted average of 2101.0 14 (1972Hu10) and 2097.5 5 (1976Sp08).<br>$\omega\gamma=1.6$ eV 5 (1972Hu10).<br>$\omega\gamma=1.4$ eV 3 (1976Sp08).<br>$\omega\gamma=0.5$ eV 3 (1976Sp08).                                                                                                                                                                                                                          |
| 8416.7 5              |                             | 2106.6 5                   | $\omega\gamma=1.0$ eV 2 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                           |
| 8430.3 5              |                             | 2120.6 5                   | $\omega\gamma=0.5$ eV 3 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                           |
| 8434.8 5              |                             | 2125.2 5                   | $\omega\gamma=0.9$ eV 5 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                           |
| 8464.3 5              |                             | 2155.6 5                   | J <sup>π</sup> : from 2001Vo24 based on proposed isobaric analog to 3/2 <sup>+</sup> in $^{35}\text{S}$ .                                                                                                                                                                                                                                                                                                                                     |
| 8484.4 5              | 3/2 <sup>+</sup>            | 2176.3 5                   | E <sub>p</sub> (lab) (keV): weighted average of 2176.4 5 (1976Sp08) and 2176 1 (2001Vo24).<br>$\omega\gamma=2.1$ eV 4 (1976Sp08).<br>$\omega\gamma=2.1$ eV 4 (2001Vo24).<br>$\Gamma_\gamma=0.36$ eV 6 (2001Vo24).                                                                                                                                                                                                                             |
| 8486.1 5              |                             | 2178.0 5                   | $\omega\gamma=1.9$ eV 4 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                           |
| 8506.5 5              |                             | 2199.1 5                   | $\omega\gamma=0.3$ eV 2 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                           |
| 8514.3 5              |                             | 2207.1 5                   | $\omega\gamma=0.5$ eV 3 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                           |
| 8533.9 5              |                             | 2227.3 5                   | $\omega\gamma=0.10$ eV 5 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                          |
| 8571.9 5              |                             | 2266.4 5                   | $\omega\gamma=2.8$ eV 6 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                           |
| 8580.5 5              |                             | 2275.3 5                   | $\omega\gamma=1.0$ eV 2 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                           |
| 8585.8 5              |                             | 2280.7 5                   | $\omega\gamma=0.10$ eV 5 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                          |
| 8590.0 5              |                             | 2285.0 5                   | $\omega\gamma=0.7$ eV 4 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                           |
| 8612.0 5              |                             | 2307.7 5                   | $\omega\gamma=1.0$ eV 2 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                           |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $^{35}\text{Cl}$  Levels (continued)

| E(level) <sup>†</sup> | J <sup>π</sup> <sup>‡</sup>            | E <sub>p</sub> (lab) (keV) | Comments                                                                                                                                                                                                           |
|-----------------------|----------------------------------------|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8613.7 5              |                                        | 2309.4 5                   | $\omega\gamma=2.4$ eV 5 (1976Sp08).                                                                                                                                                                                |
| 8618.1 5              | (3/2 <sup>+</sup> , 5/2 <sup>+</sup> ) | 2314.0 5                   | J <sup>π</sup> : most probable spin=5/2 from $\gamma(\theta)$ in 1970Ko33.                                                                                                                                         |
|                       |                                        |                            | $\omega\gamma=3.7$ eV 8 (1976Sp08).                                                                                                                                                                                |
| 8630.1 3              | 7/2                                    | 2326.3 3                   | J <sup>π</sup> : spin=3/2, 7/2 from $\gamma(\theta)$ in 1976Sp09; 1976Sp09 ruled out 3/2 based on 4687 $\gamma$ to 9/2 <sup>+</sup> , 3943 level. Other: most probable spin=3/2 from $\gamma(\theta)$ in 1970Ko33. |
|                       |                                        |                            | $\omega\gamma=2.6$ eV 5 (1976Sp08).                                                                                                                                                                                |
| 8640                  | 5/2                                    | 2336                       | J <sup>π</sup> : probable spin=5/2 from $\gamma(\theta)$ in 1970Ko33.                                                                                                                                              |
|                       |                                        |                            | E <sub>p</sub> (lab) (keV): from 1970Ko33.                                                                                                                                                                         |
| 8641.4 5              | 3/2, 5/2                               | 2338.0 5                   | J <sup>π</sup> : spin=3/2 from $\gamma(\theta)$ in 1976Sp09; spin=5/2 is also possible.                                                                                                                            |
|                       |                                        |                            | $\omega\gamma=0.9$ eV 5 (1976Sp08).                                                                                                                                                                                |
| 8686.2 5              |                                        | 2384.1 5                   | $\omega\gamma=2.2$ eV 4 (1976Sp08).                                                                                                                                                                                |
| 8688.3 5              |                                        | 2386.3 5                   | $\omega\gamma=0.7$ eV 4 (1976Sp08).                                                                                                                                                                                |
| 8696.9 5              |                                        | 2395.1 5                   | $\omega\gamma=0.6$ eV 3 (1976Sp08).                                                                                                                                                                                |
| 8706.3 5              |                                        | 2404.8 5                   | $\omega\gamma=0.2$ eV 1 (1976Sp08).                                                                                                                                                                                |
| 8717.7 5              |                                        | 2416.5 5                   | $\omega\gamma=0.5$ eV 3 (1976Sp08).                                                                                                                                                                                |
| 8750.9 5              |                                        | 2450.7 5                   | $\omega\gamma=1.7$ eV 3 (1976Sp08).                                                                                                                                                                                |
| 8767.0 5              |                                        | 2467.3 5                   | $\omega\gamma=0.3$ eV 2 (1976Sp08).                                                                                                                                                                                |
| 8772.9 5              |                                        | 2473.4 5                   | $\omega\gamma=0.6$ eV 3 (1976Sp08).                                                                                                                                                                                |
| 8779.8 5              |                                        | 2480.5 5                   | $\omega\gamma=1.7$ eV 3 (1976Sp08).                                                                                                                                                                                |
| 8787.2 5              |                                        | 2488.1 5                   | $\omega\gamma=1.2$ eV 2 (1976Sp08).                                                                                                                                                                                |
| 8798.4 5              |                                        | 2499.6 5                   | $\omega\gamma=1.7$ eV 3 (1976Sp08).                                                                                                                                                                                |
| 8820.9 5              |                                        | 2522.8 5                   | $\omega\gamma=0.7$ eV 4 (1976Sp08).                                                                                                                                                                                |
| 8824.2 5              |                                        | 2526.2 5                   | $\omega\gamma=3.6$ eV 7 (1976Sp08).                                                                                                                                                                                |
| 8829.3 5              |                                        | 2531.4 5                   | $\omega\gamma=2.1$ eV 4 (1976Sp08).                                                                                                                                                                                |
| 8833.1 5              |                                        | 2535.3 5                   | $\omega\gamma=0.7$ eV 4 (1976Sp08).                                                                                                                                                                                |
| 8838.1 5              | 7/2 <sup>-</sup>                       | 2540.5 5                   | J <sup>π</sup> : 5/2 <sup>-</sup> , 7/2 <sup>-</sup> from $\sigma(\text{E}_p, \theta)$ in 1981Bi05. Spin=7/2 from $\gamma(\theta)$ in 1981Bi05.                                                                    |
|                       |                                        |                            | $\omega\gamma=1.5$ eV 3 (1976Sp08).                                                                                                                                                                                |
|                       |                                        |                            | $\Gamma_p=4$ eV 1 (1981Bi05).                                                                                                                                                                                      |
| 8856.1 5              |                                        | 2559.0 5                   | $\omega\gamma=4.5$ eV 9 (1976Sp08).                                                                                                                                                                                |
| 8868.6 5              |                                        | 2571.9 5                   | $\omega\gamma=1.0$ eV 2 (1976Sp08).                                                                                                                                                                                |
| 8884.1 5              |                                        | 2587.9 5                   | $\omega\gamma=0.4$ eV 2 (1976Sp08).                                                                                                                                                                                |
| 8886.1 5              |                                        | 2589.9 5                   | $\omega\gamma=1.4$ eV 3 (1976Sp08).                                                                                                                                                                                |
| 8893.3 5              | (5/2, 7/2) <sup>+</sup>                | 2597.3 5                   | J <sup>π</sup> : from 2001Vo24 with no supporting arguments.                                                                                                                                                       |
|                       |                                        |                            | E <sub>p</sub> (lab) (keV): weighted average of 2597.4 5 (1976Sp08) and 2597 1 (2001Vo24).                                                                                                                         |
|                       |                                        |                            | $\omega\gamma=1.2$ eV 2 (1976Sp08).                                                                                                                                                                                |
|                       |                                        |                            | $\omega\gamma=1.2$ eV 4 (2001Vo24).                                                                                                                                                                                |
|                       |                                        |                            | $\Gamma_\gamma=0.20$ eV 6 (2001Vo24).                                                                                                                                                                              |
| 8904.8 5              |                                        | 2609.2 5                   | $\omega\gamma=1.0$ eV 2 (1976Sp08).                                                                                                                                                                                |
| 8906.8 5              | 5/2 <sup>+</sup>                       | 2611.2 5                   | J <sup>π</sup> : from 2001Vo24 with no supporting arguments.                                                                                                                                                       |
|                       |                                        |                            | E <sub>p</sub> (lab) (keV): weighted average of 2611.3 5 (1976Sp08) and 2611 1 (2001Vo24).                                                                                                                         |
|                       |                                        |                            | $\omega\gamma=1.5$ eV 3 (1976Sp08).                                                                                                                                                                                |
|                       |                                        |                            | $\omega\gamma=1.5$ eV 3 (2001Vo24).                                                                                                                                                                                |
|                       |                                        |                            | $\Gamma_\gamma=0.25$ eV 5 (2001Vo24).                                                                                                                                                                              |
| 8919.8 5              |                                        | 2624.6 5                   | $\omega\gamma=1.1$ eV 2 (1976Sp08).                                                                                                                                                                                |
| 8933.3 5              |                                        | 2638.5 5                   | $\omega\gamma=1.0$ eV 2 (1976Sp08).                                                                                                                                                                                |
| 8953.0 5              | 3/2                                    | 2658.8 5                   | J <sup>π</sup> : from $\gamma(\theta)$ in 1976Sp09.                                                                                                                                                                |
|                       |                                        |                            | $\omega\gamma=2.4$ eV 5 (1976Sp08).                                                                                                                                                                                |
| 8957.9 5              |                                        | 2663.8 5                   | $\omega\gamma=1.2$ eV 2 (1976Sp08).                                                                                                                                                                                |
| 8981.9 5              |                                        | 2688.6 5                   | $\omega\gamma=2.2$ eV 4 (1976Sp08).                                                                                                                                                                                |
| 8984.2 5              |                                        | 2690.9 5                   | $\omega\gamma=0.6$ eV 3 (1976Sp08).                                                                                                                                                                                |
| 8988.4 5              |                                        | 2695.3 5                   | $\omega\gamma=0.7$ eV 4 (1976Sp08).                                                                                                                                                                                |
| 8992.4 5              |                                        | 2699.4 5                   | $\omega\gamma=2.3$ eV 5 (1976Sp08).                                                                                                                                                                                |
| 8996.7 5              |                                        | 2703.8 5                   | $\omega\gamma=1.7$ eV 3 (1976Sp08).                                                                                                                                                                                |
| 9001.1 5              |                                        | 2708.3 5                   | $\omega\gamma=0.6$ eV 3 (1976Sp08).                                                                                                                                                                                |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $^{35}\text{Cl}$  Levels (continued)

| $E(\text{level})^\dagger$ | $J^\pi^\ddagger$ | $T_{1/2}$ | $E_p(\text{lab})$ (keV) | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------|------------------|-----------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9019.3 5                  |                  | 3.1 keV 3 | 2727.1 5                | $T_{1/2}$ : width measured from the slope of the low-energy side of the resonance peak (1976Sp08).<br>$\omega\gamma=0.8$ eV 4 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                |
| 9024.5 5                  |                  |           | 2732.4 5                | $\omega\gamma=1.1$ eV 2 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 9029.9 5                  |                  |           | 2738.0 5                | $\omega\gamma=2.5$ eV 5 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 9033.1 5                  |                  |           | 2741.3 5                | $\omega\gamma=0.4$ eV 2 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 9038.3 5                  |                  |           | 2746.6 5                | $\omega\gamma=0.8$ eV 4 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 9048.4 5                  |                  |           | 2757.0 5                | $\omega\gamma=3.1$ eV 6 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 9081.4 4                  | 5/2 <sup>+</sup> | 65 eV 20  | 2791.0 4                | $J^\pi$ : from $\sigma(E_p, \theta)$ in 1981Bi05. Spin=5/2 from $\gamma(\theta)$ in 1976Sp09.<br>$T_{1/2}$ : width $\Gamma_p=62$ eV 20, $\Gamma_\gamma=2.3$ eV 4 (1976Sp09, resonant absorption).<br>$E_p(\text{lab})$ (keV): weighted average of 2791.0 4 (1976Sp08) and 2791 1 (2001Vo24).<br>$\omega\gamma=15.7$ eV 32 (1976Sp08).<br>$\omega\gamma=16.0$ eV 4 (1996Ka21).<br>$\omega\gamma=16.0$ eV 3 (2001Vo24).<br>$\omega\gamma=13.4$ eV 22 (1976Sp09).<br>$\Gamma_p=59$ eV 7 (1981Bi05).<br>$\Gamma_\gamma=2.7$ eV 5 (2001Vo24). |
| 9088.5 5                  |                  |           | 2798.3 5                | $\omega\gamma=0.7$ eV 4 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 9099.3 5                  |                  |           | 2809.4 5                | $\omega\gamma=1.2$ eV 2 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 9100.4 5                  |                  |           | 2810.6 5                | $\omega\gamma=2.1$ eV 4 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 9107.4 5                  |                  |           | 2817.8 5                | $\omega\gamma=0.5$ eV 3 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 9110.0 5                  |                  |           | 2820.5 5                | $\omega\gamma=0.5$ eV 3 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 9124.0 5                  |                  |           | 2834.9 5                | $\omega\gamma=1.2$ eV 2 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 9135.1 5                  |                  | 2.0 keV 4 | 2846.3 5                | $T_{1/2}$ : width measured from the slope of the low-energy side of the resonance peak (1976Sp08).<br>$\omega\gamma=0.2$ eV 1 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                |
| 9138.3 5                  |                  |           | 2849.6 5                | $\omega\gamma=1.9$ eV 4 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 9155.7 5                  |                  |           | 2867.5 5                | $\omega\gamma=2.0$ eV 4 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 9157.1 5                  | 5/2              |           | 2869.0 5                | $J^\pi$ : spin=5/2 from $\gamma(\theta)$ in 2001Vo24 and 1996Ka21.<br>$\omega\gamma=6.6$ eV 13 (1976Sp08).<br>$\omega\gamma=6.3$ eV 1 (1996Ka21).                                                                                                                                                                                                                                                                                                                                                                                        |
| 9163.3 5                  |                  |           | 2875.3 5                | $\omega\gamma=0.4$ eV 2 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 9184.2 5                  |                  |           | 2896.9 5                | $\omega\gamma=0.9$ eV 5 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 9188.8 5                  |                  |           | 2901.6 5                | $\omega\gamma=0.9$ eV 5 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 9194.1 5                  |                  |           | 2907.1 5                | $\omega\gamma=3.9$ eV 8 (1976Sp08).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

<sup>†</sup> For resonances above  $S(\text{p})(^{35}\text{Cl})=6370.81$  4 (2021Wa16),  $E(\text{level})=E_{\text{c.m.}}+S(\text{p})(^{35}\text{Cl})$ , where  $E_{\text{c.m.}}=E_p(\text{lab})\times m(^{34}\text{S})/[m_p+m(^{34}\text{S})]$ . Weighted average is taken when multiple  $E_p(\text{lab})$  measurements are available. For levels below 8.27 MeV ( $E_p<1.95$  MeV),  $E_p(\text{lab})$  is mainly from 1972Hu10, 1973Fa07, and 1976Me12; for levels above 8.27 MeV ( $E_p>1.95$  MeV),  $E_p(\text{lab})$  is mainly from 1976Sp08, unless otherwise noted.

<sup>‡</sup> From the comparison of experimental angular distributions of  $\gamma$ -rays with theoretical predictions or deduced from  $\gamma$ -feedings, unless otherwise noted.

| $\gamma(^{35}\text{Cl})$ |                  |                    |               |              |                                      |                    |            | Comments                                                                                                                                                        |
|--------------------------|------------------|--------------------|---------------|--------------|--------------------------------------|--------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $E_i(\text{level})$      | $J_i^\pi$        | $E_\gamma^\dagger$ | $I_\gamma^\#$ | $E_f$        | $J_f^\pi$                            | Mult. <sup>b</sup> | $\delta^b$ |                                                                                                                                                                 |
| 1219.39                  | 1/2 <sup>+</sup> | 1219.37            | 100           | 0            | 3/2 <sup>+</sup>                     |                    |            | $E_\gamma$ : other: 1220 40 (1963Ha32).<br>$A_2=-0.03$ 3, $A_4=-0.002$ 50 (1971Wi13).<br>$A_2=-0.01$ 2, $A_4=+0.005$ 24 (1967Ta08).                             |
| 1763.14                  | 5/2 <sup>+</sup> | 543.75<br>1763.09  | <0.2<br>100   | 1219.39<br>0 | 1/2 <sup>+</sup><br>3/2 <sup>+</sup> | D+Q                | -0.42 17   | $I_\gamma$ : from 1972Hu10 and 1976Me12.<br>$I_\gamma$ : from 1972Hu10 and 1976Me12.<br>Mult., $\delta$ : from $\gamma\gamma(\theta)$ in 1967Ko20 and 1967Ko23. |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$        | $E_\gamma^\dagger$ | $I_\gamma^\#$ | $E_f$   | $J_f^\pi$        | Mult. <sup>b</sup> | $\delta^b$ | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------|------------------|--------------------|---------------|---------|------------------|--------------------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     |                  |                    |               |         |                  |                    |            | Other: $+2.8 < \delta < +4.0$ (1971Wi13).<br>$A_2 = -0.24$ 6, $A_4 = -0.08$ 7 (1971Wi13).<br>$A_2 = +0.25$ 3, $A_4 = -0.49$ 8<br>(1967Ko20,1967Ko23,1970Ko33).<br>$I_\gamma$ : weighted average of 8 1 (1972Hu10)<br>and 9.4 10 (1976Me12).                                                                                                                                                                                                                                                                                                              |
| 2645.70             | (5/2,7/2)        | 882.55             | 8.7 10        | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : weighted average of 92 1 (1972Hu10)<br>and 90.6 10 (1976Me12).<br>Mult., $\delta$ : D+Q with $\delta = -0.62 + 38-36$ from<br>$\gamma(\theta)$ for $J_1=3/2$ (1967Ko20).<br>$A_2 = +0.124$ 17, $A_4 = -0.230$ 24<br>(1967Ko20,1967Ko23).                                                                                                                                                                                                                                                                                                    |
|                     |                  | 1426.28            | <2            | 1219.39 | 1/2 <sup>+</sup> |                    |            | $E_\gamma$ : other: 933 7 (1970Ko33).<br>$I_\gamma$ : weighted average of 13 2 (1972Hu10)<br>and 13.0 10 (1976Me12).<br>$A_2 = -0.05$ 38 (1971Wi13).<br>$E_\gamma$ : other: 1475 7 (1970Ko33).<br>$I_\gamma$ : weighted average of 6 2 (1972Hu10)<br>and 8.0 10 (1976Me12). Other: 66 14<br>from 1963Ha32.                                                                                                                                                                                                                                               |
|                     |                  | 2645.59            | 91.3 10       | 0       | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : weighted average of 81 3 (1972Hu10)<br>and 79 2 (1976Me12). Other: 34 9<br>from 1963Ha32.                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 2693.9              | (3/2,5/2)        | 930.8              | 13.0 10       | 1763.14 | 5/2 <sup>+</sup> |                    |            | $\delta$ : $+0.1 < \delta < +0.26$ or $-35 < \delta < -7$ for $J=3/2$ ,<br>$\delta < -5$ for $J=5/2$ (1971Wi13).<br>$A_2 = +0.20$ 10, $A_4 = -0.16$ 11 (1971Wi13).<br>$I_\gamma$ : from 1972Hu10. Other: $I_\gamma < 2$<br>(1976Me12).<br>$I_\gamma$ : other: $I_\gamma \leq 5$ (1972Hu10).<br>$E_\gamma$ : other: 3006 7 (1970Ko33).<br>$I_\gamma$ : from 1972Hu10 and 1976Me12.<br>Mult., $\delta$ : from $\gamma\gamma(\theta)$ in 1967Ko20.<br>$A_2 = +0.45$ 18, $A_4 = +0.05$ 19 (1971Wi13).<br>$A_2 = -0.025$ 11, $A_4 = -0.084$ 16<br>(1967Ko20). |
|                     |                  | 1474.5             | 7.6 10        | 1219.39 | 1/2 <sup>+</sup> |                    |            | $I_\gamma$ : from 1971Pr11. Other: <3<br>(1976Me12).<br>$I_\gamma$ : from 1971Pr11. Others: 11 2<br>(1972Hu10), 8.9 20 (1976Me12).<br>$A_2 = +0.31$ 8, $A_4 = -0.06$ 8 (1971Wi13).<br>$I_\gamma$ : from 1971Pr11. Others: $\leq 0.2$<br>(1972Hu10), <0.2 (1976Me12).<br>Mult., $\delta$ : from $\gamma(\theta)$ in 1971Pr11, which<br>states that the anomalous $\gamma(\theta)$ can be<br>understood if it occurs by a mixture of<br>M2 and E1 radiation. In view of the<br>weakness of this transition, such a<br>mixture is not unreasonable.         |
|                     |                  | 2693.8             | 80 2          | 0       | 3/2 <sup>+</sup> | D+Q                |            | $E_\gamma$ : other: 3163 7 (1970Ko33).<br>$I_\gamma$ : from 1971Pr11. Others: 89 2<br>(1972Hu10), 91 2 (1976Me12).<br>Mult.: Q+O from $\gamma(\theta)$ in 1966Az01,<br>1966Wa09, 1967Wa22, and 1967Ta08;                                                                                                                                                                                                                                                                                                                                                 |
| 3002.75             | 5/2              | 1239.59            | $\leq 1$      | 1763.14 | 5/2 <sup>+</sup> |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                     |                  | 1783.31            | <3            | 1219.39 | 1/2 <sup>+</sup> |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                     |                  | 3002.61            | 100           | 0       | 3/2 <sup>+</sup> | D+Q                | -0.22 1    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 3162.99             | 7/2 <sup>-</sup> | 160.24             | 1.7 2         | 3002.75 | 5/2              |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                     |                  | 517.29             | 8 1           | 2645.70 | (5/2,7/2)        |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                     |                  | 1399.82            | 0.30 4        | 1763.14 | 5/2 <sup>+</sup> | E1+M2              | +0.44 12   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                     |                  | 1943.54            | <0.2          | 1219.39 | 1/2 <sup>+</sup> |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                     |                  | 3162.84            | 90 1          | 0       | 3/2 <sup>+</sup> | M2+E3              | +0.25 4    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$        | $E_\gamma^\dagger$                         | $I_\gamma^\#$          | $E_f$                                                                     | $J_f^\pi$ | Mult. <sup>b</sup> | $\delta^b$ | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------|------------------|--------------------------------------------|------------------------|---------------------------------------------------------------------------|-----------|--------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     |                  |                                            |                        |                                                                           |           |                    |            | magnetic character from $\gamma(\text{pol})$ in <b>1966Wa09</b> and <b>1967Wa22</b> .<br>$\delta$ : from <b>1966Az01</b> . Others: +0.25 5 ( <b>1967Ta08</b> ); +0.16 1 ( <b>1966Wa09,1967Wa22</b> ).<br>Asymmetry ratio $R=1.426$ 7, $N_{90}/N_0=1.25$ 11 ( <b>1971Wi13</b> ).<br>$\text{POL}=-0.75$ 20, $\Delta\pi=\text{yes}$ ( <b>1967Wa22</b> ).<br>$\text{POL}=-1.27$ 10, $\Delta\pi=\text{yes}$ ( <b>1966Wa09</b> ).<br>$0.00<\text{POL}<0.384$ ( <b>1966Az01,1967Ta08</b> ).<br>$A_2=+0.40$ 8, $A_4=-0.23$ 8 ( <b>1971Wi13</b> ).<br>$A_2=+0.25$ 3, $A_4=+0.02$ 5 ( <b>1967Ta08</b> ).<br>$A_2=+0.55$ 3, $A_4=-0.008$ 38 ( <b>1966Az01</b> ). |
| 3918.48             | 3/2 <sup>+</sup> | 1272.76<br>2155.27                         | <3<br>19.5 23          | 2645.70 (5/2,7/2)<br>1763.14 5/2 <sup>+</sup>                             |           |                    |            | $I_\gamma$ : weighted average of 17 3 ( <b>1972Hu10</b> ), 25 3 ( <b>1976Me12</b> ), and 18.0 21 ( <b>1976Sp09</b> ).<br>$I_\gamma$ : weighted average of 83 3 ( <b>1972Hu10</b> ), 75 3 ( <b>1976Me12</b> ), and 82.0 10 ( <b>1976Sp09</b> ).<br>Mult.: D+Q from $\gamma(\theta)$ in <b>1976Sp09</b> ;<br>$\delta=-0.21$ 2 or +20 +10-5 ( <b>1976Sp09</b> ).<br>E1+M2 ruled by RUL.                                                                                                                                                                                                                                                                  |
|                     |                  | 3918.25                                    | 81.5 15                | 0 3/2 <sup>+</sup>                                                        |           | M1+E2              |            | $A_2=+0.07$ 2, $A_4=-0.01$ 3 ( <b>1976Sp09</b> ).<br>$I_\gamma$ : weighted average of 7 3 ( <b>1972Hu10</b> ) and 8 2 ( <b>1976Me12</b> ).<br>$I_\gamma$ : weighted average of 93 3 ( <b>1972Hu10</b> ) and 92 2 ( <b>1976Me12</b> ).                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 3942.9              |                  | 1297.2                                     | 8 2                    | 2645.70 (5/2,7/2)                                                         |           |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                     |                  | 2179.7                                     | 92 2                   | 1763.14 5/2 <sup>+</sup>                                                  |           |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                     |                  | 2723.4 <sup>c</sup><br>3942.7 <sup>c</sup> | <6<br><10              | 1219.39 1/2 <sup>+</sup><br>0 3/2 <sup>+</sup>                            |           |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 3967.3              |                  | 1273.4                                     | 2.1 5                  | 2693.9 (3/2,5/2)                                                          |           |                    |            | $I_\gamma$ : weighted average of 5 3 ( <b>1972Hu10</b> ) and 2.0 5 ( <b>1976Me12</b> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                     |                  | 1321.6 <sup>c</sup><br>2204.1<br>2747.8    | <2<br>$\leq 1$<br>79 5 | 2645.70 (5/2,7/2)<br>1763.14 5/2 <sup>+</sup><br>1219.39 1/2 <sup>+</sup> |           |                    |            | $I_\gamma$ : from <b>1972Hu10</b> . Other: <2 ( <b>1976Me12</b> ).<br>$I_\gamma$ : weighted average of 74 5 ( <b>1972Hu10</b> ) and 83 5 ( <b>1976Me12</b> ).<br>$I_\gamma$ : weighted average of 21 3 ( <b>1972Hu10</b> ) and 15 5 ( <b>1976Me12</b> ).                                                                                                                                                                                                                                                                                                                                                                                              |
|                     |                  | 3967.1                                     | 19 3                   | 0 3/2 <sup>+</sup>                                                        |           |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 4059.2              |                  | 1365.3<br>1413.5 <sup>c</sup><br>2296.0    | 1.2 4<br><1<br>4.8 15  | 2693.9 (3/2,5/2)<br>2645.70 (5/2,7/2)<br>1763.14 5/2 <sup>+</sup>         |           |                    |            | $E_\gamma$ : other: 2296 7 ( <b>1970Ko33</b> ).<br>$I_\gamma$ : other: 3 2 ( <b>1972Hu10</b> ).<br>$E_\gamma$ : other: 2838 7 ( <b>1970Ko33</b> ).<br>$I_\gamma$ : other: 97 2 ( <b>1972Hu10</b> ). 45 5 ( <b>1971Wi13</b> ).<br>$E_\gamma$ : other: 4058 7 ( <b>1970Ko33</b> ).<br>$I_\gamma$ : other: 20 2 ( <b>1971Wi13</b> ).                                                                                                                                                                                                                                                                                                                     |
|                     |                  | 2839.7                                     | 94 2                   | 1219.39 1/2 <sup>+</sup>                                                  |           |                    |            | $I_\gamma$ : weighted average of 50 20 ( <b>1972Hu10</b> ), 44 7 ( <b>1976Me12</b> ), and 48 3 ( <b>1976Sp09</b> ).<br>Mult., $\delta$ : from $\gamma(\theta)$ in <b>1976Sp09</b> .<br>$A_2=+0.56$ 3, $A_4=+0.05$ 4 ( <b>1976Sp09</b> ).                                                                                                                                                                                                                                                                                                                                                                                                              |
|                     |                  | 4059.0                                     | <2                     | 0 3/2 <sup>+</sup>                                                        |           |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 4113.7              | 7/2              | 2350.5                                     | 47 3                   | 1763.14 5/2 <sup>+</sup>                                                  |           | D+Q                | -0.16 2    | $I_\gamma$ : weighted average of 50 20 ( <b>1972Hu10</b> ), 44 7 ( <b>1976Me12</b> ), and 48 3 ( <b>1976Sp09</b> ).<br>Mult., $\delta$ : from $\gamma(\theta)$ in <b>1976Sp09</b> .<br>$A_2=+0.56$ 3, $A_4=+0.05$ 4 ( <b>1976Sp09</b> ).                                                                                                                                                                                                                                                                                                                                                                                                              |
|                     |                  | 2894.2 <sup>c</sup><br>4113.4              | <10<br>53 3            | 1219.39 1/2 <sup>+</sup><br>0 3/2 <sup>+</sup>                            |           | Q(+O)              | 0.00 10    | $I_\gamma$ : weighted average of 50 20 ( <b>1972Hu10</b> ) and 56 7 ( <b>1976Me12</b> ), and 52 3 ( <b>1976Sp09</b> ).<br>Mult., $\delta$ : Q(+O) from $\gamma(\theta)$ in <b>1976Sp09</b> .<br>$A_2=+0.46$ 25, $A_4=-0.23$ 25 ( <b>1976Sp09</b> ).                                                                                                                                                                                                                                                                                                                                                                                                   |

Continued on next page (footnotes at end of table)



$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$ | $E_\gamma^\dagger$   | $I_\gamma^\#$ | $E_f$   | $J_f^\pi$        | Mult. <sup>b</sup> | $\delta^b$ | Comments                                                                                                                      |
|---------------------|-----------|----------------------|---------------|---------|------------------|--------------------|------------|-------------------------------------------------------------------------------------------------------------------------------|
| 4173.45             |           | 1479.5               | 26 8          | 2693.9  | (3/2,5/2)        |                    |            | $I_\gamma$ : weighted average of 27 8 (1972Hu10) and 25 8 (1976Me12).                                                         |
|                     |           | 1527.71              | <10           | 2645.70 | (5/2,7/2)        |                    |            |                                                                                                                               |
|                     |           | 2410.22              | 16 5          | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : weighted average of 18 6 (1972Hu10) and 15 5 (1976Me12).                                                         |
|                     |           | 2953.93              | $\leq 3$      | 1219.39 | 1/2 <sup>+</sup> |                    |            | $I_\gamma$ : from 1972Hu10. Other: <4 (1976Me12); 32 3 (1971Wi13).                                                            |
|                     |           | 4173.18              | 58 10         | 0       | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : weighted average of 55 10 (1972Hu10) and 60 10 (1976Me12).<br>$A_2=-0.01$ 3, $A_4=-0.006$ 40 (1967Ta08).         |
| 4177.9              | 3/2       | 1175.1               | <0.5          | 3002.75 | 5/2              |                    |            |                                                                                                                               |
|                     |           | 1484.0               | 8 2           | 2693.9  | (3/2,5/2)        |                    |            | $I_\gamma$ : weighted average of 7 2 (1972Hu10) and 10 3 (1976Me12).                                                          |
|                     |           | 1532.2               | <0.5          | 2645.70 | (5/2,7/2)        |                    |            |                                                                                                                               |
|                     |           | 2414.7               | <1            | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : other $\leq 1$ (1972Hu10).                                                                                       |
|                     |           | 2958.4               | 31 5          | 1219.39 | 1/2 <sup>+</sup> | D+Q                | +0.11 4    | $I_\gamma$ : weighted average of 31 5 (1972Hu10) and 30 5 (1976Me12).                                                         |
|                     |           | 4177.6               | 61 5          | 0       | 3/2 <sup>+</sup> | D+Q                | +0.06 3    | Mult., $\delta$ : from $\gamma(\theta)$ in 1967Wa22.<br>$I_\gamma$ : weighted average of 62 6 (1972Hu10) and 60 5 (1976Me12). |
|                     |           |                      |               |         |                  |                    |            | Mult., $\delta$ : from $\gamma(\theta)$ in 1967Wa22.<br>POL=+0.12 31 (1967Wa22).                                              |
| 4347.76             | 5/2,9/2   | 1184.75              | 70 2          | 3162.99 | 7/2 <sup>-</sup> |                    |            | $I_\gamma$ : weighted average of 70 5 (1972Hu10), 72 5 (1976Me12), and 69 2 (1976Sp09).                                       |
|                     |           | 1344.98 <sup>c</sup> | <4            | 3002.75 | 5/2              |                    |            |                                                                                                                               |
|                     |           | 1653.8 <sup>c</sup>  | <4            | 2693.9  | (3/2,5/2)        |                    |            |                                                                                                                               |
|                     |           | 1702.02              | 31 2          | 2645.70 | (5/2,7/2)        | D+Q                | -0.018 12  | $I_\gamma$ : weighted average of 30 5 (1972Hu10), 28 5 (1976Me12), and 31 2 (1976Sp09).                                       |
|                     |           |                      |               |         |                  |                    |            | Mult., $\delta$ : D+Q from $\gamma(\theta)$ in 1976Sp09.<br>$A_2=-0.34$ 3, $A_4=-0.00$ 4 (1976Sp09).                          |
|                     |           | 2584.52              | $\leq 3$      | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : from 1972Hu10. Other: <5 (1976Me12).                                                                             |
|                     |           | 3128.22 <sup>c</sup> | <4            | 1219.39 | 1/2 <sup>+</sup> |                    |            |                                                                                                                               |
|                     |           | 4347.47              | $\leq 5$      | 0       | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : from 1972Hu10. Other: <10 (1976Me12).                                                                            |
| 4624.4              |           | 4624.1               | 85 10         | 0       | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 100 (1972Hu10).                                                                                           |
| 4769.86             |           | 1606.83              | 65 10         | 3162.99 | 7/2 <sup>-</sup> |                    |            | $I_\gamma$ : other: 53 10 (1972Hu10) and 47% of $I_\gamma$ to unknown levels (1972Hu10).                                      |
|                     |           | 1767.06              | 35 10         | 3002.75 | 5/2              |                    |            |                                                                                                                               |
|                     |           | 2075.9               | <10           | 2693.9  | (3/2,5/2)        |                    |            |                                                                                                                               |
|                     |           | 3550.28              | <10           | 1219.39 | 1/2 <sup>+</sup> |                    |            |                                                                                                                               |
|                     |           | 4769.51              | <10           | 0       | 3/2 <sup>+</sup> |                    |            |                                                                                                                               |
| 4839.10             |           | 1836.30              | <7            | 3002.75 | 5/2              |                    |            |                                                                                                                               |
|                     |           | 2145.1               | <4            | 2693.9  | (3/2,5/2)        |                    |            |                                                                                                                               |
|                     |           | 2193.33              | <3            | 2645.70 | (5/2,7/2)        |                    |            |                                                                                                                               |
|                     |           | 3075.82              | 59 5          | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : weighted average of 50 20 (1972Hu10) and 60 5 (1976Me12).<br>Other: 48 4 (1976Sp09).                             |
|                     |           | 3619.51              | <7            | 1219.39 | 1/2 <sup>+</sup> |                    |            |                                                                                                                               |
|                     |           | 4838.74              | 41 5          | 0       | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : weighted average of 50 20 (1972Hu10) and 40 5 (1976Me12).<br>Other: 18 3 (1976Sp09).                             |
| 4854.4              |           | 3634.8               | 75 5          | 1219.39 | 1/2 <sup>+</sup> |                    |            |                                                                                                                               |
|                     |           | 4854.0               | 25 5          | 0       | 3/2 <sup>+</sup> |                    |            |                                                                                                                               |
| 4880.96             |           | 1878.16              | 9 3           | 3002.75 | 5/2              |                    |            |                                                                                                                               |
|                     |           | 2187.0               | <10           | 2693.9  | (3/2,5/2)        |                    |            |                                                                                                                               |

Continued on next page (footnotes at end of table)



$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$ | $E_\gamma^\dagger$   | $I_\gamma^\#$    | $E_f$   | $J_f^\pi$        | Mult. <sup>b</sup> | $\delta^b$ | Comments                                                                                                                                                                                            |
|---------------------|-----------|----------------------|------------------|---------|------------------|--------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4880.96             |           | 2235.18              | 29 5             | 2645.70 | (5/2,7/2)        |                    |            | $I_\gamma$ : other: 30 10 (1972Hu10).                                                                                                                                                               |
|                     |           | 3117.67              | 62 5             | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 70 10 (1972Hu10).                                                                                                                                                               |
|                     |           | 3661.36 <sup>c</sup> | <4               | 1219.39 | 1/2 <sup>+</sup> |                    |            |                                                                                                                                                                                                     |
|                     |           | 4880.60              | <5               | 0       | 3/2 <sup>+</sup> |                    |            |                                                                                                                                                                                                     |
| 5010.1              |           | 836.6                | <25 <sup>a</sup> | 4173.45 |                  |                    |            |                                                                                                                                                                                                     |
|                     |           | 950.9                | <10 <sup>a</sup> | 4059.2  |                  |                    |            |                                                                                                                                                                                                     |
|                     |           | 1847.1               | <50 <sup>a</sup> | 3162.99 | 7/2 <sup>-</sup> |                    |            |                                                                                                                                                                                                     |
|                     |           | 2007.3               | <20 <sup>a</sup> | 3002.75 | 5/2              |                    |            |                                                                                                                                                                                                     |
|                     |           | 2316.1               | <25 <sup>a</sup> | 2693.9  | (3/2,5/2)        |                    |            |                                                                                                                                                                                                     |
|                     |           | 2364.3               | <5               | 2645.70 | (5/2,7/2)        |                    |            |                                                                                                                                                                                                     |
|                     |           | 3246.8               | <10              | 1763.14 | 5/2 <sup>+</sup> |                    |            |                                                                                                                                                                                                     |
|                     |           | 3790.5               | <7               | 1219.39 | 1/2 <sup>+</sup> |                    |            |                                                                                                                                                                                                     |
|                     |           | 5009.7               | 100              | 0       | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : from 1976Me12 and 1972Hu10.                                                                                                                                                            |
| 5163.34             | 7/2       | 2000.29              | 40 5             | 3162.99 | 7/2 <sup>-</sup> | D+Q                | +0.44 20   | $I_\gamma$ : weighted average of 40 20 (1972Hu10), 36 5 (1976Me12), and 54 10 (1976Sp09).<br>Mult., $\delta$ : D+Q from $\gamma(\theta)$ in 1976Sp09.<br>$A_2=+0.52$ 5, $A_4=-0.02$ 6 (1976Sp09).   |
|                     |           | 2160.52              | 10 4             | 3002.75 | 5/2              |                    |            | $I_\gamma$ : weighted average of 10 5 (1972Hu10), 10 4 (1976Me12), and 10 5 (1976Sp09).                                                                                                             |
|                     |           | 3400.02              | 43 5             | 1763.14 | 5/2 <sup>+</sup> | D(+Q)              | 0.00 3     | $I_\gamma$ : weighted average of 50 20 (1972Hu10), 44 5 (1976Me12), and 36 10 (1976Sp09).<br>Mult., $\delta$ : D(+Q) from $\gamma(\theta)$ in 1976Sp09.<br>$A_2=-0.23$ 6, $A_4=-0.08$ 7 (1976Sp09). |
|                     |           | 3943.71              | <10              | 1219.39 | 1/2 <sup>+</sup> |                    |            |                                                                                                                                                                                                     |
|                     |           | 5162.93              | <10              | 0       | 3/2 <sup>+</sup> |                    |            |                                                                                                                                                                                                     |
| 5215.8              |           | 205.7                | <23 <sup>a</sup> | 5010.1  |                  |                    |            |                                                                                                                                                                                                     |
|                     |           | 1042.3               | <60 <sup>a</sup> | 4173.45 |                  |                    |            |                                                                                                                                                                                                     |
|                     |           | 1156.6               | <10 <sup>a</sup> | 4059.2  |                  |                    |            |                                                                                                                                                                                                     |
|                     |           | 2052.8               | <15 <sup>a</sup> | 3162.99 | 7/2 <sup>-</sup> |                    |            |                                                                                                                                                                                                     |
|                     |           | 2213.0               | <25 <sup>a</sup> | 3002.75 | 5/2              |                    |            |                                                                                                                                                                                                     |
|                     |           | 2521.8               | <15 <sup>a</sup> | 2693.9  | (3/2,5/2)        |                    |            |                                                                                                                                                                                                     |
|                     |           | 2570.0               | <15 <sup>a</sup> | 2645.70 | (5/2,7/2)        |                    |            |                                                                                                                                                                                                     |
|                     |           | 3452.5               | <6               | 1763.14 | 5/2 <sup>+</sup> |                    |            |                                                                                                                                                                                                     |
|                     |           | 3996.2               | <4               | 1219.39 | 1/2 <sup>+</sup> |                    |            |                                                                                                                                                                                                     |
|                     |           | 5215.4               | 100              | 0       | 3/2 <sup>+</sup> |                    |            |                                                                                                                                                                                                     |
| 5403.5              |           | 2400.7               | 25 5             | 3002.75 | 5/2              |                    |            | $I_\gamma$ : other: 25 10 (1972Hu10).                                                                                                                                                               |
|                     |           | 3640.2               | <10              | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 25 10 from 1972Hu10.                                                                                                                                                            |
|                     |           | 4183.8               | 55 10            | 1219.39 | 1/2 <sup>+</sup> |                    |            |                                                                                                                                                                                                     |
|                     |           | 5403.1               | <7               | 0       | 3/2 <sup>+</sup> |                    |            |                                                                                                                                                                                                     |
| 5586                |           | 2940                 | 40 10            | 2645.70 | (5/2,7/2)        |                    |            | $I_\gamma$ : weighted average of 40 20 (1972Hu10) and 40 10 (1976Me12).                                                                                                                             |
|                     |           | 3823                 | 60 10            | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : weighted average of 60 20 (1972Hu10) and 60 10 (1976Me12).                                                                                                                             |
|                     |           | 4366                 | <10              | 1219.39 | 1/2 <sup>+</sup> |                    |            |                                                                                                                                                                                                     |
|                     |           | 5586                 | <10              | 0       | 3/2 <sup>+</sup> |                    |            |                                                                                                                                                                                                     |
| 5599.7              |           | 3836.3               | 27 10            | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : weighted average of 20 10 (1972Hu10) and 33 10 (1976Me12).                                                                                                                             |
|                     |           | 5599.2               | 74 10            | 0       | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : weighted average of 80 10 (1972Hu10) and 67 10 (1976Me12).                                                                                                                             |
| 5646                |           | 2483                 | 100              | 3162.99 | 7/2 <sup>-</sup> |                    |            | $I_\gamma$ : from 1972Hu10, 1976Me12, 1971Pr11.                                                                                                                                                     |
|                     |           | 2643                 |                  | 3002.75 | 5/2              |                    |            | $I_\gamma$ : 80 from 2001Vo24 only; not observed by others.                                                                                                                                         |
|                     |           | 3883                 |                  | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : 6 from 2001Vo24 only; not observed by others.                                                                                                                                          |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$ | $E_\gamma^\dagger$ | $I_\gamma^\#$ | $E_f$   | $J_f^\pi$        | Mult. <sup>b</sup> | $\delta^b$ | Comments                                                                 |
|---------------------|-----------|--------------------|---------------|---------|------------------|--------------------|------------|--------------------------------------------------------------------------|
| 5646                |           | 5646               | <8            | 0       | 3/2 <sup>+</sup> |                    |            |                                                                          |
| 5655                |           | 2652               | 80 5          | 3002.75 | 5/2              |                    |            |                                                                          |
|                     |           | 3892               | 6 2           | 1763.14 | 5/2 <sup>+</sup> |                    |            |                                                                          |
|                     |           | 4435               | <6            | 1219.39 | 1/2 <sup>+</sup> |                    |            |                                                                          |
| 5683                |           | 1505               | 60 10         | 4177.9  | 3/2              |                    |            |                                                                          |
| 5723.52             |           | 3029.5             | 24 3          | 2693.9  | (3/2,5/2)        |                    |            | $I_\gamma$ : from 1976Sp09.                                              |
|                     |           | 3960.14            | 38 4          | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : from 1976Sp09.                                              |
| 5758                |           | 1580               | 30 10         | 4177.9  | 3/2              |                    |            |                                                                          |
|                     |           | 4538               | <6            | 1219.39 | 1/2 <sup>+</sup> |                    |            |                                                                          |
|                     |           | 5758               | 45 10         | 0       | 3/2 <sup>+</sup> |                    |            |                                                                          |
| 5806                |           | 5806               | 100           | 0       | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : from 1972Hu10 and 1976Me12.                                 |
| 6106.2              |           | 4342.8             | 6 2           | 1763.14 | 5/2 <sup>+</sup> |                    |            |                                                                          |
|                     |           | 4886.4             | 36 10         | 1219.39 | 1/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 40 10 (1972Hu10).                                    |
|                     |           | 6105.6             | 58 10         | 0       | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 60 10 (1972Hu10).                                    |
| 6181                |           | 6180               | 45 10         | 0       | 3/2 <sup>+</sup> |                    |            |                                                                          |
| 6492                |           | 6491               | 100           | 0       | 3/2 <sup>+</sup> |                    |            |                                                                          |
| 7066.2              |           | 1902.8             | 0.9           | 5163.34 | 7/2              |                    |            |                                                                          |
|                     |           | 2892.6             | 0.5           | 4173.45 |                  |                    |            |                                                                          |
|                     |           | 2952.4             | 0.6           | 4113.7  | 7/2              |                    |            |                                                                          |
|                     |           | 3006.9             | 1.4           | 4059.2  |                  |                    |            | $I_\gamma$ : other: 1.0 5 from 1972Hu10.                                 |
|                     |           | 3903.0             | 6             | 3162.99 | 7/2 <sup>-</sup> |                    |            | $I_\gamma$ : other: 6.0 6 from 1972Hu10.                                 |
|                     |           | 4063.2             | 3             | 3002.75 | 5/2              |                    |            | $I_\gamma$ : other: 2 1 from 1972Hu10.                                   |
|                     |           | 4372.0             | 5             | 2693.9  | (3/2,5/2)        |                    |            | $I_\gamma$ : other: 6.0 6 from 1972Hu10.                                 |
|                     |           | 4420.2             | 1.6           | 2645.70 | (5/2,7/2)        |                    |            | $I_\gamma$ : other: 3.0 15 from 1972Hu10.                                |
|                     |           | 5302.6             | 14            | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 16.0 16 from 1972Hu10.                               |
|                     |           | 5846.3             | 18            | 1219.39 | 1/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 18.0 18 from 1972Hu10.                               |
|                     |           | 7065.4             | 49            | 0       | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 48.0 48 from 1972Hu10.                               |
| 7103.4              | 3/2       | 2925.4             | 1.5           | 4177.9  | 3/2              |                    |            |                                                                          |
|                     |           | 3184.8             | 1.5           | 3918.48 | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 2 1 (1972Hu10).                                      |
|                     |           | 4100.4             | 3             | 3002.75 | 5/2              |                    |            | $I_\gamma$ : other: 3.0 15 from 1972Hu10.                                |
|                     |           | 4409.2             | 11            | 2693.9  | (3/2,5/2)        | D(+Q)              | 0.00 3     | $I_\gamma$ , Mult., $\delta$ : from 1976Me12. Other: 13.0 13 (1972Hu10). |
|                     |           |                    |               |         |                  |                    |            | $A_2=+0.34$ 7, $A_4=+0.03$ 11 (1976Me12).                                |
|                     |           | 5339.8             | 4             | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 4 2 from 1972Hu10.                                   |
|                     |           | 5883.5             | 64            | 1219.39 | 1/2 <sup>+</sup> | D(+Q)              | 0.00 3     | $I_\gamma$ , Mult., $\delta$ : from 1976Me12. Other: 67.0 67 (1972Hu10). |
|                     |           |                    |               |         |                  |                    |            | $A_2=-0.53$ 8, $A_4=+0.01$ 11 (1976Me12).                                |
|                     |           | 7102.6             | 15            | 0       | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 11.0 11 (1972Hu10).                                  |
|                     |           |                    |               |         |                  |                    |            | $A_2=+0.54$ 5, $A_4=-0.01$ 8 (1976Me12).                                 |
| 7178.6              | 1/2       | 1072.4             | 3             | 6106.2  |                  |                    |            | $I_\gamma$ : other: 4 2 (1972Hu10).                                      |
|                     |           | 1775.1             | 1.7           | 5403.5  |                  |                    |            |                                                                          |
|                     |           | 2339.4             | 1.5           | 4839.10 |                  |                    |            | $I_\gamma$ : other: 2 (2001Vo24).                                        |
|                     |           | 3005.0             | 0.8           | 4173.45 |                  |                    |            | $I_\gamma$ : other: 1 (2001Vo24).                                        |
|                     |           | 3119.3             | 22            | 4059.2  |                  |                    |            | $I_\gamma$ : others: 22 (2001Vo24), 24.0 24 (1972Hu10).                  |
|                     |           |                    |               |         |                  |                    |            | $A_2=-0.02$ 4, $A_4=+0.10$ 66 (1976Me12).                                |
|                     |           |                    |               |         |                  |                    |            | $A_2=-0.01$ 4, $A_4=+0.09$ 24 (2001Vo24).                                |
|                     |           | 3211.1             | 9             | 3967.3  |                  |                    |            | $I_\gamma$ : others: 9 (2001Vo24), 10.0 10 (1972Hu10).                   |
|                     |           | 3260.0             | 3             | 3918.48 | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : others: 3 (2001Vo24), 2 1 (1972Hu10).                       |
|                     |           | 4484.4             | 4             | 2693.9  | (3/2,5/2)        |                    |            | $I_\gamma$ : others: 4 (2001Vo24), 4 2 (1972Hu10).                       |
|                     |           | 5958.7             | 17            | 1219.39 | 1/2 <sup>+</sup> |                    |            | $I_\gamma$ : others: 17 (2001Vo24), 16.0 16 (1972Hu10).                  |
|                     |           |                    |               |         |                  |                    |            | $A_2=-0.01$ 3, $A_4=+0.06$ 7 (2001Vo24).                                 |
|                     |           |                    |               |         |                  |                    |            | $A_2=-0.01$ 5, $A_4=+0.07$ 8 (1976Me12).                                 |
|                     |           | 7177.8             | 38            | 0       | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : others: 38 (2001Vo24), 40.0 40 (1972Hu10).                  |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$        | $E_\gamma^\dagger$ | $I_\gamma^\#$ | $E_f$   | $J_f^\pi$    | Mult. <sup>b</sup> | $\delta^b$ | Comments                                                                             |
|---------------------|------------------|--------------------|---------------|---------|--------------|--------------------|------------|--------------------------------------------------------------------------------------|
|                     |                  |                    |               |         |              |                    |            | $A_2=-0.01$ 6, $A_4=-0.02$ 7 (1976Me12).<br>$A_2=-0.02$ 4, $A_4=-0.01$ 5 (2001Vo24). |
| 7185                |                  | 1969               | $<0.5^a$      | 5215.8  |              |                    |            |                                                                                      |
|                     |                  | 2175               | $<0.5^a$      | 5010.1  |              |                    |            |                                                                                      |
|                     |                  | 3011               | $<5^a$        | 4173.45 |              |                    |            |                                                                                      |
|                     |                  | 3126               | $26^a$ 6      | 4059.2  |              |                    |            |                                                                                      |
|                     |                  | 4022               | $<3^a$        | 3162.99 | $7/2^-$      |                    |            |                                                                                      |
|                     |                  | 4182               | $<3^a$        | 3002.75 | $5/2$        |                    |            |                                                                                      |
|                     |                  | 4491               | $2^a$ 1       | 2693.9  | $(3/2, 5/2)$ |                    |            |                                                                                      |
|                     |                  | 4539               | $6^a$ 2       | 2645.70 | $(5/2, 7/2)$ |                    |            |                                                                                      |
|                     |                  | 5421               | $4^a$ 3       | 1763.14 | $5/2^+$      |                    |            |                                                                                      |
|                     |                  | 5965               | $16^a$ 7      | 1219.39 | $1/2^+$      |                    |            |                                                                                      |
| 7194.6              | 1/2,3/2          | 7184               | $46^a$ 10     | 0       | $3/2^+$      |                    |            |                                                                                      |
|                     |                  | 2184.4             | 0.8           | 5010.1  |              |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                                |
|                     |                  | 2340.1             | 2             | 4854.4  |              |                    |            | $I_\gamma$ : other: 2 1 (1972Hu10).                                                  |
|                     |                  | 2355.4             | 1.8           | 4839.10 |              |                    |            |                                                                                      |
|                     |                  | 3016.6             | 16            | 4177.9  | $3/2$        |                    |            | $I_\gamma$ : other: 15.0 15 (1972Hu10).                                              |
|                     |                  | 3135.3             | 3             | 4059.2  |              |                    |            | $I_\gamma$ : other: 3.0 15 (1972Hu10).                                               |
|                     |                  | 3227.1             | 1.8           | 3967.3  |              |                    |            | $I_\gamma$ : other: 2 1 (1972Hu10).                                                  |
|                     |                  | 3276.0             | 4             | 3918.48 | $3/2^+$      |                    |            | $I_\gamma$ : other: 5.0 25 (1972Hu10).                                               |
|                     |                  | 4500.4             | 1.1           | 2693.9  | $(3/2, 5/2)$ |                    |            |                                                                                      |
|                     |                  | 5431.0             | 0.5           | 1763.14 | $5/2^+$      |                    |            |                                                                                      |
| 7225.5              | 5/2              | 5974.7             | 67            | 1219.39 | $1/2^+$      |                    |            | $I_\gamma$ : other: 72.0 72 (1972Hu10).                                              |
|                     |                  | 7193.8             | 2             | 0       | $3/2^+$      |                    |            |                                                                                      |
|                     |                  | 1119.3             | 1.8           | 6106.2  |              |                    |            | $I_\gamma$ : other: 3.0 15 (1972Hu10).                                               |
|                     |                  | 1420               |               | 5806    |              |                    |            | $I_\gamma$ : 1.0 5 (1972Hu10).                                                       |
|                     |                  | 1543               | 2.0           | 5683    |              |                    |            | $I_\gamma$ : other: 2 1 (1972Hu10).                                                  |
|                     |                  | 1571               | 0.8           | 5655    |              |                    |            |                                                                                      |
|                     |                  | 1822.0             | 0.9           | 5403.5  |              |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                                |
|                     |                  | 2009.6             | 7             | 5215.8  |              |                    |            | $I_\gamma$ : other: 6.0 30 (1972Hu10).                                               |
|                     |                  | 3047.5             | 8             | 4177.9  | $3/2$        |                    |            | $I_\gamma$ : other: 7.0 35 (1972Hu10).                                               |
|                     |                  | 3166.2             | 2             | 4059.2  |              |                    |            |                                                                                      |
| 7234.0              | 5/2 <sup>+</sup> | 3306.9             | 1.5           | 3918.48 | $3/2^+$      |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                                |
|                     |                  | 4222.5             |               | 3002.75 | $5/2$        |                    |            | $I_\gamma$ : 3.0 15 (1972Hu10).                                                      |
|                     |                  | 4531.3             | 10            | 2693.9  | $(3/2, 5/2)$ | D+Q                | +0.36 4    | $I_\gamma$ , Mult., $\delta$ : from 1976Me12. Other: 10.0 10 (1972Hu10).             |
|                     |                  |                    |               |         |              |                    |            | $A_2=+0.43$ 8, $A_4=-0.21$ 9 (1976Me12).                                             |
|                     |                  | 5461.9             | 7             | 1763.14 | $5/2^+$      | D+Q                | -0.36 7    | $I_\gamma$ , Mult., $\delta$ : from 1976Me12. Other: 8.0 8 (1972Hu10).               |
|                     |                  |                    |               |         |              |                    |            | $A_2=+0.02$ 10, $A_4=-0.09$ 17 (1976Me12).                                           |
|                     |                  | 7224.7             | 59            | 0       | $3/2^+$      | D+Q                | +0.36 4    | $I_\gamma$ , Mult., $\delta$ : from 1976Me12. Other: 58.0 58 (1972Hu10).             |
|                     |                  |                    |               |         |              |                    |            | $A_2=+0.40$ 7, $A_4=+0.04$ 12 (1976Me12).                                            |
|                     |                  | 1579               | 0.2           | 5655    |              |                    |            |                                                                                      |
|                     |                  | 3056.0             |               | 4177.9  | $3/2$        |                    |            | $I_\gamma$ : 1.0 5 (1972Hu10).                                                       |
| 7272.7              | (1/2,3/2)        | 4539.8             | 1.8           | 2693.9  | $(3/2, 5/2)$ |                    |            | $I_\gamma$ : other: 2 1 (1972Hu10).                                                  |
|                     |                  | 4588.0             | 3             | 2645.70 | $(5/2, 7/2)$ |                    |            | $I_\gamma$ : other: 2 1 (1972Hu10).                                                  |
|                     |                  | 5470.4             | 0.2           | 1763.14 | $5/2^+$      |                    |            |                                                                                      |
|                     |                  | 6014.1             | 1.8           | 1219.39 | $1/2^+$      |                    |            | $I_\gamma$ : other: 2 1 (1972Hu10).                                                  |
|                     |                  | 7233.2             | 93            | 0       | $3/2^+$      | D+Q                | -0.10      | $I_\gamma$ : other: 93.0 93 (1972Hu10).                                              |
|                     |                  |                    |               |         |              |                    |            | Mult., $\delta$ : from $\gamma(\theta)$ in 1960An06.                                 |
|                     |                  | 1515               | 1.2           | 5758    |              |                    |            | $I_\gamma$ : other: 1.5 8 (1972Hu10).                                                |
|                     |                  | 1869.2             | 0.2           | 5403.5  |              |                    |            | $I_\gamma$ : other: 0.5 3 (1972Hu10).                                                |
|                     |                  | 2262.5             | 1.4           | 5010.1  |              |                    |            | $I_\gamma$ : other: 1.5 8 (1972Hu10).                                                |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$ | $E_\gamma^\dagger$ | $I_\gamma^\#$ | $E_f$   | $J_f^\pi$        | Mult. <sup>b</sup> | $\delta^b$ | Comments                                                                                                    |
|---------------------|-----------|--------------------|---------------|---------|------------------|--------------------|------------|-------------------------------------------------------------------------------------------------------------|
| 7272.7              | (1/2,3/2) | 2433.5             | 1.0           | 4839.10 |                  |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                                                       |
|                     |           | 3099.1             | 0.4           | 4173.45 |                  |                    |            | $I_\gamma$ : other: 0.5 3 (1972Hu10).                                                                       |
|                     |           | 3213.3             | 0.9           | 4059.2  |                  |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                                                       |
|                     |           | 3305.2             | 0.8           | 3967.3  |                  |                    |            | $I_\gamma$ : other: 1.0 5 from 1972Hu10.                                                                    |
|                     |           | 4578.5             | 1.1           | 2693.9  | (3/2,5/2)        |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                                                       |
|                     |           | 6052.8             | 24            | 1219.39 | 1/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 23.0 23 (1972Hu10).                                                                     |
|                     |           | 7271.9             | 69            | 0       | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 69.0 69 (1972Hu10).                                                                     |
| 7362.0              | 3/2       | 1707               | 0.3           | 5655    |                  |                    |            | $I_\gamma$ : other: 0.5 3 (1972Hu10).                                                                       |
|                     |           | 2351.8             | 0.6           | 5010.1  |                  |                    |            | $I_\gamma$ : other: 0.5 3 (1972Hu10).                                                                       |
|                     |           | 2522.8             | 1.0           | 4839.10 |                  |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                                                       |
|                     |           | 3183.9             | 0.5           | 4177.9  | 3/2              |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                                                       |
|                     |           | 3302.6             | 2             | 4059.2  |                  |                    |            | $I_\gamma$ : others: 3.0 15 (1972Hu10); 4 (1971Wi13).                                                       |
|                     |           | 3394.5             | 3             | 3967.3  |                  |                    |            | $I_\gamma$ : others: 3.0 15 (1972Hu10); 4 (1971Wi13).                                                       |
|                     |           | 3443.3             | 0.8           | 3918.48 | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 0.5 3 (1972Hu10).                                                                       |
|                     |           | 4359.0             | 0.3           | 3002.75 | 5/2              |                    |            | $I_\gamma$ : other: 0.5 3 (1972Hu10).                                                                       |
|                     |           | 4667.8             | 0.5           | 2693.9  | (3/2,5/2)        |                    |            |                                                                                                             |
|                     |           | 5598.4             | 8             | 1763.14 | 5/2 <sup>+</sup> | D(+Q)              |            | $I_\gamma$ : others: 10.0 10 (1972Hu10); 9 (1971Wi13).                                                      |
|                     |           |                    |               |         |                  |                    |            | Mult., $\delta$ : $\delta=-0.01$ 10 or $\delta>+2.5$ (1977Ko34).                                            |
|                     |           | 6142.0             | 73            | 1219.39 | 1/2 <sup>+</sup> | D+Q                |            | $A_2=-0.20$ 24, $A_4=+0.92$ 25 (1977Ko34).                                                                  |
|                     |           |                    |               |         |                  |                    |            | $I_\gamma$ : others: 70.0 70 (1972Hu10); 72 (1971Wi13).                                                     |
|                     |           |                    |               |         |                  |                    |            | Mult., $\delta$ : $\delta=-0.05$ 3 or $+2.5$ 3 (1977Ko34); $\delta=-20$ or $+0.22<\delta<+0.26$ (1971Wi13). |
|                     |           |                    |               |         |                  |                    |            | $A_2=-0.05$ 3, $A_4=-0.04$ 5 (1971Wi13).                                                                    |
|                     |           |                    |               |         |                  |                    |            | $A_2=-0.25$ 11, $A_4=+0.31$ 11 (1977Ko34).                                                                  |
|                     |           | 7361.2             | 10            | 0       | 3/2 <sup>+</sup> | D+Q                |            | $I_\gamma$ : others: 10.0 10 (1972Hu10); 11 (1971Wi13).                                                     |
|                     |           |                    |               |         |                  |                    |            | Mult., $\delta$ : $-0.16<\delta<-0.12$ or $+7.5<\delta<+10$ (1971Wi13).                                     |
|                     |           |                    |               |         |                  |                    |            | $A_2=+0.19$ 3, $A_4=-0.08$ 5 (1971Wi13).                                                                    |
| 7396.0              | 7/2       | 3048.1             | 11            | 4347.76 | 5/2,9/2          |                    |            | $I_\gamma$ : other: 10.0 10 (1972Hu10).                                                                     |
|                     |           | 3282.1             | 8             | 4113.7  | 7/2              |                    |            | $I_\gamma$ : other: 9.0 9 (1972Hu10).                                                                       |
|                     |           | 3452.9             | 5             | 3942.9  |                  |                    |            | $I_\gamma$ : other: 8.0 8 (1972Hu10).                                                                       |
|                     |           | 4232.7             | 47            | 3162.99 | 7/2 <sup>-</sup> | D+Q                | +0.28 25   | $I_\gamma$ : other: 49.0 49 (1972Hu10).                                                                     |
|                     |           |                    |               |         |                  |                    |            | Mult., $\delta$ : from $\gamma(\theta)$ in 1977Ko35.                                                        |
|                     |           | 4393.0             | 14            | 3002.75 | 5/2              | D(+Q)              | +0.011 18  | $I_\gamma$ : other: 14.0 14 (1972Hu10).                                                                     |
|                     |           |                    |               |         |                  |                    |            | Mult., $\delta$ : from $\gamma(\theta)$ in 1977Ko35.                                                        |
|                     |           |                    |               |         |                  |                    |            | $I_\gamma$ : other: 10.0 10 (1972Hu10).                                                                     |
|                     |           | 4750.0             | 13            | 2645.70 | (5/2,7/2)        |                    |            |                                                                                                             |
|                     |           | 5632.4             | 2             | 1763.14 | 5/2 <sup>+</sup> |                    |            |                                                                                                             |
| 7451.1              | 3/2       | 3532.4             | 5.0 25        | 3918.48 | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : from 1972Hu10.                                                                                 |
|                     |           | 4448.1             | 10 1          | 3002.75 | 5/2              |                    |            | $I_\gamma$ : from 1972Hu10.                                                                                 |
|                     |           | 4756.9             | 8.0 8         | 2693.9  | (3/2,5/2)        |                    |            | $I_\gamma$ : from 1972Hu10.                                                                                 |
|                     |           | 5687.5             | 4 2           | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : from 1972Hu10.                                                                                 |
|                     |           | 6231.1             | 73 7          | 1219.39 | 1/2 <sup>+</sup> |                    |            | $I_\gamma$ : from 1972Hu10.                                                                                 |
| 7501.1              |           | 3558.0             | 25 3          | 3942.9  |                  |                    |            | $I_\gamma$ : from 1972Hu10.                                                                                 |
|                     |           | 4498.0             | 13.0 13       | 3002.75 | 5/2              |                    |            | $I_\gamma$ : from 1972Hu10.                                                                                 |
|                     |           | 4855.0             | 55 6          | 2645.70 | (5/2,7/2)        |                    |            | $I_\gamma$ : from 1972Hu10.                                                                                 |
|                     |           | 5737.5             | 7.0 7         | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : from 1972Hu10.                                                                                 |
| 7502.9              |           | 1396.7             | 7             | 6106.2  |                  |                    |            | $I_\gamma$ : from 1976Me12 containing $\gamma$ rays of 7501 and 7503 doublet initial levels.                |
|                     |           |                    |               |         |                  |                    |            | Other: 10.0 10 for 7503 level only (1972Hu10).                                                              |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)**

| $\gamma(^{35}\text{Cl})$ (continued) |                  |                    |               |         |                  |                           |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |                                                                                          |
|--------------------------------------|------------------|--------------------|---------------|---------|------------------|---------------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------------------------------------------------------------------------------------------|
| $E_i(\text{level})$                  | $J_i^\pi$        | $E_\gamma^\dagger$ | $I_\gamma^\#$ | $E_f$   | $J_f^\pi$        | Mult. <sup><i>b</i></sup> | $\delta^b$ | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |        |                                                                                          |
| 7502.9                               |                  | 1903.1             | 2             | 5599.7  |                  |                           |            | $I_\gamma$ : from <a href="#">1976Me12</a> containing $\gamma$ rays of 7501 and 7503 doublet initial levels.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |        |                                                                                          |
|                                      |                  | 2492.7             | 2             | 5010.1  |                  |                           |            | $I_\gamma$ : from <a href="#">1976Me12</a> containing $\gamma$ rays of 7501 and 7503 doublet initial levels.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |        |                                                                                          |
|                                      |                  | 3324.8             | 3             | 4177.9  | 3/2              |                           |            | $I_\gamma$ : from <a href="#">1976Me12</a> containing $\gamma$ rays of 7501 and 7503 doublet initial levels.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |        |                                                                                          |
|                                      |                  | 3443.5             | 50            | 4059.2  |                  |                           |            | $I_\gamma$ : from <a href="#">1976Me12</a> containing $\gamma$ rays of 7501 and 7503 doublet initial levels. Other: 70.0 70 for 7503 level only ( <a href="#">1972Hu10</a> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |        |                                                                                          |
|                                      |                  | 3559.8             | 2             | 3942.9  |                  |                           |            | $I_\gamma$ : from <a href="#">1976Me12</a> containing $\gamma$ rays of 7501 and 7503 doublet initial levels.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |        |                                                                                          |
|                                      |                  | 4339.6             | 4             | 3162.99 | 7/2 <sup>-</sup> |                           |            | $I_\gamma$ : from <a href="#">1976Me12</a> containing $\gamma$ rays of 7501 and 7503 doublet initial levels.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |        |                                                                                          |
|                                      |                  | 4499.8             | 4             | 3002.75 | 5/2              |                           |            | $I_\gamma$ : from <a href="#">1976Me12</a> containing $\gamma$ rays of 7501 and 7503 doublet initial levels.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |        |                                                                                          |
|                                      |                  | 4856.8             | 7             | 2645.70 | (5/2,7/2)        |                           |            | $I_\gamma$ : from <a href="#">1976Me12</a> containing $\gamma$ rays of 7501 and 7503 doublet initial levels.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |        |                                                                                          |
|                                      |                  | 5739.3             | 0.7           | 1763.14 | 5/2 <sup>+</sup> |                           |            | $I_\gamma$ : from <a href="#">1976Me12</a> containing $\gamma$ rays of 7501 and 7503 doublet initial levels.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |        |                                                                                          |
|                                      |                  | 6282.9             | 18            | 1219.39 | 1/2 <sup>+</sup> |                           |            | $I_\gamma$ : from <a href="#">1976Me12</a> containing $\gamma$ rays of 7501 and 7503 doublet initial levels. Other: 20.0 20 for 7503 level only ( <a href="#">1972Hu10</a> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |        |                                                                                          |
|                                      |                  | 7502.0             | 0.3           | 0       | 3/2 <sup>+</sup> |                           |            | $I_\gamma$ : from <a href="#">1976Me12</a> containing $\gamma$ rays of 7501 and 7503 doublet initial levels.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |        |                                                                                          |
| 7518.8                               | 7/2              | 1919.0             | 2 1           | 5599.7  |                  |                           |            | $I_\gamma$ : from <a href="#">1972Hu10</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |        |                                                                                          |
|                                      |                  | 1933               | 3.0 15        | 5586    |                  |                           |            | $I_\gamma$ : from <a href="#">1972Hu10</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |        |                                                                                          |
|                                      |                  | 3170.9             | 2 1           | 4347.76 | 5/2,9/2          |                           |            | $I_\gamma$ : from <a href="#">1972Hu10</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |        |                                                                                          |
|                                      |                  | 3575.7             | 2 1           | 3942.9  |                  |                           |            | $I_\gamma$ : from <a href="#">1972Hu10</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |        |                                                                                          |
|                                      |                  | 4355.5             | 68 7          | 3162.99 | 7/2 <sup>-</sup> | D(+Q)                     | +0.1 2     | $I_\gamma$ : from <a href="#">1972Hu10</a> .<br>Mult., $\delta$ : from <a href="#">1977Ko35</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |                                                                                          |
|                                      |                  | 4872.7             | 3.0 15        | 2645.70 | (5/2,7/2)        |                           |            | $I_\gamma$ : from <a href="#">1972Hu10</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |        |                                                                                          |
|                                      |                  | 5755.2             | 20 2          | 1763.14 | 5/2 <sup>+</sup> | D+Q                       | +0.098 22  | $I_\gamma$ : from <a href="#">1972Hu10</a> .<br>Mult., $\delta$ : from $\gamma(\theta)$ in <a href="#">1977Ko35</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |                                                                                          |
| 7548.9                               | 7/2 <sup>-</sup> | 1903               | 0.55 6        | 5646    |                  |                           |            | $I_\gamma$ : from <a href="#">1971Pr11</a> . Others: 0.6 ( <a href="#">1976Me12</a> ); 0.5 3 ( <a href="#">1972Hu10</a> ); 1 ( <a href="#">2001Vo24</a> , <a href="#">2002Vo17</a> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |                                                                                          |
|                                      |                  | 1963               | 0.1           | 5586    |                  |                           |            | $I_\gamma$ : other: 0.2 1 ( <a href="#">1972Hu10</a> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |                                                                                          |
|                                      |                  | 2778.9             | 1.4 1         | 4769.86 |                  |                           |            | $I_\gamma$ : from <a href="#">1971Pr11</a> . Others: 1.3 ( <a href="#">1976Me12</a> ); 1.5 8 ( <a href="#">1972Hu10</a> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |        |                                                                                          |
|                                      |                  | 4385.6             | 94 1          | 3162.99 | 7/2 <sup>-</sup> | M1+E2                     | -0.07 2    | $I_\gamma$ : from <a href="#">1971Pr11</a> . Others: 95 ( <a href="#">1976Me12</a> ); 95.0 95 ( <a href="#">1972Hu10</a> ); $\geq 93$ ( <a href="#">1971Wi13</a> ).<br>Mult.: D+Q from $\gamma(\theta)$ in <a href="#">1967Wa22</a> and <a href="#">1966Wa09</a> ; magnetic character from $\gamma(\text{pol})$ in <a href="#">1967Wa22</a> .<br>$\delta$ : from <a href="#">1966Wa09</a> and <a href="#">1967Wa22</a> . Others: $\delta=+0.08$ 10 ( <a href="#">1977Ko35</a> ), $\delta=-0.03$ 3 from <a href="#">2001Vo24</a> and <a href="#">2002Vo17</a> .<br>$N_{90}/N_0=0.72$ 6 ( <a href="#">1971Wi13</a> ).<br>POL=+0.76 6, $\Delta\pi=\text{no}$ ( <a href="#">1966Wa09</a> ).<br>POL=+0.93 10, $\Delta\pi=\text{no}$ ( <a href="#">1967Wa22</a> ).<br>0.021<POL<0.813 ( <a href="#">1966Az01</a> ).<br>$A_2=+0.42$ 3, $A_4=0.00$ 3 ( <a href="#">2001Vo24</a> , <a href="#">2002Vo17</a> ).<br>$A_2=+0.42$ 3, $A_4=-0.001$ 27 ( <a href="#">1966Az01</a> ).<br>$A_2=+0.42$ 3, $A_4=+0.00$ 3 ( <a href="#">1967Ta08</a> ). |        |                                                                                          |
|                                      |                  |                    |               | 4545.8  | 2.0 1            | 3002.75                   | 5/2        | D(+Q)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | +0.6 4 | $I_\gamma$ : from <a href="#">1971Pr11</a> . Others: 2 ( <a href="#">1976Me12</a> ); 1.8 |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$        | $E_\gamma^\dagger$ | $I_\gamma^\#$ | $E_f$   | $J_f^\pi$        | Mult. <sup>b</sup> | $\delta^b$ | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------|------------------|--------------------|---------------|---------|------------------|--------------------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7548.9              | 7/2 <sup>-</sup> | 4902.8             | 0.68 7        | 2645.70 | (5/2,7/2)        |                    |            | 9 (1972Hu10); 1.3 (1971Wi13).<br>Mult., $\delta$ : from $\gamma(\theta)$ in 1977Ko35.<br>I $_\gamma$ : from 1971Pr11. Others: 0.5 (1976Me12); 0.5 3 (1972Hu10); $\leq 5$ (1971Wi13).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                     |                  | 5785.3             | 0.33 4        | 1763.14 | 5/2 <sup>+</sup> |                    |            | I $_\gamma$ : from 1971Pr11. Others: 0.3 (1976Me12); 0.3 2 (1972Hu10); $\approx 0.4$ (1971Wi13).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                     |                  | 7548.0             | 0.27 3        | 0       | 3/2 <sup>+</sup> | Q(+O)              | +0.1 2     | I $_\gamma$ : from 1971Pr11. Others: 0.2 (1976Me12); 0.2 1 (1972Hu10); $\approx 0.3$ (1971Wi13).<br>Mult., $\delta$ : from $\gamma(\theta)$ in 1977Ko35.<br>I $_\gamma$ : other: 1.0 5 (1972Hu10).<br>I $_\gamma$ : other: 5.0 25 (1972Hu10).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 7561.3              | 1/2,3/2          | 3383.2             | 0.9           | 4177.9  | 3/2              |                    |            | I $_\gamma$ : other: 1.0 5 (1972Hu10).<br>I $_\gamma$ : other: 21.0 21 (1972Hu10).<br>A $_2$ =+0.04 3, A $_4$ =-0.05 3 (1976Me12).<br>I $_\gamma$ : other: 36.0 36 (1972Hu10).<br>A $_2$ =-0.02 2, A $_4$ =-0.05 3 (1976Me12).<br>I $_\gamma$ : other: 36.0 36 (1972Hu10).<br>A $_2$ =+0.01 2, A $_4$ =-0.03 3 (1976Me12).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                     |                  | 3501.9             | 6             | 4059.2  |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                     |                  | 3593.8             | 0.5           | 3967.3  |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                     |                  | 3642.6             | 1.6           | 3918.48 | 3/2 <sup>+</sup> |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                     |                  | 4867.0             | 22            | 2693.9  | (3/2,5/2)        |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                     |                  | 6341.3             | 35            | 1219.39 | 1/2 <sup>+</sup> |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                     |                  | 7560.4             | 34            | 0       | 3/2 <sup>+</sup> |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 7600.8              | 5/2              | 2437.4             | 1.9           | 5163.34 | 7/2              |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                     |                  | 2719.7             | 2             | 4880.96 |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                     |                  | 2761.6             | 2             | 4839.10 |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                     |                  | 2830.8             | 1.3           | 4769.86 |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                     |                  | 2976.3             | 1.8           | 4624.4  |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                     |                  | 3422.7             | 3             | 4177.9  | 3/2              |                    |            | I $_\gamma$ : other: 2 1 (1972Hu10).<br>I $_\gamma$ : 2 1 from 1972Hu10.<br>I $_\gamma$ : other: 1.5 8 (1972Hu10).<br>I $_\gamma$ : others: 4 2 (1972Hu10); 8 (1971Wi13).<br>Mult., $\delta$ : Q+O with $\delta=-0.25$ 15 for J $_i$ =3/2 (1977Ko34).<br>A $_2$ =+0.19 2, A $_4$ =+0.08 2 (1977Ko34).<br>A $_2$ =+0.35 32, A $_4$ =-0.44 32 (1971Wi13).<br>I $_\gamma$ : others: 9.0 9 (1972Hu10); 12 (1971Wi13).<br>Mult., $\delta$ : from 1971Wi13; $\delta$ =+0.85 42 for J $_i$ =3/2 (1977Ko34).<br>A $_2$ =+0.75 7, A $_4$ =-0.06 7 (1971Wi13).<br>A $_2$ =+0.53 21, A $_4$ =-0.04 25 (1976Me12).<br>A $_2$ =+0.42 3, A $_4$ =-0.10 3 (1977Ko34).<br>I $_\gamma$ : others: 19.0 19 (1972Hu10); 24 (1971Wi13).<br>$\delta$ : from 1971Wi13; other: $\delta=-5.3$ 9 or +0.01 5 for J $_i$ =3/2 (1977Ko34).<br>A $_2$ =+0.33 1, A $_4$ =-0.05 1 (1977Ko34).<br>A $_2$ =+0.18 4, A $_4$ =-0.02 5 (1976Me12).<br>A $_2$ =+0.26 10, A $_4$ =-0.05 10 (1971Wi13).<br>I $_\gamma$ : other: 1.5 8 (1972Hu10).<br>I $_\gamma$ : others: 20 2 (1971Wi13), 2 1 (1972Hu10); 25 (1971Wi13).<br>Mult., $\delta$ : from 1971Wi13. $\delta$ =+0.2 22 for J $_i$ =3/2 (1977Ko34).<br>A $_2$ =+0.10 7, A $_4$ =-0.10 8 (1971Wi13).<br>A $_2$ =+0.12 2, A $_4$ =-0.10 3 (1976Me12).<br>A $_2$ =+0.30 9, A $_4$ =-0.17 9 (1977Ko34).<br>I $_\gamma$ : others: 1.0 5 (1972Hu10); $\leq 1.5$ (1971Wi13). |
|                     |                  | 3427.2             |               | 4173.45 |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                     |                  | 3541.4             | 2             | 4059.2  |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                     |                  | 4437.5             | 5             | 3162.99 | 7/2 <sup>-</sup> |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                     |                  | 4597.7             | 8             | 3002.75 | 5/2              | D+Q                | +0.50 22   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                     |                  | 4906.5             | 18            | 2693.9  | (3/2,5/2)        | D+Q                | +0.32 7    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                     |                  | 4954.7             | 1.9           | 2645.70 | (5/2,7/2)        |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                     |                  | 5837.1             | 19            | 1763.14 | 5/2 <sup>+</sup> | D+Q                | +0.31 7    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                     |                  | 6380.8             | 1.1           | 1219.39 | 1/2 <sup>+</sup> |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10** (continued)

| $\gamma(^{35}\text{Cl})$ (continued) |                  |                    |               |         |                  |                    |            | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------------------------|------------------|--------------------|---------------|---------|------------------|--------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $E_i(\text{level})$                  | $J_i^\pi$        | $E_\gamma^\dagger$ | $I_\gamma^\#$ | $E_f$   | $J_f^\pi$        | Mult. <sup>b</sup> | $\delta^b$ |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 7600.8                               | 5/2              | 7599.9             | 33            | 0       | 3/2 <sup>+</sup> | D+Q                | -0.19 4    | $I_\gamma$ : others: 31.0 3I (1972Hu10); 31 (1971Wi13).<br>Mult., $\delta$ : from 1976Me12. Others: $\delta=+0.17$ 3 (1971Wi13).<br>$A_2=-0.73$ 7, $A_4=+0.04$ 6 (1971Wi13).<br>$A_2=-0.62$ 5, $A_4=-0.06$ 5 (1976Me12).<br>$A_2=-0.56$ 3, $A_4=-0.05$ 2 (1977Ko34).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 7618.7                               | 5/2              | 1438               | 0.6           | 6181    |                  |                    |            | $I_\gamma$ : other: 3.0 15 (1972Hu10).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                      |                  | 2402.8             | 2             | 5215.8  |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                      |                  | 2455.3             | 0.7           | 5163.34 | 7/2              |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                      |                  | 2779.5             | 0.7           | 4839.10 |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                      |                  | 3440.6             | 3             | 4177.9  | 3/2              |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                      |                  | 3559.3             | 2             | 4059.2  |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                      |                  | 4455.4             | 3             | 3162.99 | 7/2 <sup>-</sup> |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                      |                  | 5855.0             | 10            | 1763.14 | 5/2 <sup>+</sup> | D+Q                | +0.48 11   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                      |                  |                    |               |         |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                      |                  |                    |               |         |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 7656.6                               |                  | 1550.4             | 3.0 15        | 6106.2  |                  |                    |            | $I_\gamma$ : other: 10.0 10 (1972Hu10).<br>Mult., $\delta$ : from $\gamma(\theta)$ in 1977Ko34.<br>$A_2=+0.38$ 5, $A_4=-0.01$ 6 (1976Me12).<br>$A_2=-0.11$ 3, $A_4=-0.06$ 3 (1977Ko34).<br>$I_\gamma$ : other: 80 8 (1972Hu10).<br>Mult., $\delta$ : from $\gamma(\theta)$ in 1977Ko34.<br>$A_2=-0.46$ 4, $A_4=+0.03$ 4 (1976Me12).<br>$A_2=-0.57$ 3, $A_4=+0.27$ 3 (1977Ko34).<br>$I_\gamma$ : from 1972Hu10.<br>$I_\gamma$ : from 1972Hu10.<br>$I_\gamma$ : from 1972Hu10.<br>$I_\gamma$ : from 1972Hu10.<br>$I_\gamma$ : from 1972Hu10.<br>$I_\gamma$ : from 1972Hu10.<br>$I_\gamma$ : from 1972Hu10.<br>$I_\gamma$ : from 1972Hu10.                                                                                                                                                                                                                                                                                                                                                                                           |
|                                      |                  | 1974               | 1.0 5         | 5683    |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                      |                  | 2002               | 1.0 5         | 5655    |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                      |                  | 2817.4             | 3.0 15        | 4839.10 |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                      |                  | 3478.5             | 1.0 5         | 4177.9  | 3/2              |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                      |                  | 3597.2             | 32.0 32       | 4059.2  |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                      |                  | 3689.1             | 2.0 10        | 3967.3  |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                      |                  | 7655.7             | 57 6          | 0       | 3/2 <sup>+</sup> |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                      |                  |                    |               |         |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                      |                  |                    |               |         |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 7671.9                               | 5/2              | 2072.1             | 1.2           | 5599.7  |                  |                    |            | $I_\gamma$ : other: 4 2 (1972Hu10).<br>$I_\gamma$ : other: 1.0 5 (1972Hu10).<br><br>$I_\gamma$ : other: 6.0 6 (1972Hu10).<br>$I_\gamma$ : other: 4 2 (1972Hu10).<br>$I_\gamma$ : other: 4 2 (1972Hu10).<br>$I_\gamma$ : other: 8.0 8 (1972Hu10).<br>Mult., $\delta$ : from $\gamma(\theta)$ in 1977Ko34.<br>$A_2=-0.35$ 4, $A_4=+0.01$ 5 (1976Me12).<br>$A_2=+0.52$ 1, $A_4=-0.18$ 1 (1977Ko34).<br>$I_\gamma$ : other: 14.0 14 (1972Hu10).<br>Mult., $\delta$ : from $\gamma(\theta)$ in 1977Ko34.<br>$A_2=+0.41$ 3, $A_4=-0.03$ 4 (1976Me12).<br>$A_2=+0.33$ 8, $A_4=-0.12$ 8 (1977Ko34).<br>$I_\gamma$ : other: 57.0 57 (1972Hu10).<br>Mult., $\delta$ : from $\gamma(\theta)$ in 1977Ko34.<br>$A_2=-0.38$ 4, $A_4=-0.06$ 5 (1976Me12).<br>$A_2=-0.40$ 9, $A_4=+0.07$ 8 (1977Ko34).<br>$I_\gamma$ : other: 2 1 (1972Hu10).<br>$I_\gamma$ : 0.5 3 (1972Hu10).<br>$I_\gamma$ : other: 2 1 (1972Hu10).<br>$I_\gamma$ : other: 2 1 (1972Hu10).<br>$I_\gamma$ : other: 0.5 3 (1972Hu10).<br><br>$I_\gamma$ : other: 2 1 (1972Hu10). |
|                                      |                  | 2508.5             | 4             | 5163.34 | 7/2              |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                      |                  | 2790.8             | 1.2           | 4880.96 |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                      |                  | 2901.9             | 1.0           | 4769.86 |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                      |                  | 3324.0             | 6             | 4347.76 | 5/2,9/2          |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                      |                  | 3498.3             | 5             | 4173.45 |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                      |                  | 4508.6             | 1.8           | 3162.99 | 7/2 <sup>-</sup> |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                      |                  | 4668.8             | 9             | 3002.75 | 5/2              | D+Q                | -1.4 5     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                      |                  |                    |               |         |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                      |                  |                    |               |         |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 7685.5                               | 3/2 <sup>-</sup> | 5025.8             | 13            | 2645.70 | (5/2,7/2)        | D(+Q)              | +0.3 22    | $I_\gamma$ : other: 2 1 (1972Hu10).<br>$I_\gamma$ : 0.5 3 (1972Hu10).<br>$I_\gamma$ : other: 2 1 (1972Hu10).<br>$I_\gamma$ : other: 2 1 (1972Hu10).<br>$I_\gamma$ : other: 0.5 3 (1972Hu10).<br><br>$I_\gamma$ : other: 2 1 (1972Hu10).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                      |                  | 5908.2             | 57            | 1763.14 | 5/2 <sup>+</sup> | D+Q                | +0.86 14   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                      |                  |                    |               |         |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                      |                  |                    |               |         |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                      |                  |                    |               |         |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                      |                  |                    |               |         |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                      |                  |                    |               |         |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                      |                  |                    |               |         |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                      |                  |                    |               |         |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                      |                  |                    |               |         |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                      |                  |                    |               |         |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 7685.5                               | 3/2 <sup>-</sup> | 7671.0             | 0.8           | 0       | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 2 1 (1972Hu10).<br>$I_\gamma$ : 0.5 3 (1972Hu10).<br>$I_\gamma$ : other: 2 1 (1972Hu10).<br>$I_\gamma$ : other: 2 1 (1972Hu10).<br>$I_\gamma$ : other: 0.5 3 (1972Hu10).<br><br>$I_\gamma$ : other: 2 1 (1972Hu10).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                      |                  | 1927               |               | 5758    |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                      |                  | 2085.7             | 2             | 5599.7  |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                      |                  | 2281.9             | 2             | 5403.5  |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                      |                  | 2469.6             | 0.7           | 5215.8  |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                      |                  | 2675.3             | 0.6           | 5010.1  |                  |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                      |                  | 3507.4             | 3             | 4177.9  | 3/2              |                    |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

Continued on next page (footnotes at end of table)



$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)**

| $\gamma(^{35}\text{Cl})$ (continued) |                  |                    |               |         |                  |                    |            |                                                                                                                                                                                                                                    |
|--------------------------------------|------------------|--------------------|---------------|---------|------------------|--------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $E_i(\text{level})$                  | $J_i^\pi$        | $E_\gamma^\dagger$ | $I_\gamma^\#$ | $E_f$   | $J_f^\pi$        | Mult. <sup>b</sup> | $\delta^b$ | Comments                                                                                                                                                                                                                           |
| 7685.5                               | 3/2 <sup>-</sup> | 3511.9             | 4             | 4173.45 |                  |                    |            | $I_\gamma$ : other: 5.0 25 (1972Hu10).<br>$A_2=+0.01$ 14, $A_4=+0.23$ 16 (1977Ko34).                                                                                                                                               |
|                                      |                  | 3626.1             | 6             | 4059.2  |                  |                    |            | $I_\gamma$ : other: 6.0 6 (1972Hu10).                                                                                                                                                                                              |
|                                      |                  | 4682.4             | 10            | 3002.75 | 5/2              | D(+Q)              | -0.01 5    | $I_\gamma$ : other: 9.0 9 (1972Hu10).<br>Mult., $\delta$ : other: $\delta=-2.4$ +20-4 (1976Me12);<br>$\delta=+0.01$ 7 (1977Ko34).<br>$A_2=-0.04$ 6, $A_4=-0.09$ 7 (1976Me12).<br>$A_2=-0.06$ 12, $A_4=+0.08$ 13 (1977Ko34).        |
|                                      |                  | 5921.8             | 0.7           | 1763.14 | 5/2 <sup>+</sup> |                    |            |                                                                                                                                                                                                                                    |
|                                      |                  | 6465.5             | 1             | 1219.39 | 1/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                                                                                                                                                                              |
|                                      |                  | 7684.6             | 70            | 0       | 3/2 <sup>+</sup> | D+Q                | +6.0 7     | $I_\gamma$ , Mult., $\delta$ : from 1976Me12. Other: $I_\gamma=72.0$ 72 (1972Hu10); $\delta<-3.5$ for $J_i=3/2$ (1977Ko34).<br>$A_2=+0.25$ 1, $A_4=-0.03$ 1 (1976Me12).<br>$A_2=+0.18$ 11, $A_4=-0.04$ 11 (1977Ko34).              |
| 7693.9                               |                  | 2094.1             | 1.0 5         | 5599.7  |                  |                    |            | $I_\gamma$ : from 1972Hu10.                                                                                                                                                                                                        |
|                                      |                  | 3634.5             | 1.0 5         | 4059.2  |                  |                    |            | $I_\gamma$ : from 1972Hu10.                                                                                                                                                                                                        |
|                                      |                  | 4690.8             | 2.0 10        | 3002.75 | 5/2              |                    |            | $I_\gamma$ : from 1972Hu10.                                                                                                                                                                                                        |
|                                      |                  | 7693.0             | 96 10         | 0       | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : from 1972Hu10.                                                                                                                                                                                                        |
| 7706.4                               | 5/2 <sup>+</sup> | 2051               | 4             | 5655    |                  |                    |            | $I_\gamma$ : other: 4 2 (1972Hu10).<br>Mult., $\delta$ : D+Q with $\delta\leq-2$ or $\geq 13$ for<br>$J_f(5655)=5/2$ (1977Ko34), but $J_f(5655)=(3/2)^+$<br>from the Adopted Levels.<br>$A_2=+0.14$ 15, $A_4=-0.39$ 17 (1977Ko34). |
|                                      |                  | 2825.3             | 0.4           | 4880.96 |                  |                    |            |                                                                                                                                                                                                                                    |
|                                      |                  | 2867.2             | 1.3           | 4839.10 |                  |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                                                                                                                                                                              |
|                                      |                  | 3081.9             | 0.8           | 4624.4  |                  |                    |            |                                                                                                                                                                                                                                    |
|                                      |                  | 3528.3             | 2             | 4177.9  | 3/2              |                    |            |                                                                                                                                                                                                                                    |
|                                      |                  | 3592.5             | 2             | 4113.7  | 7/2              |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                                                                                                                                                                              |
|                                      |                  | 3787.7             | 0.6           | 3918.48 | 3/2 <sup>+</sup> |                    |            |                                                                                                                                                                                                                                    |
|                                      |                  | 4703.3             | 1             | 3002.75 | 5/2              |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                                                                                                                                                                              |
|                                      |                  | 5060.3             | 4             | 2645.70 | (5/2,7/2)        | D+Q                | -1.01 10   | $I_\gamma$ : other: 4 2 (1972Hu10).<br>Mult., $\delta$ : or $\geq 13$ (1977Ko34).<br>$A_2=-0.03$ 6, $A_4=+0.08$ 8 (1976Me12).<br>$A_2=-0.18$ 17, $A_4=+0.41$ 18 (1977Ko34).                                                        |
|                                      |                  | 5942.7             | 0.4           | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                                                                                                                                                                              |
|                                      |                  | 6486.4             | 0.5           | 1219.39 | 1/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 88.0 88 (1972Hu10).                                                                                                                                                                                            |
|                                      |                  | 7705.5             | 83            | 0       | 3/2 <sup>+</sup> | D(+Q)              | +0.1 1     | Mult., $\delta$ : from $\gamma(\theta)$ in 1977Ko34.<br>$A_2=-0.54$ 3, $A_4=-0.07$ 3 (1976Me12).<br>$A_2=-0.63$ 4, $A_4=+0.15$ 3 (1977Ko34).                                                                                       |
| 7744.8                               | 7/2              | 2863.7             | 20            | 4880.96 |                  |                    |            | $I_\gamma$ : other: 19.0 19 (1972Hu10).                                                                                                                                                                                            |
|                                      |                  | 2974.8             | 2             | 4769.86 |                  |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                                                                                                                                                                              |
|                                      |                  | 3630.9             | 6             | 4113.7  | 7/2              |                    |            | $I_\gamma$ : other: 5.0 25 (1972Hu10).                                                                                                                                                                                             |
|                                      |                  | 3801.7             | 10            | 3942.9  |                  | D(+Q)              | -0.16 17   | $I_\gamma$ : other: 12.0 12 (1972Hu10).<br>Mult., $\delta$ : from $\gamma(\theta)$ in 1977Ko35.                                                                                                                                    |
|                                      |                  | 4581.5             | 16            | 3162.99 | 7/2 <sup>-</sup> | D+Q                | +0.18 8    | Mult., $\delta$ : from $\gamma(\theta)$ in 1977Ko35.                                                                                                                                                                               |
|                                      |                  | 5098.7             | 43            | 2645.70 | (5/2,7/2)        | D(+Q)              | +0.1 30    | Mult., $\delta$ : from $\gamma(\theta)$ in 1977Ko35.                                                                                                                                                                               |
|                                      |                  | 5981.1             | 2.4           | 1763.14 | 5/2 <sup>+</sup> |                    |            |                                                                                                                                                                                                                                    |
|                                      |                  | 6524.8             | 0.6           | 1219.39 | 1/2 <sup>+</sup> |                    |            |                                                                                                                                                                                                                                    |
| 7777.1                               |                  | 2122               | 0.5           | 5655    |                  |                    |            |                                                                                                                                                                                                                                    |
|                                      |                  | 2613.7             | 1.1           | 5163.34 | 7/2              |                    |            |                                                                                                                                                                                                                                    |
|                                      |                  | 2937.9             | 2             | 4839.10 |                  |                    |            | $I_\gamma$ : other: 2 1 (1972Hu10).                                                                                                                                                                                                |
|                                      |                  | 3007.1             | 3             | 4769.86 |                  |                    |            | $I_\gamma$ : other: 2 1 (1972Hu10).                                                                                                                                                                                                |
|                                      |                  | 3599.0             | 7             | 4177.9  | 3/2              |                    |            | $I_\gamma$ : other: 7.0 7 (1972Hu10).                                                                                                                                                                                              |
|                                      |                  | 3717.7             | 1.1           | 4059.2  |                  |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                                                                                                                                                                              |

Continued on next page (footnotes at end of table)



$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$        | $E_\gamma^\dagger$ | $I_\gamma^\#$ | $E_f$   | $J_f^\pi$        | Mult. <sup>b</sup> | $\delta^b$ | Comments                                                                                                                                                     |
|---------------------|------------------|--------------------|---------------|---------|------------------|--------------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7777.1              |                  | 4613.8             | 2             | 3162.99 | 7/2 <sup>-</sup> |                    |            |                                                                                                                                                              |
|                     |                  | 4774.0             | 4             | 3002.75 | 5/2              |                    |            | $I_\gamma$ : other: 4 2 (1972Hu10).                                                                                                                          |
|                     |                  | 5082.8             | 9             | 2693.9  | (3/2,5/2)        |                    |            | $I_\gamma$ : other: 7.0 7 (1972Hu10).                                                                                                                        |
|                     |                  | 5131.0             | 1.3           | 2645.70 | (5/2,7/2)        |                    |            | $I_\gamma$ : other: 2 1 (1972Hu10).                                                                                                                          |
|                     |                  | 6013.4             | 40            | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 46.0 46 (1972Hu10).                                                                                                                      |
|                     |                  | 6557.1             | 11            | 1219.39 | 1/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 13.0 13 (1972Hu10).                                                                                                                      |
|                     |                  | 7776.2             | 18            | 0       | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 16.0 16 (1972Hu10).                                                                                                                      |
| 7781.7              | 5/2 <sup>-</sup> | 1675.5             | 1.4           | 6106.2  |                  |                    |            |                                                                                                                                                              |
|                     |                  | 2181.9             |               | 5599.7  |                  |                    |            | $I_\gamma$ : 1.0 5 from 1972Hu10.                                                                                                                            |
|                     |                  | 2378.1             |               | 5403.5  |                  |                    |            | $I_\gamma$ : 1.0 5 from 1972Hu10.                                                                                                                            |
|                     |                  | 2565.8             | 1.5           | 5215.8  |                  |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                                                                                                        |
|                     |                  | 3011.7             | 7             | 4769.86 |                  |                    |            | $I_\gamma$ : other: 7.0 7 (1972Hu10).                                                                                                                        |
|                     |                  | 3433.8             | 2             | 4347.76 | 5/2,9/2          |                    |            | $I_\gamma$ : other: 2 1 (1972Hu10).                                                                                                                          |
|                     |                  | 3608.1             | 36            | 4173.45 |                  |                    |            | $I_\gamma$ : other: 40.0 40 (1972Hu10).                                                                                                                      |
|                     |                  | 3667.8             | 6             | 4113.7  | 7/2              |                    |            | $I_\gamma$ : other: 3.0 15 (1972Hu10).                                                                                                                       |
|                     |                  | 3722.3             | 1.2           | 4059.2  |                  |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                                                                                                        |
|                     |                  | 3863.0             | 11            | 3918.48 | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 10.0 10 (1972Hu10).                                                                                                                      |
|                     |                  | 4618.4             | 5             | 3162.99 | 7/2 <sup>-</sup> |                    |            | $I_\gamma$ : other: 5.0 25 (1972Hu10).                                                                                                                       |
|                     |                  | 5135.6             | 8             | 2645.70 | (5/2,7/2)        |                    |            | $I_\gamma$ : other: 9.0 9 (1972Hu10).                                                                                                                        |
|                     |                  | 6018.0             | 18            | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 20.0 20 (1972Hu10).                                                                                                                      |
|                     |                  | 6561.7             | 1.2           | 1219.39 | 1/2 <sup>+</sup> |                    |            |                                                                                                                                                              |
|                     |                  | 7780.8             | 1.7           | 0       | 3/2 <sup>+</sup> |                    |            |                                                                                                                                                              |
| 7797.1              | 1/2              | 2039               |               | 5758    |                  |                    |            | $I_\gamma$ : 0.5 3 from 1972Hu10.                                                                                                                            |
|                     |                  | 2142               | 1.3           | 5655    |                  |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                                                                                                        |
|                     |                  |                    |               |         |                  |                    |            | $A_2=+0.03$ 8 (1988Va06).                                                                                                                                    |
|                     |                  | 2393.5             | 2             | 5403.5  |                  |                    |            | $I_\gamma$ : other: 2 1 (1972Hu10).                                                                                                                          |
|                     |                  |                    |               |         |                  |                    |            | $A_2=+0.04$ 5 (1988Va06).                                                                                                                                    |
|                     |                  | 2786.9             | 2             | 5010.1  |                  |                    |            | $I_\gamma$ : other: 2 1 (1972Hu10).                                                                                                                          |
|                     |                  |                    |               |         |                  |                    |            | $A_2=-0.07$ 5 (1988Va06).                                                                                                                                    |
|                     |                  | 2942.6             | 1.3           | 4854.4  |                  |                    |            | $I_\gamma$ : other: 0.5 3 (1972Hu10).                                                                                                                        |
|                     |                  | 2957.9             | 1.1           | 4839.10 |                  |                    |            |                                                                                                                                                              |
|                     |                  | 3737.7             | 0.3           | 4059.2  |                  |                    |            |                                                                                                                                                              |
|                     |                  | 6577.1             | 9             | 1219.39 | 1/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 9.0 9 (1972Hu10).                                                                                                                        |
|                     |                  |                    |               |         |                  |                    |            | $\delta$ : $\delta(Q/D)=+0.21$ 4 or $-1.9$ 2 for $J=3/2$ (1976Me12), which is invalidated by the $J^\pi=1/2^-$ assignment for the 7797 level in 1988Va06.    |
|                     |                  |                    |               |         |                  |                    |            | $A_2=-0.02$ 6, $A_4=-0.14$ 7 (1976Me12).                                                                                                                     |
|                     |                  |                    |               |         |                  |                    |            | $A_2=-0.03$ 3 (1988Va06).                                                                                                                                    |
|                     |                  | 7796.2             | 83            | 0       | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 85.0 85 (1972Hu10).                                                                                                                      |
|                     |                  |                    |               |         |                  |                    |            | $\delta$ : $\delta(Q/D)=-0.33$ 2 or $-19$ +5-10 for $J=3/2$ (1976Me12), which is invalidated by the $J^\pi=1/2^-$ assignment for the 7797 level in 1988Va06. |
|                     |                  |                    |               |         |                  |                    |            | $A_2=-0.06$ 2, $A_4=-0.03$ 2 (1976Me12).                                                                                                                     |
|                     |                  |                    |               |         |                  |                    |            | $A_2=+0.006$ 14 (1988Va06).                                                                                                                                  |
| 7837.2              | 3/2              | 1656               | 0.5           | 6181    |                  |                    |            | $I_\gamma$ : for 7837 and 7839 doublet initial levels (1976Me12). Other: 1 from 2001Vo24.                                                                    |
|                     |                  | 2237.4             | 0.5           | 5599.7  |                  |                    |            | $I_\gamma$ : for 7837 and 7839 doublet initial levels (1976Me12). Other: 1 (2001Vo24).                                                                       |
|                     |                  | 2621.3             | 1.4           | 5215.8  |                  |                    |            | $I_\gamma$ : for 7837 and 7839 doublet initial levels (1976Me12). Others: 1 (2001Vo24); 1.0 5 (1972Hu10).                                                    |
|                     |                  | 3659.1             | 28            | 4177.9  | 3/2              | M1+E2              | -0.05 3    | $I_\gamma$ : for 7837 and 7839 doublet initial levels (1976Me12). Others: 28 (2001Vo24); 35.0 35 (1972Hu10).                                                 |
|                     |                  |                    |               |         |                  |                    |            | Mult., $\delta$ : D+Q from $\gamma(\theta)$ in 1967Wa22; magnetic                                                                                            |

Continued on next page (footnotes at end of table)

$\gamma(^{35}\text{Cl})$  (continued)Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$      | $E_\gamma^\dagger$ | $I_\gamma^\#$ | $E_f$   | $J_f^\pi$    | Comments                                 |
|---------------------|----------------|--------------------|---------------|---------|--------------|------------------------------------------|
| 7868.7              |                | 7867.8             | 61            | 0       | $3/2^+$      | $I_\gamma$ : other: 62.0 62 (1972Hu10).  |
| 7880.8              | $3/2, 5/2$     | 2226               | 12            | 5655    |              | $I_\gamma$ : other: 9.0 9 (1972Hu10).    |
|                     |                | 3041.6             | 1.7           | 4839.10 |              | $I_\gamma$ : other: 3.0 15 (1972Hu10).   |
|                     |                | 3256.2             | 2             | 4624.4  |              | $I_\gamma$ : other: 1.0 5 (1972Hu10).    |
|                     |                | 3702.7             | 3             | 4177.9  | $3/2$        |                                          |
|                     |                | 3766.9             | 1.3           | 4113.7  | $7/2$        | $I_\gamma$ : other: 4 2 (1972Hu10).      |
|                     |                | 3821.4             | 2             | 4059.2  |              | $I_\gamma$ : other: 1.0 5 (1972Hu10).    |
|                     |                | 3962.1             | 2             | 3918.48 | $3/2^+$      | $I_\gamma$ : other: 2 1 (1972Hu10).      |
|                     |                | 4877.7             | 11            | 3002.75 | $5/2$        | $I_\gamma$ : other: 10.0 10 (1972Hu10).  |
|                     |                | 5186.5             | 25            | 2693.9  | $(3/2, 5/2)$ | $I_\gamma$ : other: 30.0 30 (1972Hu10).  |
|                     |                |                    |               |         |              | $A_2=+0.50$ 3, $A_4=+0.09$ 4 (1976Me12). |
|                     |                | 6117.1             | 30            | 1763.14 | $5/2^+$      | $I_\gamma$ : other: 31.0 31 (1972Hu10).  |
|                     |                |                    |               |         |              | $A_2=+0.08$ 3, $A_4=-0.03$ 4 (1976Me12). |
|                     |                | 7879.9             | 10            | 0       | $3/2^+$      | $I_\gamma$ : other: 9.0 9 (1972Hu10).    |
|                     |                |                    |               |         |              | $A_2=-0.06$ 4, $A_4=+0.04$ 5 (1976Me12). |
| 7899.2              | $(3/2^-, 5/2)$ | 2253               | 1.5           | 5646    |              |                                          |
|                     |                | 2735.8             | 0.9           | 5163.34 | $7/2$        |                                          |
|                     |                | 3018.1             | 0.8           | 4880.96 |              |                                          |
|                     |                | 3274.6             | 0.5           | 4624.4  |              |                                          |
|                     |                | 3721.1             | 8             | 4177.9  | $3/2$        | $I_\gamma$ : other: 8.0 8 (1972Hu10).    |
|                     |                | 3725.5             | 8             | 4173.45 |              | $I_\gamma$ : other: 6.0 6 (1972Hu10).    |
|                     |                | 3839.8             | 3             | 4059.2  |              | $I_\gamma$ : other: 2 1 (1972Hu10).      |
|                     |                | 4735.9             | 4             | 3162.99 | $7/2^-$      | $I_\gamma$ : other: 4 2 (1972Hu10).      |
|                     |                | 5204.9             | 1.3           | 2693.9  | $(3/2, 5/2)$ |                                          |
|                     |                | 6135.5             | 68            | 1763.14 | $5/2^+$      | $I_\gamma$ : other: 77.0 77 (1972Hu10).  |
|                     |                | 7898.2             | 4             | 0       | $3/2^+$      | $I_\gamma$ : other: 3.0 15 (1972Hu10).   |
| 7923.3              |                | 3084.1             | 1.4           | 4839.10 |              |                                          |
|                     |                | 3153.3             | 2             | 4769.86 |              |                                          |
|                     |                | 3298.7             | 1.9           | 4624.4  |              |                                          |
|                     |                | 3745.2             | 3             | 4177.9  | $3/2$        | $I_\gamma$ : other: 2 1 (1972Hu10).      |
|                     |                | 3749.6             | 5             | 4173.45 |              | $I_\gamma$ : other: 6.0 6 (1972Hu10).    |
|                     |                | 3863.9             | 1.7           | 4059.2  |              | $I_\gamma$ : other: 1.0 5 (1972Hu10).    |
|                     |                | 3955.8             | 1.5           | 3967.3  |              | $I_\gamma$ : other: 1.0 5 (1972Hu10).    |
|                     |                | 4004.6             | 1.2           | 3918.48 | $3/2^+$      | $I_\gamma$ : other: 1.0 5 (1972Hu10).    |
|                     |                | 4920.2             | 0.9           | 3002.75 | $5/2$        |                                          |
|                     |                | 5229.0             | 11            | 2693.9  | $(3/2, 5/2)$ | $I_\gamma$ : other: 12.0 12 (1972Hu10).  |
|                     |                | 5277.2             | 0.9           | 2645.70 | $(5/2, 7/2)$ |                                          |
|                     |                | 6159.6             | 1.5           | 1763.14 | $5/2^+$      | $I_\gamma$ : other: 1.0 5 (1972Hu10).    |
|                     |                | 6703.2             | 27            | 1219.39 | $1/2^+$      | $I_\gamma$ : other: 28.0 28 (1972Hu10).  |
|                     |                | 7922.3             | 41            | 0       | $3/2^+$      | $I_\gamma$ : other: 48.0 48 (1972Hu10).  |
| 7970.3              |                | 2324               | 3             | 5646    |              | $I_\gamma$ : other: 3.0 15 (1972Hu10).   |
|                     |                | 2384               | 3             | 5586    |              |                                          |
|                     |                | 2754.4             | 3             | 5215.8  |              | $I_\gamma$ : other: 3.0 15 (1972Hu10).   |
|                     |                | 3089.2             | 10            | 4880.96 |              | $I_\gamma$ : other: 9.0 9 (1972Hu10).    |
|                     |                | 3200.3             | 5             | 4769.86 |              | $I_\gamma$ : other: 6.0 6 (1972Hu10).    |
|                     |                | 3622.3             | 8             | 4347.76 | $5/2, 9/2$   | $I_\gamma$ : other: 8.0 8 (1972Hu10).    |
|                     |                | 3792.2             | 6             | 4177.9  | $3/2$        | $I_\gamma$ : other: 6.0 6 (1972Hu10).    |
|                     |                | 3856.4             | 6             | 4113.7  | $7/2$        | $I_\gamma$ : other: 2 1 (1972Hu10).      |
|                     |                | 4807.0             | 10            | 3162.99 | $7/2^-$      | $I_\gamma$ : other: 13.0 13 (1972Hu10).  |
|                     |                | 4967.2             | 5             | 3002.75 | $5/2$        | $I_\gamma$ : other: 5.0 25 (1972Hu10).   |
|                     |                | 5324.2             | 20            | 2645.70 | $(5/2, 7/2)$ | $I_\gamma$ : other: 22.0 22 (1972Hu10).  |
|                     |                | 6206.6             | 20            | 1763.14 | $5/2^+$      | $I_\gamma$ : other: 21.0 21 (1972Hu10).  |
|                     |                | 6750.2             | 1             | 1219.39 | $1/2^+$      |                                          |
| 7987.8              | $3/2$          | 1881.6             | 0.5           | 6106.2  |              | $I_\gamma$ : other: 1.0 5 (1972Hu10).    |
|                     |                | 2333               | 1.7           | 5655    |              | $I_\gamma$ : other: 1.0 5 (1972Hu10).    |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$               | $E_\gamma^\dagger$ | $I_\gamma^\#$ | $E_f$   | $J_f^\pi$        | Mult. <sup>b</sup> | $\delta^b$ | Comments                                                                                                                       |
|---------------------|-------------------------|--------------------|---------------|---------|------------------|--------------------|------------|--------------------------------------------------------------------------------------------------------------------------------|
| 7987.8              | 3/2                     | 3809.7             | <0.8          | 4177.9  | 3/2              |                    |            | $I_\gamma$ : total $I_\gamma=0.8$ for $\gamma$ decays to 4173 and 4178 doublet final levels (1976Me12).                        |
|                     |                         | 3814.1             | <0.8          | 4173.45 |                  |                    |            | $I_\gamma$ : total $I_\gamma=0.8$ for $\gamma$ decays to 4173 and 4178 doublet final levels (1976Me12).                        |
|                     |                         | 3928.4             | 1             | 4059.2  |                  |                    |            |                                                                                                                                |
|                     |                         | 6767.7             | 60            | 1219.39 | 1/2 <sup>+</sup> | D+Q                | +0.21 4    | $I_\gamma$ : other: 62.0 62 (1972Hu10).<br>Mult., $\delta$ : or -3.1 4 (1976Me12).<br>$A_2=-0.07$ 2, $A_4=-0.05$ 3 (1976Me12). |
|                     |                         | 7986.8             | 36            | 0       | 3/2 <sup>+</sup> | D+Q                | -0.36 2    | $I_\gamma$ : other: 36.0 36 (1972Hu10).<br>Mult., $\delta$ : or -11 3 (1976Me12).<br>$A_2=-0.10$ 2, $A_4=-0.06$ 2 (1976Me12).  |
| 7995.6              | 5/2                     | 2190               | 0.6           | 5806    |                  |                    |            |                                                                                                                                |
|                     |                         | 2395.8             | 1.1           | 5599.7  |                  |                    |            |                                                                                                                                |
|                     |                         | 2592.0             | 3             | 5403.5  |                  |                    |            | $I_\gamma$ : other: 3.0 15 (1972Hu10).                                                                                         |
|                     |                         | 2832.1             | 1.5           | 5163.34 | 7/2              |                    |            |                                                                                                                                |
|                     |                         | 3817.5             | 4             | 4177.9  | 3/2              |                    |            | $I_\gamma$ : other: 3.0 15 (1972Hu10).                                                                                         |
|                     |                         | 3936.2             | 1.1           | 4059.2  |                  |                    |            |                                                                                                                                |
|                     |                         | 4832.3             | 7             | 3162.99 | 7/2 <sup>-</sup> |                    |            | $I_\gamma$ : other: 6.0 6 (1972Hu10).                                                                                          |
|                     |                         | 4992.5             | 2             | 3002.75 | 5/2              |                    |            |                                                                                                                                |
|                     |                         | 5301.3             | 0.2           | 2693.9  | (3/2,5/2)        |                    |            |                                                                                                                                |
|                     |                         | 5349.5             | 0.6           | 2645.70 | (5/2,7/2)        |                    |            |                                                                                                                                |
|                     |                         | 6231.9             | 0.9           | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                                                                          |
|                     |                         | 7994.6             | 78            | 0       | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 87.0 87 (1972Hu10).                                                                                        |
| 8001.0              | (7/2,9/2 <sup>+</sup> ) | 2355               | 1.8           | 5646    |                  |                    |            | $I_\gamma$ : other: 3.0 15 (1972Hu10).                                                                                         |
|                     |                         | 2785.1             | 0.8           | 5215.8  |                  |                    |            |                                                                                                                                |
|                     |                         | 3119.9             | 0.7           | 4880.96 |                  |                    |            |                                                                                                                                |
|                     |                         | 3231.0             | 8             | 4769.86 |                  |                    |            | $I_\gamma$ : other: 8.0 8 (1972Hu10).                                                                                          |
|                     |                         | 3653.0             | 1.9           | 4347.76 | 5/2,9/2          |                    |            |                                                                                                                                |
|                     |                         | 3827.3             | 10            | 4173.45 |                  |                    |            | $I_\gamma$ : other: 8.0 8 (1972Hu10).                                                                                          |
|                     |                         | 4057.9             | 3             | 3942.9  |                  |                    |            | $I_\gamma$ : other: 2 1 (1972Hu10).                                                                                            |
|                     |                         | 4837.7             | 0.8           | 3162.99 | 7/2 <sup>-</sup> |                    |            |                                                                                                                                |
|                     |                         | 5354.9             | 1.6           | 2645.70 | (5/2,7/2)        |                    |            | $I_\gamma$ : other: 2 1 (1972Hu10).                                                                                            |
|                     |                         | 6237.3             | 70            | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 76.0 76 (1972Hu10).                                                                                        |
|                     |                         | 8000.0             | 1.4           | 0       | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                                                                          |
| 8005.0              | 5/2 <sup>+</sup>        | 2350               | 1.8           | 5655    |                  |                    |            |                                                                                                                                |
|                     |                         | 3123.9             | 0.6           | 4880.96 |                  |                    |            |                                                                                                                                |
|                     |                         | 3165.8             | 3             | 4839.10 |                  |                    |            |                                                                                                                                |
|                     |                         | 3826.9             | <1.3          | 4177.9  | 3/2              |                    |            | $I_\gamma$ : total $I_\gamma=1.3$ for $\gamma$ decays to 4173 and 4178 doublet final levels (1976Me12).                        |
|                     |                         | 3831.3             | <1.3          | 4173.45 |                  |                    |            | $I_\gamma$ : total $I_\gamma=1.3$ for $\gamma$ decays to 4173 and 4178 doublet final levels (1976Me12).                        |
|                     |                         | 4841.7             | 15            | 3162.99 | 7/2 <sup>-</sup> |                    |            | $I_\gamma$ : other: 17.0 17 (1972Hu10).<br>$A_2=-0.084$ 14, $A_4=-0.0041$ 15 (1981Bi05).                                       |
|                     |                         | 5001.9             | 2             | 3002.75 | 5/2              |                    |            | $I_\gamma$ : other: 3.0 15 (1972Hu10).                                                                                         |
|                     |                         | 5310.7             | 1.7           | 2693.9  | (3/2,5/2)        |                    |            |                                                                                                                                |
|                     |                         | 5358.9             | 4             | 2645.70 | (5/2,7/2)        |                    |            |                                                                                                                                |
|                     |                         | 6241.3             | 15            | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 20.0 20 (1972Hu10).<br>$A_2=-0.11$ 2, $A_4=-0.09$ 3 (1976Me12).                                            |
|                     |                         | 6784.9             | 0.6           | 1219.39 | 1/2 <sup>+</sup> |                    |            |                                                                                                                                |
|                     |                         | 8004.0             | 55            | 0       | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 60.0 60 (1972Hu10).<br>$A_2=-0.03$ 1, $A_4=-0.02$ 2 (1976Me12).                                            |
| 8035.7              |                         | 2230               | 6.0 6         | 5806    |                  |                    |            | $I_\gamma$ : from 1972Hu10.                                                                                                    |
|                     |                         | 3857.6             | 15.0 15       | 4177.9  | 3/2              |                    |            | $I_\gamma$ : from 1972Hu10.                                                                                                    |
|                     |                         | 3976.3             | 11.0 11       | 4059.2  |                  |                    |            | $I_\gamma$ : from 1972Hu10.                                                                                                    |
|                     |                         | 5341.4             | 5.0 25        | 2693.9  | (3/2,5/2)        |                    |            | $I_\gamma$ : from 1972Hu10.                                                                                                    |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$               | $E_\gamma^\dagger$ | $I_\gamma^\#$ | $E_f$   | $J_f^\pi$        | Mult. <sup>b</sup> | $\delta^b$ | Comments                                                                                                                                                                                                                                                                                        |
|---------------------|-------------------------|--------------------|---------------|---------|------------------|--------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8035.7              |                         | 6272.0             | 5.0 25        | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : from 1972Hu10.                                                                                                                                                                                                                                                                     |
|                     |                         | 6815.6             | 56 6          | 1219.39 | 1/2 <sup>+</sup> |                    |            | $I_\gamma$ : from 1972Hu10.                                                                                                                                                                                                                                                                     |
|                     |                         | 8034.7             | 2.0 10        | 0       | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : from 1972Hu10.                                                                                                                                                                                                                                                                     |
| 8038.5              | 3/2                     | 3183.9             | 0.6           | 4854.4  |                  |                    |            |                                                                                                                                                                                                                                                                                                 |
|                     |                         | 3199.2             | 1.6           | 4839.10 |                  |                    |            |                                                                                                                                                                                                                                                                                                 |
|                     |                         | 3413.9             | 3             | 4624.4  |                  |                    |            |                                                                                                                                                                                                                                                                                                 |
|                     |                         | 3860.4             | 1.6           | 4177.9  | 3/2              |                    |            |                                                                                                                                                                                                                                                                                                 |
|                     |                         | 3979.1             | 3             | 4059.2  |                  |                    |            | $I_\gamma$ : other: 4 2 (1972Hu10).                                                                                                                                                                                                                                                             |
|                     |                         | 4071.0             | 0.8           | 3967.3  |                  |                    |            |                                                                                                                                                                                                                                                                                                 |
|                     |                         | 4119.8             | 1.4           | 3918.48 | 3/2 <sup>+</sup> |                    |            |                                                                                                                                                                                                                                                                                                 |
|                     |                         | 5035.4             | 4             | 3002.75 | 5/2              |                    |            | $I_\gamma$ : other: 3.0 15 (1972Hu10).                                                                                                                                                                                                                                                          |
|                     |                         | 5344.2             | 57            | 2693.9  | (3/2,5/2)        | D+Q                | -0.20 2    | $I_\gamma$ : other: 61.0 61 (1972Hu10).<br>Mult., $\delta$ : or +19 +10-5 (1976Me12).<br>$A_2=+0.08$ 2, $A_4=-0.03$ 2 (1976Me12).<br>$I_\gamma$ : other: 17.0 17 (1972Hu10).<br>$A_2=+0.24$ 2, $A_4=+0.01$ 3 (1976Me12).                                                                        |
|                     |                         | 6274.8             | 15            | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 6.0 6 (1972Hu10).                                                                                                                                                                                                                                                           |
|                     |                         | 6818.4             | 4             | 1219.39 | 1/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 9.0 9 (1972Hu10).                                                                                                                                                                                                                                                           |
|                     |                         | 8037.5             | 8             | 0       | 3/2 <sup>+</sup> | D+Q                | -0.13 4    | Mult., $\delta$ : or +8.1 +33-18 (1976Me12).<br>$A_2=+0.20$ 3, $A_4=-0.08$ 3 (1976Me12).<br>$I_\gamma$ : other: 10.0 10 (1972Hu10).<br>$I_\gamma$ : other: 11.0 11 (1972Hu10).<br>$I_\gamma$ : other: 7.0 7 (1972Hu10).<br>$I_\gamma$ : 1.0 5 from 1972Hu10.<br>$I_\gamma$ : 2 1 from 1972Hu10. |
| 8075.9              |                         | 2430               | 11            | 5646    |                  |                    |            | $I_\gamma$ : other: 4 2 (1972Hu10).                                                                                                                                                                                                                                                             |
|                     |                         | 2490               | 11            | 5586    |                  |                    |            | $I_\gamma$ : other: 7.0 7 (1972Hu10).                                                                                                                                                                                                                                                           |
|                     |                         | 2860.0             | 8             | 5215.8  |                  |                    |            | $I_\gamma$ : other: 2 1 (1972Hu10).                                                                                                                                                                                                                                                             |
|                     |                         | 2912.4             |               | 5163.34 | 7/2              |                    |            | $I_\gamma$ : other: 2 1 (1972Hu10).                                                                                                                                                                                                                                                             |
|                     |                         | 3194.8             |               | 4880.96 |                  |                    |            | $I_\gamma$ : other: 53.0 53 (1972Hu10).                                                                                                                                                                                                                                                         |
|                     |                         | 3305.9             | 4             | 4769.86 |                  |                    |            | $I_\gamma$ : 1.0 5 from 1972Hu10.                                                                                                                                                                                                                                                               |
|                     |                         | 3897.8             | 8             | 4177.9  | 3/2              |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                                                                                                                                                                                                                                           |
|                     |                         | 3962.0             | 2             | 4113.7  | 7/2              |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                                                                                                                                                                                                                                           |
|                     |                         | 5072.8             | 2             | 3002.75 | 5/2              |                    |            | $I_\gamma$ : other: 2 1 (1972Hu10).                                                                                                                                                                                                                                                             |
|                     |                         | 6312.2             | 56            | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 53.0 53 (1972Hu10).                                                                                                                                                                                                                                                         |
|                     |                         | 8074.9             |               | 0       | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : 1.0 5 from 1972Hu10.                                                                                                                                                                                                                                                               |
| 8096.1              | (5/2,7/2 <sup>+</sup> ) | 2290               | 2             | 5806    |                  |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                                                                                                                                                                                                                                           |
|                     |                         | 2450               | 5             | 5646    |                  |                    |            | $I_\gamma$ : other: 5.0 25 (1972Hu10).                                                                                                                                                                                                                                                          |
|                     |                         | 2496.3             | 5             | 5599.7  |                  |                    |            | $I_\gamma$ : other: 6.0 6 (1972Hu10).                                                                                                                                                                                                                                                           |
|                     |                         | 2880.2             | 1.3           | 5215.8  |                  |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                                                                                                                                                                                                                                           |
|                     |                         | 2932.6             | 2             | 5163.34 | 7/2              |                    |            | $I_\gamma$ : other: 2 1 (1972Hu10).                                                                                                                                                                                                                                                             |
|                     |                         | 3256.8             | 3             | 4839.10 |                  |                    |            | $I_\gamma$ : other: 3.0 15 (1972Hu10).                                                                                                                                                                                                                                                          |
|                     |                         | 3326.1             | 2             | 4769.86 |                  |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                                                                                                                                                                                                                                           |
|                     |                         | 3918.0             | 1.9           | 4177.9  | 3/2              |                    |            |                                                                                                                                                                                                                                                                                                 |
|                     |                         | 4036.7             | 2             | 4059.2  |                  |                    |            | $I_\gamma$ : other: 2 1 (1972Hu10).                                                                                                                                                                                                                                                             |
|                     |                         | 4177.4             | 1.8           | 3918.48 | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 3.0 15 (1972Hu10).                                                                                                                                                                                                                                                          |
|                     |                         | 4932.7             | 5             | 3162.99 | 7/2 <sup>-</sup> |                    |            | $I_\gamma$ : other: 5.0 25 (1972Hu10).                                                                                                                                                                                                                                                          |
|                     |                         | 5401.8             | 5             | 2693.9  | (3/2,5/2)        |                    |            | $I_\gamma$ : other: 5.0 25 (1972Hu10).                                                                                                                                                                                                                                                          |
|                     |                         | 5449.9             | 17            | 2645.70 | (5/2,7/2)        |                    |            | $I_\gamma$ : other: 17.0 17 (1972Hu10).                                                                                                                                                                                                                                                         |
|                     |                         | 6332.4             | 8             | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 9.0 9 (1972Hu10).                                                                                                                                                                                                                                                           |
|                     |                         | 8095.1             | 39            | 0       | 3/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 40.0 40 (1972Hu10).                                                                                                                                                                                                                                                         |
| 8106.4              | 3/2                     | 2506.6             | 0.5           | 5599.7  |                  |                    |            | $I_\gamma$ : other: 0.5 3 (1972Hu10).                                                                                                                                                                                                                                                           |
|                     |                         | 3251.8             | 0.6           | 4854.4  |                  |                    |            | $I_\gamma$ : other: 0.5 3 (1972Hu10).                                                                                                                                                                                                                                                           |
|                     |                         | 3267.1             | 0.5           | 4839.10 |                  |                    |            | $I_\gamma$ : other: 0.5 3 (1972Hu10).                                                                                                                                                                                                                                                           |
|                     |                         | 3928.3             | 1.6           | 4177.9  | 3/2              |                    |            | $I_\gamma$ : other: 1.0 5 (1972Hu10).                                                                                                                                                                                                                                                           |
|                     |                         | 4047.0             | 1.3           | 4059.2  |                  |                    |            | $I_\gamma$ : other: 0.5 3 (1972Hu10).                                                                                                                                                                                                                                                           |
|                     |                         | 4138.8             | 3             | 3967.3  |                  |                    |            | $I_\gamma$ : other: 3.0 15 (1972Hu10).                                                                                                                                                                                                                                                          |
|                     |                         | 5103.3             | 9             | 3002.75 | 5/2              |                    |            | $I_\gamma$ : other: 8.0 8 (1972Hu10).                                                                                                                                                                                                                                                           |
|                     |                         | 5412.1             | 9             | 2693.9  | (3/2,5/2)        | D+Q                | +4.9 10    | $I_\gamma$ : other: 8.0 8 (1972Hu10).                                                                                                                                                                                                                                                           |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)**

| $\gamma(^{35}\text{Cl})$ (continued) |                                           |                    |                  |         |                  |                    |                                                    |                                            |
|--------------------------------------|-------------------------------------------|--------------------|------------------|---------|------------------|--------------------|----------------------------------------------------|--------------------------------------------|
| $E_i(\text{level})$                  | $J_i^\pi$                                 | $E_\gamma^\dagger$ | $I_\gamma^\#$    | $E_f$   | $J_f^\pi$        | Mult. <sup>b</sup> | $\delta^b$                                         | Comments                                   |
|                                      |                                           |                    |                  |         |                  |                    |                                                    | Mult., $\delta$ : $-0.05$ 4 (1976Me12).    |
| 8106.4                               | 3/2                                       | 6342.6             | 3                | 1763.14 | 5/2 <sup>+</sup> |                    |                                                    | $A_2=+0.32$ 4, $A_4=-0.08$ 5 (1976Me12).   |
|                                      |                                           | 6886.3             | 35               | 1219.39 | 1/2 <sup>+</sup> |                    |                                                    | $I_\gamma$ : other: 3.0 15 (1972Hu10).     |
|                                      |                                           |                    |                  |         |                  |                    |                                                    | $I_\gamma$ : other: 35.0 35 (1972Hu10).    |
|                                      |                                           | 8105.4             | 37               | 0       | 3/2 <sup>+</sup> | D+Q                | +1.2 1                                             | $A_2=-0.46$ 2, $A_4=-0.05$ 3 (1976Me12).   |
|                                      |                                           |                    |                  |         |                  |                    |                                                    | $I_\gamma$ : other: 40.0 40 (1972Hu10).    |
|                                      |                                           |                    |                  |         |                  |                    |                                                    | Mult., $\delta$ : or $-0.46$ 4 (1976Me12). |
| 8113.3                               | (1/2 <sup>+</sup> ,3/2,5/2 <sup>+</sup> ) | 1621               | 6                | 6492    |                  |                    |                                                    | $A_2=+0.83$ 3, $A_4=+0.01$ 2 (1976Me12).   |
|                                      |                                           | 2007.0             | 6                | 6106.2  |                  |                    |                                                    | $I_\gamma$ : other: 6.0 6 (1972Hu10).      |
|                                      |                                           | 2307               | 1.1              | 5806    |                  |                    |                                                    |                                            |
|                                      |                                           | 2458               | 0.9              | 5655    |                  |                    |                                                    |                                            |
|                                      |                                           | 2513.5             |                  | 5599.7  |                  |                    |                                                    | $I_\gamma$ : 3.0 15 (1972Hu10).            |
|                                      |                                           | 2897.4             | 1.8              | 5215.8  |                  |                    |                                                    |                                            |
|                                      |                                           | 3939.6             | 7                | 4173.45 |                  |                    |                                                    | $I_\gamma$ : other: 5.0 25 (1972Hu10).     |
|                                      |                                           | 4053.9             | 1.3              | 4059.2  |                  |                    |                                                    |                                            |
|                                      |                                           | 4145.7             | 5                | 3967.3  |                  |                    |                                                    | $I_\gamma$ : other: 5.0 25 (1972Hu10).     |
|                                      |                                           | 5110.2             | 1.9              | 3002.75 | 5/2              |                    |                                                    |                                            |
|                                      |                                           | 5419.0             | 13               | 2693.9  | (3/2,5/2)        |                    |                                                    | $I_\gamma$ : other: 17.0 17 (1972Hu10).    |
|                                      |                                           | 6349.5             | 5                | 1763.14 | 5/2 <sup>+</sup> |                    |                                                    | $I_\gamma$ : other: 6.0 6 (1972Hu10).      |
|                                      |                                           | 6893.2             | 32               | 1219.39 | 1/2 <sup>+</sup> |                    |                                                    | $I_\gamma$ : other: 32.0 32 (1972Hu10).    |
|                                      |                                           | 8112.3             | 25               | 0       | 3/2 <sup>+</sup> |                    |                                                    | $I_\gamma$ : other: 26.0 26 (1972Hu10).    |
|                                      |                                           | 8147.2             | 3/2 <sup>-</sup> | 2464    | 1.0              | 5683               |                                                    |                                            |
| 3137.0                               | 3                                         |                    |                  | 5010.1  |                  |                    |                                                    |                                            |
| 3969.1                               | 8                                         |                    |                  | 4177.9  | 3/2              |                    |                                                    |                                            |
| 4087.7                               | 3                                         |                    |                  | 4059.2  |                  |                    |                                                    |                                            |
| 4179.6                               | 2                                         |                    |                  | 3967.3  |                  |                    |                                                    |                                            |
| 4228.5                               | 3                                         |                    |                  | 3918.48 | 3/2 <sup>+</sup> |                    |                                                    | $I_\gamma$ : other: 3.0 15 (1972Hu10).     |
| 5452.8                               | 3                                         |                    |                  | 2693.9  | (3/2,5/2)        |                    |                                                    | $I_\gamma$ : other: 4 2 (1972Hu10).        |
| 6927.1                               | 15                                        |                    |                  | 1219.39 | 1/2 <sup>+</sup> |                    |                                                    | $I_\gamma$ : other: 20.0 20 (1972Hu10).    |
| 8146.2                               | 62                                        |                    |                  | 0       | 3/2 <sup>+</sup> |                    |                                                    | $I_\gamma$ : other: 75.0 75 (1972Hu10).    |
| 8156.8                               | (5/2 <sup>+</sup> ,7/2 <sup>-</sup> )     |                    |                  | 2511    | 4                | 5646               |                                                    |                                            |
|                                      |                                           | 2571               | 8                | 5586    |                  |                    | $I_\gamma$ : other: 7.0 7 (1972Hu10).              |                                            |
|                                      |                                           | 2940.9             | 6                | 5215.8  |                  |                    | $I_\gamma$ : other: 8.0 8 (1972Hu10).              |                                            |
|                                      |                                           | 3386.8             | 1.0              | 4769.86 |                  |                    |                                                    |                                            |
|                                      |                                           | 3978.7             | 5                | 4177.9  | 3/2              |                    | $I_\gamma$ : other: 8.0 8 (1972Hu10).              |                                            |
|                                      |                                           | 4042.9             | 7                | 4113.7  | 7/2              |                    | $I_\gamma$ : other: 7.0 7 (1972Hu10).              |                                            |
|                                      |                                           | 4097.3             | 1                | 4059.2  |                  |                    |                                                    |                                            |
|                                      |                                           | 4213.6             | 2                | 3942.9  |                  |                    |                                                    |                                            |
|                                      |                                           | 4993.4             | 5                | 3162.99 | 7/2 <sup>-</sup> |                    | $I_\gamma$ : other: 9.0 9 (1972Hu10).              |                                            |
|                                      |                                           | 5153.6             | 6                | 3002.75 | 5/2              |                    | $I_\gamma$ : other: 4 2 (1972Hu10).                |                                            |
|                                      |                                           | 5510.6             | 51               | 2645.70 | (5/2,7/2)        |                    | $I_\gamma$ : other: 50.0 50 (1972Hu10).            |                                            |
|                                      |                                           | 6393.0             | 1                | 1763.14 | 5/2 <sup>+</sup> |                    |                                                    |                                            |
|                                      |                                           | 8155.8             | 3                | 0       | 3/2 <sup>+</sup> |                    | $I_\gamma$ : other: 4 2 (1972Hu10).                |                                            |
|                                      |                                           | 8179.4             |                  | 3554.8  | 9.0 9            | 4624.4             |                                                    |                                            |
| 5016.0                               | 8.0 8                                     |                    |                  | 3162.99 | 7/2 <sup>-</sup> |                    | $I_\gamma$ : from 1972Hu10.                        |                                            |
| 5176.2                               | 4.0 20                                    |                    |                  | 3002.75 | 5/2              |                    | $I_\gamma$ : from 1972Hu10.                        |                                            |
| 5485.0                               | 3.0 15                                    |                    |                  | 2693.9  | (3/2,5/2)        |                    | $I_\gamma$ : from 1972Hu10.                        |                                            |
| 6415.6                               | 68 7                                      |                    |                  | 1763.14 | 5/2 <sup>+</sup> |                    | $I_\gamma$ : from 1972Hu10.                        |                                            |
| 8178.4                               | 8.0 8                                     |                    |                  | 0       | 3/2 <sup>+</sup> |                    | $I_\gamma$ : from 1972Hu10.                        |                                            |
| 8208.2                               | 5/2 <sup>+</sup>                          | 2553               | 0.7              | 5655    |                  |                    | $I_\gamma$ : other: 1 (2001Vo24).                  |                                            |
|                                      |                                           | 3327.1             | 0.8              | 4880.96 |                  |                    | $I_\gamma$ : other: 1 (2001Vo24).                  |                                            |
|                                      |                                           | 3368.9             | 0.4              | 4839.10 |                  |                    | $I_\gamma$ : other: 1 (2001Vo24).                  |                                            |
|                                      |                                           | 4148.7             | 1.5              | 4059.2  |                  |                    | $I_\gamma$ : others: 2 (2001Vo24); 2 1 (1972Hu10). |                                            |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$ | $E_\gamma^\dagger$ | $I_\gamma^\#$ | $E_f$   | $J_f^\pi$    | Mult. <sup>b</sup> | $\delta^b$ | Comments                                                                                                |
|---------------------|-----------|--------------------|---------------|---------|--------------|--------------------|------------|---------------------------------------------------------------------------------------------------------|
| 8208.2              | $5/2^+$   | 4289.4             |               | 3918.48 | $3/2^+$      |                    |            | $I_\gamma$ : 1.0 5 (1972Hu10).                                                                          |
|                     |           | 5205.0             | 1.1           | 3002.75 | $5/2$        |                    |            | $I_\gamma$ : others: 1 (2001Vo24); 1.0 5 (1972Hu10).                                                    |
|                     |           | 5513.8             | 0.5           | 2693.9  | $(3/2, 5/2)$ |                    |            | $I_\gamma$ : other: 1 (2001Vo24); 1.0 5 (1972Hu10).                                                     |
|                     |           | 6444.4             | 14            | 1763.14 | $5/2^+$      | D+Q                | +0.33 4    | $I_\gamma$ : other: 14 (2001Vo24); 16.0 16 (1972Hu10).                                                  |
|                     |           |                    |               |         |              |                    |            | Mult., $\delta$ : from 2001Vo24.                                                                        |
|                     |           |                    |               |         |              |                    |            | $A_2=+0.51$ 13, $A_4=-0.21$ 11, $A_6=+0.0004$ (1996Ka21).                                               |
|                     |           |                    |               |         |              |                    |            | $A_2=+0.51$ 13, $A_4=-0.21$ 11 (2001Vo24).                                                              |
|                     |           | 6988.1             | 3             | 1219.39 | $1/2^+$      |                    |            | $I_\gamma$ : other: 3 (2001Vo24); 3.0 15 (1972Hu10).                                                    |
|                     |           | 8207.2             | 78            | 0       | $3/2^+$      | D+Q                | -0.36 4    | $I_\gamma$ : other: 78 (2001Vo24); 76.0 76 (1972Hu10).                                                  |
|                     |           |                    |               |         |              |                    |            | Mult., $\delta$ : from 2001Vo24. Other: $\delta(\text{Q/D})=0.52$ 42 from 1996Ka21.                     |
| 8216.3              | $5/2$     |                    |               |         |              |                    |            | $A_2=-0.55$ 8, $A_4=+0.004$ 78, $A_6=+0.006$ 72 (1996Ka21).                                             |
|                     |           |                    |               |         |              |                    |            | $A_2=-0.55$ 8, $A_4=0.00$ 8 (2001Vo24).                                                                 |
|                     |           | 2561               | 0.8           | 5655    |              |                    |            | $I_\gamma$ : other: 1 (2001Vo24).                                                                       |
|                     |           | 3446.3             | 0.8           | 4769.86 |              |                    |            | $I_\gamma$ : other: 1 (2001Vo24).                                                                       |
|                     |           | 4038.2             | <3            | 4177.9  | $3/2$        |                    |            | $I_\gamma$ : other: 3 (2001Vo24).                                                                       |
|                     |           | 4042.6             | <3            | 4173.45 |              |                    |            |                                                                                                         |
|                     |           | 5052.9             | 41            | 3162.99 | $7/2^-$      | D+Q                | +0.11 2    | $I_\gamma$ : other: 41 (2001Vo24); 44.0 44 (1972Hu10).                                                  |
|                     |           |                    |               |         |              |                    |            | Mult., $\delta$ : from $\gamma(\theta)$ in 2001Vo24.                                                    |
|                     |           |                    |               |         |              |                    |            | $A_2=-0.32$ 7, $A_4=-0.11$ 7, $A_6=+0.05$ 6 (1996Ka21).                                                 |
|                     |           |                    |               |         |              |                    |            | $A_2=-0.32$ 7, $A_4=-0.11$ 7 (2001Vo24).                                                                |
|                     |           | 5213.1             | 5             | 3002.75 | $5/2$        |                    |            | $I_\gamma$ : other: 5 (2001Vo24); 5.0 25 (1972Hu10).                                                    |
|                     |           | 5521.9             | 3             | 2693.9  | $(3/2, 5/2)$ |                    |            | $I_\gamma$ : other: 3 (2001Vo24); 4 2 (1972Hu10).                                                       |
|                     |           | 6452.5             | 1.4           | 1763.14 | $5/2^+$      |                    |            | $I_\gamma$ : other: 1 (2001Vo24).                                                                       |
|                     |           | 8215.3             | 45            | 0       | $3/2^+$      | D+Q                | +0.06 3    | $I_\gamma$ : other: 45 (2001Vo24); 47.0 47 (1972Hu10).                                                  |
|                     |           |                    |               |         |              |                    |            | Mult., $\delta$ : from $\gamma(\theta)$ in 2001Vo24. Other: $\delta(\text{Q/D})=0.59$ 56 from 1996Ka21. |
|                     |           |                    |               |         |              |                    |            | $A_2=-0.560$ 8, $A_4=+0.017$ 71, $A_6=+0.066$ 75 (1996Ka21).                                            |
|                     |           |                    |               |         |              |                    |            | $A_2=-0.56$ 1, $A_4=+0.02$ 7 (2001Vo24).                                                                |
| 8242.4              |           | 4068.7             | 7             | 4173.45 |              |                    |            |                                                                                                         |
|                     |           | 5239.2             | 2             | 3002.75 | $5/2$        |                    |            |                                                                                                         |
|                     |           | 7022.3             | 91            | 1219.39 | $1/2^+$      |                    |            |                                                                                                         |
| 8251?               |           | 4137               | 4&            | 4113.7  | $7/2$        |                    |            |                                                                                                         |
|                     |           | 5248               | 6&            | 3002.75 | $5/2$        |                    |            |                                                                                                         |
|                     |           | 7031               | 82&           | 1219.39 | $1/2^+$      |                    |            |                                                                                                         |
|                     |           | 8250               | 8&            | 0       | $3/2^+$      |                    |            |                                                                                                         |
| 8268.9              |           | 3498.9             | 2             | 4769.86 |              |                    |            | $I_\gamma$ : from 1976Sp08.                                                                             |
|                     |           | 4090.7             | 12            | 4177.9  | $3/2$        |                    |            |                                                                                                         |
|                     |           | 4350.1             | 3             | 3918.48 | $3/2^+$      |                    |            |                                                                                                         |
|                     |           | 5105.5             | 7             | 3162.99 | $7/2^-$      |                    |            |                                                                                                         |
|                     |           | 5265.7             | 13            | 3002.75 | $5/2$        |                    |            |                                                                                                         |
|                     |           | 5574.5             | 6             | 2693.9  | $(3/2, 5/2)$ |                    |            |                                                                                                         |
|                     |           | 5622.7             | 4             | 2645.70 | $(5/2, 7/2)$ |                    |            |                                                                                                         |
|                     |           | 6505.1             | 8             | 1763.14 | $5/2^+$      |                    |            |                                                                                                         |
|                     |           | 8267.9             | 45            | 0       | $3/2^+$      |                    |            |                                                                                                         |
|                     |           | 2622               | 1.5           | 5655    |              |                    |            |                                                                                                         |
| 8277.2              |           | 3396.1             | 5             | 4880.96 |              |                    |            |                                                                                                         |
|                     |           | 3437.9             | 11            | 4839.10 |              |                    |            |                                                                                                         |
|                     |           | 3507.2             | 1.2           | 4769.86 |              |                    |            |                                                                                                         |
|                     |           | 3652.6             | 3             | 4624.4  |              |                    |            |                                                                                                         |
|                     |           | 3929.2             | 1.8           | 4347.76 | $5/2, 9/2$   |                    |            |                                                                                                         |
|                     |           |                    |               |         |              |                    |            |                                                                                                         |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10** (continued) $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$               | $E_\gamma^\dagger$ | $I_\gamma^\#$ | $E_f$   | $J_f^\pi$        | Mult. <sup>b</sup> | Comments                                                                   |
|---------------------|-------------------------|--------------------|---------------|---------|------------------|--------------------|----------------------------------------------------------------------------|
| 8277.2              |                         | 4103.5             | 2             | 4173.45 |                  |                    |                                                                            |
|                     |                         | 4217.7             | 1.5           | 4059.2  |                  |                    |                                                                            |
|                     |                         | 4358.4             | 5             | 3918.48 | 3/2 <sup>+</sup> |                    |                                                                            |
|                     |                         | 5274.0             | 10            | 3002.75 | 5/2              |                    |                                                                            |
|                     |                         | 5582.8             | 3             | 2693.9  | (3/2,5/2)        |                    |                                                                            |
|                     |                         | 5631.0             | 3             | 2645.70 | (5/2,7/2)        |                    |                                                                            |
|                     |                         | 6513.4             | 23            | 1763.14 | 5/2 <sup>+</sup> |                    |                                                                            |
|                     |                         | 8276.2             | 29            | 0       | 3/2 <sup>+</sup> |                    |                                                                            |
| 8282.4              | (3/2 <sup>-</sup> ,5/2) | 2696               | 8             | 5586    |                  |                    |                                                                            |
|                     |                         | 3066.5             | 8             | 5215.8  |                  |                    |                                                                            |
|                     |                         | 3118.9             | 3             | 5163.34 | 7/2              |                    |                                                                            |
|                     |                         | 4104.2             | 14            | 4177.9  | 3/2              |                    |                                                                            |
|                     |                         | 4222.9             | 6             | 4059.2  |                  |                    |                                                                            |
|                     |                         | 5119.0             | 26            | 3162.99 | 7/2 <sup>-</sup> |                    |                                                                            |
|                     |                         | 6518.6             | 7             | 1763.14 | 5/2 <sup>+</sup> |                    |                                                                            |
|                     |                         | 8281.4             | 28            | 0       | 3/2 <sup>+</sup> |                    |                                                                            |
| 8287.82             | 1/2 <sup>-</sup>        | 2633               | 2             | 5655    |                  |                    |                                                                            |
|                     |                         | 3277.6             | 3             | 5010.1  |                  |                    |                                                                            |
|                     |                         | 4109.7             | 9             | 4177.9  | 3/2              |                    |                                                                            |
|                     |                         | 4228.4             | 2             | 4059.2  |                  |                    |                                                                            |
|                     |                         | 4320.2             | 3             | 3967.3  |                  |                    |                                                                            |
|                     |                         | 4369.05            | 1             | 3918.48 | 3/2 <sup>+</sup> |                    |                                                                            |
|                     |                         | 5593.4             | 23            | 2693.9  | (3/2,5/2)        |                    | $A_2=-0.01$ 4, $A_4=-0.02$ 4 (1976Me12).                                   |
|                     |                         | 7067.66            | 44            | 1219.39 | 1/2 <sup>+</sup> |                    | $\delta: \delta(\text{Q/D})=+0.21$ 2 or $-3.1$ 2 for $J_i=3/2$ (1976Me12). |
|                     |                         |                    |               |         |                  |                    | $A_2=-0.08$ 2, $A_4=-0.01$ 3 (1976Me12).                                   |
|                     |                         | 8286.77            | 13            | 0       | 3/2 <sup>+</sup> |                    | $\delta: \delta(\text{Q/D})=-0.35$ 4 or $-11$ 3 for $J_i=3/2$ (1976Me12).  |
|                     |                         |                    |               |         |                  |                    | $A_2=-0.11$ 3, $A_4=-0.01$ 4 (1976Me12).                                   |
| 8298.2              | 3/2                     | 3287.9             | 3             | 5010.1  |                  |                    |                                                                            |
|                     |                         | 3417.1             | 4             | 4880.96 |                  |                    |                                                                            |
|                     |                         | 3443.6             | 5             | 4854.4  |                  |                    |                                                                            |
|                     |                         | 4124.5             | 6             | 4173.45 |                  |                    |                                                                            |
|                     |                         | 4238.7             | 10            | 4059.2  |                  |                    |                                                                            |
|                     |                         | 5134.8             | 1             | 3162.99 | 7/2 <sup>-</sup> |                    |                                                                            |
|                     |                         | 5295.0             | 1             | 3002.75 | 5/2              |                    |                                                                            |
|                     |                         | 5603.8             | 10            | 2693.9  | (3/2,5/2)        |                    | $A_2=+0.09$ 6, $A_4=-0.07$ 7 (1976Me12).                                   |
|                     |                         | 5652.0             | 6             | 2645.70 | (5/2,7/2)        |                    |                                                                            |
|                     |                         | 6534.4             | 8             | 1763.14 | 5/2 <sup>+</sup> |                    | $A_2=+0.43$ 11, $A_4=-0.51$ 12 (1976Me12).                                 |
|                     |                         | 7078.0             | 24            | 1219.39 | 1/2 <sup>+</sup> | D+Q                | Mult., $\delta$ : $-1.7$ 1 or $-0.02$ 2 (1976Me12).                        |
|                     |                         |                    |               |         |                  |                    | $A_2=-0.53$ 3, $A_4=+0.11$ 3 (1976Me12).                                   |
|                     |                         | 8297.1             | 13            | 0       | 3/2 <sup>+</sup> | D+Q                | Mult., $\delta$ : $-0.15$ 4 or $+9.5$ +48-24 (1976Me12).                   |
|                     |                         |                    |               |         |                  |                    | $A_2=+0.18$ 3, $A_4=-0.10$ 4 (1976Me12).                                   |
| 8318.1              |                         | 2512               | 0.7           | 5806    |                  |                    |                                                                            |
|                     |                         | 2672               | 0.7           | 5646    |                  |                    |                                                                            |
|                     |                         | 2718.3             | 1.5           | 5599.7  |                  |                    |                                                                            |
|                     |                         | 3548.1             | 0.3           | 4769.86 |                  |                    |                                                                            |
|                     |                         | 3693.5             | 1.8           | 4624.4  |                  |                    |                                                                            |
|                     |                         | 4139.9             | <3            | 4177.9  | 3/2              |                    |                                                                            |
|                     |                         | 4144.4             | <3            | 4173.45 |                  |                    |                                                                            |
|                     |                         | 4258.6             | 1.2           | 4059.2  |                  |                    |                                                                            |
|                     |                         | 4350.5             | 3             | 3967.3  |                  |                    |                                                                            |
|                     |                         | 5154.7             | 0.5           | 3162.99 | 7/2 <sup>-</sup> |                    |                                                                            |
|                     |                         | 5314.9             | 3             | 3002.75 | 5/2              |                    |                                                                            |
|                     |                         | 5623.7             | 1.6           | 2693.9  | (3/2,5/2)        |                    |                                                                            |
|                     |                         | 5671.9             | 0.9           | 2645.70 | (5/2,7/2)        |                    |                                                                            |
|                     |                         | 6554.3             | 1.8           | 1763.14 | 5/2 <sup>+</sup> |                    |                                                                            |
|                     |                         | 7097.9             | 2             | 1219.39 | 1/2 <sup>+</sup> |                    |                                                                            |

Continued on next page (footnotes at end of table)



$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$        | $E_\gamma^\dagger$ | $I_\gamma^\#$ | $E_f$   | $J_f^\pi$        | Mult. <sup>b</sup> | $\delta^b$ | Comments                                                       |
|---------------------|------------------|--------------------|---------------|---------|------------------|--------------------|------------|----------------------------------------------------------------|
| 8318.1              |                  | 8317.0             | 78            | 0       | 3/2 <sup>+</sup> |                    |            |                                                                |
| 8323.1              |                  | 4209.1             | 1 &           | 4113.7  | 7/2              |                    |            |                                                                |
|                     |                  | 5319.9             | 3 &           | 3002.75 | 5/2              |                    |            |                                                                |
|                     |                  | 5676.9             | 4 &           | 2645.70 | (5/2,7/2)        |                    |            |                                                                |
|                     |                  | 6559.3             | <2 &          | 1763.14 | 5/2 <sup>+</sup> |                    |            |                                                                |
|                     |                  | 7102.9             | 8 &           | 1219.39 | 1/2 <sup>+</sup> |                    |            |                                                                |
|                     |                  | 8322.0             | 82 &          | 0       | 3/2 <sup>+</sup> |                    |            |                                                                |
| 8381.8              | 5/2 <sup>+</sup> | 3611.7             |               | 4769.86 |                  |                    |            | $I_\gamma$ : 1 (2001Vo24).                                     |
|                     |                  | 3757.2             |               | 4624.4  |                  |                    |            | $I_\gamma$ : 1 (2001Vo24).                                     |
|                     |                  | 4267.8             | 8.0 8         | 4113.7  | 7/2              |                    |            | $I_\gamma$ : from 1972Hu10. Other: 7 (2001Vo24).               |
|                     |                  | 4463.0             | 24.0 24       | 3918.48 | 3/2 <sup>+</sup> | D+Q                | +0.15 3    | $I_\gamma$ : from 1972Hu10. Other: 24 (2001Vo24).              |
|                     |                  |                    |               |         |                  |                    |            | Mult., $\delta$ : from $\gamma(\theta)$ in 2001Vo24.           |
|                     |                  |                    |               |         |                  |                    |            | $A_2=+0.25$ 3, $A_4=-0.04$ 2 (2001Vo24).                       |
|                     |                  | 5378.6             | 25.0 25       | 3002.75 | 5/2              | D(+Q)              | +0.02 3    | $I_\gamma$ : from 1972Hu10. Other: 25 (2001Vo24).              |
|                     |                  |                    |               |         |                  |                    |            | Mult., $\delta$ : from $\gamma(\theta)$ in 2001Vo24. Other:    |
|                     |                  |                    |               |         |                  |                    |            | $\delta=-0.11$ 1 from 1967Ko20, 1967Ko23.                      |
|                     |                  |                    |               |         |                  |                    |            | $A_2=+0.60$ 7, $A_4=-0.02$ 7 (1967Ko20,1967Ko23).              |
|                     |                  | 5687.4             | 1.0 5         | 2693.9  | (3/2,5/2)        |                    |            | $A_2=+0.51$ 23, $A_4=-0.05$ 24 (2001Vo24).                     |
|                     |                  |                    |               |         |                  |                    |            | $I_\gamma$ : from 1972Hu10. Other: 1 (2001Vo24).               |
|                     |                  | 5735.6             | 5.0 25        | 2645.70 | (5/2,7/2)        |                    |            | $A_2=+0.22$ 9, $A_4=-0.02$ 11 (1967Ko20,1967Ko23).             |
|                     |                  |                    |               |         |                  |                    |            | $I_\gamma$ : from 1972Hu10. Others: 5 (2001Vo24).              |
|                     |                  |                    |               |         |                  |                    |            | Mult., $\delta$ : D+Q with $\delta=+0.72$ +42-20 and $J_f=3/2$ |
|                     |                  |                    |               |         |                  |                    |            | for level 2645 (1967Ko20).                                     |
|                     |                  | 6618.0             | 34.0 34       | 1763.14 | 5/2 <sup>+</sup> | D+Q                | +0.06 3    | $I_\gamma$ : from 1972Hu10. Other: 34 (2001Vo24).              |
|                     |                  |                    |               |         |                  |                    |            | Mult., $\delta$ : from $\gamma(\theta)$ in 2001Vo24. Other:    |
|                     |                  |                    |               |         |                  |                    |            | $\delta=-0.12$ 10 from $\gamma(\theta)$ in 1967Ko23.           |
|                     |                  |                    |               |         |                  |                    |            | $A_2=+0.79$ 15, $A_4=-0.08$ 15                                 |
|                     |                  |                    |               |         |                  |                    |            | (1967Ko20,1967Ko23).                                           |
|                     |                  |                    |               |         |                  |                    |            | $A_2=+0.02$ 13, $A_4=-0.08$ 13 (2001Vo24).                     |
|                     |                  | 8380.7             | 3.0 15        | 0       | 3/2 <sup>+</sup> | D+Q                | 1.65       | $I_\gamma$ : from 1972Hu10. Other: 2 (2001Vo24).               |
|                     |                  |                    |               |         |                  |                    |            | Mult., $\delta$ : from $\gamma(\theta)$ in 1967Ko23.           |
|                     |                  |                    |               |         |                  |                    |            | $A_2=-0.83$ 22, $A_4=+0.72$ 33                                 |
|                     |                  |                    |               |         |                  |                    |            | (1967Ko20,1967Ko23).                                           |
| 8402.9              |                  | 3632.8             | 4.8 24        | 4769.86 |                  |                    |            |                                                                |
|                     |                  | 4224.7             | 9 5           | 4177.9  | 3/2              |                    |            |                                                                |
|                     |                  | 4343.4             | 23 6          | 4059.2  |                  |                    |            |                                                                |
|                     |                  | 5708.5             | 10 5          | 2693.9  | (3/2,5/2)        |                    |            |                                                                |
|                     |                  | 5756.7             | 7 4           | 2645.70 | (5/2,7/2)        |                    |            |                                                                |
|                     |                  | 6639.1             | 15 4          | 1763.14 | 5/2 <sup>+</sup> |                    |            |                                                                |
|                     |                  | 8401.8             | 31 8          | 0       | 3/2 <sup>+</sup> |                    |            |                                                                |
| 8405.7              |                  | 3524.6             | 15 4          | 4880.96 |                  |                    |            |                                                                |
|                     |                  | 4232.0             | 1.6 8         | 4173.45 |                  |                    |            |                                                                |
|                     |                  | 4291.7             | 1.4 7         | 4113.7  | 7/2              |                    |            |                                                                |
|                     |                  | 5402.5             | 15 4          | 3002.75 | 5/2              |                    |            |                                                                |
|                     |                  | 6641.9             | 43 11         | 1763.14 | 5/2 <sup>+</sup> |                    |            |                                                                |
|                     |                  | 8404.6             | 24 6          | 0       | 3/2 <sup>+</sup> |                    |            |                                                                |
| 8416.7              |                  | 3406.4             | 5 &           | 5010.1  |                  |                    |            |                                                                |
|                     |                  | 4243.0             | 11 &          | 4173.45 |                  |                    |            |                                                                |
|                     |                  | 4473.5             | 3 &           | 3942.9  |                  |                    |            |                                                                |
|                     |                  | 5413.5             | 7 &           | 3002.75 | 5/2              |                    |            |                                                                |
|                     |                  | 5770.5             | 8 &           | 2645.70 | (5/2,7/2)        |                    |            |                                                                |
|                     |                  | 6652.9             | 22 &          | 1763.14 | 5/2 <sup>+</sup> |                    |            |                                                                |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10** (continued) $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$        | $E_\gamma^\dagger$ | $I_\gamma^\#$ | $E_f$   | $J_f^\pi$        | Comments                                                                                                                     |
|---------------------|------------------|--------------------|---------------|---------|------------------|------------------------------------------------------------------------------------------------------------------------------|
| 8416.7              |                  | 7196.5             | 9&            | 1219.39 | 1/2 <sup>+</sup> |                                                                                                                              |
|                     |                  | 8415.6             | 35&           | 0       | 3/2 <sup>+</sup> |                                                                                                                              |
| 8464.3              |                  | 2809               | 8 4           | 5655    |                  |                                                                                                                              |
|                     |                  | 3248.3             | 5.8 29        | 5215.8  |                  |                                                                                                                              |
|                     |                  | 3454.0             | 8 4           | 5010.1  |                  |                                                                                                                              |
|                     |                  | 3609.7             | 3.3 17        | 4854.4  |                  |                                                                                                                              |
|                     |                  | 3839.7             | 4.3 22        | 4624.4  |                  |                                                                                                                              |
|                     |                  | 4286.1             | 5.2 26        | 4177.9  | 3/2              |                                                                                                                              |
|                     |                  | 4290.6             | 5.5 28        | 4173.45 |                  |                                                                                                                              |
|                     |                  | 4496.7             | 5.2 26        | 3967.3  |                  |                                                                                                                              |
|                     |                  | 4545.5             | 2.6 13        | 3918.48 | 3/2 <sup>+</sup> |                                                                                                                              |
|                     |                  | 5461.1             | 17 4          | 3002.75 | 5/2              |                                                                                                                              |
|                     |                  | 5769.9             | 16 4          | 2693.9  | (3/2,5/2)        |                                                                                                                              |
|                     |                  | 7244.1             | 9 4           | 1219.39 | 1/2 <sup>+</sup> |                                                                                                                              |
|                     |                  | 8463.2             | 10 3          | 0       | 3/2 <sup>+</sup> |                                                                                                                              |
| 8484.4              | 3/2 <sup>+</sup> | 2829               | 5.6 28        | 5655    |                  | $I_\gamma$ : other: 6 <a href="#">2001Vo24</a> .                                                                             |
|                     |                  | 3603.2             | 7.0 35        | 4880.96 |                  | $I_\gamma$ : other: 7 <a href="#">2001Vo24</a> .                                                                             |
|                     |                  | 3859.8             | 1.0 5         | 4624.4  |                  | $I_\gamma$ : other: 1 <a href="#">2001Vo24</a> .                                                                             |
|                     |                  | 4516.8             | 1.2 6         | 3967.3  |                  | $I_\gamma$ : other: 1 <a href="#">2001Vo24</a> .                                                                             |
|                     |                  | 4565.6             | 5.2 26        | 3918.48 | 3/2 <sup>+</sup> | $I_\gamma$ : other: 5 <a href="#">2001Vo24</a> .                                                                             |
|                     |                  | 5481.2             | 7 4           | 3002.75 | 5/2              | $I_\gamma$ : other: 7 <a href="#">2001Vo24</a> .                                                                             |
|                     |                  | 5790.0             | 20 5          | 2693.9  | (3/2,5/2)        | $I_\gamma$ : other: 20 <a href="#">2001Vo24</a> .                                                                            |
|                     |                  | 5838.2             | 2.8 14        | 2645.70 | (5/2,7/2)        | $I_\gamma$ : other: 3 <a href="#">2001Vo24</a> .                                                                             |
|                     |                  | 6720.6             | 46 12         | 1763.14 | 5/2 <sup>+</sup> | $I_\gamma$ : other: 46 <a href="#">2001Vo24</a> .                                                                            |
|                     |                  | 8483.3             | 4 2           | 0       | 3/2 <sup>+</sup> | $I_\gamma$ : other: 4 <a href="#">2001Vo24</a> .                                                                             |
| 8486.1              |                  | 2728               | 1.1 6         | 5758    |                  |                                                                                                                              |
|                     |                  | 3861.5             | 1.8 9         | 4624.4  |                  |                                                                                                                              |
|                     |                  | 4307.9             | <7.2          | 4177.9  | 3/2              | $I_\gamma$ : total $I_\gamma=4.8$ 24 for $\gamma$ decays to 4173 and 4178 doublet final levels ( <a href="#">1976Sp08</a> ). |
|                     |                  | 4312.4             | <7.2          | 4173.45 |                  | $I_\gamma$ : total $I_\gamma=4.8$ 24 for $\gamma$ decays to 4173 and 4178 doublet final levels ( <a href="#">1976Sp08</a> ). |
|                     |                  | 4426.6             | 1.9 10        | 4059.2  |                  |                                                                                                                              |
|                     |                  | 4518.5             | 7 4           | 3967.3  |                  |                                                                                                                              |
|                     |                  | 5482.9             | 8 4           | 3002.75 | 5/2              |                                                                                                                              |
|                     |                  | 5791.7             | 19 5          | 2693.9  | (3/2,5/2)        |                                                                                                                              |
|                     |                  | 6722.3             | 3.6 18        | 1763.14 | 5/2 <sup>+</sup> |                                                                                                                              |
|                     |                  | 7265.9             | 6.6 33        | 1219.39 | 1/2 <sup>+</sup> |                                                                                                                              |
|                     |                  | 8485.0             | 46 12         | 0       | 3/2 <sup>+</sup> |                                                                                                                              |
| 8514.3              |                  | 3659.7             | 4.4 22        | 4854.4  |                  |                                                                                                                              |
|                     |                  | 4336.1             | 2 1           | 4177.9  | 3/2              |                                                                                                                              |
|                     |                  | 4454.8             | 27 7          | 4059.2  |                  |                                                                                                                              |
|                     |                  | 4546.7             | 23 6          | 3967.3  |                  |                                                                                                                              |
|                     |                  | 4595.5             | 6.6 33        | 3918.48 | 3/2 <sup>+</sup> |                                                                                                                              |
|                     |                  | 7294.1             | 10 3          | 1219.39 | 1/2 <sup>+</sup> |                                                                                                                              |
|                     |                  | 8513.2             | 27 7          | 0       | 3/2 <sup>+</sup> |                                                                                                                              |
| 8571.9              |                  | 2848.3             | 2.2 11        | 5723.52 |                  |                                                                                                                              |
|                     |                  | 2917               | 1.3 7         | 5655    |                  |                                                                                                                              |
|                     |                  | 2972.1             | 0.7 4         | 5599.7  |                  |                                                                                                                              |
|                     |                  | 3408.4             | 1.8 9         | 5163.34 | 7/2              |                                                                                                                              |
|                     |                  | 3690.7             | 0.9 5         | 4880.96 |                  |                                                                                                                              |
|                     |                  | 4398.2             | 0.7 4         | 4173.45 |                  |                                                                                                                              |
|                     |                  | 4457.9             | 1.1 6         | 4113.7  | 7/2              |                                                                                                                              |
|                     |                  | 4653.1             | 1.1 6         | 3918.48 | 3/2 <sup>+</sup> |                                                                                                                              |
|                     |                  | 5408.5             | 1.6 8         | 3162.99 | 7/2 <sup>-</sup> |                                                                                                                              |
|                     |                  | 5568.7             | 4.5 23        | 3002.75 | 5/2              |                                                                                                                              |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10** (continued) $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$                             | $E_\gamma$ <sup><math>\dagger</math></sup> | $I_\gamma$ <sup><math>\#</math></sup> | $E_f$   | $J_f^\pi$        | Mult. <sup><math>b</math></sup> | $\delta$ <sup><math>b</math></sup> | Comments                                                                                                           |
|---------------------|---------------------------------------|--------------------------------------------|---------------------------------------|---------|------------------|---------------------------------|------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| 8571.9              |                                       | 5877.5                                     | 0.6 3                                 | 2693.9  | (3/2,5/2)        |                                 |                                    |                                                                                                                    |
|                     |                                       | 5925.7                                     | 8 4                                   | 2645.70 | (5/2,7/2)        |                                 |                                    |                                                                                                                    |
|                     |                                       | 7351.7                                     | 5.4 27                                | 1219.39 | 1/2 <sup>+</sup> |                                 |                                    |                                                                                                                    |
|                     |                                       | 8570.8                                     | 70 7                                  | 0       | 3/2 <sup>+</sup> |                                 |                                    |                                                                                                                    |
| 8580.5              |                                       | 2088                                       | 4.4 22                                | 6492    |                  |                                 |                                    |                                                                                                                    |
|                     |                                       | 3741.2                                     | 1.3 7                                 | 4839.10 |                  |                                 |                                    |                                                                                                                    |
|                     |                                       | 4402.3                                     | 4.4 22                                | 4177.9  | 3/2              |                                 |                                    |                                                                                                                    |
|                     |                                       | 4521.0                                     | 2 1                                   | 4059.2  |                  |                                 |                                    |                                                                                                                    |
|                     |                                       | 4661.7                                     | 2.3 12                                | 3918.48 | 3/2 <sup>+</sup> |                                 |                                    |                                                                                                                    |
|                     |                                       | 5886.1                                     | 5.6 28                                | 2693.9  | (3/2,5/2)        |                                 |                                    |                                                                                                                    |
|                     |                                       | 7360.3                                     | 54 6                                  | 1219.39 | 1/2 <sup>+</sup> |                                 |                                    |                                                                                                                    |
|                     |                                       | 8579.4                                     | 26 7                                  | 0       | 3/2 <sup>+</sup> |                                 |                                    |                                                                                                                    |
| 8590.0              |                                       | 2832                                       | 4.8 24                                | 5758    |                  |                                 |                                    |                                                                                                                    |
|                     |                                       | 2935                                       | 3.7 19                                | 5655    |                  |                                 |                                    |                                                                                                                    |
|                     |                                       | 3819.9                                     | 16 4                                  | 4769.86 |                  |                                 |                                    |                                                                                                                    |
|                     |                                       | 4416.3                                     | 13 3                                  | 4173.45 |                  |                                 |                                    |                                                                                                                    |
|                     |                                       | 4476.0                                     | 3.6 18                                | 4113.7  | 7/2              |                                 |                                    |                                                                                                                    |
|                     |                                       | 5426.6                                     | 8 4                                   | 3162.99 | 7/2 <sup>-</sup> |                                 |                                    |                                                                                                                    |
|                     |                                       | 5586.8                                     | 22 6                                  | 3002.75 | 5/2              |                                 |                                    |                                                                                                                    |
|                     |                                       | 5943.8                                     | 5.5 28                                | 2645.70 | (5/2,7/2)        |                                 |                                    |                                                                                                                    |
|                     |                                       | 6826.2                                     | 14 4                                  | 1763.14 | 5/2 <sup>+</sup> |                                 |                                    |                                                                                                                    |
|                     |                                       | 8588.9                                     | 10 3                                  | 0       | 3/2 <sup>+</sup> |                                 |                                    |                                                                                                                    |
| 8612.0              |                                       | 6848.1                                     | 50 5                                  | 1763.14 | 5/2 <sup>+</sup> |                                 |                                    |                                                                                                                    |
|                     |                                       | 7391.8                                     | 17 4                                  | 1219.39 | 1/2 <sup>+</sup> |                                 |                                    |                                                                                                                    |
|                     |                                       | 8610.9                                     | 33 8                                  | 0       | 3/2 <sup>+</sup> |                                 |                                    |                                                                                                                    |
| 8613.7              |                                       | 3732.5                                     | 1.6 8                                 | 4880.96 |                  |                                 |                                    |                                                                                                                    |
|                     |                                       | 4554.2                                     | 5.4 27                                | 4059.2  |                  |                                 |                                    |                                                                                                                    |
|                     |                                       | 4670.5 <sup>c</sup>                        | 4 &                                   | 3942.9  |                  |                                 |                                    | $I_\gamma$ : from <b>1967Da12</b> only; not observed by others.                                                    |
|                     |                                       | 4694.9                                     | 7 4                                   | 3918.48 | 3/2 <sup>+</sup> |                                 |                                    |                                                                                                                    |
|                     |                                       | 5610.5                                     | 2.5 13                                | 3002.75 | 5/2              |                                 |                                    |                                                                                                                    |
|                     |                                       | 5919.3                                     | 4.3 22                                | 2693.9  | (3/2,5/2)        |                                 |                                    |                                                                                                                    |
|                     |                                       | 5967.5                                     | 23 6                                  | 2645.70 | (5/2,7/2)        |                                 |                                    |                                                                                                                    |
|                     |                                       | 6849.8 <sup>c</sup>                        | 9 &                                   | 1763.14 | 5/2 <sup>+</sup> |                                 |                                    | $I_\gamma$ : from <b>1967Da12</b> only; not observed by others.                                                    |
|                     |                                       | 7393.5 <sup>c</sup>                        | 2 &                                   | 1219.39 | 1/2 <sup>+</sup> |                                 |                                    | $I_\gamma$ : from <b>1967Da12</b> only; not observed by others.                                                    |
| 8618.1              | (3/2 <sup>+</sup> ,5/2 <sup>+</sup> ) | 8612.6                                     | 56 6                                  | 0       | 3/2 <sup>+</sup> |                                 |                                    |                                                                                                                    |
|                     |                                       | 2894.5                                     | 2.1 11                                | 5723.52 |                  |                                 |                                    |                                                                                                                    |
|                     |                                       | 2963                                       | 1.7 9                                 | 5655    |                  |                                 |                                    |                                                                                                                    |
|                     |                                       | 3763.5                                     | 1.4 7                                 | 4854.4  |                  |                                 |                                    |                                                                                                                    |
|                     |                                       | 3993.5                                     | 1.5 8                                 | 4624.4  |                  |                                 |                                    |                                                                                                                    |
|                     |                                       | 4650.5                                     | 3.0 15                                | 3967.3  |                  |                                 |                                    |                                                                                                                    |
|                     |                                       | 4699.3                                     | 2.3 12                                | 3918.48 | 3/2 <sup>+</sup> |                                 |                                    |                                                                                                                    |
|                     |                                       | 5614.9                                     | 16 4                                  | 3002.75 | 5/2              |                                 |                                    |                                                                                                                    |
|                     |                                       | 5923.7                                     | 14 4                                  | 2693.9  | (3/2,5/2)        |                                 |                                    | $A_2=+0.147$ 33, $A_4=+0.114$ 52 ( <b>1970Ko33</b> ).                                                              |
|                     |                                       | 5971.9                                     | 1.7 9                                 | 2645.70 | (5/2,7/2)        |                                 |                                    |                                                                                                                    |
|                     |                                       | 6854.2                                     | 23 6                                  | 1763.14 | 5/2 <sup>+</sup> |                                 |                                    | $A_2=+0.662$ 23, $A_4=+0.319$ 39 ( <b>1970Ko33</b> reported for 7855 $\gamma$ , 7855 is likely a misprint).        |
|                     |                                       | 8617.0                                     | 33 8                                  | 0       | 3/2 <sup>+</sup> |                                 |                                    | $A_2=-0.293$ 29, $A_4=-0.028$ 47 ( <b>1970Ko33</b> ).                                                              |
| 8630.1              | 7/2                                   | 2906.5                                     | 0.3 2                                 | 5723.52 |                  |                                 |                                    |                                                                                                                    |
|                     |                                       | 3466.6                                     | 8 4                                   | 5163.34 | 7/2              | D+Q                             | +0.60 10                           | Mult., $\delta$ : D+Q from $\gamma(\theta)$ in <b>1976Sp09</b> . $A_2=+0.60$ 3, $A_4=-0.08$ 4 ( <b>1976Sp09</b> ). |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$ | $E_\gamma^\dagger$ | $I_\gamma^\#$ | $E_f$   | $J_f^\pi$        | Mult. <sup>b</sup> | $\delta^b$ | Comments                                                                                                                                                                                                                                                                    |
|---------------------|-----------|--------------------|---------------|---------|------------------|--------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8630.1              | 7/2       | 3860.0             | 1.4 7         | 4769.86 |                  |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 4282.1             | 12 3          | 4347.76 | 5/2,9/2          | D+Q                | -0.184 14  | Mult., $\delta$ : D+Q from $\gamma(\theta)$ in <a href="#">1976Sp09</a> .<br>$A_2=+0.11$ 2, $A_4=+0.02$ 2 ( <a href="#">1976Sp09</a> ).                                                                                                                                     |
|                     |           | 4456.4             | 3.6 18        | 4173.45 |                  |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 4516.1             | 10 5          | 4113.7  | 7/2              | D+Q                | -0.06 2    | Mult., $\delta$ : D+Q from $\gamma(\theta)$ in <a href="#">1976Sp09</a> .<br>$A_2=+0.43$ 2, $A_4=-0.01$ 2 ( <a href="#">1976Sp09</a> ).                                                                                                                                     |
|                     |           | 4686.9             | 10 5          | 3942.9  |                  |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 5466.7             | 3.5 18        | 3162.99 | 7/2 <sup>-</sup> |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 5626.9             | 1.4 7         | 3002.75 | 5/2              |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 5983.9             | 2.6 13        | 2645.70 | (5/2,7/2)        |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 6866.2             | 47 12         | 1763.14 | 5/2 <sup>+</sup> | D+Q                | -0.011 3   | Mult., $\delta$ : D+Q from $\gamma(\theta)$ in <a href="#">1976Sp09</a> .<br>$A_2=+0.10$ 33, $A_4=-0.65$ 46 ( <a href="#">1970Ko33</a> ).<br>$A_2=-0.387$ 7, $A_4=+0.012$ 8 ( <a href="#">1976Sp09</a> ).<br>$A_2=+0.398$ 46, $A_4=+0.019$ 78 ( <a href="#">1970Ko33</a> ). |
|                     |           | 8629.0             | 0.5 3         | 0       | 3/2 <sup>+</sup> |                    |            | $E_\gamma$ : other: 4689 7 ( <a href="#">1970Ko33</a> ).                                                                                                                                                                                                                    |
| 8640                | 5/2       | 4697               | 23 @          | 3942.9  |                  |                    |            | $E_\gamma$ : other: 5473 7 ( <a href="#">1970Ko33</a> ).                                                                                                                                                                                                                    |
|                     |           | 5477               | 20 @          | 3162.99 | 7/2 <sup>-</sup> |                    |            | $E_\gamma$ : other: 5630 7 ( <a href="#">1970Ko33</a> ).                                                                                                                                                                                                                    |
|                     |           | 5637               | 6 @           | 3002.75 | 5/2              |                    |            | $E_\gamma$ : other: 5941 7 ( <a href="#">1970Ko33</a> ).                                                                                                                                                                                                                    |
|                     |           | 5946               | 13 @          | 2693.9  | (3/2,5/2)        |                    |            | $E_\gamma$ : other: 6874 7 ( <a href="#">1970Ko33</a> ).                                                                                                                                                                                                                    |
|                     |           | 6876               | 38 @          | 1763.14 | 5/2 <sup>+</sup> |                    |            | $A_2=-0.52$ 12, $A_4=-0.29$ 19 ( <a href="#">1970Ko33</a> ).                                                                                                                                                                                                                |
|                     |           |                    |               |         |                  |                    |            |                                                                                                                                                                                                                                                                             |
| 8641.4              | 3/2,5/2   | 2986               | 2.9 15        | 5655    |                  |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 3055               | 2.5 13        | 5586    |                  |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 3237.7             | 4.7 24        | 5403.5  |                  |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 3802.1             | 2.6 13        | 4839.10 |                  |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 3871.3             | 24 6          | 4769.86 |                  |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 4016.8             | 1.9 10        | 4624.4  |                  |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 4463.2             | <18           | 4177.9  | 3/2              |                    |            | $I_\gamma$ : total $I_\gamma=14$ 4 for $\gamma$ decays to 4173 and 4178 doublet final levels ( <a href="#">1976Sp08</a> ).                                                                                                                                                  |
|                     |           | 4467.6             | <18           | 4173.45 |                  |                    |            | $I_\gamma$ : total $I_\gamma=14$ 4 for $\gamma$ decays to 4173 and 4178 doublet final levels ( <a href="#">1976Sp08</a> ).                                                                                                                                                  |
|                     |           | 4581.9             | 9 5           | 4059.2  |                  |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 4722.6             | 5.9 30        | 3918.48 | 3/2 <sup>+</sup> |                    |            |                                                                                                                                                                                                                                                                             |
| 8686.2              |           | 5947.0             | 5.2 26        | 2693.9  | (3/2,5/2)        |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 8640.3             | 28 7          | 0       | 3/2 <sup>+</sup> |                    |            | $A_2=-0.12$ 2, $A_4=-0.01$ 2 ( <a href="#">1976Sp09</a> ).                                                                                                                                                                                                                  |
|                     |           | 3100               | 2.1 11        | 5586    |                  |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 3282.5             | 3.1 16        | 5403.5  |                  |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 4508.0             | 2 1           | 4177.9  | 3/2              |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 4626.7             | 3.5 18        | 4059.2  |                  |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 5522.7             | 3.7 19        | 3162.99 | 7/2 <sup>-</sup> |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 5991.8             | 2.8 14        | 2693.9  | (3/2,5/2)        |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 6922.3             | 4.8 24        | 1763.14 | 5/2 <sup>+</sup> |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 7466.0             | 1.9 10        | 1219.39 | 1/2 <sup>+</sup> |                    |            |                                                                                                                                                                                                                                                                             |
| 8696.9              |           | 8685.0             | 76 8          | 0       | 3/2 <sup>+</sup> |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 2891               | 2.9 15        | 5806    |                  |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 3097.1             | 4.8 24        | 5599.7  |                  |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 7476.7             | 12 3          | 1219.39 | 1/2 <sup>+</sup> |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 8695.7             | 80 8          | 0       | 3/2 <sup>+</sup> |                    |            |                                                                                                                                                                                                                                                                             |
| 8750.9              |           | 2945               | 2.2 11        | 5806    |                  |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 3347.2             | 4.9 25        | 5403.5  |                  |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 3534.9             | 2.4 12        | 5215.8  |                  |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 4572.7             | 2.1 11        | 4177.9  | 3/2              |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 4577.1             | 2.3 12        | 4173.45 |                  |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 6056.4             | 3.0 15        | 2693.9  | (3/2,5/2)        |                    |            |                                                                                                                                                                                                                                                                             |
|                     |           | 6987.0             | 30 8          | 1763.14 | 5/2 <sup>+</sup> |                    |            |                                                                                                                                                                                                                                                                             |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $E_\gamma^\dagger$  | $I_\gamma^\#$ | $E_f$   | $J_f^\pi$        | Comments                                                                                                 |
|---------------------|---------------------|---------------|---------|------------------|----------------------------------------------------------------------------------------------------------|
| 8750.9              | 7530.6 <sup>c</sup> | 2             | 1219.39 | 1/2 <sup>+</sup> | $I_\gamma$ : from 1967Da12 only; not observed by others.                                                 |
|                     | 8749.7              | 53 5          | 0       | 3/2 <sup>+</sup> |                                                                                                          |
| 8779.8              | 3179.9              | 4.3 22        | 5599.7  |                  |                                                                                                          |
|                     | 3563.8              | 6.6 33        | 5215.8  |                  |                                                                                                          |
|                     | 4601.6              | <19           | 4177.9  | 3/2              | $I_\gamma$ : total $I_\gamma=15$ 4 for $\gamma$ decays to 4173 and 4178 doublet final levels (1976Sp08). |
|                     | 4606.0              | <19           | 4173.45 |                  | $I_\gamma$ : total $I_\gamma=15$ 4 for $\gamma$ decays to 4173 and 4178 doublet final levels (1976Sp08). |
|                     | 4861.0              | 4.2 21        | 3918.48 | 3/2 <sup>+</sup> |                                                                                                          |
|                     | 5776.5              | 27 7          | 3002.75 | 5/2              |                                                                                                          |
|                     | 6085.3              | 12 3          | 2693.9  | (3/2,5/2)        |                                                                                                          |
|                     | 6133.5              | 13 3          | 2645.70 | (5/2,7/2)        |                                                                                                          |
|                     | 7015.9              | 6.7 34        | 1763.14 | 5/2 <sup>+</sup> |                                                                                                          |
|                     | 8778.6              | 11 3          | 0       | 3/2 <sup>+</sup> |                                                                                                          |
| 8787.2              | 3063.5              | 6.8 34        | 5723.52 |                  |                                                                                                          |
|                     | 3623.7              | 19 5          | 5163.34 | 7/2              |                                                                                                          |
|                     | 4017.1              | 6.7 34        | 4769.86 |                  |                                                                                                          |
|                     | 4609.0              | 9 5           | 4177.9  | 3/2              |                                                                                                          |
|                     | 5623.7              | 5.8 29        | 3162.99 | 7/2 <sup>-</sup> |                                                                                                          |
|                     | 5783.9              | 20 5          | 3002.75 | 5/2              |                                                                                                          |
|                     | 7023.3              | 25 6          | 1763.14 | 5/2 <sup>+</sup> |                                                                                                          |
|                     | 8786.0              | 8 4           | 0       | 3/2 <sup>+</sup> |                                                                                                          |
| 8798.4              | 4738.9              | 8 4           | 4059.2  |                  |                                                                                                          |
|                     | 5795.1              | 16 4          | 3002.75 | 5/2              |                                                                                                          |
|                     | 7034.5              | 7.0 35        | 1763.14 | 5/2 <sup>+</sup> |                                                                                                          |
|                     | 7578.1              | 7.0 35        | 1219.39 | 1/2 <sup>+</sup> |                                                                                                          |
|                     | 8797.2              | 62 6          | 0       | 3/2 <sup>+</sup> |                                                                                                          |
| 8820.9              | 5817.6              | 9 5           | 3002.75 | 5/2              |                                                                                                          |
|                     | 6126.4              | 18 5          | 2693.9  | (3/2,5/2)        |                                                                                                          |
|                     | 7057.0              | 21 5          | 1763.14 | 5/2 <sup>+</sup> |                                                                                                          |
|                     | 8819.7              | 52 5          | 0       | 3/2 <sup>+</sup> |                                                                                                          |
| 8824.2              | 5820.9              | 37 9          | 3002.75 | 5/2              |                                                                                                          |
|                     | 6129.7              | 6.0 3         | 2693.9  | (3/2,5/2)        |                                                                                                          |
|                     | 7060.3              | 16 4          | 1763.14 | 5/2 <sup>+</sup> |                                                                                                          |
|                     | 7603.9              | 22 6          | 1219.39 | 1/2 <sup>+</sup> |                                                                                                          |
|                     | 8823.0              | 19 5          | 0       | 3/2 <sup>+</sup> |                                                                                                          |
| 8829.3              | 3174                | 10 5          | 5655    |                  |                                                                                                          |
|                     | 3990.0              | 1.3 7         | 4839.10 |                  |                                                                                                          |
|                     | 4059.2              | 1.5 8         | 4769.86 |                  |                                                                                                          |
|                     | 4910.5              | 4.5 23        | 3918.48 | 3/2 <sup>+</sup> |                                                                                                          |
|                     | 5826.0              | 22 6          | 3002.75 | 5/2              |                                                                                                          |
|                     | 6134.8              | 13 3          | 2693.9  | (3/2,5/2)        |                                                                                                          |
|                     | 6183.0              | 33 8          | 2645.70 | (5/2,7/2)        |                                                                                                          |
|                     | 7065.4              | 9 5           | 1763.14 | 5/2 <sup>+</sup> |                                                                                                          |
|                     | 7609.0              | 2.6 13        | 1219.39 | 1/2 <sup>+</sup> |                                                                                                          |
|                     | 8828.1              | 3.8 19        | 0       | 3/2 <sup>+</sup> |                                                                                                          |
| 8833.1              | 3617.1              | 6.9 35        | 5215.8  |                  |                                                                                                          |
|                     | 4063.0              | 2.6 13        | 4769.86 |                  |                                                                                                          |
|                     | 4659.3              | 14 4          | 4173.45 |                  |                                                                                                          |
|                     | 4719.1              | 4.7 24        | 4113.7  | 7/2              |                                                                                                          |
|                     | 4773.6              | 11 3          | 4059.2  |                  |                                                                                                          |
|                     | 4914.3              | 11 3          | 3918.48 | 3/2 <sup>+</sup> |                                                                                                          |
|                     | 5669.6              | 3.7 19        | 3162.99 | 7/2 <sup>-</sup> |                                                                                                          |
|                     | 5829.8              | 6.7 34        | 3002.75 | 5/2              |                                                                                                          |
|                     | 6138.6              | 4.5 23        | 2693.9  | (3/2,5/2)        |                                                                                                          |
|                     | 6186.8              | 23 6          | 2645.70 | (5/2,7/2)        |                                                                                                          |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10** (continued) $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$              | $E_\gamma^\dagger$  | $I_\gamma^\#$ | $E_f$   | $J_f^\pi$        | Comments                                                                   |
|---------------------|------------------------|---------------------|---------------|---------|------------------|----------------------------------------------------------------------------|
| 8833.1              |                        | 7069.2              | 9 5           | 1763.14 | 5/2 <sup>+</sup> |                                                                            |
|                     |                        | 8831.9              | 2.9 15        | 0       | 3/2 <sup>+</sup> |                                                                            |
| 8838.1              | 7/2 <sup>-</sup>       | 3674.6 <sup>‡</sup> |               | 5163.34 | 7/2              |                                                                            |
|                     |                        | 3956.9 <sup>‡</sup> |               | 4880.96 |                  |                                                                            |
|                     |                        | 4068.0              | 8 4           | 4769.86 |                  |                                                                            |
|                     |                        | 4490.0 <sup>‡</sup> |               | 4347.76 | 5/2,9/2          |                                                                            |
|                     |                        | 4778.6 <sup>‡</sup> |               | 4059.2  |                  |                                                                            |
|                     |                        | 5674.6              | 29 7          | 3162.99 | 7/2 <sup>-</sup> |                                                                            |
|                     |                        | 7074.2              | 63 6          | 1763.14 | 5/2 <sup>+</sup> | $A_2=-0.20$ 2, $A_4=-0.00$ 2 ( <a href="#">1981Bi05</a> ).                 |
| 8856.1              |                        | 3974.9              | 3.4 17        | 4880.96 |                  |                                                                            |
|                     |                        | 4231.4              | 5.1 26        | 4624.4  |                  |                                                                            |
|                     |                        | 4677.9              | 3.5 18        | 4177.9  | 3/2              |                                                                            |
|                     |                        | 4937.3              | 11 3          | 3918.48 | 3/2 <sup>+</sup> |                                                                            |
|                     |                        | 5692.6              | 51 5          | 3162.99 | 7/2 <sup>-</sup> |                                                                            |
|                     |                        | 6161.6              | 2.8 14        | 2693.9  | (3/2,5/2)        |                                                                            |
|                     |                        | 6209.8              |               | 2645.70 | (5/2,7/2)        |                                                                            |
|                     |                        | 7092.2              | 7 4           | 1763.14 | 5/2 <sup>+</sup> |                                                                            |
|                     |                        | 8854.9              | 16 4          | 0       | 3/2 <sup>+</sup> |                                                                            |
| 8868.6              |                        | 4690.4              | 2.6 13        | 4177.9  | 3/2              |                                                                            |
|                     |                        | 5865.3              | 16 4          | 3002.75 | 5/2              |                                                                            |
|                     |                        | 6174.1              | 12 3          | 2693.9  | (3/2,5/2)        |                                                                            |
|                     |                        | 7104.7              | 4.4 22        | 1763.14 | 5/2 <sup>+</sup> |                                                                            |
|                     |                        | 7648.3              | 16 4          | 1219.39 | 1/2 <sup>+</sup> |                                                                            |
|                     |                        | 8867.4              | 49 12         | 0       | 3/2 <sup>+</sup> |                                                                            |
| 8886.1              |                        | 3231                | 6 3           | 5655    |                  |                                                                            |
|                     |                        | 3286.2              | 6.6 33        | 5599.7  |                  |                                                                            |
|                     |                        | 3670.1              | 2.2 11        | 5215.8  |                  |                                                                            |
|                     |                        | 3722.6              | 9 4           | 5163.34 | 7/2              |                                                                            |
|                     |                        | 4116.0              | 4.5 23        | 4769.86 |                  |                                                                            |
|                     |                        | 4707.9              | 8 4           | 4177.9  | 3/2              |                                                                            |
|                     |                        | 4967.2              | 4.1 21        | 3918.48 | 3/2 <sup>+</sup> |                                                                            |
|                     |                        | 5882.8              | 8 4           | 3002.75 | 5/2              |                                                                            |
|                     |                        | 6239.8              | 26 7          | 2645.70 | (5/2,7/2)        |                                                                            |
|                     |                        | 7122.2              | 12 3          | 1763.14 | 5/2 <sup>+</sup> |                                                                            |
|                     |                        | 8884.9              | 14 4          | 0       | 3/2 <sup>+</sup> |                                                                            |
| 8893.3              | (5/2,7/2) <sup>+</sup> | 3293.4 <sup>c</sup> |               | 5599.7  |                  | $I_\gamma$ : 1 from <a href="#">2001Vo24</a> only; not observed by others. |
|                     |                        | 3307                | 1.6 8         | 5586    |                  |                                                                            |
|                     |                        | 4779.3              | 9 4           | 4113.7  | 7/2              | $I_\gamma$ : other: 9 <a href="#">2001Vo24</a> .                           |
|                     |                        | 4950.0              | 3.9 20        | 3942.9  |                  | $I_\gamma$ : other: 4 <a href="#">2001Vo24</a> .                           |
|                     |                        | 5729.8              | 29 7          | 3162.99 | 7/2 <sup>-</sup> | $I_\gamma$ : other: 29 <a href="#">2001Vo24</a> .                          |
|                     |                        | 6198.8              | 37 9          | 2693.9  | (3/2,5/2)        | $I_\gamma$ : other: 37 <a href="#">2001Vo24</a> .                          |
|                     |                        | 7129.4              | 19 5          | 1763.14 | 5/2 <sup>+</sup> | $I_\gamma$ : other: 19 <a href="#">2001Vo24</a> .                          |
|                     |                        | 8892.1              | 1.0 5         | 0       | 3/2 <sup>+</sup> | $I_\gamma$ : other: 1 <a href="#">2001Vo24</a> .                           |
| 8904.8              |                        | 2724                | 4.3 22        | 6181    |                  |                                                                            |
|                     |                        | 3250                | 3.4 17        | 5655    |                  |                                                                            |
|                     |                        | 3894.5              | 3.9 20        | 5010.1  |                  |                                                                            |
|                     |                        | 4050.2              | 6.2 31        | 4854.4  |                  |                                                                            |
|                     |                        | 4065.5              | 4.5 23        | 4839.10 |                  |                                                                            |
|                     |                        | 4726.6              | 18 5          | 4177.9  | 3/2              |                                                                            |
|                     |                        | 4845.2              | 5.7 29        | 4059.2  |                  |                                                                            |
|                     |                        | 6210.3              | 15 4          | 2693.9  | (3/2,5/2)        |                                                                            |
|                     |                        | 7684.5              | 10 3          | 1219.39 | 1/2 <sup>+</sup> |                                                                            |
|                     |                        | 8903.6              | 29 7          | 0       | 3/2 <sup>+</sup> |                                                                            |
| 8906.8              | 5/2 <sup>+</sup>       | 3252                |               | 5655    |                  | $I_\gamma$ : 4 from <a href="#">2001Vo24</a> only; not observed by others. |
|                     |                        | 3261                | 3.7 19        | 5646    |                  | $I_\gamma$ : other: 2 <a href="#">2001Vo24</a> .                           |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$  **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$        | $E_\gamma^\dagger$  | $I_\gamma^\#$ | $E_f$   | $J_f^\pi$        | Mult. <sup>b</sup> | $\delta^b$ | Comments                                                                                                                              |
|---------------------|------------------|---------------------|---------------|---------|------------------|--------------------|------------|---------------------------------------------------------------------------------------------------------------------------------------|
| 8906.8              | 5/2 <sup>+</sup> | 3306.9              | 2.2 11        | 5599.7  |                  |                    |            |                                                                                                                                       |
|                     |                  | 3321                | 4             | 5586    |                  |                    |            | $I_\gamma$ : other: 4 2001Vo24.                                                                                                       |
|                     |                  | 3387                | 2.5 13        | 5520    |                  |                    |            |                                                                                                                                       |
|                     |                  | 3690.8              | 4.4 22        | 5215.8  |                  |                    |            |                                                                                                                                       |
|                     |                  | 4136.7              | 5.6 28        | 4769.86 |                  |                    |            | $I_\gamma$ : other: 6 2001Vo24.                                                                                                       |
|                     |                  | 4963.5              | 13 3          | 3942.9  |                  |                    |            | $I_\gamma$ : other: 15 2001Vo24.                                                                                                      |
|                     |                  | 7142.9              | 69 7          | 1763.14 | 5/2 <sup>+</sup> |                    |            | $I_\gamma$ : other: 69 2001Vo24.                                                                                                      |
| 8919.8              |                  | 3274                | 5.6 28        | 5646    |                  |                    |            |                                                                                                                                       |
|                     |                  | 4571.7              | 8 4           | 4347.76 | 5/2,9/2          |                    |            |                                                                                                                                       |
|                     |                  | 5756.3              | 57 6          | 3162.99 | 7/2 <sup>-</sup> |                    |            |                                                                                                                                       |
|                     |                  | 6273.5              | 26 7          | 2645.70 | (5/2,7/2)        |                    |            |                                                                                                                                       |
| 8953.0              | 3/2              | 7155.9              | 4 2           | 1763.14 | 5/2 <sup>+</sup> |                    |            |                                                                                                                                       |
|                     |                  | 7732.7              |               | 1219.39 | 1/2 <sup>+</sup> | D+Q                |            | Mult.: from $\gamma(\theta)$ in 1976Sp08; $\delta=-0.34$ 2 or $\delta=-5.0$ 4 (1976Sp09).                                             |
|                     |                  | 8951.8              |               | 0       | 3/2 <sup>+</sup> | D+Q                |            | $A_2=+0.12$ 3, $A_4=-0.02$ 4 (1976Sp09).<br>Mult.: from $\gamma(\theta)$ in 1976Sp08; $\delta=-0.39$ 5 or $\delta=-8.2$ 7 (1976Sp09). |
|                     |                  |                     |               |         |                  |                    |            | $A_2=-0.19$ 2, $A_4=+0.02$ 2 (1976Sp09).                                                                                              |
| 8984.2              |                  | 3260.5              | 0.8 4         | 5723.52 |                  |                    |            |                                                                                                                                       |
|                     |                  | 4144.8              | 1.2 6         | 4839.10 |                  |                    |            |                                                                                                                                       |
|                     |                  | 4214.1              | 0.5 3         | 4769.86 |                  |                    |            |                                                                                                                                       |
|                     |                  | 4359.5              | 0.5 3         | 4624.4  |                  |                    |            |                                                                                                                                       |
|                     |                  | 4806.0              | 1.5 8         | 4177.9  | 3/2              |                    |            |                                                                                                                                       |
|                     |                  | 4810.4              | 1.5 8         | 4173.45 |                  |                    |            |                                                                                                                                       |
|                     |                  | 5065.3              | 9 5           | 3918.48 | 3/2 <sup>+</sup> |                    |            |                                                                                                                                       |
|                     |                  | 5820.7              | 6 3           | 3162.99 | 7/2 <sup>-</sup> |                    |            |                                                                                                                                       |
|                     |                  | 6289.7              | 1.8 9         | 2693.9  | (3/2,5/2)        |                    |            |                                                                                                                                       |
|                     |                  | 6337.9              | 1.2 6         | 2645.70 | (5/2,7/2)        |                    |            |                                                                                                                                       |
|                     |                  | 7220.3              | 16 4          | 1763.14 | 5/2 <sup>+</sup> |                    |            |                                                                                                                                       |
|                     |                  | 8983.0              | 60 6          | 0       | 3/2 <sup>+</sup> |                    |            |                                                                                                                                       |
| 9081.4              | 5/2 <sup>+</sup> | 3357.7              | 1             | 5723.52 |                  |                    |            | $I_\gamma$ : from 2001Vo24.                                                                                                           |
|                     |                  | 4200.2              | 1             | 4880.96 |                  |                    |            | $I_\gamma$ : from 2001Vo24.                                                                                                           |
|                     |                  | 4242.0 <sup>‡</sup> |               | 4839.10 |                  |                    |            |                                                                                                                                       |
|                     |                  | 4311.3 <sup>‡</sup> |               | 4769.86 |                  |                    |            |                                                                                                                                       |
|                     |                  | 4456.7 <sup>‡</sup> |               | 4624.4  |                  |                    |            |                                                                                                                                       |
|                     |                  | 4903.1              | 2             | 4177.9  | 3/2              |                    |            | $I_\gamma$ : from 2001Vo24.                                                                                                           |
|                     |                  | 4907.6              | 2             | 4173.45 |                  |                    |            | $I_\gamma$ : from 2001Vo24.                                                                                                           |
|                     |                  | 5162.5              | 9             | 3918.48 | 3/2 <sup>+</sup> | D(+Q)              | -0.002 9   | $I_\gamma$ : from 2001Vo24.<br>Mult., $\delta$ : from $\gamma(\theta)$ in 1976Sp09.<br>$A_2=-0.43$ 3, $A_4=+0.05$ 3 (1976Sp09).       |
|                     |                  | 5917.9              | 6             | 3162.99 | 7/2 <sup>-</sup> |                    |            | $I_\gamma$ : from 2001Vo24.                                                                                                           |
|                     |                  | 6386.9              | 2             | 2693.9  | (3/2,5/2)        |                    |            | $I_\gamma$ : from 2001Vo24.                                                                                                           |
|                     |                  | 6435.1              | 1             | 2645.70 | (5/2,7/2)        |                    |            | $I_\gamma$ : from 2001Vo24.                                                                                                           |
|                     |                  | 7317.4              | 16            | 1763.14 | 5/2 <sup>+</sup> | D+Q                | +0.11 2    | $I_\gamma$ : from 2001Vo24.<br>Mult., $\delta$ : from $\gamma(\theta)$ in 1976Sp09.<br>$A_2=+0.555$ 14, $A_4=+0.03$ 2 (1976Sp09).     |
|                     |                  | 9080.1              | 60            | 0       | 3/2 <sup>+</sup> | D+Q                | +0.089 9   | $I_\gamma$ : from 2001Vo24.<br>Mult., $\delta$ : from $\gamma(\theta)$ in 1976Sp09.<br>$A_2=-0.243$ 11, $A_4=+0.05$ 2 (1976Sp09).     |
| 9157.1              | 5/2              | 3941.1              | 7.4 19        | 5215.8  |                  |                    |            |                                                                                                                                       |
|                     |                  | 4386.9              | 6.9 17        | 4769.86 |                  |                    |            |                                                                                                                                       |
|                     |                  | 4978.8              | <16           | 4177.9  | 3/2              |                    |            | $I_\gamma$ : total $I_\gamma=13$ 3 for $\gamma$ decays to 4173 and 4178 doublet final levels (1976Sp08).                              |
|                     |                  | 4983.3              | <16           | 4173.45 |                  |                    |            | $I_\gamma$ : total $I_\gamma=13$ 3 for $\gamma$ decays to 4173 and 4178 doublet final levels (1976Sp08).                              |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$  [1976Me12](#),[1976Sp08](#),[1972Hu10](#) (continued) $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $E_\gamma^\dagger$ | $I_\gamma^\#$ | $E_f$   | $J_f^\pi$        | Mult. <sup>b</sup> | $\delta^b$ | Comments                                                                                                                                                                                                                                              |
|---------------------|--------------------|---------------|---------|------------------|--------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9157.1              | 5993.6             | 9.6 24        | 3162.99 | 7/2 <sup>-</sup> | D+Q                | +0.10 2    | Mult., $\delta$ : from $\gamma(\theta)$ in <a href="#">2001Vo24</a> .<br>A <sub>2</sub> =-0.15 8, A <sub>4</sub> =-0.31 9, A <sub>6</sub> =-0.05 8 ( <a href="#">1996Ka21</a> ).                                                                      |
|                     | 6153.8             | 8.4 21        | 3002.75 | 5/2              |                    |            |                                                                                                                                                                                                                                                       |
|                     | 6462.6             | 1.3 7         | 2693.9  | (3/2,5/2)        |                    |            |                                                                                                                                                                                                                                                       |
|                     | 6510.8             | 2.9 15        | 2645.70 | (5/2,7/2)        |                    |            |                                                                                                                                                                                                                                                       |
|                     | 7393.1             | 7.5 19        | 1763.14 | 5/2 <sup>+</sup> |                    |            |                                                                                                                                                                                                                                                       |
|                     | 9155.8             | 43 4          | 0       | 3/2 <sup>+</sup> | D+Q                | +0.09 1    | Mult., $\delta$ : from $\gamma(\theta)$ in <a href="#">2001Vo24</a> . Other: $\delta(\text{Q/D})=0.47$<br>50 ( <a href="#">1996Ka21</a> ).<br>A <sub>2</sub> =-0.29 6, A <sub>4</sub> =-0.38 8, A <sub>6</sub> =-0.10 6 ( <a href="#">1996Ka21</a> ). |

<sup>†</sup> From level-energy differences, unless otherwise noted.

<sup>‡</sup>  $\gamma$  branches observed in [1981Bi05](#). Branching ratios are not given.

<sup>#</sup> For levels below 8.27 MeV ( $E_p < 1.95$  MeV),  $I_\gamma$  is mainly from [1976Me12](#) and for levels above 8.27 MeV ( $E_p > 1.95$  MeV),  $I_\gamma$  is mainly from [1976Sp08](#), unless otherwise noted.

@ From [1970Ko33](#).

& From [1967Da12](#).

<sup>a</sup> From [1963Ha32](#).

<sup>b</sup> From  $\gamma(\theta)$  and/or  $\gamma\gamma(\theta)$  data as given under comments.

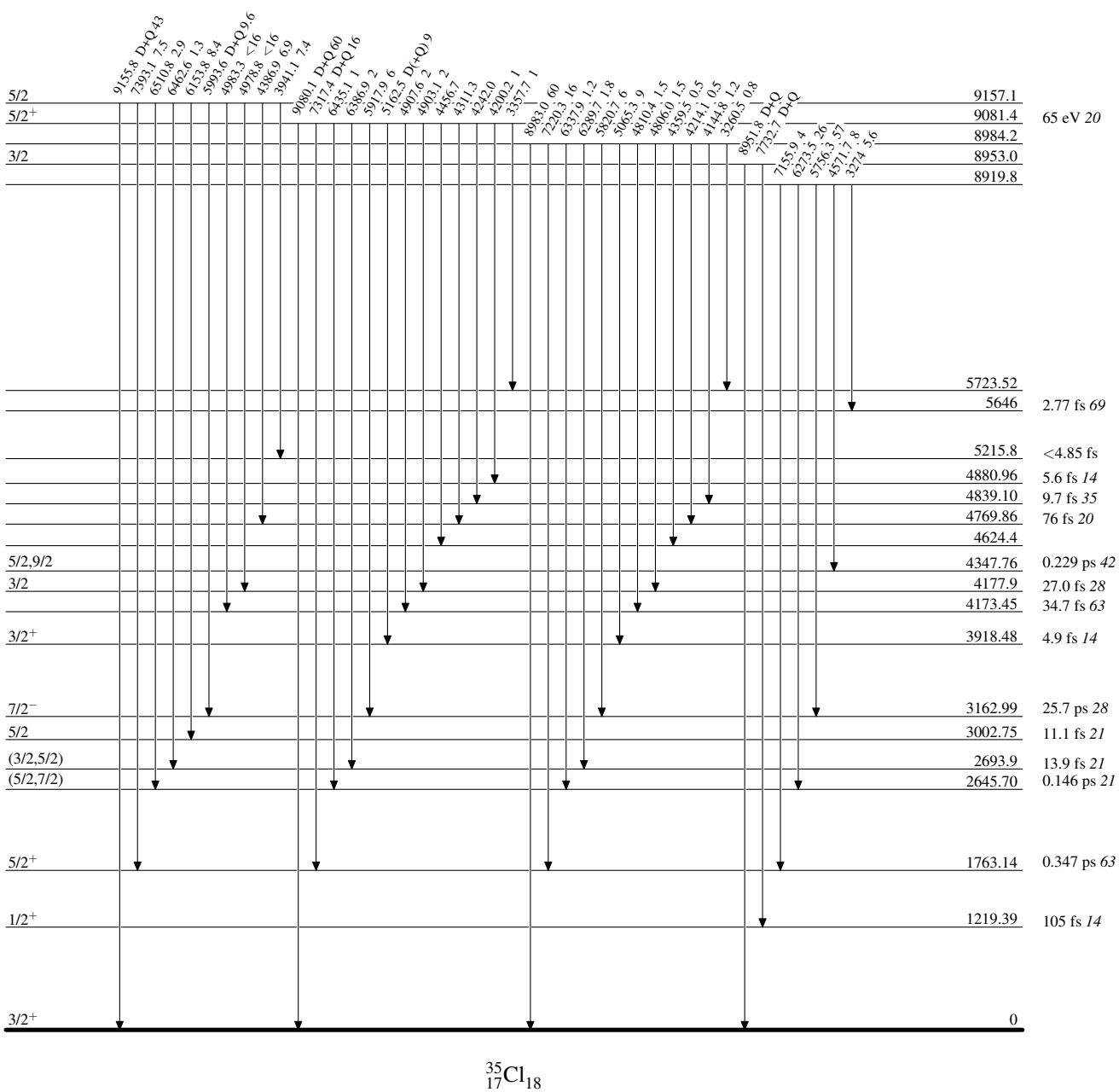
<sup>c</sup> Placement of transition in the level scheme is uncertain.



$^{34}\text{S}(\text{p},\gamma)$  1976Me12,1976Sp08,1972Hu10

## Level Scheme

Intensities: % photon branching from each level

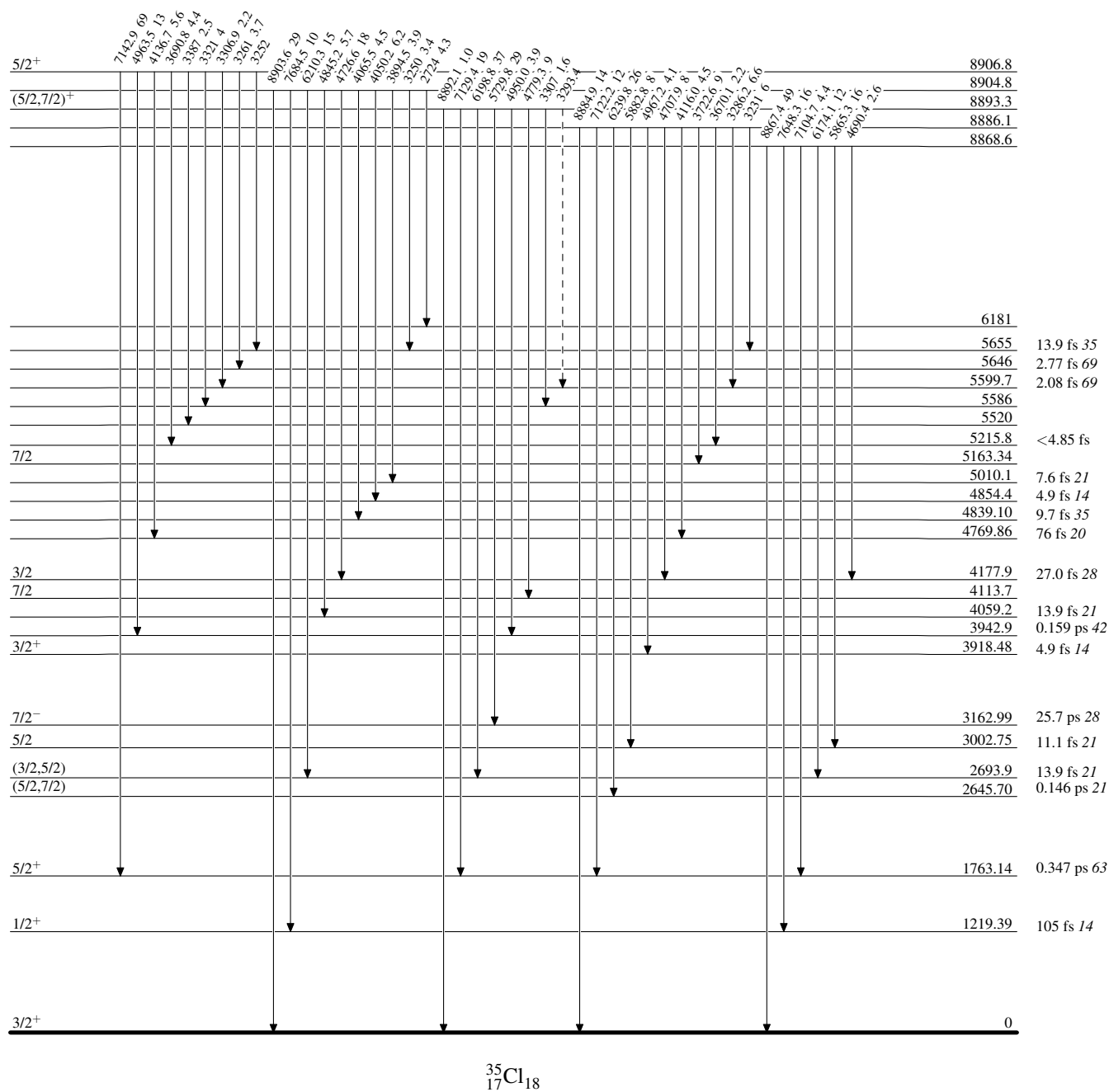


$^{34}\text{S}(\text{p},\gamma)$  1976Me12,1976Sp08,1972Hu10

Legend

## Level Scheme (continued)

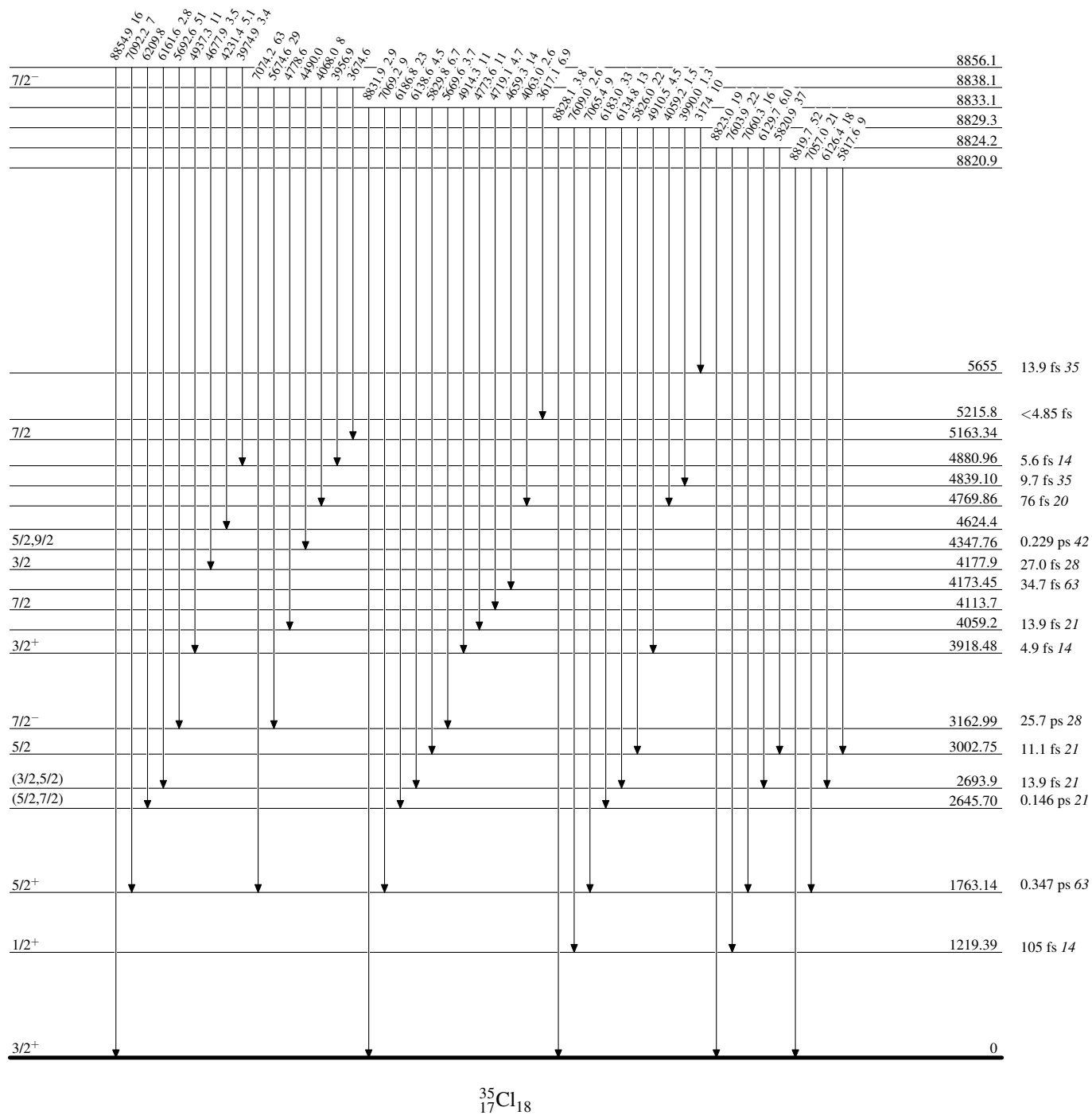
Intensities: % photon branching from each level

-----►  $\gamma$  Decay (Uncertain)

$^{34}\text{S}(\text{p},\gamma)$  1976Me12,1976Sp08,1972Hu10

## Level Scheme (continued)

Intensities: % photon branching from each level

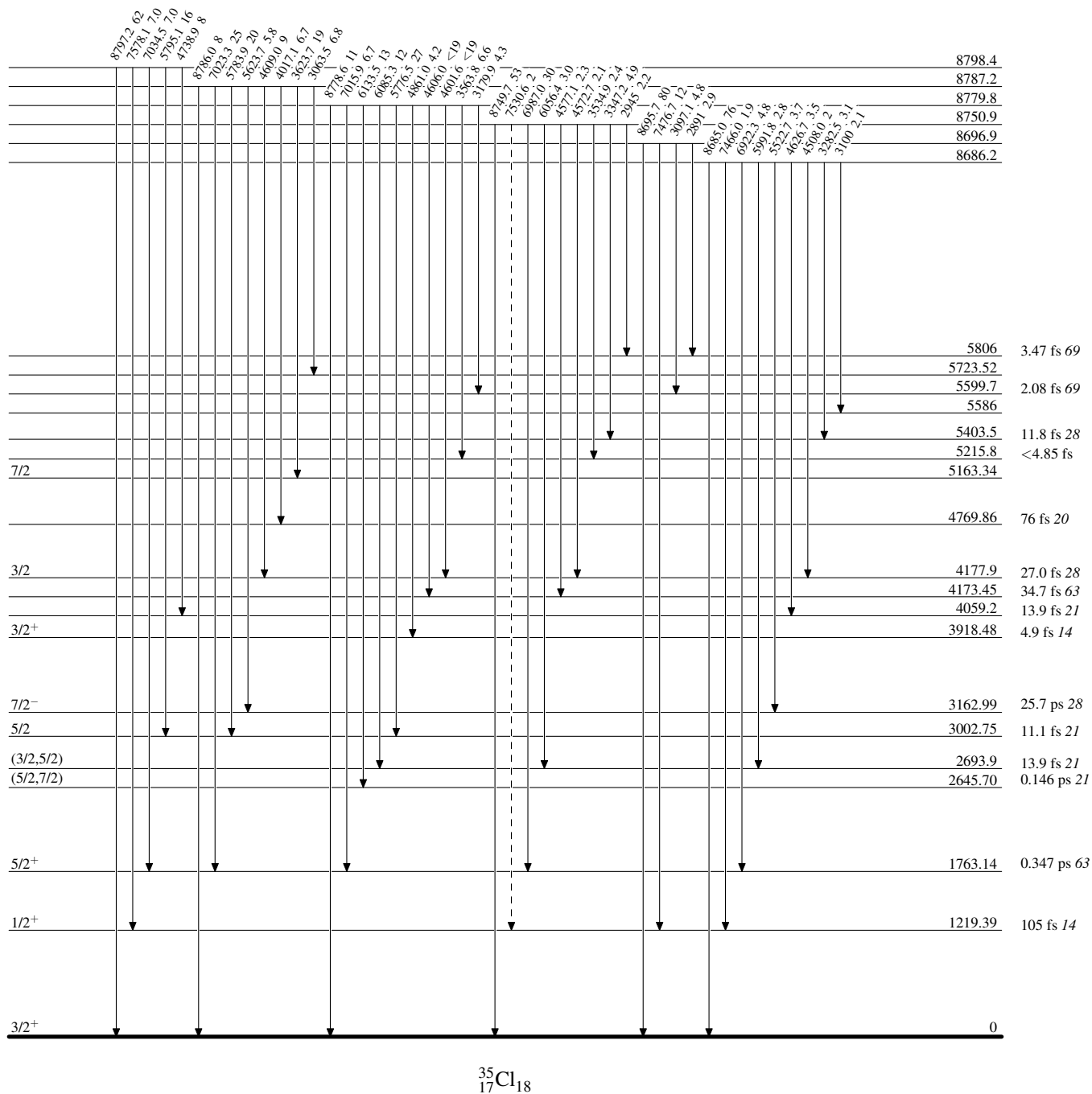


$^{34}\text{S}(\text{p},\gamma)$  1976Me12,1976Sp08,1972Hu10

Legend

## Level Scheme (continued)

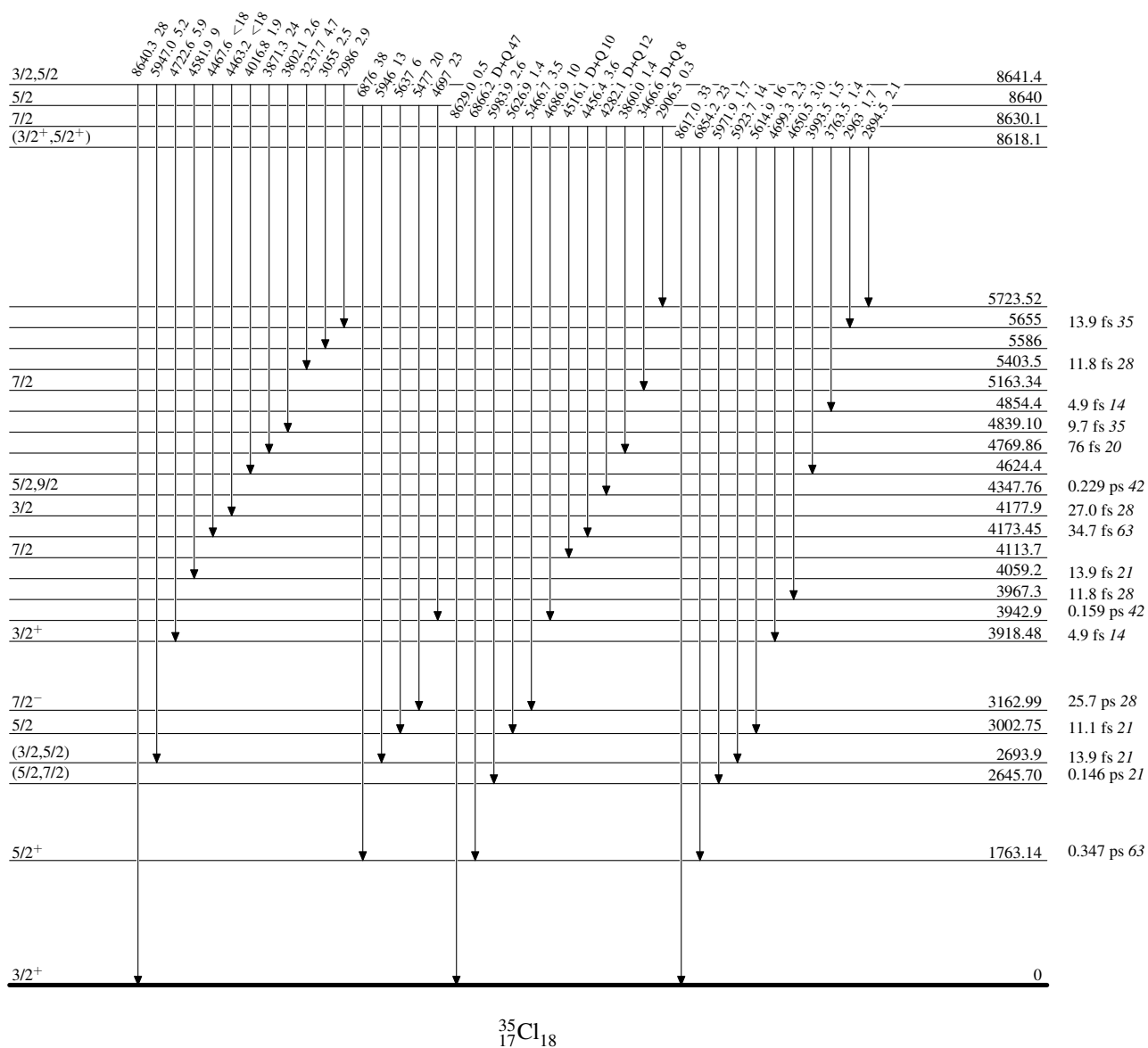
Intensities: % photon branching from each level

-----►  $\gamma$  Decay (Uncertain)

<sup>34</sup>S(p,γ) 1976Me12,1976Sp08,1972Hu10

## Level Scheme (continued)

Intensities: % photon branching from each level

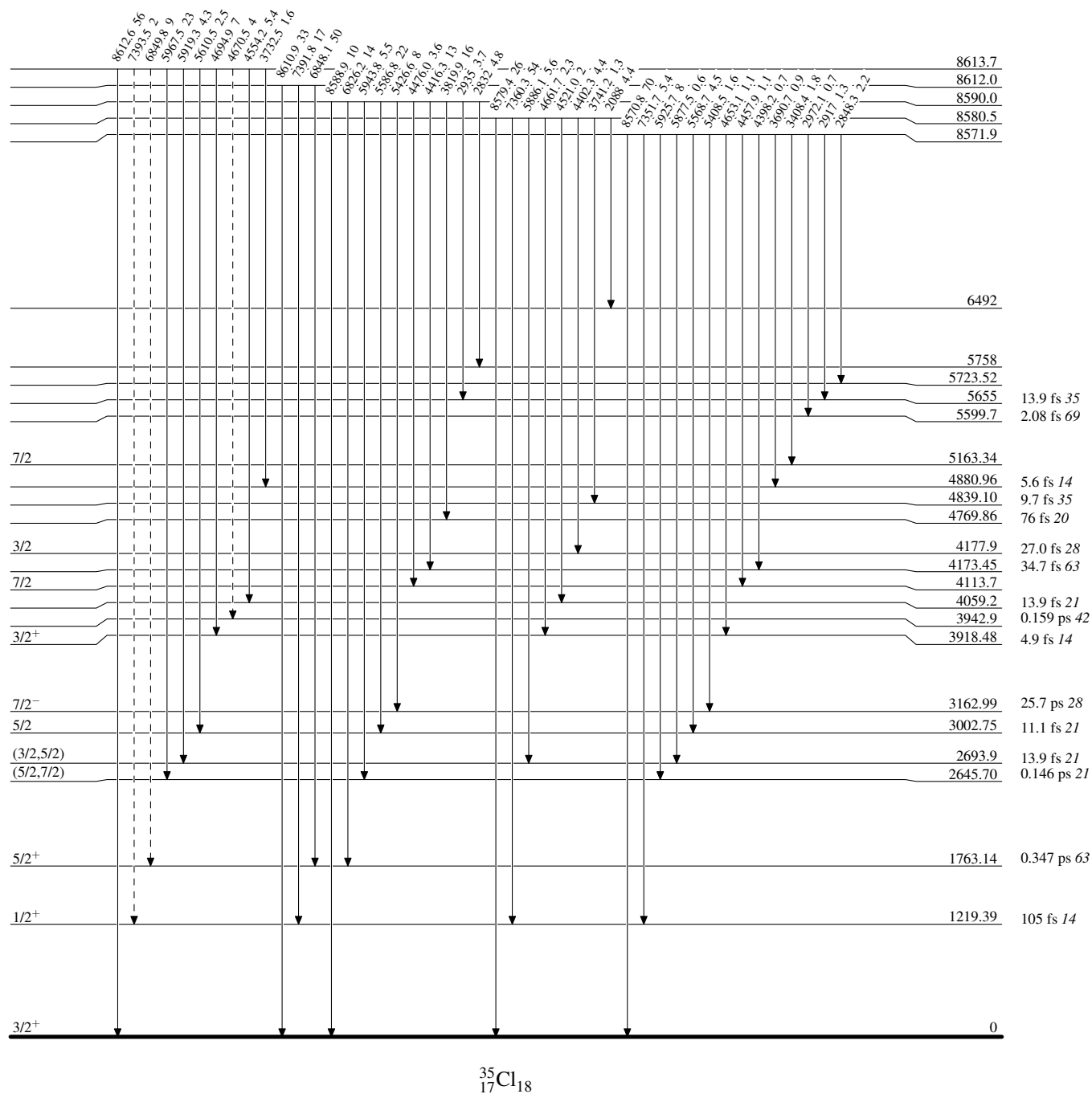


$^{34}\text{S}(p,\gamma)$  1976Me12,1976Sp08,1972Hu10

Legend

## Level Scheme (continued)

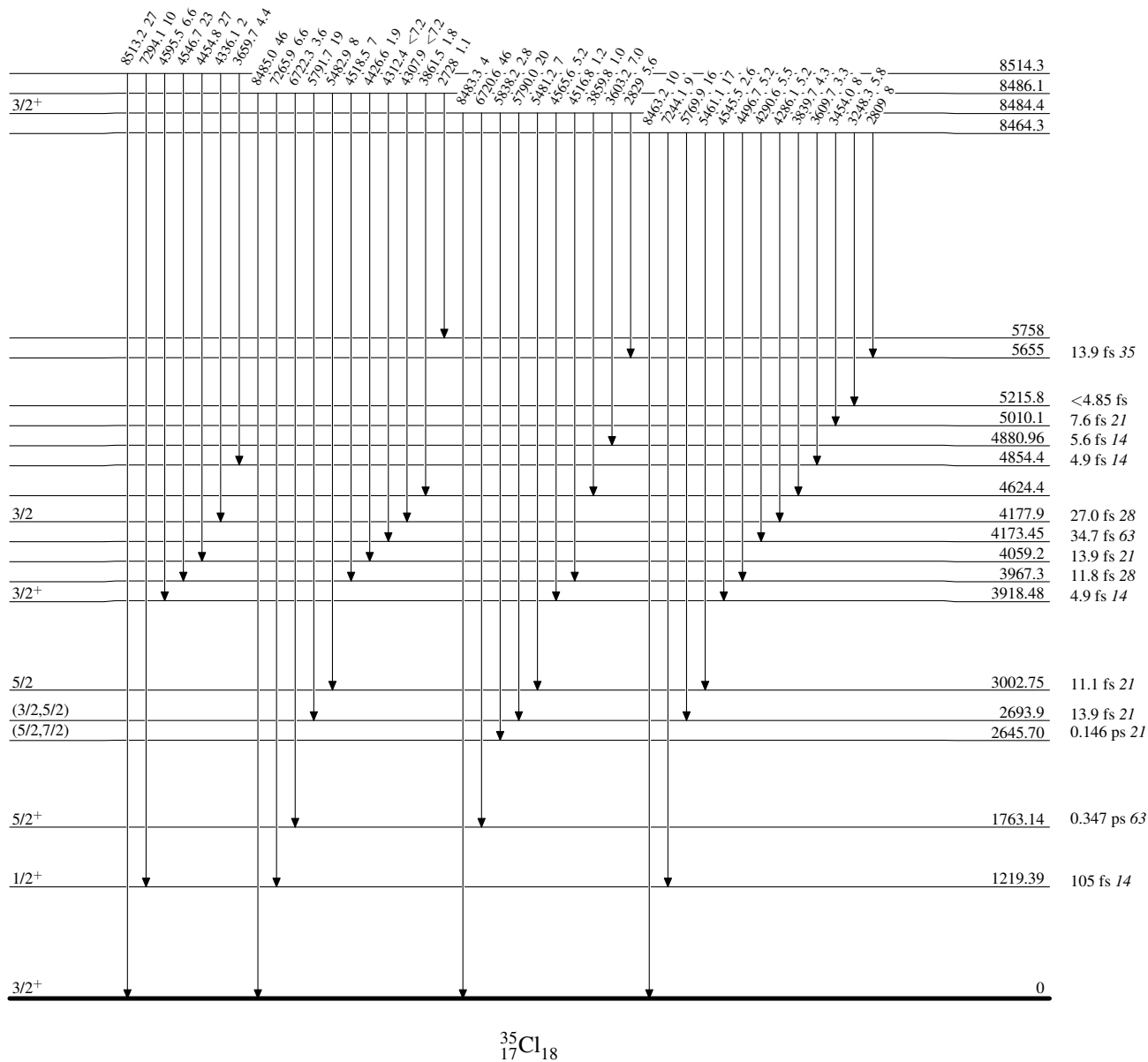
Intensities: % photon branching from each level

-----►  $\gamma$  Decay (Uncertain)


$^{34}\text{S}(\text{p},\gamma)$  1976Me12,1976Sp08,1972Hu10

## Level Scheme (continued)

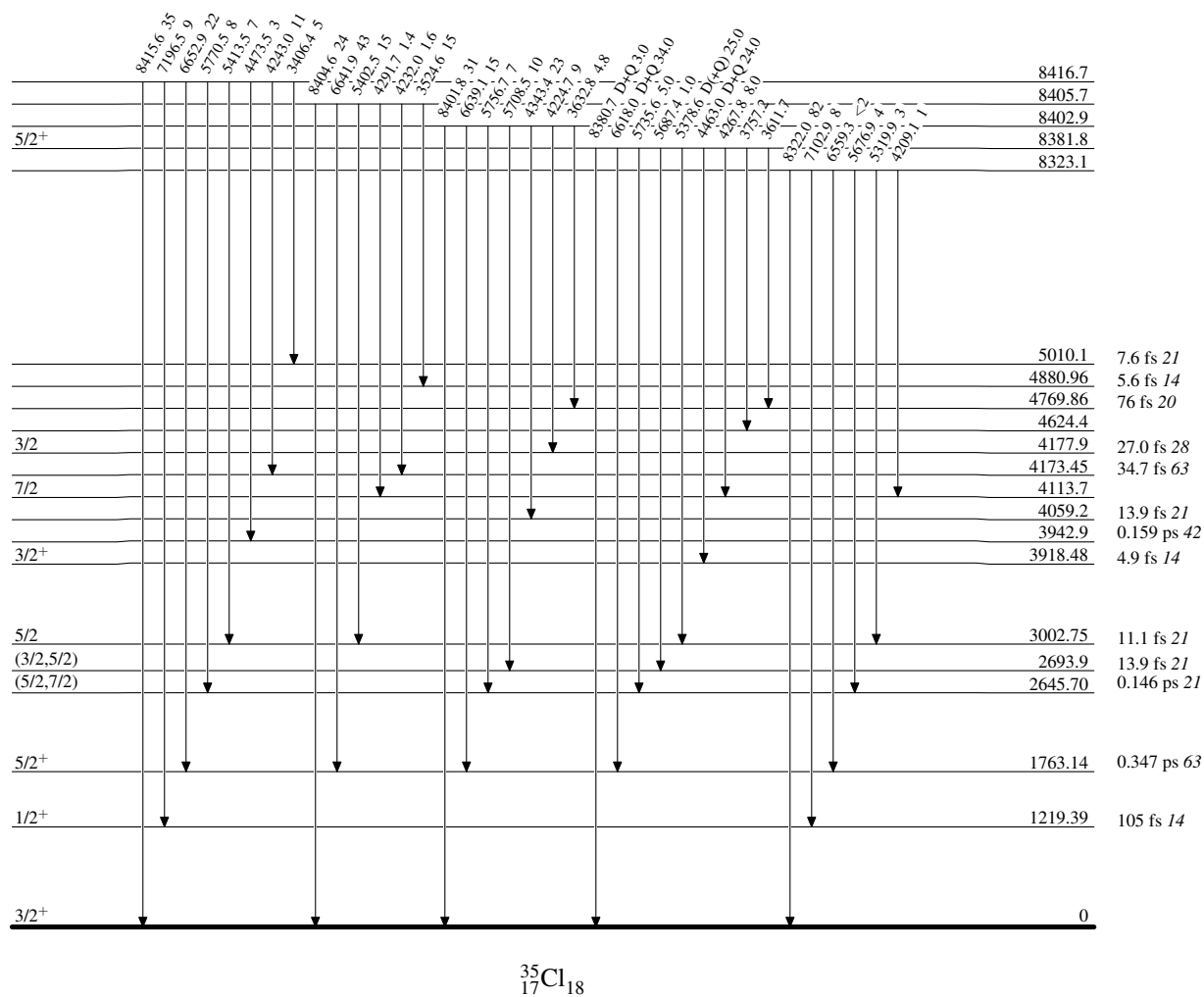
Intensities: % photon branching from each level



$^{34}\text{S}(\text{p},\gamma)$  1976Me12,1976Sp08,1972Hu10

## Level Scheme (continued)

Intensities: % photon branching from each level

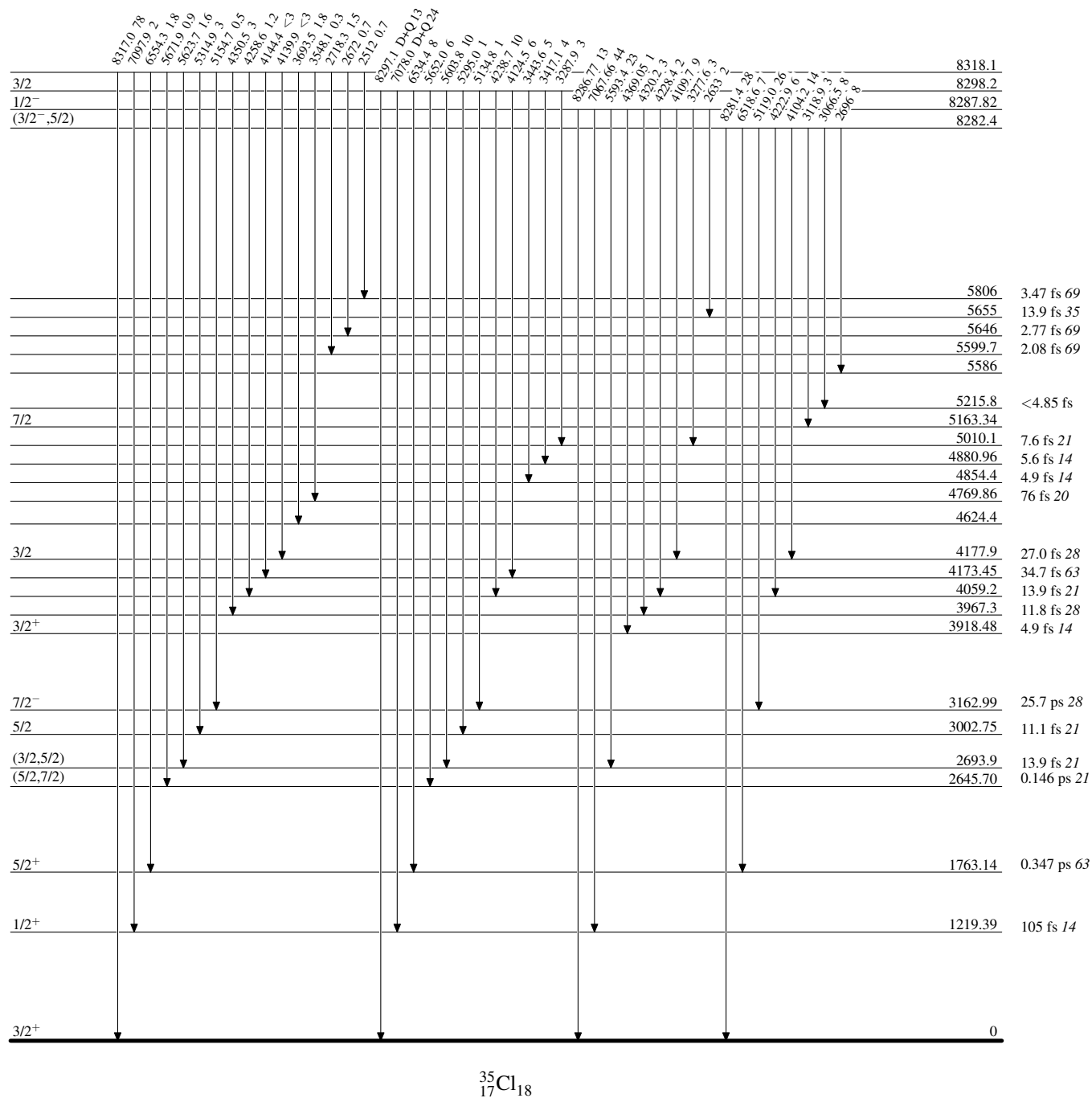
 $^{35}_{17}\text{Cl}_{18}$



$^{34}\text{S}(\text{p},\gamma)$  1976Me12,1976Sp08,1972Hu10

## Level Scheme (continued)

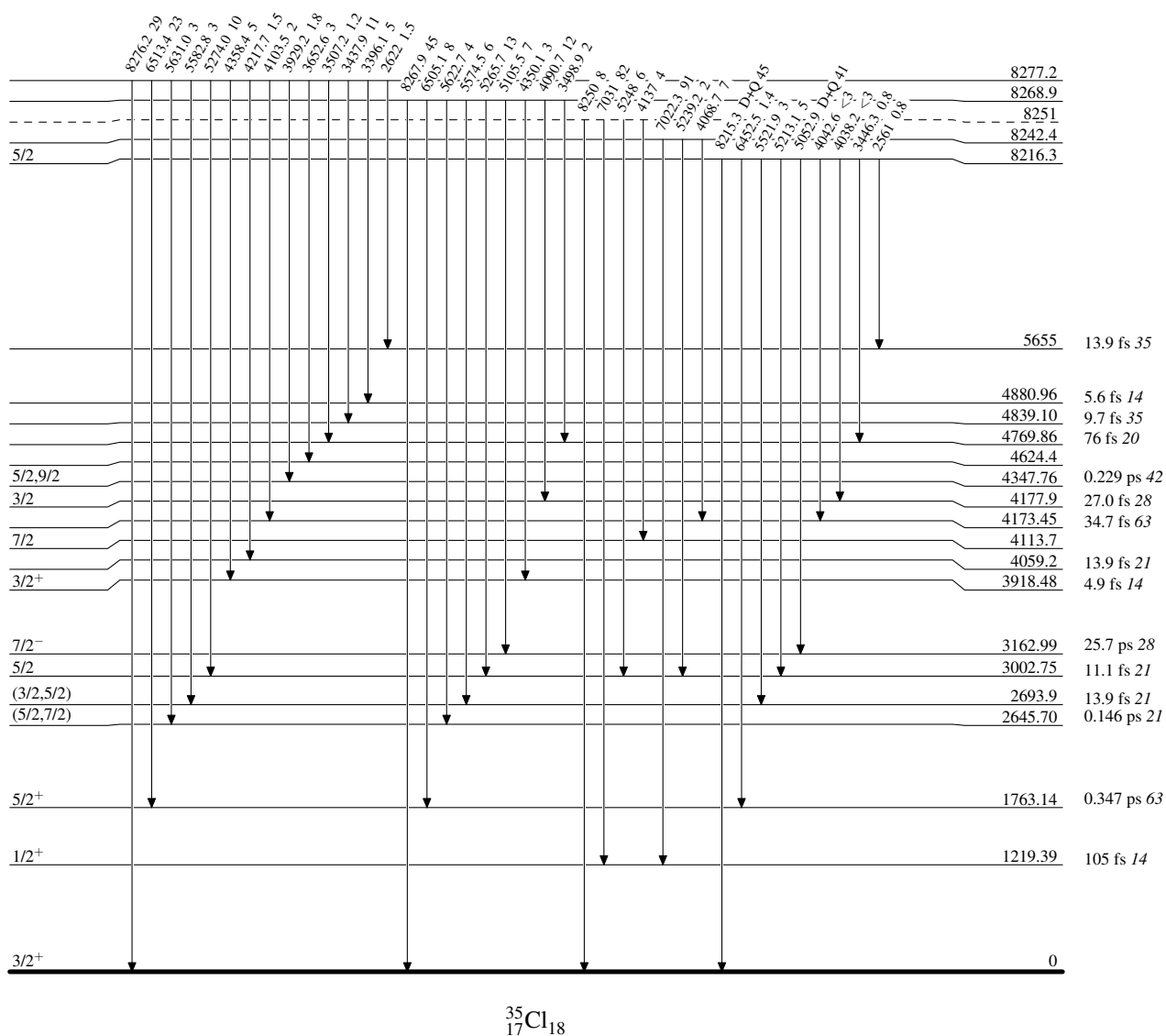
Intensities: % photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

$^{34}\text{S}(\text{p},\gamma)$  1976Me12,1976Sp08,1972Hu10

## Level Scheme (continued)

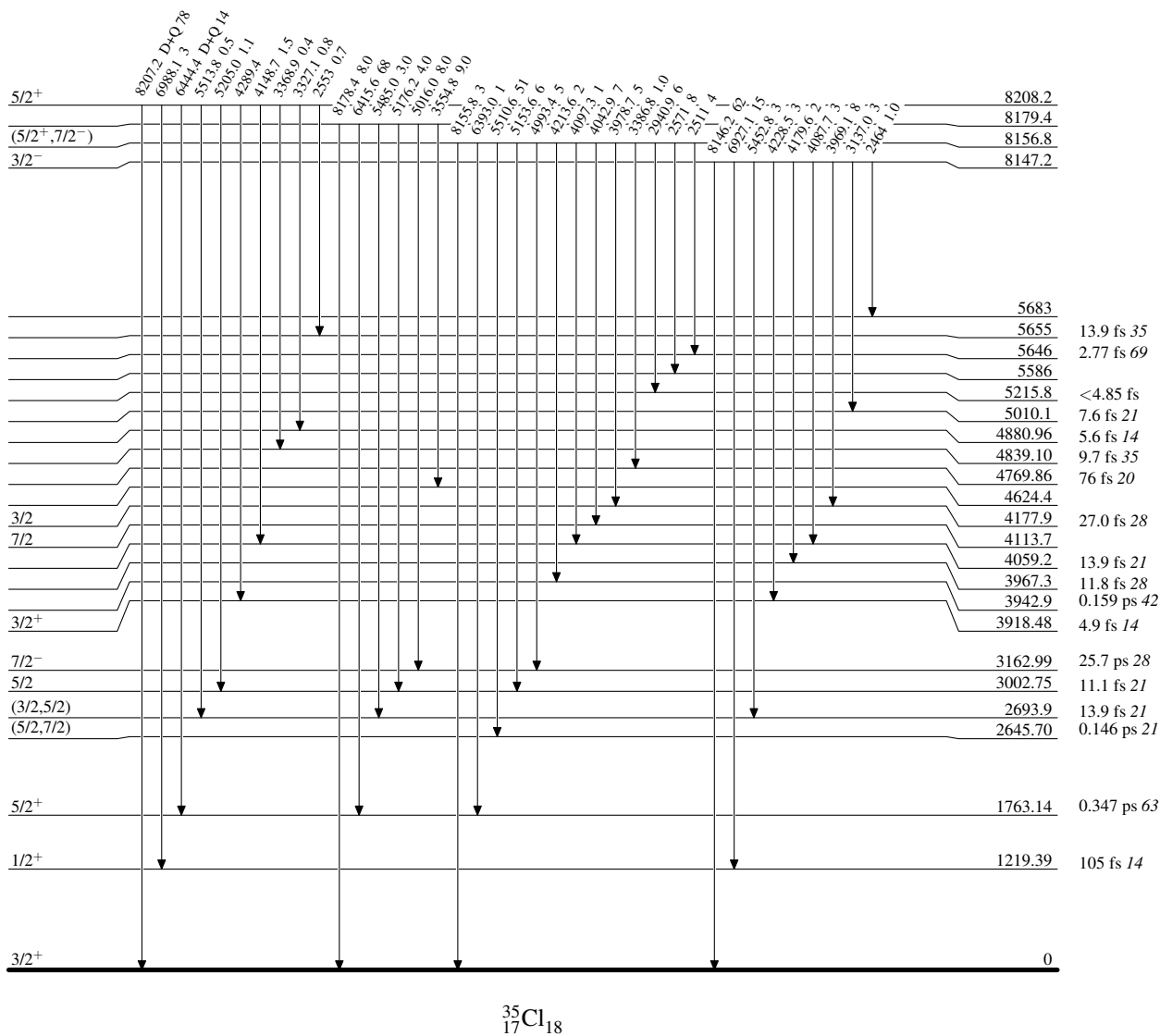
Intensities: % photon branching from each level



$^{34}\text{S}(p,\gamma)$  1976Me12,1976Sp08,1972Hu10

## Level Scheme (continued)

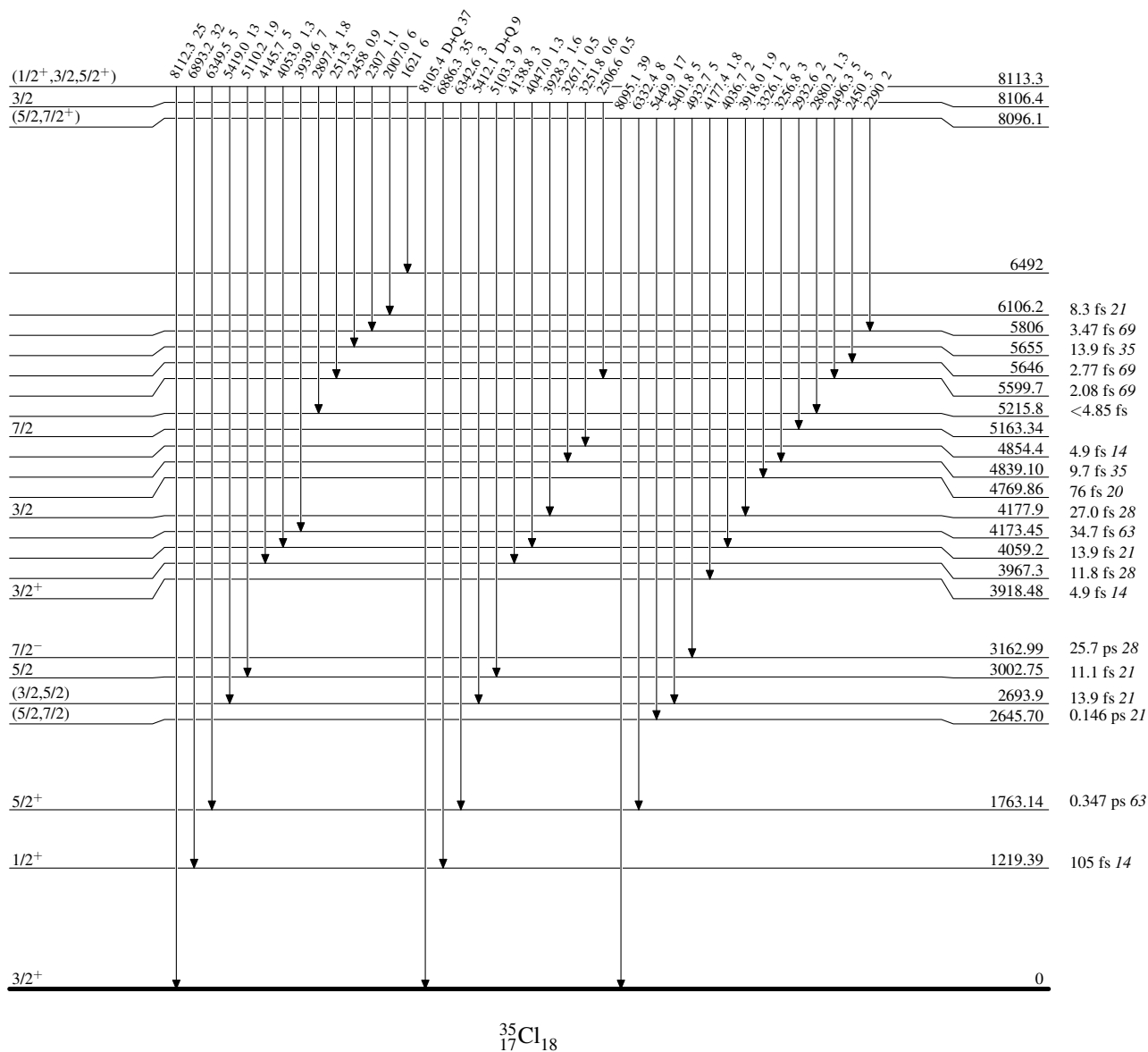
Intensities: % photon branching from each level



$^{34}\text{S}(\text{p},\gamma)$  1976Me12,1976Sp08,1972Hu10

## Level Scheme (continued)

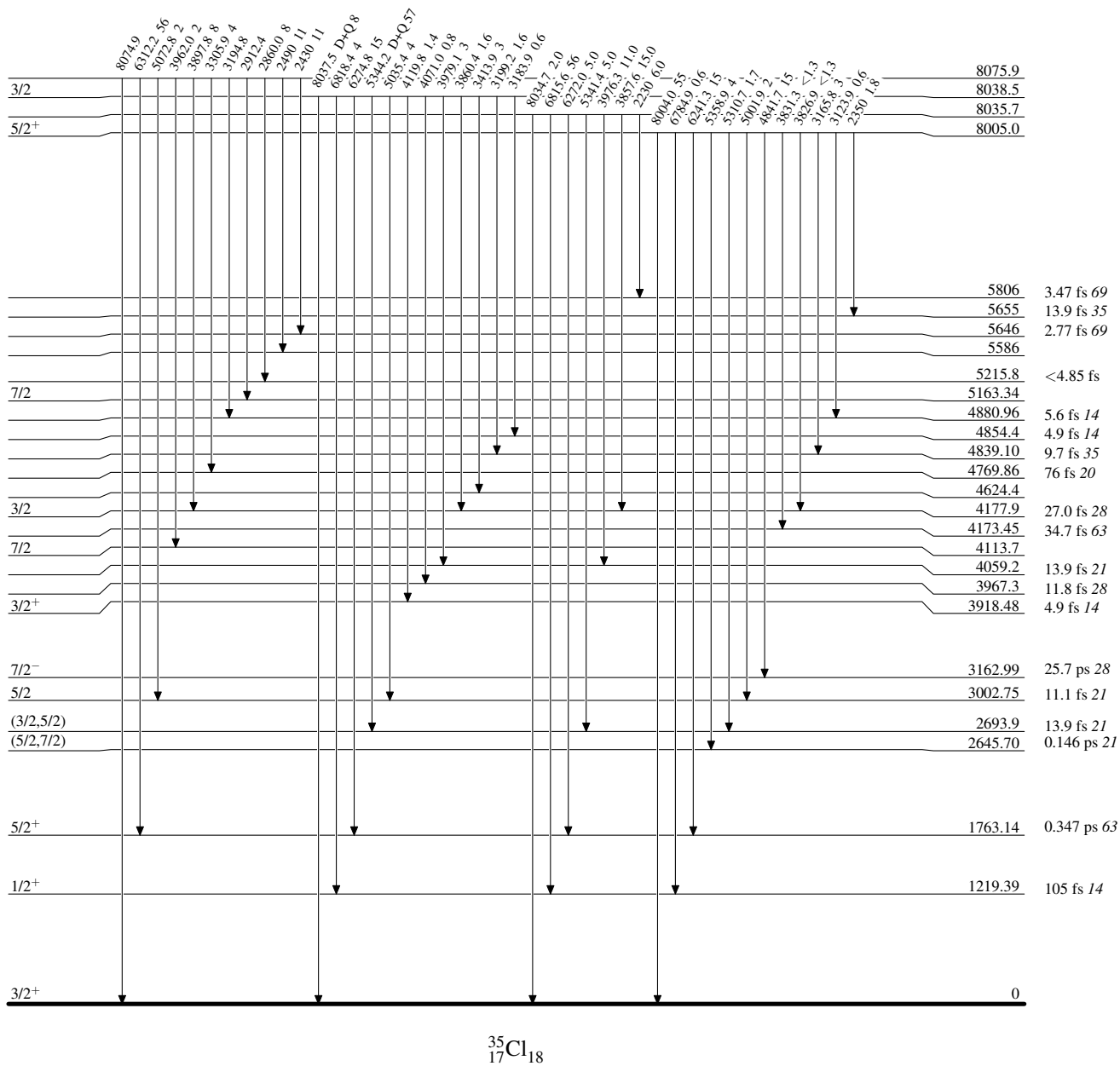
Intensities: % photon branching from each level



$^{34}\text{S}(\text{p},\gamma)$  1976Me12,1976Sp08,1972Hu10

## Level Scheme (continued)

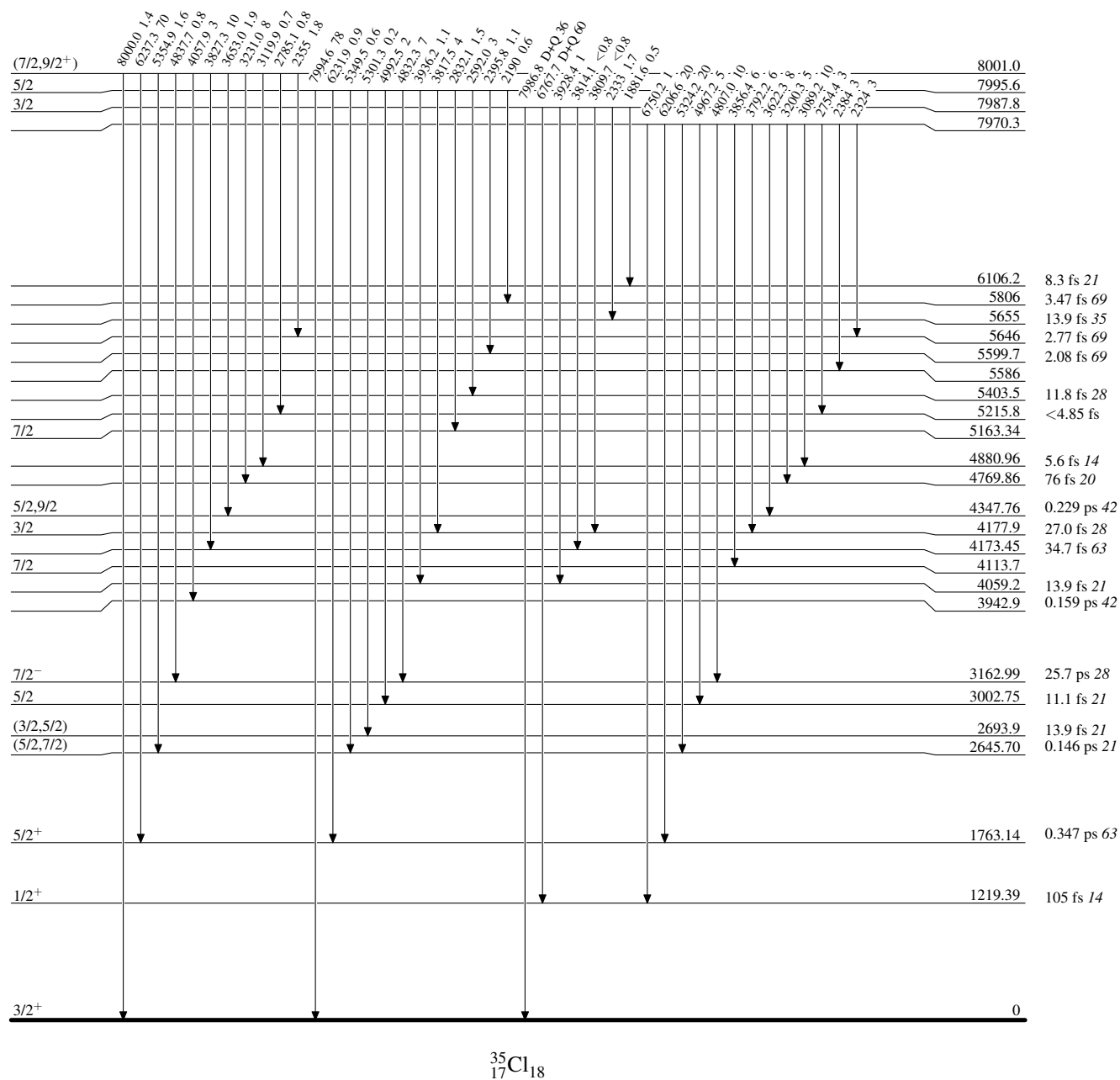
Intensities: % photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

$^{34}\text{S}(\text{p},\gamma)$  1976Me12,1976Sp08,1972Hu10

## Level Scheme (continued)

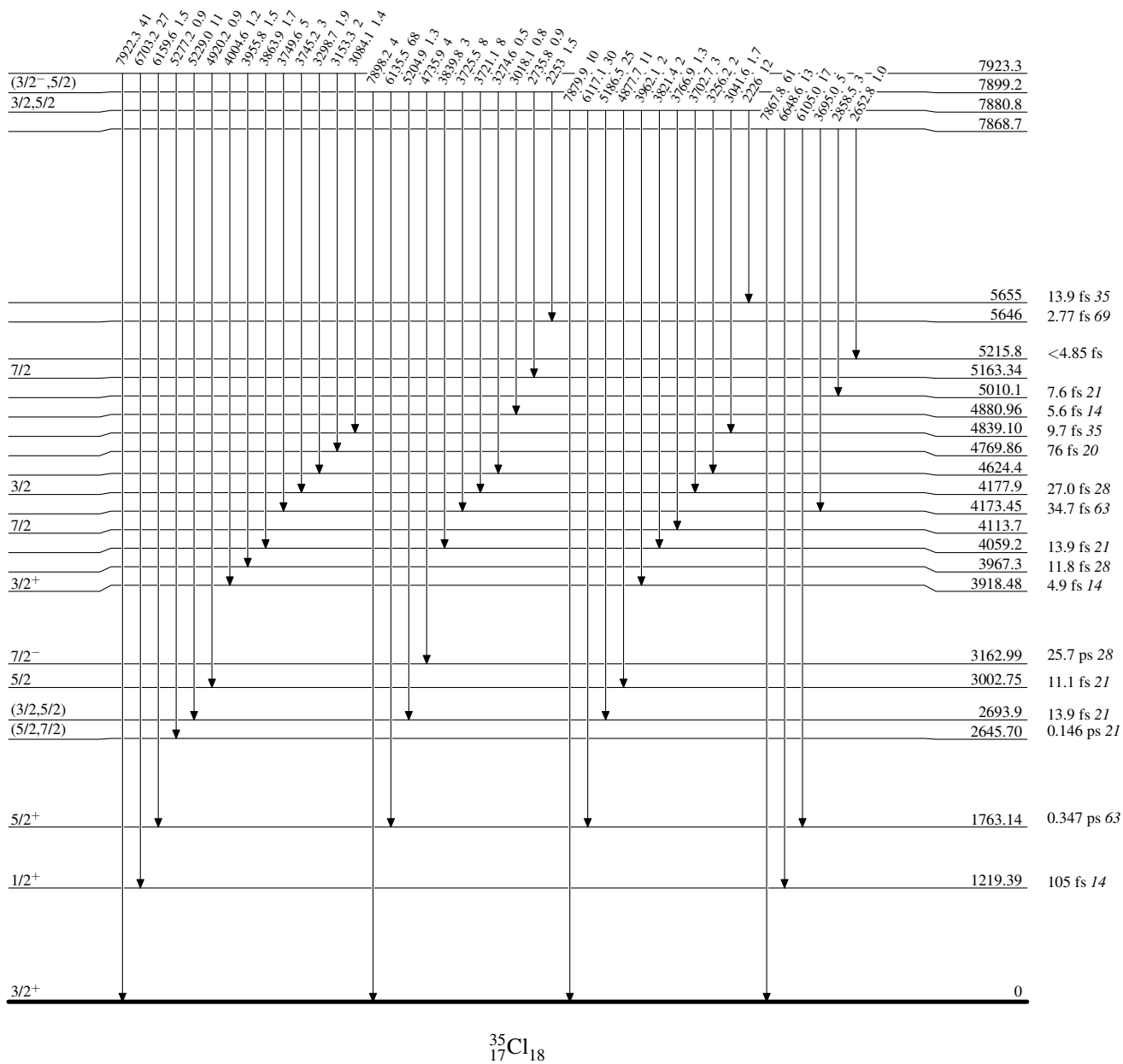
Intensities: % photon branching from each level



$^{34}\text{S}(\text{p},\gamma)$  1976Me12,1976Sp08,1972Hu10

## Level Scheme (continued)

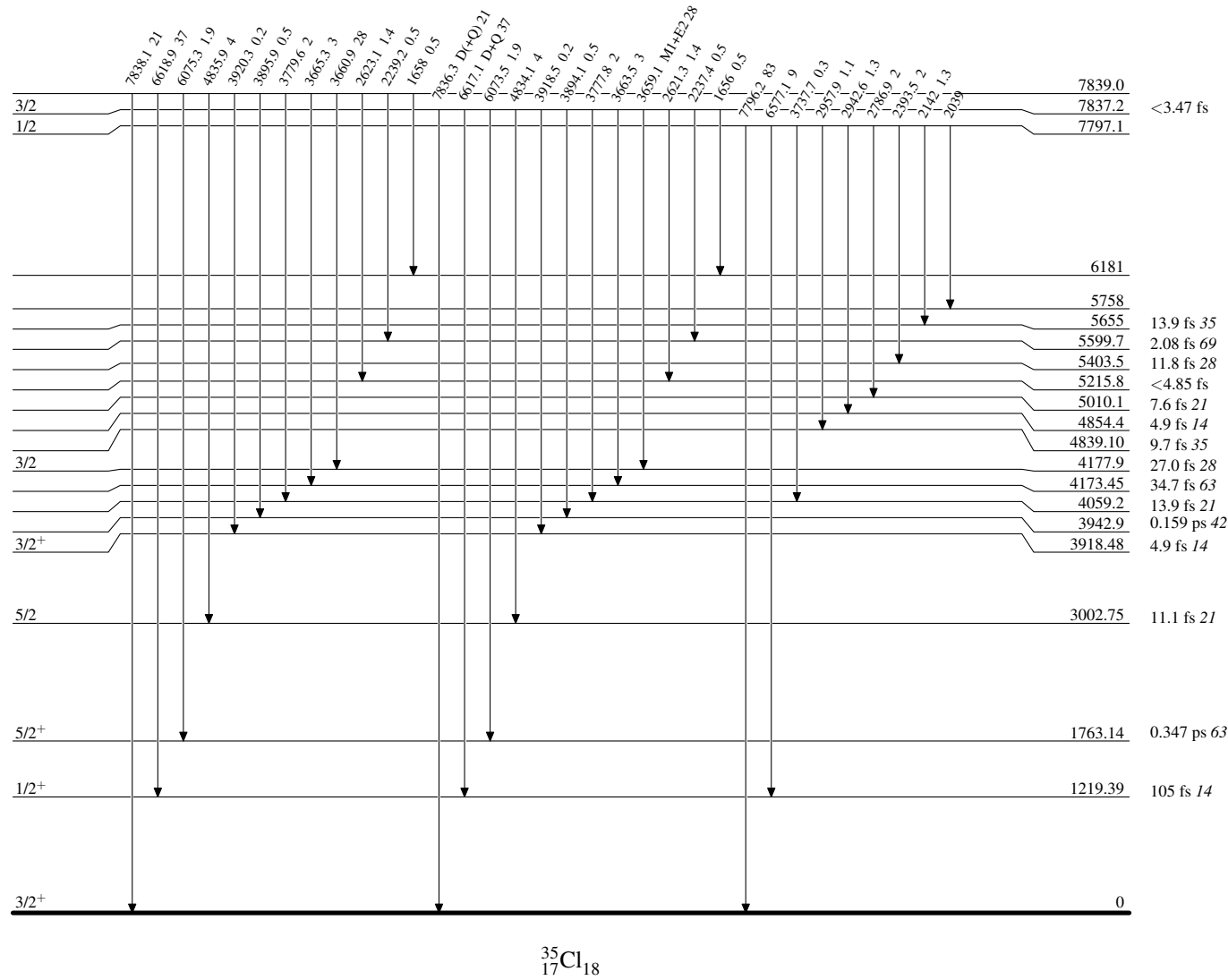
Intensities: % photon branching from each level



<sup>34</sup>S(p,γ) 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

Intensities: % photon branching from each level



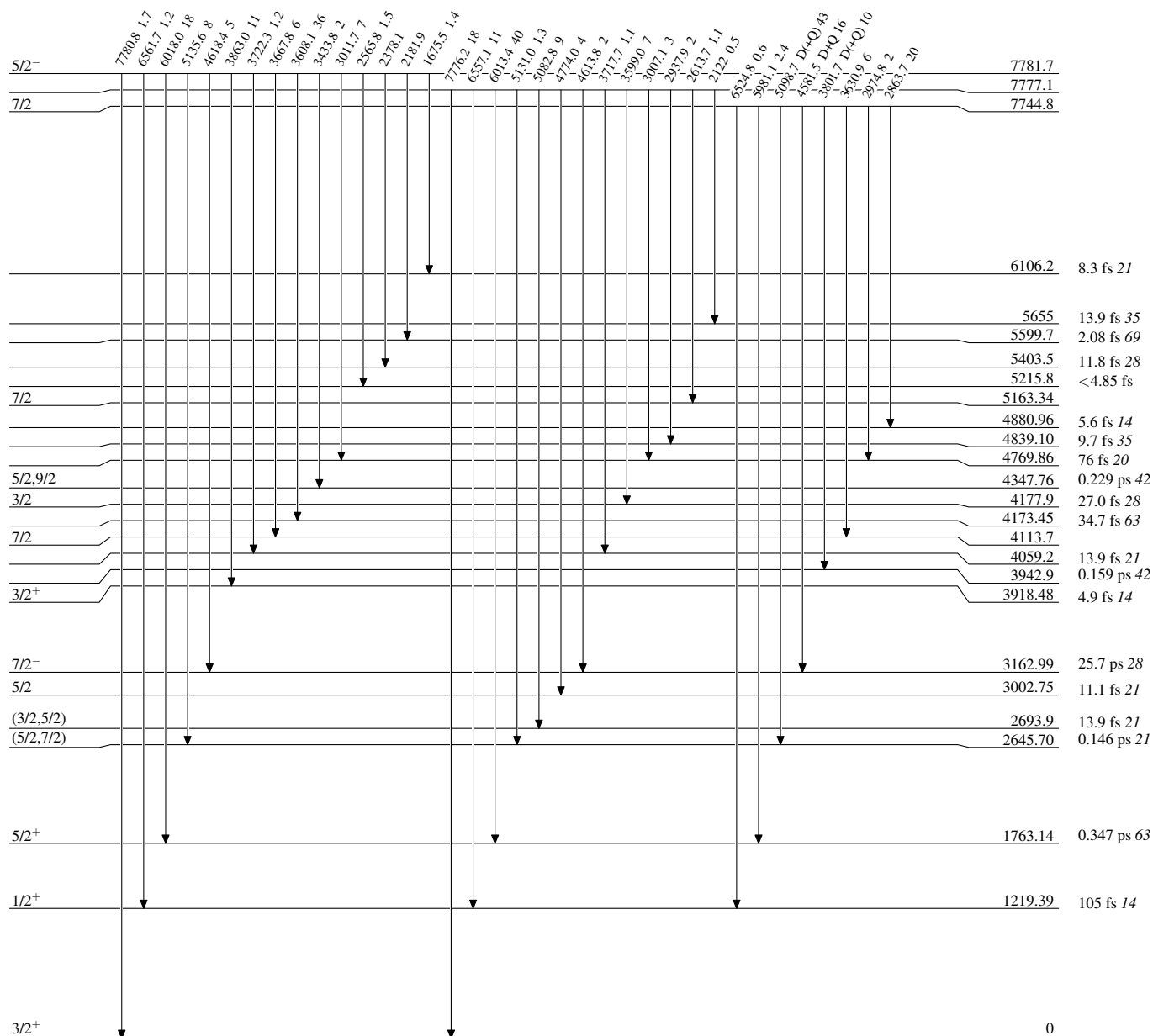
<sup>35</sup><sub>17</sub>Cl<sub>18</sub>



$^{34}\text{S}(\text{p},\gamma)$  1976Me12,1976Sp08,1972Hu10

## Level Scheme (continued)

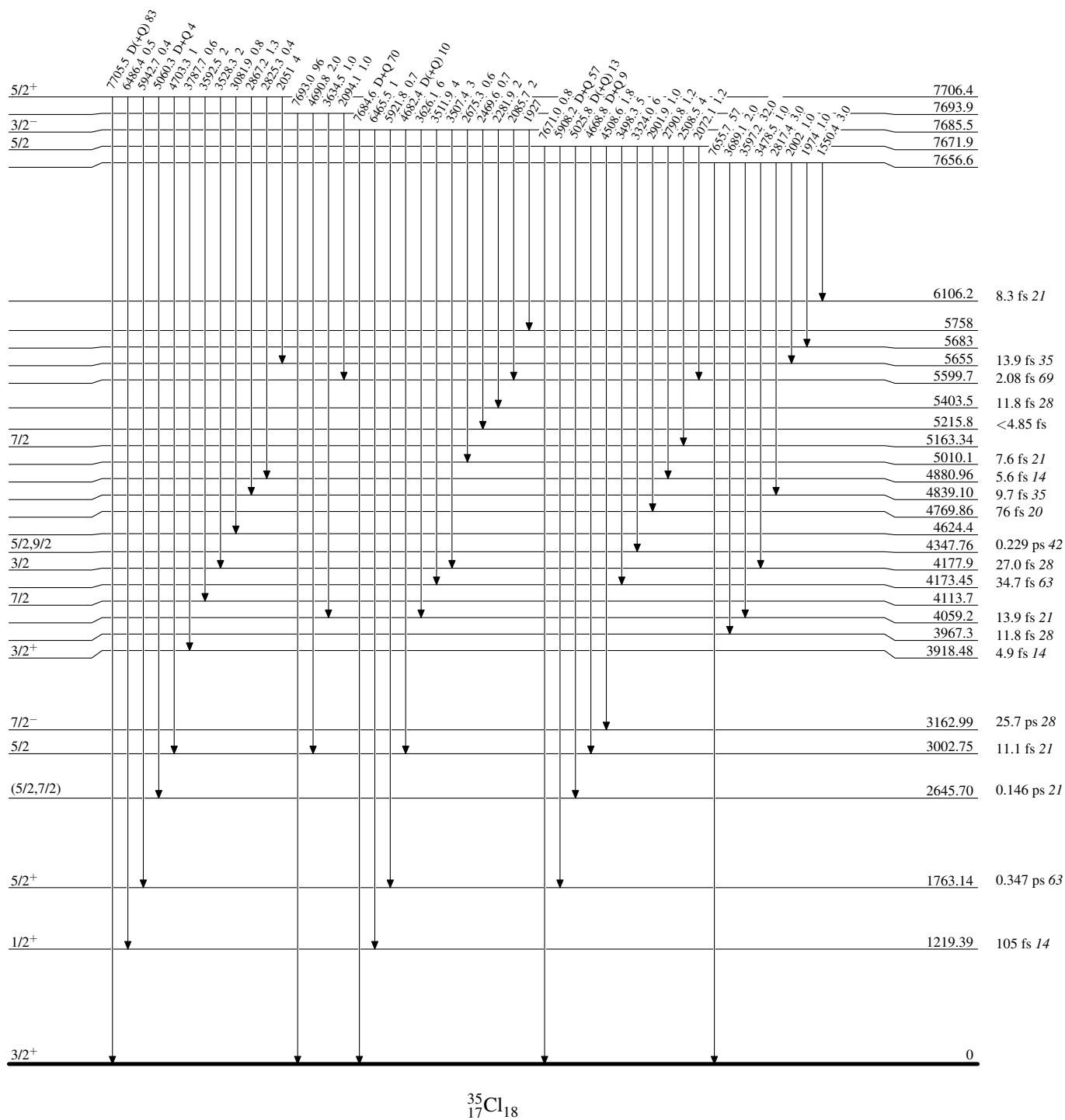
Intensities: % photon branching from each level



$^{34}\text{S}(\text{p},\gamma)$  1976Me12,1976Sp08,1972Hu10

## Level Scheme (continued)

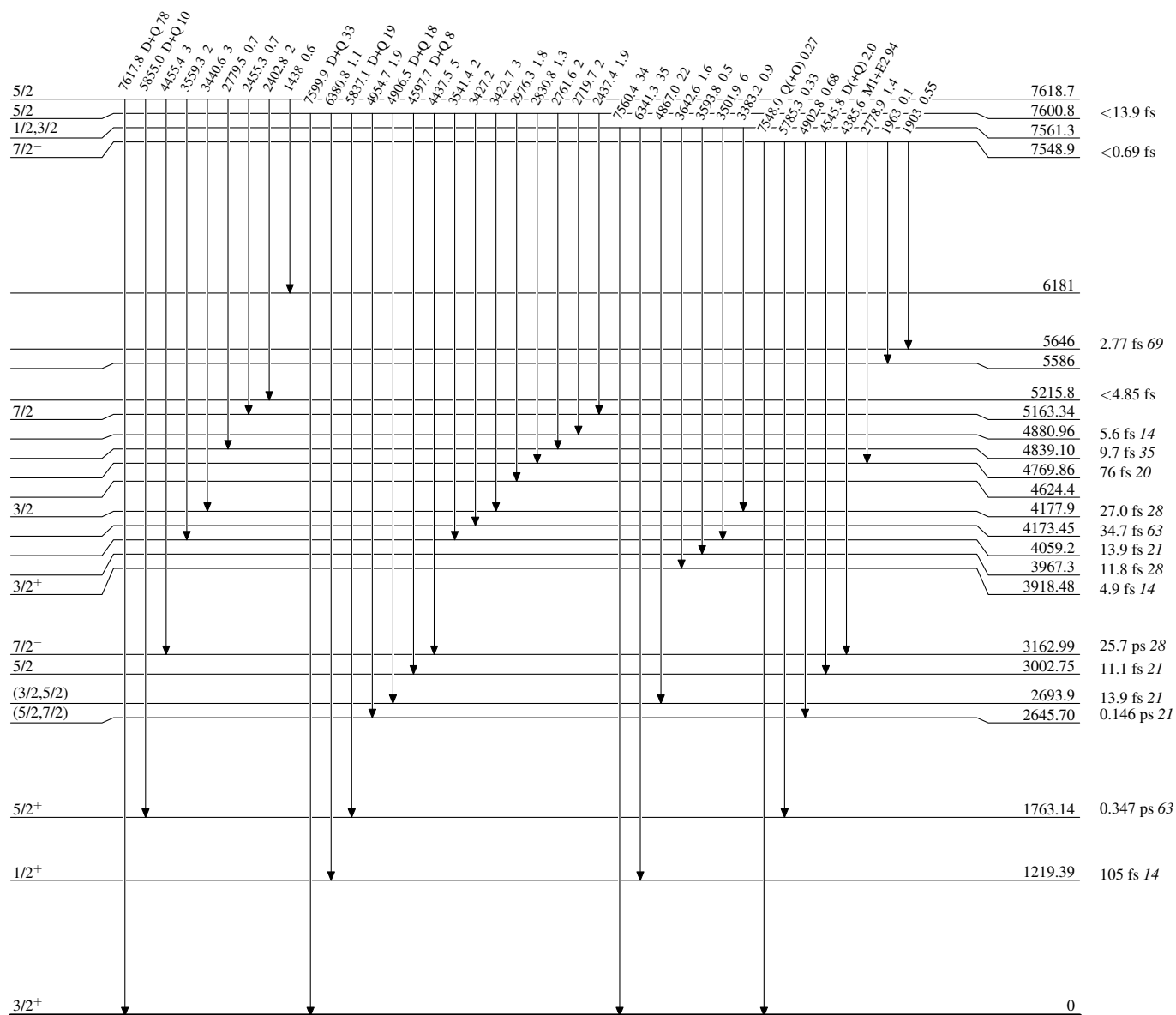
Intensities: % photon branching from each level



$^{34}\text{S}(\text{p},\gamma)$  1976Me12,1976Sp08,1972Hu10

## Level Scheme (continued)

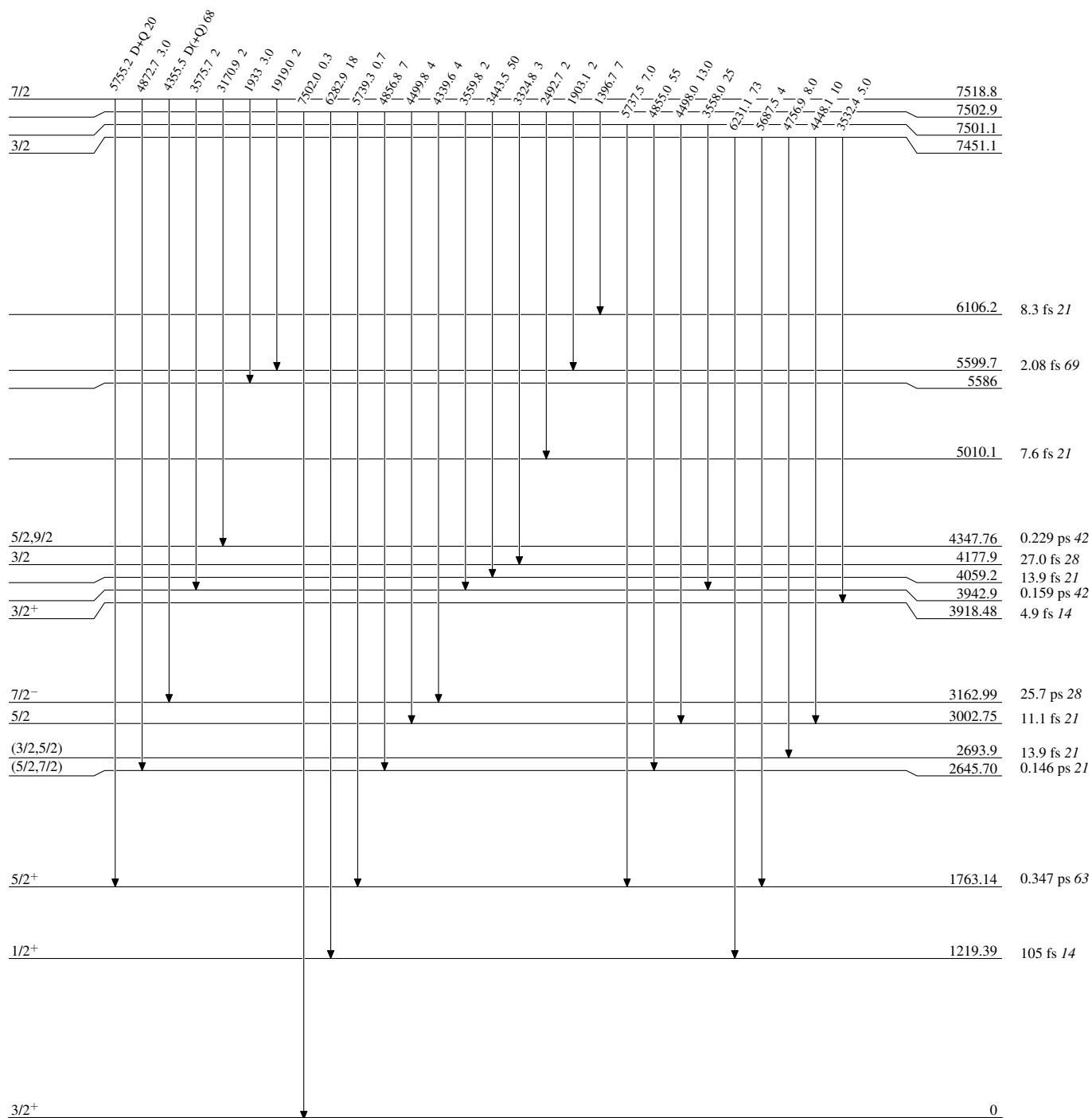
Intensities: % photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

$^{34}\text{S}(\text{p},\gamma)$  1976Me12,1976Sp08,1972Hu10

## Level Scheme (continued)

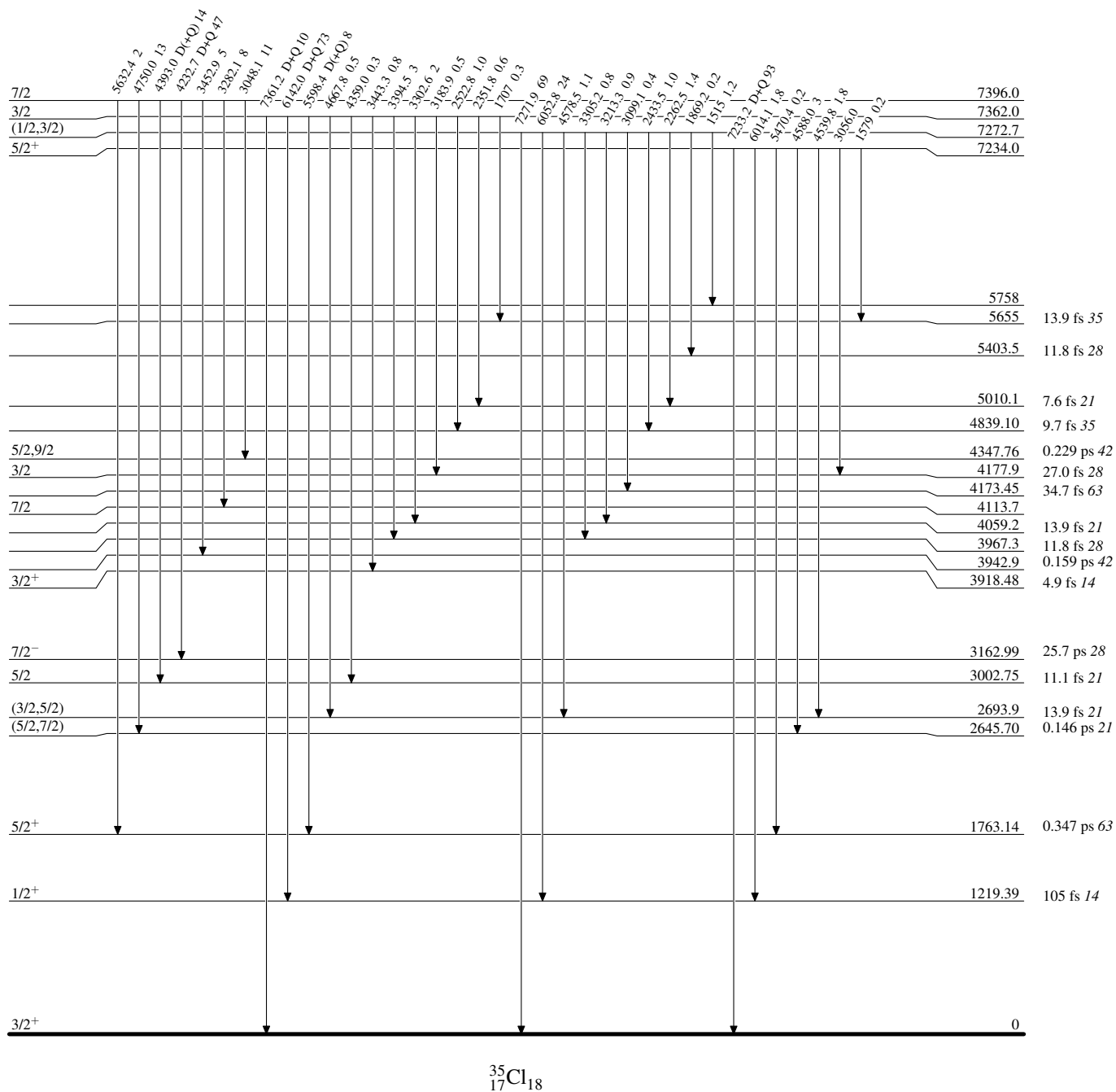
Intensities: % photon branching from each level



$^{34}\text{S}(\text{p},\gamma)$  1976Me12,1976Sp08,1972Hu10

## Level Scheme (continued)

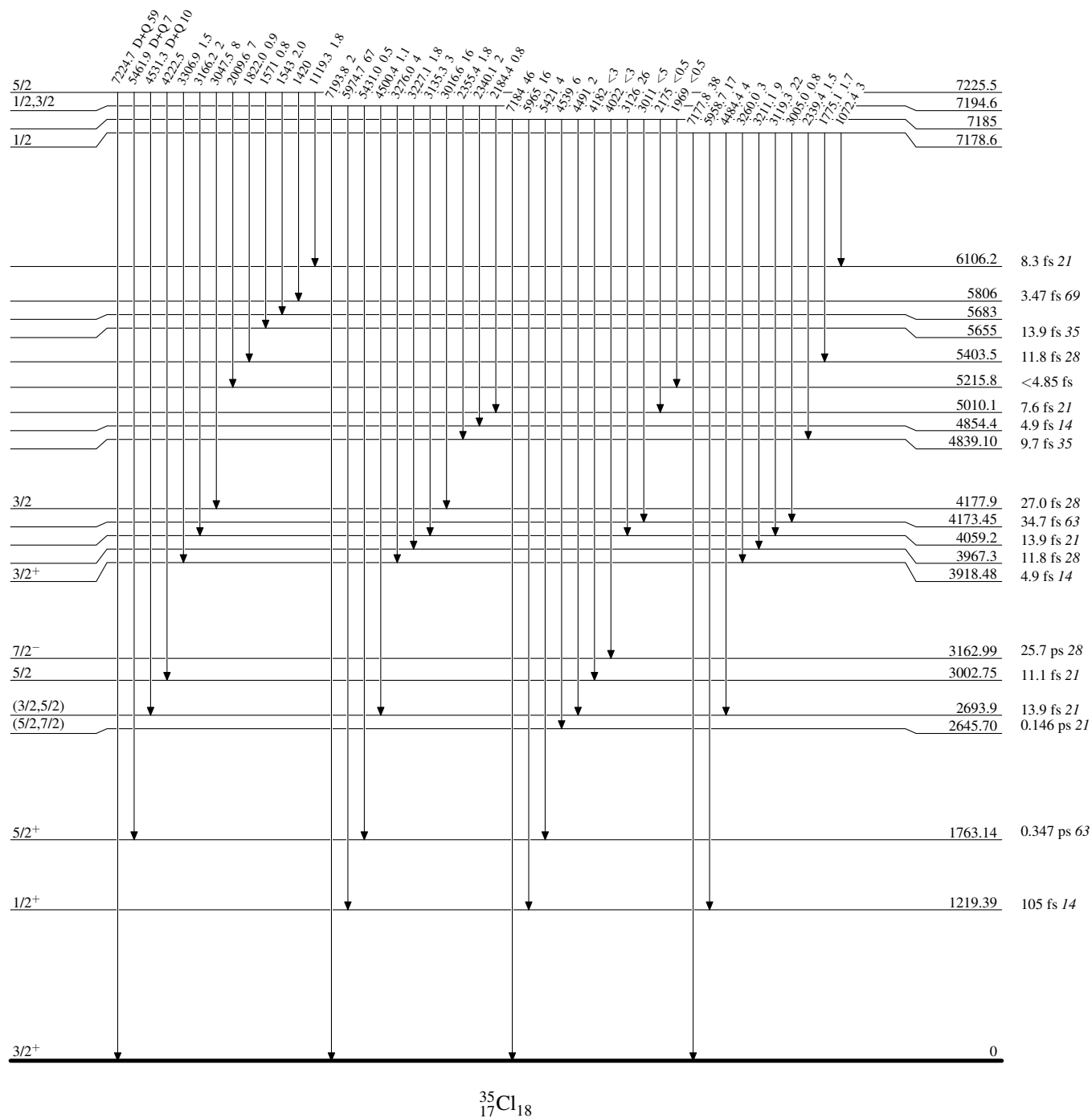
Intensities: % photon branching from each level



$^{34}\text{S}(\text{p},\gamma)$  1976Me12,1976Sp08,1972Hu10

## Level Scheme (continued)

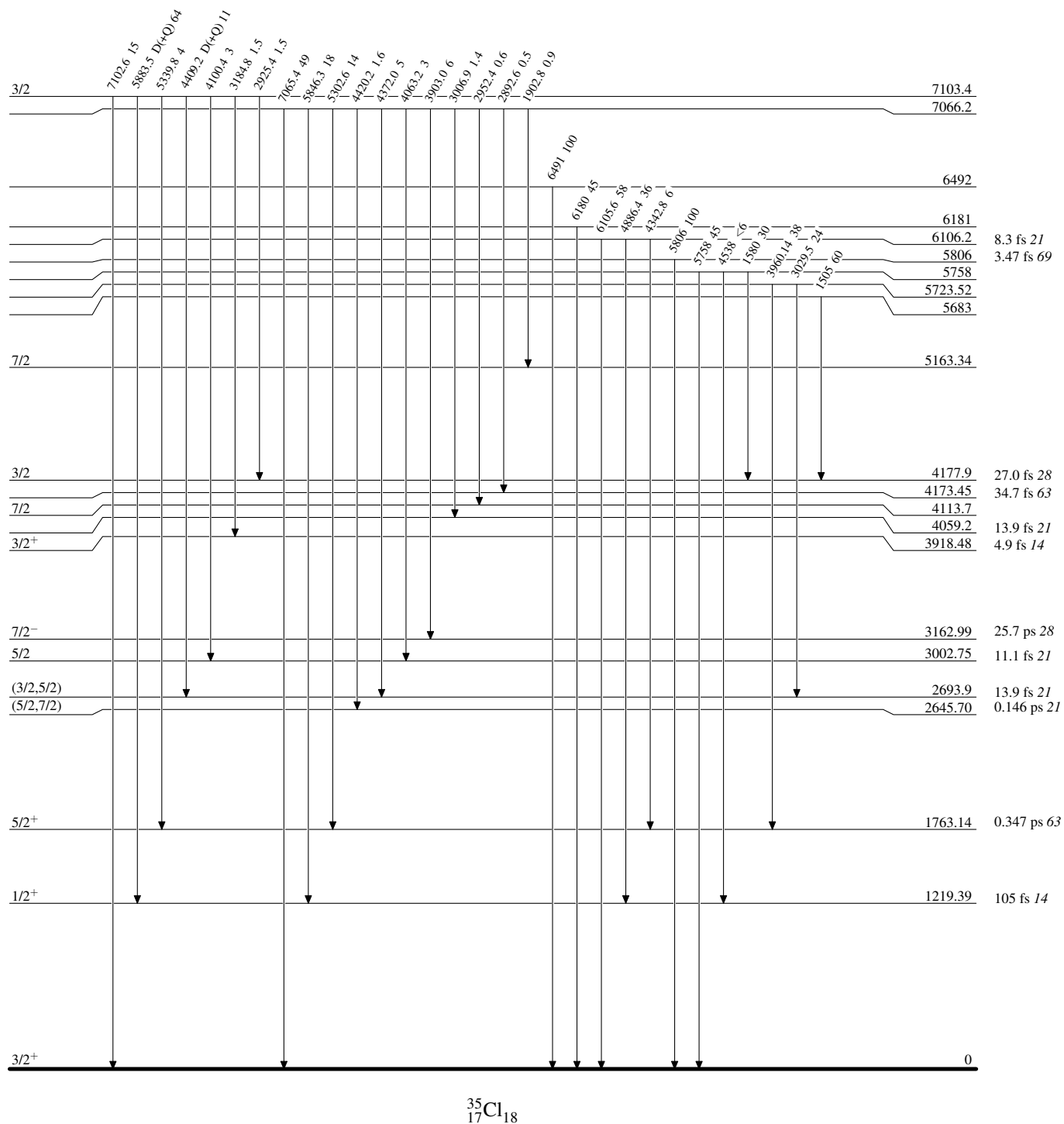
Intensities: % photon branching from each level



$^{34}\text{S}(\text{p},\gamma)$  1976Me12,1976Sp08,1972Hu10

## Level Scheme (continued)

Intensities: % photon branching from each level

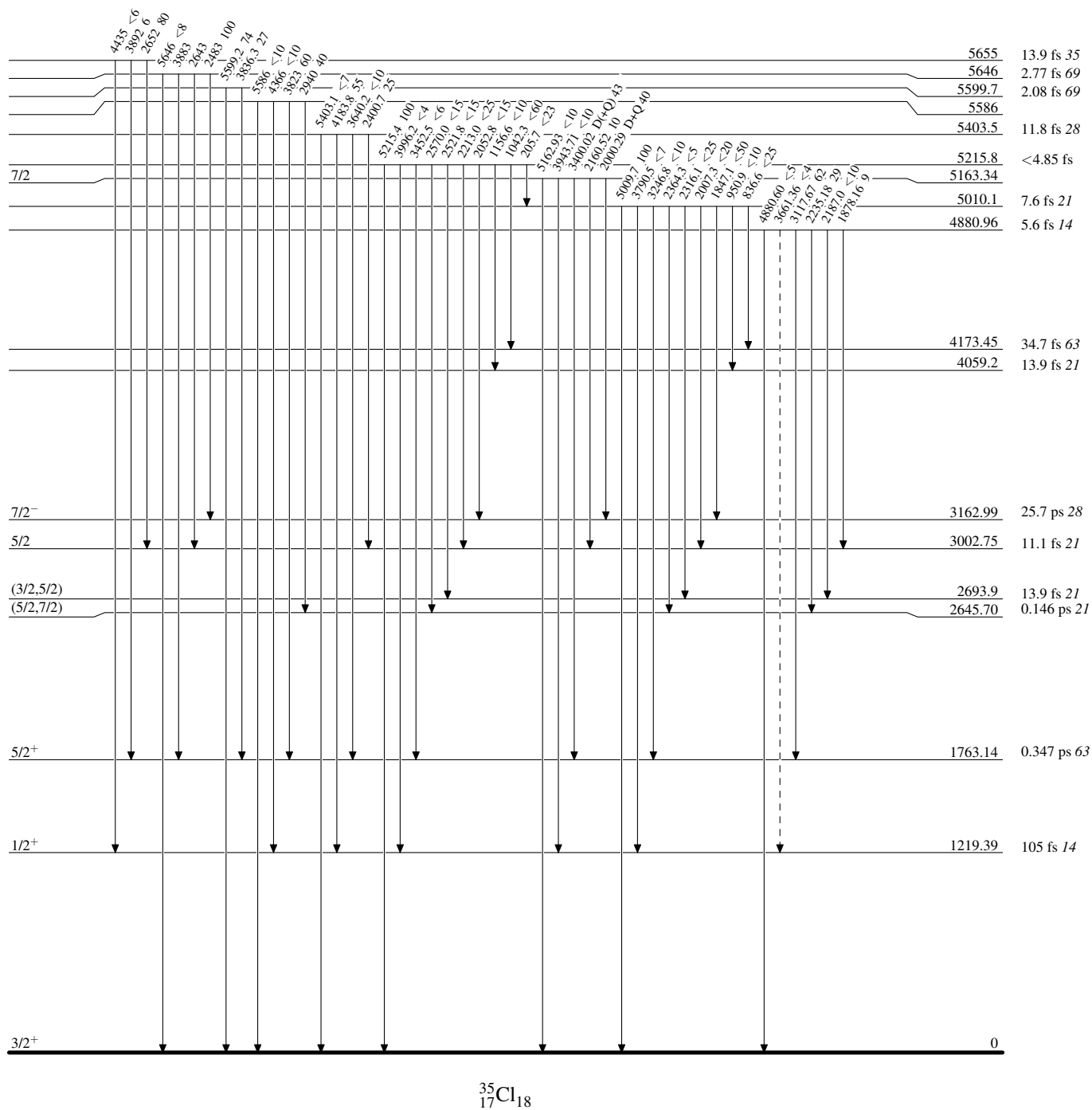


$^{34}\text{S}(\text{p},\gamma)$  1976Me12,1976Sp08,1972Hu10

Legend

## Level Scheme (continued)

Intensities: % photon branching from each level

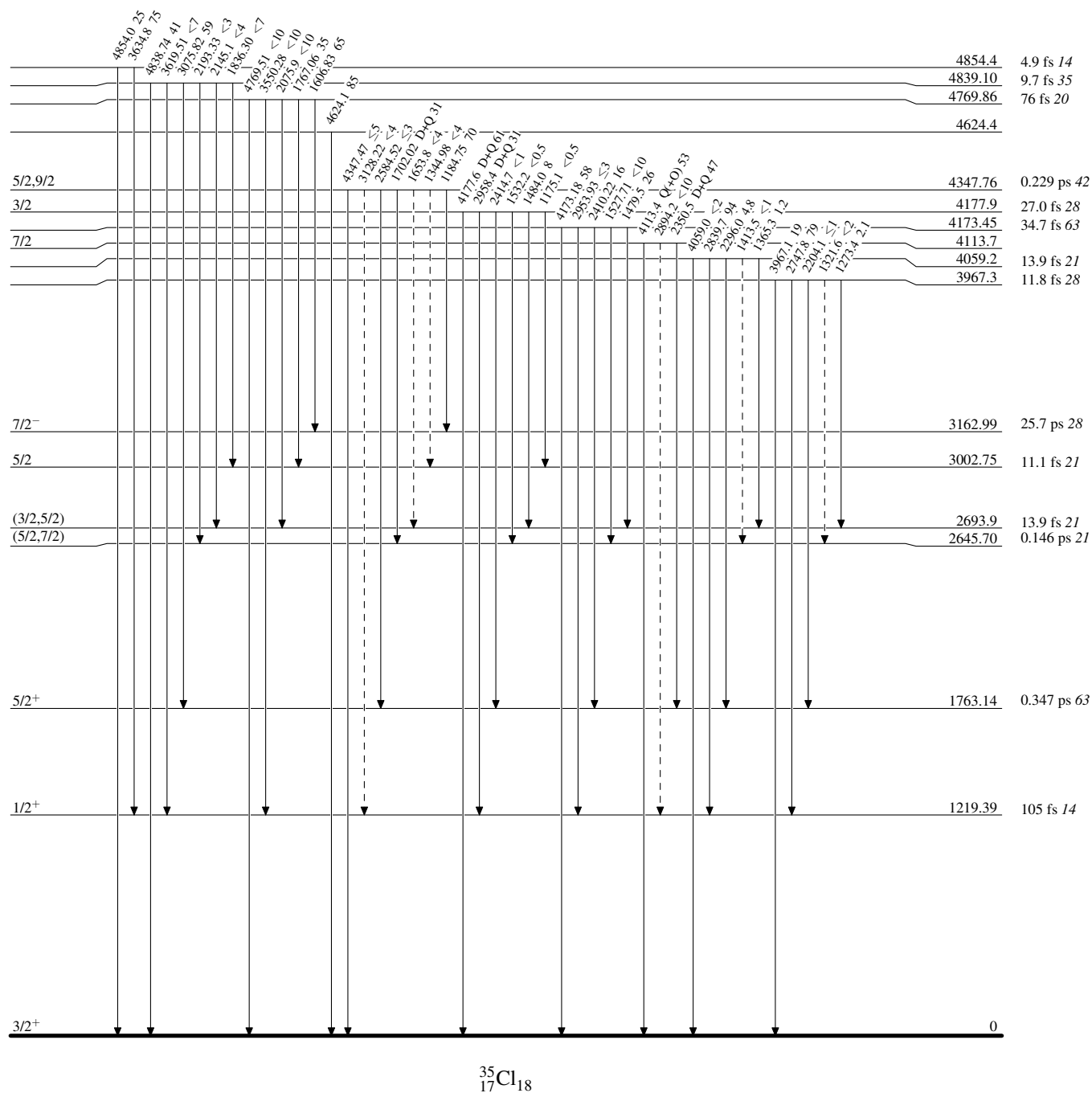
-----►  $\gamma$  Decay (Uncertain) $^{35}_{17}\text{Cl}_{18}$



### Legend

Intensities: % photon branching from each level

-----►  $\gamma$  Decay (Uncertain)

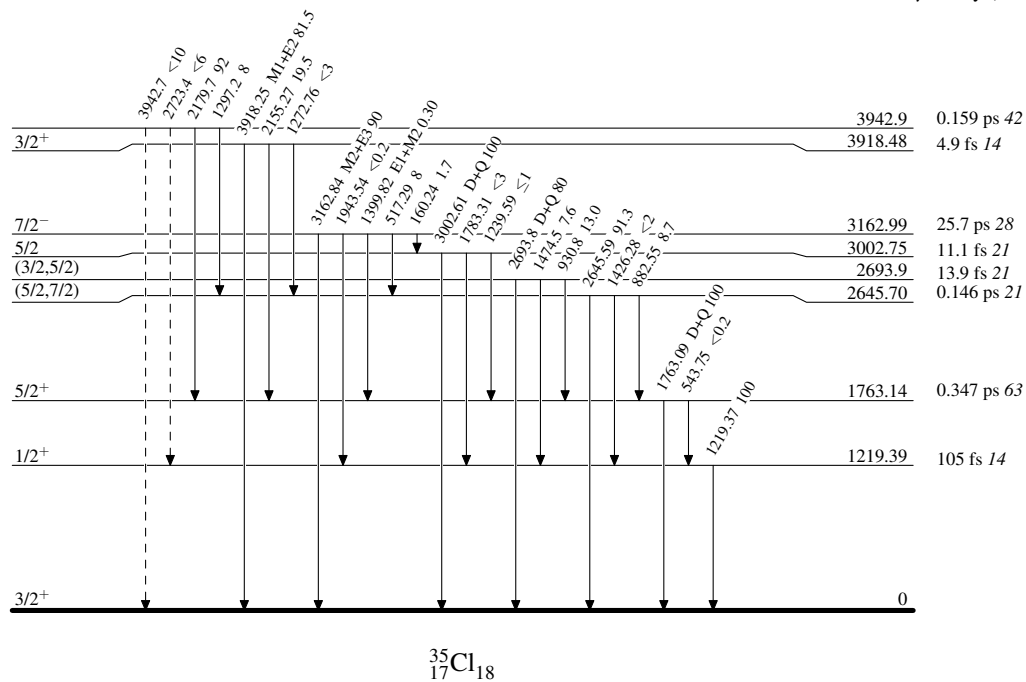


$^{34}\text{S}(\text{p},\gamma)$  1976Me12,1976Sp08,1972Hu10

Legend

## Level Scheme (continued)

Intensities: % photon branching from each level

-----►  $\gamma$  Decay (Uncertain)

$^{34}\text{S}(\text{p,p}),(\text{p,p}')\text{:resonances}$  **1967Ko19,1977Ou01,1981Bi05** $^{34}\text{S}(\text{p,p}')\gamma$  measurement:

**1967Ko19:** E=2.78-3.08-MeV proton beams were produced from the 4-MeV electrostatic accelerator of the Physical-Technical Institute of the Ukrainian SSR Academy of Sciences, in energy step of 1 keV. Target was enriched thin  $^{34}\text{S}$ .  $\gamma$  rays were detected with a 70 by 60 mm NaI(Tl) crystal for detecting the 2.13 MeV  $\gamma$ -ray of  $^{34}\text{S}$  decay, resolution of 11.5% for the 0.622 MeV  $^{137}\text{Cs}$  line. Measured  $\sigma(E_p, \theta)$  from yields of the 2.13 MeV  $\gamma$ -ray,  $\text{py}(\theta)$  of the 2.13 MeV  $\gamma$ -ray. Deduced resonance energies, J,  $\pi$ , relative resonance intensities.

 $^{34}\text{S}(\text{p,p})$  measurements:

**1977Ou01:** E=1.45-2.82 MeV proton beams were produced from the TUNL 3-MV Van de Graaff accelerator in energy steps of 100 eV for on-resonance and 400 eV for off-resonance, overall FWHM=350-450 eV. Target was CdS, with 90% enriched  $^{34}\text{S}$  evaporated onto 5-10  $\mu\text{g}/\text{cm}^2$  carbon backings. Scattered protons were detected with surface barrier detectors at 90°, 105°, 135° and 160°. Measured  $\sigma(E_p, \theta)$ . Deduced resonance energies, J,  $\pi$ ,  $\Gamma_p$  from R-Matrix analysis.

**1981Bi05:** E=1.4-2.8 MeV proton beams were produced from the Groningen 5 MV Van de Graaff accelerator. Targets were 4 and 1  $\mu\text{g}/\text{cm}^2$   $\text{Sb}_2\text{S}_3$  (90% in  $^{34}\text{S}$ ) evaporated onto 10  $\mu\text{g}/\text{cm}^2$  carbon backings. Scattered protons were detected with three silicon surface barrier detectors at 88°, 124° and 174°. Measured  $\sigma(E_p, \theta)$ . Deduced resonance energies, J,  $\pi$ ,  $\Gamma_p$ . Also measured (p, $\gamma$ ).

**1971BrXT:** Measured  $\sigma(E_p)$ . Deduced resonance energies, J,  $\pi$ , widths.

**1974Bi16:** E=1.2-2.1 MeV protons were produced from the Helsinki University 2.5 MV van de Graaff accelerator. Targets were prepared from natural sulphur (4.22%  $^{34}\text{S}$ ) with  $^{34}\text{S}$  ions embedded into carbon foils. Scattered protons were detected with four silicon surface barrier detectors. Measured  $\sigma(E_p, \theta)$ . Deduced resonance energies, J,  $\pi$ , widths from analysis of the resonances.

Others: **1978Ce02**, **1980Fa07**, **1983Ch50**, **1988Za04**, **2019Ka53**.

 $^{35}\text{Cl}$  Levels

$A_2$  and  $A_4$  values under comments are from  $\text{py}(\theta)$  of the 2.13 MeV  $\gamma$ -ray from  $^{34}\text{S}$  (**1967Ko19**).

I(res) under comments are resonance intensities relative to 1 for  $E_p(\text{lab})=2799$  resonance (**1967Ko19**).

| E(level) <sup>†</sup> | J $\pi$ <sup>‡</sup>                  | $\Gamma_p$ <sup>#</sup> | $E_p(\text{lab})(\text{keV})$ <sup>#</sup> | Comments                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------|---------------------------------------|-------------------------|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7195@                 | 1/2 <sup>-</sup> @                    |                         | 848                                        | $\Gamma=27$ eV 8 ( <b>1971BrXT</b> ).                                                                                                                                                                                                                                                                                                                                                                                              |
| 7274@                 | 1/2 <sup>-</sup> @                    |                         | 929                                        | $\Gamma=14$ eV 4 ( <b>1971BrXT</b> ).                                                                                                                                                                                                                                                                                                                                                                                              |
| 7550                  | 7/2 <sup>-</sup>                      |                         | 1214                                       | $\Gamma_p=4.6$ eV 10 and $\Gamma=5$ eV +4-2 from a $Q^2$ minimization analysis of the (p,p) excitation function ( <b>1971BrXT</b> ). $\Gamma=10.5$ eV 15, $\Gamma_p=5.8$ eV 7, $\Gamma_\gamma=4.7$ eV 17 ( <b>1971BrXT</b> ); $\Gamma=11.0$ eV 15, $\Gamma_p=7.2$ eV 15, $\Gamma_\gamma=3.8$ eV 15 ( <b>1974Bi16</b> ), which are deduced using a questionable (p, $\gamma$ ) resonance strength of 21 eV 3 from <b>1966En04</b> . |
| 7686&                 | 3/2 <sup>-</sup> &                    |                         | 1354                                       | $\Gamma_p=445$ eV 11 for $J^\pi=3/2^-$ ( <b>1981Bi05</b> ).                                                                                                                                                                                                                                                                                                                                                                        |
| 7706&                 | 3/2 <sup>+</sup> , 5/2 <sup>+</sup> & |                         | 1375                                       | $\Gamma_p=4$ eV 1 for $J^\pi=5/2^+$ , 17 eV 3 for $J^\pi=3/2^+$ ( <b>1981Bi05</b> ).                                                                                                                                                                                                                                                                                                                                               |
| 7796 3                | (1/2) <sup>-</sup>                    | 0.031 keV 10            | 1467.3 5                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 7837 3                | 3/2 <sup>-</sup>                      | 3.7 keV 4               | 1510.0 5                                   | $\Gamma=3.8$ keV 2, $\Gamma_p=3.8$ keV 2 ( <b>1974Bi16</b> ).                                                                                                                                                                                                                                                                                                                                                                      |
| 7880 3                | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>   | 0.008 keV 4             | 1554.0 5                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 8004 3                | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>   | 0.011 keV 5             | 1682.1 5                                   | $\Gamma_p=21$ keV 3 for $J^\pi=5/2^+$ , 24 eV 3 for $J^\pi=3/2^+$ ( <b>1981Bi05</b> ).                                                                                                                                                                                                                                                                                                                                             |
| 8034 3                | 1/2 <sup>-</sup> , 3/2 <sup>-</sup>   | 0.026 keV 7             | 1712.9 5                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 8038 3                | 1/2 <sup>+</sup>                      | 0.30 keV 2              | 1716.9 5                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 8148 3                | 1/2 <sup>-</sup>                      | 2.66 keV 27             | 1830.2 5                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 8150 3                | 3/2 <sup>-</sup>                      | 0.56 keV 6              | 1831.5 5                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 8209 3                | (5/2) <sup>+</sup>                    | 0.033 keV 10            | 1893.2 5                                   | $\Gamma=100$ eV 20, $\Gamma_p=120$ eV 20 ( <b>1974Bi16</b> ).                                                                                                                                                                                                                                                                                                                                                                      |
| 8211 3                | 1/2 <sup>+</sup>                      | 0.094 keV 15            | 1894.4 5                                   | $\Gamma_p=39$ eV 5 for $J^\pi=5/2^+$ , 44 eV 6 for $J^\pi=3/2^+$ , 120 eV 11 for $J^\pi=1/2^+$ ( <b>1981Bi05</b> ).                                                                                                                                                                                                                                                                                                                |
|                       |                                       |                         |                                            | $\Gamma=270$ eV 50, $\Gamma_p=270$ eV 50 ( <b>1974Bi16</b> ).                                                                                                                                                                                                                                                                                                                                                                      |
| 8216                  | 5/2 <sup>+</sup>                      | 0.014 keV 3             | 1900                                       | $\Gamma=14$ eV 4, $\Gamma_\gamma=0.78$ eV ( <b>1974Bi16</b> ).                                                                                                                                                                                                                                                                                                                                                                     |
| 8243 3                | 3/2 <sup>-</sup>                      | 0.140 keV 15            | 1928.2 5                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 8271 3                | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>   | 0.005 keV 3             | 1956.1 5                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 8279 3                | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>   | 0.006 keV 3             | 1964.6 5                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 8290 3                | (1/2) <sup>-</sup>                    | 0.04 keV 1              | 1975.7 5                                   | $\Gamma_p=49$ eV 11 for $J^\pi=3/2^-$ , 48 eV 6 for $J^\pi=1/2^-$ ( <b>1981Bi05</b> ).                                                                                                                                                                                                                                                                                                                                             |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p,p}),(\text{p,p}'\gamma):\text{resonances}$  **1967Ko19,1977Ou01,1981Bi05 (continued)** $^{35}\text{Cl}$  Levels (continued)

| E(level) <sup>†</sup> | $J^{\pi\ddagger}$                      | $\Gamma_{\text{p}}^{\#}$ | $E_{\text{p}}(\text{lab})(\text{keV})^{\#}$ | Comments                                                                                                                                                                        |
|-----------------------|----------------------------------------|--------------------------|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8300 3                | 3/2 <sup>-</sup>                       | 0.073 keV 15             | 1986.7 5                                    | $\Gamma=30$ eV 7, $\Gamma_{\text{p}}=28$ eV 6, $\Gamma_{\gamma}=2.4$ eV (1974Bi16).<br>$J^{\pi}$ : 5/2 <sup>+</sup> in 1974Bi16.                                                |
| 8383 3                | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>    | 0.023 keV 7              | 2072.2 5                                    |                                                                                                                                                                                 |
| 8404 3                | 5/2 <sup>-</sup> , 7/2 <sup>-</sup>    | 0.002 keV 1              | 2093.9 5                                    |                                                                                                                                                                                 |
| 8406 3                | (5/2 <sup>-</sup> , 7/2 <sup>-</sup> ) | 0.001 keV 1              | 2095.2 5                                    |                                                                                                                                                                                 |
| 8409 3                | 1/2 <sup>-</sup>                       | 0.125 keV 15             | 2098.7 5                                    |                                                                                                                                                                                 |
| 8418 3                | 1/2 <sup>+</sup>                       | 0.026 keV 7              | 2107.5 5                                    |                                                                                                                                                                                 |
| 8435 3                | (3/2) <sup>+</sup>                     | 0.090 keV 15             | 2125.9 5                                    |                                                                                                                                                                                 |
| 8486 3                | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>    | 0.012 keV 5              | 2178.2 5                                    |                                                                                                                                                                                 |
| 8488 3                | 3/2 <sup>-</sup>                       | 0.150 keV 15             | 2179.7 5                                    |                                                                                                                                                                                 |
| 8515 3                | 1/2 <sup>-</sup>                       | 0.150 keV 15             | 2208.1 5                                    |                                                                                                                                                                                 |
| 8573 3                | (5/2) <sup>+</sup>                     | 0.08 keV 1               | 2267.9 5                                    |                                                                                                                                                                                 |
| 8582 3                | 1/2 <sup>+</sup>                       | 0.75 keV 8               | 2276.7 5                                    |                                                                                                                                                                                 |
| 8592 3                | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>    | 0.003 keV 2              | 2287.4 5                                    |                                                                                                                                                                                 |
| 8616 3                | 5/2 <sup>+</sup>                       | 0.175 keV 20             | 2311.6 5                                    |                                                                                                                                                                                 |
| 8620 3                | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>    | 0.002 keV 1              | 2316.2 5                                    |                                                                                                                                                                                 |
| 8633 3                | (3/2, 5/2 <sup>+</sup> )               | 0.001 keV 1              | 2328.8 5                                    |                                                                                                                                                                                 |
| 8643 3                | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>    | 0.003 keV 2              | 2339.9 5                                    |                                                                                                                                                                                 |
| 8687 3                | 5/2 <sup>-</sup> , 7/2 <sup>-</sup>    | 0.001 keV 1              | 2385.0 5                                    |                                                                                                                                                                                 |
| 8689 3                | 1/2 <sup>+</sup>                       | 0.20 keV 2               | 2387.2 5                                    |                                                                                                                                                                                 |
| 8691 3                | 1/2 <sup>-</sup>                       | 6.4 keV 7                | 2388.6 5                                    |                                                                                                                                                                                 |
| 8698 3                | 3/2 <sup>-</sup>                       | 0.8 keV 1                | 2396.0 5                                    |                                                                                                                                                                                 |
| 8750 3                | 3/2 <sup>-</sup>                       | 0.30 keV 3               | 2449.5 5                                    |                                                                                                                                                                                 |
| 8775 3                | 1/2 <sup>-</sup>                       | 0.57 keV 6               | 2475.2 5                                    |                                                                                                                                                                                 |
| 8781 3                | 3/2 <sup>-</sup>                       | 0.214 keV 25             | 2481.7 5                                    |                                                                                                                                                                                 |
| 8788 3                | (3/2 <sup>+</sup> , 5/2 <sup>+</sup> ) | 0.001 keV 1              | 2489.1 5                                    |                                                                                                                                                                                 |
| 8799 3                | (3/2 <sup>+</sup> , 5/2 <sup>+</sup> ) | 0.001 keV 1              | 2500.3 5                                    |                                                                                                                                                                                 |
| 8825 3                | 1/2 <sup>+</sup>                       | 1.70 keV 17              | 2526.8 5                                    |                                                                                                                                                                                 |
| 8829 3                | 1/2 <sup>-</sup>                       | 12.2 keV 12              | 2530.8 5                                    |                                                                                                                                                                                 |
| 8830 3                | 1/2 <sup>+</sup>                       | 0.080 keV 15             | 2532.0 5                                    |                                                                                                                                                                                 |
| 8839 3                | 5/2 <sup>-</sup> , 7/2 <sup>-</sup>    | 0.001 keV 1              | 2541.7 5                                    | $\Gamma_{\text{p}}=4$ eV 1 for $J^{\pi}=7/2^{-}$ , 2 eV 1 for $J^{\pi}=5/2^{-}$ (1981Bi05).                                                                                     |
| 8858 3                | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>    | 0.010 keV 5              | 2560.8 5                                    |                                                                                                                                                                                 |
| 8870 3                | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>    | 0.027 keV 10             | 2573.2 5                                    |                                                                                                                                                                                 |
| 8909 3                | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>    | 0.002 keV 1              | 2613.2 5                                    |                                                                                                                                                                                 |
| 8955 3                | (3/2) <sup>+</sup>                     | 0.075 keV 15             | 2661.3 5                                    |                                                                                                                                                                                 |
| 8959 3                | (1/2 <sup>-</sup> , 3/2 <sup>-</sup> ) | 0.04 keV 1               | 2665.4 5                                    |                                                                                                                                                                                 |
| 8983 3                | 5/2 <sup>-</sup> , 7/2 <sup>-</sup>    | 0.003 keV 2              | 2690.0 5                                    |                                                                                                                                                                                 |
| 8985 3                | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>    | 0.025 keV 10             | 2692.1 5                                    |                                                                                                                                                                                 |
| 8998 3                | 5/2 <sup>-</sup> , 7/2 <sup>-</sup>    | 0.002 keV 1              | 2705.1 5                                    |                                                                                                                                                                                 |
| 9021 3                | 3/2 <sup>-</sup>                       | 3.50 keV 35              | 2728.4 5                                    |                                                                                                                                                                                 |
| 9031 3                | (5/2) <sup>+</sup>                     | 0.04 keV 1               | 2739.4 5                                    |                                                                                                                                                                                 |
| 9040 3                | 1/2 <sup>-</sup>                       | 0.292 keV 30             | 2748.2 5                                    |                                                                                                                                                                                 |
| 9048 3                | 5/2 <sup>-</sup> , 7/2 <sup>-</sup>    | 0.001 keV 1              | 2756.7 5                                    |                                                                                                                                                                                 |
| 9050 3                | (5/2) <sup>+</sup>                     | 0.095 keV 15             | 2758.8 5                                    |                                                                                                                                                                                 |
| 9084 3                | (5/2) <sup>+</sup>                     | 0.057 keV 10             | 2793.2 5                                    | $\Gamma_{\text{p}}=59$ eV 7 for $J^{\pi}=5/2^{+}$ , 67 eV 8 for $J^{\pi}=3/2^{+}$ (1981Bi05).<br>$I(\text{res})=1$ .<br>$E_{\text{p}}(\text{lab})(\text{keV})$ : from 1967Ko19. |
| 9089 4                |                                        |                          | 2799 4                                      |                                                                                                                                                                                 |
| 9101 3                | 3/2 <sup>-</sup>                       | 0.20 keV 2               | 2810.9 5                                    |                                                                                                                                                                                 |
| 9108 4                |                                        |                          | 2818 4                                      | $I(\text{res})=0.18$ .                                                                                                                                                          |
| 9114 4                |                                        |                          | 2825 4                                      | $I(\text{res})=0.34$ .                                                                                                                                                          |
| 9120 4                |                                        |                          | 2831 4                                      | $I(\text{res})=0.31$ .                                                                                                                                                          |
| 9127 4                |                                        |                          | 2838 4                                      | $I(\text{res})=0.16$ .                                                                                                                                                          |
| 9136 4                |                                        |                          | 2847 4                                      | $I(\text{res})=0.34$ .                                                                                                                                                          |
| 9147 4                |                                        |                          | 2859 4                                      | $I(\text{res})=0.27$ .                                                                                                                                                          |
| 9159 4                |                                        |                          | 2871 4                                      | $I(\text{res})=0.29$ .                                                                                                                                                          |
| 9168 4                | (3/2 <sup>-</sup> )                    |                          | 2880 4                                      | $I(\text{res})=3.70$ .                                                                                                                                                          |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p,p}),(\text{p,p}'\gamma):\text{resonances}$  **1967Ko19,1977Ou01,1981Bi05** (continued) $^{35}\text{Cl}$  Levels (continued)

| E(level) <sup>†</sup> | J <sup>π</sup> <sup>‡</sup> | E <sub>p</sub> (lab)(keV) <sup>#</sup> | Comments                                                                           |
|-----------------------|-----------------------------|----------------------------------------|------------------------------------------------------------------------------------|
| 9185 4                |                             | 2898 4                                 | A <sub>2</sub> =+0.19 133, A <sub>4</sub> =+0.017 1480 (1967Ko19).<br>I(res)=0.11. |
| 9196 4                |                             | 2909 4                                 |                                                                                    |
| 9207 4                | 9/2 <sup>+</sup>            | 2920 4                                 | I(res)=1.70.<br>A <sub>2</sub> =+0.09 18, A <sub>4</sub> =+0.37 12 (1967Ko19).     |
| 9222 4                |                             | 2936 4                                 | I(res)=0.25.                                                                       |
| 9227 4                | 5/2 <sup>-</sup>            | 2941 4                                 | I(res)=0.59.<br>A <sub>2</sub> =+0.59 28, A <sub>4</sub> =+0.29 32 (1967Ko19).     |
| 9241 4                | 7/2 <sup>+</sup>            | 2955 4                                 | I(res)=2.32.<br>A <sub>2</sub> =+0.24 40, A <sub>4</sub> =+0.27 53 (1967Ko19).     |
| 9255 4                |                             | 2970 4                                 | I(res)=0.45.                                                                       |
| 9271 4                |                             | 2986 4                                 | I(res)=0.34.                                                                       |
| 9276 4                |                             | 2991 4                                 | I(res)=0.32.                                                                       |
| 9284 4                | 5/2 <sup>-</sup>            | 3000 4                                 | I(res)=3.86.<br>A <sub>2</sub> =+0.72 28, A <sub>4</sub> =-0.17 29 (1967Ko19).     |
| 9294 4                |                             | 3010 4                                 | I(res)=0.68.                                                                       |
| 9299 4                |                             | 3015 4                                 | I(res)=0.45.                                                                       |
| 9325 4                | 5/2 <sup>+</sup>            | 3042 4                                 | I(res)=1.81.<br>A <sub>2</sub> =+0.81 28, A <sub>4</sub> =-0.48 35 (1967Ko19).     |
| 9336 4                | 7/2 <sup>+</sup>            | 3053 4                                 | I(res)=1.00.<br>A <sub>2</sub> =+0.17 29, A <sub>4</sub> =+0.34 40 (1967Ko19).     |
| 9345 4                |                             | 3062 4                                 | I(res)=0.50.                                                                       |
| 9349 4                |                             | 3067 4                                 | I(res)=0.50.                                                                       |
| 9355 4                |                             | 3073 4                                 | I(res)=2.23.                                                                       |

<sup>†</sup> From E(level)=E<sub>p</sub>(cm)+S(p)( $^{35}\text{Cl}$ ), where S(p)=6370.81 4 (2021Wa16) and E<sub>p</sub>(cm) deduced from E<sub>p</sub>(lab) by E<sub>p</sub>(cm)=E<sub>p</sub>(lab)\*[m( $^{34}\text{S}$ )/(m( $^{34}\text{S}$ )+m(p))].

<sup>‡</sup> From comparison of experimental differential cross sections with theoretical predictions by R-Matrix calculations (1977Ou01) for levels up to 9101 level and from  $\text{py}(\theta)$  (1967Ko19) for levels after that.

<sup>#</sup> From 1977Ou01 up to 9101 level and from 1967Ko19 after that, unless otherwise noted. 1977Ou01 state that although the absolute proton energies are accurate only to about 3 keV, the relative values of E<sub>p</sub>(lab) should be reliable to within a few hundred eV, based on which the evaluators have assigned a 0.5 keV uncertainty for E<sub>p</sub>(lab) and 3 keV uncertainty for E(level) from 1977Ou01.

@ From 1971BrXT.

& From 1981Bi05.

<sup>34</sup>S(d,n) 1969Da12

$J^\pi=0^+$  for <sup>34</sup>S ground state.  
**1969Da12:** a 4.95-MeV deuteron beam was produced from the 5.5-MeV Van de Graaff accelerator at University of Alberta and impinged on a CdS target enriched to 37% <sup>34</sup>S evaporated onto a 0.05-cm gold backing. Neutrons were detected by NE213 scintillators using the time-of-flight (tof) method with FWHM=1.0 ns. Measured  $\sigma(E_n, \theta)$ . Deduced levels, J,  $\pi$ , and L-transfers from DWUCK-DWBA analysis of the measured  $\sigma(\theta)$ .  
**1993TeZX:** <sup>34</sup>S(d,n)<sup>35</sup>Cl at E(d)=25 MeV at CYRJC, Tohoku University.

<sup>35</sup>Cl Levels

| E(level) <sup>†</sup> | L <sup>‡</sup> | Comments                                                                                                             |
|-----------------------|----------------|----------------------------------------------------------------------------------------------------------------------|
| 0                     | 2              |                                                                                                                      |
| 1220                  | 0              |                                                                                                                      |
| 1762                  | 2              |                                                                                                                      |
| 2645                  | (1)            | L: weak stripping or possibly two unresolved levels.                                                                 |
| 2695                  |                |                                                                                                                      |
| 3006                  | 3              |                                                                                                                      |
| 3163                  | 3              | L: Discrepancy: L=2 from 0 <sup>+</sup> in ( <sup>3</sup> He,d), (d, <sup>3</sup> He), and (pol d, <sup>3</sup> He). |
| 4060                  |                |                                                                                                                      |
| 4110                  |                |                                                                                                                      |
| 4170                  |                |                                                                                                                      |

<sup>†</sup> From 1969Da12.  
<sup>‡</sup> From DWBA analysis of the measured  $\sigma(\theta)$  in 1969Da12.

$^{34}\text{S}(^3\text{He},\text{d})$  2017Gi09,1969Gr23 $J^\pi=0^+$  for  $^{34}\text{S}$  g.s.

- 2017Gi09:** a 20-MeV  $^3\text{He}^{2+}$  beam was produced from the MP Van de Graaff tandem accelerator at the Maier-Leibnitz Laboratorium (MLL) and impinged on a  $50\text{-}\mu\text{g}/\text{cm}^2$   $\text{Ag}_2\text{S}$  target 99.999% enriched in  $^{34}\text{S}$  with a  $8\text{-}\mu\text{g}/\text{cm}^2$  carbon backing. Reaction products were momentum-separated by the Q3D magnetic spectrograph and focused on focal plane detectors consisting of two multiwire proportional counters (MWPCs) and a plastic scintillator. Deduced 22 levels, L-transfers, J,  $\pi$ , spectroscopic factors from FRESKO-DWBA analysis, and resonance strengths. Deduced the thermonuclear  $^{34}\text{S}(\text{p},\gamma)^{35}\text{Cl}$  reaction rate through Monte Carlo methods using the STARLIB code. Calculated the  $^{34}\text{S}/^{32}\text{S}$  isotopic ratio using the hydrodynamical nova model SHIVA.
- 1969Gr23:** a 11.0-MeV  $^3\text{He}$  beam was produced from the MIT-ONR Van de Graaff generator. The target was  $9.3\text{-}\mu\text{g}/\text{cm}^2$   $^{34}\text{S}$  made by evaporating PbS (67.6%  $^{34}\text{S}$ ) onto a formvar backing. Deuterons were detected on  $50\text{ }\mu\text{m}$  nuclear emulsions in the MIT multiple-gap spectrograph at 23 different angles from  $7.5^\circ$  to  $172.5^\circ$ . Measured  $\sigma(E_d,\theta)$ . Deduced 17 levels, J,  $\pi$ , L-transfers, spectroscopic factors from zero-range JULIE-DWBA analysis.
- 1970Mo01:** a 13-MeV  $^3\text{He}$  beam was produced from the Argonne tandem Van de Graaff accelerator. The target was  $200\text{-}\mu\text{g}/\text{cm}^2$  86% enriched  $^{34}\text{S}$ . Reaction products were detected using two telescopes of silicon surface-barrier detectors with FWHM $\approx 80$  keV and analyzed by a magnetic spectrograph with FWHM $\approx 20$  keV at  $\theta < 20^\circ$ . Measured  $\sigma(E_d,\theta)$ ,  $\theta_{\text{c.m.}} \approx 5^\circ$  to  $100^\circ$ . Deduced levels, J,  $\pi$ , L-transfers, spectroscopic factors from local, zero-range JULIE-DWBA analysis.
- 1994Ve04:** a 25-MeV  $^3\text{He}$  beam was produced from the Orsay MP Tandem Van de Graaff accelerator. The target was KCl with natural chlorine. Reaction products were momentum-analyzed by an Enge split-pole magnetic spectrograph and detected using a 50-cm long telescope of a position-sensitive proportional counter, proportional  $\Delta E$  counter, and a plastic scintillator. Measured  $\sigma(E_d,\theta)$ . Deduced J,  $\pi$ , L-transfer, and spectroscopic factor for the ground state of  $^{35}\text{Cl}$  from local, zero-range DWUCK-DWBA analysis.

 $^{35}\text{Cl}$  Levels

For levels below 6 MeV, spectroscopic factor  $(2J+1)\text{C}^2\text{S}=\sigma(\theta)_{\text{exp}}/\sigma(\theta)_{\text{DWBA}}/N$ , where N is a normalization factor. J is the spin of the populated  $^{35}\text{Cl}$  level. N=4.42 is used in 1969Gr23 and 1970Mo01 from 1966Ba54. N=4.43 is used in 1994Ve04. For levels above 6 MeV,  $\text{C}^2\text{S}$  values are from 2017Gi09 without using the factor N. 2017Gi09 also states that it is not possible to distinguish between  $J=L+1/2$  or  $J=L-1/2$  coupling given the experimental uncertainties.

Values of resonance strengths  $\omega\gamma(\text{min})$  and  $\omega\gamma(\text{max})$  are calculated by 2017Gi09 using their measured  $\text{C}^2\text{S}$  and calculated  $\Gamma_p$ , assuming the  $\Gamma_{\text{tot}}$  is dominated by  $\Gamma_\gamma$  and thus  $\omega\gamma$  depends only on  $\Gamma_p$ .

| E(level) <sup>†</sup> | L <sup>‡</sup> | Comments                                                                                                                                                                                                           |
|-----------------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0                     | 2              | S: $(2J+1)\text{C}^2\text{S}=2.88$ in 1969Gr23 for $J^\pi=3/2^+$ ; $\text{C}^2\text{S}=1.3$ in 1970Mo01 for $J^\pi=3/2^+$ ; $\text{C}^2\text{S}=1.07$ in 1994Ve04 for $J^\pi=3/2^+$ .                              |
| 1220 10               | 0              | E(level): other: 1220 12 (1970Mo01).<br>S: $(2J+1)\text{C}^2\text{S}=0.26$ in 1969Gr23 for $J^\pi=1/2^+$ ; $\text{C}^2\text{S}=0.28$ in 1970Mo01 for $J^\pi=1/2^+$ .                                               |
| 1758 10               |                |                                                                                                                                                                                                                    |
| 2645 10               |                |                                                                                                                                                                                                                    |
| 2686 10               | 2              | S: $(2J+1)\text{C}^2\text{S}=0.06$ in 1969Gr23 for $J^\pi=(3/2,5/2)^+$ .                                                                                                                                           |
| 3002 10               | 2              | E(level): weighted average of 2997 10 (1969Gr23) and 3010 12 (1970Mo01).<br>S: $(2J+1)\text{C}^2\text{S}=0.14$ in 1969Gr23 for $J^\pi=(5/2)^+$ ; $\text{C}^2\text{S}=0.04$ in 1970Mo01 for $J^\pi=5/2^+$ .         |
| 3158 10               | 3              | E(level): weighted average of 3156 10 (1969Gr23) and 3160 12 (1970Mo01).<br>S: $(2J+1)\text{C}^2\text{S}=2.88$ in 1969Gr23 for $J^\pi=7/2^-$ ; $\text{C}^2\text{S}=0.66$ in 1970Mo01 for $J^\pi=7/2^-$ .           |
| 3908 10               |                |                                                                                                                                                                                                                    |
| 3962 10               | 0              | E(level): weighted average of 3956 10 (1969Gr23) and 3970 12 (1970Mo01).<br>S: $(2J+1)\text{C}^2\text{S}=0.03$ in 1969Gr23 for $J^\pi=1/2^+$ ; $\text{C}^2\text{S}=0.06$ in 1970Mo01 for $J^\pi=1/2^+$ .           |
| 4053 10               | 1              | E(level): weighted average of 4048 10 (1969Gr23) and 4060 12 (1970Mo01).<br>S: $(2J+1)\text{C}^2\text{S}=0.38$ in 1969Gr23 for $J^\pi=(3/2)^-$ ; $\text{C}^2\text{S}=0.15$ in 1970Mo01 for $J^\pi=(3/2)^-$ .       |
| 4167 10               | 1              | E(level): weighted average of 4165 10 (1969Gr23) and 4170 12 (1970Mo01).<br>S: $(2J+1)\text{C}^2\text{S}=0.85$ in 1969Gr23 for $J^\pi=(3/2)^-$ ; $\text{C}^2\text{S}=0.28, 0.62$ in 1970Mo01 for $J^\pi=(3/2)^-$ . |
| 5010 15               |                |                                                                                                                                                                                                                    |
| 5163 15               | (2)            | S: $(2J+1)\text{C}^2\text{S}=0.11$ without specifying J (1969Gr23).                                                                                                                                                |
| 5409 12               | 1              | E(level): weighted average of 5407 15 (1969Gr23) and 5410 12 (1970Mo01).<br>L: 1 in 1969Gr23 and L=(1,2) in 1970Mo01.                                                                                              |

Continued on next page (footnotes at end of table)

$^{34}\text{S}(^3\text{He},\text{d})$  **2017Gi09,1969Gr23 (continued)** $^{35}\text{Cl}$  Levels (continued)

| E(level) <sup>†</sup> | L <sup>‡</sup> | Comments                                                                                                                                                                                                   |
|-----------------------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5660 15               | 2              | S: (2J+1)C <sup>2</sup> S=0.07 in 1969Gr23 for $J^\pi=(3/2)^-$ .<br>E(level): 5670 multiplet reported in 1970Mo01.<br>L: 2 in 1969Gr23, possibly a component of the 5670 multiplet with L=(2) in 1970Mo01. |
| 5684 12               | 1              | S: (2J+1)C <sup>2</sup> S=0.47 in 1969Gr23 for $J^\pi=3/2^+$ .<br>E(level): 5670 multiplet reported in 1970Mo01.<br>L: 1 in 1969Gr23, possibly a component of the 5670 multiplet with L=(2) in 1970Mo01.   |
| 5760 12               | (0,1)          | S: (2J+1)C <sup>2</sup> S=0.11 in 1969Gr23 for $J^\pi=(3/2)^-$ .<br>E(level): weighted average of 5760 15 (1969Gr23) and 5760 12 (1970Mo01).<br>L: 1 in 1969Gr23 and L=0 in 1970Mo01.                      |
| 6284 4                | (2)            | S: (2J+1)C <sup>2</sup> S=0.12 in 1969Gr23 for $J^\pi=(3/2)^-$ ; C <sup>2</sup> S=0.21 in 1970Mo01 for $J^\pi=1/2^+$ .                                                                                     |
| 6329 4                | (0,1)          | S: (2J+1)C <sup>2</sup> S=0.086 2.                                                                                                                                                                         |
| 6377 2                | (2,3)          | S: (2J+1)C <sup>2</sup> S=0.0030 2 for L=0, 0.0044 4 for L=1.                                                                                                                                              |
| 6427 2                | (3)            | S: (2J+1)C <sup>2</sup> S=0.0012 6 for L=2, 0.0024 8 for L=3.                                                                                                                                              |
| 6468 2                | (1)            | S: (2J+1)C <sup>2</sup> S=0.049 2.<br>$\omega\gamma(\text{min})=6.5\times10^{-24}$ eV 2, $\omega\gamma(\text{max})=2.2\times10^{-22}$ eV 1.                                                                |
| 6491 2                | (2)            | S: (2J+1)C <sup>2</sup> S=0.034 1.<br>$\omega\gamma(\text{min})=4.2\times10^{-14}$ eV 5, $\omega\gamma(\text{max})=4.7\times10^{-14}$ eV 5.                                                                |
| 6545 2                | (0,1)          | S: (2J+1)C <sup>2</sup> S=0.072 2.<br>$\omega\gamma(\text{min})=3.7\times10^{-13}$ eV 1, $\omega\gamma(\text{max})=4.1\times10^{-13}$ eV 1.                                                                |
| 6643 2                | (1)            | S: (2J+1)C <sup>2</sup> S=0.004 1 for L=0, 0.0028 4 for L=1.<br>$\omega\gamma(\text{min})=7.4\times10^{-10}$ eV 13, $\omega\gamma(\text{max})=4.1\times10^{-9}$ eV 1.                                      |
| 6674 2                | (1,2,3)        | S: (2J+1)C <sup>2</sup> S=0.0144 8.<br>$\omega\gamma(\text{min})=8.5\times10^{-6}$ eV 10, $\omega\gamma(\text{max})=8.9\times10^{-6}$ eV 10.                                                               |
| 6761 2                | (0,1)          | S: (2J+1)C <sup>2</sup> S=0.0020 4 for L=1, 0.0048 6 for L=2, 0.008 1 for L=3.<br>$\omega\gamma(\text{min})=2.4\times10^{-8}$ eV 2, $\omega\gamma(\text{max})=4.1\times10^{-6}$ eV 8.                      |
| 6778 2                | (1)            | S: (2J+1)C <sup>2</sup> S=0.0056 12 for L=0, 0.0032 4 for L=1.<br>$\omega\gamma(\text{min})=2.4\times10^{-4}$ eV 3, $\omega\gamma(\text{max})=1.5\times10^{-3}$ eV 1.                                      |
| 6823 2                | (1)            | S: (2J+1)C <sup>2</sup> S=0.0084 8.<br>$\omega\gamma(\text{min})=9.4\times10^{-4}$ eV 12, $\omega\gamma(\text{max})=1.0\times10^{-3}$ eV 1.                                                                |
| 6842 2                | (2,3)          | S: (2J+1)C <sup>2</sup> S=0.006 4.                                                                                                                                                                         |
| 6866 2                | (0,2)          | S: (2J+1)C <sup>2</sup> S=0.00216 6 for L=2, 0.035 2 for L=3.<br>$\omega\gamma(\text{min})=2.7\times10^{-3}$ eV 3, $\omega\gamma(\text{max})=2.7\times10^{-3}$ eV 3.                                       |
| 7066 2                | (1,2)          | S: (2J+1)C <sup>2</sup> S=0.0216 6 for L=2, 0.035 2 for L=3.<br>$\omega\gamma(\text{min})=3.3\times10^{-5}$ eV 1, $\omega\gamma(\text{max})=7.0\times10^{-4}$ eV 3.                                        |
| 7103 2                | (1,3)          | S: (2J+1)C <sup>2</sup> S=0.0160 2(L=0)+0.0660 6 (L=2).<br>$\omega\gamma(\text{min})=3.0\times10^{-2}$ eV 3, $\omega\gamma(\text{max})=4.7\times10^{-2}$ eV 5.                                             |
| 7178 2                | (2)            | S: (2J+1)C <sup>2</sup> S=0.0088 8 for L=1, 0.012 1 for L=2.<br>$\omega\gamma(\text{min})=2.2\times10^{-2}$ eV 1, $\omega\gamma(\text{max})=2.2\times10^{-2}$ eV 1.                                        |
| 7194 2                | (2)            | S: (2J+1)C <sup>2</sup> S=0.0184 12 for L=1, 0.025 2 for L=3.<br>$\omega\gamma(\text{min})=6.0\times10^{-2}$ eV 3, $\omega\gamma(\text{max})=1.30$ eV 7.                                                   |
| 7227 2                | (0,1)          | S: (2J+1)C <sup>2</sup> S=0.032 3.                                                                                                                                                                         |
| 7273 2                | (0,1)          | S: (2J+1)C <sup>2</sup> S=0.065 15 for L=0, 0.044 2 for L=1.                                                                                                                                               |
| 7361 2                | (1)            | S: (2J+1)C <sup>2</sup> S=0.050 12 for L=0, 0.030 1 for L=1.<br>$\omega\gamma(\text{min})=11$ eV 2, $\omega\gamma(\text{max})=34$ eV 3.                                                                    |
| 7398 2                | (2,3)          | S: (2J+1)C <sup>2</sup> S=0.0040 8.<br>$\omega\gamma(\text{min})=0.30$ eV 4, $\omega\gamma(\text{max})=3.0$ eV 6.                                                                                          |
|                       |                | S: (2J+1)C <sup>2</sup> S=0.042 2 for L=2, 0.0656 24 for L=3.<br>$\omega\gamma(\text{min})=0.18$ eV 1, $\omega\gamma(\text{max})=0.19$ eV 1.                                                               |

<sup>†</sup> From 1969Gr23 for E(level)<6 MeV and from 2017Gi09 for E(level)>6 MeV, unless otherwise noted.



---

$^{34}\text{S}(^3\text{He,d})$     [2017Gi09,1969Gr23](#) (continued)

---

$^{35}\text{Cl}$  Levels (continued)

<sup>‡</sup> From DWBA analysis of the measured  $\sigma(\theta)$  in [1969Gr23](#) for  $E(\text{level}) < 6$  MeV and in [2017Gi09](#) for  $E(\text{level}) > 6$  MeV, unless otherwise noted. Firm assignments by [2017Gi09](#) are placed in parentheses by the evaluators because they are based on very few data points in a very narrow range of angles in the DWBA fit which may not yield definitive and firm results.

$^{35}\text{Cl}(\gamma, \gamma')$     **1966Ho02, 1962Bo17, 1991Li12**

**1966Ho02**: a bremsstrahlung photon beam was produced by impinging 1.4-MeV electrons on a thin layer of lead embedded in a water-cooled aluminium disc at the Diamond Research Laboratory in Johannesburg. The target was  $\text{C}_2\text{Cl}_6$ . Scattered  $\gamma$  rays were detected using a 7.6-cm by 7.6-cm NaI crystal. Deduced the width for the 1.22-MeV first excited state using the self-absorption method of nuclear resonance fluorescence.

**1962Bo17**: an electron beam was produced from the MIT Van de Graaff generator, passed through a 76- $\mu\text{m}$  aluminum window and a 25- $\mu\text{m}$  platinum foil, and stopped in the cooling water of the electron target.  $\gamma$  rays were detected using a 5.1-cm by 5.1-cm NaI crystal. Lifetimes of  $^{35}\text{Cl}$  levels are deduced from the measured intensity of the resonance fluorescence radiation which occurs when a nucleus is bombarded by bremsstrahlung.

**1991Li12**: a bremsstrahlung photon beam was produced by focusing a 4.1-MeV electron beam onto a radiator target of a gold disc on a water-cooled copper backing at the bremsstrahlung facility at the Stuttgart Dynamitron Accelerator. The target was a stack of discs that contain  $^{13}\text{C}$ ,  $^{23}\text{Na}$ ,  $^{27}\text{Al}$ ,  $^{31}\text{P}$ ,  $^{35}\text{Cl}$ ,  $^{48}\text{Ti}$ .  $\gamma$  rays were detected using 3 Ge detectors. Deduced  $\sigma$  of photon scattering.

Other: **1972Sh07**.

 $^{35}\text{Cl}$  Levels

$\Gamma_0\Gamma_f/\Gamma$  from **1991Li12**.  $\Gamma_0$ ,  $\Gamma_f$ , and  $\Gamma$  are the decay width to the ground state, the decay width to the final state, and the total width, respectively.

| <u>E(level)<sup>†</sup></u> | <u>T<sub>1/2</sub></u> | <u>Comments</u>                                                                                                                                                                         |
|-----------------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0                           |                        |                                                                                                                                                                                         |
| 1220                        | 90 fs +21-14           | T <sub>1/2</sub> : from lifetime=0.13 ps +3-2 in <b>1966Ho02</b> with resonance fluorescence. Other: lifetime=0.30 ps 10 in <b>1962Bo17</b> with resonance fluorescence assuming J=1/2. |
| 1760                        | 0.42 ps 15             | T <sub>1/2</sub> : from lifetime=0.60 ps 21 in <b>1962Bo17</b> with resonance fluorescence assuming J=5/2.                                                                              |
| 2693.7                      |                        | $\Gamma_0\Gamma_f/\Gamma=12$ meV 3 ( <b>1991Li12</b> ).                                                                                                                                 |
| 3002.7                      |                        | $\Gamma_0\Gamma_f/\Gamma=26$ meV 11 ( <b>1991Li12</b> ).                                                                                                                                |
| 3918.6                      |                        | $\Gamma_0\Gamma_f/\Gamma=63$ meV 18 ( <b>1991Li12</b> ).                                                                                                                                |

<sup>†</sup> From  $E_\gamma$  data.

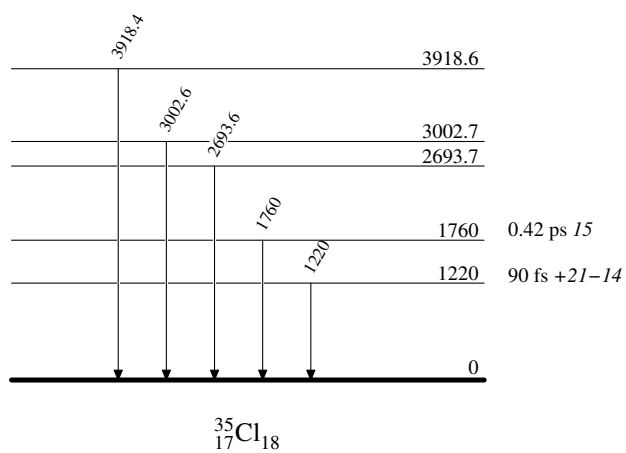
 $\gamma(^{35}\text{Cl})$ 

| <u>E<sub><math>\gamma</math></sub></u> | <u>E<sub>i</sub>(level)</u> | <u>E<sub>f</sub></u> | <u>Comments</u>                                                               |
|----------------------------------------|-----------------------------|----------------------|-------------------------------------------------------------------------------|
| 1220                                   | 1220                        | 0                    | E <sub><math>\gamma</math></sub> : from <b>1962Bo17</b> and <b>1966Ho02</b> . |
| 1760                                   | 1760                        | 0                    | E <sub><math>\gamma</math></sub> : from <b>1962Bo17</b> .                     |
| 2693.6 <sup>†</sup>                    | 2693.7                      | 0                    |                                                                               |
| 3002.6 <sup>†</sup>                    | 3002.7                      | 0                    |                                                                               |
| 3918.4 <sup>†</sup>                    | 3918.6                      | 0                    |                                                                               |

<sup>†</sup> From **1991Li12**, which quoted from **1978En02** evaluation.

$^{35}\text{Cl}(\gamma, \gamma')$  1966Ho02, 1962Bo17, 1991Li12

Level Scheme



$^{35}\text{Cl}(\text{n},\text{n}'\gamma)$  1989Ge09,1984El12,1966Ni03

1984El12,1989Ge09: fast neutrons of diameter 26 mm were produced from the IRT-2000 reactor at the Institute for Nuclear Research and Nuclear Power of the Bulgarian Academy of Sciences in Sofia. Two  $^{35}\text{Cl}$  targets were placed at  $45^\circ$  and  $135^\circ$  with a distance of 70 cm apart.  $\gamma$  rays were detected using a Ge(Li) detector. Measured  $E_\gamma$ . Deduced lifetimes for 4 levels using the Doppler Shift Attenuation Method (DSAM) and transition strengths.

1966Ni03: 2.5-4.2-MeV neutrons were produced via the T(p,n) reactions with protons from the university of Kentucky 6-MeV accelerator. Targets were a cylinder of  $\text{CCl}_4$  and a 5.6-cm-diam sphere of  $\text{C}_2\text{Cl}_6$ .  $\gamma$  rays were detected using a 6.35-cm by 6.35-cm NaI(Tl) detector and 7.5-cc Ge(Li) detector. Measured  $E_\gamma$ ,  $I_\gamma$ ,  $\gamma(\theta)$ . Deduced levels, J.

1971Fr05: neutrons up to 10 MeV were produced from a 5-Ci Pu-Be neutron source and bombarded table salt targets.  $\gamma$  rays were detected using a Ge(Li) detector. Measured 2  $\gamma$  rays and deduced 2 levels.

Others: 1955To32, 1977Ko19, 1983Gl07, 1992Be60 (cross sections).

 $^{35}\text{Cl}$  Levels

| <u><math>E(\text{level})^\dagger</math></u> | <u><math>J^\pi^\#</math></u> | <u><math>T_{1/2}^\ddagger</math></u> |
|---------------------------------------------|------------------------------|--------------------------------------|
| 0                                           | $3/2^+$                      |                                      |
| 1219                                        | $1/2$                        | 139 fs 56                            |
| 1762                                        | $5/2$                        |                                      |
| 2646                                        | $7/2$                        | 187 fs 62                            |
| 2694                                        |                              | 43 fs 6                              |
| 3002                                        |                              | 50 fs 9                              |
| 3163                                        |                              |                                      |

$^\dagger$  From a least-squares fit to  $\gamma$ -ray energies.

$^\ddagger$  From 1984El12 and 1989Ge09 using DSAM.

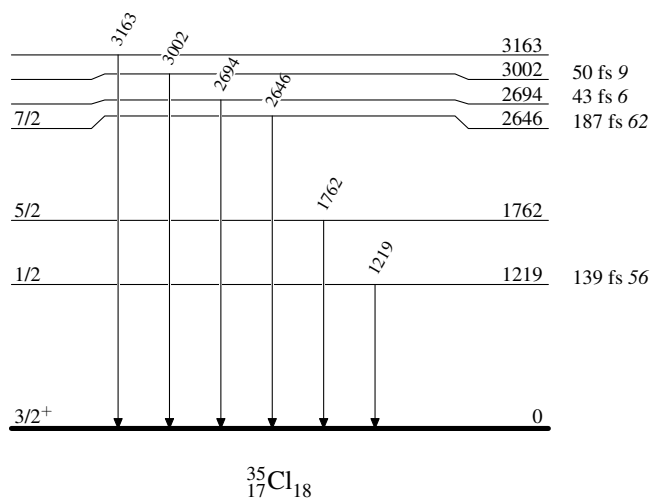
$^\#$  From  $\gamma(\theta)$  in 1966Ni03, except for the ground state from the Adopted Levels.

 $\gamma(^{35}\text{Cl})$ 

| <u><math>E_\gamma</math></u> | <u><math>E_i(\text{level})</math></u> | <u><math>J_i^\pi</math></u> | <u><math>E_f</math></u> | <u><math>J_f^\pi</math></u> | <u>Comments</u>                                                            |
|------------------------------|---------------------------------------|-----------------------------|-------------------------|-----------------------------|----------------------------------------------------------------------------|
| 1219                         | 1219                                  | $1/2$                       | 0                       | $3/2^+$                     | $E_\gamma$ : from 1966Ni03.<br>$A_0=+0.942$ 8, $A_2=-0.148$ 23 (1966Ni03). |
| 1762                         | 1762                                  | $5/2$                       | 0                       | $3/2^+$                     |                                                                            |
| 2646                         | 2646                                  | $7/2$                       | 0                       | $3/2^+$                     | $A_0=+1.078$ 5, $A_2=+0.165$ 12 (1966Ni03).                                |
| 2694                         | 2694                                  |                             | 0                       | $3/2^+$                     |                                                                            |
| 3002                         | 3002                                  |                             | 0                       | $3/2^+$                     | $E_\gamma$ : from 1966Ni03.                                                |
| 3163                         | 3163                                  |                             | 0                       | $3/2^+$                     |                                                                            |

$^{35}\text{Cl}(n,n'\gamma)$  1989Ge09,1984El12,1966Ni03

Level Scheme



$^{35}\text{Cl}(\text{p},\text{p}'\gamma)$  1969Du08,1970Ta02,1972Br45

Also include (p,p') measurements listed below.

$^{35}\text{Cl}(\text{p},\text{p}'\gamma)$  measurements:

**1969Du08:** E=4.6-6.2 MeV 0.05  $\mu\text{A}$  proton beam. Thin targets of  $\text{CdCl}_2$  (99%  $^{35}\text{Cl}$ ), 100  $\mu\text{g}/\text{cm}^2$ . A 300  $\mu\text{m}$  annular surface-barrier detector on the beam axis and a movable 390  $\mu\text{m}$  surface-barrier detector rotated from  $20^\circ$  to  $148^\circ$  relative to beam direction for detecting protons, FWHM=24 keV; a 7.62 by 7.62-cm-diam NaI(Tl) crystal and 15- and 35-cc Ge(Li) for detecting  $\gamma$ -rays. Measured  $\sigma(E_p, \theta)$ ,  $E_\gamma$ ,  $I_\gamma$ ,  $\text{py}(\theta)$ . Deduced levels,  $J^\pi$ , branchings, mixing ratios, half-lives using Doppler Shift Attenuation Method (DSAM).

**1970Ta02:** E=4.24, 4.82, 5.08 and 5.11 MeV proton beams produced from the Universite Montreal EN tandem accelerator. Targets of 16  $\mu\text{g}/\text{cm}^2$  AgCl evaporated onto 0.013 cm gold foils, 99% enriched in  $^{35}\text{Cl}$ . A 12.70-cm-diam by 15.24-cm-long NaI(Tl) and a 30  $\text{cm}^3$  Ge(Li) detector for detecting  $\gamma$ -rays. Measured  $E_\gamma$ ,  $I_\gamma$ ,  $\gamma(\theta)$ ,  $\gamma(\text{lin pol})$ . Deduced levels,  $J^\pi$ , branchings, mixing ratios.

**1972Br45:** E=3.45-6.00 MeV protons produced from the University of Auckland folded tandem accelerator. Natural chlorine targets of AgCl,  $\text{PbCl}_2$  and solid Cl (75.77%  $^{35}\text{Cl}$ , 24.33%  $^{37}\text{Cl}$ ). A 25  $\text{cm}^3$  Ge(Li) detector for detecting  $\gamma$ -rays, FWHM=2.3 keV for  $E_\gamma=1.332$  MeV. Measured  $E_\gamma$ . Deduced levels, half-lives using DSAM. **1972Br45** reports a few mixing ratios in their TABLE V, but it is unclear where they are taken from. They are definitely not from **1972Br45** since this work is only focused on lifetime measurements.

**1972Va06:** E=5.47-5.61 MeV protons from the 5.5 MV Van de Graaff accelerator of the Southern Universities Nuclear Institute. Targets of  $\text{BaCl}_2$  (99%  $^{35}\text{Cl}$ ) on gold backings. A 57  $\text{cm}^3$  Ge(Li) detector. Measured  $E_\gamma$ . Deduced half-lives using DSAM.

**1961St09:** E=2.3-3.25 MeV. NaI(Tl) detector. Targets of AgCl. Measured  $\sigma(E_p, \theta)$ ,  $\gamma(\theta)$ . Deduced levels, J for the levels of 1.22 and 1.76 MeV.

**1966Er06:** E=0.733-2.595 MeV protons produced from the Utrecht 3 MV Van de Graaff accelerator.  $\text{BaCl}_2$  target (99.5%  $^{35}\text{Cl}$ ) on a tantalum backing. Cylindrical NaI assemblies, 10-cm by 10-cm. Measured  $E_\gamma$ ,  $\gamma(\theta)$ . Deduced levels,  $J^\pi$ , mixing ratios, mainly for  $^{36}\text{Ar}$  and for the  $^{35}\text{Cl}$  level at  $E_x=1760$ .

**1970Ta09:** E=5.11 MeV proton beam. NaI(Tl) detector. Measured  $E_\gamma$ . Developed an analytical method to interpret the linear polarization measurement of two unresolved  $\gamma$ -rays: 3.00 and 3.16 MeV. Same authors as **1970Ta02**.

**1971Ca40:** E=4.15-6.15 MeV protons from the Oxford University Tandem accelerator. Targets of 20  $\mu\text{g}/\text{cm}^2$  natural  $\text{BaCl}_2$  on gold, lead, and carbon backings. A 32  $\text{cm}^3$  Ge(Li) detector. Measured  $E_\gamma$ . Deduced half-lives for the levels of 2646 and 2695 keV using DSAM.

**1974Hu09:** E=5.65 MeV protons from the pulsed beam facility of the University of Alberta Van de Graaff accelerator. Targets of natural KCl on tantalum backing and  $\text{PbCl}_2$  on gold backing. A 24  $\text{cm}^3$  coaxial Ge(Li) detector. Measured  $E_\gamma$ . Deduced half-life for the level of 3162 keV.

Other (p,p' $\gamma$ ) measurement: **1985Ki07** and **1975De16** (measured 1219 $\gamma$  yield).

(p,p') measurements:

**1969Ta13,1971Ta02:** E=3.115 and 3.345 MeV protons produced from the Universite de Montreal Model EN tandem accelerator. AgCl target (99%  $^{35}\text{Cl}$ ) of 16  $\mu\text{g}/\text{cm}^2$  on gold backing. A 12.70-cm-diam by 15.24-cm-long NaI(Tl) for detecting  $\gamma$ -rays. Measured  $E_\gamma$ ,  $I_\gamma$ ,  $\gamma(\theta)$ . Deduced levels,  $J^\pi$  and mixing ratio for the level of 1763 keV. Same authors as **1970Ta02**.

**1969An35:** E=6.03 MeV proton beams on a  $\text{CdCl}_2$  target. Inelastic protons detected by a Si(Li) detector. Measured  $\sigma(E_p, \theta)$ . Deduced J for a level at 3930.

Other (p,p') measurement: **1993Bo40** (measured  $\sigma$ ).

 $^{35}\text{Cl}$  Levels

| E(level) <sup>†</sup> | $J^\pi$ <sup>‡</sup> | $T_{1/2}$       | Comments                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------|----------------------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0                     | $3/2^+$              |                 |                                                                                                                                                                                                                                                                                                                                                                                  |
| 1219.42 10            | $1/2^+$              | 0.150 ps +42–28 | $J^\pi$ : $\text{py}(\theta)$ in <b>1969Du08</b> is consistent with spin=1/2.<br>$T_{1/2}$ : from $\tau=0.216$ ps +60–40, weighted average of 0.215 ps 75 ( <b>1972Va06</b> ), 0.21 ps +6–4 ( <b>1972Br45</b> ), 0.27 ps +19–10 ( <b>1969Du08</b> ).                                                                                                                             |
| 1763.26 10            | $5/2^+$              | 0.43 ps 20      | $J^\pi$ : $5/2^+$ from $\gamma(\theta)$ and $\gamma(\text{lin pol})$ in <b>1966Er06</b> ; spin=5/2 also from $\text{py}(\theta)$ in <b>1969Du08</b> .<br>$T_{1/2}$ : from $\tau=0.62$ ps 28, unweighted average of 0.205 ps 75 ( <b>1972Va06</b> ), 0.49 ps +9–6 ( <b>1972Br45</b> ), 1.15 ps +45–63 ( <b>1969Du08</b> ).                                                        |
| 2645.74 17            | $7/2^+$              | 0.165 ps 35     | $J^\pi$ : $5/2^+$ from $\gamma(\theta)$ and $\gamma(\text{lin pol})$ in <b>1970Ta02</b> ; spin $\geq 5/2$ from $\text{py}(\theta)$ in <b>1969Du08</b> .<br>$T_{1/2}$ : from $\tau=0.238$ ps 50, weighted average of 0.185 ps 50 ( <b>1972Va06</b> ) and 0.26 ps 6 ( <b>1972Br45</b> ), and 0.35 ps +10–7 ( <b>1969Du08</b> , using 2646 $\gamma$ ). Other: $\tau=0.29$ ps +37–13 |

Continued on next page (footnotes at end of table)

$^{35}\text{Cl}(\text{p},\text{p}'\gamma)$  **1969Du08,1970Ta02,1972Br45 (continued)** $^{35}\text{Cl}$  Levels (continued)

| $E(\text{level})^\dagger$ | $J^\pi^\ddagger$ | $T_{1/2}$ | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------|------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2694.50 28                | $3/2^+$          | 43 fs 11  | using 882 $\gamma$ also reported in 1969Du08, but is considered less reliable by authors; 0.083 ps 21 from 1971Ca40 is considered an outlier.<br>$J^\pi$ : $3/2^+$ from $\gamma(\theta)$ and $\gamma(\text{lin pol})$ in 1970Ta02.<br>$T_{1/2}$ : from $\tau=63$ fs 16, weighted average of 65 fs 20 (1972Va06) and 62 fs 16 (1969Du08, using 2695 $\gamma$ ). Other: $\tau=115$ fs +95–59 using 931 $\gamma$ also reported in 1969Du08, but is considered less reliable by authors; <120 fs (1972Br45); 11 fs 7 from 1971Ca40 is considered an outlier. |
| 3002.6 5                  | $5/2^+$          | 19 fs 9   | $J^\pi$ : $5/2^+$ from $\gamma(\theta)$ and $\gamma(\text{lin pol})$ in 1970Ta02.<br>$T_{1/2}$ : from $\tau=27$ fs 13, weighted average of 18 fs +26–15 (1972Va06) and 31 fs 13 (1969Du08). Other: <80 fs (1972Br45).                                                                                                                                                                                                                                                                                                                                    |
| 3162.75 10                | $7/2^-$          | 32 ps 4   | $J^\pi$ : spin= $7/2$ from $\gamma(\theta)$ and $\pi=-$ from $\gamma(\text{lin pol})$ in 1970Ta02.<br>$T_{1/2}$ : from $\tau=46$ ps 6 in 1974Hu09. Other: $\tau>17$ ps (1972Va06), >10 ps (1972Br45).                                                                                                                                                                                                                                                                                                                                                    |
| 3930 30                   | $3/2$            |           | $E(\text{level}), J^\pi$ : from 1969An35. Spin is determined based on J-dependence of measured proton inelastic scattering cross sections and known assignments of 3163 and 3000 levels.                                                                                                                                                                                                                                                                                                                                                                 |

$^\dagger$  From a least-squares fit to  $\gamma$ -ray energies with uncertainties, unless otherwise noted.

$^\ddagger$  From Adopted Levels. Supporting arguments from this dataset are given under comments if available.

 $\gamma(^{35}\text{Cl})$ 

| $E_i(\text{level})$ | $J_i^\pi$ | $E_\gamma^\dagger$     | $I_\gamma^\ddagger$ | $E_f$   | $J_f^\pi$ | Mult. <sup>#</sup> | $\delta^\#$ | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------|-----------|------------------------|---------------------|---------|-----------|--------------------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1219.42             | $1/2^+$   | 1219.4 1               | 100                 | 0       | $3/2^+$   |                    |             | $E_\gamma$ : weighted average of 1219.4 1 (1972Br45) and 1219.3 3 (1969Du08).<br>$\Gamma_\gamma=27.43\times 10^{-4}$ eV (1969Du08).<br>$A_2=-0.025$ 29 for $E_p=3.115$ MeV; +0.007 8 for $E_p=3.345$ MeV (1969Ta13).<br>$A_2=+0.031$ 16, $A_4=+0.014$ 20; $A_2=-0.012$ 22, $A_4=-0.003$ 30; $A_2=-0.009$ 14, $A_4=+0.014$ 6, for different beam energies (1961St09).<br>$A_2=+0.02$ 6, $A_4=+0.05$ 9 (1969Du08).<br>$E_\gamma$ : weighted average of 1763.2 1 (1972Br45), 1763.3 3 (1969Du08).<br>$\delta$ : weighted average of $-3.0$ 5 from $\gamma(\theta, \text{pol})$ in 1966Er06 and $-2.64$ 12 from $\gamma(\theta, \text{pol})$ in 1969Ta13 and 1971Ta02. Others: $\delta<-0.4$ or $\delta>+0.4$ from $\text{py}(\theta)$ in 1969Du08.<br>$A_2=-0.369$ 9, $A_4=+0.213$ 13; $A_2=-0.355$ 14, $A_4=+0.200$ 20, for different beam energies (1969Ta13).<br>$A_2=-0.02$ 2, $A_4=+0.07$ 2 (1969Du08).<br>$A_2=-0.141$ 13, $A_4=+0.107$ 16; $A_2=-0.225$ 29, $A_4=+0.083$ 35; $A_2=-0.049$ 12, $A_4=-0.036$ 14, for different beam energies (1961St09). |
| 1763.26             | $5/2^+$   | 1763.2 1               | 100                 | 0       | $3/2^+$   | M1+E2              | -2.66 12    | $I_\gamma$ : unweighted average of 8.7 22 from 1970Ta02 and 17.6 35 from 1969Du08.<br>%Branching=8 2 (1970Ta02), 15 3 (1969Du08).<br>$A_2=-0.372$ 34, $A_4=+0.034$ 38; $A_2=-0.533$ 31, $A_4=+0.041$ 34, for different beam energies (1970Ta02).<br>%Branching<1 (1970Ta02).<br>$E_\gamma$ : weighted average of 2645.7 2 (1972Br45) and 2645.6 5 (1969Du08).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 2645.74             | $7/2^+$   | 882.3 <sup>@</sup> 3   | 13 5                | 1763.26 | $5/2^+$   | M1+E2              | +0.17 5     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                     |           | 1426 <sup>&amp;a</sup> | <1 <sup>&amp;</sup> | 1219.42 | $1/2^+$   |                    |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                     |           | 2645.7 2               | 100 2               | 0       | $3/2^+$   | E2                 |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

Continued on next page (footnotes at end of table)

$^{35}\text{Cl}(\text{p},\text{p}'\gamma)$  **1969Du08,1970Ta02,1972Br45 (continued)** $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$        | $E_\gamma^\dagger$ | $I_\gamma^\ddagger$ | $E_f$   | $J_f^\pi$        | Mult. # | $\delta^\#$ | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------|------------------|--------------------|---------------------|---------|------------------|---------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     |                  |                    |                     |         |                  |         |             | $\delta$ : $\delta(\text{O}/\text{Q})=0$ from $\gamma(\theta, \text{pol})$ in 1970Ta02 and $\text{p}\gamma(\theta)$ in 1969Du08.<br>$I_\gamma$ : from 1970Ta02. Other: 100 4 from 1969Du08.<br>%Branching=92 2 (1970Ta02), 85 3 (1969Du08).<br>$A_2=+0.22$ 5, $A_4=-0.13$ 6 (1969Du08).<br>$A_2=+0.316$ 22, $A_4=+0.024$ 28; $A_2=+0.233$ 11, $A_4=+0.044$ 10; $A_2=+0.312$ 32, $A_4=-0.117$ 41; $A_2=+0.205$ 18, $A_4=-0.044$ 25, for different beam energies (1970Ta02).                                                                                                         |
| 2694.50             | 3/2 <sup>+</sup> | 931.3@ 3           | 18.6 25             | 1763.26 | 5/2 <sup>+</sup> | M1+E2   | +0.09 3     | $I_\gamma$ : weighted average of 19.0 25 (1970Ta02) and 17.7 38 (1969Du08).<br>%Branching=15 2 (1970Ta02), 14 3 (1969Du08).<br>$A_2=+0.018$ 46 (1970Ta02).                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                     |                  | 1475@ 2            | 8.3 25              | 1219.42 | 1/2 <sup>+</sup> | M1+E2   | -0.63 21    | $I_\gamma$ : weighted average of 8.0 25 (1970Ta02) and 8.9 38 (1969Du08).<br>%Branching=6 2 (1970Ta02), 7 3 (1969Du08).<br>$A_2=-0.111$ 45 (1970Ta02).                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                     |                  | 2694.1 6           | 100 3               | 0       | 3/2 <sup>+</sup> | M1+E2   | +0.26 2     | $E_\gamma$ : unweighted average of 2693.5 1 (1972Br45), 2694.6 5 (1969Du08).<br>$I_\gamma$ : from 1970Ta02. Other: 100 4 from 1969Du08.<br>%Branching=79 2 (1970Ta02), 79 3 (1969Du08).                                                                                                                                                                                                                                                                                                                                                                                            |
| 3002.6              | 5/2 <sup>+</sup> | 1239& 1            | 2.2& 11             | 1763.26 | 5/2 <sup>+</sup> |         |             | $I_\gamma$ : from 1970Ta02.<br>%Branching=2 1 (1970Ta02).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                     |                  | 1783               | 3.3 11              | 1219.42 | 1/2 <sup>+</sup> |         |             | $E_\gamma$ : from level-energy difference.<br>$I_\gamma$ : weighted average of 5.4 32 (1970Ta02) and 3.1 11 (1969Du08).<br>%Branching=5 3 (1970Ta02), 3 1 (1969Du08).                                                                                                                                                                                                                                                                                                                                                                                                              |
|                     |                  | 3002.5 5           | 100 1               | 0       | 3/2 <sup>+</sup> | M1+E2   | +0.07 2     | $E_\gamma$ : weighted average of 3002.5 5 (1972Br45) and 3002.4 10 (1969Du08).<br>$I_\gamma$ : from 1969Du08. Other: 100 2 from 1970Ta02.<br>$\delta$ : others: $-20 < \delta < -0.3$ or $+0.2 < \delta < +5$ (1969Du08, $\text{p}\gamma(\theta)$ ); $-0.02 < \delta < 0.09$ (1970Ta09, $\gamma(\text{lin pol})$ ).<br>%Branching=93 2 (1970Ta02), 97 1 (1969Du08).<br>$A_2=-0.30$ 4 (1969Du08).<br>$A_2=-0.166$ 7, $A_4=+0.004$ 10; $A_2=-0.104$ 23, $A_4=+0.050$ 40; $A_2=-0.329$ 13, $A_4=-0.001$ 28; $A_2=+0.342$ 24, $A_4=+0.022$ 43, for different beam energies (1970Ta02). |
| 3162.75             | 7/2 <sup>-</sup> | 517.0@ 8           | 19 8                | 2645.74 | 7/2 <sup>+</sup> |         |             | $I_\gamma$ : unweighted average of 11.1 22 (1970Ta02) and 26.6 63 (1969Du08).<br>%Branching=10 2 (1970Ta02), 21 5 (1969Du08).                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                     |                  | 1399&a             | <1&                 | 1763.26 | 5/2 <sup>+</sup> |         |             | %Branching<1 (1970Ta02).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                     |                  | 1943&a             | <1&                 | 1219.42 | 1/2 <sup>+</sup> |         |             | %Branching<1 (1970Ta02).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

Continued on next page (footnotes at end of table)



$^{35}\text{Cl}(\text{p},\text{p}'\gamma)$  **1969Du08,1970Ta02,1972Br45** (continued) $\gamma(^{35}\text{Cl})$  (continued)

| $E_i(\text{level})$ | $E_\gamma^\dagger$ | $I_\gamma^\ddagger$ | $E_f$ | $J_f^\pi$        | Mult. <sup>#</sup> | $\delta^\&$ | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------|--------------------|---------------------|-------|------------------|--------------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3162.75             | 3162.6 1           | 100 2               | 0     | 3/2 <sup>+</sup> | M2+E3              | -0.22 5     | <p><math>E_\gamma</math>: weighed average of 3162.6 1 (1972Br45) and 3162.5 10 (1969Du08).</p> <p><math>I_\gamma</math>: from 1970Ta02. Other: 100 6 from 1969Du08.</p> <p><math>\delta</math>: weighted average of -0.14 +8-6 (1969Du08, <math>\text{p}\gamma(\theta)</math>) and -0.25 4 (1970Ta02, <math>\gamma(\theta,\text{pol})</math>). Other: -0.31&lt;<math>\delta</math>&lt;-0.21 from <math>\gamma(\text{lin pol})</math> in 1970Ta09.</p> <p>%Branching=90 2 (1970Ta02), 79 5 (1969Du08).</p> <p><math>A_2=+0.48</math> 21, <math>A_4=-0.13</math> 73 (1969Du08).</p> <p><math>A_2=+0.281</math> 68, <math>A_4=+0.241</math> 82; <math>A_2=+0.453</math> 44, <math>A_4=+0.136</math> 63; <math>A_2=+0.475</math> 24, <math>A_4=+0.096</math> 36, for different beam energies (1970Ta02).</p> |

<sup>†</sup>  $E_\gamma$  values with uncertainties for the ground-state transitions are not explicitly listed in 1972Br45 and 1969Du08 and the evaluators have taken their listed  $E(\text{level})$  values and uncertainties (which are deduced from  $E_\gamma$  values by authors) as  $E_\gamma$  values of the corresponding ground-state transitions, considering ground-state transitions are dominant in each level.

<sup>‡</sup> Relative photon branching ratios deduced by the evaluators based on percentage branching ratios reported in 1970Ta02 and 1969Du08, which are given under comments where available.

<sup>#</sup> From  $\gamma(\theta)$  and  $\gamma(\text{lin pol})$  in 1970Ta02, unless otherwise noted.

<sup>@</sup> From 1969Du08.

<sup>&</sup>  $\gamma$  reported in 1970Ta02. Energy is from level-energy difference and intensity is from 1970Ta02. The ones with  $I_\gamma$  reported as upper limit are considered questionable by the evaluators.

<sup>a</sup> Placement of transition in the level scheme is uncertain.

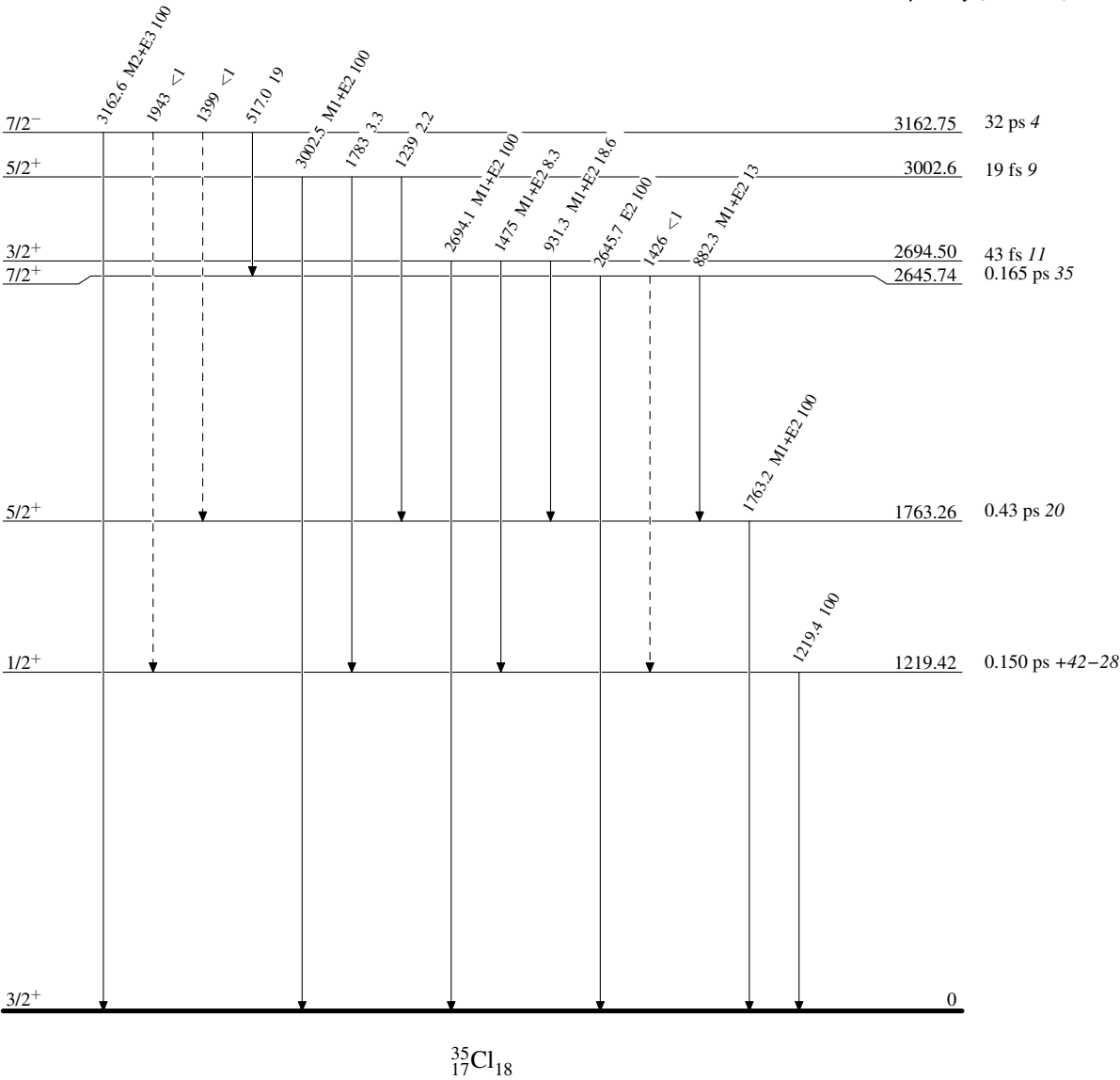
<sup>35</sup>Cl(p,p'γ) 1969Du08,1970Ta02,1972Br45

Legend

Level Scheme

Intensities: Relative photon branching from each level

-----► γ Decay (Uncertain)



$^{37}\text{Cl}(\text{p,t})$  1971Vi02,2017Ch22

$^{37}\text{Cl} \rightarrow ^{35}\text{Cl} + 2\text{n}$  from  $J^\pi = 3/2^+$   $^{37}\text{Cl}$  ground state.

1971Vi02: A 40-MeV proton beam was produced from the Grenoble variable energy cyclotron at intensities of 20-200 nA depending on the scattering angle with a 90-keV energy resolution. Targets were natural purified chlorine gas 100 mm in diameter and 25 mm in height and solid AgCl of isotopic  $^{37}\text{Cl}$ . The reaction particles were detected using two separate counter telescopes consisting of a 200- $\mu\text{m}$  phosphorous-drifted silicon  $\Delta\text{E}$  detector, a 3-mm lithium-drifted silicon E detector, and a 3-mm lithium-drifted silicon E-reject detector with FWHM=130 keV for tritons. Measured  $\sigma(E_t, \theta)$ . Deduced L-transfers for 7 levels from zero-range Nelson spin-dependent-DWBA analysis of the measured  $\sigma(\theta)$ .

2017Ch22: a 30-MeV proton beam was produced from the Holifield Radioactive Ion Beam Facility at Oak Ridge National Laboratory. The target was  $\sim 600 \mu\text{g}$   $\text{PbCl}_2$  ( $\sim 95\%$  enriched and  $\sim 160 \mu\text{g}/\text{cm}^2$  in  $^{37}\text{Cl}$ ) on a  $20 \mu\text{g}/\text{cm}^2$  parylene ( $\text{C}_8\text{H}_8$ ) backing. Tritons were detected using the annular silicon detector array (SIDAR) consisting of 100- $\mu\text{m}$ - $\Delta\text{E}$  and 1000- $\mu\text{m}$ -E telescopes with FWHM=60-100 keV. Measured  $\sigma(E_t, \theta)$ . Deduced levels, J,  $\pi$ , and L-transfers from TWOFR-DWBA analysis of the measured  $\sigma(\theta)$ . Provided the first spin-parity constraint of a 306-keV  $^{34}\text{S}(\text{p}, \gamma)$  resonance potentially important for nova nucleosynthesis.

Other: 1985Mi06.

 $^{35}\text{Cl}$  Levels

| $E(\text{level})^\dagger$ | $L^\ddagger$ | Comments                                                                                                                                                                                                                        |
|---------------------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0                         | 0            | L: from 2017Ch22. Other: 0+2 in 1971Vi02.<br>Relative peak intensity=100 (1971Vi02).                                                                                                                                            |
| 1220 40                   |              | Relative peak intensity=0 (1971Vi02).                                                                                                                                                                                           |
| 1750 40                   | 2            | Relative peak intensity=41 (1971Vi02).                                                                                                                                                                                          |
| 2650 40                   | 2            | Relative peak intensity=41 (1971Vi02). Unresolved 2645 and 2694 doublet.                                                                                                                                                        |
| 3000 40                   |              | Relative peak intensity=5 (1971Vi02).                                                                                                                                                                                           |
| 5650 50                   | 0+2          | Relative peak intensity=22 (1971Vi02).                                                                                                                                                                                          |
| 6677 15                   | (2)          | E(level): from 2017Ch22. 2017Ch22 adopts L=2 and states that combinations of L=0+2 or L=1+3 are also possible. Because L=2 and L=0 are better fits than L=1 and L=3, a positive parity assignment for the 6677 level is strong. |
| 7250 50                   |              | Relative peak intensity=9 (1971Vi02).                                                                                                                                                                                           |

$^\dagger$  From 1971Vi02, unless otherwise noted.

$^\ddagger$  From DWBA analysis of the measured  $\sigma(\theta)$  in 1971Vi02, unless otherwise noted.

$^{36}\text{Ar}(\text{n},\text{d})$  1977Pa29

$J^\pi=0^+$  for  $^{36}\text{Ar}$  ground state.

1977Pa29: 14.1-MeV neutrons were produced via the  $^3\text{H}(\text{d},\text{n})^4\text{He}$  reaction at an average rate of  $10^9$  neutrons per second into  $4\pi$  sr from the Brown University neutron generator and impinged on a  $2.85\text{-mg/cm}^2$   $^{36}\text{Ar}$  target in the form of a gas cell filled with 99.8% enriched  $^{36}\text{Ar}$ . Deuterons were detected using a time-of-flight counter telescope consisting of two proportional counters and a NaI(Tl) crystal. Measured  $\sigma(E_{\text{d}},\theta)$ . Deduced levels, L-transfers, spectroscopic factors from local zero-range DWUCK-DWBA analysis of the measured  $\sigma(\theta)$ .

 $^{35}\text{Cl}$  Levels

| <u>E(level)<sup>†</sup></u> | <u>L<sup>‡</sup></u> | <u>Comments</u>                                                                                                                                                          |
|-----------------------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0                           | 2                    | S: 3.83 48 and 3.90 47 using two sets of optical parameters.                                                                                                             |
| 1219 <sup>#</sup>           | 0 <sup>#</sup>       | S: 1.94 23 and 3.30 40 using two sets of optical parameters. Other spectroscopic factors calculated using two sets of optical parameters did not show large differences. |
| 1763 <sup>#</sup>           | (2) <sup>#</sup>     |                                                                                                                                                                          |

<sup>†</sup> From 1977Pa29.

<sup>‡</sup> From DWBA analysis of the measured  $\sigma(\theta)$  in 1977Pa29.

<sup>#</sup> 1977Pa29 state that the first and second excited states are recorded together and not resolved, but their angular distributions could be separated due to different L-transfers, however statistical errors prevent an accurate determination of the smaller L=2 component from the second excited state.

$^{36}\text{Ar}(\text{pol d}, ^3\text{He})$  1993Ma50

$J^\pi=0^+$  for  $^{36}\text{Ar}$  ground state.

1993Ma50: a 52-MeV polarized deuteron beam at 15 nA was produced from the Karlsruhe isochronous cyclotron at the Max-Planck-Institut für Kernphysik. The target was 350-mbar 99.86% enriched  $^{36}\text{Ar}$  gas. Reaction products were detected using a Si detector telescope consisting of a 250- $\mu\text{m}$   $\Delta E$ -strip detector and a 1.5-mm surface-barrier E detector with FWHM=130 keV. Measured angular distributions of cross sections  $\sigma(E(^3\text{He}), \theta)$  and angular distributions of analyzing powers  $iT_{11}(E(^3\text{He}), \theta)$ . Deduced levels,  $J$ ,  $\pi$ , L-transfers, and spectroscopic factors from DWBA analysis of  $\sigma(E(^3\text{He}), \theta)$  and  $iT_{11}(E(^3\text{He}), \theta)$ .

1993Ma50 states that spectroscopic factors were obtained by fitting the forward-angle maxima. The agreement with the analyzing powers is only qualitative. The determination of spins and parities of final states is mainly based on a comparison with measured angular distributions leading to final states with known spins. The comparison with DWBA predictions serves as additional check. The data were taken only in a relatively small angle interval. This is sufficient for the spin determination but not favorable for the measurement of spectroscopic factors.

 $^{35}\text{Cl}$  Levels

Spectroscopic factor  $C^2S = \sigma(\theta)_{\text{exp}} / \sigma(\theta)_{\text{DWBA}} / N$ , where  $N=2.95$  is a normalization factor used by 1993Ma50 from 1966Ba54.

| $E(\text{level})^\dagger$ | $J^\pi^\ddagger$    | $L^\#$ | $C^2S^\#$               | Comments                                                                                                                                                                                       |
|---------------------------|---------------------|--------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0                         | $3/2^+ @$           | 2      | 1.72 @                  |                                                                                                                                                                                                |
| 1213 4                    | $1/2^+$             | 0      | 0.78                    |                                                                                                                                                                                                |
| 1785 50                   | $(5/2^+) \&$        |        | 0.04 <sup>a</sup>       |                                                                                                                                                                                                |
| 2676 7                    | $(3/2^+) \&$        |        | 0.16 <sup>a</sup>       |                                                                                                                                                                                                |
| 3002 6                    | $5/2^+$             | 2      | 1.32                    |                                                                                                                                                                                                |
| 3159 20                   | $7/2^-$             | 3      | 0.19                    |                                                                                                                                                                                                |
| 3948 40                   | $(3/2^+, 1/2^+) \&$ | 2      | 0.03, 0.08              | $E(\text{level})$ : possible doublet of 3918 and 3968 (1990En08) suggested by Table 2 of 1993Ma50.                                                                                             |
| 4839 12                   | $(5/2^+, 1/2^+) \&$ |        | 0.08, 0.03 <sup>a</sup> |                                                                                                                                                                                                |
| 5155 11                   | $(5/2^+)$           | 2      | 0.12                    | $J^\pi$ : $(7/2^-)$ and $5/2^+$ are given in Table 2 of 1993Ma50. Evaluators adopt $(5/2^+)$ , based on DWBA fit of the measured $\sigma(\theta)$ and $iT_{11}(\theta)$ in Fig. 3 of 1993Ma50. |
| 5583 13                   | $5/2^+$             | 2      | 0.38                    |                                                                                                                                                                                                |
| 5720 11                   | $5/2^+$             | 2      | 1.10                    |                                                                                                                                                                                                |
| 6136 8                    | $5/2^+$             | 2      | 0.63                    |                                                                                                                                                                                                |
| 6745 12                   | $(1/2^+, 5/2^+) \&$ | (0, 2) | 0.09, 0.28              |                                                                                                                                                                                                |
| 6953 43                   | $5/2^+$             | 2      | 0.49                    |                                                                                                                                                                                                |
| 7181 60                   | $5/2^+$             | 2      | 0.20                    |                                                                                                                                                                                                |
| 7565 20                   | $5/2^+$             | 2      | 0.30                    |                                                                                                                                                                                                |
| 8007 36                   | $(1/2^-, 5/2^+) \&$ | (1, 2) | 0.16, 0.20              |                                                                                                                                                                                                |
| 8204 34                   | $5/2^+$             | 2      | 0.14                    |                                                                                                                                                                                                |
| 8580 20                   | $5/2^+ \&$          |        | 0.22 <sup>a</sup>       |                                                                                                                                                                                                |
| 8921 70                   | $5/2^+ \&$          |        | 0.18 <sup>a</sup>       |                                                                                                                                                                                                |
| $9.22 \times 10^3$ 12     | $5/2^+ \&$          |        | 0.12 <sup>a</sup>       |                                                                                                                                                                                                |
| 9514 20                   | $(1/2^-, 5/2^+) \&$ | (1, 2) | 0.18, 0.30              |                                                                                                                                                                                                |
| $9.82 \times 10^3$ 12     | $(1/2^-, 5/2^+) \&$ | (1, 2) | 0.10, 0.26              |                                                                                                                                                                                                |
| $1.25 \times 10^4$ 5      |                     |        |                         | $1d_{5/2}$ is given for a range of 12.0-13.0 MeV in Fig. 4 of 1993Ma50.                                                                                                                        |

<sup>†</sup> From 1993Ma50.

<sup>‡</sup>  $L+1/2$  transfer from analyzing power measurements with  $1d_{5/2}$  proton transfer assumed in DWBA calculations, unless otherwise noted.

<sup>#</sup> From DWBA analysis of the measured  $\sigma(\theta)$  and  $iT_{11}(\theta)$  in 1993Ma50.  $C^2S$  value is extracted for corresponding  $J^\pi$  assignment. Where two  $C^2S$  values are given, they correspond to respective  $J^\pi$  values.

Continued on next page (footnotes at end of table)

---

 $^{36}\text{Ar}(\text{pol d}, ^3\text{He})$  [1993Ma50](#) (continued)

---

 $^{35}\text{Cl}$  Levels (continued)

@ L-1/2 transfer from analyzing power measurements with  $1d_{3/2}$  proton transfer assumed in DWBA calculations.

& Assumed by [1993Ma50](#) for extracting spectroscopic factors.

<sup>a</sup> No data on  $\sigma(E(^3\text{He}), \theta)$ ,  $iT_{11}(E(^3\text{He}), \theta)$ , or L-transfer is given.

$^{36}\text{Ar}(\text{d}, ^3\text{He})$  1974Do12

$J^\pi=0^+$  for  $^{36}\text{Ar}$  ground state.

1974Do12: a 52-MeV deuteron beam was produced from the Karlsruhe isochronous cyclotron at the Max-Planck-Institut für Kernphysik. The target was 200-Torr 99.9% enriched  $^{36}\text{Ar}$  gas.  $^3\text{He}$  particles were detected using detector telescopes consisting of a 200- $\mu\text{m}$   $\Delta E$  and a 2-mm E-surface-barrier counter with FWHM=125 keV. Measured  $\sigma(E(^3\text{He}), \theta)$ . Deduced levels, J,  $\pi$ , L-transfer, and spectroscopic factors from DWUCK-DWBA analysis.

Other: 1975Wa17 (theory discussions).

 $^{35}\text{Cl}$  Levels

Spectroscopic factor  $C^2S=\sigma(\theta)_{\text{exp}}/\sigma(\theta)_{\text{DWBA}}/N$ , where  $N=2.95$  is a normalization factor used by 1974Do12 from 1966Ba54.

| <u>E(level)<sup>†</sup></u> | <u>J<math>\pi</math><sup>#</sup></u> | <u>L<sup>‡</sup></u> | <u>C<sup>2</sup>S<sup>‡</sup></u> | <u>E(level)<sup>†</sup></u> | <u>J<math>\pi</math><sup>#</sup></u> | <u>L<sup>‡</sup></u> | <u>C<sup>2</sup>S<sup>‡</sup></u> | <u>E(level)<sup>†</sup></u> | <u>J<math>\pi</math><sup>#</sup></u> | <u>L<sup>‡</sup></u> | <u>C<sup>2</sup>S<sup>‡</sup></u> |
|-----------------------------|--------------------------------------|----------------------|-----------------------------------|-----------------------------|--------------------------------------|----------------------|-----------------------------------|-----------------------------|--------------------------------------|----------------------|-----------------------------------|
| 0                           | 3/2 <sup>+</sup>                     | 2                    | 2.2                               | 3170 20                     | 7/2 <sup>-</sup>                     | 3                    | 0.22                              | 6750 20                     | (5/2) <sup>+</sup>                   | 2                    | 0.79                              |
| 1220 20                     | 1/2 <sup>+</sup>                     | 0                    | 1.34                              | 5150 20                     | (5/2) <sup>+</sup>                   | 2                    | 0.09                              | 6930 20                     | (5/2) <sup>+</sup>                   | 2                    | 0.46                              |
| 1763                        |                                      |                      | <0.02                             | 5600 20                     | (5/2) <sup>+</sup>                   | 2                    | 0.47 19                           | 8010 20                     | (5/2) <sup>+</sup>                   | 2                    | 0.23                              |
| 2700 20                     | 3/2 <sup>+</sup>                     | 2                    | 0.39                              | 5720 20                     | (5/2) <sup>+</sup>                   | 2                    | 0.73 29                           | 8210 20                     | (5/2) <sup>+</sup>                   | 2                    | 0.28                              |
| 3000 20                     | 5/2 <sup>+</sup>                     | 2                    | 1.49                              | 6140 20                     | (5/2) <sup>+</sup>                   | 2                    | 0.72                              | 8610 20                     | (5/2) <sup>+</sup>                   | 2                    | 0.34                              |

<sup>†</sup> From 1974Do12.

<sup>‡</sup> From DWBA analysis of the measured  $\sigma(\theta)$  in 1974Do12.

<sup>#</sup> As given in 1974Do12 for levels below 5 MeV. For levels above 5 MeV, 1974Do12 states that they tend to interpret them as 5/2<sup>+</sup> states because the summed low-lying  $C^2S(1d_{3/2})$  already exceeds the simple shell model value for  $^{36}\text{Ar}$  and the measured  $\sigma(\theta)$  resembles more closely that of the 5/2<sup>+</sup> state at 3.0 MeV than that of the 3/2<sup>+</sup> ground state. All  $C^2S$  values are given according to the J values.

$^{39}\text{K}(\gamma, \alpha)$  1987Ar08

1987Ar08:  $E_{\text{max}}=32$ -MeV  $\gamma$ -ray beam was produced from the betatron at Moscow University Nuclear Physics Research Institute and impinged on a natural metallic potassium target of mass  $\approx 200$  g. Photons from  $(\gamma, p)$ ,  $(\gamma, n)$ , and  $(\gamma, \alpha)$  from the giant dipole resonance (GDR) in  $^{39}\text{K}$  were detected using a 100-cm<sup>3</sup> Ge(Li) detector at 140°. Measured  $E_\gamma$ . Deduced partial photonuclear cross sections, levels, J,  $\pi$ .

 $^{35}\text{Cl}$  Levels

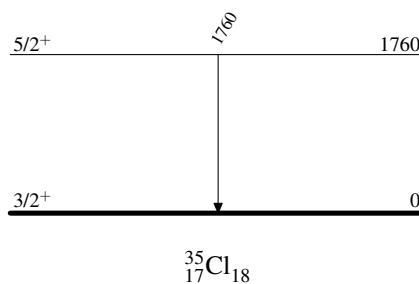
| <u>E(level)<sup>†</sup></u> | <u>J<math>\pi</math><sup>‡</sup></u> | Comments |
|-----------------------------|--------------------------------------|----------|
| 0                           | 3/2 <sup>+</sup>                     |          |
| 1760                        | 5/2 <sup>+</sup>                     | T=1/2    |

<sup>†</sup> From 1987Ar08.

<sup>‡</sup> From Adopted Levels.

 $\gamma(^{35}\text{Cl})$ 

| <u><math>E_\gamma</math></u> | <u><math>E_i(\text{level})</math></u> | <u><math>J_i^\pi</math></u> | <u><math>E_f</math></u> | <u><math>J_f^\pi</math></u> |
|------------------------------|---------------------------------------|-----------------------------|-------------------------|-----------------------------|
| 1760                         | 1760                                  | 5/2 <sup>+</sup>            | 0                       | 3/2 <sup>+</sup>            |

 $^{39}\text{K}(\gamma, \alpha)$  1987Ar08Level Scheme



$^{40}\text{Ca}(\mu^-, \nu\alpha n\gamma)$  2006Me08

**2006Me08:** Muons were generated from the decay of 90-MeV/c  $\pi^-$  beam in a 6-m, 1.2-T superconducting solenoid and were selected by a bending magnet at the M9B beamline at TRIUMF. The target was 15-g natural calcium.  $\gamma$  rays and x rays were detected using a 186-cm<sup>3</sup> HPGe detector. Charged particles caused by electrons from muon decay were tagged by a plastic scintillator in front of the HPGe detector. Measured  $E_\gamma$ ,  $I_\gamma$ ,  $E(\text{x ray})$ ,  $I(\text{x ray})$ . Deduced levels, muon capture yields.

-----  
Muonic Lyman series for natural Calcium

| $\mu$ x ray         | Energy     | Intensity in percent |
|---------------------|------------|----------------------|
| 2p-1s               | 783.659 25 | 83.8 10              |
| 3p-1s               | 940.63 10  | 6.2 2                |
| 4p-1s               | 995.48 10  | 2.0 1                |
| 5p-1s               | 1020.81 10 | 2.0 1                |
| 6p-1s               | 1034.62 10 | 1.8 1                |
| 7p-1s               | 1042.71 20 | 1.4 1                |
| (8 $^-\infty$ )p-1s | 1046-1063  | 2.8 4                |

-----  
Muonic Balmer series for natural Calcium

| $\mu$ x ray         | Energy    | Intensity in percent |
|---------------------|-----------|----------------------|
| 3d-2p               | 157.35 13 | 64.5 9               |
| 4d-2p               | 212.03 10 | 8.85 20              |
| 5d-2p               | 237.31 10 | 4.34 20              |
| 6d-2p               | 251.06 10 | 3.29 20              |
| 7d-2p               | 259.45 10 | 1.37 20              |
| (8 $^-\infty$ )d-2p | 261-277   | 1.4 3                |

 $^{35}\text{Cl}$  Levels

| <u><math>E(\text{level})^\dagger</math></u> | <u><math>J^\pi^\dagger</math></u> |
|---------------------------------------------|-----------------------------------|
| 0                                           | 3/2 <sup>+</sup>                  |
| 1219.4                                      | 1/2 <sup>+</sup>                  |
| 1763.2                                      | 5/2 <sup>+</sup>                  |
| 2645.7                                      | 7/2 <sup>+</sup>                  |
| 2694.0                                      | 3/2 <sup>+</sup>                  |

<sup>†</sup> From Adopted Levels. Level energies are rounded-off values.

 $\gamma(^{35}\text{Cl})$ 

| <u><math>E_\gamma^\dagger</math></u> | <u><math>I_\gamma^\ddagger</math></u> | <u><math>E_i(\text{level})</math></u> | <u><math>J_i^\pi</math></u> | <u><math>E_f</math></u> | <u><math>J_f^\pi</math></u> |
|--------------------------------------|---------------------------------------|---------------------------------------|-----------------------------|-------------------------|-----------------------------|
| 1219.4                               | <0.3                                  | 1219.4                                | 1/2 <sup>+</sup>            | 0                       | 3/2 <sup>+</sup>            |
| 1763.2                               | 0.5 3                                 | 1763.2                                | 5/2 <sup>+</sup>            | 0                       | 3/2 <sup>+</sup>            |
| 2645.7                               | <0.2                                  | 2645.7                                | 7/2 <sup>+</sup>            | 0                       | 3/2 <sup>+</sup>            |
| 2693.7                               | <0.2                                  | 2694.0                                | 3/2 <sup>+</sup>            | 0                       | 3/2 <sup>+</sup>            |

<sup>†</sup> Rounded-off values from Adopted Gammas.

<sup>‡</sup> Percent  $\gamma$ -ray yield per muon capture (2006Me08).

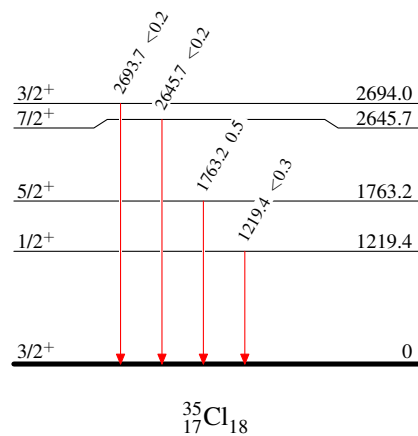
$^{40}\text{Ca}(\mu^-, \nu\alpha n\gamma)$  2006Me08

## Level Scheme

Intensities: percent  $\gamma$ -ray yield per muon capture

## Legend

- $I_\gamma < 2\% \times I_\gamma^{\max}$
- $I_\gamma < 10\% \times I_\gamma^{\max}$
- $I_\gamma > 10\% \times I_\gamma^{\max}$



**Coulomb excitation**    **1969Ha17,1977Sc36**

**1969Ha17:** 51-61-MeV  $^{35}\text{Cl}^{5,6+}$  beams of 80 nA were produced from the Chalk River MP tandem accelerator and bombarded 9.6-mg/cm<sup>2</sup> carbon and 5-mg/cm<sup>2</sup>  $^{24}\text{Mg}$ . Deexcitation  $\gamma$  rays were detected using two 40-cm<sup>3</sup> Ge(Li) detectors at 125° and at 0° or 90°. Measured  $E_\gamma$ ,  $I_\gamma$ ,  $\gamma(\theta)$  anisotropy,  $\gamma$ -ray yields. Deduced levels, mixing ratio, transition strengths,  $T_{1/2}$  for 2 levels using the Doppler Shift Attenuation Method (DSAM).

**1977Sc36:** 53-56-MeV  $^{35}\text{Cl}^{6+}$  beams of 60 nA were produced from a single MP tandem Van de Graaff accelerator at the three-stage facility of Brookhaven National Laboratory. Targets were  $^{23}\text{Na}$  and  $^{27}\text{Al}$ . Deexcitation  $\gamma$  rays were detected using a 55-cm<sup>3</sup> coaxial Ge(Li) detector at 0°. Measured  $E_\gamma$ ,  $I_\gamma(\theta)$ ,  $\gamma$ -ray yields. Deduced levels, mixing ratios, transition strengths,  $T_{1/2}$  for the levels of 1220, 1762 and 2650 keV using the Doppler Shift Attenuation Method (DSAM).

 $^{35}\text{Cl}$  Levels

Values of  $B(E2)\uparrow$  from **1969Ha17** and **1977Sc36** are from absolute cross-section measurements.

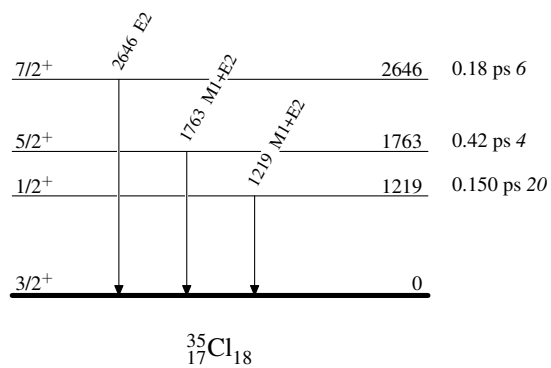
| $E(\text{level})^\dagger$ | $J^\pi^\dagger$ | $T_{1/2}$   | Comments                                                                                                                                                                                                                                                                                                                           |
|---------------------------|-----------------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0                         | $3/2^+$         |             |                                                                                                                                                                                                                                                                                                                                    |
| 1219                      | $1/2^+$         | 0.150 ps 20 | $B(E2)\uparrow=0.00076\ 7$<br>$T_{1/2}$ : Lifetime $\tau=0.216$ ps 28 from weighted average of $\tau=0.16$ ps 5 ( <b>1969Ha17</b> , DSAM) and $\tau=0.230$ ps 25 ( <b>1977Sc36</b> , DSAM).<br>$B(E2)\uparrow$ : weighted average of 0.00081 9 ( <b>1969Ha17</b> ) and 0.00073 7 ( <b>1977Sc36</b> ).<br>$B(E2)\uparrow=0.0106\ 7$ |
| 1763                      | $5/2^+$         | 0.42 ps 4   | $T_{1/2}$ : Lifetime $\tau=0.61$ ps 5 from weighted average of $\tau=0.57$ ps 7 ( <b>1969Ha17</b> , DSAM) and $\tau=0.63$ ps 5 ( <b>1977Sc36</b> , DSAM).<br>$B(E2)\uparrow$ : weighted average of 0.0117 9 ( <b>1969Ha17</b> ) and 0.0102 5 ( <b>1977Sc36</b> ).<br>$B(E2)\uparrow=0.0029\ 7$ ( <b>1977Sc36</b> )                 |
| 2646                      | $7/2^+$         | 0.18 ps 6   | $T_{1/2}$ : Lifetime $\tau=0.26$ ps 9 ( <b>1977Sc36</b> , DSAM).                                                                                                                                                                                                                                                                   |

$^\dagger$  From the Adopted Levels. Energies are rounded values.

 $\gamma(^{35}\text{Cl})$ 

| $E_\gamma^\dagger$ | $E_i(\text{level})$ | $J_i^\pi$ | $E_f$ | $J_f^\pi$ | Mult. $^\dagger$ | $\delta$ | Comments                                                                                                                                                                                                                                                                                                  |
|--------------------|---------------------|-----------|-------|-----------|------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1219               | 1219                | $1/2^+$   | 0     | $3/2^+$   | M1+E2            | 0.093 10 | $\delta$ : deduced by the evaluators from adopted $B(E2)\uparrow=0.00076\ 7$ in this dataset and $T_{1/2}=0.119$ ps 14 from Adopted Levels. Other: 0.106 7 deduced by <b>1977Sc36</b> based on their measured $B(E2)\uparrow$ and lifetime.                                                               |
| 1763               | 1763                | $5/2^+$   | 0     | $3/2^+$   | M1+E2            | -2.95 45 | $\delta$ : from $\gamma(\theta)$ anisotropy in <b>1969Ha17</b> . Other: $1.8 +10-4$ ; deduced by the evaluators from adopted $B(E2)\uparrow=0.0106\ 7$ in this dataset and $T_{1/2}=0.36$ ps 3 from Adopted Levels; -2.7 7 from <b>1977Sc36</b> deduced from their measured $B(E2)\uparrow$ and lifetime. |
| 2646               | 2646                | $7/2^+$   | 0     | $3/2^+$   | E2               |          |                                                                                                                                                                                                                                                                                                           |

$^\dagger$  From Adopted Gammas. Energies are rounded values.

**Coulomb excitation 1969Ha17,1977Sc36**Level Scheme

Adopted Levels, Gammas

$Q(\beta^-) = -11874.4$  9;  $S(n) = 12740.3$  7;  $S(p) = 5896.2$  7;  $Q(\alpha) = -6429.7$  7 [2021Wa16](#)  
 $Q(\beta^-n) = -29632$  17, from mass excesses of  $-1487$  17 for  $^{34}\text{K}$  measured by [2024Dr01](#);  $-23047.3$  7 for  $^{35}\text{Ar}$  from [2021Wa16](#). Value from [2021Wa16](#):  $Q(\beta^-n) = -29900$  200 (syst).  
 $S(2n) = 29805.6$  8,  $S(2p) = 11039.4$  7,  $Q(\epsilon) = 5966.2$  7 ([2021Wa16](#)).  
 Isotope discovery ([2012Th10](#)):  $^{32}\text{S}(\alpha,n)^{35}\text{Ar}$  at Purdue ([1940Ki12](#), [1941Ki01](#), [1941El04](#)).  
 $^{35}\text{Ar}$  production:  
[2012Zh06](#):  $^9\text{Be}$ ,  $^{181}\text{Ta}(^{40}\text{Ar}, X)$  at  $E(^{40}\text{Ar}) = 57$  MeV/nucleon at HIRFL. Measured momentum distributions and production cross sections of fragments. Observed competition between projectile fragmentation and other mechanisms. Compared with EPAX, abrasion-ablation, and HIPSE models. Studied target dependence of fragment cross sections.  
[2007No13](#):  $^{181}\text{Ta}(^{40}\text{Ar}, X)$  at  $E(^{40}\text{Ar}) = 100$  MeV/nucleon at RIKEN. Measured fragment momentum distributions and production cross sections.  
 $^{35}\text{Ar}$  decay measurements:  
 $^{35}\text{Ar}$  produced via  $^{35}\text{Cl}(p,n)^{35}\text{Ar}$  reaction ([1985Da04](#), [1984Ad01](#), [1980Wi13](#), [1979Ha04](#), [1977Az01](#), [1971Ge04](#), [1971De05](#), [1969Wi18](#), [1969Fr09](#), [1960Wa04](#), [1956Ki29](#)). Measured  $\epsilon + \beta^+$ -delayed  $\gamma$  rays, decay branching ratios, and parent  $^{35}\text{Ar}$   $T_{1/2}$ .  
 $^{35}\text{Ar}$  radius measurements:  
[2002Oz03](#):  $\text{C}(^{35}\text{Ar}, X)$  at  $E(^{35}\text{Ar}) \approx 950$  MeV/nucleon at RIKEN. Measured interaction cross sections. Deduced effective radii and proton skin features.  
[2000Ge20](#):  $^{35}\text{Ar}$  produced at ISOLDE. Measured  $\beta$  asymmetry and hyperfine structure using  $\beta$ -NMR spectroscopy. Deduced mean squared charge radii and quadrupole moments.  
[1996Ki04](#), [1995KiZZ](#):  $^{35}\text{Ar}$  produced by ISOLDE. Measured isotope shifts and hyperfine structure using collinear fast-beam laser spectroscopy. Deduced mean square charge radii and electric quadrupole moments.  
 $^{35}\text{Ar}$  mass measurement: [2011Tu09](#).  
 Theoretical calculations: [2020Ri06](#), [2020RiZX](#), [2020RiZZ](#).

 $^{35}\text{Ar}$  LevelsCross Reference (XREF) Flags

|          |                                                     |          |                                                  |          |                                       |
|----------|-----------------------------------------------------|----------|--------------------------------------------------|----------|---------------------------------------|
| <b>A</b> | $^{35}\text{K}$ $\epsilon + \beta^+$ decay (175 ms) | <b>E</b> | $^{24}\text{Mg}(^{16}\text{O}, \alpha n \gamma)$ | <b>I</b> | $^{36}\text{Ar}(p, d)$                |
| <b>B</b> | $^{36}\text{Ca}$ $\epsilon p$ decay (100.9 ms)      | <b>F</b> | $^{32}\text{S}(\alpha, n)$                       | <b>J</b> | $^{36}\text{Ar}(d, t)$                |
| <b>C</b> | $^1\text{H}(^{36}\text{Ar}, d)$                     | <b>G</b> | $^{33}\text{S}(^3\text{He}, n \gamma)$           | <b>K</b> | $^{36}\text{Ar}(^3\text{He}, \alpha)$ |
| <b>D</b> | $^{16}\text{O}(^{24}\text{Mg}, \alpha n \gamma)$    | <b>H</b> | $^{35}\text{Cl}(^3\text{He}, t)$                 |          |                                       |

| $E(\text{level})^\dagger$ | $J^\pi$   | $T_{1/2}$   | XREF         | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------|-----------|-------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0.0                       | $3/2^+$   | 1.7755 s 14 | ABCDEFGF IJK | $\% \epsilon + \% \beta^+ = 100$<br>$\mu = +0.6322$ 2 ( <a href="#">2002Ma41</a> , <a href="#">2019StZV</a> )<br>$Q = -0.084$ 15 ( <a href="#">1996Ki04</a> , <a href="#">2021StZZ</a> )<br>$\mu$ : $\beta$ -NMR ( <a href="#">2002Ma41</a> ). Others: $+0.632$ 2 ( <a href="#">1965Ca04</a> ), $+0.633$ 7 ( <a href="#">1996Ki04</a> ) using $\beta$ -NMR.<br>$Q$ : $\beta$ -NMR ( <a href="#">1996Ki04</a> ).<br>$J^\pi$ : $L=2$ from $0^+$ in $(p, d)$ , $(d, t)$ , $(^3\text{He}, \alpha)$ , and $^1\text{H}(^{36}\text{Ar}, d)$ . Allowed $\epsilon + \beta^+$ feedings to $1/2^+$ levels in $^{35}\text{Cl}$ .<br>$T_{1/2}$ : weighted average of 1.83 s 3 ( <a href="#">1956Ki29</a> ), 1.83 s 2 ( <a href="#">1959Al10</a> ), 1.79 s 1 ( <a href="#">1960Ja12</a> ), 1.84 s 10 ( <a href="#">1960Wa04</a> ), 1.76 s 3 ( <a href="#">1963Ne05</a> ), 1.770 s 6 ( <a href="#">1969Wi18</a> ), 1.787 s 12 ( <a href="#">1971Ge04</a> ), 1.774 s 3 ( <a href="#">1977Az01</a> ), and 1.7754 s 11 ( <a href="#">2006Ia05</a> ).<br>Evaluated rms nuclear charge radius $R = 3.3636$ fm 42 ( <a href="#">2013An02</a> ).<br>XREF: F(890)<br>$E(\text{level})$ : <a href="#">1963Ne05</a> ( $\alpha, n$ ) observed the first excited state in $^{35}\text{Ar}$ at 890 50 keV.<br>$J^\pi$ : $L=0$ from $0^+$ in $(p, d)$ , $(d, t)$ , and $(^3\text{He}, \alpha)$ .<br>XREF: F(2030)I(1700)J(1700) |
| 1184.08 25                | $1/2^+$   |             | ABC FG IJK   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 1750.78 22                | $(5/2)^+$ |             | A DEFG IJK   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

Continued on next page (footnotes at end of table)

**Adopted Levels, Gammas (continued)** $^{35}\text{Ar}$  Levels (continued)

| E(level) <sup>†</sup>   | J <sup>π</sup>                                            | XREF      | Comments                                                                                                                                                                                                                                                                                      |
|-------------------------|-----------------------------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         |                                                           |           | E(level): <a href="#">1963Ne05</a> ( $\alpha, n$ ) observed the second excited state in $^{35}\text{Ar}$ at 2030 80 keV.                                                                                                                                                                      |
|                         |                                                           |           | J <sup>π</sup> : L=2 from 0 <sup>+</sup> in ( $^3\text{He}, \alpha$ ). Possible mirror level of 5/2 <sup>+</sup> at 1763 keV in $^{35}\text{Cl}$ .                                                                                                                                            |
| 2603.22 28              | 7/2 <sup>(+)</sup>                                        | DE G      | J <sup>π</sup> : 2603.0 $\gamma$ Q, $\Delta J=2$ to 3/2 <sup>+</sup> .                                                                                                                                                                                                                        |
| 2638.01 26              | (3/2) <sup>+</sup>                                        | A IJK     | XREF: I(2615)K(2649)                                                                                                                                                                                                                                                                          |
|                         |                                                           |           | J <sup>π</sup> : L=2 from 0 <sup>+</sup> in (p,d) and ( $^3\text{He}, \alpha$ ) and spin=(3/2) from J-dependence in (p,d).                                                                                                                                                                    |
| 2982.79 12              | (5/2) <sup>+</sup>                                        | A C IJK   | XREF: I(2970)                                                                                                                                                                                                                                                                                 |
|                         |                                                           |           | J <sup>π</sup> : L=2 from 0 <sup>+</sup> in (p,d), (d,t), and ( $^3\text{He}, \alpha$ ) and spin=(5/2) from J-dependence in (p,d).                                                                                                                                                            |
| 3196.98 <sup>‡</sup> 26 | 7/2 <sup>-</sup>                                          | CDE G IJK | J <sup>π</sup> : L=3 from 0 <sup>+</sup> in (p,d) and ( $^3\text{He}, \alpha$ ); 3197.0 $\gamma$ Q, $\Delta J=2$ to 3/2 <sup>+</sup> .                                                                                                                                                        |
| 3882 5                  | 1/2 <sup>+</sup>                                          | K         | J <sup>π</sup> : L=0 from 0 <sup>+</sup> in ( $^3\text{He}, \alpha$ ).                                                                                                                                                                                                                        |
| 4001 3                  | 1/2 <sup>-</sup> , 3/2 <sup>-</sup>                       | K         | J <sup>π</sup> : L=1 from 0 <sup>+</sup> in ( $^3\text{He}, \alpha$ ).                                                                                                                                                                                                                        |
| 4065.0? 4               | (1/2 <sup>+</sup> , 3/2 <sup>+</sup> , 5/2 <sup>+</sup> ) | A         | XREF: A(?)                                                                                                                                                                                                                                                                                    |
|                         |                                                           |           | J <sup>π</sup> : possibly allowed $\varepsilon + \beta^+$ feeding from 3/2 <sup>+</sup> parent with log ft=5.6 +4-2.                                                                                                                                                                          |
| 4113 4                  |                                                           | K         |                                                                                                                                                                                                                                                                                               |
| 4135 4                  | 1/2 <sup>-</sup> , 3/2 <sup>-</sup>                       | K         | J <sup>π</sup> : L=1 from 0 <sup>+</sup> in ( $^3\text{He}, \alpha$ ).                                                                                                                                                                                                                        |
| 4359.0 5                | (9/2 <sup>-</sup> )                                       | DE K      | J <sup>π</sup> : 1162.0 $\gamma$ (D), $\Delta J=(1)$ to 7/2 <sup>-</sup> ; mirror level to 4348, 9/2 <sup>-</sup> level in $^{35}\text{Cl}$ proposed by <a href="#">2007De14</a> in $^{24}\text{Mg}(^{16}\text{O}, \alpha n\gamma)$ .                                                         |
| 4528.3 4                | (1/2 <sup>+</sup> , 3/2 <sup>+</sup> , 5/2 <sup>+</sup> ) | A K       | XREF: K(4515)                                                                                                                                                                                                                                                                                 |
|                         |                                                           |           | J <sup>π</sup> : possibly allowed $\varepsilon + \beta^+$ feeding from 3/2 <sup>+</sup> parent with log ft=5.4 +4-2.                                                                                                                                                                          |
| 4725.9 6                | 1/2 <sup>+</sup>                                          | A Hi K    | XREF: i(4756)K(4713)                                                                                                                                                                                                                                                                          |
|                         |                                                           |           | J <sup>π</sup> : L=0 from 0 <sup>+</sup> in ( $^3\text{He}, \alpha$ ). Other: L=0 from 0 <sup>+</sup> in (p,d) for a group at 4756 28.                                                                                                                                                        |
| 4785.8 11               | 1/2 <sup>+</sup> , 3/2 <sup>+</sup> , 5/2 <sup>+</sup>    | A Hi K    | XREF: i(4756)K(4774)                                                                                                                                                                                                                                                                          |
|                         |                                                           |           | J <sup>π</sup> : allowed $\varepsilon + \beta^+$ feeding from 3/2 <sup>+</sup> parent with log ft=5.2 2. Other: L=0 from 0 <sup>+</sup> in (p,d) for a group at 4756 28.                                                                                                                      |
| 5059 11                 |                                                           | i K       |                                                                                                                                                                                                                                                                                               |
| 5116 2                  | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>                       | Hi K      | E(level): from ( $^3\text{He}, \alpha$ ). Other: 5102 20 from (p,d).                                                                                                                                                                                                                          |
|                         |                                                           |           | J <sup>π</sup> : L=2 from 0 <sup>+</sup> in ( $^3\text{He}, \alpha$ ) in <a href="#">1973Be26</a> . Discrepancy: L=3 from 0 <sup>+</sup> in (p,d) ( <a href="#">1968Jo04</a> ). <a href="#">1973Be26</a> also considered L=3 but found L=2 gives a much better fit of the data than L=3 does. |
| 5207 3                  |                                                           | H K       | E(level): from ( $^3\text{He}, \alpha$ ).                                                                                                                                                                                                                                                     |
| 5384.2 <sup>‡</sup> 4   | (11/2 <sup>-</sup> )                                      | DE HI K   | XREF: I(5400)                                                                                                                                                                                                                                                                                 |
|                         |                                                           |           | J <sup>π</sup> : 2187.1 $\gamma$ Q, $\Delta J=2$ to 7/2 <sup>-</sup> ; band assignment.                                                                                                                                                                                                       |
| 5482 2                  | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>                       | H K       | E(level): from ( $^3\text{He}, \alpha$ ).                                                                                                                                                                                                                                                     |
|                         |                                                           |           | J <sup>π</sup> : L=2 from 0 <sup>+</sup> in ( $^3\text{He}, \alpha$ ).                                                                                                                                                                                                                        |
| 5572.67 15              | 3/2 <sup>+</sup>                                          | A G       | T=3/2                                                                                                                                                                                                                                                                                         |
|                         |                                                           |           | XREF: G(5537)                                                                                                                                                                                                                                                                                 |
|                         |                                                           |           | J <sup>π</sup> : isobaric analog state of 3/2 <sup>+</sup> $^{35}\text{K}$ g.s. with log ft=3.31 4. L=(0) from 3/2 <sup>+</sup> in ( $^3\text{He}, n$ ). $^{35}\text{Cl}(^3\text{He}, t)$ attempted to search this T=3/2 level, but did not find it.                                          |
| 5594 2                  | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>                       | C HI K    | XREF: C(5570)                                                                                                                                                                                                                                                                                 |
|                         |                                                           |           | E(level): weighted average of 5598 20 from (p,d) and 5594 2 from ( $^3\text{He}, \alpha$ ).                                                                                                                                                                                                   |
|                         |                                                           |           | J <sup>π</sup> : L=2 from 0 <sup>+</sup> in (p,d) and ( $^3\text{He}, \alpha$ ).                                                                                                                                                                                                              |
|                         |                                                           |           | Evaluators consider the 5994 level to be different from the 5572.67 T=3/2 level because (p,d) and ( $^3\text{He}, \alpha$ ) from T=0 targets should not populate T=3/2 levels.                                                                                                                |
| 5613.6 9                | (11/2 <sup>-</sup> )                                      | E         | J <sup>π</sup> : 1254.6 $\gamma$ to (9/2 <sup>-</sup> ); mirror level to 5407, 11/2 <sup>-</sup> level in $^{35}\text{Cl}$ proposed by <a href="#">2007De14</a> in $^{24}\text{Mg}(^{16}\text{O}, \alpha n\gamma)$ .                                                                          |
| 5765.8 5                | (13/2 <sup>-</sup> )                                      | DE        | J <sup>π</sup> : 381.6 $\gamma$ D, $\Delta J=1$ to (11/2 <sup>-</sup> ), 1406.9 $\gamma$ (Q), $\Delta J=(2)$ to (9/2 <sup>-</sup> ).                                                                                                                                                          |
| 5915 3                  |                                                           | H JK      | E(level): weighted average of 5913 5 from (d,t) and 5916 3 from ( $^3\text{He}, \alpha$ ).                                                                                                                                                                                                    |

Continued on next page (footnotes at end of table)

**Adopted Levels, Gammas (continued)** $^{35}\text{Ar}$  Levels (continued)

| E(level) <sup>†</sup>  | J <sup>π</sup>                                         | XREF  | Comments                                                                                                                                                                                                                                  |
|------------------------|--------------------------------------------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5991 3                 |                                                        | J     |                                                                                                                                                                                                                                           |
| 6037 3                 | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>                    | HIJK  | XREF: I(6024)<br>E(level): weighted average of 6037 3 from (d,t) and 6036 3 from ( $^3\text{He},\alpha$ ). Other:<br>6024 20 from (p,d).<br>J <sup>π</sup> : L=2 from 0 <sup>+</sup> in (p,d) and ( $^3\text{He},\alpha$ ).<br>XREF: J(?) |
| 6055? 3                |                                                        | J     |                                                                                                                                                                                                                                           |
| 6076 3                 |                                                        | J     |                                                                                                                                                                                                                                           |
| 6163 2                 |                                                        | JK    | E(level): weighted average of 6164 3 from (d,t) and 6162 2 from ( $^3\text{He},\alpha$ ).                                                                                                                                                 |
| 6253 3                 |                                                        | Jk    | XREF: k(6262)                                                                                                                                                                                                                             |
| 6273 3                 |                                                        | Jk    | XREF: k(6262)                                                                                                                                                                                                                             |
| 6302 3                 |                                                        | J     |                                                                                                                                                                                                                                           |
| 6332 3                 |                                                        | J     |                                                                                                                                                                                                                                           |
| 6345 3                 | (1/2, 3/2, 5/2)                                        | A J   | E(level): From (d,t). Other: 6348 11 from $^{35}\text{K}$ $\varepsilon$ decay.<br>J <sup>π</sup> : $\varepsilon+\beta^+$ feeding from 3/2 <sup>+</sup> parent with log ft=7.2 1.                                                          |
| 6415 2                 |                                                        | J     |                                                                                                                                                                                                                                           |
| 6439? 4                |                                                        | J     | XREF: J(?)                                                                                                                                                                                                                                |
| 6460 3                 |                                                        | J     |                                                                                                                                                                                                                                           |
| 6523 3                 |                                                        | J     |                                                                                                                                                                                                                                           |
| 6557 3                 |                                                        | J     |                                                                                                                                                                                                                                           |
| 6585 3                 |                                                        | J     |                                                                                                                                                                                                                                           |
| 6606 3                 |                                                        | iJ    | XREF: i(6620)                                                                                                                                                                                                                             |
| 6616 2                 | 1/2 <sup>+</sup>                                       | iJK   | XREF: i(6620)<br>E(level): weighted average of 6617 2 from (d,t) and 6615 3 from ( $^3\text{He},\alpha$ ). Other:<br>6620 30 from (p,d).<br>J <sup>π</sup> : L=0 from 0 <sup>+</sup> in ( $^3\text{He},\alpha$ ) and (p,d).               |
| 6644 3                 |                                                        | iJ    | XREF: i(6620)                                                                                                                                                                                                                             |
| 6651 3                 |                                                        | iJ    | XREF: i(6620)                                                                                                                                                                                                                             |
| 6673 4                 | 5/2 <sup>-</sup> , 7/2 <sup>-</sup>                    | IJ    | XREF: I(6700)<br>E(level): weighted average of 6700 20 from (p,d) and 6672 3 from (d,t).<br>J <sup>π</sup> : L=3 from 0 <sup>+</sup> in (p,d).                                                                                            |
| 6823 2                 | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>                    | I K   | E(level): from ( $^3\text{He},\alpha$ ). Other: 6820 30 from (p,d).<br>J <sup>π</sup> : L=2 from 0 <sup>+</sup> in (p,d).                                                                                                                 |
| 6948 2                 |                                                        | K     |                                                                                                                                                                                                                                           |
| 7044 4                 | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>                    | A I K | XREF: I(7030)<br>E(level): weighted average of 7053 11 from $^{35}\text{K}$ $\varepsilon$ decay, 7030 20 from (p,d),<br>and 7043 4 from ( $^3\text{He},\alpha$ ).<br>J <sup>π</sup> : L=2 from 0 <sup>+</sup> in (p,d).                   |
| 7117 10                |                                                        | K     |                                                                                                                                                                                                                                           |
| 7255 11                |                                                        | A     |                                                                                                                                                                                                                                           |
| 7289 10                |                                                        | A K   | E(level): weighted average of 7283 11 from $^{35}\text{K}$ $\varepsilon$ decay and 7293 10 from ( $^3\text{He},\alpha$ ).                                                                                                                 |
| 7427 10                |                                                        | A K   | E(level): weighted average of 7431 11 from $^{35}\text{K}$ $\varepsilon$ decay and 7423 10 from ( $^3\text{He},\alpha$ ).                                                                                                                 |
| 7509 10                | 1/2 <sup>+</sup> , 3/2 <sup>+</sup> , 5/2 <sup>+</sup> | A K   | E(level): weighted average of 7518 11 from $^{35}\text{K}$ $\varepsilon$ decay and 7502 10 from ( $^3\text{He},\alpha$ ).<br>J <sup>π</sup> : allowed $\varepsilon+\beta^+$ feeding from 3/2 <sup>+</sup> parent with log ft<5.0.         |
| 7840 10                |                                                        | K     |                                                                                                                                                                                                                                           |
| 8019 10                |                                                        | K     |                                                                                                                                                                                                                                           |
| 8109.7 <sup>±</sup> 13 | (15/2 <sup>-</sup> )                                   | E     | J <sup>π</sup> : 2725.7γ to (11/2 <sup>-</sup> ); band assignment.                                                                                                                                                                        |
| 8212.6 8               | (15/2 <sup>-</sup> )                                   | E     | J <sup>π</sup> : 2828.3γ Q, ΔJ=2 to (11/2 <sup>-</sup> ) and 2446.6γ to (13/2 <sup>-</sup> ).                                                                                                                                             |
| 8393? 20               | 1/2 <sup>+</sup> , 3/2 <sup>+</sup> , 5/2 <sup>+</sup> | A     | XREF: A(?)<br>E(level): From ( $^{35}\text{K}$ $\varepsilon$ decay).<br>J <sup>π</sup> : allowed $\varepsilon+\beta^+$ feeding from 3/2 <sup>+</sup> parent with log ft=4.6 +3-2.                                                         |

Continued on next page (footnotes at end of table)

**Adopted Levels, Gammas (continued)** $^{35}\text{Ar}$  Levels (continued)

| <u>E(level)<sup>†</sup></u> | <u>J<sup>π</sup></u> | <u>XREF</u> | <u>Comments</u>                                                              |
|-----------------------------|----------------------|-------------|------------------------------------------------------------------------------|
| 9906.0 <sup>‡</sup> 20      | (19/2 <sup>-</sup> ) | <b>E</b>    | J <sup>π</sup> : 1693.3γ Q, ΔJ=2 to (15/2 <sup>-</sup> ); band assignment.   |
| 12277.0 <sup>‡</sup> 32     | (23/2 <sup>-</sup> ) | <b>E</b>    | J <sup>π</sup> : 2370.9γ Q, ΔJ=2 γ to (19/2 <sup>-</sup> ); band assignment. |

<sup>†</sup> E(level) from a least-squares fit to γ-ray energies for levels connected with γ transitions; from particle-transfer reactions or  $^{35}\text{K}$  ε+β<sup>+</sup>-delayed proton decays for other levels.

<sup>‡</sup> Band(A): Band based on f<sub>7/2</sub> orbital.

| <u>γ(<math>^{35}\text{Ar}</math>)</u> |                                                           |                                  |                                  |                            |                                  |                          | <u>Comments</u>                                                                                                                                                                              |
|---------------------------------------|-----------------------------------------------------------|----------------------------------|----------------------------------|----------------------------|----------------------------------|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <u>E<sub>i</sub>(level)</u>           | <u>J<sub>i</sub><sup>π</sup></u>                          | <u>E<sub>γ</sub><sup>†</sup></u> | <u>I<sub>γ</sub><sup>†</sup></u> | <u>E<sub>f</sub></u>       | <u>J<sub>f</sub><sup>π</sup></u> | <u>Mult.<sup>‡</sup></u> |                                                                                                                                                                                              |
| 1184.08                               | 1/2 <sup>+</sup>                                          | 1184.1 3                         | 100                              | 0.0                        | 3/2 <sup>+</sup>                 |                          | E <sub>γ</sub> : weighted average of 1184.0 3 from $^{35}\text{K}$ ε decay and 1184.3 4 from $^{36}\text{Ca}$ εp decay.                                                                      |
| 1750.78                               | (5/2) <sup>+</sup>                                        | 1750.6 3                         | 100                              | 0.0                        | 3/2 <sup>+</sup>                 |                          | E <sub>γ</sub> : weighted average of 1750.5 3 from $^{35}\text{K}$ ε decay, 1750.7 4 from ( $^{24}\text{Mg}, \alpha n \gamma$ ), and 1750.8 5 from ( $^{16}\text{O}, \alpha n \gamma$ ).     |
| 2603.22                               | 7/2 <sup>(+)</sup>                                        | 851.9 9                          | 12.3 33                          | 1750.78 (5/2) <sup>+</sup> |                                  |                          | E <sub>γ</sub> : weighted average of 852 1 from ( $^{24}\text{Mg}, \alpha n \gamma$ ) and 851.8 9 from ( $^{16}\text{O}, \alpha n \gamma$ ).                                                 |
|                                       |                                                           | 2603.0 5                         | 100 10                           | 0.0                        | 3/2 <sup>+</sup>                 | Q                        | I <sub>γ</sub> : weighted average of 10 5 from ( $^{24}\text{Mg}, \alpha n \gamma$ ) and 13.3 33 from ( $^{16}\text{O}, \alpha n \gamma$ ).                                                  |
|                                       |                                                           |                                  |                                  |                            |                                  |                          | E <sub>γ</sub> : weighted average of 2603.0 5 from ( $^{24}\text{Mg}, \alpha n \gamma$ ) and 2602.6 15 from ( $^{16}\text{O}, \alpha n \gamma$ ).                                            |
|                                       |                                                           |                                  |                                  |                            |                                  |                          | I <sub>γ</sub> : other: 100 22 from ( $^{24}\text{Mg}, \alpha n \gamma$ ).                                                                                                                   |
| 2638.01                               | (3/2) <sup>+</sup>                                        | 886.8 <sup>#</sup> 5             | 16 <sup>#</sup> 6                | 1750.78 (5/2) <sup>+</sup> |                                  |                          |                                                                                                                                                                                              |
|                                       |                                                           | 2638.0 <sup>#</sup> 4            | 100 <sup>#</sup> 13              | 0.0                        | 3/2 <sup>+</sup>                 |                          |                                                                                                                                                                                              |
| 2982.79                               | (5/2) <sup>+</sup>                                        | 1798.9 <sup>#</sup> 5            | 3.5 <sup>#</sup> 6               | 1184.08 1/2 <sup>+</sup>   |                                  |                          |                                                                                                                                                                                              |
|                                       |                                                           | 2982.68 <sup>#</sup> 13          | 100 <sup>#</sup> 4               | 0.0                        | 3/2 <sup>+</sup>                 |                          |                                                                                                                                                                                              |
| 3196.98                               | 7/2 <sup>-</sup>                                          | 593.7 2                          | 16.4 30                          | 2603.22 7/2 <sup>(+)</sup> |                                  |                          | E <sub>γ</sub> : weighted average of 593 1 from ( $^{24}\text{Mg}, \alpha n \gamma$ ) and 593.7 2 from ( $^{16}\text{O}, \alpha n \gamma$ ).                                                 |
|                                       |                                                           |                                  |                                  |                            |                                  |                          | I <sub>γ</sub> : weighted average of 16 8 from ( $^{24}\text{Mg}, \alpha n \gamma$ ) and 16.4 30 from ( $^{16}\text{O}, \alpha n \gamma$ ).                                                  |
|                                       |                                                           | 1446.2 2                         | 100 8                            | 1750.78 (5/2) <sup>+</sup> |                                  | D                        | E <sub>γ</sub> : weighted average of 1446.2 2 from ( $^{24}\text{Mg}, \alpha n \gamma$ ), 1446.1 6 from ( $^{16}\text{O}, \alpha n \gamma$ ), and 1446.0 6 from ( $^3\text{He}, n \gamma$ ). |
|                                       |                                                           |                                  |                                  |                            |                                  |                          | I <sub>γ</sub> : other: 100 9 from ( $^{24}\text{Mg}, \alpha n \gamma$ ).                                                                                                                    |
|                                       |                                                           | 3197.0 7                         | 21 5                             | 0.0                        | 3/2 <sup>+</sup>                 | Q                        | E <sub>γ</sub> : from ( $^{24}\text{Mg}, \alpha n \gamma$ ). Other: 3197 6 from ( $^{16}\text{O}, \alpha n \gamma$ ).                                                                        |
|                                       |                                                           |                                  |                                  |                            |                                  |                          | I <sub>γ</sub> : weighted average of 18 5 from ( $^{24}\text{Mg}, \alpha n \gamma$ ) and 24 5 from ( $^{16}\text{O}, \alpha n \gamma$ ).                                                     |
| 4065.0?                               | (1/2 <sup>+</sup> , 3/2 <sup>+</sup> , 5/2 <sup>+</sup> ) | 1426.8 <sup>#</sup> 4            | 100                              | 2638.01 (3/2) <sup>+</sup> |                                  |                          |                                                                                                                                                                                              |
| 4359.0                                | (9/2 <sup>-</sup> )                                       | 1162.0 8                         | 65 24                            | 3196.98 7/2 <sup>-</sup>   |                                  | (D)                      | E <sub>γ</sub> : weighted average of 1162 1 from ( $^{24}\text{Mg}, \alpha n \gamma$ ) and 1162.0 8 from                                                                                     |

Continued on next page (footnotes at end of table)



**Adopted Levels, Gammas (continued)**

| $\gamma(^{35}\text{Ar})$ (continued) |                                                           |                        |                      |         |                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------|-----------------------------------------------------------|------------------------|----------------------|---------|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $E_i(\text{level})$                  | $J_i^\pi$                                                 | $E_\gamma^\dagger$     | $I_\gamma^\dagger$   | $E_f$   | $J_f^\pi$                                                 | Mult. <sup>‡</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                      |                                                           |                        |                      |         |                                                           | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 4359.0                               | (9/2 <sup>-</sup> )                                       | 1756 1                 | 100 15               | 2603.22 | 7/2 <sup>(+)</sup>                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                      |                                                           |                        |                      |         |                                                           | ( $^{16}\text{O}, \alpha\gamma$ ).<br>I <sub>γ</sub> : unweighted average of 41<br>11 from ( $^{24}\text{Mg}, \alpha\gamma$ ) and 88<br>18 from ( $^{16}\text{O}, \alpha\gamma$ ).<br>E <sub>γ</sub> : weighted average of 1756 1<br>from ( $^{24}\text{Mg}, \alpha\gamma$ ) and 1756.3<br>14 from ( $^{16}\text{O}, \alpha\gamma$ ).<br>I <sub>γ</sub> : from ( $^{24}\text{Mg}, \alpha\gamma$ ). Other:<br>100 53 from ( $^{16}\text{O}, \alpha\gamma$ ).                                                                  |
| 4528.3                               | (1/2 <sup>+</sup> , 3/2 <sup>+</sup> , 5/2 <sup>+</sup> ) | 4527.9 <sup>#</sup> 7  | 100                  | 0.0     | 3/2 <sup>+</sup>                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 4725.9                               | 1/2 <sup>+</sup>                                          | 3542.0 <sup>#</sup> 6  | 100 <sup>#</sup> 21  | 1184.08 | 1/2 <sup>+</sup>                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                      |                                                           | 4724.5 <sup>#</sup> 11 | 41 <sup>#</sup> 17   | 0.0     | 3/2 <sup>+</sup>                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 4785.8                               | 1/2 <sup>+</sup> , 3/2 <sup>+</sup> , 5/2 <sup>+</sup>    | 4785.4 <sup>#</sup> 11 | 100                  | 0.0     | 3/2 <sup>+</sup>                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 5384.2                               | (11/2 <sup>-</sup> )                                      | 1025.2 4               | 14 4                 | 4359.0  | (9/2 <sup>-</sup> )                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                      |                                                           |                        |                      |         |                                                           | E <sub>γ</sub> : weighted average of 1025 1<br>from ( $^{24}\text{Mg}, \alpha\gamma$ ) and 1025.2<br>4 from ( $^{16}\text{O}, \alpha\gamma$ ).<br>I <sub>γ</sub> : weighted average of 21 8<br>from ( $^{24}\text{Mg}, \alpha\gamma$ ) and 12 4<br>from ( $^{16}\text{O}, \alpha\gamma$ ).<br>E <sub>γ</sub> : weighted average of 2187.4<br>4 from ( $^{24}\text{Mg}, \alpha\gamma$ ) and<br>2186.8 4 from ( $^{16}\text{O}, \alpha\gamma$ ).<br>I <sub>γ</sub> : other: 100 13 from<br>( $^{24}\text{Mg}, \alpha\gamma$ ). |
|                                      |                                                           | 2187.1 4               | 100 6                | 3196.98 | 7/2 <sup>-</sup>                                          | Q                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 5572.67                              | 3/2 <sup>+</sup>                                          | 1044.4 <sup>#</sup> 4  | 2.5 <sup>#</sup> 8   | 4528.3  | (1/2 <sup>+</sup> , 3/2 <sup>+</sup> , 5/2 <sup>+</sup> ) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                      |                                                           | 1507.4 <sup>#</sup> 5  | 3.7 <sup>#</sup> 8   | 4065.0? | (1/2 <sup>+</sup> , 3/2 <sup>+</sup> , 5/2 <sup>+</sup> ) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                      |                                                           | 2589.8 <sup>#</sup> 1  | 100 <sup>#</sup> 4   | 2982.79 | (5/2 <sup>+</sup> )                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                      |                                                           | 2934.5 <sup>#</sup> 5  | 6.7 <sup>#</sup> 12  | 2638.01 | (3/2 <sup>+</sup> )                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                      |                                                           | 3821.7 <sup>#</sup> 7  | 6.7 <sup>#</sup> 14  | 1750.78 | (5/2 <sup>+</sup> )                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                      |                                                           | 4387.2 <sup>#</sup> 9  | 6.7 <sup>#</sup> 16  | 1184.08 | 1/2 <sup>+</sup>                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                      |                                                           | 5572.3 <sup>#</sup> 10 | 11.7 <sup>#</sup> 31 | 0.0     | 3/2 <sup>+</sup>                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 5613.6                               | (11/2 <sup>-</sup> )                                      | 1254.6 8               | 100                  | 4359.0  | (9/2 <sup>-</sup> )                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 5765.8                               | (13/2 <sup>-</sup> )                                      | 381.6 1                | 100 10               | 5384.2  | (11/2 <sup>-</sup> )                                      | D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                      |                                                           |                        |                      |         |                                                           | E <sub>γ</sub> : weighted average of 381.6<br>1 from ( $^{24}\text{Mg}, \alpha\gamma$ ) and<br>381.5 3 from ( $^{16}\text{O}, \alpha\gamma$ ).                                                                                                                                                                                                                                                                                                                                                                               |
|                                      |                                                           | 1406.9 7               | 17.2 35              | 4359.0  | (9/2 <sup>-</sup> )                                       | (Q)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 8109.7                               | (15/2 <sup>-</sup> )                                      | 2342.6 28              | 100 25               | 5765.8  | (13/2 <sup>-</sup> )                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                      |                                                           | 2725.7 14              | 50 13                | 5384.2  | (11/2 <sup>-</sup> )                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 8212.6                               | (15/2 <sup>-</sup> )                                      | 2446.6 16              | 21 7                 | 5765.8  | (13/2 <sup>-</sup> )                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                      |                                                           | 2828.3 7               | 100 18               | 5384.2  | (11/2 <sup>-</sup> )                                      | Q                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 9906.0                               | (19/2 <sup>-</sup> )                                      | 1693.3 27              | 100 20               | 8212.6  | (15/2 <sup>-</sup> )                                      | Q                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                      |                                                           | 1796.3 25              | 67 20                | 8109.7  | (15/2 <sup>-</sup> )                                      | Q                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 12277.0                              | (23/2 <sup>-</sup> )                                      | 2370.9 25              | 100                  | 9906.0  | (19/2 <sup>-</sup> )                                      | Q                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

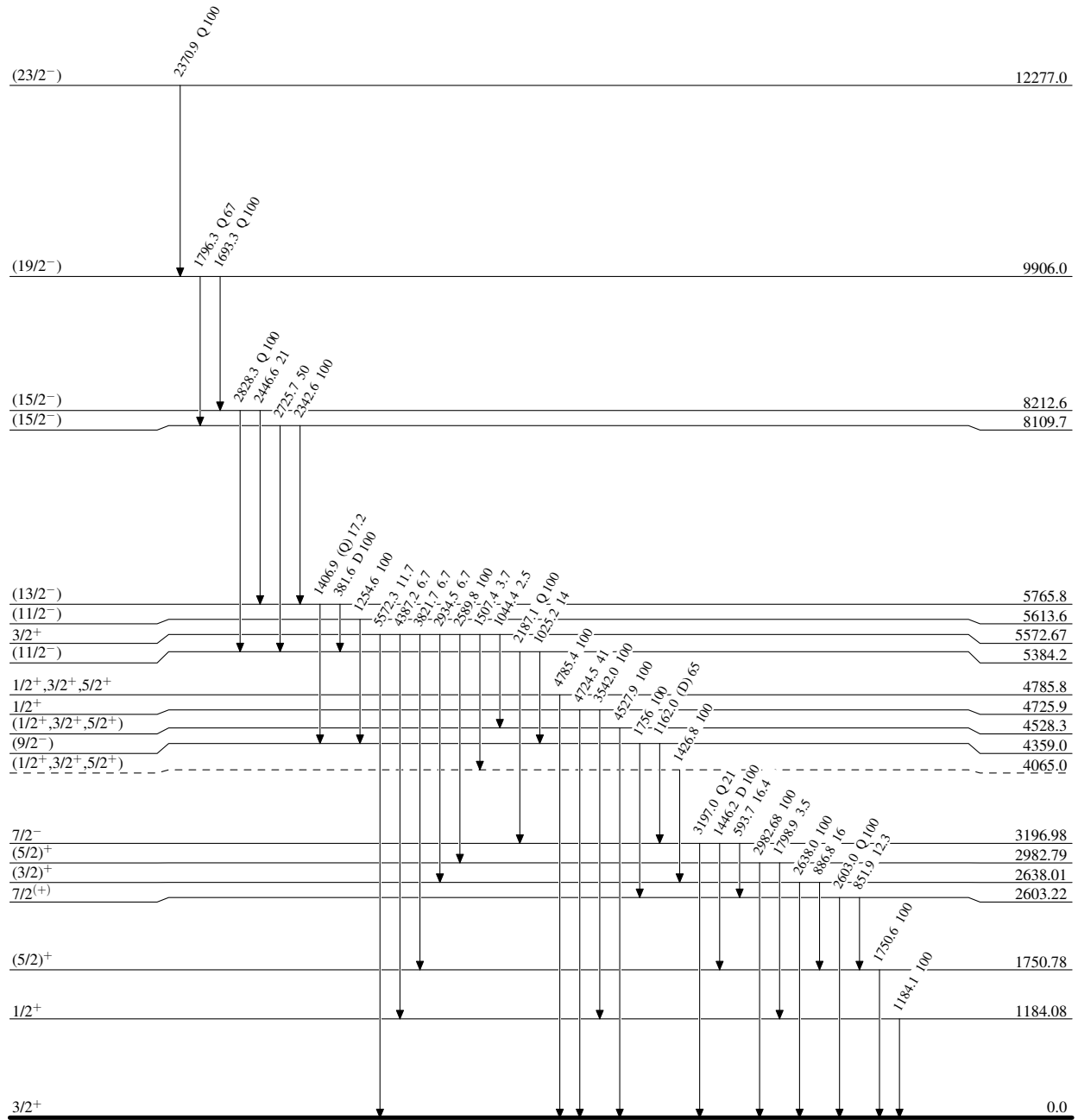
<sup>†</sup> From ( $^{16}\text{O}, \alpha\gamma$ ), unless otherwise noted.

<sup>‡</sup> From ( $^{16}\text{O}, \alpha\gamma$ ) and/or ( $^{24}\text{Mg}, \alpha\gamma$ ). Stretched quadrupole (Q,  $\Delta J=2$ ) or stretched dipole (D,  $\Delta J=1$ ) transitions are assigned by evaluators based on the measured  $\gamma\gamma(\theta)$ (ADO) and/or ratios of yields  $R(\gamma(\theta))$  compared with the expected values given by the authors.

<sup>#</sup> From  $^{35}\text{K}$   $\varepsilon$  decay.

Adopted Levels, GammasLevel Scheme

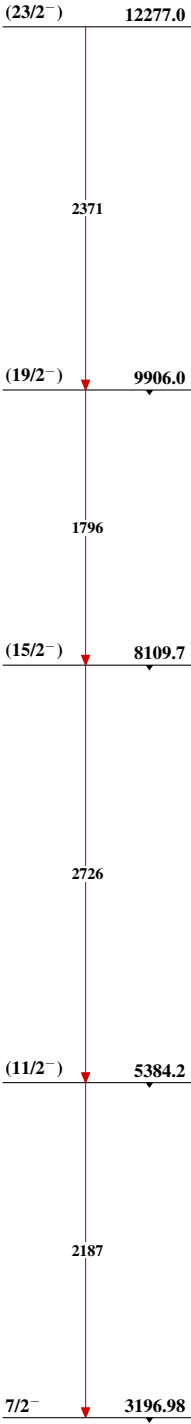
Intensities: Relative photon branching from each level



1.7755 s 14

Adopted Levels, Gammas

Band(A): Band based on f<sub>7/2</sub>  
orbital



<sup>35</sup>Ar<sub>17</sub>

**$^{35}\text{K}$   $\varepsilon+\beta^+$  decay (175 ms) 1980Ew02**

Parent:  $^{35}\text{K}$ :  $E=0$ ;  $J^\pi=3/2^+$ ;  $T_{1/2}=175$  ms 2;  $Q(\varepsilon+\beta^+)=11874.4$  9;  $\% \varepsilon+\beta^+$  decay=100

$^{35}\text{K}$ - $J^\pi$ ,  $T_{1/2}$ : From the Adopted Levels of  $^{35}\text{K}$ . Adopted  $T_{1/2}$  from weighted average of 175 ms 2 (2018Sa54), 178 ms 8 (1998Sc19), and 190 ms 30 (1980Ew02) in this dataset.

$^{35}\text{K}$ - $Q(\varepsilon+\beta^+)$ : From 2021Wa16.

1980Ew02, 1979Ca15: A 600-MeV proton beam was produced from the synchrocyclotron at CERN-ISOLDE and bombarded a  $\text{ScC}_2$  target. The  $^{45}\text{Sc}(p,8n3p)$  spallation reaction products diffused out of the target and reached a tungsten surface ionization source where potassium isotopes were selectively ionized. The beam was extracted from the ion source, separated by the ISOLDE analyzing magnet, and collected by a mylar foil for  $\gamma$ -ray measurements and then a carbon foil for proton measurements.  $\gamma$  rays were detected using a  $\text{Ge}(\text{Li})$  detector. Time for positron activities were determined using a 700- $\mu\text{m}$  thick silicon detector. Protons were detected using a 20- $\mu\text{m}$ -700- $\mu\text{m}$  thick  $\Delta E$ -E telescope of silicon surface barrier detectors with FWHM=50 keV. Measured  $E_\gamma(<5$  MeV),  $I_\gamma$ ,  $E_p(>0.9$  MeV),  $I_p$ . Deduced levels,  $J$ ,  $\pi$ , decay branching ratios,  $\log ft$ , parent  $^{35}\text{K}$   $T_{1/2}$ , and coefficients of the isobaric multiplet mass equation for  $A=35$ ,  $T=3/2$  quartets. Comparisons with shell-model calculations and the mirror nucleus  $^{35}\text{Cl}$ . Also see abstracts 1978HaYH, 1979HaZY, 1979HaZT, and 1979AnZZ.

2018Sa54: A 36-MeV/nucleon  $^{36}\text{Ar}$  primary beam was produced from the K500 cyclotron at Texas A&M University. The secondary  $^{35}\text{K}$  beam was produced via the  $^1\text{H}(^{36}\text{Ar}, ^{35}\text{K})2n$  reaction of  $^{36}\text{Ar}$  bombarding a  $\text{LN}_2$ -cooled hydrogen gas target, separated by MARS, and implanted into a 45- $\mu\text{m}$  DSSD sandwiched between a 140- $\mu\text{m}$  SSSD and a 1-mm Si-pad detector in a pulsed-beam mode.  $\varepsilon+\beta^+$ -delayed protons were detected by the implantation detector.  $\gamma$  rays were detected by two HPGe detectors. Measured  $E_p(>300$  keV),  $I_p$ ,  $E_\gamma$ ,  $I_\gamma$ ,  $p\gamma$ -coin,  $\gamma\gamma$ -coin. Deduced parent  $^{35}\text{K}$   $T_{1/2}$ .

2019ChZU: Same beam production as 2018Sa54.  $^{35}\text{K}$  was implanted into the AstroBox2 detector filled with 800-Torr P5 gas.  $\varepsilon+\beta^+$ -delayed protons were detected by the implantation detector.  $\gamma$  rays were detected by 4 Clover Ge detectors. Measured  $E_p(>100$  keV),  $I_p$ ,  $E_\gamma$ ,  $I_\gamma$ ,  $p\gamma$ -coin,  $\gamma\gamma$ -coin.

1998Sc19: A polarized  $^{35}\text{K}$  beam was produced via the fragmentation of 500-MeV/nucleon  $^{40}\text{Ca}$  impinging on a  $^9\text{Be}$  target at GSI, separated using  $\Delta E$ -tof by FRS, momentum-selected by slits, and implanted into a KBr single crystal placed in the central region of a magnet. Positrons were detected using plastic scintillators.  $\gamma$  rays were detected using a Ge detector. Measured  $\beta$ -decay asymmetry and  $\beta\gamma$ -coin. Deduced polarization and  $g$ -factor of  $^{35}\text{K}$  ground state from  $\beta$ -NMR and  $^{35}\text{K}$   $T_{1/2}$  from  $\beta\gamma$ -decay time spectra.

2006Me04: A polarized  $^{35}\text{K}$  beam was produced via the proton-pickup reaction  $^{36}\text{Ar}(^9\text{Be}, ^{10}\text{Li})^{35}\text{K}$ , separated by NSCL-A1900, and implanted into a KBr crystal. Positrons were detected using plastic scintillators. Deduced the magnetic dipole moment and  $g$ -factor of  $^{35}\text{K}$  ground state from  $\beta$ -NMR.

Theoretical studies involving  $^{35}\text{K}$  decay: shell model (1985Br29, 2003Sm02).

 $^{35}\text{Ar}$  Levels

| $E(\text{level})^\dagger$ | $J^\pi^\ddagger$        | $T_{1/2}^\ddagger$ | Comments                                     |
|---------------------------|-------------------------|--------------------|----------------------------------------------|
| 0                         | $3/2^+$                 | $1.7755$ s 14      |                                              |
| 1184.01 25                | $1/2^+$                 |                    |                                              |
| 1750.72 25                | $(5/2)^+$               |                    |                                              |
| 2637.99 26                | $(3/2)^+$               |                    |                                              |
| 2982.79 12                | $(5/2)^+$               |                    |                                              |
| 4065.0? 4                 | $(1/2^+, 3/2^+, 5/2^+)$ |                    |                                              |
| 4528.2 4                  | $(1/2^+, 3/2^+, 5/2^+)$ |                    |                                              |
| 4725.9 6                  | $1/2^+$                 |                    |                                              |
| 4785.8 11                 | $1/2^+, 3/2^+, 5/2^+$   |                    |                                              |
| 5572.66 15                | $3/2^+$                 |                    | $T=3/2$                                      |
| 6348 11                   | $(1/2, 3/2, 5/2)$       |                    | $E(p0)_{\text{c.m.}}=452$ keV 11 (2019ChZU). |

Continued on next page (footnotes at end of table)

$^{35}\text{K}$   $\varepsilon+\beta^+$  decay (175 ms) **1980Ew02** (continued) $^{35}\text{Ar}$  Levels (continued)

| E(level) <sup>†</sup> | J <sup>π</sup> <sup>‡</sup>                            | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5896+x                |                                                        | E(level): $x < 5978.2$ 5 from $Q(\varepsilon)(^{35}\text{K}) - S(p)(^{35}\text{Ar})$ , where $Q(\varepsilon) = 11874.4$ 9 and $S(p) = 5896.2$ 7 from                                                                                                                                                                                                                                                                                                     |
| 7053 11               | 3/2 <sup>+</sup> , 5/2 <sup>+</sup>                    | E(p0) <sub>c.m.</sub> = 1157 keV 11 ( <b>2019ChZU</b> ).                                                                                                                                                                                                                                                                                                                                                                                                 |
| 7255 11               |                                                        | E(p3) <sub>c.m.</sub> = 693 keV 11 ( <b>2019ChZU</b> ).                                                                                                                                                                                                                                                                                                                                                                                                  |
| 7283 11               |                                                        | E(p0) <sub>c.m.</sub> = 1387 keV 11 ( <b>2019ChZU</b> ).                                                                                                                                                                                                                                                                                                                                                                                                 |
| 7431 11               |                                                        | E(p3) <sub>c.m.</sub> = 869 keV 11 ( <b>2019ChZU</b> ).                                                                                                                                                                                                                                                                                                                                                                                                  |
|                       |                                                        | E(level): weighted average of E(level) of 7497 20, 7510 20, and 7527 11. The former two E(level) are deduced from E(p0) <sub>c.m.</sub> = 1601 20 ( <b>1980Ew02</b> ) and E(p1) <sub>c.m.</sub> = 1467 20 ( <b>1980Ew02</b> ), respectively, with the corresponding E(level)( $^{34}\text{Cl}$ ) ( <b>2012Ni10</b> ) and $S(p)(^{35}\text{Ar}) = 5896.2$ 7 ( <b>2021Wa16</b> ). The 7527 11 is from <b>2019ChZU</b> with E(p3) <sub>c.m.</sub> = 965 11. |
| 8393? 20              | 1/2 <sup>+</sup> , 3/2 <sup>+</sup> , 5/2 <sup>+</sup> | E(level): weighted average of E(level) of 8392 20, 8392 20, and 8395 20, deduced from E(p0) <sub>c.m.</sub> = 2496 20 ( <b>1980Ew02</b> ), E(p1) <sub>c.m.</sub> = 2349 20 ( <b>1980Ew02</b> ), and E(p2) <sub>c.m.</sub> = 2038 20 ( <b>1980Ew02</b> ), respectively, with the corresponding E(level)( $^{34}\text{Cl}$ ) ( <b>2012Ni10</b> ) and $S(p)(^{35}\text{Ar}) = 5896.2$ 7 ( <b>2021Wa16</b> ).                                                |

<sup>†</sup> From a least-squares fit to  $\gamma$ -ray energies in **1980Ew02** for levels connected with  $\gamma$  transitions.

<sup>‡</sup> From the Adopted Levels.

 $\varepsilon, \beta^+$  radiations

| E(decay)                             | E(level) | $I\beta^+$ <sup>‡</sup> | $I\varepsilon$ <sup>‡</sup> | Log $ft$ | $I(\varepsilon+\beta^+)$ <sup>†‡</sup> | Comments                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------|----------|-------------------------|-----------------------------|----------|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (3481 20)                            | 8393?    | 0.062 26                | $4.3 \times 10^{-4}$ 18     | 4.6 +3-2 | 0.062 26                               | $I(\varepsilon+\beta^+)$ : 0.062 26 $I(p0+p1+p2)$ from Table 3 of <b>1980Ew02</b> .                                                                                                                                                                                                                                                                                          |
| (4356 11)                            | 7518     | >0.090                  | $>3 \times 10^{-4}$         | <5.0     | >0.09                                  | $I(\varepsilon+\beta^+)$ : 0.15 6 $I(p0+p1)$ from Table 3 of <b>1980Ew02</b> . Evaluators adopted a lower limit due to unreported $I(p3)$ ( <b>2019ChZU</b> ).                                                                                                                                                                                                               |
| ( $5.5 \times 10^3$ <sup>#</sup> 55) | 5896+x   |                         |                             |          | 0.16 16                                | $I(\varepsilon+\beta^+)$ : $I(\varepsilon+\beta^+) = 0.37$ 15 for all E(p)>0.9 MeV is reported in <b>1980Ew02</b> . The protons placed from the 7518 and 8393 levels in $^{35}\text{Ar}$ sum to $I(\varepsilon+\beta^+) = 0.21$ 7. Therefore, the remaining $I(\varepsilon+\beta^+) = 0.16$ 16 is assigned by the evaluators to the unplaced protons from the 5896+x levels. |
| (5526 11)                            | 6348     | 0.0025 5                | $2.9 \times 10^{-6}$ 6      | 7.2 1    | $2.5 \times 10^{-3}$ 5                 |                                                                                                                                                                                                                                                                                                                                                                              |
| (6301.7 14)                          | 5572.66  | 36.3 24                 | 0.0265 18                   | 3.31 4   | 36.3 24                                |                                                                                                                                                                                                                                                                                                                                                                              |
| (7088.6 18)                          | 4785.8   | 1.0 4                   | $5 \times 10^{-4}$ 2        | 5.2 2    | 1.0 4                                  |                                                                                                                                                                                                                                                                                                                                                                              |
| (7148.5 15)                          | 4725.9   | 2.1 4                   | 0.0010 2                    | 4.9 1    | 2.1 4                                  |                                                                                                                                                                                                                                                                                                                                                                              |
| (7346.2 14)                          | 4528.2   | 0.7 4                   | $3 \times 10^{-4}$ 2        | 5.4 +4-2 | 0.7 4                                  |                                                                                                                                                                                                                                                                                                                                                                              |
| (7809.4 14)                          | 4065.0?  | 0.56 33                 | $2.0 \times 10^{-4}$ 12     | 5.6 +4-2 | 0.56 33                                |                                                                                                                                                                                                                                                                                                                                                                              |
| (8891.6 14)                          | 2982.79  | 26.0 22                 | 0.0060 5                    | 4.27 4   | 26.0 22                                |                                                                                                                                                                                                                                                                                                                                                                              |
| (9236.4 14)                          | 2637.99  | ≤0.4                    |                             | ≥6.2     | ≤0.4                                   |                                                                                                                                                                                                                                                                                                                                                                              |
| (10123.7 14)                         | 1750.72  | 11.9 9                  | 0.00181 14                  | 4.91 4   | 11.9 9                                 |                                                                                                                                                                                                                                                                                                                                                                              |
| (10690.4 14)                         | 1184.01  | 2.2 7                   | $2.8 \times 10^{-4}$ 9      | 5.8 +2-1 | 2.2 7                                  |                                                                                                                                                                                                                                                                                                                                                                              |
| (11874.4 17)                         | 0        | 19 4                    | 0.0018 4                    | 5.1 1    | 19 4                                   | $I(\varepsilon+\beta^+)$ : From <b>1980Ew02</b> assuming mirror log $ft$ with a small asymmetry correction.                                                                                                                                                                                                                                                                  |

<sup>†</sup> From  $\gamma$  intensity balance at each state, except for proton-emitting states. **1980Ew02** authors state that in complex decay schemes of heavy nuclides this method is known to be suspect since there is significant  $\gamma$  intensity that is unobserved because it lies in a

Continued on next page (footnotes at end of table)

$^{35}\text{K}$   $\varepsilon+\beta^+$  decay (175 ms) **1980Ew02** (continued) $\varepsilon, \beta^+$  radiations (continued)

multitude of very weak  $\gamma$ -ray peaks. In a nucleus as light as  $^{35}\text{K}$  the problem is less acute. They have generated a pandemonium test in the same spirit as in [1977Ha51](#) and find that less than one percent of the  $\gamma$  intensity from  $^{35}\text{K}$  decay should be missed for that reason.

‡ Absolute intensity per 100 decays.

# Estimated for a range of levels.

 $\gamma(^{35}\text{Ar})$ 

I $\gamma$  normalization: From  $\Sigma\%I\gamma(\gamma \text{ to g.s.})=80.6 \pm 4.0$ , deduced from  $100-\%(\varepsilon+\beta^+)_{\text{p}}-\%I(\varepsilon+\beta^+)_{\text{g.s.}}$ , where  $\%(\varepsilon+\beta^+)_{\text{p}}=0.37 \pm 1.5$  ([1980Ew02](#)) and  $\%I(\varepsilon+\beta^+)_{\text{g.s.}}=19 \pm 4$  ([1980Ew02](#)), and the latter corresponds to  $\log ft=5.07 \pm 0.5$ , which was deduced from the  $^{35}\text{S}$  (g.s.) to  $^{35}\text{Cl}$  (g.s.) mirror  $\log ft=5.01 \pm 0.2$  with a small asymmetry correction.  $\%(\varepsilon+\beta^+)_{\text{p}}=0.37 \pm 1.5$  for  $E(\text{p})>0.9$  MeV ([1980Ew02](#)). Some weak  $E(\text{p})<0.9$  MeV branches have also been observed ([2018Sa54,2019ChZU](#)).

Conversion coefficients are considered negligibly small.

| $E_{\gamma}^{\dagger}$                                                                                                                | $I_{\gamma}^{\ddagger\#}$ | $E_i(\text{level})$ | $J_i^{\pi}$             | $E_f$   | $J_f^{\pi}$             | Comments                                                                                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------|---------------------------|---------------------|-------------------------|---------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 886.8 5                                                                                                                               | 0.9 3                     | 2637.99             | $(3/2)^+$               | 1750.72 | $(5/2)^+$               | $\%I\gamma=0.46 \pm 1.9-1.7$                                                                                                                                                                                   |
| 1044.4 4                                                                                                                              | 1.3 4                     | 5572.66             | $3/2^+$                 | 4528.2  | $(1/2^+, 3/2^+, 5/2^+)$ | $\%I\gamma=0.66 \pm 2.5-2.3$                                                                                                                                                                                   |
| 1184.0 3                                                                                                                              | 14.3 7                    | 1184.01             | $1/2^+$                 | 0       | $3/2^+$                 | $\%I\gamma=7.2 \pm 5$                                                                                                                                                                                          |
| 1426.8 4                                                                                                                              | 3.0 5                     | 4065.0?             | $(1/2^+, 3/2^+, 5/2^+)$ | 2637.99 | $(3/2)^+$               | $\%I\gamma=1.5 \pm 4-3$                                                                                                                                                                                        |
| 1507.4 5                                                                                                                              | 1.9 4                     | 5572.66             | $3/2^+$                 | 4065.0? | $(1/2^+, 3/2^+, 5/2^+)$ | $\%I\gamma=0.96 \pm 2.7-2.5$                                                                                                                                                                                   |
| 1750.5 3                                                                                                                              | 28 1                      | 1750.72             | $(5/2)^+$               | 0       | $3/2^+$                 | $\%I\gamma=14.1 \pm 9$                                                                                                                                                                                         |
| 1798.9 5                                                                                                                              | 3.5 6                     | 2982.79             | $(5/2)^+$               | 1184.01 | $1/2^+$                 | $\%I\gamma=1.8 \pm 4$                                                                                                                                                                                          |
| 2589.8 1                                                                                                                              | 52 2                      | 5572.66             | $3/2^+$                 | 2982.79 | $(5/2)^+$               | $\%I\gamma=26.3 \pm 1.8$                                                                                                                                                                                       |
| 2638.0 4                                                                                                                              | 5.5 7                     | 2637.99             | $(3/2)^+$               | 0       | $3/2^+$                 | $\%I\gamma=2.8 \pm 5$                                                                                                                                                                                          |
| <sup>x</sup> 2697.7 6                                                                                                                 |                           |                     |                         |         |                         | Unplaced $\gamma$ ray, accounting for no more than 1.2% $\varepsilon+\beta^+$ -feeding ( <a href="#">1980Ew02</a> ). No $^{35}\text{Ar}$ $\gamma$ rays at this energy were observed in other reaction studies. |
| 2934.5 5                                                                                                                              | 3.5 6                     | 5572.66             | $3/2^+$                 | 2637.99 | $(3/2)^+$               | $\%I\gamma=1.8 \pm 4$                                                                                                                                                                                          |
| 2982.68 13                                                                                                                            | 100 4                     | 2982.79             | $(5/2)^+$               | 0       | $3/2^+$                 | $\%I\gamma=50.5 \pm 2.7$                                                                                                                                                                                       |
| 3542.0 6                                                                                                                              | 2.9 6                     | 4725.9              | $1/2^+$                 | 1184.01 | $1/2^+$                 | $\%I\gamma=1.5 \pm 4$                                                                                                                                                                                          |
| 3821.7 7                                                                                                                              | 3.5 7                     | 5572.66             | $3/2^+$                 | 1750.72 | $(5/2)^+$               | $\%I\gamma=1.8 \pm 5$                                                                                                                                                                                          |
| 4387.2 9                                                                                                                              | 3.5 8                     | 5572.66             | $3/2^+$                 | 1184.01 | $1/2^+$                 | $\%I\gamma=1.8 \pm 5$                                                                                                                                                                                          |
| 4527.9 7                                                                                                                              | 2.6 7                     | 4528.2              | $(1/2^+, 3/2^+, 5/2^+)$ | 0       | $3/2^+$                 | $\%I\gamma=1.3 \pm 4$                                                                                                                                                                                          |
| 4724.5 11                                                                                                                             | 1.2 5                     | 4725.9              | $1/2^+$                 | 0       | $3/2^+$                 | $\%I\gamma=0.61 \pm 3.0-2.7$                                                                                                                                                                                   |
| 4785.4 11                                                                                                                             | 1.9 7                     | 4785.8              | $1/2^+, 3/2^+, 5/2^+$   | 0       | $3/2^+$                 | $\%I\gamma=1.0 \pm 4$                                                                                                                                                                                          |
| 5572.3 10                                                                                                                             | 6.1 16                    | 5572.66             | $3/2^+$                 | 0       | $3/2^+$                 | $\%I\gamma=3.1 \pm 1.0-0.9$                                                                                                                                                                                    |
| <b>1980Ew02</b> observed the double escape peak at 4550 keV of this $\gamma$ ray. <b>2018Sa54</b> observed the photopeak at 5572 keV. |                           |                     |                         |         |                         |                                                                                                                                                                                                                |

<sup>†</sup> From [1980Ew02](#).

<sup>‡</sup> For absolute intensity per 100 decays, multiply by 0.505  $\pm$  0.29.

<sup>x</sup>  $\gamma$  ray not placed in level scheme.



**<sup>36</sup>Ca  $\varepsilon\beta$  decay (100.9 ms) 1997Tr05,2001Lo11,2015Su01**

Parent: <sup>36</sup>Ca: E=0; J<sup>π</sup>=0<sup>+</sup>; T<sub>1/2</sub>=100.9 ms 20; Q( $\varepsilon\beta$ )=9275 6; % $\varepsilon\beta$  decay=54.1 12

<sup>36</sup>Ca-J<sup>π</sup>: From the Adopted Levels of <sup>36</sup>Ca (2012Ni01).

<sup>36</sup>Ca-T<sub>1/2</sub>: Weighted average of 102 ms 2 (1995Tr02,1997Tr05), 100.1 ms 23 (2007Do17), and 100.0 ms 24 (2015Su01). Other: 100 ms +90-40 (1981Ay01).

<sup>36</sup>Ca-Q( $\varepsilon\beta$ ): Deduced by evaluators from <sup>36</sup>Ca mass excess of -6483 6; weighted average of -6483.6 56 (2015Su04), -6480 40 (2021La04), and -6450 40 (2021Wa16,1977Tr03), and <sup>35</sup>Ar mass excess of -23047.3 7 (2021Wa16). Q( $\varepsilon\beta$ ) from 2021Wa16: 9310 40.

<sup>36</sup>Ca-% $\varepsilon\beta$  decay: Unweighted average of % $\varepsilon+\beta^+$ p=56.8 13 (1997Tr05), 54.3 18 (2001Lo11), 51.2 10 (2007Do17), and 53.9 72 (2015Su01).

1997Tr05,1995Tr02: a-300 MeV/nucleon <sup>40</sup>Ca primary beam was produced by the GSI heavy-ion synchrotron. The secondary <sup>36</sup>Ca beam was produced via the projectile fragmentation of <sup>40</sup>Ca impinging on a <sup>9</sup>Be target and was selected using  $\Delta E$ -tof-B $\rho$  by FRS at GSI, Darmstadt. A total of  $2.8 \times 10^4$  <sup>36</sup>Ca ions were implanted into a 500- $\mu$ m-thick Si detector.  $\varepsilon+\beta^+$ -delayed protons were detected by the implantation detector.  $\beta$  particles were detected by the implantation detector and two 500- $\mu$ m-thick Si counters.  $\gamma$  rays were detected by two Ge detectors. Measured E<sub>p</sub>, I<sub>p</sub>, E $\gamma$ , I $\gamma$ ,  $\beta\beta$ -coin,  $\beta\gamma$ -coin, and  $\beta\gamma$ -coin. Deduced levels, decay branching ratios, log *ft*, B(F), and B(GT). Deduced parent <sup>36</sup>Ca T<sub>1/2</sub> from the time spectrum of proton events accumulated during the beam-off in the pulsed-beam mode. Comparisons with shell-model calculations.

2001Lo11: A 95-MeV <sup>40</sup>Ca primary beam was produced by the SSI facility at GANIL. The secondary <sup>36</sup>Ca beam was produced via the projectile fragmentation of <sup>40</sup>Ca impinging on a <sup>nat</sup>Ni target and was selected using  $\Delta E$ -tof by the LISE3 spectrometer and purified by a velocity filter. A total of 102407 <sup>36</sup>Ca ions were implanted into a 500- $\mu$ m-thick Si detector.  $\varepsilon+\beta^+$ -delayed protons were detected by the implantation detector.  $\beta$  particles were detected by two 500- $\mu$ m-thick Si counters.  $\gamma$  rays were detected by three Ge detectors. Measured E<sub>p</sub>, I<sub>p</sub>, E $\gamma$ , I $\gamma$ ,  $\beta\beta$ -coin, and  $\beta\gamma$ -coin. Deduced levels, decay branching ratios, log *ft*, B(F), and B(GT). Comparisons with shell-model calculations.

2015Su01: A 69.42-MeV/nucleon <sup>40</sup>Ca primary beam was produced by the Sector Focusing Cyclotron and Separated Sector Cyclotron at the Heavy Ion Research Facility in Lanzhou (HIRFL). The secondary <sup>36</sup>Ca beam was produced via the projectile fragmentation of <sup>40</sup>Ca impinging on a <sup>9</sup>Be target and was selected using  $\Delta E$ -tof-B $\rho$  by RIBLL. A total of 22890 <sup>36</sup>Ca ions were implanted into a 525- $\mu$ m-thick DSSD.  $\varepsilon+\beta^+$ -delayed protons were detected by the DSSD with a threshold of 500 keV.  $\varepsilon+\beta^+$ -delayed  $\gamma$  rays were detected by four Clover Ge detectors surrounding the DSSD chamber. Measured E<sub>p</sub>, I<sub>p</sub>, E $\gamma$ , I $\gamma$ ,  $\beta\beta$ -coin,  $\beta\gamma$ -coin, and implant-decay time correlations. Deduced levels, decay branching ratios, and parent <sup>36</sup>Ca T<sub>1/2</sub>. Also see 2016Li45.

2007Do17: A 74.5-MeV/nucleon <sup>58</sup>Ni primary beam was produced by the SSI facility at GANIL. The secondary <sup>36</sup>Ca beam was produced via the projectile fragmentation of <sup>58</sup>Ni impinging on a <sup>nat</sup>Ni target and was selected using  $\Delta E$ -tof-B $\rho$  by the ALPHA-LISE3 separator. A total of 16991 <sup>36</sup>Ca ions were implanted into a 500- $\mu$ m thick DSSD.  $\varepsilon+\beta^+$ -delayed protons were detected by the DSSD with a threshold of 60-80 keV.  $\varepsilon+\beta^+$ -delayed  $\gamma$  rays were detected by four Ge detectors surrounding the implantation array. A 5-mm thick lithium-drifted Si detector was used as a veto for implantation events and to detect  $\beta$  particles. Measured E<sub>p</sub>, I<sub>p</sub>, and implant-decay time correlations. Deduced levels, decay branching ratios, and parent <sup>36</sup>Ca T<sub>1/2</sub>.

1995Ga16: 60-keV <sup>36</sup>Ca was produced by the ISOLDE general-purpose on-line isotope separator at the CERN PS/Booster and implanted into the entrance window of a gas-Si-Si  $\Delta E$ -E-veto detector telescope. Measured E<sub>p</sub>. Deduced coefficients of the isobaric multiplet mass equation for A=36, T=2 quintets. A by-product of <sup>37</sup>Ca decay study (1995Ga03).

1981Ay01,1980AyZZ: <sup>36</sup>Ca was produced via the <sup>40</sup>Ca(<sup>3</sup>He, $\alpha$ n) reaction using a 95-MeV <sup>3</sup>He beam from the 88-inch Cyclotron at Lawrence Berkeley Laboratory.  $\beta$ -delayed protons were detected using a Si surface barrier detector telescope with FWHM=55 keV and a minimum threshold of  $\approx 1.5$  MeV. Measured E<sub>p</sub>. Deduced <sup>36</sup>Ca T<sub>1/2</sub> and coefficients of the isobaric multiplet mass equation for A=36, T=2 quintets.

Theoretical studies involving <sup>36</sup>Ca decay: shell model (1984Mu25,1990Br26), covariant density functional theory (2013Ni09).

<sup>35</sup>Ar Levels

| E(level) | J <sup>π</sup> <sup>†</sup> | T <sub>1/2</sub> <sup>†</sup> | Comments                        |
|----------|-----------------------------|-------------------------------|---------------------------------|
| 0        | 3/2 <sup>+</sup>            | 1.7755 s 14                   |                                 |
| 1184.3 4 | 1/2 <sup>+</sup>            |                               | E(level): From E $\gamma$ data. |

<sup>†</sup> From the Adopted Levels.



$^{36}\text{Ca}$   $\varepsilon$ p decay (100.9 ms) [1997Tr05](#), [2001Lo11](#), [2015Su01](#) (continued) $\gamma(^{35}\text{Ar})$ 

I $\gamma$  normalization: From 100/% $\varepsilon$ + $\beta^+$ p.

| $E_\gamma$ | $I_\gamma^\dagger$ | $E_i(\text{level})$ | $J_i^\pi$        | $E_f$ | $J_f^\pi$        | Comments                                                                                                                                                        |
|------------|--------------------|---------------------|------------------|-------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1184.3 4   | 1.2 4              | 1184.3              | 1/2 <sup>+</sup> | 0     | 3/2 <sup>+</sup> | $E_\gamma$ : weighted average of 1184.2 4 ( <a href="#">1997Tr05</a> ) and 1185 1 ( <a href="#">2001Lo11</a> ).<br>$I_\gamma$ : From <a href="#">1997Tr05</a> . |

<sup>†</sup> Absolute intensity per 100 decays.

Delayed Protons ( $^{35}\text{Ar}$ )

| $E(p)^\dagger$       | $E(^{35}\text{Ar})$ | $I(p)^\#$ | $E(^{36}\text{K})^\ddagger$ | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------|---------------------|-----------|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $1.37 \times 10^3$   | 1184.3              | 1.2 4     | 4281.7                      | $E(p), I(p)$ : From <a href="#">1997Tr05</a> .<br>$E(^{36}\text{K})$ : $J^\pi=0$ , T=2 isobaric analog state in $^{36}\text{K}$ .                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 1648 18              | 0                   | 8.4 12    | 3354                        | $E(p)$ : weighted average of 1676 39 ( <a href="#">1997Tr05</a> ), 1645 21 ( <a href="#">2001Lo11</a> ), 1660 18 ( <a href="#">2007Do17</a> ), and 1624 22 ( <a href="#">2015Su01</a> ).<br>$I(p)$ : unweighted average of 11.3 6 ( <a href="#">1997Tr05</a> ), 9.3 8 ( <a href="#">2001Lo11</a> ), 7.3 8 ( <a href="#">2007Do17</a> ), and 5.7 16 ( <a href="#">2015Su01</a> ).                                                                                                                                                                                               |
| 2549.9 22            | 0                   | 37.4 10   | 4281.7                      | $E(p)$ : weighted average of 2519 21 ( <a href="#">1981Ay01</a> ), 2550.2 22 ( <a href="#">1995Ga16</a> ), 2548 37 ( <a href="#">1997Tr05</a> ), 2551 21 ( <a href="#">2001Lo11</a> ), 2538 18 ( <a href="#">2007Do17</a> ), and 2594 30 ( <a href="#">2015Su01</a> ).<br>$I(p)$ : weighted average of 37.8 10 ( <a href="#">1997Tr05</a> ), 32.1 42 ( <a href="#">2007Do17</a> ), and 34.0 58 ( <a href="#">2015Su01</a> ). Other: 37 1 ( <a href="#">2001Lo11</a> ) without separating the weaker proton branch from the same level to $^{35}\text{Ar}$ first excited state. |
| 2713 21              | 0                   | 2.6 9     | 4449                        | $E(^{36}\text{K})$ : $J^\pi=0$ , T=2 isobaric analog state in $^{36}\text{K}$ .<br>$E(p)$ : weighted average of 2713 31 ( <a href="#">1997Tr05</a> ) and 2713 21 ( <a href="#">2001Lo11</a> ).                                                                                                                                                                                                                                                                                                                                                                                 |
| 2921 35              | 0                   | 1.3 2     | 4663                        | $I(p)$ : unweighted average of 1.7 2 ( <a href="#">1997Tr05</a> ) and 3.5 5 ( <a href="#">2001Lo11</a> ).<br>$E(p)$ : weighted average of 2937 35 ( <a href="#">1997Tr05</a> ) and 2895 44 ( <a href="#">2001Lo11</a> ).                                                                                                                                                                                                                                                                                                                                                       |
| 3484 21              | 0                   | 0.6 2     | 5242                        | $I(p)$ : weighted average of 1.4 2 ( <a href="#">1997Tr05</a> ) and 1.0 3 ( <a href="#">2001Lo11</a> ).<br>$E(p), I(p)$ : From <a href="#">2001Lo11</a> .                                                                                                                                                                                                                                                                                                                                                                                                                      |
| $3.98 \times 10^3$ 7 | 0                   | 0.9 2     | 5753                        | $E(p), I(p)$ : From <a href="#">2001Lo11</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| $4.15 \times 10^3$ 5 | 0                   | 2.2 5     | 5927                        | $E(p)$ : weighted average of 4162 45 ( <a href="#">1997Tr05</a> ) and 4135 44 ( <a href="#">2001Lo11</a> ).<br>$I(p)$ : unweighted average of 2.7 4 ( <a href="#">1997Tr05</a> ) and 1.7 3 ( <a href="#">2001Lo11</a> ).                                                                                                                                                                                                                                                                                                                                                       |
| $4.99 \times 10^3$ 7 | 0                   | 0.4 2     | 6791                        | $E(p)$ : weighted average of 4989 69 ( <a href="#">1997Tr05</a> ) and 4983 67 ( <a href="#">2001Lo11</a> ).<br>$I(p)$ : weighted average of 0.7 2 ( <a href="#">1997Tr05</a> ) and 0.3 1 ( <a href="#">2001Lo11</a> ).                                                                                                                                                                                                                                                                                                                                                         |

<sup>†</sup> In lab frame. Evaluators take a weighted average when multiple measurements of a  $E(p)$  are available. [2007Do17](#) and [2015Su01](#) reported  $E_{c.m.}$ . [1997Tr05](#) and [2001Lo11](#) reported the  $E(\text{level})$  of  $^{36}\text{K}$  proton-emitting levels. Evaluators deduced proton center-of-mass energies  $E_{c.m.}=E(\text{level})(^{36}\text{K})-S(p)(^{36}\text{K})-E(\text{level})(^{35}\text{Ar})$  using their original  $S(p)=1666$  8. Evaluators then deduced each  $E(p)_{\text{lab}}=E_{c.m.} \times m(^{35}\text{Ar})/[m(p)+m(^{35}\text{Ar})]$ .

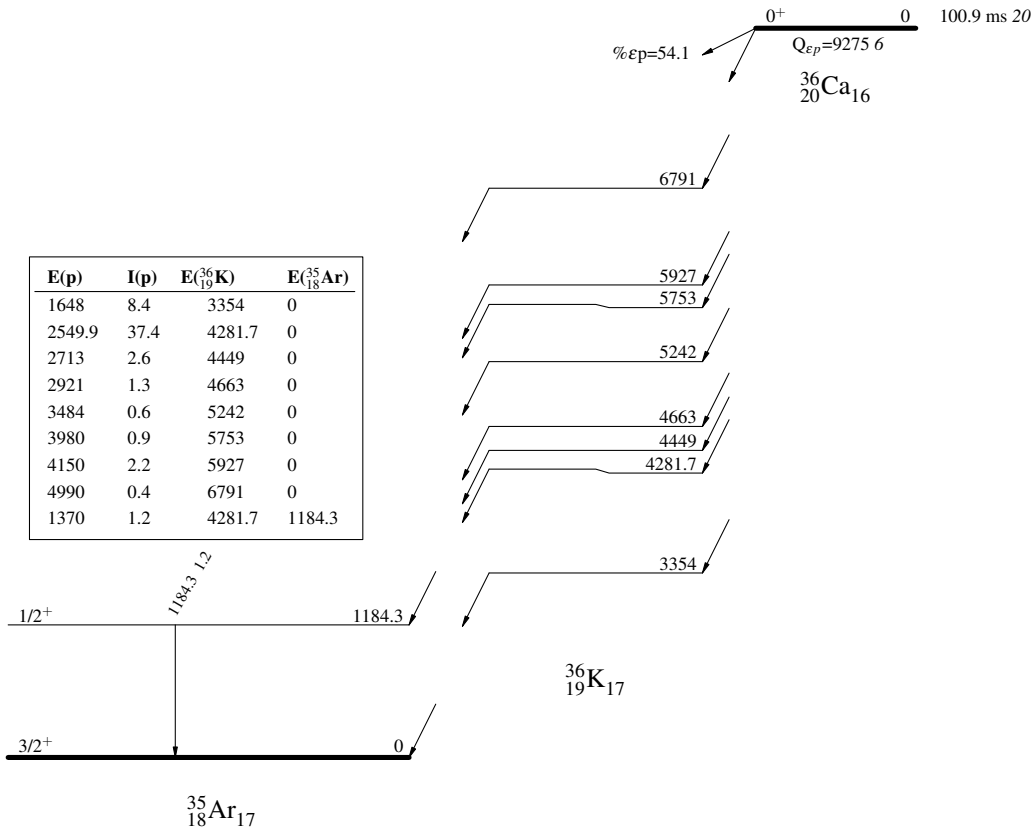
<sup>‡</sup>  $E(\text{level})(^{36}\text{K})=E(p)_{\text{lab}} \times [m(p)+m(^{35}\text{Ar})]/m(^{35}\text{Ar})+S(p)(^{36}\text{K})+E(\text{level})(^{35}\text{Ar})$ , where  $S(p)(^{36}\text{K})=1658.9$  8 ([2021Wa16](#)).

<sup>#</sup> Absolute intensity per 100 decays.

<sup>36</sup>Ca εp decay (100.9 ms) 1997Tr05,2001Lo11,2015Su01

Decay Scheme

γ Intensities: I<sub>γ</sub> per 100 parent decays  
I(p) Intensities: I(p) per 100 parent decays



$^1\text{H}(^{36}\text{Ar},\text{d})$     [2010Le03,2011Le01](#)

$J^\pi=0^+$  for  $^{36}\text{Ar}$  ground state.

[2010Le03](#), [2011Le01](#): A  $^{36}\text{Ar}$  beam at 33 MeV/u was provided at the National Superconducting Cyclotron Laboratory, MSU.

Targets were polyethylene ( $\text{CH}_2$ )<sub>n</sub>. Deuterons were detected using the High-Resolution Array (HiRA) of Si and CsI(Tl) telescope detectors in coincidence with recoil residues identified in the S800 spectrometer by the focal plane ionization chamber andToF.

Measured  $\sigma(E_d,\theta)$  in inverse kinematics. Deduced neutron spectroscopic factors from adiabatic distorted wave approximation (ADWA) analysis of the measured  $\sigma(\theta)$  using Chapel-Hill global optical potential parameters (CH89) and JLM optical potentials and geometry for transferred neutron constrained by Hartree–Fock calculations (JLM+HF). Comparisons with shell–model calculated spectroscopic factors.

Theoretical studies involving  $^1\text{H}(^{36}\text{Ar},\text{d})^{35}\text{Ar}$ : [2011Nu01](#), [2023He15](#).

 $^{35}\text{Ar}$  Levels

| <u>E(level)<sup>†</sup></u> | <u><math>J^\pi</math></u> | <u><math>L^\ddagger</math></u> | <u><math>C^2S^\ddagger</math></u> | <u>Comments</u>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------|---------------------------|--------------------------------|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0                           | $3/2^+$                   | 2                              | 2.3 2                             | C <sup>2</sup> S: others: 1.6 <i>I</i> from <a href="#">2011Le01</a> using JLM+HF calculations; 2.10 from large basis–shell model calculations ( <a href="#">2010Le03</a> ); 2.21 <i>49</i> from a reanalysis of the $\sigma(\theta)$ data using finite-range ADWA ( <a href="#">2011Nu01</a> ), including theoretical uncertainties associated with optical potentials (7%) and the approximate solution of three-body problems (19%); 2.1 +2–4 from a reanalysis of the $\sigma(\theta)$ data using ADWA within a Bayesian framework ( <a href="#">2023He15</a> ), including theoretical uncertainties associated with optical potentials. |
| 1180                        |                           | 0                              | 1.2 <i>I</i>                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 2980 <sup>#</sup>           |                           |                                |                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 3190 <sup>#</sup>           |                           |                                |                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 5570                        |                           |                                |                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

<sup>†</sup> From [2010Le03](#).

<sup>‡</sup> From ADWA analysis of measured  $\sigma(\theta)$  in [2011Le01](#) using CH89 optical potential parameters.

<sup>#</sup> Doublet in measured spectra.

$^{16}\text{O}(^{24}\text{Mg},\alpha n\gamma)$  2004Ek01,2005Ek01

**2004Ek01,2005Ek01:** A 60-MeV  $^{24}\text{Mg}$  beam was produced at the Legnaro National Laboratory, Italy. The target was 0.5-mg/cm<sup>2</sup> enriched  $^{40}\text{Ca}$  with a 7-mg/cm<sup>2</sup> tantalum backing. Oxygen was present in the target, giving rise to the fusion evaporation reactions of  $^{16}\text{O}(^{24}\text{Mg},\alpha n\gamma)^{35}\text{Ar}$  and  $^{16}\text{O}(^{24}\text{Mg},\alpha p\gamma)^{35}\text{Cl}$ .  $\gamma$  rays were detected using the GASP array of Ge detectors and 80 BGO detectors. Charged particles were detected using the ISIS array of 40 Si  $\Delta E$ -E telescopes. Neutrons were detected using a Neutron Ring replacing the six BGO elements at the most forward angles. The event trigger required one Ge detector, one BGO detector, and one neutron detector, or two Ge detectors and one BGO detector firing. Measured  $E\gamma$ ,  $I\gamma$ ,  $\gamma\gamma$ ,  $\alpha n\gamma$ -coin, and  $\alpha p\gamma$ -coin. Deduced levels, J,  $\pi$  from the measured ratios of yields  $R(\gamma(\theta))$  and comparisons with the mirror nucleus  $^{35}\text{Cl}$ .

 $^{35}\text{Ar}$  Levels

| <u>E(level)<sup>†</sup></u> | <u>J<math>\pi</math><sup>‡</sup></u> |
|-----------------------------|--------------------------------------|
| 0.0                         | 3/2 <sup>+</sup>                     |
| 1750.8 3                    | 5/2 <sup>+</sup>                     |
| 2603.2 4                    | (7/2 <sup>+</sup> )                  |
| 3197.0 4                    | 7/2 <sup>(-)</sup>                   |
| 4359.2 7                    | (9/2 <sup>-</sup> )                  |
| 5384.4 5                    | 11/2 <sup>(-)</sup>                  |
| 5766.0 5                    | 13/2 <sup>(-)</sup>                  |

<sup>†</sup> From a least-squares fit to  $\gamma$ -ray energies.

<sup>‡</sup> As given in **2004Ek01** based on known assignments of low-lying levels and mirror levels in  $^{35}\text{Cl}$  and the measured ratios of yields  $R(\gamma(\theta))$ . When considered in the Adopted Levels, the firm assignments here are placed within parentheses if there are no other strong arguments to support these firm assignments.

 $\gamma(^{35}\text{Ar})$ 

The ratios of yields  $R(\gamma(\theta))$  were measured at 35° and 81° with respect to the beam axis. Expected values are  $R(\gamma(\theta))\approx 1.2$  for stretched quadrupole ( $\Delta J=2$ ) and  $R(\gamma(\theta))\approx 0.7$  for stretched dipole ( $\Delta J=1$ ) transitions.

| <u><math>E_\gamma</math><sup>†</sup></u> | <u><math>I_\gamma</math><sup>†</sup></u> | <u><math>E_i(\text{level})</math></u> | <u><math>J_i^\pi</math></u> | <u><math>E_f</math></u> | <u><math>J_f^\pi</math></u> | <u>Mult.<sup>#</sup></u> | <u>Comments</u>                                                                                          |
|------------------------------------------|------------------------------------------|---------------------------------------|-----------------------------|-------------------------|-----------------------------|--------------------------|----------------------------------------------------------------------------------------------------------|
| 381.6 1                                  | 26 3                                     | 5766.0                                | 13/2 <sup>(-)</sup>         | 5384.4                  | 11/2 <sup>(-)</sup>         | D                        | $R(\gamma(\theta))=0.69$ 18.                                                                             |
| 593 1                                    | 12 6                                     | 3197.0                                | 7/2 <sup>(-)</sup>          | 2603.2                  | (7/2 <sup>+</sup> )         |                          |                                                                                                          |
| 852 <sup>‡</sup> 1                       | 4 2                                      | 2603.2                                | (7/2 <sup>+</sup> )         | 1750.8                  | 5/2 <sup>+</sup>            |                          |                                                                                                          |
| 1025 <sup>‡</sup> 1                      | 5 2                                      | 5384.4                                | 11/2 <sup>(-)</sup>         | 4359.2                  | (9/2 <sup>-</sup> )         |                          |                                                                                                          |
| 1162 1                                   | 11 3                                     | 4359.2                                | (9/2 <sup>-</sup> )         | 3197.0                  | 7/2 <sup>(-)</sup>          |                          |                                                                                                          |
| 1446.2 2                                 | 76 7                                     | 3197.0                                | 7/2 <sup>(-)</sup>          | 1750.8                  | 5/2 <sup>+</sup>            | D                        | $R(\gamma(\theta))=0.71$ 9.                                                                              |
| 1750.7 4                                 | 100 7                                    | 1750.8                                | 5/2 <sup>+</sup>            | 0.0                     | 3/2 <sup>+</sup>            |                          | $R(\gamma(\theta))=1.41$ 14 indicates $\Delta J=2$ , but $\Delta J=1$ from level scheme is inconsistent. |
| 1756 <sup>‡</sup> 1                      | 27 4                                     | 4359.2                                | (9/2 <sup>-</sup> )         | 2603.2                  | (7/2 <sup>+</sup> )         |                          |                                                                                                          |
| 2187.4 4                                 | 24 3                                     | 5384.4                                | 11/2 <sup>(-)</sup>         | 3197.0                  | 7/2 <sup>(-)</sup>          | Q                        | $R(\gamma(\theta))=1.60$ 36.                                                                             |
| 2603.0 5                                 | 41 9                                     | 2603.2                                | (7/2 <sup>+</sup> )         | 0.0                     | 3/2 <sup>+</sup>            |                          | $R(\gamma(\theta))=1.01$ 17.                                                                             |
| 3197.0 7                                 | 14 4                                     | 3197.0                                | 7/2 <sup>(-)</sup>          | 0.0                     | 3/2 <sup>+</sup>            | Q                        | $R(\gamma(\theta))=1.45$ 51.                                                                             |

<sup>†</sup> From **2004Ek01**.

<sup>‡</sup> Tentative placements; later confirmed by  $^{24}\text{Mg}(^{16}\text{O},\alpha n\gamma)^{35}\text{Ar}$  (**2007De14**) and adopted by evaluators.

<sup>#</sup> Not explicitly given in **2004Ek01**, but deduced by evaluators based on their measured ratios of yields  $R(\gamma(\theta))$  compared with expected values as stated in **2004Ek01**.

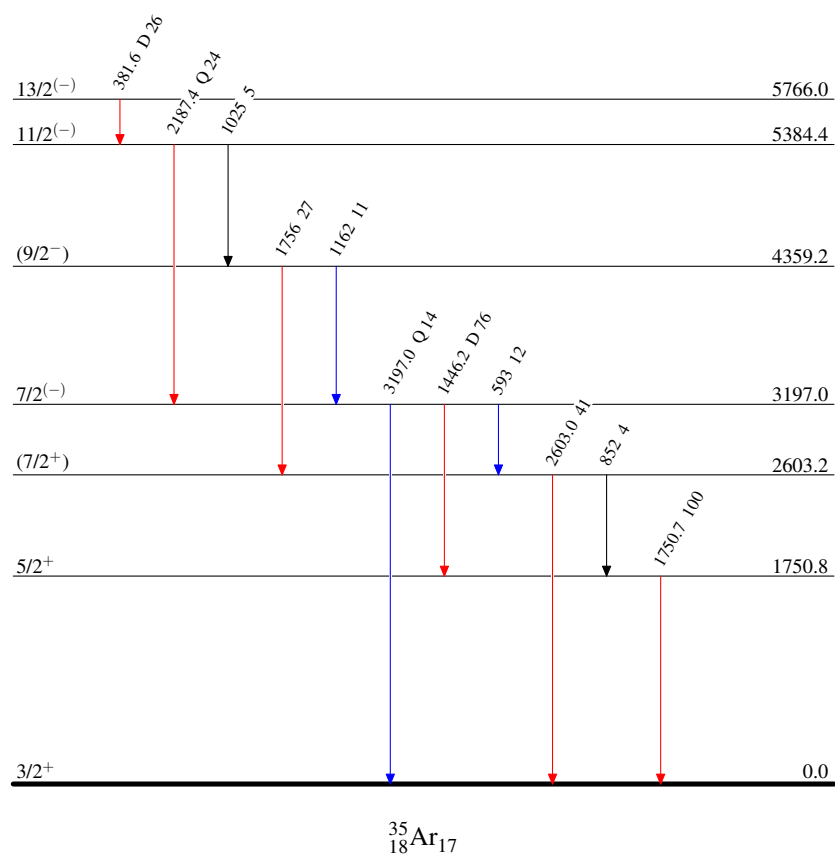
$^{16}\text{O}(^{24}\text{Mg}, \alpha n \gamma)$  2004Ek01, 2005Ek01

## Level Scheme

Intensities: Relative  $I_\gamma$ 

## Legend

- $I_\gamma < 2\% \times I_\gamma^{\max}$   
—→  $I_\gamma < 10\% \times I_\gamma^{\max}$   
—→  $I_\gamma > 10\% \times I_\gamma^{\max}$

 $^{35}_{18}\text{Ar}_{17}$

$^{24}\text{Mg}(^{16}\text{O},\alpha n\gamma)$  2007De14,2005DeZZ

**2007De14,2005DeZZ:** A 70-MeV  $^{16}\text{O}$  beam was produced by the XTU-Tandem accelerator at the Legnaro National Laboratory, Italy. The target was a 400  $\mu\text{g}/\text{cm}^2$  self-supporting target of  $^{24}\text{Mg}$ .  $\gamma$  ray from fusion evaporation reactions of  $^{24}\text{Mg}(^{16}\text{O},\alpha n\gamma)^{35}\text{Ar}$  and  $^{24}\text{Mg}(^{16}\text{O},\alpha p\gamma)^{35}\text{Cl}$  were detected using the GASP spectrometer, which consists of an array of 40 Compton-suppressed HPGe detectors and a multiplicity filter of 80 BGO scintillators. The GASP spectrometer is operated in conjunction with the  $4\pi$  charged-particle detector ISIS and a neutron ring of 6 BC501A scintillators. The events were collected when at least two Ge detectors and one BGO scintillator were fired in coincidence. Measured  $E_\gamma$ ,  $I_\gamma$ ,  $\gamma\gamma$ ,  $\alpha n\gamma$ -coin,  $\alpha p\gamma$ -coin,  $\gamma(\theta)$ ,  $\gamma\gamma(\theta)$ (ADO). Deduced levels, J,  $\pi$ , and transition multipolarities from  $\gamma$ -ray ADO ratios. Comparisons with Shell-model calculations show the multipole Coulomb interaction and the electromagnetic spin-orbit interaction contribute to the observed mirror energy differences. 2007De14 also report data on  $^{35}\text{Cl}$  and propose the level scheme of  $^{35}\text{Ar}$  based on mirror symmetry.

 $^{35}\text{Ar}$  Levels

| <u><math>E(\text{level})^\dagger</math></u> | <u><math>J^\pi^\ddagger</math></u> | <u><math>E(\text{level})^\dagger</math></u> | <u><math>J^\pi^\ddagger</math></u> | <u><math>E(\text{level})^\dagger</math></u> | <u><math>J^\pi^\ddagger</math></u> | <u><math>E(\text{level})^\dagger</math></u> | <u><math>J^\pi^\ddagger</math></u> |
|---------------------------------------------|------------------------------------|---------------------------------------------|------------------------------------|---------------------------------------------|------------------------------------|---------------------------------------------|------------------------------------|
| 0.0                                         | 3/2 <sup>+</sup>                   | 3196.7 <sup>#</sup> 7                       | 7/2 <sup>-</sup>                   | 5613.2 11                                   | (11/2 <sup>-</sup> )               | 8212.1 10                                   | 15/2 <sup>-</sup>                  |
| 1750.8 5                                    | 5/2 <sup>+</sup>                   | 4358.6 8                                    | 9/2 <sup>-</sup>                   | 5765.3 8                                    | 13/2 <sup>-</sup>                  | 9905.5 <sup>#</sup> 21                      | 19/2 <sup>-</sup>                  |
| 2603.0 7                                    | 7/2 <sup>+</sup>                   | 5383.7 <sup>#</sup> 7                       | 11/2 <sup>-</sup>                  | 8109.2 <sup>#</sup> 14                      | 15/2 <sup>-</sup>                  | 12276.5 <sup>#</sup> 33                     | 23/2 <sup>-</sup>                  |

<sup>†</sup> From a least-squares fit to  $\gamma$ -ray energies.

<sup>‡</sup> As given in 2007De14 based on known assignments of low-lying levels and mirror levels in  $^{35}\text{Cl}$ , the measured  $\gamma\gamma(\theta)$ (ADO) ratios, and the mirror symmetry with levels with similar energies in  $^{35}\text{Cl}$  with known assignments. When considered in the Adopted Levels, the firm assignments here are placed within parentheses if there are no other strong arguments to support these firm assignments.

<sup>#</sup> Band(A): Band based on  $f_{7/2}$  orbital.

 $\gamma(^{35}\text{Ar})$ 

$R_{\text{ADO}}=[I_\gamma(34^\circ)+I_\gamma(146^\circ)]/2I_\gamma(90^\circ)$ . Expected values are  $R_{\text{ADO}}\approx 1.3$  for stretched quadrupole ( $\Delta J=2$ ) and  $\text{ADO}\approx 0.8$  for stretched dipole ( $\Delta J=1$ ) transitions.

| <u><math>E_\gamma^\dagger</math></u> | <u><math>I_\gamma^\dagger</math></u> | <u><math>E_i(\text{level})</math></u> | <u><math>J_i^\pi</math></u> | <u><math>E_f</math></u> | <u><math>J_f^\pi</math></u> | <u>Mult.<sup>‡</sup></u> | <u>Comments</u>                                                                                         |
|--------------------------------------|--------------------------------------|---------------------------------------|-----------------------------|-------------------------|-----------------------------|--------------------------|---------------------------------------------------------------------------------------------------------|
| 381.5 3                              | 29 3                                 | 5765.3                                | 13/2 <sup>-</sup>           | 5383.7                  | 11/2 <sup>-</sup>           | D                        | $R_{\text{ADO}}=0.81$ 10.                                                                               |
| 593.7 2                              | 11 2                                 | 3196.7                                | 7/2 <sup>-</sup>            | 2603.0                  | 7/2 <sup>+</sup>            |                          |                                                                                                         |
| 851.8 9                              | 8 2                                  | 2603.0                                | 7/2 <sup>+</sup>            | 1750.8                  | 5/2 <sup>+</sup>            |                          |                                                                                                         |
| 1025.2 4                             | 6 2                                  | 5383.7                                | 11/2 <sup>-</sup>           | 4358.6                  | 9/2 <sup>-</sup>            |                          |                                                                                                         |
| 1162.0 8                             | 15 3                                 | 4358.6                                | 9/2 <sup>-</sup>            | 3196.7                  | 7/2 <sup>-</sup>            | (D)                      | $R_{\text{ADO}}=0.95$ 25.                                                                               |
| 1254.6 8                             | 15 5                                 | 5613.2                                | (11/2 <sup>-</sup> )        | 4358.6                  | 9/2 <sup>-</sup>            |                          |                                                                                                         |
| 1406.9 7                             | 5 1                                  | 5765.3                                | 13/2 <sup>-</sup>           | 4358.6                  | 9/2 <sup>-</sup>            | (Q)                      | $R_{\text{ADO}}=1.3$ 7.                                                                                 |
| 1446.1 6                             | 67 5                                 | 3196.7                                | 7/2 <sup>-</sup>            | 1750.8                  | 5/2 <sup>+</sup>            | D                        | $R_{\text{ADO}}=0.87$ 19.                                                                               |
| 1693.3 27                            | 15 3                                 | 9905.5                                | 19/2 <sup>-</sup>           | 8212.1                  | 15/2 <sup>-</sup>           | Q                        | $R_{\text{ADO}}=1.41$ 23.                                                                               |
| 1750.8 5                             | 100 9                                | 1750.8                                | 5/2 <sup>+</sup>            | 0.0                     | 3/2 <sup>+</sup>            |                          | $R_{\text{ADO}}=1.46$ 24 indicates $\Delta J=2$ , but inconsistent with $\Delta J=1$ from level scheme. |
| 1756.3 14                            | 17 9                                 | 4358.6                                | 9/2 <sup>-</sup>            | 2603.0                  | 7/2 <sup>+</sup>            |                          |                                                                                                         |
| 1796.3 25                            | 10 3                                 | 9905.5                                | 19/2 <sup>-</sup>           | 8109.2                  | 15/2 <sup>-</sup>           | Q                        | $R_{\text{ADO}}=1.8$ 3.                                                                                 |
| 2186.8 4                             | 49 3                                 | 5383.7                                | 11/2 <sup>-</sup>           | 3196.7                  | 7/2 <sup>-</sup>            | Q                        | $R_{\text{ADO}}=1.31$ 15.                                                                               |
| 2342.6 28                            | 8 2                                  | 8109.2                                | 15/2 <sup>-</sup>           | 5765.3                  | 13/2 <sup>-</sup>           |                          |                                                                                                         |
| 2370.9 25                            | 15 5                                 | 12276.5                               | 23/2 <sup>-</sup>           | 9905.5                  | 19/2 <sup>-</sup>           | Q                        | $R_{\text{ADO}}=1.8$ 4.                                                                                 |
| 2446.6 16                            | 6 2                                  | 8212.1                                | 15/2 <sup>-</sup>           | 5765.3                  | 13/2 <sup>-</sup>           |                          |                                                                                                         |
| 2602.6 15                            | 60 6                                 | 2603.0                                | 7/2 <sup>+</sup>            | 0.0                     | 3/2 <sup>+</sup>            | Q                        | $R_{\text{ADO}}=1.37$ 20.                                                                               |
| 2725.7 14                            | 4 1                                  | 8109.2                                | 15/2 <sup>-</sup>           | 5383.7                  | 11/2 <sup>-</sup>           |                          |                                                                                                         |
| 2828.3 7                             | 28 5                                 | 8212.1                                | 15/2 <sup>-</sup>           | 5383.7                  | 11/2 <sup>-</sup>           | Q                        | $R_{\text{ADO}}=1.7$ 6.                                                                                 |
| 3197 6                               | 16 3                                 | 3196.7                                | 7/2 <sup>-</sup>            | 0.0                     | 3/2 <sup>+</sup>            | Q                        | $R_{\text{ADO}}=1.7$ 8.                                                                                 |

Continued on next page (footnotes at end of table)

---

$^{24}\text{Mg}(^{16}\text{O},\alpha n\gamma)$  [2007De14,2005DeZZ](#) (continued)

---

$\gamma(^{35}\text{Ar})$  (continued)

<sup>†</sup> From [2007De14](#).

<sup>‡</sup> Not explicitly given in [2007De14](#), but deduced by evaluators based on their measured  $R_{\text{ADO}}$  ratios compared with expected values as stated in [2007De14](#).

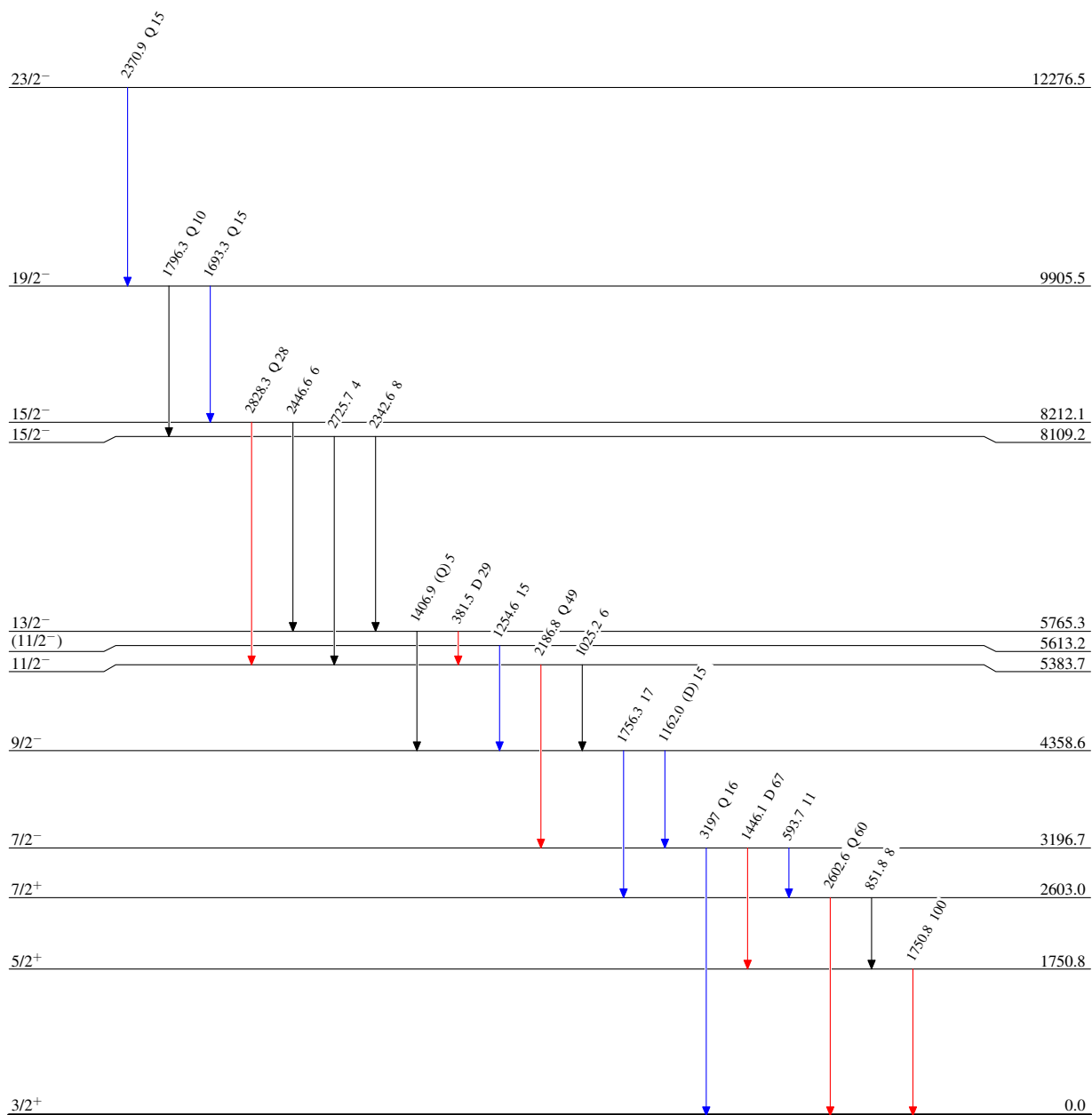
$^{24}\text{Mg}(^{16}\text{O}, \alpha n \gamma)$  2007De14, 2005DeZZ

## Level Scheme

Intensities: Relative  $I_\gamma$ 

## Legend

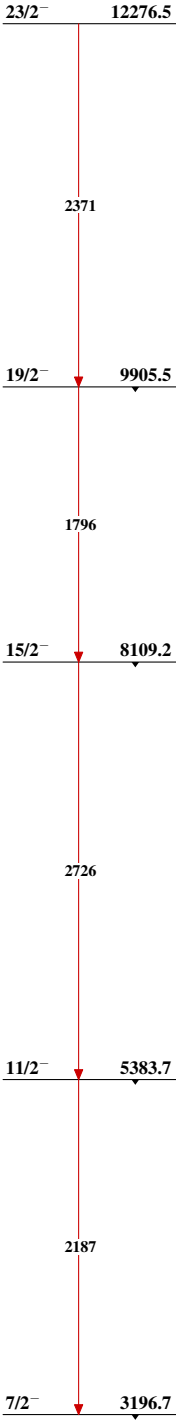
- $\longrightarrow$   $I_\gamma < 2\% \times I_\gamma^{\max}$   
 $\longrightarrow$   $I_\gamma < 10\% \times I_\gamma^{\max}$   
 $\longrightarrow$   $I_\gamma > 10\% \times I_\gamma^{\max}$

 $^{35}_{18}\text{Ar}_{17}$



<sup>24</sup>Mg(<sup>16</sup>O,αnγ) 2007De14,2005DeZZ

Band(A): Band based on f<sub>7/2</sub>  
orbital



<sup>35</sup>Ar<sub>17</sub>

$^{32}\text{S}(\alpha, n)$     **1963Ne05**

**1963Ne05**: A 12.20-MeV  $\alpha$  beam was produced by the Tandem Van de Graaff Accelerator at Florida State University. Targets were natural sulfur. Positrons were detected using a Ne102 plastic scintillator. Annihilation  $\gamma$  rays were detected using a NaI crystal. The half-life of  $^{35}\text{Ar}(\text{g.s.})$  was measured with a Technical Measurements Corporation 256 channel analyzer and multiscaler logic unit. Counting time per channel was determined by the sweep of a Tektronix Oscilloscope and measured using the standard frequency output of a Hewlett-Packard Electronic Counter Model 524-C. Changes in slope of the excitation curve are interpreted as excited-state thresholds due to excited levels in the product nucleus  $^{35}\text{Ar}$ . The threshold of  $^{35}\text{Cl}(\text{p}, n)^{35}\text{Ar}(\text{g.s.})$  using a thick AgCl target was also measured to confirm the  $^{35}\text{Ar}$  ground state mass.

$^{35}\text{Ar}$  isotope discovery:  $^{32}\text{S}(\alpha, n)^{35}\text{Ar}$  at Purdue ([1940Ki12](#), [1941Ki01](#), [1941El04](#)).

 $^{35}\text{Ar}$  Levels

| <u>E(level)<sup>†</sup></u> | <u>T<sub>1/2</sub></u> | <u>Comments</u>                                                                                                            |
|-----------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------|
| 0                           | 1.76 s 3               | T <sub>1/2</sub> : from decay curve in <a href="#">1963Ne05</a> .<br>Threshold E( $\alpha$ ) <sub>lab</sub> =9.846 MeV 20. |
| 890 50                      |                        | Threshold E( $\alpha$ ) <sub>lab</sub> =10.69 MeV 50.                                                                      |
| 2030 80                     |                        | Threshold E( $\alpha$ ) <sub>lab</sub> =11.97 MeV 80.                                                                      |

<sup>†</sup> From [1963Ne05](#).

$^{33}\text{S}(^3\text{He},n\gamma)$  1975Da14

$^{33}\text{S}+2p \rightarrow ^{35}\text{Ar}$  from  $J^\pi=3/2^+$   $^{33}\text{S}$  ground state.

**1975Da14:** a  $^3\text{He}^+$  beam was produced from the University of Alberta Van de Graaff accelerator. Targets were  $150 \mu\text{g}/\text{cm}^2$  layers of  $\text{Ag}_2\text{S}$  (59%  $^{33}\text{S}$ ) on silver backings. At  $E(^3\text{He})=6.375 \text{ MeV}$ , neutrons were detected using an NE213 liquid scintillator placed at  $\theta_{\text{lab}}=0^\circ, 10^\circ$ , and  $20^\circ$ . At  $E(^3\text{He})=6.660 \text{ MeV}$ , neutrons were detected using the NE213 scintillator placed at  $\theta_{\text{lab}}=0^\circ$  and  $20^\circ$ . At  $E(^3\text{He})=6.390 \text{ MeV}$ , neutrons were detected using the NE213 scintillator placed at  $\theta_{\text{lab}}=0^\circ$  and neutron-coincident  $\gamma$  rays were detected using a 10% efficient Ge(Li) detector placed at  $90^\circ$ . Measured time-of-flight (TOF) spectra of neutrons,  $\sigma(E_n, \theta)$ ,  $E_\gamma$ ,  $n\gamma$ -coin. Deduced Q values, levels,  $L_{2p}$ , J,  $\pi$ , and isospin.

 $^{35}\text{Ar}$  Levels

| $E(\text{level})^\dagger$ | $J^\pi$   | $L^\ddagger$ | Comments                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------|-----------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0                         | $3/2^+$   | (0)          | $J^\pi$ : mirror $3/2^+$ $^{35}\text{Cl}$ g.s.                                                                                                                                                                                                                                                                                                             |
| 1184.2 6                  |           |              |                                                                                                                                                                                                                                                                                                                                                            |
| 1749.8 9                  |           |              |                                                                                                                                                                                                                                                                                                                                                            |
| 2600.8 15                 |           |              | $J^\pi$ : <b>1975Da14</b> observed $L(^3\text{He},n)=(0)$ for a 2.60-MeV level in $^{35}\text{Ar}$ , implying the existence of a $3/2^+$ level, which possibly corresponds to the $3/2^+$ 2638-keV level in $^{35}\text{Ar}$ .                                                                                                                             |
| 3195.8 11                 |           |              | E(level): deduced by evaluators from the unplaced 1446.0 $\gamma$ .                                                                                                                                                                                                                                                                                        |
| 5537 25                   | $(3/2^+)$ | (0)          | T=3/2<br>E(level): from measured neutron energy corresponding to this level.<br>$J^\pi$ : $(\pi 1d_{3/2})^2(\nu 1d_{3/2})^1$ configuration formed by the $J^\pi=0^+$ , T=1 $(\pi 1d_{3/2})^2$ di-proton transfer from $^3\text{He}$ to $^{33}\text{S}$ of $^{32}\text{S}(\nu 1d_{3/2})^1$ configuration. Tentative first T=3/2 state in $^{35}\text{Ar}$ . |

$^\dagger$  From **1975Da14** based on measured  $E_\gamma$ , but the  $E_\gamma$  values are not provided in **1975Da14**, unless otherwise noted.

$^\ddagger$  the observed maximum at  $\theta=0^\circ$  in  $\sigma(E_n, \theta)$  implies  $L=0$ .

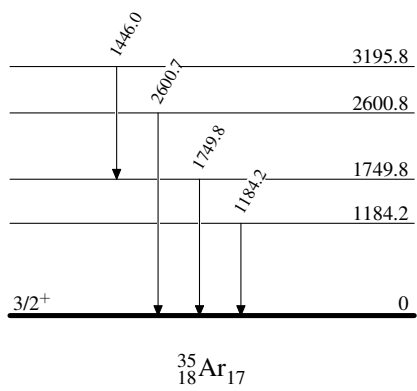
 $\gamma(^{35}\text{Ar})$ 

| $E_\gamma^\dagger$ | $E_i(\text{level})$ | $E_f$  | $J_f^\pi$ | Comments                                                                                                                                                                                                                                                              |
|--------------------|---------------------|--------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1184.2             | 1184.2              | 0      | $3/2^+$   |                                                                                                                                                                                                                                                                       |
| 1446.0 6           | 3195.8              | 1749.8 |           | Unplaced $\gamma$ in <b>1975Da14</b> . Evaluators placed it based on the Adopted Levels from $^{16}\text{O}(^{24}\text{Mg}, \alpha n\gamma)^{35}\text{Ar}$ ( <b>2004Ek01</b> ) and $^{24}\text{Mg}(^{16}\text{O}, \alpha n\gamma)^{35}\text{Ar}$ ( <b>2007De14</b> ). |
| 1749.8             | 1749.8              | 0      | $3/2^+$   |                                                                                                                                                                                                                                                                       |
| 2600.7             | 2600.8              | 0      | $3/2^+$   |                                                                                                                                                                                                                                                                       |

$^\dagger$   $E_\gamma$  values without uncertainties are deduced by evaluators from the level-energy differences from **1975Da14**.

$^{33}\text{S}(^3\text{He},n\gamma)$  1975Da14

Level Scheme



$^{35}\text{Cl}(^3\text{He},t)$     **1976Be08**

$J^\pi=3/2^+$  for  $^{35}\text{Cl}$  ground state.

**1976Be08**:  $^{35}\text{Cl}(^3\text{He},t)^{35}\text{Ar}$  is studied in an attempt to search for the member of T=3/2 isobaric quartets in  $^{35}\text{Ar}$ . A 35-MeV  $^3\text{He}$  beam from the Michigan State University cyclotron impinged on a  $200\text{ }\mu\text{g}/\text{cm}^2$  Li- $^{35}\text{Cl}$  target. Tritons were detected using a scintillator-proportional counter detector system at the focal plane of the Enge split-pole spectrograph. A T=3/2,  $3/2^+$  level at 5537 keV 25 in  $^{35}\text{Ar}$  was observed in  $^{33}\text{S}(^3\text{He},n)^{35}\text{Ar}$  (**1975Da14**) but causing an IMME breakdown. **1976Be08** measured the  $^{35}\text{K}$  g.s. mass using  $^{40}\text{Ca}(^3\text{He},^8\text{Li})^{35}\text{K}$  and predicted the T=3/2 member in  $^{35}\text{Ar}$  to be 5579 keV 14. **1976Be08** did not find new peaks in  $^{35}\text{Cl}(^3\text{He},t)^{35}\text{Ar}$  between 5484 keV 10 and 5591 keV 10 that were already known from  $^{36}\text{Ar}(^3\text{He},\alpha)^{35}\text{Ar}$  (**1973Be26**).

 $^{35}\text{Ar}$  Levels

E(level)<sup>†</sup>

4721  
4782  
5116  
5205  
5387  
5484  
5591  
5911  
6033

<sup>†</sup> Peaks are seen in triton spectrum in **1976Be08** with no energy values reported. The authors mark those peaks with numbers corresponding to those listed in TABLE I of **1973Be26** (not the same authors) in  $^{36}\text{Ar}(^3\text{He},\alpha)$ . The energy values quoted here are taken by the evaluators from that table.

$^{36}\text{Ar}(\text{p,d})$  1968Jo04,1968Ko11

$J^\pi=0^+$  for  $^{36}\text{Ar}$  ground state.

**1968Jo04**: a 27.5-MeV  $I$  proton beam was produced by the 1.3-m FFAG cyclotron at the Nuclear Physics Laboratory at the University of Colorado. Targets were natural argon gas (99.6%  $^{40}\text{Ar}$ ) and 99.0% enriched  $^{36}\text{Ar}$  gas. Deuterons were detected using a telescope of a 0.228-mm transmission surface barrier detector and a 0.37-cm lithium-drifted stopping counter with FWHM=120 keV. Measured  $\sigma(E_d, \theta)$ . Deduced levels, L-transfers, and spectroscopic factors from finite-range DWUCK-DWBA analysis of the measured  $\sigma(\theta)$ .

**1968Ko11**: a 33.6-MeV proton beam was produced by the 64-in. sector-focusing cyclotron at Michigan State University. The target was >99% isotopically enriched  $^{36}\text{Ar}$  gas. Deuterons were detected using a telescope of a 279- $\mu\text{m}$  silicon surface barrier detector and a 3-mm lithium-drifted silicon counter with FWHM=130 keV. Measured  $\sigma(E_d, \theta)$ . Deduced levels,  $J$ ,  $\pi$ , L-transfers, and spectroscopic factors from zero-range Macefield-DWBA analysis of the measured  $\sigma(\theta)$ .

 $^{35}\text{Ar}$  Levels

Spectroscopic factor  $C^2S=\sigma(\theta)_{\text{exp}}/\sigma(\theta)_{\text{DWBA}}/N$ , where  $N=1.65*3/2$  is a normalization factor adopted by **1968Jo04**. The DWBA calculation in **1968Ko11** assumed neutron-well radius parameter  $r_{0n}=1.15$  fm for  $d_{5/2}$  pickup and 1.35 fm for  $d_{3/2}$  pickup, respectively, to reproduce the forward-angle  $J$ -dependence observed in the  $L=2$  deuteron angular distributions. **1968Ko11** demonstrates that a larger radius results in a larger DWBA-calculated cross section and this gives rise to problems in extracting absolute spectroscopic factors, and therefore, the evaluators put the  $C^2S$  from **1968Ko11** in comments. **1968Ko11** states that there is some evidence that the radius parameter corresponding to the best DWBA fit to the data might also result in the most trustworthy value for the spectroscopic factor.

| $E(\text{level})^\dagger$ | $J^\pi^\#$     | $L^\ddagger$ | $C^2S^\S$  | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------|----------------|--------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0                         | $3/2^+$        | 2            | 2.92       | $C^2S$ : others: 3.03 for 1.15 fm and 1.76 for 1.35 fm (better fit) ( <b>1968Ko11</b> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 1180 20                   | $1/2^+$        | 0            | 2.50       | $E(\text{level})$ : also from <b>1968Ko11</b> .<br>$C^2S$ : others: 1.29 for 1.15 fm and 1.05 for 1.35 fm ( <b>1968Ko11</b> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 1700 30                   | $(5/2^+)$      |              | 0.1        | $C^2S$ : from <b>1968Ko11</b> for radius parameter=1.15 fm.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 2615 20                   | $(3/2^+)$      | 2            |            | $E(\text{level})$ : weighted average of 2630 20 ( <b>1968Jo04</b> ) and 2600 20 ( <b>1968Ko11</b> ).<br>$J^\pi$ : spin= $(3/2)$ based on the observed $J$ -dependence of the deuteron angular distributions ( <b>1968Ko11</b> ).<br>$L$ : from <b>1968Ko11</b> . Other: 1 from <b>1968Jo04</b> is inconsistent. <b>1973Be26</b> determined $L(n)=2$ from $^{36}\text{Ar}(^3\text{He}, \alpha)^{35}\text{Ar}$ and states that $L=1$ is probably in error as this level corresponds well with the known $3/2^+$ state at 2695 keV in $^{35}\text{Cl}$ and there are no nearby negative-parity states in $^{35}\text{Cl}$ .<br>$C^2S$ : 0.42 for 1.15 fm and 0.28 for 1.35 fm (better fit) for $L=2$ from <b>1968Ko11</b> . Other: 0.12 for $L=1$ from <b>1968Jo04</b> . |
| 2970 20                   | $(5/2^+)$      | 2            | 2.47       | $E(\text{level})$ : weighted average of 2990 20 ( <b>1968Jo04</b> ) and 2950 20 ( <b>1968Ko11</b> ).<br>$J^\pi$ : spin= $(5/2)$ based on the observed $J$ -dependence of the deuteron angular distributions ( <b>1968Ko11</b> ).<br>$C^2S$ : assuming $5/2^+$ ( <b>1968Jo04</b> ). Other: 3.10 assuming $3/2^+$ ( <b>1968Jo04</b> ). 2.31 for 1.15 fm (better fit) and 1.53 for 1.35 fm from <b>1968Ko11</b> .                                                                                                                                                                                                                                                                                                                                                        |
| 3200 20                   | $7/2^-$        | 3            | 0.63       | $E(\text{level})$ : weighted average of 3210 20 ( <b>1968Jo04</b> ) and 3190 20 ( <b>1968Ko11</b> ).<br>$C^2S$ : others: 0.64 for 1.15 fm and 0.37 for 1.35 fm (better fit) ( <b>1968Ko11</b> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 4756 28                   | $(1/2^+)$      | 0            | 0.172      | $E(\text{level})$ : weighted average of 4770 20 ( <b>1968Jo04</b> ) and 4700 40 ( <b>1968Ko11</b> ).<br>$C^2S$ : others: 0.05 for 1.15 fm and 0.04 for 1.35 fm ( <b>1968Ko11</b> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5102 20                   | $7/2^-$        | 3            | 0.46       | $E(\text{level})$ : weighted average of 5110 20 ( <b>1968Jo04</b> ) and 5070 40 ( <b>1968Ko11</b> ).<br>$L$ : From <b>1968Jo04</b> . $L$ is not reported in <b>1968Ko11</b> . <b>1973Be26</b> determined $L=2$ from $^{36}\text{Ar}(^3\text{He}, \alpha)^{35}\text{Ar}$ and states that $L=2$ gives a much better account of the data than $L=3$ does.                                                                                                                                                                                                                                                                                                                                                                                                                |
| 5400 50                   |                |              |            | $E(\text{level})$ : From <b>1968Ko11</b> only.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 5598 20                   | $3/2^+, 5/2^+$ | 2            | 2.93, 2.37 | $E(\text{level})$ : weighted average of 5610 20 ( <b>1968Jo04</b> ) and 5570 30 ( <b>1968Ko11</b> ).<br>$C^2S$ : others: 1.77 for 1.15 fm (better fit) and 1.25 for 1.35 fm, both for $J^\pi=(3/2, 5/2)^+$ ( <b>1968Ko11</b> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 6024 20                   | $3/2, 5/2^+$   | 2            | 1.58, 1.31 | $E(\text{level})$ : weighted average of 6030 20 ( <b>1968Jo04</b> ) and 6010 30 ( <b>1968Ko11</b> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

Continued on next page (footnotes at end of table)

$^{36}\text{Ar}(\text{p,d})$  **1968Jo04,1968Ko11** (continued) $^{35}\text{Ar}$  Levels (continued)

| <u>E(level)<sup>†</sup></u> | <u>J<sup>π</sup><sup>#</sup></u> | <u>L<sup>‡</sup></u> | <u>C<sup>2</sup>S<sup>‡</sup></u> | Comments                                                                                                                        |
|-----------------------------|----------------------------------|----------------------|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
|                             |                                  |                      |                                   | C <sup>2</sup> S: others: 1.18 for 1.15 fm and 0.83 for 1.35 fm (better fit), both for $J^\pi=(3/2,5/2)^+$ ( <b>1968Ko11</b> ). |
| 6620 30                     | (1/2 <sup>+</sup> )              | 0                    |                                   | E(level),L: From <b>1968Ko11</b> only.                                                                                          |
| 6700 20                     | 7/2 <sup>-</sup>                 | 3                    | 2.60                              | C <sup>2</sup> S: 0.24 for 1.15 fm and 0.19 for 1.35 fm ( <b>1968Ko11</b> ).                                                    |
| 6820 30                     |                                  | 2                    |                                   | L,C <sup>2</sup> S: From <b>1968Jo04</b> only.                                                                                  |
|                             |                                  |                      |                                   | E(level),L: From <b>1968Ko11</b> only.                                                                                          |
|                             |                                  |                      |                                   | C <sup>2</sup> S: 0.72 for 1.15 fm (better fit) and 0.51 for 1.35 fm, both for $J^\pi=(3/2,5/2)^+$ ( <b>1968Ko11</b> ).         |
| 7030 20                     | (3/2,5/2) <sup>+</sup>           | 2                    | 2.20,1.79                         |                                                                                                                                 |

<sup>†</sup> From **1968Jo04**, unless otherwise noted.

<sup>‡</sup> From DWBA analysis of measured  $\sigma(\theta)$ . Values of L-transfers are from both **1968Jo04** and **1968Ko11** and values of C<sup>2</sup>S are from **1968Jo04**, unless otherwise noted. Where two C<sup>2</sup>S values are given, they correspond to respective  $J^\pi$  values as given.

<sup>#</sup> Assumed in **1968Jo04** and/or **1968Ko11** for extracting C<sup>2</sup>S, unless otherwise noted.

$^{36}\text{Ar}(\text{d,t})$  1970Wh04,2015Fr01

$J^\pi=0^+$  for  $^{36}\text{Ar}$  ground state.

**1970Wh04**: A 21.0-MeV deuteron beam was produced by the Yale MP tandem Van de Graaff accelerator. The target was a  $^{36}\text{Ar}$  gas cell. Tritons were detected using a 140- $\mu\text{m}$ -530- $\mu\text{m}$  thick  $\Delta E$ -E telescope of silicon surface barrier detectors with FWHM=65-70 keV. Measured  $\sigma(E_t, \theta)$ . Deduced levels, L-transfers, and spectroscopic factors from JULIE-DWBA analysis of the measured  $\sigma(\theta)$ . Comparisons with shell-model calculations.

**2015Fr01**: A 22-MeV deuteron beam was produced by the MP tandem Van de Graaff accelerator at the Maier-Leibnitz Laboratorium (MLL) in Garching, Germany. Targets were fabricated by implanting 25-70-keV 3-6  $\mu\text{g}/\text{cm}^2$  of  $^{36}\text{Ar}$  ions into 30  $\mu\text{g}/\text{cm}^2$  natural abundance carbon foils. Reaction products were momentum analyzed by a Q3D magnetic spectrograph. Tritons were detected using a multiwire gas-filled proportional counter backed by a scintillator at the focal plane. Measured  $E_t$  at  $\theta_{\text{lab}}=15^\circ$ ,  $20^\circ$ ,  $25^\circ$  with FWHM $\approx$ 9 keV and at  $54^\circ$  with FWHM $\approx$ 16 keV. Deduced levels, proton resonance energies, level densities. Comparisons with shell-model calculations.

 $^{35}\text{Ar}$  Levels

Spectroscopic factor  $C^2S=\sigma(\theta)_{\text{exp}}/\sigma(\theta)_{\text{DWBA}}/N$ , where  $N=3.33$  is a normalization factor adopted by **1970Wh04** from **1966Ba54**.

| $E(\text{level})^\dagger$ | $J^\pi^\ddagger$ | $L^\#$ | $C^2S^\#$ | Comments                                                                                                                                                                |
|---------------------------|------------------|--------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0                         | $3/2^+$          | 2      | 3.4       |                                                                                                                                                                         |
| 1180 10                   | $1/2^+$          | 0      | 1.4       |                                                                                                                                                                         |
| 1700                      | $(5/2^+)$        | (2)    | <0.2      |                                                                                                                                                                         |
| 2635 20                   | $(3/2^+)$        | (2)    | 0.5       | $C^2S$ : <b>1970Wh04</b> authors state that there is a large uncertainty in the spectroscopic strength. <b>1970Wh04</b> also gives $S=0.11$ or $0.032$ assuming $L=1$ . |
| 2986 20                   | $5/2^+$          | 2      | 2.6       |                                                                                                                                                                         |
| 3200 20                   | $7/2^-$          | (3)    | 0.33,0.11 | $C^2S$ : assuming $r_{0n}=1.25$ F and $V_n\approx 60$ MeV, respectively. <b>1970Wh04</b> authors state that there is a large uncertainty in the spectroscopic strength. |
| 5913 5                    |                  |        |           |                                                                                                                                                                         |
| 5991 3                    |                  |        |           |                                                                                                                                                                         |
| 6037 3                    |                  |        |           | May be a doublet ( <b>2015Fr01</b> ).                                                                                                                                   |
| 6055? 3                   |                  |        |           | Tentative ( <b>2015Fr01</b> ).                                                                                                                                          |
| 6076 3                    |                  |        |           |                                                                                                                                                                         |
| 6164 3                    |                  |        |           |                                                                                                                                                                         |
| 6253 3                    |                  |        |           |                                                                                                                                                                         |
| 6273 3                    |                  |        |           |                                                                                                                                                                         |
| 6302 3                    |                  |        |           |                                                                                                                                                                         |
| 6332 3                    |                  |        |           |                                                                                                                                                                         |
| 6345 3                    |                  |        |           |                                                                                                                                                                         |
| 6415 2                    |                  |        |           |                                                                                                                                                                         |
| 6439? 4                   |                  |        |           | Tentative ( <b>2015Fr01</b> ).                                                                                                                                          |
| 6460 3                    |                  |        |           |                                                                                                                                                                         |
| 6523 3                    |                  |        |           |                                                                                                                                                                         |
| 6557 3                    |                  |        |           |                                                                                                                                                                         |
| 6585 3                    |                  |        |           |                                                                                                                                                                         |
| 6606 3                    |                  |        |           |                                                                                                                                                                         |
| 6617 2                    |                  |        |           |                                                                                                                                                                         |
| 6644 3                    |                  |        |           |                                                                                                                                                                         |
| 6651 3                    |                  |        |           |                                                                                                                                                                         |
| 6672 3                    |                  |        |           |                                                                                                                                                                         |

$^\dagger$  From **1970Wh04** for low-lying states up to 3200 keV and from **2015Fr01** for others.

$^\ddagger$  As given in **1970Wh04**, used for extracting  $C^2S$ .

$^\#$  From DWBA analysis of the measured  $\sigma(\theta)$  in **1970Wh04**.



<sup>36</sup>Ar(<sup>3</sup>He,α) 1973Be26,1998VoZY

$J^\pi=0^+$  for <sup>36</sup>Ar ground state.

**1973Be26:** an 18-MeV <sup>3</sup>He beam was produced by the University of Pennsylvania Tandem Van de Graaff accelerator. The target was pure argon gas enriched to 99.8% in <sup>36</sup>Ar. α particles were momentum analyzed in a multi-angle spectrograph and detected using Ilford K-1 nuclear emulsions with FWHM=35 keV. Measured  $\sigma(E_\alpha, \theta)$ . Deduced levels, J,  $\pi$ , L-transfers, and spectroscopic factors from local zero-range DWUCK-DWBA analysis of the measured  $\sigma(\theta)$ . Comparisons with shell-model calculations and the mirror nucleus <sup>35</sup>Cl. Also see **1972MiZO**.

**1998VoZY:** a 25-MeV <sup>3</sup>He beam at 0.5-μA intensity was provided from the FN tandem accelerator at the Nuclear Structure Laboratory of the University of Notre Dame. The targets were made by implanting 75-keV <sup>36</sup>Ar into 40 μg/cm<sup>2</sup> carbon foils from the University of Toledo ion source. The reaction products were momentum analyzed by the Notre Dame broad-range magnetic spectrograph. The particles were subsequently detected at the focal surface of the spectrograph by a position sensitive proportional gas detector backed with a plastic scintillator. Measured  $E_\alpha$ . Deduced levels.

<sup>35</sup>Ar Levels

Spectroscopic factor  $C^2S=(2j+1)\times\sigma(\theta)_{\text{exp}}/\sigma(\theta)_{\text{DWBA}}/N$ , where the isospin Clebsch-Gordan coefficient  $C^2$  is 1/2 in this case, j is the total angular momentum of the transferred neutron, and the normalization factor  $N=16.8$ . **1973Be26** states that the overall normalization for the (<sup>3</sup>He,α) reaction is not well determined and therefore resort to empirical means to determine N.  $N=15.5$  deduced from shell-model calculated total  $S=3.52$  for all four 1/2<sup>+</sup> states and the **1973Be26** measured  $NS=54.6$ .  $N=18.1$  deduced from the <sup>35</sup>Cl(<sup>3</sup>He,d)<sup>36</sup>Ar(g.s.)  $S=4.73$  (**1970Mo10**) and the **1973Be26** measured <sup>36</sup>Ar(<sup>3</sup>He,α)<sup>35</sup>Ar(g.s.)  $NS=85.4$ . **1973Be26** adopted the average  $N=16.8$ .

| E(level)             | $J^\pi$                | L <sup>†</sup> | $C^2S^\dagger$          | Comments                                                                                                                           |
|----------------------|------------------------|----------------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| 0                    | 3/2 <sup>+</sup>       | 2              | 2.545                   |                                                                                                                                    |
| 1175 2               | 1/2 <sup>+</sup>       | 0              | 1.19                    | E(level): weighted average of 1179 10 ( <b>1973Be26</b> ) and 1175 2 ( <b>1998VoZY</b> ).                                          |
| 1747 2               | 5/2 <sup>+</sup>       | 2              | 0.025                   | E(level): weighted average of 1738 10 ( <b>1973Be26</b> ) and 1747 2 ( <b>1998VoZY</b> ).                                          |
| 2649 2               | 3/2 <sup>+</sup>       | 2              | 0.57                    | E(level): weighted average of 2637 10 ( <b>1973Be26</b> ) and 2649 2 ( <b>1998VoZY</b> ).                                          |
| 2983 2               | 5/2 <sup>+</sup>       | 2              | 1.39                    | E(level): weighted average of 2982 10 ( <b>1973Be26</b> ) and 2983 2 ( <b>1998VoZY</b> ).                                          |
| 3197 2               | 7/2 <sup>-</sup>       | 3              | 0.39                    | E(level): weighted average of 3193 10 ( <b>1973Be26</b> ) and 3197 2 ( <b>1998VoZY</b> ).                                          |
| 3882 5               | 1/2 <sup>+</sup>       | 0              | 0.02                    | E(level): weighted average of 3884 10 ( <b>1973Be26</b> ) and 3881 5 ( <b>1998VoZY</b> ).                                          |
| 4001 3               | (3/2) <sup>-</sup>     | 1              | 0.065                   | E(level): weighted average of 4012 10 ( <b>1973Be26</b> ) and 4000 3 ( <b>1998VoZY</b> ).                                          |
| 4113 4               |                        |                |                         | E(level): weighted average of 4110 10 ( <b>1973Be26</b> ) and 4113 4 ( <b>1998VoZY</b> ).                                          |
| 4135 4               | (3/2) <sup>-</sup>     | 1              | 0.025                   | E(level): weighted average of 4142 10 ( <b>1973Be26</b> ) and 4134 4 ( <b>1998VoZY</b> ).                                          |
| 4350 6               |                        |                |                         | E(level): weighted average of 4350 10 ( <b>1973Be26</b> ) and 4350 6 ( <b>1998VoZY</b> ).                                          |
| 4515 5               |                        |                |                         | E(level): weighted average of 4530 10 ( <b>1973Be26</b> ) and 4514 3 ( <b>1998VoZY</b> ).                                          |
| 4713 6               | 1/2 <sup>+</sup>       | 0              | 0.05                    | E(level): weighted average of 4721 10 ( <b>1973Be26</b> ) and 4710 6 ( <b>1998VoZY</b> ).                                          |
| 4774 6               |                        |                |                         | E(level): weighted average of 4782 10 ( <b>1973Be26</b> ) and 4771 6 ( <b>1998VoZY</b> ).                                          |
| 5059 11              |                        |                |                         | E(level): unweighted average of 5048 10 ( <b>1973Be26</b> ) and 5069 4 ( <b>1998VoZY</b> ).                                        |
| 5116 2               | (3/2,5/2) <sup>+</sup> | 2              | 0.25,0.145 <sup>#</sup> | E(level): weighted average of 5116 10 ( <b>1973Be26</b> ) and 5116 2 ( <b>1998VoZY</b> ).                                          |
| 5207 3               |                        |                |                         | E(level): weighted average of 5205 10 ( <b>1973Be26</b> ) and 5207 3 ( <b>1998VoZY</b> ).                                          |
| 5389 8               |                        |                |                         | E(level): weighted average of 5387 10 ( <b>1973Be26</b> ) and 5391 8 ( <b>1998VoZY</b> ).                                          |
| 5482 2               | (3/2,5/2) <sup>+</sup> | 2              | 0.77,0.445 <sup>#</sup> | E(level): weighted average of 5484 10 ( <b>1973Be26</b> ) and 5482 2 ( <b>1998VoZY</b> ).                                          |
| 5594 2               | (3/2,5/2) <sup>+</sup> | 2              | 1.98,1.14 <sup>#</sup>  | E(level): weighted average of 5591 10 ( <b>1973Be26</b> ) and 5594 2 ( <b>1998VoZY</b> ).                                          |
| 5916 3               |                        |                |                         | E(level): weighted average of 5911 10 ( <b>1973Be26</b> ) and 5916 3 ( <b>1998VoZY</b> ).                                          |
| 6036 3               | (3/2,5/2) <sup>+</sup> | 2              | 1.3,0.755 <sup>#</sup>  | E(level): weighted average of 6033 10 ( <b>1973Be26</b> ) and 6036 3 ( <b>1998VoZY</b> ).                                          |
| 6162 2               |                        |                |                         | E(level): weighted average of 6153 10 ( <b>1973Be26</b> ) and 6162 2 ( <b>1998VoZY</b> ).                                          |
| 6262 10              |                        |                |                         | E(level): weighted average of 6258 10 ( <b>1973Be26</b> ) and 6267 12 ( <b>1998VoZY</b> ).                                         |
| 6615 3               | 1/2 <sup>+</sup>       | 0              | 0.36                    | E(level): weighted average of 6631 10 ( <b>1973Be26</b> ) and 6614 2 ( <b>1998VoZY</b> ).<br>Probable doublet in <b>1973Be26</b> . |
| 6823 2               |                        |                |                         | E(level): weighted average of 6827 10 ( <b>1973Be26</b> ) and 6823 2 ( <b>1998VoZY</b> ).                                          |
| 6948 2               |                        |                |                         | E(level): weighted average of 6959 10 ( <b>1973Be26</b> ) and 6948 2 ( <b>1998VoZY</b> ).                                          |
| 7043 4               |                        |                |                         | E(level): weighted average of 7055 10 ( <b>1973Be26</b> ) and 7042 3 ( <b>1998VoZY</b> ).                                          |
| 7117 <sup>†</sup> 10 |                        |                |                         |                                                                                                                                    |
| 7293 <sup>†</sup> 10 |                        |                |                         |                                                                                                                                    |

Continued on next page (footnotes at end of table)

$^{36}\text{Ar}(^3\text{He},\alpha)$     **1973Be26,1998VoZY (continued)** $^{35}\text{Ar}$  Levels (continued)E(level)7423<sup>†</sup> 107502<sup>†</sup> 107840<sup>†</sup> 108019<sup>†</sup> 10

<sup>†</sup> From **1973Be26**. L and C<sup>2</sup>S values are extracted from DWBA analysis of measured  $\sigma(\theta)$ . Quoted values of C<sup>2</sup>S are converted from the S values in **1973Be26** with C<sup>2</sup>=1/2.

<sup>‡</sup> As given in **1973Be26**, used for extracting C<sup>2</sup>S.

# **1973Be26** states that the differences for j=3/2 and 5/2 are small in the DWBA-calculated L=2 shapes. It is not possible to differentiate between the two allowed j values for L=2 transitions. Both C<sup>2</sup>S values are given for each level with two spin values. Assuming that all four levels have spins of 3/2 would lead to a summed L=2 C<sup>2</sup>S that exceeds the simple shell-model sum rule limit of 8 for combined 1d<sub>3/2</sub> and 1d<sub>5/2</sub> pickup, which suggests that all four of these levels probably have 5/2<sup>+</sup>.

Adopted Levels

$Q(\beta^-) = -1.595 \times 10^4$  11;  $S(n) = 17757$  17;  $S(p) = 83.6$  5;  $Q(\alpha) = -6563$  3 [2021Wa16](#)  
 $Q(\beta^-), S(n)$ : Deduced by evaluators using mass excesses of 4777 105 for  $^{35}\text{Ca}$  measured by [2023La09](#), and  $-1487$  17 for  $^{34}\text{K}$  measured by [2024Dr01](#);  $-11172.9$  5 for  $^{35}\text{K}$  from [2021Wa16](#). Values from [2021Wa16](#):  $Q(\beta^-) = -16360$  200 (syst),  $S(n) = 18020$  200 (syst).  $S(2n) = 34860$  200 (syst),  $S(2p) = 4747.5$  6,  $Q(\varepsilon) = 11874.4$  9,  $Q(\varepsilon p) = 5978.2$  5 ([2021Wa16](#)).  
 Isotope discovery ([2012Th10](#)):  $^{40}\text{Ca}(^3\text{He}, ^8\text{Li})^{35}\text{K}$  at Michigan State ([1976Be08](#)).  
 $^{35}\text{K}$  decay measurements:  
[1980Ew02, 1979Ca15](#):  $^{35}\text{K}$  produced via  $^{45}\text{Sc}(p, 8n3p)$  spallation at CERN. Measured  $T_{1/2}$  and  $\varepsilon + \beta^+$ -delayed protons and  $\gamma$  rays.  
[2018Sa54, 2019ChZU](#):  $^{35}\text{K}$  produced via  $^1\text{H}(^{36}\text{Ar}, ^{35}\text{K})2n$  at Texas A&M. Measured  $T_{1/2}$  and  $\varepsilon + \beta^+$ -delayed protons and  $\gamma$  rays.  
[1998Sc19](#): Polarized  $^{35}\text{K}$  produced via fragmentation of  $^{40}\text{Ca}$  on  $^9\text{Be}$  target at GSI. Measured  $T_{1/2}$  and  $g$ -factor of  $^{35}\text{K}$  ground state from  $\beta$ -NMR.  
[2006Me04](#): Polarized  $^{35}\text{K}$  produced via  $^{36}\text{Ar}(^9\text{Be}, ^{10}\text{Li})^{35}\text{K}$  at NSCL, MSU. Measured  $g$ -factor of  $^{35}\text{K}$  ground state from  $\beta$ -NMR.  
 $^{35}\text{K}$  mass measurements: [2023Zh10, 2007Ya08, 1976Be08](#).  
 Theoretical calculations (binding energies, moments, radii, levels,  $J$ ,  $\pi$ , mass,  $T_{1/2}$ , reaction rates, etc.): [2025Sh19, 2023Bo17, 2023Fo05, 2022Zo01, 2020Ma25, 2016Si02, 2016Me17, 2003Sm02](#).

 $^{35}\text{K}$  LevelsCross Reference (XREF) Flags

- A**  $^{35}\text{Ca}$   $\varepsilon + \beta^+$  decay (25.7 ms)  
**B**  $^9\text{Be}(^{36}\text{Ca}, ^{35}\text{K})$   
**C**  $^{40}\text{Ca}(^3\text{He}, ^8\text{Li})$

| <u>E(level)<sup>†</sup></u> | <u><math>J^\pi</math><sup>‡</sup></u> | <u><math>T_{1/2}</math></u> | <u>XREF</u>  | <u>Comments</u>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------------|---------------------------------------|-----------------------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0.0                         | $3/2^+$                               | 175 ms 2                    | <b>A B C</b> | $\% \varepsilon + \% \beta^+ = 100$ ; $\% \varepsilon p = 0.37$ 15<br>$\mu = (+)0.392$ 7 ( <a href="#">2006Me04, 2019StZV</a> )<br>$J^\pi$ : $L(^{36}\text{Ca}, ^{35}\text{K}) = 2$ from $0^+$ and allowed $\varepsilon + \beta^+$ feeding to 4725.9, $1/2^+$ level in $^{35}\text{Ar}$ . Possible mirror level of $3/2^+$ $^{35}\text{S}$ g.s.<br>$T_{1/2}$ : weighted average of 175 ms 2 ( <a href="#">2018Sa54</a> ), 178 ms 8 ( <a href="#">1998Sc19</a> ), and 190 ms 30 ( <a href="#">1980Ew02</a> ).<br>$\mu$ : From $\beta$ -NMR spectroscopy ( <a href="#">2006Me04</a> ). Other: 0.36 3 ( <a href="#">1998Sc19, \beta-NMR spectroscopy). The positive sign is based on the mirror <math>^{35}\text{S}</math> g.s.<br/> <math>\% \varepsilon p</math>: from <a href="#">1980Ew02</a> for <math>E(p) &gt; 0.9</math> MeV. <math>E(p) &lt; 0.9</math> MeV has also been observed (<a href="#">2018Sa54, 2019ChZU</a>). </a> |
| 1553 5                      | $(1/2)^+$                             |                             | <b>A C</b>   | E(level): other: 1560 40 from $(^3\text{He}, ^8\text{Li})$ .<br>$J^\pi$ : allowed $\varepsilon + \beta^+$ feeding from $3/2^+$ parent; possible mirror level of 1572, $1/2^+$ level in $^{35}\text{S}$ ( <a href="#">1976Be08</a> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 2690 50                     |                                       |                             | <b>C</b>     | E(level): from $(^3\text{He}, ^8\text{Li})$ only.<br>Possible mirror level of 2717, $5/2^+$ level in $^{35}\text{S}$ ( <a href="#">1976Be08</a> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 3781 26                     | $1/2^+, 3/2^+$                        |                             | <b>A</b>     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 4018 37                     | $1/2^+, 3/2^+$                        |                             | <b>A</b>     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 4788 49                     | $1/2^+, 3/2^+$                        |                             | <b>A</b>     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 4982 13                     | $1/2^+, 3/2^+$                        |                             | <b>A</b>     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 5249 73                     | $1/2^+, 3/2^+$                        |                             | <b>A</b>     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 5533 49                     | $1/2^+, 3/2^+$                        |                             | <b>A</b>     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 5710 49                     | $1/2^+, 3/2^+$                        |                             | <b>A</b>     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 5865 38                     | $1/2^+, 3/2^+$                        |                             | <b>A</b>     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 6089 62                     | $1/2^+, 3/2^+$                        |                             | <b>A</b>     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 6335 73                     | $1/2^+, 3/2^+$                        |                             | <b>A</b>     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 9168 23                     | $1/2^+$                               |                             | <b>A</b>     | $T = 5/2$<br>$J^\pi, T$ : isobaric analog state of $1/2^+$ $^{35}\text{Ca}$ g.s. ( <a href="#">1985Ay01, 1999Tr04</a> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

Continued on next page (footnotes at end of table)

---

**Adopted Levels (continued)**

---

 **$^{35}\text{K}$  Levels (continued)**

<sup>†</sup> From  $^{35}\text{Ca}$   $\varepsilon+\beta^+$  decay, unless otherwise noted.

<sup>‡</sup> Allowed  $\varepsilon+\beta^+$  feeding from  $1/2^+$   $^{35}\text{Ca}$  parent, unless otherwise noted.

**$^{35}\text{Ca}$   $\varepsilon+\beta^+$  decay (25.7 ms) 1999Tr04,1985Ay01**

Parent:  $^{35}\text{Ca}$ :  $E=0$ ;  $J^\pi=1/2^+$ ;  $T_{1/2}=25.7$  ms 2;  $Q(\varepsilon+\beta^+)=1.595\times 10^4$  11; % $\varepsilon+\beta^+$  decay=100

$^{35}\text{Ca}$ - $J^\pi$ ,  $T_{1/2}$ : From the Adopted Levels of  $^{35}\text{Ca}$ .

$^{35}\text{Ca}$ - $Q(\varepsilon+\beta^+)$ : 15950 105 deduced by evaluators from mass excesses of 4777 105 for  $^{35}\text{Ca}$  measured by 2023La09 and  $-11172.9$  5 for  $^{35}\text{K}$  from 2021Wa16.  $Q(\varepsilon)$  from 2021Wa16: 16360 200 (syst).

1999Tr04, 1998Le45: A secondary  $^{35}\text{Ca}$  beam at 0.3 ions/s and 98% purity was produced via the projectile fragmentation of a 95-MeV/nucleon  $^{40}\text{Ca}^{20+}$  primary beam impinging on a rotating  $^{nat}\text{Ni}$  target, selected using  $\Delta E$ -ToF by the GANIL LISE3 spectrometer, and implanted into a 500- $\mu\text{m}$  thick silicon detector sandwiched between two 500- $\mu\text{m}$  thick silicon detectors for detecting  $\beta^+$  particles.  $3.5\times 10^4$   $^{35}\text{Ca}$  ions were stopped at a depth of 300  $\mu\text{m}$  with FWHM=70  $\mu\text{m}$  (setting 1) and  $2.5\times 10^4$   $^{35}\text{Ca}$  ions were stopped at a depth of 450  $\mu\text{m}$  (setting 2).  $\varepsilon+\beta^+$ -delayed protons were detected by the implantation detector.  $\gamma$  rays were detected by three Ge detectors and two NaI detectors. Measured  $E_p$ ,  $I_p$ ,  $E_\gamma$ ,  $I_\gamma$ ,  $E_{2p}$ ,  $I_{2p}$ ,  $\beta p$ -coin,  $p\gamma$ -coin. Built the decay scheme consisting of 1p-emitting states in  $^{35}\text{K}$ , a 2p-emitting state ( $T=5/2$  IAS) in  $^{35}\text{K}$ , 1p daughter states in  $^{34}\text{Ar}$ , and a 2p daughter state in  $^{33}\text{Cl}$ . Deduced decay branching ratios,  $B(\text{GT})$  and  $B(\text{F})$ , and parent  $^{35}\text{Ca}$   $T_{1/2}$  from implant-decay correlations.

1985Ay01:  $^{35}\text{Ca}$  isotope discovery.  $^{35}\text{Ca}$  was produced by bombarding a natural calcium target using a 135-MeV  $^3\text{He}$  beam from the 88-inch Cyclotron at Lawrence Berkeley Laboratory. Recoiling products were slowed down, transported, and collected on a slowly rotating catcher wheel. The  $\varepsilon+\beta^+$ -delayed protons were detected using Si detector telescopes. Measured  $E_p$ ,  $I_p$ ,  $pp$ -coin. Built the decay scheme consisting of a 2p-emitting state ( $T=5/2$  IAS) in  $^{35}\text{K}$ , sequential 2p intermediate states in  $^{34}\text{Ar}$ , and 2p daughter states in  $^{33}\text{Cl}$ . Deduced  $^{35}\text{Ca}$   $T_{1/2}$  and  $^{35}\text{Ca}$  mass using the known members of  $A=35$ ,  $T=5/2$  sextuplets IMME.

Theoretical studies involving  $^{35}\text{Ca}$  decay: 2003Sm02, 1991De26, 1990Br26.

The decay scheme is considered relatively complete as all observed individual I(1p) and I(2p) add up to almost 100% in 1999Tr04.

 $^{35}\text{K}$  Levels

| E(level) <sup>†</sup> | $J^\pi$ <sup>#</sup> | $T_{1/2}$ <sup>#</sup> | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------|----------------------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0                     | $3/2^+$              | 175 ms 2               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 1553 5                | $(1/2)^+$            |                        | $E(p0)_{\text{lab}}=1427$ 5, proton line 1 in 1999Tr04.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 3781 26               | $1/2^+, 3/2^+$       |                        | $E(p0)_{\text{lab}}=3592$ 25, proton line 5 in 1999Tr04.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 4018 37               | $1/2^+, 3/2^+$       |                        | $E(p0)_{\text{lab}}=3822$ 36, proton line 6 in 1999Tr04.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 4520 <sup>‡</sup>     |                      |                        | $E(p1)_{\text{lab}}=1909$ -2647, proton group 2 in 1999Tr04, corresponding to $E(\text{level})=4140$ -4900.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 4788 49               | $1/2^+, 3/2^+$       |                        | $E(p0)_{\text{lab}}=4570$ 48, proton line 9 in 1999Tr04.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 4982 13               | $1/2^+, 3/2^+$       |                        | $E(p0)_{\text{lab}}=4754$ 38, proton line 10 in 1999Tr04, corresponding to $E(\text{level})=4977$ 39.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                       |                      |                        | $E(p1)_{\text{lab}}=2727$ 13, proton line 3 in 1999Tr04, corresponding to $E(\text{level})=4982$ 13.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                       |                      |                        | $E(\text{level})$ : weighted average of the two $E(\text{level})$ values of 4977 39 (p0) and 4982 13 (p1).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 5249 73               | $1/2^+, 3/2^+$       |                        | $E(p0)_{\text{lab}}=5018$ 71, proton line 11 in 1999Tr04.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 5493 <sup>‡</sup>     |                      |                        | $E(p1)_{\text{lab}}=2947$ -3500, proton group 4 in 1999Tr04, corresponding to $E(\text{level})=5208$ -5778.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5533 49               | $1/2^+, 3/2^+$       |                        | $E(p0)_{\text{lab}}=5294$ 48, proton line 12 in 1999Tr04.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 5710 49               | $1/2^+, 3/2^+$       |                        | $E(p0)_{\text{lab}}=5466$ 48, proton line 13 in 1999Tr04.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 5716 <sup>‡</sup>     |                      |                        | $E(p2)_{\text{lab}}=1909$ -2647, proton group 2 in 1999Tr04, corresponding to $E(\text{level})=5336$ -6096.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5865 38               | $1/2^+, 3/2^+$       |                        | $E(p0)_{\text{lab}}=5616$ 37, proton line 14 in 1999Tr04.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 6089 62               | $1/2^+, 3/2^+$       |                        | $E(p0)_{\text{lab}}=5834$ 60, proton line 15 in 1999Tr04.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 6302 <sup>‡</sup>     |                      |                        | $E(p3)_{\text{lab}}=1909$ -2647, proton group 2 in 1999Tr04, corresponding to $E(\text{level})=5922$ -6681.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 6335 73               | $1/2^+, 3/2^+$       |                        | $E(p1)_{\text{lab}}=4041$ 71, proton line 7 in 1999Tr04.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 6585 <sup>‡</sup>     |                      |                        | $E(p0)_{\text{lab}}=5983$ -6649, proton group 16 in 1999Tr04, corresponding to $E(\text{level})=6243$ -6928.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 7813 <sup>‡</sup>     |                      |                        | $E(p0)_{\text{lab}}=7131$ -7887, proton group 18 in 1999Tr04, corresponding to $E(\text{level})=7424$ -8203.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 9168 23               | $1/2^+$              |                        | $T=5/2$<br>$E(\text{level})$ : weighted average of the three $E(\text{level})$ values of 9144 92 (p0), 9157 23 (p1), and 9186 27 (2p0).<br>$J^\pi, T$ : isobaric analog state of $1/2^+$ $^{35}\text{Ca}$ g.s. (1985Ay01, 1999Tr04).<br>$E(p0)_{\text{lab}}=8802$ 89, proton line 19 in 1999Tr04, corresponding to $E(\text{level})=9144$ 92.<br>$E(p1)_{\text{lab}}=6783$ 22, proton line 17 in 1999Tr04, corresponding to $E(\text{level})=9157$ 23.<br>$E(2p0)_{\text{lab}}=4305$ 26, proton line 8 in 1999Tr04, corresponding to $E(\text{level})=9186$ 27 with $S(2p)(^{35}\text{K})=4747.5$ 6 (2021Wa16) and adding a +7-keV correction for the difference in the recoil effect between 1p and 2p emissions (1999Tr04). |

Continued on next page (footnotes at end of table)

$^{35}\text{Ca}$   $\varepsilon+\beta^+$  decay (25.7 ms) 1999Tr04,1985Ay01 (continued) $^{35}\text{K}$  Levels (continued)

| <u>E(level)<sup>†</sup></u>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <u>J<sup>π</sup><sup>#</sup></u> | <u>T<sub>1/2</sub><sup>#</sup></u> | Comments |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|------------------------------------|----------|
| Others: E(2p0) <sub>lab</sub> =4089 30, E(2p0) <sub>c.m.</sub> =4311 40 (1985Ay01), corresponding to E(level)=9059 41 in $^{35}\text{K}$ with S(2p)( $^{35}\text{K}$ )=4747.5 6 (2021Wa16). 1985Ay01 observed E(2p1) <sub>lab</sub> =3287 30 and proposed both 2p0 and 2p1 proceed via a sequential decay mechanism with the first proton E(p) <sub>lab</sub> =2213 keV, corresponding to an intermediate state in $^{34}\text{Ar}$ at 6807 keV. However, 2p1 has been ruled out in 1999Tr04 due to the nonobservation of expected py coincidences. 1999Tr04 also states that the observed ratio I(2p0)/I(p)=0.98 9 agrees with the calculated branching ratio I(2p)/I(p)=1 for the IAS (1991De26). |                                  |                                    |          |

<sup>†</sup> Evaluators deduced E(level)( $^{35}\text{K}$ )=E(p)<sub>lab</sub>×[m(p)+m( $^{34}\text{Ar}$ )]/m( $^{34}\text{Ar}$ )+S(p)( $^{35}\text{K}$ )+E(level)( $^{34}\text{Ar}$ ), with S(p)( $^{35}\text{K}$ )=83.6 5 (2021Wa16), E(level)( $^{34}\text{Ar}$ ) from 2012Ni10, and E(p)<sub>lab</sub> from 1999Tr04. For a  $^{35}\text{K}$  proton-emitting level with multiple proton branches, evaluators take the weighted average for E(level)( $^{35}\text{K}$ ) values deduced from each proton branch. 1999Tr04 used S(p)( $^{35}\text{K}$ )=78 20 from 1993Au07, which causes a small difference between the original E(level)( $^{35}\text{K}$ ) in 1999Tr04 and the deduced E(level)( $^{35}\text{K}$ ) here.

<sup>‡</sup> Unresolved proton-emitting levels corresponding to a group of unresolved protons populating one daughter state in  $^{34}\text{Ar}$ , which are not included in the Adopted Levels.

<sup>#</sup> From the Adopted Levels.

 $\varepsilon, \beta^+$  radiations

| <u>E(decay)</u>            | <u>E(level)</u> | <u>I<math>\beta^+</math><sup>#</sup></u> | <u>I<math>\varepsilon</math><sup>#</sup></u> | <u>Log ft</u> | <u>I(<math>\varepsilon+\beta^+</math>)<sup>‡#</sup></u> | Comments                                                                |
|----------------------------|-----------------|------------------------------------------|----------------------------------------------|---------------|---------------------------------------------------------|-------------------------------------------------------------------------|
| (6.78×10 <sup>3</sup> 11)  | 9168            | 8.4 4                                    | 0.0056 9                                     | 3.3 1         | 8.4 4                                                   | I( $\varepsilon+\beta^+$ ): %I(p0)=0.41 6, %I(p1)=3.8 2, %I(2p0)=4.2 3. |
| (8.14×10 <sup>3</sup> 11)  | 7813            | 1.1 2                                    | 4.0×10 <sup>-4</sup> 9                       | 4.6           | 1.1 <sup>‡</sup> 2                                      |                                                                         |
| (9.37×10 <sup>3</sup> 11)  | 6585            | 1.08 17                                  | 2.5×10 <sup>-4</sup> 5                       | 4.9           | 1.08 <sup>‡</sup> 17                                    |                                                                         |
| (9.62×10 <sup>3</sup> 13)  | 6335            | 2.9 3                                    | 6.1×10 <sup>-4</sup> 10                      | 4.6 1         | 2.9 3                                                   |                                                                         |
| (9.65×10 <sup>3</sup> 11)  | 6302            | 2.0 7                                    | 4×10 <sup>-4</sup> 2                         | 4.7           | 2.0 <sup>‡</sup> 7                                      |                                                                         |
| (9.86×10 <sup>3</sup> 13)  | 6089            | 1.39 19                                  | 2.7×10 <sup>-4</sup> 5                       | 4.9 1         | 1.39 19                                                 |                                                                         |
| (1.009×10 <sup>4</sup> 12) | 5865            | 1.42 17                                  | 2.6×10 <sup>-4</sup> 5                       | 5.0 1         | 1.42 17                                                 |                                                                         |
| (1.023×10 <sup>4</sup> 11) | 5716            | 1.0 4                                    | 1.7×10 <sup>-4</sup> 7                       | 5.2           | 1.0 <sup>‡</sup> 4                                      |                                                                         |
| (1.024×10 <sup>4</sup> 12) | 5710            | 0.61 15                                  | 1.1×10 <sup>-4</sup> 3                       | 5.4 +2-1      | 0.61 15                                                 |                                                                         |
| (1.042×10 <sup>4</sup> 12) | 5533            | 0.72 18                                  | 1.2×10 <sup>-4</sup> 3                       | 5.4 +2-1      | 0.72 18                                                 |                                                                         |
| (1.046×10 <sup>4</sup> 11) | 5493            | 2.2 3                                    | 3.6×10 <sup>-4</sup> 7                       | 4.9           | 2.2 <sup>‡</sup> 3                                      |                                                                         |
| (1.070×10 <sup>4</sup> 13) | 5249            | 3.9 3                                    | 5.9×10 <sup>-4</sup> 9                       | 4.7 1         | 3.9 3                                                   |                                                                         |
| (1.097×10 <sup>4</sup> 11) | 4982            | 10.1 7                                   | 0.0014 2                                     | 4.32 6        | 10.1 7                                                  | I( $\varepsilon+\beta^+$ ): %I(p0)=4.2 4, %I(p1)=6.0 5.                 |
| (1.116×10 <sup>4</sup> 12) | 4788            | 2.9 3                                    | 3.8×10 <sup>-4</sup> 6                       | 4.9 1         | 2.9 3                                                   |                                                                         |
| (1.143×10 <sup>4</sup> 11) | 4520            | 5.4 9                                    | 7×10 <sup>-4</sup> 2                         | 4.7           | 5.4 <sup>‡</sup> 9                                      |                                                                         |
| (1.193×10 <sup>4</sup> 12) | 4018            | 3.8 3                                    | 4.1×10 <sup>-4</sup> 6                       | 4.9 1         | 3.8 3                                                   |                                                                         |
| (1.217×10 <sup>4</sup> 11) | 3781            | 3.0 3                                    | 3.0×10 <sup>-4</sup> 5                       | 5.1 1         | 3.0 3                                                   |                                                                         |
| (1.440×10 <sup>4</sup> 11) | 1553            | 48.2 13                                  | 0.0030 4                                     | 4.26 3        | 48.2 13                                                 |                                                                         |

<sup>†</sup> Deduced from the I(p) values in 1999Tr04 multiplied by 0.994. The original % $\Sigma$ I(p)=96.4 18 and % $\Sigma$ I(2p)=4.2 3 in 1999Tr04 lead to % $\Sigma$ I(p)+% $\Sigma$ I(2p)=100.6. The evaluators perform a renormalization of 100.6 to 100, which yields the factor of 0.994. The original I(p) in 1999Tr04 was determined from the number of observed proton events and the total number of implants, with simulated proton-detection efficiencies.

<sup>‡</sup> Feeding to a group of unresolved levels in  $^{35}\text{K}$ . The corresponding Log ft values are displayed without uncertainties.

<sup>#</sup> Absolute intensity per 100 decays.

$^9\text{Be}(^{36}\text{Ca}, ^{35}\text{K})$     **2012Sh21**

$J^\pi=0^+$  for  $^{36}\text{Ca}$  ground state.

**2012Sh21:** A  $^{36}\text{Ca}$  secondary beam was produced via the projectile fragmentation of a 140-MeV/nucleon  $^{40}\text{Ca}$  primary beam impinging on a  $^9\text{Be}$  target at NSCL, MSU. The  $^{36}\text{Ca}$  nuclei were selected by the A1900 separator to a purity of 8%. The ground states of  $^{35}\text{K}$  and  $^{35}\text{Ca}$  were populated by the one-proton/neutron knockout reactions, respectively, from the  $^{36}\text{Ca}$  beam at a midtarget energy of  $\approx 70$  MeV/nucleon on a 188-mg/cm $^2$   $^9\text{Be}$  secondary target. Knockout residues were identified from their energy loss measured by an ionization chamber at the focal plane of the S800 spectrometer and from their ToF measured between two scintillators at the object position and at the focal plane of the S800 spectrometer. The CsI(Na)  $\gamma$ -ray spectrometer CAESAR was placed around the Be target position of the S800 to search for  $\gamma$  decay of any excited states of the residuals formed in knockout reactions. Measured the knockout cross sections for producing  $^{35}\text{K}$  and  $^{35}\text{Ca}$  from  $^{36}\text{Ca}$  and the longitudinal momentum distribution of residuals. Deduced  $J$ ,  $\pi$ , orbital angular momenta of the nucleons removed from  $^{36}\text{Ca}$ , and spectroscopic factors. Calculated single-particle cross sections ( $\sigma_{\text{sp}}$ ) for proton removal and longitudinal momentum distributions using eikonal models.

 $^{35}\text{K}$  Levels

| <u>E(level)</u> | <u><math>J^\pi</math></u> | <u>L</u> | <u><math>C^2S^\dagger</math></u> | <u>Comments</u>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------|---------------------------|----------|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0.0             | $3/2^+$                   | 2        | 3.19 16                          | <p><math>J^\pi</math>: interpreted as knockout from proton <math>1d_{3/2}</math> orbital.</p> <p>L: <b>2012Sh21</b> deduced L by comparing the measured and eikonal-calculated longitudinal momentum distributions of residuals.</p> <p><math>C^2S</math>: from eikonal/Hartree-Fock model with <math>R_s=0.82</math> 4, uncertainty only include experimental contributions. Others: 4.37 24 from eikonal/Strong Absorption model with <math>R_s=1.14</math> 6; 3.62 from shell model calculations.</p> <p><math>\sigma_{\text{exp}}=51.1</math> mb 26. Calculations: <math>\sigma_{\text{sp}}=16.2</math> mb from eikonal/Hartree-Fock model; <math>\sigma_{\text{sp}}=11.7</math> mb from eikonal/Strong Absorption model.</p> |

$^\dagger$  Spectroscopic factor  $C^2S=\sigma_{\text{exp}}/\sigma_{\text{sp}}$ .

<sup>40</sup>Ca(<sup>3</sup>He, <sup>8</sup>Li)    [1976Be08](#)

[1976Be08](#): <sup>35</sup>K isotope discovery. <sup>40</sup>Ca(<sup>3</sup>He, <sup>8</sup>Li)<sup>35</sup>K is studied using 73.7- and 75.8-MeV <sup>3</sup>He beams produced by the Michigan State University cyclotron. Targets were 370 and 190 μg/cm<sup>2</sup> enriched <sup>40</sup>Ca on a 20 μg/cm<sup>2</sup> natural carbon backing. <sup>8</sup>Li particles were detected using a scintillator-proportional counter detector system at the focal plane of the Enge split-pole spectrograph. Measured the cross section for producing <sup>35</sup>K g.s. and the mass excess of <sup>35</sup>K. Also see [1976BeXJ](#) and [1976BeZJ](#).

<sup>35</sup>K Levels

| <u>E(level)</u> | <u>Comments</u>                                                                                                                  |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------|
| 0               |                                                                                                                                  |
| 1560 40         | Assigned by <a href="#">1976Be08</a> as the mirror state of 1/2 <sup>+</sup> level almost at the same energy in <sup>35</sup> S. |
| 2690 50         | Assigned by <a href="#">1976Be08</a> as the mirror state of 5/2 <sup>+</sup> level almost at the same energy in <sup>35</sup> S. |



Adopted Levels

$Q(\beta^-) = -21910$  syst;  $S(n) = 17770$  syst;  $S(p) = 1.03 \times 10^3$  11;  $Q(\alpha) = -8.94 \times 10^3$  11 2021Wa16

$S(p), Q(\alpha)$ : Deduced by evaluators using mass excesses of 4777 105 for  $^{35}\text{Ca}$  measured by 2023La09,  $-1487$  17 for  $^{34}\text{K}$  measured by 2024Dr01, and 11290 16 for  $^{31}\text{Ar}$  measured by 2024Yu13. Values from 2021Wa16:  $S(p) = 880$  280 (syst),  $Q(\alpha) = -8560$  280 (syst).  $\Delta Q(\beta^-) = 450$ ,  $\Delta S(n) = 360$  (syst, 2021Wa16).

$S(2p) = 417$  105,  $Q(\varepsilon) = 15950$  105,  $Q(\varepsilon p) = 15866$  105, from mass excesses of 4777 105 for  $^{35}\text{Ca}$  measured by 2023La09;  $-9384.3$  4 for  $^{33}\text{Ar}$ ,  $-11172.9$  5 for  $^{35}\text{K}$ , and  $-18378.29$  8 for  $^{34}\text{Ar}$  from 2021Wa16. Values from 2021Wa16:  $S(2p) = 00$  200 (syst),

$Q(\varepsilon) = 16360$  200 (syst),  $Q(\varepsilon p) = 16280$  200 (syst).

$S(2n) = 41980$  450 (syst) (2021Wa16).

Isotope discovery (2011Am01):  $^{40}\text{Ca}(^3\text{He}, \alpha n)^{35}\text{Ca}$  at Berkeley (1985Ay01).

$^{35}\text{Ca}$  production:

1986La17, 1986AnZV:  $\text{Ni}(^{40}\text{Ca}, X)$  at GANIL.

$^{35}\text{Ca}$  decay measurements:

1999Tr04, 1998Le45:  $\text{Ni}(^{40}\text{Ca}, X)$  at GANIL. Measured  $T_{1/2}$  and  $\varepsilon + \beta^+$ -delayed 1p/2p emissions.

$^{35}\text{Ca}$  mass measurements: 2023La09, 1985Ay01.

Theoretical calculations: 2003Sm02, 1998Co30, 1997Co19, 1991De26, 1990Br26.

 $^{35}\text{Ca}$  LevelsCross Reference (XREF) Flags

A  $^1\text{H}(^{37}\text{Ca}, t)$   
 B  $^9\text{Be}(^{36}\text{Ca}, ^{35}\text{Ca})$   
 C  $^9\text{Be}(^{37}\text{Ca}, X)$

| E(level)              | $J^\pi$ | $T_{1/2}$ | XREF | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------|---------|-----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0.0                   | $1/2^+$ | 25.7 ms 2 | AB   | $\% \varepsilon + \% \beta^+ = 100$ ; $\% \varepsilon p = 95.8$ 3; $\% \varepsilon 2p = 4.2$ 3<br>$J^\pi$ : $L(^{36}\text{Ca}, ^{35}\text{Ca}) = 0$ from $0^+$ .<br>$T_{1/2}$ : From implant-decay correlation in 1999Tr04. Other: 50 ms 30 estimated by comparison with the $^{22}\text{Al}$ yield from 1985Ay01.<br>$\% \varepsilon p$ and $\% \varepsilon 2p$ are derived from the renormalization of $\% \Sigma I(1p) + \% \Sigma I(2p) = 100.6$ in 1999Tr04 to 100. The original decay branching ratios in 1999Tr04:<br>$\% \Sigma I(1p) = 96.4$ 18 and $\% \Sigma I(2p) = 4.2$ 3. |
| $2.09 \times 10^3$ 10 | $3/2^+$ |           | A C  | E(level): weighted average of $2.24 \times 10^3$ 33 from $^1\text{H}(^{37}\text{Ca}, t)$ and $2.08 \times 10^3$ 10 from $(^{37}\text{Ca}, X)$ .<br>$J^\pi$ : $L = 0$ from $3/2^+$ in $^1\text{H}(^{37}\text{Ca}, t)$ .                                                                                                                                                                                                                                                                                                                                                                  |

$^1\text{H}(^{37}\text{Ca},\text{t})$  2023La09

$J^\pi=3/2^+$  for  $^{37}\text{Ca}$  ground state.

2023La09: A 50-MeV/nucleon  $^{37}\text{Ca}$  secondary beam was produced via the fragmentation of a 95-MeV/nucleon  $^{40}\text{Ca}^{20+}$  primary beam impinging on a  $^9\text{Be}$  target, selected by the LISE3 spectrometer at GANIL, and then impinged on a liquid hydrogen cryogenic target (CRYPTA). Beam ions before target were tracked using two low-pressure multiwire proportional chambers (CATS). Heavy residuals after target were detected by a zero-degree detection system consisting of an ionization chamber, a set of two XY drift chambers, and a plastic scintillator. Tritons from 2n-transfer were detected using a set of six MUST2 DSSD-CsI telescopes in coincidence with the heavy residues of Ca or Ar. Measured  $\sigma(E_t, \theta)$  in inverse kinematics. Deduced levels, J,  $\pi$ , L-transfers from FRESKO-DWBA analysis of measured  $\sigma(\theta)$ .

First measurement of  $^{35}\text{Ca}$  mass excess: 4777 keV 105.

First observation of excited states in  $^{35}\text{Ca}$ .

Evidence of the magicity of N=16 close to the proton drip line.

 $^{35}\text{Ca}$  Levels

| E(level)              | $J^\pi$   | L | Comments                                                                                                                                                                                                                                   |
|-----------------------|-----------|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0.0                   | $(1/2)^+$ | 2 | $J^\pi$ : mirror level: $1/2^+$ $^{35}\text{P}$ g.s. and shell model calculations.<br>L: removal of one neutron from the $2s_{1/2}$ orbital and the other from the $1d_{3/2}$ orbital, leaving a single neutron in the $2s_{1/2}$ orbital. |
| $2.24 \times 10^3$ 33 | $3/2^+$   | 0 | $J^\pi$ : mirror level: $3/2^+$ $^{35}\text{P}$ first excited state at 2386.9 11 and shell model calculations.<br>L: removal of two neutrons from the $2s_{1/2}$ orbital, leaving a single neutron in the $1d_{3/2}$ orbital.              |

$^9\text{Be}(^{36}\text{Ca}, ^{35}\text{Ca})$  2012Sh21

$J^\pi=0^+$  for  $^{36}\text{Ca}$  ground state.

**2012Sh21:** A  $^{36}\text{Ca}$  secondary beam was produced via the projectile fragmentation of a 140-MeV/nucleon  $^{40}\text{Ca}$  primary beam impinging on a  $^9\text{Be}$  target at NSCL, MSU. The  $^{36}\text{Ca}$  nuclei were selected by the A1900 separator to a purity of 8%. The ground states of  $^{35}\text{K}$  and  $^{35}\text{Ca}$  were populated by the one-proton/neutron knockout reactions, respectively, from the  $^{36}\text{Ca}$  beam at a midtarget energy of  $\approx 70$  MeV/nucleon on a 188-mg/cm $^2$   $^9\text{Be}$  secondary target. Knockout residues were identified from their energy loss measured by an ionization chamber at the focal plane of the S800 spectrometer and from their ToF measured between two scintillators at the object position and at the focal plane of the S800 spectrometer. The CsI(Na)  $\gamma$ -ray spectrometer CAESAR was placed around the Be target position of the S800 to search for  $\gamma$  decay of any excited states of the residuals formed in knockout reactions. Measured knockout cross sections for producing  $^{35}\text{K}$  and  $^{35}\text{Ca}$  from  $^{36}\text{Ca}$  and the longitudinal momentum distribution of residuals. Deduced  $J$ ,  $\pi$ , orbital angular momenta of the nucleons removed from  $^{36}\text{Ca}$ , and spectroscopic factors. Calculated single-particle cross sections ( $\sigma_{\text{sp}}$ ) for neutron removal and longitudinal momentum distributions using eikonal models.

 $^{35}\text{Ca}$  Levels

| <u>E(level)</u> | <u><math>J^\pi</math></u> | <u>L</u> | <u><math>C^2S^\dagger</math></u> | <u>Comments</u>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------|---------------------------|----------|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0.0             | $1/2^+$                   | 0        | 0.45 8                           | <p><math>J^\pi</math>: from <math>L(n)=0</math> from <math>0^+</math>, interpreted as knockout from neutron <math>2s_{1/2}</math> orbital.</p> <p>L: <b>2012Sh21</b> deduced L by comparing the measured and eikonal-calculated longitudinal momentum distributions of residuals.</p> <p><math>C^2S</math>: from eikonal/Hartree-Fock model with <math>R_s=0.24</math> 2, uncertainty only include experimental contributions. Others: 0.49 8 from eikonal/Strong Absorption model with <math>R_s=0.26</math> 2 and also from transfer-to-continuum model; <math>R_s=0.26</math> 2; 1.80 from shell model.</p> <p><math>\sigma_{\text{exp}}=5.03</math> mb 46. Calculated values: <math>\sigma_{\text{sp}}=11.1</math> mb from eikonal/Hartree-Fock model; <math>\sigma_{\text{sp}}=10.2</math> mb from eikonal/Strong Absorption model; <math>\sigma_{\text{sp}}=10.3</math> mb from transfer-to-continuum model.</p> |

$^\dagger$  Spectroscopic factor  $C^2S=\sigma_{\text{exp}}/\sigma_{\text{sp}}$ .

<sup>9</sup>Be(<sup>37</sup>Ca,X)    [2024Dr01](#)

$J^\pi=3/2^+$  for <sup>37</sup>Ca ground state.  
[2024Dr01](#): a 72-MeV/nucleon <sup>37</sup>Ca secondary beam was produced via the fragmentation of a 140-MeV/nucleon <sup>40</sup>Ca<sup>20+</sup> primary beam impinging on a 0.5-mm-thick Be target. Experimental setup includes the CAESium-iodide scintillator ARray (CAESAR) for detecting  $\gamma$  rays, a DSSD-CsI(Tl)  $\Delta E$ -E Ring Telescope for detecting protons, a Scintillating-Fiber Array (SFA) and the S800 spectrograph for detecting heavy residuals. Measured  $E_p$ ,  $E_\gamma$ , and  $E(\text{residual})$ . Deduced the first excited state in <sup>35</sup>Ca via the  $2p+^{33}\text{Ar}$  exit channel from total decay-energy spectra using the invariant-mass method. Comparisons with shell-model calculations with the USDC Hamiltonian.

<sup>35</sup>Ca Levels

| <u>E(level)</u>         | <u>J<sup><math>\pi</math></sup></u> | <u>Comments</u>                                                                                                                          |
|-------------------------|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| 2.08×10 <sup>3</sup> 10 | (3/2 <sup>+</sup> )                 | E(level): From total decay energy of $2p+^{33}\text{Ar}$ $E_T=1667$ keV 20.<br>J <sup><math>\pi</math></sup> : shell-model calculations. |

## REFERENCES FOR A=35

- 1919As01 F.W.Aston - Nature(London) 104, 393 (1919).  
*The Constitution of the Elements.*
- 1936An01 E.B.Andersen - Z.Phys.Chem., Abt. B32, 237 (1936).  
*Ein radioaktives Isotop des Schwefels.*
- 1940Ki12 L.D.P.King, D.R.Elliott - Phys.Rev. 58, 846 (1940).  
*Short-Lived Radioactivities of  $^{14}\text{Si}^{27}$ ,  $^{16}\text{S}^{31}$ , and  $^{18}\text{A}^{35}$ .*
- 1941El04 D.R.Elliott, L.D.P.King - Phys.Rev. 59, 403 (1941).  
*Radioactive  $^{21}\text{Sc}^{41}$ ,  $^{18}\text{A}^{35}$  and  $^{16}\text{S}^{31}$ .*
- 1941Ka01 M.D.Kamen - Phys.Rev. 60, 537 (1941).  
*Production and Isotopic Assignment of Long-Lived Radioactive Sulfur.*
- 1941Ki01 L.D.P.King, D.R.Elliott - Phys.Rev. 59, 108 (16) (1941).  
*Short-Lived Radioactivities  $^{14}\text{Si}^{27}$ ,  $^{16}\text{S}^{31}$ ,  $^{18}\text{A}^{35}$ , and  $^{21}\text{Sc}^{41}$ .*
- 1943He01 R.H.Hendricks, L.C.Bryner, M.D.Thomas, J.O.Ivie - J.Phys.Chem. 47, 469 (1943).  
*Measurement of the activity of radiosulfur in barium sulfate.*
- 1948To10 C.H.Townes, A.N.Holden, F.R.Merritt - Phys.Rev. 74, 1113 (1948); See Also 47To16 (N,Br), 49To17 (Cl).  
*Microwave Spectra of Some Linear X Y Z Molecules.*
- 1949Co12 V.W.Cohen, W.S.Koski, T.Wentink, Jr. - Phys.Rev. 76, 703 (1949).  
*Nuclear Spin and Quadrupole Coupling of  $\text{S}^{35}$ .*
- 1949Da14 L.Davis, Jr., B.T.Feld, C.W.Zabel, J.R.Zacharias - Phys.Rev. 76, 1076 (1949).  
*The Hyperfine Structure and Nuclear Moments of the Stable Chlorine Isotopes.*
- 1951We11 T.Wentink, Jr., W.S.Koski, V.W.Cohen - Phys.Rev. 81, 948 (1951); See Also 49Co12.  
*The Mass of  $\text{S}^{35}$  from Microwave Spectroscopy.*
- 1954Bi40 G.R.Bird, C.H.Townes - Phys.Rev. 94, 1203 (1954); Calculated Using Data of 51We11.  
*Sulfur Bonds and the Quadrupole Moments of O, S, and Se Isotopes.*
- 1954Bu05 B.F.Burke, M.W.P.Strandberg, V.W.Cohen, W.S.Koski - Phys.Rev. 93, 193 (1954).  
*The Nuclear Magnetic Moment of  $\text{S}^{35}$  by Microwave Spectroscopy.*
- 1954En19 P.M.Endt, J.C.Kluyver - Rev.Mod.Phys. 26, 95 (1954).  
*Energy Levels of Light Nuclei ( $Z = 11$  to  $Z = 20$ ).*
- 1955Pa54 C.H.Paris, W.W.Buechner, P.M.Endt - Phys.Rev. 100, 1317 (1955).  
*Energy Levels of  $\text{S}^{33}$ ,  $\text{S}^{35}$ ,  $\text{Cl}^{36}$ ,  $\text{Cl}^{38}$ , and  $\text{Ba}^{139}$ .*
- 1955To32 L.A.Toller, J.R.Patterson, H.W.Newson - Phys.Rev. 99, 620 (1955).  
*Neutron Resonances in the Kilovolt Region:  $\text{Cl}^{35}$ ,  $\text{Cl}^{37}$ ,  $\text{K}^{39}$ , and  $\text{K}^{41}$ .*
- 1956Ki29 O.C.Kistner, A.Schwarzschild, B.M.Rustad - Phys.Rev. 104, 154 (1956).  
*Decay of  $\text{A}^{35}$ .*
- 1957En01 P.M.Endt, C.M.Braams - Rev.Mod.Phys. 29, 683 (1957).  
*Energy Levels of Light Nuclei ( $Z = 11$  to  $Z = 20$ ). II.*
- 1958En51 P.M.Endt, C.H.Paris - Phys.Rev. 110, 89 (1958).  
*Nuclear Levels in  $\text{P}^{30}$ ,  $\text{S}^{33}$ , and  $\text{S}^{35}$ .*
- 1958Se49 H.H.Seliger, W.B.Mann, L.M.Cavallo - J.Research Natl.Bur.Standards 60, 447 (1958).  
*Average Energy of Sulfur-35 Beta Decay.*
- 1959Al10 J.S.Allen, R.L.Burman, W.B.Herrmannsfeldt, P.Stahelm, T.H.Braid - Phys.Rev. 116, 134 (1959).  
*Determination of the Beta-Decay Interaction from Electron-Neutrino Angular Correlation Measurements.*
- 1959Ca12 J.P.Cali, L.F.Lowe - Nucleonics 17, No.10, 86 (1959).  
*Direct Determination of the Half-Lives of Nine Nuclides.*
- 1959Co56 R.D.Cooper, E.S.Cotton - Science 129, 1360 (1959).  
*Half-Life of Sulfur-35.*
- 1960An06 Yu.P.Antufev, A.K.Valter, V.Yu.Gonchar, E.G.Kopanets et al. - Izvest.Akad.Nauk SSSR, Ser.Fiz. 24, 877 (1960);  
Columbia Tech.Transl.24, 878 (1961).
- 1960Ja12 J.Janecke - Z.Naturforsch. 15a, 593 (1960).  
*The Half Lives of Some Positron Emitters with Superpermitted Transitions.*
- 1960Wa04 R.Wallace, J.A.Welch, Jr. - Phys.Rev. 117, 1297 (1960).  
*Beta Spectra of the Mirror Nuclei.*
- 1961Oz01 Y.Ozias, H.Roux, P.Boivin, E.Calvet - Compt.rend. 253, 2944 (1961).  
*Determination Microcalorimetrique des Caracteristiques d'un Element Radioactif.*
- 1961St09 R.S.Storey, L.W.Oleksiuk - Can.J.Phys. 39, 917 (1961).  
*Low-Lying Energy Levels in  $\text{Cl}^{35}$ .*
- 1961Wy01 E.I.Wyatt, S.A.Reynolds, T.H.Handley, W.S.Lyon, H.A.Parker - Nucl.Sci.Eng. 11, 74 (1961).  
*Half-Lives of Radionuclides. II.*
- 1962Bo17 E.C.Booth, K.A.Wright - Nuclear Phys. 35, 472 (1962).  
*Nuclear Resonance Scattering of Bremsstrahlung.*
- 1962Du05 J.Dubois - Arkiv Fysik 22, 400 (1962).  
*(p, $\gamma$ ) Reactions on Nuclei in the Mass Region  $A = 30$ -50.*

## REFERENCES FOR A=35(CONTINUED)

- 1962En05 P.M.Endt, C.van der Leun - Nucl.Phys. 34, 1 (1962).  
*Energy Levels of Light Nuclei. III Z = 11 to Z = 20.*
- 1963Du08 J.Dubois - Trans. Chalmers Univ.Technol., Gothenburg, No. 266 (1963).  
*An Investigation of the  $S^{34}(p,\gamma)Cl^{35}$  Reaction.*
- 1963Ha32 N.Hazewindus, W.Lourens, A.Scheepmaker, A.H.Wapstra - Physica 29, 681 (1963).  
*Gamma Ray Transitions in  $Cl^{35}$ .*
- 1963Ne05 J.W.Nelson, E.B.Carter, G.E.Mitchell, R.H.Davis - Phys.Rev. 129, 1723 (1963).  
*Thresholds Observed in the  $C^{12}(\alpha,n)O^{15}$ ,  $Si^{28}(\alpha,n)S^{31}$ ,  $S^{32}(\alpha,n)Ar^{35}$ ,  $S^{34}(\alpha,n)Ar^{37}$ , and  $Cl^{35}(p,n)Ar^{35}$  Reactions.*
- 1964Hy01 A.K.Hyder, Jr., W.A.Anderson, G.I.Harris - Bull.Am.Phys.Soc. 9, No.4, 440, FA11 (1964).  
*Resonances in the  $^{32,33,34}S(p,\gamma)^{33,34,35}Cl$  Reactions.*
- 1964Ku12 J.Kuperus - Physica 30, 899 (1964).  
*Protons from the alpha-particle bombardment of  $^{31}P$ .*
- 1965Ca04 F.P.Calaprice, E.D.Commins, D.A.Dobson - Phys.Rev. 137, B1453 (1965).  
*Beta-Decay Asymmetry and Nuclear Magnetic Moment of Argon-35.*
- 1965Fl02 K.F.Flynn, L.E.Glendenin, E.P.Steinberg - Nucl.Sci.Eng. 22, 416 (1965).  
*Half-Life Determinations by Direct Decay.*
- 1966Az01 R.E.Azuma, L.W.Oleksiuk, J.D.Prentice, P.Taras - Phys.Rev.Letters 17, 659 (1966).  
*Properties of the 3.16-MeV Level of  $Cl^{35}$ .*
- 1966Ba54 R.H.Bassel - Phys.Rev. 149, 791 (1966).  
*Normalization of and Finite-Range Effects in  $(^3He,d)$  and  $(t,d)$  Reactions.*
- 1966En04 G.A.P.Engelbertink, P.M.Endt - Nucl.Phys. 88, 12(1966).  
*Measurements of  $(p,\gamma)$  Resonance Strengths in the s-d Shell.*
- 1966Er06 F.C.Erne - Nucl.Phys. 84, 241(1966).  
*Gamma-Ray Angular Correlation and Polarization Measurements in the Reaction  $^{35}Cl(p,\gamma)^{36}Ar$ .*
- 1966Ho02 J.H.Hough, W.L.Mouton - Nucl.Phys. 76, 248(1966).  
*Nuclear Resonant Scattering of Bremsstrahlung from Low-Lying Levels in  $^{27}Al$ ,  $^{31}P$  and  $^{35}Cl$ .*
- 1966Ko15 E.G.Kopanets, Y.S.Korda, A.A.Koval, L.N.Sukhotin, S.P.Tsytko - Ukr.Fiz.Zh. 11, 929 (1966).  
*Investigation of the  $^{34}S(p,\gamma)^{35}Cl$  Reaction.*
- 1966Ko22 E.G.Kopanets, A.A.Koval, Y.S.Korda, L.N.Sukhotin, S.P.Tsytko - Izv.Akad.Nauk SSSR, Ser.Fiz. 30, 407 (1966);  
Bull.Acad.Sci.USSR, Phys.Ser. 30, 414 (1967).  
*Investigation of the  $S^{34}(p,\gamma)Cl^{35}$  Reaction at the 1904 and 2008keV Proton Resonance Energies.*
- 1966Ni03 D.B.Nichols, B.D.Kern, M.T.McEllistrem - Phys.Rev. 151, 879 (1966).  
*Cross Sections of  $^{35,37}Cl(n,n'\gamma)$  Reactions.*
- 1966Sc09 J.P.Schiffer, L.L.Lee, Jr., A.Marinov, C.Mayer-Boricke - Phys.Rev. 147, 829 (1966).  
*Dependence of the Angular Distribution of the  $(d,p)$  Reaction on the Total Angular-Momentum Transfer. II.*
- 1966Wa09 D.D.Watson - Phys.Letters 22, 183 (1966).  
*Spin and Parity of the 3.16 MeV and 7.54 MeV Levels in  $^{35}Cl$ .*
- 1967Da12 D.A.E.Darwish, S.A.El-Kateb, L.M.El-Nadi, O.E.Badawy, A.N.Lvov - Z.Physik 205, 8 (1967).  
*The  $S^{34}(p,\gamma)Cl^{35}$  Reaction and Energy Levels of the  $Cl^{35}$  Nucleus.*
- 1967En05 P.M.Endt, C.Van der Leun - Nucl.Phys. A105, 1 (1967); Erratum Nucl.Phys. A115, 697 (1968).  
*Energy Levels of Z=11-21 Nuclei (IV).*
- 1967Ko19 E.G.Kopanets, A.A.Koval, Y.S.Korda, L.N.Sukhotin, S.P.Tsytko - Izv.Akad.Nauk SSSR, Ser.Fiz. 31, 626 (1967);  
Bull.Acad.Sci.USSR, Phys.Ser. 31, 621 (1968).  
*Investigation of the Gamma Rays from Inelastic Scattering of Protons on  $^{34}S$ .*
- 1967Ko20 E.G.Kopanets, A.A.Koval, L.N.Sukhotin, S.P.Tsytko - Izv.Akad.Nauk SSSR, Ser.Fiz. 31, 631 (1967);  
Bull.Acad.Sci.USSR, Phys.Ser. 31, 626 (1968).  
*Spin and Parity Assignments for the 3.01 and 2.65 MeV Levels of  $^{35}Cl$ .*
- 1967Ko23 E.G.Kopanets, Y.S.Korda, A.A.Koval, L.N.Sukhotin, S.P.Tsytko - Yadern.Fiz. 6, 233 (1967); Soviet J.Nucl.Phys. 6, 170 (1968).  
*p- $\gamma$ - $\gamma$  Triple Angular Correlations from the Reactions  $S^{34}(p,\gamma)Cl^{35}$ .*
- 1967Ma48 J.C.Manthuruthil, F.D.Lee, D.D.Watson - Nucl.Sci.Appl. 3, No.2, 44 (1967).  
*Analog States Through the  $S^{34}(p,\gamma)Cl^{35}$  Reaction.*
- 1967Ta08 P.Taras, L.W.Oleksiuk, R.E.Azuma, J.D.Prentice - Phys.Rev. 164, 1386 (1967).  
*Properties of Some Levels of  $Cl^{35}$ .*
- 1967Wa22 D.D.Watson, J.C.Manthuruthil, F.D.Lee - Phys.Rev. 164, 1399 (1967).  
*Isobaric Analog States in  $Cl^{35}$ .*
- 1968Az02 R.E.Azuma, N.Anyas-Weiss, A.M.Charlesworth - Nucl.Phys. A109, 577(1968).  
*Lifetime of the 3.163 MeV Level of  $^{35}Cl$ .*
- 1968Jo04 R.R.Johnson, R.J.Griffiths - Nucl.Phys. A108, 113(1968).  
*The  $^{40}Ar(p,d)^{39}Ar$  and  $^{36}Ar(p,d)^{35}Ar$  Reactions at 27.5 MeV.*
- 1968Ko11 R.L.Kozub - Phys.Rev. 172, 1078 (1968).

## REFERENCES FOR A=35(CONTINUED)

- (p,d) Reaction on  $N = Z$  Nuclei in the 2s-1d shell.  
 1968Te06 L.K.ter Veld, T.W.van der Mark - Phys.Rev. 173, 1101 (1968).  
*First Excited State of  $^{35}\text{S}$ .*  
 1968Th07 W.J.Thompson, J.L.Adams, D.Robson - Phys.Rev. 173, 975 (1968); Erratum Phys.Rev. C7, 1273 (1973).  
*Neutron Spectroscopic Factors from Isobaric Analog States.*  
 1968Wo06 E.J.Woodhouse, T.H.Norris - J.Inorg.Nucl.Chem. 30, 1373 (1968).  
*Half-Life of Sulphur-35.*  
 1969Ah02 H.Ahlgren, J.Dubois, H.Linusson - Arkiv Fysik 38, 585 (1969).  
*A Study of the  $^{34}\text{S}(p,\gamma)^{35}\text{Cl}$  Reaction.*  
 1969An35 A.E.Antropov, P.P.Zarubin, P.D.Ioannu, B.N.Orlov et al. - Izv.Akad.Nauk SSSR, Ser.Fiz. 33, 1721 (1969);  
 Bull.Acad.Sci.USSR, Phys.Ser. 33, 1576 (1970).  
*Elastic and Inelastic Proton Scattering on  $^{33}\text{S}$ ,  $^{34}\text{S}$  and  $^{35}\text{Cl}$ .*  
 1969Da12 N.E.Davison, J.T.Sample, W.J.McDonald, W.J.Wiesehahn et al. - Phys.Rev. 182, 1168 (1969).  
*Levels in  $^{35}\text{Cl}$  Populated by  $^{34}\text{S}(d,n)^{35}\text{Cl}$ .*  
 1969Du08 D.D.Duncan, K.H.Buerger, R.L.Place, B.D.Kern - Phys.Rev. 185, 1515 (1969).  
*Low-Lying Levels of  $^{35}\text{Cl}$  and  $^{37}\text{Cl}$ .*  
 1969Fr09 J.M.Freeman, D.C.Robinson, G.L.Wick - Phys.Letters 30B, 240 (1969).  
*Magnitude of the Vector Coupling Constant Deduced from Recent Beta Decay Measurements.*  
 1969Gr23 A.Graue, L.H.Herland, J.R.Lien, G.E.Sandvik et al. - Nucl.Phys. A136, 577 (1969).  
*A Study of the  $^{35}\text{Cl}$  Levels by Means of the  $^{34}\text{S}(^3\text{He},d)^{35}\text{Cl}$  Reaction.*  
 1969Ha17 O.Hausser, D.Pelte, T.K.Alexander, H.C.Evans - Can.J.Phys. 47, 1065 (1969).  
*Coulomb Excitation of  $^{35}\text{Cl}$  Projectiles.*  
 1969In04 F.Ingebreetsen, T.K.Alexander, O.Hausser, D.Pelte - Can.J.Phys. 47, 1295(1969).  
*Lifetime Measurements of the Six Lowest Lying Levels in  $^{35}\text{Cl}$ .*  
 1969Ko26 E.G.Kopanets - Izv.Akad.Nauk SSSR, Ser.Fiz. 33, 711 (1969); Bull.Acad.Sci.USSR, Phys.Ser. 33, 655 (1970).  
*Analog States in  $^{35}\text{Cl}$ .*  
 1969KrZW M.Kregar, P.Kump, V.Ramsak, M.Vakselj et al. - Bull.Am.Phys.Soc. 14, No.12, 1222, CC12 (1969).  
*Properties of Some Excited States in  $^{35}\text{Cl}$ .*  
 1969La34 F.Lagoutine, J.Legrand, Y.Le Gallic - Intern.J.Appl.Radiation Isotopes 20, 868 (1969).  
*Periodes de Quelques Radionucleides.*  
 1969Mi23 V.Y.Migalenyia, E.G.Kopanets, A.M.Lvov, A.P.Gaidamaka et al. - Ukr.Fiz.Zh. 14, 1198 (1969).  
*The 3.95 MeV Level and Decay Scheme of the 8.38 MeV Isobar-Analog State in  $\text{Cl}^{35}$ .*  
 1969Mo12 G.E.Moss - Nucl.Phys. A131, 235 (1969).  
*Excitation Energies of Levels in  $^{35}\text{S}$ .*  
 1969Ta13 P.Taras, J.Matas - Can.J.Phys. 47, 1605 (1969).  
*Properties of the 1.763 MeV Level in  $^{35}\text{Cl}$ .*  
 1969Wi18 G.L.Wick, D.C.Robinson, J.M.Freeman - Nucl.Phys. A138, 209 (1969).  
*A New Measurement of the ft Value for the Mirror Decay  $^{35}\text{Ar}(\beta^+)^{35}\text{Cl}$  and a Calculation of the Vector Coupling Constant.*  
 1970Bu18 K.S.Burton, L.C.McIntyre, Jr. - Nucl.Phys. A154, 551 (1970).  
*Mean Lives and Angular Correlations in  $^{35}\text{S}$  with the  $^{34}\text{S}(d,py)^{35}\text{S}$  Reaction.*  
 1970Ho09 B.W.Hooton, O.Hausser, F.Ingebreetsen, T.K.Alexander - Can.J.Phys. 48, 1259 (1970).  
*Level Structure of  $^{35}\text{Cl}$  from the  $^{32}\text{S}(\alpha,py)^{35}\text{Cl}$  Reaction.*  
 1970In01 F.Ingebreetsen, C.Broude, J.S.Forster - Phys.Lett. 31B, 297 (1970).  
*Properties of the 3942 keV Level in  $^{35}\text{Cl}$  and Evidence for a Ground State Rotational Band.*  
 1970Ko33 E.G.Kopanets, V.Y.Migalenyia, A.N.Lvov, A.A.Koval et al. - Izv.Akad.Nauk SSSR, Ser.Fiz. 34, 2101 (1970);  
 Bull.Acad.Sci. USSR, Phys.Ser. 34, 1873 (1971).  
*Search for a  $2p_{1/2}$  Analog State in  $^{35}\text{Cl}$ .*  
 1970Mo01 R.A.Morrison - Nucl.Phys. A140, 97 (1970).  
*A Study of Nuclei with Masses Near 32 with the  $(^3\text{He},d)$  Reaction.*  
 1970Mo10 M.A.Moinester, W.P.Alford - Nucl.Phys. A145, 143 (1970).  
*The Structure of  $^{36}\text{Ar}$  and  $^{38}\text{Ar}$  via the  $(^3\text{He},d)$  Reaction on  $^{35}\text{Cl}$  and  $^{37}\text{Cl}$ .*  
 1970Ta02 P.Taras, J.Matas - Can.J.Phys. 48, 603 (1970).  
*Properties of the 2.645, 2.693, 3.001, and 3.162 MeV Levels in  $^{35}\text{Cl}$ .*  
 1970Ta09 P.Taras, J.Matas - Nucl.Instrum.Methods 77, 10 (1970).  
*Analysis of Linear Polarization Measurements of Unresolved Gamma Rays.*  
 1970Um01 C.J.Umbarger, K.W.Kemper, J.W.Nelson, H.S.Plendl - Phys.Rev. C2, 1378 (1970).  
*Excitation Functions for the Reactions  $^{34}\text{S}(p,n)^{34}\text{Cl}$  and  $^{31}\text{P}(\alpha,n)^{34}\text{Cl}$ .*  
 1970Wh04 C.A.Whitten, Jr., M.C.Mermaz, D.A.Bromley - Phys.Rev. C1, 1455 (1970).  
*Study of the (d,t) Reaction on  $\text{Si}^{28}$ ,  $\text{S}^{32}$ , and  $\text{Ar}^{36}$ .*  
 1971Ar32 A.G.Artukh, V.V.Avdeichikov, G.F.Gridnev, V.L.Mikheev et al. - Nucl.Phys. A176, 284 (1971).



## REFERENCES FOR A=35(CONTINUED)

- New Isotopes  $^{29,30}\text{Mg}$ ,  $^{31,32,33}\text{Al}$ ,  $^{33,34,35,36}\text{Si}$ ,  $^{35,36,37,38}\text{P}$ ,  $^{39,40}\text{S}$  and  $^{41,42}\text{Cl}$  Produced in Bombardment of a  $^{232}\text{Th}$  Target with 290 MeV  $^{40}\text{Ar}$  Ions.*  
1971Au07 L.S.August, P.Shapiro, L.R.Cooper, C.D.Bond - Phys.Rev. C4, 2291 (1971).  
*J Dependence for  $L = 2$  ( $\alpha, p$ ) Transitions in  $s$ - $d$  Shell Nuclei.*  
1971Ba23 R.D.Barton, J.S.Wadden, A.L.Carter, H.L.Pai - Can.J.Phys. 49, 971 (1971).  
*Lifetime of the 3.163 MeV Level of  $^{35}\text{Cl}$ .*  
1971BrXT W.Bruynesteyn - Drukkerij Elinkwijk-Utrecht (1971).  
*Elastische Protonenverstrooiing aan  $^{34}\text{S}$  en  $^{35}\text{Cl}$ .*  
1971Ca40 J.M.G.Caraca, R.D.Gill, P.B.Johnson, H.J.Rose - Nucl.Phys. A176, 273 (1971).  
*Transition Strengths in  $^{37}\text{Ar}$  and  $^{35}\text{Cl}$ .*  
1971De05 C.Detraz, C.E.Moss, C.S.Zaidins - Phys.Lett. 34B, 128 (1971).  
*Beta Decay of  $^{23}\text{Mg}$ ,  $^{27}\text{Si}$ ,  $^{31}\text{S}$ ,  $^{35}\text{Ar}$  and  $^{39}\text{Ca}$ .*  
1971Fr05 W.R.French, Jr., W.M.Wehrbein, D.Goss, J.K.Gasper - Amer.J.Phys. 39, 131 (1971).  
*Determination of Energy Levels in Stable Nuclei by Neutron Inelastic Scattering.*  
1971Ge04 J.S.Geiger, B.W.Hooton - Can.J.Phys. 49, 663 (1971).  
*Beta Decay of  $^{35}\text{Ar}$ .*  
1971Gr53 W.Grimm, W.Herzog - Z.Naturforsch. 26a, 1933 (1971).  
*Phosphor-35 – ein neues Nuklid.*  
1971Ko32 E.G.Kopanets, A.N.Lvov, V.Y.Migaleny, V.Y.Kostin et al. - Izv.Akad.Nauk SSSR, Ser.Fiz. 35, 1676 (1971);  
Bull.Acad.Sci.USSR, Phys.Ser. 35, 1526 (1972).  
*Proton Radiative Capture and Isobaric-Analog States in  $^{27}\text{Al}$ ,  $^{35}\text{Cl}$ , and  $^{41}\text{K}$ .*  
1971Ko33 R.P.Kolalis, Y.A.Nemilov, V.S.Romanov, V.S.Sadkovskii et al. - Izv.Akad.Nauk SSSR, Ser.Fiz. 35, 1684 (1971);  
Bull.Acad.Sci.USSR, Phys.Ser. 35, 1533 (1972).  
*( $d, p$ ) and ( $d, \alpha$ ) Reactions on  $^{34}\text{S}$  for 6.6 MeV Deuterons.*  
1971Ko46 E.G.Kopanets, V.Y.Migaleny, A.N.Lvov, A.A.Koval et al. - Izv.Akad.Nauk SSSR, Ser.Fiz. 35, 1682 (1971);  
Bull.Acad.Sci.USSR, Phys.Ser. 35, 1531 (1972).  
*Behavior of ( $p, \gamma$ )-Reaction Excitation Functions Near Isobaric Analog Resonances.*  
1971Mc23 W.R.McMurray, D.M.Holz, I.J.van Heerden, G.Wiechers - Z.Phys. 247, 453 (1971).  
*Study of the Reactions  $^{31}\text{P}(\alpha, p)^{34}\text{S}$  and  $^{29}\text{Si}(\alpha, n)^{32}\text{S}$ .*  
1971Me12 M.C.Mermaz, C.A.Whitten, Jr., J.W.Champlin et al. - Phys.Rev. C4, 1778 (1971).  
*Study of the ( $d, p$ ) Reaction on  $^{28}\text{Si}$ ,  $^{32}\text{S}$ , and  $^{36}\text{Ar}$  at  $E = 18.00$  MeV.*  
1971Pr11 F.W.Prosser, Jr., G.I.Harris - Phys.Rev. C4, 1611 (1971).  
*Gamma-Ray Transitions in  $A = 35$  Nuclei.*  
1971Ta02 P.Taras - Can.J.Phys. 49, 328 (1971).  
*Gamma-Ray Linear Polarization and Angular Distribution Formulas in Terms of Phase-Defined Reduced Matrix Elements.*  
1971Va18 J.G.van der Baan, H.G.Leighton - Nucl.Phys. A170, 607 (1971).  
*Investigation of the  $^{34}\text{S}(d, p)^{35}\text{S}$  Reaction at  $E = 10$  MeV.*  
1971VaZG JOUR NTNAA 37 396, T W van der Mark.  
1971Vi02 B.Vignon, J.F.Burandet, N.Longequeue, I.S.Towner - Nucl.Phys. A162, 82 (1971).  
*Analysis of the  $^{35,37}\text{Cl}(p, t)$  and ( $p, ^3\text{He}$ ) Reactions to Mirror and Analogue States.*  
1971Wi13 W.J.Wieschahn - Can.J.Phys. 49, 2396 (1971).  
*Levels in  $^{35}\text{Cl}$ : Experimental.*  
1972Ap01 K.E.Apt, J.D.Knight - Phys.Rev. C6, 842 (1972).  
*Decay of  $^{35}\text{P}$ .*  
1972Br33 C.Broude, J.S.Forster, F.Ingebretsen - Nucl.Phys. A192, 291 (1972).  
*Electromagnetic Transition Strengths in  $^{35}\text{Cl}$ .*  
1972Br45 J.E.Brock, I.A.Luketina, A.R.Poletti - Phys.Rev. C6, 1298 (1972).  
*Lifetime Measurements in  $^{35}\text{Cl}$  and  $^{37}\text{Cl}$ .*  
1972Dz13 J.D.Jafar, A.A.Abdullah, N.K.Al-Kuraishi, M.S.Alvash et al. - Yad.Fiz. 15, 1093 (1972); Sov.J.Nucl.Phys. 15, 605 (1972).  
*Spectra of  $\gamma$  Rays Produced in  $\text{Si}^{30}$  and  $\text{S}^{34}$  Capture of Thermal Neutrons.*  
1972Fr11 R.M.Freeman, R.Faerber, M.Toulemonde, A.Gallmann - Nucl.Phys. A197, 529 (1972).  
*Etude de Transitions  $\gamma$  dans le Noyau  $^{35}\text{S}$ .*  
1972Go31 D.R.Goosman, D.E.Alburger - Phys.Rev. C6, 820 (1972).  
*Mass and Decay of a Highly Neutron-Rich Isotope of Phosphorus: 48-sec  $^{35}\text{P}$ .*  
1972Hu10 P.Hubert, M.M.Aleonard, D.Castera, F.Leccia, P.Mennrath - Nucl.Phys. A195, 485 (1972).  
*Etude des Etats Excites du Noyau  $^{35}\text{Cl}$  (I). Courbe d'Excitation de la Reaction  $^{34}\text{S}(p, \gamma)^{35}\text{Cl}$ , Energie et Rapports d'Embranchement des Niveaux du  $^{35}\text{Cl}$ .*  
1972Hu11 P.Hubert, M.M.Aleonard, D.Castera, F.Leccia, P.Mennrath - Nucl.Phys. A195, 502 (1972).  
*Etude des Etats Excites du  $^{35}\text{Cl}$  (II). Vies Moyennes.*  
1972LeYW F.D.Lee, D.D.Watson - Bull.Amer.Phys.Soc. 17, No.10, 892, AB9 (1972).  
*Gamma Decay of Excited States of  $\text{Cl}^{35}$ .*



## REFERENCES FOR A=35(CONTINUED)

- 1972Mi13 D.W.Miller, F.Everling, D.A.Outlaw, T.G.Dzubay et al. - Phys.Rev. C6, 869 (1972).  
*Decay of  $^{36}\text{K}$ .*
- 1972MiZO JOUR BAPSA 17 892,R Middleton,10/25/72.
- 1972Sh07 N.Shikazono, Y.Kawarasaki - Nucl.Phys. A188, 461 (1972).  
*Study of Odd-A Nuclei in the 2s-1d Shell by Means of the  $(\gamma, \gamma')$  Reaction.*
- 1972St38 R.M.Sternheimer - Phys.Rev. A6, 1702 (1972).  
*Quadrupole Shielding and Antishielding Factors for Several Atomic Ground States.*
- 1972Va06 R.J.Van Reenen, W.J.Naude, W.L.Mouton - Nucl.Phys. A183, 651 (1972).  
*Lifetimes of the First Six Excited States of  $^{35}\text{Cl}$ .*
- 1972Va07 T.W.van der Mark, L.K.ter Veld - Nucl.Phys. A181, 196 (1972).  
*Angular Correlations and Decay of Excited States in  $^{35}\text{S}$ .*
- 1973Al22 N.G.Alenius, E.Wallander - Phys.Scr. 8, 129 (1973).  
*High Spin States in  $^{35}\text{Cl}$  from the  $^{32}\text{S}(\alpha, p)^{35}\text{Cl}$  Reaction.*
- 1973An01 N.Anyas-Weiss, R.Griffiths, N.A.Jelley, W.Randolph et al. - Nucl.Phys. A201, 513 (1973).  
*Lifetimes in  $^{22}\text{Ne}$ ,  $^{22}\text{Na}$ ,  $^{30}\text{Si}$ ,  $^{30}\text{P}$  and  $^{35}\text{Cl}$  Using the Recoil-Distance Method.*
- 1973Be26 R.R.Betts, H.T.Fortune, R.Middleton - Phys.Rev. C8, 660 (1973).  
*Neutron Pickup from  $^{36}\text{Ar}$ .*
- 1973Br26 F.Brandolini, M.De Poli, C.Rossi Alvarez - Lett.Nuovo Cim. 8, 342 (1973).  
*Lifetimes of the First  $7/2^-$  Levels in  $^{35}\text{Cl}$  and  $^{37}\text{Cl}$ .*
- 1973Co25 M.N.H.Comsan, M.A.Farouk, A.Z.El-Behay - Atomkernenergie 21, 203 (1973).  
*Fluctuation Study of the  $^{34}\text{S}(d, p)^{35}\text{S}$  Reaction.*
- 1973EnVA P.M.Endt, C.Van der Leun - Nucl.Phys. A214, 1 (1973); Erratum Nucl.Phys. A248, 153 (1975).  
*Energy Levels of  $A = 21-44$  Nuclei (V).*
- 1973Fa07 B.Fant, J.Keinonen, A.Anttila, M.Bister - Z.Phys. 260, 185 (1973).  
*Investigation of Gamma Ray Transitions and Lifetimes in  $^{35}\text{Cl}$ .*
- 1973Go16 J.D.Goss, H.Stocker, N.A.Detorie, C.P.Browne, A.A.Rolfeson - Phys.Rev. C7, 1871 (1973).  
*States of  $^{35}\text{Cl}$  via the  $^{32}\text{S}(\alpha, p)^{35}\text{Cl}$  Reaction.*
- 1973Wa10 E.K.Warburton, J.W.Olness, G.A.P.Engelbertink, T.K.Alexander - Phys.Rev. C7, 1120 (1973).  
*Doppler-Shift Attenuation Measurements of Lifetimes for (2s-1d)-Shell Nuclei Excited by Heavy-Ion Bombardment of  $^2\text{H}$ .*
- 1974Al04 M.M.Aleonard, C.Boursiquot, P.Hubert, P.Mennrath - Phys.Lett. 49B, 40 (1974).  
*Strengths of  $(p, \gamma)$  Resonances in  $^{33}\text{Cl}$ ,  $^{35}\text{Cl}$  and  $^{28}\text{Si}$ .*
- 1974Bi16 M.Bister, A.Anttila, J.Keinonen, B.Fant - Phys.Fenn. 9, 23 (1974).  
*Isobaric Analog Resonances in  $^{35}\text{Cl}$ .*
- 1974Do12 P.Doll, H.Mackh, G.Mairle, G.J.Wagner - Nucl.Phys. A230, 329 (1974).  
*Comparative Spectroscopy with the  $(d, \tau)$  Reaction on Even Ar Isotopes.*
- 1974Hu09 D.A.Hutcheon, D.C.S.White, W.J.McDonald, G.C.Neilson - Can.J.Phys. 52, 1090 (1974).  
*Lifetimes of  $7/2^-$  States in  $^{35}\text{Cl}$ ,  $^{37}\text{Cl}$ , and  $^{39}\text{K}$ .*
- 1974Il01 L.V.Ilenko, E.A.Rudak - Yad.Fiz. 19, 1181 (1974); Sov.J.Nucl.Phys. 19, 603 (1975).  
*Gamma Decay of Resonance States from the Reaction  $^{34}\text{S}(p, \gamma)^{35}\text{Cl}$ .*
- 1974Lo17 P.R.G.Lornie, E.M.Jayasinghe, G.D.Jones, H.G.Price et al. - J.Phys.(London) A7, 1977 (1974).  
*High Spin Negative Parity States in  $^{35}\text{Cl}$ .*
- 1974Ma34 H.Mackh, G.Mairle, G.J.Wagner - Z.Phys. 269, 353 (1974).  
*Proton Shell Structure of Mass 28-32 Nuclei.*
- 1974Os02 A.Osman, M.Zaky - Nuovo Cim. 22A, 345 (1974).  
*Short-Range Correlations in  $(d, p)$  Stripping Reactions.*
- 1974Va13 M.A.Van Driel, H.H.Eggenhuisen, J.A.J.Hermans, D.Bucurescu et al. - Nucl.Phys. A226, 326 (1974).  
*Investigation of the 3.46 MeV Isomeric Level of  $^{38}\text{K}$  with the  $^{24}\text{Mg}(^{16}\text{O}, p n \gamma)^{38}\text{K}$  Reaction.*
- 1975Ca27 X.Campi, H.Flocard, A.K.Kerman, S.Koonin - Nucl.Phys. A251, 193 (1975).  
*Shape Transition in the Neutron Rich Sodium Isotopes.*
- 1975Da14 J.M.Davidson, T.Taylor, D.A.Hutcheon, D.M.Sheppard, W.C.Olsen - Nucl.Phys. A250, 221 (1975).  
*Study of  $^{35}\text{Ar}$  from  $^{33}\text{S}(^3\text{He}, n)^{35}\text{Ar}$  and  $^{33}\text{S}(^3\text{He}, n \gamma)^{35}\text{Ar}$ .*
- 1975De16 G.Deconninck, G.Debras - Radiochem.Radioanal.Lett. 20, 175 (1975).  
*Analysis of Chlorine by Prompt Gamma Rays Induced by Protons.*
- 1975Gu15 A.Guichard, H.Nann, B.H.Wildenthal - Phys.Rev. C12, 1109 (1975).  
 *$^{37}\text{Cl}(p, ^3\text{He})^{35}\text{S}$  Reaction.*
- 1975JeZX REPT LBL-4000,P93.
- 1975Ke11 J.Keinonen, M.Riihonen, A.Anttila - Phys.Scr. 12, 280 (1975).  
*Strengths of Analogue Resonances in  $(p, \gamma)$  Reactions on Sulphur Isotopes.*
- 1975Na18 H.Nann, W.S.Chien, A.Saha, B.H.Wildenthal - Phys.Lett. 60B, 32 (1975).  
*The Strongest  $L=6$  Transitions Observed in the  $(\alpha, d)$  Reaction on Odd-Mass  $A=33-41$  Target Nuclei.*
- 1975Sc40 W.A.Schier, B.K.Barnes, G.P.Couchell, J.J.Egan et al. - Nucl.Phys. A254, 80 (1975).

## REFERENCES FOR A=35(CONTINUED)

- 1975VaYG *Spin and Parity Assignments for  $^{35}\text{Cl}$  Levels from the  $^{31}\text{P}(\alpha,p)^{34}\text{S}$  Reaction.*  
T.W.van der Mark - Thesis, Groningen Univ. Netherlands (1975).  
*Studies on the nuclei  $^{33}\text{S}$ ,  $^{33}\text{Cl}$  and  $^{35}\text{S}$ .*
- 1975Wa17 G.J.Wagner, P.Doll, K.T.Knopfle, G.Mairle - Phys.Lett. 57B, 413 (1975).  
*Lifetime of Nuclear Hole States Caused by Phonon-Hole Coupling.*
- 1976AbZX R.Abegg, S.K.Datta, J.D.Hutton - Bull.Am.Phys.Soc. 21, No.8, 990, CF9 (1976).  
*Study of the Reaction  $^{34}\text{S}(d(\text{pol}),p)^{35}\text{S}$  with Vector-Polarized Deuterons.*
- 1976Be08 W.Benenson, A.Guichard, E.Kashy, D.Mueller, H.Nann - Phys.Rev. C13, 1479 (1976).  
*Mass of  $^{35}\text{K}$ .*
- 1976BeXJ REPT MSUCL 74/76 Annual,P3,Benenson.
- 1976BeZJ REPT MSUCL-199,W Benenson.
- 1976Fr03 M.J.Fritts - Phys.Rev. C13, 331 (1976).  
*Decay of  $^{36}\text{K}$ .*
- 1976Ke02 J.Keinonen, R.Rascher, M.Uhrmacher, N.Wust, K.P.Lieb - Phys.Rev. C14, 160 (1976).  
*M2 Transitions between the  $1f_{7/2}$  and the  $1d_{3/2}$  Orbit.*
- 1976Me12 M.A.Meyer, I.Venter, W.F.Coetzee, D.Reitmann - Nucl.Phys. A264, 13 (1976).  
*Energy Levels of  $^{35}\text{Cl}$ .*
- 1976Sp08 R.J.Sparks - Nucl.Phys. A265, 416 (1976).  
*Investigation of the Reaction  $^{34}\text{S}(p,\gamma)^{35}\text{Cl}$  (I). Resonances at Proton Energies Above 2 MeV.*
- 1976Sp09 R.J.Sparks - Nucl.Phys. A265, 429 (1976).  
*Investigation of the Reaction  $^{34}\text{S}(p,\gamma)^{35}\text{Cl}$  (II). Angular Distributions and Resonant Absorption.*
- 1976Va24 M.A.Van Driel, H.H.Eggenhuisen, G.A.P.Engelbertink, L.P.Ekstrom, J.A.J.Hermans - Nucl.Phys. A272, 466 (1976).  
*Model Independent  $J^\pi$  Assignments in  $^{38}\text{Ar}$  from  $\gamma$ -Ray Experiments in Heavy-Ion Induced Reactions.*
- 1976Wa11 E.K.Warburton, J.W.Olness, A.R.Poletti, J.J.Kolata - Phys.Rev. C14, 996 (1976).  
*Decay Schemes for High-Spin States in  $^{35,36,37}\text{Cl}$  and  $^{37}\text{Ar}$  from Heavy-Ion Fusion-Evaporation.*
- 1976We29 E.A.Wesolowski, M.Kicinska - Acta Phys.Pol. B7, 633 (1976).  
*Shell and Excitation Energy Effects in the Potential Radii Deduced from (d,p) Reactions on s-d Shell Nuclei at Low Deuteron Energy.*
- 1977Ab07 R.Abegg, S.K.Datta - Nucl.Phys. A287, 94 (1977).  
*Study of the Reaction  $^{34}\text{S}(d,p)^{35}\text{S}$  with Vector-Polarized Deuterons.*
- 1977Az01 G.Azuolos, J.E.Kitching, K.Ramavataram - Phys.Rev. C15, 1847 (1977).  
*Half-Lives and Branching Ratios of Some  $T = 1/2$  Nuclei.*
- 1977Ha51 J.C.Hardy, L.C.Carraz, B.Jonson, P.G.Hansen - Phys.Lett. B 71, 307 (1977).  
*The essential decay of pandemonium: A demonstration of errors in complex beta-decay schemes.*
- 1977Ko19 L.Koester, K.Knopf, W.Waschkowski - Z.Phys. A282, 371 (1977).  
*Scattering of Slow Neutrons by the Isotopes of Chlorine.*
- 1977Ko34 E.G.Kopanets, A.A.Koval, V.Y.Kostin, L.P.Korda et al. - Izv.Akad.Nauk SSSR, Ser.Fiz. 41, 1688 (1977);  
Bull.Acad.Sci.USSR, Phys.Ser. 41, No.8, 128 (1977).  
*Angular Distributions of  $\gamma$ -Rays Emitted in the  $^{34}\text{S}(p,\gamma)^{35}\text{Cl}$  Reaction.*
- 1977Ko35 A.A.Koval, E.G.Kopanets, V.Y.Kostin, L.P.Korda et al. - Izv.Akad.Nauk SSSR, Ser.Fiz. 41, 1658 (1977);  
Bull.Acad.Sci.USSR, Phys.Ser. 41, No.8, 104 (1977).  
*The Fine Structure of the  $7/2^-$  Isobar-Analog States of the  $^{35}\text{Cl}$  Nucleus.*
- 1977Na10 H.Nann, W.S.Chien, A.Saha, B.H.Wildenthal - Phys.Rev. C15, 1959 (1977).  
*High-Spin States of (fp) $^2$  Character in  $^{34,35,36}\text{Cl}$  and  $^{37,39}\text{Ar}$ .*
- 1977Os07 A.Osman - Nuovo Cim. 41A, 199 (1977).  
*Deuteron Stripping Reactions with the Tabakin Potential.*
- 1977Ou01 D.A.Outlaw, G.E.Mitchell, E.G.Bilpuch - Nucl.Phys. A284, 1 (1977).  
*High Resolution Study of the  $^{34}\text{S}(p,p)$  Reaction.*
- 1977Pa29 D.C.Palmer, D.R.Maxson, J.R.Bading - J.Phys.(London) G3, 1363 (1977).  
 *$^{35}\text{Cl}$ ,  $^{37}\text{Cl}$  and  $^{36}\text{Ar}(n,d)$  Reactions at 14.1 MeV.*
- 1977Sc36 D.Schwalm, E.K.Warburton, J.W.Olness - Nucl.Phys. A293, 425 (1977).  
*Coulomb Excitation of sd Shell Nuclei: A Self-Contained Set of  $B(E2)$  Values and Lifetime Measurements.*
- 1977Tr03 R.E.Tribble, J.D.Cossairt, R.A.Kenefick - Phys.Rev. C15, 2028 (1977).  
*Mass of  $^{36}\text{Ca}$ .*
- 1978Az01 G.Azuolos, G.R.Rao, P.Taras - Phys.Rev. C17, 443 (1978).  
 *$^{35}\text{Cl}(p,n)^{35}\text{Ar}$  Threshold Energy and its Relation to the Vanishing Cabibbo Angle.*
- 1978Ce02 M.Cenja, M.Duma, C.Hategan, M.Tanase - Nucl.Phys. A307, 65 (1978).  
*The (p,n) Threshold Anomaly in Proton Elastic Scattering on  $^{23}\text{Na}$ ,  $^{27}\text{Al}$ ,  $^{31}\text{P}$ ,  $^{35}\text{Cl}$  and  $^{34}\text{S}$ .*
- 1978En02 P.M.Endt, C.van der Leun - Nucl.Phys. A310, 1 (1978).  
*Energy Levels of  $A = 21-44$ .*
- 1978Es01 M.A.Eswaran, H.E.Gove - Phys.Rev. C18, 1528 (1978).  
*Systematics of Ground State  $\alpha$ -Cluster Spectroscopic Strengths for Odd-A sd-Shell Nuclei.*

## REFERENCES FOR A=35(CONTINUED)

- 1978HaYH REPT AECL-6366,p29,Hardy.  
 1979AnZZ G.Andersson, E.Hagberg, B.Jonson, S.Mattsson, G.Nyman - NEANDC(OR)-152L, p.7 (1979).  
*On-Line Studies of Very Unstable Nuclei.*  
 1979Ca15 L.C.Carraz, G.T.Ewan, E.Hagberg, J.C.Hardy et al. - Phys.Lett. B 85, 212 (1979).  
*A precise determination of the excitation energy of the lowest  $T=3/2$  state in  $^{35}\text{Ar}$ .*  
 1979Es05 M.A.Eswaran, H.E.Gove, R.Cook, B.Sikora - Nucl.Phys. A325, 269 (1979).  
*The  $\alpha$ -Transfer Reactions  $^{27}\text{Al}(^6\text{Li},d)^{31}\text{P}$ ,  $^{29}\text{Si}(^6\text{Li},d)^{33}\text{S}$  and  $^{31}\text{P}(^6\text{Li},d)^{35}\text{Cl}$  at 36 MeV.*  
 1979Ha04 E.Hagberg, J.C.Hardy, B.Jonson, S.Mattsson et al. - Nucl.Phys. A313, 276 (1979).  
*The Superaligned Decay of  $^{35}\text{Ar}$ : A Persistent Anomaly.*  
 1979HaZT E.Hagberg, J.C.Hardy, L.C.Carraz, P.G.Hansen et al. - Bull.Am.Phys.Soc. 24, No.4, 613, DM11 (1979).  
*A Precise Determination of the Excitation Energy of the Lowest  $T=3/2$  State in  $^{35}\text{Ar}$ .*  
 1979HaZY E.Hagberg, J.C.Hardy - AECL-6452, p.14 (1979).  
*The decays of  $^{35}\text{K}$  and  $^{36}\text{K}$ .*  
 1979Pa16 B.M.Paine, D.G.Sargood - Nucl.Phys. A331, 389 (1979).  
*(p, $\gamma$ ) Resonance Strengths in the s-d Shell.*  
 1979So01 L.M.Solin, Y.A.Nemilov, V.N.Kuzmin, K.I.Zherebtsova - Yad.Fiz. 29, 289 (1979); Sov.J.Nucl.Phys. 29, 143 (1979).  
*New Levels of  $^{37}\text{S}$ .*  
 1979Sy01 T.J.M.Symons, Y.P.Viyogi, G.D.Westfall, P.Doll et al. - Phys.Rev.Lett. 42, 40 (1979).  
*Observation of New Neutron-Rich Isotopes by Fragmentation of 205-MeV/Nucleon  $^{40}\text{Ar}$  Ions.*  
 1979Ta03 Tai Kuang-Hsi, T.Dossing, C.Gaarde, J.S.Larsen - Nucl.Phys. A316, 189 (1979).  
*Angular Correlations in Compound Nucleus Decay Following the Reactions  $^{16}\text{O} + ^{27}\text{Al}$ ,  $^{40}\text{Ca}$  and  $^{58}\text{Ni}$ .*  
 1980AyZZ REPT JYFL 1980 Ann,P26,Aysto.  
 1980Es01 K.Eskola, M.Riihonen, K.Vierinen, J.Honkanen et al. - Nucl.Phys. A341, 365 (1980).  
*Beta-Delayed Particle Emission of  $^{36}\text{K}$ .*  
 1980Ew01 G.T.Ewan, E.Hagberg, J.C.Hardy, B.Jonson et al. - Nucl.Phys. A337, 189 (1980).  
*Beta-Delayed Proton and Alpha Emission from  $^{36}\text{K}$ .*  
 1980Ew02 G.T.Ewan, E.Hagberg, J.C.Hardy, B.Jonson et al. - Nucl.Phys. A343, 109 (1980).  
*The Decay of  $^{35}\text{K}$ .*  
 1980Fa07 E.Fabrizi, S.Micheletti, M.Pignanelli, F.G.Resmini et al. - Phys.Rev. C21, 844 (1980).  
*Proton Elastic Scattering on Light Nuclei. II. Nuclear Structure Effects.*  
 1980Wi13 H.S.Wilson, R.W.Kavanagh, F.M.Mann - Phys.Rev. C22, 1696 (1980).  
*Gamow-Teller Transitions in Some Intermediate-Mass Nuclei.*  
 1981Ay01 J.Aysto, M.D.Cable, R.F.Parry, J.M.Wouters et al. - Phys.Rev. C23, 879 (1981).  
*Decays of the  $T(z) = -2$  Nuclei  $^{20}\text{Mg}$ ,  $^{24}\text{Si}$ , and  $^{36}\text{Ca}$ .*  
 1981Bi05 W.Biesiot, Ph.B.Smith, J.L.Stavast, P.B.Goldhoorn, S.Van der Hoek - Nucl.Phys. A359, 149 (1981).  
*Parameters of Selected Levels in  $^{35}\text{Cl}$ .*  
 1981Va02 C.van der Leun, G.J.L.Nooren - Phys.Lett. B98, 26 (1981).  
*Spin  $J=1/2$  Assignments to Proton-Capture Resonances.*  
 1982Ch10 D.R.Chakrabarty, M.A.Eswaran - Phys.Rev. C25, 1933 (1982).  
*Cluster Representation of  $^6\text{Li}$  and the Finite-Range Distorted-Wave Born Approximation Analysis of  $^6\text{Li}$ -Induced Alpha Transfer Reactions.*  
 1982Le04 K.T.Lesko, D.-K.Lock, A.Lazzarini, R.Vandenbosch et al. - Phys.Rev. C25, 872 (1982).  
*Energy Dependence of Fusion Cross Section for  $^{28}\text{Si} + ^{12}\text{C}$  by Evaporation Residue Measurements.*  
 1982Ra25 R.A.Racca, F.W.Prosser, C.N.Davids, D.G.Kovar - Phys.Rev. C26, 2022 (1982).  
*Search for Structure in the Fusion of  $^{16}\text{O} + ^{24}\text{Mg}$ .*  
 1983Ch50 M.Chenzha, M.Duma, C.Hategan, M.Tanase et al. - Izv.Akad.Nauk SSSR, Ser.Fiz. 47, 2143 (1983).  
*Experimental Study of Threshold Anomalies in the Elastic Scattering of Protons by Nuclei with  $A \leq 30$ .*  
 1983GI07 H.Glatli, J.Coustham - J.Phys.(Paris) 44, 957 (1983).  
*Spin-Dependent Scattering and Absorption of Thermal Neutrons on Dynamically Polarized Nuclei.*  
 1983La12 M.Langevin, C.Detraz, D.Guillemaud-Mueller, A.C.Mueller et al. - Phys.Lett. 125B, 116 (1983).  
 *$^{35}\text{Na}$ : A new neutron-rich Sodium isotope.*  
 1983Ra04 S.Raman, E.T.Jurney, D.A.Outlaw, I.S.Towner - Phys.Rev. C27, 1188 (1983).  
 *$^{34}\text{Cl}$  Superaligned  $\beta$  Decay.*  
 1983Wi08 B.H.Wildenthal, M.S.Curtin, B.A.Brown - Phys.Rev. C28, 1343 (1983).  
*Predicted Features of the Beta Decay of Neutron-Rich sd-Shell Nuclei.*  
 1984Ad01 E.G.Adelberger, J.L.Osborne, H.E.Swanson, B.A.Brown - Nucl.Phys. A417, 269 (1984).  
*The I-Forbidden Gamow-Teller Decay of  $^{39}\text{Ca}$ .*  
 1984Ca14 R.F.Carlton, W.M.Good, J.A.Harvey, R.L.Macklin, B.Castel - Phys.Rev. C29, 1980 (1984).  
*Determination of Unbound States in  $^{35}\text{S}$  from Neutron Total and Capture Cross-Section Measurements on  $^{34}\text{S}$ .*  
 1984El12 D.Elenkov, D.Lefterov, G.Toumbev - Nucl.Instrum.Methods 228, 62 (1984).  
*Two-Target DSAM following the  $(n,n'\gamma)$  Reaction with Fast Reactor Neutrons.*

## REFERENCES FOR A=35(CONTINUED)

- 1984La03 M.Langevin, C.Detraz, D.Guillemaud-Mueller, A.C.Mueller et al. - Nucl.Phys. A414, 151 (1984).  
 *$\beta$ -Delayed Neutrons from very Neutron-Rich Sodium and Magnesium Isotopes.*
- 1984Ma49 W.A.Mayer, W.Henning, R.Holzwarth, H.J.Korner et al. - Z.Phys. A319, 287 (1984).  
*Mass Excess and Excited States of Neutron-Rich Silicon, Phosphorus and Sulphur Isotopes.*
- 1984Mu25 W.Muller, B.C.Metsch, W.Knupfer, A.Richter - Nucl.Phys. A430, 61 (1984).  
*Calculation of the  $\beta^+$  Decay of Proton-Rich sd-Shell Nuclei into a Gamow-Teller Resonance.*
- 1984Pi03 S.Piskor, P.Franc, J.Kremenek, W.Schaferlingova - Nucl.Phys. A414, 219 (1984).  
*Spectroscopic Information on  $^{35}\text{S}$  and  $^{37}\text{S}$  from the (d,p) Reaction.*
- 1984Th08 C.E.Thorn, J.W.Olness, E.K.Warburton, S.Raman - Phys.Rev. C30, 1442 (1984).  
 *$^{36}\text{S}(d,p)^{37}\text{S}$  and  $^{34,36}\text{S}(d,^3\text{He})^{33,35}\text{P}$  Reactions.*
- 1985Al11 T.Altzitzoglou, F.Calaprice, M.Dewey, M.Lowry et al. - Phys.Rev.Lett. 55, 799 (1985).  
*Experimental Search for a Heavy Neutrino in the Beta Spectrum of  $^{35}\text{S}$ .*
- 1985Ap01 A.M.Apalikov, S.D.Boris, A.I.Golutvin, L.P.Lapin et al. - Pisma Zh.Eksp.Teor.Fiz. 42, 233 (1985); JETP Lett.(USSR) 42, 289 (1985).  
*Search for Heavy Neutrinos in  $\beta$  Decay.*
- 1985Ay01 J.Aysto, D.M.Moltz, X.J.Xu, J.E.Reiff, J.Cerny - Phys.Rev.Lett. 55, 1384 (1985).  
*Observation of the First  $T(z) = -5/2$  Nuclide,  $^{35}\text{Ca}$  via Its  $\beta$ -Delayed Two-Proton Emission.*
- 1985Br29 B.A.Brown, B.H.Wildenthal - At.Data Nucl.Data Tables 33, 347 (1985).  
*Experimental and Theoretical Gamow-Teller Beta-Decay Observables for the sd-Shell Nuclei.*
- 1985Da04 W.W.Daehnick, R.D.Rosa - Phys.Rev. C31, 1499 (1985).  
*Weak Branches in  $^{42}\text{Sc}$ ,  $^{35}\text{Ar}$ , and  $^{27}\text{Si}$   $\beta^+$  Decay.*
- 1985Dr06 P.V.Drumm, L.K.Fifield, R.A.Bark, M.A.C.Hotchkis et al. - Nucl.Phys. A441, 95 (1985).  
*Ground-State Mass Excesses and Low-Lying Levels of the Neutron-Rich Nuclei  $^{36}\text{P}$ ,  $^{35}\text{P}$  and  $^{34}\text{P}$ .*
- 1985Ke08 T.J.Kennett, W.V.Prestwich, J.S.Tsai - Z.Phys. A322, 121 (1985).  
*Radiative Thermal Neutron Capture by Natural Sulfur.*
- 1985Kh04 S.Khan, Th.Kihm, K.T.Knopfle, G.Mairle et al. - Phys.Lett. 156B, 155 (1985).  
*The Mass of  $^{35}\text{P}$  and Spin-Parity Assignments for Excited  $^{35}\text{P}$  States.*
- 1985Ki07 A.Z.Kiss, E.Koltay, B.Nyako, E.Somorjai et al. - J.Radioanal.Nucl.Chem. 89, 123 (1985).  
*Measurements of Relative Thick Target Yields for PIGE Analysis on Light Elements in the Proton Energy Interval 2.4-4.2 MeV.*
- 1985Kr21 P.Kristiansson, J.P.Bondorf, L.Carlen, H.-A.Gustafsson et al. - Nucl.Phys. A447, 577c (1985).  
*Large-Angle Correlations Observed in Intermediate Energy Heavy Ion Collisions.*
- 1985Ly02 Yu.S.Lyutostansky, V.K.Sirotkin, I.V.Panov - Phys.Lett. 161B, 9 (1985).  
*The  $\beta$ -Delayed Multi-Neutron Emission.*
- 1985Ma59 J.Markey, F.Boehm - Phys.Rev. C32, 2215 (1985).  
*Search for Admixture of Heavy Neutrinos with Masses between 5 and 55 keV.*
- 1985Mi06 K.Miura, Y.Hirate, T.Shoji, T.Suehiro et al. - Nucl.Phys. A436, 221 (1985).  
*Ground-State Transition Strengths of the (p,t) Reactions for Nuclei with A=23-238.*
- 1985Oh06 T.Ohi, M.Nakajima, H.Tamura, T.Matsuzaki et al. - Phys.Lett. 160B, 322 (1985).  
*Search for Heavy Neutrinos in the Beta Decay of  $^{35}\text{S}$ . Evidence Against the 17 keV Heavy Neutrino.*
- 1985Ra15 S.Raman, R.F.Carlton, J.C.Wells, E.T.Jurney, J.E.Lynn - Phys.Rev. C32, 18 (1985).  
*Thermal Neutron Capture Gamma Rays from Sulfur Isotopes: Experiment and theory.*
- 1986AnZV R.Anne, M.Bernas, C.Detraz, J.Galin et al. - Univ.Paris, Inst.Phys.Nucl., 1985 Ann.Rept., pE3 (1986).  
*Premiere Observation de la Serie  $T(Z) = -5/2$ :  $^{23}\text{Si}$ ,  $^{27}\text{S}$ ,  $^{31}\text{Ar}$  et  $^{35}\text{Ca}$ .*
- 1986Br26 M.E.Brandan, A.Menchaca-Rocha, A.Szanto de Toledo, N.Carlin Filho et al. - J.Phys.(London) G12, 391 (1986).  
*Detection of  $^8\text{Be}$  Nuclei from the Reaction  $^{16}\text{O} + ^{27}\text{Al}$  at 65 MeV.*
- 1986Du07 J.P.Dufour, R.Del Moral, A.Fleury, F.Hubert et al. - Z.Phys. A324, 487 (1986).  
*Beta Decay of  $^{17}\text{C}$ ,  $^{19}\text{N}$ ,  $^{22}\text{O}$ ,  $^{24}\text{F}$ ,  $^{26}\text{Ne}$ ,  $^{32}\text{Al}$ ,  $^{34}\text{Al}$ ,  $^{35-36}\text{Si}$ ,  $^{36-37-38}\text{P}$ ,  $^{40}\text{S}$ .*
- 1986El09 M.Elbel, R.Quad - Z.Naturforsch. 41a, 15 (1986).  
*The Quadrupole Moments of the Stable Nuclei of Sulphur and Chlorine.*
- 1986Fi06 L.K.Fifield, C.L.Woods, W.N.Catford, R.A.Bark et al. - Nucl.Phys. A453, 497 (1986).  
*The Mass of  $^{35}\text{Si}$ .*
- 1986HuZW F.Hubert, J.P.Dufour, R.del Moral, H.Emmermann et al. - CENBG-8605 (1986).  
*Studies of New Radioactive Nuclei at GANIL Energies.*
- 1986La17 M.Langevin, A.C.Mueller, D.Guillemaud-Mueller, M.G.Saint-Laurent et al. - Nucl.Phys. A455, 149 (1986).  
*Mapping of the Proton Drip-Line up to Z = 20: Observation of the  $T(z) = -(5/2)$  series  $^{23}\text{Si}$ ,  $^{27}\text{S}$ ,  $^{31}\text{Ar}$  and  $^{35}\text{Ca}$ .*
- 1986Sm05 R.J.Smith, P.J.Woods, R.Chapman, J.L.Durell et al. - Z.Phys. A324, 283 (1986).  
*Use of the  $^{64}\text{Ni}(^{36}\text{S}, ^{34,35}\text{Si})^{66,65}\text{Zn}$  Reactions to Measure the Masses of  $^{34}\text{Si}$  and  $^{35}\text{Si}$ .*
- 1986Wa22 E.K.Warburton, D.E.Alburger, J.A.Becker, B.A.Brown, S.Raman - Phys.Rev. C34, 1031 (1986).  
*Probe of the Shell Crossing at A = 40 via Beta Decay: Experiment and theory.*
- 1986Wo02 C.L.Woods - Nucl.Phys. A451, 413 (1986).  
*Shell-Model Calculations for Neutron-Rich Nuclei with A = 35-41.*



## REFERENCES FOR A=35(CONTINUED)

- 1987Ar08 U.R.Arzibekov, A.S.Gabelko, M.Kh.Zhalilov, K.M.Irgashev et al. - *Izv.Akad.Nauk SSSR, Ser.Fiz.* 51, 134 (1987); *Bull.Acad.Sci.USSR, Phys.Ser.* 51, No.1, 124 (1987).  
*Decay Characteristics of Dipole Giant Resonance in the Nucleus  $^{39}\text{K}$ .*
- 1987BaZI D.Bazin, R.Anne, D.Guerreau, D.Guillemaud-Mueller et al. - *Contrib.Proc. 5th Int.Conf.Nuclei Far from Stability*, Rosseau Lake, Canada, K7 (1987).  
*The Production Mechanisms of Nuclei Far from Stability at Ganil Energies and their Application to New  $\beta$ -Neutron Emitters.*
- 1987DuZU J.P.Dufour, R.Del Moral, F.Hubert, D.Jean et al. - *Contrib.Proc. 5th Int.Conf.Nuclei Far from Stability*, Rosseau Lake, Canada, D1 (1987).  
*Spectroscopic Measurements with a New Method: The projectile-fragments isotopic separation.*
- 1987Ge04 T.Genka, K.Kobayashi, S.Hagiwara - *Appl.Radiat.Isot.* 38, 845 (1987).  
*A Calorimeter for the Measurement of the Activity of Tritium and Other Pure Beta Emitters.*
- 1987Gi05 A.Gillibert, W.Mittig, L.Bianchi, A.Cunsolo et al. - *Phys.Lett.* 192B, 39 (1987).  
*New Mass Measurements Far From Stability.*
- 1987Na22 R.Nathuram, G.Subrahmanian, S.R.Thontadarya - *Indian J.Phys.* 61A, 209 (1987).  
*Range of Beta Particles in  $4\pi$  Geometry.*
- 1987SaZQ L.Satpathy, R.C.Nayak - *Contrib.Proc. 5th Int.Conf.Nuclei Far from Stability*, Rosseau Lake, Canada, A15 (1987).  
*Masses of Nuclei in the Infinite Nuclear Matter Model.*
- 1987Wa10 E.K.Warburton, J.A.Becker - *Phys.Rev.* C35, 1851 (1987).  
*Shell-Model Description of the Beta Decay of the  $N = 21$  Isotones  $^{35}\text{Si}$  and  $^{36}\text{P}$ .*
- 1988BaYZ D.Bazin, R.Anne, D.Guerreau, D.Guillemaud-Mueller et al. - *Proc. 5th Int.Conf.Nuclei Far from Stability*, Rosseau Lake, Canada 1987, Ed., I.S.Towner, p.722 (1988), *AIP Conf.Proc.* 164 (1987).  
*On the Production Mechanisms of Nuclei Far from Stability at GANIL Energies and Their Application to New  $\beta$ -Delayed Neutron Emitters.*
- 1988Ch45 Y.Charon, J.C.Cuzon, H.Tricoire, L.Valentin - *Nucl.Instrum.Methods Phys.Res.* A273, 748 (1988).  
*A High Resolution  $\beta^-$  Detector.*
- 1988Co03 F.Corvi, A.Prevignano, H.Liskien, P.B.Smith - *Nucl.Instrum.Methods Phys.Res.* A265, 475 (1988).  
*An Experimental Method for Determining the Total Efficiency and the Response Function of a Gamma-Ray Detector in the Range 0.5-10 MeV.*
- 1988DuZS J.P.Dufour - *Priv.Comm.* (1988).
- 1988DuZT J.P.Dufour, R.Del Moral, F.Hubert, D.Jean et al. - *Proc. 5th Int.Conf.Nuclei Far from Stability*, Rosseau Lake, Canada 1987, Ed., I.S.Towner, p.344 (1988), *AIP Conf.Proc.* 164 (1987).  
*Spectroscopic Measurements with a New Method: The projectile-fragments isotopic separation.*
- 1988Ga02 J.D.Garnett, E.D.Commins, K.T.Lesko, E.B.Norman - *Phys.Rev.Lett.* 60, 499 (1988).  
 *$\beta$ -Decay Asymmetry Parameter for  $^{35}\text{Ar}$ : An anomaly resolved.*
- 1988Mu08 A.C.Mueller, D.Bazin, W.D.Schmidt-Ott, R.Anne et al. - *Z.Phys.* A330, 63 (1988).  
 *$\beta$ -Delayed Neutron Emission of  $^{15}\text{B}$ ,  $^{18}\text{C}$ ,  $^{19}\text{N}$ ,  $^{20}\text{N}$ ,  $^{34}\text{Al}$  and  $^{39}\text{P}$ .*
- 1988MuZY A.C.Mueller, D.Bazin, W.D.Schmidt-Ott, R.Anne et al. - *Univ.Paris, Inst.Phys.Nucl.*, 1987 *Ann.Rept.*, p.E7 (1988).  
*New Light Beta-Delayed Neutron Emitters.*
- 1988Or01 N.A.Orr, W.N.Catford, L.K.Fifield, T.R.Ophel et al. - *Nucl.Phys.* A477, 523 (1988); *Erratum Nucl.Phys.* A485, 734 (1988).  
*Heavy-Ion Reaction Studies of  $^{35}\text{P}$ .*
- 1988Va06 O.P.van Pruissen, G.J.L.Nooren, C.van der Leun - *Nucl.Phys.* A480, 77 (1988).  
*Primary Gamma-Ray Transitions of Mixed  $E1 + M2$  Character.*
- 1988Wa04 E.K.Warburton, J.A.Becker - *Phys.Rev.* C37, 754 (1988).  
*Shell-Model Description of the  $\beta^-$  Decay of the  $N = 21$  and 22 Isotones  $^{34}\text{Al}$ ,  $^{36}\text{Si}$ , and  $^{37}\text{P}$ .*
- 1988Za04 P.P.Zarubin, V.P.Gusev, A.E.Antropov - *Izv.Akad.Nauk SSSR, Ser.Fiz.* 52, 923 (1988); *Bull.Acad.Sci.USSR, Phys.Ser.* 52, No.5, 86 (1988).  
*Compound-Nucleus Proton Elastic Scattering Cross Sections at  $E(p)$  6 MeV.*
- 1989Ba04 N.M.Badiger, N.Umakantha - *Phys.Rev.* C39, 216 (1989).  
*Internal Bremsstrahlung Spectrum of  $^{35}\text{S}$  in the Range from 10 to 100 keV.*
- 1989De02 P.A.DeYoung, M.S.Gordon, Xiuqin Lu, R.L.McGrath et al. - *Phys.Rev.* C39, 128 (1989).  
*Emission Times in Low-Energy Heavy-Ion Reactions by Particle-Particle Correlations.*
- 1989Ge09 M.K.Georgieva, D.V.Elenkov, D.P.Lefterov, G.H.Toumbev - *Fiz.Elem.Chastits At.Yadra* 20, 930 (1989); *Sov.J.Part.Nucl.* 20, 393 (1989).  
*Measurement of the Lifetimes of Excited Nuclear States by the Doppler Shift Attenuation Method Using the Reaction  $(n,n'\gamma)$  with Two Targets.*
- 1989Gu03 D.Guillemaud-Mueller, Yu.E.Penionzhkevich, R.Anne, A.G.Artukh et al. - *Z.Phys.* A332, 189 (1989).  
*Observation of New Neutron Rich Nuclei  $^{26}\text{F}$ ,  $^{35}\text{Mg}$ ,  $^{38}\text{Al}$ ,  $^{40}\text{Si}$ ,  $^{43}\text{P}$ ,  $^{45}\text{S}$ ,  $^{46}\text{Cl}$ , and  $^{49}\text{Ar}$  from the Interactions of 55 MeV/u  $^{48}\text{Ca} + \text{Ta}$ .*
- 1989Le16 M.Lewitowicz, Yu.E.Penionzhkevich, A.G.Artukh, A.M.Kalinin et al. - *Nucl.Phys.* A496, 477 (1989).  
 *$\beta$ -Delayed Neutron Emission of the Isotopes  $^{20}\text{C}$ ,  $^{40}\text{P}$ ,  $^{41}\text{S}$ ,  $^{42}\text{S}$ .*
- 1989Ly01 Yu.S.Lyutostansky, M.V.Zverev, I.V.Panov - *Izv.Akad.Nauk SSSR, Ser.Fiz.* 53, 849 (1989); *Bull.Acad.Sci.USSR, Phys.Ser.* 53, No.5, 29 (1989).

## REFERENCES FOR A=35(CONTINUED)

- New Region of Deformation of Neutron-Rich Nuclei and  $\beta$ -Delayed Neutron Emission.*  
1989MuZU A.C.Mueller, M.Lewitowicz, R.Anne, A.G.Artukh et al. - Univ.Paris, Inst.Phys.Nucl., 1988 Ann.Rept., p.25 (1989).
- Etude de l'Emission de Neutrons Retardés par des Isotopes Légers très Riches en Neutrons.*  
1989Ta08 M.Takiue, T.Natake, H.Fujii - Nucl.Instrum.Methods Phys.Res. A274, 345 (1989).
- Nuclide Identification of  $\beta$ -Emitter by a Double Ratio Technique Using a Liquid Scintillation Counter.*  
1990Br26 B.A.Brown - Phys.Rev.Lett. 65, 2753 (1990).
- Isospin-Forbidden  $\beta$ -Delayed Proton Emission.*  
1990En08 P.M.Endt - Nucl.Phys. A521, 1 (1990); Errata and Addenda Nucl.Phys. A529, 763 (1991); Errata Nucl.Phys. A564, 609 (1993).
- Energy Levels of A = 21-44 Nuclei (VII).*  
1991De26 C.Detraz - Z.Phys. A340, 227 (1991).
- The Branching Ratio of  $\beta$ -Delayed Two-Proton Emission.*  
1991Ja11 H.C.Jain, S.Chattopadhyay, Y.K.Agarwal, M.L.Jhingan et al. - Pramana 37, 269 (1991).
- A Plunger Set-Up for Measuring Picosecond Nuclear Half-Lives.*  
1991Li12 S.Lindenstruth, A.Degener, R.D.Heil, A.Jung et al. - Nucl.Instrum.Methods Phys.Res. A300, 293 (1991).
- Measurements and Simulations of Low Energy, Thick Target Bremsstrahlung Spectra.*  
1991Mi18 M.A.Miller, P.A.Voytas, A.D.Roberts, P.A.Quin, W.Haeblerli - Phys.Rev. C44, 1995 (1991).
- Polarization Transfer in the  $^{23}\text{Na}(p(\text{pol}), ^{23}\text{Mg}(\text{pol}))n$ ,  $^{31}\text{P}(p(\text{pol}), ^{31}\text{S}(\text{pol}))n$ , and  $^{35}\text{Cl}(p(\text{pol}), ^{35}\text{Ar}(\text{pol}))n$  Reactions.*  
1991Na05 O.Naviliat-Cuncic, T.A.Girard, J.Deutsch, N.Severijns - J.Phys.(London) G17, 919 (1991).
- Left-Right Symmetry Breaking Sensitivity of  $\beta$ -Asymmetry Measurements.*  
1991Or01 N.A.Orr, W.Mittig, L.K.Fifield, M.Lewitowicz et al. - Phys.Lett. 258B, 29 (1991).
- New Mass Measurements of Neutron-Rich Nuclei Near N = 20.*  
1991Pa19 S.K.Patra, C.R.Praharaj - Phys.Lett. 273B, 13 (1991).
- Relativistic Mean Field Study of 'Island of Inversion' in Neutron-Rich Ne, Na, Mg Nuclei.*  
1991Pa21 S.K.Patra, C.R.Praharaj - Pramana 37, L445 (1991).
- Deformed Relativistic Mean Field Study of Binding Energy Anomaly in Neutron-Rich Ne, Na, Mg Nuclei.*  
1991Zh24 X.G.Zhou, X.L.Tu, J.M.Wouters, D.J.Vieira et al. - Phys.Lett. 260B, 285 (1991).
- Direct Mass Measurements of the Neutron-Rich Isotopes of Fluorine Through Chlorine.*  
1992Be60 M.Belgaid, M.Siad, M.Allab - J.Radioanal.Nucl.Chem. 166, 493 (1992).
- Measurement of 14.7 MeV Neutron Cross Sections for Several Isotopes.*  
1992Ch27 M.Chen, D.A.Imel, T.J.Radcliffe, H.Henrikson, F.Boehm - Phys.Rev.Lett. 69, 3151 (1992).
- New Limits on the 17 keV Neutrino.*  
1992Hi06 K.H.Hicks, D.E.Alburger - Phys.Lett. 276B, 423 (1992).
- Search for Light Neutral Boson Associated with Beta Decay.*  
1993Ab11 H.Abele, G.Helm, U.Kania, C.Schmidt et al. - Phys.Lett. 316B, 26 (1993).
- On the Origin of the 17 keV Neutrino Signals, and a Loss-Free Measurement of the  $^{35}\text{S}$   $\beta$ -Spectrum.*  
1993Au07 G.Audi, A.H.Wapstra - Nucl.Phys. A565, 66 (1993).
- The 1993 Atomic Mass Evaluation. (II). Nuclear-Reaction and Separation Energies.*  
1993Be21 G.E.Berman, M.L.Pitt, F.P.Calaprice, M.M.Lowry - Phys.Rev. C48, R1 (1993).
- New Evidence Against 17-keV Neutrino Emission in the  $\beta$  Decay Momentum Spectrum of  $^{35}\text{S}$ .*  
1993Bo40 I.Bogdanovic, S.Fazinic, M.Jaksic, T.Tadic et al. - Nucl.Instrum.Methods Phys.Res. B79, 524 (1993).
- Proton Elastic Scattering from Fluorine, Chlorine, Zinc, Selenium and Bromine in the Energy Region from 2.5 to 4.8 MeV.*  
1993Co07 A.Converse, M.Allet, W.Haeblerli, W.Hajdas et al. - Phys.Lett. 304B, 60 (1993).
- Measurement of the Asymmetry Parameter for  $^{35}\text{Ar}$   $\beta$ -Decay as a Test of the CVC Hypothesis.*  
1993Gi08 T.A.Girard, R.P.Henriques, J.I.Collar, M.Godinho et al. - Nucl.Instrum.Methods Phys.Res. A334, 645 (1993).
- Response of a Metastable Superconducting Grains Suspension to Irradiation by  $^{35}\text{S}$  Decay Electrons.*  
1993Gr07 A.Grau Carles, L.Rodriguez Barquero, A.Grau Malonda - Nucl.Instrum.Methods Phys.Res. A325, 234 (1993).
- Standardization of  $^{14}\text{C}$  and  $^{35}\text{S}$  Mixtures.*  
1993Ma50 G.Mairle, M.Seeger, H.Reinhardt, T.Kihm et al. - Nucl.Phys. A565, 543 (1993).
- The Distribution of Proton-Hole Strengths in the (2s,1d)-Shell of  $^{35}\text{Cl}$ ,  $^{37}\text{Cl}$  and  $^{39}\text{Cl}$ .*  
1993Mo01 J.L.Mortara, I.Ahmad, K.P.Coulter, S.J.Freedman et al. - Phys.Rev.Lett. 70, 394 (1993).
- Evidence Against a 17 keV Neutrino from  $^{35}\text{S}$  Beta Decay.*  
1993Su36 D.Sundholm, J.Olsen - J.Chem.Phys. 98, 7152 (1993).
- Finite Element Multiconfiguration Hartree-Fock Determination of the Nuclear Quadrupole Moments of Chlorine, Potassium, and Calcium Isotopes.*  
1993TeZX A.Terakawa, T.Tohei, T.Nakagawa, A.Sato et al. - Cyclotron Rad.Center, Tohoku Univ., Ann.Rept., 1992, p.14 (1993).
- Proton Occupation Numbers in the Ground State of sd-Shell Nuclei II.*  
1994Bo33 M.G.Bowler, N.A.Jelley - Phys.Lett. 331B, 193 (1994).
- The Origin of the Spurious 17 keV Neutrino Signal Observed in  $^{35}\text{S}$  Beta-Decay.*  
1994Fo04 B.Fornal, R.H.Mayer, I.G.Bearden, Ph.Benet et al. - Phys.Rev. C49, 2413 (1994).
- Gamma Ray Studies of Neutron-Rich sdf Shell Nuclei Produced in Heavy Ion Collisions.*  
1994Gr04 A.Grau Carles - Appl.Radiat.Isot. 45, 83 (1994).
- A New Linear Spectrum Unfolding Method Applied to Radionuclide Mixtures in Liquid Scintillation Spectrometry.*

## REFERENCES FOR A=35(CONTINUED)

- 1994Ho35 Y.Homma, Y.Murase, K.Handa - J.Radioanal.Nucl.Chem. 187, 367 (1994).  
*Absolute Liquid Scintillation Counting of  $^{35}\text{S}$  and  $^{45}\text{Ca}$  using a Modified Integral Counting Method.*
- 1994Mo37 G.Mouze - Nuovo Cim. 107A, 1565 (1994).  
*On Some Properties of the Valence Bosons.*
- 1994Mu26 S.Muller, S.Chen, H.Daniel, O.Dragoun et al. - Z.Naturforsch. 49a, 874 (1994).  
*Search for an Admixture of a 17 keV Neutrino in the  $\beta$  Decay of  $^{35}\text{S}$ .*
- 1994Po05 A.Poves, J.Retamosa - Nucl.Phys. A571, 221 (1994).  
*Theoretical Study of the Very Neutron-Rich Nuclei Around  $N = 20$ .*
- 1994Ve04 J.Vernotte, G.Berrier-Ronsin, J.Kalifa, R.Tamisier, B.H.Wildenthal - Nucl.Phys. A571, 1 (1994).  
*Spectroscopic Factors from One-Proton Stripping Reactions on sd-Shell Nuclei: Experimental Measurements and Shell-Model Calculations.*
- 1995Bo43 M.G.Bowler, N.A.Jelley - Z.Phys. C68, 391 (1995).  
*Investigations into the Origin of the Spurious 17 keV Neutrino Signal Observed in  $^{35}\text{S}$  Beta Decay.*
- 1995Ga03 A.Garcia, E.G.Adelberger, P.V.Magnus, H.E.Swanson et al. - Phys.Rev. C51, R439 (1995).  
 *$\beta$ -Delayed  $\gamma$ -Ray Emission in  $^{37}\text{Ca}$  Decay.*
- 1995Ga16 A.Garcia, E.G.Adelberger, P.V.Magnus, H.E.Swanson et al. - Phys.Rev. C51, 3487 (1995).  
 *$^{36}\text{Ca}$   $\beta$  Decay and the Isobaric Multiplet Mass Equation.*
- 1995KIZZ A.Klein, M.Keim, U.Georg, R.Neugart et al. - Proc.Intern.Conf on Exotic Nuclei and Atomic Masses, Arles, France, June 19-23, 1995, p.139 (1995).  
*Moments and Mean Square Charge Radii of Short-Lived Argon Isotopes.*
- 1995Mo17 S.Moriyama, M.Minowa - Phys.Lett. 347B, 152 (1995).  
*The 17 keV Neutrino and the Search for Anomalous  $\gamma$  Rays in  $^{35}\text{S}$  Decay.*
- 1995ReZZ P.L.Reeder, Y.Kim, W.K.Hensley, H.S.Miley et al. - Proc.Intern.Conf on Exotic Nuclei and Atomic Masses, Arles, France, June 19-23, 1995, p.587 (1995).  
*Beta Decay Half-Lives and Delayed Particle Emission from TOFI Measurements.*
- 1995Tr02 W.Trinder, E.G.Adelberger, B.A.Brown, Z.Janas et al. - Phys.Lett. 348B, 331 (1995).  
*Gamow-Teller Strength in the  $\beta$ -Decay of  $^{36}\text{Ca}$ .*
- 1996II02 C.Iliadis, R.E.Azuma, L.Buchmann, J.Chow et al. - Nucl.Phys. A609, 237 (1996).  
*Beta-Delayed Particle Decay of  $^{36}\text{K}$ .*
- 1996Je06 V.Jeudy, J.I.Collar, T.A.Girard, D.Limagne, G.Waysand - Nucl.Instrum.Methods Phys.Res. A373, 65 (1996).  
 *$^{35}\text{S}$  Beta Irradiation of a Tin Strip in a State of Superconducting Geometrical Metastability.*
- 1996Ka21 A.S.Kachan, A.N.Vodin, V.M.Mishchenko, R.P.Slabospitsky - Yad.Fiz. 59, No 5, 775 (1996); Phys.Atomic Nuclei 59, 739 (1996).  
*Fine Structure of the M1 Resonance in  $^{35}\text{Cl}$ .*
- 1996K104 A.Klein, B.A.Brown, U.Georg, M.Keim et al. - Nucl.Phys. A607, 1 (1996).  
*Moments and Mean Square Charge Radii of Short-Lived Argon Isotopes.*
- 1996Re10 Z.Ren, Z.Y.Zhu, Y.H.Cai, G.Xu - Phys.Lett. 380B, 241 (1996).  
*Relativistic Mean-Field Study of Mg Isotopes.*
- 1996So21 M.Soltani-Farshi, J.D.Meyer, P.Misaelides, K.Bethge - Nucl.Instrum.Methods Phys.Res. B 113, 399 (1996).  
*Cross section of the  $^{32}\text{S}(\alpha, p)^{35}\text{Cl}$  nuclear reaction for sulphurdetermination.*
- 1996Vo20 A.N.Vodin, A.S.Kachan, V.M.Mishchenko, R.P.Slabospitsky - Bull.Rus.Acad.Sci.Phys. 60, 805 (1996).  
*Isobar-Analogue  $d_{5/2}$  State in the  $^{35}\text{Cl}$  Nucleus.*
- 1997Be42 H.Beer, C.Coceva, R.Hofinger, P.Mohr et al. - Nucl.Phys. A621, 235c (1997).  
*Measurement of Direct Neutron Capture by Neutron-Rich Sulfur Isotopes.*
- 1997Co19 B.J.Cole - Phys.Rev. C56, 1866 (1997).  
*Systematics of Proton and Diproton Separation Energies for Light Nuclei.*
- 1997Fo01 B.Fornal, R.Broda, W.Krolas, T.Pawlat et al. - Phys.Rev. C55, 762 (1997).  
 *$\gamma$ -Ray Studies of Neutron-Rich  $N = 18, 19$  Nuclei Produced in Deep-Inelastic Collisions.*
- 1997II03 C.Iliadis, R.E.Azuma, J.Chow, J.D.King et al. - Nucl.Phys. A621, 211c (1997).  
*Decay Studies of Importance to Explosive Hydrogen Burning.*
- 1997Mo25 P.Moller, J.R.Nix, K.-L.Kratz - At.Data Nucl.Data Tables 66, 131 (1997).  
*Nuclear Properties for Astrophysical and Radioactive-Ion-Beam Applications.*
- 1997Tr05 W.Trinder, E.G.Adelberger, B.A.Brown, Z.Janas et al. - Nucl.Phys. A620, 191 (1997).  
*Study of the  $\beta$  Decays of  $^{37}\text{Ca}$  and  $^{36}\text{Ca}$ .*
- 1997Vo03 H.Vonach, A.Pavlik, A.Wallner, M.Drosg et al. - Phys.Rev. C55, 2458 (1997).  
*Spallation Reactions in  $^{27}\text{Al}$  and  $^{56}\text{Fe}$  Induced by 800 MeV Protons.*
- 1998Co30 B.J.Cole - Phys.Rev. C58, 2831 (1998).  
*Proton and Two-Proton Drip Lines in the sd Shell.*
- 1998En04 P.M.Endt - Nucl.Phys. A633, 1 (1998).  
*Supplement to Energy Levels of  $A = 21-44$  Nuclei (VII).*
- 1998Le45 M.Lewitowicz, W.Trinder, M.Bhattacharya, E.G.Adelberger et al. - Nuovo Cim. 111A, 835 (1998).  
*Beta-Decay of Light Nuclei Close to the Proton Drip-Line:  $^{40}\text{Ti}$  and  $^{35}\text{Ca}$ .*

## REFERENCES FOR A=35(CONTINUED)

- 1998Sc19 M.Schafer, W.-D.Schmidt-Ott, T.Dorfler, T.Hild et al. - Phys.Rev. C57, 2205 (1998).  
*Polarization in Fragmentation, g Factor of  $^{35}\text{K}$ .*
- 1998VoZY S.Vouzoukas - Thesis, University of Notre Dame (1998).  
*A Study of the Levels of Astrophysical Importance in  $^{32}\text{Cl}$  and  $^{35}\text{Ar}$ .*
- 1999Ai02 N.Aissaoui, N.Added, N.Carlin, G.M.Crawley et al. - Phys.Rev. C60, 034614 (1999).  
*Strong Absorption Radii from Reaction Cross Section Measurements for Neutron-Rich Nuclei.*
- 1999Du05 J.Dufo, A.P.Zuker - Phys.Rev. C59, R2347 (1999).  
*The Nuclear Monopole Hamiltonian.*
- 1999GI02 Yu.M.Gledenov, V.I.Salatsky, P.V.Sedyshev, M.Stempinsky, P.J.Szalansky - Yad.Fiz. 62, No 5, 877 (1999); Phys.Atomic Nuclei 62, 818 (1999).  
*Measurement of the Cross Section for the Reaction  $^{35}\text{Cl}(n,p)^{35}\text{S}$  Induced by Thermal Neutrons and Investigation of Resonances at 398 and 4249 eV.*
- 1999GI09 Yu.M.Gledenov, R.Machrafi, A.I.Oprea, V.I.Salatski et al. - Nucl.Phys. (Supplement) A654, 943c (1999).  
*A Search for P-Odd and P-Even Correlations in the  $^{35}\text{Cl}(n,p)^{35}\text{S}$  Reaction.*
- 1999Ib01 R.W.Ibbotson, T.Glasmacher, P.F.Mantica, H.Scheit - Phys.Rev. C59, 642 (1999).  
*Coulomb Excitation of Odd-A Neutron-Rich  $\pi(s-d)$  and  $\nu(f-p)$  Shell Nuclei.*
- 1999Oh09 D.Ohsawa, R.Katano, Y.Isozumi - Phys.Rev. C60, 044604 (1999).  
*K-Shell Internal Ionization and Excitation in  $\beta$  Decay of  $^{35}\text{S}$ .*
- 1999Pa18 L.Palermo, E.Holzschuh, W.Kundig, P.Wenk, R.Alberto - Nucl.Instrum.Methods Phys.Res. A423, 337 (1999).  
*Preparation of a  $^{35}\text{S}$ -Source for  $\beta$ -Spectroscopy.*
- 1999Tr04 W.Trinder, J.C.Angelique, R.Anne, J.Aysto et al. - Phys.Lett. 459B, 67 (1999).  
 *$\beta$ -Decay of  $^{35}\text{Ca}$ .*
- 1999YoZW K.Yoneda, H.Sakurai, N.Aoi, N.Fukuda et al. - RIKEN-98, p.78 (1999).  
*The First Measurements of Half Lives and Neutron Emission Probabilities of Very Neutron Rich Nuclides of Mg, Al, Si, and P.*
- 2000Ge20 W.Geithner, U.Georg, S.Kappertz, M.Keim et al. - Hyperfine Interactions 129, 271 (2000).  
*Measurement of nuclear moments and radii by collinear laser spectroscopy and by  $\beta$ -NMR spectroscopy.*
- 2000Ha64 H.Haas, H.M.Petrilli - Phys.Rev. B61, 13588 (2000).  
*Quadrupole Moments of the Halogen Nuclei.*
- 2000Ho13 E.Holzschuh, L.Palermo, H.Stussi, P.Wenk - Phys.Lett. 482B, 1 (2000).  
*The  $\beta$ -Spectrum of  $^{35}\text{S}$  and Search for the Admixture of Heavy Neutrinos.*
- 2000Ka30 Ye.A.Karelin, V.N.Efimov, V.T.Filimonov, R.A.Kuznetsov et al. - Appl.Radiat.Isot. 53, 825 (2000).  
*Radionuclide Production using a Fast Flux Reactor.*
- 2000PrZX B.V.Pritychenko - Thesis, Michigan State University (2000).  
*Intermediate-Energy Coulomb Excitation of the Neutron-Rich Radioactive Isotopes  $^{26,28}\text{Ne}$ ,  $^{28-31}\text{Na}$ ,  $^{30-34}\text{Mg}$ ,  $^{34,35}\text{Al}$ ,  $^{33}\text{Si}$  and  $^{34}\text{P}$ .*
- 2000Re01 R.Reifarth, K.Schwarz, F.Kappeler - Astrophys.J. 528, 573 (2000).  
*The Stellar Neutron-Capture Rate of  $^{34}\text{S}$ : The origin of  $^{36}\text{S}$  challenged.*
- 2000Sa21 F.Sarazin, H.Savajols, W.Mittig, F.Nowacki et al. - Phys.Rev.Lett. 84, 5062 (2000).  
*Shape Coexistence and the  $N = 28$  Shell Closure Far from Stability.*
- 2000Vo21 A.N.Vodin, A.S.Kachan - Bull.Rus.Acad.Sci.Phys. 64, 327 (2000).  
 *$\gamma$ -Transitions between Compound States in  $1d_{2s}$ -Shell Nuclei.*
- 2001Ka69 A.S.Kachan, A.N.Vodin, V.M.Mishchenko, R.P.Slabospitsky - Bull.Rus.Acad.Sci.Phys. 65, 725 (2001).  
 *$M1$  Resonance in Odd  $sd$ -Shell Nuclei.*
- 2001Lo11 M.J.Lopez-Jimenez, M.G.Saint-Laurent, J.C.Angelique, R.Anne et al. - Eur.Phys.J. A 10, 119 (2001).  
*New Transitions in the  $\beta$ -Decay of  $^{36}\text{Ca}$ .*
- 2001Nu01 S.Nummela, P.Baumann, E.Caurier, P.Dessagne et al. - Phys.Rev. C63, 044316 (2001).  
*Spectroscopy of  $^{34,35}\text{Si}$  by  $\beta$  Decay:  $sd$ - $fp$  Shell gap and single-particle states.*
- 2001Pi10 S.Pirrone, G.Politi, G.Lanzalone, S.Aiello et al. - Phys.Rev. C64, 024610 (2001).  
*Fusion and Competing Processes in the  $^{32}\text{S} + ^{12}\text{C}$  Reaction at  $E(^{32}\text{S}) = 19.5$  MeV/nucleon.*
- 2001Sa72 F.Sarazin, H.Savajols, W.Mittig, F.Nowacki et al. - Hyperfine Interactions 132, 147 (2001).  
*Shape Coexistence and the  $N = 28$  Shell Closure Far from Stability.*
- 2001Vo24 A.N.Vodin, A.S.Kachan, V.M.Mishchenko - Bull.Rus.Acad.Sci.Phys. 65, 746 (2001).  
*Isobar-Analogue Resonances in the  $^{35}\text{Cl}$  Nucleus.*
- 2001ZhZZ C.Zhou, R.B.Firestone - INDC(CPR)-054 (2001).  
*Thermal-Neutron Capture Data for  $A = 26-35$ .*
- 2002LuZT S.M.Lukyanov, Yu.E.Penionzhkevich - JINR-E7-2002-237 (2002).  
*Peculiarities of O-Mg Isotopes at the Neutron Drip Line.*
- 2002Ma41 K.Matsuta, T.Tsubota, C.Ha, T.Miyake et al. - Nucl.Phys. A701, 383c (2002).  
*Precise Magnetic Moment of Short-Lived Beta Emitter  $^{35}\text{Ar}$ .*
- 2002Nu02 S.Nummela, P.Baumann, E.Caurier, S.Courtin et al. - Nucl.Phys. A701, 410c (2002).  
*Study of the Neutron-Rich Nuclei with  $N = 21$ ,  $^{35}\text{Si}$  and  $^{33}\text{Mg}$ , by  $\beta$  Decay of  $^{35}\text{Al}$  and  $^{33}\text{Na}$ .*



## REFERENCES FOR A=35(CONTINUED)

- 2002Oz03 A.Ozawa, T.Baumann, L.Chulkov, D.Cortina et al. - Nucl.Phys. A709, 60 (2002); Erratum Nucl.Phys. A727, 465 (2003).  
*Measurements of the Interaction Cross Sections for Ar and Cl Isotopes.*
- 2002Sa08 C.Samanta, S.Adhikari - Phys.Rev. C65, 037301 (2002); Comment Phys.Rev. C 69, 049803 (2004).  
*Extension of the Bethe-Weizsacker Mass Formula to Light Nuclei and Some New Shell Closures.*
- 2002Vo17 A.N.Vodin, A.S.Kachan, V.M.Mishchenko - Bull.Rus.Acad.Sci.Phys. 66, 40 (2002).  
*Isobar-Analogue  $f_{7/2}$  Resonances in Odd 1d<sub>2s</sub>-Shell Nuclei.*
- 2003Gr22 S.Grevy, J.Mrazek, J.C.Angelique, P.Baumann et al. - Nucl.Phys. A722, 424c (2003).  
*Beta-decay studies at the  $N = 28$  shell closure.*
- 2003Sm02 N.A.Smirnova, C.Volpe - Nucl.Phys. A714, 441 (2003).  
*On the asymmetry of Gamow-Teller  $\beta$ -decay rates in mirror nuclei in relation with second-class currents.*
- 2004Al08 R.E.Alonso, A.Svane, C.O.Rodriguez, N.E.Christensen - Phys.Rev. B 69, 125101 (2004); Phys.Rev. B 70, 119901 (2004).  
*Nuclear quadrupole moment determination of  $^{35}\text{Cl}$ ,  $^{79}\text{Br}$ , and  $^{127}\text{I}$ .*
- 2004Ek01 J.Ekman, D.Rudolph, C.Fahlander, A.P.Zuker et al. - Phys.Rev.Lett. 92, 132502 (2004).  
*Unusual Isospin-Breaking and Isospin-Mixing Effects in the  $A = 35$  Mirror Nuclei.*
- 2004Ge02 L.S.Geng, H.Toki, A.Ozawa, J.Meng - Nucl.Phys. A730, 80 (2004).  
*Proton and neutron skins of light nuclei within the relativistic meanfield theory.*
- 2004Gr20 S.Grevy, J.C.Angelique, P.Baumann, C.Borcea et al. - Phys.Lett. B 594, 252 (2004).  
*Beta-decay half-lives at the  $N=28$  shell closure.*
- 2004Kh16 S.Khaled, M.Ramdhane, F.Benrachi - Eur.Phys.J. A 22, 17 (2004).  
*Application of the weak-coupling model for calculating the binding energies of neutron-rich  $A \sim 32$  nuclei.*
- 2004Lu10 Yu.S.Lutostansky - Nucl.Phys. A734, E69 (2004).  
*Properties of nuclei near the neutron-drip line in  $O - Si$  region.*
- 2005Ch71 J.-G.Chen, X.-Z.Cai, T.-T.Wang, Yu.-G.Ma et al. - Chinese Physics 14, 2444 (2005).  
*Investigation on the deformation of Ne and Mg isotope chains within relativistic mean-field model.*
- 2005DeZZ F.Della Vedova, S.M.Lenzi, M.Ionescu-Bujor, N.Marginean et al. - Proc.Nuclei at the Limits, Argonne, Illinois, D.Seweryniak and T.L.Khoo, eds., p.205 (2005); AIP Conf. Proc 764 (2005).  
*Isospin Symmetry Along The  $N=Z$  Line In The  $sd$  Shell.*
- 2005Ek01 J.Ekman, L.-L.Andersson, C.Fahlander, E.K.Johansson et al. - Eur.Phys.J. A 25, Supplement 1, 363 (2005).  
*News on mirror nuclei in the  $sd$  and  $fp$  shells.*
- 2005Ti11 C.Timis, J.C.Angelique, A.Buta, N.L.Achouri et al. - J.Phys.(London) G31, S1965 (2005).  
*Spectroscopy around  $N = 20$  shell closure:  $\beta$ -n decay study of  $^{35}\text{Al}$ .*
- 2006AnZW J.C.Angelique, C.Timis, S.Pietri, N.L.Achouri et al. - Proc.Frontiers in Nuclear Structure, Astrophysics, and Reactions, Isle of Kos, Greece, 12-17 Sept. 2005, S.V.Harissopulos, P.Demetriou, R.Julin, Eds., p. 134 (2006); AIP Conf.Proc. 831 (2006).  
*Spectroscopy Near the  $N=20$  Shell Closure:  $\beta$ -n Decay Studies of  $^{33}\text{Mg}$  and  $^{35}\text{Al}$ .*
- 2006FuZX T.Fukui, S.Ota, S.Shimoura, N.Aoi et al. - CNS-REP-69, p.19 (2006).  
*High Resolution In-beam Gamma Spectroscopy of  $N \sim 20$  Neutron-rich Nuclei.*
- 2006Ia05 V.E.Iacob, J.C.Hardy, J.F.Brinkley, C.A.Gagliardi et al. - Phys.Rev. C 74, 055502 (2006).  
*Precise half-life measurements for the superallowed  $\beta^+$  emitters  $^{34}\text{Ar}$  and  $^{34}\text{Cl}$ .*
- 2006Kh08 A.Khouaja, A.C.C.Villari, M.Benjelloun, D.Hirata et al. - Nucl.Phys. A780, 1 (2006).  
*Reaction cross-section and reduced strong absorption radius measurements of neutron-rich nuclei in the vicinity of closed shells  $N = 20$  and  $N = 28$ .*
- 2006Me04 T.J.Mertzimekis, P.F.Mantica, A.D.Davies, S.N.Liddick, B.E.Tomlin - Phys.Rev. C 73, 024318 (2006).  
*Ground state magnetic dipole moment of  $^{35}\text{K}$ .*
- 2006Me08 D.F.Measday, T.J.Stocki - Phys.Rev. C 73, 045501 (2006).  
 *$\gamma$  rays from muon capture in natural Ca, Fe, and Ni.*
- 2006Ro34 B.T.Roeder, K.W.Kemper, N.Aoi, D.Bazin et al. - Phys.Rev. C 74, 034602 (2006).  
*Production cross sections for heavy-ion fragmentation reactions on a liquid deuterium target at intermediate energies.*
- 2006Zh19 Q.Zhi, Z.Ren - Phys.Lett. B 638, 166 (2006).  
*Systematic calculations on the ground state properties of Mg isotopes by the macroscopic-microscopic model.*
- 2007Ch82 G.Chaudhuri, S.Das Gupta, W.G.Lynch, M.Mocko, M.B.Tsang - Phys.Rev. C 76, 067601 (2007).  
*Cross sections of neutron-rich nuclei from projectile fragmentation: Canonical thermodynamic model estimates.*
- 2007ChZX H.D.Choi, R.B.Firestone, R.M.Lindstrom, G.L.Molnar et al. - IAEA-STI/PUB/1263 (2007).  
*Database of Prompt Gamma Rays from Slow Neutron Capture for Elemental Analysis.*
- 2007De14 F.Della Vedova, S.M.Lenzi, M.Ionescu-Bujor, N.Marginean et al. - Phys.Rev. C 75, 034317 (2007).  
*Isospin symmetry breaking at high spin in the mirror nuclei  $^{35}\text{Ar}$  and  $^{35}\text{Cl}$ .*
- 2007Do17 C.Dossat, N.Adimi, F.Aksouh, F.Becker et al. - Nucl.Phys. A792, 18 (2007).  
*The decay of proton-rich nuclei in the mass  $A = 36$ -56 region.*
- 2007GeZX M.Gelin - Thesis, Universite de Caen (2007).  
 *$\gamma$ -Spectroscopy of Neutron-Rich Nuclei around  $N=20$ .*
- 2007Ha53 I.Hamamoto - Phys.Rev. C 76, 054319 (2007).

## REFERENCES FOR A=35(CONTINUED)

- 2007Ju03 Nilsson diagrams for light neutron-rich nuclei with weakly-bound neutrons.  
B.Jurado, H.Savjols, W.Mittig, N.A.Orr et al. - Phys.Lett. B 649, 43 (2007).  
Mass measurements of neutron-rich nuclei near the  $N = 20$  and 28 shellclosures.
- 2007Ks01 R.Kshetri, M.S.Sarkar, I.Ray, P.Banerjee et al. - Nucl.Phys. A781, 277 (2007).  
High spin structure of  $^{35}\text{Cl}$  and the  $sd$ - $fp$  shell gap.
- 2007Na31 P.Napolitani, K.-H.Schmidt, L.Tassan-Got, P.Armbruster et al. - Phys.Rev. C 76, 064609 (2007).  
Measurement of the complete nuclide production and kinetic energies of the system  $^{136}\text{Xe} + \text{hydrogen}$  at 1 GeV per nucleon.
- 2007Ne14 G.Neyens, P.Himpe, D.L.Balabanski, P.Morel et al. - Eur.Phys.J. Special Topics 150, 149 (2007).  
The "island of inversion" from a nuclear moments perspective and the  $g$  factor of  $^{35}\text{Si}$ .
- 2007No13 M.Notani, H.Sakurai, N.Aoi, H.Iwasaki et al. - Phys.Rev. C 76, 044605 (2007).  
Projectile fragmentation reactions and production of nuclei near the neutron drip line.
- 2007Ts09 M.B.Tsang, W.G.Lynch, W.A.Friedman, M.Mocko et al. - Phys.Rev. C 76, 041302 (2007).  
Fragmentation cross sections and binding energies of neutron-rich nuclei.
- 2007Ya08 C.Yazidjian, G.Audi, D.Beck, K.Blaum et al. - Phys.Rev. C 76, 024308 (2007).  
Evidence for a breakdown of the isobaric multiplet mass equation: A study of the  $A = 35$ ,  $T = 3/2$  isospin quartet.
- 2008A139 A.N.Aleynikov, V.A.Rabotkin, A.P.Shumeyko - Bull.Rus.Acad.Sci.Phys. 72, 1556 (2008); Izv.Akad.Nauk RAS, Ser.Fiz. 72, 1642 (2008).  
Study of the  $^{35}\text{S}$  internal bremsstrahlung spectra by the threshold detector method.
- 2008FiZZ R.B.Firestone, M.Krticka, D.P.McNabb, B.Sleasford et al. - Proc.2007 International Workshop on Compound-Nuclear Reactions and Related Topics, Yosemite Nat.Park, Ca., 22-26 OCT. 2007, J.Escher, F.S.Dietrich, T.Kawano, I.J.Thompson, Eds. p.26 (2008); AIP CONF.PROC. 1005 (2008).  
New Methods for the Determination of Total Radiative Thermal Neutron Capture Cross Sections.
- 2008Ki07 T.Kibedi, T.W.Burrows, M.B.Trzhaskovskaya, P.M.Davidson et al. - Nucl.Instrum.Methods Phys.Res. A589, 202 (2008).  
Evaluation of theoretical conversion coefficients using  $\text{BrIcc}$ .
- 2008Py02 P.Pyykko - Molecular Physics 106, 1965 (2008).  
Year-2008 nuclear quadrupole moments.
- 2008ReZZ P.L.Reeder - Priv.Com. (2008).
- 2008Wi09 M.Wiedeking, E.Rodriguez-Vieitez, P.Fallon, M.P.Carpenter et al. - Phys.Rev. C 78, 037302 (2008).  
Intruder excitations in  $^{35}\text{P}$ .
- 2008Wi11 M.T.Win, K.Hagino - Phys.Rev. C 78, 054311 (2008).  
Deformation of  $\Lambda$  hypernuclei.
- 2009Ly01 Yu.S.Lyutostansky - Bull.Rus.Acad.Sci.Phys. 73, 176 (2009); Izv.Akad.Nauk RAS, Ser.Fiz. 73, 187 (2009).  
Disturbance of the shell structure of neutron-rich nuclei in the oxygen-magnesium region.
- 2009No01 F.Nowacki, A.Poves - Phys.Rev. C 79, 014310 (2009).  
New effective interaction for  $0(h\text{-bar})\omega$  shell-model calculations in the  $sd$ - $pf$  valence space.
- 2009Yo05 K.Yoshida - Phys.Rev. C 79, 054303 (2009).  
Core polarization for the electric quadrupole moment of neutron-rich aluminum isotopes.
- 2010Le03 J.Lee, M.B.Tsang, D.Bazin, D.Coupland et al. - Phys.Rev.Lett. 104, 112701 (2010).  
Neutron-Proton Asymmetry Dependence of Spectroscopic Factors in Ar Isotopes.
- 2010Ro23 C.Rodriguez-Tajes, H.Alvarez-Pol, T.Aumann, E.Benjamim et al. - Phys.Rev. C 82, 024305 (2010); Publishers Note  
Phys.Rev. C 82, 029910(2010).  
One-neutron knockout from light neutron-rich nuclei at relativistic energies.
- 2010Wa12 Z.M.Wang, R.Chapman, X.Liang, F.Haas et al. - Phys.Rev. C 81, 054305 (2010).  
 $\gamma$ -ray spectroscopy of neutron-rich  $^{40}\text{S}$ .
- 2010Wa20 Z.M.Wang, R.Chapman, X.Liang, F.Haas et al. - Phys.Rev. C 81, 064301 (2010).  
Intruder negative-parity states of neutron-rich  $^{33}\text{Si}$ .
- 2010WaZT Z.Wang - Thesis, Univ.of West Scotland (2010).  
 $\gamma$ -ray Spectroscopy of Neutron-rich Silicon and Sulphur Nuclei using Grazing Reactions.
- 2010WiZZ M.Wiedeking - Priv.Comm. (2010).
- 2011Am01 S.Amos, J.L.Gross, M.Thoennessen - At.Data Nucl.Data Tables 97, 383 (2011).  
Discovery of the calcium, indium, tin, and platinum isotopes.
- 2011Ch48 J.Chen, J.Cameron, B.Singh - Nucl.Data Sheets 112, 2715 (2011).  
Nuclear Data Sheets  $A = 35$ .
- 2011FuZZ N.Fukuda, T.Kubo, T.Ohnishi, N.Inabe et al. - RIKEN Accelerator Progress Report 2010, p.17 (2011).  
Production cross sections of neutron-rich isotopes produced by the fragmentation of a  $^{48}\text{Ca}$  beam at 345 MeV/nucleon.
- 2011Ga15 A.Gade, D.Bazin, B.A.Brown, C.M.Campbell et al. - Phys.Rev. C 83, 044305 (2011).  
In-beam  $\gamma$ -ray spectroscopy of  $^{35}\text{Mg}$  and  $^{33}\text{Na}$ .
- 2011Ka01 R.Kanungo, A.Prochazka, W.Horiuchi, C.Nociforo et al. - Phys.Rev. C 83, 021302 (2011).  
Matter radii of  $^{32-35}\text{Mg}$ .
- 2011Ka03 K.Kaneko, Y.Sun, T.Mizusaki, M.Hasegawa - Phys.Rev. C 83, 014320 (2011).  
Shell-model study for neutron-rich  $sd$ -shell nuclei.
- 2011Ki12 M.Kimura - Int.J.Mod.Phys. E20, 893 (2011).  
Systematic study of the many-particle and many-hole states in and around the island of inversion.

## REFERENCES FOR A=35(CONTINUED)

- 2011Le01 J.Lee, M.B.Tsang, D.Bazin, D.Coupland et al. - Phys.Rev. C 83, 014606 (2011).  
*Neutron spectroscopic factors of  $^{34}\text{Ar}$  and  $^{46}\text{Ar}$  from (p,d) transfer reactions.*
- 2011Nu01 F.M.Nunes, A.Deltuva, J.Hong - Phys.Rev. C 83, 034610 (2011).  
*Improved description of  $^{34,36,46}\text{Ar}$  (p,d) transfer reactions.*
- 2011Tu09 X.L.Tu, M.Wang, Yu.A.Litvinov, Y.H.Zhang et al. - Nucl.Instrum.Methods Phys.Res. A654, 213 (2011).  
*Precision isochronous mass measurements at the storage ring CSRe in Lanzhou.*
- 2012BoZT M.Bouhelal, F.Haas, E.Caurier, F.Nowacki - Proc.Intern.Conf.on Nuclear Structure and Dynamics,12, Opatija, Croatia, 9-13 July, 2012, T.Niksic, M.Milin, D.Vretenar, S.Szilner, Eds., p.38 (2012);AIP Conf.Proc.1491 (2012).  
*0 and 1  $\hbar\omega$  shell model description of phosphorus isotopes with A= 30 to 35.*
- 2012Fo27 H.T.Fortune, R.Sherr - Phys.Rev. C 86, 064322 (2012).  
*Matter radii of  $^{29-35}\text{Mg}$ .*
- 2012Ho19 W.Horiuchi, T.Inakura, T.Nakatsukasa, Y.Suzuki - Phys.Rev. C 86, 024614 (2012).  
*Glauber-model analysis of total reaction cross sections for Ne, Mg, Si, and S isotopes with Skyrme-Hartree-Fock densities.*
- 2012IdZZ E.Ideguchi, S.Go, H.Miya, K.Kisamori et al. - CNS-REP-88, Ann.Report 2010, p.5 (2012).  
*High-Spin Level Structure in  $^{35}\text{S}$ .*
- 2012Im01 N.Imai, Y.Hirayama, Y.X.Watanabe, T.Teranishi et al. - Phys.Rev. C 85, 034313 (2012).  
*Isobaric analog resonances of the N=21 nucleus  $^{35}\text{Si}$ .*
- 2012Kw02 E.Kwan, D.J.Morrissey, D.A.Davies, M.Steiner et al. - Phys.Rev. C 86, 014612 (2012).  
*Systematic studies of light neutron-rich nuclei produced via the fragmentation of  $^{40}\text{Ar}$ .*
- 2012Ni01 N.Nica, J.Cameron, B.Singh - Nucl.Data Sheets 113, 1 (2012).  
*Nuclear Data Sheets for A = 36.*
- 2012Ni10 N.Nica, B.Singh - Nucl.Data Sheets 113, 1563 (2012).  
*Nuclear Data Sheets for A = 34.*
- 2012No05 C.Nociforo, A.Prochazka, R.Kanungo, T.Aumann et al. - Phys.Rev. C 85, 044312 (2012).  
*One-neutron removal reactions on Al isotopes around the N=20 shell closure.*
- 2012Sh21 R.Shane, R.J.Charity, L.G.Sobotka, D.Bazin et al. - Phys.Rev. C 85, 064612 (2012).  
*Proton and neutron knockout from  $^{36}\text{Ca}$ .*
- 2012Th10 M.Thoennessen - At.Data Nucl.Data Tables 98, 933 (2012).  
*Discovery of the Isotopes with  $11 \leq Z \leq 19$ .*
- 2012Zh06 X.H.Zhang, Z.Y.Sun, R.F.Chen, Z.Q.Chen et al. - Phys.Rev. C 85, 024621 (2012).  
*Projectile fragmentation reactions of  $^{40}\text{Ar}$  at 57 MeV/nucleon.*
- 2013An02 I.Angeli, K.P.Marinova - At.Data Nucl.Data Tables 99, 69 (2013).  
*Table of experimental nuclear ground state charge radii: An update.*
- 2013Bi10 A.Bisoi, M.S.Sarkar, S.Sarkar, S.Ray et al. - Phys.Rev. C 88, 034303 (2013).  
*Superdeformation and  $\alpha$ -cluster structure in  $^{35}\text{Cl}$ .*
- 2013Ch31 R.Chatterjee, R.Shyam, K.Tsushima, A.W.Thomas - Nucl.Phys. A913, 116 (2013).  
*Structure and Coulomb dissociation of  $^{23}\text{O}$  within the quark-meson coupling model.*
- 2013GoZW S.Go, E.Ideguchi, R.Yokoyama, M.Kobayashi et al. - JAEA-Review 2013-057, p.13 (2013).  
*Study of high spin states in  $^{35}\text{S}$ .*
- 2013Ja17 M.Jaszunski, M.Repisky, T.B.Demissie, S.Komorovsky et al. - J.Chem.Phys. 139, 234202 (2013).  
*Spin-rotation and NMR shielding constants in HCl.*
- 2013Li39 H.Li, Z.Ren - J.Phys.(London) G40, 105110 (2013).  
*Shell model calculations for the  $\beta^-$ -decays of Z = 9-13 nuclei.*
- 2013Ni09 Z.M.Niu, Y.F.Niu, Q.Liu, H.Z.Liang, J.Y.Guo - Phys.Rev. C 87, 051303 (2013).  
*Nuclear  $\beta^+$ /EC decays in covariant density functional theory and the impact of isoscalar proton-neutron pairing.*
- 2013Sh05 M.K.Sharma, M.S.Mehta, S.K.Patra - Int.J.Mod.Phys. E22, 1350005 (2013).  
*Nuclear reaction cross-section for drip-line nuclei in the framework of Glauber model using relativistic and nonrelativistic densities.*
- 2013StZY K.Steiger - Fakultat fur Physik der Technischen Universitat Munchen (2013).  
*Decay spectroscopy of neutron-rich nuclei around  $^{37}\text{Al}$ .*
- 2014Ay01 S.Aydin, M.Ionescu-Bujor, F.Recchia, S.M.Lenzi et al. - Phys.Rev. C 89, 014310 (2014).  
*High-spin level structure of  $^{35}\text{S}$ .*
- 2014Bu01 G.Burgunder, O.Sorlin, F.Nowacki, S.Giron et al. - Phys.Rev.Lett. 112, 042502 (2014).  
*Experimental Study of the Two-Body Spin-Orbit Force in Nuclei.*
- 2014Ca21 E.Caurier, F.Nowacki, A.Poves - Phys.Rev. C 90, 014302 (2014).  
*Merging of the islands of inversion at N = 20 and N = 28.*
- 2014ChZZ S.Chakraborty, s306 Collaboration - Int.Nuclear Physics Conf. 2013,(IUPAP),Firenze,Italy, June 2-7,2013, S.Lunardi, P.G.Bizzeti, W.S.Kabana, C.Bucci,et al.Eds.p.02019 (2014); EPJ web of Conf.v.66, (2014).  
*Ground-state configuration of neutron-rich Aluminum isotopes through Coulomb Breakup.*
- 2014Do05 P.Doornenbal, H.Scheit, S.Takeuchi, Y.Utsuno et al. - Prog.Theor.Exp.Phys. 2014, 053D01 (2014).  
*Rotational level structure of sodium isotopes inside the "island of inversion".*
- 2014Ga13 M.K.Gaidarov, P.Sarriguren, A.N.Antonov, E.Moya de Guerra - Phys.Rev. C 89, 064301 (2014).

## REFERENCES FOR A=35(CONTINUED)

- 2014GoZZ *Ground-state properties and symmetry energy of neutron-rich and neutron-deficient Mg isotopes.*  
S.Go, E.Ideguchi, R.Yokoyama, M.Kobayashi et al. - CNS-REP-92, p.5 (2014).  
*Study of high spin states in  $^{35}\text{S}$ .*
- 2014Pr09 B.Pritychenko, E.Betak, B.Singh, J.Totans - Nucl.Data Sheets 120, 291 (2014).  
*Nuclear Science References Database.*
- 2014Si21 A.Singh, A.S.Dhaliwal - Appl.Radiat.Isot. 94, 44 (2014).  
*Studies of internal bremsstrahlung spectrum of  $^{35}\text{S}$  beta emitter in the photon energy region of 1-100 keV.*
- 2014St18 S.R.Stroberg, A.Gade, J.A.Tostevin, V.M.Bader et al. - Phys.Rev. C 90, 034301 (2014).  
*Single-particle structure of silicon isotopes approaching  $^{42}\text{Si}$ .*
- 2014To14 J.A.Tostevin, A.Gade - Phys.Rev. C 90, 057602 (2014).  
*Systematics of intermediate-energy single-nucleon removal cross sections.*
- 2014Wa14 S.Watanabe, K.Minomo, M.Shimada, S.Tagami et al. - Phys.Rev. C 89, 044610 (2014).  
*Ground-state properties of neutron-rich Mg isotopes.*
- 2015Ch56 R.Chapman, A.Hodsdon, M.Bouhelal, F.Haas et al. - Phys.Rev. C 92, 044308 (2015).  
*Spectroscopy of neutron-rich  $^{34,35,36,37,38}\text{P}$  populated in binary grazing reactions.*
- 2015Fr01 C.Fry, C.Wrede, S.Bishop, B.A.Brown et al. - Phys.Rev. C 91, 015803 (2015).  
*Discovery of  $^{34g.m}\text{Cl}(p,\gamma)^{35}\text{Ar}$  resonances activated at classical nova temperatures.*
- 2015GoZY S.Go, E.Ideguchi, R.Yokoyama, M.Kobayashi et al. - CNS-REP-93, Ann.Report 2013, p.21 (2015).  
*Superdeformation in  $^{35}\text{S}$ .*
- 2015Mo17 S.Momota, I.Tanihata, A.Ozawa, M.Notani et al. - Phys.Rev. C 92, 024608 (2015).  
*Velocity-dependent transverse momentum distribution of fragments produced from  $^{40}\text{Ar}+^9\text{Be}$  at 95 MeV/nucleon.*
- 2015MuZY A.Mutschler - Thesis, Univ. Paris-Sud (2015).  
*Le noyau-bulle de  $^{34}\text{Si}$ : Un outil experimental pour etudier l'interaction spin-orbite.*
- 2015Sh21 M.K.Sharma, R.N.Panda, M.K.Sharma, S.K.Patra - Chin.Phys.C 39, 064102 (2015).  
*Nuclear structure study of some bubble nuclei in the light mass region using mean field formalism.*
- 2015St06 S.R.Stroberg, A.Gade, J.A.Tostevin, V.M.Bader et al. - Phys.Rev. C 91, 041302 (2015).  
*Neutron single-particle strength in silicon isotopes: Constraining the driving forces of shell evolution.*
- 2015Su01 L.-J.Sun, C.-J.Lin, X.-X.Xu, J.-S.Wang et al. - Chin.Phys.Lett. 32, 012301 (2015).  
*Experimental Study of Beta-Delayed Proton Emission of  $^{36,37}\text{Ca}$ .*
- 2016Ba59 H.-B.Bai, Z.-H.Zhang, X.-W.Li - Chin.Phys.C 40, 114101 (2016).  
*Investigation of the Mg isotopes using the shell-model-like approach in relativistic mean field theory.*
- 2016Li45 C.Lin, X.Xu, J.Wang, L.Sun et al. - Yuan.Wul.Ping. 33, 160 (2016); Nucl.Phys.Rev. 33, 160 (2016).  
*Proton and Two-proton Emissions from Proton-rich Nuclei with  $10 \leq Z \leq 20$ .*
- 2016Me17 T.J.Mertzimekis - Phys.Rev. C 94, 064313 (2016).  
*Prediction and evaluation of magnetic moments in  $T = 1/2, 3/2$ , and  $5/2$  mirror nuclei.*
- 2016Mu03 A.Mutschler, O.Sorlin, A.Lemasson, D.Bazin et al. - Phys.Rev. C 93, 034333 (2016).  
*Spectroscopy of  $^{35}\text{P}$  using the one-proton knockout reaction.*
- 2016Sa46 F.Sammarruca, Y.Nosyk - Phys.Rev. C 94, 044311 (2016).  
*Impact of the neutron matter equation of state on neutron skins and neutron drip lines in chiral effective field theory.*
- 2016Sh05 M.K.Sharma, R.N.Panda, M.K.Sharma, S.K.Patra - Phys.Rev. C 93, 014322 (2016).  
*Search for halo structure in  $^{37}\text{Mg}$  using the Glauber model and microscopic relativistic mean-field densities.*
- 2016Si02 J.Simonis, K.Hebel, J.D.Holt, J.Menendez, A.Schwenk - Phys.Rev. C 93, 011302 (2016).  
*Exploring sd-shell nuclei from two- and three-nucleon interactions with realistic saturation properties.*
- 2017Ch22 K.A.Chipps, S.D.Pain, R.L.Kozub, D.W.Bardayan et al. - Phys.Rev. C 95, 045808 (2017).  
*First spin-parity constraint of the 306 keV resonance in  $^{35}\text{Cl}$  for nova nucleosynthesis.*
- 2017Ch36 S.Chakraborty, U.Datta, T.Aumann, S.Beceiro Novo et al. - Phys.Rev. C 96, 034301 (2017).  
*Ground-state configuration of neutron-rich  $^{35}\text{Al}$  via Coulomb breakup.*
- 2017Ga20 A.T.Gallant, M.Alanssari, J.C.Bale, C.Andreoiu et al. - Phys.Rev. C 96, 024325 (2017).  
*Mass determination near  $N=20$  for Al and Na isotopes.*
- 2017Gi09 S.A.Gillespie, A.Parikh, C.J.Barton, T.Faestermann et al. - Phys.Rev. C 96, 025801 (2017).  
*First measurement of the  $^{34}\text{S}(p,\gamma)^{35}\text{Cl}$  reaction rate through indirect methods for presolar nova grains.*
- 2017Ha23 R.Han, X.Q.Li, W.G.Jiang, Z.H.Li et al. - Phys.Lett. B 772, 529 (2017).  
*Northern boundary of the "island of inversion" and triaxiality in  $^{34}\text{Si}$ .*
- 2017Mo26 S.Momiyama, P.Doornenbal, H.Scheit, S.Takeuchi et al. - Phys.Rev. C 96, 034328 (2017).  
*In-beam  $\gamma$ -ray spectroscopy of  $^{35}\text{Mg}$  via knockout reactions at intermediate energies.*
- 2018MuZY S.Mughabghab - Atlas of Neutron Resonances, 6th ed. Vol.1, Resonance Properties and Thermal Cross Sections Z = 1-60, Elsevier, Amsterdam (2018).
- 2018Py01 P.Pyykko - Molecular Physics 116, 1328 (2018).  
*Year-2017 nuclear quadrupole moments.*
- 2018Sa54 A.Saastamoinen, G.J.Lotay, A.Kankainen, B.T.Roeder et al. - J.Phys.:Conf.Ser. 940, 012004 (2018).  
*Study of excited states of  $^{35}\text{Ar}$  through  $\beta$ -decay of  $^{35}\text{K}$  for nucleosynthesis in novae and X-ray bursts.*
- 2018Yo06 S.Yoshida, Y.Utsuno, N.Shimizu, T.Otsuka - Phys.Rev. C 97, 054321 (2018); Erratum Phys.Rev. C 109, 029904 (2024).



## REFERENCES FOR A=35(CONTINUED)

- 2019Ah07 *Systematic shell-model study of  $\beta$ -decay properties and Gamow-Teller strength distributions in A = 40 neutron-rich nuclei.*  
D.S.Ahn, N.Fukuda, H.Geissel, N.Inabe et al. - Phys.Rev.Lett. 123, 212501 (2019).
- 2019As04 *Location of the Neutron Dripline at Fluorine and Neon.*  
P.Ascher, N.Althubiti, D.Atanasov, K.Blaum et al. - Phys.Rev. C 100, 014304 (2019).
- 2019Ba16 *Mass measurements of neutron-rich isotopes near N=20 by in-trap decay with the ISOLTRAP spectrometer.*  
J.C.Batchelder, S.-A.Chong, J.Morrell, M.A.Unzueta et al. - Phys.Rev. C 99, 044612 (2019).
- 2019ChZU *Possible evidence of nonstatistical properties in the  $^{35}\text{Cl}(n, p)^{35}\text{S}$  cross section.*  
R.Chyzyh - Thesis, Texas A and M University (2019).
- 2019Gr08 *Measurement of  $\beta$ -delayed Protons from  $^{35}\text{K}$  Relevant to the  $^{34}\text{Cl}^{\text{g.m.}}(p, \gamma)^{35}\text{Ar}$  Reaction.*  
L.Grocutt, R.Chapman, M.Bouhelal, F.Haas et al. - Phys.Rev. C 100, 064308 (2019).
- 2019Ka53 *Lifetime measurements of N = 20 phosphorus isotopes using the AGATA  $\gamma$ -ray tracking spectrometer.*  
A.S.Kachan, I.V.Kurhuz, V.M.Mischenko, S.N.Utenkov - Bull.Rus.Acad.Sci.Phys. 83, 1545 (2019).
- 2019Mo01 *Structure and Total Strength of the Magnetic Dipole Resonance on Excited States in sd-Shell Nuclei.*  
P.Moller, M.R.Mumpower, T.Kawano, W.D.Myers - At.Data Nucl.Data Tables 125, 1 (2019).
- 2019Se09 *Nuclear properties for astrophysical and radioactive-ion-beam applications (II).*  
K.Setoodehnia, J.H.Kelley, C.Marshall, F.Portillo Chaves, R.Longland - Phys.Rev. C 99, 055812 (2019).
- 2019StZV *Experimental study of  $^{35}\text{Cl}$  excited states via  $^{32}\text{S}(\alpha, p)$ .*  
N.J.Stone - INDC(NDS)-0794 (2019).
- 2020Ch21 *Table of Recommended Nuclear Magnetic Dipole Moments: Part I – Long-lived States.*  
Q.Z.Chai, J.C.Pei, N.Fei, D.W.Guan - Phys.Rev. C 102, 014312 (2020).
- 2020Ku17 *Constraints on the neutron drip line with the newly observed  $^{39}\text{Na}$ .*  
S.A.Kuvin, H.Y.Lee, T.Kawano, B.DiGiovine et al. - Phys.Rev. C 102, 024623 (2020).
- 2020Ma25 *Nonstatistical fluctuations in the  $^{35}\text{Cl}(n, p)^{35}\text{S}$  reaction crosssection at fast-neutron energies from 0.6 to 6 MeV.*  
A.Magilligan, B.A.Brown - Phys.Rev. C 101, 064312 (2020).
- 2020Mi15 *New isospin-breaking “USD” Hamiltonians for the sd shell.*  
T.Miyagi, S.R.Stroberg, J.D.Holt, N.Shimizu - Phys.Rev. C 102, 034320 (2020).
- 2020Ra14 *Ab initio multishell valence-space Hamiltonians and the island of inversion.*  
R.Rahaman, A.Bisoi, Y.Sapkota, A.Adhikari et al. - Phys.Rev. C 102, 024315 (2020).
- 2020Ri06 *Spectroscopic study of  $^{38}\text{K}$  above the 31.67  $\mu\text{s}$  isomer.*  
W.A.Richter, B.A.Brown, R.Longland, C.Wrede et al. - Phys.Rev. C 102, 025801 (2020).
- 2020RiZX *Shell-model studies of the astrophysical rp-process reactions  $^{34}\text{S}(p, \gamma)^{35}\text{Cl}$  and  $^{34}\text{g.m.}\text{Cl}(p, \gamma)^{35}\text{Ar}$ .*  
W.A.Richter, B.A.Brown, R.Longland, C.Wrede et al. - 27th Int.Nuclear Physics Conference (INPC2019) 29 July – 2 August 2019, Glasgow, UK, p.012064 (2020), J. Phys.:Conf.Ser.1643 (2020).
- 2020RiZZ *Shell-model studies of the astrophysical rp-process reactions  $^{34}\text{S}(p, \gamma)^{35}\text{Cl}$  and  $^{34}\text{g.m.}\text{Cl}(p, \gamma)^{35}\text{Ar}$ .*  
G.Rimpault, G.Noguere, C.de Saint Jean - Proc.Intern.Conf.Nuclear Data for Science and Technology (ND2019), Beijing, China, May 19-24, 2019, Z.Ge, et al. Eds., p.13002 (2020); EPJ Web of Conf.Vol.239 (2020).
- 2020Sa44 *Trends on major actinides from an integral data assimilation.*  
M.Salathe, H.L.Crawford, A.O.Macchiavelli, B.P.Kay et al. - Phys.Rev. C 102, 064317 (2020).
- 2020StZV *Search for the  $1/2^+$  intruder state in  $^{35}\text{P}$ .*  
N.Stone - INDC(NDS)-0816 (2020).
- 2020Ts03 *Table of Recommended Nuclear Magnetic Dipole Moments: Part II, Short-Lived States.*  
N.Tsunoda, T.Otsuka, K.Takayanagi, N.Shimizu et al. - Nature(London) 587, 66 (2020).
- 2021Bh12 *The impact of nuclear shape on the emergence of the neutron dripline.*  
A.Bhattacharyya, U.Datta, A.Rahaman, S.Chakraborty et al. - Phys.Rev. C 104, 045801 (2021).
- 2021Go09 *Neutron capture cross sections of light neutron-rich nuclei relevant for r-process nucleosynthesis.*  
S.Go, E.Ideguchi, R.Yokoyama, N.Aoi et al. - Phys.Rev. C 103, 034327 (2021).
- 2021JoZZ *High-spin states in  $^{35}\text{S}$ .*  
S.Jongile - Thesis, University of Stellenbosch (2021).
- 2021Ka07 *The Structure of  $^{33}\text{Si}$ ,  $^{35}\text{S}$  and the magicity of the N = 20 gap at Z = 14, 16.*  
H.Kasuya, K.Yoshida - Prog.Theor.Exp.Phys. 2021, 013D01 (2021).
- 2021La04 *Hartree-Fock-Bogoliubov theory for odd-mass nuclei with a time-odd constraint and application to deformed halo nuclei.*  
L.Lalanne, O.Sorlin, M.Assie, F.Hammache et al. - Phys.Rev. C 103, 055809 (2021).
- 2021Lo09 *Evaluation of the  $^{35}\text{K}(p, \gamma)^{36}\text{Ca}$  reaction rate using the  $^{37}\text{Ca}(p, d)^{36}\text{Ca}$  transfer reaction.*  
M.Lovely, A.Lennarz, D.Connolly, M.Williams et al. - Phys.Rev. C 103, 055801 (2021).
- 2021Mi17 *Proton capture on  $^{34}\text{S}$  in the astrophysical energy regime of O-Ne novae.*  
F.Minato, T.Marketin, N.Paar - Phys.Rev. C 104, 044321 (2021).
- 2021StZZ  *$\beta$ -delayed neutron-emission and fission calculations within relativistic quasiparticle random-phase approximation and a statistical model.*  
N.J.Stone - INDC(NDS)-0833 (2021).
- 2021Su04 *Table of Nuclear Electric Quadrupole Moments.*  
J.Surbrook, G.Bollen, M.Brodeur, A.Hamaker et al. - Phys.Rev. C 103, 014323 (2021).
- First Penning trap mass measurement of  $^{36}\text{Ca}$ .*

## REFERENCES FOR A=35(CONTINUED)

- 2021Wa16 M.Wang, W.J.Huang, F.G.Kondev, G.Audi, S.Naimi - Chin.Phys.C 45, 030003 (2021).  
*The AME 2020 atomic mass evaluation (II). Tables, graphs and references.*
- 2021WaZY J.N.Warren - Thesis, Ohio University (2021).  
 *$^{35}\text{Cl}(n,p)$  reactions in a  $^6\text{Li}$  enhanced CLYC detector.*
- 2022Ah02 D.S.Ahn, J.Amano, H.Baba, N.Fukuda et al. - Phys.Rev.Lett. 129, 212502 (2022).  
*Discovery of  $^{39}\text{Na}$ .*
- 2022Cr03 H.L.Crawford, V.Tripathi, J.M.Allmond, B.P.Crider et al. - Phys.Rev.Lett. 129, 212501 (2022).  
*Crossing  $N = 28$  Toward the Neutron Drip Line: First Measurement of Half-Lives at FRIB.*
- 2022Gr07 L.Grocutt, R.Chapman, M.Bouhelal, F.Haas et al. - Phys.Rev. C 106, 024314 (2022).  
*Lifetime measurements of states of  $^{35}\text{S}$ ,  $^{36}\text{S}$ ,  $^{37}\text{S}$ , and  $^{38}\text{S}$  using the AGATA  $\gamma$ -ray tracking spectrometer.*
- 2022Ot01 T.Otsuka, N.Shimizu, Y.Tsunoda - Phys.Rev. C 105, 014319 (2022).  
*Moments and radii of exotic Na and Mg isotopes.*
- 2022Zo01 Y.Y.Zong, C.Ma, M.Q.Lin, Y.M.Zhao - Phys.Rev. C 105, 034321 (2022).  
*Mass relations of mirror nuclei for both bound and unbound systems.*
- 2023Ba27 R.Barman, S.Sarkar, R.Chatterjee - Phys.Rev. C 108, 025812 (2023).  
*Structure effects of exotic nuclei in the  $A = 30$  mass region relevant for determining abundance patterns in explosive nucleosynthesis.*
- 2023Bo17 I.N.Borzov, S.V.Tolokonnikov - Physics of Part.and Nuclei 54, 586 (2023).  
*Self-Consistent Calculation of Nuclear Charge Radii in K Isotopes.*
- 2023De34 S.Dede, S.D.Essenmacher, P.Gastis, K.V.Manukyan et al. - Nucl.Instrum.Methods Phys.Res. A1055, 168472 (2023).  
*Electrospraying deposition and characterization of potassium chlorid targets for nuclear science measurements.*
- 2023Fo05 J.M.R.Fox, C.W.Johnson, R.N.Perez - Phys.Rev. C 108, 054310 (2023).  
*Uncertainty quantification of transition operators in the empirical shell model.*
- 2023He15 C.Hebborn, F.M.Nunes, A.E.Lovell - Phys.Rev.Lett. 131, 212503 (2023).  
*New Perspectives on Spectroscopic Factor Quenching from Reactions.*
- 2023KoZS J.Kopecky - INDC(NDS)0886 (2023).  
*Thermal Neutron Capture in the Low Mass Region.*
- 2023La09 L.Lalanne, O.Sorlin, A.Poves, M.Assie et al. - Phys.Rev.Lett. 131, 092501 (2023).  
 *$N = 16$  Magicity Revealed at the Proton Drip Line through the Study of  $^{35}\text{Ca}$ .*
- 2023Lu07 R.S.Lubna, S.N.Liddick, T.H.Ogunbeku, A.Chester et al. - Phys.Rev. C 108, 014329 (2023).  
 *$\beta$  decay of  $^{36}\text{Mg}$  and  $^{36}\text{Al}$ : Identification of a  $\beta$ -decaying isomer in  $^{36}\text{Al}$ .*
- 2023Ra22 A.Ravlic, E.Yuksel, T.Niksic, N.Paar - Phys.Rev. C 108, 054305 (2023).  
*Influence of the symmetry energy on the nuclear binding energies and the neutron drip line position.*
- 2023Zh10 M.Zhang, X.Zhou, M.Wang, Y.H.Zhang et al. - Eur.Phys.J. A 59, 27 (2023).  
*B $\rho$ -defined isochronous mass spectrometry and mass measurements of  $^{58}\text{Ni}$  fragments.*
- 2023Zh15 K.Y.Zhang, P.Papakonstantinou, M.-H.Mun, Y.Kim et al. - Phys.Rev. C 107, L041303 (2023).  
*Collapse of the  $N=28$  shell closure in the newly discovered  $^{39}\text{Na}$  nucleus and the development of deformed halos towards the neutron dripline.*
- 2024Co04 A.L.Conley, B.Kelly, M.Spieker, R.Aggarwal et al. - Nucl.Instrum.Methods Phys.Res. A1058, 168827 (2024).  
*The CeBrA demonstrator for particle- $\gamma$  coincidence experiments at the FSU Super-Enge Split-Pole Spectrograph.*
- 2024Dr01 N.Dronchi, R.J.Charity, L.G.Sobotka, B.A.Brown et al. - Phys.Rev. C 110, L031302 (2024).  
*Evolution of shell gaps in the neutron-poor calcium region from invariant-mass spectroscopy of  $^{37,38}\text{Sc}$ ,  $^{35}\text{Ca}$ , and  $^{34}\text{K}$ .*
- 2024Ha18 K.Hanselman, S.A.Kuvin, H.Y.Lee, T.Kawano et al. - Phys.Rev. C 110, 024609 (2024).  
*Improved modeling of neutron-induced reactions on chlorine isotopes aided through new  $(n,p)$  and  $(n,\alpha)$  measurements at LANSCE.*
- 2024Ku24 A.N.Kuchera, C.R.Hoffman, G.Ryan, I.B.D'Amato et al. - Eur.Phys.J. A 60, 176 (2024).  
*Single-neutron adding on  $^{34}\text{S}$ .*
- 2024Lu06 Y.-X.Luo, Q.Liu, J.-Y.Guo - Chin.Phys.C 48, 044103 (2024).  
*Exploration of the ground state properties of neutron-rich sodium isotopes using the deformed relativistic mean field theory in complex momentum representations with BCS pairings.*
- 2024Yu13 Y.Yu, Y.M.Xing, Y.H.Zhang, M.Wang et al. - Phys.Rev.Lett. 133, 222501 (2024).  
*Nuclear Structure of Dripline Nuclei Elucidated through Precision Mass Measurements of  $^{23}\text{Si}$ ,  $^{26}\text{P}$ ,  $^{27,28}\text{S}$ , and  $^{31}\text{Ar}$ .*
- 2025Ku15 T.Kubo - Nucl.Phys. A1062, 123154 (2025).  
*Neutron dripline search for fluorine, neon and sodium and the discovery of  $^{39}\text{Na}$  at RIKEN RIBF.*
- 2025Ly01 E.M.Lykiardopoulou, C.Walls, J.Bergmann, M.Brodeur et al. - Phys.Rev.Lett. 134, 052503 (2025).  
*Refined Topology of the  $N=20$  Island of Inversion with High Precision Mass Measurements of  $^{31-33}\text{Na}$  and  $^{31-35}\text{Mg}$ .*
- 2025Sh19 S.Shukla, P.C.Srivastava, Ch.Sarma - Nucl.Phys. A1061, 123139 (2025).  
*Shell model study of isobaric analog states for  $T_z \neq 3/2$  nuclei using isospin-breaking interactions.*
- 2026Jo01 S.Jongile, M.Wiedeking, O.Sorlin, R.Neveling et al. - Phys.Rev. C 113, 024323 (2026).  
*Probing the  $j$  dependence of angular distributions and  $N=20$  shell rigidity via the  $^{36}\text{S}(p,d)^{35}\text{S}$  reaction.*